FOUR-DIMENSIONAL SU(2) YANG–MILLS THEORY AN OUTLINE OF THE PROOF FOR THE $\mathrm{tr}(F^2)$ OBSERVABLE

Abstract. This manuscript outlines the proof of the continuum construction of the renormalized $\mathrm{tr}(F^2)$ observable in four-dimensional Wilson SU(2) theory. The argument connects multiscale estimates, Euclidean correlation limits, and scalar reconstruction.

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Here $\mathrm{tr}(F^2)$ denotes the renormalized scalar curvature-squared observable, with the Wilson normalization in Section 1. At each lattice cutoff, we use $\mathcal{O}(f) - \mathbb{E}\mathcal{O}(f)$: the expectation of the smeared observable is subtracted before taking the continuum limit.

Theorem 0. For sufficiently small reference coupling, the correlations of $\mathrm{tr}(F^2)$ in four-dimensional Wilson SU(2) theory have nontrivial volume-first subsequential continuum limits. The full moment functionals $G_{\xi,n}$ of every canonical component, normalized by $G_{\xi,0} = 1$, satisfy the following Osterwalder–Schrader axioms.

Let $\mathcal{S}_{\neq,n}$ be the Schwartz functions on $(\mathbb{R}^4)^n$ whose every derivative vanishes whenever two arguments coincide. Let $\mathcal{S}_{+,n}$ be the subspace also flat outside the region where all time coordinates are positive. Set $\mathcal{S}_{\neq,0} = \mathcal{S}_{+,0} = \mathbb{C}$, and write

$$|\varphi|_r = \max_{|\alpha|\le r} \sup_x (1 + |x|^2)^{r/2} |\partial^\alpha \varphi(x)|.$$

(i) **Regularity and growth.** Each $G_{\xi,n}$ is continuous on $\mathcal{S}_{\neq,n}$. There are constants $A, B > 0$, $p \ge 0$ and an integer $k \ge 1$, independent of n and ξ, such that

$$|G_{\xi,n}(\varphi)| \le A B^n (n!)^p |\varphi|_{kn}, \qquad n \ge 1.$$

(ii) **Euclidean invariance.** The functionals are invariant under simultaneous translations of all arguments and every rotation or reflection in O(4).

(iii) **Reflection positivity.** Let $\theta(x_1, x_2, x_3, x_4) = (x_1, x_2, x_3, -x_4)$ and put

$$(\Theta\overline{\varphi})(x_1, \dots, x_n) = \overline{\varphi(\theta x_1, \dots, \theta x_n)}.$$

For every finite sequence of tests $\varphi_n \in \mathcal{S}_{+,n}$, $0 \le n \le N$,

$$\sum_{m,n=0}^{N} G_{\xi,m+n}(\Theta\overline{\varphi}_m \otimes \varphi_n) \ge 0.$$

(iv) **Symmetry and reality.** The functionals are invariant under permutations of the arguments, and

$$G_{\xi,n}(\overline{\varphi}) = \overline{G_{\xi,n}(\varphi)}.$$

(v) **Clustering.** Two groups of observables factorize when their spatial separation tends to infinity. Explicitly, for $\varphi \in \mathcal{S}_{+,m}$ and $\psi \in \mathcal{S}_{+,n}$,

$$G_{\xi,m+n}(\Theta\overline{\varphi} \otimes \tau_{(v,0)}\psi) \longrightarrow \overline{G_{\xi,m}(\varphi)}\, G_{\xi,n}(\psi) \qquad (|v| \longrightarrow \infty),$$

where $v \in \mathbb{R}^3$ and $\tau_{(v,0)}$ translates all arguments of ψ by the same spatial vector.

The precise construction and component statements are Theorems 0 and 0' in Section 15; the test-space conventions are given in Section 54.

Here volume-first means taking the infinite-volume limit at each fixed lattice spacing before sending the spacing to zero along the stated admissible weak-coupling sequence. A canonical component is one phase of the limiting translation-invariant state. It retains the reference subtraction used at the cutoffs, so its one-point function need not vanish. The conclusion concerns the scalar sector generated by $\mathrm{tr}(F^2)$.

Proof sketch. The multiscale construction supplies source bounds uniform in cutoff and volume. Compactness and quantitative extension give the Euclidean correlations and their factorial growth on diagonal-flat Schwartz tests. The mirror, angular and phase arguments establish reflection positivity, Euclidean invariance and clustering in each component; symmetry and reality pass from the lattice correlations.

Plan of the proof

This manuscript outlines the proof of the continuum construction of the renormalized $\mathrm{tr}(F^2)$ observable in four-dimensional Wilson SU(2) theory. Its lattice mean is subtracted before taking the continuum limit. The resulting theory is the scalar sector generated by this observable. Theorems 0 and 0' give the full statements.

The outline presents the main constructions, estimates, and connections between the steps. It develops the central mechanisms in detail and gives shorter arguments for supporting results; full derivations are reserved for the fuller proof. Definitions and detailed technical statements appear at the points where they enter the argument.

The multiscale construction. Block averaging provides the coordinates for the proof. Local pivot charts straighten the average near regular configurations and give the coarea and determinant formulas. The classical flat problem supplies the constrained gap, curl-energy comparison, average lift, and exponential kernels used in the many-level background construction.

The regular integration combines local Gaussian comparison with a polymer expansion. The pointed Hessian and nonlinear window estimates control the extracted coefficients and the remainder. The large-field stage sums histories while retaining the Wilson weight and the reserve needed for source estimates. Together these steps lead to ordinary source stability and the partition floor on the localization geometry of the construction.

Sources, limits, and short distance. Mixed source differentiation selects connected contributions meeting every source support. Cauchy estimates and geometric counting give bounds uniform in cutoff and volume. Compactness then extracts continuum correlation forms on separated compact tests. The quantitative extension passes to Schwartz tests flat on all coincidence diagonals.

For the volume-first construction, the infinite-volume limit is taken at each fixed lattice spacing through compatible periodic tori. The spacing then tends to zero along the stated admissible weak-coupling sequence. The two-point extraction and running normalization give the controlled short-distance law, whose lower bound establishes nontriviality.

Euclidean structure and reconstruction. Lattice mirror inequalities give reflection positivity. Uniform insertion estimates, plate densities, angular kernels, and ultraviolet harmonic elimination yield Euclidean covariance. The phase decomposition identifies the canonical components and their clustering properties.

The growth estimates, covariance, positivity, and clustering enter the analytic-continuation step on the diagonal-flat test domain. Reconstruction gives, for each canonical component, a local Hermitian scalar Wightman theory on a separable Hilbert space, with Poincaré covariance, forward-cone spectrum, and a unique vacuum. Its cyclic sector is generated by the renormalized $\mathrm{tr}(F^2)$ observable.

Reading the outline. Codes such as "4.8", "6c.2", and "8a.1.11" track arguments across sections. The argument index locates their statements and uses; the notation appendix fixes the multiscale objects. The dependency catalogue collects the detailed analytic statements, and Section 54 gives the reconstruction step.

1. Finite-cutoff foundations

This section establishes the finite-dimensional identities and compactness arguments used throughout the construction. None of its statements assumes a continuum Yang–Mills measure. A constant depending on the number of plaquettes is not a bound uniform in the cutoff. We retain this distinction even when every integral under discussion is finite.

1.1. **The configuration space and its probability measure.** Let $G = \mathrm{SU}(2)$, the compact group of complex 2×2 matrices U with $U^*U = 1$ and $\det U = 1$. Its normalized Haar probability measure is denoted by dg. Fix a finite oriented graph obtained from a cubic lattice on a four-dimensional torus. Choose one orientation of each geometric bond; write B_+ for this set and impose $U_{\bar{b}} = U_b^{-1}$ for the reverse orientation. The configuration space is the compact space $\mathcal{V} = G^{B_+}$, with product probability measure dU.

A path is an ordered list of consecutive oriented bonds. Its transport is the ordered product of their matrices. For each geometrical elementary square choose one boundary orientation and one starting vertex, and denote its transport by U_p. Define

$$a_p(U) = 1 - \tfrac{1}{2} \operatorname{Re} \operatorname{tr} U_p,$$
$$A(U) = \sum_{p \in P} a_p(U),$$
$$N_P = |P|.$$

Changing the starting vertex conjugates U_p, and reversing orientation inverts it. Both changes preserve a_p.

Proposition 1.1 (Finite-cutoff existence and elementary bounds)**.** *For every real $\beta \geq 0$,*

$$e^{-2\beta N_P} \leq Z(\beta) := \int e^{-\beta A(U)} dU \leq 1.$$

Consequently $Z(\beta)^{-1} e^{-\beta A(U)} dU$ is a well-defined probability measure. Every bounded Borel observable has a finite expectation.

Proof. The eigenvalues of an element of G are e^{it}, e^{-it} for a real t. Thus $\frac{1}{2} \operatorname{Re} \operatorname{tr} U_p = \cos t$ lies in $[-1, 1]$, so $0 \leq a_p \leq 2$ and $0 \leq A \leq 2N_P$. Exponentiate these inequalities and integrate against a probability measure. The lower bound is strictly positive and supplies the normalization. □

The proof uses only compactness and positivity. Its constants degenerate when $N_P \to \infty$. In particular it supplies neither tightness of renormalized local fields nor a useful estimate for a continuum limit.

A gauge transformation is a family (h_x) of group elements indexed by vertices. It acts by $U_b \mapsto h_{b_-} U_b h_{b_+}^{-1}$. Successive factors cancel along a path, so a closed transport based at x is conjugated by h_x. Haar measure on each bond is unchanged by its independent left and right multiplications. This proves gauge invariance of the action and of the probability measure. A permutation of vertices and bonds coming from a lattice isometry also preserves the product measure and permutes the plaquettes. Therefore the finite torus measure has every isometry of that torus. This conclusion concerns these finite torus measures; it is not a proof that every infinite-volume periodic Gibbs state has every reflection.

1.2. **The continuum normalization is a local expansion.** Represent the Lie algebra by Hermitian traceless matrices and set $|F|^2 = \frac{1}{2}\operatorname{tr}(F^2)$. For a fixed such matrix,

$$1 - \tfrac{1}{2}\operatorname{Re}\operatorname{tr} e^{itF} = \tfrac{t^2}{4}\operatorname{tr}(F^2) + O(t^4|F|^4) = \tfrac{1}{2}t^2|F|^2 + O(t^4|F|^4).$$

The expansion follows by the matrix exponential series: the linear trace vanishes and the odd powers have zero real trace. If a smooth connection has elementary holonomy $U_p = \exp(ia^2 F_{\mu\nu}(x_p) + O(a^3))$, with all required derivatives uniformly bounded, it follows that

$$a_p = \tfrac{1}{2}a^4|F_{\mu\nu}(x_p)|^2 + O(a^5).$$

Hence, for a fixed continuous test function f supported in a compact region, summing $f(x_p)a_p$ over plaquettes gives the corresponding Riemann integral of $\frac{1}{2}f\sum_{\mu<\nu}|F_{\mu\nu}|^2$ as $a \downarrow 0$.

This is a deterministic smooth-field calculation. Typical configurations under a quantum measure have not been shown to satisfy its hypotheses. The calculation fixes the intended normalization of the observable; it does not prove convergence of that observable under the lattice measures.

1.3. **Sources, cumulants, and centering.** For bounded functions f_i on the plaquette barycentres put $O_i = \sum_p f_i(x_p)a_p$ and

$$Z(z) = \int \exp\{-\beta A(U) + \sum_{i=1}^{n} z_i O_i(U)\}\, dU,$$
$$z \in \mathbb{C}^n.$$

The estimate $|O_i| \le 2\sum_p |f_i(x_p)|$ gives an integrable uniform majorant on every compact source set. Termwise differentiation of the exponential series is therefore valid, and Z is entire. Since $Z(0) > 0$, continuity gives a sufficiently small polydisc about zero where Z has no zeros. On that polydisc there is a unique holomorphic logarithm W with $W(0) = \log Z(0)$.

The joint cumulant is

$$\kappa(O_1, \dots, O_n) = \partial_{z_1} \cdots \partial_{z_n} W(z)|_{z=0}\,.$$

Its expression in moments is

$$\kappa(O_1, \dots, O_n) = \sum_{\pi \in \mathrm{Part}(n)} (|\pi| - 1)!(-1)^{|\pi|-1} \prod_{J \in \pi} \mathbb{E} \prod_{i \in J} O_i.$$

Indeed, write $Z(z)/Z(0) = 1 + u(z)$ and use $\log(1+u) = \sum_{m \ge 1}(-1)^{m-1}u^m/m$ near zero. The coefficient of $z_1 \cdots z_n$ in u^m sums over ordered partitions into m nonempty blocks. Dividing by m turns their $m!$ orderings into $(m-1)!$. Conversely, expanding $e^{W(z)-W(0)}$ proves the moment–cumulant relation

$$\mathbb{E}\prod_i O_i = \sum_{\pi \in \mathrm{Part}(n)} \prod_{J \in \pi} \kappa((O_i)_{i \in J}).$$

Replacing O_i by $O_i - c_i$ changes the logarithm only by $-\sum_i z_i c_i$. Thus every cumulant of order at least two is unchanged. If $c_i = \mathbb{E}O_i$, every singleton cumulant vanishes. The centered moment functional is consequently

$$\mathbb{E}\prod_i (O_i - \mathbb{E}O_i) = \sum_{\pi \in \mathrm{Part}_{\geq 2}(n)} \prod_{J \in \pi} \kappa((O_i)_{i \in J}).$$

There are finitely many partitions at fixed n. Convergence of every separated cumulant of order at least two therefore implies convergence of the centered moments on the same tuples. Positivity must be applied to these moments, not to the cumulants separately.

The source is exactly a local coupling shift:

$$-\beta A + \sum_i z_i O_i = -\sum_p (\beta - w(x_p)) a_p,$$

$$w = \sum_i z_i f_i.$$

For a constant source c, $Z_\beta(c + w) = Z_{\beta - c}(w)$. This algebraic identity holds before any renormalization procedure is chosen.

1.4. **What a mass-preserving transformation preserves.** Let T be a fixed linear map on finite measures on a measurable space, with $(T\nu)(1) = \nu(1)$ for all measures in its domain. If $\nu_z = e^{\sum_i z_i O_i}\nu$ belongs to that domain, then

$$(T\nu_z)(1) = \nu_z(1).$$

This proves exact equality of sourced partition functions, and hence of their derivatives when the analytic dependence can be justified. It does not identify $T\nu$ as the distribution of a predetermined block variable. For that stronger conclusion one needs a measurable map M and the identity $(T\nu)(F) = \nu(F \circ M)$ for every bounded measurable F.

For example, on the two-point set $\{0, 1\}$ let $T\nu = \nu(\{0, 1\})\delta_0$. This positive linear map preserves total mass but changes the expectation of the coordinate from $\nu(\{1\})$ to zero when ν is a probability measure. Thus mass preservation alone contains no statement about observable expectations. Sources must be transported through the exact integral, with the choices of the transformation fixed throughout the source family.

1.5. **A fixed-spacing infinite-volume limit.** Consider the countable cubic lattice at a fixed positive spacing. Its configuration space is $\mathcal{V}_\infty = G^{B_+}$ with the product topology. This is a compact metrizable space: enumerate the bonds and use, for example, the sum of 2^{-j} times bounded compatible distances on their group coordinates. The space of Borel probability measures is sequentially compact in the weak topology.

For a finite set of bonds D, define the conditional specification

$$(\gamma_D F)(U) = \frac{\int F(V_D, U_{D^c}) \exp[-\beta A_D(V_D, U_{D^c})]\, dV_D}{\int \exp[-\beta A_D(V_D, U_{D^c})]\, dV_D},$$

$$A_D = \sum_{p : p \cap D \neq \varnothing} a_p.$$

The sum is finite and the denominator is bounded below by $e^{-2\beta N(D)}$, where $N(D)$ is the number of plaquettes meeting D. For continuous cylinder F, numerator and denominator depend continuously on only finitely many boundary bonds; dominated convergence proves that $\gamma_D F$ is another continuous cylinder function.

Proposition 1.2 (DLR existence at fixed lattice parameters)**.** *Take torus Wilson measures at fixed spacing and fixed β, and extend their bond configurations periodically to the infinite lattice. Every sequence of torus side lengths tending to infinity has a weakly convergent subsequence. Each limit ν satisfies $\nu(\gamma_D F) = \nu(F)$ for every finite D and bounded Borel cylinder function F. It is invariant under all lattice translations and all lattice isometries preserved by the finite torus measures.*

Proof. Compactness gives the subsequence. Fix D and a continuous cylinder F. For sufficiently large tori the finite set consisting of D, all adjacent plaquettes and the support of F embeds without identifications. The finite-volume conditional expectation identity then reads $\mu_\ell(\gamma_D F) = \mu_\ell(F)$ after periodic extension. Both integrands are continuous cylinder functions, so the identity passes to the weak limit. Continuous functions on the finite-dimensional compact coordinate spaces determine measures. The identity therefore extends first to bounded Borel cylinder functions, and, if desired, to all bounded Borel functions by the monotone class theorem. Invariance passes to the limit by the same continuous-cylinder argument, since a fixed lattice isometry acts continuously on the product space. □

This proposition does not change the spacing or the coupling. It provides no bounds uniform as the spacing tends to zero and gives no uniqueness or extremality of the limiting state. A theorem about all periodic Gibbs states is stronger than a theorem about these torus limit points.

1.6. **Compactness of separated correlation forms.** Let $\mathbb{E}$ be $\mathbb{R}^4$ or a fixed torus. A separated tuple consists of finitely many compactly supported C^1 functions with pairwise disjoint supports. Their compactness makes the minimum distance between nonempty supports positive. Fix the norm $\|f\|_1 = \max(\|f\|_\infty, \|\nabla f\|_\infty)$.

Proposition 1.3 (A diagonal extraction principle)**.** *Suppose $T_{n,j}$ are symmetric multilinear forms on separated tuples and for every fixed n, $R < \infty$ and $d > 0$ there is a finite $C(n, R, d)$, independent of all sufficiently large j, such that*

$$|T_{n,j}(f_1, \dots, f_n)| \le C(n, R, d) \prod_i \|f_i\|_1$$

whenever the supports lie in $B(0, R)$ and are separated by at least d. Then a subsequence converges at every separated tuple, simultaneously for all n. The limits obey the same bounds.

Proof. Choose all finite families of finite unions of rational closed boxes with positive mutual distance. The C^1 functions supported in each union form a separable normed space, being a subspace of a finite product of spaces of continuous functions on a compact set. Choose countable dense subsets. There are countably many finite tuples drawn from these subsets. Each sequence of values is eventually bounded by the assumed estimate. Bolzano–Weierstrass and a successive diagonal extraction give a common subsequence on which all these values converge.

Given any separated tuple, enclose its supports in pairwise separated finite unions of rational boxes, still inside a fixed bounded region. Approximate each function in C^1 by elements of the chosen dense sets. The telescoping multilinear identity

$$T(f_1, \dots, f_n) - T(g_1, \dots, g_n) = \sum_{i=1}^{n} T(g_1, \dots, g_{i-1}, f_i - g_i, f_{i+1}, \dots, f_n)$$

and the uniform bound make the approximation error uniformly small along the extracted subsequence. This proves the Cauchy property for the original tuple. Multilinearity, symmetry and the bound pass to the limit on each product of spaces with separated supports. □

The difficult Yang–Mills step, which we must prove, is the estimate supplying $C(n, R, d)$. Finite-cutoff boundedness with a constant involving the number of plaquettes does not meet it. An exponentially large dependence on the physical torus volume also does not supply a bound uniform on $\mathbb{R}^4$. These are different compactness problems.

1.7. **Reflection positivity and closed limits.** Let θ be a reflection. Real reflection-invariant centering preserves reflection positivity: replacing each $O_j(f)$ by $O_j(f) - m_j(f)$ maps every positive-half polynomial to another such polynomial and commutes with reflection when $m_j(\theta f) = m_j(f)$. Suppose for each cutoff a centered moment functional σ_j satisfies

$$\sum_{r,s=1}^{m} \overline{c_r} c_s \, \sigma_j(\theta\overline{X_r} \cdot X_s) \geq 0$$

for every finite collection of complex test tuples X_r in the positive half-space, where $\overline{X_r}$ conjugates every test and θ acts by pullback. Within each X_r the supports are separated; the reflected negative tuple and the positive tuple are automatically separated from one another. Overlap between supports from X_r and X_s on the same side causes no problem, because these supports do not occur together in the same concatenated pair $\theta\overline{X_r} \cdot X_s$.

If all the entries converge, the finite sum converges and remains nonnegative. Thus the limiting moment functional is reflection positive on this vector space of separated monomials. This elementary closedness statement is the precise sense in which positivity passes to limits. Passing from that domain to a Schwartz test space requires a density result and a continuous bound for the limiting distributions; it cannot be justified by closedness alone.

The algebraic form $\langle F, G\rangle = \sigma(\theta\overline{F} \cdot G)$ is positive semidefinite. Positivity of $\langle F + zG, F + zG\rangle$ for every complex z gives $|\langle F, G\rangle|^2 \leq \langle F, F\rangle\langle G, G\rangle$. Consequently a null vector is orthogonal to every vector. The quotient by the null space is an inner-product space, and its completion is a Hilbert space. Defining translations on this quotient additionally requires that they preserve the null space and satisfy the appropriate bounds. Strong continuity and a nonnegative generator are later conclusions with their own hypotheses.

1.8. **Two limits of what scalar correlations prove.** Let X be a real standard Gaussian variable and put $Y = X^2 - 1$. For $t < 1/2$ near zero,

$$\log \mathbb{E}e^{tY} = -t - \tfrac{1}{2}\log(1 - 2t) = \sum_{n\geq 2} \frac{2^{n-1}}{n} t^n.$$

It follows that $\kappa_n(Y) = 2^{n-1}(n-1)!$ for every $n \geq 2$. In particular a nonzero connected four-point function of a quadratic observable is compatible with a Gaussian underlying theory. For distinct centered jointly Gaussian variables the same conclusion is expressed by the connected cycle contractions of their Wick squares.

Accordingly a non-Gaussian action-density field is an important property of that field, but is not by itself a characterization of an interacting Yang–Mills gauge theory. Reconstruction from the action-density correlations produces the cyclic sector generated by that observable. Identifying a richer gauge-invariant algebra requires additional observables and compatibility arguments.

A polynomial cluster estimate likewise gives no positive mass gap. In a Hilbert space with a nonnegative self-adjoint H, a connected time correlation has the spectral form $\langle v, e^{-tH}v\rangle = \int_0^\infty e^{-tE}\, d\mu_v(E)$ for a vector orthogonal to the vacuum. If $d\mu_v(E) = Ee^{-E}\, dE$, the integral is $(1 + t)^{-2}$. It decays, yet its spectral measure meets every interval $(0, \varepsilon)$. The support condition required for a gap would instead exclude all such intervals below one fixed positive threshold.

These examples clarify the scope of the construction and the additional estimates needed for richer observable algebras and a positive spectral gap. The ultraviolet estimates enter with their own stated hypotheses.

2. Block normalization and a uniform scalar gap

2.1. **Block means, adjoints, and a uniform scalar Poincaré inequality.** An adjoint depends on the inner products of both its domain and its range. This matters when a block mean maps a fine lattice to a coarse lattice. The following finite-dimensional calculation fixes the convention and gives the scalar estimate used in a gauge-fixed Gaussian comparison. It applies to real or complex scalar functions; a finite number of color components are handled by summing the same estimate component by component.

Fix an integer $m \geq 1$ and put $\eta = m^{-1}$. A block consists of the sites

$$B = \{0, \eta, \dots, (m-1)\eta\}^4.$$

Its associated physical cell has side $m\eta = 1$ and volume one. Translation of the block, including translation by half a fine spacing to use cell centers, does not change the calculation. Let $\mathcal{B}$ be a finite disjoint family of such blocks, for example the blocks of a periodic torus whose side is an integer number of coarse cells. Write Λ_f for their union and Λ_c for the set indexing the blocks. A coarse function has one value per block. We count every fine bond once, with positive orientation. The block Neumann form uses the intrinsic open rectangular graph. In a chosen lift $B = x_B + \eta\{0, \dots, m-1\}^4$, include exactly the bonds $x_B + \eta j \longrightarrow x_B + \eta(j + e_\mu)$ with $j \in \{0, \dots, m-1\}^4$ and $j_\mu \leq m-2$. Thus there are no internal bonds if $m = 1$. Exclude opposite-face periodic wrap bonds even when their two torus endpoints belong to the same block. The ambient periodic d_η retains all fine bonds, with their usual positive-orientation multiplicities.

The physical inner products are

$$\langle f, g\rangle_f = \eta^4 \sum_{x \in \Lambda_f} \overline{f(x)} g(x),$$

$$\langle u, v\rangle_c = \sum_{B \in \mathcal{B}} \overline{u(B)} v(B),$$

$$\langle a, b\rangle_e = \eta^4 \sum_{e} \overline{a(e)} b(e).$$

The last sum can mean all fine bonds of the periodic lattice or just the internal bonds of the blocks; the operator being considered determines which set is used. The same weight is used on fine sites and fine bonds. The physical difference is $d_\eta f(x, x + \eta e_\mu) = \eta^{-1}[f(x + \eta e_\mu) - f(x)]$. The unscaled combinatorial difference is $d_0 = \eta d_\eta$. These two operators must not be interchanged in an estimate with a mesh-independent constant.

Proposition 2.1 (The mean, its adjoint, and its projection)**.** *Define the block mean $Q : H_f \to H_c$ by*

$$(Qf)(B) = \eta^4 \sum_{x \in B} f(x) = m^{-4} \sum_{x \in B} f(x).$$

For the physical inner products just specified, its adjoint is $(Q^* u)(x) = u(B)$ for $x \in B$. Consequently $QQ^* = 1$, and $P = Q^* Q$ is the orthogonal projection onto the functions constant in each block. In particular,

$$\|f\|_f^2 = \|(1-P)f\|_f^2 + \|Qf\|_c^2.$$

Proof. Directly from the definitions,

$$\langle Qf, u\rangle_c = \sum_B \eta^4 \sum_{x\in B} \overline{f(x)} u(B) = \langle f, Q^* u\rangle_f.$$

Since a block contains m^4 sites and $\eta^4 m^4 = 1$, its mean applied to a constant is that constant. Thus $QQ^* = 1$. It follows that $P^* = P$ and $P^2 = Q^*(QQ^*)Q = P$. Its range is precisely the block-constant functions, and the displayed norm identity is the orthogonal decomposition by that projection. No limiting argument is involved. □

Proposition 2.2 (A block Neumann estimate)**.** *Let $d_{\eta,N}$ denote the physical difference restricted to internal bonds of each block. If $m \geq 2$, then*

$$\|(1-P)f\|_f^2 \leq \frac{\eta^2}{4\sin^2(\pi\eta/2)} \|d_{\eta,N} f\|_e^2 \leq \tfrac{1}{4}\|d_{\eta,N} f\|_e^2.$$

For $m = 1$ the left side is zero. If d_η includes bonds between blocks, the same bound holds with d_η on the right. With the combinatorial difference the exact first bound is instead

$$\|(1-P)f\|_f^2 \leq \frac{1}{4\sin^2(\pi/(2m))} \|d_{0,N} f\|_e^2.$$

Proof. First consider a one-dimensional path with m vertices indexed by $j = 0, \ldots, m-1$. Its Neumann graph Laplacian is the matrix associated with the quadratic form $\sum_{j=0}^{m-2} |f_{j+1} - f_j|^2$. For an interior vertex it acts as $2f_j - f_{j-1} - f_{j+1}$; at the two endpoints it acts as $f_0 - f_1$ and $f_{m-1} - f_{m-2}$. For $k = 0, \ldots, m-1$, set

$$v_k(j) = \cos\bigl(\pi k(j + \tfrac{1}{2})/m\bigr).$$

The cosine addition formula gives, in the interior,

$$2v_k(j) - v_k(j-1) - v_k(j+1) = 4\sin^2(\pi k/(2m)) v_k(j).$$

At the left endpoint,

$$\cos(\theta/2) - \cos(3\theta/2) = 4\sin^2(\theta/2)\cos(\theta/2),$$
$$\theta = \pi k/m.$$

The right endpoint follows by reflection. Thus these are eigenvectors with eigenvalues $4\sin^2(\pi k/(2m))$. The eigenvalues are distinct because sine is strictly increasing on $[0, \pi/2)$; the nonzero eigenvectors therefore form an orthogonal basis. The $k = 0$ vector is constant.

On the four-dimensional block the graph Laplacian is the sum of the four one-dimensional matrices acting on separate tensor factors. Tensor products of the v_k therefore form an eigenbasis. The eigenvalues of the physical Neumann Laplacian $d_{\eta,N}^* d_{\eta,N}$ are

$$4\eta^{-2} \sum_{\mu=1}^{4} \sin^2(\pi k_\mu/(2m)),$$

$$0 \le k_\mu < m.$$

Its only zero mode is the constant vector, and the smallest positive eigenvalue is $\lambda_N = 4\eta^{-2}\sin^2(\pi/(2m))$. Multiplication of the site and bond inner products by the common constant η^4 does not change this matrix or these eigenvalues. Expanding $f - Pf$ in the eigenbasis gives $\lambda_N\|f - Pf\|_f^2 \le \|d_{\eta,N} f\|_e^2$ on one block and hence on their disjoint union.

Finally, concavity of sine on $[0, \pi/2]$ gives $\sin t \ge 2t/\pi$ there. At $t = \pi/(2m)$ this yields $\sin(\pi/(2m)) \ge 1/m = \eta$, so $\lambda_N \ge 4$. Adding the nonnegative contributions of the other ambient lattice bonds, including periodic wraps whose endpoints may lie in the same block, can only increase the Dirichlet form. Substitution of $d_{0,N} = \eta d_{\eta,N}$ gives the combinatorial version. □

These estimates immediately give a uniform scalar gap on a periodic fine lattice. If $\mathcal{L}_\eta = d_\eta^* d_\eta$ and $\mathcal{A} = \mathcal{L}_\eta + Q^*Q$, then

$$\langle f, \mathcal{A}f\rangle_f = \|d_\eta f\|_e^2 + \|Qf\|_c^2 \ge \|(1-P)f\|_f^2 + \|Qf\|_c^2 = \|f\|_f^2.$$

Thus $\mathcal{A}$ is invertible and $\|\mathcal{A}^{-1}\| \le 1$, independently of the mesh and number of blocks. This proves a scalar block-constrained gap. It supplies neither a vector-field coercivity theorem nor a mass gap for a physical Hamiltonian; those are different assertions.

A uniform inverse for a coarse Gram matrix. The same convention also makes a useful projection formula precise. On the periodic lattice put $G' = \mathcal{A}^{-1}$, with $\mathcal{A}$ as above. There is a numerical constant $c > 0$, independent of m and the number of blocks, such that

$$QG'^2Q^* \ge c1.$$

Here is a finite-dimensional proof, including a choice of the test functions. For $m \ge 2$ define on a block indexed by $j = (j_1, \dots, j_4)$

$$\phi_m(j) = 16\prod_{\mu=1}^{4} \sin^2\big(\pi(j_\mu + \tfrac{1}{2})/m\big).$$

The sum of the m roots $e^{2\pi i(j+1/2)/m}$ is zero. Taking real parts in $\sin^2 t = (1 - \cos 2t)/2$ shows that the one-dimensional arithmetic mean of each sine-square factor is $1/2$. Thus the mean of ϕ_m on a block is one. For a coarse function u, define $Tu(x) = u(B)\phi_m(x)$ on the block B. Then $QT = 1$ and $\|T\| \le 16$.

There is also a bound $\|\mathcal{L}_\eta Tu\|_f \le C\|u\|_c$ uniform in m. To see it, the second physical difference of $\sin^2(\pi t)$ at interior sites is bounded by the supremum $2\pi^2$ of its second derivative. At an end site, its value is $\sin^2(\pi\eta/2) \le \pi^2\eta^2/4$. A second difference that crosses a block boundary therefore contributes at most a constant times the absolute coarse values in the two neighboring blocks: the factor η^{-2} is canceled by this endpoint bound. The other three sine-square factors are bounded by one. After summing the four coordinate differences, a pointwise bound is $C\sum_{B' \sim B} |u(B')|$, where the sum includes B and its face neighbors. Squaring, summing with weight η^4, and using the bounded number of neighbors proves the norm estimate. For $m = 1$, set $T = 1$; the unit-spacing Laplacian has norm at most 16, so the same conclusion holds. Consequently $\|\mathcal{A}Tu\|_f \le C\|u\|_c$ for all m.

Self-adjointness and $G'\mathcal{A} = 1$ now give

$$\|u\|_c^2 = \langle Q^*u, Tu\rangle_f = \langle G'Q^*u, \mathcal{A}Tu\rangle_f \le C\|G'Q^*u\|_f\|u\|_c.$$

This proves the claimed positive lower bound for QG'^2Q^*.

The map $V = G'Q^*$ is injective, its Gram matrix is $V^*V = QG'^2Q^*$, and elementary linear algebra shows that

$$R = 1 - G'Q^*(QG'^2Q^*)^{-1}QG'$$

is the orthogonal projection onto $\ker(QG')$. This kernel equals $\mathcal{L}_\eta \ker Q$. Indeed, if $Qv = 0$ then $\mathcal{A}v = \mathcal{L}_\eta v$, so $QG'\mathcal{L}_\eta v = 0$. Conversely, if $QG'h = 0$, put $v = G'h$. Then $Qv = 0$ and $h = \mathcal{A}v = \mathcal{L}_\eta v$. The expression for R is therefore well defined with a uniformly invertible coarse Gram matrix. Every adjoint in this argument uses the physical inner products fixed above.

Translation to plain Euclidean coordinates. There is an equivalent convention with unweighted coordinate inner products, but the averaging matrix has to change. Let $S_f f = \eta^2 f$ on fine sites, $S_e a = \eta^2 a$ on fine bonds, and $S_c u = u$ on coarse sites. These maps are unitaries from the physical spaces just defined to the spaces with plain coordinate sums. The correctly conjugated averaging matrix is

$$\widetilde{Q} = S_c Q S_f^{-1},$$
$$(\widetilde{Q}v)(B) = \eta^2 \sum_{x \in B} v(x),$$
$$(\widetilde{Q}^* u)(x) = \eta^2 u(B).$$

Each row has m^4 entries of size $\eta^2 = m^{-2}$, so its squared Euclidean length is one. Rows have disjoint supports, proving $\widetilde{Q}\widetilde{Q}^* = 1$. The conjugated projection is $\widetilde{P} = \widetilde{Q}^*\widetilde{Q} = S_f P S_f^{-1}$. Because sites and bonds have the same physical weight, $S_e d_\eta S_f^{-1}$ is the same physical difference matrix $\eta^{-1}(\text{shift} - 1)$. Therefore the entire Poincaré estimate survives unitary conjugation with its exact constant.

In contrast, suppose that one retains the arithmetic mean matrix $\overline{Q}v = \eta^4 \sum_{x \in B} v(x)$ while using plain inner products on both lattices. Its plain adjoint is $(\overline{Q}^* u)(x) = \eta^4 u(B)$, and

$$\overline{Q}\overline{Q}^* = \eta^4 1,$$
$$P = \eta^{-4}\overline{Q}^*\overline{Q}.$$

Thus $\overline{Q}^*\overline{Q}$ itself is not the projection. For a nonzero constant fine function and $m > 1$, the expression $\|f - \overline{Q}^*\overline{Q}f\|^2$ is positive whereas $d_\eta f = 0$. This gives an immediate check against accidentally combining the arithmetic mean with the wrong adjoint. Either the physical weighted convention or the conjugated unweighted convention is consistent; mixing them is not.

Consequences for cochains and Gaussian formulas. For the centered exact-cochain Gaussian that is standard on the exact-form subspace in the plain inner product (Section 5), its covariance is the orthogonal projection onto that subspace. After changing to physical coordinates, transform the random vector and the covariance together. For a linear random vector F with covariance C in plain coordinates and a linear coordinate change T, the covariance of TF is TCT^*. An orthogonal projection is carried to an orthogonal projection when the coordinate change is interpreted as the unitary between the two specified Hilbert spaces; an arbitrary rescaling of its coordinate entries without the corresponding inner product change has a different covariance.

For a Hilbert-space operator $B : H_1 \to H_2$, unitary changes of coordinates S_1, S_2 give

$$\widetilde{B} = S_2 B S_1^{-1},$$

$$\widetilde{B}^* = S_1 B^* S_2^{-1}.$$

In particular products, orthogonal projections, positive inverses and Moore–Penrose inverses commute with this conjugation. A conditional Gaussian covariance written as $C - CB^*(BCB^*)^+BC$ remains the same Hilbert-space operator after every factor is conjugated. This observation permits a proof to begin in plain finite cochain coordinates and then use physical block coordinates. The change must be announced before the first adjoint, constraint, or Poincaré estimate that uses the latter convention. Every discrete derivative must also say whether it is d_0 or d_η. These declarations prevent a missing power of the mesh from being hidden inside an adjoint symbol.

2.2. **A scalar block resolvent with uniform off-diagonal decay.** This finite-dimensional lemma supplies a scalar part of the propagator input. All distances are physical distances. The argument uses a relative form bound and a Neumann series; the exponentially conjugated operator is generally not self-adjoint.

Let a four-dimensional periodic lattice of spacing $\eta = 1/m$, $m \geq 1$, be partitioned into blocks of m^4 sites and physical side one. Its torus has an integer number of blocks in each direction. On site functions use $\langle u, v\rangle = \eta^4 \sum_x \overline{u_x} v_x$. On positively oriented fine bonds use the same weight. Put

$$(d_\eta u)(x, x + \eta e_\mu) = \eta^{-1}(u_{x+\eta e_\mu} - u_x),$$
$$\mathcal{L}_\eta = d_\eta^* d_\eta.$$

The block mean Q takes the arithmetic mean of the m^4 fine values. Its coarse inner product is the unweighted sum, so Q^* embeds a coarse value as a constant on its block. Thus $P = Q^*Q$ is an orthogonal projection and its matrix entries, as an operator acting on coordinate vectors, are $P_{xy} = \eta^4$ when x, y are in the same block and zero otherwise.

Write dist_η for shortest-path distance on the periodic fine graph, with length η per bond. The distance includes wrapping paths. The diameter of a single block in this metric is at most 4. The block Poincaré inequality proves

$$A_\eta := \mathcal{L}_\eta + P \geq 1,$$
$$\|u\| \leq \|A_\eta^{1/2} u\|,$$
$$\|d_\eta u\| \leq \|A_\eta^{1/2} u\|.$$

These statements also hold when $m = 1$, since then $P = 1$.

Proposition 2.3 (A mesh-independent Combes–Thomas bound)**.** *Let w be a real function with $|w_x - w_y| \leq \eta$ on every neighboring pair. For real t let M_t multiply by e^{tw_x} and set $A_t = M_t A_\eta M_t^{-1} = A_\eta + E_t$. There is a numerical constant C, independent of the mesh, volume and function w, such that, for $|t| \leq 1$,*

$$\|A_\eta^{-1/2} E_t A_\eta^{-1/2}\| \leq C(|t| + t^2).$$

Consequently one may choose a fixed $t_0 > 0$, independent of those parameters, so that $\|A_t^{-1}\| \leq 2$ for $|t| \leq t_0$. For arbitrary nonempty sets of sites X, Y this gives

$$\|1_X A_\eta^{-1} 1_Y\| \leq 2 \exp[-t_0 \,\mathrm{dist}_\eta(X, Y)].$$

Proof. Set $A = A_\eta$. First consider the contribution $E_t^{\mathcal{L}}$ from the Laplacian. On a positively oriented bond with endpoints x, y, put $\delta = w_x - w_y$, $a = e^{t\delta} - 1$ and $b = e^{-t\delta} - 1$. The diagonal entries of the conjugated edge Laplacian do not change. Its off-diagonal contribution to $\langle u, E_t^{\mathcal{L}} v\rangle$ is therefore

$$-\eta^4 \eta^{-2}(a\overline{u_x}v_y + b\overline{u_y}v_x).$$

The exact identity

$$a\overline{u_x}v_y + b\overline{u_y}v_x = a(\overline{u_x} - \overline{u_y})v_y + b\overline{u_y}(v_x - v_y) + (a+b)\overline{u_y}v_y$$

exposes a difference of u or v in the terms linear in t. Since $|\delta| \le \eta \le 1$ and $|t| \le 1$,

$$|a|, |b| \le e|t|\eta,$$
$$|a + b| = 2|\cosh(t\delta) - 1| \le et^2\eta^2.$$

Each vertex occurs as the head of at most four positively oriented bonds. Cauchy–Schwarz applied to the first term, summed over the bonds, consequently gives $2e|t|\|d_\eta u\|\|v\|$. The second term is bounded by $2e|t|\|u\|\|d_\eta v\|$. The last term is bounded by $4et^2\|u\|\|v\|$. In particular,

$$|\langle u, E_t^{\mathcal{L}} v\rangle| \le 2e|t|(\|d_\eta u\|\|v\| + \|u\|\|d_\eta v\|) + 4et^2\|u\|\|v\|.$$

The two powers of η^{-1} in the Laplacian have been canceled by the Lipschitz difference and the difference of a test vector. Estimating the off-diagonal matrix entries directly would miss this cancellation.

For the block term, the neighbor condition implies $|w_x - w_y| \le 4$ for sites in one block. Its conjugation error has entries

$$(E_t^P)_{xy} = \eta^4 1_{\{x,y \text{ in one block}\}}(e^{t(w_x - w_y)} - 1).$$

Their absolute row sums and column sums are at most $4e^4|t|$, because a block has m^4 sites and $m^4\eta^4 = 1$. The Schur test therefore yields $\|E_t^P\| \le 4e^4|t|$. The common site weight η^4 does not alter this Schur estimate. Using the two energy bounds preceding the proposition gives

$$|\langle u, E_t v\rangle| \le C(|t| + t^2)\|A^{1/2}u\|\|A^{1/2}v\|,$$

where, for example, $C = 8e + 4e^4$ suffices. Substitute $u = A^{-1/2}f$ and $v = A^{-1/2}g$ and take the supremum over unit vectors f, g. This proves the asserted relative operator bound.

Choose $t_0 = \min\{1, (4C)^{-1}\}$. Then $B_t = A^{-1/2}E_t A^{-1/2}$ has norm at most $1/2$ when $|t| \le t_0$. The exact factorization and its inverse are

$$A_t = A^{1/2}(1 + B_t)A^{1/2},$$
$$A_t^{-1} = A^{-1/2}(1 + B_t)^{-1}A^{-1/2}.$$

The Neumann series gives $\|(1 + B_t)^{-1}\| \le 2$ and hence $\|A_t^{-1}\| \le 2$. This step requires no assertion that A_t is self-adjoint or ordered below by a positive scalar.

Finally take $w_x = \operatorname{dist}_\eta(x, Y)$ and $t = t_0$. The triangle inequality makes w admissible, and $w = 0$ on Y. If $D = \operatorname{dist}_\eta(X, Y)$, then $w \ge D$ on X. Similarity of the two inverses gives

$$1_X A^{-1} 1_Y = 1_X M_t^{-1} A_t^{-1} M_t 1_Y.$$

The rightmost multiplication is the identity on Y, while the norm of the leftmost multiplication is at most $e^{-t_0 D}$. Combining these statements with $\|A_t^{-1}\| \le 2$ proves the final estimate. □

The estimate concerns operator norms between sets in the weighted lattice Hilbert space. It does not assert a mesh-independent bound for a Green function kernel normalized relative to the physical counting measure $\eta^4 \sum_x$. Converting a coordinate matrix to that kernel introduces its own power of η. The scalar lemma likewise does not establish the non-Abelian vector-field or multiscale background estimates. Those require their additional gauge constraints and response bounds.

Dirichlet restrictions and the distance being estimated. The same estimate holds for a Dirichlet compression of this particular operator, with constants independent of the shape of the region. To state it without ambiguity, let D be any set of sites of the periodic fine lattice. Let H_D consist of the functions supported in D, with the inherited weighted inner product, and let $J_D : H_D \to H$ be extension by zero. Define

$$A_D = J_D^* A_\eta J_D.$$

In particular the block term is the compression of the *original* projection P. On a block intersected by the boundary it is not a new mean taken over just the surviving sites. Replacing it by that new mean would define a different operator and is not justified by this corollary.

For $u \in H_D$, the quadratic form is

$$\langle u, A_D u \rangle = \|d_\eta J_D u\|^2 + \|Q J_D u\|_c^2 \ge \|u\|^2.$$

It controls the physical gradient of the zero extension, including boundary bonds. Restrict an admissible ambient weight w to D and conjugate by its exponential on H_D. Because multiplication by this exponential commutes with the site indicator 1_D, the new error form is exactly the restriction of E_t to zero-extended vectors. The estimates in the proof therefore give

$$\|A_D^{-1/2}(M_t A_D M_t^{-1} - A_D) A_D^{-1/2}\| \le C(|t| + t^2).$$

The same Neumann argument implies, for $X, Y \subset D$,

$$\|1_X A_D^{-1} 1_Y\| \le 2e^{-t_0 \operatorname{dist}_\eta(X,Y)}.$$

Here the distance is computed in the ambient periodic lattice. It may use a path outside D. No estimate in the possibly much larger intrinsic distance of a region with holes has been asserted.

For comparison with continuum notation, the shortest-path distance is the periodic physical Manhattan distance, and hence is no smaller than the periodic Euclidean distance. The displayed bound therefore implies the same bound with Euclidean distance in its exponent. One can alternatively take w to be Euclidean distance to Y directly, since its change across a fine bond is at most η. Both formulations concern physical distances; using the number of bonds without the factor η would incorrectly strengthen the assertion as the mesh is refined.

The calculation explains the uniform mechanism: the scalar energy controls one physical difference, the weight varies by only $O(t\eta)$ across an edge, and its genuinely quadratic error is $O(t^2\eta^2)$. The block projection has range at most one physical cell and row mass one. Consequently the conjugation cost depends on the dimension and physical block size, with no factor proportional to the number of fine sites in a block or the torus.

3. A polymer expansion with a complete convergence proof

The small-field integration must produce activities satisfying a summability criterion. Once that estimate is available, the passage from activities to a local logarithm is a combinatorial theorem. We prove that theorem here, so its convergence mechanism is independent of the particular gauge-field coordinates. The estimate on the activities remains a separate analytic task.

3.1. **Finite polymer systems.** A finite polymer system consists of a finite set $\mathcal{P}$ and a symmetric incompatibility relation $\not\sim$ on it. Every polymer is incompatible with itself. Give each $X \in \mathcal{P}$ a complex activity z_X. A compatible family contains no two incompatible polymers. Its partition function is

$$Z(z) = \sum_{\substack{\mathcal{F}\subset\mathcal{P}\\ \mathcal{F}\text{ compatible}}} \prod_{X\in\mathcal{F}} z_X.$$

The empty family contributes one. There are no questions of convergence in this polynomial. The problem is to construct its logarithm and bound it in terms of local activities without a loss exponential in $|\mathcal{P}|$.

For a list $\mathbf{X} = (X_1, \dots, X_n)$, let $H(\mathbf{X})$ be the simple graph on the labeled vertices $1, \dots, n$ whose edges are the pairs with $X_i \not\sim X_j$. Repeated polymers are permitted in lists; their distinct vertices are adjacent by self-incompatibility. Define

$$\phi(\mathbf{X}) = \sum_{\substack{F\subset E(H(\mathbf{X}))\\ ([n],F)\text{ connected}}} (-1)^{|F|},$$

$$n \geq 2,$$

$$\phi(X_1) = 1.$$

If $H(\mathbf{X})$ is disconnected the sum is zero.

Lemma 3.1 (The connected graph identity)**.** *As a formal power series at $z = 0$,*

$$\log Z(z) = \sum_{n\geq 1} \frac{1}{n!} \sum_{X_1,\dots,X_n\in\mathcal{P}} \phi(\mathbf{X}) \prod_{i=1}^{n} z_{X_i}.$$

Proof. The ordered-list expression for Z is

$$Z(z) = \sum_{n\geq 0} \frac{1}{n!} \sum_{\mathbf{X}\in\mathcal{P}^n} \prod_i z_{X_i} \prod_{i<j} (1 - \mathbf{1}_{\{X_i \not\sim X_j\}}).$$

The product is one on pairwise compatible lists and zero otherwise. Such lists contain no repetition, and their $n!$ orderings give exactly the original polynomial. Expand the product as a sum over edge sets. Each graph decomposes uniquely into its connected components. The weight factors over the components, including the activity at each vertex. Expanding the exponential of the proposed connected-graph series enumerates a partition of the labeled vertices into these components. The factor for ordering the components is canceled by the exponential's factorial, and the factorials within the components cancel their vertex orderings. This reconstructs the ordered-list expression coefficient by coefficient. Its constant term is one, so its formal logarithm is the connected series. □

3.2. **The tree bound, including the cancellation.**

Lemma 3.2 (A hard-core tree inequality)**.** *For a finite simple graph H on $n \geq 1$ labeled vertices,*

$$\left| \sum_{\substack{F \subset E(H) \\ ([n],F) \text{ connected}}} (-1)^{|F|} \right| \leq \#\{\textit{spanning trees of } H\}.$$

For $n = 1$ the empty tree is counted once.

Proof. Fix a total order of the edges of H. Call an edge e eligible for an edge set F if the endpoints of e can be joined by a path in F consisting only of edges smaller than e. This definition does not require $e \in F$. If at least one edge is eligible, choose the smallest eligible edge e and toggle its membership in F.

This is an involution. Eligibility of e depends only on edges smaller than e, which were not changed; the same holds for every smaller edge, so the smallest eligible edge stays the same. Toggling e preserves the connected components, because its endpoints already have a path using smaller edges. It reverses the parity of $|F|$. Thus all connected edge sets with an eligible edge cancel in pairs in the alternating sum.

A surviving edge set has no cycle: if it contained one, the largest edge of that cycle would be eligible. Consequently every surviving connected edge set is a spanning tree. Each contributes the same sign $(-1)^{n-1}$, and there can be no more survivors than spanning trees. This proves the bound. The same argument also proves that the alternating sum has sign $(-1)^{n-1}$ when it is nonzero. □

In particular,

$$|\phi(X_1, \dots, X_n)| \leq \sum_{T \text{ tree on } [n]} \prod_{\{i,j\} \in E(T)} \mathbf{1}_{\{X_i \not\sim X_j\}}.$$

The cancellation in this inequality is essential. Taking absolute values before summing over connected graphs would count all connected subgraphs, which need not admit the rooted-tree bound used next.

3.3. **The rooted-tree estimate.** Let $a_X \geq 0$ be real numbers and suppose

$$\sum_{Y : Y \not\sim X} |z_Y| e^{a_Y} \leq a_X \quad \text{for every } X \in \mathcal{P}. \tag{3.1}$$

This is the polymer criterion used here [KP86]. For a fixed root polymer X, let $\mathcal{T}_X$ be the total nonnegative weight of all rooted labeled trees, with the root fixed to have polymer X, every nonroot vertex of polymer Y carrying weight $|z_Y|$, and every edge requiring incompatibility of its endpoint polymers. A tree on m vertices has the factor $1/(m-1)!$ for the nonroot vertex labels.

One can define this sum without assuming its convergence. Start with $T_X^{(0)} = 1$ and set

$$T_X^{(h+1)} = \exp\left(\sum_{Y : Y \not\sim X} |z_Y| T_Y^{(h)} \right).$$

The quantity $T_X^{(h)}$ sums trees of height at most h. To see this, delete the root. Each child becomes the root of a tree of height at most $h-1$, and an unordered collection of r child trees contributes the factor $1/r!$ in the exponential. Expanding those exponentials and assigning the

nonroot labels gives exactly $1/(m-1)!$: the multinomial coefficient for assigning labels to the child subtrees cancels their factorials and the ordering of the r children. This also verifies the formula for every fixed degree before any infinite sum is taken.

The iterates increase with h. Moreover $T_X^{(h)} \le e^{a_X}$ for every h: this is true at $h = 0$, and the induction step is precisely (3.1). Monotone convergence proves

$$\mathcal{T}_X = \lim_{h\to\infty} T_X^{(h)} \le e^{a_X}. \tag{3.2}$$

Theorem 3.3 (Absolute convergence and a nonzero partition function)**.** *Under* (3.1) *the connected series converges absolutely and defines a number $L(z)$ with $e^{L(z)} = Z(z)$. In particular $Z(z) \neq 0$. For each root polymer X,*

$$\sum_{n\ge 1} \frac{1}{(n-1)!} \sum_{X_2,\dots,X_n} |\phi(X, X_2, \dots, X_n)| \prod_{i=2}^{n} |z_{X_i}| \le e^{a_X}.$$

Also $|L(z)| \le \sum_X |z_X| e^{a_X}$. If the criterion holds throughout a complex activity domain with a uniform majorant, the series converges locally normally there and gives the branch of $\log Z$ *that equals zero at zero.*

Proof. The tree inequality bounds the root series by $\mathcal{T}_X$, proving its estimate by (3.2). Rooting each nonempty connected list at its first vertex bounds the absolute unrooted series by $\sum_X |z_X| e^{a_X}$: replacing $1/n!$ by $1/(n-1)!$ only enlarges it.

For $|t| < 1$ the same majorant applies to the activities tz_X, and the connected series is analytic in t there. Near zero its exponential is $Z(tz)$ by the formal identity and absolute convergence. The analytic identity theorem gives $e^{L(tz)} = Z(tz)$ on the entire unit disc. The absolute majorant is summable also at $|t| = 1$, so the connected series is continuous as t tends radially to one. Taking that limit proves $Z(z) = e^{L(z)} \neq 0$.

On a domain with a uniform summable majorant, the Weierstrass criterion gives normal convergence on smaller domains and hence holomorphy. The branch statement follows by agreement near zero and continuation within the connected component of that domain containing zero. This statement does not assert that an arbitrary disconnected activity domain has a single preferred logarithm. □

For a countable polymer set the same statements hold in a finite region whenever $\sum_X |z_X| e^{a_X} < \infty$ there, by exhaustion and the displayed absolute majorant. In an infinite lattice the total unrooted sum is generally extensive and infinite. The rooted bound remains meaningful if the incompatibility sums satisfy the criterion, and it is the form used to define local connected expansions. No finite infinite-volume partition function is asserted.

3.4. **Application to fluctuation activities.** If a polymer activity depends only on field variables in an enlargement of that polymer, each connected term depends only on the union of those enlargements. This is a direct consequence of its product formula. Extra exponential weights can be incorporated in the activity majorant before the criterion is checked. The root estimate then retains those weights. Holomorphic dependence on external fields or sources passes to the connected expansion on every domain where the same absolute majorant is valid.

For gauge-field fluctuation activities, apply the theorem after verifying (3.1) with the covariance, localization-domain count, source radii, boundary costs and required derivatives included in its majorant. Large-field histories use their separate weighted label estimate. These are the analytic conditions that we must verify in the multiscale application of the combinatorial convergence theorem.

4. The free curvature model: an exact calculation

This section computes the noncoincident correlations of the action density of a free curvature in four Euclidean dimensions. The calculation fixes a normalization and proves the cycle identities used in the short-distance comparison. It does not identify an interacting lattice measure with this model. That comparison, including its error, remains a separate estimate.

4.1. **A Gaussian field defined without a pointwise potential.** Fix a positive number g and an integer $d_G \geq 1$; for the adjoint color space of SU(2), $d_G = 3$. Coordinates in $\mathbb{R}^4$ are denoted by $x_1, \dots, x_4$. Our Fourier transform is

$$\widehat{f}(k) = \int_{\mathbb{R}^4} e^{-ik\cdot x} f(x)\, dx,$$
$$f(x) = (2\pi)^{-4} \int e^{ik\cdot x}\, \widehat{f}(k)\, dk.$$

Let G be the tempered distribution with Fourier transform $|k|^{-2}$. In four dimensions this multiplier is locally integrable at zero. Its inverse transform, away from zero, is

$$G(x) = \frac{1}{4\pi^2 |x|^2}.$$

One direct verification uses $|k|^{-2} = \int_0^\infty e^{-t|k|^2}\, dt$ and the product Gaussian integral. The inverse transform of $e^{-t|k|^2}$ is $(4\pi t)^{-2} e^{-|x|^2/(4t)}$. Integrating in t, with $u = |x|^2/(4t)$, gives $1/(4\pi^2|x|^2)$. The same identity holds as tempered distributions by Fubini: the integral $\int |\varphi(x)|\, |x|^{-2} dx$ is finite for every Schwartz function φ, controlling the absolute integral of the heat-kernel formula.

For real Schwartz functions define

$$(f, h)_G = (2\pi)^{-4} \int_{\mathbb{R}^4} \frac{\overline{\widehat{f}(k)}\widehat{h}(k)}{|k|^2}\, dk.$$

This is a positive real bilinear form. The integral is finite near zero because $\int_0^1 r^3 r^{-2} dr < \infty$, and is finite at infinity by Schwartz decay. Quotient by its null space and complete to a real Hilbert space $\mathcal{H}_G$. It is separable: the map $f \mapsto |k|^{-1}\widehat{f}(k)$ embeds it into a separable real form of $L^2(\mathbb{R}^4)$.

Choose an orthonormal basis (e_j) and independent real standard normal variables (N_j) on their countable product probability space. For $f \in \mathcal{H}_G$, the series

$$W(f) = \sum_j (f, e_j)_G N_j$$

converges in L^2 and defines a centered jointly Gaussian family with $\mathbb{E}[W(f)W(h)] = (f, h)_G$. Finite partial sums prove the covariance formula and joint Gaussianity; convergence of their characteristic functions proves it for the limit. Independent copies give families $A_\mu^a(f) = gW_\mu^a(f)$ for $1 \leq \mu \leq 4$, $1 \leq a \leq d_G$. Their covariance is

$$\mathbb{E}[A_\mu^a(f) A_\nu^b(h)] = g^2 \delta_{ab} \delta_{\mu\nu} (f, h)_G.$$

No value $A_\mu^a(x)$ at a point is used in this definition.

Define the curvature as the Gaussian distribution

$$F^a_{\mu\nu}(f) = -A^a_\nu(\partial_\mu f) + A^a_\mu(\partial_\nu f),$$
$$\mu < \nu.$$

There are six independent coordinate labels $P = (\mu, \nu)$, although the six curvature components need not be independent random variables. Differentiating the covariance as a distribution gives

$$\mathbb{E}[F^a_{\mu\nu}(x)F^b_{\rho\sigma}(y)] = g^2\delta_{ab}K_{\mu\nu,\rho\sigma}(x-y),$$

where the notation on the left denotes a distributional kernel and

$$K_{\mu\nu,\rho\sigma}(x) = -\delta_{\nu\sigma}\partial_\mu\partial_\rho G(x) + \delta_{\nu\rho}\partial_\mu\partial_\sigma G(x) + \delta_{\mu\sigma}\partial_\nu\partial_\rho G(x) - \delta_{\mu\rho}\partial_\nu\partial_\sigma G(x).$$

The signs follow from differentiating one argument at x and the other at y: $\partial_{y_\rho}G(x-y) = -\partial_{x_\rho}G(x-y)$. At nonzero separation this is an ordinary smooth matrix-valued function.

4.2. **The six-dimensional curvature matrix.** Give $\Lambda^2\mathbb{R}^4$ the Euclidean inner product for which $(e_\mu \wedge e_\nu)_{\mu<\nu}$ is an orthonormal basis. Put $r = |x|$, $u = x/r$, and $R_u = I - 2u \otimes u$. This R_u is the orthogonal reflection normal to u. Write $\Lambda^2 R_u$ for its induced operator on two-forms.

Proposition 4.1 (The curvature coefficient)**.** *For $x \neq 0$,*

$$K(x) = \frac{1}{\pi^2 r^4}\Lambda^2 R_u,$$
$$K(x)^{\mathsf{T}}K(x) = \frac{1}{\pi^4 r^8}I_6,$$
$$\operatorname{tr}(K(x)^{\mathsf{T}}K(x)) = \frac{6}{\pi^4 r^8}.$$

Proof. Twice differentiating G gives

$$\partial_\mu\partial_\rho G(x) = \frac{1}{2\pi^2 r^4}(4u_\mu u_\rho - \delta_{\mu\rho}).$$

At $u = e_4$ this Hessian is diagonal with entries $(-1,-1,-1,3)/(2\pi^2 r^4)$. Inserting it in the component formula for K shows that K is diagonal: it has eigenvalue $1/(\pi^2 r^4)$ on $e_1 \wedge e_2, e_1 \wedge e_3, e_2 \wedge e_3$, and eigenvalue $-1/(\pi^2 r^4)$ on $e_1 \wedge e_4, e_2 \wedge e_4, e_3 \wedge e_4$. This is exactly the action of $\Lambda^2 R_{e_4}$ times the stated scalar. Both the differentiated radial kernel and the claimed expression transform by conjugation under orthogonal changes of coordinates, proving the first identity for every u. Since $\Lambda^2 R_u$ is an orthogonal involution on a six-dimensional space, its squared norm matrix is I_6, giving the other two identities. □

Fix the standard orientation and let $J = *$ be the Hodge operator on two-forms. It satisfies $J^2 = I$. For an orthogonal map R,

$$J(\Lambda^2 R) = (\det R)(\Lambda^2 R)J.$$

To check the formula, use the defining identity $\alpha \wedge *\beta = \langle \alpha, \beta \rangle\, dx_1 \wedge \cdots \wedge dx_4$ and the fact that the top exterior power of R multiplies the oriented volume by $\det R$. Since $\det R_u = -1$, $JK(x) = -K(x)J$. Every product of an odd number of matrices $K(x_i)$ therefore anticommutes with J and has trace zero: conjugation by J preserves its trace while changing its sign.

4.3. **Regularized densities and connected cycle contractions.** Take a real smooth even function ρ supported in the unit ball with $\int \rho = 1$ and put $\rho_\varepsilon(x) = \varepsilon^{-4}\rho(x/\varepsilon)$. Smear every curvature component with its translate to define $F^a_{\varepsilon,P}(x)$. This is a Gaussian variable with a smooth covariance kernel. Its centered quadratic density is

$$q_\varepsilon(x) = \frac{1}{2}\sum_{a=1}^{d_G}\sum_{P=1}^{6}\big(F^a_{\varepsilon,P}(x)^2 - \mathbb{E}[F^a_{\varepsilon,P}(x)^2]\big).$$

On any compact set its covariance is continuous and bounded. The integral $q_\varepsilon(f) = \int f(x)q_\varepsilon(x)dx$ for compactly supported smooth f is defined by L^2 limits of Riemann sums. The continuous covariance makes those sums Cauchy. The Gaussian moment formula gives convergence of their moments of each fixed order as well: all the integrands are finite sums of products of bounded continuous covariances on a fixed compact set.

For clarity, first fix distinct points $x_1, \dots, x_n$ at a fixed ε. For one color let C_{ij} be the 6×6 covariance block of $F_\varepsilon(x_i)$ and $F_\varepsilon(x_j)$, including the factor g^2. Put $Z = \operatorname{diag}(z_1 I_6, \dots, z_n I_6)$ and let C be the full block covariance matrix. The Gaussian integral yields, near $z = 0$,

$$\log \mathbb{E}\exp\big(\textstyle\sum_i z_i q_\varepsilon(x_i)\big) = -\frac{d_G}{2}\log\det(I - CZ) - \frac{d_G}{2}\sum_i z_i \operatorname{tr} C_{ii}.$$

Here C is positive semidefinite; write the Gaussian vector as $C^{1/2}N$ and diagonalize the real symmetric quadratic form for small real z. The one-dimensional Gaussian integral proves the determinant identity there. Holomorphic continuation proves it in a complex neighborhood of zero. Sylvester's determinant identity identifies $\det(I - C^{1/2}ZC^{1/2})$ with $\det(I - CZ)$, including singular C.

The last term only subtracts the first cumulants. Expand the logarithm of the determinant by its convergent matrix power series. For $n \geq 2$ the joint cumulant is consequently

$$\kappa_n(q_\varepsilon(x_1), \dots, q_\varepsilon(x_n)) = \frac{d_G}{2n}\sum_{\sigma\in S_n}\operatorname{tr}\big(C_{\sigma(1)\sigma(2)}\cdots C_{\sigma(n)\sigma(1)}\big).$$

Indeed only the term of degree n in $\frac{d_G}{2}\sum_{m\geq 1}\operatorname{tr}((CZ)^m)/m$ contributes to the mixed derivative. Each permutation records the order of the n distinct vertices of the cycle. The factor $1/n$ removes its cyclic repetitions; the factor $1/2$ remains even when $n = 2$.

4.4. **The noncoincident limit and its consequences.** Let the test functions $f_1, \dots, f_n$ have pairwise disjoint compact supports, and let $d > 0$ be their minimum separation. The regularized off-diagonal kernel is $C_{ij} = g^2(K * \rho_\varepsilon * \rho_\varepsilon)(x_i - x_j)$. For $\varepsilon < d/4$ the convolution samples K only away from zero. It converges uniformly, with every fixed derivative, on the compact set of relevant differences. The cycle formula and ordinary dominated convergence therefore give a finite limit of the smeared connected correlations. Its kernel is

$$C_n(x_1, \dots, x_n) = \frac{d_G g^{2n}}{2n}\sum_{\sigma\in S_n}\operatorname{tr}\big(K(x_{\sigma(1)} - x_{\sigma(2)})\cdots K(x_{\sigma(n)} - x_{\sigma(1)})\big),$$
$$x_i \neq x_j.$$

This defines separated correlations. It does not assert convergence of $q_\varepsilon(f)$ in L^2 when the same test is used twice; coincidence singularities require a different distributional treatment.

Corollary 4.2 (Two points, odd cycles, and a positive four-point kernel)**.** *The limiting connected two-point kernel is*

$$C_2(x,y) = \frac{3d_G g^4}{\pi^4 |x-y|^8}.$$

Every odd connected kernel C_n with $n \geq 3$ vanishes at noncoincident arguments. If four distinct points are collinear, then

$$C_4(x_1,x_2,x_3,x_4) = \frac{6d_G g^8}{\pi^8} \sum_{\text{three unoriented cycles}} \prod_{(i,j) \text{ in the cycle}} |x_i - x_j|^{-4} > 0.$$

For four sufficiently small disjoint neighborhoods of such points the four-point kernel remains strictly positive throughout their product.

Proof. For two points, the permutation formula is $d_G g^4 \operatorname{tr}(K(x-y)K(y-x))/2$. The kernel is even and symmetric, so the curvature coefficient proposition proves the stated value. For odd n, every trace in the cycle sum vanishes by the Hodge anticommutation identity. For collinear points all difference directions give the same reflection $R_u = R_{-u}$. The product of four curvature matrices is the positive scalar product of their radial factors times $(\Lambda^2 R_u)^4 = I$. Every trace is therefore six times that scalar. The 24 permutations divide into three unoriented cycles, each represented eight times, canceling the factor $2n = 8$. Positivity on small neighborhoods follows from continuity and the strictly positive value at the chosen configuration. □

For nonnegative nonzero separated tests this gives

$$C_2(f,h) = \frac{3d_G g^4}{\pi^4} \iint \frac{f(x)h(y)}{|x-y|^8}\, dx\, dy > 0.$$

Thus the coefficient $3(N^2-1)/\pi^4$ used for SU(N) is the exact free coefficient in this normalization. For SU(2) it is $9/\pi^4$. Recovering this coefficient from the interacting lattice theory still requires its ultraviolet comparison and uniform remainder estimate. The four-point calculation also shows why non-Gaussianity of the scalar density alone cannot certify an interacting underlying gauge theory: this density already has a positive connected four-point kernel in an explicitly Gaussian model.

5. A uniform comparison for the free lattice curvature

This section proves a free-field input of the construction. The centered Gaussian exact lattice two-form that is standard on the exact-form subspace has covariance equal to the orthogonal projection onto that subspace. After multiplication by the physical factor a^{-4}, its entries converge away from coincidence to the continuum curvature kernel. We prove an error estimate uniform in the lattice spacing and the torus side, and compute the resulting separated quadratic-density coefficient. No interacting measure is compared with this Gaussian model here.

5.1. **Cochains, Fourier conventions, and the statement.** Let $a > 0$, $\ell > 0$, and $N = \ell/a \in \mathbb{N}$. The lattice is $\Lambda_{a,\ell} = a\mathbb{Z}^4/\ell\mathbb{Z}^4$. A one-form has coordinates $A_\mu(x)$, one per positive bond based at x. A two-form has coordinates $F_{\mu\nu}(x)$ for $\mu < \nu$, based at the lower vertex of its positively oriented plaquette. Both cochain spaces use the plain coordinate inner products. Define

$$(d_1 A)_{\mu\nu}(x) = a^{-1}\{A_\nu(x + ae_\mu) - A_\nu(x) - A_\mu(x + ae_\nu) + A_\mu(x)\}.$$

The factor a^{-1} does not affect the range. Let P_{ex} be the orthogonal projection onto $\operatorname{Ran} d_1$. It is the covariance of a centered Gaussian vector which is standard on that subspace. Potential gauge directions need not be inverted to define this curvature covariance. The constant two-forms are excluded because they are not in $\operatorname{Ran} d_1$.

Write $P = (\mu, \nu)$, $\mu < \nu$, and put $c_P = (e_\mu + e_\nu)/2$. The centre of the plaquette (x, P) is $x + ac_P$. The physical lattice kernel is

$$\mathsf{K}^{PQ}_{a,\ell}(x, y) = a^{-4} P_{\mathrm{ex}}((x, P), (y, Q)).$$

For a displacement on the torus, its length means distance to $\ell\mathbb{Z}^4$. Representatives are chosen in $[-\ell/2, \ell/2]^4$ when needed.

The continuum Fourier transform on $\mathbb{R}^4$ is

$$\widehat{f}(k) = \int e^{-ik\cdot x} f(x)\, dx,$$

$$f(x) = (2\pi)^{-4} \int e^{ik\cdot x} \widehat{f}(k)\, dk.$$

Let $G(x) = 1/(4\pi^2|x|^2)$ and define the continuum curvature distribution by

$$\begin{aligned}\mathsf{K}_{\mu\nu,\rho\sigma}(x) = {} & -\delta_{\nu\sigma}\partial_\mu\partial_\rho G(x) + \delta_{\nu\rho}\partial_\mu\partial_\sigma G(x) \\ & + \delta_{\mu\sigma}\partial_\nu\partial_\rho G(x) - \delta_{\mu\rho}\partial_\nu\partial_\sigma G(x).\end{aligned}$$

At $x \neq 0$ this is an ordinary smooth matrix. Its contact terms at zero are retained in its Fourier definition and are not squared in the statements below.

Theorem 5.1 (Free lattice curvature comparison)**.** *There is a numerical constant C such that, for every a, ℓ as above and every pair of plaquettes with centre displacement*

$$u = x - y + a(c_P - c_Q),$$

$$10a \le r = \operatorname{dist}(u, \ell\mathbb{Z}^4) \le \ell/10,$$

one has

$$\left|\mathsf{K}^{PQ}_{a,\ell}(x, y) - \mathsf{K}_{PQ}(u)\right| \le C r^{-4} \left\{(a/r)^{1/2} + (r/\ell)^4\right\}. \tag{5.1}$$

The continuum kernel on the right is evaluated at the short representative of u. The constants do not depend on the orientation labels, the lattice spacing, the number of sites, or the torus side.

The exponent $1/2$ is sufficient for convergence and is not claimed optimal. Keeping the barycentre phases in a sharper symbol expansion can improve the lattice power. The proof here deliberately keeps a periodic raw symbol and moves to barycentres only after the Fourier estimate.

5.2. **The exact projection symbol.** The torus frequencies are $k \in \delta\mathbb{Z}^4$, where $\delta = 2\pi/\ell$. Choose one Brillouin representative of each frequency modulo $(2\pi/a)\mathbb{Z}^4$ and denote this set by $\mathcal{Q}_a$. Its coordinates lie between $-\pi/a$ and π/a, with one endpoint omitted when necessary. Define

$$t_{a,\mu}(k) = a^{-1}(e^{iak_\mu} - 1),$$

$$\omega_a(k) = \sum_\mu |t_{a,\mu}(k)|^2 = 4a^{-2} \sum_\mu \sin^2(ak_\mu/2).$$

Let $B_a(k) : \mathbb{C}^4 \to \Lambda^2\mathbb{C}^4$ be

$$(B_a(k)v)_{\mu\nu} = t_{a,\mu}(k)v_\nu - t_{a,\nu}(k)v_\mu.$$

This is exactly the Fourier symbol of d_1. Direct multiplication gives

$$B_a^* B_a = \omega_a I - t_a t_a^*,$$
$$B_a t_a = 0.$$

Consequently $(B_a B_a^*)^2 = \omega_a B_a B_a^*$. For $k \neq 0$ in the Brillouin zone,

$$P_a(k) = B_a(k)B_a(k)^*/\omega_a(k)$$

is the orthogonal projection onto $\operatorname{Ran} B_a(k)$. Set $P_a(0) = 0$. Thus $\|P_a(k)\| \leq 1$ and its rank at nonzero frequency is three. At frequency zero the exact-form subspace is zero.

The unitary finite Fourier transform and its inverse are

$$\widehat{F}(k) = N^{-2} \sum_{x \in \Lambda_{a,\ell}} e^{-ik\cdot x} F(x),$$
$$F(x) = N^{-2} \sum_{k \in \mathcal{Q}_a} e^{ik\cdot x} \widehat{F}(k).$$

Therefore

$$a^{-4} P_{\mathrm{ex}}((x,P),(y,Q)) = \ell^{-4} \sum_{k \in \mathcal{Q}_a} e^{ik\cdot(x-y)} [P_a(k)]_{PQ}. \tag{5.2}$$

Indeed the projection entry before multiplication by a^{-4} has prefactor N^{-4}, and $a^{-4}N^{-4} = \ell^{-4}$.

For the continuum symbol replace t_a by $t_0(k) = ik$. Put

$$P_0(k) = B_0(k)B_0(k)^*/|k|^2$$
$$(k \neq 0),$$
$$P_0(0) = 0.$$

It is again a norm-one rank-three projection. With the specified Fourier convention its inverse transform is precisely the distribution K defined above. On the torus define the distribution

$$\mathsf{K}_\ell(z) = \ell^{-4} \sum_{k \in \delta\mathbb{Z}^4} e^{ik\cdot z} P_0(k). \tag{5.3}$$

The bounded Fourier coefficients define a periodic distribution. Below we show that it is smooth away from the origin and identify its finite-volume correction.

5.3. **Uniform derivative and counting estimates.** For each multi-index α there is a constant C_α such that, in the Brillouin zone away from zero,

$$\|\partial^\alpha P_a(k)\| \leq C_\alpha |k|^{-|\alpha|}. \tag{5.4}$$

The same estimate holds for P_0. To prove it, use $2|\sin(t/2)| \geq 2|t|/\pi$ for $|t| \leq \pi$. Hence $c|k|^2 \leq \omega_a(k) \leq |k|^2$ throughout the Brillouin zone. Differentiating the exponential gives $|\partial^\alpha t_a(k)| \leq C_\alpha |k|^{1-|\alpha|}$; for positive derivative order this uses $a|k| \leq 2\pi$. The product rule gives the analogous bounds of order $2-|\alpha|$ for ω_a and $B_a B_a^*$. Induction in the identity $\omega_a \omega_a^{-1} = 1$ bounds derivatives of the reciprocal. The quotient rule proves (5.4). All functions involved are periodic; near a Brillouin face the same bounds hold in periodic coordinates, where $|k|$ is comparable to a^{-1}.

For $a|k| \le c_0$, with a fixed small $c_0 > 0$, Taylor's formula gives $t_a(k) = ik + O(a|k|^2)$. The preceding denominator bounds imply

$$\|P_a(k) - P_0(k)\| \le Ca|k|. \tag{5.5}$$

For example, the numerator difference is $O(a|k|^3)$ and the denominator difference is $O(a|k|^3)$, while both denominators are comparable to $|k|^2$. Both symbols have been assigned zero at $k = 0$, so that frequency contributes no error.

We will use two elementary grid bounds. If $R \ge C\delta$ and $M > 4$, then

$$\ell^{-4}\#\{k \in \delta\mathbb{Z}^4 : |k| \le 2R\} \le CR^4, \tag{5.6}$$

$$\ell^{-4} \sum_{k \in \delta\mathbb{Z}^4} (R + |k|)^{-M} \le C_M R^{4-M}. \tag{5.7}$$

For a proof, place a cube of side δ about each grid point. The cubes are disjoint up to boundaries. If $R \ge C\delta$, the values of $R + |k|$ and $R + |q|$ on such a cube differ by bounded factors. The second sum is bounded by a constant times $(2\pi)^{-4} \int (R + |q|)^{-M} dq$, which is $C_M R^{4-M}$ after scaling. The first bound follows by enclosing the cubes in a ball of radius $3R$.

5.4. **Summation by parts on the frequency grid.** Fix a smooth cutoff χ equal to one on $|k| \le 1$ and zero on $|k| \ge 2$. Suppose

$$R \ge 2M\delta,$$

$$2R < \pi/(2a).$$

Extend $\chi(k/R)$ periodically on the frequency torus of side $2\pi/a$. It is smooth because its support stays away from the Brillouin faces. The high-frequency lattice symbol

$$H_a(k) = (1 - \chi(k/R))P_a(k)$$

is smooth on that frequency torus, including at zero. Write

$$\rho_a(k) = \operatorname{dist}(k, (2\pi/a)\mathbb{Z}^4).$$

The derivative estimates and the cutoff product rule give

$$\|\partial_\mu^M H_a(k)\| \le C_M (R + \rho_a(k))^{-M}.$$

Let $\Delta_\mu H(k) = H(k + \delta e_\mu) - H(k)$. The iterated fundamental theorem of calculus yields

$$\Delta_\mu^M H(k) = \delta^M \int_{[0,1]^M} \partial_\mu^M H(k + \delta(t_1 + \cdots + t_M)e_\mu)\, dt.$$

The torus distance ρ_a changes by at most $M\delta$ along this segment. Since $R \ge 2M\delta$, $R + \rho_a$ remains comparable to its value at k. Thus

$$\|\Delta_\mu^M H_a(k)\| \le C_M \delta^M (R + \rho_a(k))^{-M}. \tag{5.8}$$

Now let $z \in a\mathbb{Z}^4/\ell\mathbb{Z}^4$, $z \ne 0$, and choose a short representative. Its length r_z is at most ℓ, and some coordinate satisfies $|z_\mu| \ge r_z/2$. On $|z_\mu| \le \ell/2$,

$$|e^{-i\delta z_\mu} - 1| = 2|\sin(\pi z_\mu/\ell)| \ge 4|z_\mu|/\ell \ge 2r_z/\ell.$$

The phase $e^{ik\cdot z}$ is periodic on the frequency grid because z/a is integral. Finite summation by parts therefore gives the exact identity

$$(e^{-i\delta z_\mu}-1)^M \sum_{k\in\mathcal{Q}_a} e^{ik\cdot z}H_a(k) = \sum_{k\in\mathcal{Q}_a} e^{ik\cdot z}\Delta_\mu^M H_a(k).$$

There is no Brillouin boundary term. Apply (5.8) and (5.7), using $\rho_a(k)=|k|$ on the chosen representatives. The result is

$$\left\| \ell^{-4} \sum_{k\in\mathcal{Q}_a} e^{ik\cdot z}H_a(k) \right\| \le C_M r_z^{-M} R^{4-M}. \tag{5.9}$$

The powers of the torus side cancel: the inverse difference factor contributes $(\ell/r_z)^M$ and the frequency difference contributes $\delta^M=(2\pi/\ell)^M$.

The continuum torus tail obeys the same estimate. Here $H_0(k)=(1-\chi(k/R))P_0(k)$ on the infinite reciprocal grid. Its derivatives satisfy the same bounds with $|k|$ in place of ρ_a. The Mth differences are absolutely summable for $M>4$. Multiplication of a Fourier distribution by $(e^{-i\delta z_\mu}-1)^M$ shifts its Fourier coefficients and hence gives the same identity distributionally. The right side is an absolutely convergent continuous Fourier series. On the open set where the multiplying factor is nonzero, division by that smooth factor identifies the tail with an ordinary continuous function and proves

$$\|\mathsf{K}_\ell^{\mathrm{high}}(z)\| \le C_M r_z^{-M} R^{4-M}. \tag{5.10}$$

This distributional argument avoids assuming convergence of an undifferentiated order-zero Fourier series.

5.5. **Comparison with the continuum torus kernel.** On the common low-frequency support, the finite and infinite reciprocal grids agree. Equations (5.5) and (5.6) give

$$\left\| \ell^{-4} \sum_{k} e^{ik\cdot z}\chi(k/R)(P_a(k)-P_0(k)) \right\| \le CaR^5.$$

Combining this with the two high-frequency bounds yields, for $M>4$,

$$\|\mathsf{K}_{a,\ell}^{\mathrm{raw}}(z)-\mathsf{K}_\ell(z)\| \le C_M\{aR^5+r_z^{-M}R^{4-M}\}. \tag{5.11}$$

Here $\mathsf{K}_{a,\ell}^{\mathrm{raw}}$ is the matrix Fourier sum in (5.2) at the common plaquette-origin displacement z.

Take $M=9$ and choose

$$R=\frac{D}{r_z}(r_z/a)^{1/10},$$

where D is a fixed sufficiently large numerical constant. Because $r_z\le\ell$, this ensures $R\ge 18\delta$. For r_z/a larger than a fixed numerical threshold C_0, it also ensures $2aR<c_0$ and $2R<\pi/(2a)$. Both terms in (5.11) are then at most a constant times

$$r_z^{-4}(a/r_z)^{1/2}.$$

Indeed $aR^5=D^5r_z^{-4}(a/r_z)^{1/2}$ and $r_z^{-9}R^{-5}=D^{-5}r_z^{-4}(a/r_z)^{1/2}$.

For $1\le r_z/a\le C_0$, the lattice Fourier sum has norm at most a^{-4} because $\|P_a\|\le 1$. The continuum bound $\|\mathsf{K}_\ell(z)\|\le Cr_z^{-4}$, established in the next section, shows that their difference is

at most $C'a^{-4}$. Increasing the numerical constant by a factor depending only on C_0 bounds this by $Cr_z^{-4}(a/r_z)^{1/2}$. Consequently, for every nonzero raw lattice displacement,

$$\|\mathsf{K}_{a,\ell}^{\mathrm{raw}}(z) - \mathsf{K}_\ell(z)\| \le Cr_z^{-4}(a/r_z)^{1/2}. \tag{5.12}$$

All thresholds used here are fixed numbers. They impose no relation between the physical volume and the ultraviolet cutoff beyond $\ell/a \in \mathbb{N}$.

5.6. **The finite-volume correction and contact terms.** The continuum torus heat kernel is

$$p_t^\ell(x) = \sum_{m\in\mathbb{Z}^4} (4\pi t)^{-2} e^{-|x+m\ell|^2/(4t)} = \ell^{-4} \sum_{k\in\delta\mathbb{Z}^4} e^{-t|k|^2} e^{ik\cdot x}.$$

For completeness, the first expression is an absolutely convergent periodic sum for $t > 0$. Its Fourier coefficient at k is the ordinary Gaussian Fourier integral $\ell^{-4}e^{-t|k|^2}$, which proves the second expression. Define the mean-zero torus Green distribution by

$$G_\ell(x) = \int_0^\infty (p_t^\ell(x) - \ell^{-4})\, dt.$$

Its nonzero Fourier coefficient is $\ell^{-4}|k|^{-2}$ and its zero coefficient is zero. Applying the same second-derivative combination as in the definition of K gives (5.3).

We record the uniform local correction. At $\ell = 1$ and $|x| \le 1/4$,

$$\begin{aligned} G_1(x) - G(x) = {} & \int_0^1 \sum_{m\neq 0} (4\pi t)^{-2} e^{-|x+m|^2/(4t)}\, dt - 1 \\ & + \int_1^\infty (p_t^1(x) - 1)\, dt - \int_1^\infty (4\pi t)^{-2} e^{-|x|^2/(4t)}\, dt. \end{aligned}$$

Every fixed derivative of the first integral is uniformly bounded: differentiating a Gaussian gives a polynomial times a Gaussian, while $|x+m| \ge c|m|$ and $\sum_{m\neq 0} e^{-c|m|^2/t} \le Ce^{-c'/t}$ for $0 < t \le 1$. The resulting power of t^{-1} is integrable against this last exponential. For the second integral, its differentiated Fourier series is bounded by a constant times e^{-ct} for $t \ge 1$. Derivatives of the last integral are bounded by $C_\alpha t^{-2-|\alpha|/2}$ on this ball and are integrable. Thus $G_1 - G$ extends smoothly across zero with bounded derivatives of every fixed order.

Scaling gives $G_\ell(x) = \ell^{-2}G_1(x/\ell)$ and the corresponding homogeneity for G. In particular,

$$\|\mathsf{K}_\ell(z) - \mathsf{K}(z)\| \le C\ell^{-4} \quad (0 < |z| \le \ell/4). \tag{5.13}$$

The same argument gives an extra factor ℓ^{-1} for each additional derivative of the difference. On the remainder of the unit torus, away from zero, the heat-kernel formula and its derivatives converge uniformly on compact sets. Scaling and compactness therefore also give the global bounds

$$\|\partial^\alpha \mathsf{K}_\ell(z)\| \le C_\alpha r_z^{-4-|\alpha|}, \qquad z \neq 0.$$

These justify the estimate used in the bounded r_z/a regime above.

All these statements concern the kernels away from coincidence. The symbols have not discarded their order-zero distributional terms. For example, the trace of $P_0(k)$ is three for $k \neq 0$. Accordingly the trace of the torus kernel as a distribution is $3(\delta_{\mathrm{per}} - \ell^{-4})$; away from zero it is $-3\ell^{-4}$. The trace of the free noncoincident kernel is zero. This is consistent with (5.13); it would

be inconsistent to replace the full distribution by its noncoincident formula before taking Fourier coefficients.

5.7. **Moving to barycentres and proving the theorem.** Let u be the centre displacement in (5.1) and let $z = u - a(c_P - c_Q)$ be its raw-origin displacement. The shift has length at most a. Choose representatives so $|u| = r \le \ell/10$ and $|z - u| \le a$. Then

$$0.9r \le |z| \le 1.1r < \ell/4.$$

Use (5.12) for the raw kernel and (5.13) for the continuum torus kernel at z. The free kernel is homogeneous of degree -4 and smooth away from zero, so $\|\nabla \mathsf{K}(v)\| \le C|v|^{-5}$. The segment from z to u stays at distance at least $0.9r$ from zero. Hence

$$|\mathsf{K}_{PQ}(z) - \mathsf{K}_{PQ}(u)| \le Car^{-5}.$$

Combining the three differences proves

$$|\mathsf{K}_{a,\ell}^{PQ}(x,y) - \mathsf{K}_{PQ}(u)| \le C\{r^{-4}(a/r)^{1/2} + \ell^{-4} + ar^{-5}\}.$$

Since $a/r \le 1$, the last term is absorbed by the first, and $\ell^{-4} = r^{-4}(r/\ell)^4$. This proves (5.1).

The argument also proves convergence to K_ℓ uniformly on any fixed separated subset of the torus, when plaquette centres are used in both arguments. In this statement use the global derivative bound for K_ℓ to move from origins to centres. No assumption that a chosen separation is small relative to ℓ is needed for this fixed-torus convergence.

5.8. **The separated quadratic-density coefficient.** Let ξ_p^b, $1 \le b \le d_G$, be independent color copies of the Gaussian exact two-form, each with covariance $g^2 P_{\mathrm{ex}}$. For a test f on the torus define the finite-dimensional centered density

$$Q_a(f) = \frac{1}{2} \sum_{b=1}^{d_G} \sum_p f(x_p)\{(\xi_p^b)^2 - \mathbb{E}(\xi_p^b)^2\}.$$

Here x_p is the plaquette barycentre. Equivalently, $a^{-2}\xi_p^b$ is the physical curvature and the displayed sum includes its a^4 Riemann weight. For two tests of positive mutual separation, the Gaussian identity $\mathrm{Cov}(X^2, Y^2) = 2\,\mathrm{Cov}(X,Y)^2$ gives

$$\mathrm{Cov}(Q_a(f), Q_a(h)) = \frac{d_G g^4}{2}\, a^8 \sum_{p,q} f(x_p)h(x_q)\,|\mathsf{K}_{a,\ell}(p,q)|^2.$$

For real kernels the square here is the ordinary real square of the scalar entry. The uniform separated convergence just proved, followed by ordinary shifted-grid Riemann sums, gives

$$\lim_{a \downarrow 0} \mathrm{Cov}(Q_a(f), Q_a(h)) = \frac{d_G g^4}{2} \iint f(x)h(y) \sum_{P,Q} \mathsf{K}_{\ell,PQ}(x-y)^2\, dx\, dy.$$

For continuous compactly supported separated tests the Riemann-sum integrand is continuous after setting it to zero in a neighbourhood of the diagonal; for C^1 tests it is C^1 there. There is therefore no use of an integral of $|x-y|^{-8}$ through coincidence.

Now keep the tests in a fixed compact subset of $\mathbb{R}^4$ and let $\ell \to \infty$. The correction estimate makes $\mathsf{K}_\ell \to \mathsf{K}$ uniformly on the relevant separated differences. Direct differentiation of G shows

$$\mathsf{K}(x) = \frac{1}{\pi^2|x|^4}\Lambda^2(I - 2\widehat{x}\otimes\widehat{x}).$$

One can check the coefficient at $\widehat{x} = e_4$: the six eigenvalues are three copies of $1/(\pi^2|x|^4)$ and three copies of its negative. Orthogonal covariance proves the identity in every direction. The induced two-form reflection is orthogonal, and therefore

$$\sum_{P,Q} \mathsf{K}_{PQ}(x)^2 = \frac{6}{\pi^4|x|^8}.$$

It follows that the infinite-volume separated covariance is exactly

$$\frac{3d_G g^4}{\pi^4} \iint f(x)h(y)|x-y|^{-8}\,dx\,dy. \tag{5.14}$$

For SU(2), $d_G = 3$, so the coefficient is $9g^4/\pi^4$ in this convention. Higher separated Gaussian cumulants can likewise be passed through their finite cycle formulas using the same kernel convergence.

There is also a relative finite-volume statement for nonnegative pairs. Suppose every cross-distance between the supports lies between d and $C_{\mathrm{sh}}d$, and $C_{\mathrm{sh}}d \le \ell/10$. The correction estimate and the Frobenius norm of the free kernel give pointwise

$$\left|\sum_{P,Q} \mathsf{K}_{\ell,PQ}(v)^2 - \frac{6}{\pi^4|v|^8}\right| \le C\left(\frac{|v|}{\ell}\right)^4 |v|^{-8}.$$

Indeed the difference of the squares is bounded by $C|v|^{-4}\ell^{-4} + C\ell^{-8}$. Multiplication by $f(x)h(y) \ge 0$ and integration show that the limiting torus covariance differs from (5.14) by a relative error at most $C'_{\mathrm{sh}}(d/\ell)^4$, provided both tests are nonzero. The mesh error tends to zero for each such fixed test pair by the uniform kernel estimate and the Riemann-sum argument above. A rate relative to a varying family of tests would require their norms and nondegeneracy to be specified.

This proves the free covariance comparison and the free density coefficient. It asserts neither an L^2 limit of $Q_a(f)$ with the same test repeated nor an interacting short-distance asymptotic. Establishing the latter requires a comparison of the interacting measure with this Gaussian model and a remainder smaller than the leading term.

6. Periodic Gibbs states: exact layer elimination and link reflections

This section concerns one fixed lattice and one fixed positive coupling. No continuum limit or renormalization estimate is used. The conclusion quantifies over every periodic DLR state. In particular, the state is not assumed to arise as a limit of torus measures. The finite-dimensional transfer calculation below, together with a local elimination identity, is what permits that quantifier.

Set the lattice spacing equal to one, let $G = \mathrm{SU}(2)$, and use normalized Haar measure. On one orientation of the bonds of $\mathbb{Z}^4$ assign variables $U_b \in G$; reverse orientation means $U_{\bar{b}} = U_b^{-1}$. For an elementary plaquette set

$$a_p = 1 - \tfrac{1}{2}\operatorname{tr} U_p,$$
$$0 \le a_p \le 2.$$

The trace is real for $G = \mathrm{SU}(2)$. Fix $\beta > 0$. A DLR state is a probability measure whose conditional law on any finite bond set D, given all other bonds, has density proportional to $\exp[-\beta \sum_{p:p\cap D\neq\varnothing} a_p]$. A state is periodic when it is invariant under a subgroup of finite index of $\mathbb{Z}^4$. A link reflection is reflection in a plane $\{x_\mu = i + \frac{1}{2}\}$, with $i \in \mathbb{Z}$.

Theorem 6.1 (Link symmetry of every periodic Wilson state)**.** *Every periodic DLR state of the four-dimensional* SU(2) *Wilson interaction at $\beta > 0$ is invariant under every link reflection. Consequently every such state is invariant under $2\mathbb{Z}^4$.*

The proof uses positivity and strict positivity of different objects. The one-bond convolution operator is positive as an operator, with strictly positive Peter–Weyl eigenvalues. Its original kernel is strictly positive pointwise. An auxiliary operator inserted between two half-kernels will be a contraction, and need not be self-adjoint or positive. Confusing these conditions would invalidate the variational step.

A lower tangent valid for an arbitrary periodic state. Let ϕ be a real continuous local observable and let $m\mathbb{Z}^4$ be a subgroup of periods of a DLR state ν. For torus sides s divisible by m, write

$$W_s = \sum_{v\in m\mathbb{Z}^4/s\mathbb{Z}^4} \tau_v\phi,$$

$$Z_s(\phi) = \int e^{W_s - \beta\sum_p a_p}\, dU.$$

Here local observables are evaluated on the periodic extension of the torus configuration. Suppose

$$Z_s(\phi) \le Z_s(0)$$

(GL1) for all sufficiently large multiples s of m.

Then

(GL2) $$\nu(\phi) \le 0.$$

To prove this implication, take a cube D_s of sites of side s, and let B_s be the positively oriented bonds whose initial sites belong to the cube. Choose the representatives of the source anchors inside D_s, and denote their sum, evaluated with an arbitrary outside configuration, by $\widetilde{W}_s$. This is a finite local function; terms near the boundary can involve outside bonds. The number of anchors is exactly $M_s = (s/m)^4$. The DLR equation and Jensen's inequality give

$$\log \gamma_{B_s}(e^{\widetilde{W}_s})(U') \ge \gamma_{B_s}(\widetilde{W}_s)(U').$$

The conditional partition-function ratio on the left can be compared with $Z_s(\phi)/Z_s(0)$ uniformly in U'. Interior terms are identical under the identification of B_s with the torus bond coordinates. Only plaquettes and translates of ϕ in a fixed-width boundary layer can differ. Their number is at most $C_{\phi,m}s^3$, and each difference is bounded by $C(\beta + \|\phi\|_\infty)$. Thus

$$\log \frac{Z_s(\phi)}{Z_s(0)} \ge \nu(\widetilde{W}_s) - C_{\phi,m,\beta}s^3 = M_s\nu(\phi) - C_{\phi,m,\beta}s^3.$$

In the first inequality we integrated the conditional inequality against ν and used the DLR equation. The equality uses only $m\mathbb{Z}^4$ invariance. Divide by M_s and let $s \to \infty$ through multiples of m. Assumption (GL1) proves (GL2).

This argument needs neither differentiability of an infinite-volume pressure nor representation of ν by finite torus measures. The local source is fixed during the limit $s \to \infty$. If the source later depends on a parameter tending to zero, that parameter is kept fixed until the surface error has disappeared. Constant parts of ϕ cause no difficulty with the anchored definition of $\widetilde{W}_s$: they occur with exactly the same multiplicity in the box and torus source sums.

The one-bond convolution and its smooth conjugations. Define

$$c(g) = e^{-\beta(1-\frac{1}{2}\operatorname{tr} g)},$$

$$(CF)(w) = \int_G c(wv^{-1})F(v)\,dv.$$

The kernel is real, symmetric, and bounded below by $e^{-2\beta}$. Let Π_l be the Peter–Weyl projection corresponding to the irreducible representation of spin $l \in \frac{1}{2}\mathbb{N}_0$ and dimension $d_l = 2l+1$. Schur's lemma and the Weyl integration formula give

$$C = \sum_l c_l \Pi_l,$$

$$c_l = \frac{2e^{-\beta}}{\beta} I_{2l+1}(\beta) > 0. \tag{GL3}$$

For completeness, write g with eigenvalues e^{it}, e^{-it} and use $\chi_l(g) = \sin((2l+1)t)/\sin t$. The character coefficient is

$$d_l c_l = \frac{2e^{-\beta}}{\pi} \int_0^\pi e^{\beta\cos t} \sin t \sin(d_l t)\,dt = e^{-\beta}(I_{2l} - I_{2l+2})(\beta).$$

The Bessel recurrence gives (GL3). Its series

$$I_n(\beta) = \sum_{k\ge 0} \frac{(\beta/2)^{n+2k}}{k!(n+k)!}$$

has positive terms and implies factorial decay in n. Moreover,

$$\frac{I_{n+1}}{I_n} \le \frac{\beta}{2(n+1)},$$

$$\frac{I_n}{I_{n+1}} = \frac{2(n+1)}{\beta} + \frac{I_{n+2}}{I_{n+1}} \le C_\beta(1+n). \tag{GL4}$$

Consequently both ratios between neighboring c_l grow at most polynomially in l. Factorial decay also shows that C is trace class and that $C^{1/2}$ has a real symmetric smooth kernel.

For a finite block $\mathcal{B}$ of spatial bonds put

$$\mathcal{C} = C_{\mathcal{B}} = \bigotimes_{b\in\mathcal{B}} C,$$

$$c_{\boldsymbol{l}} = \prod_{b\in\mathcal{B}} c_{l_b},$$

$$\Pi_{\boldsymbol{l}} = \bigotimes_{b\in\mathcal{B}} \Pi_{l_b}.$$

Its kernel is $\mathcal{C}(u,v) = \prod_{b\in\mathcal{B}} c(u_b v_b^{-1})$. In particular it is strictly positive on the compact product space. It commutes with the simultaneous vertex gauge action on its two arguments.

Let f be a polynomial in matrix coordinates and their conjugates on $G^{\mathcal{B}}$. Define on $C^\infty(G^{\mathcal{B}})$

$$\widehat{f}F = \sum_{\boldsymbol{l},\boldsymbol{l}'} \frac{c_{\boldsymbol{l}}}{c_{\boldsymbol{l}'}} \Pi_{\boldsymbol{l}'} M_f \Pi_{\boldsymbol{l}} F, \tag{GL5}$$

where M_f denotes multiplication by f. This is a continuous map of the smooth Frechet space into itself. Indeed, if the degree in coordinate b is d_b, the Clebsch–Gordan rule makes the summand zero unless $|l_b - l'_b| \le d_b/2$. By (GL4), the remaining ratios are bounded by a fixed polynomial in the indices. Smooth functions have Peter–Weyl coefficients decreasing faster than every polynomial, as follows by applying powers of the Casimir. The finite-dimensional reproducing-kernel estimate bounds the supremum norm on a block by a polynomial times its L^2 norm. The same holds after any fixed number of derivatives. These facts give convergence of (GL5) with all derivatives and the asserted continuity.

The notation $\widehat{f} = \mathcal{C}^{-1} M_f \mathcal{C}$ is therefore legitimate on this smooth domain when interpreted by (GL5). It does not assert that $\mathcal{C}^{-1}$ acts on every smooth function or that $\widehat{f}$ is bounded on L^2. The formal L^2 adjoint on smooth functions is given by the same finite-band series with the coefficient ratios reversed and f replaced by $\overline{f}$. Those ratios also have polynomial growth, so this formal adjoint maps smooth functions continuously to smooth functions. In particular no unbounded inverse is applied outside its justified spectral domain in the following identity:

$$\mathcal{C}\widehat{f}F = M_f \mathcal{C} F. \tag{GL6}$$

It follows by multiplying (GL5) by $\mathcal{C}$ and summing the projections. In kernel form, for every w,

$$f(w) \int \mathcal{C}(v,w) F(v)\, dv = \int \mathcal{C}(v,w) (\widehat{f}F)(v)\, dv. \tag{GL7}$$

The equality is pointwise, since the series and both integrals define smooth functions. All these assertions also hold for smooth families depending on extra variables on a finite compact product: differentiation in those variables commutes with the series. The operator $\widehat{f}$ commutes with multiplication by functions of variables on which it does not act, and operators acting on disjoint variable sets commute.

A transfer contraction inequality. Fix a direction, called time, and take a cubic torus of side s. Let $\mathcal{E}$ be its spatial bonds in one layer. On $L^2(G^{\mathcal{E}})$, let P be vertex gauge averaging and let V be β times the spatial plaquette action in that layer. The operator

$$L = P C_{\mathcal{E}}^{1/2} e^{-V} C_{\mathcal{E}}^{1/2} P$$

is positive and trace class. The convolution, its square root, and e^{-V} all commute with P. Integrating the time-like bonds in the torus integral inserts P between successive layers. Using cyclicity of the trace gives

(GL8) $$Z_s(0) = \operatorname{Tr} L^s.$$

Here normalized Haar measure makes the gauge average an orthogonal projection without an additional volume factor.

Let Ξ be a real bounded operator on $L^2(G^{\mathcal{B}})$ commuting with the block vertex gauge action, and suppose $\|\Xi\| \le 1$. Suppose the kernel

$$k_\Xi = \text{kernel of } \mathcal{C}^{1/2}\Xi\mathcal{C}^{1/2}$$

is continuous and strictly positive pointwise. In two adjacent layers, use transported spatial variables u, v and set

(GL9) $$\phi_\Xi(u,v) = \log \frac{k_\Xi(u,v)}{\mathcal{C}(u,v)}.$$

This is a real continuous gauge-invariant local observable.

Place translates of the block so far apart that blocks in a single time step have disjoint bond sets. Multiplication by e^{ϕ_Ξ} replaces the corresponding factor $\mathcal{C}$ by $\mathcal{C}^{1/2}\Xi\mathcal{C}^{1/2}$. Thus the full kernel in step j is $C_{\mathcal{E}}^{1/2}\Xi_j C_{\mathcal{E}}^{1/2}$, where Ξ_j is a tensor product of copies of Ξ and identities and has norm at most one. Gauge averaging the time-like bonds gives the trace of the product of $e^{-V}C_{\mathcal{E}}^{1/2}\Xi_j C_{\mathcal{E}}^{1/2}P$. Moving the first half-kernel and layer weight cyclically to the end, and using commutation with P, rewrites this trace as

$$\operatorname{Tr}(\Xi_1 L \Xi_2 L \cdots \Xi_s L).$$

The unperturbed steps have $\Xi_j = 1$. Schatten Holder gives

(GL10) $$|\operatorname{Tr}(\Xi_1 L \cdots \Xi_s L)| \le \prod_{j=1}^{s} \|\Xi_j L\|_s \le \|L\|_s^s = \operatorname{Tr} L^s.$$

The last equality uses $L \ge 0$. The modified integral is positive because every replaced kernel is positive pointwise. Therefore (GL10) is precisely the partition-function inequality (GL1) for a sufficiently sparse periodic array of ϕ_Ξ.

Given an arbitrary periodic state ν, choose m large enough that $m\mathbb{Z}^4$ is a subgroup of its periods and the copies are disjoint. Equations (GL1)–(GL2) imply

(GL11) $$\nu(\phi_\Xi) \le 0.$$

The norm contraction and the lower tangent are the only ingredients of this conclusion. No self-adjointness of Ξ and no differentiability of a limiting pressure has been used.

Two adjacent layers cannot carry an antisymmetric mean. Let $Y(u,v)$ be a real continuous function on $G^{\mathcal{B}} \times G^{\mathcal{B}}$ that is invariant under simultaneous vertex gauge transformations and antisymmetric: $Y(v,u) = -Y(u,v)$. Then every periodic DLR state satisfies

(GL12) $$\nu(Y(U_i, U_{i+1})) = 0.$$

To prove this, first suppose $K(u,v) = Y(u,v)\mathcal{C}(u,v)$ is a polynomial. Its integral operator has finite Peter–Weyl support. Antisymmetry makes K a real skew-adjoint operator. On that finite-dimensional support define

$$D = \mathcal{C}^{-1/2} K \mathcal{C}^{-1/2}.$$

Every eigenvalue $c_{\boldsymbol{l}}$ is strictly positive, so the definition is valid. Extend D by zero on the orthogonal complement. It is a finite-rank real skew-adjoint operator, commutes with gauge transformations, and $\Xi_\varepsilon = e^{\varepsilon D}$ is a real orthogonal operator for real ε of either sign. In particular $\|\Xi_\varepsilon\| = 1$.

The kernel difference from $\mathcal{C}$ has finite Peter–Weyl support and admits, uniformly in (u, v),

$$k_{\Xi_\varepsilon}(u, v) = \mathcal{C}(u, v) + \varepsilon K(u, v) + O(\varepsilon^2).$$

Since $\mathcal{C} \geq e^{-2\beta|\mathcal{B}|}$, this kernel is strictly positive for sufficiently small $|\varepsilon|$. Consequently $\phi_{\Xi_\varepsilon} = \varepsilon Y + O(\varepsilon^2)$ uniformly. Apply (GL11), first at each fixed nonzero ε after taking the thermodynamic limit in the lower-tangent argument. For both signs it gives $\varepsilon\nu(Y) \leq C\varepsilon^2$. Letting $\varepsilon \to 0$ proves (GL12) in this case.

For general Y, approximate $Y\mathcal{C}$ uniformly by polynomials on the compact product group. Antisymmetrize each approximant and average it over the simultaneous gauge group. These operations preserve polynomiality, because a finite matrix-coordinate degree spans a finite-dimensional representation, and preserve the uniform approximation. Divide by the strictly positive kernel $\mathcal{C}$. The resulting functions Y_n converge uniformly to Y and have the already proved property $\nu(Y_n) = 0$. This proves (GL12). The argument also specifies why injectivity of $\mathcal{C}$, rather than merely its operator positivity, is needed.

The exact one-layer DLR elimination identity. Fix a finite stack of layers $[a, b]$. Write $T_{j,y}$ for the time-like bond from (j, y) to $(j + 1, y)$ and $\mathring{U}_{j,e}$ for a spatial bond in layer j. Define

$$g_{j,y} = T_{a,y} \cdots T_{j-1,y},$$

$$U_{j,\langle y,y'\rangle} = g_{j,y} \mathring{U}_{j,\langle y,y'\rangle} g_{j,y'}^{-1}. \tag{GL13}$$

These are temporal-gauge coordinates on the stack, with the empty product at $j = a$. A gauge-invariant local function of the stack is exactly a function of the U_j invariant under one common vertex gauge action in all layers. This follows by setting the time-like bonds to the identity successively. For any fixed finite observable only finitely many vertices are involved, so only finite-support gauge transformations are needed.

Let $a < j < b$ and resample the spatial bonds $\{j\} \times \mathcal{B}$, leaving every other original bond conditioned. The matrices $g_{j,y}$ use only time-like bonds and hence are fixed during this resampling. The substitution $\mathring{U}_j \mapsto v = U_j|_{\mathcal{B}}$ preserves Haar measure. The two time-like plaquettes attached to each resampled spatial bond produce the factors below; all spatial plaquettes that meet the block produce the finite potential $V_j(v; \eta)$, where η denotes the other spatial variables of layer j. Thus the conditional density is exactly

$$\frac{1}{z(w, w', \eta)} \mathcal{C}(w, v) e^{-V_j(v;\eta)} \mathcal{C}(v, w')\, dv,$$

$$w = U_{j-1}|_{\mathcal{B}},$$

$$w' = U_{j+1}|_{\mathcal{B}}. \tag{GL14}$$

No plaquette meeting the resampled bonds is omitted: it is either spatial in layer j or one of the two time-like plaquettes of that bond. The potential can involve spatial neighbors outside $\mathcal{B}$, but they lie in the same layer and in a finite enlargement of the block.

Let f be a real polynomial on $G^{\mathcal{B}}$ and let Φ be smooth in finitely many transported variables from layers j and above, and independent of the spatial variables of layer $j-1$. Define

$$\Phi^\flat = \frac{e^{V_j(v;\eta)}}{\mathcal{C}(v,w')}\widehat{f}_v\Big[e^{-V_j(v;\eta)}\mathcal{C}(v,w')\Phi\Big]. \tag{GL15}$$

All factors are smooth on a finite compact product, and the denominator is strictly positive. Equation (GL5) therefore proves that $\Phi^\flat$ is smooth. Its new dependence is confined to layer j and the already present higher layers; it acquires no spatial variable in layer $j-1$.

Every DLR state, periodic or not, satisfies the exact identity

$$\nu\big(f(U_{j-1}|_{\mathcal{B}})\Phi\big) = \nu(\Phi^\flat). \tag{GL16}$$

Indeed, condition on all bonds except the specified block in layer j. The conditional numerator on the left is

$$f(w)\int \mathcal{C}(v,w)e^{-V_j(v;\eta)}\mathcal{C}(v,w')\Phi\,dv.$$

Apply (GL7) to the smooth bracketed function. Substitution of (GL15) identifies the result with the numerator of the conditional expectation on the right. The same normalizer $z(w,w',\eta)$ occurs on both sides; it is not differentiated or canceled through an operator. Integration of this conditional identity gives (GL16). Reflection of the construction yields the corresponding identity that eliminates an upper outer layer.

Removing two outer layers and preserving reflection. Let θ be reflection about the middle of a stack with $2k+2$ layers, numbered $i-k,\dots,i+k+1$. Let X be a real gauge-invariant polynomial observable in that stack. In temporal coordinates, a finite sum of products writes it as

$$X = \sum_\alpha f_\alpha(w)h_\alpha(z)\Phi_\alpha,$$

where $w = U_{i-k}|_{\mathcal{B}}$ and $z = U_{i+k+1}|_{\mathcal{B}}$ are the outer layers and Φ_α depends on the inner layers. Take one finite block containing all the variables that occur in this decomposition. The potentials used during elimination can enlarge this block spatially, but only finitely and only in the inner layers.

Use (GL16) to remove $f_\alpha(w)$ through the first inner layer $u = U_{i-k+1}|_{\mathcal{B}}$, and then its reflected version to remove $h_\alpha(z)$ through the last inner layer $v = U_{i+k}|_{\mathcal{B}}$. The result is

$$\nu(X) = \nu(EX),$$
$$EX = \sum_\alpha \Upsilon^{-1}\widehat{f}_{\alpha,u}\widehat{h}_{\alpha,v}(\Upsilon\Phi_\alpha). \tag{GL17}$$

When the two inner layers are not adjacent, Υ is the product of their two factors e^{-V} and their bond kernels toward the remaining interior. When $k=1$ they are adjacent and

$$\Upsilon = e^{-V(u;\eta)}e^{-V(v;\eta')}\mathcal{C}(u,v). \tag{GL18}$$

The common bond kernel occurs once, not twice. To see the cancellation directly, the first elimination introduces $e^{V(u)}/\mathcal{C}(u,v)$ outside its conjugated operator. In the upper elimination the conditional numerator contains $e^{-V(v)}\mathcal{C}(u,v)$, canceling that denominator before the second operator acts. The operators act on distinct variables and commute. Moving the remaining

independent factors through them gives exactly (GL18). This calculation is also a check that normalizing factors from different conditional laws have not been interchanged.

Let R denote literal reversal of the array of transported layer coordinates. An actual reflection of the original bond configuration induces R followed by a common vertex gauge transformation arising from the change of temporal-gauge base. The spatial potential is the same function of its own layer variables at both ends, and $\mathcal{C}(u,v)$ is symmetric. Reflection interchanges u and v, including all their spatial neighbors, and preserves Υ. It also sends $\widehat{f}_u$ to $\widehat{f}_v$. Consequently coordinate elimination commutes with literal reversal:

$$E(RX) = R(EX). \tag{GL19}$$

The output need not be a polynomial, but it is a smooth function of finitely many inner-layer variables.

Every DLR state is invariant under finite-support vertex gauge transformations. This follows by applying its conditional law to the finite set of bonds incident to the transformed vertices and changing variables using Haar invariance and plaquette conjugacy. We may therefore replace EX in expectation by its average under the common temporal-gauge action; denote it by $\mathcal{A}$. This averaging commutes with R. Once averaged, the choice of the lowest layer in (GL13) disappears: changing the base from $i-k$ to $i-k+1$ conjugates all remaining transported variables by one common vertex gauge transformation. The averaged function is invariant under that transformation, even when its values are the actual time-like transport matrices. This last statement is a pointwise identity of invariant functions, and does not invoke invariance of the state under a field-dependent change of variables. Thus the averaged output is a continuous gauge-invariant observable of the smaller stack of $2k$ layers. Since X is gauge invariant, its actual reflection is represented by RX in temporal coordinates. After averaging the common gauge factor on the output also disappears. Thus the identity needed on the original bond space is $\mathcal{A}E(\theta X) = \theta\mathcal{A}EX$.

Proof of the link-reflection theorem. Fix the reflection θ in $x_\mu = i + \frac{1}{2}$. We prove by induction on k that every real gauge-invariant polynomial in the stack $[i-k+1, i+k]$ has equal expectation to its reflection.

For $k=1$, temporal gauge writes the observable as $\Psi(U_i, U_{i+1})$. Reflection interchanges the two arguments, up to a common gauge transformation, which does not affect Ψ. The difference from its reflection is the function $Y(u,v) = \Psi(u,v) - \Psi(v,u)$, to which (GL12) applies. Hence the expectations agree.

Assume the assertion for k. For a polynomial X in the stack with $2k+2$ layers, (GL17)–(GL19) and gauge averaging reduce the comparison of $\nu(X)$ and $\nu(\theta X)$ to the same comparison for a continuous gauge-invariant function of the smaller stack. Invariant polynomials are uniformly dense in the invariant continuous functions on a finite product of compact matrix groups: first apply Stone–Weierstrass to matrix coordinates and their conjugates, then average the approximants over the finite vertex gauge group. The induction hypothesis therefore applies to this continuous output. It gives $\nu(\mathcal{A}EX) = \nu(\theta\mathcal{A}EX)$ and proves the next step.

Every gauge-invariant continuous local function fits in a finite symmetric stack and is uniformly approximated as above. We have proved equality of its expectation and that of its reflection. For a general continuous local F, gauge average over the finite set of vertices relevant to its support. Gauge invariance of both ν and its reflected state shows that both expectations of F equal the expectations of that gauge average. Thus $\nu(F) = \nu(\theta F)$ for all continuous local F, which determine the probability measure on the compact product configuration space. The state is reflection invariant. The direction μ and integer i were arbitrary. Composing reflections in two successive link planes gives translation by $2e_\mu$, proving the stated $2\mathbb{Z}^4$ invariance. □

Passage to all mirrors. The common period is the input to the independent fold-and-separation proof in Section 7. That section supplies adapted diagonal split tori, finite crossing kernels, the full chessboard inequality, and the arbitrary-state separation argument. The link theorem above does not require those later ingredients.

7. All lattice mirrors for every periodic Wilson state

The result in this section concerns the complete algebra of bonds on a closed side of a mirror. Its proof does not identify an arbitrary Gibbs state with a limit of unperturbed torus measures. Instead, a pressure inequality produces, for each separating observable, a suitable limit of *perturbed* torus measures. The common period supplied by link reflection invariance is what makes the separation argument work.

We use the following result proved in the preceding link-symmetry argument: every periodic DLR state of the four-dimensional SU(2) Wilson specification, at $\beta > 0$, is invariant under all coordinate link reflections and consequently under $P = 2\mathbb{Z}^4$. No uniqueness or extremality is assumed here.

Let $\Omega = \mathrm{SU}(2)^{\mathcal{B}}$, with one group variable for each positively oriented nearest-neighbour bond; reversing a bond inverts its variable. Put

$$a_p(U) = 1 - \tfrac{1}{2}\operatorname{tr} U_p,$$

$$c(g) = \exp[-\beta(1 - \tfrac{1}{2}\operatorname{tr} g)].$$

The trace is real for SU(2), and $0 \le a_p \le 2$. The product topology makes Ω compact and metrizable. A state is periodic if it is invariant under some finite-index subgroup of $\mathbb{Z}^4$. Denote by $\mathcal{G}_\beta^P$ the compact convex set of Wilson DLR states invariant under P.

The normals considered below are

$$\mathcal{N} = \{e_\mu : 1 \le \mu \le 4\} \cup \{e_\mu + e_\lambda, e_\mu - e_\lambda : 1 \le \mu < \lambda \le 4\}.$$

For $n \in \mathcal{N}$ write $q = |n|^2$ and

$$\theta_c x = x - 2(n \cdot x - c)n/q.$$

The admissible positions are $c \in \frac{1}{2}\mathbb{Z}$ when $q = 1$ and $c \in \mathbb{Z}$ when $q = 2$. These and only these positions in the stated families give lattice reflections. Let $\mathcal{A}_c^+$ be the bounded Borel functions of bonds both of whose endpoints satisfy $n \cdot x \ge c$, and define $\mathcal{A}_c^-$ similarly. The antilinear reflection is $\Theta_c F = \overline{F \circ \theta_c^*}$.

Theorem 7.1 (All-mirror positivity of periodic states)**.** *Every periodic Wilson DLR state at $\beta > 0$ is invariant under every admissible reflection θ_c in the sixteen families above. For every such mirror,*

$$\nu(\Theta_c F\, F) \ge 0$$

$$(F \in \mathcal{A}_c^+ \text{ or } F \in \mathcal{A}_c^-). \tag{AM1}$$

In particular, the conclusion applies to every bounded observable on any finite stack contained in either closed half-space, regardless of the number of layers in that stack.

Composing the reflections in $x_\mu = 0$ and $x_\mu = 1/2$ gives translation by one lattice unit. Coordinate reflections give sign changes, and reflection in $x_\mu - x_\lambda = 0$ exchanges two coordinates. Thus every such state is invariant under all lattice translations and the full signed-permutation

group. At $\beta = 0$, the DLR equations directly force every finite-bond marginal to be product Haar; that unique product state has the same properties.

Positive crossing kernels on split tori. Write an element of SU(2) as a unit quaternion $s(g) \in S^3 \subset \mathbb{R}^4$. Then

$$\tfrac{1}{2}\operatorname{tr}(gh^{-1}) = s(g) \cdot s(h).$$

Consequently

$$c(gh^{-1}) = e^{-\beta} \sum_{m=0}^{\infty} \frac{\beta^m}{m!} \sum_{i_1,\dots,i_m=1}^{4} \prod_{j=1}^{m} s_{i_j}(g) \prod_{j=1}^{m} s_{i_j}(h). \tag{AM2}$$

Every coefficient is nonnegative. Each feature has absolute value at most one, and the sum of the coefficients is $e^{3\beta}$, so the series converges absolutely and uniformly. Tensor products of finitely many such expansions are again uniformly convergent positive Gram expansions. The coarse total coefficient bound is immaterial; each torus used below is finite.

A torus is called split for the mirror normal n if its period lattice has the form

$$\Gamma = \Gamma_{\parallel} \oplus \mathbb{Z}(2zn/q),$$
$$\Gamma_{\parallel} \subset n^{\perp}, \tag{AM3}$$

where $\Gamma_{\parallel}$ has rank three, $2zn/q$ is a lattice vector, and z is a sufficiently large integer. The reflection has two fixed cuts, at heights c and $c + z$, on this torus. The two intervening closed halves are exchanged. All tori below have arbitrarily large shortest nonzero period, so that no small-quotient ambiguity affects plaquettes.

We verify reflection positivity on these tori for all bounded half-space functions. Partition the bond variables into the interiors of the two halves, the bonds lying on the cuts, and, when present, the bonds crossing a cut. A coordinate site cut has no plaquette with vertices strictly on both sides. Conditional on its boundary bonds, the action is a sum of a positive-side action, its reflected action, and an action on the boundary. Its reflection form is thus an integral of absolute squares.

At a diagonal cut, signed permutations reduce the geometry to $n = e_1 + e_2$, $c \in \mathbb{Z}$. A bond changes the height $x_1 + x_2$ by 0 or 1 in absolute value, so no bond crosses from a strictly negative height to a strictly positive one. The only plaquettes meeting both strict sides are in the $(1, 2)$ plane, with heights $c-1, c, c, c+1$. Each is cut into two two-bond paths between the two vertices at height c. Orienting both paths in the same direction expresses its Wilson factor as $c(P_+P_-^{-1})$. Reflection interchanges the paths, and (AM2) supplies the required Gram expansion. Plaquettes in all other planes lie in a closed half. The bonds on the mirror, in directions 3 and 4, are fixed pointwise. Conditioning on them and multiplying the crossing expansions again gives an integral of squares. The same argument applies at the second cut.

For a coordinate link cut there are crossing bonds. Integrate them before evaluating a half-space function, which cannot depend on them. Let X and Y be the collections of bonds parallel to the cut in its two adjacent layers. The product crossing kernel before integration is

$$C(X, Y) = \prod_b c(X_b Y_b^{-1}).$$

It is a positive kernel by (AM2). If $T = (T_x)$ is a collection of group elements at the vertices of one boundary layer, set $(T \cdot X)_{xy} = T_x X_{xy} T_y^{-1}$. Let P_0 be Haar averaging under this action. Simultaneous gauge covariance gives $C(T \cdot X, T \cdot Y) = C(X, Y)$, since each plaquette trace changes

by conjugation. Hence the associated integral operator commutes with the orthogonal projection P_0. Integration of the crossing bonds gives the kernel of

$$P_0C = CP_0 = P_0CP_0 \geq 0. \tag{AM4}$$

Equivalently, in each Gram feature of the product expansion one may Haar-average both its X argument and its Y argument. Absolute uniform convergence justifies this operation. The crossing bonds at the two cuts are disjoint, so the two constructions multiply. This argument places no gauge-invariance condition on the test function.

In all three geometries the resulting reflection form has the form

$$\int \Theta_c F' F\, e^{-\beta \sum_p a_p}\, dU = \sum_\gamma b_\gamma \int \overline{I_\gamma(F';Y)} I_\gamma(F;Y)\, d\lambda(Y),$$

$$b_\gamma \geq 0, \tag{AM5}$$

where Y denotes the fixed boundary variables; in the link case Y is absent. The functions I_γ integrate over one half, with its Boltzmann factor and the corresponding Gram feature. The boundary Boltzmann factor is included in the positive measure λ. Fubini's theorem follows from the coefficient summability above. This proves the Cauchy–Schwarz inequality for the reflection form and reflection positivity for all bounded half-space functions. Haar invariance, orientation reversal, and invariance of the plaquette trace also prove reflection invariance of the finite measure.

If E is a bounded real function on one closed half, multiplying the density by $e^{E+\theta_c E}$ preserves both properties: insert Fe^E in (AM5). This observation will apply to folded perturbations.

A finite one-direction chessboard proof. Fix parallel slab faces at heights $c + kH$, where $H \geq 2$ is an integer. Use a split torus with normal height period $2MH$, $M \geq 1$. Its $2M$ slabs are indexed by $\mathbb{Z}/(2M)$. The group generated by reflection in the slab faces acts simply transitively on the slabs: its translations change the index by an even integer and its reflections send k to $2j - 1 - k$. Thus a function on any one slab has a well-defined copy on every other slab obtained using this group. Boundary bonds may belong to two closed slabs; this causes no problem, since the products are functions of the same boundary variables and (AM5) permits those variables in both half-algebras.

For positive bounded slab functions $f_0, \ldots, f_{2M-1}$ with strictly positive infima, denote by $Z^{\mathrm{dis}}(f_j)$ the partition function obtained by copying f_j into every slab. We claim

$$\int \prod_{j=0}^{2M-1} f_j\, e^{-\beta \sum_p a_p}\, dU \leq \prod_{j=0}^{2M-1} Z^{\mathrm{dis}}(f_j)^{1/(2M)}. \tag{AM6}$$

Here is a proof valid for every M, without a power-of-two restriction. Transport the $2M$ functions to one reference slab and retain their labels. For any filling $\sigma : \mathbb{Z}/(2M) \to \{0, \ldots, 2M-1\}$, let $Z(\sigma)$ be the partition function obtained by placing profile $\sigma(k)$ in slab k. Put

$$w_j = Z^{\mathrm{dis}}(f_j)^{1/(2M)},$$
$$T(\sigma) = Z(\sigma) \Big/ \prod_k w_{\sigma(k)}.$$

All these numbers are positive, and every constant filling has $T = 1$. Reflection Cauchy–Schwarz at any pair of opposite slab faces gives

(AM7) $$T(\sigma)^2 \le T(\sigma_+)T(\sigma_-),$$

where σ_+ retains one semicircle of labels and reflects it into the other, and σ_- does the reverse. Indeed, the unnormalized inequality is (AM5); the two denominators together contain each original denominator factor twice.

There are finitely many fillings. If σ maximizes T, (AM7) forces both reflected fillings to maximize it as well. Take a cyclic constant run of length l in a maximizing filling and reflect at the face at the start of that run. If $l \le M$, retaining the half beginning there gives a maximizing filling with a run of length $2l$. If $l \ge M$, retain a semicircle inside the run and reflect it; the result is constant. Iterating therefore produces a constant maximizing filling. The maximum is one. Applying this to the original filling proves (AM6).

This argument uses reflection positivity only for finite tori. It does not infer the positivity of an arbitrary DLR state from the positivity of one adjacent-layer kernel.

A lower tangent for an arbitrary periodic DLR state. Let Q be a fixed finite-index period subgroup, let A be a real continuous local observable, and form the locally finite interaction array $\mathbb{A} = (\tau_v A)_{v\in Q}$. Write $\kappa = [\mathbb{Z}^4 : Q]^{-1}$. Let $\Gamma_i \subset Q$ be rectangular adapted period lattices with every side tending to infinity. On the corresponding tori put

$$Z_i(t\mathbb{A}) = \int \exp\left[t \sum_{v\in Q/\Gamma_i} (\tau_v A)^{\Gamma_i} - \beta \sum_p a_p\right] dU.$$

For every Q-invariant Wilson DLR state ν and every fixed real t,

(AM8) $$\frac{1}{|D_i|} \log \frac{Z_i(t\mathbb{A})}{Z_i(0)} \ge t\kappa\nu(A) - o(1).$$

The error may depend on A, Q, β, t, but is uniform in ν.

For completeness, choose a fundamental parallelepiped D_i and integrate the positively oriented bonds anchored in it, using arbitrary external bonds as boundary conditions. Let W_i sum the source copies whose entire support is interior to this bond set. Conditional Jensen and the DLR equation give

$$\log \frac{Z^t_{D_i}(\eta)}{Z^0_{D_i}(\eta)} \ge t\gamma_{D_i} W_i(\eta),$$

$$\int \gamma_{D_i} W_i \, d\nu = \nu(W_i).$$

The open and torus Hamiltonians differ only in a boundary layer of fixed thickness, determined by the range of A and the plaquettes. Each energy term is bounded; each bond is met by only a fixed finite number of array terms. Thus their logarithmic partition ratios differ by at most $C_A(\beta + |t|\|A\|)|\partial_R D_i|$, uniformly in η. The number of omitted array anchors is at most $C_A|\partial_R D_i|$. Since the remaining Q-translates all have expectation $\nu(A)$, integrating the Jensen bound proves (AM8), with an error bounded by a constant times $|\partial_R D_i|/|D_i|$. This ratio tends to zero for the adapted rectangular tori, including the diagonal ones. No differentiability of infinite-volume pressure is used.

Fold domination with a fixed common period. Fix one admissible mirror (n, c). Choose an even positive integer h and set

$$w = hn \in P,$$
$$H = n \cdot w = hq,$$
$$P_{\parallel} = P \cap n^{\perp},$$
$$Q = P_{\parallel} \oplus \mathbb{Z}w,$$
$$Q_f = P_{\parallel} \oplus \mathbb{Z}(2w). \tag{AM9}$$

Both Q and Q_f are finite-index subgroups of P and $[Q : Q_f] = 2$. The tangent lattice $P_{\parallel}$ has an orthogonal integer basis: for a coordinate normal use the three other vectors $2e_j$; for $e_1 \pm e_2$ use $2(e_1 \mp e_2), 2e_3, 2e_4$, with the corresponding signed permutations for the other cases. Thus

$$\Gamma_{M,J} = JP_{\parallel} \oplus \mathbb{Z}(2Mw) \tag{AM10}$$

is an adapted rectangular split torus. Its shortest period tends to infinity as $M, J \to \infty$.

Assume for the moment that A is supported in the slab $c \le n \cdot x \le c + H$. The unfolded array contains all its translates by Q. The folded array contains translates of A in the even slabs and the reflected copies $\theta_{c+H}A$ in the odd slabs, with period Q_f. It is invariant under reflection at every face $c + kH$. In (AM6) take

$$f_k = \exp\left[t \sum_{v \in P_{\parallel}/JP_{\parallel}} (\tau_{v+kw}A)^{\Gamma_{M,J}}\right],$$
$$t > 0.$$

Their product is the unfolded perturbation. Their disseminated partition functions are identical: normal translation by kw carries one dissemination to another, permutes the slab faces, and preserves the unperturbed Wilson measure. Each is the folded partition function. Hence

$$Z_{M,J}(t\mathbb{A}^{\mathrm{u}}) \le Z_{M,J}(t\mathbb{A}^{\mathrm{f}}). \tag{AM11}$$

Let $\mu_{M,J,t}$ be the normalized folded measure, and define

$$p_{M,J}(t) = |D_{M,J}|^{-1} \log[Z_{M,J}(t\mathbb{A}^{\mathrm{f}})/Z_{M,J}(0)].$$

It is convex, $p_{M,J}(0) = 0$, and therefore $p_{M,J}(t) \le tp'_{M,J}(t)$ for $t > 0$. The folded measure is invariant under Q_f and under θ_{c+H}. All its source terms consequently have the same expectation as A. There are $\kappa|D_{M,J}|$ terms, where $\kappa = [\mathbb{Z}^4 : Q]^{-1}$, so

$$p'_{M,J}(t) = \kappa\mu_{M,J,t}(A).$$

Combining this identity, (AM8), and (AM11) gives, for every $\nu \in \mathcal{G}^P_\beta$,

$$t\kappa\nu(A) - o(1) \le p_{M,J}(t) \le t\kappa\mu_{M,J,t}(A). \tag{AM12}$$

The finite folded measure is invariant and reflection positive at every face $c + kH$. Indeed, its perturbation is $E + \theta_{c+kH}E$, where E is the sum of its copies lying in one closed half of the torus.

Copies on a boundary can be assigned half weight to each side. The observation following (AM5) applies.

First fix $t > 0$ and let $M, J \to \infty$. Compactness gives a limit ρ_t. It is a DLR state for the folded, finite-range perturbed interaction, is Q_f-periodic, is invariant and reflection positive at all its slab faces, and satisfies $\nu(A) \le \rho_t(A)$. Here DLR passage is local and exact: any fixed collection of bonds, its interaction neighbourhood, and the support of a cylinder test embed injectively in all sufficiently large tori. The finite torus conditional equation is then exactly the required local DLR equation on that test. Its kernel is continuous, so the equation passes to a weak limit. A finite test on one half-space eventually lies in a single torus half; hence its reflection inequality passes to the limit as well.

Next let $t \downarrow 0$ along a subsequence for which ρ_t converges to ρ_0. On any fixed finite set, only finitely many source copies meet that set. Their bounded total contribution is $O(t)$ uniformly in the exterior configuration. The perturbed conditional densities thus converge uniformly to the Wilson conditional density. It follows that ρ_0 is a Wilson DLR state. It retains Q_f periodicity, reflection invariance and positivity at the slab faces, and $\nu(A) \le \rho_0(A)$.

Now apply the previously proved link-symmetry theorem to ρ_0. Its finite-index period Q_f is enough for that theorem; the result upgrades its period to P. We have proved the following domination:

$$\begin{aligned} &A \text{ supported in } [c, c+H], \quad \nu \in \mathcal{G}_\beta^P \quad \Longrightarrow \quad \exists \rho_0 \in \mathcal{G}_\beta^P : \\ &\rho_0 \text{ is invariant and positive at } (n, c), \qquad \nu(A) \le \rho_0(A). \end{aligned} \tag{AM13}$$

The order of limits is essential: the surface error in (AM12) is divided by $t\kappa$, so volume tends to infinity before t tends to zero.

Separation closes the full half-space statement. Let $K = \mathcal{G}_\beta^P$ and let $K_c \subset K$ consist of the states invariant under θ_c and positive on every continuous cylinder function in the positive half-space. The set K is compact and convex: the DLR kernels preserve continuous cylinder functions, and all its defining equations and period equations are closed affine conditions. The set K_c is also closed and convex. It is nonempty, for example by the folded construction with $A = 0$.

For any real continuous local A, choose $v \in P$ so that $\tau_v A$ is supported strictly above the fixed mirror c. Then choose the even h large enough that this support lies below $c + H$. Such a v exists because $n \cdot (2ke_j) = 2k$ for a coordinate j with $n_j = 1$. Apply (AM13) to $\tau_v A$. Both ν and the resulting ρ_0 are P-invariant, so

$$\begin{aligned} \nu(A) = \nu(\tau_v A) \le \rho_0(\tau_v A) = \rho_0(A), \\ \rho_0 \in K_c. \end{aligned} \tag{AM14}$$

If some $\nu \in K$ were outside K_c, separation of a point from a compact convex set of measures would supply a real continuous function A with $\nu(A) > \sup_{\rho \in K_c} \rho(A)$. Continuous cylinder functions are uniformly dense in $C(\Omega)$, so A could be chosen local while retaining a strict inequality. This contradicts (AM14). Therefore $K = K_c$.

This step explains why the common period cannot be omitted. A state produced by folding initially has only the normal period $2w$. Such a large period need not move a test straddling the mirror into one slab. The link-symmetry theorem supplies $2\mathbb{Z}^4$ for both states in (AM14), independently of the observable and slab width. No claim that periodic DLR states are torus-selected is needed.

The mirror was arbitrary, so every periodic state is invariant and positive at all the stated mirrors on continuous local half-space functions. Finally, on the compact metrizable product of the positive half-space bonds, continuous cylinder functions are dense in $L^2(\nu)$. For bounded

Borel F choose such F_j with $\|F_j - F\|_2 \to 0$. Reflection invariance makes Θ_c an isometry of $L^2(\nu)$, and Cauchy–Schwarz gives

$$|\nu(\Theta_c F_j F_j) - \nu(\Theta_c F F)| \le \|F_j - F\|_2(\|F_j\|_2 + \|F\|_2) \longrightarrow 0.$$

This proves (AM1) on the complete positive-half algebra. Replacing a negative-half function by its reflected conjugate proves the other half. In particular, arbitrary finite stacks and their bounded Borel tests are covered. □

As a further consequence, every periodic state is invariant under all unit lattice translations: in direction e_μ the product of the site reflection at 0 and the link reflection at $1/2$ is translation by e_μ. It is also invariant under the hypercubic group. The coordinate reflections at zero generate coordinate sign changes, and the diagonal reflections with normal $e_\mu - e_\lambda$ interchange coordinates. These transformations generate all signed permutations.

Scope of the implication. Positive semidefinite contractions of an adjacent-layer kernel can produce a one-layer positivity test, just as skew contractions produce an antisymmetric invariance test. That observation alone supplies no identity reducing arbitrary odd or even stacks to a single layer. The proof above uses a different route: exact finite-torus positivity on the full half-algebra, a finite chessboard inequality, a lower tangent for each arbitrary periodic DLR state, and separation after the common period is established. Thus the all-stack and diagonal conclusions use no layer-elimination positivity assumption. The result is at a fixed lattice spacing and coupling; it makes no continuum-limit or interacting renormalization-group assertion.

8. A uniform lattice plateau estimate from reflection positivity

The estimate in this section is a consequence of translation invariance and reflection positivity in coordinate and diagonal directions. No infrared bound, mass gap, or differentiability of a pressure is assumed. The Gibbs equations are not needed in the proof. Their role in the Wilson application is to place every periodic Wilson Gibbs state in the reflection-positive class by Theorem 7.1, proved independently in Section 7. The state here is a measure on the infinite lattice, rather than a finite-torus law lifted periodically.

By dilation it suffices initially to take lattice spacing one. Let ν be a translation-invariant probability measure on lattice configurations. A local function means a bounded Borel function of finitely many bonds. The algebra $\mathfrak{A}(S)$ contains the functions of bonds with both endpoints in S. Assume reflection positivity on the closed side of every coordinate mirror $x_\mu = c$, $c \in \frac{1}{2}\mathbb{Z}$, and every diagonal mirror $x_\mu \pm x_\lambda = c$, $c \in \mathbb{Z}$. These are sixteen families. Reflections act linearly on functions by pullback, without complex conjugation. Write θ_μ for reflection in $x_\mu = 0$.

For $r \ge 1$ an integer put $Q_r = [-r, r]^4$. For $A \in \mathfrak{A}(Q_r)$ define

$$\mathcal{N}_{\nu,r}(A) = \max_{\mu,\gamma} \nu\big(\overline{\theta_\mu \gamma A}\, \tau_{2re_\mu} \gamma A\big), \tag{PL1}$$

where γ ranges over the sixteen coordinate sign changes. The terms in this maximum are nonnegative squared reflection norms. Let $\mathcal{I}$ be the completed invariant σ-field for all lattice translations and set

$$\mathrm{Pl}_\nu(A, B) = \nu\big(\overline{E_\nu[A \mid \mathcal{I}]} E_\nu[B \mid \mathcal{I}]\big). \tag{PL2}$$

Theorem 8.1 (Uniform plateau bound)**.** *There is a numerical constant C such that, for the measures just specified, $A, B \in \mathfrak{A}(Q_r)$, and $y \in \mathbb{Z}^4$ with $n = |y|_\infty - 2r \ge 1$,*

(PL3) $$\left|\nu(\overline{A}\tau_y B) - \mathrm{Pl}_\nu(A,B)\right| \le C\left[(r/n)^2 + (r/n)^3\right]\mathcal{N}_{\nu,r}(A)^{1/2}\mathcal{N}_{\nu,r}(B)^{1/2}.$$

The constant is independent of the measure, the support size, and any coupling. At lattice spacing a, the same statement holds with $Q_r = a[-r,r]^4$ and $n = |y|_\infty/a - 2r$.

We use the spectral theorem for commuting bounded normal operators and the mean ergodic theorem for a unitary operator. All estimates supplying the power in (PL3), including the momentum-density estimate, are proved below.

8.1. **Reflection Hilbert spaces and their spectral carriers.** Reflection positivity and translation invariance first imply invariance under each of the indicated reflections. On one closed side, positivity makes $\langle F, G\rangle_\theta = \nu(\overline{\theta F}G)$ Hermitian. Testing it against 1 shows $\nu(F) = \nu(\theta F)$ for functions supported on that side. Any local function can be translated into the side by a lattice vector normal to the mirror. Reflecting reverses that translation; translation invariance therefore proves the equality for the original function. The reflections consequently act unitarily on $L^2(\nu)$.

In the coordinate picture μ, complete the quotient of local functions on $x_\mu \ge 0$ for the positive semidefinite form

$$\langle F, G\rangle_\mu = \nu(\overline{\theta_\mu F}G).$$

Denote the class of F by $[F]_\mu$. Its norm is at most $\|F\|_\infty$. Translation by e_μ induces a positive self-adjoint contraction T_μ, and tangential coordinate translations induce commuting unitaries $U_{\mu,\lambda}$, $\lambda \ne \mu$.

Here are the details of this elementary reconstruction. The normal translation is symmetric on the pre-Hilbert space by translation invariance and $\theta_\mu \tau_{e_\mu} = \tau_{-e_\mu}\theta_\mu$. Its quadratic form is nonnegative by reflection positivity at the mirror $x_\mu = -1/2$. Repeated Cauchy–Schwarz gives

$$\|T_\mu[F]\| \le \|[F]\|^{1-2^{-k}}\|T_\mu^{2^k}[F]\|^{2^{-k}}.$$

The last norm is at most $\|F\|_\infty$. Letting $k \to \infty$ proves contraction and preservation of null vectors. The bounded symmetric extension is self-adjoint and positive. Tangential translations preserve the form and have inverses of the same kind, so are unitaries. Their commutations on local functions pass to the completion. In particular, for integers $s \ge 0$ and tangential $w \in \mathbb{Z}^4$,

(PL4) $$\langle [F]_\mu, T_\mu^s U_\mu^w [G]_\mu\rangle = \nu(\overline{\theta_\mu F}\tau_{se_\mu + w}G).$$

There is a joint spectral measure with variables $(E,\eta) \in [0,\infty] \times (-\pi,\pi]^3$ for which $T_\mu = e^{-E}$ and $U_{\mu,\lambda} = e^{i\eta_\lambda}$. The value $E = \infty$ represents a possible kernel of T_μ. The following carrier bound is essential:

$$\cosh E + \cos\eta_\lambda \ge 2$$
$$(\lambda \ne \mu),$$
$$E \ge \kappa|\eta|_\infty,$$

(PL5) $$\kappa = \frac{2}{\pi}\log(1+\sqrt{2}).$$

We now derive it from the diagonal mirrors.

A polynomial-measure lemma. Suppose a finite positive measure m on the closed unit disc satisfies

$$\int |p(z)|^2\, dm(z) \le C_0 \sup_{[0,1]} |p| \sup_{|z|=1} |p| \tag{PL6}$$

for every polynomial. Then it gives zero mass to $\{|1+z|^2 < 4|z|\}$.

To prove this without invoking an extremal-function theorem, set

$$L(\zeta) = \operatorname{Log} \frac{1+\zeta}{1-\zeta},$$

$$f_s(z) = \frac{\cos(sL(\sqrt{z}))}{\cosh(\pi s/2)},$$

$$s > 0.$$

The Cayley transform has positive real part in the disc. Thus L is holomorphic, odd, and satisfies $|\operatorname{Im} L| < \pi/2$. The cosine is even in $\sqrt{z}$, so f_s is a single-valued holomorphic function of z in the disc. Put $\varphi(z) = |\operatorname{Im} L(\sqrt{z})|$. The identity $|\cos(u+iv)|^2 = \cos^2 u + \sinh^2 v$ gives

$$|f_s| \le 1,$$

$$\sup_{[0,1)} |f_s| \le \cosh(\pi s/2)^{-1},$$

$$|f_s(z)| \ge \frac{\sinh(s\varphi(z))}{\cosh(\pi s/2)}.$$

Apply (PL6) to Taylor approximants of $f_s(qz)$, $0 < q < 1$, which converge uniformly on the closed disc. The resulting inequality, followed by Fatou as $q \uparrow 1$, is

$$\int \sinh^2(s\widetilde{\varphi}(z))\, dm(z) \le C_0 \cosh(\pi s/2).$$

Here $\widetilde{\varphi} = \varphi$ inside the disc, equals $\pi/2$ on the boundary except at 1, and may be set to zero at 1. Indeed the radial limits of φ at the other boundary points equal $\pi/2$. For each $\epsilon > 0$ this inequality forces $m(\{\widetilde{\varphi} \ge \pi/4 + \epsilon\}) = 0$ by letting $s \to \infty$. Finally, if $z = \zeta^2$ inside the disc,

$$\tan \varphi(z) = \frac{2|\operatorname{Im} \zeta|}{1 - |\zeta|^2},$$

$$|1+\zeta^2|^2 - 4|\zeta|^2 = (1-|\zeta|^2)^2 - 4(\operatorname{Im}\zeta)^2.$$

Thus $\varphi > \pi/4$ is precisely the excluded set. The same assertion holds on the boundary, where only $z = 1$ is not excluded. This proves the lemma.

Fix $\lambda \ne \mu$ and a local A on $x_\mu \ge 0$. Choose an integer $R \ge 1$ enclosing its support in $0 \le x_\mu \le R$, $|x_\lambda| \le R$. Let θ' be reflection in $x_\mu + x_\lambda = R$, put $B = \theta'\theta_\mu A$, $A_0 = \tau_{2Re_\mu} A$, and $d_\pm = e_\mu \pm e_\lambda$. Both B and A_0 lie on the positive side of this diagonal mirror. Reconstruction at that mirror gives a positive contraction T_+ for d_+ and a unitary V_- for d_-. The preceding reconstruction proof applies: positivity of T_+ uses the admissible diagonal mirror one height unit behind, and its contraction uses the bounded local representative. For integers $k, l \ge 0$,

$$\langle [A]_\mu, T_\mu^{2R+k+l} U_{\mu,\lambda}^{k-l} [A]_\mu \rangle = \langle [B]', T_+^k V_-^l [A_0]' \rangle. \tag{PL7}$$

Both sides are the same lattice correlation: their displacement is $2Re_\mu + kd_+ + ld_-$. No analytic continuation is used here.

Push the spectral measure of $[A]_\mu$, multiplied by t^{2R}, forward by $z = t\zeta$, where $t = e^{-E}$ and $\zeta = e^{i\eta_\lambda}$. Expanding a polynomial and its conjugate in (PL7) gives

$$\int |p|^2 \, dm = \langle [B]', p(T_+)\overline{p}(V_-)[A_0]' \rangle,$$

$$\overline{p}(z) = \sum_k \overline{p_k} z^k.$$

The spectral theorem bounds this by the right side of (PL6), with $C_0 = \|[B]'\|\|[A_0]'\|$. The lemma excludes $|1 + t\zeta|^2 < 4t$. Since $t^{2R} > 0$ for $t > 0$, the joint spectral measure of $[A]_\mu$ itself vanishes there. The local classes are dense, so the operator spectral measure vanishes there. At $t = 0$ the equivalent operator inequality is automatically satisfied. We have proved

$$(1 - T_\mu)^2 \geq T_\mu (2 - U_{\mu,\lambda} - U_{\mu,\lambda}^*).$$

For $0 < t \leq 1$ this is equivalent to the first assertion of (PL5). The curve is $E \geq 2 \operatorname{arcsinh} |\sin(\eta_\lambda/2)|$. Concavity of $x \mapsto \operatorname{arcsinh} \sin x$ on $[0, \pi/2]$ puts it above its endpoint chord, giving the constant κ. Intersect the three carriers to obtain (PL5).

8.2. **The zero-energy plateau is the invariant conditional field.** Write $\Pi_\mu = 1_{\{E=0\}}$; spectral dominated convergence gives $T_\mu^s \to \Pi_\mu$ strongly as $s \to \infty$. By (PL5), every tangential unitary equals the identity on this zero-energy space. Let $\mathcal{I}_\mu$ denote the invariant field for the single translation τ_{e_μ}. The mean ergodic theorem on $L^2(\nu)$ and (PL4) give

$$\langle [F]_\mu, \Pi_\mu [G]_\mu \rangle = \nu(\overline{\theta_\mu F} E_\nu[G \mid \mathcal{I}_\mu]). \tag{PL8}$$

Reflection acts as a unitary involution on $L^2(\mathcal{I}_\mu, \nu)$. That involution is positive. Indeed, approximate an invariant function J in L^2 by bounded local G_k and translate each approximant into $x_\mu \geq 0$. The error is unchanged because J is invariant. Reflection positivity then passes to the L^2 limit and gives $\langle \theta_\mu J, J \rangle \geq 0$. A positive self-adjoint involution has only the eigenvalue $+1$, so reflection fixes J. Consequently (PL8) equals

$$\nu\big(\overline{E_\nu[F \mid \mathcal{I}_\mu]} E_\nu[G \mid \mathcal{I}_\mu]\big).$$

The map $\Pi_\mu[F]_\mu \mapsto E_\nu[F \mid \mathcal{I}_\mu]$ is therefore an isometry onto a dense subspace of $L^2(\mathcal{I}_\mu, \nu)$. Density follows by the same shifted local approximations. It intertwines the tangential translations. Since those translations are the identity on Π_μ, every $\mathcal{I}_\mu$-measurable function is invariant under all four coordinate translations. Hence $\mathcal{I}_\mu = \mathcal{I}$ modulo ν, and

$$\langle [F]_\mu, \Pi_\mu [G]_\mu \rangle = \nu\big(\overline{E_\nu[F \mid \mathcal{I}]} E_\nu[G \mid \mathcal{I}]\big). \tag{PL9}$$

This also proves that $E_\nu[\theta_\mu F \mid \mathcal{I}] = E_\nu[F \mid \mathcal{I}]$. The possible T_μ kernel has energy infinity and does not affect any of these zero-energy identities.

8.3. **A positive time filter and a bounded momentum density.** For $A \in \mathfrak{A}(Q_r)$ define its cube vector $\Phi_A^\mu = [\tau_{re_\mu} A]_\mu$. Translation invariance shows $\|\Phi_A^\mu\|^2 = \nu(\overline{\theta_\mu A}\tau_{2re_\mu} A)$. Thus (PL1) is the maximum of the corresponding squared norms of γA. In particular $\mathcal{N}^{1/2}$ is a seminorm, is invariant under the sign-change group, and is at most the supremum norm.

Let b be the nonnegative periodic triangle function supported on $[-\pi/4, \pi/4]$, normalized to have mean one. Its Fourier coefficients are

$$b_m = \left(\frac{\sin(m\pi/8)}{m\pi/8}\right)^2,$$
$$b_0 = 1.$$

This follows by writing the triangle as the convolution of two interval indicators and normalizing their zeroth coefficients. Let z_m be the even period-four sequence $(0, 1, -2, 1)$ and set

$$\widehat{q}(u) = \sum_{m\in\mathbb{Z}} z_m b_m e^{imu},$$
$$\Psi(q) = 2\sum_{m\geq 1} z_m b_m q^{m-1},$$
$$0 \leq q \leq 1. \tag{PL10}$$

Both series converge absolutely. The identity $z_m = \frac{1}{2}i^m - (-1)^m + \frac{1}{2}(-i)^m$ gives

$$\widehat{q}(u) = \tfrac{1}{2}b(u + \pi/2) - b(u + \pi) + \tfrac{1}{2}b(u - \pi/2).$$

In particular $\widehat{q}$ vanishes whenever $\operatorname{dist}(u, 2\pi\mathbb{Z}) \leq \pi/4$. The crucial energy property is

$$0 \leq \Psi \leq C,$$
$$\Psi(1) = 0,$$
$$\Psi(e^{-x}) \geq c \tanh x$$
$$(x > 0). \tag{PL11}$$

To verify its nontrivial lower bound, use the Poisson kernel

$$P_x(u) = \sum_m e^{-|m|x} e^{imu} = \frac{\sinh x}{\cosh x - \cos u}.$$

Since $z_0 = 0$, the Poisson integral yields

$$\Psi(e^{-x}) = \frac{e^x}{2\pi}\int_{|v|\leq\pi/4} b(v)[\tfrac{1}{2}P_x(v - \pi/2) + \tfrac{1}{2}P_x(v + \pi/2) - P_x(v - \pi)]\, dv.$$

Writing $C_x = \cosh x$, the bracket is

$$\sinh x\left[\frac{C_x}{C_x^2 - \sin^2 v} - \frac{1}{C_x + \cos v}\right] \geq \frac{\tanh x}{\sqrt{2}(C_x + 2^{-1/2})}.$$

For this inequality first replace the first fraction by $1/C_x$, then use $\cos v \geq 2^{-1/2}$ and monotonicity of $q/(C_x + q)$. The factor $e^x/(C_x + 2^{-1/2})$ has a positive lower bound, and b has mean one.

This proves the lower bound. The upper bound follows from absolute convergence of (PL10), and $\Psi(1) = \widehat{q}(0) = 0$. These statements extend continuously to $q = 0$, where $\Psi(0) = 2b_1 > 0$.

First assume $\theta_\mu A = \epsilon A$, $\epsilon \in \{1, -1\}$, and put $\Phi = \Phi^\mu_A$. For $s \in \mathbb{Z}$ and tangential $w \in \mathbb{Z}^3$, define

$$g_A(s, w) = \epsilon\nu(\overline{A}\tau_{se_\mu + w}A).$$

For $s \geq 2r$, the cube-crossing identity from (PL4) gives

$$g_A(s, w) = \langle \Phi, T_\mu^{s-2r} U_\mu^w \Phi\rangle,$$

$$|g_A(s, w)| \leq \mathcal{N}(A). \tag{PL12}$$

Reflection in $x_\mu = 0$ and translation invariance give $g_A(-s, w) = g_A(s, w)$ and $g_A(s, -w) = \overline{g_A(s, w)}$. Define

$$K_A(w) = \sum_{m\in\mathbb{Z}} z_m b_m g_A(2rm, w).$$

The term at $m = 0$ vanishes. Every other time separation has absolute value at least $2r$, so (PL12) bounds this sum by $C\mathcal{N}(A)$, uniformly in w. By the same identities,

$$K_A(w) = \langle \Phi, \Psi(T_\mu^{2r}) U_\mu^w \Phi\rangle = \int e^{i\eta\cdot w}\, dM_A(\eta), \tag{PL13}$$

where M_A is the momentum pushforward of the positive measure

$$\Psi(e^{-2rE})\, d\langle \Phi, \mathsf{E}_\mu(E, \eta)\Phi\rangle.$$

Thus M_A is positive; the signs in z_m do not compromise positivity because (PL11) controls their spectral sum.

For $|w|_\infty \geq 2r$, choose $\lambda \neq \mu$ at a maximal coordinate. Replace w by $-w$ if necessary, which only conjugates $K_A(w)$, and suppose $w_\lambda \geq 2r$. In the coordinate picture λ, for every integer s the same correlation is

$$g_A(s, w) = \epsilon \left\langle \Phi^\lambda_{\theta_\lambda A}, T_\lambda^{w_\lambda - 2r} U^s_{\lambda,\mu} \prod_{\lambda'\neq\lambda,\mu} U^{w_{\lambda'}}_{\lambda,\lambda'} \Phi^\lambda_A \right\rangle.$$

Sum this identity at $s = 2rm$ against $z_m b_m$. In this picture the filter becomes the bounded spectral function $\widehat{q}(2r\eta'_\mu)$ of a unitary translation. Whenever it is nonzero, $|\eta'_\mu| > \pi/(8r)$, since the gap around zero modulo 2π includes the ordinary interval about zero. Aliases at other multiples of 2π do not invalidate this implication. The carrier (PL5) consequently gives $E' \geq \alpha = \kappa\pi/(8r)$ on the filter support. Both cube vectors have norm at most $\mathcal{N}(A)^{1/2}$. Hence

$$|K_A(w)| \leq C\mathcal{N}(A) e^{-\alpha(|w|_\infty - 2r)}$$

$$(|w|_\infty \geq 2r). \tag{PL14}$$

There are $O(r^3)$ points with $|w|_\infty < 2r$ and $O(k^2)$ points on the shell $|w|_\infty = k$. Since α is a fixed constant times r^{-1}, (PL14) and the previous bound give

$$\sum_w |K_A(w)| \le C\mathcal{N}(A)\left[r^3 + \sum_{j\ge 0}(r^2+j^2)e^{-\alpha j}\right] \le Cr^3\mathcal{N}(A).$$

Therefore $h_A(\eta) = (2\pi)^{-3}\sum_w K_A(w)e^{-i\eta\cdot w}$ is continuous and bounded by $Cr^3\mathcal{N}(A)$. It has the same Fourier coefficients as M_A. Uniqueness of a finite measure from its Fourier coefficients follows by density of trigonometric polynomials in the continuous functions on the torus. Consequently

$$M_A = h_A(\eta)\,d\eta,$$

$$0 \le h_A \le Cr^3\mathcal{N}(A). \tag{PL15}$$

This is a derived bound for an energy-weighted momentum measure. It is not an assumed bound for the unweighted spectral measure.

8.4. **Removing the filter and proving the plateau estimate.** Let $m_\Phi = \langle \Phi, \mathsf{E}_\mu(\cdot)\Phi\rangle$. Since M_A has a density, it has no atom at $\eta = 0$. By (PL11), its weight is strictly positive at every finite $E > 0$. Thus m_Φ gives zero mass to $\{0 < E < \infty, \eta = 0\}$. The energy-zero contribution is precisely the plateau and was not removed by this inference: its weight is zero. The point $E = \infty$ contributes zero to T_μ^n for $n \ge 1$.

For $E > 0$ the function $e^{-nE}/\tanh(2rE)$ decreases in E. Using (PL11), then (PL5), and then (PL15), gives

$$\begin{aligned}\langle \Phi, T_\mu^n(1-\Pi_\mu)\Phi\rangle &\le C\int_{\mathbb{T}^3} \frac{e^{-n\kappa|\eta|_\infty}}{\tanh(2r\kappa|\eta|_\infty)} h_A(\eta)\,d\eta \\ &\le Cr^3\mathcal{N}(A)\int_0^\infty u^2 e^{-n\kappa u}\left(1+\frac{1}{2r\kappa u}\right)du \\ &\le C\mathcal{N}(A)\big[(r/n)^3 + (r/n)^2\big]. \qquad \text{(PL16)}\end{aligned}$$

In the second line we used $1/\tanh v \le 1 + 1/v$ and the three-dimensional cube shell volume $24u^2du$. The apparent singularity at $u = 0$ is integrable. The pushforward step can include $E = \infty$ and other unused spectral mass, since the integrand is nonnegative; it does not add an atom at $\eta = 0$ by (PL15).

For an arbitrary complex A, write $A = A_+ + A_-$ with $A_\pm = (A \pm \theta_\mu A)/2$. The seminorm property and sign-change invariance imply $\mathcal{N}(A_\pm) \le \mathcal{N}(A)$. The operator $D_n = T_\mu^n(1-\Pi_\mu)$ is positive, so its quadratic form at a sum is at most twice the sum of its two quadratic forms. Applying (PL16) to A_+ and A_- proves (PL16), with an enlarged numerical constant, for every A.

Finally suppose $y_\mu = |y|_\infty > 0$. The cube-crossing identity is

$$\nu(\overline{A}\tau_y B) = \langle \Phi^\mu_{\theta_\mu A}, T_\mu^n U_\mu^{y_\perp}\Phi^\mu_B\rangle.$$

The Π_μ term is (PL2) by (PL9), conditional invariance under translations and reflection, and $U_\mu^{y_\perp}\Pi_\mu = \Pi_\mu$. For the remaining term, Cauchy–Schwarz in the positive form of D_n bounds the modulus by the geometric mean of the two quantities in (PL16). The tangential unitary commutes with D_n, so does not change that form. Also $\mathcal{N}(\theta_\mu A) = \mathcal{N}(A)$. This proves (PL3). For a negative maximal coordinate use $\nu(\overline{A}\tau_{-y}B) = \overline{\nu(\overline{B}\tau_y A)}$ and conjugate symmetry of the plateau. Dilation restores spacing a and replaces distances by their numbers of lattice steps, proving the stated scaling. □

For an ergodic state the invariant conditional expectations are constants, so (PL3) is clustering. For a mixture its generally nonzero invariant plateau is retained. When local continuum representatives have large supremum norms, the reflection sizes in (PL1) can still be bounded by separated reflected moments. This is the reason (PL3), rather than a supremum-norm substitute, is the useful estimate for a continuum phase construction. No spectral gap above the invariant subspace has been proved: the filter weight vanishes linearly near zero energy, and removing it is exactly what produces the inverse-square term.

Corollary 8.2 (Closedness of the ergodic reflection-positive class)**.** *At fixed spacing, the translation-invariant measures satisfying these mirror-positivity hypotheses form a weakly closed set, and so does its subset of ergodic measures.*

Proof. Translation invariance and each positivity inequality on continuous local functions pass to a weak limit. Reflection invariance passes as well, and approximation in L^2 extends positivity to bounded Borel local functions. Thus the full reflection-positive class is closed.

Suppose now that ν_j are ergodic in this class and converge weakly to ν. For fixed continuous local A, B in a common cube, their plateaux under ν_j are $\overline{\nu_j(A)}\nu_j(B)$. Passing to the limit in (PL3) gives

$$|\nu(\overline{A}\tau_y B) - \overline{\nu(A)}\nu(B)| \le C[(r/n)^2 + (r/n)^3]\|A\|_\infty \|B\|_\infty.$$

In particular the correlation converges to the product along $y = ke_\mu$, $k \to \infty$. Its Cesaro means, with $A = B$, identify the squared L^2 norm of $E_\nu[A \mid \mathcal{I}_\mu]$ by the mean ergodic theorem. That norm is $|\nu(A)|^2$, so the conditional expectation is constant. Continuous local functions are dense in $L^2(\nu)$; hence $\mathcal{I}_\mu$, and therefore $\mathcal{I}$, is trivial. This is ergodicity of ν. □

9. Supporting analytic results and proof dependencies

Supporting results and their use in the proof. This catalogue fixes the estimates and their domains used by the construction. The labels identify proof tasks. The regular Gaussian matrices and response, the hereditary substitution room, the weighted history sum and the finite ordinary-source induction are developed below for the actual construction. The interacting extraction then uses its clean conditional law and the uniform finite-reference comparison. The one-loop calculation is developed in Section 35.1.

The main scalar route uses D1 in its actual rooted geometry, G1 and F2–F3. The proof outlines supplying these inputs are in Sections 29, 32, 36, 43.1–43.3 and 44.3. The later bridge sections derive compactness, D2, H1, Hilbert evolution and G2 from those inputs. Existence of an arbitrarily fine admissible sequence uses the determining coupling law. Weighted and matched-run comparisons and typed Ward estimates are required only for the stronger statements invoking them. Every use of a package retains its stated uniformity and scope.

Here are conventions shared by the packages. For numerical coefficients written in terms of N, the group is $\mathrm{SU}(N)$ and here $N = 2$. Fix an odd integer $L > 1$, a compact simple matrix group G, and dimension four. The principal example is $G = \mathrm{SU}(2)$. The lattice spacing is $a_K = L^{-K}$, and a finite torus has physical side ℓ. There is one group variable U_b per positively oriented bond, $U_{b^{-1}} = U_b^{-1}$, and normalized product Haar measure dU. Write

$$a_p(U) = 1 - \tfrac{1}{2}\operatorname{Re}\operatorname{tr} U_p,$$

$$A(w, U) = \sum_p w(x(p))a_p(U),$$

$$Z_{K,\ell}(w) = \int e^{-\beta A(1,U)+A(w,U)} dU.$$

The trace convention in this display is for SU(2); another group requires its own normalization. Let $\mathcal{O}_K(f) = A(f,U)$. The connected functions are the mixed derivatives at zero of $\log Z_{K,\ell}(\sum_i z_i f_i)$. Test functions used together have pairwise disjoint compact supports, separation at least $d > 0$, and individual supports in balls of radius R. Set $\|f\|_\sharp = \max(\|f\|_\infty, c_M\|\nabla f\|_\infty)$, with c_M fixed. All uniform constants below must be independent of K, the torus, the history, the positions of the tests, and the relevant Gibbs state. Dependence on g, n, d, R must be displayed. A lower bound on admissible ℓ may depend on g and on constants fixed earlier. Replacing this uniformity by a bound at one fixed cutoff does not discharge an obligation.

A. Background fields and propagators: four obligations. **Obligation A1: the constrained critical orbit.** We must prove that, for every allowed nested family of small-field regions and every regular coarse datum V, there exists a regular critical orbit $[U(V)]$ for the Wilson action with the prescribed block averages, unique among regular critical orbits in a specified larger neighborhood, and that a chosen gauge gives a representative which is Borel in all real data and holomorphic in a complex neighborhood of each regular real datum. Both neighborhoods have stated radii on the local scale; their radii and constants are uniform in the number of scales. Existence of an unrestricted global minimizer is insufficient: the needed assertion concerns the constrained regular orbit and its gauge representative. Compatibility with restriction to subregions, with seam constraints, and with changing the torus must be part of the statement. This is the background-orbit obligation behind the local assertions numbered 1a.3.6, 1a.3.8 and the background theorems. For the pointed window deduction, the actual restrictions and substitutions must remain in their prescribed pair cores. These cores contain the usable Taylor chart ball and obey the curvature and own tent-action bounds on the whole admitted core, including the finitely many young ages. Section 17 proves a local chart for the actual one-step average near a pure-gauge double block, including the pivot derivative, exact straightening and measure factors. Its explicit chart radius and nonwrapping premise do not imply existence, uniformity or all-substitution room for the many-level critical orbit.

Obligation A2: coercivity on a fluctuation fiber. Let Q be the derivative of the chosen block average, and let $\mathcal{E}$ be the gauge-fixed Hessian on the fluctuation variables at a regular background. The chosen coordinates must remove all residual gauge directions. On each conditioning set and with its specified boundary conditions we must prove

$$\mathrm{Re}\langle B, \mathcal{E}B\rangle \geq c\|B\|_2^2,$$
$$QB = 0,$$

with the norm in the units of that fluctuation scale. The same estimate, with $c/2$, must hold throughout the claimed complex neighborhoods. Zero modes, torus holonomies, and node constraints must either be absent from this space or treated separately. A mass for this constrained operator is not a mass gap for the physical continuum Hamiltonian. This is the missing quantitative content of the conditioning and covariance assertions numbered 1c.6 and related claims. For the actual unweighted classical flat reference, Sections 18 and 19 prove $c(L)I \leq \mathsf{A}_k^{\mathrm{flat}} \leq CL^8 I$, uniformly in the number of previous classical blocking steps and in compatible tori, including real flat holonomies. They use the actual plain constraint and the soft internal-contour gauge. The curl-energy comparison is $\|da\|^2 \leq E_k(a) \leq 256\|da\|^2$. This supplies the flat classical case, not the non-Abelian, nonzero-current, signed-profile, layered or enlarged-complex-domain versions of A2.

Obligation A3: local inverse and response bounds. We must prove that the inverses and response operators actually used, including conditional covariances and derivatives of $U(V)$, have kernels bounded by $Ce^{-c\,\mathrm{dist}(x,y)}$ in scale units. The bound must also control the stated first differences and the stated Hölder differences throughout the full complex datum polydisc, not only at the identity or at real data. For the pointed window this is the product bond-log domain (PW1), with the scale-weighted potential, gradient and current response in (PW2)–(PW3). For a region with several local scales, use a precisely defined weighted path distance and the corresponding powers of the local spacings. Holomorphic derivatives in each decoupling parameter must satisfy Cauchy bounds on a fixed enlarged complex disc. A derivative joining distant sets must retain the decay needed by the polymer norm. The mesh-uniform scalar estimate and its Dirichlet compressions are proved in the block-resolvent section. The corresponding non-Abelian background and multiscale response estimates remain to be proved. This includes the decoupled response assertion numbered 1c.10; decay of one real inverse alone does not establish it. Section 20 proves exponential kernels for the actual unweighted classical flat precision and its constrained covariance, uniformly in depth and compatible volume. It derives the exact alias symbol, cancels its apparent gauge denominator on the full complex zero cone, proves a uniform strip, and then inverts using the flat gap. Periodization and unitary flat holonomies preserve the bounds. This establishes the stated flat reference kernels, not the nonlinear response or decoupling bounds required on the full interacting domains. The finite-box comparison also requires compatible actual linearized equations and boundary rows, including exponentially local gauge rows when present. Cutting off an affine solution must leave polynomially bounded boundary forcing, or an exponentially decaying boundary tail. The one-pillar inverse bound is in the natural force norms, with the fine-cell weights included. These properties, used in (PW12), do not follow from bounds on unrelated inverse kernels.

Obligation A4: averaging schemes and locality. We must prove that any transfer from one averaging scheme to another provides a uniform intertwining map, including its coarse gauge factor. On fluctuation fibers the transferred Hessians, inverses and constraints must obey A1–A3 with the same allowed losses, for every number of blocking steps. Restrictions to a region must depend only on an explicitly bounded enlargement of that region. Equality of two linearized averages only at the identity is insufficient. A proof must bound the accumulation of their differences at nonflat backgrounds and must state the gauge in which the comparison is made.

B. Cluster expansion and power counting: four requirements. **Requirement B1: a polymer majorant.** Let polymers be finite connected sets of scale cubes, with incompatibility meaning intersecting supports. A specified positive size $a(X)$ and activities $z(X)$ must satisfy, uniformly in all external data,

$$\sum_{Y:Y\not\sim X} |z(Y)|e^{a(Y)} \le a(X).$$

The same inequality, with strict room, must hold for complex sources and decoupling parameters on the domains used for differentiation. Require the absolutely convergent connected-cluster formula for the logarithm, including termwise differentiation and a uniform majorant for each derivative used later. The finite-volume identity must be exact before taking limits. This is the content needed from the convergence statement numbered 1d.4.1; a formal graph expansion is not sufficient. Section 26 derives the local unmarked activity and union-filed logarithm bounds from its actual enlarged Gaussian matrix/core hypotheses, sharp-cutoff structure and local potential budgets. The Mayer-parameter discs are established at the Gaussian integral, before zero-freeness is inferred. Gaussian volume costs, failed bonds, actual neighbor-layer closures and the spare potential slot are all paid for in the rate allocation. This also supplies SI5 under those primitive

hypotheses. We must still verify them for the actual complex interacting matrices and all allowed cores; convergence of an unrelated model gas would not supply this.

Requirement B2: pointed marginal extraction. Sections 23 and 24 prove the following implication for an actual regular member with no stored color spectators. Its pointed total and differentiated sums must converge normally; it is locally gauge invariant, and the individual member's pointed totals are covariant under unit translations and the full hypercubic group. Use the actual free response, with gauge covariance and affine constant-curvature reproduction modulo gauge, with zero linear current. The usable pair core, localized response and entropy gap must give the absolute Hessian moments in (PH5). Then (PH18) defines its coefficient and (PH23) proves cancellation by that same member's own midpoint tent action. Different members may be chosen at different sites; their coefficients cannot be interchanged.

On physical radius a, the readout retains $|b(E)| + \|T[E]\| \le Ca^{-2}\|E\|$. Section 14 supplies its stated near-zero inclusion and normalization. For the nonlinear remainder also require A1–A3 throughout the complex datum domain, the actual core inclusions, spatial Hölder regularity, compatible boundary equations and the whole-core bound on the own tent action. Put $\sigma = L^{-(n-j)}$, $\rho = \max(1, a_{\mathrm{out}}/a)$ and $0 < \alpha_H \le 1$. Under the window domain conditions the near-domain sum at one base point, after its vacuum and own counterterm subtraction, has bound

$$C\|E\| \left[\rho^2\sigma^{4+\alpha_H} + \rho^3\sigma^6 + (1+\rho^2\sigma^4)e^{-c/\sigma}\right].$$

The final exponential term includes remote domains without a nonwrapping chart. No factor proportional to the actual curvature is asserted for that tail. Under the ordinary age bound $\rho^2 \le C(1+n-j)$, the finite young-age core fallback and the $O(\sigma^{-4})$ site count give the summable weight $C\|E\|(1+n-j)^{3/2}L^{-\alpha_H(n-j)}$.

These deductions use the full member and field-domain hypotheses above. In the actual RG application we must prove those hypotheses. Torus-sensitive or field-sensitive tails, source derivatives of the formation map, and the fluctuation disintegration laws in F2 use their separate estimates. Stronger assertions in the arguments coded 1e.1.13, 1e.1.15 or 1e.1.16 need their corresponding estimates, which we must also prove.

Obligation B3: the one-loop law with a summable lattice correction and controlled nonlinear errors. For the chosen normalization let $c_0 = 11N^2 \log L/(12\pi^2)$. We must prove that the Gaussian contribution b_j^0 is independent of the bare coupling and obeys $|b_j^0 - c_0| \le CL^{-\sigma j}$, or a stated summable substitute, and that the full increment has a decomposition

$$\begin{aligned} b_j &= b_j^0 + f_j + q_j, \\ |f_j| &\le C x_{j-1}^{-1/2} (\log x_{j-1})^p, \\ |q_j| &\le C x_{j-1}^{-(\kappa_0-4)/2}, \\ \kappa_0 &> 4. \end{aligned}$$

If a later deduction needs a stronger error, that stronger estimate must be proved separately. We must obtain c_0 from an actual lattice trace calculation with the stated zero-mode convention. A continuum perturbative coefficient quoted without this comparison does not discharge this obligation.

Obligation B4: the ordinary step and its finer derivatives. The ordinary source box allows a separate Euclidean regular member at each base point, its own scalar counterterm, a variable Wilson weight, and scale-small noncovariant defects. We must prove that the step is holomorphic

on this product box with a majorant independent of the number of labels. It is not necessary that the whole spatial array be translation invariant. The ordinary regular-output estimate does not by itself prove every marked or large-field derivative.

For the stronger comparisons actually invoked, we must also prove the potential derivative bound (11.1) and the one-zone complex-mass bound (11.2), on the domains stated in that section, with constants independent of the levels and source positions. The regular irrelevance weight, squared-radius weights for defects, and large-field survival weights have different roles. In particular the survival factor belongs in the bound on a zone's output; it does not enlarge its input radius by its reciprocal. Joint marked derivatives and typed tensor counter-insertions must satisfy their corresponding estimates. We must prove these finer estimates, with their stated complex domains and uniform bounds, for the stronger comparison and Ward routes. The ordinary-source insertion and angular proofs below do not obtain their operator estimates by assuming these typed Ward derivatives. Section 33 proves ordinary old-level summations, scalar action-load and reserve implications. The finite Gaussian, insertion and renewed-pointing sections derive the local first response under their local Gaussian and bond-split hypotheses; the unmarked section derives the needed majorant from the stronger primitive matrix/core and potential package. Section 29 derives the ordinary formation bound using one near window member at each old base point before its bracket is applied. The linear bootstrap retains the source norm and gradient gain under these inputs, the constant-source slopes and the old-remainder bounds. Actual nonlinear background/core inclusions and history formation remain to be verified; these implications do not supply them.

C. Large-field histories and closed induction: four obligations. **Obligation C1: exact measurable operations.** We must construct every move as a specified Borel kernel, change of variables, or partition of unity on an explicitly defined space, and prove the following. The tests are total, including at configurations where logarithms or minimizers fail to exist. Every stored variable is integrated exactly once or retained exactly once. For every bounded gauge-invariant insertion F, the sourced construction preserves the total integral of $Fe^{-\beta A}$. An operation preserving only the integral without F cannot be used to transport that observable. For real inputs every summand used as a probability weight must be nonnegative.

Obligation C2: geometric margins and determination. We must give the allowed histories a constructive definition and prove the following. At level j, tests, local backgrounds and retained terms depend on at most a stated enlargement of the determining regions. Every enlargement lies inside the margin supplied by the next region. Healing, restriction, joining and continuation must preserve these properties, including near holes, seams, and waiting windows. On an admissible torus no object in these operations wraps around the torus. The number of histories with a specified size must have a proved entropy bound, with the same size function used in the rarity estimate.

Obligation C3: rarity with entropy and insertions paid for. For each created large-field component X at level j, we must prove a mass bound $\mathfrak{M}_j(X) \le \exp[-cp_0(g_j) - \kappa|X|_j]$ with the stated supremum over external complex data inside its internal integral when component majorants are multiplied. Here $p_0(g) = (\log g^{-2})^p$, and κ exceeds the entropy rate of C2. The bound must include the Wilson factor actually available to pay for that component, all stored Jacobians and gauge factors, and the designated insertion weights. A failure of existence of a background, as opposed to a quantitative large-field inequality, must also be covered. Drop-outs without their own Wilson factor need a separate bound after attachment to their carrying term.

Obligation C4: invariance of the inductive class. We must construct one nonempty class $\mathfrak{C}_j$ of complete presentations, with the same constants at every level, containing the bare measure, and prove that under the increment and tube conditions the exact step maps $\mathfrak{C}_j$ into $\mathfrak{C}_{j+1}$. Its

conclusions include the analytic polymer bounds, C1–C3, boundary bounds, covariance, and the required stability floor. All constants must admit one simultaneous order of choice, without a parameter needing to be both smaller and larger than quantities chosen later. This is the full obligation represented by the step assertion numbered 2.11; listing its clauses does not discharge it.

D. Sources and local correlation bounds: four requirements. **Obligation D1: the ordinary localized source bound and stability.** We must prove the precise torus conditions (ST5)–(ST6) of Section 38: holomorphic point counterterms and localized source energies, a strict entropy gap, source dependence only on the specified regions, and independence of the representation of a source function by a list of tests. On the ordinary source ball, we must also prove the normalized upper stability bound and unsourced partition-function floor there, with constants uniform in cutoff and allowed torus. The source class must include the global mirror-periodized Lipschitz profiles used in the chessboard construction. Fixed-cutoff bounds for point counterterms may depend on the cutoff but must be uniform in the torus.

For the graded deduction, this obligation also specifies the actual domain indexing of Section 39. Take $M = L^{m_0}$ with fixed integer $m_0 \geq 0$, and the cofinal compatible tori $\ell = 2L^m$, $m \geq m_0$. At scale $a_s = L^{-s}$ the localization cubes have side Ma_s; their counts in every direction are even integers. Every fixed power-of-L chessboard tiles all sufficiently large members of this family. All actual scales $0 \leq s \leq K$ are retained. A domain X is indexed by its union of these cubes, with $m(X)$ the minimum number of edges of a finite connected cubical edge tree meeting every cube. Its source dependence is contained in a fixed number of neighboring cube layers. The source-energy bound uses $e^{-\kappa m(X)}$, with κ above the rooted entropy rate for this actual indexing. If extra domain types are present, their multiplicity must be included in that stated entropy gap.

The closed-walk count in (GS1), the ball-root count in (GS2), and the global weighted count (GS3) are then proved geometric consequences. In particular, the sharp local count is $C(1 + u/a_s)^4 e^{-ct}$, and the global count is $C\ell^4 a_s^{-4}$. The weaker abstract (ST6) estimate alone does not imply these sharper bounds and is insufficient for the graded conclusion below. Checking that the localized RG energies use this specified geometry is part of D1, not supplied by writing an abstract domain sum.

These are raw partition-function bounds. A zero-free logarithm on the whole global source ball is not part of this obligation. The fixed-volume normalized zero-free lemma gives a suitable neighborhood when needed; the all-state transfer theorem instead constructs common holomorphic cell normalizers and obtains the derivatives at zero directly. The elementary transfer does not prove the ordinary RG source estimates in this obligation. Section 37 proves explicit sufficient criteria for the upper bound and floor from the normalized positive label/history representation: total-variation bounds with a weighted label sum, and a regular all-small-field label with its own action bound. We must still verify those criteria, including the reserve and allocation of Wilson factors. A pointwise polymer logarithm cannot replace that history representation without an exact common integral formula.

Requirement D2: graded all-state bounds derived from D1 and G1. Theorem 39.1 uses D1 in its actual rooted setting above and the proved chessboard domination from G1. It yields one fixed finite integer k and $B(R) = C(R+1)^4$ such that, for periodic Gibbs states ρ, ν at the same datum,

$$\left| \rho \prod_{i=1}^{n} [\mathcal{O}_K(f_i) - \nu(\mathcal{O}_K(f_i))] \right| \leq B(R)^n (n!)^k \max\{1, d^{-5n}\} \prod_i \|f_i\|_\sharp.$$

For $n \geq 2$ the cumulants under ρ satisfy the same estimate with $[2B(R)]^n$ in place of $B(R)^n$. Each support lies in its own radius-R ball, placed anywhere, and different supports have distance at least d. The first-order case uses the one-point difference bound. All constants are uniform in cutoff, states and support positions.

The proof gives a polynomial cell-normalizer loss $Q_i \leq Cn^p$ and $k = \lceil p \rceil + 5$. It uses a genuine product source polydisc from separate chessboard factors and common univariate normalizers. The separation singularity is explicitly d^{-5n}, not a single d^{-5} prefactor independent of n. Thus D1, its specified geometry and G1 imply D2. We must retain the actual D1 construction of Section 36; the present counting argument does not supply it independently.

Obligation D3: age-resolved uniform rarity. We must prove that the probabilities or positive weights of live regions persisting from a given level to a later level have a summable bound uniform in volume and in the location of a test. Witness selection, replication by reflections, and pattern counting must preserve the insertion and source weights. State the resulting tail function $r(m)$ with $\sum_m r(m) < \infty$, together with every additional moment in m used later. A rarity estimate for one newly created component is insufficient for this persistence assertion. This obligation represents the age and survival arguments wherever actual history persistence or weighted comparisons are used. The ordinary graded all-state estimate D2 is obtained by its GS route above and does not require this separate history-persistence obligation.

For deductions from a fixed-bump Riemann sum, the cutoff is required to be sufficiently fine.

Requirement D4: one-point transfer derived from D1. Theorem 38.1 proves, under D1, that periodic Gibbs states ν, ρ at the same datum satisfy

$$|\nu(\mathcal{O}_K(f)) - \rho(\mathcal{O}_K(f))| \leq C_1(R)\|f\|_\sharp.$$

If both states are invariant under all plaquette isometries, the bound implies $|\nu(a_p) - \rho(a_p)| \leq Ca_K^4$ by testing a fixed nonnegative bump of nonzero integral. The uniform bound on differences follows from a common holomorphic normalizer; neither individual uncentered mean is required to remain bounded. This is essential when a pure component is centered with respect to a mixture.

E. Flow, tuning and comparison of cutoffs: four obligations. **Obligation E1: a consistent flow.** We must prove that the recursion $x_j = \beta - \sum_{i \leq j} b_i$ is defined up to first passage, with $b_i \leq B_\sharp$ and $\lambda(j-i) - C \leq x_i - x_j \leq B_\sharp(j-i)$, $\lambda > 0$, and that its increments are independent of the admissible torus and of the unused finer cutoff. The elementary first-passage argument is applied to a recursion whose analytic formation and domain conditions we must establish through every preceding step of the run. Together with B3, these properties imply the leading logarithmic relation between β and the cutoff.

Obligation E2: a frozen calibration and tuning map. For auxiliary choices fixed independently of β, we must prove that the terminal map $\beta \mapsto x_K^\natural(\beta)$ is holomorphic on a stated complex neighborhood of a real interval, that $0 < c \leq \partial_\beta x_K^\natural \leq C$ on that interval, with endpoint values bracketing the prescribed renormalized inverse coupling, and that all intermediate values remain in the allowed tubes. This then supplies existence and local uniqueness of a tuned bare coupling by elementary analysis. The holomorphy assertion is not valid merely because the original finite-lattice partition function is entire. The flow section proves the slope implication both from the stronger independent coupling polydiscs and from the weaker marked bare-disc route. In the latter, the marked forcing is of size $C\epsilon_i'/\varrho$, not $C\epsilon_i'/x_{i-1}$, and is retained in the discrete Gronwall estimate. Its summability needs the stated exponent restriction. Canonical tuning uses the admitted real interval; broader presentation gluing additionally needs an admitted real overlap. Overlapping geometric complex discs alone do not establish that admission.

Obligation E3: a summable matched-cutoff comparison. At matched terminal coupling, we must prove for each fixed torus and separated tuple

$$|S^T_{n;K+1}(X) - S^T_{n;K}(X)| \le C(n,d,g,\ell)\epsilon_K \prod_i \|f_i\|_\sharp,$$

$$\sum_K \epsilon_K < \infty.$$

The floor in K may depend on the fixed torus. To compare two nearby terminal couplings, add a uniform Lipschitz term proportional to their difference. Neither assertion implies infinite-volume uniqueness. This is the actual analytic obligation behind the matched-run claims numbered 8a and the tuned Cauchy sequence; a bounded difference at each cutoff does not suffice.

Obligation E4: compatibility of calibrations. When moving between a self-calibrated first-passage run and a frozen run, we must prove that the maps represent the same bare integral and satisfy a stated uniform bound on their coupling discrepancy, small enough for every comparison domain and source radius used above. Exact dilation identifies the same finite lattice in two units, but supplies no estimate on this calibration discrepancy. Uniqueness statements must state which terminal coupling and which calibration they use.

F. Short-distance normalization and nontriviality: four requirements. **Requirement F1: the proved free-kernel comparison.** The exact continuum coefficient is proved in Section 4. For the remaining lattice comparison, let $a = \ell/n$, with integer n, and use plain coordinate inner products on site, bond and positively oriented plaquette cochains of the periodic lattice. Let d_1 be its unscaled circulation matrix and P_{ex} the orthogonal projection onto $\operatorname{ran} d_1$. A standard Gaussian on that range has covariance P_{ex}. Its physical curvature kernel is the 6×6 matrix of components

$$K_{a,\ell;P,Q}(p,q) = a^{-4} P_{\mathrm{ex}}(p_P, q_Q),$$

with each component sampled at its own plaquette barycentre. Section 5 proves the following comparison with the continuum matrix K, with $\gamma = 1/2$:

$$|K_{a,\ell;P,Q}(p,q) - K_{P,Q}(x(p) - x(q))| \le C r^{-4}\big[(a/r)^\gamma + (r/\ell)^4\big],$$
$$a \le r/10,$$
$$r \le \ell/10,$$
$$r = |x(p) - x(q)|,$$

for some fixed $\gamma > 0$, uniformly in a, ℓ, plaquette positions and orientations. Periodic differences are represented by the short displacement in this range. The same section proves separated smearing convergence and the resulting quadratic-density coefficient. This closes the free exact-cochain Gaussian requirement; the interacting extraction F2 must still be proved separately.

The projection convention in this requirement is distinct from physical weighted block means. When passing to unit physical blocks use the adjoints and physical difference d_η established in the block-normalization section. Under plain fine and coarse inner products an arithmetic block mean does not have an isometric adjoint; no projection identity may silently interchange these conventions. Neither normalization proves the interacting extraction F2.

Obligation F2: extraction of the leading two-point term. For the prescribed short-distance test pairs we must prove the finite-cutoff identity

$$S^T_{2;K,\ell}(f,h) = C_G \zeta_K(d)^2 g_K(d)^4 \mathcal{K}(f,h)(1 + r_{K,\ell}(f,h)),$$

where $C_G > 0$, the real-data error satisfies $|r| \le 1/2$, and all lower cutoff requirements are explicit. For the angular deduction this obligation must cover the full class $\mathfrak{T}(d;\vartheta_*)$ in (TU9)–(TU13) of Section 49, with $\vartheta_* = 1/12000$, or a broader class. These are nonnegative C^1 tests in radius-d balls, with support distance between d and $6d$, the stated lower integral and upper Lipschitz bounds, and arbitrary positions. A law proved only for a narrower shape class is insufficient without a further inclusion argument.

We must prove the same bound for $S_2^T[\rho]$ in every periodic Gibbs state used to approximate the limit or any component. At fixed d its eventual cutoff threshold and constants must be uniform in those states and support positions. This uniformity in the state and support positions is part of F2. On each fixed torus any asserted $r \to 0$ needs a separate uniform estimate; it must not be silently promoted to a statement uniform in volume. This is the substantive Gaussian-extraction obligation behind the short-distance law, including all interaction and large-field remainders. The finite source insertion and renewed-pointing proofs retain the $\alpha^2 s$ potential norm under the actual unmarked-majorant, leg-path and complex-domain hypotheses. The window formation and linear bootstrap propagate it under their explicit ordinary inputs; the resulting near-field estimate requires the complete field to have size δ, not merely a δ-sized increment on an arbitrary α-background. Remote and holonomy tails retain their separate bounds. These deductions do not prove the background and history obligations for the interacting construction.

The actual clean-history normal form must still control every old activity's localized Hessian, the conditional exterior force and pointwise deletion with the same exterior field and correct flag mass. The actual nonlinear averaging-fibre comparison is an additional obligation for that particular route; gauge fixing the actual run provides a route that avoids it. That route has its own normal-form, exterior-force and pointwise clean-class conditions, which we must prove. Section 44 proves the covariance arithmetic and common-normalizer transfer once their inputs hold. Section 31 derives the old-activity localized Hessian from the actual all-substitution window room, window-first value bounds and holomorphy on a Gaussian-coordinate ball of radius $c\alpha/g$. The resulting scale-small bound has factor $(g/\alpha)^2$. The room mechanism is in Section 29; the conditional exterior-force and pointwise clean-class laws are outlined in Sections 43.1–43.3. The local one-step chart or a value bound without a complex domain does not supply those conditions.

The transfer of a Gaussian reference law has two different scopes. D1's common cell normalizer compares any two periodic infinite-volume states; an infinite reference obtained with a Gaussian estimate uniform along arbitrarily large tori therefore needs no finite-volume vacuum comparison. A law proved only on one prescribed finite reference torus instead requires a comparison with those states. Compatible local counterterms, the floor and vacuum-density comparison give one sufficient route. The local companion criterion (ET8)–(ET10) proves that vacuum comparison from explicit nonwrapping and normalization data; it is not required in the infinite-reference route. Per-torus estimates whose distance thresholds decrease with the volume do not establish the latter reference law. Section 44.3 supplies the finite-reference route directly: upper stability is used on the reference torus, and all larger tori contribute only their unsourced floors and the common vacuum-density bound.

Obligation F3: the running normalization at the witness scale. The witness depth must be defined without reference to a desired conclusion, and its lattice separation must lie within the permitted comparison range. We must prove

$$g_K(d)^{-2} = c_* \log(1/d)(1 + o(1)),$$

$$\zeta_K(d) = 1 + o(1),$$

$$c_* > 0,$$

with the precise uniformity needed by F2. A derivative of a running coupling with respect to a constant source must be shown to equal this normalization; it is not an independent choice of field renormalization. F2–F3 give a nonzero two-point limit by a closed lower bound. Appendix A proves summability of the normalization slopes from the actual increment estimates on coupling-sized independent complex polydiscs and control of the large-field remainder loss. The flow section preserves the alternative with an actual marked remainder on a fixed bare-coupling disc and its separate summable forcing. A sharp $O(g \log^p g^{-2})$ slope rate on this weaker route needs $p_R \geq 3/2$; summability alone needs $p_R > 1$. Analyticity on an ordinary source disc alone does not provide the required marked size and history bounds.

Here is their sufficient ultraviolet consequence, with its size estimate included. Use

$$\mathcal{K}(f,h) = \iint f(x)h(y)|x-y|^{-8}\,dx\,dy.$$

If the nonnegative supports lie in radius-d balls at distance at least d, and v_4 is the volume of the unit ball in $\mathbb{R}^4$, then

$$0 \leq \mathcal{K}(f,h) \leq d^{-8}\|f\|_{L^1}\|h\|_{L^1} \leq v_4^2\|f\|_\infty\|h\|_\infty.$$

The uniform F3 bounds imply $|\zeta_K(d)| \leq Z$ and $g_K(d)^4 \leq C[\log(1/d)]^{-2}$ for small d and all permitted sufficiently fine cutoffs. Thus F2 gives

$$0 \leq S_2^T[\rho](f,h) \leq C'[\log(1/d)]^{-2}\|f\|_\infty\|h\|_\infty.$$

The lower inequality uses $1 + r \geq 1/2$; the upper inequality uses $1 + r \leq 3/2$. At each fixed pair these closed inequalities pass to every limit and to the all-state phase realizations. Taking $u(d) = C'[\log(1/d)]^{-2}$ proves precisely (TU10), with $u(d) \to 0$. This deduction does not verify the state-uniform extraction or running normalization required in F2–F3.

Obligation F4: interpretation of the observable sector. The conclusions of F1–F3 concern the scalar curvature-density sector. To identify a reconstructed theory as the full Yang–Mills theory, we must specify a sufficient gauge-invariant observable family, its consistency relations, renormalization, and its relation to this scalar sector, and prove the needed continuum gauge identities or an equivalent identification. A nonzero scalar Wightman field alone does not provide this identification. Until this is proved, the conclusion must be stated as existence of the conditional scalar observable theory, with full gauge-theory identification left open.

G. Symmetries, rotations and phases: four requirements. **Requirement G1: proved finite-lattice symmetries and positivity.** Section 6 proves link-reflection invariance and the common period $2\mathbb{Z}^4$ for every periodic SU(2) Wilson DLR state at fixed $\beta > 0$. Section 7 then proves invariance and reflection positivity at every coordinate site, coordinate link, and diagonal lattice mirror, on the full bounded Borel half-space bond algebra. No restriction to torus-selected states is made. The compositions give all unit translations and signed permutations. At $\beta = 0$ the unique product Haar state has the same properties. Adapted split tori are used for diagonal mirrors; an invalid diagonal split of an ordinary cubic torus is not assumed.

Requirement G2: scalar rotations derived from ordinary sources and the ultraviolet law. Consider limits through periodic Gibbs states, including the canonical components with their all-state realizations. G1 and the compactness inheritance argument provide their coordinate and

diagonal reflection positivity; Section 42 provides the exact coordinate cube cone. A component with mean $c\int f$ is first centered at its own mean. Its self-centered cutoff moments converge to this centered functional by the finite subset expansion and the convergent one-point differences; the same expansion restores the invariant mean after the proof. Under D1 in its actual rooted geometry, Theorem 46.1 proves a Hilbert insertion bound uniform in the number of surrounding fields. This operator estimate does not follow from D2 alone.

Theorem 47.1 converts that bound to an analytic L^1 plate density, with the explicit loss of five powers of the time margin. Section 48 constructs the holomorphic product kernels and vectors, glues their frames, and proves the finite angular band and continued Gram positivity. Theorem 49.1 uses the exact (TU10) shape law to remove every nonconstant two-point harmonic, including the critical frequency; its diagonal measure and cone support are constructed in the proof. Theorem 50.1 proves polygon divisibility, integer convexity, removal of exceptional distributional support by common cuts, and full O(4) covariance. QF transfers this covariance to the diagonal-flat Schwartz domain.

Together with the preceding deduction from F2–F3 to (TU10), this is a conditional proof of G2 for this state class. It assumes neither typed Ward counter-insertions nor vanishing distant-shell flux. It does not verify D1, F2 or F3. An arbitrary growing ordinary cubic-torus limit must separately satisfy the inherited mirror and cone interface before this route applies; being a separated-point limit alone supplies no such identification.

The retained stronger Ward route still has its own contract: an exact lattice identity must pass to $\delta_\omega S_n(X) = -\lim_{R\to\infty}\lim_j \langle \mathfrak{B}_{\omega,R}; X\rangle_j$, with every bulk and discretization error controlled and the claimed boundary limit existing and vanishing. Wherever that route is invoked, we must prove these limits in that order, with the typed estimates they need. The angular proof does not establish those separate comparison estimates.

Requirement G3: the proved finite-spacing plateau law. For a spacing $a > 0$, let $\Omega_a = \mathrm{SU}(2)^{\mathcal{B}_a}$, where $\mathcal{B}_a$ is the countable set of positively oriented bonds of $a\mathbb{Z}^4$. Give this space its product topology. Let $\mathcal{G}_{a,\beta}$ be the translation-invariant probability measures satisfying the Wilson DLR equations at inverse coupling β. The finite-bond conditional kernel integrates the bonds in W against the normalized density $\exp[-\beta\sum_{p:p\cap W\neq\varnothing} a_p]$ and holds the other bonds fixed. These kernels take continuous local functions to continuous local functions. Thus $\mathcal{G}_{a,\beta}$ is a compact convex set. Its reflection invariance and reflection positivity are proved under G1.

Let $\mathcal{I}$ be the completed σ-field of events invariant, modulo the measure, under all translations in $a\mathbb{Z}^4$. For bounded local functions A, B define

$$\mathrm{Pl}_\nu(A,B) = \nu\big(\overline{E_\nu[A\mid\mathcal{I}]}E_\nu[B\mid\mathcal{I}]\big).$$

Theorem 8.1 proves the following law, with one numerical constant C independent of a,β,ν,r,n,A,B. If $r\geq 1$ is an integer, A,B depend only on bonds with both endpoints in $a[-r,r]^4$, and $y\in a\mathbb{Z}^4$ has $n = |y|_\infty/a - 2r \geq 1$, then

$$|\nu(\overline{A}\,\tau_y B) - \mathrm{Pl}_\nu(A,B)| \leq C\left[(r/n)^2 + (r/n)^3\right]\mathcal{N}_{\nu,r}(A)^{1/2}\mathcal{N}_{\nu,r}(B)^{1/2},$$

where

$$\mathcal{N}_{\nu,r}(A) = \max_{\mu=1,\dots,4;\,\gamma\in\Gamma_{16}} \nu\big(\overline{\theta_\mu\gamma A}\tau_{2rae_\mu}\gamma A\big) \leq \|A\|_\infty^2.$$

Here θ_μ reflects x_μ to $-x_\mu$, and Γ_{16} is the group of coordinate sign changes. Nonnegativity of these reflection sizes follows from G1. The law holds for every $\nu\in\mathcal{G}_{a,\beta}$, including mixtures. It is a statement before any continuum component is constructed.

Theorem 51.1 constructs the required conditional probabilities on this compact metric space, including Borel versions and simultaneous probability-valued uniqueness. Applied to $\mathcal{I}$, this theorem, the DLR equations, and the law above yield a unique decomposition into ergodic DLR states. The argument and the required closedness of the pure state set are proved in the phase-interface section. Compactness of that pure set and existence of continuum components are conclusions, not assumptions.

Requirement G4: all-state moments derived from ordinary sources. Choose $a_j = L^{-K_j} \downarrow 0$, couplings β_j, and centering states $\nu_j \in \mathcal{G}_{a_j,\beta_j}$. Write $\mathcal{O}_j(f) = \sum_p f(x(p))a_p$. Once the precise ordinary source obligation D1 is proved, Theorem 38.1 gives for every $\rho \in \mathcal{G}_{a_j,\beta_j}$ that

$$|\rho(\mathcal{O}_j(f)) - \nu_j(\mathcal{O}_j(f))| \le C_1(R)\|f\|_\sharp,$$
$$|\kappa_n^\rho(\mathcal{O}_j(f_1), \dots, \mathcal{O}_j(f_n))| \le C_n(d;R) \prod_i \|f_i\|_\sharp$$
$$(n \ge 2).$$

The tests are real C^1 functions; each support is contained in a ball of radius R, and different supports have distance at least $d > 0$ in the second bound. Here $\|f\|_\sharp = \max(\|f\|_\infty, c_M\|\nabla f\|_\infty)$. The constants are independent of j, ρ and support positions. They also apply to the separated reflected copies used in G3. For the chosen sequence, we take the moments centered at ν_j to converge on every separated tuple, passing to a subsequence by the preceding compactness theorem if necessary. The per-order transfer alone would not establish factorial growth. In the actual rooted setting of D1, Theorem 39.1 also provides D2, and Theorem 41.1 gives H1. The bounds displayed here and G3 suffice for the canonical decomposition constructed below. They do not assume that decomposition or clustering of its future components.

H. Test spaces and reconstruction: four requirements. **Requirement H1: quantitative diagonal-flat extension derived from D2.** Let $\mathcal{S}_{\neq}(\mathbb{R}^{4n})$ be the Schwartz functions flat, to all orders, whenever two arguments coincide. Under the D2 moment bound, passed to a separated-tuple limit, Theorem 41.1 constructs its unique continuous extension to this space and proves

$$|\mathsf{S}_n(\varphi)| \le C_{\mathrm{E}}^n (n!)^{\kappa_{\mathrm{E}}} \sup_{|\alpha| \le k_{\mathrm{E}} n, x} (1+|x|^2)^{k_{\mathrm{E}} n/2} |\partial^\alpha \varphi(x)|,$$
$$k_{\mathrm{E}} = 20,$$
$$\kappa_{\mathrm{E}} = k + 32.$$

The constants are independent of n. The same uniform statement applies to component limits obtained from the all-state bounds. Only the position-uniform radius-one moment bound is used; a unit partition already gives an effective $B(R) \le CB(1)(1+R)^4$. The explicit dyadic/product-box proof includes tensor-relation independence, all diagonal cutoffs, infinity cutoffs, and the positive-time boundary. Its constrained tensor density transfers reflection positivity to finite positive-time Schwartz sequences. Thus D1, its actual GS geometry and G1 imply H1.

A distribution only on the complement of the diagonals with unspecified constants would not establish this result. Diagonal-flat functions are not dense in the full Schwartz space; an extension across diagonals or contact-term identities still need a separate renormalization statement.

Requirement H2: positivity and continuous Euclidean evolution. Section 42 proves the compact-tuple Hilbert construction from invariant separated moments, full reflection positivity and position-independent per-ball bounds. It proves strong continuity, time contraction and the

cube spectral cone for the stated lattice approximation. For the full positive-time test sequences in the precise reconstruction theorem, the required finite quadratic inequalities are $\sum_{p,q} \mathsf{S}_{p+q}(\Theta\overline{\varphi}_p \otimes \varphi_q) \geq 0$. All tensors in this expression must belong to the declared domain. Under H1 and the constrained tensor-density lemma, those full sequence inequalities and the same Hilbert evolution follow from the compact-tuple construction. The proof bounds all translated matrix elements before iterating Cauchy–Schwarz. Its position-independent per-ball hypotheses are supplied by D1 for the ordinary phase moment bounds. The compact argument does not by itself give quantitative Schwartz growth; that additional conclusion is now the GS/QF deduction recorded as H1 above.

Imported theorem H3: the reconstruction theorem in its applicable form. The reconstruction step uses an Osterwalder–Schrader theorem whose test spaces and growth condition match H1–H2. Its other hypotheses are Euclidean invariance, permutation symmetry, and the required cluster property for a unique vacuum. G2 supplies the invariance for the periodic-state limits and components once the ordinary-source and ultraviolet obligations D1 and F2–F3 are proved. The reconstruction theorem itself remains external. Its conclusion is a scalar Wightman theory with tempered Minkowski distributions, locality, a positive Hilbert metric, and the forward spectral cone. Here it is used as an imported theorem. Contact terms invisible on diagonal-flat Euclidean tests must not be assigned arbitrary values and then treated as consequences of the reconstruction.

H4: conclusions and scope. To complete the proof on its declared route for periodic-state limits, the actual D1 and F2–F3 proof outlines must be completed and a sequence of admissible data constructed. The supporting propositions then supply the downstream scalar conditions, and H3 is applied in its matching test spaces and growth class. D1 and the interacting short-distance law are thus obligations of this route, not assumptions. The conclusions that use weighted, matched-cutoff or Ward comparisons require those comparisons to be proved as well. Identification with the full gauge theory additionally requires F4. A mass-gap conclusion requires a separate spectral estimate, for example one placing the spectrum outside zero in $[m, \infty)$ for some $m > 0$. The conclusions are therefore taken in stages: a finite lattice measure, a subsequential separated-point limit, an axiomatic scalar theory, and its Yang–Mills identification, each with its own obligations.

10. From local source bounds to continuum limits

10.1. **From local source estimates to separated correlation limits.** This section shows that the local analytic energies and residual finite-volume integral bounds stated below, which we must prove, yield the source-disc, cumulant and limit conclusions of this section. Exact measure transport and geometric forgetting of older scales enter through their own stated conditions, which we must also prove.

10.1.1. *Finite-dimensional starting point.* Let $\mathcal{P}$ be a finite set of plaquettes, let $G = \mathrm{SU}(2)$, and assign one variable $U_b \in G$ to each positively oriented bond. Opposite orientations carry $U_{-b} = U_b^{-1}$. The product of normalized Haar measures is denoted by dU. Write

$$a_p(U) = 1 - \tfrac{1}{2} \operatorname{Re} \operatorname{tr} U_p,$$

$$A(U) = \sum_{p \in \mathcal{P}} a_p(U).$$

Since $0 \leq a_p \leq 2$, the probability measure

$$d\mu_\beta(U) = Z_\beta^{-1} e^{-\beta A(U)} dU,$$

$$Z_\beta = \int e^{-\beta A(U)} dU$$

is well defined for every real β. Given bounded real functions f_i on plaquette barycentres, set $X_i = \sum_p f_i(x_p) a_p(U)$ and

$$\Gamma(z) = \int e^{\sum_i z_i X_i} d\mu_\beta.$$

The bound $|X_i| \le 2 \sum_p |f_i(x_p)|$ implies uniform domination on every compact subset of $\mathbb{C}^n$. Differentiation under the integral therefore proves that Γ is entire. Since $\Gamma(0) = 1$, it has a holomorphic logarithm in a neighbourhood of the origin, and the joint cumulant is defined by

$$\langle X_1; \dots; X_n \rangle = \partial_{z_1} \cdots \partial_{z_n} \log \Gamma(0).$$

The neighbourhood obtained this way can depend on the number of plaquettes.

If T is a fixed linear operation on finite measures which preserves their total mass, then $T(e^{\sum z_i X_i} \mu_\beta)$ has total mass $\Gamma(z)$. For a finite sequence of such operations the same conclusion follows by induction. If T is specified only on gauge-invariant densities, all insertions in this assertion must be gauge invariant. Neither linearity nor mass preservation supplies a uniform analytic bound on the resulting density.

10.1.2. *A zero-free disc from a normalized bound.*
Lemma. Let $c_i > 0$ and put $\|z\|_c = \sum_i c_i |z_i|$. Suppose h is holomorphic on $\|z\|_c < 1$, $h(0) = 1$, and $|h| \le Q$, where $Q \ge 1$. Then

$$|h(z) - 1| \le (Q+1) \|z\|_c.$$

In particular h is nonzero on $\|z\|_c < (Q+1)^{-1}$. On the smaller ball $\|z\|_c < [2(Q+1)]^{-1}$ its principal logarithm satisfies $|\operatorname{Log} h| \le \log 2$.

Proof. For $z \neq 0$ write $z = tv$, where $t = \|z\|_c$ and $\|v\|_c = 1$. The function $(h(\zeta v) - 1)/(Q+1)$ maps the unit disc into its closed unit disc and vanishes at zero. The Schwarz lemma gives the first estimate. Strictness on the stated open balls follows by using slightly smaller compact discs and taking limits. For $|u| \le 1/2$, the logarithmic series gives

$$|\operatorname{Log}(1+u)| \le \sum_{k \ge 1} \frac{|u|^k}{k} = -\log(1 - |u|) \le \log 2.$$

The same series defines a single holomorphic branch on that ball. □

Corollary. Suppose $\Gamma = e^{\mathcal{E}} h$ near zero, where $\mathcal{E}(0) = 0$ and the lemma applies to h. Then

$$|\partial_{z_1} \cdots \partial_{z_n} (\log \Gamma - \mathcal{E})(0)| \le \log 2 \, [2(Q+1)n]^n \prod_i c_i.$$

Proof. Use Cauchy's integral formula on a polydisc with radii strictly smaller than $[2(Q+1)nc_i]^{-1}$. The sum of the weighted radii is less than $[2(Q+1)]^{-1}$. Each variable is differentiated once, so no factorial occurs in this application of the formula. Let the radii increase to their stated values. □

The dependence on Q is material. For example, $\cosh(Vz)$ is the moment-generating function of a random variable taking the values V and $-V$ with equal probabilities. It is at most e^V on the unit disc, but has a zero at $i\pi/(2V)$. Thus an exponential-in-volume estimate alone cannot give a volume-independent zero-free disc.

10.1.3. *A precise local-energy hypothesis.* Fix $L > 1$, a finite physical volume V, and a scale index $s \geq 0$, with physical scale $a_s = L^{-s}$. A local domain D has an integer size $m(D) \geq 0$ and a dependence region D^+. Assume there are constants $C_0, C_1, \kappa_{\mathrm{ent}}$ such that

$$\operatorname{diam} D^+ \leq C_0 a_s (m(D) + 1),$$

and, for every fixed compact test support B, the number of domains of size m whose dependence region meets B is at most

$$C_1 V a_s^{-4} e^{\kappa_{\mathrm{ent}} m}.$$

The factor V is a finite-volume allowance; a more local counting estimate can replace it when available.

For cutoff K let $e_{K,s,D}(w)$, $0 \leq s < K$, be holomorphic on the source ball

$$\sum_i |z_i| \|f_i\|_* / r < 1,$$

$$w = \sum_i z_i f_i,$$

where $r > 0$ is independent of K. Suppose

$$e_{K,s,D}(0) = 0,$$

$$|e_{K,s,D}(w)| \leq A e^{-\kappa m(D)},$$

$$\kappa > \kappa_{\mathrm{ent}}.$$

Finally suppose $e_{K,s,D}$ depends on w only through its values in D^+. These statements are the local-energy hypothesis. They include analyticity, an entropy gap, and geometric control of the dependence region.

Proposition. Let $n \geq 2$ and suppose the supports of $f_1, \ldots, f_n$ have pairwise distance at least $d > 0$. Under the local-energy hypothesis,

$$\left| \partial_{z_1} \cdots \partial_{z_n} \sum_{s<K} \sum_D e_{K,s,D}(w) \Big|_{z=0} \right| \leq CAV \max\{1, d^{-4}\} (n/r)^n \prod_i \|f_i\|_*.$$

The constant is independent of K; it can depend on L, C_0, C_1 and the entropy gap.

Proof. If D^+ misses one support, the energy is independent of the corresponding variable, and the mixed derivative vanishes. Every contributing domain therefore meets both the first and second supports. Consequently

$$m(D) \geq d/(C_0 a_s) - 1.$$

Cauchy's formula with radii approaching $r/(n\|f_i\|_*)$ bounds each derivative by

$$A(n/r)^n \prod_i \|f_i\|_* \, e^{-\kappa m(D)}.$$

A zero test function makes the assertion immediate, so these radii may be assumed finite. The entropy estimate and a geometric series in m bound the sum at scale s by a constant times

$$AVa_s^{-4}\exp(-cd/a_s),$$

with a larger harmless constant when d/a_s is bounded. It remains to estimate $\sum_{s\ge 0} L^{4s}e^{-cdL^s}$.

For $0 < d \le 1$, let s_* be the largest integer with $dL^{s_*} \le 1$. The sum over $s \le s_*$ is at most $CL^{4s_*} \le Cd^{-4}$. Writing $s = s_* + j$ in the remaining sum bounds it by

$$Cd^{-4}\sum_{j\ge 1} L^{4j}e^{-c'L^j},$$

which converges by the ratio test. For $d > 1$, replace d by 1 in the exponential. This proves the stated uniform bound. At each fixed cutoff the domain sums converge normally on smaller source balls, permitting termwise differentiation there. Only the separated mixed-derivative estimate is asserted uniformly as the number of scales grows; the undifferentiated energy sums need not converge in that limit. □

If in addition $\Gamma_K = e^{\mathcal{E}_K}h_K$, where $\mathcal{E}_K = \sum_{s,D} e_{K,s,D}$ and $|h_K| \le Q(V)$ with $h_K(0) = 1$, the corollary above gives a cutoff-independent bound on every separated cumulant. This conclusion remains a finite-volume conclusion when Q depends on V.

10.1.4. *What a comparison of cutoffs must prove.* Uniform boundedness gives compactness only after equicontinuity and a suitable countable test class have been supplied. Convergence along a prescribed whole sequence requires more. One sufficient condition is a summable comparison: for every fixed separated tuple,

$$|S_{K+1}(f_1,\dots,f_n) - S_K(f_1,\dots,f_n)| \le C(f_1,\dots,f_n)\epsilon_K,$$
$$\sum_K \epsilon_K < \infty.$$

The triangle inequality then makes the sequence Cauchy. Its tail is bounded by $C\sum_{k\ge K}\epsilon_k$.

A local proof of this comparison can split the scales at $K/2$. At $s \le K/2$, it is enough to compare paired local energies with a bound

$$|e_{K+1,s,D} - e_{K,s,D}| \le C(1+s)\epsilon_K e^{-\kappa m(D)}.$$

At $s > K/2$, the individual energy bounds already give a tail controlled by $\sum_{s>K/2} L^{4s}e^{-cdL^s}$. This tail is summable in K. The factor $1 + s$ in the paired bound is harmless because $\sum_s (1 + s)L^{4s}e^{-cdL^s} < \infty$ for each $d > 0$. The residual integrals must also be compared on a common zero-free source ball. If both residuals are within $1/2$ of 1, then the logarithmic difference is bounded by twice the difference of the residuals, using the line segment between their values and the derivative $1/z$ of the logarithm. Cauchy's formula converts this bound into a cumulant comparison.

These are consequences of an actual paired estimate. A bound for two covariant constant-source lists does not automatically prove the estimate for position-dependent arrays. Similarly, ordinary holomorphy in an unweighted norm does not create geometric factors for the age of an inherited activity. Those factors must be present in a separately established weighted estimate or in the domain and bound of a weighted holomorphic extension.

10.1.5. *A conditional tuning lemma.* Let F_K be a real differentiable renormalized-coupling coordinate on an interval I_K, and suppose

$$2/3 \le F_K'(\beta) \le 4/3$$
$$(\beta \in I_K).$$

If F_K takes values on both sides of x_*, the intermediate value theorem gives a unique $\beta_K \in I_K$ with $F_K(\beta_K) = x_*$. Moreover,

$$\tfrac{2}{3}|\beta - \beta'| \le |F_K(\beta) - F_K(\beta')| \le \tfrac{4}{3}|\beta - \beta'|.$$

Both inequalities follow from the mean value theorem on the interval between the two points. The inverse coordinate is therefore Lipschitz with constant $3/2$ on its image. If $F_K(\beta) = \beta - \sum_{j\le K} b_j(\beta)$, the derivative hypothesis follows from $\sum_j \sup_{I_K} |b_j'| \le 1/3$. We must establish the existence of the interval, the crossing of the target, and the derivative estimate through the construction.

The conclusions of this section concern separated correlation functionals. They do not, without additional estimates, provide extensions across coincident points, reflection-positive distributions on the full test-function algebra, rotation covariance, or a characterization of the resulting scalar observable sector as a Yang–Mills theory. Those are additional mathematical steps with their own hypotheses.

11. Weighted analytic domains and a local test for scale forgetting

A multiscale argument needs several analyticity statements with different meanings. Holomorphy of a finite-dimensional partition function is the weakest of them. Holomorphy of a renormalization step on a class of effective interactions is stronger. Geometric forgetting of old interactions is stronger still. The latter can follow from an analytic extension with appropriate domains and bounds, but the weights in that extension must themselves be justified.

There is an additional distinction for large-field regions. A small probability of retaining an old region can appear as a small factor in the bound on an analytic function. It does not in general permit the analytic input radius to grow by the reciprocal of that probability. This section makes these distinctions precise, reconstructs a suitable input space, and isolates local inequalities that can be tested before undertaking a full continuum reconstruction. It does not assert that the required inequalities are false for the Yang–Mills step.

11.1. **The ordinary input space.** Fix a cutoff, a reference sequence of inverse couplings $\theta_0, \dots, \theta_k$, a complete set of labels and tests, and hereditary complex domains $\mathcal{U}_j(X; r_{0,j}, r_{1,j})$. A point of such a domain is a pair $(\mathbf{U}, \mathbf{J})$; the current $\mathbf{J}$ is an independent complex variable. A regular background gives the special pair $(U, J(U))$. An identity verified at these special pairs is not automatically an identity at all pairs in the complex domain.

One convenient Banach space of pointed local families has the norm

$$\|E\|_{j,\kappa} = \sup_{X,z} e^{\kappa d_j(X)} \sup_{\mathbf{p}\in\mathcal{U}_j(X;r_{0,j},r_{1,j})} |E(X,\mathbf{p},z)|.$$

When families have infinite-lattice companions, include their norms in the maximum. Gauge invariance, the prescribed translation covariance, and compatibility with restriction are closed linear conditions. Holomorphy is preserved under convergence in this supremum norm on the declared domains. These facts specify a closed space of analytic families once the domains themselves have been fixed. We must still prove the existence and uniform size of those domains.

A regular Euclidean member comprises a family G, its companion, and its scalar quadratic coefficient $b(G)$. It has covariant pointed totals on the prescribed regular backgrounds and carries its own counterterm. A spatially varying input can assign a different such member $G_z^{(j)}$ to every base point z. This does not require the array $z \mapsto G_z^{(j)}$ to be translation invariant. Its norm is $\sup_z \|G_z^{(j)}\|_{j,\kappa}$. Each assigned member must retain its own coefficient and the required pointed covariance; covariance of the sum of all base points is a weaker condition.

The complete input also contains a bare plaquette weight u, defect families $D^{(j)}$, remainders $H^{(j)}$, boundary terms B, and excess exponents Ψ in stored large-field operations. A useful product box has inequalities of the following form:

$$\|u\|_\infty + \sum_{j\le k} \sup_z |b(G_z^{(j)}) - b_j^{\mathrm{ctr}}| \le C_u, \qquad \sup_z \|G_z^{(j)}\|_{j,\kappa} \le E_0,$$
$$\|D^{(j)}\|_{j,\kappa} \le L^{-(m-j)}, \qquad \|H^{(j)}\|_{j,\kappa} \le C_H \theta_j^{-\kappa_0/2}, \qquad m \in \{k, k+1\}.$$

Here b_j^{ctr} denotes the coefficient of the chosen real centre. Boundary terms are measured against specified positive majorants $\overline{B}_{j,h}(X)$:

$$\|B\|_{\mathrm{bdry}} = \sup_{j,h,X} |B_{j,h}(X)|/\overline{B}_{j,h}(X).$$

For a stored history ω let $N(\omega)$ count its stored sites, each at its own scale. Excess exponents require both a bound on $|\Psi_\omega|/N(\omega)$ and a carrier-by-carrier oscillation bound. If $\Psi_\omega = \Psi_\omega^0 + \sum_f \psi_f$, the latter can be expressed as

$$\sup_{\omega,R} \overline{M}_\omega(R)^{-1} \sum_{f: R_f = R} \operatorname{osc} \psi_f < \infty.$$

The carrier majorant and the history size are part of the definition, not consequences of holomorphy.

A source trajectory fits this format by freezing the source separately at each base point. Schematically,

$$G_z^{(j)} = E^{(j),\zeta}\big|_{\zeta = w(p_z)},$$
$$D^{(j)}(X,\cdot,z) = E^{(j),w}(X,\cdot,z) - E^{(j),w(p_z)}(X,\cdot,z).$$

For a Lipschitz source, a local freezing estimate can give $\|D^{(j)}\| \le CL^{-(K-j)}\|w\|_{\mathrm{Lip}}$. This smallness is a property of the source trajectory. An arbitrary defect in an ordinary box has only the size stipulated by that box.

This organization resolves a possible misconception. Global noncovariance does not by itself invalidate a construction with sources: Euclidean members can be frozen site by site. The remaining task is to prove the asserted bounds for the full array, including its noncovariant defects, its variable Wilson weight, and its stored-history operations. Scalar marginal cancellation for a globally covariant list alone does not complete that task.

11.2. **Three kinds of weights.** Let e_j measure a difference of regular families and let r_j measure a difference of remainders. The regular irrelevance weights have the form

$$\omega_\nu \le C\nu^4 L^{-\sigma(\nu-1)},$$

$$\sigma > 0.$$

They represent decay after the vacuum and the applicable marginal quadratic terms have been removed. In contrast, an unrenormalized remainder can enter with a squared ratio of analyticity radii,

$$w_{n,j} = \left(\frac{r_{0,n}}{r_{0,j}}\right)^2 \asymp \frac{\theta_j}{\theta_n}\left(\frac{\log \theta_n}{\log \theta_j}\right)^{2q}.$$

These weights are not geometric forgetting factors. A small size of the remainder, rather than an extra symmetry cancellation, pays for their contribution.

For a typed tangent list, let z_j and b_j be nonnegative bounds on the absolute marginal amplitude and increment of the direction. Here b_j is a size bound, distinct from the signed running increment in the flow; the weights and the size entries e_j, r_j in the load below are also nonnegative. Define

$$z_n^\sharp = \sum_{j\le n} L^{-(n-j)}(z_j + z_{j-1}),$$

$$z_{-1} = 0,$$

and a load

$$p_n(\delta) = \sum_{j\le n} \omega_{n+1-j} e_j + \sum_{j\in \mathcal{J}_n} w_{n,j} r_j + \gamma_n \left(z_n^\sharp + \sum_{j\le n} L^{-c(n-j)} b_j \right),$$

$$\gamma_n \asymp \theta_n^{-1} (\log \theta_n)^C.$$

The set $\mathcal{J}_n$ consists of remainder slots still present at level n in the chosen filing convention. If a remainder is re-filed into a regular family, it must be removed from this set and counted in that family instead. A recent-level window can have a separate term without geometric age decay. This bookkeeping cannot be suppressed by changing notation.

The survival age of a large-field zone has a different role. For $0 < q_a < 1$ the desired integrated bound involves

$$\Sigma_k(\delta) = \sum_{n\le k} (k - n + 1) q_a^{(k-n-1)_+} p_n(\delta).$$

The factor $k - n + 1$ counts the steps at which the zone can be used. The survival factor $q_a^{(k-n-1)_+}$ belongs to an output estimate on the relevant histories. It need not, and generally cannot, be put into a single enlarged analytic input radius.

11.3. **What weighted holomorphy actually proves.**

Proposition 11.1 (A segment estimate)**.** *Let X, Y be complex Banach spaces, let $C \subset X$ be convex, and let p be a continuous seminorm. Suppose F is holomorphic on*

$$\mathcal{T} = \{a + h : a \in C,\ p(h) < \epsilon\},$$

$$\epsilon > 0,$$

and, for every $a \in C$ and $p(h) < \epsilon$, one has $\|F(a + h) - F(a)\|_Y \le B$. Then, for $a, b \in C$,

$$\|F(b) - F(a)\|_Y \le (B/\epsilon) p(b - a).$$

Proof. Put $h = b - a$. At every $a_t = a + th$, $0 \le t \le 1$, the map $z \mapsto F(a_t + zh) - F(a_t)$ is holomorphic on $|z| < \epsilon/p(h)$ and has norm at most B. The Banach-valued Cauchy formula gives $\|DF(a_t)h\| \le Bp(h)/\epsilon$. Integrating over t proves the inequality. If $p(h) = 0$, the line is entire and bounded in oscillation. Every bounded linear functional on Y makes it a bounded scalar entire function, so Liouville's theorem and separation by the dual imply that it is constant. □

The seminorm in this proposition is an assumption. Applying it with $p(h) = \sum_j \omega_{k+1-j}\|h_j\|$ produces the displayed weights because they already occur in the domain. It does not derive them from an ordinary product box.

A differential bound is a weaker sufficient input. If $\|DF(a)h\| \le Cp(h)$ holds on the segment joining two actual trajectories, integrating this inequality proves the same comparison without constructing a full weighted tube around every off-trajectory point. This is useful when formulating a minimal proof obligation.

Proposition 11.2 (A marked slot)**.** *Suppose $F(a, m)$ is jointly holomorphic when $p(a - a_0) < r_a$ and $\|m\| < r_m$, is bounded there by B, and satisfies $F(a, 0) = 0$. For $p(a - a_0) \le r_a/2$ and $\|m\| \le r_m/2$,*

$$\|D_a F(a, m)h\| \le \frac{CB}{r_a r_m}\, p(h)\|m\|.$$

Proof. Cauchy's formula first in the complex line $a + zh$, and then in the line tm, bounds the mixed derivative by $CBp(h)\|m\|/(r_a r_m)$ on the reduced domains. Integrate that derivative from $m = 0$ to m. The derivative in a vanishes at $m = 0$ because $F(a, 0)$ is identically zero. The argument uses two complex lines and introduces no factor equal to the dimension of either input space. □

This explains how a small marked-slot size can multiply the sensitivity to other data. It does not establish the required radii, the marked-slot vanishing identity, or a uniform bound on the enlarged domain.

11.4. **Exact examples separating the assertions.**

Proposition 11.3 (Ordinary analyticity does not imply forgetting)**.** *Fix $0 < q < 1$ and any positive numbers s_k. There are entire scalar functions F_k bounded by s_k on the unit polydisc of $\mathbb{C}^{k+1}$ for which no constant C independent of k satisfies*

$$|F_k(x) - F_k(y)| \le C s_k \sum_{j=0}^{k} q^{k-j}|x_j - y_j|$$

on the polydisc of radius $1/4$.

Proof. Take $F_k(x) = s_k x_0$, $x = 0$, and $y = (1/4, 0, \dots, 0)$. The inequality would give $1 \le Cq^k$. This is impossible as $k \to \infty$. The same example is holomorphic on every weighted tube, but its supremum on a tube of fixed weighted radius grows like $s_k q^{-k}$. Thus the missing issue can be a uniform bound, even when the extension itself is entire. □

This example is a logical separation, not a model satisfying the gauge-theory power-counting identities. It shows exactly why analyticity and a small one-step factor alone do not establish those identities.

Proposition 11.4 (Rarity does not enlarge the complex radius)**.** *Let $0 < q < 1$, let a be a nonnegative integer, and let an event have probability q^a. For $\delta > 0$ its excess generating function can be*

$$R_a(\lambda) = q^a(e^{\lambda\delta} - 1).$$

On $|\lambda|\delta < c$ it has a bound $C_c q^a$, and $R_a'(0) = q^a\delta$. There is no age-independent bound $C_c q^a$ on the larger disc $|\lambda| q^a \delta < c$.

Proof. The first bound follows from $|e^z - 1| \le e^{|z|} + 1$. At the positive point $\lambda = cq^{-a}/(2\delta)$ the ratio $R_a(\lambda)/q^a$ equals $e^{cq^{-a}/2} - 1$, which tends to infinity. The event model is a genuine two-point probability space, and its generating function is entire. □

The correct large-field strategy is therefore a bounded per-zone radius together with a rare-history factor in the bound. A demand for a bounded extension on a single tube measured only by the aged load would be stronger than what the argument needs and can fail even in this elementary example.

There is a separate reason for the Euclidean-member and marginal-slot conditions. Let

$$H = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$
$$M_\varepsilon = I + \varepsilon H.$$

For real $|\varepsilon| < 1$, take independent centered Gaussian vectors $X_1, \dots, X_m$ with precision matrix M_ε. The normalized block average $m^{-1/2}\sum_i X_i$ has exactly the same covariance M_ε^{-1}, hence exactly the same precision matrix. The traceless quadratic perturbation $\varepsilon x_1 x_2$ survives this blocking unchanged, although its scalar trace coefficient is zero. The Gaussian integrals are holomorphic and uniformly coercive on each fixed complex disc $|\varepsilon| < r < 1$.

This example explains why a scalar counterterm cannot remove an arbitrary anisotropic quadratic perturbation. It does not contradict the use of a scalar counterterm in the Euclidean class. It also explains why a typed extension must supply all of its prescribed tensor counter-insertions before claiming that the remaining part is irrelevant.

Finally, identities on backgrounds do not determine an arbitrary extension to independent currents. The polynomial $P(z, j) = j - z^2$ vanishes on every pair (z, z^2) and is nonzero elsewhere. An analogous cancellation known only on a graph of background pairs needs an additional argument before it can be used throughout a domain where the current is independent.

11.5. **Two local inequalities to establish.** The first test concerns the small-field potentials before exponentiation. Let $\dot{V}_{k,j}$ be the variation of the actual localized fluctuation potential caused by one regular family at level j. Include its own scalar counterterm, or the complete tensor counter-insertions for a typed family. A suitable test is

$$\|\dot{V}_{k,j}[\delta E]\|_{\mathrm{pot}} \le C\varkappa_k \omega_{k+1-j} \|\delta E\|_{j,\kappa}. \tag{11.1}$$

The supremum in the potential norm must range over the complex pairs and decoupling parameters actually used inside the fluctuation integral, not just real backgrounds. One possible norm is the least M such that

$$|\dot{V}(Y; \mathbf{p}, B)| \le M e^{-\lambda d_k(Y)} \left(1 + M_0^{-4} \sum_{b \subset Y} |B(b)|^2\right).$$

Here M_0 is the fixed block-size parameter. The complex field domains, the allowed fluctuation cut, and the enlarged decoupling discs must be specified. Defects and remainders have their own test with squared-radius weights in place of ω. A variable Wilson weight is a marginal slot; it does not receive an irrelevance weight merely by being written as a local family.

For ordinary site-frozen Euclidean members, a regular-output dependence estimate already has this intended structure after the cluster expansion. The additional typed claim requires the same cancellation at the complex pairs in the integrand, with the full tensor removed. A background extraction formula alone is insufficient for that extension. Proving (11.1) for these typed inputs is a concrete local task.

The second test concerns a single zone in a large-field operation. Fix a zone n and a step k' at which it is used, where $n \le k' \le k$. Let $\lambda_{n,k'}$ multiply only that use of a direction of load $p_n(\delta)$. Isolate the normally convergent subseries of histories affected by this dial, and denote the resulting output by $R_{k;n,k'}(\lambda)$. The required axis estimate has the form

$$\sup_{|\lambda| p_n(\delta) < c} \|R_{k;n,k'}(\lambda)\|_{k,\kappa} \le C \epsilon_k q_a^{(k-n-1)_+}. \tag{11.2}$$

The constants c, C are independent of the levels, the history, and the positions of the sources. The source is allowed to be complex on the declared source ball. The factor ϵ_k is the small mass assigned to the output under consideration. For an unintegrated boundary term it must be replaced by that term's actual majorant; its residual dependence is subsequently paid under the relevant mass, rather than being assigned a false termwise survival bound.

Cauchy's formula applied to (11.2) gives

$$\|\partial_\lambda R_{k;n,k'}(0)\|_{k,\kappa} \le (C/c) \epsilon_k q_a^{(k-n-1)_+} p_n(\delta).$$

Summing over the $k-n+1$ uses of that zone, and then over n, gives $(C/c)\epsilon_k \Sigma_k(\delta)$. This proves the differentiated age estimate from a precise bounded complex-mass statement. The proof needs no age-dependent increase of the analytic radius.

Use (11.2) under the moduli, suprema and reserve weights of the actual operation. Together with the needed first and mixed derivative estimates on its trajectories, this controls the perturbed integrand on rare histories. These direct axis estimates provide the required hypothesis; a full off-trajectory analytic tube with the required uniform bounds is an alternative sufficient input.

11.6. **Ordinary sources and the stronger comparison routes.** The ordinary product-box theorem, a weighted dependence estimate for its regular output, and the finer marked and zone-wise statements are different inputs. Coefficient readout uses the physical radii of the chain, as proved in Section 14: fixed radii give a uniform norm, and running radii carry the inverse-square factor already recorded by the radius weights. Near-zero core inclusion is supplied by the stated tower bounds. The finer assertions concern actual marked and typed evaluations in complex windows, with their proper radius normalization. Contraction estimates derived from a linearized class retain all its hypotheses. The weighted and zone-wise estimates are additional inputs to the ordinary complex-box theorem.

For a prescribed whole-sequence continuum limit, the smallest useful replacement is an actual comparison of the two tuned trajectories: a step inequality, domination of the observable evaluations by the comparison distance, and a terminal-integral inequality with summable defects. These three estimates can be proved directly. Establishing a large weighted analytic class is one route to them, not a logical necessity.

Finite-cutoff source identities do not require this finer statement. Fixed-spacing DLR compactness and closedness of reflection positivity do not require it either. Subsequential separated-correlation compactness follows from its own cutoff- and volume-uniform source bounds. Those bounds remain substantial analytic requirements; their volume-free proof can use undifferentiated age-resolved rarity without proving the differentiated zone estimates above. Similarly, tuning at one fixed cutoff can follow from its own holomorphic slope and target-crossing estimates without establishing a comparison of consecutive cutoffs.

The summable matched-cutoff comparison and the resulting whole-sequence tuned convergence use the stronger comparison route. The typed tangent contractions, scale-uniform memory estimates, and differentiated anisotropy method for rotations also require their corresponding finer inputs. Estimates for several insertions at nonzero source require joint bounds beyond a single first derivative at zero. An independent vanishing-flux theorem would provide a different route to rotations and should be assessed on its own hypotheses.

The potential test and the one-zone complex-mass test specify the uniform estimates on the actual inputs used by these stronger routes. Their stated domains, reserve weights and derivative bounds are retained in each application.

12. A reserve test for a single large-field zone

The one-zone estimate uses the reserve lemma and the local evaluation-map norm bound stated below. The reserve argument and the direct Wilson-weight channel apply on regular zones. For the formed-copy application, use the evaluation-map bound for every actual formed copy of the direction, with the stated complex perturbation and reserve weights.

12.1. **The integral before taking a logarithm.** Keep the reference sequence, tests, history labels, changes of variables and reference kernels fixed. A local operation is then represented on a disjoint union of stored-variable spaces by

$$I(\lambda,p)=\sum_{\omega}\int_{S_\omega} e^{\Phi_\omega(\lambda,\xi;p)}\,\mu_\omega(d\xi),$$

where each μ_ω is positive and independent of λ. A harmless finite frame average can be included in these measures. The parameter p ranges over the external complex-pair domain. A single zone-use parameter has an index (n,k'): it multiplies the chosen direction where the stored zone n is used at step k'. Only histories containing that use are affected.

Before re-exponentiation, the substitution of an input list by a line of lists is affine in the exponent when its evaluation maps have been frozen. Its perturbation has the concrete form

$$L_{\omega,n,k'}[\delta](\xi,p)=\sum_{e\in\mathcal{E}(\omega,n,k')}\epsilon_e\,\delta F_e\big(\kappa_{\omega,e}(\xi,p)\big)+\delta c_\omega(\xi). \tag{12.1}$$

The index e records the entry, its counter-insertions, its carrier, and its particular occurrence. The signs or fixed coefficients are ϵ_e. The formed argument $\kappa_{\omega,e}$ must lie in that entry's original analytic domain. The last term records scalar parts, including the values subtracted at the trivial field and their contributions to the vacuum bookkeeping. Constants cannot be omitted from this map merely because the observable ultimately uses a connected derivative.

For this affine use one has $\Phi_\omega(\lambda)=\Phi_\omega(0)+\lambda L_{\omega,n,k'}[\delta]$. After local histories or polymer clusters have been summed, the resulting effective families are generally nonlinear in λ. They should be controlled by a convergent marked expansion, rather than by pretending that their logarithms remain affine. If the formed arguments themselves are changed with the input, their derivatives are additional terms in (12.1); they require their own domain and norm bounds.

12.2. **What an exponential reserve proves.**

Proposition 12.1 (A one-zone reserve lemma)**.** *Let the fixed affected event E_a consist of histories of age at least a in a measurable disjoint union of history spaces with positive reference measure. Assume that the integrands are measurable in the stored variables and holomorphic in λ for almost every stored configuration. Suppose $W_\omega(\xi)\ge 0$ is measurable, and assume a measurable majorant M_ω satisfies*

$$\sup_p e^{\operatorname{Re}\Phi_\omega(0,\xi;p)} \le M_\omega(\xi),$$

$$\sum_\omega \int_{E_a} M_\omega e^{cW_\omega}\, d\mu_\omega \le B\varepsilon q^a,$$

where $c, C > 0$, $0 < q < 1$ and the fixed load satisfies $p_n \ge 0$. Put $R = c/(2Cp_n)$, interpreted as infinity when $p_n = 0$. Suppose the perturbation is defined and holomorphic on $|\lambda| < R$, and throughout that disc

$$\sup_p |\Phi_\omega(\lambda,\xi;p) - \Phi_\omega(0,\xi;p)| \le C|\lambda| p_n W_\omega(\xi). \tag{12.2}$$

Then on $|\lambda| p_n < c/(2C)$ the integral over E_a is bounded in modulus by $B\varepsilon q^a$, and is holomorphic. Its difference from its value at zero is bounded by $2B\varepsilon q^a$. Its first derivative at zero is at most

$$(2CB/c)\varepsilon q^a p_n.$$

The same conclusions hold with an additional insertion which is measurable in the stored variables, holomorphic in λ on the same disc, and uniformly bounded there, multiplied by that uniform bound. A fixed bounded insertion is a special case.

Proof. The modulus of the perturbed integrand is at most $M_\omega e^{cW_\omega/2}$. The assumed positive majorant is integrable over the disjoint union of histories. Dominated integration, or Morera's theorem and Fubini, proves holomorphy on the open disc. The mass estimate follows from the same domination. The difference is bounded by the sum of the two masses. Cauchy's formula on circles of radius approaching $c/(2Cp_n)$ gives the derivative bound. If $p_n = 0$, the hypothesis makes the perturbation independent of λ, and the derivative is zero. □

Finite upper integrals can replace the measurable majorants in this proposition: choose measurable majorants with integrals arbitrarily close to their upper integrals, apply the argument, and take the infimum. The supremum over external pairs must remain inside the history integral.

A reserve can also produce the age factor. If $W \ge a$ on E_a and

$$\sum_\omega \int M_\omega e^{(c_{\mathrm{pert}}+c_{\mathrm{age}})W}\, d\mu_\omega \le B\varepsilon,$$

then a perturbation paid by $c_{\mathrm{pert}}W$ leaves the estimate $B\varepsilon e^{-c_{\mathrm{age}}a}$. Thus a per-zone radius independent of the age is compatible with geometric forgetting. The perturbation budget and age budget must be allocated before the coupling ceiling is selected.

This argument is effective only after (12.2) has been proved. An oscillation bound for the unperturbed factors does not establish it.

12.3. **A common reserve for many zone uses.** One-zone bounds do not by themselves give a common polydisc for an arbitrarily large number of zone uses. The reserve also needs a capacity bound. The following precise version distinguishes the two issues.

Proposition 12.2 (A capacity criterion for joint parameters)**.** *Let $\mathcal{Z}$ be a finite set of zone-use indices and let i label finitely many directions at each index. Their nonnegative loads are $p_{z,i}$. Use the measurable stored spaces and fixed reference measures of the preceding proposition, and assume that all phase coefficients are measurable in the stored variables. Fix nonnegative measurable history charges $W_{\omega,z}$, a finite family of nonnegative measurable action pieces $S_{\omega,b}$, and nonnegative coefficients $a_{\omega,z,b}$, with*

$$\sum_{z\in\mathcal{Z}} W_{\omega,z} \le W_\omega,$$

$$\sum_{z\in\mathcal{Z}} a_{\omega,z,b} \le 1$$

for every b.

Suppose the affine perturbation is

$$\Phi_\omega(\vec{\lambda}) - \Phi_\omega(0) = \sum_{z,i} \lambda_{z,i} L_{\omega,z,i},$$

$$|L_{\omega,z,i}| \le p_{z,i} \left(C W_{\omega,z} + \sum_b a_{\omega,z,b} S_{\omega,b} \right).$$

Assume a majorant at zero obeys, on a fixed event E,

$$\sum_\omega \int_E M_\omega \exp\left(cW_\omega + \sigma \sum_b S_{\omega,b} \right) d\mu_\omega \le B_E,$$

$$\sup_p e^{\operatorname{Re}\Phi_\omega(0,\xi;p)} \le M_\omega,$$

where $c, \sigma > 0$. *Choose* $0 < r < \min\{c/(2C), \sigma/2\}$. *Then the integral over* E *is jointly holomorphic and bounded by* B_E *on the product of weighted balls*

$$\mathcal{D}_r = \left\{ \vec{\lambda} : \sum_i |\lambda_{z,i}| p_{z,i} < r \text{ for every } z \right\}.$$

All statements use fixed reference measures, fixed events and fixed evaluation maps; if additional parameter dependence is present, the integrands must instead be given holomorphically on this domain with the same phase bound.

Proof. For $\vec{\lambda} \in \mathcal{D}_r$, the triangle inequality and the two capacity inequalities give

$$|\Phi_\omega(\vec{\lambda}) - \Phi_\omega(0)| \le Cr \sum_z W_{\omega,z} + r \sum_b \left(\sum_z a_{\omega,z,b} \right) S_{\omega,b} \le (c/2) W_\omega + (\sigma/2) \sum_b S_{\omega,b}.$$

Thus a single integrable majorant works on the entire parameter domain, independently of $|\mathcal{Z}|$ and the number of directions. Dominated integration proves separate holomorphy and local boundedness, and the finite-dimensional Hartogs theorem gives joint holomorphy. Equivalently, one can apply Morera's theorem successively on each compact polydisc. □

If a designated parameter affects only an event E_a whose majorant has $B_{E_a} \le B\varepsilon q^a$, setting that parameter to zero changes the whole integral by at most $2B\varepsilon q^a$, uniformly in the other parameters on $\mathcal{D}_r$. At their common zero, Cauchy's formula gives the first derivative bound $B\varepsilon q^a p_{z,i}/r$. To see this without subtracting a large unaffected part, differentiate only the integral on E_a; outside E_a the integrand is independent of the designated parameter. For m distinct directions at the same index, Cauchy's formula on circles with radii tending to $r/(mp_{z,i})$ gives

$$\left|\partial_{\lambda_{z,1}}\cdots\partial_{\lambda_{z,m}} I_E(0)\right| \le B_E(m/r)^m \prod_{i=1}^{m} p_{z,i}.$$

Here zero loads mean that the corresponding affine perturbations vanish, and those derivatives are zero. Higher mixed bounds require an appropriate majorant for the history event on which all marked uses occur; they do not acquire a product of rarity factors from holomorphy alone.

The capacity assumptions have content. Without them, let a single history have $W = 1$ and let each of N indices contribute $L_z = p_z = 1$. Every separate perturbation is bounded by $|\lambda_z|W$, but choosing all $\lambda_z = r/2$ gives a total exponent change $Nr/2$. A fixed reserve in W cannot pay this uniformly in N. A proof for the actual formation must therefore assign all simultaneous uses to a common site and action budget, in addition to bounding each separate direction.

12.4. **A channel which can be bounded directly.** For $U \in \mathrm{SL}(2,\mathbb{C})$, the determinant identity

$$\det(U - I) = \det U - \operatorname{tr} U + 1 = 2 - \operatorname{tr} U$$

gives $a(U) = \frac{1}{2}\det(U - I)$. With the operator matrix norm,

$$|a(U)| \le \tfrac{1}{2}\|U - I\|^2. \tag{12.3}$$

For real $U \in \mathrm{SU}(2)$ this is an equality, since the two singular values of $U - I$ coincide.

Consider a regular zone expressed in its own units, with bare mesh η_n, containing N_n unit cells. Suppose the configurations used in the actual substitutions satisfy $\|U_p - I\| \le \alpha_n \eta_n^2$ on that zone. There are at most $C\eta_n^{-4}$ plaquettes per unit cell. Hence

$$|A(\delta c, U)| \le C\alpha_n^2 \|\delta c\|_\infty N_n. \tag{12.4}$$

Consequently, if the zone load includes $p_n \ge \alpha_n^2\|\delta c\|_\infty$, this direct Wilson-weight contribution to (12.1) is bounded by Cp_nN_n. If the reserve volume dominates the number of such stored cells, it satisfies (12.2).

The usual scale $\alpha_n^2 \asymp \theta_n^{-1}(\log\theta_n)^{2q}$ has precisely the form of the coefficient multiplying a marginal amplitude in the load. This calculation uses only the stated curvature regularity at the actual formed arguments. It does not prove that every substitution has that regularity; that is a separate inclusion requirement.

12.5. **An unrestricted bare zone needs action reserve.** There is a genuine limitation to a reserve measured only by stored sites. On a bare region where the real field is unrestricted, a plaquette can have $U_p = -I$, and then $a_p = 2$. A direction which changes a local Wilson weight by one has an exponent variation of order one at such a configuration. If its proposed load is

$$p_0 = \beta^{-1}(\log\beta)^C$$

and the site reserve is of order $W = N\log\beta$ on a fixed finite region, then

$$\frac{|\delta\Phi|}{p_0 W} \ge c\,\frac{\beta}{(\log\beta)^{C+1}}$$

for an appropriate fixed configuration and fixed N. This diverges as $\beta \to \infty$. Such configurations are obtained, for example, by setting one bond to $-I$ and the other bonds of a small patch to I. The number of altered plaquettes remains bounded. Thus (12.2) with this site reserve cannot hold pointwise on the entire unrestricted bare-field domain.

This does not refute the complex mass estimate. The Wilson factor strongly suppresses those configurations. For a bare insertion $\mathfrak{b}$ satisfying a pointwise action bound

$$|\mathfrak{b}| \le C_b \|k\|_\infty A_{\text{available}}(U),$$

one instead has

$$|\lambda\mathfrak{b}| \le \frac{C_b|\lambda|\|k\|_\infty}{\beta} S_{\text{available}}(U),$$
$$S_{\text{available}} = \beta A_{\text{available}}.$$

At a radius $|\lambda|p_0 < c$, with $p_0 \ge C_b\beta^{-1}(\log\beta)^C\|k\|_\infty$, this spends at most $c(\log\beta)^{-C}$ of the available Wilson action. It is a fractional action estimate, not a site-volume estimate.

A correct reserve lemma can include both budgets. If the majorant has the stronger reserve $e^{cV+\sigma S_{\text{available}}}$ and

$$|\Phi(\lambda) - \Phi(0)| \le C|\lambda|p_n V + \eta(\lambda) S_{\text{available}},$$
$$C|\lambda|p_n \le c/2,$$
$$|\eta(\lambda)| \le \sigma/2,$$

the mass proof above applies on the intersection of these parameter domains. A derivative estimate proportional to p_n requires this intersection to contain a disc of radius c'/p_n with a fixed $c' > 0$. One sufficient condition is $|\eta(\lambda)| \le C_{\text{act}}|\lambda|p_n$; otherwise the Cauchy radius is the smaller action-controlled radius. To use this statement one must identify action genuinely left available after the failed-test payments, gauge fixing, and other insertions. It cannot be spent twice. For several active zone-use parameters, their fractions must have a bounded sum on each shared action piece. The existence of a site-volume reserve alone proves neither of these action-allocation facts.

12.6. **What the carrier bounds do not control.** Write a standing exponent as

$$\Phi_\omega(\xi; p) = \Phi^0_\omega(\xi) + \sum_f \phi_{\omega,f}(\xi; p|_{R_f}).$$

The carrier condition bounds the oscillation of each factor in its external pair argument. Choose a permitted anchor $p^0_\omega(\xi)$ and split

$$\phi_{\omega,f}(\xi; p) = \phi_{\omega,f}(\xi; p^0_\omega) + \{\phi_{\omega,f}(\xi; p) - \phi_{\omega,f}(\xi; p^0_\omega)\}.$$

Oscillation controls the braced expression. It controls neither the scalar anchor value nor the dependence of either expression on an input-list parameter. Cauchy's estimate in the pair variables is not Cauchy's estimate in that new input parameter.

A simple exact example separates these statements. Let $\pi_a \downarrow 0$ be a sequence of putative loads, let one stored history have volume $W_a \ge a$, and let its formed exponent be

$$\Phi_a(x; p) = \log\varepsilon - (c + s + 1)W_a + xW_a,$$
$$s > 0.$$

It is entire, independent of p, and its carrier oscillation is zero. At $x = 0$ it has the prepaid mass bound $e^{\Phi_a(0)+cW_a} \le \varepsilon e^{-sa}$. The same bound, with room, holds throughout a fixed ordinary disc $|x| \le 1/2$. It fails on a disc $|x|\pi_a < c'$: at $x = c'/(2\pi_a)$ the positive exponent eventually overwhelms every fixed reserve coefficient.

This is a counterexample to deriving the enlarged-radius estimate from carrier oscillation and an ordinary reserve alone. It is not a counterexample to the Wilson operation. Even if the incoming direction has size π_a, the example can be produced by an abstract local formation map which multiplies that direction by π_a^{-1}. All input sizes are then small, while the output exponent has the displayed variation. Locality and holomorphy do not exclude this map; a uniform norm bound on formation must exclude it. Exact preservation of a represented integral, by itself, also supplies no such bound on a particular decomposition into exponents and vacuum factors.

12.7. **A sufficient structural bound on the actual formation map.** The needed additional statement is precise. For the actual index set and formed arguments in (12.1), prove

$$|L_{\omega,n,k'}[\delta](\xi,p)| \le Cp_n(\delta)V_\omega(\xi) \quad \text{for every admissible } (\xi,p), \tag{12.5}$$

with a separate fractional-action term when the bare zone is unrestricted. The norm must include scalar anchors, not only oscillations. The constant must be independent of the age, the level, and the history. If a parameter changes the arguments as well, prove the corresponding derivative bound throughout its proposed disc and integrate it to obtain (12.2).

One verifiable route to (12.5) is a carrier incidence estimate. Let the charged stored cells be a finite set $\mathcal{C}_\omega(\xi)$ with $|\mathcal{C}_\omega(\xi)| \le V_\omega(\xi)$, and decompose the scalar part as $\delta c_\omega = \sum_{c\in\mathcal{C}_\omega} \delta c_{\omega,c}$. Assign each formed occurrence to these cells and give nonnegative weights $w_{e,c}$ with $\sum_c w_{e,c} = 1$. Suppose the actual evaluations stay in their original domains and

$$\sum_e |\epsilon_e| w_{e,c} \, \|\delta F_e\|_{\text{evaluation}} + |\delta c_{\omega,c}| \le Cp_n(\delta)$$

at every stored cell c. Summing over the cells proves (12.5). The evaluation norm here is the norm after the specified unit change and counter-insertions. Using a norm at an earlier level without those transformations can lose exactly the scale weights under investigation.

A condition on the sum of sizes of the original direction terms in a unit cell is helpful, but it is not yet this incidence statement. The proof must show that substitutions and repeated occurrences preserve that bound. The formation rules distinguish copies still anchored in a live component, copies whose last anchor is removed now, and copies inside older already formed terms. The first kind is not a small new remainder. The last kind must be summed with its own norm and history mass. Counting its occurrence does not permit its bound to be reused without payment. In particular a factor obtained only after summation of histories is not a pointwise bound on the unsummed exponent.

If "size in a unit cell" is defined to mean the sum of the evaluated sizes of all formed occurrences, including their scalar parts and the indicated unit changes, then the incidence estimate is already part of that definition. Under that stronger interpretation the reserve implication is proved. The substantive remaining task is to show that directions generated by the actual renormalization step meet that definition with the proposed load. A condition on the unformed entries cannot be silently replaced by this stronger condition.

12.8. **Passing the bound through a logarithmic sum.** Suppose that a finite polymer system has activities $z_X(\lambda)$ with a common Kotecký–Preiss majorant r_X, and that the corresponding pinned bound is e^{a_X}. Choose the logarithm defined by the connected expansion, equal to zero at the zero activity vector. Then a useful marked estimate is

$$|\log Z(z(\lambda)) - \log Z(z(0))| \le \sum_X e^{a_X} |z_X(\lambda) - z_X(0)|,$$

To prove it, join the two activity vectors by their straight segment. Every point on that segment has modulus at most r_X coordinatewise. The absolutely convergent connected expansion gives

$|\partial_{z_X} \log Z| \le e^{a_X}$ by the pinned bound. Differentiate along the segment and integrate. Thus a summable one-zone age bound on the changed activities passes to the logarithm. No logarithm of an individual possibly zero history integral is required.

For a countably infinite system, apply the same estimate first to finite restrictions of the same activity and incompatibility system. If $\sum_X e^{a_X}|z_X(\lambda) - z_X(0)| < \infty$, the integrated pinned series defines a relative cluster logarithm and satisfies this bound, independently of the finite exhaustion by dominated convergence of the rooted expansions. Writing two separate infinite-volume logarithms requires convergence of both unmarked global series; a common local majorant alone does not ensure that. This deduction has two inputs: the activity majorant must remain valid on the complex parameter disc, and the changed activities must satisfy the marked bound after the actual formation and copy bookkeeping. A convergence theorem for polymers does not prove those inputs. The exponential reserve can pay a controlled perturbation of an integrand; it cannot manufacture control of the perturbation map.

A recursive treatment of old copied terms can be sufficient instead of a single global phase estimate. Fix $0 < q_0 < q < 1$ and $C > 0$. For example, suppose an already justified local operation estimate gives nonnegative bounds, for $j \ge n + 1$,

$$A_{j,n} \le C q_0^{j-n} + C \sum_{n<e<j} \epsilon_e q_0^{j-e} A_{e,n},$$
$$0 \le \epsilon_e \le \epsilon.$$

Divide by q_0^{j-n}. A discrete Gronwall induction gives

$$A_{j,n} \le C q_0^{j-n} \prod_{n<e<j} (1 + C\epsilon_e) \le C\{q_0(1 + C\epsilon)\}^{j-n}.$$

The induction follows by expanding the product, or using $\prod_{e<j}(1 + C\epsilon_e) = 1 + \sum_{e<j} C\epsilon_e \prod_{r<e}(1 + C\epsilon_r)$. Thus choosing $q_0 \le q/(1 + C\epsilon)$ preserves the desired rate q. No small sum of the ϵ_e is needed. This pays growth of output bounds after local integration. It does not pay an exponentially amplified perturbation inside an unsummed exponent. To apply this alternative one must establish the local parameterized operation bound at each new loss of an anchor and the stated copying recurrence. The sufficient formation test above is not claimed to be the only possible proof route.

The present test therefore proves the reserve implication and the direct regular-zone Wilson channel. It shows why an unrestricted bare channel requires available action, and isolates the additional uniform formation estimate for the remaining entries. The specified carrier oscillation, ordinary presentation bounds, and end-mark definitions do not alone establish that estimate. A proof of it for the actual local substitutions, with their scalar parts and copy incidence, would complete the proposed one-zone route rather than assume its conclusion.

13. The objects carried by the renormalisation argument

This section fixes the types, domains and norms in the conditional renormalisation argument. A presentation contains functions and measures, as well as numerical estimates. In particular, an array of numbers satisfying the estimates below does not constitute a presentation. Statements about the existence of backgrounds, the inclusion of substituted fields in complex domains, and the reproduction of the presentation by a step must be proved separately. No conclusion about those statements follows from the definitions here.

13.1. **Lattices, currents, and auxiliary geometric data.** The normalization in this section is that of $G = \mathrm{SU}(2)$. Its Lie algebra is represented by traceless Hermitian matrices, with $\langle A, B\rangle = \frac{1}{2}\operatorname{tr}(AB)$ and complexification $\mathfrak{g}^{\mathbb{C}}$. Fix an odd integer $L > 1$. The level-j lattice, denoted $\mathbb{T}^{(j)}$, consists of the sites retained after j blockings of a periodic bare lattice. Its bare spacing, expressed in level-j units, is $\eta_j = L^{-j}$. A bond configuration satisfies $U(\bar{b}) = U(b)^{-1}$. Products along oriented paths are ordered in path order. Positive coordinate bonds suffice to specify a configuration. All suprema over bonds below use these bonds. In the left multiplication chart, reversal is $J(\bar{b}) = -\operatorname{Ad}(U_b^{-1})J(b)$. The complex configuration group is $\mathrm{SL}(2,\mathbb{C})$.

For a positively oriented plaquette p, based at its chosen corner, write U_p for its holonomy and set

$$a_p(U) = 1 - \tfrac{1}{2}\operatorname{tr} U_p,$$
$$A(U) = \sum_p a_p(U),$$
$$A(c, U) = \sum_p c(x(p))a_p(U).$$

Here $x(p)$ is the plaquette midpoint. On real configurations this is the real Wilson action; the same formula is its holomorphic extension. For definiteness the current on a lattice of spacing η is specified by its variational normalization:

$$\frac{d}{dt}A(e^{i\eta tB}U)\bigg|_{t=0} = \eta^4 \sum_b \langle J_\eta(U)(b), B(b)\rangle. \tag{R1}$$

The sum is over positive bonds and B is arbitrary. This identity defines every component of J_η, also for complex U by analytic continuation. It fixes the transports and factors implicit in the notation $D_U^* F_U$. One must use the plaquette divergence associated with (R1) in that notation; the adjoint of the site gradient alone has a different domain. At $U = \mathbf{1}$, differentiation in the potential coordinate $U(t) = e^{i\eta tB}$ gives $J'_\eta(\mathbf{1})B = d_\eta^* d_\eta B$, where d_η here is the oriented plaquette difference on one-forms divided by η. If the numerical mesh changes from η to $\eta' = L^m\eta$ while the group configuration stays fixed, potentials are multiplied by L^{-m} and currents by L^{-3m}. This follows directly from (R1) and the chart $e^{i\eta A}$, and specifies the direction of the unit change. The current variable of a complex pair, introduced below, is independent of this particular function of U.

Fix powers of L called $\mathfrak{M}, M_1', M$, a gap $\rho = R_1\mathfrak{M}$, and constants $C_{\mathrm{box}}, C_{\mathrm{pot}}$. They are part of the auxiliary data, selected before domains or norms are formed. The first two sizes concern backgrounds and walks; M is the localization cube size. They are not interchangeable. The partition π_j consists of the centered, half-open M-cubes in level-j units. Boundary conventions are fixed once on every lattice. For a cubical region Y, $Y^{\sim n}$ means addition of n layers in the partition currently named; $Y^{\sim -n}$ means removal of those layers. A hull is the smallest union of cubes of the named partition containing the set. These operations include their periodic versions.

The geometric data include two erosion assignments. Given a level-p cubical subdomain Z and an integer $0 \le n \le p$, each assignment returns a finite nested sequence of cubical site sets, cut at depth n. In the first assignment the sets are unions of $\mathfrak{M}$-cubes in their own units, with consecutive sets separated by a gap greater than ρ in the outer unit. In the second they are unions of M_1'-cubes, with gap at least M_1' in the inner unit. An assignment specifies the actual sets, not just these inequalities. It must be a single translation-compatible rule on all tori, commuting with common lifts of non-wrapping regions. If more than one erosion has these properties, either a particular

assignment is supplied or the core clauses below are imposed on all admissible assignments. A choice cannot be recovered from a gap inequality alone.

Here is the determining set of a nested sequence. Work temporarily in level-k units, so the spacing of level j is $\xi_j = L^{j-k}$. Let $\Omega_0 \supset \cdots \supset \Omega_k$ and $\Omega_{k+1} = \varnothing$. The set Γ_j consists of bonds of $\mathbb{T}^{(j)}$ whose endpoints lie in Ω_0, at least one endpoint lies in Ω_j, and neither lies in Ω_{j+1}. Thus inward bonds crossing into the next layer are excluded. Put

$$\mathbf{B} = \bigcup_j \Gamma_j,$$

$$M_{\mathbf{B}}(U) = (\mathcal{M}^j U|_{\Gamma_j})_j. \tag{R2}$$

The endpoints of these bonds are the nodes. They are the tops of the blocks, or pillars, partitioning the determining region. A residual gauge function equals $\mathbf{1}$ at all nodes. In particular, imposing additional inward-bond constraints changes this problem.

The averaging map $\mathcal{M}$ used in (R2) is the symmetric contour average. For group elements V_i of small diameter their mean is the unique nearby V satisfying $\sum_i \log(V_i V^{-1}) = 0$. The contour transport $U(y, x)$ is the mean over the shortest paths from y to x in its centered block that traverse the coordinate directions in one of their orders. Set $U(x, y) = U(y, x)^{-1}$. For a coarse bond c, $x' = x + (c_+ - c_-)$, and straight transports $U[c]$ and $U([x, x'])$,

$$\mathcal{M}(U)(c) = \exp\left(L^{-4} \sum_{x \in B(c_-)} \log\{U(c_-, x)U([x, x'])U(x', c_+)U[c]^{-1}\} \right) U[c]. \tag{R3}$$

All logarithms in regular neighborhoods are their near-identity branches. Away from those neighborhoods a measurable extension of the contour means and logarithms is auxiliary input. Exact identities using that extension require its gauge covariance; small-field analyticity concerns only the displayed branches. Existence of a globally symmetric extension with every desired property is not asserted merely by writing (R3).

For every determining sequence and real regular datum $\mathbf{V}$, the background input is a regular critical orbit for $\sum_{p \subset \Omega_0} a_p$ on $\{U : M_{\mathbf{B}}(U) = \mathbf{V}\}$, together with a chosen analytic continuation $\Phi_{\mathbf{V}}(B)$ at datum $e^{iB}\mathbf{V}$. The representatives, gauge, analytic domain and transport under local charts are included in this input. Uniqueness, existence and uniform estimates for these maps are separate background theorems. The word "background" does not substitute for them.

13.2. **Hereditary domains and their shifts.** A triple of level j on X is defined on the bare mesh η_j when lengths are expressed in level-j units. It is $P = (U, A', \mathbf{J})$, with U real and $A', \mathbf{J}$ complex Lie-algebra-valued bond forms. Its pair is $\mathbf{p}(P) = (\mathbf{U}, \mathbf{J})$, where $\mathbf{U} = e^{i\eta_j A'} U$. A triple is a splitting of its pair. Fix positive radii a_0, a_1, a_2. The simple conditions are

$$\begin{aligned} &|\partial U - \mathbf{1}|, |\partial \mathbf{U} - \mathbf{1}| < a_0 \eta_j^2, \qquad |A'|, |\nabla_U A'| < a_1, \qquad |\mathbf{J}| < a_2; \\ &|A|, |\nabla A| < C_{\mathrm{pot}} M a_0 \quad \text{in the real gauges described below.} \end{aligned} \tag{R4}$$

For each cubical box contained in X with side at most $C_{\mathrm{box}} LM$, such a real gauge must exist, with $U = e^{i\eta_j A}$. The norms are maximum matrix norms in the named local units. For a one-form component A_μ based at its initial site, fix the forward convention

$$(\nabla_\nu^U A)_\mu(x) = \eta_j^{-1}\{\mathrm{Ad}(U_\nu(x))A_\mu(x + \eta_j e_\nu) - A_\mu(x)\}.$$

The maximum of these component norms defines $|\nabla_U A|$. In the usual two-radius notation $a_2 = a_0$.

For every subdomain $Z \subset X$ at every level $p \le j$, for both erosion assignments, and every truncation depth $1 \le n \le p$, form its determining set $\mathbf{B}'$. Define its tower configuration by

$$U_n(Z;P) = \Phi_{M_{\mathbf{B}'}(U)}\left(\frac{1}{i}\log\{M_{\mathbf{B}'}(\mathbf{U})M_{\mathbf{B}'}(U)^{-1}\}\right). \tag{R5}$$

This expression is required to exist in the chosen continuation domain. On Z with two layers of the M-cubes of level $p-1$ removed, require

$$\begin{aligned} |\partial U_n(Z;P) - \mathbf{1}| &< a_0 L^{-(j-p)}\eta_p^2, \\ |J(U_n(Z;P))| &< a_0 L^{-(j-p)} L^{2(n-p)}. \end{aligned} \tag{R6}$$

The current in the second line is formed in level-n units. Impose the same conditions with $\mathbf{U}$ replaced by U. These are quantifiers over every allowed Z, p, n, not just the domain on which an activity is eventually evaluated. The hereditary core $\mathcal{U}_j^\circ(X; a_0, a_1, a_2)$ is the set of triples satisfying (R4)–(R6). Its pair image is the analytic pair domain; holomorphy means holomorphy in $(A', \mathbf{J})$ on every open slice with U fixed. The inequalities are strict. Requiring the tower maps to exist in their open continuation domains is part of membership.

An activity on this core must have the tie: its values at any two splittings of the same complex pair agree. Gauge invariance means invariance under simultaneous transformation of both entries of the pair; the current at a bond is based at its initial site and transforms by conjugation there. The core itself is not saturated by complex gauge transformations. Extension of a function to the union of complex orbits requires equality on all intersections and holomorphic descent; it is an additional statement when used.

Restriction ρ_X discards bonds and variables outside X without extending them across the boundary. If an evaluation changes its numerical mesh from η_j to $\eta' = L^m \eta_j$, the coordinate change is explicitly

$$\begin{aligned} (U, A', \mathbf{J}) &\longmapsto (U, L^{-m}A', L^{-3m}\mathbf{J}), \\ m &\ge 0. \end{aligned} \tag{R7}$$

For a shift $(\mathbf{H}, \mathsf{j})$ on the same lattice define

$$\begin{aligned} P \dot{+} (\mathbf{H}, \mathsf{j}) &= (U, A' \boxplus \mathbf{H}, \mathbf{J} + \mathsf{j}), \\ e^{i\eta_j(A' \boxplus \mathbf{H})} &= e^{i\eta_j \mathbf{H}} e^{i\eta_j A'}. \end{aligned} \tag{R8}$$

The logarithm defining $\boxplus$ is the small branch; inclusion in its domain is required. The independent current is shifted additively. Replacing it by $J(e^{i\eta_j \mathbf{H}}\mathbf{U})$ at every complex argument would change the construction.

A decoupled response input on a cubical region Y is a pair of maps $\mathfrak{h}_Y(s; P, B') = (\mathbf{H}, \mathsf{j})$, holomorphic in complex cube parameters $|s(\Delta)| < e^{\bar{\kappa}}$, the complex triple variables, and the datum B'. It vanishes at $B' = 0$. Its first component is the solution of the actual background equations with the propagators decoupled by these parameters:

$$\begin{aligned} \mathbf{H} &= \mathbf{a} - \mathsf{H}(s)\mathsf{D}, \qquad & \mathsf{D} &= C_k(\mathbf{a} - \mathsf{H}(s)\mathsf{D}), \\ \mathbf{a} &= \mathsf{H}_1(s)B' + \mathbf{A}_1, \qquad & \mathbf{A}_1 &+ \mathfrak{G}(s)\mathsf{W}_s(\mathbf{a}) = 0. \end{aligned} \tag{R9}$$

Here C_k is the nonlinear part of the straightened constraint map; H, H_1 are the constrained linear minimizers; $\mathfrak{G}$ is the inverse on the simultaneous Landau and zero-average slice; and W_s is the nonlinear gradient remainder. As data for (R9), these are specified linear maps between the finite-dimensional bond and constraint spaces, and specified holomorphic maps C_k, W_s on open balls of those spaces. They include their walk coefficients and convergence domains. The reduced current is a second component of this input, obtained from the constraint multiplier. At one change of units its value is $L^{-3}\mathbf{J} + \mathsf{j}$; the multiplier construction and its uniform bounds are required independently. Bounds on first derivatives of $\mathbf{H}$ alone do not define this component.

The response hypothesis requires realness, gauge covariance, compatibility under restriction to wall-components, and transport by common lifts. A wall-component is a component of cubes sharing three-dimensional faces. When s vanishes on separating cubes, the equations split over the remaining components, and the response on a component with no datum is zero. In local first-order norms its size is at most $Ce^{C\bar{\kappa}} \sup |B'|$; a response viewed $k - j$ levels below has the additional potential factor $L^{-(k-j)}$. At $s = 1$ and real background data its shifted pair is the old background and its own current, up to the recorded node gauge factor. Existence, uniqueness and these properties of (R9) are substantive response inputs.

For a term F on X let X_0 be its hull in the cubes of the decoupling scale. The interpolation is

$$\Phi_Y[F](s, \nu) = F\big((\rho_X P) \dot{+} \nu \mathfrak{h}_Y(s; P, B')\big). \tag{R10}$$

Every occurrence of (R10) requires a domain-inclusion hypothesis: for all the indicated s and $|\nu| < \varrho_F$, its triple lies in the core of F. One sufficient quantitative hypothesis, with a fixed constant $c_F > 0$, is inclusion throughout

$$|\nu| < c_F L^{k-j} a_{0,j} \delta(\theta_k)^{-1} e^{-C_F \bar{\kappa}} =: \varrho_F > 1. \tag{R11}$$

In inner and removed substitutions separate inclusion bounds and their own radii are required. Equation (R11) is not a consequence of the formal shift law (R8).

For disjoint finite cube sets $\mathcal{K}, \mathcal{N}$ define

$$F[\mathcal{K} \mid \mathcal{N}] = \sum_{\mathcal{K}' \subset \mathcal{K}} (-1)^{|\mathcal{K}| - |\mathcal{K}'|} F(\mathbf{1}_{\mathcal{K}' \cup \mathcal{N}}). \tag{R12}$$

This equals the integral of $\prod_{\Delta \in \mathcal{K}} \partial_{s(\Delta)} F(s + \mathbf{1}_{\mathcal{N}})$ over the unit cube, by iterating the one-variable fundamental theorem of calculus. Summing (R12) over subsets gives its exact resummation identity. For the rooting assertion, use the wall-components of $\mathcal{K} \cup \mathcal{N} \cup X_0$. If such a component contains a cube of the differenced set $\mathcal{K}$ and misses the base X_0, then locality and splitting make the function independent of that component's parameters, so the alternating bracket is zero. An additional component made only of fixed $\mathcal{N}$ cubes need not make a bracket zero. For $\Phi_Y[F](s, \nu) - \Phi_Y[F](s, 0)$ the expression vanishes if the component of the base has no datum for the shifted response, since that response is then zero on the base. These conclusions require the stated splitting and support properties of the actual response. If $|\Phi_Y[F]| \le M_F$ on the stated polydisc, Schwarz's lemma gives $|\Phi_Y[F](s, 1) - \Phi_Y[F](s, 0)| \le 2M_F/\varrho_F$. Cauchy's formula on the s discs then bounds the bracket of this excess by $2M_F \varrho_F^{-1} (e^{\bar{\kappa}} - 1)^{-|\mathcal{K}|}$. For $\bar{\kappa} \ge 1$ this is at most $2M_F \varrho_F^{-1} e^{-(\bar{\kappa}-1)|\mathcal{K}|}$. When $\mathcal{K}$ is nonempty, this also bounds the bracket of $\Phi_Y[F](s, 1)$ itself: its value at $\nu = 0$ is the s-independent constant $F(\rho_X P)$, which the bracket annihilates. For an empty differenced set, an un-subtracted constant does not acquire ϱ_F^{-1}. Thus the algebraic bracket bound is proved once the response and inclusion hypotheses have been supplied.

13.3. **Pointed families, marginal coefficients, and list norms.** Let $\mathcal{D}_j$ be the connected unions of cubes of π_j. Set $d_j(X) = M^{-1}$ times the length of a shortest tree in X meeting each cube; it is zero for one cube. A pointed family is the collection $E(X, \mathbf{p}, z)$ for $X \in \mathcal{D}_j$ and unit sites $z \in X \cap \mathbb{T}^{(j)}$, with each term analytic on its core, gauge invariant and tied. Its norm is

$$\|E\|_{j,\kappa} = \sup_{\text{admissible tori and their companion}} \sup_{X, z \in X} e^{\kappa d_j(X)} \sup_{P \in \mathcal{U}_j^\circ(X)} |E(X, \mathbf{p}(P), z)|. \tag{R13}$$

The supremum in z is not a sum over base points. For an unpointed family omit z from this formula. Sums over domains require the separate lattice-animal summability bound at the chosen decay rate.

A companion is a single family $\widehat{E}$ on $\mathbb{Z}^4$, agreeing with every torus term on every common lift of a non-wrapping domain with the necessary collar. For a fixed level, fixed frame prefix and fixed initial weight, the same companion serves every cutoff on which those input data coincide. Its existence is compatibility between tori, not a property that can be tested on just one torus. Individual terms are required to be covariant under the translations preserving π_j. The pointed total has the stronger property

$$\mathcal{P}_z^{\mathsf{T}}(W) = \sum_{X \ni z} E(X, \mathbf{p}_W, z),$$

$$\mathcal{P}_{rz}^{\mathsf{T}}(rW) = \mathcal{P}_z^{\mathsf{T}}(W) \tag{R14}$$

for every lattice isometry r, including unit translations. Here

$$\mathbf{p}_W = (U_j(W), J(U_j(W))).$$

The covariance class consists of the W for which these backgrounds exist, every restriction lies in the required core, and

$$L^{2j} \max_p |\partial U_j(W)(p) - \mathbf{1}| < c \min\{\varepsilon(\theta_{j-1}), \varepsilon(\theta_j)\}. \tag{R15}$$

Both W and rW must be in this class. Invariance of the sum over z alone is weaker than (R14).

The site tent is $h_z(x) = \prod_{\mu=1}^4 \max\{1 - |x_\mu - z_\mu|, 0\}$. Evaluate it at bond and plaquette midpoints. Then $\sum_z h_z(x) = 1$ and $h_{rz}(rx) = h_z(x)$. A bond-rooted term is pointed by $E(X, \cdot, z) = \sum_b h_z(x_b) E(X, b; \cdot)$; its collar must contain the supports of these tents. The pointed Wilson family has terms $A(h_z, \mathbf{U})$ on the cube of z enlarged by one layer, and zero elsewhere. This convention distributes a coupling counterterm by the same base points as the regular family.

To define the coefficient, use the chart $\mathsf{e}_X(A, J) = (e^{i\eta_j A}, J)$ at $(\mathbf{1}, 0)$. Directions are pairs of complex bond forms, with norm $\max\{\sup|A|, \sup|\nabla A|, \sup|J|\}$. Let H_∞ be the derivative of the infinite-lattice background map at its trivial datum, in its fixed gauge, and put

$$\mathfrak{h}_\infty B = (\mathsf{H}_\infty B, d^* d \mathsf{H}_\infty B).$$

This linear response is an input from the background theory; it must include its domain and decay estimates. Choose an orthonormal color basis t_1, t_2, t_3, and let δ_c^a be t_a on the unit bond c and zero elsewhere. With Hessians taken in the displayed chart,

$$\Pi^E(c,c';z)=\tfrac{1}{3}\sum_{a=1}^{3}\sum_{\widehat{X}\ni z}\operatorname{Hess}\widehat{E}(\widehat{X},\cdot,z)[\mathfrak{h}_\infty\delta_c^a,\mathfrak{h}_\infty\delta_{c'}^a],$$

$$\Pi^E(c,c')=\sum_z\Pi^E(c,c';z)=:\Pi^E_{\mu\nu}(c'_--c_-), \tag{R16}$$

$$b[\widehat{E}]=\sum_{u\in\mathbb{Z}^4}\Pi^E_{\mu\nu}(u)u_\mu u_\nu,\qquad \mu\neq\nu.$$

The coefficient is defined only when all these sums converge absolutely, the middle expression depends just on directions and displacement, and the last value is independent of the distinct directions chosen. Gauge independence, the Ward identities and the reduction of the second-moment tensor to this scalar require proof. They are not incorporated by declaring an arbitrary Hessian to have a coefficient. In particular, a claim that the Wilson tent family has coefficient one requires the linear-response identities with the normalization (R1).

A member is a triple $(E,\widehat{E},e)$ satisfying (R13)–(R16), with $e=b[\widehat{E}]$. Activity bounds and coefficient bounds use their declared physical radii, including the inverse-square readout factor of Section 14. A list at step $k+1$ contains such members for $j\le k$ and unpointed slot families $H^{(j)}$ with no coupling counterterm. Its action is

$$\begin{aligned}S[\mathfrak{f}](U)=&\sum_{j\le k}\left[\sum_z\sum_{X\ni z}\{E^{(j)}(X,U,z)-E^{(j)}(X,\mathbf{1},z)\}-e_jA(U)\right]\\&+\sum_{j,X}\{H^{(j)}(X,U)-H^{(j)}(X,\mathbf{1})\}.\end{aligned} \tag{R17}$$

Every evaluation uses the background and units appropriate to its level. Slot families have companions, term covariance under the localization translations, and invariant totals on the stated background class, but no requirement of a pointed coefficient. For $0<\alpha_{\mathrm H}<1$ their list size is

$$N_{\mathfrak a}(\mathfrak f)=\sum_{j\le k}(k+2-j)^{3/2}L^{-\alpha_{\mathrm H}(k-j)}\|E^{(j)}\|_{j,\kappa}+\sum_j\varrho_j^2\|H^{(j)}\|_{j,\kappa},$$

$$\varrho_j=\max_\iota\frac{(1+\beta_{\mathrm s})a_{\iota,k+1}}{a_{\iota,j}}. \tag{R18}$$

All three radii, including the independent-current radius, enter this maximum. Admissible radius systems require $CMa_{0,j}<a_{1,j}$, $a_{1,j}^2\le Ca_{0,j}$, $a_{\iota,j}\le(1+\beta_{\mathrm s})a_{\iota,k+1}\le C(k+2-j)^{1/2}a_{\iota,j}$, and a fixed small upper bound on the absolute radii. They also require $\mathsf{s}_{\mathfrak a}:=\delta(\theta_k)/a_{0,k+1}\le c$. The constants and the admissible systems are specified before taking (R18). For the running system use $a_{\iota,j}=\check\alpha_{\iota,j}:=\min\{\alpha_{\iota,j-1},\alpha_{\iota,j}\}$ at $j\le k$ and $a_{\iota,k+1}=\alpha_{\iota,k}$. For the absolute system use fixed α_ι at $j\le k$ and $a_{\iota,k+1}=\alpha_\iota/(1+\beta_{\mathrm s})$.

13.4. **Frames, histories, structures, and stored variables.** A frame is a real sequence $\theta_0,\dots,\theta_k\ge\gamma^{-2}$, with $0<\gamma<e^{-1/2}$, such that, for $i\le j$,

$$\underline{\lambda}(j-i)-c_5\le\theta_i-\theta_j\le\bar B(j-i),$$

$$\check R_j=R(\min_{i\le j}\theta_i),$$

$$\check R_j\le L\check R_{j+1}, \tag{R19}$$

where $R(y)$ is the smallest power of L not below $(\log y)^r$. Set $\bar g_j = \theta_j^{-1/2}$, $p_0(g) = A_0(\log g^{-2})^{\mathfrak{p}_0}$, $p_1(g) = (\log g^{-2})^{\mathfrak{p}_1}$, $\varepsilon_j = \bar g_j p_0(\bar g_j)$, $\delta_j = (A_1/A_0)\varepsilon_j$ and $\alpha_{\iota,j} = C_\iota \theta_j^{-1/2}(\log \theta_j)^q$. Here $\mathfrak{p}_0 > \mathfrak{p}_1$, $A_1 < A_0$, and all constants are fixed. The waiting depth is $\mathsf{N}_j = \check R_j$. The barred couplings belong only to the auxiliary frame. The actual running couplings $x_j^{-1/2}$ are separate. The weights x_j are separate entries, with $x_0 = \beta$ and $x_j = x_{j-1} - b_j$. A tube requires

$$
\begin{aligned}
|\theta_j - x_j| &\le \mathsf{t}_j,\\
\mathsf{t}_j &\le c\theta_{j+1}(\log \theta_{j+1})^{-2q_{\mathrm{t}}},\\
q_{\mathrm{t}} &\ge q,\\
2q_{\mathrm{t}} &> q + \mathfrak{p}_0.
\end{aligned}
\tag{R20}
$$

Its radii are non-increasing. At a terminal frame with no next member the radius must instead be supplied explicitly; for a one-member frame require $\mathsf{t}_0 \le 2c\theta_0(\log\theta_0)^{-2q_{\mathrm{t}}}$. All tests, regions and kernels are formed from θ, not from the unknown entries x_j.

A history is a finite record of every outcome of the tests of the steps and of the operations inside large-field components, including coordinate frames in the signed-permutation group W_4. The formation data assign to each record its regions, stored bonds and integration variables. In particular they give total tests $\chi, 1-\chi$ as Borel functions on the actual variable space, with each value in $\{0,1\}$. A plaquette test is $\mathbf{1}\{|\partial U_{j,\square} - \mathbf{1}| < \varepsilon_j \eta_j^2\}$; a fluctuation test is $\mathbf{1}\{|\mathcal{A}_j| < \delta_j\}$. The local background map, the neighborhoods of each test, and the collar and rim tests with their sharper thresholds are part of the formation data. Listing the names of tests does not define them.

The label is $\mathcal{S} = (\Omega_j, \Lambda_j, S_j)_{j\le k}$ with $T_\ell \supset \Omega_1 \supset \Lambda_1 \supset \cdots \supset \Lambda_k$; $S_j \subset \Omega_j \setminus \Lambda_j$ records strips and failed fluctuation regions. Write $Z_j = \Lambda_j^c$. The step creates unions of $M\check R_j$-cubes, with gaps at least $cM\check R_j$ in local units. Nests inside an operation may have gaps of only one M-cube of the later level. The formation data must state the actual enlargements and test neighborhoods; bounds described by a fixed number of layers leave that number as a required datum.

An operation at level t acts on the prescribed class of components which are small, quiet for N_t steps, and satisfy the waiting and quarantine conditions. It first integrates stored fields in N_t stages, then tests extension of the exterior field, integrates the inner box in an axial gauge with frame in W_4, restores block variables through normalized reference laws, localizes the exponent, and performs its last test. A healed component leaves the label; an island passes the earlier tests and fails the last; a drop-out fails after $n \le \mathsf{N}_t$ stages and keeps those stages. These alternatives and every operation kernel must be given as measurable maps and positive measures. Their exactness is a separate equality of represented measures.

An operation can use its waiting window only when $t_\flat = t - \mathsf{N}_t \ge 0$. Before then it is not eligible; no negative-level history is introduced. A drop-out C records its former regions $\Omega_l^\circ \cap C$ and its stage boxes $Z_l''(C)$. For each integer $0 \le \nu(C) \le n(C)$ define

$$
\Omega_l^{\boldsymbol\nu} \cap C = \begin{cases} C \setminus Z_l''(C), & t_\flat < l \le t_\flat + \nu(C),\\ \Omega_l^\circ \cap C, & t_\flat + \nu(C) < l \le t_\flat + n(C). \end{cases}
\tag{R21}
$$

Outside the affected levels retain the label regions. The choice $\boldsymbol\nu = n$ is the determining sequence; it governs backgrounds. The choice $\boldsymbol\nu = 0$ is the action sequence; it governs the sums and the coupling profiles. Standing marks record the analogous Λ_l, S_l data. Their waiting windows are required to be disjoint at a point, and old marks disappear when swallowed by a healed component or an island. These geometric facts require verification for the actual operation.

A structure $\sigma = (\mathcal{S}, \boldsymbol{\nu}; t, \phi)$ also records a phase. Its phase weight is 0 at output, 1 at the step input, and $1 - n/(2\mathsf{N}_t)$ after n stages. A structure precedes another if it counts no more completed stages and has lower level, or the same level with larger phase weight. At an index $0 \le m \le t$ the radius coefficient is

$$c_\sigma(x \mid m) = 1 - \beta_{\mathrm{s}}(1 - 2^{-(t-m)}) + \varpi_\phi \mathsf{g}_t(m) - \beta_{\mathrm{R}} \sum_{l \in \mathrm{Re}_\sigma(x)} 2^{-|m-l|},$$

$$\mathsf{g}_t(m) = \begin{cases} \theta_{\mathrm{s}}\beta_{\mathrm{s}} 2^{-(t-m)}, & m < t, \\ \beta_{\mathrm{s}}, & m = t. \end{cases} \tag{R22}$$

Here $0 < \theta_{\mathrm{s}} < 1$ and $\beta_{\mathrm{R}} = c_{\mathrm{R}}\beta_{\mathrm{s}}$ are fixed fractions; $\mathrm{Re}_\sigma(x)$ consists exactly of the levels of the first $\nu(C)$ stages of each drop-out containing x. The layer-radius formation data give the original regularity layer and the layer after each stage. Their nonnegative difference is $e_\sigma(x)$. The lift is $L_\flat^{2e_\sigma(x)}$ for a fixed $L_\flat \ge 1$. The layer-m radii are multiplied by this lift and, except for the real-potential gauge bounds, by (R22). Unfinished structures here have $t \ge 1$: at layer zero use level-one radii; at the unfinished top use level-$t-1$ radii. The initial $t = 0$ base structure is separate input with its own specified radii. The data must ensure $c \le c_\sigma \le 1 + \beta_{\mathrm{s}}$.

The space $\widetilde{U}^c_\sigma(X)$ imposes the simple, tower and small-field-core clauses with those layer radii, with one splitting, on every subdomain and every preceding structure. The layer assignments, erosion rules, continuation maps and recorded remade levels are thus indispensable parts of its definition. The claims that restriction or a stage substitution maps one such space into another are analytic inclusion hypotheses. They require the specified radii and constants; a formal partial order by itself does not prove them.

Stored fluctuation variables are real bond forms $\mathcal{A}^{\mathrm{s}}_l(b)$ belonging to the kernels outside the exponent. They transform by $\mathrm{Ad}\, u(b_-)$. Near bonds are the bonds in collars, rims and the nonanalytic cutoff zone; all other stored bonds are far. Fix $\lambda^{\mathrm{st}}_l = (\log \theta_l)^{6r+1} > 1$. A stored-field function is Borel in the near variables and is the value at $\mu_b = 1$ of a function jointly analytic in the pair, the far variables, and switches $|\mu_b| < \lambda^{\mathrm{st}}_l$. It is independent of a near variable when its switch is zero, and independent of that switch when the variable is zero. For far variables the analytic ball has radius λ^{st}_l times their specified real range. That range is part of the kernel data. The pair tie is required at every switch value.

13.5. **Presentations, standing operations, and their bounds.** A real presentation consists of the frame and formation data above, weights x_j, regular members $(E^{(j),\mathrm{u}}, E^{(j),\mathrm{m}}, \widehat{E}^{(j)}, b^{\mathrm{u}}_j, b^{\mathrm{m}}_j)$, remainder families $(R^{(j)}, \widehat{R}^{(j)})$, boundary functions $B_{\mathcal{S}}(j, X, o)$, and the standing-operation exponents below. Here $E = E^{\mathrm{u}} + E^{\mathrm{m}}$, $b_j = b^{\mathrm{u}}_j + b^{\mathrm{m}}_j = x_{j-1} - x_j = b[\widehat{E}^{(j)}]$. Boundary origins o range over five types: the step, the last layer of a regular family, a step remainder beside an island, a stage, and exponentiation. A slot is the tuple (j, X, o) together with the structure-domain on which the function was formed. It retains that domain at later steps.

For a finite exempt family H of nonempty regions, define $d^\natural_{j,\mathsf{H}}(X)$ by minimizing M^{-1} times tree length also over subfamilies $\mathsf{F} \subset \mathsf{H}$ contained in X: the tree meets each cube outside $\bigcup \mathsf{F}$ and each member of F, but need not fill those members. For a label use the level-j hulls of its standing islands as H. Denote the result by $d^\natural_j(\mathcal{S}; X)$. It is at most $d_j(X)$ and asserts no decay across an exempt island. The real class requires

$$\|E^{(j)}\|_{j,\kappa} \le E_0,$$

$$\|R^{(j)}\|_{j,\kappa} \le \bar{g}_j^{\kappa_0},$$

$$|B_{\mathcal{S}}(j,X,o)| \le \mathsf{B} e^{-\kappa d_j^{\natural}(\mathcal{S};X)}. \tag{R23}$$

Boundary functions have the stored-field property. All entries are real on real arguments and invariant under the fixed simultaneous action on pairs and stored data. For regular families (R23) is evaluated at $\check{\alpha}_{\iota,j}$; for the unmarked family also require a member of size at most E_0 on fixed absolute radii. The whole-lattice step is additional formed data: its positive tame kernel is $\mathbf{T}^{\mathrm{sf}}_{j-1}[\theta_{j-1}]$, with its non-pivot bond coordinates and sharp fluctuation box. For a list action S its step functional is

$$\begin{aligned}\mathcal{E}^{(j)}[\theta;x;S](W) = {}& \log\big(\mathbf{T}^{\mathrm{sf}}_{j-1}[\theta]e^{-xA(U_{j-1})+S(U_{j-1})}\big)(W)\\ &+ xA(U_j(W)) - S(U_j(W)).\end{aligned}$$

The kernel, the fibre coordinates, the normalization and the near-real logarithm branch must be given for this definition. The localization rule producing a pointed family from this functional is supplied once, for every torus and list; it is not determined by its total. The identification clause requires $E^{(j)}$ to be exactly this localized output from the earlier entries. Its unmarked version is the same output using only their unmarked entries.

The formed data also contain pointed Gaussian families $\mathbf{N}^{\circ}_j$ and $\mathbf{G}_{j,\tau}$ on those absolute cores, the first localizing the Gaussian logarithmic normalizer and its trivial-field local numbers, and the second localizing the change of its quadratic form when $x/\theta = \tau$. Their finite-dimensional covariance matrices, localization maps and logarithm branches are typed input data, so the following conditions have definite meanings. Set $b^0_j = b[\mathbf{N}^{\circ}_j]$ and $\tau_{j-1} = x_{j-1}/\theta_{j-1}$. Require, for fixed constants $C_{\mathrm{m}}, C_{\mathrm{u}}, C_b$,

$$\begin{aligned}\|E^{(j),\mathrm{u}} - \mathbf{N}^{\circ}_j - \mathbf{G}_{j,\tau_{j-1}}\|_{j,\kappa} &\le C_{\mathrm{u}}\bar{g}_{j-1}(\log\theta_{j-1})^{\mathfrak{p}_0},\\ \|E^{(j),\mathrm{m}}\|_{j,\kappa} &\le C_{\mathrm{m}}\bar{g}_{j-1}^{\kappa_0-C_{\mathrm{m}}},\\ |b^{\mathrm{m}}_j| \le C_b\bar{g}_{j-1}^{\kappa_0-4}, &\qquad |b^{\mathrm{u}}_j - b^0_j| \le C_b\bar{g}_{j-1}(\log\theta_{j-1})^{\mathfrak{p}_0}.\end{aligned}$$

No universality formula or asymptotic-freedom estimate for b^0_j is implied by calling these families Gaussian. We must prove their covariance construction and coefficient estimates separately.

A standing operation at an island (t,Y) has a countable or finite local-history index ω, a Borel subset S_ω of a finite product of group variables, real bond variables and finite frame sets, and a positive Borel measure μ_ω on that set. These sets and measures are formed independently of the entries. Its exponents are Borel in $\xi \in S_\omega$, holomorphic in the pair, real at real arguments, and have the form

$$\mathbf{T}'_t(Y)F = \frac{1}{|W_4|}\sum_{\omega}\int_{S_\omega} F(\xi)e^{\Phi_\omega(\xi;\mathbf{p})}\mu_\omega(d\xi),$$

$$\Phi_\omega = \Phi^{\circ}_\omega(\xi) + \sum_f \Phi_{\omega,f}(\xi;\mathbf{p}|_{R_f}). \tag{R24}$$

Each carrier $R_f \subset Y^{\sim}$ supports both dependencies of its factor. The finite factor index set, the measurable coordinate projections onto stored variables in each carrier, and any nested island variables are part of the data. No variable is integrated twice: swallowed-operation variables retain their earlier labels inside the product assigned to the swallowing component.

If (s,D) ranges over the forest of nested stored components, set $v(\omega) = \sum_{(s,D)} \log\theta_s \,\#_s(D)$, where $\#_s$ counts sites in their own level. For a fixed creation-time complex domain $\mathcal{K}_{t,Y}$ require

(R25)
$$\frac{1}{|W_4|}\sum_{\omega} e^{c_{\mathrm{r}} v(\omega)} \int_{S_\omega}^{*} \sup_{\mathbf{p}\in\mathcal{K}_{t,Y}} |e^{\Phi_\omega(\xi;\mathbf{p})}| \mu_\omega(d\xi) \le C e^{-2p_0(\bar g_t)/(1+\beta_0)},$$
$$\sum_{f:R_f=R} \operatorname{osc}\Phi_{\omega,f} \le \mathsf{B}_\Phi (\log\theta_t)^{2\mathfrak{p}_0} e^{-\kappa d_t^\flat(\mathcal{S};R)}.$$

The first integral is an upper integral, allowing a nonmeasurable supremum. The oscillation is the supremum of $|\Phi_{\omega,f}(\xi;p)-\Phi_{\omega,f}(\xi;p')|$ over ξ and both pairs in that fixed domain. The domain is supplied from the data at creation and must contain every later evaluation; this last inclusion is an independent requirement. $c_{\mathrm{r}} \ge 0$ is fixed before the coupling ceiling, and $\beta_0 > 0$ satisfies $p_0(\bar g_m) \ge (1+\beta_0)^{-1} p_0(\bar g_n)$ for $m < n$. The supremum remains inside the integral and sum. There is no asserted small factor in the size of Y. A drop-out has no new free entry of type (R24): it keeps its partially formed operation, and its carrying Wilson factor is included in its separate majorant. Such a majorant must be supplied to use a drop-out estimate.

The effective exponent additionally requires finite index maps and vacuum numbers, which are formed data. At label $\mathcal{S}$ the map $\mathcal{I}_j^E(\mathcal{S})$ is a subset of $\{(z,X) : z \in \Lambda_j \cap \mathbb{T}^{(j)},\ X \in \mathcal{D}_j, z \in X\}$; it gives the actual enlarged-region rule for the regular terms. The enlargement width and treatment of boundary-crossing domains must be included in that map. The remainder index map is $\mathcal{I}_j^R(\mathcal{S}) = \{X \in \mathcal{D}_j : X \subset \Lambda_j^{\sim -1}\}$. The boundary index map selects the slots meeting the one-layer enlargement of Z_j. These maps use the action sequence in (R21). Profiles φ_j are supplied real functions on plaquette midpoints, equal to zero outside their prescribed small-field regions and to one on their prescribed interior regions. The interpolation rule in the intervening layers is part of the data. With $x_k(p) = \beta - \sum_{j\le k} b_j \varphi_j(p)$, the exponent is

$$\begin{aligned} A_{k,\mathcal{S}}(U) = & -A(x_k(\cdot),U) + \sum_{j,(z,X)\in\mathcal{I}_j^E(\mathcal{S})} \{E^{(j)}(X,U,z) - E^{(j)}(X,\mathbf{1},z)\} \\ & - \sum_{j\le k} b_j A(\varphi_j,U) + \sum_{j,X\in\mathcal{I}_j^R(\mathcal{S})} \{R^{(j)}(X,U) - R^{(j)}(X,\mathbf{1})\} \\ & + \sum_{(j,X,o)\text{ in boundary index map}} B_{\mathcal{S}}(j,X,o;U,\text{stored data}) - \mathcal{E}_{k,\mathcal{S}}. \end{aligned}$$

The real scalar $\mathcal{E}_{k,\mathcal{S}}$ is supplied with its vacuum bookkeeping rule, including the stage numbers retained by drop-outs. The pointwise identity $x_k(p) + \sum_{j\le k} b_j\varphi_j(p) = \beta$ is exact. It does not authorize changing the index maps or the vacuum rule. Thus these objects have explicit mathematical types, even where their actual formation and their uniform estimates remain upstream input.

Erase every healed component together with its ancestral cluster from a history h; the result is its standing part $h^\flat$. Erasure being an admissible history with the same label, and the hard-core decomposition used to sum healed clusters, are structural requirements on the formation data. They are not assumed merely from the word "erase." The never-large standing part has $\Omega_j = \Lambda_j = T_\ell$. The presentation gives finite positive measures $\widehat\tau_{h^\flat}$ on V_k, stored fields and coordinate frames, with marginals $\tau_{h^\flat}$ on V_k. It also retains the individual-history measures $\widehat\tau_h$ produced by the exact kernels. Their identities are

$$\rho_k(V_k)dV_k = \sum_{h^\flat} \tau_{h^\flat} = \sum_h \tau_h,$$

$$\sum_{h} \tau_h(\mathcal{V}_k) = Z. \tag{R26}$$

The first equality uses the summed healed operations; the exponent of an individual h need not have the effective-action form.

The reading at X records the standing regions, outcomes, marks, stored test bonds and frames within the hull of X enlarged by one level-l cell at every scale l. A chart is a translation by top-level cells, a common lift between tori, or their composite. Coherence requires a chart matching readings to carry the terms, carriers, Borel sets and measures to their counterparts. Factorwise,

$$\Phi_{\iota\omega,\iota f}(\iota\xi;\iota p) = \Phi_{\omega,f}(\xi;p). \tag{R27}$$

Unit translations not preserving the partitions are covered by the pointed-total requirement (R14), not by this chart definition. The simultaneous lattice symmetries preserving level-j cells must also transport the measures with their variables.

For a datum at a just below a pillar top y, let $\tau[U](a) = (\mathcal{M}^{m-1}U)(y,a)$ be its contour transport. An adapted background satisfies $\tau[U](a) = \mathbf{1}$ at every stored datum site. To define $\widehat{\tau}$ require that all backgrounds exist on its support, every argument belongs to its entry's domain, and the exponential has one value at all adapted representatives. The last clause is independent of simultaneous gauge invariance: an orbit transformation need not fix every datum site. These are part of class membership, not formal consequences of the norm bounds.

Finally let $\mathcal{Z}_k(h)$ be the live region after healing. For a live component X, define its lineage downwards by $\mathsf{Z}_k(X) = X$ and by taking at level j all live components contained in $\mathsf{Z}_{j+1}(X)$. Write $|C|_j$ for its number of level-j cells and $d'_j(W)$ for tree length in those cells. The created sets in a lineage are the components W of the step's creation set there; operation marks do not enter its deposit. With one fixed deposit constant $c > 0$, set

$$\begin{aligned}
\mathrm{Vol}_k(X;h) &= \sum_{(j,C)\text{ in its lineage}} 2M^4 \check{R}_j^5 |C|_j, \\
\mathrm{Dep}_k(X;h) &= \sum_{(j,W)\text{ created in its lineage}} c p_0(\bar{g}_j)^2 (d'_j(W)+1), \\
\Theta^{(k)}(h) &= \sum_{X\text{ live at }k} \{w\,\mathrm{Dep}_k(X;h) + c_a\,\mathrm{Vol}_k(X;h)\}, \quad 0 \le w < 1, \quad c_a \ge 0.
\end{aligned} \tag{R28}$$

The end mark is $\Theta^{(K)}$. Label marks and this numerical weight are different objects. The prepaid-stability input is $\sum_h e^{-E}\tau_h(\mathcal{V}_k)e^{\Theta^{(k)}(h)} \le e^{E_+\ell^4}$, with the stipulated normalization and dials. Its use requires the creation-deposit and operation-majorant bounds; (R28) defines the numerical weight.

Thus the class $\mathfrak{C}_k(\theta)$ consists of the complete typed presentations satisfying (R13)–(R28), their structural and positivity clauses, and the specified analytic domains. The induction uses the step-closure input: formation of the next presentation, preservation of the same constants and margins, and equality of the represented measures under its kernels. The background, response, substitution, island-mass, drop-out-majorant and closure estimates are the supporting analytic estimates used at this step, which we must prove.

14. Field coordinates and the radius of quadratic extraction

The quadratic coefficient of an effective interaction is defined in the physical block field. The field used to normalize a fluctuation Gaussian is a different coordinate. Keeping these coordinates distinct explains both the fixed radii of the unmarked chain and the squared radius ratios in the

marked chain. The required inclusion near the trivial field follows from the background and tower estimates already used to define the analytic field domains. It is not an additional loss caused by the number of blocking steps.

All constants below may depend on the fixed blocking and localization parameters. They are independent of the level, torus size and number of earlier levels. The non-Abelian background and tower estimates are explicit hypotheses in the statements that use them.

14.1. **Physical fields, Gaussian fields and the coefficient.** At level j, write the configuration in the trivial-field chart as

$$\mathbf{U} = e^{i\eta_j A'},$$

$$v = (A', J),$$

$$\|v\|_{1,j} = \max\{\|A'\|_\infty, \|\nabla_j A'\|_\infty, \|J\|_\infty\}. \tag{ER1}$$

Here $0 < \eta_j \le 1$ is the bare spacing in level-j units and ∇_j is a difference quotient in those units. The current J is an independent analytic variable. In particular, A' in this chart is a physical potential; no coupling multiplies it by definition.

The block datum of the background is $V = e^{iB}$. Let $p_j(B)$ be its configuration-current pair in a fixed gauge near zero. Derivatives of its configuration entry are taken in the chart (ER1). Suppose

$$p_j(0) = (\mathbf{1}, 0), \qquad Dp_j(0)B = \mathfrak{h}_j B = (H_j B, D^* D H_j B). \tag{ER2}$$

The datum B is again physical. Its continuation radius r_{B} is a radius in e^{iB}, chosen independently of the coupling. For a gauge-invariant local interaction E, the first derivative at the trivial pair vanishes: invariance under constant gauge transformations makes it an invariant linear functional on the semisimple Lie algebra, and that functional is zero. Consequently

$$D^2(E \circ p_j)(0)[B_1, B_2] = \operatorname{Hess} E(0)[\mathfrak{h}_j B_1, \mathfrak{h}_j B_2]. \tag{ER3}$$

The possible second derivative of p_j contributes nothing because it is multiplied by $DE(0)$.

The polarisation kernel is the kernel of (ER3). Its second spatial moments define the curvature tensor $T[E]$, and, in the invariant scalar channel, the action coefficient $b(E)$. These definitions use derivatives with respect to B. For the Wilson functional of the block datum itself, use the normalization

$$b(A_{\mathrm{block}}) = 1, \tag{ER4}$$

whose quadratic part is $\frac{1}{2}\|dB\|^2$. For a Wilson family evaluated after the background map, the analogous normalization requires the linear-response identities in that map. Likewise $T[\mathsf{A}[c]] = c$ for a constant allowed curvature tensor c is the prescribed counter-insertion normalization. Neither identity for pulled-back families is proved merely by the coordinate change below.

In a fluctuation integral one instead sets $\mathcal{A}_j = g_j \mathcal{A}_j^{\mathrm{s}}$. If $\widetilde{E}_j(B^{\mathrm{s}}) = E(p_j(g_j B^{\mathrm{s}}))$, the chain rule gives

$$D^2 \widetilde{E}_j(0) = g_j^2 D^2(E \circ p_j)(0),$$

$$T_{\mathrm{s}}[E] = g_j^2 T[E],$$

$$b_{\mathrm{s}}(E) = g_j^2 b(E). \tag{ER5}$$

Thus a fixed probe in the Gaussian variable may fit inside a shrinking physical domain. It extracts the rescaled coefficient in (ER5). Recovering the coefficient of the physical action requires division by g_j^2.

For example, impose the response-decay and localization hypotheses of the amplitude proposition below, and suppose that the usable physical radius is a_j and the family norm is at most M_j there. Its radius in B^{s} is comparable to a_j/g_j. The Cauchy radius factor in the amplitude bound then transforms as

$$|g_j^2 b(E)| \le C M_j (g_j/a_j)^2, \qquad |b(E)| \le C M_j a_j^{-2}. \tag{ER6}$$

The coupling cancels from the coordinate conversion. The same calculation holds if the physical radius is written g_j^{γ}: the canonical radius is $g_j^{\gamma-1}$, and the physical coefficient bound remains $C M_j g_j^{-2\gamma}$. The change of variables creates no extra loss and removes no loss already present in the physical radius.

14.2. **A near-zero ball lies in the hereditary core.** We state precisely the background information needed for this local inclusion. A core at radii (a_0, a_1) is described by the following simple conditions and their tower counterparts. A triple is (U, A', J), with U real and $\mathbf{U} = e^{i\eta_j A'} U$. The real and complex curvatures are smaller than $a_0 \eta_j^2$; the real factor has the prescribed local regular gauges; $|A'|, |\nabla_U A'| < a_1$; and $|J| < a_0$. The tower counterparts require the corresponding curvature and current bounds for every allowed restricted tower inside the domain.

Assume the following uniform tower estimate. If the simple conditions hold at sufficiently small radii (ϵ_0, ϵ_1), the tower configurations exist holomorphically. Their curvature and current, in the local units of their layer, satisfy the defining tower bounds with radius

$$K_{\mathrm{T}}(\epsilon_0 + \epsilon_1^2). \tag{ER7}$$

The estimate includes the prescribed change-of-unit factors and holds for both tower rules, every restricted domain and every depth. This is the substantive background estimate used here. It is stronger than existence of a background for one real datum.

Proposition 14.1 (Near-zero core inclusion)**.** *Under (ER7), for sufficiently small a_0, a_1 there is a constant $c > 0$, independent of j, such that*

$$\|v\|_{1,j} < r_j,$$
$$r_j = c \min\{a_0, a_1, 1\}$$
$$\Longrightarrow$$
$$(\mathbf{1}, A', J) \text{ belongs to the core at } (a_0, a_1). \tag{ER8}$$

Proof. The plaquette product in the exponential chart obeys

$$|\partial e^{i\eta_j A'} - \mathbf{1}| \le C\eta_j^2 (\|\nabla_j A'\|_\infty + \|A'\|_\infty^2) \le C\eta_j^2 (r_j + r_j^2). \tag{ER9}$$

This follows by expanding a plaquette product: the linear term is the discrete curl and each nonlinear term contains at least two bond potentials. On a fixed small chart the remaining exponential series is bounded by a uniform multiple of the quadratic term. In particular, there is no inverse power of η_j in (ER9). This is the reason for including the first difference quotient in (ER1).

Choose

$$\epsilon_0 = \frac{a_0}{4K_{\mathrm{T}}},$$
$$\epsilon_1 = \min\{a_1/2, \sqrt{a_0/(4K_{\mathrm{T}})}\}.$$

Taking c smaller if necessary ensures $C(r_j + r_j^2) < \epsilon_0$ and $r_j < \min\{\epsilon_0, \epsilon_1\}$. The real configuration is $\mathbf{1}$, so its curvature and local gauge potential vanish. All simple conditions therefore hold at these reduced radii. Moreover, $K_{\mathrm{T}}(\epsilon_0 + \epsilon_1^2) \le a_0/2$. The tower estimate gives every hereditary clause at the desired radii, and $|J| < r_j < a_0$. This proves (ER8). □

A bound only on $\|A'\|_\infty$ would be different: arbitrary oscillation on the bare lattice can give $\|\nabla_j A'\|_\infty$ of order $\eta_j^{-1}\|A'\|_\infty$. The ball relevant to (ER8) is the C^1 ball, not that weaker ball.

For completeness, suppose the continued background in (ER2) is holomorphic for $\|B\|_\infty < r_{\mathrm{B}}$ and satisfies

$$\|\mathcal{H}_j(B)\|_{C^1} + \|J(e^{i\eta_j \mathcal{H}_j(B)})\|_\infty \le C_{\mathrm{B}}\|B\|_\infty. \tag{ER10}$$

Then its triple belongs to the core whenever

$$\|B\|_\infty < \min\{r_{\mathrm{B}}/2, r_j/(2C_{\mathrm{B}})\}. \tag{ER11}$$

The same conclusion for a first-order response uses the derivative of (ER10). Equations (ER8)–(ER11) give a fixed datum radius when (a_0, a_1) are fixed, and a radius comparable to $\min\{a_0, a_1\}$ when they run. The hypotheses are exactly the background continuation, its uniform first-order bounds and the hereditary tower estimate. No analyticity radius in the normalized Gaussian variable is substituted for this physical datum radius.

14.3. **Cauchy's estimate, response decay and entropy.** Work first with the infinite-lattice companion. Its torus versions use the uniform response bounds and the lift convergence supplied by the background hypotheses. Consider pointed local families satisfying

$$|E(X, p, z)| \le Me^{-\kappa d(X)}$$

on the core, and assume that the near-zero ball there has usable radius a. Assume also the uniform response decay

$$\|\mathfrak{h}_j \delta_c\|_{1,j;X} \le Ce^{-\delta \operatorname{dist}(c,X)}. \tag{ER12}$$

The distances in this section are measured in the unit of level j. For a fixed root cube, suppose the number of domains with tree length between m and $m+1$ is at most $Ce^{\kappa_{\mathrm{D}} m}$, while the number of pointed sites in such a domain is at most $C(m+1)^4$. These are the usual localization entropy and volume bounds.

Proposition 14.2 (The amplitude bound with its radius)**.** *If $\kappa > \kappa_{\mathrm{D}}$ with a fixed positive margin, the polarisation sums converge absolutely and*

$$|\Pi_{\mu\nu}(u)| \le Ca^{-2}Me^{-\delta_1|u|},$$
$$|b(E)| + \|T[E]\| \le Ca^{-2}M, \tag{ER13}$$

whenever the scalar coefficient or the stated summed tensor is defined by the prescribed covariance conditions. The constants are uniform in the level and torus volume. A tensor at a fixed base point satisfies the same bound before any isotropy assertion is made.

Proof. The two-variable Cauchy formula on a bidisc inside the ball gives

$$|\operatorname{Hess} E(X,z)[v,w]| \le Ca^{-2}Me^{-\kappa d(X)}\|v\|_{1,j;X}\|w\|_{1,j;X}.$$

Combining this with (ER12) bounds the contribution of one term by

$$Ca^{-2}Me^{-\kappa d(X)}e^{-\delta(\operatorname{dist}(c,X)+\operatorname{dist}(c',X))}. \tag{ER14}$$

If the localization cubes have fixed side M_0, then

$$|c_- - c'_-| \le \operatorname{dist}(c,X) + \operatorname{dist}(c',X) + CM_0(d(X)+1).$$

A sufficiently small fixed fraction of the exponents in (ER14) supplies $e^{-\delta_1|c_- - c'_-|}$ for a fixed $\delta_1 > 0$. Choose that fraction so that the remaining localization rate stays strictly above κ_{D}. The remainder is summable. Indeed, root a domain at a cube closest to c. Summing its pointed sites costs at most a polynomial in $m+1$, and summing its shapes costs $e^{\kappa_{\mathrm{D}}m}$. These costs are absorbed by the remaining $e^{-\kappa_{\mathrm{rem}}m}$, where $\kappa_{\mathrm{rem}} > \kappa_{\mathrm{D}}$. Summing the root location is bounded by the remaining exponential in its distance from c. This proves the first inequality in (ER13) and absolute convergence.

The second moment defining $b(E)$ is bounded by $Ca^{-2}M\sum_u |u|^2 e^{-\delta_1|u|}$, a finite constant times $a^{-2}M$. For a tensor at a fixed base point, the two moment factors are bounded by polynomials in the domain diameter and the two response distances. Those polynomials are absorbed by a further fixed fraction of the same exponentials. This proves the tensor bound. Identifying an invariant tensor with the scalar action coefficient uses the separate Ward and lattice-symmetry identities; it is not needed for the amplitude estimate. □

In particular, the extraction norm is uniform on fixed physical radii. On the running radii

$$a_j \asymp \theta_j^{-1/2}(\log\theta_j)^q,$$
$$q \ge 0,$$

the correct general bound has the factor $a_j^{-2} \asymp \theta_j(\log\theta_j)^{-2q}$. One can express uniformity instead as $a_j^2|b(E)| \le C\|E\|$, provided that this normalization is retained in all later estimates.

14.4. **What the squared radius ratios pay for.** Let a remainder born at level j have size r_j on physical radius a_j and have its vacuum value removed. There is no linear term. Assume a gauge-invariant window estimate in which the quadratic curvature factors each scale by $L^{-2(n-j)}$, while a unit volume at level n contains $L^{4(n-j)}$ sites of level j. After these geometric factors cancel, evaluating the quadratic part on a background of radius a_n costs

$$w_{n,j}r_j,$$
$$w_{n,j} = (a_n/a_j)^2. \tag{ER15}$$

This is the relative quadratic evaluation weight. Extracting the physical coefficient from the resulting family at level n costs a_n^{-2}, so

$$a_n^{-2}w_{n,j}r_j = a_j^{-2}r_j. \tag{ER16}$$

Charging another factor a_j^{-2} on top of the left side would count the same shrinking input radius twice. Omitting the displayed readout factor a_n^{-2} would instead confuse an evaluated potential

with its physical quadratic coefficient. The full nonlinear remainder estimate also requires its actual backgrounds to lie in the original analytic domains and the separate control of large domains. Equation (ER15) records the radius factor in that estimate; it provides no continuation outside those domains.

For example, let $r_j \le C\theta_j^{-\kappa_0/2}$, $\kappa_0 > 4$, and suppose $\theta_j \ge \theta_n + \lambda(n-j) - C_0$ with $\lambda > 0$ and a sufficiently large common lower bound on θ_n. Integral comparison gives

$$\begin{aligned} \sum_{j\le n} w_{n,j} r_j &\le C a_n^2 \sum_{j\le n} \theta_j^{1-\kappa_0/2} (\log \theta_j)^{-2q} \le C\theta_n^{1-\kappa_0/2}, \\ a_n^{-2} \sum_{j\le n} w_{n,j} r_j &\le C\theta_n^{2-\kappa_0/2} (\log \theta_n)^{-2q}. \end{aligned} \tag{ER17}$$

The first loss of one power of θ comes from summing the older levels; the second comes from quadratic extraction. This is the usual g^{κ_0-4} bound, with its logarithmic improvement. The two losses are already present in (ER17).

The unmarked chain uses fixed physical radii, so (ER13) preserves its small interaction bound. The marked chain uses running radii, and its stronger smallness pays the readout loss in (ER17). These are the radius estimates used under the stated background, response and entropy hypotheses. The complex-window, tensor-marginal, weighted-sensitivity and large-field-zone derivative propositions use their additional stated estimates.

15. The construction and the plan of proof

Proof outline. The supporting analytic results used in this section, including the obligations that we must still prove, are listed under A–H in the dependency catalogue. Their hypotheses and precise forms are retained at the points where the argument uses them.

A dagger marks auxiliary assertions outside the central target statements. Their hypotheses and dependencies are read in the same way as those of the main argument.

0. Objects and proof strategy. This section states the two theorems and explains the structure of the proof. It identifies the methods adapted from Bałaban, the classical tools, and the additional arguments, and shows how the multiscale construction addresses the difficulties of a direct approach. Each later step contributes to a clause of (0a).

Objects. We take $G = \mathrm{SU}(2)$ and $N = 2$. On the torus T_ℓ, with $\ell \in \mathfrak{L}$, we consider the following objects: the lattice $\Lambda_{K,\ell}$, whose spacing is $a_K = L^{-K}$; the configurations U on it; the quantity $a_p(U) = 1 - \frac{1}{2}\operatorname{Re}\operatorname{tr} U_p$; Wilson's measure; the smeared curvature density; and the connected correlation functions of that density:

$$\mu_{K,\ell,\beta}(dU) = Z_{K,\ell,\beta}^{-1}\, e^{-\beta \sum_p a_p(U)}\, dU,$$

$$\mathcal{O}(f) = \sum_p f(x(p))\, a_p(U),$$

$$S^T_{n;K,\ell,\beta}(f_1, \dots, f_n) = \langle \mathcal{O}(f_1); \dots; \mathcal{O}(f_n) \rangle_{K,\ell,\beta}.$$

In view of $a_p \approx \frac{1}{2} a_K^4 |F_U(p)|^2$, the sum $\mathcal{O}(f)$, which carries no prefactor involving the spacing, is a Riemann-sum approximation of $\frac{1}{2}\int f \sum_{\mu<\nu} |F_{\mu\nu}|^2$. It is therefore the action density $\operatorname{tr} F \cdot F$. The test functions are restricted to separated tuples $(f_i) \in \mathfrak{S}_n(\mathbb{E})$, which are tuples of real C_c^1 functions with pairwise disjoint supports. The reason is given under "The frame" in (0a).

The cutoff is tied to the coupling. The renormalisation group of steps 1 and 2 assigns to $\beta = g_0^{-2}$ a sequence of running inverse couplings $x_0 = \beta$, $x_k = x_{k-1} - b_k$, whose increments satisfy $b_k \le B_\sharp$. These increments are independent of the torus (3b), and also of the cutoff $K' \ge k$ used to carry out the steps (3d). We fix a small reference coupling g. As the unit of length we take the scale at which the flow reaches, for the first time, the reference window $g^{-2} \le x \le g^{-2} + B_\sharp$ belonging to g:

$$K(\beta, g) = \min\{k : \, x_k(\beta) \le g^{-2} + B_\sharp\},$$

$$\mathcal{W}_K(g) = \{\beta \ge g^{-2} + B_\sharp : \, K(\beta, g) = K\};$$

a pair (K, β) such that $\beta \in \mathcal{W}_K(g)$ is called a first-passage datum.

† The remaining pairs with $\beta \ge g^{-2} + B_\sharp$ are accounted for as follows. Suppose that $K \ge K' = K(\beta, g)$, put $b = K - K'$, and let $f^{[b]}(x) = f(L^{-b}x)$ on $T_{L^b\ell}$. Then

$$\langle \mathcal{O}(f_1); \dots; \mathcal{O}(f_n) \rangle_{K,\ell,\beta} = \langle \mathcal{O}(f_1^{[b]}); \dots; \mathcal{O}(f_n^{[b]}) \rangle_{K',\, L^b\ell,\, \beta}$$

holds exactly, for all bounded f_i. To see this, note that the dilation $x \mapsto L^b x$ maps $\Lambda_{K,\ell}$ onto $\Lambda_{K',L^b\ell}$, plaquettes to plaquettes and barycentres to barycentres. Wilson's weight depends only on the plaquette variables and on Haar measures, and no power of the spacing enters $\mathcal{O}$. The same identity holds between a periodic Gibbs state at β on $\mathbb{L}_K$ and its image on $\mathbb{L}_{K'}$ (3d). Consequently a pair with $K > K(\beta, g)$ amounts to the first-passage datum of β on a dilated torus, on which $\|f^{[b]}\|_\sharp \le \|f\|_\sharp$ and every distance is multiplied by L^b. The restriction to first passage therefore loses none of these lattice measures; its only effect is to fix the unit of length. (When $K < K(\beta, g)$ the identity is used in the opposite direction, on a contracted torus. This requires a lower bound on ℓ, and the norms grow by at most the factor $L^{K(\beta,g)-K}$ (3d).)

The final value $x_K(\beta)$ lies in the window but is not fixed at g^{-2}. This is why (a)(ii) passes to subsequences, and it is the purpose of clause (d). On each torus, provided the conditions of (d)(1) are satisfied, that clause dispenses with the subsequences as soon as the renormalised coupling, defined below, converges along the sequence. On $\mathbb{R}^4$, in (a)(iii), a subsequence is taken in every case, including the case in which the renormalised coupling converges: the fixed-torus statement does not assert uniqueness of the infinite-volume limit. The renormalised inverse coupling $x_K^\natural(\beta)$ is the final value of a particular run of K steps starting at β. In that run, the auxiliary choices of level k that depend on the coupling are made with $\theta_k = g^{-2} + c_0(K-k)$ instead of x_k, where $c_0 = \frac{11N^2}{12\pi^2}\log L$. Here and in all steps, log denotes the natural logarithm. A second kind of run is needed for the following reason. Continuity and holomorphy in the bare coupling are established by setting the constructions at two bare couplings side by side and comparing them entry by entry, and by letting a single fixed linear operation act on a whole family of bare densities. Each of these two devices needs the choices to coincide at the two couplings. In a run made with its own couplings, however, the choices vary with β. Some of them, namely the cube sizes $\check{R}_k$, change through integer values (3c). Freezing the choices leaves $Z_{K,\ell,\beta}$ and S_n^T unchanged, since these are quantities of the measure. At first-passage data, and under the conditions of (d)(1), the shift that freezing produces in the couplings is at most g times a power of $\log g^{-2}$ (3c), (8b). By a continuum limit at g we mean a family $S = (S_n)_{n\ge 2}$ with

$$S_n(X) = \lim_{j\to\infty} S^T_{n;K_j,\ell_j,\beta_j}(X),$$
$$\beta_j \in \mathcal{W}_{K_j}(g),$$
$$K_j \to \infty,$$

taken either on T_ℓ (with all $\ell_j = \ell$) or on $\mathbb{R}^4$ (with $\ell_j \to \infty$). Limits on a torus along the tuned couplings of (a)(i) are also counted as continuum limits. A volume-first limit at g is a limit $\lim_j S_n^T[\nu_j]$, taken along first-passage data (K_j, β_j) with $K_j \to \infty$. In it each $\nu_j \in \mathcal{G}^{\mathrm{tor}}_{K_j,\beta_j}$ is a limit point of the periodically extended torus measures as $\ell \to \infty$, with spacing and coupling held fixed. Thus the volume is removed first and the spacing only afterwards.

The two theorems in brief. The exponents are fixed first. Then L is taken large, then the remaining constants are chosen in their order, and finally g is taken small. Theorem 0 states the following. The two assertions below hold uniformly in the first-passage datum, and the left-hand one holds also uniformly in the torus as soon as $\ell \ge Cn(R+1)$:

$$\big|S^T_{n;K,\ell,\beta}(f_1,\dots,f_n)\big| \le C_n^\infty(\mathsf{d};R)\prod_i \|f_i\|_\sharp,$$

$$S_2(f,h) = \frac{3(N^2-1)}{\pi^4}\Big(\frac{12\pi^2}{11N^2}\Big)^2 \frac{\mathcal{K}_{\mathrm{free}}(f,h)}{(\log(1/\mathsf{d}))^2}\,(1+\eta) > 0.$$

In the left-hand bound, $n \geq 2$, and the supports lie in balls of radius R and are pairwise at distance $\geq \mathsf{d}$. In the right-hand formula, S is a limit along first-passage data, $f, h \geq 0$ are bumps whose size and separation are of the order of a small d, and $\mathcal{K}_{\mathrm{free}}(f,h) = \iint f(x)h(y)|x-y|^{-8}\,dx\,dy$. The factor $1+\eta$ is bounded above and below by positive constants, and on each torus $\eta \to 0$ as $\mathsf{d} \to 0$; on $\mathbb{R}^4$ the clause consists of the bounds by the two constants and nothing more. The five clauses of this theorem are the following:

- limits exist on the tori and on $\mathbb{R}^4$ (a);
- they inherit translation and hypercubic invariance, reflection positivity, symmetry and regularity; the factorial growth assertion in (b) is supplied for limits formed from periodic Wilson DLR states;
- they are non-trivial and have, at short distance, the normalisation of asymptotic freedom (c);
- on a torus they depend only on the renormalised coupling (d);
- on $\mathbb{R}^4$ they are rotation invariant if and only if a flux through far shells vanishes (e).

Theorem 0' states that a volume-first limit S is O(4)-invariant. Its centred functional $\sigma_S = \sum_{\pi \in \mathrm{Part}_{\geq 2}} \prod_{J \in \pi} S_{|J|}$ is precisely the system of moments obtained when the one-point function is set to zero, which is to say the moments of the centred density. This functional admits a canonical decomposition, which consists of a probability measure λ, carried by a compact set $\mathsf{X} = \operatorname{supp}\lambda$, together with a continuous injective map $\xi \mapsto \sigma_\xi$ such that

$$\int_{\mathsf{X}} \prod_{i=1}^{q} \sigma_\xi(X^i)\,\lambda(d\xi) = \lim_{|v_q|\to\infty} \cdots \lim_{|v_2|\to\infty} \sigma_S\big(X^1 \cdot \tau_{v_2} X^2 \cdots \tau_{v_q} X^q\big)$$

$$(q \geq 1),$$

so that

$$\sigma_S = \int_{\mathsf{X}} \sigma_\xi\,\lambda(d\xi).$$

Once the stated analytic obligations are proved, each component σ_ξ has the Euclidean properties listed in Theorem 0'. The reconstruction implication additionally uses the quantitative growth condition and the external analytic-continuation theorem specified in Section 54.

(0a) The theorem, stated clause by clause. There exist a shape constant $\vartheta_0 \in (0,1]$ and, for each source radius $\varrho > 0$ (step 4) and each shape $\vartheta \in (0,1]$, thresholds $0 < \gamma_{U'} \leq \gamma_U \leq \gamma_J$ and $0 < \gamma_W(\vartheta) \leq \gamma_J$ such that Theorem 0' and the clauses below hold. Throughout, $0 < g \leq \gamma_J$, $\ell \in \mathfrak{L}(g)$, and (K,β) is a first-passage datum. Apart from dependences that are displayed, the constants do not vary with K, β, ℓ or with positions. The ceilings $\gamma_{U'}, \gamma_U, \gamma_J$ depend only on the constants fixed before them, among which is ϱ, and $\gamma_W(\vartheta)$ depends only on these and on ϑ. Each constant, threshold and function introduced by a clause of the two theorems may in addition depend on g and on the quantities displayed with it. The brackets after a clause give the steps that yield that clause, together with some of what those steps take from other steps.

The frame is adapted to the tools. When the side is $\ell = 2L^m$ with $m \in \mathbb{N}$, each axis carries an even number $2L^{m+K}$ of sites. Each mirror pair $\{y_\mu = c\} \cup \{y_\mu = c + \ell/2\}$ therefore cuts the torus into two halves, and the chessboard cells come in pairs (4d). Because L is odd, the mirrors that bisect bonds at successive cutoffs are nested: $a_K(\mathbb{Z} + \frac{1}{2}) \subset a_{K+1}(\mathbb{Z} + \frac{1}{2})$. Because barycentres are used, the identity $\mathcal{O}(f) \circ \mathsf{r}^* = \mathcal{O}(\mathsf{r}f)$ is exact for each lattice isometry r, where r^* denotes its action on configurations. For the same reason $\mathcal{O}(f)$ stays on the same side of such a mirror as f. In the exact-cochain Gaussian reference of Section 5, the mean of the quadratic approximation to $\mathcal{O}(f)$ is $\frac{N^2-1}{4\beta}(1 - V_K^{-1}) \sum_p f(x(p))$, where $V_K = (\ell/a_K)^4$ is the number of sites.

For a fixed smooth f of nonzero integral on a fixed torus, its leading Riemann-sum behavior is $\frac{3(N^2-1)}{2\beta}a_K^{-4}\int f$, which diverges in magnitude along the stated first-passage scaling. Moreover $|x-y|^{-8}$ is not locally integrable. For these reasons only $n \geq 2$ and only separated tuples are considered, and positivity is stated for the centred σ_S. This is the diagonal-flat test domain used in Section 54. Flatness removes sensitivity to contact terms; it does not supply the growth bound, continuity or analytic-continuation hypotheses needed for reconstruction. By a mirror Σ of $\mathbb{R}^4$ we mean a hyperplane $\{u \cdot x = c\}$ with either $u = e_\mu$, the unit vector of the μ-th coordinate (coordinate), or $u = (e_\mu \pm e_\nu)/\sqrt{2}$ for some $1 \leq \mu < \nu \leq 4$ (diagonal). Its reflection is θ_Σ and its open side is $\mathbb{E}_+(\Sigma)$. A torus has only the coordinate pairs as mirrors, and there the open side is a slab.

Theorem 0 (a) [existence; 3, 5, 8]. Each run started from $\beta \geq g^{-2} + B_\sharp$ reaches first passage. For such a run, $c(\beta - g^{-2}) - C \leq K(\beta, g) \leq C(\beta - g^{-2}) + C$, and since $b_k \leq B_\sharp$ also $g^{-2} < x_K(\beta) \leq g^{-2} + B_\sharp$. Hence $K(\beta, g) \to \infty$ as $\beta \to \infty$, and so $\mathcal{W}_K(g) \neq \varnothing$ for infinitely many K. The values $x_k(\beta)$ are the same whatever torus the steps are carried out on, and $\beta/\log(1/a_K) \to \frac{11N^2}{12\pi^2}$ uniformly on $\mathcal{W}_K(g)$. (i) Let $g \leq \gamma_U$. Then for each K there is a tuned $\beta_K^\natural$, that is, one with $x_K^\natural(\beta_K^\natural) = g^{-2}$, and it satisfies $\frac{1}{2}c_0 K - c_5 \leq \beta_K^\natural - g^{-2} \leq \frac{3}{2}c_0 K + c_5$. Moreover $S^T_{n;K,\ell,\beta_K^\natural} \to S_n^{\natural,\ell}$ as $K \to \infty$ along the whole sequence. This convergence holds on each torus of the family and on tuples of Lipschitz functions whose supports are pairwise disjoint. (ii) On a fixed torus, from every sequence of first-passage data with $K_j \to \infty$ one can extract a subsequence that converges on $\mathfrak{S}(T_\ell)$. (iii) The same holds on $\mathfrak{S}(\mathbb{R}^4)$ when in addition $\ell_j \to \infty$. The limits obtained in this way along tori, with defining sequence (K_j, ℓ_j, β_j), are what the phrase "a limit of (a)(iii)" means. The same also holds for $S_n^T[\nu_j]$ with $\nu_j \in \mathcal{G}^{\mathrm{tor}}_{K_j,\beta_j}$, so volume-first limits exist.

(b) [inheritance; 6; the bounds from (4c)–(4e), those of $S^{\natural,\ell}$ through (8a), the last sentence from (5c) and (10a)]. Each continuum limit has the following properties. (i) It is invariant under W_4 and under the translations of $\mathbb{E}$. (ii) It is reflection positive at each coordinate mirror: $\sum_{i,j}\bar{c}_i c_j\,\sigma_S(\theta_\Sigma X_i \cdot X_j) \geq 0$ whenever $X_i \in \mathfrak{S}(\mathbb{E}_+(\Sigma))$. (iii) It is symmetric. (iv) It is regular: there is a unique distribution of locally finite order on $\mathbb{E}^n_{\neq}$ of which S_n is the restriction. On C^1 functions, and with the same constants, it satisfies (c)(1) on T_ℓ. The per-order bound labelled (c)(2) passes to limits of (a)(ii)–(iii), with its displayed volume condition retained in the fixed-torus case. The factorial bound displayed after it passes to limits formed from periodic Wilson DLR states. (v) On $\mathbb{R}^4$, a limit formed from periodic Wilson DLR states satisfies the growth condition

$$|\mathsf{S}_n^{\sigma_S}(\varphi)| \leq c_{\mathrm{E}} C_{\mathrm{E}}^n (n!)^{\varkappa} |\varphi|_{k_{\mathrm{E}} n}$$

on $\mathcal{S}_{\neq}(\mathbb{R}^{4n})$. For every $\mathbb{R}^4$ limit, take the form $\langle F, G\rangle = \sigma_S(\theta\bar{F} \cdot G)$ on $\mathbb{C}\mathfrak{S}(\{x_4 > 0\})$. The separable space $\mathcal{H}(S)$ then carries three structures:

- $\mathcal{T}_t[F] = [\tau_{te_4}F]$, $\mathcal{T}_t = e^{-tH}$, which form a strongly continuous semigroup of self-adjoint contractions, where $H \geq 0$ and $H\Omega = 0$;
- a strongly continuous unitary representation $e^{iv\cdot P}$ of $\mathbb{R}^3$ that leaves Ω fixed, the P_j commuting strongly with H;
- a unitary representation of W_3 that leaves Ω fixed and commutes with H.

Already for $g \leq \gamma_J$, each volume-first limit is a limit of (a)(iii) and is, in addition, reflection positive at each diagonal mirror.

(c) [bounds, non-triviality; 4, 7; the envelope of $g(\mathsf{d})$ from (3a), the periodic Gibbs states through (10a)]. Let $n \geq 2$, let the f_i be Lipschitz with supports pairwise at distance $\geq \mathsf{d} > 0$, and in (2) let the supports also lie in balls of radius R. Then

$$(1)\,|S_n^T| \le C_n(\mathsf{d})\, e^{nc_V\ell^4} \prod_i \|f_i\|_\sharp;$$

$$(2)\,|S_n^T| \le C_n^\infty(\mathsf{d};R) \prod_i \|f_i\|_\sharp \ (\ell \ge Cn(R+1)),$$

$$|S_n^T[\nu]| \le [2B(R)]^n (n!)^k \max\{1, \mathsf{d}^{-5n}\} \prod_i \|f_i\|_\sharp;$$

Here $C_n^\infty(\mathsf{d};R)$ does not increase as d grows. The last bound is for periodic Wilson DLR states ν and real C^1 tests, under D1 in its actual rooted geometry and on its compatible source tori, together with G1. One may take $B(R) = C(R+1)^4$ and one fixed finite integer k; see Theorem 39.1. A finite-torus factorial bound, or a single polynomial separation prefactor of degree independent of n, requires an additional estimate. The bounds (2) also hold, with no condition on a volume, for $S_n^T[\nu]$, where ν is an arbitrary periodic Gibbs state on $\mathbb{L}_K$ of Wilson's weight at β, and where (K,β) is a first-passage datum.

(3) The class $\mathfrak{T}_{\mathbb{E}}(\mathsf{d};\vartheta)$ consists of the pairs of Lipschitz functions $f, h \ge 0$, neither of which is identically zero, that satisfy the following conditions. Each is supported in a ball of radius d, and $\mathsf{d} \le \operatorname{dist}(\operatorname{supp} f, \operatorname{supp} h) \le 6\mathsf{d}$. In addition, $\int f \ge \vartheta \|f\|_\infty \mathsf{d}^4$ and $\operatorname{Lip} f \le \|f\|_\infty/(\vartheta\mathsf{d})$, and the same two conditions hold for h. Assume $g \le \gamma_W(\vartheta)$, $\mathsf{d} \le \mathsf{d}_0$ and $(f,h) \in \mathfrak{T}_{T_\ell}(\mathsf{d};\vartheta)$. Assume also $K - s(\mathsf{d}) \ge n_0$, so that at least n_0 steps lie between the scale d and the cutoff. Then

$$\langle \mathcal{O}(f); \mathcal{O}(h)\rangle_{K,\ell,\beta} = \zeta_K(\mathsf{d})^2\, \tfrac{3(N^2-1)}{\pi^4}\, g(\mathsf{d})^4\, \mathcal{K}_{\mathrm{free}}(f,h)\,(1+\varepsilon),$$

$$|\varepsilon| \le \min\{\tfrac{1}{2}, \bar\varepsilon_\ell(\mathsf{d})\}.$$

In this formula $g(\mathsf{d}) = g_{K-s(\mathsf{d})}(\beta)$, where $L^{-s(\mathsf{d})} \approx \mathsf{d}$ up to logarithmic factors. Also $\underline{x}_s \le g(\mathsf{d})^{-2} \le \bar{x}_s$, and both bounds are affine in $s = s(\mathsf{d})$. The factor $\zeta_K(\mathsf{d}) = \zeta'_{K-s(\mathsf{d})}$ is the running normalisation, that is, it measures how $x_{K-s(\mathsf{d})}$ responds when β is shifted by a constant (4b). One has $|\zeta_K(\mathsf{d}) - 1| \le \bar\eta(\mathsf{d})$ and $|\frac{11N^2}{12\pi^2} g(\mathsf{d})^2 \log\frac{1}{\mathsf{d}} - 1| \le \bar\eta(\mathsf{d})$. Here $\mathsf{d}_0 \in (0,1)$, and the two functions $\bar\eta : (0,\mathsf{d}_0] \to [0,\frac{1}{3}]$ and $\bar\varepsilon_\ell : (0,\mathsf{d}_0] \to [0,\frac{1}{2}]$ are non-decreasing; moreover $\bar\eta, \bar\varepsilon_\ell \to 0$ as $\mathsf{d} \to 0$. Hence, on each torus, $\varepsilon \to 0$ uniformly with respect to the cutoff, to the bare coupling as it ranges over its window, to the pair, and to where the pair is placed. For periodic Gibbs states the identity remains true, but only with $|\varepsilon| \le \frac{1}{2}$.

(4) Consequently the limits of (a)(ii)–(iii) obey the law stated at the opening of the text, for those pairs in $\mathfrak{T}_{\mathbb{E}}(\mathsf{d};\vartheta)$ in which f, h are of class C^1; the tuned $S^{\natural,\ell}$ are not included. The error satisfies $(1-\bar\varepsilon)(1-\bar\eta)^4 \le 1+\eta \le (1+\bar\varepsilon)(1+\bar\eta)^4$, with $\bar\varepsilon = \bar\varepsilon_\ell(\mathsf{d})$ on T_ℓ and $\bar\varepsilon = \frac{1}{2}$ on $\mathbb{R}^4$, where in addition $\dim \mathcal{H}(S) \ge 2$.

(5) The class $\mathfrak{T}_{\mathbb{E}}^{(4)}(\mathsf{d};\vartheta)$ consists of the quadruples whose pairs all belong to $\mathfrak{T}_{\mathbb{E}}(\mathsf{d};\vartheta)$ and for which $\operatorname{supp} f_i \subset B(c_i, \mathsf{d}/16)$, the four centres c_i lying on one line. On the tori there exists $\vartheta_\square \in (0,1]$ with the following property. Let $\vartheta \le \vartheta_\square$ and $g \le \gamma_W(\vartheta)$. Then there are constants $0 < c_\square \le C_\square$, which may depend on g and ϑ, such that the bound below holds whenever $K - s(\mathsf{d}) \ge n_4$, where $n_4 \ge n_0$ is an integer, d is smaller than a threshold, which may depend on ℓ, and the quadruple belongs to $\mathfrak{T}_{T_\ell}^{(4)}(\mathsf{d};\vartheta)$:

$$c_\square (N^2-1)\, g(\mathsf{d})^8 \prod_i \|f_i\|_\infty \le S_{4;K,\ell,\beta}^T(f_1,\dots,f_4) \le C_\square (N^2-1)\, g(\mathsf{d})^8 \prod_i \|f_i\|_\infty :$$

on these quadruples the connected four-point function is therefore strictly positive. On rescaled quadruples it decreases as the fourth power of the reciprocal of the logarithm of the scale.

(d) [uniqueness, $g \le \gamma_U$; 8]. Take $(\beta_K^\natural)$ and $S^{\natural,\ell}$ as in (a)(i). (1) Let ϱ be large and $g \le \gamma_{U'}$. Then the first-passage windows lie close to the tuned couplings: $\mathcal{W}_K(g) \subset [\beta_K^\natural - \frac{1}{4}\varrho,\ \beta_K^\natural + \frac{1}{4}\varrho]$ for each K. (2) For $|\beta - \beta_K^\natural| \le \frac{1}{4}\varrho$ the reference runs of length K are well defined, and $\beta \mapsto x_K^\natural(\beta)$ is bi-Lipschitz uniformly in K. As $K \to \infty$, the functions $t \mapsto x_K^\natural(\beta_K^\natural - t)$ converge uniformly on $|t| \le \frac{1}{4}\varrho$. The limit function has a compact interval I as its range, and this interval contains $[g^{-2} - c'\varrho, g^{-2} + c'\varrho]$. (3) For each torus of the family, one has a family $(S^{(\ell,x)})_{x\in I}$. Each of its members is a family of connected functions, defined on tuples of Lipschitz functions whose supports are pairwise disjoint. It satisfies $S^{(\ell,g^{-2})} = S^{\natural,\ell}$, and $x \mapsto S_n^{(\ell,x)}$ is continuous on I. Moreover, whenever $K_j \to \infty$, $|\beta_j - \beta_{K_j}^\natural| \le \frac{1}{4}\varrho$ and $x_{K_j}^\natural(\beta_j) \to x$, it follows that $x \in I$ and $S^T_{n;K_j,\ell,\beta_j} \to S_n^{(\ell,x)}$, with no passage to a subsequence.

Consequently, take two sequences of bare data close to the tuned couplings whose renormalised couplings converge to the same value. They have the same limit on T_ℓ. Under the conditions of (1) this applies to every sequence of first-passage data, and every limit of (a)(ii) is equal to some $S^{(\ell,x)}$. On each torus, the renormalised coupling by itself determines the limit theory.

(e) [rotations; 9]. Let $\omega \in \mathfrak{so}(4)$, and let χ be a cutoff whose support lies in $(-\ell/2+2a_K, \ell/2-2a_K)^4$. Consider the rotational Ward identity satisfied by $\mu_{K,\ell,\beta}$ with respect to the deformation field $\chi(y)\,\omega y$. The shell flux is the boundary term of this identity:

$$\mathfrak{B}_{\omega,\chi} = \beta \sum_p a_p\,(\nabla^{\mathrm{s}}_\omega \chi)(x(p)) - \beta \sum_p \mathfrak{h}_{\omega,\chi}(p) - \tfrac{3}{2} \sum_p a_p \sum_{\rho\in\{\mu,\nu\}} (\omega x(p))_\rho\,(\nabla^{\mathrm{c}}_\rho \chi)(x(p)),$$

where $\mu < \nu$ are the two directions spanning p and $\nabla^{\mathrm{s}}_\omega, \nabla^{\mathrm{c}}_\rho$ are lattice differences, and where

$$\mathfrak{h}_{\omega,\chi}(p) = a_K^{-1} \sum_{b\subset\partial p} \sum_\rho [\chi(x_b) - \chi(x(p))]\,(\omega x_b)_\rho\,\mathfrak{d}_{b,p,\rho}$$

, the $\mathfrak{d}_{b,p,\rho}$ being bounded, gauge-invariant local densities. Each term contains a difference of χ. Therefore $\mathfrak{B}_{\omega,\mathcal{R}} := \mathfrak{B}_{\omega,\chi_\mathcal{R}}$ is supported on the shell $\{\mathcal{R} - a \le |x(p)| \le 2\mathcal{R} + a\}$, $a = a_K$, which surrounds the shell $\mathcal{R} \le |y| \le 2\mathcal{R}$ where $\chi_\mathcal{R}$ falls from 1 to 0. A limit of (a)(iii) is O(4)-invariant if and only if the following holds for one $\omega \ne 0$ (equivalently, for all ω), for all $n \ge 2$ and for all separated tuples with smooth entries:

$$\liminf_{\mathcal{R}\to\infty} \limsup_{j\to\infty} \left|\langle \mathfrak{B}_{\omega,\mathcal{R}}; \mathcal{O}(f_1); \dots; \mathcal{O}(f_n)\rangle_{K_j,\ell_j,\beta_j}\right| = 0.$$

In that case $\mathcal{H}(S)$ carries a strongly continuous unitary representation of O(3). This representation leaves Ω fixed, commutes with H, and extends the representation of W_3. This equivalence singles out the one infrared input, and for volume-first limits Theorem 0' settles it.

What the equivalence uses. When $\chi = 1$ near the supports, the identity takes the following form (Theorem 9.3 [to be proved; Package G], in (9a)):

$$\sum_{i=1}^n \big\langle \mathcal{O}(f_1); \dots; \mathcal{O}(\nabla^{\mathrm{s}}_\omega f_i); \dots; \mathcal{O}(f_n)\big\rangle_{K,\ell,\beta} = \mathfrak{U}_{\omega,\chi}(f_1, \dots, f_n)$$
$$+ \big\langle \mathfrak{B}_{\omega,\chi}; \mathcal{O}(f_1); \dots; \mathcal{O}(f_n)\big\rangle_{K,\ell,\beta} :$$

besides the flux, it contains a bulk term $\mathfrak{U}_{\omega,\chi}$. This term is the correlation between the $\mathcal{O}(f_i)$ and lattice artefacts of dimension six, distributed over $\operatorname{supp}\chi$. Once $\mathfrak{U}_{\omega,\chi_\mathcal{R}} \to 0$ along every defining

sequence, the rotation defect of S, namely the derivative $\frac{d}{dt} S_n(e^{t\omega} f_1, \dots, e^{t\omega} f_n)|_{t=0}$, equals minus the limit of the correlation between the flux and the $\mathcal{O}(f_i)$, and clause (e) follows.

This vanishing (Theorem 9.1 [to be proved; Package G]) is a consequence of three propositions about an insertion that does not have the form of the action, followed to first order through the K steps (Propositions 9.7 [to be proved; Package G], 9.8, 9.9), together with a fourth. The first three yield the vanishing for slowly growing volumes as well as on a fixed torus. The fourth is a moment bound in which several insertions occur simultaneously (Proposition 9.11 [to be proved; Package G]); by means of it, the transfer in the volume (Proposition 9.10 [to be proved; Package G]) extends the vanishing to every sequence with $\ell_j \to \infty$. All of these propositions are in (9b). Along either route, the vanishing depends on a single number.

One marginal operator remains in $\mathfrak{U}$: a traceless diagonal anisotropy of Wilson's weight. Its coefficient is of the size $g_0^{-1} \approx K^{1/2}$, and it disappears only through its own decay K^{-c_1} along the flow. What is needed is $c_1 > \frac{1}{2}$.

This number is expressed through one-loop data. Put the weight $1 + \mathsf{e}_{\pi(p)}$ on each a_p, where $\pi(p)$ is the coordinate plane of p. Differentiating the one-loop coefficient of step l plane by plane at $\mathsf{e} = 0$ gives a derivative that commutes with W_4. Let $\mathsf{m}(l)$ denote the eigenvalue of this derivative on the traceless diagonal weights $\mathsf{e}_{\{\mu,\nu\}} = \mathsf{e}_\mu + \mathsf{e}_\nu$, $\sum_\mu \mathsf{e}_\mu = 0$. Then $c_1 \in (\frac{1}{2}, 1]$ is an exponent such that $\sum_{i<l\le j} \mathsf{m}(l) \ge c_1 c_0 (j-i) - C$. The one-loop law in this channel, $|\sum_{l\le k} \mathsf{m}(l) - c_0 k| \le C$, gives $c_1 = 1$.

The law comes from a telescoping sum. By the law of total covariance, the sum of the one-loop responses of the steps $l \le k$ is a single trace on the bare lattice. The logarithmic part of this trace is $\int_1^{L^{2k}} d\sigma/\sigma = 2k \log L$, in proper time σ, multiplied by the heat coefficient on a constant abelian background. It equals $k c_0$, with the same c_0 as the coupling itself. In the continuum, an anisotropy of the weight amounts, to first order, to a change of coordinates, and it does not run (9b). Rotation invariance in Theorem 0' (1) is obtained along a different route, without any Ward identity (10b), and does not depend on c_1.

Theorem 0' [10] (Let

$$g \le \min\{\gamma_W(\vartheta), \gamma_W(\vartheta_0)\}$$

, and let S be a volume-first limit with defining sequence (K_j, β_j, ν_j).) (1) Every ν_j is a periodic Gibbs state and is invariant under all translations of the lattice; moreover S is a limit of (a)(iii). In addition to (b) and (c)(4), it satisfies $H \ge |P_j|$ for $j = 1, 2, 3$, is O(4)-invariant, and is reflection positive at the diagonal mirrors. (2) Its canonical decomposition exists and is unique up to isomorphism. Here an isomorphism is a homeomorphism of the spaces X that carries λ and the σ_ξ along with it. Let P_0 be the projection onto $\ker H$. The map $P_0[F] \mapsto (\xi \mapsto \sigma_\xi(F))$ extends to a unitary map $\ker H \to L^2(\mathsf{X}, \lambda)$ with $\Omega \mapsto 1$, so that $\dim \ker H = \#\mathsf{X}$ when X is finite, while $\ker H$ is infinite-dimensional when X is infinite. The components are permuted, and λ is preserved, by the isometries that leave S invariant (translations, W_4, rotations). (3) Each σ_ξ has the following properties. (i) It has the same invariances and positivity properties, and it is symmetric and regular. (ii) Its one-point function is $S_1^{\sigma_\xi}(f) = \mathfrak{c}_\xi \int f$, where $\mathfrak{c}_\xi$ is real and continuous in ξ and $\int \mathfrak{c}_\xi \, \lambda(d\xi) = 0$. With the same constants, it satisfies the growth condition and the bounds (c)(2); it also satisfies the law (c)(4) in its form on $\mathbb{R}^4$. (iii) On $\mathcal{H}(\sigma_\xi)$ it has the operators of (b)(v) and a unique vacuum, $\ker H_{\sigma_\xi} = \mathbb{C}\Omega_{\sigma_\xi}$. Its joint spectrum lies in the closed forward cone $\{E \ge |q|\}$. It also has a strongly continuous unitary representation of O(3) that leaves Ω_{σ_ξ} fixed and commutes with H_{σ_ξ}. (iv) It has the cluster property:

$$|\sigma_\xi(X \cdot \tau_v Y) - \sigma_\xi(X)\, \sigma_\xi(Y)| \le C(X, Y)\, |v|^{-2}$$
$$(|v| \ge r(X, Y)),$$

where $C(X, Y)$ and $r(X, Y)$ are the same for all ξ and for all volume-first limits at g. (4) Under the analytic hypotheses of this theorem, σ_ξ has the stated growth, invariance under $\mathbb{R}^4 \rtimes \mathrm{O}(4)$, reflection positivity, symmetry and clustering. Applying the external reconstruction theorem with the quantitative hypotheses of Section 54 then identifies these Schwinger functions with those of a Wightman theory. This theory has a single Hermitian scalar field on a separable Hilbert space and a unique vacuum. It is covariant under the proper orthochronous Poincaré group, its spectrum lies in the closed forward cone, and it is local.

16. Block averages and gauge fixing

Proof outline. The supporting analytic results used in this section, including the obligations that we must still prove, are listed under A and C in the dependency catalogue. Their hypotheses and precise forms are retained at the points where the argument uses them.

(1a) Block averages, gauge fixing, and the renormalisation group of this construction. Setting and objects. *Requirements on an average.* We work in level-k units, with L odd and centred blocks $B(y) = \{x : |x - y|_\infty \le \frac{1}{2}(L-1)\}$. We put $\varpi(W) = \sup_p |\partial W(p) - \mathbf{1}|$, and ϖ_c denotes the same supremum taken over the double block $B(c_-) \cup B(c_+)$. For kernels we use the norms $\|K\|_{\mathrm{row}} = \sup_c \sum_b |K(c, b)|$ and $\|K\|_{\mathrm{col}} = \sup_b \sum_c |K(c, b)|$. Elements of the group cannot be added, and transports joining different pairs of points cannot even be compared. A block average $\mathcal{M} : \mathcal{V}_k \to \mathcal{V}_{k+1}$ is required to have the following properties: gauge covariance; locality; linearity at first order; analyticity near flat fields, on G^c as well; symmetry under the isometries of the coarse lattice; and Borel measurability on the whole of $\mathcal{V}_k$, because $\mathbf{T}$ integrates over all configurations. For the mean of group elements Bałaban gives axioms (109 (1987), (0.5)–(0.9)). Such a mean is any analytic G^c-valued function M on families of small diameter that satisfies

$$M(\{\mathbf{U}_j^{-1}\}) = M(\{\mathbf{U}_j\})^{-1},$$
$$M(\{u\mathbf{U}_j v\}) = u\, M(\{\mathbf{U}_j\})\, v,$$
$$M(\pi\{\mathbf{U}_j\}) = M(\{\mathbf{U}_j\}),$$

$$\tfrac{1}{i}\log M(\{e^{iA_j}\}) = \tfrac{1}{n}\sum_j A_j + (\text{higher orders}),$$
$$\mathbf{U}_j \in G \text{ for all } j \;\Rightarrow\; M(\{\mathbf{U}_j\}) \in G.$$

According to Bałaban, the results of his 1987 paper and of the earlier papers hold for every mean of this kind. For the map $\mathcal{M}$, the second of the two symmetric block averages that he introduces, he states that every result of the paper on averaging operations (Comm. Math. Phys. 98) holds, with proofs that are for the most part unchanged. This statement concerns that single paper; the broader sentence, which refers to "this, and previous papers", is the one about means. Of the two symmetric averages together he states that they are equally good, and that many other averages can also be used. He also states that the block average admits an axiomatisation as well. Already in 1985 he states that the analysis of the paper on averaging operations, together with its propositions, extends easily to other averages that satisfy the two requirements set out in its introduction, namely gauge covariance and agreement with the linear block average at first order. No proof accompanies any of these statements. For the theorems of 1984–85 on averaging, gauge fixing, propagators and backgrounds, the proofs are carried out for an average that is not symmetric. Proposition 1a.1.13 states which properties of an average the proofs use, and it compares the linear parts (1a.1).

The mean, the contours, the map. The *mean* $\mathrm{av}(V_\bullet)$ (Federbush) is the solution, close to the family, of $\sum_i \log(V_i V^{-1}) = 0$. By $\mathcal{G}(y,x)$ we denote the family of shortest paths from y to $x \in B(y)$ that complete the displacement in one direction before passing to the next. There is one such path for each ordering of the directions, so the family has at most 4! members and does not contain every shortest path. The *comb* $\Gamma_{y,x}$ is the member with the order $4,3,2,1$. The averaged contour variable is $U(y,x) := \mathrm{av}(U(\Gamma))_{\Gamma\in\mathcal{G}(y,x)}$, and we set $U(x,y) = U(y,x)^{-1}$ and $U(y,y) = \mathbf{1}$. With $x' = x + (c_+ - c_-)$, one defines

$$W_x = U(c_-, x)\, U([x,x'])\, U(x', c_+)\, U[c]^{-1},$$
$$\mathcal{M}(U)(c) = \exp\Big(L^{-4}\sum_{x\in B(c_-)} \log W_x\Big) U[c]$$
$$\text{on } \mathrm{Dom}(c) = \{\varpi_c < cL^{-2}\} :$$

the W_x are closed transports based at c_- and lying near $\mathbf{1}$, so their logarithms can be added. Where $U(y,x) = \mathbf{1}$, only straight segments are left, so the form is adapted to the block axial gauge. Off $\mathrm{Dom}(c)$ the map is given by two cruder rules, both covariant. When none of the elements $Y_{x,\Gamma,\Gamma'} = U(\Gamma)U([x,x'])U(\Gamma')^{-1}U[c]^{-1}$ equals $-\mathbf{1}$, one replaces $\log W_x$ by the arithmetic mean, over (Γ,Γ'), of the intrinsic logarithms of the $Y_{x,\Gamma,\Gamma'}$; this is Bałaban's first symmetrised average. In the remaining case one sets $\mathcal{M}(U)(c) = U[c]$. This third rule is required because Bałaban defines his average on all configurations by means of the spectral logarithm, which satisfies $\log(-\mathbf{1}) = i\pi\mathbf{1} \notin i\mathfrak{g}$ and so leads out of SU(2) (1a.1). By $\mathcal{M}^k$ we mean the iterate only, and not a mean taken in a single stroke over B^k. In Bałaban's paper of 1987 the lattices consist of centres of cubes and have coordinates in $a_j(\mathbb{Z}+\frac{1}{2})$. The single translation $x \mapsto x + \frac{1}{2}L^m \sum_\mu e_\mu$ of T_ℓ,

$\ell = 2L^m$, maps them onto the $\Lambda_{j,\ell}$ at every level simultaneously, since $\frac{1}{2}L^m = \frac{1}{2}L^{m+j}a_j$ and L^{m+j} is odd. This is the second place where the oddness of L is used, the first being the centring of the blocks (1a.1). Bałaban also remarks that all the results were obtained first for the two-dimensional chiral models, in which the fields sit on the sites and the action is $\sum_b[1 - \frac{1}{2}\operatorname{Re}\operatorname{tr} U(b_-)U(b_+)^{-1}]$. In those models block averages are means of group elements, and neither contours nor a gauge group enter.

Linear averages of bond functions a, which enter in the form $e^{i\mathsf{a}}W$ without any length factor, are defined as follows, where $\bar{\mathsf{a}}(y,x)$ denotes the arithmetic mean of $\mathsf{a}(\Gamma)$ over $\mathcal{G}(y,x)$:

$$(\mathsf{Q}^{\mathrm{s}}\mathsf{a})(c) = L^{-4}\sum_{x\in B(c_-)} \mathsf{a}([x,x']),$$

$$\sigma(y) = L^{-4}\sum_{x\in B(y)} \bar{\mathsf{a}}(y,x),$$

$$(\mathsf{Q}^{\mathrm{pl}}\mathsf{a})(c) = \sigma(c_-) + (\mathsf{Q}^{\mathrm{s}}\mathsf{a})(c) - \sigma(c_+),$$

$$(q\lambda)(y) = L^{-4}\sum_{x\in B(y)} \lambda(x);$$

here Q^{pl} is the derivative of $\mathcal{M}$ at $\mathbf{1}$. The operator $\mathsf{S}(W_0)$ is Q^{s} formed with parallel transport to c_-, namely

$$(\mathsf{S}(W_0)\mathsf{a})(c) = L^{-4}\sum_{x\in B(c_-)}\sum_{b\in[x,x']} \operatorname{Ad}\big(W_0(c_-,x)W_0([x,b_-])\big)\mathsf{a}(b)$$

.

The straightened average of $\mathbf{W} = e^{i\mathsf{a}}W_0$ relative to W_0 and its *block factor* u (written $\mathsf{n}_{\mathbf{W}}$ in 1a.3) are given by

$$\mathsf{u}(y) = \exp\Big(L^{-4}\sum_{x\in B(y)} \log\big[\mathbf{W}(y,x)W_0(y,x)^{-1}\big]\Big),$$

$$e^{i\mathcal{Q}^1(W_0,\mathsf{a})(c)}\mathcal{M}(W_0)(c) = \mathsf{u}(c_-)^{-1}\mathcal{M}(\mathbf{W})(c)\,\mathsf{u}(c_+).$$

The right-hand side equals $\mathcal{M}(\mathbf{W}^v)$, where v is the gauge transformation with $\mathbf{W}^v(y,x) = W_0(y,x)$ (the block axial gauge relative to W_0) and with trivial average $\bar{R}_{W_0}v = \mathbf{1}$. Together these two conditions imply $v = \mathsf{u}^{-1}$ at the coarse sites. Bałaban gives two reasons for the straightening. First, the variational problems of (1b) involve composing the average with the passage from the axial gauge to the Landau gauge. Second, only the straight linear average admits the explicit representations of the propagators (1c); this average is compatible with composition, has an explicit Fourier symbol and does not change under $q\lambda = 0$. The change of gauge has to preserve this form, and the non-linear counterpart of $q\lambda = 0$ is $\bar{R}_{W_0}v = \mathbf{1}$. A third reason appears in the proof of Theorem 1a.1.6.

Order k; the maps denoted by Q. Along the sequence $W_{0,n} = \mathcal{M}^n U_0$ we set $\mathsf{a}_0 = \eta_k A$, $\mathsf{a}_{n+1} = \mathcal{Q}^1(W_{0,n},\mathsf{a}_n)$, $v_0 = \mathbf{1}$, $v_{n+1} = v_n\,\mathsf{u}(W_{0,n},\mathsf{a}_n)$, and define

$$\mathcal{Q}_k^\sharp(U_0;A) := \mathsf{a}_k = Q_k^\sharp(U_0)A + C_k(U_0,A),$$

$$\mathcal{Q}_k(U_0;A) = \tfrac{1}{i}\log\big[\mathcal{M}^k(e^{i\eta_k A}U_0)\mathcal{M}^k(U_0)^{-1}\big] = Q_k(U_0)A + C_k^{\mathrm{pl}}(U_0,A).$$

A sharp indicates the straightened objects, which are the ones that enter the propagators; Bałaban's $Q_k(U_0, \eta A)$ is the same object built with combs. Without the sharp, $\mathcal{Q}_k$ and Q_k refer to the plain average, whose remainder is denoted C_k^{pl}, while C_k denotes the remainder of the straightened average. The product $v_k = v_k[A]$ of the k block factors is the *node factor*, and $\dot v_k$ denotes the linear part of $\frac{1}{i}\log v_k$. The letter Q is used for maps of four kinds (1a.1). (1) The non-linear constraint map $\mathcal{Q}_k$ of the plain iterate $\mathcal{M}^k$, and next to it its straightened form $\mathcal{Q}_k^\sharp$. (2) $Q_k(U) = \partial_A \mathcal{Q}_k(U;A)|_{A=0}$, together with $Q = Q_1(\mathbf{1})$ and $Q_V = Q_1(V)$; this is the operator in the delta function of a step. (3) $Q_k^\sharp(U)$, the operator that enters the propagators and the Landau gauge. Its leading term is the *straight* average with parallel transport, $Q_k^{\mathrm{st}}(U) = \eta_k \mathsf{S}(W_{0,k-1})\cdots\mathsf{S}(W_{0,0})$. The two differ by $O(\alpha_0)$ in row norm, and they are equal at $U = \mathbf{1}$. There, after the translation of Proposition 1a.1.13, each of them is Bałaban's Q_k of 1984. (4) $Q_k'(U)$, which averages site functions. In a background, the linear average in Bałaban's papers (99 (1985) 389, (3.15)) is $Q_j^{\mathrm{B}}(U) := Q(\bar U^{j-1})\cdots Q(\bar U)Q(U)$. It is the linear part of his gauge-corrected average built with combs, and each of its factors equals the straight covariant average up to $O(L^2\alpha_0)$. Theorem 1a.1.6 and Proposition 1a.1.13 provide two comparisons, that of (2) with (3) and that of Q_k^{B} with $Q_k^\sharp$, and (1c) relies on both.

Gauge transformations are averaged by means of the same contours,

$$(\bar R_W v)(y) = v(y)\exp\Big(L^{-4}\sum_{x\in B(y)} \log\big[v(y)^{-1}\,\mathrm{Ad}(W(y,x))v(x)\big]\Big),$$

and the iterates of this map give $\bar R_U^{(j)}$. The linear part, $Q_j'(U)\lambda$, is the mean over $B^j(y)$ of λ after transport to y.

Isometries on 1*-forms.* An isometry r acts by $(\mathsf{r}U)(b) = U(\mathsf{r}^{-1}b)$. Bond functions are specified on the positively oriented bonds, whereas a reflection reverses orientation. Since $(e^{i\mathsf{a}}W_0)(\bar b) = \exp\big(-i\,\mathrm{Ad}(W_0(b)^{-1})\mathsf{a}(b)\big)W_0(\bar b)$, the action relative to W_0 must include a twist:

$$(\mathsf{r}_{W_0}\mathsf{a})(b') = \mathsf{a}(b) \ \text{ if } \mathsf{r}^{-1}b' = b,$$
$$(\mathsf{r}_{W_0}\mathsf{a})(b') = -\,\mathrm{Ad}\big(W_0(b)^{-1}\big)\mathsf{a}(b) \ \text{ if } \mathsf{r}^{-1}b' = \bar b;$$

on pairs the action is $\mathsf{r}(\mathbf{U},\mathbf{J}) = (\mathsf{r}\mathbf{U}, \mathsf{r}_{\mathbf{U}}\mathbf{J})$ (1a.1).

Lift, regions, determining sets. The *lift* is defined by $(Q_k^{s*}V)(b) = V(c)$ when $b_\pm \in B^k(c_\pm)$; on every bond contained in a single block of order k its value is $\mathbf{1}$. For a bond function B on $\mathbb{T}^{(k)}$, its linear lift is the 1-form that takes the value $\eta_k^{-1}B(c)$ on these crossing bonds and the value 0 on bonds lying within a block. The value $\mathcal{M}^q(W)(c)$, and also which rule applies, depend on W only in $B^q(c_-)\cup B^q(c_+)$. Averages are therefore defined for a field given on part of a lattice; they remain covariant and obey the semigroup law $\mathcal{M}^{q+1}W(c') = \mathcal{M}(\mathcal{M}^qW)(c')$. Understood in this way, $M_{\mathbf{B}}(U) := (\mathcal{M}^jU|_{\Gamma_j})_j$ is defined on configurations given on a region, and the problems that define the localised backgrounds $U_{k,\square}$ are set up near a cube without reference to the torus (1a.1).

The transformation. Here δ denotes the Dirac distribution at $\mathbf{1}\in G$ with respect to Haar measure. It is a class function, and it is the limit $\mathfrak{a}\to\infty$ of the normalised weights $\propto e^{-\mathfrak{a}(1-\frac12\,\mathrm{Re\,tr}\,W)}$. The transformation sends a complex Borel measure ν to its image $\mathbf{T}\nu = \mathcal{M}_*\nu$, which is written with the kernel $\delta(\mathcal{M}(U)V^{-1})$. Thus a coarse configuration receives the total weight of the fine configurations whose average it is. Wilson and Kadanoff introduced the block-spin transformations $(\mathbf{T}_t\rho)(V) = \int dU\, t(V,U)\rho(U)$, with $t\ge 0$, $\int dV\, t(V,U) = 1$ and $t(V^{u|_{\mathrm{coarse}}}, U^u) = t(V,U)$; the transformation here is the member of this family with a hard constraint. Decompositions of unity and the factor $\mathfrak{G}_k$ are inserted into the integral. They are built with the auxiliary choices

$\varepsilon_k, \delta_k, \alpha_{i,k}, \check{R}_k$ and with the weight of $\mathfrak{G}_k$. All of these depend on a single number t, which is x_k in the run and θ_k in a reference run. Only $\check{R}_k$ is formed with the minimum of these numbers taken over all levels $i \le k$, which makes it non-increasing in k. A history h records outcomes, and these yield the term τ_h, or the term $\tau_h^{\mathfrak{w}}$ when ρ_0 is multiplied by a bounded complex weight $\mathfrak{w}(U)$. In addition to χ_k and $\chi^{(k)}$ there is the approximate fluctuation test $\chi'_k(\square) = \mathbf{1}\{|V_k(V_{k,\square}^{\mathrm{bg}})^{-1} - \mathbf{1}| < 2\delta_k \text{ near } \square\}$, which is built with the local averaged background $V_{k,\square}^{\mathrm{bg}} = \mathcal{M}^k U_{k+1,\square}$. Braces denote a difference between backgrounds: $\{F\} := F(U_k(V_k)) - F(U_{k+1})$.

The block algebra. The block field of depth s, $V^{(s)} := \mathcal{M}^{K-s}U$, is defined on $\Lambda_{s,\ell}$, a lattice that does not depend on the cutoff $K \ge s$. The *block algebra* $\mathfrak{A}$ is the set of gauge-invariant functions F, bounded and continuous, that depend on only finitely many block variables $V^{(s)}(b)$. The composite $F \circ \mathcal{M}^{K-s}$ is defined for every $K \ge s$, so these observables exist at all cutoffs simultaneously (1a.2). The block algebra is a notion separate from Theorem 0 and Theorem 0', whose observables are the smeared curvature evaluated at the bare scale; neither theorem refers to $\mathfrak{A}$.

Two operations (1a.2). Let $\Delta = \mathsf{C}^{\mathsf{T}}\Delta^{(k)}\mathsf{C}$, let $z^{(k)}$ be a product of Jacobians, and let $\chi(S_{k+1})$ denote the tests on $\Omega_{k+1} \setminus \Lambda_{k+1}$. To obtain $\zeta(\Omega_{k+1}^c)$, form for each outcome that determines this same Ω_{k+1} the product of the tests and gauge-fixing factors localised in $\Omega_k \setminus \Omega_{k+1}$, and add these products over all such outcomes. The one-step operation collects everything that step $k+1$ does not carry out:

$$\mathbf{T}^{(k)}(Z_{k+1})F = \int dV_k\big|_{\Omega_{k+1}^c} \delta(\mathcal{M}(V_k)V_{k+1}^{-1})\,\zeta(\Omega_{k+1}^c)\; z^{(k)} \int d\mathcal{A}_k^{\mathrm{s}}\big|_{\Omega_{k+1}\setminus\Lambda_{k+1}} \chi(S_{k+1})\, e^{-\frac{1}{2}\langle \mathcal{A}_k^{\mathrm{s}}, [\Delta - \Delta C^{(k)}(\Lambda_{k+1})\Delta]\mathcal{A}_k^{\mathrm{s}}\rangle} F,$$

where, more precisely, the Gaussian exponent of the operation is a part of this quadratic form, namely a part that couples no two components of $\Omega_{k+1}\setminus\Lambda_{k+1}$; the remainder of the form is assigned to the boundary terms $B^{(k+1)}$ (2b). As long as $\mathbf{R}$ does not intervene, $\mathbf{T}'_{k+1} = \mathbf{T}^{(k)}(Z_{k+1})\mathbf{T}'_k$. For $x \in \mathbb{C}$ and a function S of the bare fields, we define the pure small-field integral and then the regular part of step $k+1$ by

$$(\mathbf{T}_k^{\mathrm{sf}}[\mathsf{t}]F)(W) = \int dV\, \delta(\mathcal{M}(V)W^{-1})\, \mathfrak{G}_k(V)\, \chi^{(k)}(T_\ell)\, F(V),$$

$$\mathfrak{E}_{k+1}[\mathsf{t}; x; S](W) = \log\big(\mathbf{T}_k^{\mathrm{sf}}[\mathsf{t}]\, e^{-xA(U_k(\cdot))+S(U_k(\cdot))}\big)(W) + xA(U_{k+1}(W)) - S(U_{k+1}(W)).$$

Regular configurations. We consider sequences $\Omega_0 \supset \cdots \supset \Omega_k$, with $\Omega_{k+1} = \emptyset$, in which each Ω_j is a union of M_1-cubes of level j and $\mathrm{dist}(\Omega_j^c, \Omega_{j+1}) \ge CM_1L^j\eta_k$. The small-field case is the constant sequence $\Omega_j = T_\ell$. Layer j is the set $\Omega_j \setminus \Omega_{j+1}$. It has the local scale $\xi_j = L^j\eta_k$ and the tops $\mathbb{Y}_j = \Lambda_j(\mathbf{B})$, which are its sites in $\mathbb{T}^{(j)}$. Its pillars are the $B^j(y)$, $y \in \mathbb{Y}_j$, and over all $j \le k$ the pillars partition the bare sites of Ω_0. We define

$$\mathfrak{U}_k(\Omega_\bullet, \alpha) = \big\{U :\, |\partial U - \mathbf{1}| < \alpha\,\eta_k^2\xi_j^{-2},\; |J(U)| < \alpha\,\xi_j^{-3} \text{ on layer } j\big\},$$

$$|f|_{(-s)} = \sup_j \sup_{\text{layer } j} \xi_j^s |f|,$$

together with

$$\|A\|_1 = \max\{|A|_{(-1)}, |\nabla_{U_0}A|_{(-2)}\}$$

and

$$\|\lambda\|_0 = \max\{\sup|\lambda|, |D_{U_0}\lambda|_{(-1)}\}$$

. Both conditions are invariant under gauge transformations. Combined with the Bianchi identity, the second condition is an elliptic system for the field strength F_U, and all bounds on derivatives come from this system; minimisers satisfy the second condition. The complex neighbourhood of radius α_1 is the set of $\mathbf{U} = e^{i\eta_k A'}U$ with U real and regular and $\|A'\|_1 < \alpha_1$. Since $\|A'\|_1$ bounds only first differences, it does not control the current of such a $\mathbf{U}$, which is a second difference. For this reason the current enters the domains $U_j^c(X, \alpha_0, \alpha_1)$ as an independent variable $\mathbf{J}$ with $|\mathbf{J}| < \alpha_0$. The real centre U is required to have small curvature and small local potentials, but not a small current. The backgrounds of the averages of $\mathbf{U}$ are required to satisfy what the condition on the current provided, and U_j^c is the set of all such pairs. This set is not saturated with respect to the G^c-valued u, which act by

$$(\mathbf{U}, \mathbf{J}) \to (\mathbf{U}^u, \mathrm{Ad}(u)\mathbf{J})$$

. The extent to which a term is invariant under these transformations is given in 1a.3.10 and in (1d).

Gauges. Because G is compact, the gauge is not fixed in order to obtain finiteness. The reason for fixing it is that the integrand of $\mathbf{T}$ does not change under the block group $\mathcal{G}_{\mathrm{bl}} = \{u : u = \mathbf{1} \text{ on } \mathbb{T}^{(k+1)}\}$. Consequently, in each block there are $3(L^4 - 1)$ directions $D_V\lambda$ along which the Hessian at the minimum vanishes, and there is then no Gaussian measure, no covariance and no decay. Similarly, the residual group $\mathcal{G}^{\mathrm{res}}$ leaves the constraints $\mathcal{M}^j U = V_j$ on Γ_j unchanged. This group consists of the bare u that equal $\mathbf{1}$ at the *nodes*, which are the end points of the bonds of $\mathbf{B}[\Omega_\bullet]$. Hence a background is an orbit and not a point. No gauge has all of the following properties at once: locality (on a torus, no global gauge turns small curvature into a small potential), compatibility with the averages, covariance under the remaining gauge group, symmetry under the hypercubic group W_4, and regularity uniform in k. Axial gauges are algebraic and strictly local and have Jacobian one; they appear in the integrals and serve to select representatives. *Box axial:* the condition $V = \mathbf{1}$ on the tree of combs that issues from the base point y_P of a box. *Block axial:* the condition $V(y, x) = \mathbf{1}$ on the averaged contour variables. It is imposed softly on a set P'' of blocks, at those pairs (y, x) for which the mean $V(y, x)$ is defined (no factor is inserted at the other pairs), by means of

$$\mathfrak{G}_k(V; P'') = \prod_{y \in P''} \prod_{x \in B(y) \setminus y} z_{\mathfrak{G}}^{-1}\, \mathbf{1}\{|V(y,x) - \mathbf{1}| < \varepsilon_k\}\, e^{-x_k(1 - \frac{1}{2}\operatorname{Re}\operatorname{tr} V(y,x))},$$

$$\mathrm{GF}_k(V) = \sum_{y,x} \left[1 - \tfrac{1}{2}\operatorname{Re}\operatorname{tr} V(y,x)\right] \geq 0.$$

Tower axial relative to U_0, written $\mathrm{Ax}(\Omega_\bullet, U_0)$: in every pillar and at every level $n < j$, the contour variables of $\mathcal{M}^n U$ coincide with those of $\mathcal{M}^n U_0$. When the tops are the only nodes, solving this condition from the top down selects one representative in each orbit of $\mathcal{G}^{\mathrm{res}}$. An example, for $U_0 = \mathbf{1}$, is the representative of the orbit $U_k(\mathbf{V})$ of the regular critical configurations of (1b). Suppose that an inward bond of layer $j - 1$ is retained as an additional constraint (see (1b), *Inward bonds*; by the Notation, §4, the determining set contains no such bond). Its end point then lies inside a pillar of layer j. That end point is a node but not a top, and it is called a *junction node*; in (1b) it is the seam node. The transport from the top to a junction node is invariant under $\mathcal{G}^{\mathrm{res}}$, so the condition cannot be achieved there. Leaving out these conditions gives the gauge $\mathrm{Ax}^\circ(\Omega_\bullet, U_0)$, which meets in exactly one point every orbit where the contour variables

are defined. Representatives are covariant. Given a gauge transformation v on the nodes, its lift is the bare transformation $\tilde{v}$ whose value at a bare site is the value of v at the first node reached when one ascends from that site through the centres of its blocks. The map $W \mapsto W^{\tilde{v}}$ then sends the constraint set of $\mathbf{V}$ onto the constraint set of $\mathbf{V}^v$; it also sends orbits to orbits and maps $\mathrm{Ax}^\circ(\Omega_\bullet, \mathbf{1})$ onto itself. Hence, for the representative taken at the reference $U_0 = \mathbf{1}$, $U_k(\mathbf{V}^v) = U_k(\mathbf{V})^{\tilde{v}}$. For nests of boxes whose lattices are of different levels (2b), the tree consists of shell trees, connected by one entrance bond in each box. Trees depend on the choice of a frame in W_4, and averaging over the $2^4\,4!$ frames restores the symmetry (1a.3).

The estimates are carried out in the elliptic Landau gauge, and this gauge comes with a determinant. We put $Q^\sharp = (Q_j^\sharp|_{\Gamma_j})_j$, with each component normalised to total weight $1 + O(\alpha_0)$, so that $\mathcal{Q}_j^\sharp(U_0; A) = \xi_j Q_j^\sharp A + C_j(A)$. Further, $Q' = (Q'_j|_{\mathbb{Y}_j})_j$, and its null space is denoted $N(Q')$. On layer j we set $\xi = \xi_j$ and $a = \xi_j^{-2}$. We also put $G' = (\Delta_{U_0} + Q'^* a Q')^{-1}$, and R is the orthogonal projection onto $\Delta_{U_0} N(Q')$. Relative to U_0, a *Landau pair* (A, u) of U is a pair satisfying

$$U^{u^{-1}} = e^{i\eta_k A} U_0,$$
$$R\, D^*_{U_0} A = 0,$$
$$\bar{R}^{(j)}_{U_0} u = \mathbf{1} \ \text{ on } \mathbb{Y}_j,$$

in which conditions on averages take the place of the condition $u = \mathbf{1}$ at the nodes, since that condition cannot be handled analytically. This has a cost. Transformations u of this kind do not form a group: $\bar{R}u_1 = \bar{R}u_2 = \mathbf{1}$ does not imply $\bar{R}(u_1 u_2) = \mathbf{1}$. The word "orbit" therefore has no meaning, and the set that such transformations generate depends on its base point. For this reason the operations come in a fixed order. First, the tower axial gauge Ax° selects a base point in each orbit of $\mathcal{G}^{\mathrm{res}}$. Next, the set is generated from that base point. Finally, the set is intersected with the slice. The average that belongs to this gauge is the *reduced average*

$$\mathcal{M}_{\mathrm{red}}(W; U_0) = \mathcal{M}(W)^{\mathsf{n}_W^{-1}} \mathcal{M}(U_0)^{-1} = e^{i\mathcal{Q}^1(U_0, \mathsf{a})}$$

; here n_W stands for the block factor $\mathsf{u}(U_0, \mathsf{a})$ taken at $W = e^{i\mathsf{a}} U_0$. The reduced average is the straightened average, under the name it bears in the theory of gauges. The contour variable is required to be the same in all four objects $\mathcal{M}$, n_W, $\bar{R}$ and Ax: it must be the mean in each of them, or the comb in each of them.

Statements. Lemma 1a.1.1 (the mean). Consider families $V_\bullet = (V_1, \dots, V_n)$ in G^c of small diameter $\operatorname{diam} V_\bullet := \max_{i,i'} |V_i V_{i'}^{-1} - \mathbf{1}|$, where the radii do not depend on n. For such families the mean near the family is unique. It is holomorphic, with bounds independent of n (by the contraction principle), and it is symmetric. On families in G it takes values in G, and there it is the only mean lying in G. Here a mean of $V_\bullet$ is an element V with $\max_i |V_i V^{-1} - \mathbf{1}| < 1$ and $\sum_i \log(V_i V^{-1}) = 0$; the logarithm is the series logarithm, which has a meaning because of the first condition. Next, $\operatorname{av}(T V_\bullet T') = T \operatorname{av}(V_\bullet) T'$ holds for $T, T' \in G$. It also holds for $T \in G^c$ as long as $\operatorname{cn}(T) \operatorname{diam} V_\bullet$ remains below the radius, where $\operatorname{cn}(T) := |T|\,|T^{-1}|$ is the condition number. The identity $\operatorname{av}(V_\bullet^{-1}) = \operatorname{av}(V_\bullet)^{-1}$ holds in G. It holds in G^c when $\operatorname{diam} V_\bullet$ is below half the radius and $\operatorname{cn}(V_i) \le 2$ for all i. Finally, $\frac{1}{i} \log \operatorname{av}(e^{iA_\bullet}) = \frac{1}{n} \sum_j A_j + O(\max_j |A_j|^2)$.

Lemma 1a.1.2 (small curvature: nearby transports, comb gauge). Consider a box that contains at most s sites along each direction and has plaquette size ϖ. Suppose the m steps of a path are rearranged so that the rearranged paths stay inside the box. Then the transport along the path changes by at most $(1 + \varpi)^{m(m-1)/2} - 1$. Moreover, with $h(x) = W(\Gamma_{y,x})$,

$$|W^h(b)-\mathbf{1}|\le |b_- - y|_1\,\varpi,$$
$$W^h = e^{i\mathsf{a}^0},$$
$$|\mathsf{a}^0|\le Cs\varpi.$$

Consequently the contour families on $\mathrm{Dom}(c)$ are of diameter $\le CL^2\varpi_c$, and one has $\mathcal{M}(W)(c)=e^{i\mathsf{Y}}W[c]$ where $|Y|\le CL^2\varpi_c$.

Proposition 1a.1.3 (the averaging map). Let U, u be arbitrary, and let r be any isometry that preserves $\mathbb{T}^{(k+1)}$ (translations, W_4, coordinate and diagonal reflections). Then

$$\mathcal{M}(U^u)=\mathcal{M}(U)^{u|_{\mathrm{coarse}}},$$
$$\mathcal{M}(\mathsf{r}U)=\mathsf{r}\,\mathcal{M}(U);$$

and $\mathcal{M}^n(U)(c)$ depends on U only in $B^n(c_-)\cup B^n(c_+)$. The averaging map has four further properties.

- It is Borel. Each of the three rules applies on a Borel set, and $\mathcal{M}$ is continuous on each of these sets but not across their boundaries. Outside $\mathrm{Dom}(c)$ no further regularity is required.
- It is holomorphic at $e^{i\mathsf{a}}W_0$ for $\varpi_c(W_0), |\mathsf{a}|<c_L$.
- It is onto: $\mathcal{M}^k(Q_k^{s*}V)=V$.
- On small fields of one colour it is linear: $\mathcal{M}(e^{ifT})=e^{i(\mathsf{Q}^{\mathrm{pl}}f)T}$ whenever $T\in\mathfrak{g}$, $|T|=1$, and f is real with $\sup|f|<c_L$.

For a G^c-valued g, covariance is obtained at the expense of room in the domain, and the amount of room lost is $\mathrm{cn}(g)$. Precisely, consider the graded domains $\mathrm{Dom}^c_\theta(c)$, $0<\theta\le 1$, on which the contour families and the W_x lie within θ times the radii. If $\mathrm{cn}(g(c_\pm))\le\varkappa$ and $\varkappa\theta\le 1$, then $\mathbf{W}^g\in\mathrm{Dom}^c_{\varkappa\theta}(c)$ and

$$\mathcal{M}(\mathbf{W}^g)(c)=g(c_-)\,\mathcal{M}(\mathbf{W})(c)\,g(c_+)^{-1}.$$

Proposition 1a.1.4 (one averaging multiplies curvature by L^2).

$$|\partial\mathcal{M}(W)(p')-\mathbf{1}|\le L^2\varpi+C(L^2\varpi)^2;$$
$$|\partial U-\mathbf{1}|\le\alpha_0\eta_k^2,\ \alpha_0\le c$$
$$\Rightarrow$$
$$\varpi(\mathcal{M}^nU)\le 2\alpha_0(L^n\eta_k)^2$$
$$(n\le k),$$

and $\mathcal{M}^nU(c)$ differs from the straight bare transport by at most $C\alpha_0(L^n\eta_k)^2$. Consequently the first rule is applicable on every level, and on $\{|\partial U-\mathbf{1}|<c\,\eta_k^2\}$ the map $\mathcal{M}^k$ is real-analytic.

Proposition 1a.1.5 (one step relative to a reference). Let $\varpi=\varpi(W_0)\le c_L$, and consider complex $|\mathsf{a}|<c_L$. Then $\mathcal{Q}^1$ and $\frac{1}{i}\log\mathsf{u}$ depend holomorphically on a, are covariant and do not exceed $C|\mathsf{a}|$ in size; moreover

$$\mathcal{Q}^1(W_0,\mathsf{a})=\mathsf{Q}(W_0)\mathsf{a}+C^1(W_0,\mathsf{a}),$$

$$\mathsf{Q}(\mathbf{1}) = \mathsf{Q}^{\mathrm{s}},$$
$$\|\mathsf{Q}(W_0) - \mathsf{S}(W_0)\|_{\mathrm{row},\,\mathrm{col}} \le C\varpi,$$

$$\|\mathsf{S}(W_0)\|_{\mathrm{row}} \le L,$$
$$\|\mathsf{S}(W_0)\|_{\mathrm{col}} \le L^{-3},$$
$$|C^1| \le C|\mathsf{a}|^2.$$

Theorem 1a.1.6 (k steps, uniformly in k). Assume $|\partial U_0 - \mathbf{1}| \le \alpha_0\eta_k^2$ with $\alpha_0 \le c$, and let A be complex with $\sup|A| < c$. Here c, C depend only on L. The following hold holomorphically in A and in the complex shifts $\mathbf{U} = e^{i\eta_k A'}U_0$ of U_0 with $\sup|A'| < \alpha_1 \le c$. The bounds are uniform in A'; in the bounds on the row norm and on the functional derivative, $\alpha_0 + \alpha_1$ then replaces α_0.

$$\mathcal{M}^n(e^{i\eta_k A}U_0) = \big(e^{i\mathsf{a}_n}\mathcal{M}^n U_0\big)^{v_n},$$
$$|\mathsf{a}_n| \le 2L^n\eta_k \sup|A|,$$
$$|v_n - \mathbf{1}| \le CL^n\eta_k \sup|A|,$$

$$Q_k^\sharp(U_0) = \eta_k\, \mathsf{Q}(W_{0,k-1}) \cdots \mathsf{Q}(W_{0,0}),$$
$$\|Q_k^\sharp(U_0)\|_{\mathrm{row}} \le 1 + C\alpha_0,$$
$$|C_k| \le C \sup|A|^2,$$

$$Q_k(U_0)A = Q_k^\sharp(U_0)A - D_{\mathcal{M}^k U_0}\dot{v}_k(A),$$
$$|Q_k A| \le C \sup|A|,$$
$$|C_k^{\mathrm{pl}}| \le C \sup|A|^2 :$$

thus a coarse gradient separates the two linear parts, and this gradient does not become small as the curvature does. More generally, let h be G^c-valued with $\mathrm{cn}(h)\,\mathrm{cn}(v_n) \le 2$. Then $(e^{i\eta_k A}U_0)^h$ has well-defined averages, and $\mathcal{M}^n\big((e^{i\eta_k A}U_0)^h\big) = \big(e^{i\mathsf{a}_n}\mathcal{M}^n U_0\big)^{hv_n}$. This ball is much larger than the regular class, because the plaquettes of $e^{i\eta_k A}U_0$ are $O(\eta_k)$ rather than $O(\eta_k^2)$. Accordingly, the validity of the first rule on every level and the analyticity are consequences of the factorisation, not of 1a.1.4. The remainder decomposes as $C_k = C_k^{(2)} + C_k^{(3)} + \cdots$, where

$$C_k^{(2)}(U_0, A) = \sum_{n<k} \mathsf{Q}(W_{0,k-1}) \cdots \mathsf{Q}(W_{0,n+1})\, C^{1,(2)}\big(W_{0,n}, \mathsf{a}_n^{\mathrm{lin}}\big),$$
$$|n\text{-th term}| \le CL^{n-k} \sup|A|^2 :$$

the non-linearity consists of commutators and is concentrated at the coarsest of the k scales. The derivative $K_A(c,b) := \partial\mathcal{Q}_k^\sharp(c)/\partial A(b)$ satisfies $\|K_A\|_{\mathrm{col}} \le 2\eta_k^4$ and the Lipschitz bound $\|K_A - K_{A'}\|_{\mathrm{col}} \le C\eta_k^4 \sup|A - A'|$. Consequently $1 + C\alpha_0 + C|A|$ is a bound for the functional derivative $\eta_k^{-4}K_A$, the adjoint K_A^* is a bounded operator for the sup norm, and K_A acts boundedly from

L^2 to ℓ^2. The Lagrange equations of (1b) involve K_A^*. Complex regular fields $\mathbf{U} = e^{i\eta_k A'} U_0$ with $\sup|A'| < \alpha_1 \le c$ have averages that are regular, each on its own scale:

$$|\partial\mathbf{U} - \mathbf{1}| \le \alpha_0'\eta_k^2,\ \alpha_0' \le c \Rightarrow |\partial\mathcal{M}^n\mathbf{U} - \mathbf{1}| \le 2\alpha_0'(L^n\eta_k)^2$$

. From $|\nabla_{U_0}A'| < \alpha_1$ one obtains only $\alpha_0' = \alpha_0 + C\alpha_1$. That is why the domains U_j^c require, as a separate condition, that the complex plaquettes be small; averaging reproduces exactly this condition.

All of this holds on determining sets, layer by layer. At the bare bonds of layer j the local scale is $\xi(b) = \xi_j = L^j\eta_k$; it is the local scale of layer j in the sense of "Regular configurations". On determining sets the ceilings on α_0 and on $\sup_b \xi(b)|A(b)|$ are lowered by the factors L^{-2} and L^{-1}. At a bond of Γ_j that lies astride a seam, one factor is lost. Writing $K_A(c,b) = \partial\mathsf{a}_j(c)/\partial A(b)$ for $c \in \Gamma_j$, one has

$$|\partial\mathcal{M}^j U_0 - \mathbf{1}| \le 2L^2\alpha_0,$$
$$|\mathsf{a}_j| \le 2L \sup_b \xi(b)|A(b)|,$$
$$\sup_c \sum_b \frac{|K_0(c,b)|}{\xi(b)} \le 2L,$$
$$\sum_j \sum_{c\in\Gamma_j} |K_A(c,b)| \le \frac{C\eta_k^4}{\xi(b)^3},$$

and this factor is lost once, without compounding; it cannot be removed (1a.1.14). The last bound is a sum over the levels at which a given bare bond enters.

Propositions 1a.1.7, 1a.1.8 (linear averages; gauge transformations). The operator

$$Q_k^{\mathrm{st}} := Q_k^{\mathrm{st}}(\mathbf{1}) = Q_k^\sharp(\mathbf{1}) = \eta_k(\mathsf{Q}^{\mathrm{s}})^k$$

is the straight average taken over the blocks of order k, and

$$(Q_k^{\mathrm{st}}A)(c) = \eta_k^5 \sum_{x\in B^k(c_-)} A([x,x']),$$
$$u_k(p) = \prod_\mu \frac{\eta_k(e^{ip_\mu} - 1)}{e^{i\eta_k p_\mu} - 1},$$
$$c \le Q_k^{\mathrm{st}} Q_k^{\mathrm{st}*} \le 1,$$

where u_k coincides with the Fourier symbol of Q_k' up to a phase factor. The normalisation is as follows. Constants are mapped to the same constants, and $\|Q_k^{\mathrm{st}}\|_{\mathrm{row}} = 1$. Composing Q_k^{st} with the linear lift gives the identity, and $Q_k'Q_k'^* = 1$. The adjoint does not coincide with the lift. The operator $Q_k^{\mathrm{st}}Q_k^{\mathrm{st}*}$ is not a multiple of the identity: it is a convolution that couples each coarse bond to its two neighbours on the same line. This is why the display gives only a two-sided bound. In a background one has $|Q_k^\sharp(U_0)A| \le Q_k^{\mathrm{st}}|A| + C\alpha_0 Q_k''|A|$, where Q_k'' takes the mean of $|A|$ over every bond in the double block of order k. With gradients $(d\lambda)(b) = \lambda(b_+) - \lambda(b_-)$, the two averages intertwine gradients in different ways:

$$\mathsf{Q}^{\mathrm{pl}} d\lambda = d(\lambda|_{\mathrm{coarse}}),$$

$$\mathsf{Q}^{\mathrm{s}} d\lambda = d(q\lambda),$$

$$Q_k^\sharp(U) D_U \lambda = D_{\mathcal{M}^k U}\, Q_k'(U)\lambda,$$

the last of these relations being exact and valid for the straightened average only. In a background the plain average satisfies, likewise exactly, $Q_k(U) D_U \lambda = D_{\mathcal{M}^k U}(\lambda|_{\mathbb{T}^{(k)}})$, where the restriction stands in place of Q_k'. For the main term $Q_k^{\mathrm{st}}(U)$, by contrast, the relation involving Q_k' is valid only up to $O(\alpha_0)$.

It follows that there are two residual invariances. The constraint $\mathcal{M}^k U = V$ is preserved by every transformation with $u = \mathbf{1}$ on $\mathbb{T}^{(k)}$. The straightened constraint is preserved by the w with $r_k = \mathbf{1}$, where r_k is defined below. In linearised form this condition is $Q_k'\lambda = 0$. When $A = 0$, and likewise at a field tower axial relative to U_0, these transformations are precisely the w whose averages are trivial. Of the two sets, only the first forms a group. The set generated by the second depends on its base point, and the tower axial gauge selects that base point in each orbit of the first. The two invariances correspond to each other through $v_k[A]$.

Now let A be real with $\sup|A| \le c$. Put $w = e^{i\lambda}$ and $e^{i\eta_k A^w} U_0 := (e^{i\eta_k A} U_0)^w$, and for a site function assume $\|\lambda\|_1 := \max\{\sup|\lambda|, \sup|D_{U_0}\lambda|\} \le c$. This norm is taken in level-k units, without local scales, and it differs from the norm $\|\cdot\|_1$ of 1-forms. It equals $\|\lambda\|_0$ for the constant sequence $\Omega_j = T_\ell$, whose only layer is k, with $\xi_k = 1$. The straightened averages change by r_n, where $r_0 = w$ and $r_{n+1} = \mathsf{u}(W_{0,n}, \mathsf{a}_n(A^w))^{-1}\, r_n\, \mathsf{u}(W_{0,n}, \mathsf{a}_n(A))$:

$$e^{i\mathsf{a}_n(A^w)} W_{0,n} = \big(e^{i\mathsf{a}_n(A)} W_{0,n}\big)^{r_n},$$

$$w|_{\mathbb{T}^{(n)}}\, v_n[A] = v_n[A^w]\, r_n,$$

$$r_n|_{A=0} = \bar{R}_{U_0}^{(n)} w,$$

$$\tfrac{1}{i}\log r_k = Q_k'(U_0)\lambda + O\big((\sup|A| + \|\lambda\|_1)\|\lambda\|_1\big).$$

We say that u is of class α when each $\bar{R}^{(j)}u$ is within α of $\mathbf{1}$ and its variation inside any block is less than $\alpha L^{j+1}\eta_k$. If u_1, u_2 are of class $\alpha \le c$, the averages are multiplicative up to second order,

$$\bar{R}^{(j)}(u_1 u_2) = \bar{R}^{(j)}u_1\, \bar{R}^{(j)}u_2\, e^{i\mathsf{z}_j},$$

$$|\mathsf{z}_j| \le C(\alpha L^j \eta_k)^2,$$

and $u_1 u_2$ is of class $2\alpha + C\alpha^2$. The averages of products depend holomorphically on complex λ. Precisely, let $u' = e^{i\lambda}$ with $\|\lambda\|_1 \le \alpha' \le c$, and let u_1 be of class α. Then the quotients $\tilde{u}^{(j)} := \bar{R}^{(j)}(u'u_1)(\bar{R}^{(j)}u_1)^{-1}$ have four properties: they are holomorphic in λ; they lie within $C\alpha'$ of $\mathbf{1}$; their covariant differences are $\le C\alpha' L^j \eta_k$; and $\frac{1}{i}\log\tilde{u}^{(k)} = Q_k'(U_0)\lambda + O((\alpha+\alpha')\alpha')$. 1a.3 uses this to show that the Landau gauge depends analytically on the fields.

Lemma 1a.1.9 (rigidity of one step). Suppose that W, W_0 have plaquette size $\le \varpi$, that their contour variables agree up to a defect δ, and that $|\mathcal{M}(W)\mathcal{M}(W_0)^{-1} - \mathbf{1}| \le \alpha_1$. Then

$$|W(b)W_0(b)^{-1} - \mathbf{1}| \le \alpha_1 + C\varpi + 2\delta$$

at every bond of the double block. Put differently, modulo the curvature, each bond variable is a function of the contour variables together with the average:

$$\big|W(b) - W(y, b_-)^{-1} W(y, b_+)\big| \le C\varpi,$$

$$\big|W(b') - W(c_-, b'_-)^{-1}\, \mathcal{M}(W)(c)\, W(c_+, b'_+)\big| \le C\varpi,$$

for b inside $B(y)$ with $y = c_\pm$, and for b' in the corridor of c, that is, among the bonds parallel to c joining $B(c_-)$ to $B(c_+)$. The defect accounts for two things. One is what the soft gauge fixing leaves, which gives $|V(y,x) - \mathbf{1}| < \varepsilon_k$ rather than equality. The other is what a seam leaves (1a.3.11).

Proposition 1a.1.10 (the fibre is curved; the current enters). Let $|\partial U - \mathbf{1}| \le \alpha_0 \eta_k^2$ with $\alpha_0 \le c$ (the range of Theorem 1a.1.6). Let moreover $U \in \mathfrak{U}_k(\Omega_\bullet, \alpha_0)$ at the constant sequence $\Omega_j = T_\ell$; this membership brings in the further condition $|J(U)| < \alpha_0$. Suppose that M_1, $M\alpha_0$ and the number of sites of $\mathbb{T}^{(k)}$ satisfy the hypotheses of Proposition 1a.3.5 [to be proved; Package A], which provides G and H_k. Finally, let A be small. On $Q_k^\sharp(U)A + C_k(U,A) = 0$, in the Landau gauge, one has $A = A_1 - \mathsf{H}_k C_k^{(2)}(A_1) + \dots$ with $Q_k^\sharp(U)A_1 = 0$, and

$$A(e^{i\eta_k A}U) = A(U) + \langle J(U), A_1\rangle + \tfrac{1}{2}\langle A_1, \Delta_U^{\mathrm{W}} A_1\rangle - \langle \mathsf{H}_k C_k^{(2)}(A_1), J(U)\rangle + O(|A_1|^3).$$

In this expansion H_k denotes the right inverse $\mathsf{H} = GQ^*(QGQ^*)^{-1}$ of $Q = Q_k^\sharp(U)$, built from the G of (1c); with this choice the relation $RD^*GQ^* = 0$ is exact. For the derivative of $V \mapsto U_k(V)$ the corresponding objects are G_1, H_1. The term $O(|A_1|^3)$ collects all terms of order ≥ 3 in A_1. Indeed, the constraint is solved up to second order, and the solution is substituted into the Taylor expansion of the action.

Proposition 1a.1.11 (isometries and linearised objects). Suppose that r preserves every lattice involved. Then $\mathsf{r}(e^{i\mathsf{a}}W_0) = e^{i\mathsf{r}W_0\,\mathsf{a}}\,\mathsf{r}W_0$, and

$$\mathcal{Q}^1(\mathsf{r}W_0, \mathsf{r}_{W_0}\mathsf{a}) = \mathsf{r}_{\mathcal{M}(W_0)}\mathcal{Q}^1(W_0, \mathsf{a}),$$
$$Q_k^\sharp(\mathsf{r}U)\,\mathsf{r}_U = \mathsf{r}_{\mathcal{M}^k U}\,Q_k^\sharp(U),$$
$$J(\mathsf{r}U) = \mathsf{r}_U J(U),$$
$$\Delta^{\mathrm{W}}_{\mathsf{r}U}\mathsf{r}_U = \mathsf{r}_U \Delta^{\mathrm{W}}_U,$$

and the corresponding relations hold for Q_k, Q'_k, $\bar{R}$, u. For real W_0, the map r_{W_0} is an isometry both for the sup norm and for $\langle\cdot,\cdot\rangle$. These maps compose according to $(\mathsf{r}\mathsf{r}')_{W_0} = \mathsf{r}_{\mathsf{r}'W_0}\mathsf{r}'_{W_0}$. If the twisted action is replaced by the plain one (the action at $W_0 = \mathbf{1}$), these relations in general fail for a reflection reversing bonds. The plain and the twisted action are equal where $\mathrm{Ad}(W_0(b))$ acts as the identity on the reversed bonds b. Consequently, propagators, H_k, $\Delta^{(k)}$ and $C^{(k)}$ are covariant under all of W_4 only in the twisted sense, and (1c), (1d) and (1e) use the symmetry in this form.

Proposition 1a.1.12 (the fundamental equality; the transformation to the tower axial gauge). Let U_0, A be as in 1a.1.6. There exists exactly one u, defined on the bare sites, for which $(e^{i\eta_k A}U_0)^u$ is tower axial relative to U_0 and $\bar{R}^{(k)}_{U_0} u = \mathbf{1}$. Its averages are given by $\bar{R}^{(n)}_{U_0} u = u\, v_n$, so that $u = v_k^{-1}$ on $\mathbb{T}^{(k)}$, and

$$\mathcal{M}^k\big((e^{i\eta_k A}U_0)^u\big) = e^{i\mathcal{Q}_k^\sharp(U_0;A)}\,\mathcal{M}^k U_0 :$$

once the gauge has been changed in this way, averaging of order k amounts to composing the straightened one-step averages k times. The transformation can be written explicitly. Let $x = x_0, \dots, x_k = y$ be the tower of block centres. Let $\mathsf{w}_j^{\mathrm{rel}}(z,x) = (e^{i\mathsf{a}_j}W_{0,j})(z,x)W_{0,j}(z,x)^{-1}$ be the relative contour variables of the straightened averages, let u_j be the block factors, and let T_{U_0} be the transport down the tower. Then

$$u(x) = \prod_{j=k-1}^{0} \operatorname{Ad}\big(\mathsf{T}_{U_0}(x_{j+1}, x)\big)^{-1} \Big[\mathsf{u}_j(x_{j+1})^{-1}\, \mathsf{w}_j^{\mathrm{rel}}(x_{j+1}, x_j)\Big],$$

$$|u - \mathbf{1}| \le C \sup |A|.$$

Consequently u depends locally and holomorphically on A and is of class $C \sup|A|$. At the bare scale, however, it fails to be regular: it jumps by $O(L^j \eta_k \sup|A|)$ between neighbouring bare sites belonging to different blocks of order j.

Proposition 1a.1.13 (two schemes; the linear parts compared; what carries over). Bałaban's theorems of 1984–85 are proved for blocks that have their site at a corner and for single combs; this is scheme c. The map $\mathcal{M}$ uses centred blocks and means; this is scheme m. Since L is odd, the levelwise translations by $\mathsf{e}_j = \frac{1}{2}(L^j - 1)a_K(1,1,1,1)$ carry the cornered blocks of level j onto centred blocks. Compatibility of the operators across levels is part of the comparison below. Of the contour variable, the proofs of 1a.1.2–1a.1.9 use only the following properties:

- there is one contour variable for each pair (y, x), and it is covariant;
- its paths stay in the block and make fewer than L steps in each direction;
- coarse translations carry the contour variables into one another;
- it is G-valued on G;
- it is holomorphic near $\mathbf{1}$, with a mean of the $\mathsf{a}(\Gamma)$ as its linear part;
- where the bond variables commute with one another, it is the exponential of that mean.

This list is one way of carrying out the axiomatisation that Bałaban mentions. The results of 1a.1.2–1a.1.9 therefore hold in both schemes, with two exceptions. The first exception is covariance under isometries other than coarse translations. It fails for combs, since under an element of W_4 a comb goes over into a comb of a different order, and a lowest corner into a different corner; this failure is precisely why the mean is used. The second exception is the three rules, which are formulated for $\mathcal{M}$ only. When the background is trivial, the two schemes have the same straightened linear part, $\mathsf{Q}_\star(\mathbf{1}) = \mathsf{Q}^{\mathrm{s}}$, but different plain linear parts. In a background, let t denote a transport of U_0 joining the two base sites of a block. Then

$$Q_k^\sharp(U_0) = \operatorname{Ad}(t)\big(Q_k^{\mathrm{B}}(U_0) + \mathsf{D}_k\big),$$

$$\|\mathsf{D}_k\|_{\mathrm{row}} \le C\alpha_0,$$

$$\|\mathsf{D}_k\|_{\mathrm{col}} \le C\alpha_0 \eta_k^4,$$

and the same holds for Q_k'. Moreover, $\mathsf{D}_k(c, b) = 0$ unless b lies under c. In each scheme, the linear part and the straight average of that scheme differ by a kernel of row norm $C\alpha_0$. In this leading part, a bare bond b parallel to c receives as its weight η_k^5 multiplied by the count of segments $[x, x']$, $x \in B^k(c_-)$, passing through b (a tent along the direction of c). Every such segment enters with the transport of U_0 taken down a chain of contours and segments. The exact identity $Q_k = Q_k^\sharp - D\dot{v}_k$ is supplied in the present text. The identity involving D_k takes the place of the assertion of 1987 (109, p. 254) that the new average satisfies all the results of the paper on averaging operations. For real U_0 the factor $\operatorname{Ad}(t)$ is orthogonal, so it cancels from $Q_k^\sharp(U_0)^*\, a\, Q_k^\sharp(U_0)$. Consequently, in the two schemes the forms defining G', R, G_k, H_k differ from each other by local forms whose relative size is $C\alpha_0$. Through this comparison, the operators built from $Q_k^{\mathrm{B}}(U_0)$ are used in (1b). The operators are built afresh from $Q_k^\sharp(U_0)$, and their bounds are derived from the case $U_0 = \mathbf{1}$, in which, after the blocks have been aligned, the two linear parts are equal (1c).

Proposition 1a.1.14 (sharpness: fields of one colour). Let $T \in \mathfrak{g}$ with $|T| = 1$. For $U = e^{ifT}$, with f real and small (1a.1.3), everything is linear and explicitly computable. A constant curvature φ gives $\partial\mathcal{M}U(p') = e^{iL^2\varphi T}$, so the factor L^2 of 1a.1.4 is attained.

Now consider a seam. Let Ω_1 be, locally, a slab $\{x_1 \geq \text{const}\}$. Let $z \in \mathbb{T}^{(1)}$ be the last coarse site outside it, and let p' be the coarse plaquette with corners $z, z + Le_1, z + Le_1 + Le_2, z + Le_2$. Set $f(\langle x, x + e_2\rangle) = \mathsf{f}^{\Sigma}(x_1)$ and $f = 0$ on the other bonds, where $\mathsf{f}^{\Sigma}(t) := \sum_{t'<t} \mathsf{f}(t')$. Here $\mathsf{f} = \varphi$ on the $L - 1$ bonds in direction 1 that lie in the block of z; on the $L - 1$ bonds preceding them the value is $-\varphi$, and everywhere else it is 0. The curvature has size at most φ and vanishes in Ω_1, while $\partial\mathcal{M}U(p') = e^{i\Theta T}$ with $\Theta = \frac{1}{2}L(L-1)\varphi$. Thus no bound $C\alpha_0$ with C independent of L can take the place of the bound $2L^2\alpha_0$ of 1a.1.6.

At $U_0 = \mathbf{1}$, take a bond $c \in \Gamma_j$ astride a seam with outer end c_-. Such a bond has one end, its outer end, outside Ω_j; in (1b) it is called an outward bond. The weighted row sum equals $\sum_b \frac{|K_0(c,b)|}{\xi(b)} = \frac{1}{2}L(1 - L^{-j}) + \frac{1}{2}(1 + L^{-j})$, and its limit as $j \to \infty$ is $\frac{1}{2}(L + 1)$. Indeed, along a segment $[x, x']$, a bare bond contained in $B^j(c_-)$ has $\xi(b) = L^{j-1}\eta_k$, while the remaining ones extend into $B^j(c_+)$, where the scale equals $L^j\eta_k$.

For a bond of Γ_j astride a seam, an additional block factor arises at its outer end. Moreover, the datum entering the straightened constraint (the reduced datum of (1b)) is not the same as the plain datum. The factor is genuinely present. At $U_0 = \mathbf{1}$, take $\eta_k A = tT$ on the single bare bond $\langle z, z + e_\nu\rangle$, where z is the outer end of a bond of Γ_1, and $A = 0$ elsewhere. Then $\mathsf{u}(z) = \exp\left(it\frac{L^4-1}{8L^4}T\right) \neq \mathbf{1}$. Finally, $\|\mathsf{Q}^{\mathrm{pl}}\|_{\mathrm{row}} > 2L$.

Proposition 1a.2.1 (the exact identities). The transformation $\mathbf{T}$ preserves mass, positivity ($|\mathbf{T}\nu| \leq \mathbf{T}|\nu|$) and gauge invariance. It commutes with the isometries that map $\mathbb{T}^{(k+1)}$ onto itself, and

$$\mathbf{T}^n\nu = (\mathcal{M}^n)_*\nu,$$
$$\langle F(\mathcal{M}^k U)\rangle_{K,\ell,\beta} = \int F\, d(\mathbf{T}^k\rho_0) \Big/ \int d(\mathbf{T}^k\rho_0).$$

Consequently, whenever $F \in \mathfrak{A}$ has largest depth s, each of the numbers $\langle F \circ \mathcal{M}^{K-s}\rangle_{K,\ell,\beta}$, $K \geq s$, is the integral of one fixed function with respect to the normalised measures $\mathbf{T}^{K-s}\rho_0$, all on the same compact space $\mathcal{V}(\Lambda_{s,\ell})$; this idea stands alongside the two theorems.

Proposition 1a.2.2 (soft and hard constraints). Each kernel t of the kind described above preserves mass, positivity and gauge invariance. In the case of linear site fields and of block means Q, the Gaussian kernels $t_{\mathfrak{a}} \propto e^{-\frac{\mathfrak{a}}{2}|\Psi - Q\Phi|^2}$, taken after the canonical rescaling by $L^{(d-2)/2}$, form a family that is closed under composition. In these kernels Φ, Ψ are site functions in dimension d, Q is the rescaled block mean, and Q_k is the mean over B^k, rescaled in the same way; the composition law reads

$$t_{\mathfrak{a}} * t_{\mathfrak{a}_k} = t_{\mathfrak{a}_{k+1}},$$
$$\mathfrak{a}_{k+1}^{-1} = \mathfrak{a}^{-1} + L^{-2}\mathfrak{a}_k^{-1},$$
$$\mathfrak{a}_k \to \mathfrak{a}(1 - L^{-2}),$$

and these kernels give rise to the propagators $(\Delta + \mathfrak{a}_k Q_k^* Q_k)^{-1}$, where $\Delta = D^*D$ acts on site functions. For the delta kernel the fibres are nested, and over regular data, up to gauge,

$$\{\mathcal{M}^{k+1}U = W\} = \bigsqcup_{\mathcal{M}(V)=W} \{\mathcal{M}^k U = V\}$$

$$\Rightarrow$$

$$U_k(V_k^{\mathrm{bg}}) = U_{k+1}(W),$$

$$V_k^{\mathrm{bg}} = \mathcal{M}^k U_{k+1}(W),$$

$$U_j(\mathcal{M}^j U_k(V)) = U_k(V)\ (j \le k):$$

every background therefore coincides with the background of its own averages, and V_k^{bg}, with $\mathcal{M}^k U$ taken in the block axial gauge, is in that gauge the critical point of $V \mapsto A(U_k(V))$ on $\{\mathcal{M}(V) = W\}$. Suppose that $\Delta \ge 0$ is positive on $\ker Q$ and that Q is surjective. Then, under the constraint $QA = B$, the form $\frac{1}{2}\langle A, \Delta A\rangle$ attains its minimum at $\mathsf{H}B$, where $\mathsf{H} = \mathsf{G}Q^*(Q\mathsf{G}Q^*)^{-1}$ and $\mathsf{G} = (\Delta + aQ^*Q)^{-1}$. This holds for every $a > 0$. *Remark.* Let W be real and lie in a neighbourhood of $\mathbf{1}$; this neighbourhood may depend on k and on the torus. Then V_k^{bg} is, in that gauge, the unique point of minimum of $V \mapsto A(U_k(V))$ on $\{\mathcal{M}(V) = W\}$.

Proposition 1a.2.3 (the steps with their insertions are exact). Each operation of $\mathbf{T}$ and of $\mathbf{R}$ consists of finitely many positive kernels, one for each outcome of the operation, whose sum is a probability kernel. These kernels depend on the auxiliary choices. Once the choices are fixed, the kernels depend neither on β nor on the density. When the run makes its own choices, the operations themselves vary with β. For a bounded, gauge-invariant, complex-valued $\mathfrak{w}$,

$$\tau_h^{\mathfrak{w}}(dV) = \int dU\, \mathfrak{w}(U)\rho_0(U)\, \mathsf{K}_h(U, dV),$$

$$\sum_h \mathsf{K}_h(U, \mathcal{V}_k) = 1.$$

Consequently $\tau_h \ge 0$ and $|\tau_h^{\mathfrak{w}}| \le \sup|\mathfrak{w}|\,\tau_h$. Moreover $\int \rho_k = Z_{K,\ell,\beta}$ and $\int \rho_k^w = Z(w)$ at every level, on a torus that is large compared with the cubes of side $M\check{R}_j$ of every level. This holds irrespective of the gauges and representatives chosen. It also holds for every real sequence of numbers t among those admitted in (2e). These are the sequences whose terms are all $\ge \gamma^{-2}$, whose differences obey two-sided bounds of the same type as the flow inequalities of 1a.2.7, and for which $\check{R}_j \le L\check{R}_{j+1}$. The sum over outcomes of the operations of $\mathbf{T}$ is $\mathcal{M}_*$. By contrast, $\mathbf{R}$ preserves only those integrals of F that do not depend on the components on which it acts; in general, therefore, $\rho_k \ne \mathbf{T}^k\rho_0$. *Exact insertion:* for every gauge-invariant F,

$$\langle F(\mathcal{M}^k U)\rangle_{K,\ell,\beta} = \sum_h \int F^{(h)} d\tau_h \Big/ \sum_h \int d\tau_h,$$

$$|F^{(h)}| \le \sup|F|,$$

where $F^{(h)}$ denotes the conditional expectation, under the joint law constructed from the kernels, of $F(\mathcal{M}^k U)$ given (h, V_k). If F depends only on fields away from those components, then $F^{(h)} = F$. *Covariance:* $\mathsf{r}_*\tau_h = \tau_{\mathsf{r}h}$ for every isometry that preserves all the partitions involved. An individual $E^{(j)}(X, \cdot)$ is covariant under the translations that preserve π_j. An individual $R^{(j)}(X, \cdot)$ is covariant under the translations that preserve the $M\check{R}_j$-cubes. Covariance under all isometries r of $\mathbb{T}^{(j)}$ holds only for the unrestricted pointed totals $\sum_{X \ni z} E^{(j)}(X, U_j, z)$. These totals are taken at $U_j = U_j(V)$, where V and $\mathsf{r}V$ both belong to a class of real block fields of level j lying near

the regular ones (1d). Wherever the covariance of pointed totals is invoked below, it has this meaning.

Lemmas 1a.2.4–1a.2.6 (units; the delta function resolved; regrouping). When the plaquette variables are close to $\mathbf{1}$, $\xi^{d-4}a_p = \frac{1}{2}\xi^d\langle F_U, F_U\rangle(1 + O(\xi^4|F_U|^2))$. Hence, for $d = 4$, Wilson's action is a Riemann sum for $\frac{1}{2}\int|F|^2$ whatever the unit of length. A change of unit leaves A, ρ_k and x_k unchanged. On small fields, $\mathcal{M}(e^{i\mathcal{A}}V_k^{\mathrm{bg}})V_{k+1}^{-1} = \exp i\mathcal{Q}_1(L\mathcal{A})$ and $\mathcal{Q}_1(L\mathcal{A}) = LQ\mathcal{A} + C^{\mathrm{pl},(2)}(\mathcal{A}) + \dots$. Here $\mathcal{A}$ carries no factor of length. Up to the scaling of its argument by L, the map is $\mathcal{Q}_1(\cdot) = \mathcal{Q}_1(V_k^{\mathrm{bg}};\cdot)$, and this scaling accounts for the factor L in front of its linear part. The average here is the plain $Q = Q_{V_k^{\mathrm{bg}}}$, and $C^{\mathrm{pl},(2)}$ is the second-order part of the plain remainder $C_1^{\mathrm{pl}}(V_k^{\mathrm{bg}}, L\mathcal{A})$. The symbol $C_k^{(2)}$ is reserved for the straightened average.

An analytic substitution $\mathcal{A} = \psi^{\mathrm{lin}}(\mathcal{A}') = \mathcal{A}' - \mathsf{h}\tilde{D}(\mathcal{A}')$ changes only the variables on the pivot bonds. It turns the delta function into the linear one, $\delta(LQ\mathcal{A}')$, which is solved for a single pivot bond $b_0(c)$ for each coarse bond. The substitution comes from the implicit function theorem. In it, h inverts LQ on the pivot bonds, $LQ\mathsf{h} = 1$. The linearising correction $\tilde{D}$ is of second order and is written $\widetilde{\mathsf{D}}$ in (1d). It is the unique small solution of $\mathcal{Q}_1\big(L(\mathcal{A}' - \mathsf{h}\tilde{D}(\mathcal{A}'))\big) = LQ\mathcal{A}'$, obtained as the fixed point of a contraction. For F supported in the analytic chart just constructed, the local coarea identity is

$$\int_{\mathrm{chart}} dV_k\, \delta(\mathcal{M}(V_k)V_{k+1}^{-1})F(V_k) = z^{(k)} \int \prod_{b\neq b_0(c)} d\mathcal{A}'(b)\; e^{\mathrm{Tr}\log(1-\mathsf{h}\,\partial\tilde{D})+\sum_b \log\sigma_{\mathrm{H}}(\mathcal{A}(b))}\, F\big(e^{i\psi^{\mathrm{lin}}(\mathsf{C}\mathcal{A}')}V_k^{\mathrm{bg}}\big),$$

where the right-hand integral is over the preimage of that chart in the nonpivot coordinates, and σ_{H} is the density of Haar measure in exponential coordinates. A decomposition into charts or the remaining large-field terms is required for the unrestricted integral. The explicit normalization of this local identity is given in Section 17. Once this density is divided by its value $\sigma_{\mathrm{H}}(0)$ at $\mathbf{1}$, the Jacobians are $1 + O(\delta_k)$ for each bond. They are local and analytic, and they are included in $\mathbb{P}^{(k)}$. The factor $\sigma_{\mathrm{H}}(0)$, taken once for each non-pivot bond, is absorbed into the constant of the step. For every number b,

$$-x_kA + \sum_X E^{(k+1)}(X,\cdot) = -(x_k - b)A + \Big(\sum_X \big[E^{(k+1)}(X,\cdot) - E^{(k+1)}(X,\mathbf{1})\big] - bA\Big) + \sum_X E^{(k+1)}(X,\mathbf{1}),$$

and, summed over the levels, $-A(x_k(\cdot), U) - \sum_{j\le k} b_j A(\varphi_j, U) = -\beta A(U)$. The constant of a step depends on the regions of that step. Let $|\Omega^{(k)*}|$ be the number of non-pivot bonds of $\mathbb{T}^{(k)}$ in Ω, and $\#_{k+1}(\Omega)$ the corresponding number of sites of $\mathbb{T}^{(k+1)}$. Let $z_{\mathfrak{G}} = \mathfrak{z}(x_k, \varepsilon_k)$ be the normaliser of $\mathfrak{G}_k$ for each block site (1a.3), and $\nu_{\mathfrak{g}} = \log\int_{\mathfrak{g}} e^{-|Y|^2/2}dY$. Then

$$\mathsf{E}^{(k)}(\Omega,\Lambda) = \big(\log\sigma_{\mathrm{H}}(0) - \tfrac{1}{2}(N^2-1)\log x_k\big)|\Omega^{(k)*}| - (L^4-1)\log z_{\mathfrak{G}}\,\#_{k+1}(\Omega) + \nu_{\mathfrak{g}}|\Lambda^{(k)*}| + \sum_{X\subset\Lambda}\big[E^{(k+1)} + R^{(k+1)}\big](X,\mathbf{1}),$$

together with $-\mathcal{E}_k = \sum_{j<k}\mathsf{E}^{(j)}(\Omega_{j+1},\Lambda_{j+1})$ and $E = \sum_{j<K}\mathsf{E}^{(j)}(T_\ell, T_\ell)$. Here $\mathcal{E}_k$ depends on the label, and at the all-small-field label the vacuum constant E equals $-\mathcal{E}_K$.

Theorem 1a.2.7 (the invariant form: the programme). Let $\mathfrak{F}_k$ be the class of measures of the form below. In it the $E^{(j)}$ are pointed families, the pointed totals of which are covariant with respect to every isometry of $\mathbb{T}^{(j)}$, and b_j is their coefficient (1e). A step does not preserve the form and the bounds alone (2e). The measures of the class, with their bounds, are

$$\sum_{\mathcal{S}} \mathbf{T}'_k(\mathcal{S}) \big[\chi_{k,\mathcal{S}}\, e^{A_{k,\mathcal{S}}(U_k(\mathbf{V}))}\big],$$

$$\|E^{(j)}\|_{j,\kappa} \le E_0,$$

$$\|R^{(j)}\|_{j,\kappa} \le g_j^{\kappa_0},$$

$$|B^{(j)}(X,\cdot)| \le \mathsf{B}\, e^{-\kappa d_j^\sharp(\mathcal{S};X)},$$

$$\mathsf{M}(i,X) \le e^{-(2-c')\,p_0(g_i)},$$

where B is a constant chosen before γ ((2d), (2e)), and where the last inequality, the *mass bound*, concerns those components X that $\mathbf{R}$ has acted on at some level $i \le k$. Such a component was healed, or left standing as an island, or else it dropped out at a failed test, either of a stage or of the main step of $\mathbf{R}$. The proposed induction must derive the drop-out bound from the lower-level island bounds while retaining the Wilson weight of the carrying term over X. Until that derivation is verified for the actual operations, the drop-out majorant remains a required input, as in Section 13. Let the constants be chosen in order, and let the torus be large compared with the $M\check{R}_j$-cubes at each level. If, in addition, $g_j \le \gamma$ and

$$\lambda_\sharp(j-i) - c_5 \le x_i - x_j \le B_\sharp(j-i)$$
$$(i \le j \le k+1),$$

hold, then $\rho_0 \in \mathfrak{F}_0$ and $\mathbf{RT}\mathfrak{F}_k \subset \mathfrak{F}_{k+1}$ *with the same constants*. No family of a level $j \le k$ is altered; only the background at which they are evaluated changes, $U_k \to U_{k+1}$. The new families come out of $\mathbf{T}$ with a surplus, and $\mathbf{R}$ uses it up: radii $(1+\beta_{\mathrm{s}})\alpha_{i,k} \ge \alpha_{i,k+1}$ and rate $(1+4\beta_{\mathrm{s}})\kappa$, where β_{s} is a fixed fraction. For given $(x_0,\dots,x_j)$, the activities $E^{(j)}(X,\cdot)$, $R^{(j)}(X,\cdot)$ do not depend on the label. When X does not wrap around the torus, they depend neither on the torus nor on the cutoff. Further, $x_{k+1} = x_k - b_{k+1}(x_0,\dots,x_k)$ with $|b_{k+1}| \le B_\sharp$. When the label has no large fields, $|\mathbf{E}_k(U_k)| + |\mathbf{R}_k(U_k)| \le C|\mathbb{T}^{(k)}|$. For a general label, every layer is measured in its own scale, and for each $j \le k$ one adds the sites of level j that lie in a rim around the large-field region Z_j ((1e), (3a)). Finally, stability holds if the flow inequalities hold for all $i \le j \le K$:

$$\chi_k\, e^{-x_k A(U_k) - E_- |\mathbb{T}^{(k)}|} \le \bar\rho_k,$$

$$\int \bar\rho_k \le e^{E_+ |\mathbb{T}^{(k)}|},$$

$$E_+ \le \tfrac{3}{2}(N^2-1)\log x_k + O(1).$$

The large-field factor (clause (f) of 1a.2.7; (2c)). The operation $\mathbf{T}_k(Z)$ contains delta functions, tests and conditional Gaussian weights. Apart from the core factors of the islands in Z, it contains no Wilson weight. Hence, for a component produced by failed plaquette tests, $\mathbf{T}_k(Z)1$ carries no small factor. If no $\mathbf{R}$ has acted on Z, then under the tests of the term it obeys only the bound $\le \exp[C\sum_{t\le k} \#_t(Z_t \cap Z)]$. The small factor lies in three places. They depend on pairwise disjoint sets of plaquettes, factors and bonds, so no part of the small factor is used twice. The first is the Wilson weight: near a failed plaquette test of step j, $x_j A \ge c\, p_0(g_j)^2$. The second is the kept gauge-fixing exponents; together with the Wilson sum, they pay for a failed approximate fluctuation test. The third is the Gaussian forms of the fluctuation fields, which pay for a failed fluctuation test.

(1) Fix a history, and a component X of Z_k that has not been acted on by any $\mathbf{R}$. Let $(j_i, \mathfrak{c}_i)$ be the creation cells of the history in X, namely those cells of the creation set of step j_i, which are $M\check{R}_{j_i}$-cubes, that meet a failed test (2c). Then, for all values of the exterior variables, and with the norm taken as the total variation of that part which the term of the history carries,

$$\big\|\mathbf{T}_k(X)\big[\chi_{k,\mathcal{S}}\, e^{-A(x_k(\cdot),U_k)}\big]\big\| \le \prod_i e^{-c\,p_0(g_{j_i})^2}\, \exp\Big[C\sum_{t\le k}\#_t(Z_t\cap X)\Big]\,:$$

there is one factor for each creation, taken at the coupling of its own step, and one constant for each site.

(2) Bounds of this kind do not multiply over the components, because the background couples the components to one another.

(3) What does multiply, piece by piece, is the pointwise majorant on which (1) rests. A piece (t, D) consists of a component D of the large-field region formed at step t, whose enlargement $D^{\sim}$ shares no cube with any component that $\mathbf{R}$ has acted on at a level in $[t, k]$, together with the stored variables $\xi_{t,D}$ belonging to it and the weight $\mathsf{w}_{t,D}$ of those variables. Throughout the support of the history,

$$A(x_k(\cdot), U_k) \ge A_{\mathrm{cr}},$$

$$e^{-A_{\mathrm{cr}}}\prod_{(t,D)}\mathsf{w}_{t,D} \le \prod_{\text{creations}} e^{-c\,p_0(g_t)^2}\prod_{(t,D)}\bar{\mathsf{w}}_{t,D}(\xi_{t,D}),$$

$$\int \bar{\mathsf{w}}_{t,D} \le e^{C\#_t(D)},$$

where A_{cr} denotes the Wilson sums near the creations in these pieces. Only those creations are counted on the right-hand side. Each $\bar{\mathsf{w}}_{t,D}$ is a function of the variables of its own piece only.

(4) After integration, take a whole label together with the list of those outermost components (i, X) that $\mathbf{R}$ has acted on and that still stand in the label. For some history having this label and this list, the bound carries one factor for each of its creation cells lying outside the components that $\mathbf{R}$ has acted on and that still stand in the label (islands and drop-outs; the creations inside such a component are spent in its mass), and one mass for each outermost one among these components:

$$\int \mathbf{T}'_k(\mathcal{S})\big[\chi_{k,\mathcal{S}}\, e^{-A(x_k(\cdot),U_k)}\big]dV_k \le \exp\Big[C\sum_j(1+\log x_j)\#_j(Z_j)\Big]\prod_{\text{creation cells }(t,\mathfrak{c})} e^{-c\,p_0(g_t)}\prod_{(i,X)}\mathsf{M}(i,X),$$

$$\mathsf{M}(i,X) = \frac{1}{|W_4|}\sum_\omega e^{c\,\mathsf{v}(\omega)}\int \sup_{\mathbf{p}}\big|e^{\Phi_\omega(\xi;\mathbf{p})}\big|\,\mu_\omega(d\xi),$$

where ω ranges over the local histories of X, each equipped with a hypercubic frame in W_4 of the gauge tree, and ξ denotes the stored variables. The variable $\mathbf{p} = (\mathbf{U}, \mathbf{J})$ ranges over the complex pairs on $X^{\sim}$ at which a term evaluates its factor e^{Φ_ω}. The exponent has the form $\Phi_\omega = \Phi^\circ_\omega(\xi) + \sum_f \Phi_{\omega,f}(\xi; \mathbf{p}|_{R_f})$ with $R_f \subset X^{\sim}$, and $\mathsf{v}(\omega)$ is a space-time volume. The supremum is taken inside the integral and the sum. This is the form of the mass for a healed component or an island. In the case of a drop-out, the Wilson weight of the carrying term over X appears inside the integral, and the supremum extends over the evaluations at ξ. These are the real configurations that are compatible with the stored variables and regular layer by layer, and at which every test listed by ω takes its recorded value, with the exception of the test whose failure made X a drop-out. They form a set fibred over ξ (2c). The induction carries the mass bound at

the coupling of level i of the action. In this bound the c of $2-c$ is small, whatever the c in the weight $e^{c\,\mathsf{v}(\omega)}$. The mass evaluated at a later level does not exceed the mass at level i.

(5) The integral operation $\mathbf{T}'_k(X)$ on a component on which $\mathbf{R}$ acts contains, except in the case of a drop-out, a Wilson factor of its own, namely that of the core. When the component is healed or remains as an island, the total mass that is bounded is that of this operation, not that of $\mathbf{T}_k(Z)$:

$$\mathbf{T}'_k(X)\,1 \le \sup \exp\Big\{ \sum_{j\le k} C M^4 \check{R}_j^5\, d'_j(Z_j)\Big\} \prod_{j\le k}\prod_{i} \exp\Big(- c\, p_0(g_j)^2 \big(d'_j(Z_j^{(i)})+1\big) - 2p_0(g_j)\Big),$$

where the supremum extends over every admissible choice of the domains Z_j, $Z_j^{(i)}$ (Bałaban, Comm. Math. Phys. 122 (1989) 355, (1.79)), and the sums over the admissible sequences are counted among the costs. The inner product is taken over the regions $Z_j^{(i)}$ created at step j. For a component Z of Z_j which has been neither healed nor marked, the factor is written $\exp(-\kappa_j(Z) - 2p_0(g_{j(Z)}))$; here $j(Z)$ is the level at which the first creation in Z occurs. The *credit* satisfies $\kappa_j(Z) \ge \sum_{n=j+1}^{j+\mathsf{k}} C M^4 \check{R}_n^5\, d'_n(S^{n-j}Z)$, where the sum runs over the k further steps at which nothing is created. Here S denotes the hull in the next partition with a margin added. The number of $M\check{R}_n$-cubes that meet the region takes the place of the linear size d'_n, because a single cube has $d'=0$ and still carries a cost. The credit covers the volume costs of every step until $\mathbf{R}$ acts on Z. Once no further step is owed, $\mathbf{T}'_k(X)1 \le e^{-(2-c)\,p_0(g_k)}$. Moreover, in the weighted sum over the labels, every component C of Z_k keeps a factor $\exp[-c\,p_0(g_k) - \kappa_1 |C|_k]$, where $|C|_k$ is its number of cells.

Proposition 1a.2.8 (the small-field step as a map on actions). Suppose that the tests and $\mathfrak{G}_k$ are formed at t, where t is large, and let $x \in \mathbb{C}$ with $|x-\mathsf{t}| \le c\,\mathsf{t}$. Suppose that the analyticity radii (α_0, α_1) of the levels $j \le k+1$ are small, with $CM\alpha_0 < \alpha_1$ and $\alpha_1^2 \le C\alpha_0$ at each level. Suppose also that they are comparable: the quotient of the radii of level $k+1$ by those of any level $j \le k$ lies between $(1+\beta_{\mathrm{s}})^{-1}$ and $(1+\beta_{\mathrm{s}})(k+2-j)^{1/2}$. These are the admissible systems of (1d). Let S be local, analytic and *renormalised* in the same way as $\mathbf{E}_k + \mathbf{R}_k$:

$$S = \sum_{j\le k} \Big[\sum_{z} \sum_{X \ni z} \big(G^{(j)}(X,\cdot,z) - G^{(j)}(X,\mathbf{1},z)\big) - e_j A\Big] + \sum_{j\in\mathcal{J}} \sum_{X} \big(H^{(j)}(X,\cdot) - H^{(j)}(X,\mathbf{1})\big),$$

$$N(S) = \sum_{j\le k} \omega_{k+1-j} \|G^{(j)}\|_{j,\kappa} + \sum_{j\in\mathcal{J}} \mathsf{q}_j^2 \|H^{(j)}\|_{j,\kappa},$$

$$\omega_n \le C n^C L^{-\alpha_{\mathrm{H}} n},$$

where the pointed totals of $G^{(j)}$ are covariant under all isometries of $\mathbb{T}^{(j)}$, and e_j is the coefficient (1e) of $G^{(j)}$. The totals of $H^{(j)}$ are invariant under the isometries of $\mathbb{T}^{(k+1)}$. Finally, q_j is the ratio of the analyticity radii of level $k+1$ to those of level j. Geometric weights occur only in the first sum. Both symmetries are part of the definition of "renormalised". The covariance of the pointed totals is what makes the bracket left after the subtraction of $e_j A$ irrelevant (1e). Consider W in a class of real block fields, namely those at which every background formed in the step exists and yields pairs in the domains of the terms. Then, on the enlarged radii and uniformly in k,

$$\mathfrak{E}_{k+1}[\mathsf{t}; x; S] = \log Z^{(k)} + \sum_{X\in\mathcal{D}_{k+1}} \mathsf{F}(X, U_{k+1}) + \text{const}$$

, where

$$|\mathsf{F}(X,\cdot)| \le C\big(|1-x/\mathsf{t}| + \delta(\mathsf{t}) + \mathsf{s}\,N(S) + e^{-c\,\mathsf{t}\delta(\mathsf{t})^2}\big)\, e^{-(1+4\beta_{\mathrm{s}})\kappa d_{k+1}(X)},$$

with holomorphic dependence on x and on the parameters of S. The map $S \mapsto \mathsf{F}$ satisfies a Lipschitz bound with constant $C\mathsf{s}$; here $\mathsf{s} \ll 1$ denotes the quotient of the fluctuation threshold by the analyticity radius. Everything above holds under the condition $\mathsf{s}\,(1+N(S)) \le c$. Along the run, $G^{(j)} = E^{(j)}$ and $e_j = b_j$, while the $H^{(j)}$ are the aged $R^{(j)}$ of 1a.2.9.

Proposition 1a.2.9 (anatomy of step $k+1$). The insertions are made in the following order. First, the plaquettes of the local backgrounds of V_{k+1} are tested. Next, on the blocks P'' that lie away from failures and from Z_k, where the $V(y,x)$ are defined, the factor $\mathfrak{G}_k(\cdot;P'')$ is inserted. Then the approximate fluctuation $V_k(V_{k,\square}^{\mathrm{bg}})^{-1}$ is tested, and this determines Ω_{k+1}. On that region V_k^{bg} is defined and $V_k = e^{i\mathcal{A}_k} V_k^{\mathrm{bg}}$ is set. Finally, $\mathcal{A}_k$ is tested, and this determines Λ_{k+1}. The insertions and tests just described, together with the regions that they determine, are those of Bałaban's form, whereas the sharper tests on collar and rim, together with their outcomes, are given in (2a). As a formula,

$$(\mathbf{T}\rho_k)(V_{k+1}) = \sum_{P,Q,R} \int dV_k\, \delta(\mathcal{M}(V_k)V_{k+1}^{-1})\; \chi_{k+1}\chi_{k+1}^{c}(P)\; \mathfrak{G}_k(V_k)\; \chi_k'\chi_k'^{c}(Q)\; \chi^{(k)}\chi^{(k)c}(R)\; \rho_k(V_k).$$

For the node-free small-field step, apply 1a.2.5 and scale $\mathcal{A}_k = g_k B$. The Taylor expansion about the critical point then has the following form. On a label with seams, this is the interior expansion only: the additional node and seam contributions, including their constant and linear terms, remain in the boundary exponent and require its separate estimates.

$$x_k\big[A(U_k(V_k)) + \mathrm{GF}_k(V_k)\big] = x_k A(U_{k+1}) + \tfrac{1}{2}\langle B, \Delta^{(k)} B\rangle + \sum_b O(g_k|B|^3).$$

Completing the square in $B|_{\Lambda_{k+1}}$ produces $C^{(k)}(\Lambda_{k+1})$, together with the conditional Gaussian that is kept in $\mathbf{T}^{(k)}(Z_{k+1})$:

$$(\mathbf{T}\rho_k)(V_{k+1}) = \sum_{\mathcal{S}'} \chi_{k+1}(\Omega_{k+1})\, \mathbf{T}^{(k)}(Z_{k+1})\mathbf{T}_k'(\mathcal{S})\Big[e^{A_{k,\mathcal{S}}(U_{k+1}) + E_0^{(k)}(\Omega_{k+1})}\,\mathsf{I}\Big],$$

$$\mathsf{I} = Z^{(k)}(\Lambda_{k+1}) \int d\mu_{C^{(k)}(\Lambda_{k+1})}(B)\, \chi^{(k)}(\Lambda_{k+1})\, e^{\mathbb{P}^{(k)} + \{A_{k,\mathcal{S}}\}' + \mathsf{W}^{(k)}},$$

where $E_0^{(k)}(\Omega)$ consists of the first two terms of $\mathsf{E}^{(k)}(\Omega,\Lambda)$. The measure $\mu_{C^{(k)}(\Lambda_{k+1})}$ is the Gaussian measure conditioned on the exterior fluctuation field, and its mean is linear in that field. The prime indicates that the quadratic part of $-x_k A$ has been absorbed into the measure. The term $\mathsf{W}^{(k)}$ is left over from a change of variables on the strip, that is, on the part of S_{k+1} lying along $\partial\Lambda_{k+1}$. Before any expansion, the logarithm is divided into three parts by interpolation:

$$\log \mathsf{I} = \int_0^1 dt_1\, \big\langle \{\mathbf{B}_k - A(x_k(\cdot) - x_k, \cdot)\} + \mathsf{W}^{(k)} \big\rangle_{t_1,1} + \int_0^1 dt_2\, \big\langle \{\mathbf{R}_k''\} \big\rangle_{0,t_2} + \log \mathsf{I}\big|_{t_1=t_2=0},$$

where $\mathbf{R}_k''$ contains the young $R^{(j)}$, namely those whose cubes are larger than a block. The logarithm is expanded (1d) and regrouped (1a.2.6, with b_{k+1}, φ_{k+1} and $\mathsf{E}^{(k)}(\Omega_{k+1},\Lambda_{k+1})$). This gives the form of level $k+1$:

$$(\mathbf{T}\rho_k)(V_{k+1}) = \sum_{\mathcal{S}'} \chi_{k+1}\, \mathbf{T}'_{k+1}(\mathcal{S}') \exp\Big[A_{k,\mathcal{S}}(U_{k+1}) + \sum_{X\subset\Lambda_{k+1}} \big[E^{(k+1)} + R^{(k+1)}\big](X, U_{k+1}) + \sum_{X\cap Z^{\sim}_{k+1}\neq\emptyset} B^{(k+1)}(X; U_{k+1}, \underline{\mathfrak{f}}) + E_0^{(k)}(\Omega_{k+1}) + \nu_{\mathfrak{g}}|\Lambda^{(k)*}_{k+1}|\Big]$$

$$= \sum_{\mathcal{S}'} \chi_{k+1}\, \mathbf{T}'_{k+1}(\mathcal{S}') \exp A_{k+1,\mathcal{S}'}(U_{k+1});$$

after this, $\mathbf{R}$ acts. The terms with $X \subset \Lambda_{k+1}$ coincide with the terms of the pure small-field integral (among the $R^{(k+1)}$, here and in $\mathsf{E}^{(k)}$, a label keeps those whose X lies one cell inside Λ_{k+1}, and the others are counted among the $B^{(k+1)}$); this coincidence is the locality stated in 1a.2.7.

Proposition 1a.2.10 (a model case: the free field; the renormalisation group as nested orthogonal projections). Let the *free curvature* F be the standard Gaussian vector in the Euclidean space $\Omega^2_{\mathrm{ex}} = \operatorname{ran} d$ of exact 2-cochains on the bare lattice. Here d is the coboundary taking real 1-cochains to real 2-cochains on the bare lattice, and $d_{(j)}$ is the coboundary of $\mathbb{T}^{(j)}$. This vector stands for one colour of the gauge field, taken in the quadratic approximation and up to closed forms. The plain linear average and the straight linear average induce the same average of curvatures, called the *plaquette average*:

$$d_{(j)}(\mathsf{Q}^{\mathrm{pl}})^j = d_{(j)}(\mathsf{Q}^{\mathrm{s}})^j = \mathsf{A}^{(j)} d,$$
$$(\mathsf{A}^{(j)}F)(P) = L^{-4j} \sum_{z\in B^j(y)} \sum_{p\in \mathrm{Sq}^{(j)}(z;P)} F(p),$$

where $\mathrm{Sq}^{(j)}(z;P)$ is the set of L^{2j} bare plaquettes which together cover the plaquette P of $\mathbb{T}^{(j)}$ with corner y, once it is translated to z. Put $\mathsf{W}_j := \Omega^2_{\mathrm{ex}} \ominus (\Omega^2_{\mathrm{ex}} \cap \ker \mathsf{A}^{(j)})$, and let P_{W} denote the orthogonal projections. Then the spaces are nested, $\mathsf{W}_0 \supset \mathsf{W}_1 \supset \cdots \supset \mathsf{W}_K$. Given the block field of level k, the conditional expectation of F is $P_{\mathsf{W}_k}F$. This is the curvature of the linear background, namely the exact cochain of smallest norm that has the prescribed plaquette averages. The covariances of the background and of the fluctuation are two complementary orthogonal projections, $\mathbb{E}\big[P_{\mathsf{W}_k}F(p)\, P_{\mathsf{W}_k}F(q)\big] = P_{\mathsf{W}_k}(p,q)$ and $P_{\mathrm{ex}} = P_{\mathsf{W}_k} + P_{\mathsf{W}_k^{\perp}}$, where P_{ex} is the projection onto Ω^2_{ex}. Finally,

$$F = P_{\mathsf{W}_K}F + \sum_{j=1}^{K} \mathcal{C}_{(j)}F,$$
$$\mathcal{C}_{(j)} = P_{\mathsf{W}_{j-1}} - P_{\mathsf{W}_j} = \mathcal{H}_{j-1} C_j^{\mathrm{tr}} \mathcal{H}^*_{j-1},$$
$$|\mathcal{C}_{(j)}(p,q)| \le C\eta_{j-1}^4 e^{-\delta|x(p)-x(q)|/a_{K-j+1}},$$

where the summands are independent and the covariance of each summand is its own projection. Here, for a block field B of level $j-1$, $\mathcal{H}_{j-1}B$ denotes the curvature of its linear background, namely the exact 2-cochain on the bare lattice of smallest norm satisfying $\mathsf{A}^{(j-1)}\mathcal{H}_{j-1}B = d_{(j-1)}B$; and C_j^{tr} is the fluctuation covariance, in a tree gauge, of step j. Thus step j integrates out precisely $\mathcal{C}_{(j)}F$, and the amplitude and range of each slice are those of its own scale. In this model the semigroup law is the nesting of the spaces, and the delta constraint is conditioning. The nesting of backgrounds takes the form $P_{\mathsf{W}_j}P_{\mathsf{W}_k} = P_{\mathsf{W}_k}$ $(j \le k)$. Three features of the full theory are absent from the model: the fibre is flat, no coupling runs, and no field is large. The current

of a linear background is not zero: $d^* P_{\mathsf{W}_k} F = ((\mathsf{Q}^{\mathrm{s}})^k)^* d^*_{(k)} \phi$, which is the image of a Lagrange multiplier. What is missing is the term $\langle \mathsf{H}_k C_k^{(2)}, J \rangle$ of 1a.1.10, because $C_k^{(2)} = 0$ on a flat fibre. In the non-linear theory the current and the multiplier do vanish, at $U = \mathbf{1}$. For this reason the linearisation at that point of the minimiser of A subject to $\mathcal{M}^k U = V$ is the minimiser of $\frac{1}{2}\|d\mathsf{a}\|^2$ subject to the linearised constraint, and the second-order part of the averages makes no contribution. Hence the model is the linearisation at $\mathbf{1}$. It serves as the template for (7a), (7b).

Lemmas 1a.3.1, 1a.3.2 (axial gauges). Consider a box of side D on which $|\partial V - \mathbf{1}| \le \epsilon$. Among the representatives $V^{\psi'}$ with $\psi'(y_P) = \mathbf{1}$, the configuration $V^{\mathrm{ax}} = V^{\psi}$, $\psi(w) = V(\Gamma_{y_P, w})$, is the only one that equals $\mathbf{1}$ on the tree. The non-abelian Stokes theorem expresses $V^{\mathrm{ax}}(b)$ as a product of conjugated plaquette variables, at most $|b_- - y_P|_1$ in number:

$$V^{\mathrm{ax}}(b) = \prod_{i=1}^{m} \psi(w_i)\, \partial V(p_i)^{\pm 1} \psi(w_i)^{-1},$$

$$|V^{\mathrm{ax}}(b) - \mathbf{1}| \le |b_- - y_P|_1\, \epsilon \le 4D\epsilon;$$

Suppose that $D\epsilon \le c$ and that $F = \frac{1}{i} \log \partial V$ is Hölder along combs; it is distinct from the field strength F_V introduced in the Notation, §2, being the logarithm of each plaquette variable. Then $\frac{1}{i} \log V_\mu^{\mathrm{ax}}(x) = \sum_{\rho<\mu} (x_\rho - y_\rho) F_{\rho\mu}(y_P) + O(|x - y_P|^2) \times (\text{Hölder constant} + \epsilon^2)$, which is the expansion used in (1e) and (2b). Now suppose that $U_0, U \in \mathfrak{U}_k(\Omega_\bullet, \alpha_0)$, that U is tower axial, and that the averages at the tops differ by an amount $< \alpha_1$. Then $|\mathcal{M}^j U (\mathcal{M}^j U_0)^{-1} - \mathbf{1}| < C\alpha_0 + \alpha_1$ at every level, uniformly in k. This bound holds bond by bond and is coarser than what scaling requires. On a single level (compare the rigidity Lemma 1a.1.9), $\mathcal{A}_k$ is defined, and is $O(\varepsilon_k)$, only after the gauge has been fixed.

Proposition 1a.3.3 (Faddeev–Popov identities with Jacobian one). Let a compact group H act, and let $\mathsf{k} \ge 0$ satisfy $\int_H \mathsf{k}(h \cdot x)\, dh = 1$. Then for every finite complex measure ν and every bounded invariant F,

$$\int \nu(dx) \int_H dh\, \mathsf{k}(h \cdot x)\, F(h \cdot x) = \int F\, d\nu;$$

The measure obtained in this way is constructed as follows: one chooses h with law $\mathsf{k}(h \cdot x) dh$ and replaces x by $h \cdot x$. This measure equals $\mathsf{k}\,\bar{\nu}$, where $\bar{\nu}$ is the gauge average of ν. When ν is invariant, it is the product $\mathsf{k}\,\nu$. If $u = \mathbf{1}$ on the coarse sites, then $V^u(y, x) = V(y, x) u(x)^{-1}$. Where $V(y, x)$ is undefined, no factor is inserted at the pair (y, x), and with this convention

$$\int \prod_x du(x)\, \mathfrak{G}_k(V^u; P'') = 1,$$

$$(\mathbf{T}\rho_k)(V_{k+1}) = \int dV\, \delta(\mathcal{M}(V) V_{k+1}^{-1})\, \mathfrak{G}_k(V; P'')\, \rho_k(V).$$

In addition, $\mathfrak{G}_k(V^v) = \mathfrak{G}_k(V)$ whenever v is constant on blocks. The same holds for delta functions on spanning trees. Let T be a tree whose root lies outside the sites of the group, and let V_0 be any configuration on T. Then a unique u satisfies $(u \cdot V)|_T = V_0$, and the measure dV splits into factors along the tree:

$$\int f\, dV = \int du \int \prod_{b \notin T} dS(b)\, f\big(u \cdot (S \sqcup V_0)\big),$$

so

$$\int dV\, \delta_T(VV_0^{-1})\, \rho F = \int dV\, \rho F$$

for invariant ρ, F. Finally, the delta function can be removed: the gauge average of $\Psi\, \delta_T(VV_0^{-1})dV$ is $(\Psi \circ \pi_T)\, dV$, where π_T is the projection onto the slice along the orbits.

Lemma 1a.3.4, Proposition 1a.3.5 [to be proved; Package A] (the Landau gauge). Let U be tower axial and let $W = U^{u^{-1}}$. Then $\mathsf{n}_W = u^{-1}\bar{R}_{U_0}u$, so a u with trivial averages leaves the straightened averages exactly unchanged. In linearised form this reads $Q_j^\sharp(U_0)D\lambda = D_{\mathcal{M}^j U_0}Q_j'(U_0)\lambda$. Hence $Q^\sharp A$ does not change under $A \to A - D\lambda$, $\lambda \in N(Q')$. Up to terms that contain $J(U_0)$, gradients are null directions of Δ^{W}. Indeed $D^*J(U_0) = 0$, and, writing λ_+ for $\lambda(b_+)$ transported to b_-,

$$\langle A - D\lambda, \Delta^{\mathrm{W}}(A - D\lambda)\rangle = \langle A, \Delta^{\mathrm{W}}A\rangle - \big\langle i[\lambda, A] - i[A, \lambda_+] - i[\lambda, D\lambda],\, J(U_0)\big\rangle.$$

The projection is given by

$$R = 1 - G'Q'^*(Q'G'^2Q'^*)^{-1}Q'G',$$
$$\operatorname{ran} R = \{f : Q'G'f = 0\},$$
$$\inf_{\lambda \in N(Q')} \|D^*A - \Delta\lambda\|^2 = \|(1-R)D^*A\|^2,$$

and $R\Delta\lambda = \Delta\lambda = G'^{-1}\lambda$ for $\lambda \in N(Q')$. The map $\lambda \mapsto (R\Delta\lambda, Q'\lambda)$ is a bijection onto $\operatorname{ran} R \times \{\text{functions on tops}\}$. Its inverse is $(f, \nu) \mapsto G'f + H'\nu$, where ν is a function on the tops and H' is a right inverse of Q', constructed from G', with $R\Delta H' = 0$. Let $U \in \mathfrak{U}_k(\Omega_\bullet, \alpha_0)$ and let M_1 be large. Assume that $M\alpha_0$ (and therefore $M_1\alpha_0$) lies below a number that depends only on L, and that $\mathbb{T}^{(k)}$ has at least CM_1 sites in each coordinate direction. Then the form

$$\langle A, \Delta_U^{\mathrm{W}}A\rangle + \|RD_U^*A\|^2 + \langle Q^\sharp A, aQ^\sharp A\rangle$$

is positive definite. Its inverse $\mathsf{G}_k(U)$ maps $|\cdot|_{(-3)}$ into $\|\cdot\|_1$, with a norm that depends only on L. On the set $\{RD^*A = 0,\ \xi Q^\sharp A = B\}$ the form differs from the Hessian by a constant. Consequently $\mathsf{H}_k B = \mathsf{G}Q^{\sharp *}(Q^\sharp \mathsf{G} Q^{\sharp *})^{-1}\xi^{-1}B$ minimises the Hessian itself under the constraint, and $RD^*\mathsf{G}Q^{\sharp *} = 0$. Here G stands for Bałaban's G, and not for the inverse of the displayed form, which is his G_0. The formula for $\mathsf{H}_k B$ and the identity just stated hold exactly only when the Hessian in the form is replaced by its invariant extension Δ_π. This extension is defined by $\langle A, \Delta_\pi A\rangle = \langle A - D\lambda, \Delta^{\mathrm{W}}(A - D\lambda)\rangle$, where $\lambda \in N(Q')$ is the one with $RD^*(A - D\lambda) = 0$. The replacement is a perturbation of size $CM\alpha_0$, which is small together with the current because α_0 is chosen after M. The bounds persist under this perturbation, and it vanishes where $J = 0$. In the symbols of the Notation, §5, this operator equals $\mathsf{H}\,\xi^{-1}$. There H denotes the linear minimiser constructed from G, and it satisfies $Q^\sharp \mathsf{H} = 1$. Where $J = 0$ and $\xi = 1$, the operator coincides with the H_k of the Notation. It carries the normalisation for the constraint $\xi Q^\sharp A = B$. With his Q^{B}, Bałaban uses both normalisations: the minimiser under $QA = B$ and the minimiser under $L^j \eta\, Q_j A = B$. On layer j the two differ by the factor $\xi^{-1} = (L^j\eta_k)^{-1}$. The normalisation with ξ is the one taken in his paper on the variational problem.

Theorem 1a.3.6 (from the axial to the Landau gauge). Work in the scheme c, in which Bałaban's average of 1985 replaces $\mathcal{M}$ in every object, and let c_1, B_1 depend only on L. Assume the hypotheses of Lemma 1a.3.2 [to be proved; Package A], with the full gauge $\mathrm{Ax}(\Omega_\bullet, U_0)$ and not Ax°, and assume $\alpha_0 + \alpha_1 \le c_1$. Then U has exactly one Landau pair that satisfies

$$|A|_{(-1)},\, |\nabla_{U_0} A|_{(-2)},\, |D^*DA|_{(-3)},\, |\Delta_{U_0} A|_{(-3)} \le B_1(\alpha_0+\alpha_1),$$
$$\sup|u-\mathbf{1}| \le CB_1(\alpha_0+\alpha_1),$$

that satisfies Hölder bounds on ∇A (these bounds give rise to the α_{H} below), and that has the straightened averages of U,

$$\mathcal{Q}^\sharp_j(U_0;A) = B \ \text{ on the bonds at } \mathbb{Y}_j,$$
$$B = \tfrac{1}{i}\log\big[\mathcal{M}^j U(\mathcal{M}^j U_0)^{-1}\big],$$
$$|B| \le C\alpha_1$$

(with an explicit correction entering on bonds that join two layers). For $\mathcal{M}$ itself, under the same hypotheses with $\alpha_0+\alpha_1 \le c$, there exists a pair (A,u) in the slice of (1b) that has these averages and satisfies $\|A\|_1,\ |D^*DA|_{(-3)} \le C(\alpha_0+\alpha_1)$. That slice has the same form as the Landau gauge but a different projection, whose site averages run along combs from the block corners.

Proposition 1a.3.7 (a priori bounds on the slice). Assume that $U_0, e^{i\eta_k A}U_0 \in \mathfrak{U}_k(\Omega_\bullet,\alpha_0)$, that $RD^*A=0$, and that the averages are as above. Assume further that the only bound given is $|A|_{(-1)} < \alpha_2$, with $\alpha_0,\alpha_1,\alpha_2 \le c$ (c depending only on L) and $C\alpha_2^2 \le \alpha_0+\alpha_1$. Then, in the scheme c, all the bounds of the theorem on A hold, with $B_1 = CL\times$ the norms of $\mathsf{G}_k(U_0)$, $\mathsf{H}_k(U_0)$. For $\mathcal{M}$, the bounds on $\|A\|_1$ and $|D^*DA|_{(-3)}$ hold, in the slice of (1b).

Proposition 1a.3.8 (regular means small potential, locally). Let $U \in \mathfrak{U}_k(\Omega_\bullet,\alpha_0)$, take a cube of side $M\xi_j$ in Ω_j, and assume $CM\alpha_0 \le c_1$. Then for some u

$$U^{u^{-1}} = e^{i\eta_k A},$$
$$\xi_j|A|,\ \xi_j^2|\nabla A|,\ \xi_j^3|D^*_{\mathbf{1}}D_{\mathbf{1}}A|,\ \xi_j^3|\Delta A| \le CM\alpha_0,$$

and, in the scheme c, the box axial potential of the averaged field gives the averages of A: $\mathcal{Q}^\sharp_j(\mathbf{1};A) = \frac{1}{i}\log(\mathcal{M}^j U)^{\mathrm{ax}}$. Conversely, let $U_0 \in \mathfrak{U}_k(\Omega_\bullet,\alpha_0)$ and $\|A\|_1, |D^*DA|_{(-3)} < \alpha_2$, with α_0,α_2 small. Then $e^{i\eta_k A}U_0 \in \mathfrak{U}_k(\Omega_\bullet,\alpha_0+3\alpha_2)$. Now take a complex $\mathbf{U} = e^{i\eta_k A'}U$ with $\xi_j|A'| < \alpha_1$ whose own plaquettes are small, $|\partial\mathbf{U}-\mathbf{1}| < \alpha_0\eta_k^2\xi_j^{-2}$ on the cube. (From $\|A'\|_1 < \alpha_1$ this follows only with $\alpha_0+C\alpha_1$ in place of α_0, 1a.1.6.) Assume also $CM(\alpha_0+\alpha_1) \le c$. Then the averages are V^v. Here V is what the averages of $\mathbf{U}$ become in the comb gauge of the cube, and $v = v'v_{\mathrm{r}}$ is a G^c-valued function on the nodes. The factor v_{r} is G-valued and does not depend on A', the factor v' is holomorphic in A', and

$$|V-\mathbf{1}| \le CM\alpha_0,$$
$$\mathrm{cn}(v) = |v|\,|v^{-1}| \le e^{CM\alpha_1},$$
$$|v'-\mathbf{1}| \le CM\alpha_1.$$

The third bound of Bałaban, $|v-\mathbf{1}| \le CM\alpha_1$, holds for v'. For v it holds only at pairs normalised by $U(\Gamma_{y_\square,x}) = \mathbf{1}$. Indeed, for $U=\mathbf{1}^g$, $A'=0$ one has $v = g(\cdot)g(y_\square)^{-1}$, which is arbitrary, whereas $\mathrm{cn}(v)=1$.

Propositions 1a.3.9, 1a.3.10 (transversality, locality, node factor; the choices disappear). The slice is transversal. Let $\tilde{A}$ be real with $\|\tilde{A}\|_1 \le c$, and let a G-valued y near $\mathbf{1}$ be prescribed on the tops. Then some $w = e^{i\lambda}$ brings $\tilde{A}$ into the slice with $r_j = \mathsf{y}$ on $\mathbb{Y}_j$, and

$\|\lambda\|_0 \le C(|D^*\tilde A|_{(-2)} + \sup|\mathsf{y}-\mathbf{1}|)$. Here r_j is defined as in 1a.1.8 and formed at $\tilde A$; at $\tilde A = 0$ it is $\bar R^{(j)}_{U_0} w$. The pair (A,u) depends locally on the data $(J(U),B)$, with tails $Ce^{-\delta d(y,y')}$ and $\delta < \delta_0$. Distances are measured in local scales, $J(U)$ in $|\cdot|_{(-3)}$, B in the sup norm and A in $\|\cdot\|_1$. Along differentiable families of U, this locality follows from the relation

$$\delta A = \mathsf{G}_k(U_0)\big[\operatorname{Ad}(u^{-1})\delta J(U) + \mathcal{L}\delta A\big] + \mathsf{G}_k(U_0)Q^{\sharp *}\xi^{-3}\big[\delta B - \partial_A C(A)\delta A\big],$$

and, for backgrounds over a real datum $\mathbf{V}$ that is regular in the sense of (1b) (its test loops lie within a small ε of $\mathbf{1}$, and, where inward bonds are kept, so do the two-scale loops at the seams), the pair depends holomorphically on the datum within a ball of radius c, the representative at $e^{iB}\mathbf{V}$ being $(e^{i\eta_k A(B)}U_k(\mathbf{V}))^{h(B)}$. None of these statements depends on k. The plain averages of $e^{i\eta_k A}U_0$ are the datum transformed by the *node factor*: $\mathcal{M}^j(e^{i\eta_k A}U_0)(c) = \mathsf{n}(c_-)V_j(c)\mathsf{n}(c_+)^{-1}$, where $\mathsf{n} = u^{-1}|_{\text{nodes}}$. At the tops the node factor is the iterated n_W, namely $v_j[A]$. At a junction node it contains one further factor, the quotient of the datum by the reduced average on one bond ending at that node. Every ingredient of a step is gauge covariant, and the map $(A,u) \to (\operatorname{Ad}(v)A, vuv^{-1})$ takes the set of Landau pairs of U with respect to U_0 onto the set of Landau pairs of U^v with respect to U_0^v. Consequently $E^{(k+1)}(X,\cdot)$ is invariant under

$$(\mathbf{U},\mathbf{J}) \to (\mathbf{U}^u, \operatorname{Ad}(u)\mathbf{J})$$

for G-valued u. It is therefore also invariant for the G^c-valued u that are connected to these within the domain, and it can be evaluated in any gauge. A function F constructed on a slice is invariant under the subgroup that preserves the slice, and it has a unique invariant extension, $F^\natural(\mathfrak{z}) = F(\mathsf{c}[\mathfrak{z}]\cdot\mathfrak{z})$. Here $\mathsf{c}[\mathfrak{z}]$ is the transformation that takes $\mathfrak{z} = (U, A', \mathbf{J})$ to the slice. The extension retains locality, the bounds, and holomorphy in $(A',\mathbf{J})$ at fixed real U.

Lemma 1a.3.11 (the block axial gauge outside the nodes). Near a seam, the block of a coarse site contains nodes, and every element of $\mathcal{G}^{\text{res}}$ is pinned at these nodes. In that case the gauge condition is $W(y,x) = \mathbf{1}$ for $y \in P''$ and for those $x \in B(y)\setminus y$ other than nodes. For a pair of nodes, the value $(\mathcal{M}^i U)(y,x)$ is invariant on the orbit, and the datum does not contain it. If P'' consists of nodes, the gauge is met once by every orbit of $\mathcal{G}^{\text{res}}$, meaning that the averages $\mathcal{M}^i U$ on the bonds within $B(P'')$ and the nodes depend only on the orbit. Hence $V_k^{\text{bg}} = \mathcal{M}^k U$, with U in the orbit $U_{k+1}(\mathbf{V})$ and in the gauge, is a function of the datum; this is the case $i = k$, with $k+1$ in place of k. In this situation

$$\operatorname{GF}_i(\mathcal{M}^i U; P'') = \sum_{y\in P''} \sum_{x\in B(y)\setminus y,\, x \text{ a node}} \big[1 - \tfrac12 \operatorname{Re}\operatorname{tr}(\mathcal{M}^i U)(y,x)\big],$$

which is constant on the orbit. Near a seam, this is what is left of the identity $\operatorname{GF}_k(V_k^{\text{bg}}) = 0$.

Proposition 1a.3.12 (a generalisation of Theorem 1a.3.6, in the scheme c). Fix $\gamma' > 0$. Where the constant B_1 of the theorem enters, γ' is not large; $\gamma' = 1$ is an example. Assume the hypotheses of the theorem, with c_1 depending on γ' as well; for arbitrary γ', the constant B_1 must also depend on γ'. Let $f \in \operatorname{ran} R$ with $|f|_{(-2)} < \gamma'(\alpha_0+\alpha_1)$. Then the contraction in $\|\lambda\|_0$ from the proof of the theorem produces, for λ in a ball of fixed radius, a unique u whose averages are trivial and for which $R\,D^*_{U_0}A = f$. Nothing more than this enters the consequence below. Bałaban's Theorem 8 [to be proved; Package E] also asserts, without proof, all the bounds of the theorem, with the same B_1. The second-order and Hölder bounds require, in addition, a bound on Df in $|\cdot|_{(-3)}$. As a consequence, the set $\{A : \mathcal{Q}^\sharp(U_0;A) = B,\ RD^*A = f \text{ for some small } f\}$ consists of pieces of gauge orbits, and each piece meets the slice $f = 0$ in exactly one point. On each piece the action takes a single value. Hence a point of the slice that is critical there is critical

on the thickened set as well, so that the Lagrange equations no longer involve the gauge condition. Bałaban makes use of this a single time, to obtain the second-order bounds on the minimiser. In (1c) the corresponding step, which passes from the slice to a multiplier attached to the averages only, is carried out with gauge directions the images of which under RD^* fill all of $\operatorname{ran} R$. The domains U_j^c have real centres that are regular, though not nearly critical, and at these centres the construction demands small local potentials outright.

Proposition 1a.3.13 (a seam in dimension one). Take all definitions on $\mathbb{Z} \supset L\mathbb{Z}$, and perform one step on the weight 1 on two blocks, the second of which is centred at y; the third block, $B(y')$, remains fine. The seam bond s is kept in the datum, p is the transport from s to y', and $f_{\mathfrak{G}}$ is the one-site factor of $\mathfrak{G}_k$. Let $V^+(c)$ be the coarse variable on $c = \langle y, y' \rangle$. Then the integral of the step can be computed exactly:

$$\Phi = f_{\mathfrak{G}}\big(V^+(c)\,\mathsf{p}^{-1} V(s)^{-1}\big).$$

Under the node-wise gauge group of the datum, this integral fails to be invariant. Let a transformation be g at the centre y and $\mathbf{1}$ at all remaining end points of bonds of the datum, the end of s among them. It changes the integral into $f_{\mathfrak{G}}(gV^+(c)\mathsf{p}^{-1}V(s)^{-1})$. The integral is invariant under the action in which the end of s moves with the centre of its block, and with respect to this action the tree insertion is a valid identity. Under the node-wise action, Φ has gauge average equal to the constant 1. The procedure of 1a.3.3 therefore produces the bare slice measure, not Φ times that measure, and in this case the insertion is not an identity. If s is integrated together with its block, then $\int dV(s)\Phi = 1$, and nothing remains to vary.

Proofs. *Averaging.* Every covariance property follows from the identity $\log(TXT^{-1}) = T(\log X)T^{-1}$. Each expression under a logarithm is conjugated at a single site (c_- for $\log W_x$). The norms are gauge invariant, and so therefore is the choice of rule. For 1a.1.1, in the chart $V = e^Y V_1$ the means are exactly the fixed points of the map $Y \mapsto Y + \frac{1}{n}\sum_i \log(V_i V_1^{-1} e^{-Y})$. This map is a contraction at rate $O(\operatorname{diam} + r)$, with bounds independent of n. Under a conjugation that is not unitary, diameters change by at most $\operatorname{cn}(T)$. The isometry r permutes the centred blocks and the families $\mathcal{G}$, but it carries a comb to a comb in a different order. This is why Bałaban's first average, built on cornered blocks and combs, needed to be symmetrised. Lemma 1a.1.2 (lattice homotopy): each interchange of two steps in a path costs one plaquette. The contours sweep out $O(L^2)$ plaquettes, and this determines $\operatorname{Dom}(c)$. In the comb gauge the bound increases with the size of the box, so small curvature yields a small potential only on boxes of bounded size. In the comb gauge one has $\mathcal{M}(e^{i\mathsf{a}}) = \exp(i\mathsf{Q}^{\mathrm{pl}}\mathsf{a} + O(|\mathsf{a}|^2))$. Since the contour terms σ cancel at the corners of p', the lattice Stokes formula yields

$$\sum_{c \in \partial p'} \pm(\mathsf{Q}^{\mathrm{pl}}\mathsf{a})(c) = L^{-4} \sum_{x \in B(y)} \sum_{p \subset S_x} \mathsf{a}(\partial p),$$

$$S_x = p', \text{ with corner } y, \text{ translated to } x,\ L^2 \text{ plaquettes.}$$

Consequently $\hat{\varpi}_n = \varpi(\mathcal{M}^n U)/\alpha_0(L^n\eta_k)^2$ satisfies $\hat{\varpi}_{n+1} \le \hat{\varpi}_n(1 + C\alpha_0 L^{-2(k-n)})$. Measured in its own scale, the curvature of level n is summable, and this outweighs the accumulation of k errors.

A single step (Proposition 1a.1.5) is proved by universality and Cauchy bounds, whereas Bałaban computes $\mathsf{Q}(W_0)$ in closed form from the Campbell–Hausdorff formula. In the comb gauge everything reduces to finitely many holomorphic functions that depend on L alone, and hence $\mathcal{Q}^1 = \mathsf{Q}(W_0)\mathsf{a} + O(|\mathsf{a}|^2)$. At $W_0 = \mathbf{1}$ one has $\frac{1}{i}\log \mathsf{u} = \sigma + O(|\mathsf{a}|^2)$. Thus the block factors take away precisely the contour terms of Q^{pl}, and $\mathsf{Q}(\mathbf{1}) = \mathsf{Q}^{\mathrm{s}}$. The kernels of $\mathsf{Q}(W_0)$ and $\mathsf{S}(W_0)$ are holomorphic and coincide at $\mathbf{1}$, so they differ by $O(\varpi)$. The norms of S reduce to counting: there

are $L^4 \cdot L$ pairs (x, b) for each coarse bond, each fine bond is contained in L segments, and every pair carries the weight L^{-4}. For k steps, the factorisation follows from covariance by induction. At a complex 1-form A, covariance under the G^c-valued v_n uses up part of the room in the domain, and it is applied with $\mathrm{cn}(v_n) = |v_n|\,|v_n^{-1}| \le 2$. Setting $t_n = \sup |\mathsf{a}_n|/L^n\eta_k$ and $\varpi_n = \varpi(W_{0,n})$, one has

$$t_{n+1} \le t_n\big(1 + C\varpi_n + CL^n\eta_k t_n\big),$$

$$\sum_n \varpi_n \le C\alpha_0,$$

$$\sum_{n<k} L^n\eta_k \le 2L^{-1}.$$

The recursion closes because $\|\mathsf{S}(W_{0,n})\|_{\mathrm{row}}$ does not exceed L, the ratio of the spacings, with no loss. This is the third gain of straightening. Since $\|\mathsf{Q}^{\mathrm{pl}}\|_{\mathrm{row}} \ge L + \frac{3(L^2-1)}{2L} > 2L$, a product of one-step bounds for the plain average grows geometrically in k. The derivative is bounded through column sums and Schur's test. When the plaquettes are complex, the real comb gauge does not work: at every level its error is as large as the bound itself, which gives only $1 + Ck\alpha_1$. The G^c-valued comb gauge of $\mathcal{M}^n\mathbf{U}$, whose condition number is $1 + C\alpha_1 L^n\eta_k$, gives $\varpi^c_{n+1} \le L^2\varpi^c_n(1 + C\alpha_1 L^n\eta_k)(1 + CL^2\varpi^c_n)$. The price of the non-unitary conjugation is therefore one factor per level, and these factors are summable. The determining sets are obtained by folding with reflections. The double block of order j beneath a bond of Γ_j lies in Ω_{j-1}. There the fields are, in the worst case, regular only in the next finer scale, and this costs one factor, a single time.

Intertwining: at a pure gauge $e^{i\mathsf{a}^w}W_0 = W_0^w$ one has $\mathsf{u}(W_0, \mathsf{a}^w) = w\,(\bar R_{W_0} w)^{-1}$. It is for the sake of this identity that $\bar R$ is defined as it is. 1a.1.8 rests on a coupled recursion for the size of r_n and for its covariant difference, which the same geometric sums close. For products one has $|\mathsf{z}_{j+1}| \le |\mathsf{z}_j| + C(|\mathsf{z}_j| + \alpha L^{j+1}\eta_k)^2$, because the defect added at each step is quadratic in the variation within a block. This is the origin of the second condition in the definition of a class. 1a.1.9: up to an error $C\varpi$, a bond inside a block is the quotient of two contour transports, and the average then determines the crossing bonds. The comparison is made factor by factor. 1a.1.10: the constraint is solved to second order by means of the right inverse H_k. At a critical point under the constraint, $J(U) = Q_k^\sharp(U)^*\omega$ is in general not zero. It is the image of a Lagrange multiplier ω, whose size is that of the curvature of V. The extra term does not occur in abelian theories, and it gives rise to G_1, H_1 in (1c). 1a.1.11 follows from the first identity together with $\mathcal{M}(\mathsf{r}U) = \mathsf{r}\mathcal{M}(U)$ and $(\mathsf{r}U)(\mathsf{r}y, \mathsf{r}x) = U(y, x)$. 1a.1.12 is purely algebraic. Writing $\hat u_n = u\,v_n$, the axial condition at level n takes the form

$$\hat u_n(x) = \mathrm{Ad}(W_{0,n}(z, x))^{-1}[\hat u_{n+1}(z)\mathsf{u}_n(z)^{-1}\mathsf{w}_n^{\mathrm{rel}}(z, x)]$$

. This fixes u level by level, starting at the top and going down, and it yields $\bar R_{W_{0,n}}\hat u_n = \hat u_{n+1}$. 1a.1.13: two chains of transports with common end points differ by the curvature that they enclose. At level n this is $O(\alpha_0 L^{-2(k-n)})$, so the differences form a geometric series. The transport t is unavoidable, because in each block the two schemes see a gauge transformation at two different sites. 1a.1.14 follows from linearity and the Stokes formula, which give $\Theta = L^{-4}\sum_{x\in B(z)} L[\mathsf{f}^\Sigma(x_1 + L) - \mathsf{f}^\Sigma(x_1)]$, together with a count.

The transformation. 1a.2.1 combines the definition of an image measure with a property of $\mathcal{M}$. The name "Group" is inaccurate. All that holds is the semigroup law $\mathbf{T}^n\nu = (\mathcal{M}^n)_*\nu$, which relates different spaces. In physical units the symmetry group becomes smaller as k grows; in the units of the level it is the symmetry group of the unit lattice. Finite range does not survive, since

$\mathbf{T}\rho_0$ couples all bonds. Reflection positivity is required of the bare measure only (4d), (6b), (10a). 1a.2.2: the variances are additive (taking a block mean divides by L^d, the rescaling multiplies by L^{d-2}, and the new kernel contributes $\mathfrak{a}^{-1}$). Nested backgrounds: backgrounds exist only over regular data, so minimising in two stages is not enough. The argument is instead as follows. A point that is critical on the larger fibre is critical on the smaller one, regularity improves in finer units, and uniqueness (1b) completes the proof. Because the argument involves critical points alone, it extends to complex data. The minimiser $\mathsf{H}B$ is the one given by the method of Lagrange multipliers. The delta kernel secures four things:

- closure under composition. Since $\mathcal{M}$ is non-linear, the composite of two soft kernels on G is not a function of W and $\mathcal{M}^2U$ alone.
- a single large term. The critical point does not depend on the coupling, the leading term is exactly $x_kA(U_{k+1})$, and terms produced at U_j can be evaluated at U_k at every later level.
- exact averages. Since $V_k = \mathcal{M}^kU$, a test on the block fields is a test on the bare field.
- no analytic loss. The mass $aQ^{\sharp *}Q^{\sharp}$ remains in G_k as a device, with $a > 0$ arbitrary.

The cost is that the fibre is curved. For the gauge fixing the opposite choice is made: it enters through a soft weight.

1a.2.3 (stochastic kernels, balance of masses). The mass, summed over the outcomes, is preserved by operations of three kinds. *Tests:* at any given state exactly one outcome takes the value 1. A test is counted as failed whenever the object it examines (a mean, a logarithm, a minimal orbit, a minimiser) is undefined, so every test is defined on all configurations. The actual test maps, including the selections of representatives and the failure sets of partially defined objects, must be Borel. Definability can establish this when it is checked for those maps and their selections; it is not supplied by the convention that an undefined object fails its test. *Draws:* the kernel $\kappa_{\mathfrak{G}}(V,\cdot) = \int du\, \mathfrak{G}_k(V^u)\delta_{V^u}$. *Maps:* adjoining $\mathcal{M}(V_k)$, and forgetting a variable. No smallness is involved in any of this. The operation $\mathbf{R}$ draws the variables of a component afresh, and after a Haar draw a plaquette trace in that component has integral 0. Because of the partitions, single terms have only restricted symmetry. Only the totals at a base point are covariant under the full group W_4, which (1e) requires. Base points are required because the sum $\sum_{X\subset\Lambda_j}$ is not invariant and the cancellation against $b_jA(\varphi_j,\cdot)$ must be made at each point separately. When z lies one layer of $M\check{R}_j$-cubes inside Λ_j, the restricted pointed total differs from the unrestricted one by $O(e^{-\frac{1}{2}\kappa\check{R}_j})$.

1a.2.4: a compact group admits no dilation. Hence x_k *is* the normalisation of the kinetic term. Gauge invariance links the quadratic, cubic and quartic parts of the marginal term, so the renormalisations of the field strength and of the coupling are given by a single number. The substitution $\mathcal{A}_k = g_k\mathcal{A}_k^{\mathrm{s}}$ takes the place of the rescaling. Consequently there is no scaling transformation, no fixed point and no eigen-direction. The word "irrelevant" can then mean only a bound $O(\eta_{k-j}^{4+\alpha_{\mathrm{H}}})$ at regular backgrounds, where $\alpha_{\mathrm{H}} < 1$ is a Hölder exponent (1e). Level k is a problem on a unit lattice that carries a fine lattice of spacing $\eta_k \to 0$. Every estimate in (1b), (1c) must therefore hold uniformly in η_k. 1a.2.6: the density itself does not change. Its exponent is always $-\beta A(U_k)$ together with the terms in their unrenormalised form. The point is which grouping of the terms admits uniform bounds:

- without grouping, the bound $\|E^{(j)}\| \le E_0$ yields no more than $CE_0\eta_k^{-4}|\mathbb{T}^{(k)}|$;
- the number $b = b_{k+1}$ by itself leaves a geometric sum over j;
- if the marginal terms are left among the corrections, they add up to c_0kA to leading order;
- once absorbed into x_k, they fix the next thresholds and g_{k+1}, the width of the following Gaussian.

The profiles φ_j keep track of the place at which step j was performed. In the constant, the contribution $-\frac{1}{2}(N^2-1)\log x_k$ for each free bond is the Jacobian of $\mathcal{A}_k = g_k B$. Laplace's method gives $z_{\mathfrak{G}} \approx C x_k^{-(N^2-1)/2}$. With four bonds per site, less one gauge direction, this gives $\mathsf{E}^{(j)}(T_\ell, T_\ell) = -(\frac{3}{2}(N^2-1)\log x_j + O(1))(|\mathbb{T}^{(j)}| - |\mathbb{T}^{(j+1)}|)$. The term $\log x_k$ in E_+ arises from the constants $\mathsf{E}^{(j)}(T_\ell, T_\ell)$, $k \le j < K$, which belong to the steps $k+1, \dots, K$ and remain owed in $\bar\rho_k = e^{-E}\rho_k$ (when $k = K$ no constant remains owed, and $E_+ = O(1)$). Over a large-field region, the constants differ by a volume factor $e^{C \log x_j \#_j(Z_{j+1})}$, and the small factors of (2c) must outweigh it.

1a.2.7 (the division of the work among the parts: 1a.2). Each part supplies the following:

- (1b): the backgrounds;
- (1c): the propagators;
- (1d): the new $E^{(k+1)}$, by a polymer expansion under a criterion of Kotecký–Preiss type;
- (1e): the coupling and the bounds on the sums;
- (2a): tests, regions and histories;
- (2b): $\mathbf{R}$;
- (2c): the large-field factor and stability;
- (2d): the terms that $\mathbf{R}$ leaves behind;
- (2e): the order of the constants and the inductive form;
- (3a): the flow;
- (3b): independence from the label, the torus and the cutoff.

The last item rests on locality. A term localised in X is assembled from walks, potentials and tests within X. It therefore does not depend on the label, and, when X does not wrap around the torus, it depends on neither the torus nor the cutoff. The small-field half requires $g_j \le \gamma$. It also requires $-C \le x_i - x_j \le C(j-i)$ for $i \le j \le k$, because the analyticity radii of level j run with g_j and the radii of two levels must be comparable. For radii that do not run, no relation between the couplings is required; Bałaban's remark that asymptotic freedom is not assumed refers to this case. For the complete model, what is used is the lower half of asymptotic freedom. At the levels of creation $j \le k$ the small factors were deposited, and since $x_j \ge x_k - C$ one has $p_0(g_j) \ge c\, p_0(g_k)$. The terms $R^{(j)}$ carry the vacuum subtraction but no counterterm for the coupling, and they are marginal. On the support of the tests, the following bound holds for every $X \in \mathcal{D}_j$ that contains a site z of Ω_k and remains within cML^{k-j} of z in level-j units (a longer X satisfies $d_j(X) \ge cL^{k-j} - C$ and has only the bound $2g_j^{\kappa_0} e^{-\kappa d_j(X)}$; around a site of $\Omega_n \setminus \Omega_{n+1}$, $j \le n < k$, the right-hand side holds with n in place of k):

$$|R^{(j)}(X, U_k) - R^{(j)}(X, \mathbf{1})| \le C(d_j(X)+3)^2 (x_j/x_k)\, \eta_{k-j}^4\, g_j^{\kappa_0} e^{-\kappa d_j(X)},$$

$$\sum_{j \le k} (x_j/x_k)\, g_j^{\kappa_0} \le C g_k^{\kappa_0 - 2} :$$

these terms do not decrease as $k-j$ grows. Only the linear growth of x_j makes them summable, and it does so for $\kappa_0 > 4$. Here x_j/x_k is, up to a bounded logarithmic factor, the squared ratio of the analyticity radii of levels k and j, that is, the relative quadratic evaluation weight of (ER15). If one assumes no more than $x_j \ge x_k - C$, the sum can grow to order $k g_k^{\kappa_0}$. It is this summability that makes a counterterm for them unnecessary. It is also why aged R-terms change b_j by no more than $C\gamma^{\kappa_0 - C}$, so that $\lambda_\sharp > 0$ is preserved (3a). The upper half ensures that the thresholds of adjacent levels are comparable: $\varepsilon_{k+1} \le C\varepsilon_k$.

The large-field factor: the reasons for these forms. The fields at which a plaquette test fails do not form a small set for the measures contained in $\mathbf{T}_k(Z)$. They do form a small set for the

Wilson weight, since $e^{-x_j A} \le e^{-c\,x_j \varepsilon_j^2} = e^{-c\,p_0(g_j)^2}$. Therefore, if no $\mathbf{R}$ has acted on a component X, the small factor that a failed plaquette test carries there is obtained from $\mathbf{T}_k(X)$ applied to $\chi\, e^{-A(x_k(\cdot),U_k)}$, and never from $\mathbf{T}_k(X)1$. This weight is evaluated at the background, which depends on the variables of all the components simultaneously. For this reason, bounds that are uniform in the exterior variables do not combine multiplicatively. The phenomenon already occurs in measure theory. Take Lebesgue measure on $[0,\mathsf{l}]$ in each of two variables, $0 < \epsilon \ll \mathsf{l}$, and $F = \mathbf{1}\{|\xi_1 - \xi_2| < \epsilon\}$. Then

$$\sup_{\xi_2} \int F\, d\xi_1 = 2\epsilon,$$

$$\iint F\, d\xi_1 d\xi_2 = 2\epsilon\,\mathsf{l} - \epsilon^2 \gg (2\epsilon)^2.$$

The remedy is a pointwise bound by a product of functions, each depending only on its own group of variables, the groups being disjoint, combined with Tonelli's theorem. The small factors come out as numbers from the Wilson sums near the creations, joined with a fixed fraction both of the Gaussian forms and of the kept gauge-fixing exponents. What remains of the weights is bounded by functions that no longer depend on the background. The supremum in M is placed under the integral for the same reason. The factor of X then becomes a function of ξ only, and the masses multiply. When a component has dropped out, the supremum is taken over the evaluations at ξ. These are the real configurations that are compatible with the stored variables and regular layer by layer, and at which every test listed by ω takes its recorded value, with one exception: the test whose failure turned X into a drop-out. They form a set fibred over ξ. A product set contains a flat $\mathbf{U}$, at which $e^{-A} = 1$, so taking the supremum over it eliminates the Wilson weight. For a live component, a bound $e^{-c\,p_0(g_k)}$ does not reproduce itself, since a step without a creation costs $e^{+CM^4 \check{R}_k^5 (d'_k(Z)+1)}$. What is carried along is the credit. As long as nothing is created, linear sizes counted in cells decrease geometrically. At a drop of $\check{R}$ the two partitions coincide and nothing shrinks, and no two consecutive steps are both drops. The waiting lasts at most $\check{R}_j$ steps, unless two components join or a mark is set. In that case the waiting begins anew. In the count of (2c), every joining is charged to a separate created cell belonging to it alone, and a mark sets one factor $e^{-p_0(g_t)}$ of its mass against the same $\check{R}_t^6$. Hence the costs still owed add up to at most $CM^4 L^5 \check{R}_j^6 (d'_j(Z)+1)$. The $p_0(g_j)^2$ per cube deposited at the creation covers $\check{R}_j^6$ per cube, and the powers of $\log g^{-2}$ are chosen in this order. After $\mathbf{R}$ has acted, the mass bound is what is carried forward.

Three losses are made good:

- The expansions use up part of κ and of the radii. Cubes that are L times larger give back a factor of order L in the rate, and the curvature decreases by L^{-2} at each level. $\mathbf{T}$ reserves a surplus of this gain for $\mathbf{R}$, whose re-expansions consume that surplus and restore exactly κ, α_i.
- The marginal term is absorbed into x_k.
- $e^{-p_0(g_j)}$ is smaller than any power of g_j, but it is not a power of the spacing when $d = 4$, so it cannot pay for all later steps. This is the reason for $\mathbf{R}$.

In the presence of large fields, $\log \rho_k$ is not a sum of local terms, and this is why the construction uses densities and sums over labels. Analyticity serves as a device. On a smaller complex domain, a supremum bound supplies derivative bounds by Cauchy's formula. Its transport through substitution, integration and logarithms uses the stated domain inclusions, integrable majorants and zero-free cluster estimates.

1a.2.8 is (1d) carried out with parameters. The size of $\mathbb{P}^{(k)}$ is $g|B|^3$, and under the test $g|B| < \delta(\mathsf{t})$ this is $\le \delta(\mathsf{t})|B|^2$. The bracket of level j in $\{S\}$ is $O(\mathsf{q}_j^3\, \eta_{k-j}^{4+\alpha_{\mathrm{H}}})$, because every Cauchy estimate taken at a later background loses a power of q_j, up to the third (this loss does not occur on radii that do not run). Applied to a shift of size δ, Cauchy's estimate produces s, and a cube of level k contains η_{k-j}^{-4} cubes of π_j. What is left at the age $n = k+1-j$ is $\mathsf{q}_j^3\, \eta_{k-j}^{\alpha_{\mathrm{H}}} \le C(n+1)^{3/2} L^{-\alpha_{\mathrm{H}}(n-1)}$, and these are the weights ω_n. The $H^{(j)}$ are marginal, and their sum is controlled only through their norms. If the raw local terms are taken without the counterterms $b_j A$, then $\{S\}$ contains $\sum_{j\le k} b_j\{A\} \approx c_0 k\{A\}$. In that case no bound holds uniformly in k, and the statement does not hold. Keeping t (which governs the tests, the gauge-fixing weight and the scaling) separate from x, the coefficient of A, comes at the price of a quadratic term proportional to $1 - x/\mathsf{t}$. In return it allows frozen choices (3c) and complex weights (4a), (8a). Once the choices are fixed, the steps are fixed operations that are linear in the measure, and β enters only through the weight of the action.

1a.2.9: on blocks with no additional node, the block axial representative satisfies $\mathrm{GF}_k(V_k^{\mathrm{bg}}) = 0$. At a seam, the node terms described in 1a.3.11 can remain constant on the orbit and must be retained with the boundary terms. The following expansion uses the gauge-fixing factors on the blocks where the vanishing holds. The linear term $x_k\langle \mathsf{H}_k \mathcal{A}', J(U_{k+1})\rangle$ vanishes on $Q_{V_k^{\mathrm{bg}}}\mathcal{A}' = 0$, since the current is the image of a Lagrange multiplier. Because the fibre is curved, the term $\langle \mathsf{H}_k \mathsf{h} C^{\mathrm{pl},(2)}(B), J\rangle$ remains, and it is included in $\Delta^{(k)}$. In this node-free interior expansion, $x_k A(U_{k+1})$ is the only term with a negative power of g_k. This assertion does not discard the seam contribution $x_k \mathrm{GF}_k(V_k^{\mathrm{bg}})$ or its fluctuation variations. They must be retained in the boundary exponent, whose estimates we must prove separately. The choices are made in the following order. b_{k+1} is obtained on the infinite lattice from the choices of level k, so x_{k+1}, ε_{k+1}, $\check{R}_{k+1}$ are fixed in advance of testing V_{k+1}. *The new tests imply the old:* on Ω_{k+1}, wherever the new functions are supported, the old ones take the value 1. By the nesting, $U_{k,\square} = (e^{i\eta_k \mathbf{H}} U_{k+1,\square'})^u$. Here $\mathbf{H}$ denotes the response of (1b) when the datum is shifted, the shift being $O(\delta_k)$ everywhere but in a boundary layer at a distance of order $M_2\check{R}_k$. Therefore

$$|\mathbf{H}|, |\nabla \mathbf{H}| \le C\big(\delta_k + e^{-\check{R}_k}\varepsilon_k\big) \le c\,\varepsilon_k,$$

$$|\partial U_{k,\square} - \mathbf{1}| \le 2\varepsilon_{k+1}\eta_{k+1}^2 + c\,\varepsilon_k \eta_k^2 < \varepsilon_k \eta_k^2 :$$

so χ_k passes into χ_{k+1}. This holds because the new threshold is smaller by the factor $L^{-2}\varepsilon_{k+1}/\varepsilon_k \le CL^{-2}$ (by the upper half of the flow inequalities, thresholds of neighbouring levels remain comparable), and because $\delta_k = (A_1/A_0)\varepsilon_k \ll \varepsilon_k$ (both carry the same power of the logarithm, and their ratio is small and fixed). Likewise every indicator in $\mathfrak{G}_k$ is equal to 1, and χ_k' follows from $\chi^{(k)}$. Since the local backgrounds are so close to the global ones, the tests can be carried out on local objects, which factorise, while the expansions are made around global ones. *The split:* an aged $R^{(j)}$ has each of its cubes inside a single block, its total transforms covariantly under the isometries of $\mathbb{T}^{(k+1)}$, and it is counted together with $\mathbf{E}_k$. For a young one this is not so, and a young one may not be admitted to the family from which b_{k+1} is computed. The t_1-integral, which comprises boundary terms, the part of the coupling that depends on position, and the strip term $\mathsf{W}^{(k)}$, contributes only to $B^{(k+1)}$. The t_2-integral takes the expectation of the young terms in a measure which itself contains them. Consequently at least one factor $g_j^{\kappa_0}$ appears in every term of its expansion, and this integral contributes to both $R^{(k+1)}$ and $B^{(k+1)}$. The third term contains only regular terms. A further interpolation makes it independent of the exterior fluctuation field, and the resulting contribution goes to $B^{(k+1)}$. Its cluster expansion then produces, for $X \subset \Lambda_{k+1}$, the pure small-field integral's terms, $E^{(k+1)}$, and in addition boundary terms. b_{k+1} is computed from this symmetric remainder. Thus a contribution to

$R^{(k+1)}$ comes from **T** already, ahead of the one that **R** adds. 1a.2.10: iterating the Stokes formula of 1a.1.4 gives the first display. When a standard Gaussian vector is conditioned on the value of a linear map, its conditional expectation projects orthogonally onto the complement of the kernel of that map, and its components in mutually orthogonal subspaces are independent. The slice is a Schur complement, and $\mathcal{H}_{j-1}$ transports it. The only property of it that has to be analysed is its decay, and this is the analysis of (1c) at **1**.

Gauges. 1a.3.1 is Stokes' formula on a tree. Because of the factor $|b_- - y_P|$, boxes have side $O(M)$ and $M\varepsilon_k$ has to be small; that is, the coupling is taken small after M. 1a.3.2 is a descent through the levels, with increments $C\alpha_0 \sum_i L^{-2i}$. 1a.3.3 follows from Fubini's theorem and the invariance of Haar measure. No determinant arises, because the gauge function is a group variable translated by $u(x)$. The mean is used because single combs break W_4. The constraint is taken soft because fixing the mean by a delta function imposes a non-linear restriction which couples all the bonds of the paths to one another and has no explicit solution. The weight is x_k. Let P'' be a family of blocks with no additional node, on which every background contour variable is **1**. Restricting the gauge sum to these blocks gives

$$x_k \,\mathrm{GF}_k(e^{ig_k B} V_k^{\mathrm{bg}}; P'') = \tfrac{1}{2} \sum_{y \in P''} \sum_{x \in B(y) \setminus \{y\}} \big[|B(y,x)|^2 + O(g_k |B|^3) \big].$$

The quadratic coefficient is of order one in these Gaussian variables. The omitted seam-node factors are retained, with their constant and linear contributions, in the conditional boundary estimates; this formula provides no bound for them. Here $g_k B(y,x)$ denotes the linear part in B of $\frac{1}{i}\log$ applied to the mean $(e^{ig_k B} V_k^{\mathrm{bg}})(y,x)$, and when $V_k^{\mathrm{bg}} = \mathbf{1}$, $B(y,x)$ coincides with the mean of $B(\Gamma)$ taken over $\mathcal{G}(y,x)$. This quadratic gauge form is strictly positive on each nonzero residual gauge direction $D\lambda$ supported in the blocks P'', with λ zero at their coarse sites: there $|B(y,x)| = |\lambda(x)|$. This local calculation does not identify the full layered null space. Its removal and the coercivity of the complete fluctuation form remain the separate requirements of A2 in the dependency catalogue. The weight itself is auxiliary: the first identity of 1a.3.3 yields an identity for every choice of weight, so the weight may be frozen (3c) and need not depend on the source (4a). *Procedure and product:* the insertion, taken as a draw, is exact on any measure, and on invariant measures it agrees with the written product. A term with stored fields is invariant. Indeed, the distance from P'' to Z_k is $\geq cM\check{R}_k$. Hence every transformation over P'' can be lifted to a bare transformation which is constant on pillars and equal to **1** near Z_k, the region where all stored quantities enter. In the case of the trees of (2b), the stored fields are acted on by the group via the tops of their pillars. The exact model 1a.3.13 shows that this verification is not idle: the group that acts is decided by whether or not the seam bonds lie in the determining set. The removal identity shows that dropping δ_T from the density gives an equivalent density, not an equal one: the two coincide when integrated against invariant functions.

1a.3.4: in the pillars, U and U_0 have the same contour variables. Hence $W(y,x)U_0(y,x)^{-1} = u(y)^{-1}\,\mathrm{Ad}(U_0(y,x))u(x)$, and the block factor n_W becomes the definition of $u^{-1}\bar{R}_{U_0}u$. This is Bałaban's deeper reason for imposing the axial gauge first and for keeping a single contour variable throughout (with two of them, circulations of A survive in the linear part). *Origin of R:* the term sought is Feynman's $\|\partial^* A\|^2$, but only the $\lambda \in N(Q')$ leave $\delta(B - Q^\sharp A)$ unchanged. Over these, the Faddeev–Popov quotient with gauge parameter α_{g} is

$$e^{-\frac{1}{2\alpha_{\mathrm{g}}}\|\partial^* A\|^2} \Big/ \int d\lambda\, \delta(Q'\lambda)\, e^{-\frac{1}{2\alpha_{\mathrm{g}}}\|\partial^* A - \Delta\lambda\|^2} = \mathrm{const} \cdot e^{-\frac{1}{2\alpha_{\mathrm{g}}}\|R\,\partial^* A\|^2}.$$

In a background the constants turn into determinants that depend on U_0:

$$e^{-\frac{1}{2}\langle A,\Delta_\pi A\rangle} = \big|\det\big(\Delta_{U_0}|_{N(Q')}\big)\big| \int d\lambda\, \delta(Q'\lambda)\, \delta_R(RD^*A - \Delta\lambda)\, e^{-\frac{1}{2}\langle A-D\lambda, \Delta^{\mathrm{W}}(A-D\lambda)\rangle} :$$

the determinant here is the Landau gauge's Faddeev–Popov determinant, and axial gauges have no such determinant. In normalised expressions such as $\mathsf{H}_k B$ it cancels; in Gaussian normalisers it does not. In the one-loop coefficient (1e), (3a) it enters together with the determinant of the gauge-fixed Hessian. Positivity (1a.3.5): consider $U = \mathbf{1}$ (the remaining cases are perturbations of it). A gradient $\partial\lambda$ ($\partial = D_{\mathbf{1}}$) is detected by $Q^\sharp$ if $\partial Q'\lambda \neq 0$, and by $R\,\partial^*$ otherwise, since in that case $\lambda \in N(Q')$ once a constant is subtracted. The full DD^* is positive as well, but admissible transformations cannot reach $D^*A = 0$, and with the full operator $\mathsf{G}Q^{\sharp *}(Q^\sharp \mathsf{G} Q^{\sharp *})^{-1}$ is no longer the minimiser of the Hessian. $RD^*\mathsf{G}Q^{\sharp *} = 0$ requires the invariant extension Δ_π, which is a perturbation of size $CM\alpha_0$. (Bałaban's G_0 is defined by the form of the statement, and his G goes with the identity. Adding the term of 1a.1.10 gives G_1. The three satisfy the same bounds, with the exception of the one with the covariant Laplacian, and they coincide where $J = 0$.)

1a.3.7 (elliptic regularity through $\mathsf{G}_k(U_0)$): both currents, each smaller than α_0, give $D^*DA = \mathcal{J}$ with $|\mathcal{J}|_{(-3)} \leq 2\alpha_0 + C\alpha_2 |\nabla A|_{(-2)} + \dots$. The averages give $\xi Q^\sharp A = B - C(A) =: \mathsf{b}$ with $|C(A)| \leq C\alpha_2^2$ (1a.1.6 leaves a remainder that is of second order), and this accounts for the α_1 in the result and for the condition on α_2^2. On the slice,

$$A = \mathsf{G}_k(U_0)\mathcal{J} + \mathsf{G}_k(U_0) Q^{\sharp *} \xi^{-3} \mathsf{b},$$
$$\|A\|_1 \leq B_0\big(|\mathcal{J}|_{(-3)} + \sup|\mathsf{b}|\big),$$

and absorption of the gradient term takes place as soon as $B_0 C\alpha_2 \leq \frac{1}{2}$. 1a.3.6 (Bałaban's argument, written out for the scheme c) is proved by induction on k. On Ω_k the pair (A, u_1) of depth $k-1$ is a factor L worse; here u_1 denotes the explicit product from 1a.1.12. One seeks $u = u_1 e^{i\lambda}$. By Campbell–Hausdorff, once the non-linear average has been straightened to $Q'\lambda = 0$,

$$\lambda = -G'RD^*A + G'R\,\mathfrak{N}_4(\lambda, D\lambda, A, D^*A),$$
$$|R\mathfrak{N}_4|_{(-2)} \leq CB_1(\alpha_0 + \alpha_1)\|\lambda\|_0,$$

and this is a contraction on complex λ in $\|\lambda\|_0$. (The bound is valid on the ball of radius $CB_1(\alpha_0+\alpha_1)$, which contains the solution; within a ball of fixed radius the solution is unique.) Curvature and mismatch thus have to beat only norms that depend on L. The new constant is of order LB_1, and 1a.3.7 resets it to B_1; without this reset its growth in k is geometric. Uniqueness holds among small A, so no Gribov ambiguity arises. In (1b) the passage to the slice is redone for $\mathcal{M}$. There Brouwer's theorem replaces the contraction. It provides a pair, though not its uniqueness, and with it the bounds on $\|A\|_1$ and $|D^*DA|_{(-3)}$. For a sequence with layers, the induction is replaced by a single preliminary gauge followed by an ascent through the truncations $(\Omega_0, \dots, \Omega_j)$. 1a.3.8: the cube is surrounded by a tower that descends to the bare scale. After passing to tower and box axial gauges, one applies the theorem to $(\mathbf{1}, U)$ within the scheme c. Only propagators at $\mathbf{1}$ enter, and this breaks the circularity with (1c). At complex fields we have $v = w^{-1}$ and $w(x) = \mathbf{U}(\Gamma_{y_\square, x}) = \Pi\, v_{\mathrm{r}}^{-1}$. Here Π is a product of factors $\exp(i\eta_k\, tA'(b)t^{-1})$, where each t is a unitary transport of U, so $\mathrm{cn}(v) = \mathrm{cn}(\Pi)$. The potential of the background inside a large-field box is obtained in (2b) by the same sequence of steps: the box axial gauge, used for the average, beneath it the tower axial gauge, and then Landau. Every change of its gauge is compensated on the stored variables. Local gauges $\hat{k}_y$ on overlapping boxes are glued together by $k(x) = \mathcal{N}\big(\sum_y \chi_y(x)\hat{k}_y(x)\big)$, $\mathcal{N}(X) = X/\sqrt{\det X}$, using real weights $\chi_y(x) \geq 0$ with $\sum_y \chi_y(x) = 1$. A convex combination of sufficiently close elements of SU(2) is nonzero and is a positive scalar

times a group element. On a complex neighborhood of these combinations, $\mathcal{N}$ is holomorphic for the local square-root branch continuing the positive real root.

1a.3.9: with weighted norms, or on complex balls, the induction incurs a loss at every level, and there is no reset. Instead one differentiates the identity of 1a.3.7, with G_k bounded in the sup norms that carry the weight $e^{\delta d(\cdot,y')}$. This is a single linear estimate and needs no induction. That the averages are trivial does not force u to be trivial at the nodes, and this gives rise to the node factor. Invariant functions do not detect it. At sites that carry stored fields but have ceased to be tops, n^{-1} is continued into each pillar by means of a smooth bump, and in this way the stored data are not acted on. 1a.3.10: the objects are words in transports and logarithms, each typed at a site, and fixed points are covariant by uniqueness. 1a.3.6 gives the uniqueness of its Landau pairs only in the scheme c; the set of these pairs is nevertheless covariant, since every one of the three defining conditions of a pair is covariant. The case of complex u follows by the identity theorem, because the G-valued u form a totally real submanifold of full dimension. 1a.3.11 is a matter of covariance only: $(\mathcal{M}^i U^w)(y,x) = w(y)(\mathcal{M}^i U)(y,x)w(x)^{-1}$ with $w = \mathbf{1}$ at the nodes. 1a.3.12: the equation becomes $\lambda = G'(f - RD^*A) + G'R\,\mathfrak{N}_4$, whose inhomogeneity is of the size of the other terms. Bałaban states that the remaining proofs carry over with only minor changes. However, $\Delta_{U_0}A - D^*DA$ equals $Df + D(1-R)D^*A$ up to curvature terms, so the bounds of the theorem force $|Df|_{(-3)} \le C(\alpha_0 + \alpha_1)$. On the other hand, $|Df|$ is of order $\eta_k^{-1}|f|$ for $f = \Delta\lambda$ with $\lambda \in N(Q')$ oscillating at the bare spacing. 1a.3.13: the contour variables of a block are coordinates that preserve Haar measure. All the integrals except one give $\int f_{\mathfrak{G}} = 1$, and the remaining one, $\int dt\, f_{\mathfrak{G}}(t)\, \delta(tV(s)\mathsf{p}V^+(c)^{-1})$, is the display. In four dimensions the same occurs at each coarse site whose block contains an end of a seam bond.

Routine and delicate. Routine are the algebra of means, of logarithms and of covariance, the exact identities, Stokes' formula and the Haar identities. The delicate points are the following:

- uniformity in k, which is reached solely via the straightened average, and only because curvature and potentials are measured on the scale of each level;
- complex arguments, where each use of covariance costs room in the domain;
- the two averages, together with the invariances of the constraints they impose, which have to be kept apart;
- normalisation by averages instead of pointwise normalisation, the group property that is lost, and the node factor that is left over;
- the junctions of layers;
- the invariance of whatever an insertion encounters after fields have been stored.

For the construction as a whole, the difficulty lies in the words "with the same constants".

17. The nonlinear block average and its pivot chart

The finite calculation here concerns the actual centered one-step average. There are two means in its definition, of different kinds. An implicit group mean defines each internal contour transport. An explicit exponential of an arithmetic mean of closed logarithms then defines the block bond. We retain both operations. The calculation establishes the local pivot and quadratic-remainder estimates. The many-level background and exterior-force estimates enter through their separate analytic propositions.

The exact finite map. Work on a unit-spacing lattice in dimension four, with odd $L \ge 3$. The block at a coarse site y is $B(y) = \{x : |x - y|_\infty \le (L-1)/2\}$. For a positive coarse bond $c = (y, y + Le_\mu)$, put $y' = y + Le_\mu$ and $D_c = B(y) \cup B(y')$. On a torus we require that this double block and its one-bond neighborhood have an embedded rectangular lift preserving their unit bond graph, with no periodic edge joining opposite faces of the patch. A sufficient condition

for a cubic torus is that its period exceeds $4L+4$. All bonds used below have both endpoints in D_c, with the rectangular bond graph of this lift.

For $x \in B(y)$, let $\mathcal{G}(y,x)$ be the shortest paths which complete one coordinate displacement before starting the next. There are at most 24 such paths, each of length at most $2L$. Repeated paths, if retained, are counted with their prescribed equal weights. The transport along an oriented path is the ordered product of its bond matrices; reversed bonds use the inverse matrix. For nearby matrices $T_1, \dots, T_n$, define their contour mean as the solution near them of

$$\frac{1}{n}\sum_{j=1}^{n} \log(T_j T^{-1}) = 0. \tag{QC1}$$

All logarithms below are the local analytic logarithm near the identity. Set

$$U(y,x) = \operatorname{av}\big(U(\Gamma) : \Gamma \in \mathcal{G}(y,x)\big), \qquad U(x,y) = U(y,x)^{-1}. \tag{QC2}$$

With $x' = x + Le_\mu$, the actual block map is

$$\begin{aligned} Z_x(U) &= U(y,x)U([x,x'])U(x',y')U[c]^{-1}, \\ \mathcal{M}(U)(c) &= \exp\left[L^{-4}\sum_{x\in B(y)} \log Z_x(U)\right] U[c]. \end{aligned} \tag{QC3}$$

Here $[x,x']$ and $[c]$ are straight segments of length L. In particular, the outer operation in (QC3) is not the implicit mean (QC1). The cruder Borel definitions of the block map away from this analytic neighborhood do not enter the calculation.

For a background V, let $W = \mathcal{M}(V)$. In physical bond logarithms, with no scale factor in the exponential, put

$$q_{V,c}(a) = \frac{1}{i}\log\left[\mathcal{M}(e^{ia}V)(c)\mathcal{M}(V)(c)^{-1}\right]. \tag{QC4}$$

The matrices $a(b)$ are complex traceless 2 by 2 matrices, and their norm is the supremum of the operator norms over positive bonds. On real variables they are Hermitian. All constants denoted C, c below are universal; their values may be enlarged or decreased at successive occurrences.

The contour mean, with quantitative bounds. The elementary matrix inequalities

$$\|e^X - I - X\| \le \tfrac{1}{2}e^{\|X\|}\|X\|^2, \qquad \|\log(I+H) - H\| \le \frac{\|H\|^2}{2(1-\|H\|)} \tag{QC5}$$

hold for $\|H\| < 1$, by the power series and submultiplicativity. They show, on a fixed small ball, that

$$\begin{aligned} \log(e^X e^{-Z}) &= X - Z + E(X,Z), \\ \|E(X,Z)\| &\le C(\|X\| + \|Z\|)^2, \\ \|D_Z E(X,Z)\| &\le C(\|X\| + \|Z\|). \end{aligned} \tag{QC6}$$

For the derivative estimate one can differentiate the convergent logarithm series term by term and use $De^Z[H] = \int_0^1 e^{(1-t)Z} H e^{tZ}\, dt$. The derivative of each remainder in (QC5) is bounded by a constant times the radius. Thus the constants in (QC6) depend on no number of contour paths.

Write $T_j = e^{X_j}$, with $\max_j \|X_j\| \le t$, and seek $T = e^Z$. Equation (QC1) is equivalent to

$$Z = \overline{X} + \frac{1}{n}\sum_j E(X_j, Z), \qquad \overline{X} = \frac{1}{n}\sum_j X_j. \tag{QC7}$$

For a fixed sufficiently small t, the right side maps $\|Z\| \le 2t$ into itself and has Lipschitz constant at most $1/2$. Iteration from $\overline{X}$ proves existence and uniqueness there. It also proves $Z = \overline{X} + O(t^2)$,

uniformly in n. The iterates are holomorphic in all X_j, and convergence is locally uniform on a slightly smaller fixed ball. Their limit is therefore holomorphic. Trace zero is preserved in (QC7). For anti-Hermitian X_j, anti-Hermiticity is preserved as well, so the mean lies in SU(2) on real group inputs.

Uniqueness proves permutation symmetry and left-right covariance. Indeed under $T_j \mapsto uT_jv$, with $T \mapsto uTv$, every logarithm in (QC1) is conjugated by u, whenever the transformed families remain in their analytic domains. The inverse property follows from $\log(T_j^{-1}T) = -\operatorname{Ad}(T^{-1})\log(T_jT^{-1})$. Thus the locally constructed mean is the mean in (QC2), including families centered at a unitary matrix other than I.

Local gauges do not constrain coarse holonomies. For a real background V_0 on D_c, put $\varpi_c = \sup_{p\subset D_c} \|\partial V_0(p) - I\|$. There is a unitary gauge s_c on this rectangle such that

$$\sup_{b\subset D_c} \|V_0^{s_c}(b) - I\| \le CL\varpi_c. \tag{QC8}$$

Here $V^s(b) = s(b_-)V(b)s(b_+)^{-1}$. To prove (QC8), fix a coordinate-order comb from y to every site of the rectangle and set $s_c(x) = V_0(\Gamma_{y,x})$. For any bond b, the product $s_c(b_-)V_0(b)s_c(b_+)^{-1}$ is its comb loop. Slide the additional bond past the later coordinate segments of the comb. Each slide exchanges two adjacent perpendicular steps and inserts one plaquette, conjugated by a partial transport. There are at most CL slides, since every coordinate interval in the rectangle has length at most $2L$. All these conjugations are unitary. The norm of a product of such plaquette factors minus I is at most the sum of their deviations, proving (QC8).

The same gauge, fixed from the real background, works for $V = e^{iE}V_0$: its distance from I is at most $C(L\varpi_c + \|E\|_\infty)$, when these quantities are small. More generally we can start directly with a unitary local gauge in which

$$\sup_{b\subset D_c} \|V^{s_c}(b) - I\| \le \epsilon. \tag{QC9}$$

Different double blocks may use different gauges. We never require one gauge close to I on an entire observation window.

The group mean and (QC3) are gauge covariant. The internal transport transforms with its two endpoint factors; each Z_x transforms by conjugation at y. Consequently

$$q_{V^{s_c},c}(a^{s_c}) = \operatorname{Ad}(s_c(y))q_{V,c}(a), \qquad a^{s_c}(b) = \operatorname{Ad}(s_c(b_-))a(b). \tag{QC10}$$

All these adjoint actions have norm one. A coarse bond value $\mathcal{M}(V)(c)$ in the original gauge may be anywhere in SU(2). It is the ratio in (QC4), rather than that coarse bond itself, whose logarithm is taken near I. Nontrivial coarse holonomies therefore cause no loss in these local estimates.

Uniform Taylor estimates for the actual average. We first work in the gauge (QC9), so $V = e^{iv}$ with $\|v\|_\infty \le C\epsilon$. If $\|v\|_\infty, \|a\|_\infty < c_0/L$, every path product entering (QC2) stays in a fixed small neighborhood of I. For example, a product of N factors, each within t of I, is within $e^{CNt} - 1$ of I. Its reversed factors satisfy the same bound. Here the internal paths have $N \le 2L$. The contour-mean estimate just proved applies, and the closed products in (QC3), of total path length at most $6L$, are also in a fixed logarithm neighborhood. The outer logarithm average is normalized by L^{-4}; its size is bounded by its largest summand, without a factor L^4.

It follows that

$$F_c(v,a) = \frac{1}{i}\log\left[\mathcal{M}(e^{ia}e^{iv})(c)\mathcal{M}(e^{iv})(c)^{-1}\right] \tag{QC11}$$

is holomorphic and bounded by a universal constant on the joint supremum balls of radius c_0/L. It vanishes at $a = 0$. This establishes the domain before any derivative estimate. On balls a fixed

fraction smaller, two scalar Cauchy formulas give

$$\|D_a F_c\|_{\infty\to\mathrm{op}} \le CL, \qquad \|D_a^2 F_c\| + \|D_v D_a F_c\| \le CL^2. \tag{QC12}$$

For example, for unit supremum directions u, w, apply Cauchy to $F_c(v, a + zu + tw)$, with both scalar circles of radius c/L. This proves the mixed bilinear bound without dividing the radius by the number of bond coordinates.

Let $K_V = Dq_V(0)$ and $R_V(a) = q_V(a) - K_V a$. Taylor's formula with integral remainder gives

$$\begin{aligned} \|K_V\|_{\infty\to\infty} &\le CL, \\ \|R_V(a)\|_\infty &\le CL^2 \|a\|_\infty^2, \\ \|DR_V(a)\|_{\infty\to\infty} &\le CL^2 \|a\|_\infty. \end{aligned} \tag{QC13}$$

for $\|a\|_\infty < c/L$ and $\epsilon < c/L$. The estimates transfer to the original frame by (QC10). They are uniform over all coarse bonds, hence over the volume. In addition, varying the background in (QC12) gives

$$\|Dq_{e^{iv}}(0) - Dq_I(0)\|_{\infty\to\infty} \le CL^2 \|v\|_\infty. \tag{QC14}$$

These are estimates for the explicit nonlinear average (QC3), not for a model linear average.

The pivot coefficient and finite elimination. Write $h = (L-1)/2$. The pivot of $c = (y, y + Le_\mu)$ is $b_0(c) = (y + he_\mu, y + (h+1)e_\mu)$. It crosses from $B(y)$ to $B(y')$. No internal contour path contains it. Among all coarse pivots, only $b_0(c)$ occurs in the support of $q_{V,c}$. Indeed internal contours stay inside one block; each straight segment in (QC3) crosses only the face between these two blocks and is parallel to e_μ. The only coarse pivot on that face is $b_0(c)$. This support assertion, rather than a claim about all bonds joining two vertices of a periodic double block, is what is used below.

At the identity background define $\overline{a}(y, x)$ to be the arithmetic mean of the oriented path sums of a over $\mathcal{G}(y, x)$, and $\sigma_a(y) = L^{-4} \sum_{x\in B(y)} \overline{a}(y, x)$. The first derivative of the contour mean is this arithmetic mean. Differentiating (QC3), the negative derivative of $U[c]$ in each closed logarithm cancels its final positive derivative. Thus

$$(K_I a)(c) = \sigma_a(y) + L^{-4} \sum_{x\in B(y)} a([x, x']) - \sigma_a(y'). \tag{QC15}$$

The two σ_a terms contain no pivot. Exactly L segments in the remaining sum contain $b_0(c)$: their transverse coordinates equal those of y, and their longitudinal starting coordinate can be any of the L coordinates in the block. All occurrences have positive orientation. Therefore

$$J_c(I) := D_{a(b_0(c))} q_{I,c}(0) = L^{-3} I_3. \tag{QC16}$$

For a pure gauge $V^s = I$, the coefficient in its original frame is

$$J_c(V) = L^{-3} \operatorname{Ad}(s(y)^{-1} s(b_0(c)_-)). \tag{QC17}$$

The unitary matrix in this formula is the prefix transport from y to the start of the pivot. Replacing it by I in the original gauge would be incorrect.

From (QC14), after the unitary local change of frame, we have $\|J_c(V^{s_c}) - L^{-3} I_3\| \le CL^2\epsilon$. Consequently, under the conservative condition

$$\epsilon \le cL^{-5}, \qquad \text{in particular if} \quad \|E\|_\infty + L\varpi_c \le cL^{-5}, \tag{QC18}$$

every J_c is invertible and $\|J_c^{-1}\| \le 2L^3$. There is no off-diagonal pivot coefficient. Set

$$(h_V \nu)(b_0(c)) = J_c^{-1} \nu(c), \qquad (h_V \nu)(b) = 0 \ (b \text{ nonpivot}), \qquad \mathsf{C}_V = \iota - h_V K_V \iota, \tag{QC19}$$

where ι extends the nonpivot coordinates by zero. Then

$$K_V h_V = 1, \quad K_V \mathsf{C}_V = 0, \quad (\mathsf{C}_V u)|_{\mathrm{free}} = u, \qquad \|h_V\| \le 2L^3, \quad \|\mathsf{C}_V\| \le CL^4. \tag{QC20}$$

These are supremum operator norms. The same powers bound column norms. Indeed a fine bond belongs to at most a fixed number of double blocks, and (QC12) bounds the coefficient in each of their rows. The maps are local: a pivot row uses only its own double block. They are holomorphic in the background, by the Neumann series for J_c^{-1}.

For comparison with a convention which inserts a/L in the one-step exponential, its linear average is $Q_V = K_V/L$, and its pivot coefficient is

$$\mathsf{m}_c = J_c/L, \qquad \mathsf{m}_c(I) = L^{-4} I_3, \qquad \det J_c = L^3 \det \mathsf{m}_c. \tag{QC21}$$

This accounts for the factor L^{-3} in a real elimination Jacobian; it is not a change in the number of color coordinates.

The nonlinear graph, constructed rather than assumed. For a full physical bond variable t, solve

$$d = R_V(t - h_V d), \qquad \Psi_V(t) = t - h_V d(t). \tag{QC22}$$

On $\|t\|_\infty < r_1 = cL^{-5}$, (QC13) maps the ball $\|d\|_\infty \le CL^2\|t\|_\infty^2$ into itself, after choosing its fixed prefactor large enough. Its Lipschitz constant in d is at most $CL^5 r_1 < 1/2$. Iteration proves the unique solution, its holomorphic dependence, and

$$\begin{aligned} q_V(\Psi_V(t)) &= K_V t, & \|d(t)\| &\le CL^2\|t\|^2, \\ \|D_t d(t)\| &\le CL^2\|t\|, & \|D\Psi_V(t) - 1\| &\le CL^5\|t\|. \end{aligned} \tag{QC23}$$

The derivative estimate follows by differentiating the contraction and inverting $1 + DR_V h_V$ by its convergent Neumann series. Only the pivot values are changed. There is even an explicit inverse, on a smaller neighborhood: $\Psi_V^{-1}(a) = a + h_V R_V(a)$. Substitution in (QC22) verifies this formula by uniqueness.

Each equation for $d(c)$ uses only its own pivot value and the nonpivot data in D_c. The equations for different coarse bonds are independent once the nonpivot data are fixed. Thus the graph is exactly finite range, not merely exponentially local. In particular

$$a(u) = \Psi_V(\mathsf{C}_V u), \quad q_V(a(u)) = 0, \quad a(u)|_{\text{free}} = u, \qquad \|u\|_\infty < cL^{-9} \tag{QC24}$$

is the actual small fibre chart. It obeys $\|a(u)\|_\infty + \|D_u a(u)\|\|u\|_\infty \le CL^4\|u\|_\infty$. Conversely, any sufficiently small solution of $q_V(a) = 0$ has this form: $t = a + h_V R_V(a)$ satisfies $K_V t = 0$, hence $t = \mathsf{C}_V(a|_{\text{free}})$, and (QC23) gives uniqueness.

For the canonical Gaussian coordinate $u = gy$, the physical chart therefore contains $\|y\|_\infty < cL^{-9}/g$. At weak coupling it contains any radius $c_*\alpha/g$ with fixed c_* and sufficiently small α. This establishes the one-step primitive radius; inclusion in all the older background domains remains a different estimate.

Real coarea normalization and complex continuation. We record the normalization only on the real positive chart first. Let $\mathcal{E}, \mathcal{C}, \mathcal{F}$ be the fine positive bonds, coarse positive bonds and nonpivots, respectively. Thus $|\mathcal{F}| = |\mathcal{E}| - |\mathcal{C}|$. In real Lie algebra coordinates Haar measure is $d(e^{ia}) = \sigma(a) d^3a$, where $\sigma(a) = \sigma_0(\sin|a|/|a|)^2$ and $\sigma_0 > 0$. Group delta in the same chart satisfies $\delta_G(e^{iq}) = \sigma_0^{-1}\delta_{\mathbb{R}^3}(q)$.

In the linear variables $t = \mathsf{C}_V u + h_V v$, free coordinates are u and $K_V t = v$. Its determinant is $\prod_c \det J_c^{-1}$. The full straightening $a = \Psi_V(t)$ has derivative close to the identity by (QC23). At real backgrounds, the determinants $\det J_c$ are positive: in the local unitary frame J_c is within half of the positive scalar matrix $L^{-3} I_3$, and the real adjoint frame transformations have determinant one. Also $\det D\Psi_V > 0$, by continuity from its value one at zero and the uniform derivative

bound. The ordinary real change of variables therefore gives, on the fibre,

$$\begin{aligned}&\prod_{b\in\mathcal{E}} dU_b \prod_{c\in\mathcal{C}} \delta_G(\mathcal{M}(U)(c)W(c)^{-1})\\ &\quad = \sigma_0^{|\mathcal{F}|}\prod_c (\det J_c)^{-1}\det D\Psi_V(\mathsf{C}_V u)\prod_{b\in\mathcal{E}}\frac{\sigma(a_b(u))}{\sigma_0}\prod_{b\in\mathcal{F}} d^3u_b.\end{aligned}\tag{QC25}$$

This is an identity of local fibre measures, with the group delta interpreted by this change of coordinates. The straightening derivative is block triangular: its free block is the identity, and its pivot diagonal has one three by three block per coarse bond. Hence its log determinant is a sum of local trace logarithms. Substituting $u = gy$ in (QC25) supplies $(\sigma_0 g^3)^{|\mathcal{F}|}$, while $(\det J_c)^{-1} = L^{-3}(\det \mathsf{m}_c)^{-1}$.

For complex backgrounds the coefficient functions in (QC25) have their coherent holomorphic continuations on the chart just constructed. In particular $(\det J_c)^{-1}$ and the branch of $\log\det J_c$ continuing its real value are used, not an absolute value of a complex determinant. The Haar density is continued by the entire power series $\sigma_0[\sin\sqrt{z}/\sqrt{z}]^2$, $z = \frac{1}{2}\operatorname{tr} a^2$, rather than by a complex operator norm. No complex measure or complex delta function is being asserted by this continuation.

Dependence on the remaining scales. The average, its raw Taylor constant CL^2, and the local pivot construction contain no M, observation radius or number of free variables. If the supplied local near-pure-gauge bound is only $\|V^s - I\| \le C_b M a$, (QC18) requires $a \le c/(C_b M L^5)$. Alternatively the direct real condition $\varpi_c \le cL^{-6}$, with a small complex perturbation, suffices. The input-to-chart constants are conservative fixed powers of L; no claim of optimal powers is needed.

If only a complex gauge with condition numbers at most $K \ge 1$ is available, conjugation charges one factor K per input direction and one for the output. Thus safe bounds are CK^3L^2 for the quadratic residual, $2K^2L^3$ for h_V, and CK^4L^4 for C_V. The conservative physical nonpivot radius becomes $cK^{-9}L^{-9}$. A condition number growing with a larger window must therefore be charged; it cannot be discarded by calling the one-step chart local.

The calculation discharges the one-step pivot and Taylor primitives in these actual physical coordinates. It does not bound the many-level response or a further coordinate transformation on a large run window. In particular, a polynomial loss in that window radius in the background/core inclusion must still be retained in a later Cauchy radius. The free precision gap, the all-window core and tower estimates, the conditional exterior forces and the clean-class law remain separate assertions.

18. A gap for the classical flat fluctuation precision

The gauge form in this section is the actual soft contour gauge. It is not a divergence projector. At the flat field its quadratic form is the sum of the squares of the arithmetic internal contour path sums. The constraint is the plain derivative of the nonlinear block average, rather than the straight average alone. With these operators the classical flat precision has a gap uniform in the torus volume and in the number of finer averaging levels.

The statement concerns the unweighted Wilson action at a flat background, with its prescribed positive soft gauge coefficient. Signed counterterms, nonconstant layer weights, nonzero-current constraint corrections and complex decoupling matrices are not included in the conclusion.

The actual quadratic forms and their normalization. Use raw bond logarithms on a unit-spacing four-dimensional lattice. For a scalar site function f and a positively oriented bond field

a, set

$$(df)_\mu(x) = f(x+e_\mu) - f(x),$$
$$(da)_{\mu\nu}(x) = a_\mu(x) + a_\nu(x+e_\mu) - a_\mu(x+e_\nu) - a_\nu(x). \tag{CF1}$$

The same convention, with step n, is written d_n on a lattice of spacing n. Norms are counting ℓ^2 norms of sites, positive bonds or plaquettes, with each unordered direction pair counted once. One color suffices for the proof; the flat SU(2) forms are three identical copies.

The Wilson convention $A(U) = \sum_p [1 - \frac{1}{2}\operatorname{Re}\operatorname{tr} U_p]$ gives $A(e^{ita}) = \frac{1}{2}t^2\|da\|^2 + O(t^3)$. For the actual soft gauge,

$$\mathrm{GF}(V) = \sum_y \sum_{x \in B(y)\setminus\{y\}} [1 - \tfrac{1}{2}\operatorname{Re}\operatorname{tr} V(y,x)]. \tag{CF2}$$

Here $V(y,x)$ is the implicit mean of the transports along the coordinate-order internal paths. Its first derivative at the identity is the arithmetic mean of the corresponding oriented path sums. Denote this linear functional by $\gamma_{y,x}(a)$, and put $(Ga)_{y,x} = \gamma_{y,x}(a)$. Consequently

$$\mathrm{GF}(e^{ita}) = \tfrac{1}{2}t^2\|Ga\|^2 + O(t^3). \tag{CF3}$$

Internal paths contain no crossing pivot. Thus the gauge form on nonpivot variables is local to each block and has no pivot coordinate. For a gradient, $\gamma_{y,x}(df) = f(x) - f(y)$.

The physical step uses the scale $g = \theta^{-1/2}$ and gauge weight θ. For the real unweighted step the action coefficient is also θ. Hence $\theta g^2 = 1$, and the two quadratic forms above have coefficient one in the canonical Gaussian variables. Normalizing constants and the Haar Jacobian belong to other parts of the step; they do not change this definition of its free precision.

Plain and straight averaging. For odd n, let $B_n(y)$ be the centered block of n^4 sites. The raw straight bond average is

$$(S_n a)_\mu(y) = n^{-4} \sum_{x \in B_n(y)} \sum_{t=0}^{n-1} a_\mu(x + te_\mu). \tag{CF4}$$

At one step define $\sigma(a)(y) = L^{-4}\sum_{x\in B_L(y)} \gamma_{y,x}(a)$. Direct differentiation of the actual outer exponential of closed logarithms gives the plain derivative

$$P_L a = S_L a - d_L \sigma(a). \tag{CF5}$$

The derivative of its reference straight segment cancels its negative occurrence in every closed logarithm. This explains why no extra reference-segment term is present in (CF5).

Let $q_L f$ be the arithmetic site average over $B_L(y)$, and $r_L f = f|_{\text{block centers}}$. Telescoping along paths proves

$$S_L df = d_L q_L f, \qquad \sigma(df) = q_L f - r_L f, \qquad P_L df = d_L r_L f. \tag{CF6}$$

In particular it is restriction at the centers, not a site average, that intertwines with the plain constraint.

Write $P^{[k]}$ for the derivative at the identity of the plain k-fold nonlinear average, with raw logarithms at both ends. For $n = L^k$,

$$P^{[k]} a = S_n a + d_n \psi_k(a) \tag{CF7}$$

for a linear site functional ψ_k. Here is a direct verification. Composition of the straight averages equals S_{nL}: in each coordinate the centered digits $nr + u$, with $r \in \{-(L-1)/2, \dots, (L-1)/2\}$ and $u \in \{-(n-1)/2, \dots, (n-1)/2\}$, tile the larger centered block, and the segment indices $nt + j$, $0 \le t < L$, $0 \le j < n$, tile the longer straight segment. All weights multiply to $(nL)^{-4}$. Applying the next plain step to the gradient in (CF7) gives another coarse gradient by (CF6).

Its application to $S_n a$ differs from the next straight average by a gradient, by (CF5). This proves (CF7) inductively, starting with (CF5).

Curvature averaging is a contraction in dimension four. Expanding the boundary of an n by n square gives

$$(d_n S_n a)_{\mu\nu}(y) = n^{-4} \sum_{x \in B_n(y)} \sum_{r,s=0}^{n-1} (da)_{\mu\nu}(x + re_\mu + se_\nu). \tag{CF8}$$

Every interior bond cancels in this expansion; its four remaining sides are exactly the four straight averages in $d_n S_n$. Call the positive averaging operator on plaquettes on the right T_n. Each row has sum n^2. Each fine plaquette has total column weight n^{-2}: for each pair (r, s) its starting point determines exactly one block $B_n(y)$, since those blocks partition the fine sites. There are n^2 such pairs, each with weight n^{-4}.

For nonnegative weights t_{Pp}, weighted Cauchy gives

$$\sum_P \left| \sum_p t_{Pp} F_p \right|^2 \le \sum_P \left(\sum_p t_{Pp} \right) \left(\sum_p t_{Pp} |F_p|^2 \right) \le n^2 n^{-2} \sum_p |F_p|^2. \tag{CF9}$$

Thus $\|T_n\|_{2\to 2} \le 1$. Counting with multiplicities makes the same proof valid on a periodic lattice. The product of row and column sums in dimension d would be n^{4-d}; its cancellation here uses dimension four. Taking curls of (CF7) therefore proves the exact uniform bound

$$\|d_n P^{[k]} w\| \le \|dw\|. \tag{CF10}$$

No bound on a background propagator was used.

The classical constrained pullback. For current bond data a, on the level-k lattice, define the classical flat quadratic value

$$E_k(a) = \min\{\|dw\|^2 : P^{[k]} w = a\}. \tag{CF11}$$

The affine set is nonempty. For instance place the value of a coarse bond on every fine bond crossing its block face and put zero on fine bonds internal to a block of side $n = L^k$. The nonlinear group lift with internal bonds I has exactly the prescribed average: all parallel segments cross the same face value and all closed logarithms vanish. Applying this observation successively through the centered subdivisions gives the same assertion at k levels. Its derivative is a right inverse for $P^{[k]}$.

Existence of the minimum in (CF11) needs no uniform inverse theorem. The image under d of the finite-dimensional affine constraint set is a closed affine subspace; its point nearest zero exists. Equivalently, quotient $\ker P^{[k]} \cap \ker d$, restrict to its orthogonal complement, and solve the positive finite-dimensional normal equations there. This also shows that the minimum is a quadratic form in a. Its construction at this stage has no claimed uniform inverse bound. Equation (CF10), applied to every admissible lift, gives

$$E_k(a) \ge \|d_n a\|^2. \tag{CF12}$$

We henceforth express the current lattice in units of its own spacing, so the curl on the right is written da.

This is the classical action part of the actual flat precision. Indeed the first variation of the bare action at the identity is zero. A twice differentiable exact background lift has a tangent w with $P^{[k]} w = a$; its action Hessian is $\|dw\|^2$, because its second derivative is multiplied by that zero first variation. Linearized constrained stationarity makes this tangent minimize the quadratic form in (CF11), modulo its gauge null space. The same observation eliminates the second-order one-step constraint correction at a flat point. In particular, no nonzero-current correction has

been dropped in establishing (CF11) at the flat point. The finite Gaussian form (CF11) is defined without assuming a nonlinear background theorem; when that background is used, this argument identifies its flat quadratic pullback. Even an arbitrary exact differentiable lift would suffice for the lower bound (CF12).

Let C be the actual flat pivot inclusion, with identity on nonpivot coordinates. The physical flat free precision is characterized by

$$\langle u, A_k^{\mathrm{flat}} u\rangle = E_k(Cu) + \|GCu\|^2 \geq \|dCu\|^2 + \|GCu\|^2. \tag{CF13}$$

This formula is for the actual contour gauge and the actual plain fibre. It is not obtained by substituting a Landau projection for G.

A fixed-cell Fourier calculation. Group the current lattice into blocks of side L. In one cell let $R = \{-(L-1)/2, \dots, (L-1)/2\}^4$. A bond is indexed by its starting site $r \in R$ and direction. There are $4L^4$ bond coordinates and four pivot coordinates. The free space consequently has dimension $4L^4 - 4$, independently of the volume. For a coarse Bloch parameter $p \in (\mathbb{R}/2\pi\mathbb{Z})^4$, extend a cell field by

$$a(Lm + r, \mu) = e^{ip\cdot m} a(r, \mu). \tag{CF14}$$

The curl matrix d_p has entries which are constants or $e^{\pm ip_\mu}$. The internal-contour matrix G uses only edges within its cell, hence has no p dependence. The plain constraint matrix $P(p)$, obtained from (CF5), is also a finite matrix with such phase entries.

The middle pivot of direction μ is the crossing bond from he_μ to $(h+1)e_\mu$, $h = (L-1)/2$. At the identity its coefficient in the raw constraint is L^{-3}, since exactly L of the L^4 parallel segments contain it. There is no other pivot in that constraint. It is the own pivot in the cell of the coarse bond's start, so this coefficient has no Bloch phase. Thus the pivot block of $P(p)$ is $L^{-3} I_4$ for every p. If h_0 inserts L^3 on those pivots and ι inserts free coordinates with zero pivots, then

$$C(p) = \iota - h_0 P(p)\iota, \quad P(p)C(p) = 0, \quad C(p)|_{\mathrm{free}} = 1, \quad \operatorname{Ran} C(p) = \ker P(p). \tag{CF15}$$

The last equality follows by solving each pivot equation. It also proves the rank $4L^4 - 4$ for all p. All entries of $C(p)$ are continuous Laurent polynomials in the Bloch phases. This is the same finite-range pivot map as in the physical chart, evaluated at the flat field.

Define the Hermitian finite matrix

$$B(p) = C(p)^* \big(d_p^* d_p + G^* G\big) C(p). \tag{CF16}$$

We prove that it has no zero vector for every p, including $p = 0$. Suppose $a = C(p)u$ and $d_p a = 0$, $Ga = 0$. Extend it by (CF14). Within a block, two coordinate-order paths to the same site differ by plaquette swaps. Zero curl makes their sums equal. The same swaps express a comb loop with one additional internal bond as a sum of internal plaquette curls. Consequently zero G forces every internal path sum, and then every internal bond, to vanish.

Across a face perpendicular to e_μ, a plaquette with one transverse step compares two neighboring crossing values. Its other two bonds are internal to the two blocks, hence zero. The crossing values are therefore equal. The transverse face grid is connected, so all crossing bonds on that face have a common value $\beta_{y,\mu}$. Every straight segment from $x \in B(y)$ to $x + Le_\mu$ then has sum $\beta_{y,\mu}$, and all internal contour sums are zero. The actual constraint gives $(P_L a)(y, \mu) = \beta_{y,\mu}$. It vanishes by (CF15), killing every crossing bond as well. Thus $a = 0$, and $u = 0$ by its free restriction. The argument is valid for every real Bloch parameter, with complex fiber vectors.

One can also describe the possible closed fields explicitly. On the cell, the L^4 functions $L^{-2}\exp(i\xi_\ell \cdot r)$, $\xi_\ell = (p + 2\pi\ell)/L$, form an orthonormal basis, where each component of ℓ runs from 0 to $L-1$. Orthogonality is the finite geometric-series identity. Fine differences multiply a mode by $t_\mu = e^{i\xi_{\ell,\mu}} - 1$. The closedness equations are $t_\mu a_\nu - t_\nu a_\mu = 0$. If one $t_\mu \neq 0$, they say exactly $a = t\phi$, a gradient. The only remaining case is $p = 0, \ell = 0$, giving a constant harmonic

vector. The local argument above kills that case as well; no zero-frequency mode has been silently omitted.

The entries of $B(p)$ are continuous on the compact Bloch torus. Its least Rayleigh quotient therefore has a positive uniform minimum $c(L) > 0$. Explicitly, if the infimum were zero, a sequence of parameters and unit free vectors would have a convergent subsequence. Their limit would be a unit zero vector for the limiting $B(p)$, contradicting the previous paragraph. Thus

$$B(p) \geq c(L)1 \qquad \text{for every } p. \tag{CF17}$$

On a torus with N_μ coarse blocks in each direction, the allowed parameters are $p_\mu = 2\pi j_\mu/N_\mu$. The normalized discrete Fourier transform in the block index is unitary, again by the same geometric-series identity. It converts the comparison form on the right of (CF13) into the sum of (CF16) over these parameters. Parseval and (CF17) therefore prove

$$A_k^{\text{flat}} \geq c(L)1 \tag{CF18}$$

on the actual free nonpivot coordinates, uniformly in the torus volume and in k. Local formulas can first be used on the admissible large tori; no small-torus identification of different pivots is needed. The three color copies have the same constant.

Flat holonomies and the limit of the conclusion. The same conclusion holds for a real flat background in its prescribed block axial representative. Flatness makes internal transports path independent. Since their means in this representative are I, every internal bond is I. Flat plaquettes make the crossing value constant on each block face. The background is therefore the lift of a flat coarse connection. A gauge constant on each block flattens it on the universal cover and preserves the soft contour gauge norm. Only twisted periodicity remains.

The four torus holonomies commute: their commutator is the product of the plaquettes of a fundamental rectangle, all equal to I. Commuting unitary matrices have a common orthonormal eigenbasis; choose an eigenspace of any noncentral holonomy and use its invariance under the others, or any basis if all are central. Their adjoint actions are consequently commuting unitaries on the complexified color space. Decomposing into their common eigenspaces gives characters $e^{i\chi_\mu}$. The allowed block momenta in that character are $(2\pi j_\mu + \chi_\mu)/N_\mu$, up to the sign convention for the boundary twist. The local matrices and the curvature averaging proof are unchanged: the boundary phases have modulus one, so the absolute row and column counts in (CF9) still apply. Since (CF17) holds for every real Bloch parameter, all these shifted sets satisfy the same gap. No smallness of global holonomies is required.

The result proves the classical flat gap for the actual contour gauge and plain constraint. It is a lower bound, and does not prove a uniform upper bound or a decay estimate for a dressed precision. Nonzero-current terms from curved constraints, signed layer profiles, counterterm arrays, non-Abelian perturbations and enlarged complex decoupling parameters still need their own quantitative comparison with this positive flat form. No full background or interacting Gaussian-law assertion follows from (CF18) alone.

19. A constructive upper bound for the flat classical action

This section concerns the same unweighted classical Gaussian value as the flat-gap calculation. Let $n = L^k$, with L odd, and let $P^{[k]}$ be the raw derivative of the actual plain iterated average. For a coarse bond field a, put

$$E_k(a) = \min\{\|dw\|^2 : P^{[k]}w = a\}. \tag{WI1}$$

We construct an exact lift with energy at most $256\|d_n a\|^2$, uniformly in k and in the volume. No bound on a non-Abelian background response is used.

Discrete interpolation on the actual centered grids. Fine sites are integers and coarse sites are nm, $m \in \mathbb{Z}^4$. Write

$$\varphi_n(t) = \max\{1 - |t|/n, 0\}, \qquad \beta_n(t) = \mathbf{1}_{\{0 \le t < n\}}, \quad t \in \mathbb{Z}. \tag{WI2}$$

Periodize these functions on a finite torus. The identities $\sum_m \varphi_n(x - nm) = 1$, $\sum_m \beta_n(x - nm) = 1$, and $\sum_{x \in \mathbb{Z}} \varphi_n(x) = n$ follow by summing the two linear pieces. Also

$$\varphi_n(t+1) - \varphi_n(t) = n^{-1}\big[\beta_n(t+n) - \beta_n(t)\big]. \tag{WI3}$$

Define interpolation of sites, bonds and plaquettes by

$$\begin{aligned}
(I_0 f)(x) &= \sum_m f(m) \prod_\lambda \varphi_n(x_\lambda - nm_\lambda), \\
(I_1 a)_\mu(x) &= n^{-1} \sum_m a_\mu(m) \beta_n(x_\mu - nm_\mu) \prod_{\lambda \ne \mu} \varphi_n(x_\lambda - nm_\lambda), \\
(I_2 F)_{\mu\nu}(x) &= n^{-2} \sum_m F_{\mu\nu}(m) \beta_n(x_\mu - nm_\mu) \beta_n(x_\nu - nm_\nu) \\
&\qquad \times \prod_{\lambda \notin \{\mu,\nu\}} \varphi_n(x_\lambda - nm_\lambda).
\end{aligned} \tag{WI4}$$

The coarse arrays in these formulas are indexed by their starting node m. Their values are raw bond or plaquette values; no length factor has been included in the coarse difference d_n. Equation (WI3), followed by reindexing m, proves

$$dI_0 = I_1 d_n, \qquad dI_1 = I_2 d_n. \tag{WI5}$$

For example, differentiating a transverse nodal hat in I_1 replaces it by the difference of two interval indicators divided by n; their coefficients are precisely the two terms of the corresponding coarse plaquette boundary. This proves the second identity with the same orientation as $(da)_{\mu\nu} = a_\mu + \nabla_\mu a_\nu - a_\mu(\cdot + e_\nu)$.

For a fixed plaquette direction, I_2 has nonnegative weights. Their row sum is n^{-2}, by the two partition identities. Their column sum is n^2: each interval contributes n fine positions, each of the remaining two nodal hats has sum n, and the prefactor is n^{-2}. Weighted Cauchy therefore gives

$$\|I_2 F\|^2 \le n^{-2} n^2 \|F\|^2 = \|F\|^2. \tag{WI6}$$

These counting norms are those of the four-dimensional Wilson quadratic energy. As for curvature averaging, dimension four is used in the cancellation of the powers of n.

The exact averaging symbol. The raw straight average is

$$(S_n w)_\mu(y) = n^{-4} \sum_{x \in B_n(y)} \sum_{j=0}^{n-1} w_\mu(x + je_\mu),$$

with centered blocks. We compute $S_n I_1$ exactly, rather than using an approximate right inverse. Put $h = (n-1)/2$. In a transverse direction the coefficient of the neighboring coarse index m is $n^{-1} \sum_{x=-h}^{h} \varphi_n(x - nm)$. In the component direction it is

$$n^{-2} \#\{(x,j) : -h \le x \le h,\ 0 \le j < n,\ nm \le x + j < n(m+1)\}.$$

Both expressions are zero unless $m = 0, \pm 1$, and both have the same three values:

$$a_0 = 1 - \frac{h(h+1)}{n^2} = \frac{3}{4} + \frac{1}{4n^2}, \qquad a_1 = \frac{h(h+1)}{2n^2} = \frac{1 - n^{-2}}{8} \quad (m = \pm 1). \tag{WI7}$$

For the longitudinal count, the negative and positive overshoots each number $1 + \cdots + h = h(h+1)/2$; the remaining $n^2 - h(h+1)$ pairs give the central coefficient. For the transverse hat, its positive-side and negative-side linear sums give the same two triangular numbers.

Let T_λ translate a coarse array by one node in direction λ. The tensor product in (WI4) and the identical one-dimensional coefficients just computed show

$$S_n I_1 = \mathcal{A}_n, \qquad \mathcal{A}_n = \prod_{\lambda=1}^{4} \left[a_0 I + a_1 (T_\lambda + T_\lambda^{-1})\right]. \tag{WI8}$$

Crucially, this is the same scalar convolution on all four bond components. Each one-dimensional symbol is

$$a_n(t) = \frac{3}{4} + \frac{1}{4n^2} + \frac{1 - n^{-2}}{4} \cos t, \qquad \frac{1}{2} \le a_n(t) \le 1. \tag{WI9}$$

Fourier transform on the coarse torus, or the quadratic-form bound $a_0 - 2a_1 \ge 1/2$ for a unitary shift, proves

$$\|\mathcal{A}_n^{-1}\|_{2\to 2} \le 16. \tag{WI10}$$

All shifts commute. Their scalar convolution commutes with every coarse difference, on sites, bonds and plaquettes. Its inverse has the same property.

An exact lift for the plain constraint. For prescribed coarse a, take $w_0 = I_1 \mathcal{A}_n^{-1} a$. Equations (WI5), (WI6), (WI8) and (WI10) imply

$$S_n w_0 = a, \qquad \|dw_0\| \le \|d_n \mathcal{A}_n^{-1} a\| \le 16 \|d_n a\|. \tag{WI11}$$

This is an exact straight lift. The actual constraint is plain. For completeness its two relevant identities are

$$P^{[k]} w = S_n w + d_n \psi_k(w), \qquad P^{[k]} df = d_n (r_n f), \tag{WI12}$$

where r_n restricts a fine site function to the centers $n\mathbb{Z}^4$. At one step these follow by differentiating the actual closed-logarithm formula: $P_L = S_L - d_L \sigma$, and on gradients an internal contour is $f(x) - f(y)$, so $P_L df = d_L r_L f$. Iterating proves the second identity. Composition of straight averages is S_n; applying a plain average to a gradient again gives a gradient, proving the first by induction.

Nodal interpolation satisfies $r_n I_0 f = f$. Set

$$f = -\psi_k(w_0), \qquad w = w_0 + d I_0 f. \tag{WI13}$$

Then (WI11)–(WI12) give $P^{[k]} w = a$ exactly, while $dw = dw_0$. The gradient correction has no classical curvature cost. The energy estimate itself needs no amplitude bound; such a bound is proved below. We have proved

$$E_k(a) \le 256 \|d_n a\|^2. \tag{WI14}$$

The complementary lower bound follows from the exact curvature average: $d_n P^{[k]} w = T_n dw$, where T_n has row sum n^2 and total column weight n^{-2}. The same weighted Cauchy argument gives $\|T_n\| \le 1$. Thus

$$\boxed{\|d_n a\|^2 \le E_k(a) \le 256 \|d_n a\|^2}. \tag{WI15}$$

All statements include $n = 1$, where the interpolations and $\mathcal{A}_1$ are identities. No zero-frequency inverse is taken: $\mathcal{A}_n$ itself is strictly positive even on constants.

Consequences and scope. For the actual flat free nonpivot precision, $\langle u, A_k^{\mathrm{flat}} u\rangle = E_k(Cu) + \|GCu\|^2$. The local pivot basis has counting ℓ^2 norm at most CL^4, from its row and column bounds. Also $\|d\|^2 \le 24$ on one-forms in four dimensions. The internal contour matrix has row sum at most CL and column sum at most CL^4, hence $\|G\|^2 \le CL^5$. It contains no pivot, so $GC = G\iota$, where ι inserts free coordinates. Equations (WI14) and these bounds give the conservative uniform upper bound

$$A_k^{\mathrm{flat}} \le CL^8 I. \tag{WI16}$$

Together with the separate flat-gap theorem this bounds the same classical precision above and below independently of the number of finer levels and the volume.

Supremum norms of the constructed linear lift. The same scalar inverse is bounded in the supremum norm. For one direction the off-diagonal part of $a_0 I + a_1(T + T^{-1})$, divided by a_0, has supremum operator norm at most $2a_1/a_0 < 1$. Its inverse Neumann series therefore has norm at most $(a_0 - 2a_1)^{-1} \le 2$. Multiplying the four commuting direction inverses gives

$$\|\mathcal{A}_n^{-1}\|_{\infty\to\infty} \le 16, \qquad \|w_0\|_\infty \le \frac{16}{n}\|a\|_\infty. \tag{WI17}$$

The second inequality uses the row sum $1/n$ of I_1.

There is an explicit choice of the scalar in (WI12). Let $\psi_0 = 0$. When the current spacing is L^j, apply $P_L = S_L - d_L\sigma_L$ to $P^{[j]}w_0 = S_{L^j}w_0 + d_{L^j}\psi_j$. The next plain step sends the last gradient to the gradient of the center restriction. Thus

$$\psi_{j+1} = r_L\psi_j - \sigma_L(S_{L^j}w_0). \tag{WI18}$$

All fields in this recurrence are raw bond or site values on their current lattices. There is no missing spacing factor. Internal paths have at most $2(L-1)$ steps, so $\|\sigma_L\|_{\infty\to\infty} \le 2(L-1)$. Also $\|S_{L^j}\|_{\infty\to\infty} = L^j$, whereas restriction has norm one. The geometric sum yields $\|\psi_k\|_\infty \le 2(L-1)\sum_{j<k} L^j\|w_0\|_\infty = 2(n-1)\|w_0\|_\infty \le 32\|a\|_\infty$.

Since $dI_0 f = I_1 d_n f$, the correcting gradient in (WI13) has norm at most $2\|f\|_\infty/n$. Consequently the particular exact lift constructed above obeys

$$\|dI_0 f\|_\infty \le \frac{64}{n}\|a\|_\infty, \qquad \|w\|_\infty \le \frac{80}{n}\|a\|_\infty. \tag{WI19}$$

The scalar convolution and its inverse commute with curl. The row sum n^{-2} of I_2 and (WI17) also give

$$\|dw\|_\infty = \|dw_0\|_\infty \le \frac{16}{n^2}\|d_n a\|_\infty. \tag{WI20}$$

These are norms of one explicitly defined linear right inverse. They are not estimates for a minimizing response.

The interpolation identities also hold with commuting unitary boundary twists. Work on the universal cover and periodize with the corresponding character; absolute row and column sums are unchanged. Twisted shifts remain unitary and commuting, so (WI9)–(WI10) retain their lower bound for every shifted Bloch parameter. Thus the same energy comparison applies to real flat holonomies in the block axial gauge.

The construction supplies energy and supremum estimates for the specified linear lift of the unweighted classical flat form. The exponential flat-precision kernels are supplied by Section 20. Applications to a minimizing or nonlinear background, complex cores and tower substitutions, signed layer profiles, nonzero currents, counterterms and complex decoupling arrays use the corresponding actual-operator comparison propositions.

20. Exponential kernels of the classical flat precision

We compute the classical flat quadratic form of the actual plain iterated average, and then the covariance on its next-step nonpivot fiber with the actual soft contour gauge. A spectral gap by itself does not imply kernel decay. The proof below supplies a uniform complex strip for the exact symbols before using the flat gap. All fields are in raw bond coordinates, indexed by their starting sites. No midpoint phase is suppressed.

Let $n = L^k$, with L odd. The current lattice is in units of its own spacing; its finer lattice has n subdivisions per direction. We first use one scalar color. At the identity the three SU(2) colors give identical copies. The normalization of the action is $A(e^{ita}) = \frac{1}{2}t^2\|da\|^2 + O(t^3)$. After the canonical scaling $g = \theta^{-1/2}$, the real unweighted step has $\theta g^2 = 1$. Thus the matrices calculated here are precisely its classical quadratic precision matrices.

The two constrained minima are equal. Write $P^{[k]}$ for the raw derivative of the actual plain k-fold average. The straight average is

$$(S_n w)_\mu(ny) = n^{-4} \sum_{r\in B_n(0)} \sum_{j=0}^{n-1} w_\mu(ny + r + je_\mu). \tag{FX1}$$

Here $B_n(0) = \{-(n-1)/2, \dots, (n-1)/2\}^4$. The actual one-step map is that of Section 17. Let $\mathcal{G}_L(y,x)$ be its coordinate-order shortest-path multiset from y to $x \in B_L(y)$, with its prescribed multiplicities and equal weights. In positive bond coordinates set

$$\begin{aligned} \gamma_{y,x}(a) &= \frac{1}{\#\mathcal{G}_L(y,x)} \sum_{\Gamma\in\mathcal{G}_L(y,x)} \sum_{b\in\Gamma} \varepsilon_\Gamma(b)a(b), \\ (Ga)_{y,x} &= \gamma_{y,x}(a) \quad (x \neq y), \qquad \sigma(a)(y) = L^{-4} \sum_{x\in B_L(y)} \gamma_{y,x}(a). \end{aligned} \tag{FX1a}$$

Here the signs record the path orientation and $\gamma_{y,y} = 0$. Differentiation gives $P_L = S_L - d_L\sigma$. On gradients the plain map intertwines with restriction at the centers, whereas the straight map intertwines with the site average. Iteration gives

$$P^{[k]}w = S_n w + d_n\psi_k(w), \qquad P^{[k]}df = d_n r_n f, \qquad S_n df = d_n q_n f. \tag{FX2}$$

Both site maps are onto: for r_n assign the values at the centers, and for q_n use a function constant on each centered block. These are statements about finite linear cochains; no regularity bound for those corrections is required.

Suppose $P^{[k]}w = a$. Then $S_n w = a - d_n\psi_k(w)$. Choose $q_n f = \psi_k(w)$. The field $w + df$ has straight average a and the same curl. Conversely, for $S_n w = a$, choose $r_n f = -\psi_k(w)$ to correct the plain average. Consequently the actual classical quadratic value is

$$E_n(a) := \min_{P^{[k]}w=a} \|dw\|^2 = \min_{S_n w=a} \|dw\|^2. \tag{FX3}$$

Nonemptiness also follows from the filter calculation below: its central alias is nonzero in every component at $p \neq 0$, and a constant fine field lifts a constant coarse field. The two gradient corrections transfer feasibility between the constraints. Finite dimensional projection after quotienting the curl-null directions then gives both minima. The equality does not change the actual soft gauge, which is added later on the current one-step fiber.

Fourier normalization and the auxiliary covariance. Take a coarse torus with s sites per direction and a fine torus with $n\mathsf{s}$. Use the unitary transforms

$$\widehat{w}(\xi) = (n\mathsf{s})^{-2} \sum_x e^{-i\xi\cdot x} w(x), \qquad \widehat{a}(p) = \mathsf{s}^{-2} \sum_y e^{-ip\cdot y} a(ny). \tag{FX4}$$

Fine aliases of a coarse momentum p are $\xi_m = (p + 2\pi m)/n$, with $m \in I_n^4$, $I_n = \{-(n-1)/2, \dots, (n-1)/2\}$. Put

$$b_n(z) = \frac{\sin(nz/2)}{n\sin(z/2)}, \qquad B_n(\xi) = \prod_\nu b_n(\xi_\nu), \qquad h = (n-1)/2. \tag{FX5}$$

The centered block sum is B_n, and the straight segment sum is $ne^{ih\xi_\mu} b_n(\xi_\mu)$. Thus the exact filter of component μ in (FX1) is

$$s_{\mu,n}(\xi) = ne^{ih\xi_\mu} B_n(\xi) b_n(\xi_\mu), \qquad \widehat{a}_\mu(p) = n^{-2} \sum_{m \in I_n^4} s_{\mu,n}(\xi_m) \widehat{w}_\mu(\xi_m). \tag{FX6}$$

The factor n^{-2} is the square root of the ratio of the coarse and fine numbers of sites. In particular its square, not an unnormalized alias sum, enters a covariance.

The auxiliary fine Feynman form is $\|dw\|^2 + \|d^*w\|^2$. Its symbol is component-diagonal, with $\lambda(\xi) = 4\sum_\nu \sin^2(\xi_\nu/2)$. For $p \neq 0$, every alias has positive λ. The coarse covariance of this auxiliary form is diagonal:

$$D_{\mu,n}(p) = n^{-4} \sum_m \frac{|s_{\mu,n}(\xi_m)|^2}{\lambda(\xi_m)}. \tag{FX7}$$

This formula can equally be obtained by minimizing the positive fine Feynman form under its linear constraint; no probability measure on the fine constant mode is needed.

For complex continuation define

$$\begin{aligned} q_{\nu,m}(p) &= \frac{\sin((p_\nu + 2\pi m_\nu)/2)}{n\sin((p_\nu + 2\pi m_\nu)/(2n))}, \\ f_{\mu,m}(p) &= \left(\prod_\nu q_{\nu,m}(p)^2 \right) q_{\mu,m}(p)^2, \\ s_{n,m}(p) &= 4n^2 \sum_\nu \sin^2((p_\nu + 2\pi m_\nu)/(2n)). \end{aligned} \tag{FX8}$$

Removable ratios are filled in at zero. On real momenta the phase $e^{ih\xi_\mu}$ in (FX6) cancels with its conjugate. Its analytic version is the product of the filters at ξ and $-\xi$, giving exactly

$$D_{\mu,n}(p) = \sum_{m \in I_n^4} \frac{f_{\mu,m}(p)}{s_{n,m}(p)}. \tag{FX9}$$

Indeed $s_{\mu,n}(\xi) s_{\mu,n}(-\xi) = n^2 B_n(\xi)^2 b_n(\xi_\mu)^2$, and $s_{n,m} = n^2 \lambda(\xi_m)$. This checks every power of n. Squared absolute values in (FX7) are not used as holomorphic functions.

Recovering the gauge-invariant energy. For real p, let $t_\mu^+(p) = e^{ip_\mu} - 1$ and $t_\mu^-(p) = e^{-ip_\mu} - 1$. The latter is the conjugate of the former only on real momenta. In analytic formulas below $(t^-)^{\mathsf{T}}$ is a holomorphic row, not a conjugation of a complex variable.

At nonzero fine frequency, the elementary Hodge identity is $\|d\widehat{w}\|^2 + \|d^*\widehat{w}\|^2 = \lambda \|\widehat{w}\|^2$. Subtracting a fine gradient makes a field divergence free without changing its curl. Its straight average changes by a coarse gradient, by (FX2). Conversely any coarse gradient can be corrected by a fine gradient, since q_n is onto. It follows that the curl minimum in (FX3) equals the minimum of the Feynman quadratic value over coarse gauge changes:

$$E_n(p; a) = \min_{z \in \mathbb{C}} (a + t^+ z)^* D_n(p)^{-1} (a + t^+ z). \tag{FX10}$$

For clarity, the two inequalities use different simple steps. A divergence-free representative of a curl-minimizing field has Feynman energy equal to its curl energy, with coarse value $a + t^+ z$. In

the other direction a Feynman minimizer at that coarse value can be gradient-corrected back to a, and its curl energy is no greater than its Feynman energy.

Solving the scalar minimum gives the precision symbol

$$\mathcal{E}_n = D_n^{-1} - \frac{D_n^{-1} t^+ (t^-)^{\mathsf{T}} D_n^{-1}}{(t^-)^{\mathsf{T}} D_n^{-1} t^+}. \tag{FX11}$$

Here $D_n = \operatorname{diag}(D_{\mu,n})$. At the real zero momentum the classical value is zero on every constant coarse one-form: its constant fine lift is a/n. We accordingly set $\mathcal{E}_n(0) = 0$. This zero mode is retained, not removed by a mass term. We next prove that this definition is the value of a uniformly holomorphic symbol.

Uniform noncentral alias bounds. Fix a small $\rho > 0$ and allow $|\operatorname{Re} p_\nu| < \pi + \rho$, $|\operatorname{Im} p_\nu| < \rho$. For $m_\nu \neq 0$,

$$|n \sin((p_\nu + 2\pi m_\nu)/(2n))| \geq c(1 + |m_\nu|). \tag{FX12}$$

To see this, the real part of the sine argument stays in a fixed subinterval of $(-\pi, \pi)$, and its absolute value is at least $c|m_\nu|/n$. On that interval $|\sin x| \geq c|x|$, and $|\sin(x + iy)|^2 = \sin^2 x + \sinh^2 y$. The numerator in (FX8) is bounded. For $m_\nu = 0$ the ratio is a quotient of two sinc functions on a fixed compact zero-free neighborhood. Hence, uniformly in n,

$$|q_{\nu,m}(p)| \leq \frac{C}{1 + |m_\nu|}. \tag{FX13}$$

For $m \neq 0$ the denominator in (FX8) has a uniform positive real part. In fact $\operatorname{Re} \sin^2(x+iy) \geq \sin^2 x - \sinh^2 y$. After multiplication by $4n^2$, the imaginary displacement costs at most $C\rho^2$, whereas the real sum is at least $c|m|^2$. Decrease ρ to obtain

$$\operatorname{Re} s_{n,m}(p) \geq c(1 + |m|^2) \qquad (m \neq 0). \tag{FX14}$$

The constants do not grow as the alias set grows.

Write $s_n = s_{n,0}$, $q_\nu = q_{\nu,0}$, $Q_n = \prod_\nu q_\nu^2$, and $f_\mu = Q_n q_\mu^2$. Then

$$D_{\mu,n} = \frac{f_\mu}{s_n} + R_{\mu,n}, \qquad R_{\mu,n} = \sum_{m \neq 0} \frac{f_{\mu,m}}{s_{n,m}}. \tag{FX15}$$

Each summand of R is bounded by the absolutely summable majorant

$$\frac{C}{1 + |m|^2} \prod_{\nu=1}^{4} (1 + |m_\nu|)^{-2}. \tag{FX16}$$

The extra factor $(1+|m_\mu|)^{-2}$ may be discarded. Thus R is holomorphic and uniformly bounded on the stated box, independently of n. Cauchy on a smaller box bounds all its fixed-order derivatives uniformly as well. For $n = 1$ the sum is empty. Every noncentral summand vanishes at $p = 0$, so $R_{\mu,n}(0) = 0$; only boundedness is needed below.

Cancellation of the apparent gauge denominator. Near $p = 0$, uniformly in n, $q_\nu = 1 + O(|p|^2)$, $s_n = O(|p|^2)$, and $f_\mu = 1 + O(|p|^2)$. Define

$$h_\mu = (f_\mu + s_n R_{\mu,n})^{-1}, \qquad H_n = \operatorname{diag}(h_\mu). \tag{FX17}$$

On a sufficiently small fixed complex polydisc these are holomorphic and uniformly bounded. The exact filter identity

$$\sum_\mu \frac{t_\mu^+ t_\mu^-}{f_\mu} = \frac{4n^2 \sum_\mu \sin^2(p_\mu/(2n))}{Q_n} = \frac{s_n}{Q_n} \tag{FX18}$$

is essential. It follows directly by canceling the numerator $\sin^2(p_\mu/2)$ against q_μ^2, including removable values. It is not an approximation to a gauge denominator.

Since $h_\mu = f_\mu^{-1} - s_n R_{\mu,n} h_\mu / f_\mu$, equation (FX18) gives

$$\begin{aligned} F_n &:= \sum_\mu t_\mu^+ t_\mu^- h_\mu = s_n J_n, \\ J_n &= Q_n^{-1} - \sum_\mu \frac{t_\mu^+ t_\mu^- R_{\mu,n} h_\mu}{f_\mu}. \end{aligned} \tag{FX19}$$

Here $J_n(0) = 1$ and $|J_n - 1| \le C|p|^2$ uniformly. It is therefore nonzero on a smaller fixed polydisc. On its meromorphic domain $D_n^{-1} = s_n H_n$, and the denominator in (FX11) equals $s_n F_n = s_n^2 J_n$. The actual precision consequently has the holomorphic form

$$\mathcal{E}_n(p) = s_n H_n - \frac{H_n t^+ (t^-)^{\mathsf{T}} H_n}{J_n}. \tag{FX20}$$

This removes the whole complex zero cone of s_n, not merely the value at the real origin. The expression is uniformly bounded there, is $O(|p|^2)$, and gives $\mathcal{E}_n(0) = 0$. It also obeys the exact Ward identity $\mathcal{E}_n t^+ = 0$, by (FX19). For $n = 1$, it reduces to $\lambda(p) I - t^+ (t^-)^{\mathsf{T}}$, the ordinary curl quadratic form. This checks the normalization and phases.

A uniform strip and its Fourier kernels. We supply the argument away from zero rather than infer a complex domain from real positivity alone. On the real Brillouin cube the central q_ν are positive and bounded above and below uniformly in n. Also $c|p|^2 \le s_n(p) \le C|p|^2$. For real $|p| \ge r > 0$, the central term in (FX15) and the nonnegative alias terms give

$$c_D \le D_{\mu,n}(p) \le C_r, \qquad (t^-)^{\mathsf{T}} D_n^{-1} t^+ \ge c_r > 0. \tag{FX21}$$

The second inequality uses $\sum_\mu |t_\mu^+|^2 \ge c|p|^2$ and the upper bound on D.

On a sufficiently thin complex neighborhood of this real compact set, $\operatorname{Re} s_n \ge cr^2 - C|\operatorname{Im} p|^2$. Thus the central term is holomorphic and bounded there; (FX12)–(FX16) do the same for the aliases. Cauchy bounds the first derivatives uniformly in n. Shrinking the neighborhood by a fixed amount depending on r, (FX21) therefore keeps every D_μ and its scalar gauge denominator nonzero. Formula (FX11) is holomorphic and uniformly bounded on that neighborhood. This is a quantitative perturbation of uniformly separated real denominators.

Choose r inside the origin polydisc of (FX20), then reduce the common width once. The two formulas agree wherever their denominators are nonzero, by their exact algebraic derivation, and hence on their overlaps by continuity. Together they give constants $\eta > 0, M_E < \infty$, independent of n, such that

$$\mathcal{E}_n \text{ is holomorphic and } \sup_{|\operatorname{Im} p_\nu| < \eta} \|\mathcal{E}_n(p)\| \le M_E. \tag{FX22}$$

The symbol is 2π-periodic. In (FX9), changing a coarse momentum by 2π reindexes the aliases; an endpoint alias changes its fine frequency by 2π, under which the actual filter and denominator are periodic. The same is true of (FX11), and its analytic continuation has that periodicity.

Define its infinite-lattice kernel by

$$\mathcal{K}_n(z) = (2\pi)^{-4} \int_{[-\pi,\pi]^4} e^{ip\cdot z} \mathcal{E}_n(p)\, dp. \tag{FX23}$$

For $0 < \eta' < \eta$, shift the ν-th contour to $\operatorname{Im} p_\nu = \eta' \operatorname{sgn} z_\nu$. Periodicity cancels the vertical faces. The exponential then gives

$$\|\mathcal{K}_n(z)\| \le M_E e^{-\eta' |z|_1}. \tag{FX24}$$

All diagonal terms are retained in this Fourier definition. The kernel and its weighted absolute row sums are bounded uniformly in k. Its zero-frequency value and Ward identity are those already proved; no contact term is added afterward.

The actual next-step precision and its inverse. Now group the current lattice into the fixed blocks of side L used by the next RG step. For block Bloch momentum P, let $r \in B_L(0)$, $j \in I_L^4$, and $\zeta_j = (P + 2\pi j)/L$. The finite cell transform and its analytic inverse are

$$U(P)_{rj} = L^{-2} e^{i\zeta_j \cdot r}, \qquad V(P)_{jr} = L^{-2} e^{-i\zeta_j \cdot r}. \tag{FX25}$$

Finite geometric-series orthogonality proves $VU = UV = 1$. For real P, $V = U^*$. For complex P the displayed inverse, not literal conjugation, is used. Their norms are bounded on a fixed strip, since their extra diagonal factors are $e^{\pm iP\cdot r/L}$ and $|r_\nu|/L < 1/2$. The folded classical precision is

$$\mathcal{E}_n^{\mathrm{cell}}(P) = U(P)\,\mathrm{diag}_j(\mathcal{E}_n(\zeta_j))V(P), \tag{FX26}$$

with the identity on component indices understood. It is periodic and holomorphic on a fixed block strip, uniformly in n. Periodicity again follows by the alias permutation.

Let $C(P)$ be the actual flat middle-pivot basis for the plain one-step constraint. Its free restriction is the identity and its pivot block solves the coefficient $L^{-3}I_4$; hence its entries are fixed Laurent polynomials in the block phases. Let G be the internal contour matrix defined in (FX1a), with rows (y, x), $x \neq y$. Define $C^\sharp(P) = C(-P)^{\mathsf{T}}$; on real P this is $C(P)^*$. The exact classical free nonpivot precision is

$$\mathsf{A}_n(P) = C^\sharp(P)[\mathcal{E}_n^{\mathrm{cell}}(P) + G^{\mathsf{T}}G]C(P). \tag{FX27}$$

This is the action pullback plus the actual soft gauge. The Feynman form was used only to compute the gauge-invariant minimum in (FX3); it has not replaced the soft gauge in (FX27).

For clarity the uniform real gap used next can be obtained independently of this symbol calculation. Expanding the curl of (FX1) gives an average of n by n fine plaquette sheets. Its nonnegative row weights sum to n^2 and its total column weights to n^{-2}. Weighted Cauchy gives $\|d_n S_n w\| \le \|dw\|$, hence $E_n(a) \ge \|da\|^2$ in current units. Consequently (FX27) dominates the fixed-cell form $C(P)^*[d_P^* d_P + G^{\mathsf{T}}G]C(P)$. This form has no zero vector: zero curl makes internal path sums independent of the path, so zero G kills every internal bond. A plaquette crossing a block face then makes its crossing values equal tangentially. The plain constraint kills that common face value. The free restriction of C is the identity, so the original free vector is zero. The matrices in this comparison are continuous on the compact block Bloch torus, with fixed dimension and fixed rank. Their least Rayleigh quotient on the unit sphere therefore has a positive minimum $c(L)$. This proves $\mathsf{A}_n(P) \ge c(L)1$ uniformly in n, including $P = 0$. Equation (FX22), the finite transform and the fixed polynomial matrices in (FX27) give a uniform bound for A_n on a complex strip. Cauchy on a smaller strip bounds its first derivatives by a constant $M_A(L)$. Choose its width so that

$$\|\mathsf{A}_n(P + iY) - \mathsf{A}_n(P)\| \le 4M_A(L)|Y|_\infty < c(L)/2. \tag{FX28}$$

The Neumann series around the real positive matrix then proves

$$\mathsf{A}_n^{-1} \text{ is holomorphic}, \qquad \|\mathsf{A}_n(P + iY)^{-1}\| \le 2/c(L) \tag{FX29}$$

on one strip independent of n. This step uses both the proved analytic bounds and the gap.

Contour shifting as in (FX23)–(FX24) now bounds the Fourier coefficients of the block precision and its inverse by $C(L)e^{-c(L)|m|_1}$. The cell has fixed size, so converting its indices back to current nonpivot bonds only changes the constants. We obtain

$$|\mathsf{A}_k^{\mathrm{flat}}(b, b')| + |\mathsf{C}_k^{\mathrm{flat}}(b, b')| \le C_L e^{-\mu_L d(b,b')}, \qquad \mathsf{C}_k^{\mathrm{flat}} = (\mathsf{A}_k^{\mathrm{flat}})^{-1}. \tag{FX30}$$

Here $d(b, b')$ is bond distance in current lattice units; using starting-site distance or midpoint distance only changes the fixed constants. The symbols $\mathsf{C}_k^{\mathrm{flat}}$ in this conclusion are covariances; the undecorated pivot map $C(P)$ is a different rectangular matrix. Exponential weighted row and column sums follow by summing the fixed-dimensional lattice shells.

Finite tori, holonomies, and scope. For a periodic lattice the exact quadratic minimization is diagonal on its allowed Fourier momenta. Its symbols are the samples calculated above, with no volume correction. If $K_n(m)$ is any infinite block kernel just obtained and the block period is s_{b}, discrete Fourier orthogonality and absolute convergence give

$$K_{n,\mathsf{s}_{\mathrm{b}}}(m) = \sum_{v \in \mathbb{Z}^4} K_n(m + \mathsf{s}_{\mathrm{b}} v). \tag{FX31}$$

The image sum preserves an exponential bound in periodic distance, uniformly over the admissible periods. It identifies the infinite-lattice kernel determined by the same symbol, with exponentially small image errors on fixed nonwrapping neighborhoods. It does not assert exact equality of a full torus kernel and an infinite kernel on a lift. The same argument applies to the inverse symbol, since its sampled matrix is the inverse of each sampled positive precision.

For real flat holonomies in the prescribed block axial representative, a unitary gauge constant on each block flattens the connection on the universal cover and preserves the contour gauge norm. The commuting adjoint holonomies split into unitary characters. Each character shifts the allowed block momenta by a real amount; the infinite symbols are unchanged. Equivalently the image sum (FX31) receives unit-modulus factors. All absolute kernel bounds remain valid, without requiring small global holonomies.

The conclusion is the exponential precision and covariance bound for the actual unweighted classical flat quadratic problem. The auxiliary D_n itself is massless and is not asserted to have an exponential kernel. The gap and decay in (FX30) concern the next-step constrained nonpivot variables with the soft gauge, not a physical Yang–Mills mass gap. No signed profile, counterterm, nonzero-current, non-Abelian or enlarged complex-decoupling comparison has been proved by this calculation. Those require their actual operators and uniform perturbation estimates.

21. Constrained backgrounds and their regularity

Proof outline. The supporting analytic results used in this section, including the obligations that we must still prove, are listed under A in the dependency catalogue. Their hypotheses and precise forms are retained at the points where the argument uses them.

(1b) The background field: the regular critical orbit with prescribed averages. A renormalisation step expands about the background field. This field is the regular critical point of the Wilson action among bare fields whose block averages are prescribed. At each level the regions of large field are left behind (Step 2 of the outline). The averages are therefore prescribed on a lattice whose scale varies: on the j-th layer of a decreasing sequence of regions they are prescribed at level j, and every constant is required to be independent of the regions and of the number of levels.

(i) The setting and the objects involved.

Sequence, scales, distance. The *depth* of a sequence is the index $k \le K$ of its last member. Every depth in this part is the depth of a sequence: the depth $K - k$ of a level (Notation §1) does not occur here. We work in level-k units: $\eta = L^{-k}$, and the spacing of $\mathbb{T}^{(j)}$ is $\xi_j := L^{j-k}$. Take regions $\Omega_0 \supset \cdots \supset \Omega_k$, with $\Omega_{k+1} = \emptyset$ and each Ω_j a union of cubes of side $\mathfrak{M}\xi_j$. They are said to be *separated* when

$$\operatorname{dist}_\infty(\Omega_j^c, \Omega_{j+1}) > \rho\, \xi_j.$$

Both the *cube size* $\mathfrak{M}$ and the *gap* $\rho = R_1\mathfrak{M}$ are large powers of L; here R_1 is the gap number of the Notation. The cube size is the quantity M_1, which gives the size of the walks both in (1c) and in the Notation. The cube size $\mathfrak{M}$ and the side M of the localisation cubes, which is fixed at a later stage, are different quantities, and so are the gap ρ and the densities ρ_k. A member may be T_ℓ or empty, and no condition is imposed on the shape or the connectivity of the members. The case $\Omega_j \equiv T_\ell$ is the problem on a single lattice. By the *seams* we mean the boundaries $\partial\Omega_{j+1}$ that separate consecutive layers. The *tops* are the sites y of $\mathbb{T}^{(j)}$ that lie in the *layer* $\Omega_j \setminus \Omega_{j+1}$, and for such a site we set $\xi_y := \xi_j$. The blocks $B^j(y)$ of the tops are called *pillars*, and they form a partition of Ω_0. Each site x of a pillar other than its top has a *level* $n(x) < j$, given by $x \in \mathbb{T}^{(n)} \setminus \mathbb{T}^{(n+1)}$. It also has a *parent* $z(x) \in \mathbb{T}^{(n(x)+1)}$, characterised by $x \in B(z(x))$, and its *ascent* $x, z(x), z(z(x)), \dots$ rises to the top. For a bare bond or plaquette $\mathfrak{e}$, $\mathsf{L}(\mathfrak{e})$ is the largest ξ_j attached to its corners; the bond or plaquette is called *inner* when all its corners lie in Ω_0. Each layer is measured in its own unit, by the distance

$$d_{\mathbf{B}}(x,x') := \inf_{\Gamma \ni x,x'} \Big[\sum_{j\geq 1} \xi_j^{-1} \operatorname{length}\big(\Gamma \cap (\Omega_j \setminus \Omega_{j+1})\big) + \eta^{-1} \operatorname{length}(\Gamma \setminus \Omega_1) \Big]$$

(the infimum runs over connected paths Γ in the torus, and the first member of the sequence does not enter; the index $\mathbf{B}$ refers to the determining set of the sequence, defined below).

Variational problem. The set Γ_j consists of those bonds of $\mathbb{T}^{(j)}$ in Ω_0 that have an end point in Ω_j and no end point in Ω_{j+1}. Such a bond is either *interior*, joining two tops of level j, or *outward*. For an outward bond, the end $z \notin \Omega_j$, called the *outer end*, is a top of level $j-1$, and every bond lying within $B(z)$ belongs to Γ_{j-1} as one of its interior bonds. The *nodes* are the end points of the bonds of the *determining set* $\mathbf{B} = \bigcup_j \Gamma_j$, and they are the tops. No constraint is placed on the *inward* bonds, which lead from the layer j into Ω_{j+1}; the variant in which some of them are kept is treated in the next paragraph. In terms of these sets we define

$$M_{\mathbf{B}}(U) = (\mathcal{M}^j U|_{\Gamma_j})_j,$$
$$C(\mathbf{B},\mathbf{V}) = \{U \text{ on } \Omega_0 : M_{\mathbf{B}}(U) = \mathbf{V}\},$$
$$A_{\Omega_0} = \sum_{p \subset \Omega_0} a_p,$$
$$\mathcal{G}^{\mathrm{res}} = \{u : u = \mathbf{1} \text{ at the nodes}\}.$$

A configuration U is called *critical* if A_{Ω_0} is stationary along the curves in $C(\mathbf{B},\mathbf{V})$ that pass through U. It is called *regular at the radius* ε if, in the local units, its curvature and its current are of size ε:

$$|\partial U(p) - \mathbf{1}| < \varepsilon\, \eta^2 \mathsf{L}(p)^{-2},$$
$$|J(U)(b)| < \varepsilon\, \mathsf{L}(b)^{-3}.$$

A datum $\mathbf{V}$ is called ε_1-*regular* if the values $\partial\mathbf{V}(\tau)$ at its *test loops* τ (also called *tests*) lie within ε_1 of $\mathbf{1}$. These values are the plaquette products $\partial\mathbf{V}(p')$ of V_j, in which $\mathcal{M}(V_{j-1})$ is used on those bonds of p' that lie in the finer layer.

Inward bonds. If an arbitrary set of inward bonds is kept as additional constraints, Theorems 1, 2, 3 and the first half of Theorem 4 remain valid, though Theorem 4 then holds with a rate specific to that case. The problem without inward bonds is the setting of the complex pairs of Theorem 4 and of Theorem 5. It is also the setting of Theorem 6, although its restriction identity

also holds when they are kept. Every inward bond of level j that is kept has as its end point a *seam node* $x \in \Omega_{j+1}$, that is, a node that is not a top. In that case the tests also contain two-scale loops at the seams, namely the *pinch holonomies* $\mathsf{f}_{\mathbf{V}}(b)\mathsf{f}_{\mathbf{V}}(\tilde{b})^{-1}$, where $b, \tilde{b}$ is a pair of kept inward bonds of level j ending at a common site $x \in \Omega_{j+1}$, such that the plaquette they span again has its fourth corner in Ω_{j+1}. The *frame* $\mathsf{f}_{\mathbf{V}}(b) = V_{j+1}(c_b)\, V_j(y_b, b_-)\, V_j(b)$ transports from y, the centre of the block containing x, to x itself, along a route that stays outside Ω_{j+1}. In it, y_b denotes the centre of the block of b_-, $c_b = \langle y, y_b \rangle$, and $V_j(y_b, b_-)$ is the averaged contour variable (defined in the paragraph *Chart*). Bałaban's "bonds of Λ_j", where his Λ_j is the site set $\Lambda_j(\mathbf{B})$ of the Notation, §4, admit two readings, which differ by the inward bonds. They are either the bonds having at least one end point in that site set, or the bond set of Ω_j minus the bond set of Ω_{j+1}. In Commun. Math. Phys. 96 (1984) 223, (2.3), the difference is explicitly taken on bonds, and with this reading no inward bond occurs at $j \geq 1$. At level 0, the set $\Lambda_0 = \Omega_1^c$ of that paper and the set $\Gamma_0 = \Omega_1^c$ of Commun. Math. Phys. 119 (1988) 243, (2.2), contain the inward bonds. The set $\Omega_0^{(0)} \setminus \Omega_1^{(0)}$ of Commun. Math. Phys. 99 (1985) 75, (1.5), understood as a difference, does not contain them. By convention, the present construction omits the inward bonds at every level. The reason for this convention is an incompatibility that arises at a single level. Take a sequence of depth $k+1$ with $\emptyset \neq \Omega_{k+1} \neq T_\ell$ on which the inward bonds of level k are kept. Then for some regular data, every element of the critical orbit (Theorem 1) has a k-th average that fails to be in block axial gauge somewhere on $\mathbb{T}^{(k+1)} \cap \Omega_{k+1}$. Hence imposing constraints on these bonds is incompatible with imposing the gauge by which Theorem 6 defines V_k^{bg}. There are two remedies, and both of them are used. A step discards these bonds. The strip of Step 2 of the outline, in which nodes lie inside the blocks for a different reason, retains its set and imposes the axial gauge except at the nodes.

Chart. Fix a real U_0 with $\mathbf{V}_0 = M_{\mathbf{B}}(U_0)$ and curvature radius α_0, meaning that $|\partial U_0(p) - \mathbf{1}| \leq \alpha_0 \eta^2 \mathsf{L}(p)^{-2}$. About it we write $U = e^{i\eta A} U_0$ and $F(A) = A_{\Omega_0}(e^{i\eta A} U_0)$. The gradient is $F'(A) = F'(0) + \Delta^{\mathrm{W}}_{U_0} A + \mathcal{V}(A)$, where $\Delta^{\mathrm{W}}_{U_0} = D^*D + \Delta'$. The operator Δ' is local and of the size of the curvature of U_0, and $\mathcal{V}$ is of second order. Moreover

$$\mathcal{M}^j(e^{i\eta A} U_0)(c) = v_j(c_-)\, e^{i\mathcal{Q}^{\mathrm{st}}(A)(c)}\, \mathbf{V}_0(c)\, v_j(c_+)^{-1},$$
$$K_A := (\mathcal{Q}^{\mathrm{st}})'(A),$$

where $\mathcal{Q}^{\mathrm{st}} = \mathcal{Q}^{\mathrm{st}}_j(U_0; \cdot)$ on Γ_j is the *straightened average*. The site factors $v_j = v_j[A] = v_{j-1}\mathsf{v}_{j-1}$ are products of one-step *block factors* $\mathsf{v}_m(y)$, each of which is the mean over a block of the change in the contour variables of $\mathcal{M}^m U_0$. For $x \in B(y)$, the *averaged contour variable* $W(y, x)$ is the mean of the transports of W along the contour family of (y, x); it is the variable that enters the definition of $\mathcal{M}$ (Notation, §3). The *comb* $\Gamma_{y,x}$ is the member of this family that goes through the coordinate directions one after another, always in the same order. To each single bond c of $\mathbf{B}$ belongs a functional of A, its *row*. When $c \in \Gamma_j$, it depends only on the values of A in the *double block* $B^j(c_-) \cup B^j(c_+)$. The superscript st abbreviates "straightened"; the *straight* linear average $Q^{\mathrm{st}}_k(U)$ of the Notation, §3, carries the same superscript but denotes a different operator. The Notation's plain map $\mathcal{Q}_j(U_0; A) = \frac{1}{i}\log[\mathcal{M}^j(e^{i\eta A} U_0)\mathbf{V}_0^{-1}]$ is expressed in level-j units, and the site factors account for the difference between it and $\mathcal{Q}^{\mathrm{st}}$. Its derivative $\xi_j Q_j(U_0)$, in which the length ξ_j of the coarse bond appears, differs from K_0 by a gauge term on the coarse lattice. This term does not become small with the curvature of U_0. The Landau gauge is the condition $RD^*A = 0$, where R is the orthogonal projection onto $\Delta_{U_0}\{\lambda : Q'\lambda = 0\}$. In it, $(Q'\lambda)(y) = L^{-4j} \sum_{x \in B^j(y)} \mathrm{Ad}(\mathsf{T}(y, x))\lambda(x)$ is the *pillar average* at a top y of level j. The transport $\mathsf{T}(y, x)$, a function of two sites, is obtained by multiplying, down the pillar from y to x, the averaged contour variables of the successive $\mathcal{M}^n U_0$; the superscript ^{T}, by contrast, marks

a transpose. In the norms, each derivative carries one power of the local scale. The first norm is $\|A\|_X = \sup \max\{\mathsf{L}|A|, \mathsf{L}^2|\nabla_{U_0} A|\}$, the norm of 1-forms in the gauge slice, also written $\|A\|_1$ in (1a); the subscript X in it has nothing to do with the domain X of the paragraph *Complex fields*. The second is $|f|_{(-s)} = \sup \mathsf{L}^s |f|$. For complex fields every pairing is bilinear and every adjoint is taken to be a transpose ^{T}, so that each identity is complex-linear.

Bałaban's scheme. Bałaban's own average (Commun. Math. Phys. 98 (1985) 17) takes the lowest corner of each block as its base site and uses one comb contour for each site. Moved onto the centred blocks by a translation through half a block, it has linear rows Q on the bonds of $\mathbf{B}$ and pillar averages Q', the latter formed from comb transports that issue from the block corners. On Γ_j, once the length ξ_j of the coarse bond is divided out, $\xi_j^{-1}\mathsf{Q}$ is the operator written $Q_j^{\mathrm{B}}(U_0)$ in the Notation, §3. The projection R^{c} is built in the same way as R, with Q' replacing Q', and the condition $R^{\mathsf{c}} D^* A = 0$ defines the *comb slice*. Bałaban's propagators are built from these operators.

Reduced datum. An outer end is a node shared by two levels. For small $B : \mathbf{B} \to \mathfrak{g}^c$ we set $\mathsf{b}(B) = B$ at the interior bonds ($\mathsf{b}(B)$ and the coefficient $\mathsf{b}[\Phi]$ of the Notation, §6, are different objects, distinguished by their arguments). At an outward bond $c \in \Gamma_j$ whose outer end is $z = c_+$ we put the following (the formula for $z = c_-$ is the mirror image):

$$e^{i\mathsf{b}(B)(c)} = e^{iB(c)}\ \mathrm{Ad}(\mathbf{V}_0(c))\, \mathsf{v}_z(B),$$

where $\mathsf{v}_z(B)$ is the one-step block factor at z. It is built from $\mathcal{M}^{j-1}U_0$ and from the values of B on the bonds inside the block $B(z)$. The argument of the map $B \mapsto \mathsf{b}(B)$ is a family on $\mathbf{B}$; the coefficient $\mathsf{b}(f)$ of (1e), like $\mathsf{b}[\Phi]$ of the Notation, §6, has an argument of another kind. The map is holomorphic and local. It is triangular, because v_z depends only on interior bonds, and hence invertible. It satisfies $\sup|\mathsf{b}(B)| \le C \sup|B|$.

Complex fields. On a domain X at level j, a *triple* $(U, A', \mathbf{J})$ consists of a real U, the complex configuration $\mathbf{U} = e^{i\eta_j A'} U$ built from it, and an independent current $\mathbf{J}$. A *tower* of $Z \subset X$ is the separated sequence obtained by eroding Z level by level according to one of two rules. The first uses cubes of side $\mathfrak{M}$ in the unit of their own level, with a gap exceeding ρ in the unit of the outer member. The second uses cubes of side M_1' and requires a gap no smaller than M_1', measured in the unit of the inner member. The *tower configuration* $U_n(Z; \mathbf{U})$ of the tower is obtained by cutting the tower off at depth n, taking the background of the truncated tower, whose determining set is $\mathbf{B}'$, and continuing it analytically to $M_{\mathbf{B}'}(\mathbf{U})$, which is a complex datum. Explicitly, it is $\Phi_{M_{\mathbf{B}'}(U)}(B)$ of Theorem 3 evaluated at $B = \frac{1}{i}\log[M_{\mathbf{B}'}(\mathbf{U}) M_{\mathbf{B}'}(U)^{-1}]$. By this definition the tower configuration is a function of the decomposition (U, A'); the definition does not make it a function of the product $\mathbf{U}$ alone. The *core* $\mathcal{U}_j^\circ(X; \alpha_0, \alpha_1)$ is defined by two sets of requirements. The first are the *simple conditions*:

- $|\partial U - \mathbf{1}|, |\partial \mathbf{U} - \mathbf{1}| < \alpha_0 \eta_j^2$;
- on cubes of side $O(LM)$ there exist real gauges in which $|A|, |\nabla A| < CM\alpha_0$;
- $|A'|, |\nabla_U A'| < \alpha_1$;
- $|\mathbf{J}| < \alpha_0$.

The second are the *tower clauses*. Every tower configuration exists, for $Z \subset X$ of every level $p \le j$, for every $n \le p$ and for both rules. Let Z have two layers of M-cubes removed (the cubes of π_{p-1}, Notation §1; $1 \le n \le p$). On that set, $|\partial U_n(Z; \mathbf{U}) - \mathbf{1}| < \alpha_0 L^{-(j-p)} \eta_p^2$ and $|J(U_n(Z; \mathbf{U}))| < \alpha_0 L^{-(j-p)} L^{2(n-p)}$, with the current formed in level-n units. The same holds with U in place of $\mathbf{U}$. A term extends to the union of the complex gauge orbits of the pairs $(\mathbf{U}, \mathbf{J})$ of the core to the extent that it takes one value at any two pairs of the core lying on one orbit. This holds at two radii (1d). Complex gauge orbits do not saturate the core itself. Of the domain of

analyticity $U_j^{c}(X,\alpha_0,\alpha_1)$ of the Notation, the core is the hereditary form: the regularity that this domain demands of the backgrounds of the averages of $\mathbf{U}$, the core demands on every $Z\subset X$ of every level $p\le j$. On a domain meeting several layers, this space takes the form of the *layered space* $\tilde U_j^{c}$, whose radii are assigned layer by layer. Radii written (α_0',α_1') are radii of the simple conditions only, and the $\alpha_{i,j}$ are the running radii of the Notation, §7.

In the sub-parts some objects carry other symbols as well. In 1b.1, 1b.2, 1b.4 and 1b.6 the depth is denoted k; in 1b.3, 1b.5 and 1b.7–1b.9 it is denoted n. The gap is denoted g in 1b.1 and $\rho_{\mathrm g}$ in 1b.4; in 1b.6, 1b.7 and 1b.9 it is denoted $R_1\mathfrak{M}$, ρ_Ω and $\rho_{\mathrm s}$ respectively. In 1b.4 and 1b.6 the letter ρ without decoration denotes a length on which the thresholds depend, and $L^2\rho$ does not exceed the gap. The straightened average keeps the symbol $\mathcal{Q}^{\mathrm{st}}$ in 1b.1–1b.4. In 1b.5 it is denoted $a(A)$, in 1b.7 $\mathcal{Q}$, and in 1b.6 $\mathcal{Q}^{\natural}$. In 1b.3 and 1b.4 the same symbol also denotes the family obtained by adjoining the rows of the kept inward bonds. In 1b.2 the reduced datum $\mathsf{b}(B)$ is denoted $\tilde B$; it is denoted $B^{\natural}[B]$ in 1b.4, and $B^{\natural}$, the row datum, in 1b.6. In 1b.4, however, the words "reduced datum" refer to $\mathbf{V}^{\flat}$: this is what remains of a datum with inward bonds after the gauge group at the seam nodes has been divided out. In 1b.6 and 1b.7 the symbol $\mathscr{H}$ stands for $\mathcal{H}$, the Landau potential of Theorem 3. In 1b.3–1b.6 the projection $R^{\mathsf c}$ is denoted $\mathsf R$ or R. In 1b.6, A stands for the potential $A^{\mathsf c}$ of the comb slice. In 1b.8 and 1b.9 the radii of a core are denoted b_0,b_1.

(ii) The principal statements. The constants C and the rates depend on L alone, and in Hölder members also on $\alpha_{\mathrm H}$. There are two exceptions. The first is $K^{\sharp}$ and the Lipschitz constant of (b) (Theorem 5); the dependence of this Lipschitz constant is on L together with the radii $\bar\varepsilon$, $r_{\mathrm B}$ from Theorems 1 and 3. The second is the constant and the threshold of Theorem 6 (c), and their dependence is shown in that theorem. Thresholds and radii depend in addition on the lengths chosen before them (end of (iii)), and also on M where this is shown. Above three floors on the torus, there is no dependence on K, ℓ, k, on the regions, on the number of layers or on the datum. The first floor requires at least $O(\rho)$ sites of $\mathbb{T}^{(k)}$ in every direction. It ensures that the double blocks beneath the rows, and likewise the cubes on which gauges are chosen in the proofs, are shorter than half the torus. The first half of Theorem 2 and Theorems 1, 3, 4, 6 (b), (c) require no other floor than this one. The second floor is that each cube on which a statement chooses a gauge stays shorter than half the torus after it is enlarged by $O(\rho)$ of its units. In the second half of Theorem 2 these cubes have side m; in Theorems 5 and 6 (d) they have side $O(LM)$, measured at the level of the space. The third floor is that a tower or a localisation is formed only down to depths p at which M_1' spacings of $\mathbb{T}^{(p)}$ do not exceed one half of the side length of the torus. For the tori $\ell\in\mathfrak{L}(g)$ introduced in the Notation, these depths are all $p\le K$.

Theorem 1 (one regular critical orbit, linear bound). Let $\mathbf{V}$ be real and ε_1-regular, with $\varepsilon_1\le\bar\varepsilon$. The critical configurations of $C(\mathbf{B},\mathbf{V})$ that are regular at a fixed small radius then form precisely one orbit of $\mathcal{G}^{\mathrm{res}}$, and this orbit is non-empty. It is the background $U_k(\mathbf{V})$, also written $U_{\mathbf{B}}(\mathbf{V})$ when more than one sequence is involved, and

$$|\partial U_k(\mathbf{V})(p)-\mathbf{1}|<C\varepsilon_1\eta^2\mathsf{L}(p)^{-2},$$
$$|J(U_k(\mathbf{V}))(b)|<C\varepsilon_1\mathsf{L}(b)^{-3}.$$

Exactly one element $U^{\circ}(\mathbf{V})$ of the orbit lies in the *axial gauge outside the nodes*. This gauge is defined as follows. Let x be a bare site that is not a node, let n be its level ($x\in\mathbb{T}^{(n)}\setminus\mathbb{T}^{(n+1)}$), and let $z\in\mathbb{T}^{(n+1)}$ be the centre of its block; then $(\mathcal{M}^nU)(\Gamma_{z,x})=\mathbf{1}$. Hence, all the way down each pillar, the comb transports of the successive averages equal $\mathbf{1}$. When $\mathbf{V}=\mathbf{1}$ the orbit is $\{\mathbf{1}^u\}$. Conversely, if U is any configuration whose plaquettes are regular at a small radius $\varepsilon\le cL^{-2}$,

then the tests of $M_{\mathbf{B}}(U)$ are within 2ε of $\mathbf{1}$ in the interior of a layer and within $CL^2\varepsilon$ across a seam.

Remark (the gauge with the contour means). Replacing the comb transport by the averaged contour variable $(\mathcal{M}^n U)(z,x)$ gives a second gauge of the same kind, and each orbit again has exactly one element in it. The proof uses only three properties of the variable:

- it depends on the bonds inside $B(z)$;
- it is covariant, $W^h(z,x) = h(z)W(z,x)h(x)^{-1}$;
- $W(z,z) = \mathbf{1}$ and $\mathbf{1}(z,x) = \mathbf{1}$.

The two representatives are related by an element of $\mathcal{G}^{\mathrm{res}}$, and both equal $\mathbf{1}$ over $\mathbf{V} = \mathbf{1}$. In 1b.1 the gauges are developed with the means. Theorems 3 and 5 concern the comb representative, and 1b.8 uses only properties that the two representatives share. In both cases the block axial gauge of a single level, through which V_k^{bg} is defined in Theorem 6, is formed with the mean.

Theorem 2 (local regularity). Let U be critical and regular at a fixed small radius. Then at each $\mathfrak{e}$ with corner x,

$$\eta^{-2}\mathsf{L}^2|\partial U - \mathbf{1}|,\ \mathsf{L}^3|J(U)| \ \le\ C \max_{\tau} e^{-\delta\, d_{\mathbf{B}}(x,\tau)}\, |\partial \mathbf{V}(\tau) - \mathbf{1}|,$$

and consequently, for thresholds $\varepsilon^{(q)}$ given layer by layer, the same quantities are $\le\ C \sup_q \varepsilon^{(q)} e^{-\delta d_{\mathbf{B}}(x,\text{layer } q)}$. Now let $\square$ be a cube of side $m\xi_j$, where m is a multiple of ρ. Assume that its enlargement $\tilde{\square}$ by $O(\rho)\xi_j$ is contained in Ω_j and is shorter than half the torus. Take *any* U that is regular at the radius ε on $\tilde{\square}$, where $m\varepsilon \le c$. Then near $\square$ there is a gauge $U^u = e^{i\eta A}$ in which, on $\square$,

$$\xi_j|A|,\ \xi_j^2|\nabla A|,\ \xi_j^3|\partial^*\partial A|,\ \xi_j^3|\Delta A| \le Cm\varepsilon,$$
$$|\nabla A(x) - \nabla A(x')| \le Cm\varepsilon\, \xi_j^{-2-\alpha_{\mathrm{H}}}|x - x'|^{\alpha_{\mathrm{H}}}\ (\alpha_{\mathrm{H}} < 1),$$

where the last C depends on α_{H} as well and comes with no bound as $\alpha_{\mathrm{H}} \to 1$. The differences are taken with respect to the configuration $\mathbf{1}$, and $\Delta = \sum_\nu \nabla_\nu^{\mathsf{T}}\nabla_\nu$ acts componentwise on A.

Theorem 3 (continuation in the datum; Landau pair). Let $\mathbf{V}_0$ be real and ε_1-regular. Let ε_1 and r_{B} be small enough that $e^{iB}\mathbf{V}_0$ is $\bar{\varepsilon}$-regular for every real B with $\sup|B| < r_{\mathrm{B}}$. Put $U_0 = U^\circ(\mathbf{V}_0)$ and $\mathcal{B} = \{B : \mathbf{B} \to \mathfrak{g}^c,\ \sup|B| < r_{\mathrm{B}}\}$.

There is exactly one holomorphic map $\Phi = \Phi_{\mathbf{V}_0}$ on $\mathcal{B}$ that agrees with $U^\circ(e^{iB}\mathbf{V}_0)$ for real B. Throughout $\mathcal{B}$ its complex averages are defined and $M_{\mathbf{B}}\Phi(B) = e^{iB}\mathbf{V}_0$, and $\Phi(B)$ is regular at the radius $C(\varepsilon_1 + \sup|B|)$.

Assume in addition that $\bar{\varepsilon} \le c(L)/\mathfrak{M}$; this is a ceiling on the radius, chosen after $\mathfrak{M}$. Then exactly one holomorphic pair $(\mathcal{H}, u)$ on $\mathcal{B}$ takes the value $(0, \mathbf{1})$ at $B = 0$ and satisfies

$$\Phi(B) = \big(e^{i\eta\mathcal{H}(B)}U_0\big)^{u_B},$$
$$RD^{\mathsf{T}}\mathcal{H}(B) = 0,$$
$$u_B\, v_j[\mathcal{H}(B)] = \mathbf{1} \text{ at the tops of level } j.$$

For real B this pair is real, and there is a holomorphic multiplier $\omega(B)$ on $\mathbf{B}$ such that

$$F'(\mathcal{H}(B)) = K^{\mathsf{T}}_{\mathcal{H}(B)}\,\omega(B),$$
$$\mathcal{Q}^{\mathrm{st}}(\mathcal{H}(B)) = \mathsf{b}(B),$$

$$\|\mathcal{H}(B)\|_X,\ |D^{\mathsf{T}}D\mathcal{H}(B)|_{(-3)} \le C\sup|B|,$$
$$|u_B - \mathbf{1}| \le C\sup|B| \text{ at the tops.}$$

Let v be G-valued on the nodes, and let $\tilde{v}$ be its extension that is constant on pillars (when inward bonds are kept, the extension is constant along the ascent of each site as far as the first node). Then $\Phi_{\mathbf{V}_0^v}(\mathrm{Ad}(v)B) = \Phi_{\mathbf{V}_0}(B)^{\tilde{v}}$.

Every one of the continued maps is real for the conjugation $^{\mathfrak{c}}$ of the Notation, §11. This conjugation is $A^{\mathfrak{c}} = A^*$ on $\mathfrak{g}^c$ and $g^{\mathfrak{c}} = (g^*)^{-1}$ on G^c, applied value by value. Thus $\Phi(B^{\mathfrak{c}}) = \Phi(B)^{\mathfrak{c}}$, $\mathcal{H}(B^{\mathfrak{c}}) = \mathcal{H}(B)^{\mathfrak{c}}$ and $u_{B^{\mathfrak{c}}} = (u_B)^{\mathfrak{c}}$. At $B = B^{\mathfrak{c}}$ this is reality at real data.

Remark (the comb slice). The comb slice carries a second holomorphic pair (A^{c}, h) on $\mathcal{B}$. It equals $(0, \mathbf{1})$ at $B = 0$, is real in the same sense, and satisfies

$$\Phi(B) = \left(e^{i\eta A^{\mathsf{c}}(B)}U_0\right)^{h(B)},$$
$$R^{\mathsf{c}}D^{\mathsf{T}}A^{\mathsf{c}}(B) = 0,$$
$$h(B)\, v_j[A^{\mathsf{c}}(B)] = \mathbf{1} \text{ at the tops of level } j.$$

This pair satisfies the two equations $F' = K^{\mathsf{T}}\omega$, $\mathcal{Q}^{\mathrm{st}} = \mathsf{b}(B)$ as well, and its potential obeys the same bounds. In addition, $\sup_x |h(B)(x) - \mathbf{1}| \le C\sup|B|$ at *all* bare sites.

R^{c} is the projection that belongs with Bałaban's propagators, and Φ is constructed by means of this pair (1b.6). For R, one has $K_0 D\lambda = 0$ whenever the pillar averages of λ vanish, and $(\mathcal{H}, u)$ is constructed on the basis of Φ (1b.7). The two potentials are gauge transforms of each other: $e^{i\eta A^{\mathsf{c}}(B)}U_0 = (e^{i\eta\mathcal{H}(B)}U_0)^{w_B}$, where $w_B = h(B)^{-1}u_B$ is holomorphic. The two pairs are equal exactly when $RD^{\mathsf{T}}A^{\mathsf{c}}(B) = 0$ near 0. This happens whenever the restriction of U_0 to each pillar is a pure gauge, and so in particular at $\mathbf{V}_0 = \mathbf{1}$. Uniqueness as a pair concerns $(\mathcal{H}, u)$, while the bound at all bare sites concerns h.

Theorem 4 (exponential decay of the response). Let $\bar{\varepsilon} \le c(L)/\mathfrak{M}$, with $\bar{\varepsilon}$ chosen after $\mathfrak{M}$. Put $\mathsf{h} = \partial_s\mathcal{H}(B + se_c)|_{s=0}$, where $e \in \mathfrak{g}^c$ sits at the bond c and is placed at its top y_c. Let δJ be the same derivative of $J(e^{i\eta\mathcal{H}}U_0)$. Then on the whole polydisc, over the top y,

$$\xi_y|\mathsf{h}|,\ \xi_y^2|\nabla\mathsf{h}|,\ \xi_y^3|D^{\mathsf{T}}D\mathsf{h}|,\ \xi_y^3|\Delta_{U_0}\mathsf{h}|,\ \xi_y^3|\delta J| \ \le\ Ce^{-c\delta_0 d_{\mathbf{B}}(y,y_c)}|e|,$$

together with a Hölder member $\xi_y^{-2-\alpha_{\mathrm{H}}}$ for $\nabla\mathsf{h}$. For these variables the bounds used are the value estimates $|\dot{\omega}(c')| \le Ce^{-c\delta_0 d_{\mathbf{B}}(y_{c'},y_c)}|e|$ and $|\dot{u}(y)| \le Ce^{-c\delta_0 d_{\mathbf{B}}(y,y_c)}|e|$, respectively. Here $\dot{\omega}(c') = \partial_s\omega(B + se_c)(c')|_{s=0}$ for a determining bond c', with assigned top $y_{c'}$, and $\dot{u}(y) = \partial_s u_{B+se_c}(y)|_{s=0}$ for a top y; both are measured in the unweighted colour or matrix value norm. No bare 1-form difference or Laplacian is applied to these coarse-row or node variables. Here $\Delta_{U_0} = \sum_\nu \nabla_\nu^{\mathsf{T}}\nabla_\nu$ denotes the covariant Laplacian acting on 1-forms. The member containing this Laplacian holds at every bond whose neighbours all belong to Ω_0, and the member containing the current holds at every bond where the current is defined.

At a regular complex pair $(\mathbf{U}, \mathbf{J})$, critical or not, the corresponding system has a single small solution, and this solution is analytic in the pair. Its response satisfies the bounds on h, $\nabla\mathsf{h}$ and $\dot{\omega}$, and the Hölder member over distances $\le \xi_y$. The second-order members are then $(D^{\mathsf{T}}D + DRD^{\mathsf{T}})\mathsf{h}$, $D(1-R)D^{\mathsf{T}}\mathsf{h}$ and $\Delta_{\mathbf{U}}\mathsf{h}$, because away from criticality the slice condition is not available.

At $\mathbf{V}_0 = \mathbf{1}$ on a single lattice, $\mathsf{H}_k = \partial_B\mathcal{H}(0)$ is the periodisation of a $\mathbb{Z}^4$-invariant kernel. At the lifts of two bonds lying within half the torus of each other, it differs from that kernel by $Ce^{-c\delta_0\ell_k/2}$, with ℓ_k denoting the side length of the torus measured in level-k units.

Theorem 5 (spaces of complex fields). (a) Suppose that the simple conditions of level p hold on Z at small radii (α_0', α_1'). Then the tower configurations exist and depend holomorphically on A'; well inside Z they satisfy

$$|\partial U_n(Z;\mathbf{U}) - \mathbf{1}| < K^\sharp(\alpha_0' + \alpha_1'^2)\,\eta_p^2,$$
$$|J(U_n(Z;\mathbf{U}))| < K^\sharp(\alpha_0' + \alpha_1'^2)\,L^{2(n-p)} \text{ (level-}n\text{ units)}.$$

(Here $K^\sharp$ depends on L and ρ.) Consequently, once $K^\sharp(\alpha_0' + \alpha_1'^2) \le \alpha_0$, a triple of level j that satisfies the simple conditions belongs to the core $\mathcal{U}_j^\circ(X;\alpha_0,\alpha_1)$, $\alpha_1 \ge \alpha_1'$. The weight $L^{-(j-p)}$ in the clauses comes from the change of unit. The triple $(\mathbf{1},0,0)$ belongs to every core.

(b) Suppose that the tower lies, member by member, in an ambient separated sequence, which may be deeper. Let $l \ge \xi'$ be the local scales of the ambient sequence and the tower, and assume that $\eta^{-2}l^2|\partial U - \mathbf{1}|$, $l|A'|$, $l^2|\nabla_U A'|$ stay below a threshold a_T. Then the tower configuration exists, and it depends holomorphically on A'. Fix U, and at an *element* at x take the tower configuration's curvature and current. Such an element is a bare plaquette, or a bond together with its six plaquettes. Take these quantities in the scales ξ' and multiply them by $(l/\xi')^2$, that is, use the factors $\eta^{-2}l^2$ and $l^2\xi'$. At the *settled* elements, those at which the tower has reached the greatest height available to it, these quantities are Lipschitz functions of A' in first-order norms, with a weight $e^{-c\delta\, d^\flat(x,x_b)}$. Here $d^\flat$ is the weighted distance of the ambient sequence. The rate δ is that of Theorem 2; it is the smaller of the two rates involved, the other being the rate of Theorem 4.

(c) The change of unit $(U, L^{-m}A', L^{-3m}\mathbf{J})$, $m = j' - j \ge 0$, sends each core of level j' into a core of level j, the radii becoming $L^{-m}\alpha_i$. The *outer substitution* is $\mathbf{U} \mapsto e^{i\eta_j \mathbf{A}}\mathbf{U}$, with $|\mathbf{A}|, |\nabla_\mathbf{U}\mathbf{A}| < c\alpha_0$ and $c \sim 1/K^\sharp$. For radii with $\alpha_0 \le \alpha_1$, $\alpha_1^2 \le C\alpha_0$, it takes half the radii to the full radii.

By the *inner substitution* we mean the background over $\mathbf{1}$, at the datum $\varsigma B + B'$ with $\varsigma \in [0,1]$, of a box $\square_0$ made up of $O(1)$ cubes. Its ingredients are as follows:

- e^{iB} are the averages of $e^{i\eta_k \mathbf{A}} e^{i\eta_{k+1}\mathbf{H}_{k+1}}$;
- $\mathbf{H}_{k+1}$ denotes the Landau potential belonging to the box background of $\mathbf{U}$;
- $\mathbf{A}$ is arbitrary subject to $|\mathbf{A}| < c\,\alpha_{0,k}$, measured in level-$k$ units; in the application one takes for it the cut-off response to the fluctuation field;
- B' is arbitrary subject to $\sup|B'| < c\min\{\alpha_{0,j-1}, \alpha_{0,j}\}$, where j is the level to be reached.

Suppose that (U, A') satisfies, on some region containing $\square_0$ and at radii larger than $\alpha_{i,k}$ by a fixed factor, both the simple conditions of level $k+1$ and, at that same level, the tower clause of $\square_0$. Suppose further that the couplings $g_q = x_q^{-1/2}$, $q \le k$, are small and that $-C \le x_q - x_{q'} \le C(q'-q)$ for $q \le q'$. Then, on the domains well inside $\square_0$, the inner substitution in Landau form takes values in the cores of every level $1 \le j \le k$, with the running radii $\alpha_{i,j}$. In detail:

- once the tower of $\square_0$ (first rule) is cut off at depth j, the background takes the Landau form $\Phi(\varsigma B + B') = (e^{i\eta_j \mathbf{H}})^u$, with $\mathbf{H} = \mathcal{H}(\varsigma B + B')$, where $(\mathcal{H}, u)$ denotes the pair of Theorem 3 over $\mathbf{1}$;
- the triple $(\mathbf{1}, \mathbf{H}, J(e^{i\eta_j\mathbf{H}}))$ belongs to the core of level j;
- the pair formed by $\Phi(\varsigma B + B')$ and its current, expressed in level-j units, lies in the complex gauge orbit of the pair given by that triple.

(d) Consider the tower of $\square_0$, with determining set $\mathbf{B}'$, and assume the simple conditions at (α_0, α_1). Then $M_{\mathbf{B}'}(\mathbf{U}) = V^v$ with $|V - \mathbf{1}| < CM\alpha_0$ and $|v|\,|v^{-1}| < e^{CM\alpha_1}$, and both factors depend only on $\mathbf{U}|_{\square_0}$.

Theorem 6 (the sequences of the construction; restriction; comparison). (a) Each of the following sequences is separated: the sequences (Ω_j) of the labels (Notation, §9), together

with the sequences of labels to which the domain of the next step has been added; the healed sequences (which arise once a component has been absorbed into $\mathbf{R}$); the prepared sequences and the stage sequences of $\mathbf{R}$, which belong respectively to its preparatory steps and to its stages; the towers; the joined sequences of collared regions (for their definition see (c)), including those of the strip and those of the glued sets of $\mathbf{R}$; and the localisations, which are cube-wise meets of the tower of a region with a separated sequence. (b) Let a separated sequence with $\Omega'_q \subset \Omega_q$ be given, together with a real ε_1-regular $\mathbf{V}$ such that $CL^2\varepsilon_1 \le \bar\varepsilon$ (by the bound and the converse in Theorem 1, the datum of the restriction is then $\bar\varepsilon$-regular). Then

$$U_{\mathbf{B}'}\big(M_{\mathbf{B}'}(U_{\mathbf{B}}(\mathbf{V})|_{\Omega'_0})\big) = U_{\mathbf{B}}(\mathbf{V})|_{\Omega'_0}$$
$$\text{modulo } \mathcal{G}^{\mathrm{res}}(\mathbf{B}').$$

For a sequence of depth $k+1$, no block $B(y)$, $y \in \mathbb{T}^{(k+1)} \cap \Omega_{k+1}$, contains a node other than y. Hence the orbit $U_{k+1}(\mathbf{V})$ contains some U for which $\mathcal{M}^k U$ is block axial there. Moreover, $V_k^{\mathrm{bg}} := \mathcal{M}^k U$, restricted to the bonds of $\mathbb{T}^{(k)}$ that have an end point in Ω_{k+1}, does not depend on which such U is chosen, as a count of nodes shows. Applied to the truncation, the restriction identity then gives $U_k(V_k^{\mathrm{bg}}) = U_{k+1}(V_{k+1})$, that is, $U_k(\mathbf{V}') = U_{k+1}(\mathbf{V})$, up to gauge. Here $\mathbf{V}' = (V', (V_q)_{q<k})$, where $V' = V_k^{\mathrm{bg}}$ on these bonds and V_k on the remaining bonds of level k. (c) Let $\mathbf{V}$ be real, and let $\mu \ge \rho$ be a power of L. A *μ-collared region* W is a cube-wise meet of *parts* A_a. In statements on regions the letter W denotes a set of points, and in statements on blocks and boxes it denotes a configuration; its arguments distinguish the two. Each part is a union of cubes of side $\mu\xi_{j_a}$, and two conditions hold. First, the $O(1)$ outermost layers of cubes of A_a, which form its *collar*, lie within the single layer j_a of the sequence. Second, the boundaries of distinct parts A_a, $A_{a'}$ are separated by at least $O(1)$ cubes of side $\mu \max\{\xi_{j_a}, \xi_{j_{a'}}\}$. Its *joined sequence* coincides with the given sequence well inside W and with the tower of A_a in the collar. The *joined datum* equals $\mathbf{V}$ on the old bonds. On the new bonds, the *collar bonds*, whose levels are $q < j_a$, it is the *lift* of V_{j_a}. The lift takes the value $\mathbf{1}$ at a bond lying inside a block of level j_a, and the value $V_{j_a}(c)$ at a bond leading from the block of c_- into the block of c_+. The *localised background* U°_W is defined as the representative of the background attached to this problem. Take one part on one lattice, with $\mu = M'_1$. The problem is then the one posed on the tower of the part, with values taken from the lift of V. This is the problem, set on a neighbourhood of a cube $\square$, that defines the localised background $U_{k,\square}(V)$ of the Notation, §4. If $\varepsilon_1 \le \mathsf{a}(L,\rho,\mu)$, then $U|_W = (e^{i\eta f} U^\circ_W)^w$, where

$$\xi_y|f|,\ \xi_y^2|\nabla f|,\ \xi_y^3|D^*Df| \le C_\mu\, \varepsilon_1 e^{-\delta_W d_{\mathbf{B}}(y, W^c)},$$

over each top y of the joined sequence. Here ξ_y is its scale in that sequence: well inside W it is the scale of the given sequence, and in the collar it is the scale of the tower, which can be finer than ξ_{j_a}. The rate δ_W is a fixed fraction of a single rate chosen below those of Theorems 2 and 4. The constant C_μ depends on L, ρ and μ, and $C_\mu \sim e^{O(\mu)\delta_0}$. The reason is that the mismatch is located on the collar bonds, which lie up to $O(\mu)$ from W^c in $d_{\mathbf{B}}$, so that the decay sets in from the inner edge of the collar onward. The gauge transformation w is determined only at the tops lying away from the collars, where $w = v_q[f]^{-1}$; at every other point it is subject to no restriction. Two backgrounds whose sequences and whose data coincide on W differ there to this extent. (d) The *profile* of the datum at the rate δ of Theorem 2 is the function $x \mapsto \max_\tau e^{-\delta\, d_{\mathbf{B}}(x,\tau)}\, |\partial\mathbf{V}(\tau) - \mathbf{1}|$, that is, the right-hand side of the first inequality of that theorem without the constant C. Let e_X be its supremum over the neighbourhood of X whose width, in level-j units, is $O(\rho)$, and suppose that Me_X is small. If this neighbourhood is contained in Ω_j, then $(U, 0, J(U))|_X$ belongs

to the core of X at level j for all radii $\alpha_0 \ge Ce_X$, $\alpha_0, \alpha_1 > 0$. In the opposite case three further conditions are imposed. First, the gaps are wider by a fixed factor, $\operatorname{dist}_\infty(\Omega_q^c, \Omega_{q+1}) > C\rho\,\xi_q$ for every q; this holds for the sequences of (a) listed before the towers. Second, X is a union of cubes of side $M_1'\xi_j$. Third, the profile multiplied by M is small also near each cube of side $O(M\xi_r)$ in Ω_r, $r \le j$, meeting X. Under these conditions the restriction belongs to the layered space of X, with radii that, layer by layer, dominate C times the profile.

The later parts rely on three properties of the orbit: criticality, uniqueness among regular configurations, and regularity. Over **1** the orbit consists of the absolute minima. At other data "minimiser" stands for "regular critical point", which is the notion that extends to complex data.

(iii) Plan of the proofs.

Chain. 1b.1 contains the geometry of seams, the gauges and the definitions given above. 1b.2 begins with the mean of group elements that lie close to one another. It then treats $\mathcal{M}$ on its domain (analytic, local, covariant) together with its holomorphic extension, and the plaquettes of averages. Next come $\mathcal{Q}^{\mathrm{st}}$, v_j, the gauge functional $\mathcal{R}$ ($w\,v[A] = v[A^w]\mathcal{R}$), rigidity and the section of one step, all on a single lattice. Finally, everything except the section is extended by folding onto boxes, one layer after another. The folding lemma belongs to 1b.2; 1b.3 adopts it and states it once more so as to apply the statements of 1b.2 on a double block, and 1b.4 constructs a section of one step at a seam. 1b.3 contains the linear operators of mechanisms 2 and 4, the calculus of mechanism 6 and the gauge slice. 1b.4 (Theorem 1) treats first a single lattice and then sequences, carrying the datum down through the truncations and the configurations upward. Its induction is closed by an improvement of the radius that is specific to it: the global halving $\varepsilon_0 \to \max\{C\varepsilon_1, \frac{1}{2}\varepsilon_0\}$, following Proposition 8 [cited from Bałaban; Package E] of Commun. Math. Phys. 102 (1985) 277–309, with operators taken at a reference configuration cut off to **1** near a site. Over **1**, flatness follows from a separate energy inequality in scale-invariant ℓ^2 norms. 1b.5 (Theorem 2) gives that halving in local form. It concerns an arbitrary regular critical configuration and uses only operators at **1**. It uses neither existence nor uniqueness, and it yields the global improvement as a corollary. 1b.6 (Theorem 3: the map Φ, constructed by way of the comb slice) gives the complexification of the system of 1b.4. 1b.7 (Theorem 4; the pair $(\mathcal{H}, u)$ and its uniqueness) carries out the argument once more in weighted norms, with R and with K_0, which on Γ_j equals ξ_j times the $Q_j^\sharp(U_0)$ of the Notation, §3, that is, the linear part of the straightened average. 1b.8 (Theorem 5) uses Theorems 1, 2 at real points only, and Theorems 3, 4 on the whole polydisc. From the relation $\Phi_{\mathbf{V}_0} = (e^{i\eta\mathcal{H}}U_0)^u$ it takes only the real B near 0, and it carries out the continuation itself, with an explicit gauge factor. 1b.9 (Theorem 6) combines geometry of cubes, exact algebra of averages and Theorems 1 (one orbit, which drives the restriction), 2 (profile, local gauges), 3, 4 and 5 (cores).

Interplay with (1c). (1b) draws on four groups of results of (1c), and (1c) draws on Theorems 3 and 4. Three of these groups depend only on 1b.1–1b.2. They are: (a) on separated sequences, the flat operators at **1**; (b) at complex configurations, the linearised averages together with their gauge identities; and (c) at *regular pairs*, the propagators, together with the rows of $\mathcal{M}$. The regular pairs form a class defined by inequalities (curvature, gauges on cubes, complex part, independent current) that do not mention any critical point. The fourth group, (d), concerns G', $(Q'G'^2Q'^*)^{-1}$ and G at a real regular U_0, with Bałaban's rows Q, Q'. On a single lattice, (d) applies the theorems of Commun. Math. Phys. 99 (1985) 389–434 to the sequence in which every member is the torus. In that paper the first member carries Dirichlet conditions outside it, and members equal to the torus are explicitly admitted in the companion papers (volume 96 after (2.4), volume 99, 75–102, after (1.6)). This group uses these theorems for his own average, which is moved onto the centred blocks by a translation by half a block. For a separated sequence with an arbitrary first member Ω_0, the group derives the same bounds and the positivity by a separate argument. Where that paper uses an expansion whose cover covers the whole torus, this

argument uses a parametrix made from one flat operator on the torus, dressed with local gauges. Group (d) depends on those theorems, on (a), on 1b.2, and on the lemmas of 1b.3 that make no use of a bound on a propagator: the algebra of $\mathsf{Q}', \mathsf{Q}, R^{\mathsf{c}}$, pillars and rows, folding, sums over tops, Combes–Thomas, and the second variation of a single plaquette. Its operators are different from those of (c), and the two kinds of rows coincide at $\mathbf{1}$. The proposed proof must establish these results in the following order: 1b.1, 1b.2; (a); the geometric and algebraic half of 1b.3; (d); the remainder of 1b.3 and 1b.4–1b.6, which depend on (d); and then 1b.7. 1b.7 uses G_0, G_1, H_1 of (c) at the pair $(U_0, F'(0))$, which is a regular pair of small radius as soon as $\mathfrak{M}\bar{\varepsilon} \le c(L)$. This condition is a ceiling on the radius $\bar{\varepsilon}$, which is chosen after $\mathfrak{M}$; it places no upper bound on $\mathfrak{M}$.

On a sequence, 1b.3 and 1b.4 need less than (d) provides. They use only the positivity and the four bounds that involve no distance: $|G'f| \le C\xi^2 \sup|f|$, $|DG'f| \le C\xi \sup|f|$ and the corresponding bounds for G, ∇G, where ξ is the local scale. These are needed at an auxiliary first member that closely surrounds Ω_1, namely the one described in that paper on p. 394, which is Ω_1 together with a layer of big blocks around it. Theorems 3.1 and 3.3 of that paper are formulated for a first member of this kind, and they also state these bounds; the paper proves them by an expansion that is set up for a cover covering the whole torus. In the following form, which includes the decay, the positivity and these bounds hold as well. Take a separated sequence with $\Omega_1 \neq \emptyset$ and an arbitrary first member. Its cube size $\mathfrak{M}$ exceeds a threshold that depends on L alone, and its gap is at least $L\mathfrak{M}$ and exceeds a second threshold of the same kind. Take also a real configuration with small gauges on cubes and with curvature small in the local units. Then the three operators are positive definite. The third of them, and the bounds on G and ∇G, require in addition that the curvature be small in the local units at those bare plaquettes which protrude from Ω_0 while having a corner in it. For f supported over the top y', one has $|G'f|_y \le C\xi_y^2 e^{-\delta_0 d_{\mathbf{B}}(y,y')} \sup|f|$. The same bound holds with ξ_y for DG', with ξ_y^{-4} for $(Q'G'^2Q'^*)^{-1}$, and with ξ_y^2, ξ_y for G, ∇G. Here C and δ_0 depend on L alone, and the proof uses the parametrix built from a single flat operator on the torus and dressed with local gauges. The auxiliary sequence is a sequence of this kind. Its cube size and gap number, both functions of L, replace $\mathfrak{M}$ and the gap number of the given sequence. Three consequences follow (1b.3, 1b.4): the bound on $(Q'G'^2Q'^*)^{-1}$, the decay in $d_{\mathbf{B}}$ (mechanism 6), and the passage to the given Ω_0. The last of these is obtained by a partition of unity in layer 0, where the rows act as a mass η^{-2}. The required smallness is the slope of the cut-offs, and the gap pays for it.

A lemma of 1b.7 states the following. Let U_0 be real and regular at the radius ε, and let $\mathfrak{M}\varepsilon \le c$. Then $(U_0, F'(0))$ is a regular pair of radius $C\mathfrak{M}\varepsilon$, since the potential is $\sim$ side $\times$ curvature. Only the gauges on cubes require an argument. In layer 0 the comb gauge is used. Above it, the gauge on a cube is given by Proposition 6 of Commun. Math. Phys. 99 (1985) 75–102; the second half of Theorem 2 above is the counterpart of that proposition for sequences. From that proposition the lemma needs the conclusions (1.135)–(1.136) and the condition (1.7)–(1.9), the latter with (side of the cube) $\times \alpha_0$ small. Neither of these involves an average; (1.137)–(1.138) involve one and are not used, while the paper proves the proposition by means of his average. In the paper the proposition is formulated for configurations defined on the whole torus and regular at all that touches Ω_0. In the lemma, the configuration is given on Ω_0 only, and it is regular at the inner plaquettes and at the bonds where the current is defined. When Ω_0 is not the torus, the lemma extends the configuration by $\mathbf{1}$. It then applies the proposition to a sequence truncated at the level of the cube. The first member of that sequence is Ω_0 with the layer of cubes of side $\mathfrak{M}\eta$ that touch Ω_0^c removed, so the first gap is $\rho - \mathfrak{M}$. Everything that touches this member, together with the plaquettes at it, lies inside Ω_0, so (1.7)–(1.9) hold at every place where the paper checks them. For the cubes, which are of the levels $j \ge 1$, the enlargements are contained both in Ω_{j-1} and in the smaller first member, and their length is less than half that of the torus. The blocks of the paper are brought into line with the centred ones by the translation by half a block.

The decoupled response of (1c) invokes Theorems 3 and 4 only once 1b.7 is complete, and it does so in order to identify its solution, at the value 1 of all parameters, with $\mathcal{H}$; thus this dependency order does not by itself introduce a cycle. The uniform estimates and the applicability of each comparison still have to be proved. The pairs of (1c) are taken from the spaces of Theorem 5 and from the sequences of Theorem 6. These spaces and sequences depend on (1c) only through Theorems 1–4, which is to say only through those results of (1c) that come before these theorems in the order fixed in (1c) ("The order of (1b) and (1c): no circle").

The principal mechanisms.

1. **Straightening, and geometric series in the level.** Once the block factors are removed, the average of a perturbation becomes a perturbation of the average. For a single step the row sums are $\le L$ and the column sums are $\le L^{-3}$, up to relative errors $\mathsf{e}_m \le C(\alpha_0 L^{-2(j-m)} + L^{m-j}\xi_j \sup|A|)$, and these errors are summable. Consequently, when α_0 and $|A|_{(-1)}$ are small, and for complex A as well, the rows $c \in \Gamma_j$ satisfy

$$\sum_b |K_A(c,b)| \le 2\xi_j,$$

$$\sum_c |K_A(c,b)| \le 2\eta^4 \xi_j^{-3},$$

$$v_j = \prod_m e^{iY_m},\ \sum_m |Y_m| \le C|A|_{(-1)}.$$

   It follows that K_A^* is bounded from sup into $|\cdot|_{(-3)}$, and that the gauge factors obey bounds uniform in k. Non-unitary gauge transformations do not preserve the domain of the complex average. The identity $\mathcal{M}(\mathbf{W}^g)(c) = g(c_-)\mathcal{M}(\mathbf{W})(c)g(c_+)^{-1}$ remains valid provided that the condition number $|g|\,|g^{-1}|$ at the two coarse end points, multiplied by the field size in the double block, remains small. In the opposite direction, one application of the average multiplies plaquettes by exactly L^2:

$$|\partial\mathcal{M}(W)(p') - \mathbf{1}| \le L^2\varpi + C\varpi^2$$

$$(\varpi :\ \text{the largest } |\partial W - \mathbf{1}| \text{ in the four blocks at } p'),$$

   because around p' the contour terms cancel at the corners (lattice Stokes). The factor L^2 is absorbed by the change of unit, and the errors add up as a geometric series. Hence, if $|\partial U - \mathbf{1}| \le \alpha_0\eta^2$ with α_0 small, then $|\partial\mathcal{M}^n U - \mathbf{1}| \le 2\alpha_0(L^n\eta)^2$ for every $n \le k$. Beneath a corner of p' that lies outside Ω_j, the curvature is small only when it is measured in the unit of level $j-1$, and this is the loss of L^2 across a seam.

2. **A single system with a multiplier.**

$$F'(A) + DRD^*A = K_A^*\omega,$$

$$\mathcal{Q}^{\mathrm{st}}(A) = \mathsf{b}(B).$$

   The constraint stays non-linear and is imposed through the multiplier ω, which takes the place of an analytic change of variables bringing the constraint to the linear form $QA = B$. We use Bałaban's rows Q of (i), from which the propagators are constructed, and accordingly R^{c} in place of R. This holds in mechanisms 2–4, while mechanism 7 uses the form with K_0 and R. Up to a rotation of the rows, which is omitted from the notation, $\mathcal{Q}^{\mathrm{st}} = \mathsf{Q} + \mathsf{C}$. Besides its quadratic part, C has a *linear* part of relative size α_0, the radius

of U_0. With $\mathsf{G}_0 = (\Delta^{\mathrm{W}}_{U_0} + DRD^* + \mathsf{Q}^*\mathsf{Q})^{-1}$, $\mathsf{H}_0 = \mathsf{G}_0\mathsf{Q}^*(\mathsf{Q}\mathsf{G}_0\mathsf{Q}^*)^{-1}$ and $\tilde{\mathsf{G}}_0 = \mathsf{G}_0 - \mathsf{H}_0\mathsf{Q}\mathsf{G}_0$, the system takes the form of a fixed-point equation $(A,\omega) = \Psi(A,\omega)$:

$$A = \tilde{\mathsf{G}}_0\mathsf{N} + \mathsf{H}_0\big(\mathsf{b}(B) - \mathsf{C}(A)\big),$$
$$\omega = \mathsf{H}_0^*\big[F'(A) + DRD^*A - \mathsf{C}'(A)^*\omega\big],$$
$$\mathsf{N} = -F'(0) - \mathcal{V}(A) + \mathsf{C}'(A)^*\omega.$$

The current $F'(0)$ of U_0 acts as an inhomogeneous term alongside B, and every bound is linear in $|F'(0)|_{(-3)} + \sup|B|$. When U_0 is critical, the current's only effect is to produce the multiplier ω_0 of U_0. After summation by parts, $\mathsf{H}_0^*D^*DA = (D\mathsf{H}_0)^*DA$, only first differences of A remain. In terms of size the Lipschitz matrix is triangular. Its entries are $O(\alpha_0 + r)$, where r is the radius of the ball, except for one entry $O(1)$, and the norm $\max\{\|A\|_X, \Theta^{-1}\sup|\omega|\}$ overcomes that entry. (The system with the multiplier is solved in this way on one lattice, in 1b.6 and in 1b.7. For sequences the multiplier is instead eliminated, in 1b.4. Since $1 + \mathsf{H}_0^*\mathsf{C}'(A)^*$ is invertible, the second equation by itself forces $\omega = \mathsf{w}(A)$. The contraction is then carried out in $\|\cdot\|_X$ alone: the Lipschitz constant of w is not small, but it enters only through the small operator $\mathsf{C}'(A)^*$. Uniqueness then imposes no condition on ω.) Two balls are used: a fixed ball for uniqueness, and a ball of radius $C(|F'(0)|_{(-3)} + \sup|B|)$ for the linear bound. Pairing with the gauge directions that fix $\mathcal{Q}^{\mathrm{st}}$ yields $(RD^*A, RD^*\delta_\lambda A) = 0$, and transversality then yields the Landau condition a posteriori.

3. **Each regular configuration has a representative in the slice; the three resets.** For uniqueness in a ball to yield *one orbit*, it is necessary that every regular U whose averages lie within α_1 of $\mathbf{V}_0$ can be written as $U^g = e^{i\eta A}U_0$ with $RD^*A = 0$, with $g = v_k[A]$ at the nodes, and with A in that ball. Within this mechanism α_1 denotes this distance. On one lattice the proof is an induction on the level. A gauge transformation brings the averaged contour variables of $\mathcal{M}^{k-1}U$ into agreement with those of $\mathcal{M}^{k-1}U_0$. The rigidity of one step then transfers the closeness to the level below. Rigidity states that if the contour variables are equal, the averages lie within α_1 and the plaquettes are $\le \varpi$, then the bonds lie within $\alpha_1 + C\varpi$, with coefficient 1. Rescaling the representative obtained there loses a factor L^2. A gauge correction brings back the slice and the normalisation, the last factor of v_k being prescribed at the nodes; it is obtained by Brouwer's theorem and so comes with bounds only. The loss is recovered by an a priori estimate in the slice. This estimate requires no criticality. It requires only that U_0 and $e^{i\eta A}U_0$ be regular at a small radius α_0, and that $|A|_{(-1)}$ lie below a fixed threshold, which the constant B_1 below does not depend on:

$$\mathsf{G}_0^{-1}A = D^*DA + \Delta'A + \mathsf{Q}^*\big(\mathcal{Q}^{\mathrm{st}}(A) - \mathsf{C}(A)\big),$$
$$\|A\|_X,\ |D^*DA|_{(-3)} \le \mathsf{B}_1\big(\alpha_0 + \sup|\mathcal{Q}^{\mathrm{st}}(A)|\big);$$

here the currents of U_0 and U bound D^*DA, and the constant B_1, which is a multiple of the norm of G_0, does not change from level to level. On a sequence, the statement that every such regular configuration has a representative in the slice takes the same form, now with the weighted norms and with one gauge condition at each top, but the induction that proves it is a different one. The induction above rests on the gain $\alpha_0 \to L^{-2}\alpha_0$ in passing one level down. With a single radius this gain is absent whenever an outer layer is non-empty, since the gain is located on Ω_k alone. Instead, a single *preliminary gauge* $\hat{g}$, which is $\mathbf{1}$ at the tops, brings the contour variables of every average into line at once. Rigidity

with a defect then carries the closeness downwards. In this form one adds the mismatch between the contour variables at the two end points of a bond, while the averages keep the coefficient 1. Let j be the largest index for which both end points of the bond lie in Ω_j. The increments $C\alpha_0 L^{2(n-j)}$ are then summable because of the position of the bond, while the factor $2L-1$ is incurred a single time, at a seam, the gauge being frozen there in the outer block:

$$|\mathcal{M}^n(U^{\hat{g}})(b)\,\mathcal{M}^n U_0(b)^{-1} - \mathbf{1}| \le (2L-1)\alpha_1 + C\alpha_0$$

$$(n \le k,\ b \text{ a bond of } \mathbb{T}^{(n)} \text{ with an end in } \Omega_n),$$

and this family of bounds passes to truncations. The representative is then constructed by ascending along the truncations $(\Omega_0, \dots, \Omega_j)$, $j = 0, \dots, k$. Each stage consists of a rescaling (with loss L^2), a gauge correction on the tops and the estimate in the slice, and the output of the estimate does not retain the size of its input. It therefore suffices that, after the loss, the potential remains under the fixed thresholds required by the gauge correction and by the estimate; this condition determines how small $\alpha_0 + \alpha_1$ must be. Given two critical configurations, take one of them as U_0. Then both (A, ω) and $(0, \omega_0)$ solve the system with $B = 0$, hence $A = 0$ and $g = \mathbf{1}$ at the nodes. Of the three resets, each of which overcomes a loss that grows with the level, the a priori estimate is the first. The improvement of the radius (mechanism 5) is the second reset. These two devices go back to Bałaban (see (iv)). The third prevents the descent of the datum from compounding its losses. Away from the seam, the section of a single step is exact; near $\partial\Omega_n$ it loses a factor $C_\downarrow$, multiplied by the tests in a collar, and the gap keeps this collar apart from whatever the earlier sections have spoilt. In a class with two radii, ε in the collars and $C_\downarrow\varepsilon$ elsewhere, at each level the datum descends with $C_\downarrow\varepsilon$, and not with $C_\downarrow^k\varepsilon$.

4. **A lower bound uniform in the levels.** The bound $\mathsf{Q}\mathsf{G}_0\mathsf{Q}^* \ge \gamma_0$ follows from the variational formula for this form, together with a right inverse of all the rows built from bumps at the local scales. The Dirichlet energy of such a bump, $\xi^4(\xi^{-2})^2 = 1$, is scale invariant, and no bound on the kernel of G_0 enters. On one lattice the bumps are placed across the face shared by the two blocks of a row, and it suffices to use diagonal dominance. At a seam, rows of two neighbouring levels overlap. There each row carries a single bump lying well inside *one* pillar, and the profiles are biorthogonal, exactly, to both chain counts that neighbouring rows assign as weights to a bare bond. The coupling between neighbouring levels is not assumed small. The interlevel part of the bump argument uses two inputs: the nilpotent structure of the coupling and a bound on its finite elimination polynomial that is uniform in the depth. The first makes the elimination finite; the second yields a depth-independent γ_0. The bound also holds for rows that the mass term of G_0^{-1} does not contain, namely the inward rows and their differences. Conjugation in the manner of Combes–Thomas yields the decay of the inverse.
5. **A critical configuration has the regularity of its datum.** One trivialises on a cube of side σ, applies a cutoff, aligns the Landau slice with a comb gauge of the datum and splits into near and far parts. This gives

$$\mathsf{r}(x_0) \le C\,\mathsf{v}_R(x_0) + \big(C\sigma^2 e^{-c\delta_*\sigma} + C\sigma^4 \mathsf{r}_R(x_0)\big)\mathsf{r}_R(x_0),$$

$$C, c \text{ independent of } \sigma$$

(here r, v denote the local radii of U and the sizes of the tests, and the index R indicates the maximum over a $d_{\mathbf{B}}$-ball of radius $R \sim L\sigma$). The decay dominates the linear growth of

the comb, and for this reason the constant C in front of v_R does not depend on σ. Taking σ large then makes $e^{-c\delta_*\sigma}$ dominate σ^2, the side of the cube times the length of a comb. After that, a regularity radius $\lesssim \sigma^{-4}$ compensates σ^4. Theorem 2 is then obtained from a maximum principle whose weight varies slowly, at rate $\delta \sim (L\sigma)^{-1}$. In 1b.4 the inequality, in its global form, closes at every level the cycle of radii $C\varepsilon \to L^3 C\varepsilon \to C' L^3 C\varepsilon \to C\varepsilon$, made up of the change of level, the solution and the improvement. The radius that comes out is therefore the radius that went in.

6. **Each crossing of a layer costs ρ.** For every rate a that occurs, with ρ chosen so that $a\rho \geq C \log L$,

$$d_{\mathbf{B}}(y, y') \geq \rho(|j - j'| - 1)_+,$$
$$\sum_{y'} e^{-a\, d_{\mathbf{B}}(y,y')} \leq \mathsf{S}(a) :$$

the factor $e^{-a\rho}$ outweighs the entropy, L^4 for each level, and a fraction of the rate pays for powers of $\xi_y/\xi_{y'}$. The distance $d_{\mathbf{B}}$ itself is too rough to serve as a conjugation weight for the propagators. In 1b.3 and 1b.4 one therefore works with a *regularised distance* $\mathsf{f} = \mathsf{f}_{y'}$, whose first and second differences are controlled at the local scale $\xi(x)$:

$$|\mathsf{f} - d_{\mathbf{B}}(\cdot, y')| \leq C,$$
$$|\mathsf{f}(x') - \mathsf{f}(x)| \leq C \frac{\eta}{\xi(x)},$$
$$|\mathsf{f}(x + \eta e_\mu) + \mathsf{f}(x - \eta e_\mu) - 2\mathsf{f}(x)| \leq C \frac{\eta^2}{\xi(x)^2}.$$

Estimated term by term, the scale-normalised Laplacian perturbation from conjugation by $e^{\theta \mathsf{f}}$ has an apparent factor $\theta\, \xi/\eta$. Pairing the directions $\pm e_\mu$ and separating a field from its neighbouring increments gives, for small θ,

$$|(e^{\theta \mathsf{f}} \Delta e^{-\theta \mathsf{f}} - \Delta) v| \leq C\theta \big(\xi^{-2} |v| + \xi^{-1} |\nabla v| \big),$$

with the field and gradient norms taken on the local stencil. Thus the no-distance bounds on G and ∇G stated above bound the composed perturbation $(e^{\theta \mathsf{f}} \Delta e^{-\theta \mathsf{f}} - \Delta) G$ by $C\theta$ in the corresponding force supremum norms. This is a relative inverse estimate, not a uniform bound for the bare Laplacian perturbation on arbitrary bounded fields. Together with the corresponding relative bounds for the other operator terms, it is the estimate used by the Combes–Thomas argument to obtain decay in $d_{\mathbf{B}}$.

7. **Decay from the same contraction.** In 1b.7 the map Ψ of mechanism 2 is formed with K_0 itself as the linear rows, with the projection R of the Landau gauge, and with $\mathsf{H} = G_1 K_0^{\mathsf{T}} (K_0 G_1 K_0^{\mathsf{T}})^{-1}$ and $\tilde{\mathsf{G}} = G_1 - \mathsf{H} K_0 G_1$. Since K_0 carries the length ξ_c of its row c, this H is the H_1 of the Notation, §5, multiplied by ξ_c^{-1}. At the critical U_0, the operator G_1^{-1} contains the Hessian, the gauge term and the mass term of the rows, and in addition the local term $-\mathsf{W}_0$. Here $\mathsf{W}_0 \delta = \frac{d}{dt} K_{t\delta}^{\mathsf{T}} \omega_0 \big|_{t=0}$ is obtained by pairing the second-order part of the constraint with the multiplier ω_0, and accordingly N contains here the further term $-\mathsf{W}_0 A$. Because the constraint is kept non-linear, every non-linear term is local. The non-locality is confined to H, $\tilde{\mathsf{G}}$, H^{T} and $1 - R$, and the last of these decays like the others. The projection R enters through the term $\mathsf{H}^{\mathsf{T}} D R D^{\mathsf{T}} A$ of the equation for ω, and at the critical U_0 this term vanishes, because $R^{\mathsf{T}} = R$ and $R D^{\mathsf{T}} G_1 K_0^{\mathsf{T}} = 0$. Consequently $\|\partial \Psi\, Z\|_{\theta, y_0} \leq \frac{1}{2} \|Z\|_{\theta, y_0}$ in the norms weighted by $e^{\theta d_{\mathbf{B}}(\cdot, y_0)}$. This holds for every centre y_0,

for every θ from 0 up to a fixed fraction of the decay rate of H, $\tilde{\mathsf{G}}$, and at every point of a fixed small ball around $(0,\omega_0)$. Denote by $\mathbf{A}(\mathsf{b})$ the solution of the system in which the right-hand side b of the constraint is left free. Its derivative with respect to $\mathsf{b}(c)$, together with the derivative of the multiplier, satisfies

$$(1-\partial\Psi)(\mathsf{h},\dot{\omega}) = (\mathsf{H}e_c, 0).$$

Then $\mathcal{H} = \mathbf{A}\circ\mathsf{b}$, and Theorem 4 follows by the chain rule through b, since each bond enters $O(1)$ rows within $O(L)$. The second-order members are obtained from the differentiated equation, not from a kernel.

8. **Passing from real points to the polydisc; the covariant curl is small.** The iterates are holomorphic, and by Weierstrass's theorem their uniform limit is holomorphic as well. The identities that hold at real B (the slice, criticality, covariance) extend to complex arguments by the identity theorem, and the Schwarz lemma makes the bounds linear. Let ϕ be holomorphic and bounded on a ball of radius r, and let $w = w_{\mathrm{R}} + iw_{\mathrm{I}}$, where $w_{\mathrm{R}}, w_{\mathrm{I}}$ are real and $\|w_{\mathrm{R}}\|, \|w_{\mathrm{I}}\| \le \|w\| \le r/2$. Then

$$|\phi(w)| \le C\big(|\phi(0)| + |\phi(w_{\mathrm{R}})| + |\phi(w_{\mathrm{I}})|\big) + C\sup|\phi|\,\|w\|^2/r^2 :$$

the first-order Taylor term is complex linear, so the crude supremum appears only in the quadratic remainder. We apply this with ϕ the rescaled plaquette or the rescaled current of a tower configuration, and with $w = A' = A'_{\mathrm{R}} + iA'_{\mathrm{I}}$. If one uses only $|A'|, |\nabla_U A'| < \alpha'_1$, the curvature of the real fields $e^{i\eta A'_{\mathrm{R}}}U$, $e^{i\eta A'_{\mathrm{I}}}U$ is no better than $O(\alpha'_0 + \alpha'_1)$. However, the two bounds $|\partial U - \mathbf{1}|, |\partial\mathbf{U} - \mathbf{1}| < \alpha'_0\eta^2$, together with

$$\partial(e^{i\eta A'}U) = \big(\mathbf{1} + i\eta^2 D_U A' + O(\eta^2|A'|^2 + \eta|A'|\,|\partial U - \mathbf{1}|)\big)\,\partial U$$

yield $|D_U A'| \le C(\alpha'_0 + \alpha'^2_1)$. Since $\mathrm{Ad}(U)$ commutes with taking adjoints, the real part and the imaginary part satisfy the same bound, and therefore both real fields have curvature $C(\alpha'_0 + \alpha'^2_1)$. Theorem 1 is linear in the datum, and it bounds ϕ at the three real points. This gives the bound $K^\sharp(\alpha'_0 + \alpha'^2_1)$ without any bound on $\nabla A'$. The square is essential because of homogeneity. Measured in units of level n, the radii equal $L^{-2(p-n)}\alpha'_0$ and $L^{-(p-n)}\alpha'_1$. Hence α'^2_1 has the same scaling as α'_0 and as the two bounds of the clause, which in level-n units are $\eta_p^2 = L^{-2(p-n)}\eta_n^2$ and $L^{2(n-p)}$ (Theorem 5 (a); the weight $L^{-(j-p)}$ enters through the radius and is independent of n). A bound that is linear in these two radii, with a constant that does not depend on the level (the Schwarz lemma provides such a bound on the ball), costs a factor L^{p-n}. In these units $|\nabla_U A'|$ is bounded by $L^{-2(p-n)}\alpha'_1$, which is smaller than the radius by a factor $L^{-(p-n)}$. Consequently, when $\alpha'_0 + \alpha'_1 \le c$, the inequality for ϕ together with Theorem 1, applied to the two real fields, whose curvature is $O(\alpha'_0 + \alpha'_1)$, yields $C(\alpha'_0 + \alpha'_1)\eta_p^2$ and $C(\alpha'_0 + \alpha'_1)L^{2(n-p)}$, and these bounds do not incur that factor. Compared with this bound there is a second gain. The condition $K^\sharp(\alpha'_0 + \alpha'^2_1) \le \alpha_0$ allows $\alpha'_1 \gg M\alpha_0$, and therefore allows $\alpha'_1 \ge c\alpha_1$ together with $M\alpha_0 \ll \alpha_1$, the relation that holds between the two radii in the application. A condition linear in α'_1 instead forces $\alpha'_1 \le \alpha_0/K^\sharp$.

9. **The coarser constraints are frozen by the finer ones.** Let $U' \in C(\mathbf{B}', \mathbf{V}')$ be extended by U. Induction on the level then gives $M_{\mathbf{B}}(\widetilde{U'}) = \mathbf{V}$ and $A_{\Omega_0}(\widetilde{U'}) - A_{\Omega_0}(U) = A_W(U') - A_W(U|_W)$, so restriction preserves criticality. By mechanism 1 the restricted datum is $CL^2\varepsilon$-regular. It therefore has an orbit when $CL^2\varepsilon \le \bar{\varepsilon}$, and the restriction identity follows from uniqueness. For the comparison one performs a gauge alignment on

the collar; it turns the mismatch into a real B_1 with $|B_1| \le C\varepsilon_1$, supported on the bonds of the collar, and $f = \mathcal{H}(B_1)$ decays by Theorem 4. The alignment relies on 1b.9, the lemma on the *defect of lifts*, whose step from one level to the next is the reconstruction bound of 1b.2. After a gauge transformation fixing the sites of level j, each lower average of U agrees with the lift of the level-j average up to the total, over the intermediate averages, of their plaquette sizes, the coefficient being 1 at each passage from one level to the next. When the collar lies inside one layer j, this sum is the geometric series $\sum_{q<j} 2\varepsilon L^{2(q-j)} \le C\varepsilon$, uniformly in j. It is here that the argument needs the boundary to run within one layer.

Constants are chosen in the following order, each one large or small enough in terms of those chosen before it: first L; then the one-step constants of $\mathcal{M}$; then δ_0 together with the propagator constants; then γ_0, with the derived rate δ_*; then the length σ; next, the constant $\delta \sim (L\sigma)^{-1} \ll \delta_0$ that appears in Theorem 2 and, at the same step, the regularity radius $\lesssim \sigma^{-4}$; then the cube size $\mathfrak{M}$, together with the gap $\rho = R_1\mathfrak{M} \gg \sigma$, subject to (rate) $\times\rho \ge C \log L$ for each rate in use; then the remaining radii ($\bar\varepsilon$, r_{B}), with $\mathfrak{M}\bar\varepsilon$ required to be small; then M_1' and M; then the threshold of Theorem 6 (c), chosen after μ (in the application $\mu = M_1'$); finally the field strengths, with $M\alpha$ small. In the order of choice fixed in the Notation, everything listed here before M_1' comes after L and before M_1', and $\mathfrak{M}$, the cube size, stands where the Notation has M_1. For Theorem 1, and for Theorem 3 in its first half (the map $\Phi_{\mathbf{V}_0}$, together with the pair of the comb slice), the thresholds involve the gap only through one length. That length may be fixed at its least admissible value, which is a function of L; with this choice the thresholds depend on L alone, and so do the constants that the Schwarz lemma derives from them (crude bound divided by radius). The pair $(\mathcal{H}, u)$ of Theorem 3 requires in addition $\mathfrak{M}\bar\varepsilon \le c(L)$, and so, along with it, does Theorem 4; for both, the threshold on the datum therefore depends also on $\mathfrak{M}$. No upper bound is placed on $\mathfrak{M}$, ρ, M.

The reduced datum has an explicit local inverse. We now detail the two finite inversions used in the construction. In (BGD1)–(BGD4), B and b are raw row logarithms. Their per-length versions are $\ell^{-1}B$ and $\ell^{-1}b$, where $\ell = \mathrm{diag}(\xi_c)$ is the row-length matrix. Fix a real reference field U_0 whose used averages lie on the regular branches of the actual finite mean. For an interior row the reduced datum is $b(B)(c) = B(c)$. On an outward row oriented toward its outer endpoint z, it is

$$e^{ib(B)(c)} = e^{iB(c)}\, Z_c(B_{\mathrm{int}}), \qquad Z_c(B_{\mathrm{int}}) = \mathrm{Ad}(V_0(c))\, v_z(B_{\mathrm{int}}). \tag{BGD1}$$

The factor v_z is the actual one-step block factor made from the interior data in the finer block centered at z. The determining-set geometry above shows that these are prescribed interior rows, not another set of unknown outward rows. For the reverse orientation use the inverse group identity.

There is therefore a direct inverse. Recover $B_{\mathrm{int}} = b_{\mathrm{int}}$, then set

$$B(c) = \frac{1}{i} \log\left(e^{ib(c)} Z_c(b_{\mathrm{int}})^{-1}\right) \quad \text{on outward rows.} \tag{BGD2}$$

All logarithms are on the common small datum chart. Conjugation by the real $V_0(c)$ has norm one; it imposes no smallness of that coarse edge or of a global holonomy. The finite mean construction gives a radius $r_0(L) > 0$ on its admitted local reference class. On a fixed smaller ball, its block factors and their first two derivatives are bounded by constants depending on that finite construction. Thus (BGD1)–(BGD2) are admitted on balls $\|B\|_\infty < r_0/C_L$, respectively $\|b\|_\infty < r_0/C_L'$, and satisfy

$$\|b(B)\|_\infty \le C_L \|B\|_\infty, \qquad \|b^{-1}(b)\|_\infty \le C_L' \|b\|_\infty. \tag{BGD3}$$

The uniform statement is over local reference patches on which that same finite chart and its radius are admitted. Establishing those patches for a multilevel background is part of its regular-domain estimate; it is not inferred just from the existence of U_0.

It is useful to see exactly why the inverse does not contain a long product of triangular matrices. Define the right logarithmic differential $\mathcal{E}_X(H) = (1/i)(d/dt)e^{i(X+tH)}|_{t=0}e^{-iX}$. Differentiating (BGD1) gives the outward diagonal $D_c = \mathcal{E}_{b(c)}^{-1}\mathcal{E}_{B(c)}$, not the identity in general. Let D be this block diagonal matrix, with identity on interior rows, and let T contain all derivatives of block factors. Its only nonzero blocks have interior columns and outward rows. Consequently

$$Db = D + T, \qquad TD^{-1}T = 0, \qquad (Db)^{-1} = D^{-1} - D^{-1}TD^{-1}. \tag{BGD4}$$

Multiplying the last two matrices verifies both inverse identities. This nilpotence is a property of the actual reduced datum. An arbitrary lower triangular coupling between successive levels would not have it.

For the row count, one outward factor reads the $O(L^4)$ interior data of its single finer block. For the column count, an interior bond belongs to one assigned finer block, and that block can be the outer endpoint block of at most eight coarse incident rows. Thus both row and column counts are bounded at fixed L, independently of the number of layers. After inserting the actual cell weights and amplitudes, an interaction changes scale only from $j-1$ to j; its weight ratios are fixed powers of L. Equations (BGD3)–(BGD4) therefore also give two-sided weighted Schur bounds in those declared norms. Higher derivatives follow from the same finite group formula and its fixed datum disc.

This local datum calculation is different from constructing a right inverse of all the field constraint rows. The latter must also lift fields and control their energy and gauge slice at a seam. That primitive linear task is retained in the precision section. The exact derivative Db, including its nonidentity outward diagonal, is the datum forcing in the response equations.

Eliminating the multiplier. We next write the multiplier as a function of the potential. Here we use the per-length constraint $q(A) = \ell^{-1}\mathcal{Q}^{\mathrm{st}}(A)$ and put $Q_0 = Dq(0)$. The field and row pairings are the declared cell-weighted pairings, and $\sharp$ denotes their transpose. With these pairings fixed, the corresponding multiplier is $\omega_{\mathrm{len}} = \ell\omega_{\mathrm{raw}}$; we write it as ω in (BGD5)–(BGD8). The field space X carries the norm $\|A\|_X = \sup_y \max\{\xi_y|A|, \xi_y^2|\nabla A|\}$, and Z is the multiplier space with its row weights. Here X^* denotes the designated physical force space, with norm $\|f\|_{X^*} = \sup_y \xi_y^3|f|$, paired with fields using the declared cell measures; it is not the Banach dual of X. No a priori multiplier bound enters the uniqueness argument. Fix $Q_0E_0 = 1$, and write $Dq(A) = Q_0 + C'(A)$. At a critical field, apply the transpose of this right inverse to the multiplier equation. This gives

$$[1 + (C'(A)E_0)^\sharp]\omega = E_0^\sharp DF(A). \tag{BGD5}$$

If $\|(C'(A)E_0)^\sharp\|_{Z\to Z} < 1/2$, the bracket has its convergent Neumann inverse, and the multiplier is the explicit function

$$\omega(A) = [1 + (C'(A)E_0)^\sharp]^{-1}E_0^\sharp DF(A). \tag{BGD6}$$

The required transpose bound is a consequence of the column as well as row estimate of the actual lift, with the declared force and multiplier amplitudes. It does not follow from an unweighted supremum bound on E_0.

Use the prescribed propagator lift E_0 and put $F_0(A) = E_0^\sharp DF(A)$. Here γ controls the paired force throughout the chosen field ball, including its variation away from $A = 0$. The bounds needed on the field ball are $\|F_0(A)\|_Z \le C_E\gamma$ and $\|F_0(A) - F_0(A')\|_Z \le C_EC_F\|A - A'\|_X$. They concern the paired force: in the quadratic part, $E_0^\sharp D^{\mathsf{T}}DA = (DE_0)^{\mathsf{T}}DA$, so the weighted derivative bounds on the lift leave only first differences of A. No pointwise second-difference

bound on all fields of the ball is needed. Let the bracket's perturbation have Lipschitz constant C_N. The inverse difference identity gives

$$\begin{aligned} \|\omega(A)\|_Z &\le 2C_E\gamma, \\ \|\omega(A)-\omega(A')\|_Z &\le (2C_EC_F + 4C_E\gamma C_N)\|A-A'\|_X. \end{aligned} \tag{BGD7}$$

The second constant need not be small. However a field fixed-point equation reads it through the small factor $C'(A)^\sharp$. If $\|C'(A)^\sharp\| \le cr$ and its Lipschitz constant is c, then

$$\|C'(A)^\sharp\omega(A) - C'(A')^\sharp\omega(A')\|_{X^*} \le \{cr\,\mathrm{Lip}(\omega) + 2cC_E\gamma\}\|A-A'\|_X. \tag{BGD8}$$

The coefficient is small with $r+\gamma$. This is the field-only form of the coupled contraction. It shows why a non-small derivative of the multiplier is not an obstruction.

Alternatively suppose the two-component iteration has a Lipschitz matrix bounded by $\left(\begin{smallmatrix}\epsilon & \epsilon \\ C & \epsilon\end{smallmatrix}\right)$. In the norm $\max(\|A\|_X, \Theta^{-1}\|\omega\|_Z)$ its row sums are at most $\epsilon+\epsilon\Theta$ and $C/\Theta+\epsilon$. Choose a fixed $\Theta \ge 4C$ first, then $\epsilon(1+\Theta) \le 1/4$. Both row sums are less than one half. The admitted multiplier radius must be at least Θ times the field radius used by this ball. These choices are made with the actual physical norms, not by suppressing the multiplier dimension in notation.

22. Constrained propagators and local decay

Proof outline. The supporting analytic results used in this section, including the obligations that we must still prove, are listed under A in the dependency catalogue. Their hypotheses and precise forms are retained at the points where the argument uses them.

(1c) Propagators in a background field: exponential decay produced by the averages. Every object produced by a renormalisation step is assembled from inverses of "Hessian of the action + gauge term + mass made of block averages". These inverses are taken around a background on a decreasing sequence of regions, in which each layer has a scale of its own. The construction uses five properties of these inverses: they exist; they depend analytically on the background; in each layer they behave like massive Green's functions measured in the unit of that layer; they decay exponentially; and they are sums of terms that are local both in the field and in the regions. A single requirement governs the whole construction. Since the regions are supplied by the large-field analysis, none of the constants may vary with their shape or their number, nor with the level, the cutoff or the torus. Two numbers are exceptions, and neither of them enters any localised bound, rate, weight or sum of weights. (Bałaban's papers are referred to by the volume of Commun. Math. Phys. in bold. Where two papers must be told apart, the first page is added: **99**, 75 and **99**, 389; **102**, 277; **122**, 175. The references are collected in the bibliography. The principal statements supplied in (ii) are numbered 0–5. The mechanism descriptions also refer to auxiliary hypotheses 6–8, whose required operator data are listed before (iii). The mechanisms in (iii) carry the letters (a)–(k).)

(i) The setting and the objects.

Geometry. The integer L is odd and large. The cube size of the walk expansions, M_1, and the gap number, R_1, are both powers of L, with $R_1 \ge L$. We work in level-k units, in which $\mathbb{T}^{(j)}$ has spacing $\xi_j := L^{j-k}$; moreover $\eta = \eta_k = \xi_0$. A sequence $\Omega_0 \supset \cdots \supset \Omega_k$ is called *separated* when every Ω_j is a union of M_1-cubes at level j and

$$\mathrm{dist}_\infty(\Omega_j^c, \Omega_{j+1}) > R_1M_1\xi_j.$$

No further condition is placed on the regions. In particular Ω_0 is arbitrary, and Dirichlet conditions are imposed outside it. *Layers* are the sets $\mathcal{L}_j = \overline{\Omega_j \setminus \Omega_{j+1}}$. The *tops* y are the sites of $\mathbb{T}^{(j)}$ lying in $\mathcal{L}_j$. Such a top has local scale $\xi_y = \xi_j$ and block $\Delta(y)$, which is the cube of side ξ_y with centre y. The *rows* c are the bonds of $\mathbb{T}^{(j)}$ that have one end point in $\mathcal{L}_j$ and the other end

point either in $\mathcal{L}_j$ (the *interior* rows) or, when $j \geq 1$, outside Ω_j (the *outward* rows); for them $\xi_c = \xi_j$. The *inward* bonds, those that enter Ω_{j+1}, carry no row. The distance

$$d_{\mathbf{B}}(y, y') = \inf_{\Gamma \ni y, y'} |\Gamma|_{\mathbf{B}},$$

$$|\Gamma|_{\mathbf{B}} = \sum_{j \geq 1} \xi_j^{-1} \operatorname{length}(\Gamma \cap \mathcal{L}_j) + \xi_0^{-1} \operatorname{length}(\Gamma \setminus \Omega_1)$$

(Γ connected)

measures each layer in the unit of that layer. Outside Ω_1, and hence also outside Ω_0, length is counted in the bare unit. Therefore $d_{\mathbf{B}}$ is independent of Ω_0, and a path that leaves Ω_0 finds no cheaper unit of length out there. This distance differs from the distance (2.46) of **96**, whose contours are made up, in each layer, of bonds of the lattice of that layer. A larger class of paths is allowed here, and so $d_{\mathbf{B}}$ is the smaller of the two.

Regular pairs $(\mathbf{U}, \mathbf{J})$ consist of $\mathbf{U} = e^{i\eta A'} U$, with U real, together with an **independent** $\mathfrak{g}^c$-valued 1-form $\mathbf{J}$ that represents the current. Such a pair is required to satisfy, on $\mathcal{L}_j$,

$$|\partial U - \mathbf{1}|, |\partial \mathbf{U} - \mathbf{1}| < \alpha_0 \eta^2 \xi_j^{-2},$$

$$\xi_j |A'|,\ \xi_j^2 |\nabla_U A'| < \alpha_1,$$

$$|\mathbf{J}| < \gamma_0 \xi_j^{-3},$$

and, on every cube of side $O(M_1 \xi_j)$ contained in $\mathcal{L}_j \cup \mathcal{L}_{j+1}$, to admit a real gauge $U = e^{i\eta A}$ with $\xi_j |A|, \xi_j^2 |\nabla A| \leq \alpha_0$. In this condition the bound is α_0 on the entire cube, and not $O(M_1)\alpha_0$. Pairs in which the current is a separate variable, with a radius γ_0 of its own, are due to Bałaban (**109**, p. 262, (1.10)–(1.14)). In the present setting they are formed on a sequence, with the local scales. Throughout, $J(\mathbf{U})$ is the first variation of the action of the *inner* plaquettes, namely those inside Ω_0. At every bond whose six plaquettes are all inner, it coincides with the current $D^*_{\mathbf{U}} F_{\mathbf{U}}$; when $\Omega_0 = T_\ell$, every bond is of this kind. *On shell*: the pair satisfies $\mathbf{J} = J(\mathbf{U})$ at the bonds that touch Ω_1. *Real*: $A' = 0$, and $\mathbf{J}$ is real. *Local class on* W: the same conditions, required inside W.

Operators. Relative to $\mathbf{U}$, the *straightened* j-fold average of $e^{i\eta A}\mathbf{U}$ takes, at a row c of level j, the value $\exp i\mathcal{Q}(A)(c)$. Here $\mathcal{Q} = \mathsf{m}\,(Q + C^{(2)} + \dots)$, and m is multiplication by ξ_c. Thus Q, which maps 1-forms to rows, is the linear part per unit length, and $C^{(2)}$ is the quadratic part per unit length. When the level has to be displayed, the straightened average is written $\mathcal{Q}_j^{\mathrm{st}}(\mathbf{U}; A)$. In the units of level j, its linear part is the operator $Q_j^\sharp(\mathbf{U})$ of the Notation, §3, which is built from the means over the contour families of $\mathcal{M}$. The index st stands for "straightened"; the *straight* linear average $Q_j^{\mathrm{st}}(\mathbf{U})$ of the Notation is a different operator. Consequently, in this part the unadorned letters $\mathcal{Q}$ and Q differ from those of §3 of the Notation: they denote neither the logarithm $\mathcal{Q}_j(\mathbf{U}; A)$ of $\mathcal{M}^j(e^{i\eta A}\mathbf{U})\,\mathcal{M}^j(\mathbf{U})^{-1}$ nor its linear part $Q_j(\mathbf{U})$. The straightened and the plain averages are related by *site factors* v_n. These are group-valued functions on the sites of $\mathbb{T}^{(n)}$, and they accumulate from one step to the next (statement 3: $\mathcal{M}^n(e^{i\eta A}\mathbf{U}) = \big(e^{i\mathcal{Q}_n^{\mathrm{st}}(A)}\mathcal{M}^n\mathbf{U}\big)^{v_n}$). The linear parts of the straightened and of the plain average already differ at $\mathbf{1}$, by a gauge term on the coarse lattice. For the plain averages, the identity $QD = \bar{D}Q'$ fails, and so does the bound $1 + C(\dots)$ on the derivative in statement 3. The operator Q' maps site functions to tops. It is the linearisation of the average of a gauge transformation, taken with the same contour variables. The symbol ^{T} denotes the transpose with respect to bilinear pairings, so that everything remains holomorphic. The operator a acts on rows as multiplication by $a_0 \xi_c^{-2}$, where a_0 is a fixed number (equal to 1 in the sub-parts). On tops it acts as multiplication by a comparable weight $\approx \xi_y^{-2}$. We

write $D = D_{\mathbf{U}}$. Finally, $\Delta^{\mathrm{W}}_{\mathbf{U}}$ is the Hessian of the action in Ω_0. In statements 3–8 it is the Hessian of the action of the inner plaquettes, which is an exactly invariant function. In statements 1, 2 it is obtained by taking the Hessian on the entire lattice and compressing it to the bonds of Ω_0.

$$G' = (\Delta_{\mathbf{U}} + Q'^{\mathsf{T}} a Q')^{-1},$$
$$R = 1 - G'Q'^{\mathsf{T}}(Q'G'^2Q'^{\mathsf{T}})^{-1}Q'G',$$
$$\mathsf{S} = 1 - DG'RD^{\mathsf{T}},$$

$$\mathsf{G}_0 = (\Delta^{\mathrm{W}}_{\mathbf{U}} + DRD^{\mathsf{T}} + Q^{\mathsf{T}} a Q)^{-1},$$
$$\mathsf{G} = (\mathsf{G}_0^{-1} - \Delta'_\pi[\mathbf{J}])^{-1},$$
$$\mathsf{G}_1 = (\mathsf{G}^{-1} - \mathsf{S}^{\mathsf{T}}\Delta^{(2)}[\mathsf{H}^{\mathsf{T}}\mathbf{J}]\mathsf{S})^{-1},$$
$$\mathsf{H}_{(1)} = \mathsf{G}_{(1)}Q^{\mathsf{T}}(Q\mathsf{G}_{(1)}Q^{\mathsf{T}})^{-1}$$

(where a bracketed index occurs, the formula holds both with that index and without it). The operators are defined in this order: G, then H from G, then G_1, then H_1 from G_1. The operator $\Delta'_\pi[\mathbf{J}]$ is linear in $\mathbf{J}$; at the actual current it restores the gauge invariance of the Hessian, as in statement 3. Moreover, $(A, \Delta^{(2)}[\mathsf{j}]A) = 2(C^{(2)}(A), \mathsf{j})$. The *six propagators* are

$$\mathsf{G}, \mathsf{G}_1, \mathsf{H}, \mathsf{H}_1, (Q\mathsf{G}Q^{\mathsf{T}})^{-1}, (Q\mathsf{G}_1Q^{\mathsf{T}})^{-1}$$

.

The current correction is defined by the gauge identity of [B85c, (3.117)–(3.120)], with the current treated as an independent variable as in [B87, (1.8)–(1.14)]. All operators are formed at the same reference $\mathbf{U}$, with the site map, slice and pairings fixed above. Put

$$\lambda_A = G'RD^{\mathsf{T}}A, \qquad \lambda^+_A(b) = \mathrm{Ad}(\mathbf{U}(b))\lambda_A(b_+), \qquad \mathsf{S}A = A - D\lambda_A.$$

In every bracket below, λ_A without a superscript is evaluated at b_-, so all factors are based at the negative endpoint. The symmetric bilinear form is

$$\begin{aligned}\langle A, \Delta'_\pi[\mathbf{J}]C\rangle_{\mathrm{f}} = \tfrac{1}{2}\big\langle & i[\lambda_A, C] + i[\lambda_C, A] - i[A, \lambda^+_C] - i[C, \lambda^+_A] \\ & - i[\lambda_A, D\lambda_C] - i[\lambda_C, D\lambda_A], \mathbf{J}\big\rangle_{\mathrm{f}}.\end{aligned} \tag{22.1}$$

The fine pairing includes its cell volume η^4. For complex fields the transpose and colour pairing are bilinear. On the physical current, for the invariant inner-plaquette action, the second variation of gauge invariance gives

$$\Delta^{\mathrm{W}} - \Delta'_\pi[J(\mathbf{U})] = \mathsf{S}^{\mathsf{T}}\Delta^{\mathrm{W}}\mathsf{S}.$$

Equation (22.1) specifies the continuation away from that physical current. The physical identity alone would not specify it. A compressed whole-lattice Hessian retains the boundary comparison described below.

In addition we set $\mathfrak{G} = \mathsf{G}_1 - \mathsf{G}_1 DRD^{\mathsf{T}}\mathsf{G}_1 - \mathsf{H}_1 Q\mathsf{G}_1$. For step $k+1$ we work on the bonds of $\mathbb{T}^{(k)}$ that touch Ω_{k+1}, with $V = \mathcal{M}^k\mathbf{U}$. Let C_V be the elimination of the pivot bonds, and let $\mathsf{G}^{(2)}_V$ be the quadratic part of the soft axial gauge fixing. Let $\mathsf{E}^{(2)}$ be the quadratic part of the one-step constraint, which is imposed through the plain $\mathcal{M}$ at V and whose linear part is the Q_V of the Notation, §3. We set

$$\Delta_k = (Q\mathsf{G}_1Q^{\mathsf{T}})^{-1} - a - \mathsf{E}^{(2)}[\mathsf{H}_1^{\mathsf{T}}\mathbf{J}],$$

$$\mathsf{A} = \mathsf{C}_V^{\mathsf{T}}(\mathsf{G}_V^{(2)} + \Delta_k)\mathsf{C}_V,$$
$$C^{(k)}(Y) = (\chi_Y \mathsf{A} \chi_Y)^{-1}.$$

At the pair $(U_{k+1}, J(U_{k+1}))$ of the step, one has $\mathsf{A} = \mathsf{C}^{\mathsf{T}}\Delta^{(k)}\mathsf{C}$. In this identification (1c.1), the elimination and the gauge-fixing term are obtained directly from the definitions. The only step of substance is the quadratic term of the action of the background, regarded as a function of its datum. This is Bałaban's expansion **102**, 277, (174), together with **109**, (2.4)–(2.8), (2.11). There the derivation is carried out on the whole lattice, using Bałaban's own rows and his averages. For $\mathcal{M}$ and the straightened rows, on the sequence of a label, the expansion is obtained from the system with a multiplier of (1b). That multiplier coincides with the gradient of the action of the background with respect to its datum, as one sees by differentiating the constraint and pairing the result with the Euler–Lagrange equation. A single differentiation of the system yields the derivative of the multiplier. For the critical pair, the operator G_1^{-1} equals the sum of the Hessian of the Lagrangian, the gauge term, and the mass term carried by the rows. Hence this derivative is obtained by applying $(Q\mathsf{G}_1 Q^{\mathsf{T}})^{-1} - a$ to the datum. When the datum moves along a curve contained in the fibre of the one-step average, it picks up a second-order term, and this term is determined by the quadratic part of the one-step constraint. Paired with the gradient, this term yields $-\mathsf{E}^{(2)}[\mathsf{H}_1^{\mathsf{T}}\mathbf{J}]$. Bałaban also gives a second expression for the first two terms of Δ_k, namely $\mathsf{H}_1^{\mathsf{T}}[\mathsf{G}_1^{-1} - DRD^{\mathsf{T}} - Q^{\mathsf{T}}aQ]\mathsf{H}_1$. It differs from them by $-\mathsf{H}_1^{\mathsf{T}}DRD^{\mathsf{T}}\mathsf{H}_1$, a term that vanishes on shell. The bounds and the positivity in statement 6 concern the form displayed above.

Walks. A *cell* is a union of $O(1)$ cubes of side $O(M_1\xi_j)$ lying in two consecutive layers. In a *walk* $\omega = (X_0, \dots, X_m)$, any two consecutive cells overlap. In a *tree walk*, each X_i overlaps an earlier cell, called its parent, and the cells are enumerated depth first. With arbitrary parent maps, one sequence of length m is counted as many as $m!$ times. The *hull* $[\![\omega]\!]$ enlarges the walk by $O(M_1)$ local units and is not cut off at Ω_0. The *carrier* is $[\omega] = [\![\omega]\!] \cap \Omega_0$. Finally, $d_{\mathbf{B}}(\omega; y, y')$ is the smallest weighted length of a tree that passes through y, y' and meets the cells $X_1, \dots, X_m$. The tree is not required to meet X_0.

Several letters have a different meaning in the sub-parts 1c.1–1c.10, as follows. In 1c.1–1c.5, in all sections of 1c.8, and in the section of 1c.9 that expands A over sets of cubes, the cube size of the walks is denoted $\mathfrak{M}$. In those places M_1 denotes $M_1' \geq R_1 M_1$, which is the second cube size introduced in the Notation, the larger one. Within 1c.5 and 1c.9, the product $R_1 M_1$ appears as ρ and as w_0. The cubes $\square$ that carry parameters (statement 7) have side ℓ_{P}. This side is denoted w in 1c.9, and in 1c.10 the cubes themselves are denoted Δ. In 1c.2, 1c.4 and 1c.10, Γ_j is the set of rows of level j. In 1c.6, 1c.7 it also contains the inward bonds, and the rows are its bonds that are not inward. In 1c.10, Q stands for $\mathsf{m}Q$, and slanted H, H_1 stand for $\mathsf{H}\mathsf{m}^{-1}, \mathsf{H}_1\mathsf{m}^{-1}$. In the section that treats the averaging operators at complex configurations, the operator $Q_k^{\mathrm{st}}(\mathbf{U}) = \eta\, \mathbf{Q}(W_{k-1}) \cdots \mathbf{Q}(W_0)$, where $\mathbf{Q}(W)$ denotes the linear part of a single step, stands everywhere for the linear part of the straightened average, namely the $Q_k^{\sharp}(\mathbf{U})$ of §3 of the Notation; it is not the straight average defined in that section of the Notation.

In comparison with the Notation:

- R_1 denotes a number (as in **116**, p. 3) and not a region R_t.
- A cell means a cell of a walk, as opposed to a level-k cell of §9.
- The hull of a walk differs from the hull in a coarser partition.
- $\mathfrak{G}$ denotes an operator, not the gauge-fixing factor $\mathfrak{G}_k$.
- A denotes the fluctuation operator; in §5 of the Notation the same operator is written K.
- $\Delta(y)$ is the block $B^j(y)$, regarded as a cube.
- $s(\square)$ is a parameter, not a depth.

- $\bar\kappa$ denotes, as in **116**, the exponent that enters the radius of the parameters, and not the rate at which the large-field factor decays. In that paper the exponent is written κ_1; it is the $\bar\kappa$ that appears in the Notation's order of the constants.
- $\gamma, \gamma^{\mathrm{pos}}$ are lower bounds, not ceilings on couplings.
- γ_0 is the radius of the current, which is also written α_0 in §11.
- λ denotes a site function (an infinitesimal gauge transformation); in statement 8 and in mechanisms (j), (k), however, it denotes a multiplier, that is, a function on rows.
- X denotes a localisation domain in statement 0, X_i is a cell, X in (I) is a function on rows, and X_χ is a gauge direction.
- A' denotes the complex factor of the triple $(U, A', \mathbf{J})$; in the system of clause 1 of the theorem of 1c.10, however, it denotes the unknown $H_1(s)B' + \mathbf{A}$.
- S denotes, in statement 6, a set of coarse cubes and, in (II), an operator; S denotes the operator $1 - DG'RD^{\mathsf{T}}$.

In 1c.10 the following further conventions hold:

- θ_0 denotes the decay rate with respect to $d_{\mathbf{B}}$; it is not one of the members of the sequence θ of §6.
- $\mathsf{r} = v[\mathbf{H}]$ is a function on the nodes with values in G, and not an isometry.
- β and $\beta_\natural$ denote a small real datum, not the bare inverse coupling.
- $\omega(\cdot)$ is the map that sends a field to its multiplier; it is not a walk.
- $\mathsf{E}_1 = G_1 D R D^{\mathsf{T}} G_1$ denotes an operator rather than an event.
- $P_1(s)$ is the operator built from the walk data of $D(1-R)D^{\mathsf{T}}$, and neither a region nor a momentum.

(ii) The principal statements. All constants depend only on L (and on Hölder exponents). There are two exceptions, and neither enters any localised bound, rate, weight or sum of weights. The first is the bound of statement 7, which serves only for convergence. The second is the entrywise bound of mechanism (i), which makes the family normal. The numbers γ, γ^{pos}, δ_0, δ_*, δ' are positive constants of this kind.

In statement 2 the side of the torus, measured in top units, is at least a small fixed multiple of $R_1 M_1$; in the sentence of that statement which concerns a single lattice, it is at least C. From statement 4 onwards the side is at least CM_1. In 5, the single assignment of terms valid for all tori needs room for three cubes of side $O(M_1)$ in each direction. Statements 7, 8 need room for three cubes of side ℓ_{P}. These are lower bounds on the volume, and no constant depends on the volume except the bound of statement 7 just mentioned.

In what follows:

- $x \in \Delta(y)$, and the sources f lie near $\Delta(y')$;
- $\mathsf{e} = e^{-\delta_0 d_{\mathbf{B}}(y,y')}$, up to a fixed fraction of the rate;
- $\|\cdot\|_\varepsilon$ denotes a Hölder seminorm;
- kernels are taken with respect to the counting measure;
- on 1-forms, the covariant Laplacian is $\Delta_{\mathbf{U}} = \sum_\nu \nabla_\nu^{\mathsf{T}} \nabla_\nu$, where ∇_ν are the covariant differences of the components;
- $\Delta = \Delta_{\mathbf{1}}$ is the full vector Laplacian $\partial^*\partial + \partial\partial^*$, with $\partial = D_{\mathbf{1}}$. It acts componentwise and is not the $\Delta = \Delta_U^{\mathrm{W}}$ of the Notation, §5.

0. The domains of analyticity are contained in the classes. Consider a pair in the domain of an activity on $X \in \mathcal{D}_j$. Such a pair satisfies $|\partial U - \mathbf{1}|, |\partial \mathbf{U} - \mathbf{1}| < \alpha_0 \eta^2$, $|A'|, |\nabla_U A'| < \alpha_1$ and $|\mathbf{J}| < \gamma_0$, and on cubes of side $O(LM)$ it admits real gauges with $|A|, |\nabla A| < O(LM)\alpha_0$. It is a regular pair for the tower of regions obtained from X by erosion (1b.8), and the radii with which it is regular are

$$\big(O(LM)\,\alpha_0,\,\alpha_1,\,\gamma_0\big).$$

Let a label be produced by steps in which each region Ω_n, $n \ge 1$, is a union of LM-cubes of level n. By a *layered domain* we mean a domain defined by the layered bounds of (2a), *Structures*, whose radii do not exceed the running radii of the level of each layer. Only the following of these bounds enter. On the layer $\mathcal{L}_n$ of the label, $n \ge 1$, one has $|\partial U - \mathbf{1}|, |\partial \mathbf{U} - \mathbf{1}| < \alpha_{0,n}\eta^2\xi_n^{-2}$, $|\mathbf{J}| < \alpha_{0,n}\xi_n^{-3}$ and $\xi_n|A'|, \xi_n^2|\nabla_U A'| < \alpha_{1,n}$. The cubes of level n of the domain are the closed cubes of side $CM\xi_n$ that are contained in $\mathcal{L}_n \cup \mathcal{L}_{n+1}$ and meet Ω_{n+1}^c; on the part in X of each of them there is a real gauge with $\xi_n|A|, \xi_n^2|\nabla A| < CM\alpha_{0,n}$. Then on $X \cap \Omega_1$ a pair of a layered domain belongs to the local class of the *label sequence*, which is the sequence formed by the Ω_n of the label. The radii are $O(M)\max_n \alpha_{0,n}$ and $\max_n \alpha_{1,n}$, together with $\max_n \alpha_{0,n}$ for the current. In that class the gauge clause is derived for those cubes of level-n size which meet $\mathcal{L}_n$. For $n = 0$, on the cubes inside $\mathcal{L}_1$, it is derived from the gauge of a cube of level 1, which can be used at the index 0 since $\xi_0 < \xi_1$. The same remark applies to a cube of level-n size inside $\mathcal{L}_{n+1}$, because some cube of level $n+1$ of the domain contains it. This smaller cube is contained in Ω_{n+1}, intersects Ω_{n+2}^c, and its side is below $M\xi_{n+1}$. The derivation of the clause on the cubes meeting $\mathcal{L}_n$ uses no other property of a cube; hence repeating the same steps with the grids of level $n+1$ produces the larger cube. The factors $O(LM)$, $O(M)$ are the cost of the gauge clause, since on a cube of side $O(M)$ a gauge has $|A| \le O(M)\alpha_0$. The field strengths are fixed after M.

1. The flat theorem. Let $\mathbf{U} = \mathbf{1}$, $\mathbf{J} = 0$, and let Q, Q' be the straight averages taken over the centred blocks. Then on every separated sequence

$$|\mathsf{G}_0 f|, |\nabla \mathsf{G}_0 f|, |\mathsf{G}_0 \nabla^* f|, |\Delta \mathsf{G}_0 f|\,(x) \le C[\xi_y^2, \xi_y, \xi_y, 1]\,\mathsf{e}\,|f|,$$
$$|\nabla \mathsf{G}_0 \nabla^* f(x)| \le C_\varepsilon\,\mathsf{e}\,\big(\xi_{y'}^{\varepsilon}\|f\|_\varepsilon + |f|\big),$$

together with matching bounds in Hölder and L^2 norms. The same bounds hold for G', and

$$|(Q'G'^2Q'^*)^{-1}(y,y')| \le C\xi_y^{-4}\mathsf{e},$$
$$|(Q\mathsf{G}_0Q^*)^{-1}(c,c')| \le C\xi_c^{-2}\mathsf{e},$$
$$|\mathsf{H}(b,c)|, |\nabla \mathsf{H}(b,c)| \le C[1, \xi_y^{-1}]\mathsf{e},$$

$$Q\mathsf{G}_0Q^* \ge \gamma\,\xi_c^2,$$
$$\big\langle B, \big[(Q\mathsf{G}_0Q^*)^{-1} - a\big]B\big\rangle \;\ge\; c\,L^{-5}\|B\|^2$$

for every B with three properties: it lives on the bonds of $\mathbb{T}^{(k)}$ that touch a union of L-blocks lying well inside Ω_k; its straight one-step average vanishes; and it vanishes on the axial trees of the blocks. Each block has one such tree. It consists of the contours joining the site $y \in \mathbb{T}^{(k+1)}$ of the block to each of its points x, along which the coordinates are changed successively, always in the same fixed order. The gauge so defined is the strict axial gauge, which is a hard gauge fixing. It is not the block axial gauge (Notation, §3), where the means $V(y,x)$ over contour families are all equal to $\mathbf{1}$. For the plain one-step constraint, in that gauge or on trees, the lower bound is proved again in 1c.8. If the gauge condition is dropped, the form has null directions.

2. Real backgrounds with Bałaban's average. Let Q, Q' be formed with one contour per site and moved to centred blocks by a translation. Let the Hessian be the compressed one.

Let U be real, with local gauges and with small plaquettes; for G_0 this includes the plaquettes that stick out of Ω_0. Then $\Delta_U + Q'^* a Q'$, $Q' G'^2 Q'^*$ and G_0^{-1} are positive. The operators $G', \nabla G', \mathsf{G}_0, \nabla \mathsf{G}_0, (Q' G'^2 Q'^*)^{-1}$ satisfy the bounds of 1 for every Ω_0, including a collar of Ω_1 and a concentric cube. On a single lattice, and for U with $|\partial U - \mathbf{1}|, |D_U^* \partial U| < \alpha\eta^2$, where α is small, also $Q \mathsf{G}_0 Q^* \geq \gamma > 0$, and $\mathsf{G}_0 Q^* (Q \mathsf{G}_0 Q^*)^{-1}$ is a right inverse of the linearised constraint; this right inverse and its gradient decay like $e^{-\delta_* |y-y'|}$. The rows of $\mathcal{M}$ agree with these rows up to a transport and an error $O(\alpha_0)$ in the row and column norms, uniformly in the level. Statements 3–8 concern different operators, built from the rows of $\mathcal{M}$ and the Hessian of the inner plaquettes. At $\mathbf{1}$ the two kinds of rows coincide.

3. The averages and the gauge identities. The operators Q, Q' are local and holomorphic in $\mathbf{U}$. Their row sums are $\leq 1 + C(\alpha_0 + \alpha_1)$ and their entries are $\leq C(\eta/\xi_c)^4$, independently of the number of levels. At each row the non-linear average is holomorphic in (A', A) on a polydisc $\sup|A'|, \sup|A| < r_L$, where r_L is determined by L and measured in units of the row's level. For the functional-derivative bounds, work on a single lattice of depth k in its top units. Let b be a bare bond and c a row of level k, so $\xi_c = 1$, and write $\delta/\delta A_b = \eta^{-4} \partial/\partial A(b)$. In the first two inequalities below, $\mathcal{Q} = \mathcal{Q}_k^{\mathrm{st}}$. Let $\mathcal{Q}_n^{\mathrm{st}}$ be the n-fold straightened average, and let v_n be its accumulated site factor. Then on this polydisc

$$\Big|\frac{\delta \mathcal{Q}(A)(c)}{\delta A_b}\Big| \leq 1 + C(\alpha_0 + \sup|A'| + \sup|A|),$$

$$\Big|\frac{\delta(\mathcal{Q} - \mathsf{m}Q)(A)(c)}{\delta A_b}\Big| \leq C \sup|A|,$$

$$\mathcal{M}^n(e^{i\eta A}\mathbf{U}) = \big(e^{i\mathcal{Q}_n^{\mathrm{st}}(A)} \mathcal{M}^n \mathbf{U}\big)^{v_n}.$$

Let $\bar{D}$ be the covariant difference of $\mathcal{M}^j \mathbf{U}$ on $\mathbb{T}^{(j)}$. At a row of level j, let $Q'\lambda$ denote the j-fold average taken at both ends of the row. For an outward row, the outer end is a top of level $j-1$, not a top of level j. Then

$$QD\lambda = \bar{D} Q' \lambda,$$

$$\Delta_{\mathbf{U}}^{\mathrm{W}} - \Delta'_\pi[J(\mathbf{U})] = \mathsf{S}^{\mathsf{T}} \Delta_{\mathbf{U}}^{\mathrm{W}} \mathsf{S},$$

$$\mathsf{G}_{(1)} DR = DG'R + \mathsf{G}_{(1)} \Delta'_\pi[\mathbf{J} - J(\mathbf{U})] DG'R.$$

On shell, the defect is zero, $RD^{\mathsf{T}} \mathsf{H}_{(1)} = 0$, and $\mathsf{H}B$ is a critical point of $\frac{1}{2}(A, \Delta_{\mathbf{U}}^{\mathrm{W}} A)$ on $\{QA = B,\ RD^{\mathsf{T}} A = 0\}$, with critical value $\frac{1}{2}(B, [(Q \mathsf{G} Q^{\mathsf{T}})^{-1} - a]B)$.

4. The six propagators at regular pairs. The ceiling on the radii $\alpha_0, \alpha_1, \gamma_0$ is fixed before M_1 and is **independent of** M_1. At every regular pair all the inverses exist, they are holomorphic in $(\mathbf{U}, \mathbf{J})$, and they are real at real pairs. At real pairs $\mathsf{G}_0, \mathsf{G}, \mathsf{G}_1 > 0$, both on shell and off shell. The six propagators and $(Q' G'^2 Q'^{\mathsf{T}})^{-1}$ satisfy the upper bounds of 1 with covariant differences, and G and G_1 satisfy those of G_0. The exception is the bound on $\Delta_{\mathbf{U}} \mathsf{G}_{(1)}$, and the Hölder quotients are taken only over $|x - x'| \leq \xi_y$. For G' at complex pairs the bounds reduce to the supremum bounds on $G'f$ and $DG'f$. Moreover $|D(1-R)D^{\mathsf{T}}(b, b')| \leq C(\eta/\xi_{y'})^4 \xi_y^{-2} \mathsf{e}$.

5. Walk expansions. Let O be any of the following operators:

- one of the six propagators;
- $1 - R$ or one of its derivatives;
- $\mathfrak{G}$;
- Bałaban's $H_0, \tilde{G}$, namely $\mathsf{H}\mathsf{m}^{-1}$ and $\mathsf{G} - \mathsf{H}Q\mathsf{G}$ at $\mathbf{J} = 0$;

- $C^{(k)}(Y)$ or $\chi_Y \mathsf{A} \chi_Y$, these two taken at the pairs that are regular for the longer sequence $(\Omega_j)_{j \le k+1}$, where M_1 is larger and the ceiling on the radii is smaller than in 4.

For $\tilde{G}$ the indices are the walks together with the pairs of walks in which the second walk starts at a cell that a row joins to some cell of the first walk, its cells being placed after those of the first. In the display, $|\omega|$ is then the total length of the two walks, and the step that joins them contributes no small factor. Each such operator is then a sum $\sum_\omega \mathsf{O}_\omega$, where, in the norms in which O is bounded,

$$\|\mathsf{O}_\omega\| \le (\text{bound of } \mathsf{O}) \cdot (C M_1^{-1/2})^{|\omega|} M_1^{-|\omega|/2},$$
$$d_{\mathbf{B}}(y, y') \text{ replaced by } d_{\mathbf{B}}(\omega; y, y').$$

The first factor makes the sum over the walks converge. The second factor is left over, and it absorbs the factor $|\omega| + 1$ of mechanism (i). The bounds of O are as follows:

- For $\mathfrak{G}$ and $\tilde{G}$ there are exactly four: the bounds on $|\mathsf{O} f|$, $|\nabla \mathsf{O} f|$, $|\mathsf{O} \nabla^{\mathsf{T}} f|$ and on the Hölder seminorm of $\nabla \mathsf{O} f$ that 4 provides for G, G_1. For $\tilde{G}$ they hold with a smaller fraction of δ_0.
- For $1 - R$ and for its derivatives $D(1-R)$, $(1-R)D^{\mathsf{T}}$, $D(1-R)D^{\mathsf{T}}$, the relevant bounds are the kernel bounds $C(\eta/\xi_{y'})^4 [1, \xi_y^{-1}, \xi_y^{-1}, \xi_y^{-2}]\, \mathsf{e}$. The last of these is displayed in 4.
- For $\chi_Y \mathsf{A} \chi_Y$, the bound is the kernel bound that 6 states for $C^{(k)}(Y)$.

Each O_ω has the following properties:

- it is holomorphic on the local class on $[\omega]$;
- its dependence on the regions is through their traces on $[\![\omega]\!]$ alone;
- it is gauge covariant and real at real pairs;
- it is **the same function** for any two sequences, tori or cutoffs that a translation in $M_1 \xi_k \mathbb{Z}^4$ relates on the hull. For $C^{(k)}(Y)$, $\chi_Y \mathsf{A} \chi_Y$ the translation is in $M_1 \xi_{k+1} \mathbb{Z}^4$ and must also carry Ω_{k+1} and Y.

The operators are covariant under every isometry that preserves $\mathbb{T}^{(k)}$ and the regions, with the twist $-\operatorname{Ad} \mathbf{U}(b)^{-1} A(b)$ at reversed bonds. For $C^{(k)}(Y)$, $\chi_Y \mathsf{A} \chi_Y$ the isometry must also preserve $\mathbb{T}^{(k+1)}$ and Ω_{k+1}, and Y is carried along. Pivot bonds go to pivot bonds, since each pivot bond is the middle bond of its path. For single terms the covariance holds only under those isometries which in addition preserve the M_1-cubes of the top level. That level is $k+1$ for the covariance, and in the strict axial gauge the isometry must also preserve the axial trees. This covariance holds at the pairs whose image again lies in the class.

The additional operator estimates. The fluctuation matrix, its conditional inverses, the independent complex decoupling and the nonlinear response are the additional statements numbered 6–8. Their estimates are obtained from the walk data and the core bounds. The response construction is given below; the actual Gaussian comparison is in Section 25. The substitutions use the hereditary room estimate in Section 29. The layered and nonzero-current versions retain the same weighted operator bounds and the assigned determining structures.

The response and its independent current. The decoupled response is formed from the operators of the walk expansions. Put $\widehat{Q} = \mathsf{m} Q$, where m is the length of a row, and write the nonlinear straightened average as $\mathcal{Q}(A) = \widehat{Q} A + \mathcal{C}(A)$. System (R9) is then

$$X = \mathcal{C}(A_\bullet - H(s) X), \qquad A_\bullet = H_1(s) B' + \mathbf{A},$$
$$\mathbf{A} + \mathfrak{G}(s) \mathsf{W}_s(A_\bullet) = 0, \qquad \mathbf{H} = A_\bullet - H(s) X,$$

where $H = \mathsf{H}\mathsf{m}^{-1}$ and $H_1 = \mathsf{H}_1\mathsf{m}^{-1}$. The reduced expression W_s is obtained by cancelling the quadratic part before the parameters are inserted. At arbitrary s it is this expression that defines the map; identities between the undecoupled operators are not imposed on their decoupled sums.

For a positive weight ϖ with $\varpi(y) \le e^{\theta d_{\mathbf{B}}(y,y')}\varpi(y')$, use the norms

$$\|A\|_\varpi = \sup_y \varpi(y)\max\{\xi_y|A|, \xi_y^2|\nabla_{\mathbf{U}}A|\}, \qquad |f|_\varpi^\sharp = \sup_y \varpi(y)\xi_y^3|f|.$$

Each decoupled operator has norm at most $\widehat{b} = Ce^{16\bar{\kappa}}$ in these norms. On a ball $\|A_i\| \le r < c/\widehat{b}$, the nonlinear expression obeys

$$|\mathsf{W}_s(A_1) - \mathsf{W}_s(A_2)|_\varpi^\sharp \le C\widehat{b}^2 r\|A_1 - A_2\|_\varpi.$$

The fixed-point Lipschitz constant is bounded by $C\widehat{b}^4 \sup|B'|$. Thus $\sup|B'| \le c\widehat{b}^{-4}$ gives the same contraction in every such weight. Its solution is holomorphic in the parameters, the datum, the complex potential and the independent current, and

$$\xi_y|\mathbf{H}| + \xi_y^2|\nabla_{\mathbf{U}}\mathbf{H}| \le C\widehat{b}\sup|B'|.$$

Choosing an exponential weight gives the first-response decay. The same contraction, by uniqueness, gives consistency and splitting over the wall-components. It also transports the solution whenever the assigned walk data are transported.

The current is a separate entry of the pair. Its increment is formed from the multiplier, so that the second-order part of the gradient enters only through $(D^{\mathsf{T}}DA, He_c) = (DA, DHe_c)$. It is therefore bounded in $|\cdot|_\varpi^\sharp$ by the same first-order response norms. No level-independent bound on the full Laplacian of an arbitrary decoupled response is needed.

For completeness, a second system provides the other second-order bound at physical parameters. Let

$$S = (\widehat{Q}\mathsf{G}\widehat{Q}^{\mathsf{T}})^{-1}, \quad H_0 = \mathsf{G}_0\widehat{Q}^{\mathsf{T}}(\widehat{Q}\mathsf{G}_0\widehat{Q}^{\mathsf{T}})^{-1}, \quad \widetilde{G} = \mathsf{G}_0 - H_0\widehat{Q}\mathsf{G}_0,$$

and let a satisfy $\widehat{Q}^{\mathsf{T}}\mathsf{a}\widehat{Q} = Q^{\mathsf{T}}aQ$. In the action chart set $\mathcal{N}(T) = F'(T) - F'(0) - \Delta_{\mathbf{U}}^{\mathrm{W}}T$. The reduced system for (T, λ) is

$$\begin{aligned} T &= H_0(s)(B' - \mathcal{C}(T)) - \widetilde{G}(s)[\mathcal{N}(T) - \mathcal{C}_T'^{\mathsf{T}}\lambda], \\ \lambda &= H(s)^{\mathsf{T}}\mathbf{J} + (S(s) - \mathsf{a})(B' - \mathcal{C}(T)) \\ &\qquad + H(s)^{\mathsf{T}}[\mathcal{N}(T) - \mathcal{C}_T'^{\mathsf{T}}\lambda]. \end{aligned}$$

The terms quadratic in the unknowns are contractions for $\widehat{b}^2\sup|B'| \ll 1$. The terms linear in T that contain the current require the additional ceiling $\widehat{b}^2\gamma_{\mathrm{cur}} \ll 1$, where γ_{cur} is the ceiling on the independent current in this response problem. With these choices the system has one small holomorphic solution in $(T, \lambda - H(s)^{\mathsf{T}}\mathbf{J})$, with the same weighted Lipschitz estimates. This second system is used on its stated global domains and, in local form, on strip sequences. It is not substituted for the first system on arbitrary local cores.

At $s = 1$ over a critical real background, the first solution satisfies

$$\mathcal{Q}(\mathbf{H}) = B', \qquad RD^{\mathsf{T}}\mathbf{H} = 0, \qquad F'(\mathbf{H}) = K_{\mathbf{H}}^{\mathsf{T}}\lambda.$$

For real data near zero, invariance gives all missing gauge directions transverse to the slice; hence the slice equation has a multiplier for the averages alone. Holomorphy extends the identities to the full polydisc. They characterize the background response there. On the common physical polydisc, the second system follows from these equations and uniqueness identifies its field with $\mathbf{H}$. The first system then gives the bound on $DD^{\mathsf{T}}\mathbf{H}$, and the second gives the bound on $D^{\mathsf{T}}D\mathbf{H}$, both in local units. Away from critical physical pairs the two systems define separate functions. In particular no constraint identity is asserted at arbitrary complex decoupling parameters.

(iii) The strategy.

The chain. The parts have the following content.

- 1c.1 (the operators and the statements about them) fixes the operators, the classes and the form of the statements. It counts the walks and derives statement 0. It also shows that the identification of A at the step reduces to Bałaban's expansion of the background action.
- 1c.2 (the flat case, I) states the flat theorem and compares it clause by clause with the published statements. It also contains the exact algebra at $\mathbf{1}$: existence, identities with boundary terms, and the independence of R and H from Ω_0.
- In 1c.3 (the flat case, II) the one-scale operator is treated uniformly in the spacing; then come G', $(Q'G'^2Q'^*)^{-1}$ and R.
- In 1c.4 (the flat case, III) the local two-scale operators are treated, followed by G_0, $(Q\mathsf{G}_0Q^*)^{-1}$ and H. With this, statement 1 is complete.
- 1c.5 (Bałaban's theorems in a background) yields statement 2. On the whole lattice it comes from the published theorems; on sequences it comes from the flat case, by a shortened form of mechanism (f).
- 1c.6 (the averaging operators at complex pairs; the algebraic identities) yields statement 3.
- 1c.7 (the six propagators) uses mechanisms (f), (g) to derive, from the flat bounds and 1c.6, statement 4 and the expansions of 5.
- In 1c.8 (the covariance; positivity) these two mechanisms are applied to a larger system. This yields statement 6, except for the expansion of A over sets of coarse cubes.
- In 1c.9 (walk expansions as data) an expansion is regarded as a datum enjoying the short list of properties of 7. Statement 7 is deduced from these properties together with the geometry. Moreover, 1c.9 *makes* terms that are real, covariant and identical on all tori with room; these properties do not follow from the list.
- 1c.10 (the decoupled response) takes the operators as data and solves the systems (I), (II). Next it inserts the local maps and operators of 1c.6–1c.9; for $\mathfrak{G}$ and $D(1-R)D^{\mathsf{T}}$ the inserted data are those named at the end of 7. Last, it treats the case $s \equiv 1$.

The order of (1b) and (1c), which involves no circle. Statements 1–7 refer to no minimiser, since the classes are defined by inequalities. The results are obtained in the following order:

- the geometry and the real calculus of $\mathcal{M}$ (1b.1–1b.2);
- the flat case;
- the lemmas of 1b.3 that are geometric or algebraic (folding, cells and rows, the algebra of Q', Q, R, sums over tops, weights for a general metric), and none of them uses any bound on a propagator;
- 1c.5;
- the rest of 1b.3; then existence, regularity (which provides the backgrounds of (1b) with their local gauges) and continuation (1b.4–1b.6);
- 1c.6–1c.9. Their inputs are the flat case, the alignment of the blocks in 1c.5, and 1b.1–1b.2, together with, besides these, geometric facts involving neither propagators nor minimisers (meets of sequences; the covariance of $\mathcal{M}$ with respect to the isometries of the lattice);
- the decay of the response, and the membership in the classes of the backgrounds of (1b) (1b.7–1b.9);
- finally, the identification at $s \equiv 1$.

Mechanisms.

(a) **The metric matches the mass.** The form satisfies $\langle \lambda, (\Delta + Q'^* a Q')\lambda\rangle \geq \gamma \sum_j \xi_j^{-2} \|\lambda\|^2_{\mathcal{L}_j}$, and on each layer $d_{\mathbf{B}}$ has slope ξ_j^{-1}. A single conjugation by $e^{\varsigma\varphi}$, with φ Lipschitz with respect to $d_{\mathbf{B}}$ (Combes–Thomas), therefore serves for all layers at once. It costs $O(\varsigma L^2)$, and this cost stays

below γ under a ceiling on the rate, $\varsigma \le c\gamma L^{-2}$, where L is chosen first. Every layer that has to be traversed in full costs the gap, $d_{\mathbf{B}}(y,y') \ge R_1 M_1 (|j-j'|-1)_+$. Let a rate $\tau > 0$ and a power $q \ge 0$ be given, and choose the product $R_1 M_1$ after them so that $\tau R_1 M_1 \ge C(1+q)\log L$. Then

$$\sum_{y'} e^{-\tau d_{\mathbf{B}}(y,y')} \le C(\tau, L),$$

$$(\xi_{y'}/\xi_y)^q \le L^q e^{\tau d_{\mathbf{B}}(y,y')} :$$

the factor $e^{-c\tau R_1 M_1}$ for each level outweighs the growth L^4 in the number of sites. Conjugation and walk expansions do different work. Conjugation yields decay of the inverse as a whole, with a rate given by a fraction of the gap, and it requires no cube size to be large. A walk expansion gives, in addition, a sum of terms, each of which depends on the background and on the regions only near its own walk. This property is what statements 5–8, the comparison of tori and (1d) make use of. Both methods are used. Conjugation serves for the scalar, one-scale and local operators and for the inverses on tops and rows. Walks serve for the assembly, namely of G_0 in (d) and of the propagators in a background in (f). In **95**, p. 36, Bałaban describes conjugation as "probably the simplest proof" of the decay; he builds a walk representation instead, since "such representations will be our basic tool in investigation of more complicated operators".

(b) **No Dirichlet condition on a lattice finer than the blocks.** Because of the gap, $\partial\Omega_0$ lies in the bare layer $\mathcal{L}_0$, and the boundary cubes and their cut-off collars are chosen inside its all-bare collar. In the local model on such a cube $Q_\square = Q'_\square = 1$, hence $R_\square = 0$; the global R need not vanish throughout $\mathcal{L}_0$ near its interface with Ω_1. The local operator $G_\square$, which inverts this all-bare form (see (d)), obeys $a_0 \le \eta^2 G_\square^{-1} \le a_0 + 16$. Finite range together with a mass comparable with the norm gives decay for an arbitrary set of bonds. The following examples show that this placement of the boundary matters. Let G be a one-scale operator (spacing η, unit blocks) whose order is ≥ 1, with Dirichlet conditions imposed outside a union of *whole* blocks, so that the lattice is finer than the blocks and no block is cut. Two examples, a flat face and a re-entrant edge, rule out constants uniform in the order, by powers of η^{-1}. This applies to the L^2 bounds on second differences, to the bounds on ΔG, to those on ∇G and to its Hölder norm. It applies equally to the pointwise bound and to the Hölder bound involving $\nabla G \nabla^*$, taken in the form of statement 1, the powers being those that occur for ∇G and for its Hölder norm. Indeed, on the region every bounded f supported in a block can be written as $\nabla^* F$ with $F = \nabla \Delta^{-1} f$, where Δ^{-1} denotes the free inverse on $\eta\mathbb{Z}^4$ (on a torus, the inverse on functions of mean zero, after f has been modified on a block lying outside the region); moreover $|F|, \|F\|_\varepsilon \le C_\varepsilon \sup|f|$ uniformly in η, and the sum over the blocks is controlled by the exponential factor in the bound. For the L^2 bound involving $\nabla G \nabla^*$, by contrast, the two examples rule out no uniform constant. On a separated sequence no operator of this kind arises.

(c) **One scale, two scales.** Uniformity in η comes from two sources. The first is the representation $(\Delta+1)^{-1} = \int_0^\infty e^{-t} e^{-t\Delta} dt$, combined with heat-kernel bounds that hold both below and above $t = \eta^2$. The second is a factorisation of $\partial(1-R)\partial^*$ through the unit lattice. At a seam one makes an exact change of gauge $A = A_2 + H_j B - \partial H'_j \vartheta$. Here $A_2 \in \ker Q_j$, and B is a 1-form on the coarse lattice whose values on the axial trees of the L-blocks of the coarse region are zero. Further, ϑ is a site function on the coarse lattice, supported in the coarse region, whose mean over every one of those blocks is zero; for these two classes the representation is unique. Finally, H_j, H'_j are the one-scale minimisers, with H'_j the minimiser of $\|\Delta\lambda\|^2$ rather than of $\|\partial\lambda\|^2$. This change of gauge splits the form orthogonally into three parts: a form in A_2, a form in B and a form in $\vartheta - \vartheta_1$, where ϑ_1 is linear in A_2 and B. If the coarse region is all of the torus, the two-scale representation so obtained becomes a recursion $G_j \to G_{j+1}$, where G_j denotes the one-scale propagator of order j. It is applied for a single step and is never iterated. In the scalar theory,

recursions of this kind add up to a series over the scales, and they served as the method of proof of regularity (**89**, (2.34)). Here G_j occurs multiplied by other operators. Iteration then leads to long products whose estimation is very difficult, since the gauge conditions are not the same at different scales (a remark of Bałaban, **96**, pp. 246–247). For gauge propagators, accordingly, no series over the scales is employed; instead the one-scale G_j has to be known simultaneously for all j (1c.3). Three gaps on the unit lattice take the place of a small parameter. The sharp one is stated for B on a unit lattice that vanishes on the axial trees. In it, Q_1 and ∂_1 are the straight one-step average and the exterior derivative; the $Q_1(\mathbf{1})$ of the Notation, §3, also contains site terms. The sharp gap reads

$$L^2 \sum_c |(Q_1 B)(c)|^2 + \sum_p |(\partial_1 B)(p)|^2 \geq cL^{-5}\|B\|^2.$$

(d) **Many scales at 1.** No expansion of G' is required. For $u = G'(f + \nabla^* F)$ the conjugation of (a) yields

$$\|\xi^{-1} e^{\varsigma\varphi} u\| + \|e^{\varsigma\varphi}\nabla u\| \leq C\gamma^{-1}(\|\xi e^{\varsigma\varphi} f\| + \|e^{\varsigma\varphi} F\|)$$

, where ξ denotes multiplication by the local scale. From this one passes to local L^2 norms, and then to interior estimates at the scale ξ_y, with the mass term treated as a source. The walk expansions are obtained afterwards, as data. For G_0 we take $\sum_\square h_\square^2 = 1$, with $\zeta_\square = 1$ near $\operatorname{supp} h_\square$, together with the $G_\square$ of (c) on periodised cubes. For the cubes of level 0 that extend beyond Ω_0, the operators of (b) are taken instead. For a cube of level ≥ 1 the periodic two-scale model is retained, irrespective of where Ω_0^c and the finer layers lie; this is legitimate since across the two seams to $\mathcal{L}_j \cup \mathcal{L}_{j+1}$ the function $h_\square$ is cut off. With $P = 1 - R$ and $\mathcal{P} = \partial P \partial^*$ one has

$$\mathsf{G}_0^{-1} \sum_\square h_\square G_\square h_\square = 1 - \mathcal{R},$$

$$\mathcal{P}h - h\mathcal{P}_\square = \zeta[\mathcal{P}_\square, h] + \zeta(\mathcal{P} - \mathcal{P}_\square)h + (1 - \zeta)\mathcal{P}h :$$

of all these operators, only the non-local $\mathcal{P}$ cannot be transplanted. Each commutator costs M_1^{-1}, because $|\nabla h| \leq C(M_1\xi_j)^{-1}$ stands against $|\nabla G_\square| \leq C\xi_j$. The change of domain costs e^{-cM_1}; this follows from resolvent identities on the two domains, recombined by means of $1 - P = R$. The tail is likewise exponentially small, and it is turned into a step $\square \to \square'$ by means of $\sum h_{\square'}^2 = 1$. The factor M_1^{-1} for each step outweighs the number $C(\tau, L)$ of neighbours.

Lower bounds independent of the number of levels underlie the inverses on tops and rows. The bound $Q'G'^2Q'^* \geq \gamma\xi_y^4$ follows by duality with test functions ψ_y on the blocks, with $Q'\psi_y = \delta_y$, whose second differences are $\leq C\xi_y^{-2}$, across the faces as well. The bound $Q\mathsf{G}_0Q^* \geq \gamma\xi_c^2$, with γ independent of L, follows from a right inverse E for all rows of all levels at once. This right inverse has scale-invariant energy, $\|\nabla \mathsf{E} B\|^2 \leq C\|\mathsf{m}^{-1}B\|^2$. (Choosing on each level test forms biorthogonal to the rows of that level, one obtains $Q\mathsf{E}_0 = 1 + \mathsf{N}$. The operator N goes up by one level only, and it goes in one direction: from rows that do not cross a seam to rows, one level higher, that do cross one. Hence $\mathsf{N}^2 = 0$ and $\mathsf{E} = \mathsf{E}_0(1 - \mathsf{N})$. The weights of N are each $\leq L^{-4}$ and have sum ≤ 1. Set against $\xi_{j+1}^2 = L^2\xi_j^2$, they give $\|\mathsf{m}^{-1}\mathsf{N}\mathsf{m}\| \leq CL^{-1}$, and from this the constant is independent of L.) In 2, where there is a single lattice, the test forms are placed on the face separating the two blocks of a bond; there the share $\theta_0 > \frac{1}{2}$ of its own chains outweighs $1 - \theta_0$. Neither matrix is bounded below by a number independent of the scales ξ. They are therefore rescaled, to $\Sigma = \xi^{-2}Q'G'^2Q'^*\xi^{-2}$ or $\xi^{-1}Q\mathsf{G}_0Q^*\xi^{-1}$. A matrix $\Sigma \geq \gamma$ with $|\Sigma| \leq Ce^{-\tau d_{\mathbf{B}}}$ then satisfies $|\Sigma^{-1}| \leq 2\gamma^{-1}e^{-\tau' d_{\mathbf{B}}}$, where τ' is small compared with the spectral

gap γ (Combes–Thomas, Schur). Thus both inverses decay without any expansion, and H decays column by column. A sequence is compared with the same sequence cut down around a cube. The cut-down sequence is again separated and coincides with the given one on the doubled cube, and its $Q\mathsf{G}_0Q^*$ provides the local operators that enter the walk expansion of $(Q\mathsf{G}_0Q^*)^{-1}$. The comparison uses Green's identity with a cut-off, by way of $(\partial^*\partial + Q^*aQ)\,\mathsf{G}_0Q^* = Q^*$, and the gauge term vanishes on the columns.

(e) **Gauge algebra without Gaussian integrals.** On a pure gauge, take the derivative of the covariance of the straightened average; the site factors then convert point values into block means: $QD = \bar{D}Q'$. Consequently the rows annihilate $D\ker Q'$. Combine this with the invariance of the action to second order, which is exact for the inner plaquettes; this accounts for that Hessian. The result is $\mathsf{G}^{-1}D\lambda = D\Delta_{\mathbf{U}}\lambda - \Delta'_\pi[\mathbf{J} - J(\mathbf{U})]D\lambda$ on $\ker Q'$. The propagators are taken at pairs whose current is an independent variable carrying a bound of its own, as in the spaces of **109**, p. 262. The reason is that bounds on the plaquettes, on the potential and on the first differences do not control $J(U)$ uniformly in the level: a staggered field satisfies all of them with $|J(U)| \geq \alpha\eta^{-1}$. In **99**, 389, by contrast, $J = J(U)$, and a condition on second differences is imposed in addition ((3.36), (3.38)). The price of a free current is the defect in 3. Uniformity in the level holds for the following reason. Row norms and column norms are both submultiplicative, and for a single step they lie near L and L^{-3}. The relative errors $O(\alpha_0(L^n\eta)^2 + \alpha_1 L^n\eta)$ of the steps form geometric series that are dominated by the top level.

(f) **Unfolding and parametrix.** With auxiliary unknowns

$$\Psi = (A; v, w, \sigma; p, q, r; \nu)$$

, the operators R, Δ'_π, $\mathsf{S}^{\mathsf{T}}\Delta^{(2)}\mathsf{S}$ and the constraint are combined into a single system $\mathbb{K}$ of finite range. The inverse of this system has the propagators as blocks. Let $\varkappa$ be an M_1-cube; these cubes have side M_1 in the local unit and tile Ω_0, whereas the $\square$ of (b), (d) are the cubes of the flat expansions. On such a cube, in the gauge of the class, $\mathbb{K}_\varkappa$ equals the flat system plus $O(\alpha_0 + \alpha_1 + \gamma_0)$, and this perturbation need only be small compared with flat constants. Let $\theta_\varkappa$ be a partition of unity, let the cut-offs $\tilde{h}_\varkappa = 1$ near $\operatorname{supp}\theta_\varkappa$ be such that $\mathbb{K} = \mathbb{K}_\varkappa$ on $\operatorname{supp}\tilde{h}_\varkappa$, and let δ denote the rate of the local inverses. Then

$$\mathbb{K}\sum_\varkappa \tilde{h}_\varkappa\mathbb{K}_\varkappa^{-1}\theta_\varkappa = 1 - \sum_\varkappa \mathbb{R}_\varkappa,$$

$$\mathbb{R}_\varkappa = -[\mathbb{K}_\varkappa, \tilde{h}_\varkappa]\mathbb{K}_\varkappa^{-1}\theta_\varkappa = O\big(M_1^{-1}e^{-c\delta M_1/L}\big),$$

where the power comes from the commutator, and the exponential from how far $\operatorname{supp}\theta_\varkappa$ lies from $\operatorname{supp}\nabla\tilde{h}_\varkappa$. This bound outweighs the $O(1)$ neighbours. Three features keep the perturbation and the commutators from losing a derivative: scaling weights, one component of the source that admits a divergence, and the absence of any derivative on p.

(g) **Trees.** The system $\mathbb{K}$ depends affinely on the non-local coefficient $\mathsf{H}^{\mathsf{T}}\mathbf{J}$. The part of walk length zero stays in the local operators. The remainder, of size $O(\gamma_0 M_1^{-1}e^{-cM_1/L})$, forms a vertex that carries a walk of H. Thus vertices are small because of M_1 and not because of the field, and each term is indexed by a trunk with branches. When the cells are listed depth first, the number of such trees grows geometrically: for each first cell, there are C^m tree walks of length m starting from it. Both this count and the size of M_1 needed for the sum over trees to converge coincide with the ones in 1c.9. In 1c.7 and 1c.8 a branch is listed as soon as its cut is reached, and the trunk is resumed afterwards. For the data of 1c.9 the listing first continues at every cell along the trunk (within a branch, along the branch). So the whole trunk comes first, and then the branches come in the reverse order of their cuts, beginning with the last cut and ending with

the first. Each of the two orders is a depth-first order. The order of listing changes neither the cells nor their parents, and so it changes neither the hull, nor the tree length, nor the bounds.

(h) **Multiplier and positivity.** Adding the equation "QA = fluctuation field" brings in $(Q\mathsf{G}_1Q^{\mathsf{T}})^{-1}$ as a Lagrange multiplier. Consequently $C^{(k)}(Y)$ occurs as a block in the inverse of a system of finite range. In the flat case, positivity follows from $\sum_{p'} |\bar\partial QA(p')|^2 \le \eta^4 \sum_p |\partial A(p)|^2$, with $\bar\partial$ taken on $\mathbb{T}^{(k)}$, together with a definite form on double blocks. In a background, a homotopy to $\mathbf{1}$ preserves $C_\varkappa > 0$ for the local inverses. The eigenvalues z of $\chi_Y \mathsf{A} \chi_Y \sum h_\varkappa C_\varkappa h_\varkappa$ satisfy $|1 - z| \le C M_1^{-1}$.

(i) **Length against parameters.** A tree of length t comes close to at most $2^4 \max\{1, Ct/\ell_{\mathsf{P}}\}$ cubes $\square$ of the partition of side ℓ_{P} in statement 7; these are not the cubes $\square$ of (b), (d). Hence $\bar\kappa \# \mathsf{P}(\omega) \le 2^4 \bar\kappa + (C\bar\kappa/\ell_{\mathsf{P}})|\Gamma|_{\mathbf{B}}$ for every tree Γ admitted in $d_{\mathbf{B}}(\omega; y, y')$, provided that X_0, which Γ need not meet, lies within $O(1)$ top units of y (no other term enters a functional near y, since the left argument of the kernel of O_ω touches X_0). Consequently, for $|s(\square)| \le e^{\bar\kappa}$ and $\delta \ge C\bar\kappa/\ell_{\mathsf{P}}$,

$$|s^\omega|\, e^{-\delta d_{\mathbf{B}}(\omega;y,y')} \le e^{2^4\bar\kappa}\, e^{-\bar\kappa \#(\mathsf{P}(\omega)\setminus\sigma_0)}\, e^{-(\delta - C\bar\kappa/\ell_{\mathsf{P}}) d_{\mathbf{B}}(y,y')}.$$

Each cube beyond the first 2^4 adds to the tree a length $\ge c\ell_{\mathsf{P}}$. Since ℓ_{P} is chosen last, $e^{-c\delta\ell_{\mathsf{P}}}$ outweighs $e^{\bar\kappa}$. The factor $e^{2^4\bar\kappa}$ is left over and cannot be avoided, as a cell at a corner of the grid shows. The middle factor amounts to $e^{-\bar\kappa}$ for each cube that is crossed without a parameter. The estimate is Bałaban's (**116**, p. 5, after (1.11)); in his setting as well, the exponential by itself does the paying, and $e^{16\bar\kappa}$ remains. He keeps the factor $M_1^{-|\omega|/2}$ in reserve together with the exponential (p. 3, after (1.7)). This factor is not used up on the parameters, and here it is still at our disposal for the factor $|\omega|+1$ that occurs below. Since the uncut hull, which is his localisation domain, is connected, $\mathsf{P}(\omega)$ is connected through walls; this yields the splitting. The carrier is needed because the fields are given only on Ω_0, and it need not be connected when $\Omega_0 \ne T_\ell$.

The properties of single terms are produced by construction; they are not deduced. The sum is itself covariant and real. Therefore the Haar mean of the terms, made symmetric under the conjugation fixing the real pairs, is once more an expansion of the sum. By convexity it satisfies the same bounds, provided that the family of bounds is invariant under the real gauge transformations and under this conjugation; the family of localised bounds has this invariance. A single assignment for all tori is obtained in three steps: a mean over translations; a push-forward from covering tori; and a passage to the limit by Montel's theorem, along a subsequence fixed once, at the given top level and units. In the push-forward, a walk wrapping around the small torus gains at most $|\omega| + 1$ blocks; the weight therefore loses a factor $|\omega| + 1$, and nothing else is lost. The family is made normal by an entrywise bound, which is one and the same number for all tori with room. This number is allowed to depend on the level, since the entries of $(Q\mathsf{G}Q^{\mathsf{T}})^{-1}$ are $O(\eta^{-2})$, and none of the norms, rates, weights or sums of weights involves it. If a family of terms is already carried by all shifts and compatible with all coverings, it is not changed.

(j) **Operators as data.** The principle goes back to Bałaban, who observes that in the equations for the response "we can replace the propagators by arbitrary operators having the same regularity properties and satisfying the same bounds" (**116**, p. 3, (1.5)). In its published form, however, the second equation relies on the identities $RD^{\mathsf{T}}\mathsf{H} = 0$ and $\mathsf{G}^{-1}\mathsf{H} = Q^{\mathsf{T}}(Q\mathsf{G}Q^{\mathsf{T}})^{-1}$, and therefore on their consequences $Q\mathsf{H} = 1$ and $\Delta_\pi \mathsf{H} = Q^{\mathsf{T}}[(Q\mathsf{G}Q^{\mathsf{T}})^{-1} - a]$ ($\Delta_\pi = \Delta_{\mathbf{U}}^{\mathrm{W}} - \Delta'_\pi$). None of these is available for the sums $\sum_\omega s^\omega \mathsf{O}_\omega$: on the cubes around a row where s vanishes, every term is absent, so that $Q\mathsf{H}(s)$ equals 0 there and not 1. The first identity, moreover, holds only for the pairs on shell, even at $s \equiv 1$. Consequently no identity enters the argument. The identities are applied earlier, where they hold. With their help the gradient of the non-quadratic part of the action is put into a shape from which its second-order term $\Delta_\pi \mathsf{H}$ is absent, and this expression

defines W_s for every s and every pair (it is a gradient only at $s \equiv 1$ and on shell). From then on only bounds are used, in the norms

$$\|A\|_{\varpi} = \sup_y \varpi(y) \max\{\xi_y |A|,\, \xi_y^2 |\nabla_{\mathbf{U}} A|\},$$

$$|f|^{\sharp}_{\varpi} = \sup_y \varpi(y)\, \xi_y^3 |f|,$$

$$\varpi(y) \le e^{\theta d_{\mathbf{B}}(y,y')} \varpi(y') :$$

the first measures a potential together with its first derivative, the second a force. The operator $\mathfrak{G}(s)$ takes the second norm back to the first, with norm $\hat{\mathsf{b}} \sim e^{2^4 \bar{\kappa}}$, which is the size of every one of the operators; this closes the fixed-point argument. Whenever $\|A_i\| \le r < c/\hat{\mathsf{b}}$, one has, in every weight whose θ is at most a fixed fraction of δ_0,

$$|\mathsf{W}_s(A_1) - \mathsf{W}_s(A_2)|^{\sharp}_{\varpi} \le C \hat{\mathsf{b}}^2 r \, \|A_1 - A_2\|_{\varpi};$$

in the only term containing $\mathbf{J}$, the factor r is supplied by the Schwarz lemma, applied to the full ball of radius $c/\hat{\mathsf{b}}$. The Lipschitz constant of the fixed-point map is $\hat{\mathsf{b}} \cdot \hat{\mathsf{b}}^2 \cdot \hat{\mathsf{b}} \sup |B'|$, with contributions from $\mathfrak{G}$, W_s and H_1, and this is why the radius is $\hat{\mathsf{b}}^{-4}$. Decay amounts to choosing the weight $\varpi = e^{\theta d_{\mathbf{B}}(\cdot, y')}$. Locality, splitting, covariance and transport follow from exhibiting a candidate and from uniqueness in the balls. When $s \not\equiv 1$, bounds of first order on the operators do not yield a bound on $\Delta_{\mathbf{U}} \mathbf{H}$ independent of the level; an oscillating kernel belonging to the class of data is an example. The reduced current takes its place. It is built from the multiplier, and in it the second-order part of the gradient occurs only in paired form, $(D^{\mathsf{T}} D A, \mathsf{H} e_c) = (D A, D \mathsf{H} e_c)$, where e_c is an element of $\mathfrak{g}^c$ placed at the row c. As a result, only first-order norms occur.

System (II) is handled by the same method, but a single map replaces the two nested ones. In each of its two components one operator of norm $\hat{\mathsf{b}}$ acts, on a ball of radius $2\hat{\mathsf{b}} r$, on terms vanishing to second order $(\mathcal{C}, \mathcal{N}, \mathcal{C}'^{\mathsf{T}}(\lambda - H(s)^{\mathsf{T}} \mathbf{J}))$. Hence $\hat{\mathsf{b}}^2 r$ must be small, and this gives the radius $\hat{\mathsf{b}}^{-2}$ in B'. Two of the terms are of first order in the unknown. One is $\tilde{G}(s)\, \mathcal{C}_T'^{\mathsf{T}} H(s)^{\mathsf{T}} \mathbf{J}$ in the first component, whose linear part at $s \equiv 1$ is $\tilde{G} \Delta^{(2)} T$. The other is $-H(s)^{\mathsf{T}} \mathcal{C}_T'^{\mathsf{T}} H(s)^{\mathsf{T}} \mathbf{J}$ in the second component. Each of them involves two operator norms and can be made small only through the current, which requires $\hat{\mathsf{b}}^2 \gamma_0$ to be small. Under these conditions the map is a contraction with constant $\frac{1}{2}$ in all the weighted norms $\max\{\|T\|_{\varpi}, |\lambda|_{\varpi}\}$ simultaneously. Since the member $-\mathsf{a}$ is diagonal and is carried by no proper set, the splitting is built around this member.

(k) **From the slice to a multiplier.** Let $s \equiv 1$, and fix a critical real U_0 such that $(U_0, J(U_0))$ is a regular pair whose radii lie below the ceiling. Then (I) is equivalent to the constraint, the Landau condition and $\mathsf{z} := F'(\mathbf{H}) - K_{\mathbf{H}}^{\mathsf{T}} \lambda \in D R(\text{site functions})$, where λ is the multiplier formed from $\mathbf{H}$. In other words, (I) amounts to stationarity on the gauge slice alone. If B' is real and close to 0, then $\mathbf{H}(B')$ is real and lies in a neighbourhood of 0 whose size is not uniform. On that neighbourhood the invariance of the action and the factorisation in 3 provide gauge directions X_χ with $(F'(\mathbf{H}), X_\chi) = 0$ and $K_{\mathbf{H}} X_\chi = 0$, and the $R D^{\mathsf{T}} X_\chi$ fill out the range of R. Therefore $\mathsf{z} = 0$ on this neighbourhood, and by the identity theorem also on the whole polydisc, so the neighbourhood enters no constant. The factorisation of the averages holds only up to site factors and, at rows that leave their layer, up to seam factors; this is the reason for the row datum. For (II) the corresponding passage is organised by means of two defects. Take arbitrary small T and λ, and set $B' = \mathcal{Q}(T)$, $\mathsf{z} = F'(T) - K_T^{\mathsf{T}} \lambda$ and $\phi = R D^{\mathsf{T}} T$. Then both equations of (II) have residuals that depend linearly on (z, ϕ), the residual of the first being $\tilde{G}(\mathsf{z} + D\phi)$. Consequently (II) follows from the three equations and is tied to them.

Constants are chosen in the following order, each large or small enough relative to those chosen before it. First comes L, and with it, as a single group, the numbers determined by L alone: the flat constants and the flat rate, the lower thresholds that the flat case imposes on M_1 and on R_1M_1, the radius of the local perturbation theory, the rates δ_0, δ_*, δ', and the ceiling on $\alpha_0, \alpha_1, \gamma_0$. Next come M_1 and R_1M_1; the ceiling does not depend on them, and they do not depend on the ceiling. Then comes $\bar\kappa$. It does not depend on M_1, R_1, and they do not depend on it, so it may just as well be chosen before them. After that come $\ell_{\mathsf P}$, the radius in B', and the second ceiling on γ_0, which concerns (II). The final choices are M and the field strengths of the construction, taken so small that $O(LM)\alpha_0$ is below the ceiling. Two numbers are exceptions: among the bounds of statement 7, the one bound that is used for convergence alone; and, in mechanism (i), the entrywise bound by which the family is normal. Neither of them appears in a localised bound, a rate, a weight or a sum of weights. With the exception of these two, every constant, rate and ceiling, and the radius in B', is independent of k, K, ℓ, of the sequence (in particular of Ω_0), of Y and of the pair, and also of M_1, R_1 once these are large enough. The sizes $\ell_{\mathsf P}$ and M, and the side of the coarse cubes of 6, are each at least a multiple of M_1. The field strengths are small compared with $(LM)^{-1}$. Where statement 8 makes the identification with the response of (1b), the datum of the background is small compared with M_1^{-1}.

22.1. **Layered row maps and physical Hessians.** We make explicit the endpoint readouts, the three successive Hessians, and the row extensions used by the propagator construction. The lattices, separated layers and determining rows are those fixed above. In this subsection every transpose uses the physical bilinear pairing, with fine volume η^4 and row volume ξ_c^4; real lower bounds use the corresponding Hermitian form. The row map Q is per unit length. Its linearization is taken after the accumulated node factors of the average have been removed.

Write ξ_b for the larger pillar scale at the endpoints of a fine bond. We use

$$\|u\|_{\mathrm{mass}}^2 = \sum_b \eta^4 \xi_b^{-2} |u(b)|^2, \qquad \|\psi\|_{\mathrm{row}}^2 = \sum_c \xi_c^2 |\psi(c)|^2.$$

Thus $\|\psi\|_{\mathrm{row}} = \|\Xi^{-1}\psi\|_{2,\mathrm{rows}}$ when Ξ multiplies row c by ξ_c. All constants in this calculation may depend on the fixed block factor L. Their uniformity is in the depth, fine spacing, volume, and admitted separated regions.

Write $d = d_{\mathbf B}$ for the row distance inherited from the layer metric. For a complex variation put $U_E = e^{i\eta E}U$; h bounds the physical first-order quantities $\xi|E|$ and $\xi^2|\nabla_U E|$ on the determining neighborhoods in the regular-pair domain. The real curvature and local real-gauge bounds have a common ceiling ϵ. The Wilson Hessian at U_E means its holomorphic continuation, with the bilinear transpose convention retained.

On site functions let $Q' = Q'_{\mathrm{pill}}$ be the own-pillar mean and set

$$G' = (D_U^{\mathsf T} D_U + Q'^{\mathsf T} a Q')^{-1}, \quad R = 1 - G'Q'^{\mathsf T}(Q'G'^2Q'^{\mathsf T})^{-1}Q'G', \quad \mathsf S = 1 - D_U G' R D_U^{\mathsf T}.$$

Here R is the Landau projection, as in the propagator definitions above. The site identities $Q'G'R = 0$ and $\Delta_U G'R = R$ imply that $\mathsf S$ projects to the slice $RD_U^{\mathsf T}u = 0$. The field-row intertwining proved below gives $Q\mathsf S = Q$. On the physical graph the completed Hessian is $\Delta_\pi = \mathsf S^{\mathsf T}\Delta_U^{\mathrm W}\mathsf S$. We use the positive difference convention $\eta D_U\lambda = \mathrm{Ad}(U_b)\lambda(b_+) - \lambda(b_-)$.

22.1.1. *Cell norms retain the ultraviolet scale.* Work on a finite compatible torus, with a separated aligned sequence and the spacings of Section 21. Let $\Delta(y)$ be the pillar of a top y, of side ξ_y. Outside the action region one can add bare cells and extend the local scale to η; a compressed operator is zero on that added space. The point scale $\xi(x)$ is the scale of its pillar. Use the exact

path metric

$$d(x,z) = \inf_{\gamma: x \to z} \int_\gamma \frac{ds_\infty}{\xi(s)}. \tag{PX1}$$

The half-open convention only assigns a measure-zero cell boundary; paths and their infima can be approximated on either side. The corresponding lattice metric is comparable after the fixed collar constants have been inserted.

For each field species s, declare a positive amplitude $a_s(y)$ and use $\|f\|_{s,y} = a_s(y)^{-1}\|f\|_{\infty,\Delta(y)}$. For example a physical potential has amplitude ξ_y^{-1}, a physical force density ξ_y^{-3}, and a raw datum logarithm amplitude one. The site gauge parameter also has amplitude one. A raw multiplier uses counting measure and amplitude one; its per-length density has amplitude ξ_y^{-3}. The conversion is the weighted dual conversion of the domain section.

If P_y is restriction to a cell, define the normalized cell entry of an operator $K : t \to s$ by

$$k_{st}(y,z) = \|P_y K P_z\|_{\ell^\infty(t,z)\to\ell^\infty(s,y)}. \tag{PX2}$$

All physical cell measures in the action of K are included before taking this norm. It is a block operator norm, not a pointwise bound on the continuum kernel. A Green kernel with its fine Coulomb singularity can have a bounded cell norm. Replacing it by a pointwise kernel bound independent of η would be a stronger and generally false assertion.

For compatible species, the cell partition and triangle inequality give

$$(k\ell)_{su}(y,z) \le \sum_w k_{st}(y,w)\ell_{tu}(w,z). \tag{PX3}$$

This follows first for finite lattices by inserting $\sum_w P_w = 1$. In an infinite-volume application the same formula holds when the row sums below give absolute convergence. Thus a two-sided exponential cell norm is an algebra norm, without discarding the fine singularity inside a block.

22.1.2. *The growth of a ball in the layer metric.* The layer metric is not assumed to have polynomial volume growth. A small-cell region can contain many more pillars than a nearby large-cell region. Separation controls that growth quantitatively.

Suppose x has layer index j and z has index i. Any path between them crosses each complete intervening layer. For a layer r, a crossing from Ω_{r+1} to Ω_r^c has an arc whose interior lies in $\Omega_r \setminus \Omega_{r+1}$. The separation makes its physical length greater than $\rho\xi_r$. On that arc the metric density is ξ_r^{-1}, so its length in (PX1) is at least ρ. Arcs belonging to distinct layers have disjoint interiors. Consequently

$$d(x,z) \ge \rho(|i-j|-1)_+. \tag{PX4}$$

One layer is lost because an endpoint can already lie next to its neighboring layer. The same proof applies to an intermediate point of a path, so a path of metric length at most $R+1$ starting in layer j never uses a scale larger than

$$\xi_{\max} \le L\xi_j L^{(R+1)/\rho}. \tag{PX5}$$

If the top index would exceed the depth, this is simply an overestimate.

Its physical length is at most $(R+1)\xi_{\max}$. Every pillar center in the metric R-ball therefore has a lift in the physical sup-norm ball of radius

$$H_R = L(R+1)\xi_j L^{(R+1)/\rho}. \tag{PX6}$$

For each fixed level i the centers lie on its grid of spacing ξ_i. The number of grid points in that physical ball is at most $C(1 + H_R/\xi_i)^4$. Equation (PX4) restricts the possible indices to

$|i-j| \le 1 + (R+1)/\rho$. Using the smallest such ξ_i in the grid count and then summing the possible levels gives

$$\#\{z : d(y,z) \le R\} \le C_L(1+R)^5 \exp\left(\frac{8\log L}{\rho} R\right). \tag{PX7}$$

Indeed $H_R/\xi_i \le C_L(R+1)L^{2(R+1)/\rho}$; its fourth power gives the exponential in (PX7), and the number of levels costs one more polynomial power. Since $\rho \ge 1$, the harmless factors $L^{8/\rho}$ and the bound on the number of levels can be included in C_L uniformly in ρ.

The argument works on a torus without a nonwrapping-ball assumption. For every reached center choose the lift reached by its path. Distinct periodic centers give distinct lifted grid points. Counting all grid points in the cover ball is therefore an upper bound, even if that ball crosses many fundamental domains. Restricting the allowed regions only decreases the number counted.

An entry bound of the form

$$k_{st}(y,z) \le C \left(\frac{\xi_y}{\xi_z}\right)^p e^{-md(y,z)} \tag{PX8}$$

now implies a weighted row and column sum, provided the rate pays for scale growth. By (PX4) the scale ratio on an annulus of metric radius R is at most $L^{|p|}e^{|p|(\log L)(R+1)/\rho}$. Break the space into unit annuli and use (PX7). Thus

$$\sup_y \sum_z e^{\delta d(y,z)} k_{st}(y,z) + \sup_z \sum_y e^{\delta d(y,z)} k_{st}(y,z) \le C' \tag{PX9}$$

whenever

$$m - \delta > \frac{(8+|p|)\log L}{\rho}. \tag{PX10}$$

For the column sum use the same ball argument centered at z; the absolute scale exponent is unchanged. The convergent scalar sum is bounded by $C\sum_{n\ge 0}(n+2)^5 e^{-[m-\delta-(8+|p|)\log L/\rho]n}$. This proves the claim with its precise uniformity.

If the amplitude normalization has already removed the scale ratio, take $p = 0$. For several species use the largest of their finitely many exponents. An exponential bound with an arbitrarily small m does not automatically imply (PX9). The geometry and rate must satisfy (PX10), or a stronger actual row estimate must be proved directly. This is a numerical condition in the order of choice of separation and kernel rates.

22.1.3. *The site readout at an outward row.* The site average used in the Landau projector has one value at each top, taken at that top's assigned level. The endpoint readout of a row is a different list when the row is outward. We now give their exact relation.

Let $\mathsf{T}_j(y,x)$ be the product of the admitted internal mean transports from a level-j center y down its pillar to a bare site x. For a top of level j define

$$(Q'_{\mathrm{pill}}\lambda)(y) = L^{-4j} \sum_{x \in B^j(y)} \mathrm{Ad}(\mathsf{T}_j(y,x))\lambda(x). \tag{PX31}$$

These pillars are disjoint. There is therefore an explicit right inverse on node fields:

$$(E_{\mathrm{pill}}\psi)(x) = \mathrm{Ad}(\mathsf{T}_j(y,x))^{-1}\psi(y) \quad (x \in B^j(y)), \qquad Q'_{\mathrm{pill}}E_{\mathrm{pill}} = 1. \tag{PX32}$$

This proves rank without an elliptic estimate. For real transports the right inverse has supremum norm one. With fine site measure η^4 and top measure ξ_y^4, it is also an isometry in L^2. Indeed a pillar contains L^{4j} bare sites, so $\eta^4 L^{4j} = \xi_y^4$. The invariant bilinear color pairing gives

$$Q'_{\mathrm{pill}} = E_{\mathrm{pill}}^{\mathsf{T}}, \qquad Q'_{\mathrm{pill}}(Q'_{\mathrm{pill}})^{\mathsf{T}} = 1. \tag{PX33}$$

The algebraic identities hold for admitted complex transports as well. Their Hermitian norm bounds then include the actual condition numbers of those transports. A counting kernel must include the ratio ξ_y^4/η^4 in the transpose; an ordinary unweighted array transpose would not be (PX33).

For any level-j center z, write $Q'_j\lambda(z)$ for its full j-fold transported site average, whether or not its assigned top level is j. If z is the outer endpoint of a determining level-j row, its assigned level is $j-1$. The entire block $B^j(z)$ lies in that outer layer. Decompose it into its L^4 level-$(j-1)$ pillars. The product definition of the transport gives

$$Q'_j\lambda(z) = L^{-4} \sum_{w\in B(z)\cap\mathbb{T}^{(j-1)}} \mathrm{Ad}(T_{z,w}(\mathcal{M}^{j-1}U))(Q'_{\mathrm{pill}}\lambda)(w). \tag{PX34}$$

To verify the coefficient, each smaller average has weight $L^{-4(j-1)}$; multiplying it by L^{-4} gives L^{-4j}. The two transport factors multiply in their path order. This proves (PX34) exactly, without an asymptotic or small-field approximation.

Define $\mathcal{E}_{j,z}\psi$ to be $\psi(z)$ at an assigned level-j top, and to be the finite average on the right of (PX34) at an outer endpoint. Then define the extended coarse difference by

$$(\bar{D}_{\mathrm{ext}}\psi)(c) = \xi_c^{-1}\{\mathrm{Ad}((\mathcal{M}^jU)(c))\mathcal{E}_{j,c_+}\psi - \mathcal{E}_{j,c_-}\psi\}, \qquad c\in\Gamma_j. \tag{PX35}$$

For an interior row it is the ordinary covariant difference of its two top values. For an outward row it first averages the outer finer tops. There is no need to duplicate that site as a new independent top in the site Gram matrix.

Here is a direct verification of the field intertwining. Under an infinitesimal gauge $e^{-it\lambda}$ the derivative of the level-j node factor is

$$\frac{1}{i}\frac{d}{dt}v_j[tD_U\lambda](y)\Big|_{t=0} = Q'_j\lambda(y) - \lambda(y). \tag{PX36}$$

At one step the change in each relative contour has derivative $\mathrm{Ad}(T_{y,x})\lambda(x) - \lambda(y)$. The implicit mean at the identity has differential the arithmetic mean of its inputs, so this gives the one-step formula. Iterating and using the product rule for the accumulated node factors gives (PX36), with the hierarchical site average (PX31).

Differentiate the actual straightening identity $\mathcal{M}^j(e^{i\eta A}U) = v_j[A]\,e^{iq_j^{\mathrm{st}}(A)}\mathcal{M}^jU\,v_j[A]^{-1}$ on a row. The plain coarse variation is $\mathrm{Ad}(\mathcal{M}^jU(c))\lambda(c_+) - \lambda(c_-)$. Subtract the two node derivatives (PX36), with their endpoint transports. The remaining variation is

$$Dq_j^{\mathrm{st}}(0)\,D_U\lambda = \mathrm{Ad}(\mathcal{M}^jU(c))Q'_j\lambda(c_+) - Q'_j\lambda(c_-). \tag{PX37}$$

Divide by ξ_c to obtain the per-length row Q. Using (PX34) proves the intertwining identity on interior and outward rows

$$QD_U = \bar{D}_{\mathrm{ext}}Q'_{\mathrm{pill}}. \tag{PX38}$$

This supplies the intertwining needed in the slice identities above. There $\bar{D}$ means this extended row difference and Q' means the own-pillar site map. Using a single unextended value of the outer top would give the wrong formula.

Each endpoint extension reads at most L^4 tops in one adjacent layer. Its real row sum is at most one, since the transports are unitary and the weights add to one. A finer top occurs in one such outer block and at most eight incident coarse rows. Thus its two-sided normalized cell counts are bounded at fixed L, without a factor depending on the depth. For complex pairs the same statement uses the actual bounded transport condition numbers. The latter are bounds on the local transports in the regular-pair domain; the algebraic intertwining does not prove them.

22.1.4. *Finite real existence of the site operators.* There is also a real finite positivity statement independent of a quantitative inverse bound. Assume the actual mean transports are admitted and unitary, and that every active fine site lies in a connected pillar with its top. Let a be strictly positive on the top space. Then

$$L_{\mathrm{site}} = D_U^{\mathsf{T}} D_U + (Q'_{\mathrm{pill}})^{\mathsf{T}} a Q'_{\mathrm{pill}} \quad \text{is strictly positive.} \tag{PX39}$$

To prove it, a vector of zero Hermitian energy must obey $D_U\lambda = 0$ and $Q'_{\mathrm{pill}}\lambda = 0$. Split into Hermitian real and imaginary parts; it suffices to treat real λ. The first identity means that $e^{it\lambda(x)}$ leaves every active bond unchanged:

$$e^{it\lambda(b_-)} U_b e^{-it\lambda(b_+)} = U_b. \tag{PX40}$$

Gauge covariance of each actual mean transport therefore gives $\mathrm{Ad}(\mathsf{T}_j(y,x))\lambda(x) = \lambda(y)$. Average over the pillar: $Q'_{\mathrm{pill}}\lambda(y) = \lambda(y)$. Its zero value makes $\lambda(y) = 0$. Parallel transport along the active graph from the top to every site makes $\lambda = 0$. This proves (PX39).

It follows that $G' = L_{\mathrm{site}}^{-1}$ exists on every finite real problem of this type. The site Gram matrix

$$Q'_{\mathrm{pill}} G'^2 (Q'_{\mathrm{pill}})^{\mathsf{T}} \tag{PX41}$$

is strictly positive as well. For a nonzero node vector ψ, surjectivity makes $(Q'_{\mathrm{pill}})^{\mathsf{T}}\psi \neq 0$; invertibility of G' then makes its squared norm positive. Thus the formulas for R and S are genuine finite real matrices before their uniform bounds are established.

The quantitative site gap is proved next.

22.1.5. *Transport control on one pillar.* The site gap has a quantitative proof from the actual averages. Let a pillar have side $\xi = n\eta$, where $n = L^j$, and assume

$$|\partial U(p) - 1| \le \epsilon\, \eta^2 \xi^{-2} \tag{PX42}$$

on its determining neighborhood. The fixed path neighborhood has an embedded rectangular lift. If an extra fine collar is needed, the regularity bounds in the adjacent layer give the same inequality with a fixed L-dependent factor. We include that factor in the constants below.

Take the fine coordinate comb from the pillar center and put $g(x) = U(\Gamma_{y,x})$. For adjacent $x, x + \eta e_\mu$, the loop made by the two root combs and that bond is filled by strips of width one fine cell. Commuting the extra μ-step through the later coordinate segments uses at most Cn plaquettes, not Cn^2. The unitary Stokes product inequality therefore gives

$$|U^g(b) - 1| \le C\epsilon/n \quad \text{on internal fine bonds.} \tag{PX43}$$

Here $g(y) = 1$. Every curvature norm is unchanged. The gauge need not be close to one, and its derivatives are not estimated in this step.

Write $\xi_r = L^r\eta$, and let a_r be the supremum of $|\mathcal{M}^r(U^g)(c) - 1|$ on the level-r internal edges used in the hierarchical transport. Let ϖ_r be the corresponding coarse plaquette bound. The determining neighborhoods remain in the rectangular pillar neighborhood: each parent block is a union of its child blocks, and each contour stays in its assigned block. The real curvature iteration proved in the background section gives

$$\varpi_r \le C_L \epsilon (\xi_r/\xi)^2. \tag{PX44}$$

The same finite comparison shows that one actual averaged edge differs from its central straight transport by at most $C_L\varpi_r$. Indeed every closed logarithm in its definition is a comparison loop with that central transport. The unitary strip estimate bounds those logarithms, and then their exponential average. The central route has L edges, so

$$a_{r+1} \le L a_r + C_L \varpi_r. \tag{PX45}$$

The coefficient of a_r is exactly L, by the unitary product inequality.

Iterating (PX45), using $a_0 \le C\epsilon\eta/\xi$, gives

$$\begin{aligned} a_r &\le L^r a_0 + C_L\epsilon \sum_{s<r} L^{r-1-s}(\xi_s/\xi)^2 \\ &\le C_L\epsilon\, \xi_r/\xi. \end{aligned} \tag{PX46}$$

For the sum, insert $\xi_s = \eta L^s$; its geometric factor is bounded by a constant times $(\xi_r/\xi)^2$, which is no larger than ξ_r/ξ. The constants do not multiply with the number of levels.

The internal mean on level r is a finite smooth mean of paths of $O(L)$ edges. On its regular near-identity chart it differs from one by at most $C_L a_r$. The hierarchical transport has one such factor at each level below j. All are unitary, hence

$$|\mathsf{T}_j(U^g; y, x) - 1| \le C_L \sum_{r<j} a_r \le C_L\epsilon \quad (x \in B^j(y)). \tag{PX47}$$

The last sum uses $\sum_{r<j} \xi_r/\xi \le 1/(L-1)$. The fixed smallness of ϵ also admits every mean used in this argument. Thus (PX47) is a uniform-depth property of the actual iterated transports, not a hypothesis of such a property.

22.1.6. *A real weighted site gap.* We first prove the flat block inequality with its scale. Let $\bar\lambda$ be the arithmetic mean on the n^4 sites of a block. The exact variance identity is

$$\sum_x |\lambda(x) - \bar\lambda|^2 = \frac{1}{2n^4} \sum_{x,z} |\lambda(x) - \lambda(z)|^2. \tag{PX48}$$

Connect each pair by the coordinate-order path. Its length is at most $4n$. A fixed oriented edge lies in at most n^5 of these paths: its two longitudinal endpoint coordinates give at most n^2 choices, and the three other coordinates give at most n^3. Cauchy on the path and then (PX48) give a constant times n^2 times the sum of squared edge differences. Multiplying by $\eta^4\xi^{-2}$ proves

$$\xi^{-2}\|\lambda\|_{2,B}^2 \le C\big\{\|\partial_{\mathrm{int}}\lambda\|_{2,B}^2 + \xi^2|\bar\lambda|^2\big\}. \tag{PX49}$$

Here the fine norm uses η^4, and ∂ divides a bond difference by η. The mean part is $\eta^4 n^4 \xi^{-2}|\bar\lambda|^2 = \xi^2|\bar\lambda|^2$. For $n = 1$ the variance vanishes and the same inequality holds.

Apply the real comb gauge of (PX43) in each pillar. Only internal bonds are used in the block inequality, so the gauges of neighboring pillars may be different. They define a unitary site gauge globally, and all covariant norms are unchanged. In this frame

$$\|(\partial_{\mathrm{int}} - D_{U,\mathrm{int}})\lambda\|_{2,B} \le C_L\epsilon\, \xi^{-1}\|\lambda\|_{2,B}. \tag{PX50}$$

The estimate follows from $|\operatorname{Ad}(U_b) - 1| \le C_L\epsilon\eta/\xi$ and the at-most-eight incidences at a site.

Let Q'_0 be arithmetic block average. Equation (PX47), Cauchy for the finite average and $\xi^4 = n^4\eta^4$ give

$$\xi^4|(Q'_{\mathrm{pill}} - Q'_0)\lambda|^2 \le C_L\epsilon^2\|\lambda\|_{2,B}^2. \tag{PX51}$$

Use (PX50)–(PX51) on the right of (PX49). The result is the same two terms with D_U and Q'_{pill}, plus $C_L\epsilon^2\xi^{-2}\|\lambda\|_{2,B}^2$. Choose the fixed curvature ceiling so that this coefficient is at most one half of the left coefficient. Absorption proves

$$\xi^{-2}\|\lambda\|_{2,B}^2 \le C_L\big\{\|D_{U,\mathrm{int}}\lambda\|_{2,B}^2 + \xi^2|Q'_{\mathrm{pill}}\lambda|^2\big\}. \tag{PX52}$$

Sum over the disjoint pillars. Every internal fine bond is counted at most once; all other gradient terms in the full energy are nonnegative. Suppose $a(y) = c_y\xi_y^{-2}$ with $c_y \ge a_* > 0$. Then

$$\begin{aligned} \sum_x \eta^4\xi(x)^{-2}|\lambda(x)|^2 &\le C_{L,a_*}\big\{\|D_U\lambda\|_2^2 + \langle Q'_{\mathrm{pill}}\lambda, aQ'_{\mathrm{pill}}\lambda\rangle_{\mathrm{tops}}\big\} \\ &= C_{L,a_*}\langle\lambda, L_{\mathrm{site}}\lambda\rangle. \end{aligned} \tag{PX53}$$

This is the actual site weighted gap, uniform in depth, volume, region shape and real holonomy. It uses the small real curvature and actual transport construction; no physical current or criticality is used. Crossing-edge energy is kept, not replaced by a clipped block mean.

For a complex site vector split into Hermitian real and imaginary parts. Since the real unitary connection is a real matrix in an orthonormal Lie-algebra basis, the same inequality holds for its Hermitian energy. If the top scale is at most one, it in particular gives an ordinary finite L^2 gap independent of the bare mesh. In other physical units its smallest unweighted gap is scaled by the inverse square of the largest local scale.

22.1.7. *The complex site domain uses the actual iterated means.* The preceding transport argument has a complex version with a fixed physical radius, independent of the number of fine sites. We give it to avoid assuming a uniform radius for the iterated site average.

Consider one step on a fixed two-block neighborhood. Suppose all its edge matrices satisfy $|W_b - 1| \le a$, and let $\varpi = \sup|\partial W - 1|$. There are constants r_L, C_L for which the actual means and logarithms are admitted if $a, \varpi < r_L$, and

$$\begin{aligned} a_{\mathrm{next}} &\le La + C_L a^2 + C_L \varpi, \\ \varpi_{\mathrm{next}} &\le L^2(1 + C_L a)\varpi + C_L \varpi^2. \end{aligned} \tag{PX54}$$

For the first line, a central straight product differs from one by at most $La + C_L a^2$. Plaquette sliding bounds each closed comparison logarithm by $C_L\varpi$: all prefix transports have condition number at most $1 + C_L a$ on this finite neighborhood. The defining exponential average therefore changes the central product by at most that amount.

For the second line, take the local complex comb gauge. Its values and inverses are $1+O(C_L a)$. It sets the tree bonds to one; each remaining bond is then $1 + O(C_L\varpi)$ by the finite plaquette sliding identity. In that gauge the Taylor/flux argument for the actual one-step map gives the coefficient L^2 and a quadratic error $C_L\varpi^2$. Conjugating the fine and coarse plaquettes through the small complex gauge changes their norms by $1 + C_L a$. This proves (PX54). Complex covariance here follows directly from the conjugation identity for the small logarithm, or from uniqueness of the admitted finite mean. No absolute-value formula is continued holomorphically.

These constants can be bounded before an iteration is run. The finite mean construction has a joint raw-edge holomorphic ball of radius c/L, with universal supremum bounds; two scalar Cauchy circles bound its second derivative by CL^2. A comb on the one-step patch sweeps at most CL^2 plaquettes, so its non-tree logarithms have size $CL^2\varpi$. The quadratic Taylor error in the coarse plaquette is therefore at most $CL^2(L^2\varpi)^2 = CL^6\varpi^2$. The fine logarithmic curl error times its L^2 flux factor has the same upper order. Prefix conjugations cost $1 + CLa$. For example $r_L = cL^{-3}$ and $C_L = CL^8$, with fixed universal shrinkages and enlargements, suffice in (PX54). This count uses directional Cauchy on the proved full supremum ball, not one coordinate estimate summed over all field coordinates.

On a pillar let $t_r = \xi_r/\xi$, and normalize $A_r = a_r/t_r$, $V_r = \varpi_r/t_r^2$. Equation (PX54) implies, with an increased fixed constant,

$$Z_{r+1} \le Z_r + C_L t_r(Z_r + Z_r^2), \qquad Z_r = A_r + V_r, \qquad \sum_{r<j} t_r \le \frac{1}{L-1}. \tag{PX55}$$

For $Z_r \le 1$, the factor is at most $1 + 2C_L t_r$. Its product is at most $\exp(2C_L/(L-1))$. Choose Z_0 below one half the minimum one-step chart radius times the inverse of this number. Induction then gives $Z_r \le C_L Z_0$ and keeps every step in its chart. The constants depend on L, not j. With the preceding finite choices a sufficient common small input radius has the explicit form $cL^{-3}\exp(-CL^8)$. The precise numerical constants are not needed below, but this gives a positive lower radius rather than an asserted upper cap. The product of the internal means and its inverse

have norm at most $\exp(C_L \sum_r a_r)$. Hence

$$|\mathsf{T}_j(W;y,x)-1| \le C_L Z_0, \qquad |\mathsf{T}_j(W;y,x)| + |\mathsf{T}_j(W;y,x)^{-1}| \le C_L. \tag{PX56}$$

This proves the existence and boundedness of the whole finite hierarchical formula on that domain.

Now start with a real field satisfying (PX42), put it in the real comb gauge (PX43), and let $W = e^{i\eta E}U$. Assume the actual physical first-order input satisfies

$$h = \max\{\|\xi E\|_\infty, \|\xi^2 \nabla_U E\|_\infty\} \le h_L. \tag{PX57}$$

In the comb frame the ordinary first difference of E differs from its covariant one by $O(\epsilon\xi^{-1}|E|)$. The elementary plaquette product then gives $a_0 \le C(\epsilon+h)\eta/\xi$ and $\varpi_0 \le C(\epsilon+h+h^2)(\eta/\xi)^2$. Thus the starting Z_0 in (PX55) is at most $C(\epsilon+h)$.

Choose a fixed $h_L > 0$ so that the same argument holds with E replaced by sE for $|s| < h_L/h$, and assume $h \le h_L/4$. Reduce the real curvature ceiling at the same time so that $\epsilon \le h_L$ and the resulting starting Z_0 satisfies (PX55). The transport is holomorphic in s on this disc, with a uniform bound. Cauchy's formula on a smaller concentric disc, or the geometric sum of its Taylor coefficients, yields

$$|\mathsf{T}_j(e^{i\eta E}U;y,x) - \mathsf{T}_j(U;y,x)| \le C_L h. \tag{PX58}$$

The bound in (PX56) is uniform in j and in the number of coordinates. This is a single directional Cauchy argument on an actual admitted physical disc. It does not assert an entrywise absolute Hessian sum for an arbitrary many-variable functional.

For the complex site average write Q'_E and for the real reference write Q'_U. The transported finite-average calculation used in (PX51) gives

$$\|a^{1/2}(Q'_E - Q'_U)\lambda\|_{2,\mathrm{tops}} \le C_L h\, \|\xi^{-1}\lambda\|_2. \tag{PX59}$$

Here the real row weights satisfy $a(y) \le a^* \xi_y^{-2}$. The proof squares the average difference in each disjoint pillar, uses (PX58), and sums its fine volume. There is no factor equal to the number of fine sites. The same argument applies to the outer endpoint extension (PX34): its neighborhood uses one adjacent scale, so the input constants change by fixed powers of L.

22.1.8. *Relative forms give the complex site inverse.* It remains to compare the actual site operators, not their unscaled matrix coefficients. Let L_U be the real site operator and let $L_E = D^{\mathsf{T}}_{e^{i\eta E}U} D_{e^{i\eta E}U} + Q'^{\mathsf{T}}_E a Q'_E$. The transposes in L_E are bilinear analytic transposes. At a real reference they are the usual Hermitian adjoints.

For a fine bond the difference of the two site derivatives is bounded by $C|E_b||\lambda(b_+)|$. Adjacent pillar scales differ by at most the fixed factor L; the region gaps exclude a bare bond jumping over another layer. The finite incidence count therefore gives

$$\|(D_{e^{i\eta E}U} - D_U)\lambda\|_2 \le C_L h \|\xi^{-1}\lambda\|_2. \tag{PX60}$$

Keeping derivatives on the two arguments, expand the kinetic difference. Use (PX60), then (PX59) for the mass difference, and finally the real weighted site gap (PX53). For arbitrary complex vectors v, w, this gives

$$|\langle v, (L_E - L_U)w\rangle_{\mathrm{H}}| \le C_L h\, \langle v, L_U v\rangle_{\mathrm{H}}^{1/2} \langle w, L_U w\rangle_{\mathrm{H}}^{1/2}. \tag{PX61}$$

For example the quadratic mass term is $(Q'_E - Q'_U)^{\mathsf{T}} a (Q'_E - Q'_U)$; the two cross terms contain only one difference. On replacing the left argument by its complex conjugate, the same absolute bounds apply to the analytic bilinear transpose. No Hermitian conjugation of the complex coefficients is inserted into their definition.

Let $B_E = L_U^{-1/2}(L_E - L_U)L_U^{-1/2}$. Equation (PX61) proves $\|B_E\|_{2\to 2} \le C_L h$. For $C_L h \le 1/2$ one has

$$G'_E = L_U^{-1/2}(1+B_E)^{-1}L_U^{-1/2}, \qquad \mathrm{Re}\langle\lambda, L_E\lambda\rangle_{\mathrm{H}} \ge \tfrac{1}{2}\langle\lambda, L_U\lambda\rangle_{\mathrm{H}}. \tag{PX62}$$

The inverse is holomorphic on the whole admitted physical ball. It has the same weighted coercivity, up to a fixed factor, uniformly in depth and volume. This is a relative-form argument. The unscaled operator norm of a coefficient perturbation need not be bounded as $\eta \downarrow 0$.

22.1.9. *Exponential decay from the weighted site form.* Let V denote multiplication by $\xi(x)^{-2}$ on sites. The real gap (PX53) is $L_U \geq cV$. We prove a weighted inverse bound before converting it to pointwise regularity. This avoids estimating a bare kinetic operator by its unscaled norm.

Let ψ be bounded and one-Lipschitz for (PX1), and let $M_s = e^{s\psi}$ on sites. Use the value $\psi(b_-)$ on an edge and $\psi(y)$ on a top. Put

$$D_s = M_s^{\mathrm{edge}} D_U M_s^{-1}, \qquad Q'_s = M_s^{\mathrm{top}} Q'_U M_s^{-1}. \tag{PX63}$$

On an edge the two endpoint values of ψ differ by at most $C_L\eta/\xi_b$. Inside a pillar their difference from the top value is at most a fixed constant. The difference from the unweighted row therefore satisfies

$$\|(D_s - D_U)\lambda\|_2 + \|a^{1/2}(Q'_s - Q'_U)\lambda\|_{2,\mathrm{tops}} \leq C_L|s|\|V^{1/2}\lambda\|_2 \tag{PX64}$$

for $|s| \leq s_L$. The edge estimate follows from $|e^{st} - 1| \leq C|s||t|$ on the stated bounded interval. For the site row use the same estimate in its finite transported average, then Cauchy with the actual pillar volume. Its constant is independent of $\|\psi\|_\infty$.

The conjugated form has the exact factorization

$$M_s L_U M_s^{-1} = D_{-s}^{\mathsf{T}} D_s + (Q'_{-s})^{\mathsf{T}} a Q'_s. \tag{PX65}$$

Expanding it and using (PX64), (PX53) proves a relative form error at most $C_L|s|$. With a complex field perturbation in the ball of (PX62), the same calculation gives error $C_L(h + |s|)$ relative to the real L_U. Choose both small enough for that number to be at most one half. All these forms are invertible.

For any such inverse $\mathcal{G}_s$, testing the equation $L_s\lambda = V^{1/2}f$ against λ gives

$$\|V^{1/2}\mathcal{G}_s V^{1/2}\|_{2\to 2} \leq 2/c. \tag{PX66}$$

Indeed its real part is at least $(c/2)\|V^{1/2}\lambda\|_2^2$, whereas the right pairing is at most $\|V^{1/2}\lambda\|_2\|f\|_2$. This proof uses neither a normal matrix nor a pointwise positive complex kernel.

Choose $\psi(x) = \min\{d(x, \Delta(z)), N\}$. It is one-Lipschitz. The diameter of a pillar in (PX1) is bounded, so on $\Delta(y)$ its value is at least $d(y, z) - C$ after N is large enough. Equation (PX66) then yields

$$\|P_y G'_E P_z\|_{2\to 2} \leq C_L \xi_y \xi_z e^{-md(y,z)} \tag{PX67}$$

for a fixed $m = m(L) > 0$. On a finite torus take the periodic distance. On the cover first use bounded N and then increase it. The constants are independent of this truncation.

In particular the normalized operator $V^{1/2}G'_E V^{1/2}$ has exponentially decaying cell L^2 entries. Its two-sided cell Schur norm follows from (PX7)–(PX10) when the separation is chosen with

$$\frac{8\log L}{\rho} < m/2. \tag{PX68}$$

The rate m above depends only on the fixed one-step constants and the site gap; it does not depend on ρ. Thus this is a genuine order of choice: increase the separation after those constants, or verify that the already chosen separation satisfies this finite inequality. It is not an appeal to unspecified polynomial volume growth.

22.1.10. *A scalar subsolution estimate at the fine mesh.* We use a local estimate whose proof also explains why a fine Coulomb singularity causes no loss in (PX67). On the infinite scalar fine lattice in four dimensions, let $\mathcal{L} = -\Delta_\eta$. The positive heat kernel gives a positive Green kernel

$$G_0(x) = \int_0^\infty p_t^\eta(x)\,dt, \qquad |G_0(x)| \le C(|x|+\eta)^{-2}, \quad |\nabla G_0(x)| \le C(|x|+\eta)^{-3}. \tag{PX69}$$

Here is the pointwise heat-kernel bound used in this integration. Put $s_t = \max(\eta, \sqrt{t})$. The Fourier formula (BGH2)–(BGH4) gives $\|D^r p_t^\eta\|_\infty \le C_r s_t^{-4-r}$ for $r = 0, 1$: for $t \ge \eta^2$ integrate $|\theta|^r e^{-ct|\theta|^2/\eta^2}$ and rescale, and for smaller t use the bounded Fourier multiplier. Write the half-time convolution with the derivative on one fixed factor, $D^r p_t^\eta(x) = \sum_y \eta^4 (D^r p_{t/2}^\eta)(y) p_{t/2}^\eta(x-y)$. Split the sum into $|y| \ge |x|/2$ and its complement. On the first set apply (BGH5) of order r to the first factor and the order-zero supremum bound to the second. On the complement $|x-y| \ge |x|/2$; apply the order-zero tail bound to the second factor and the order-r supremum bound to the first. Since $s_{t/2}$ and s_t are comparable, for every fixed N this gives

$$|D^r p_t^\eta(x)| \le C_{r,N} s_t^{-4-r} (1 + |x|/s_t)^{-N}, \qquad r = 0, 1.$$

Integrating these inequalities gives the asserted Green bounds. For example split at $t = |x|^2$, use a polynomial off-diagonal tail before the split, and Ct^{-2}, respectively $Ct^{-5/2}$, afterwards. At $|x| \le \eta$ the bounds are the corresponding lattice scaling bounds. Also $\mathcal{L}G_0$ is the normalized lattice delta, by integrating $-\partial_t p_t^\eta$.

Suppose $w \ge 0$ and $\mathcal{L}w \le F_0$ on a ball of radius $2r$, $r \ge 8\eta$. Choose a nonnegative cutoff χ, equal to one on the r-ball, supported in the $2r$-ball, with its first two differences bounded by Cr^{-1}, Cr^{-2}. At the center x, summation by parts gives

$$w(x) = \langle \chi G_0(x - \cdot), \mathcal{L}w \rangle - \langle [\mathcal{L}, \chi] G_0(x - \cdot), w \rangle. \tag{PX70}$$

All sums use η^4. This identity follows by moving $\mathcal{L}$ off w and using $\mathcal{L}G_0 = \delta_x$. There is no derivative of w in its last term. Since the first test is nonnegative, its contribution is at most $CF_0 r^2$. Indeed the lattice sum of G_0 on that ball is at most Cr^2, by dyadic shells.

The commutator is supported on the cutoff annulus. Writing it as $(\mathcal{L}\chi)G_0$ plus products of first differences shows that its absolute value is at most Cr^{-4} there, by (PX69). Cauchy and the annulus volume therefore prove

$$w(x) \le C\{r^{-2}\|w\|_{2,B_{2r}} + r^2 F_0\}. \tag{PX71}$$

If r/η is bounded, the point-value estimate $|w(x)| \le \eta^{-2}\|w\|_2$ gives the same inequality with a changed constant. Thus the local estimate is uniform at coarse meshes as well.

For a real unitary connection, let λ be Lie-algebra valued and put $w = |\lambda|$. For $\lambda(x) \ne 0$, pair each term $\lambda(x) - \mathrm{Ad}(U_{x,x'})\lambda(x')$ with $\lambda(x)/|\lambda(x)|$. Unitarity gives

$$-\Delta_\eta |\lambda|(x) \le |D_U^{\mathsf{T}} D_U \lambda(x)|. \tag{PX72}$$

At a zero value of λ the left side is nonpositive, so the same inequality holds. This proves the covariant scalar subsolution inequality without differentiating a gauge. It is used only on a patch with the full covariant Laplacian stencil. Bare boundary cells will use the point-value fallback instead.

22.1.11. *Pointwise site Green bounds.* First take a real U and $\lambda = G'_U f$. At a site in pillar $\Delta(y)$, the mass part of its equation obeys

$$|(Q_U'^{\mathsf{T}} a Q'_U \lambda)(x)| \le C_L \xi_y^{-4} \|\lambda\|_{2,\Delta(y)}. \tag{PX73}$$

In fact it is the transported vector $a(y)Q'_U\lambda(y)$, whose norm is at most $a^* \xi_y^{-2} |Q'_U \lambda(y)|$. Cauchy in the average gives the additional ξ_y^{-2}.

Take a local ball of radius comparable to ξ_y. It meets a bounded L-dependent number of pillars, all with scales in $[\xi_y/L, L\xi_y]$. The gaps exclude a second intervening layer in that ball once $\rho > C_L L$. Call the corresponding whole-pillar neighborhood $\mathcal{N}(y)$. Its cells are at bounded d-distance from y. Apply (PX71)–(PX73) on a lift of this ball. This proves

$$\|\lambda\|_{\infty,\Delta(y)} \le C_L \xi_y^{-2} \sum_{w\in\mathcal{N}(y)} \|\lambda\|_{2,\Delta(w)} + C_L \xi_y^2 \|f\|_{\infty,\mathcal{N}(y)}. \tag{PX74}$$

Near a boundary of the action region, every pillar lacking this full patch is bare: the nonbare regions have the prescribed gap from that boundary. The single-cell fallback in (PX71) proves (PX74) there directly. On a torus the local ball is lifted to the periodic cover. Its fixed radius can meet only a fixed number of copies of a cell, since every period contains at least two top blocks. This changes only the constant.

If f is supported in $\Delta(z)$, use (PX67) and $\|f\|_{2,\Delta(z)} \le \xi_z^2 \|f\|_\infty$ on the right side. The far contribution is $C\xi_z^3 \xi_y^{-1} e^{-md(y,z)} \|f\|_\infty$. The local forcing term has $d(y,z) \le C_L$. Equation (PX4) absorbs the scale ratio:

$$\|G'_U f\|_{\infty,\Delta(y)} \le C_L \xi_y^2 e^{-m_1 d(y,z)} \|f\|_\infty, \qquad 0 < m_1 < m - \frac{3\log L}{\rho}. \tag{PX75}$$

Choosing the separation large enough leaves a positive fixed m_1. This bound concerns a bounded cell source.

The actual equation also gives

$$\|D_U^{\mathsf{T}} D_U G'_U f\|_{\infty,\Delta(y)} \le C_L e^{-m_1 d(y,z)} \|f\|_\infty. \tag{PX76}$$

Use $D_U^{\mathsf{T}} D_U \lambda = f - Q_U'^{\mathsf{T}} a Q'_U \lambda$, and bound its mass term by the supremum on its same pillar. The cell diameter in the layer metric is bounded.

22.1.12. *Site gradients and their complex continuation.* For differentiated pointwise bounds use the actual local gauge clause of the regular-pair class. On the required patch around a cell it supplies a unitary gauge with

$$U = e^{i\eta A}, \qquad \xi|A| + \xi^2|\nabla A| \le a_{\mathrm{loc}}, \qquad a_{\mathrm{loc}} \le a_L. \tag{PX77}$$

The constant a_L is fixed small. Near one interface the patch may be described in the finer adjacent scale; converting to ξ changes these bounds by fixed powers of L. The region gaps and the fixed patch collar are chosen to include no third layer. This gauge clause is an input defining the regular-pair class. Deriving it for an actual background from weaker tests is the separate background estimate.

In this local gauge the site equation is a scalar componentwise Laplace equation with first- and zeroth-order errors. Expanding the transported differences gives

$$|\Delta_\eta \lambda| \le |f - Q'^{\mathsf{T}} a Q' \lambda| + C_L a_{\mathrm{loc}} \{\xi^{-1}|\nabla\lambda| + \xi^{-2}|\lambda|\}. \tag{PX78}$$

Each difference of a link exponential costs $\eta|\nabla A|$; each undifferentiated link error costs $\eta|A|$. After division by the two powers of η in the Laplacian the terms have exactly the displayed dimensions.

Here is the interior absorption that supplies the gradient. On a slightly larger ball of radius $C\xi$, put $M = \|\lambda\|_\infty$ and $F = \|f - Q'^{\mathsf{T}} a Q' \lambda\|_\infty$. Let d_b be distance of a fine edge to the ball boundary and take the finite maximum of $d_b|\nabla\lambda(b)|$. If d_b is only a fixed multiple of η, the point-value bound on a difference controls it by CM. Otherwise apply the scalar interior gradient estimate to a ball of radius $d_b/4$ around an edge where this maximum is approached. Distances in that smaller ball are comparable to d_b. Its gradient supremum is therefore at most the weighted maximum divided by cd_b. Using (PX78) in the scalar estimate gives

$$\sup_b d_b|\nabla\lambda(b)| \le C_L(M + \xi^2 F) + C_L a_{\mathrm{loc}} \sup_b d_b|\nabla\lambda(b)|. \tag{PX79}$$

Choose $C_L a_{\mathrm{loc}} \le 1/2$. Absorption, (PX75)–(PX76) and the scalar interior Hölder estimate then give, for a bounded source in $\Delta(z)$,

$$\begin{aligned} \|\nabla_U G'_U f\|_{\infty,\Delta(y)} &\le C_L \xi_y e^{-\delta d(y,z)} \|f\|_\infty, \\ [\nabla_U G'_U f]_{\alpha,\Delta(y)} &\le C_{L,\alpha} \xi_y^{1-\alpha} e^{-\delta d(y,z)} \|f\|_\infty, \qquad 0<\alpha<1. \end{aligned} \tag{PX80}$$

The Hölder comparison is made in the assigned local frame, or with its declared transport to a common base. The bounded potential and pointwise gradient give the transported version. The constant is not asserted bounded as $\alpha \uparrow 1$. Cells with a bounded number of fine sites use the direct difference bound; near the action boundary they are bare, as already noted.

These pointwise estimates continue to the actual complex site operator by a stronger norm argument than a pointwise matrix coefficient estimate. For a fixed source cell z, let

$$\begin{aligned} \|\lambda\|_{\mathcal{Y}_z} &= \sup_y e^{\delta d(y,z)} \max\{\xi_y^{-2}\|\lambda\|_{\infty,\Delta(y)}, \xi_y^{-1}\|\nabla_U \lambda\|_{\infty,\Delta(y)}\}, \\ \|f\|_{\mathcal{F}_z} &= \sup_y e^{\delta d(y,z)} \|f\|_{\infty,\Delta(y)}. \end{aligned} \tag{PX81}$$

Use a rate δ strictly below the cell Green rate, with its spare rate larger than the ball-growth rate in (PX7). Summing the bounded-source cell estimates then proves $\|G'_U\|_{\mathcal{F}_z \to \mathcal{Y}_z} \le C_L$, uniformly in z.

For the actual left perturbation $U_E = e^{i\eta E} U$, expand the two covariant site differences and the two site averages. The kinetic difference has a coefficient bounded by $|E|$ on a first difference of λ, and a coefficient bounded by $|\nabla_U E| + |E|^2$ on λ. The mass difference uses (PX58), has size $C_L h \xi^{-2}$, and reads only the same pillar. Hence

$$\|(L_E - L_U)\lambda\|_{\mathcal{F}_z} \le C_L h \|\lambda\|_{\mathcal{Y}_z}. \tag{PX82}$$

No second difference of λ occurs: the difference of two connections in a site Laplacian is lower order. The first-order input on E is used here; its supremum alone would not bound the zeroth-order coefficient. The metric diameter of a pillar is bounded, so the profile weight passes through the mass average with a fixed factor.

For h below the resulting fixed threshold, the series

$$G'_E = \sum_{n \ge 0} [-G'_U (L_E - L_U)]^n G'_U \tag{PX83}$$

converges from $\mathcal{F}_z$ to $\mathcal{Y}_z$. At finite volume it coincides with the inverse already constructed in (PX62), by uniqueness. It therefore proves the complex versions of the pointwise potential and gradient estimates. The equation supplies the complex covariant Laplacian bound. In the same real local frame, $e^{i\eta E} e^{i\eta A}$ has a physical logarithm with the first-order bounds $C(a_{\mathrm{loc}} + h)$. The scalar interior estimate then gives the complex Hölder member as well.

The argument establishes the site estimates for bounded cell sources. A mixed gradient with a divergence source has a Calderón–Zygmund singularity and requires its specified cancellation or Hölder input. No uniform bound for that mixed operator on all bounded raw sources has been inferred.

22.1.13. *The site Gram inverse has its own lower-bound proof.* Invertibility of G' does not by itself give a uniform inverse for $Q' G'^2 Q'^{\mathsf{T}}$. We prove that estimate using actual site bumps, rather than assuming it as part of the site Green theorem. Let Ξ multiply a top value by ξ_y.

Choose a fixed smooth scalar bump χ_y, supported in the middle portion of its pillar and equal to one on a smaller middle portion. Its first two differences are bounded by $C\xi_y^{-1}$ and $C\xi_y^{-2}$. The arithmetic average $\beta_y = Q'_0 \chi_y$ has a positive lower bound independent of the fine mesh: count the grid points in that smaller cube. The finitely coarse cases use the center point.

Use the real local gauge (PX77) on a neighborhood of the pillar and its one-fine-cell collar. In this gauge the initial open bond bound is $Ca_{\mathrm{loc}}\eta/\xi_y$. The recurrence (PX45), with the same real curvature bound, therefore puts the hierarchical transports within $C_L(a_{\mathrm{loc}}+\epsilon)$ of one. Consequently the three by three matrix $m_y\psi = Q'_U(\chi_y\psi)$ equals $\beta_y I + O(C_L(a_{\mathrm{loc}}+\epsilon))$. For the fixed small class ceiling it is invertible with uniform bound.

Define a site lift on that pillar by $\chi_y m_y^{-1}\psi(y)$ in the local gauge and undo the gauge. At the node use its transformed frame as well. The resulting operator E_{sm} satisfies

$$Q'_U E_{\mathrm{sm}} = 1, \qquad \|E_{\mathrm{sm}}\psi\|_2 \le C_L\|\psi\|_{2,\mathrm{tops}}, \qquad \|L_U E_{\mathrm{sm}}\psi\|_2 \le C_L\|\Xi^{-2}\psi\|_{2,\mathrm{tops}}. \tag{PX84}$$

The first equality holds pillar by pillar, since the site averages have disjoint supports. For the last inequality, the covariant Laplacian of one bump is bounded by $C_L\xi_y^{-2}|\psi(y)|$. Expand it in the local gauge: the second cutoff differences, the connection times first differences, and the differentiated connection all have that bound by (PX77). Its mass term is the transported vector $a(y)\psi(y)$ on the same pillar. The volume is ξ_y^4, giving the last L^2 scale in (PX84). One fine collar of different bumps has bounded overlap, since each fine site has a fixed number of neighbors. For a bare pillar, take the single site bump; its Laplacian is bounded by $C\eta^{-2}|\psi|$, which is exactly the same scale. Thus no missing local gauge at the outer bare boundary is used.

For a real top vector ψ, use (PX84) with $\Xi^4\psi$ and the equation $L_U G'_U = 1$:

$$\begin{aligned} \|\Xi^2\psi\|_{2,\mathrm{tops}}^2 &= \langle (Q'_U)^{\mathsf{T}}\psi, E_{\mathrm{sm}}\Xi^4\psi\rangle \\ &= \langle G'_U(Q'_U)^{\mathsf{T}}\psi, L_U E_{\mathrm{sm}}\Xi^4\psi\rangle \\ &\le C_L\|G'_U(Q'_U)^{\mathsf{T}}\psi\|_2\|\Xi^2\psi\|_{2,\mathrm{tops}}. \end{aligned} \tag{PX85}$$

Divide by the last node norm, with the zero case immediate. This proves the operator lower bound

$$Q'_U G'^2_U (Q'_U)^{\mathsf{T}} \ge c_L\Xi^4. \tag{PX86}$$

The Ξ^4 test in (PX85) is essential. Testing only with $E_{\mathrm{sm}}\psi$ would give a weaker ratio of two norms and would not prove the stated weighted lower bound at different local scales.

For the upper bound and locality, set

$$\mathcal{F}_U = G'_U(Q'_U)^{\mathsf{T}}\Xi^{-2}, \qquad M_U = \Xi^{-2}Q'_U G'^2_U (Q'_U)^{\mathsf{T}}\Xi^{-2} = \mathcal{F}_U^{\mathsf{T}}\mathcal{F}_U. \tag{PX87}$$

The site injection $(Q'_U)^{\mathsf{T}}$ is an isometry between node L^2 and its transported constant pillar fields. Equation (PX67) therefore bounds the normalized cell entry of $\mathcal{F}_U$ by

$$C_L(\xi_y/\xi_z)e^{-md(y,z)}. \tag{PX88}$$

The same estimate holds for the complex $\mathcal{F}_E$, with the bounded transport condition numbers already proved. The cell summability calculation with $p=1$ gives a uniform exponential two-sided Schur norm of these operators, after increasing the fixed separation if needed. Their products therefore have the same type of norm. In particular $c_L I \le M_U \le C_L I$ at a real pair.

The complex perturbation is uniform as well. The finite matrix $\mathcal{F}_E$ is holomorphic on the actual physical first-order ball constructed above. Its operator norm is uniformly bounded there by (PX88). Apply one scalar Cauchy circle to the line through E. This gives $\|\mathcal{F}_E - \mathcal{F}_U\| \le C_L h$, independent of the number of coordinates. Hence

$$\|M_E - M_U\|_{2\to 2} \le C_L h, \qquad \|M_E^{-1}\|_{2\to 2} \le 2/c_L \tag{PX89}$$

on a fixed smaller ball. Here $M_E = \mathcal{F}_E^{\mathsf{T}}\mathcal{F}_E$ uses the analytic bilinear transpose. Its norm difference is bounded by $(\|\mathcal{F}_E\| + \|\mathcal{F}_U\|)\|\mathcal{F}_E - \mathcal{F}_U\|$. The transpose has the same Hilbert operator norm in the real weighted orthonormal bases. No Hermitian positivity of the complex Gram matrix is assumed.

An exponentially weighted Schur norm also controls its conjugation. For a bounded node function ψ that is one-Lipschitz in d, its cell entries change by the factor $e^{s(\psi(y)-\psi(z))}-1$. The inequality

$$|e^{st}-1| \le |s|\,|t|e^{|s||t|} \tag{PX90}$$

and a spare fraction of the exponential rate bound this perturbation by $C_L|s|$ in operator norm. Indeed $\sup_{r\ge 0} re^{-\delta r/2} < \infty$. Choose s small and use (PX89). The inverse of the conjugated M_E is uniformly bounded. The distance-weight argument used for (PX67) gives exponential entries of M_E^{-1}.

In the physical top pairing the weighted orthonormal basis at y is ξ_y^{-2} times the color basis. If B_{yz} are the entries of M_E^{-1} in those orthonormal coordinates, then $\|B_{yz}\| \le C_L e^{-m_2 d(y,z)}$. Undoing both Ξ^{-2} factors and the orthonormal bases gives the counting kernel

$$\left|(Q'_E G'^2_E (Q'_E)^{\mathsf{T}})^{-1}(y,z)\right| \le C_L \xi_y^{-4} e^{-m_2 d(y,z)}. \tag{PX91}$$

Explicitly the unnormalized entries of M_E^{-1} are $(\xi_z^2/\xi_y^2)B_{yz}$. Multiplying on the left and right by ξ_y^{-2}, ξ_z^{-2} leaves $\xi_y^{-4}B_{yz}$. This explains the asymmetric fourth power in a counting-kernel formula, despite symmetry in the physical weighted pairing.

Finally the actual site projector has the representation

$$R_E = 1 - \mathcal{F}_E M_E^{-1} \mathcal{F}_E^{\mathsf{T}}. \tag{PX92}$$

It agrees with its original Gram formula by (PX87). The smoothing part has an exponential cell L^2 Schur norm, and the matrix is holomorphic on the whole smaller physical ball. At a real field it is the orthogonal projector already identified. This completes the site Gram and projector part of the package. Fine vector coercivity, mixed derivative source norms and the current constraint terms are not consequences of (PX92).

22.1.14. *The Hessian read from the constrained minimum.* We next identify the three quadratic forms in the physical calculation. This is a finite-dimensional identity on any admitted chart; it precedes a uniform inverse estimate. Let U be the reference critical field, j_U its action current, Q the derivative of the actual straightened constraint, and ω its multiplier. Take a fixed reference slice, so that its linear condition is $RD_U^{\mathsf{T}} a = 0$.

Criticality on that slice already gives the full stationarity equation

$$j_U = Q^{\mathsf{T}}\omega. \tag{PX93}$$

To see this, first note that $Q\mathsf{S} = Q$, $\mathsf{S}a$ is in the slice, and $a - \mathsf{S}a = D_U G' R D_U^{\mathsf{T}} a$. For $a \in \ker Q$, criticality annihilates $\mathsf{S}a$; the Ward identity annihilates its remaining gauge part. Thus j_U annihilates $\ker Q$. The finite-dimensional annihilator identity gives (PX93). When the row space is the actual independent datum space, Q is onto and ω is unique. Equivalently it is unique whenever the row Gram inverse exists.

Because $Q\mathsf{H} = 1$, (PX93) implies $\mathsf{H}^{\mathsf{T}} j_U = \omega$. Define the actual Lagrangian Hessian

$$K_{\mathrm{L}} = \Delta_U^{\mathrm{W}} - \Delta^{(2)}[\omega]. \tag{PX94}$$

The quadratic constraint term here is the second derivative of the same straightened constraint as Q, with the same per-length pairing. It cannot be replaced by the quadratic part of a different row map.

For the invariant extension of the Wilson Hessian, the physical second variant is therefore

$$\mathsf{G}_1[j_U]^{-1} = \mathsf{S}^{\mathsf{T}} K_{\mathrm{L}} \mathsf{S} + D_U R D_U^{\mathsf{T}} + Q^{\mathsf{T}} a Q. \tag{PX95}$$

Let $W_1 = Q\mathsf{G}_1[j_U]Q^{\mathsf{T}}$ and $H_1 = \mathsf{G}_1[j_U]Q^{\mathsf{T}}W_1^{-1}$. The physical gauge identity gives $RD_U^{\mathsf{T}}H_1 = 0$. Consequently $\mathsf{S}H_1 = H_1$, and direct multiplication of (PX95) gives

$$H_1^{\mathsf{T}}K_{\mathrm{L}}H_1 = W_1^{-1} - a. \tag{PX96}$$

For example, $H_1^{\mathsf{T}}\mathsf{G}_1^{-1}H_1 = W_1^{-1}$, the gauge contribution is zero, and the row mass contributes a, since $QH_1 = 1$. This proves all signs in (PX96). It does not prove the existence of either inverse.

Here and in (PX97)–(PX100), B and b are per-length row coordinates. With $\ell = \mathrm{diag}(\xi_c)$, they are related to the raw data of (BGD1) by $b(B) = \ell^{-1}b_{\mathrm{raw}}(\ell B)$. The multiplier ω is dual to this per-length convention. Let the actual local critical branch be parametrized by a coarse variation B, with reduced datum $b(B)$. Its derivative h_B at zero is taken in the fixed reference slice. Differentiated stationarity gives the constrained linear equations for K_{L}, with row value $Db(0)\dot{B}$. They imply

$$h_B = H_1Db(0), \qquad D^2\mathcal{A}_{\mathrm{eff}}(0) = Db(0)^{\mathsf{T}}(W_1^{-1} - a)Db(0) + \langle\omega, D^2b(0)\rangle. \tag{PX97}$$

Here $\mathcal{A}_{\mathrm{eff}}$ is the action evaluated on that branch. For the first assertion, the differentiated equation tested against slice vectors in $\ker Q$ says $\mathsf{S}^{\mathsf{T}}K_{\mathrm{L}}h_B$ lies in the range of Q^{T}. Adding $Q^{\mathsf{T}}aQh_B$ and using (PX95) then solves the row problem by its defining inverse. For the second, differentiate the value function twice. The second derivative of the branch contracts with $j_U = Q^{\mathsf{T}}\omega$; differentiating its constraint replaces that contraction by $\omega D^2b - \omega D^2q[h_B, h_B]$. The latter term is exactly the subtraction in K_{L}.

22.1.15. *Which coarse coordinates remove the datum curvature.* The statement $b(B) = B$ has a specific coordinate meaning. In the present chart the actual coarse group field is

$$V(B)(c) = e^{i\xi_c B(c)}V_0(c). \tag{PX98}$$

The fine straightened row logarithm is relative to the same reference V_0. On an interior row the exact node identity then gives $b(B)(c) = B(c)$. It holds for nontrivial $V_0(c)$ as well. No assertion about the absolute logarithm of $V(B)(c)$ is involved.

An outward row instead has its prescribed factor at the finer outer endpoint. In group form its equation is

$$e^{i\xi_c b(B)(c)} = e^{i\xi_c B(c)}\,\mathrm{Ad}_{V_0(c)}\,v_z((\ell B)_{\mathrm{int}}). \tag{PX99}$$

Here v_z is the raw endpoint factor in (BGD1). It can have both linear and quadratic datum corrections. If only the highest active level is varied, the lower-level data B_{int} in this factor are fixed to zero. Then $v_z(0) = 1$ and the factor is constant. More generally the next-step support mask must be checked against every outward carrier. Deep interior rows give an immediate instance; a mask by itself does not change a logarithmic coordinate.

For contrast, if an absolute coordinate $W_c = \log(V_0(c))/i$ is used and the variable is still a relative increment D, its new absolute logarithm is $\log(e^{iD}e^{iW_c})/i$. Its second derivative in D generally does not vanish. Equation (PX97) then retains its datum-curvature term. This distinction is independent of the support mask.

Let ι inject the actual active next-step coarse bond variables into the row space. When the literal map (PX98)–(PX99) satisfies

$$Db(0)\iota = \iota, \qquad D^2b(0)[\iota\,\cdot, \iota\,\cdot] = 0, \tag{PX100}$$

its coarse action precision on those variables is $\iota^{\mathsf{T}}(W_1^{-1} - a)\iota$. Its coarse current is $\iota^{\mathsf{T}}H_1^{\mathsf{T}}j_U$. The restriction ι^{T} and the physical dual weights are part of the readout, even when notation suppresses them.

22.1.16. *The next curved fibre inserts a different multiplier.* At the next lattice, write raw bond increments as D, so that the group field is $e^{iD}V$. The actual plain constraint logarithm is

$$\mathfrak{q}_V(D) = \log\bigl(\mathcal{M}(e^{iD}V)\mathcal{M}(V)^{-1}\bigr)/i. \tag{PX101}$$

Its local pivot graph has derivative C_V, and the pivot injection h_V satisfies $D\mathfrak{q}_V(0)h_V = 1$. Twice differentiating the graph equation gives

$$D^2\Phi_V(0)[u,v] = -h_V D^2\mathfrak{q}_V(0)[C_V u, C_V v]. \tag{PX102}$$

This is the actual pivot graph, not a free extension chosen after the Hessian.

Let $\widehat{K}$ and $\widehat{J}$ be the coarse effective action Hessian and gradient in these raw counting coordinates. The chain rule on that graph is

$$\begin{aligned} D^2(\mathcal{A}_{\mathrm{eff}} \circ \Phi_V)(0)[u,v] = {} & \langle C_V u, \widehat{K} C_V v\rangle \\ & - \langle h_V^{\mathsf{T}}\widehat{J}, D^2\mathfrak{q}_V(0)[C_V u, C_V v]\rangle. \end{aligned} \tag{PX103}$$

Thus the next constraint multiplier is $h_V^{\mathsf{T}}\widehat{J}$. It is not $\widehat{J}$ on the new row space unless the notation already includes this transpose. Define $E_V^{(2)}[\widehat{J}]$ by the last contraction in (PX103). Let $\widehat{J}_{\mathrm{GF}}$ and $\widehat{K}_{\mathrm{GF}}$ be the gradient and Hessian at $D = 0$ of the actual function $D \mapsto \mathrm{GF}(e^{iD}V)$, in the same raw counting coordinates. The chain rule for the complete next free precision gives

$$C_V^{\mathsf{T}}\bigl(\widehat{K} + \widehat{K}_{\mathrm{GF}} - E_V^{(2)}[\widehat{J} + \widehat{J}_{\mathrm{GF}}]\bigr)C_V. \tag{PX104}$$

The physical step chooses $V = V_k^{\mathrm{bg}} = \mathcal{M}^k U$ in block axial gauge. The actual nonnegative GF_k vanishes there, so its full first derivative is zero and $\widehat{J}_{\mathrm{GF}} = 0$. This recovers the shorter physical formula with $E_V^{(2)}[\widehat{J}]$. At an arbitrary reference, both gauge derivatives in (PX104) must be retained. An absolute contour gauge and its relative version can have different derivatives away from the axial locus; the formula uses the actual gauge definition of the model being considered.

For a uniform next-lattice spacing ξ, the raw and physical conversions are

$$\widehat{K} = \xi^2 K_{\mathrm{phys,count}}, \qquad \widehat{J} = \xi^3 J_{\mathrm{phys}}. \tag{PX105}$$

Indeed $\xi^4 \sum B(KB) = \xi^2 \sum D(KD)$ when $D = \xi B$, and $\xi^4 \sum J_{\mathrm{phys}} B = \sum \xi^3 J_{\mathrm{phys}} D$. At nonuniform row lengths the corresponding diagonal length matrices must be kept on both sides; their commutation is not assumed. The coupling coefficients in the original action remain inside K and J.

Combining (PX96), (PX100) and (PX103) gives the physical diagram: fine Lagrangian Hessian, coarse value Hessian, then next-fibre Hessian. Each arrow has its own row map, its own dual current, and its own gauge term. The physical identity proves none of those three objects interchangeable.

22.1.17. *A flat Feynman comparison on all the layers.* There is a useful vector estimate which does not involve the Landau projection or an independent current. It uses the actual flat straightened averages. We prove it to specify exactly what is available before those two additional operations.

First take the full periodic fine lattice as the ambient graph. Let $a_\mu(x)$ be a fine vector one-form, with physical L^2 measure η^4. For a positive level-j row of length $\xi_j = n\eta$, $n = L^j$, its flat per-length average is

$$(Q_j^0 a)_\mu(y) = n^{-5} \sum_{u \in \mathcal{B}_n} \sum_{t=0}^{n-1} a_\mu(y + \eta u + \eta t e_\mu). \tag{PX107}$$

Here $\mathcal{B}_n$ is the centered set of n^4 fine block sites. The formula is the flat straightened map, not the plain path map. At $j = 0$ it is the identity. The coefficient of each fine bond is nonnegative,

the coefficients sum to one, and each is at most n^{-4}. The last fact follows because at most n pairs (u_μ, t) give a fixed longitudinal position, while the three other coordinates are fixed.

Its support lies in the union of the two endpoint blocks. Consequently

$$|Q_j^0 a(c)| \le C\xi_j^{-2}\|a_\mu\|_{2,\mathcal{D}_c}. \tag{PX108}$$

For example Cauchy gives $\sum_b w_b^2 \le n^{-4}$ from $\sum_b w_b = 1$ and $w_b \le n^{-4}$; converting counting L^2 to physical L^2 contributes η^{-2}. Thus the resulting coefficient is $\eta^{-2}n^{-2} = \xi_j^{-2}$. This bound is specific to this flat row formula.

For a fine bond b, put ξ_b equal to the larger of its endpoint pillar scales and define

$$\|a\|_{\mathrm{mass}}^2 = \sum_b \eta^4 \xi_b^{-2}|a(b)|^2, \qquad \mathcal{E}_{\mathrm{F}}^0(a) = \|\partial a\|_2^2 + \sum_{c\in\Gamma} c_c \xi_c^2 |Q^0 a(c)|^2, \quad 0 < c_* \le c_c \le c^*. \tag{PX109}$$

The component gradient contains all fine coordinate differences on the ambient torus. The second term is exactly the physical row pairing with $a(c) = c_c\xi_c^{-2}$. In particular it has neither an extra fine volume factor nor an unweighted counting transpose.

For each nonbare top y and each direction μ, one of the adjacent level-j rows has y as an endpoint and is a determining row. Indeed the excluded higher region is a union of aligned blocks of side at least $L\xi_j$. Along that coordinate line, the excluded block containing an exterior point has at least $L \ge 3$ level-j sites. Both neighboring sites cannot belong to the higher region. Choose one that does not. If the other endpoint is outside Ω_j, this is precisely an outward row, which is retained. At the highest level there is no higher region to exclude.

Let $m_{y,\mu}$ be the arithmetic mean of a_μ on the pillar enlarged by one fine cell. A rectangular carrier containing that enlarged pillar and the chosen row's two endpoint blocks has side length at most $C\xi_j$. For $j \ge 1$ it lies inside the ambient fine graph by the nesting gap. The ordinary box Poincare inequality, proved by the coordinate-path count used in the scalar block inequality, gives

$$|Q_j^0 a(c(y,\mu)) - m_{y,\mu}| \le C\xi_j^{-1}\|\partial a_\mu\|_{2,\mathcal{D}_{y,\mu}}. \tag{PX110}$$

To verify this without a pointwise trace estimate, subtract the carrier's arithmetic mean. Both functionals annihilate that constant in their difference and have L^2 norm at most $C\xi_j^{-2}$. Poincare then supplies one factor ξ_j.

A second use of Poincare on the enlarged pillar gives

$$\xi_j^{-2}\|a\|_{2,\mathcal{B}_y^+}^2 \le C\sum_\mu \|\partial a_\mu\|_{2,\mathcal{D}_{y,\mu}}^2 + C\xi_j^2 \sum_\mu |Q_j^0 a(c(y,\mu))|^2. \tag{PX111}$$

Every fine bond with a nonbare endpoint can be assigned to a pillar of its larger endpoint scale and is included in that pillar's enlargement. If both endpoints are bare, its level-zero row is the identity and controls it directly.

The overlap in the gradient sum is uniform. For a fixed level, all carrier centers are on its grid and their diameters are $C\xi_j$, so a fine point belongs to a fixed number of them. A carrier of an outward level-j row is inside Ω_{j-1}, and its endpoint blocks are outside Ω_{j+1}. Its one-fine-cell collar can enter only an adjacent level. Thus a fixed fine point is counted at only a fixed number of levels, not at every level of the sequence. Each chosen row has at most its two endpoints as selecting tops. Summing (PX111) therefore proves

$$\|a\|_{\mathrm{mass}}^2 \le C_L \mathcal{E}_{\mathrm{F}}^0(a). \tag{PX112}$$

The constants are independent of depth, fine mesh, volume and the shapes of the separated regions. Periodic carriers which wrap are lifted to their rectangles; the compatible torus has at least two top blocks, so the multiplicity of that lift is bounded.

There is also the upper row estimate

$$\sum_{c\in\Gamma}\xi_c^2|Q^0a(c)|^2\le C_L\|a\|_{\mathrm{mass}}^2. \tag{PX113}$$

Use (PX108) on each row carrier. All fine endpoint scales there are comparable to ξ_c, within the fixed adjacent scale factor. The preceding overlap count then applies. Equations (PX112) and (PX113) are estimates for the flat component Feynman form.

For a proper active region, extend the sequence by bare cells on the ambient torus and compress this full positive form by zero extension of its active bond coordinates. Its lower bound survives exactly: apply (PX112) to the extended vector. This is the Dirichlet compression of the component Laplacian. An inner-plaquette Maxwell Hessian has a different boundary form. Nor has the nonlocal subtraction $D_U(1-R)D_U^{\dagger}$ been included. Those comparisons are separate in the vector precision problem.

22.1.18. *The Wilson Hessian at small curvature.* The Wilson part admits a relative form estimate without assuming a matrix norm bound. We derive it directly from the plaquette product. Use $\langle X,Y\rangle_{\mathrm{col}}=\frac12\operatorname{Tr}(XY)$ on Hermitian traceless matrices. For a plaquette with oriented bonds $b_1,\dots,b_4$, put $P=U(b_1)\cdots U(b_4)$ and $T_{r-1}=U(b_1)\cdots U(b_{r-1})$. A left variation $e^{i\eta v(b)}U(b)$ inserts $X_r=\eta\operatorname{Ad}(T_{r-1})v(b_r)$. On reversed bonds the variation is $v(\bar b)=-\operatorname{Ad}(U(b)^{-1})v(b)$, as follows by inverting the group field.

The varied plaquette is $\prod_{r=1}^4 e^{iX_r}P$. At $P=1$, the trace of the mixed second derivative is the trace of the product of the two sums of insertions. Cyclicity of the trace cancels their commutators. For general real P, subtract that calculation and use the factor $P-1$ in every remaining term. Thus the plaquette action $1-\frac12\operatorname{Tr}P$ satisfies

$$\begin{aligned} D^2\mathcal{A}_p(U)[v,w]&=\left\langle\sum_r X_r,\sum_r Y_r\right\rangle_{\mathrm{col}}+\mathcal{R}_p(v,w),\\ |\mathcal{R}_p(v,w)|&\le C\|P-1\|\left(\sum_r|X_r|^2\right)^{1/2}\left(\sum_r|Y_r|^2\right)^{1/2}. \end{aligned} \tag{PX114}$$

There are only sixteen mixed insertion terms. Unitarity of the prefixes makes $\sum|X_r|^2=\eta^2\sum|v(b_r)|^2$. The same polynomial trace expression, with no real-part operation, is its holomorphic continuation.

Define the plaquette linearized curl by

$$(\mathfrak{d}_Uv)(p)=\eta^{-1}\sum_{r=1}^4\operatorname{Ad}(T_{r-1})v(b_r). \tag{PX115}$$

Then $\sum_r X_r=\eta^2\mathfrak{d}_Uv(p)$. If $\|P-1\|\le\epsilon\eta^2\xi_p^{-2}$, with ξ_p the largest corner scale, summing (PX114) over the actual plaquettes gives

$$\bigl|\langle v,\Delta_U^{\mathrm{W}}w\rangle-\langle\mathfrak{d}_Uv,\mathfrak{d}_Uw\rangle_2\bigr|\le C_L\epsilon\|v\|_{\mathrm{mass}}\|w\|_{\mathrm{mass}}. \tag{PX116}$$

Every bond is in at most six unoriented plaquettes. Each error has the factor $\eta^2\epsilon\xi_p^{-2}$ from $P-1$ and another η^2 from the two insertions. Their product is the physical η^4 measure times the natural inverse scale squared. There is no loss $\epsilon\eta^{-2}$ in this form estimate.

This curl is close to the ordinary forward covariant curl $d_Uv=\nabla_{U,\mu}v_\nu-\nabla_{U,\nu}v_\mu$. For example on a $\mu\nu$ plaquette the difference of their numerator sums is

$$(1-\operatorname{Ad}P)\bigl(\operatorname{Ad}U_\nu(x)v_\mu(x+\eta e_\nu)+v_\nu(x)\bigr). \tag{PX117}$$

Direct multiplication of the four prefixes proves this identity. It gives $\|\mathfrak{d}_U v - d_U v\|_2 \le C_L\epsilon\|v\|_{\mathrm{mass}}$, since $\eta \le \xi_p$. In particular

$$\langle v, \Delta_U^{\mathrm{W}} v\rangle \ge (1 - C_L\epsilon)\|d_U v\|_2^2 - C_L\epsilon\|v\|_{\mathrm{mass}}^2 \tag{PX118}$$

for a real field, after decreasing the fixed curvature ceiling. Positive bounded plaquette weights give the same statement with their lower and upper bounds. Signed layer profiles or additional current subtractions must be estimated separately.

On the full periodic graph, the covariant Hodge calculation is also explicit:

$$\left| \|d_U v\|_2^2 + \|D_U^{\mathsf{T}} v\|_2^2 - \sum_{\mu} \|\nabla_{U,\mu} v\|_2^2 \right| \le C_L\epsilon\|v\|_{\mathrm{mass}}^2. \tag{PX119}$$

Expand both squares. The diagonal terms agree because each covariant shift is unitary. For different directions the uncancelled terms are commutators of the two shifts, or of a shift and an inverse shift. Their coefficients are conjugates of $\operatorname{Ad} P - 1$. The two difference denominators contribute η^{-2}; the plaquette bound leaves $C\epsilon\xi^{-2}$. There are a fixed number of neighboring terms at each bond. Adjacent scales are comparable, so Cauchy proves (PX119). On a proper inner graph the summation by parts has additional boundary terms.

The full Feynman gauge uses $D_U D_U^{\mathsf{T}}$. The actual fine Landau form instead uses $D_U R D_U^{\mathsf{T}}$. Put $\mathcal{F}_U = G' Q'^{\mathsf{T}}$ and $M_U = Q' G'^2 Q'^{\mathsf{T}}$. The difference of the two gauge terms is the nonnegative real smoothing form

$$D_U(1 - R)D_U^{\mathsf{T}} = (D_U\mathcal{F}_U)M_U^{-1}(D_U\mathcal{F}_U)^{\mathsf{T}}. \tag{PX120}$$

The projected estimate uses the constrained local models and their parametrix, as explained after (PX158). This retains the subtraction together with the row mass. Its boundary model is the all-bare local block, where $Q_\square = Q'_\square = 1$ and $R_\square = 0$; no vanishing of the global R at an interface is asserted.

For the admitted complex perturbation E, the actual plaquette Taylor formula gives $\|P(U_E) - P(U)\| \le C_L h\eta^2\xi_p^{-2}$. Indeed the first order boundary sum of E is its covariant curl; the remaining terms contain either two increments or one increment times a real connection on a local gauge patch. The first-order core bounds control each by that scale. Also $\|d_{U_E} v - d_U v\|_2 \le C_L h\|v\|_{\mathrm{mass}}$. Subtracting (PX114) therefore proves

$$|\langle v, (\Delta_{U_E}^{\mathrm{W}} - \Delta_U^{\mathrm{W}})w\rangle| \le C_L h(\|d_U v\|_2 + \|v\|_{\mathrm{mass}})(\|d_U w\|_2 + \|w\|_{\mathrm{mass}}). \tag{PX121}$$

This uses a physical first-order perturbation radius, not an unscaled matrix norm for the fine precision. It is a genuine part of the complex comparison. The field-row comparison below supplies the corresponding row estimate. We must join it to the projected-current and boundary comparisons for the full precision.

22.1.19. *Compact dual bumps for the actual flat field rows.* We can also prove a uniform right inverse for the flat layered field rows. This is a different statement from a right inverse on site fields. The proof uses two longitudinal moments and a strictly triangular correction between adjacent levels.

Fix a top y of level $j \ge 1$, put $n = L^j = 2m + 1$, and work on its fine block $\eta\mathcal{B}_n + y$. Choose a field with just its μ-component nonzero there:

$$a_\mu(y + \eta u) = \psi\,\phi(u_\mu)\prod_{\nu\ne\mu}\chi(u_\nu). \tag{PX125}$$

It is zero outside the block and on bonds whose other endpoint is outside. For sufficiently large n, take $\chi(u)$ from a fixed even smooth profile of u/n, supported in $(-1/4, 1/4)$ and positive on a

smaller interval. Write

$$\beta = n^{-1}\sum_u \chi(u), \qquad \sigma = n^{-1}\sum_u (u/n)^2\chi(u). \tag{PX126}$$

Both have positive lower and finite upper bounds independent of n.

For a fine longitudinal position u in this block, the right coarse row in (PX107) counts that bond $u+m+1$ times. The left row counts it $m-u$ times. No other coarse row sees it. After the transverse sums, their two response functionals are

$$\ell_+(\phi) = \beta^3 n^{-2}\sum_u (u+m+1)\phi(u), \qquad \ell_-(\phi) = \beta^3 n^{-2}\sum_u (m-u)\phi(u). \tag{PX127}$$

Define

$$\begin{aligned} \phi_+(u) &= \beta^{-4}\chi(u) + \frac{m}{n}\frac{\beta^{-3}}{\sigma}\frac{u}{n}\chi(u), \\ \phi_-(u) &= \beta^{-4}\chi(u) - \frac{m+1}{n}\frac{\beta^{-3}}{\sigma}\frac{u}{n}\chi(u). \end{aligned} \tag{PX128}$$

Evenness gives $\sum u\chi(u) = 0$. Substitution into (PX127) now gives

$$(\ell_+(\phi_+), \ell_-(\phi_+)) = (1,0), \qquad (\ell_+(\phi_-), \ell_-(\phi_-)) = (0,1). \tag{PX129}$$

For example the sum of the two functionals is $\beta^3 n^{-1}\sum\phi$. For either profile this is one; the odd term in ϕ_+ sets the left functional to zero. The right one is then one. The calculation for ϕ_- is the same with the other moment. The signed profiles are uniformly bounded and their first differences are $O(n^{-1})$.

The finitely many odd values below this smooth-profile range need no limiting argument. Put the three transverse profiles at their center point and use longitudinal bonds with positions $-1, 0$. Both endpoints of these bonds are in the block for $n \geq 3$. Their two response columns form

$$n^{-5}\begin{pmatrix} m & m+1 \\ m+1 & m \end{pmatrix}. \tag{PX130}$$

Its determinant is $-n^{-9}$. Its inverse therefore gives the required two exact dual profiles, with a bound depending only on the fixed cutoff between this finite set and the smooth range. That is a universal constant, not a depth dependence.

Assign a determining row c to one of its level-j top endpoints y. An outward row has its interior endpoint as that top. Use ϕ_+ or ϕ_- according to the chosen endpoint. Its seed $E_c^0\psi$ is supported on fine bonds strictly inside that pillar and satisfies, on the full regular level-j grid,

$$Q_j^0 E_c^0 \psi = \delta_c \psi. \tag{PX131}$$

The transverse support lies in a single coarse transverse block, and (PX129) removes the only other longitudinal row. For a bare row take its single fine bond value. It has both endpoints in the bare layer.

Let $E^0\psi$ be the sum of these seeds at every level. There are at most eight assigned seeds at a top. Their supports belong to distinct pillars, and the collars in their gradient norm have bounded overlap. The amplitude and first-difference bounds just proved give

$$\|E^0\psi\|_{\mathrm{mass}}^2 + \|\partial E^0\psi\|_2^2 \leq C\sum_{c\in\Gamma} \xi_c^2 |\psi(c)|^2. \tag{PX132}$$

The factor on the right is forced by the dimensions: the supporting volume is ξ_c^4, and a first difference or mass factor is ξ_c^{-1}. For a bare seed the same calculation is $\eta^4\eta^{-2}|\psi|^2$.

22.1.20. *The correction between adjacent levels.* A seed of level j has zero value under every lower determining row. Indeed its fine support is in Ω_j, whereas a lower row's entire carrier is outside the next region of that lower level. For $i < j$, that next region contains Ω_j. It also has zero value under every row of level $i \geq j+2$: such a carrier is inside $\Omega_{i-1} \subset \Omega_{j+1}$, whereas the seed pillar is outside Ω_{j+1}. The same argument uses the two bare endpoints for $j = 0$. Thus only the next level can receive an unwanted value.

The full flat averages compose exactly:

$$Q_{j+1}^{0,\mathrm{full}} = Q_L^0 Q_j^{0,\mathrm{full}}. \tag{PX133}$$

In the defining sums, the transverse digits are $nr + u$, where r is in the centered L-range and u in the centered n-range. They tile the centered Ln-range. The longitudinal digits are $nt + s$, with $0 \leq t < L$, $0 \leq s < n$. They tile $0, \ldots, Ln-1$. The product normalization is $L^{-5}n^{-5}$. This proves (PX133), including the per-length normalization.

Let the row Hilbert norm be $\|\psi\|_{\mathrm{row}}^2 = \sum_c \xi_c^2 |\psi(c)|^2$. A regular L-step average has row coefficient sum one and column coefficient sum L^{-4}. The column identity can be checked directly: the two longitudinal counts add to L, while the three transverse coordinates select one block. The ordinary counting ℓ^2 operator norm is therefore at most L^{-2} by Schur. The row norm at the next level has one extra factor L in its square root. Hence the map between these row norms has norm at most L^{-1}.

Writing T for the next-level leakage, (PX131)–(PX133) give

$$Q^0 E^0 = 1 + T, \qquad T_{ij} = 0 \ (i \neq j+1), \qquad \|T\|_{\mathrm{row}\to\mathrm{row}} \leq L^{-1}. \tag{PX134}$$

Restriction to actual determining rows and injection from their subset of the regular grid cannot increase the norm. Different level blocks of T have different input and output levels, so the whole direct-sum norm is the supremum of those block norms. No sum over the number of levels is introduced.

We have proved the explicit flat row lift

$$E_{\mathrm{row}}^0 = E^0(1+T)^{-1}, \qquad Q^0 E_{\mathrm{row}}^0 = 1, \qquad \|E_{\mathrm{row}}^0 \psi\|_{\mathrm{mass}}^2 + \|\partial E_{\mathrm{row}}^0 \psi\|_2^2 \leq \frac{C}{(1-L^{-1})^2} \|\psi\|_{\mathrm{row}}^2. \tag{PX135}$$

At finite depth the inverse is the finite triangular series; its norm is bounded by the displayed geometric series. The same series converges in the row Hilbert space for an infinite compatible sequence. The assertion is a right inverse with an energy bound, not an estimate of a nonlinear background map.

At a real flat connection the same construction uses a flat unitary frame on each pillar. The response at the other coarse endpoint differs by its fixed unitary transport. The two scalar moments still annihilate the unwanted row; the desired row is normalized in its own color frame. Schur uses the same positive scalar coefficient majorants, so its constants are unchanged. All profile supports lie inside their pillars. This handles periodic flat holonomies without choosing a single globally small logarithm. Under this gauge transfer the gradient in the energy bound is ∇_U, or equivalently the ordinary component gradient in its flat parallel frame. An ordinary gradient in an arbitrary original gauge is not bounded by that statement. No analogous bare L^2 field-row estimate at a general nonflat field has been assumed.

22.1.21. *The field-row Gram bounds.* The last proof gives the flat instance of a general useful duality reduction. Let K be a real symmetric positive fine precision and $G = K^{-1}$. Let Ξ multiply each field row by its length. Suppose the actual row map has an extension E_{row} such that

$$QE_{\mathrm{row}} = 1, \qquad \langle E_{\mathrm{row}}\psi, K E_{\mathrm{row}}\psi \rangle \leq C \|\Xi^{-1}\psi\|_{2,\mathrm{rows}}^2. \tag{PX136}$$

The node pairing here has volume ξ_c^4; its last norm is just $\sum_c \xi_c^2|\psi(c)|^2$. Testing with $E_{\mathrm{row}}\Xi^2\psi$ gives

$$\|\Xi\psi\|_{2,\mathrm{rows}}^2 = \langle Q^{\mathsf{T}}\psi, E_{\mathrm{row}}\Xi^2\psi\rangle \le C^{1/2}\langle\psi, QGQ^{\mathsf{T}}\psi\rangle^{1/2}\|\Xi\psi\|_{2,\mathrm{rows}}. \tag{PX137}$$

Thus $QGQ^{\mathsf{T}} \ge C^{-1}\Xi^2$. No pointwise delta-function test in the fine field was used.

An upper bound needs the separate actual trace estimate

$$\|\Xi^{-1}Qa\|_{2,\mathrm{rows}}^2 \le C\langle a, Ka\rangle. \tag{PX138}$$

Indeed apply it to $a = G^{1/2}f$. It proves $\|\Xi^{-1}QG^{1/2}\| \le C^{1/2}$ and therefore $QGQ^{\mathsf{T}} \le C\Xi^2$. For the flat Feynman form, (PX113), (PX112) and (PX135) prove both premises. For the flat Landau form, (PX135) proves the upper energy cost in (PX136), since $0 \le R \le 1$; the needed fine coercivity is still a distinct step.

Normalize the row covariance as $N = \Xi^{-1}QGQ^{\mathsf{T}}\Xi^{-1}$. If its exponential cell Schur bound has also been proved, the exponentially weighted conjugation argument applies to N. Conjugating N by $e^{\delta d(\cdot,c_0)}$ changes it by $O(\delta)$ in the two-sided Schur norm: use the exponential weight already present in its kernel bound and $|e^{\delta t} - 1| \le \delta|t|e^{\delta|t|}$. For fixed sufficiently small δ, the real lower bound keeps the conjugated inverse bounded. Undoing the conjugation gives exponential entries of N^{-1}. In physical weighted orthonormal row coordinates, undoing normalization gives the counting kernel estimate

$$|(QGQ^{\mathsf{T}})^{-1}(c,d)| \le C\,\xi_c^{-3}\xi_d\,e^{-m_3 d(c,d)}. \tag{PX139}$$

To check the powers, an orthonormal entry of N^{-1} becomes (ξ_d^2/ξ_c^2) times that entry on unnormalized values. The two exterior Ξ^{-1} factors leave $\xi_c^{-3}\xi_d$. This differs from the site-Gram counting weight ξ_y^{-4}: the site Gram contains G'^2, whereas this row covariance contains G.

Finally a nonflat row extension is reduced to a definite error estimate. Starting from a specified flat or locally flat extension E_0, set $D_Q = QE_0 - 1$. If the actual primitive comparison proves

$$\|D_Q\|_{\mathrm{row}\to\mathrm{row}} \le \vartheta < 1, \tag{PX140}$$

then $E_{\mathrm{row}} = E_0(1 + D_Q)^{-1}$ is an actual right inverse, with its energy cost multiplied by $(1-\vartheta)^{-2}$, provided that cost for E_0 has been proved in the actual precision. We establish the row error and row trace estimates for the admitted domain below.

22.1.22. *The actual straightened row is a finite analytic recursion.* We now obtain the needed row coefficient bounds from the actual formula, rather than from a proposed inverse. Use raw one-step link logarithms on a lattice whose spacing has been set to one. Let $T_{yx}(W)$ be the internal mean transport of the block chart, with its specified coordinate-order path multiset. For a reference W and a raw increment a, put

$$\begin{aligned} u_W(a)(y) &= \exp\left[L^{-4}\sum_{x\in B_L(y)} \log\big(T_{yx}(e^{ia}W)T_{yx}(W)^{-1}\big)\right], \\ F_W(a)(c) &= \frac{1}{i}\log\big[u_W(a)(c_-)^{-1}\mathcal{M}(e^{ia}W)(c)u_W(a)(c_+)\mathcal{M}(W)(c)^{-1}\big]. \end{aligned} \tag{PX141}$$

All logarithms are the regular branch. The first line is an exponential of averaged relative logarithms; it is not another unspecified mean. The internal transports themselves use the actual Federbush mean defined in the block chart. This distinction fixes both levels of averaging.

For $\|\log W/i\|_\infty$ and $\|a\|_\infty$ below c/L, these are holomorphic finite functions. Every path has at most CL bonds. Products of their exponentials remain in a fixed logarithmic neighborhood; the mean contraction proved in the block chart is uniform in the number of paths. After a fixed decrease of c, the expression inside the second logarithm remains in that same neighborhood. Thus $\|F_W(a)\|_\infty \le C$ on this joint raw ball, with universal C.

Differentiating at $W = 1, a = 0$ cancels the two internal path sums from the plain average against the two factors u. The remaining sum is exactly the straight segment average S_L, with raw row sum L and raw column sum L^{-3}. It is essential to use both factors in (PX141); the plain average has a different derivative.

Let $A_W(a) = D_a F_W(a)$. Two scalar Cauchy circles, in arbitrary sup-norm directions of the reference logarithm and of a, give

$$\|A_W(a)\|_{\mathrm{row}} \le L(1 + C_L\delta), \quad \|A_W(a)\|_{\mathrm{col}} \le L^{-3}(1 + C_L\delta),$$
$$\sup_b \sum_{c,b'} \|D_b D_{b'} F_W(a)(c)\| \le C_L, \qquad \delta = \|\log W/i\|_\infty + \|a\|_\infty. \tag{PX142}$$

Here and below the norm of a color entry is its finite matrix norm. We explain why the derivative estimate is independent of a field-coordinate count. The two circles bound a second directional derivative by CL^2. For a fixed output color row, its operator norm from bounded input colors controls the sum of its entry norms, up to the fixed color dimension: sum the dual norms of its three scalar rows. A fixed input bond affects only the coarse rows at the endpoints of the one or two blocks containing it. Their number is fixed. Apply the same argument with one input bond fixed to obtain the last line. Dividing the row and column errors by their flat values costs only powers of L. For definiteness, all one-step constants in (PX142) may be enlarged to CL^{12}. This deliberately loose polynomial bound is before any number of levels is introduced.

The multilevel recursion is literal. Write $W_r = \mathcal{M}^r(W_0)$, $a_0 = \eta A$, and

$$a_{r+1} = F_{W_r}(a_r), \qquad v_{r+1} = v_r|_{\mathrm{next}}\, u_{W_r}(a_r). \tag{PX143}$$

Equivariance of the plain mean gives, by induction,

$$\mathcal{M}^r(e^{i\eta A} W_0) = (e^{i a_r} W_r)^{v_r}. \tag{PX144}$$

Indeed apply M to the previous factorization, restrict its gauge to the next nodes, and then use (PX141). The gauge factors multiply in the order in (PX143). They do not alter the recursion for a_r.

22.1.23. *Summability of the raw radii gives a depth-uniform row bound.* For the row estimates from this point on, ϵ is a common ceiling for the real curvature and the local real-gauge bounds. If the curvature and local-gauge ceilings are supplied separately, take their maximum for ϵ. Existence of those local gauges remains the primitive regular-pair hypothesis. No bound for a_{loc} is inferred from curvature alone.

Fix a row of length $\xi = n\eta$, $n = L^j$. On its determining carrier, the regular-pair gauge clause gives a real local potential and its first difference at scale ξ, with at most the fixed adjacent-layer factor at a seam. In the local gauge write $\alpha_r = \|\log W_r/i\|_\infty$ and $\omega_r = \|\partial W_r - 1\|_\infty$. The one-step product, plaquette-sliding and Taylor estimates give

$$\alpha_{r+1} \le L\alpha_r + C_L(\alpha_r^2 + \omega_r), \qquad \omega_{r+1} \le L^2(1 + C_L\alpha_r)\omega_r + C_L\omega_r^2.$$

For the first estimate a straight product contributes $L\alpha_r$ and its quadratic error, while each closed contour comparison contributes $C_L\omega_r$. For the second, put the one-step patch in its complex comb gauge. Its non-tree bonds have size $C_L\omega_r$; the linear flux through a coarse plaquette is L^2 times the fine flux, and the Taylor error is quadratic. Conjugating back costs $1 + C_L\alpha_r$. These are finite one-step estimates, with L fixed.

At the bottom of the carrier, $\alpha_0 \le C_L(\epsilon + h)/n$ and $\omega_0 \le C_L(\epsilon + h)/n^2$. Divide the two recurrences by t_r and t_r^2, respectively. Since $\sum_{r<j} t_r \le (L-1)^{-1}$, a first-exit argument bounds their normalized values at a fixed multiple of the initial ceiling. After the fixed weak-field shrinkage this proves

$$\|\log W_r/i\|_\infty \le C_L(\epsilon + h)t_r, \qquad t_r = L^r/n, \qquad r < j. \tag{PX145}$$

These estimates are used only on the dependence cone of that row. They also follow by extending its local small potential with a cutoff on a fixed larger cube and then using locality. The cutoff keeps its first difference at the same scale. The torus room required for nonflat local classes permits this cube; the flat periodic formulas need no such extension. No gauge bound growing with M_1 is used.

Allow an arbitrary fine increment with $\xi\|A\|_\infty \le r$. It need not have small fine derivatives. On the recursion ball, (PX142) and $F_W(0) = 0$ give

$$\|a_{r_0}\|_\infty \le 2r\, t_{r_0} \quad (0 \le r_0 \le j) \tag{PX146}$$

when $\epsilon + h + r$ is below a fixed threshold depending on L. To avoid confusing the index with the radius, write $p_s = \|a_s\|_\infty / t_s$. As long as $p_s \le 2r$, Taylor's formula and (PX142) imply

$$p_{s+1} \le p_s[1 + C_L t_s(\epsilon + h + p_s)]. \tag{PX147}$$

The sum of t_s is below $(L-1)^{-1}$. Choose the fixed ceiling so that $C_L(\epsilon + h + 2r)/(L-1) < \log 2$. Multiplying the factors proves (PX146) by first exit. The raw radii also stay strictly inside the c/L one-step balls. Finally $\|u_{W_s}(a_s) - 1\| \le C_L\|a_s\|_\infty$; their condition numbers have product at most $\exp(C_L \sum_s \|a_s\|_\infty)$. A further fixed decrease makes this at most two, supplying the room for the complex covariance in (PX144).

Put $A_s = D_a F_{W_s}(a_s)$ and $P_{b,a} = A_{b-1} \cdots A_a$. The same summable radii in (PX142) give

$$\|P_{b,a}\|_{\mathrm{row}} \le CL^{b-a}, \qquad \|P_{b,a}\|_{\mathrm{col}} \le CL^{-3(b-a)}. \tag{PX148}$$

The constant is bounded by the product of $1 + C_L(\|\log W_s/i\|_\infty + \|a_s\|_\infty)$. It is independent of a, b, j. Multiplying only uniform bounds larger than one at each level would miss this point.

The per-length row logarithm is $q_j(A) = a_j/\xi$. Its derivative with respect to the physical fine coordinates is $(\eta/\xi)P_{j,0} = n^{-1}P_{j,0}$. Consequently

$$\|Dq_j(A)\|_{\mathrm{row}} \le C, \qquad \|Dq_j(A)\|_{\mathrm{col}} \le Cn^{-4}, \qquad |D_b q_j(A)(c)| \le C(\eta/\xi)^4. \tag{PX149}$$

These are counting-coordinate derivatives. A functional derivative uses the additional η^{-4}. Telescoping the products against their flat factors also gives row error $C_L(\epsilon + h + r)$ and column error $C_L(\epsilon + h + r)n^{-4}$, in the common local unitary frames.

An actual row carrier has only its two endpoint blocks. A fixed fine bond in a layer can occur in carriers of only adjacent levels: the outer block is inside the previous region and both endpoint blocks are outside the next. At each of those levels the number of rows is fixed. Thus (PX149) implies for the full layered field map

$$\|\Xi^{-1}Qa\|_{2,\mathrm{rows}} \le C_L\|a\|_{\mathrm{mass}}. \tag{PX150}$$

For a single row, its coefficient sum is bounded and each entry is at most $C(\eta/\xi_c)^4$. Cauchy therefore gives the analogue of (PX108); the carrier overlap argument of (PX113) then proves (PX150). This estimate has been obtained from the actual finite recursion, not by treating an infinity norm bound alone as an L^2 bound.

22.1.24. *The two-legged constraint coefficient has a uniform moment.* The same computation gives the coefficient estimate needed for the quadratic current term. Let $n_s = L^s$ and $m_s = L^{j-s-1}$. Twice differentiating the recursion gives

$$D^2_{a_0} a_j = \sum_{s<j} P_{j,s+1} D^2 F_{W_s}(a_s)[P_{s,0}\,\cdot, P_{s,0}\,\cdot]. \tag{PX151}$$

Fix the first fine bond b and sum absolutely over the second. For the term indexed by s, the largest entry of the outer product is at most Cm_s^{-3}, by its column sum. The sum over the first

inner leg is at most Cn_s^{-3}. The second inner row sum is at most Cn_s. The remaining sum over the one-step Hessian, with its first input fixed, is bounded by (PX142). Thus

$$\sum_{b'} \|D_{a_0(b)} D_{a_0(b')} a_j(c)\| \le C_L \sum_{s<j} m_s^{-3} n_s^{-2}. \tag{PX152}$$

This step uses a column sum on the fixed leg and a row sum on the summed leg. Using two row sums would lose the fine density.

Physical input derivatives contribute η^2; the output per-length normalization contributes ξ^{-1}. Since $m_s^{-3} n_s^{-2} = L^3 n_s / n^3$ and $\sum_{s<j} n_s \le n/(L-1)$, we obtain

$$\sup_b \sum_{b'} \|D_b D_{b'} q_j(A)(c)\| \le C_L\, \xi\, (\eta/\xi)^4. \tag{PX153}$$

There is no number-of-levels factor. The geometric sum is dominated by its final scales. Symmetry of the second derivative gives the corresponding other-leg sum.

Let $\omega(c)$ be an arbitrary row current with $|\omega(c)| \le \gamma_\omega \xi_c^{-3}$. Pair the Hessian at zero with the physical row volume. Equation (PX153), Schur and the same bounded carrier overlap give

$$\left| \sum_c \xi_c^4 \langle \omega(c), D^2 q_c(0)[u,v] \rangle_{\mathrm{col}} \right| \le C_L \gamma_\omega \|u\|_{\mathrm{mass}} \|v\|_{\mathrm{mass}}. \tag{PX154}$$

For clarity, the row prefactor $\xi_c^4 |\omega(c)|$ is bounded by $\gamma_\omega \xi_c$. Multiplying it by the leg moment $C_L \xi_c (\eta/\xi_c)^4$ leaves $C_L \gamma_\omega \eta^4 \xi_c^{-2}$ on each fine leg. Adjacent scales replace ξ_c by the fine bond scale at only a fixed cost.

The holomorphic ball was used to extract these coefficients. The quadratic form in (PX154) then extends to every finite fine vector by its coefficient formula. It does not require a Gaussian sample to lie in that holomorphic ball. For the second variant one must still bound the actual induced current $\mathsf{H}[J]^{\mathsf{T}} J$ in this row norm, and bound S in the weighted fine energy spaces. Those are separate operator readouts.

22.1.25. *The nonflat field-row lift follows from the row comparison.* The coefficient estimates allow one to complete (PX140) at a regular pair. Keep the real local gauges of its center. Construct the profiles (PX125) in each pillar in such a gauge, and undo the gauge. For a seed assigned to the far endpoint of a row, normalize its color by the fixed unitary transport of that row. Denote this operator again by E^0. It is fixed at the real center while the complex increment varies.

Two local real gauges used on the same pillar differ there by a nearly constant unitary matrix. To prove this, write their connection equation on each fine bond and telescope along a coordinate path of length $C\xi_j$. Both gauge potentials have size at most $C_L \epsilon / \xi_j$. The transition therefore differs from its root value by at most $C_L \epsilon$. The first-order class on a fixed larger cube supplies both gauges. Only a path of size $C_L \xi_j$ is used, not the full $M_1 \xi_j$ cube.

It follows from (PX149) and this transition bound that the same-level seed response is the identity with an error of row and column norm $C_L(\epsilon + h)$ in the row energy norm. Only a fixed number of same-level rows meet a seed pillar. The next-level response has a leading part with the positive scalar majorant of Q_L^0, multiplied by fixed unitary color matrices, and an error. There are at most CL^4 lower-level seeds in one such carrier; a seed affects at most a fixed number of next-level rows. An error entry of size $C_L(\epsilon + h)$ therefore has row sum $C_L L^5 (\epsilon + h)$ and column sum $C_L L (\epsilon + h)$ after the row-energy normalization. Schur bounds this error by $C_L L^3 (\epsilon + h)$. Absorb this fixed power into C_L.

All other level blocks are zero by the carrier argument preceding (PX133). The leading next-level part still has norm at most L^{-1}; Schur uses only the scalar majorants, not a compatible

global choice of flat color frames. We have proved

$$\|Q_{\mathbf{U}}E^0 - 1\|_{\mathrm{row}\to\mathrm{row}} \le L^{-1} + C_L(\epsilon + h) =: \vartheta < 1. \tag{PX155}$$

The ceiling needed here depends on L and not on depth, volume or M_1. The actual right inverse is

$$E_{\mathrm{row}} = E^0(Q_{\mathbf{U}}E^0)^{-1}. \tag{PX156}$$

Both its squared mass norm and its covariant gradient energy are bounded by $C_L(1-\vartheta)^{-2}\|\psi\|^2_{\mathrm{row}}$. For the gradient of a seed, expand in its local gauge: the profile derivative is of size $C\xi_j^{-1}|\psi|$ and the connection term has the same bound. Supports and their fine collars have bounded overlap. The small complex perturbation adds at most $C_L h$ times the mass norm. This proves the stated energy bound, including bare seeds.

There is a corresponding actual Feynman Poincare inequality. Repeat (PX110) in a common local real gauge for a pillar and its chosen row carriers. The row comparison after (PX149) has both row and column error bounds. Schur therefore bounds its local L^2 functional norm by $C_L\epsilon\xi_j^{-2}$. Ordinary gradients and covariant gradients differ there by $C_L\epsilon\xi_j^{-1}$ times the field. Consequently (PX111), summed over the carriers, gives

$$\|a\|^2_{\mathrm{mass}} \le C_L\big(\|\nabla_U a\|_2^2 + \|a^{1/2}Q_U a\|^2_{2,\mathrm{rows}}\big) + C_L\epsilon^2\|a\|^2_{\mathrm{mass}}. \tag{PX157}$$

The repeated letter a in $a^{1/2}$ is the fixed positive row weight, whereas the arguments are fine fields. Equivalently that term is $\sum_c c_c\xi_c^2|Q_U a(c)|^2$. Absorb the last term by the fixed weak-field ceiling. Only carrier differences in the interior are used; the bare boundary fields are controlled by their identity rows. Thus the same argument applies to an active component-gradient form containing all those interior differences.

The full complex Feynman form is stable on a fixed physical ball. The actual row comparison gives

$$\|a^{1/2}(Q_{\mathbf{U}} - Q_U)v\|_2 + \|(\nabla_{\mathbf{U}} - \nabla_U)v\|_2 \le C_L h\|v\|_{\mathrm{mass}}. \tag{PX158}$$

For the first term use the row and column differences, the physical row volumes and the bounded carrier overlap. For the second use $\mathrm{Ad}(e^{i\eta E}) - 1 = O(\eta E)$. Expanding the two bilinear squares and using (PX157) gives a relative form perturbation $C_L h$. Its inverse exists by the normalized Neumann series. This completes the component Feynman estimate. The full projected estimate can be completed by the same local construction. For the Landau precision use the constrained one- and two-scale models of (c), rather than subtracting (PX120) from a Feynman lower bound. The decomposition $A = A_2 + H_j B - \partial H_j'\vartheta$ splits the flat form orthogonally with its row mass. The three terms in (d) comparing $\partial(1-R)\partial^{\mathsf{T}}$ with its local model cost CM_1^{-1} for commutators and Ce^{-cM_1} for changes of domain and separated tails. Their parametrix therefore has a convergent error series. Near the outer boundary use the massive all-bare local blocks, whose flat inverses satisfy $a_0 \le \eta^2 G_\square^{-1} \le a_0 + 16$ on arbitrary bond sets. The small local perturbation is treated in the same way; no periodic inner-Hodge identity at the boundary is required.

Include the current corrections in their defining order:

$$\mathsf{G} = \mathsf{G}_0 \sum_{n\ge 0} (\Delta'_\pi[\mathbf{J}]\mathsf{G}_0)^n,$$

$$\mathsf{G}_1 = \mathsf{G} \sum_{n\ge 0} (\mathsf{S}^{\mathsf{T}}\Delta^{(2)}[\mathsf{H}^{\mathsf{T}}\mathbf{J}]\mathsf{S}\mathsf{G})^n.$$

These are the expansions of [B85c, (3.130)–(3.138)], with the assigned row and site maps. Transfer each differential factor to the neighboring inverse on the left or right; no derivative of the independent current is needed. The row and column estimates and (PX153) give the constraint term its scale factors. Expand H inside the induced multiplier: its local part stays in the local

inverse and its remaining walks give the small tree vertices of (g). The local coefficients are small with the regularity and current radii, so the same construction gives the projected inverse and its potential and gradient bounds. We must check the species weights, the derivative allocation and the adjacent-level endpoint equations when identifying these products with the actual blocks. The covariant Laplacian member excluded in [B85c, Theorem 3.12] is not asserted here.

Differentiating the physical response. We give the calculations behind the response and current readouts in the preceding construction. Every matrix below is the derivative of the assigned nonlinear equation. The constraint-curvature term and the outward datum factors remain present throughout.

The actual equations and their differentiated right-hand sides. Use the real reference field U_0 and write the varied fine field as $e^{i\eta A}U_0$. For the multilevel determining set $\mathbf{B}$, the straightened average is denoted by $\mathcal{Q}^{\mathrm{st}}(A)$ and its derivative by K_A. This is the nonlinear straightening of the group average, rather than the straight linear average. In (BGR1)–(BGR9), $\mathcal{Q}^{\mathrm{st}}$, B and b use raw row logarithms, as in (BGD1); the fine field A still has the physical normalization $e^{i\eta A}U_0$. The row multiplier is dual to this raw convention. With the reduced datum $\mathsf{b}(B)$, the physical critical pair satisfies

$$\begin{aligned} F'(A) &= K_A^{\mathsf{T}}\omega, & \mathcal{Q}^{\mathrm{st}}(A) &= \mathsf{b}(B), \\ RD^{\mathsf{T}}A &= 0, & F(A) &= A_{\Omega_0}(e^{i\eta A}U_0). \end{aligned} \tag{BGR1}$$

Here R and D are formed at the reference field. The first equation is the physical stationarity identity proved above by testing the remaining gauge directions; its restriction to the slice suffices for the following linear algebra. When the reference is changed with the datum, differentiating the slice also contributes $(\dot{R}D^{\mathsf{T}} + R\dot{D}^{\mathsf{T}})A$. We keep the reference fixed in (BGR1). A calculation that moves the reference and then omits this contribution does not differentiate (BGR1).

The datum forcing is $Db(B)\dot{B}$, where b is the exact reduced datum in (BGD1). Its diagonal logarithmic differentials and its finite two-step inverse were computed in (BGD4). The outward block factors therefore enter this forcing with their actual interior-row derivatives.

Let $\mathcal{V}$ be the fixed linear slice, after the residual node gauge has been fixed. Its fine pairing and the datum pairing are the actual cell-weighted pairings. Write $q = K_A|_{\mathcal{V}}$ and let $q^\sharp$ be its transpose for these pairings. The constrained second variation is

$$\mathcal{K} = F''(A) - \sum_a \mu_a \omega_a (\mathcal{Q}_a^{\mathrm{st}})''(A) \quad \text{on } \mathcal{V}. \tag{BGR4}$$

The weights μ_a in this coordinate expression are the datum cell weights. Equivalently the second term is the Hessian of the scalar pairing $\langle \omega, \mathcal{Q}^{\mathrm{st}}(A)\rangle$. This latter description fixes the convention if a different coordinate basis is used. The curvature term in (BGR4) cannot be replaced by the Hessian of F alone. It is the usual second fundamental form of the constraint fibre.

Differentiating (BGR1) at a fixed reference gives the exact system

$$\mathcal{K}\dot{A} - q^\sharp\dot{\omega} = 0, \qquad q\dot{A} = D\mathsf{b}(B)\dot{B}. \tag{BGR5}$$

More generally, for explicit parameters p in the action or constraints, with the slice fixed, put $\mathcal{C}(A,p) = 0$ and $\mathcal{E}(A,\omega,p) = F_p'(A) - \mathcal{C}_A^\sharp\omega$. The right-hand sides are then

$$f = -D_p\mathcal{E}\,\dot{p}, \qquad b = -D_p\mathcal{C}\,\dot{p}, \qquad \mathcal{K}\dot{A} - q^\sharp\dot{\omega} = f, \quad q\dot{A} = b. \tag{BGR6}$$

Every differentiation in D_p is made before substituting the solution $A(p), \omega(p)$. This distinction prevents an implicit extra response term on the right-hand side.

An independent-current extension is an additional specified operator. Formula (BGR4) identifies the physical constrained Hessian. It does not identify every separately defined $\Delta_\pi'[J]$ with that Hessian. Likewise, an absolute contour gauge functional and its relative version have the

same quadratic gauge block at an axial field, but generally have different blocks away from that field. The complete gauge and current normal form must be matched before an independently named propagator can be inserted into (BGR5).

A complete block inverse, including the multiplier. The following identity recovers the multiplier along with the field. It is finite-dimensional and exact. It also applies to bounded operators when the displayed inverses and compositions exist on the stated spaces. Every transpose in the complex calculation is bilinear, with no conjugation.

Proposition 22.1 (Response from a primitive constrained inverse)**.** *Suppose $q : \mathcal{V} \to \mathcal{D}$ is onto, $E : \mathcal{D} \to \mathcal{V}$ satisfies $qE = I$, and $P : \mathcal{X} \to \mathcal{V}$ is an isomorphism onto* $\ker q$*. Assume*

$$H = P^{\sharp}\mathcal{K}P \quad \textit{is invertible}, \qquad L = PH^{-1}P^{\sharp}, \qquad T = E - L\mathcal{K}E. \tag{BGR7}$$

Then the unique solution of (BGR6) is

$$\begin{aligned} \dot{A} &= Lf + Tb, \\ \dot{\omega} &= E^{\sharp}(\mathcal{K}L - I)f + E^{\sharp}\mathcal{K}Tb. \end{aligned} \tag{BGR8}$$

If $\mathcal{K}^{\sharp} = \mathcal{K}$, this simplifies to

$$\dot{\omega} = -T^{\sharp}f + Mb, \qquad M = E^{\sharp}\mathcal{K}E - E^{\sharp}\mathcal{K}L\mathcal{K}E, \qquad M^{\sharp} = M. \tag{BGR9}$$

Proof. Write $\dot{A} = Eb + Pv$. Applying $P^{\sharp}$ to the first equation of (BGR6), and using $qP = 0$, gives $Hv = P^{\sharp}(f - \mathcal{K}Eb)$. This determines v and gives the first formula of (BGR8). The residual $\mathcal{K}\dot{A} - f$ annihilates $\ker q$. It therefore belongs to the range of $q^{\sharp}$. Since $E^{\sharp}q^{\sharp} = I$, its unique multiplier is the second formula of (BGR8). Conversely these formulas satisfy both equations, so there is no lost multiplier equation. For a homogeneous solution, the same argument gives $v = 0$, then $\dot{A} = 0$ and $\dot{\omega} = 0$.

For the symmetric assertion, $H^{\sharp} = H$ and $L^{\sharp} = L$. Thus $T^{\sharp} = E^{\sharp} - E^{\sharp}\mathcal{K}L$, which gives (BGR9). Finally $E^{\sharp}\mathcal{K}T$ equals the displayed expression for M; both its terms are symmetric. Positivity of the full $\mathcal{K}$ was not used. Invertibility was needed on the actual constrained slice. □

The inverse L acts on fine forces; T lifts the constraint datum after the energetic correction; M is the multiplier response. Bounding L alone leaves the products $L\mathcal{K}E$ and $E^{\sharp}\mathcal{K}T$ to be controlled. The size and locality of E also matter. A right inverse with a norm growing in the number of layers would not prove a uniform response theorem.

Before applying per-length row estimates to these formulas, convert the datum and its dual together. Put $\ell = \operatorname{diag}(\xi_c)$. With the cell pairings unchanged,

$$\begin{gathered} q_{\mathrm{len}} = \ell^{-1}q_{\mathrm{raw}}, \qquad \omega_{\mathrm{len}} = \ell\omega_{\mathrm{raw}}, \qquad E_{\mathrm{len}} = E_{\mathrm{raw}}\ell, \\ \mathsf{b}_{\mathrm{len}}(B_{\mathrm{len}}) = \ell^{-1}\mathsf{b}_{\mathrm{raw}}(\ell B_{\mathrm{len}}), \qquad D\mathsf{b}_{\mathrm{len}}(B_{\mathrm{len}})\dot{B}_{\mathrm{len}} = \ell^{-1}D\mathsf{b}_{\mathrm{raw}}(\ell B_{\mathrm{len}})\ell\dot{B}_{\mathrm{len}}. \end{gathered}$$

These conversions preserve both $qE = I$ and the scalar Lagrangian, so they preserve $\mathcal{K}$. At the identity in unit-spacing one-step coordinates, division of the raw coarse logarithm by L changes its pivot derivative with respect to a raw fine-bond angle from L^{-3} to L^{-4}; the right inverse and multiplier acquire the reciprocal factors above. For fine spacing η and a physical field A, the raw derivative instead contains ηL^{-3}, and division by the coarse length $L\eta$ again gives L^{-4}. The cell weights and amplitudes below are inserted after this coordinate choice.

Cell weights and a kernel algebra. We describe the norm in which the preceding formulas can be estimated without a bare-mesh counting loss. Let each component i have a positive cell weight μ_i and an amplitude $s_i > 0$. A vector is measured by $\|v\|_s = \sup_i |v_i|/s_i$. A kernel is written as

$$(Av)_i = \sum_j \mu_j A_{ij} v_j, \qquad \widehat{A}_{ij} = s_i^{-1} A_{ij} s_j. \tag{BGR10}$$

For different input and output species their respective amplitudes and cell weights are used. The identity kernel is δ_{ij}/μ_j. Thus even the identity is not represented by δ_{ij} when it is used with a weighted summation convention.

For a metric d common to the component locations define

$$\|A\|_m = \max\Big\{ \sup_i \sum_j \mu_j |\widehat{A}_{ij}| e^{md(i,j)}, \sup_j \sum_i \mu_i |\widehat{A}_{ij}| e^{md(i,j)} \Big\}. \tag{BGR11}$$

Matrix norms in a colour or local finite-dimensional fibre are operator norms. The two sums use the correct weights for their two species. The triangle inequality for d and absolute convergence give

$$\|AB\|_m \le \|A\|_m \|B\|_m. \tag{BGR12}$$

Indeed, insert the intermediate sum with weight μ_k, use $d(i,j) \le d(i,k) + d(k,j)$ and cancel the intermediate amplitudes. For the row bound take the supremum of the inner B row sum; for the column bound use the corresponding A column sum. These are distinct calculations and neither counts the number of fine bonds.

The first row sum bounds A on the normalized supremum norm. The two sums together bound it on the normalized, cell-weighted ℓ^2 space by the Schur estimate. To see the latter directly, apply weighted Cauchy–Schwarz to each row with weights $\mu_j|\widehat{A}_{ij}|$, then sum in i and use the column bound. The result is $\|Av\|_{2,s} \le \|A\|_0 \|v\|_{2,s}$.

For the natural response components above a top of scale ξ_y, the amplitudes are

$$s_H(y) = \xi_y^{-1}, \qquad s_{\nabla H}(y) = \xi_y^{-2}, \qquad s_J(y) = \xi_y^{-3}. \tag{BGR13}$$

Equivalently the output is $(\xi_y H, \xi_y^2 \nabla H, \xi_y^3 J)$. The cell weights are fixed by the actual fine, coarse and pillar pairings, rather than chosen to make a desired estimate true. Each constraint length factor belongs in the matrix before (BGR10) is formed. The transpose is determined by the two declared pairings. The input and output amplitudes must be inserted before applying (BGR12) to a product that changes species at a seam.

For a raw logarithm $a = gy$, a physical radius α becomes α/g in the canonical Gaussian variable y. A derivative with respect to y contributes g times the derivative with respect to a. Thus the second derivative of a scalar activity in Gaussian coordinates is g^2 times its raw-logarithm Hessian. This conversion occurs in addition to the length and cell-weight conversions in (BGR10)–(BGR13).

Corollary 22.2 (Uniform response after the primitive bounds)**.** *Suppose the actual normalized kernels in (BGR7) satisfy*

$$\|L\|_m \le a, \quad \|E\|_m \le e, \quad \|\mathcal{K}E\|_m \le k, \quad \|E^\sharp \mathcal{K}\|_m \le k'. \tag{BGR14}$$

Then

$$\|T\|_m \le e + ak, \qquad \|M\|_m \le k'(e + ak). \tag{BGR15}$$

In the nonsymmetric case the force-to-multiplier block has norm at most $k'a + \|E^\sharp\|_m$. In the symmetric case it is $-T^\sharp$. Its norm equals the norm of T with reversed species if the dual amplitudes are chosen reciprocally for the same cell-weighted pairings. For other force and current amplitudes its weighted transpose bound must be checked separately.

The last qualification is necessary. A raw transpose and a transpose after conjugation by amplitude weights are different matrices. If $\widehat{T}_{ij} = s_i^{-1} T_{ij} s_j$, its transposed normalized kernel is obtained by giving the dual components amplitudes s_i^{-1} and s_j^{-1}. A natural force weight of a

different physical dimension does not satisfy this rule automatically. Its conversion belongs in the kernel before a Schur norm is claimed.

For the actual row map, (PX156) and (PX150) give the lift and trace estimates. The other bounds in (BGR14) come from the field and multiplier blocks of the projected construction after (PX158). Retain the row equation and transfer derivatives to neighboring inverse or lift factors before estimating the energetic products. The conversions after (BGR9) give their datum and multiplier the same physical pairings. The remaining identification consists of recording each block's species weight and outward datum derivative. Then (BGR15) follows by the displayed kernel products.

Invertibility on the whole complex ball. On the fixed slice $\mathcal{V}$, let

$$\mathcal{S}(h,\nu) = (\mathcal{K}h - q^{\sharp}\nu,\, qh).$$

Its domain consists of slice fields and row multipliers; its range consists of the paired field forces and row data in (BGR6)–(BGR9). Fix a reference operator $\mathcal{S}_0$ on these same spaces, with the same pairings and datum convention. If the primitive estimates give

$$\|\mathcal{S}_0^{-1}\|_m \le C_0, \qquad \|\mathcal{S}(B) - \mathcal{S}_0\|_m \le \varepsilon, \qquad C_0\varepsilon < 1, \tag{BGR16}$$

then the Neumann series converges in the kernel algebra and proves

$$\|\mathcal{S}(B)^{-1}\|_m \le \frac{C_0}{1 - C_0\varepsilon}. \tag{BGR17}$$

It is essential that the second bound in (BGR16) holds on the whole retained complex ball. A bound only along the real solutions does not give the analytic disc needed later for Cauchy estimates.

If $\mathcal{S}_0$ is unbounded in the chosen norm, the correct statement uses the relative perturbation $\mathcal{S}_0^{-1}(\mathcal{S} - \mathcal{S}_0)$ between its actual graph spaces. The same series applies there. In particular an elliptic second-order operator cannot be treated as a bounded supremum-norm matrix with a mesh-independent norm merely because its inverse is bounded.

If the coefficients of $\mathcal{S}(B)$ are holomorphic in the indicated operator norm, holomorphy follows term by term on a strictly smaller disc, or by the resolvent identity

$$\mathcal{S}(B')^{-1} - \mathcal{S}(B)^{-1} = -\mathcal{S}(B')^{-1}[\mathcal{S}(B') - \mathcal{S}(B)]\mathcal{S}(B)^{-1}. \tag{BGR18}$$

The identity also proves continuity and the derivative formula wherever the coefficient derivative exists in the indicated operator norm. No complex positivity assertion is used. Real coercivity may construct $\mathcal{S}_0^{-1}$; the complex inverse then requires a genuine norm comparison or another explicitly convergent construction.

For the actual response make this comparison on the common complex ball for (BGR1), with its reference and slice fixed. Use the local analytic coefficients of K_A, the full constraint-curvature and prescribed gauge blocks, and the outward factors of (BGD4). Equations (PX121) and (PX158), together with the projected-current expansions and cutoff commutators above, give the relative comparison for the assembled system. To identify it with the actual $\mathcal{S}$, we must match the auxiliary equations, including each seam's datum and multiplier normalization. These are local coefficient checks of the same construction. After this identification, the estimates verify (BGR16) in its graph-space form on a smaller ball and supply the uniform continuation radius.

From local forcing to first response. Suppose a change in one datum coordinate $B(c)$ gives a forcing column F_c in the enlarged system. The column consists of both the field force and the constraint force in (BGR6). If it is supported within distance r_0 of the top y_c and has normalized

supremum norm at most b_0, then (BGR17) gives

$$|D_{B(c)}Z_y[e]|/s_y \le \frac{C_0 b_0 e^{mr_0}}{1-C_0\varepsilon} e^{-md(y,y_c)}|e|. \tag{BGR19}$$

Here Z contains the output species actually controlled by the inverse, with amplitude s_y from (BGR10). Equivalently apply the bound to the normalized readout Z_y/s_y; subsequent Cauchy estimates use normalized readouts. The proof is the row estimate: for each contributing component j, $d(y,y_c) \le d(y,j) + r_0$, so multiply by $e^{md(y,y_c)}$ and sum the kernel against F_c. The reduced-datum derivative in (BGD4) changes b_0, r_0 by its actual block bounds. It must be included in F_c.

For an exponentially local forcing column, a bound $|F_c(j)|/s_j \le b_0 e^{-ad(j,y_c)}$ gives (BGR19) with any $0 < m' < \min(m,a)$ and m' in place of m. Indeed the triangle inequality leaves a factor $e^{-(a-m')d(j,y_c)} \le 1$, and the remaining sum is a weighted row sum at rate m'. No uncontrolled number of sites appears.

This proves response decay from three smaller tasks: construct the actual primitive inverse, verify its weighted row norm, and compute the actual datum forcing. If the output inverse controls only H and the multiplier, the estimate has not yet been proved for ∇H or the current. Those outputs require their own readout maps.

For example the physical current is a specified nonlinear function $J(U)$. In a fixed logarithmic chart its derivative is

$$\dot J = D_A[J(e^{i\eta A}U_0)]\dot A. \tag{BGR20}$$

One may express this derivative by the multiplier and lower-order terms through the differentiated field equation. Such a replacement must be derived from (BGR1) and the definition of J. On an independent-current core the variable $\mathbf{J}$ and the computed current $J(\mathbf{U})$ are distinct until the physical pair is imposed.

Spatial differences of the response. The differentiated equation supplies spatial regularity in addition to the inverse-kernel bound.

There is a useful exact identity for that estimate. On a translation patch let T shift by one mesh step, $\nabla = (T-I)/\eta$ and $\mathcal{K}_T = T\mathcal{K}T^{-1}$. If $\mathcal{K}v = f$, then

$$\mathcal{K}_T \nabla v = \nabla f - \frac{\mathcal{K}_T - \mathcal{K}}{\eta} v. \tag{BGR21}$$

To prove it, write $T(\mathcal{K}v) = \mathcal{K}_T T v$, subtract $\mathcal{K}v$ and divide by η. At a boundary this calculation must be made on the common interior patch; it does not shift an inadmissible boundary row into the equation.

For a scalar divergence-form example $\mathcal{K} = D^\sharp a D + V$, with constant mesh difference D, the coefficient commutator in (BGR21) is

$$\frac{\mathcal{K}_T - \mathcal{K}}{\eta} = D^\sharp(\nabla a)D + \nabla V. \tag{BGR22}$$

If the energy norm is $\|v\|_{\mathcal{E}}^2 = \|Dv\|_2^2 + \|v\|_2^2$, the bilinear pairing and Cauchy–Schwarz imply

$$\big\|[(\mathcal{K}_T - \mathcal{K})/\eta]v\big\|_{\mathcal{E}^*} \le (\|\nabla a\|_\infty + \|\nabla V\|_\infty)\|v\|_{\mathcal{E}}. \tag{BGR23}$$

Thus a bounded inverse from $\mathcal{E}^*$ to $\mathcal{E}$, together with coefficient regularity, bounds ∇v in energy norm. This does not assert a supremum bound; passing to a supremum or Hölder bound requires the corresponding local elliptic estimate.

In the gauge system, shifting the covariant differences adds transport commutators, the average rows add contour terms, and a seam adds its own boundary comparison. The required finite list is: the translated interior equation, the gauge-row commutator, the constraint-row commutator,

and each seam or exterior boundary row. Each term must be bounded between its actual force and output norms. The uniform primitive regularity estimate is precisely this list of bounds and the associated local elliptic estimate. It is not supplied by exponential decay of H^{-1} alone.

Joint Cauchy estimates with joining distances. Once (BGR19) has been established uniformly on a complex product ball, several useful higher response bounds require no further model estimate. We give the argument with all radius losses visible.

Let $\mathcal{B}_r$ be a ball in the supremum of the dimensionless bond logarithms, with a fixed norm on the three colour coordinates. Suppose a holomorphic map Z_y on $\mathcal{B}_r$ satisfies

$$|\partial_c Z_y(B)[e]| \le Ce^{-md(y,y_c)}|e|, \qquad B \in \mathcal{B}_r. \tag{BGR24}$$

Fix $0 < \theta < 1$. There is a colour constant c_* such that the coordinate polydisc of radius $c_*(1-\theta)r$ about every $B \in \mathcal{B}_{\theta r}$ lies in $\mathcal{B}_r$. The number of bonds does not enter this assertion, since the norm is a supremum over bonds; simultaneous variation of three colour components changes only c_*. Cauchy's formula applied to the first derivative gives

$$|\partial_{c_1} \cdots \partial_{c_q} Z_y(B)| \le C(q-1)![c_*(1-\theta)r]^{-(q-1)} e^{-m \max_j d(y,y_{c_j})}. \tag{BGR25}$$

Repeated coordinates are allowed. The relevant multi-index factorial is at most $(q-1)!$. Holomorphic derivatives commute, so each one of the q coordinates can be selected as the first derivative in (BGR24); taking the minimum of the resulting exponential bounds proves the maximum in (BGR25).

Let $\mathcal{T}(y, c_1, \dots, c_q)$ be the minimum spanning-tree length in the underlying metric. The star through y is a competing tree, and therefore

$$\mathcal{T} \le \sum_{j=1}^{q} d(y, y_{c_j}) \le q \max_j d(y, y_{c_j}). \tag{BGR26}$$

Consequently (BGR25) gives a joining-tree exponential with rate m/q. For the second derivative it also gives the more explicit bound

$$|\partial_c \partial_{c'} Z_y(B)| \le \frac{C}{(1-\theta)r} e^{-\frac{m}{2}[d(y,y_c)+d(y,y_{c'})]}. \tag{BGR27}$$

This is the estimate used in the pointed Taylor remainder. It costs one datum radius. Formula (BGR25) is not a uniform-in-q tree bound at a fixed rate: its rate deteriorates with q. If an arbitrary-order polymer estimate needs a fixed rate, one must use the local nonlinear equations and their tree expansion, with the resulting factorial and entropy costs. A fixed-order application may use (BGR25) directly.

The same Cauchy argument applies to an output Hölder seminorm if the first response is already bounded in that seminorm on the larger ball. Cauchy's formula can be applied to every normalized difference $[Z_y - Z_{y'}]/|y - y'|^\gamma$ and then a supremum taken. It cannot create spatial Hölder regularity from a pointwise bound.

Finite boxes and compatible boundary forcing. Let $\mathcal{S}$ be the actual linearized saddle operator and $\mathcal{S}_\Lambda$ its prescribed finite-box version. To compare solutions, their equations must be put on common variables and rows in an interior collar. Write the resulting exact difference equation as

$$\mathcal{S}_\Lambda \, \delta X = f_\Lambda. \tag{BGR28}$$

The force includes the difference of boundary rows, the datum restriction, the gauge-restoration term and any cutoff commutator. The identity (BGR28) is checked before its right-hand side is bounded. Replacing an actual box boundary by a convenient Dirichlet boundary changes this force and requires a new comparison.

Suppose the inverse has weighted row bound C at rate m and the force is supported in a collar $\mathcal{C}$. Let R be the physical size of the box and $p \geq 0$ a fixed exponent. If $\|f_\Lambda\|_s \leq AR^p$, then for a component at y

$$|\delta X_y|/s_y \leq CAR^p e^{-m\,\mathrm{dist}(y,\mathcal{C})}. \tag{BGR29}$$

This follows by extracting the exponential distance from the weighted row sum. In particular no factor equal to the number of bare sites in the collar is necessary. Such a factor would generally destroy uniformity as the mesh tends to zero.

The same conclusion, at any rate $m' < \min(m, a)$, holds if

$$|f_\Lambda(j)|/s_j \leq AR^p e^{-a\,\mathrm{dist}(j,\mathcal{C})}. \tag{BGR30}$$

For each j, the distance from y to the collar is at most $d(y, j) + \mathrm{dist}(j, \mathcal{C})$. Multiplying the summand by $e^{m'\,\mathrm{dist}(y,\mathcal{C})}$ leaves a weighted row sum and a remaining exponential at most one. This proves the claim.

A cutoff must be made in physical coordinates. For a smooth function $\chi(x/R)$ on a mesh η, its first difference divided by η is $O(R^{-1})$ and its second difference divided by η^2 is $O(R^{-2})$, by the integral form of the mean-value formula. The unnormalized difference itself has an extra factor η. These are different quantities. A cutoff of an affine potential of size $O(R)$ can therefore have bounded first derivatives and polynomially bounded forcing, rather than a bare-mesh singularity.

For a nonlinear equation a cutoff of a solution is not a solution even where the cutoff is a constant strictly between zero and one. All such points belong to the transition collar. On that collar the force is computed from the actual nonlinear formula or its integral Taylor formula. The admissibility of the cutoff field, its averages, and its tower backgrounds is a separate domain check. A bounded formal commutator cannot replace it.

For two nonlinear backgrounds, subtracting their equations gives an averaged Jacobian along a common admissible coordinate segment. The inverse estimate for that averaged operator must be justified, for example by a uniform comparison of every Jacobian with the same reference as in (BGR16). An arbitrary average of invertible complex matrices need not be invertible. The first-step background section uses just such a common comparison; a multilevel application needs it on the actual multilevel ball.

The full substitution and the test for staying inside a core. An old-origin source is evaluated on data of the form

$$V_j(\nu) = \mathcal{M}^j\big(e^{i\eta(t+\nu\zeta)H}U\big)|_{\Gamma_j}. \tag{BGR31}$$

The parameters on the fine bonds in ζ may vary independently. The inner analytic argument uses $|\nu| < c/s$, not merely the physical segment between zero and one. There are also outer and removed substitutions. Each must satisfy the same list of field, independent current, node, gauge and tower conditions that defines its assigned core. A proof only of the curvature clause is incomplete.

Here is a precise useful criterion. Express the core as inequalities $\|\mathcal{R}_a(X)\| < r_a$ for its actual clauses, including all hereditary tower clauses. Let X_0 be the complete reference pair and assume margins $\|\mathcal{R}_a(X_0)\| \leq (1 - \epsilon_a)r_a$. If a common analytic curve satisfies

$$\sup_{\nu \text{ on the retained disc}} \|\mathcal{R}_a(X_\nu) - \mathcal{R}_a(X_0)\| < \epsilon_a r_a \quad \text{for every } a, \tag{BGR32}$$

then it stays in the core. The proof is the triangle inequality for every defining clause. Its substantive content in an application is the uniform bound for the infinite hereditary list, not that triangle inequality.

If the actual response and average derivatives imply

$$\frac{\|\mathcal{R}_a(X_\nu) - \mathcal{R}_a(X_0)\|}{r_a} \le C_a|\nu|\,\delta_k\sigma_j/\alpha_j, \tag{BGR33}$$

then the inner disc is admitted whenever $c\sup_a\{C_a\delta_k\sigma_j/(\epsilon_a s\alpha_j)\} < 1$. For the window maps, Section 29 gives the normalized displacement in (BGR33). Its average derivative controls the datum clauses, the reduced-current response controls the current clause, and the real-to-complex curvature calculation controls the entire tower list. For a layer gap ϵ_a decreasing as 2^{-n}, use the quotient C_a/ϵ_a rather than a positive lower bound on all the gaps. The response estimate supplies the factor $e^{-c\delta_0 ML(n-1)}$; thus the relevant ratio is bounded by

$$C2^n\sqrt{1+n}e^{-c\delta_0 ML(n-1)},$$

which is uniformly bounded after M is chosen. With $s = \delta_k/\alpha_k$, the other quotient in the admission condition is $\sigma_j\alpha_k/\alpha_j$. The slow-radius inequality gives $\sigma_j\alpha_k/\alpha_j \le C\sqrt{1+k-j}L^{-(k-j)}$, whose supremum is finite. Choose the common amplification constant c from these two suprema and the retained margin. Together with the common continuation bound below, these strict inequalities admit the disc simultaneously for all old levels and tower clauses. The powers of σ_j here come from the physical length conversion in the nonlinear average.

To use the displacement estimate for admission, continue the common analytic solution branch up to a possible first exit. Before exit, the local inverse construction following (BGR18) applies on the same scaled ball and gives a uniform continuation radius. Equations (BGR32)–(BGR33), with the tower estimate of Section 29, keep every clause strictly inside its allotted margin. The branch therefore continues past any proposed first exit. The finite address check is that the outer, inner and removed substitutions use their assigned tower, current and node maps; the normalized margin estimate already handles their full hereditary lists.

The unamplified reference value must also belong to the core. At the finitely many young ages where the critical half-datum chart is replaced by the whole-core estimate, use that estimate for the complete reference value before applying the displacement bound. The amplification argument then spends its remaining margin. This separates admission of X_0 from admission of its increment.

For removed substitutions the current and the background may change together. The path used in the comparison must then remain in the joint core. Separate coordinate Cauchy estimates at the two endpoints do not imply that the segment connecting them is admitted, especially when a hereditary condition is not convex. One can instead use the actual common analytic chart and the strict clause margins in (BGR32).

The mesh-uniform Hölder estimate used by the response. The response system needs spatial regularity as well as decay. We prove the particular scalar estimate that converts a bounded potential and a bounded componentwise Laplacian into a Hölder bound for its first differences. The estimate is uniform in the mesh. It also applies in a parallel frame for an unweighted flat unitary connection, including arbitrary flat holonomies on a compatible torus. We then apply the estimate in the local real gauges of the background construction, with the actual covariant Laplacian as its source.

Fix $0 < \alpha < 1$. On $\eta\mathbb{Z}^4$ define

$$D_i^+u(x) = \frac{u(x+\eta e_i) - u(x)}{\eta}, \quad D_i^-u(x) = \frac{u(x) - u(x-\eta e_i)}{\eta}, \quad \Delta_\eta = \sum_{i=1}^4 D_i^-D_i^+. \tag{BGH1}$$

The Laplacian has nonpositive Fourier symbol. The counting measure in all convolution kernels is η^4 times counting measure. We use the Euclidean distance, allowing dimension-dependent constants when a coordinate path is used. A derivative of order r means an arbitrary product of r of the forward or backward differences in (BGH1).

Heat kernels with enough derivative and tail control. Start on the unit lattice. For $T \geq 0$ set

$$q_T(n) = \frac{1}{(2\pi)^4} \int_{[-\pi,\pi]^4} e^{-T\omega(\theta)} e^{in\cdot\theta}\, d\theta, \qquad \omega(\theta) = 4\sum_i \sin^2(\theta_i/2). \tag{BGH2}$$

This is a nonnegative probability kernel. Indeed, if A is the sum of the eight nearest-neighbour translations, then $\Delta_1 = \mathsf{A} - 8I$ and

$$e^{T\Delta_1} = e^{-8T} \sum_{k=0}^{\infty} \frac{T^k \mathsf{A}^k}{k!}. \tag{BGH3}$$

Every coefficient is nonnegative, and applying the operator to the constant function gives one. Fourier transformation gives (BGH2), so the two constructions agree. The physical heat-kernel density is

$$p_t^\eta(\eta n) = \eta^{-4} q_{t/\eta^2}(n). \tag{BGH4}$$

It follows directly that convolution with p_t^η is a contraction on bounded functions.

We need polynomial tails rather than a sharp Gaussian bound. The following estimates suffice for the entire argument.

Lemma 22.3 (Uniform differentiated heat-kernel tails)**.** *For $\eta > 0$, $a \geq 0$ and integers $r, N \geq 0$ there is $C_{r,N}$ independent of η, t, a such that*

$$\begin{aligned} &\sum_{|x|\geq a} \eta^4 |D^r p_t^\eta(x)| \leq C_{r,N} t^{-r/2} (1 + a/\sqrt{t})^{-N}, && t \geq \eta^2, \\ &\sum_{|x|\geq a} \eta^4 |D^r p_t^\eta(x)| \leq C_{r,N} \eta^{-r} (1 + a/\eta)^{-N}, && 0 \leq t \leq \eta^2. \end{aligned} \tag{BGH5}$$

Proof. A product of r unit differences has Fourier multiplier $m_r(\theta)$, a product of factors $e^{\pm i\theta_i} - 1$. For every multi-index b,

$$|\partial^b m_r(\theta)| \leq C_{r,b} |\theta|^{(r-|b|)_+}. \tag{BGH6}$$

The bound follows by assigning the derivatives to factors. A factor that has not been differentiated vanishes to first order at zero; all its positive-order derivatives are bounded. If more than r derivatives occur the asserted power is zero.

For $\theta \in [-\pi,\pi]^4$, $\omega(\theta) \geq c|\theta|^2$. Its first derivatives are $O(|\theta|)$ and its higher derivatives are bounded. Differentiating $e^{-T\omega}$ for $T \geq 1$ gives

$$|\partial^b e^{-T\omega(\theta)}| \leq C_b T^{|b|/2} (1 + |\sqrt{T}\theta|)^{M_b} e^{-c_b T|\theta|^2}. \tag{BGH7}$$

For completeness, each term of the repeated product rule consists of $e^{-T\omega}$ times factors $T\partial^{b_j}\omega$, with $b_j \neq 0$ and $\sum b_j = b$. A first derivative contributes $\sqrt{T}\, O(|\sqrt{T}\theta|)$. A factor of order at least two is at most a constant times $T^{|b_j|/2}$ because $T \geq 1$. Multiplying proves (BGH7); the finite polynomial can be absorbed by a smaller Gaussian exponent when integrating.

Leibniz' rule, (BGH6), and (BGH7) now imply

$$\int_{[-\pi,\pi]^4} |\partial^b (m_r e^{-T\omega})|^2\, d\theta \leq C_{r,b} T^{|b|-r-2}, \qquad T \geq 1. \tag{BGH8}$$

When at most r derivatives hit the multiplier, scale $\theta = T^{-1/2} v$ in that term. When more than r hit it, its bound is smaller than the displayed power since $T \geq 1$. The four-dimensional volume factor is T^{-2}. This explains every power in (BGH8).

Let δ^r denote the corresponding unit differences. Parseval's identity and differentiation of the periodic Fourier multiplier give, for every integer $M \ge 0$,

$$\sum_n (1+|n|^2/T)^M |\delta^r q_T(n)|^2 \le C_{r,M} T^{-r-2}, \qquad T \ge 1. \tag{BGH9}$$

Expand the weight into coordinate monomials. The term $|n^b|^2 T^{-|b|}$ is bounded by (BGH8) times $T^{-|b|}$. All multiplier functions are smooth and periodic, so there are no boundary terms at $\pm\pi$.

Choose $M > N+2$, increasing it by one if necessary. Comparing the lattice sum with the radial integral in four dimensions gives

$$\sum_{|n|\ge A} (1+|n|^2/T)^{-M} \le C_{M,N} T^2 (1+A/\sqrt{T})^{-2N}, \qquad T \ge 1. \tag{BGH10}$$

One way to check the comparison is to cover the lattice by unit cubes. For $T \ge 1$ the values of $1+|x|/\sqrt{T}$ at any two points of a cube are comparable by a fixed dimensional factor. The integral of $(1+|x|^2/T)^{-M}$ outside a ball of radius comparable with A is T^2 times the corresponding integral in scaled coordinates. That integral decays at least as $(1+A/\sqrt{T})^{-2N}$ for this choice of M. Small values of A are absorbed by the full integral bound.

Cauchy–Schwarz with (BGH9) and (BGH10) proves $\sum_{|n|\ge A} |\delta^r q_T(n)| \le C T^{-r/2} (1+A/\sqrt{T})^{-N}$. For $0 \le T \le 1$, all derivatives of $m_r e^{-T\omega}$ of any fixed order are uniformly bounded. The same Parseval argument, with weight $(1+|n|^2)^M$, gives instead $C(1+A)^{-N}$. Finally (BGH4) contributes η^{-r} to the weighted ℓ^1 norm. Substitution of $T = t/\eta^2$ proves (BGH5). □

The constants depend only on the fixed derivative order and tail power. The small-time estimate is needed for the discrete second and third differences.

A massive resolvent and its first-difference modulus. For $\mu > 0$ define

$$G_\mu^\eta(x) = \int_0^\infty e^{-\mu^2 t} p_t^\eta(x)\, dt, \qquad R_\mu f = G_\mu^\eta * f. \tag{BGH11}$$

Its kernel is nonnegative and has total mass μ^{-2}. For fixed η, Δ_η is a bounded operator on bounded functions. Integrating the derivative of $e^{-\mu^2 t} e^{t\Delta_\eta}$ in operator norm shows

$$(-\Delta_\eta + \mu^2) R_\mu f = f, \qquad \|R_\mu f\|_\infty \le \mu^{-2} \|f\|_\infty. \tag{BGH12}$$

Conversely, any bounded solution of this equation equals $R_\mu f$. Indeed integrate the same identity applied to the bounded solution; the terminal term tends to zero because of $e^{-\mu^2 t}$ and the heat contraction. This also proves uniqueness without a maximum attainment argument on an infinite lattice.

Assume $\mu\eta \le 1$. Integrating (BGH5) for $r = 1$, first on $[0, \eta^2]$ and then on $[\eta^2, \infty)$, gives

$$\sum_x \eta^4 |D G_\mu^\eta(x)| \le C\mu^{-1}. \tag{BGH13}$$

The first interval contributes $C\eta$ and the second is bounded by $C\int_0^\infty e^{-\mu^2 t} t^{-1/2}\, dt = C'\mu^{-1}$.

For a lattice displacement h, telescoping along a coordinate path and using (BGH5) with $r = 1, 2$ gives, for $t \ge \eta^2$,

$$\|\tau_h D p_t^\eta - D p_t^\eta\|_1 \le C \min(t^{-1/2}, |h| t^{-1}). \tag{BGH14}$$

The second bound follows by summing the second difference over the path of physical length at most $2|h|$ in four dimensions; the first is twice the first-difference norm. For $t \le \eta^2$ use $C\eta^{-1}$ instead. Every nonzero lattice displacement has $|h| \ge \eta$.

Suppose first that $\eta \le |h| \le \mu^{-1}$. Integrate (BGH14) on $[\eta^2, |h|^2]$, on $[|h|^2, \mu^{-2}]$ and on the remaining tail. Together with the small-time interval, this gives

$$\|\tau_h DG^\eta_\mu - DG^\eta_\mu\|_1 \le C|h|\big(1 + \log(1/(\mu|h|))\big). \tag{BGH15}$$

On the middle interval the integral is $|h| \int_{|h|^2}^{\mu^{-2}} t^{-1} dt$. The first interval contributes $O(|h|)$, and the exponentially damped last interval does as well. If $|h| > \mu^{-1}$ use twice (BGH13). Since for $0 < x \le 1$

$$x^{1-\alpha}(1 + \log(1/x)) \le 1 + \frac{1}{e(1-\alpha)}, \tag{BGH16}$$

both ranges imply

$$\|\tau_h DG^\eta_\mu - DG^\eta_\mu\|_1 \le C_\alpha \mu^{\alpha-1}|h|^\alpha. \tag{BGH17}$$

The value at $h = 0$ is zero. It follows for bounded f that

$$\|DR_\mu f\|_\infty \le C\mu^{-1}\|f\|_\infty, \qquad [DR_\mu f]_\alpha \le C_\alpha \mu^{\alpha-1}\|f\|_\infty. \tag{BGH18}$$

The seminorm compares lattice points, using their physical distance. The logarithmic modulus (BGH15) explains why these estimates do not give a mesh-uniform Lipschitz bound for $DR_\mu f$ from bounded f alone. The constants are permitted to diverge as α tends to one.

Off-diagonal resolvent derivatives. Let $R \ge \eta$ and $\mu = R^{-1}$. For every fixed $c > 0$ and $r = 0, 1, 2, 3$,

$$\sum_{|x| \ge cR} \eta^4 |D^r G^\eta_{1/R}(x)| \le C_{r,c} R^{2-r}. \tag{BGH19}$$

Choose $N > \max(0, r-2)$. On the time interval $[\eta^2, R^2]$, the large-time part of (BGH5) is bounded by $Ct^{-r/2}(\sqrt{t}/R)^N$. Its integral is $O(R^{2-r})$. On $[R^2, \infty)$ the exponential e^{-t/R^2} gives the same bound by scaling. The interval $[0, \eta^2]$ contributes at most $C\eta^{2-r}(\eta/R)^N$, which is also $O(R^{2-r})$. This proves (BGH19), including the second and third derivatives for which the uncut global norms would have a worse small-scale behaviour.

There is no assertion here that the second resolvent derivative has a uniformly bounded global absolute sum. Its off-diagonal bound is what makes the interior cutoff argument work.

An interior estimate with no unknown cutoff gradient. Let Q_R be the cube of side $2R$ centred at a lattice point. The following formulation uses a fixed ratio of cubes; changing that ratio changes only the constants.

Theorem 22.4 (Interior first-difference Hölder bound)**.** *Let $R \ge 64\eta$. Suppose u is defined on Q_{2R} and $\Delta_\eta u = f$ at every point needed by the equation there. Put $M = \|u\|_{\infty, Q_{2R}}$ and $F = \|f\|_{\infty, Q_{2R}}$. On $Q_{R/2}$, with all first differences and comparisons taken inside its one-mesh enlargement,*

$$\begin{aligned} \|Du\|_\infty &\le C(R^{-1}M + RF), \\ [Du]_\alpha &\le C_\alpha(R^{-1-\alpha}M + R^{1-\alpha}F). \end{aligned} \tag{BGH20}$$

The constants are independent of R, η, u.

Proof. Choose the restriction of a smooth cutoff $\chi(x/R)$, equal to one on Q_R and supported in $Q_{3R/2}$. It satisfies $|D\chi| \le C/R$ and $|\Delta_\eta \chi| \le C/R^2$. Its difference supports lie in a collar at distance at least cR from the evaluation cube and its one-mesh enlargement. The condition $R \ge 64\eta$ makes this separation uniform after every difference used below. Extend u arbitrarily outside Q_{2R}; the value of $v = \chi u$ and all forces below are independent of that extension.

With $\mu = 1/R$ the exact product identity is

$$(-\Delta_\eta + \mu^2)v = a - [\Delta_\eta, \chi]u, \qquad a = \chi(-f + \mu^2 u). \tag{BGH21}$$

Both sides are extended by their actual compactly supported formulas. The equation for u is used only where χ or its differences are nonzero, inside the given domain. By (BGH12), $v = R_\mu a - R_\mu[\Delta_\eta, \chi]u$. The first term has force norm at most $F + R^{-2}M$, so (BGH18) gives exactly its contributions to (BGH20).

To treat the commutator one must avoid estimating the unknown Du in the cutoff collar. Move the commutator onto the kernel. For scalar multiplication by χ and the constant cell-weighted pairing,

$$[\Delta_\eta, \chi]^{\mathsf{T}} = [\chi, \Delta_\eta] = -[\Delta_\eta, \chi]. \tag{BGH22}$$

For $j = 1, 2$, apply this identity to $K_j(y) = D_x^j G_\mu^\eta(x-y)$ in the convolution. The discrete product rule shows that its absolute value is bounded by

$$CM\left(R^{-2}\|1_{|\cdot|\geq cR} D^j G_\mu^\eta\|_1 + R^{-1}\|1_{|\cdot|\geq cR} D^{j+1} G_\mu^\eta\|_1\right). \tag{BGH23}$$

To check the rule explicitly, $\Delta(\chi K) - \chi\Delta K - K\Delta\chi$ is the sum over i of $(D_i^+\chi)(D_i^+ K) + (D_i^-\chi)(D_i^- K)$. All factors containing a difference of χ are supported in the separated collar. Changes of the evaluation point by at most the fixed number of mesh steps in a derivative merely change c. The kernel is absolutely summable on that collar by (BGH19), which also justifies the summation by parts.

For $j = 1$, (BGH19) bounds (BGH23) by CM/R. For $j = 2$ it bounds it by CM/R^2. Let $w = R_\mu[\Delta_\eta, \chi]u$. A coordinate path between two points of $Q_{R/2}$ stays in that cube; telescoping the first differences of w gives $|Dw(x) - Dw(x')| \leq CMR^{-2}|x - x'|$. Its supremum bound is CM/R. Since $|x - x'| \leq CR$ in the cube, this gives $[Dw]_\alpha \leq C_\alpha M R^{-1-\alpha}$. Adding the contribution of $R_\mu a$ proves both estimates. On the evaluation cube $v = u$, including all the differences used, so the estimates apply to u itself. □

The result holds componentwise for vectors in any fixed finite-dimensional normed space. Equivalent coordinate norms contribute a constant depending on that fixed dimension.

Tori and flat unitary holonomies. On a torus with periods ℓ_i, the scalar heat kernel and massive resolvent are periodizations of their infinite-lattice kernels. Their absolute derivative sums are bounded by the corresponding infinite sums. For a set at periodic distance at least cR from the origin, every lift has Euclidean distance at least cR. Thus (BGH19) holds for periodic distance as well. If the cubes and their collars in the interior theorem embed without wrapping, the same cutoff proof gives (BGH20) with the same constants. For example $\min_i \ell_i \geq 8R$ suffices for the stated cubes.

A flat unitary connection can be made trivial on the lifted lattice. The deck transformations act by its commuting unitary holonomies. The kernel on the torus is the image sum with the corresponding unitary factors. On a common parallel frame in the embedded cube, differences act on the lifted kernel. Taking the norm and using unitarity bounds the image sum by the scalar sum already estimated. Hence all the preceding local bounds remain uniform in the holonomies. Their distance from the identity is irrelevant.

The argument requires flatness. Curved parallel transports have additional commutators. Nor does it give the same estimate for an arbitrary nonlocal or layered positive operator: the proof used the componentwise constant-coefficient Laplacian in (BGH1).

Covariant differences and the response equation. Here is how (BGH20) enters the background and response estimates. Suppose a local real gauge has supplied a potential on a cube enlarged by a fixed multiple of its local scale ξ, with

$$\|A\|_\infty \leq C_0 m\varepsilon\xi^{-1}, \qquad \|\Delta_\eta A\|_\infty \leq C_1 m\varepsilon\xi^{-3}. \tag{BGH24}$$

Apply the theorem on cubes of radius a fixed fraction of ξ, covering the interior. It proves

$$[DA]_\alpha \le C_\alpha m\varepsilon\xi^{-2-\alpha}, \qquad \|DA\|_\infty \le Cm\varepsilon\xi^{-2}. \tag{BGH25}$$

For pairs farther apart than a fixed fraction of ξ, the supremum bound gives the same Hölder estimate. Thus local cubes give the full indicated interior seminorm. This proves the Hölder implication once (BGH24) is known; it does not construct the gauge or prove the nonlinear Laplacian estimate in (BGH24).

There is a finite coarse-mesh case in this application. If the chosen fraction of ξ is less than 64η, the ratio ξ/η is bounded by a fixed constant. Directly, $|DA| \le 2\|A\|_\infty/\eta$ and $[DA]_\alpha \le 4\|A\|_\infty\eta^{-1-\alpha}$. They give (BGH25) with the same powers of ξ and a larger fixed constant. The same fallback applies to the response application below. Thus no unmentioned lower bound on ξ/η is needed.

For the physical constrained-background response we use the differentiated Euler–Lagrange equation to estimate the positive covariant componentwise Laplacian $\mathcal{L}_{U_0} = \sum_i \nabla_{U_0,i}^{\mathsf{T}}\nabla_{U_0,i}$. Thus $\mathcal{L}_{\mathbf{1}} = -\Delta_\eta$ with the heat-generator sign in (BGH1). Its conversion can also be made quantitative. In a local gauge write $U_0(b) = e^{i\eta A_0(b)}$ and let $T_b = \mathrm{Ad}(U_0(b))$ in the chosen unitary colour representation. If $\eta|A_0|$ stays in a fixed small exponential chart, then

$$\|(T_b - I)/\eta\| \le C|A_0(b)|. \tag{BGH26}$$

Differencing T_b in a transverse direction uses the integral derivative of the exponential. It gives $|DT_b| \le C\eta|DA_0|$, with the adjacent bond values interpreted in the same local frame. The inverse transport satisfies the same bound. Expanding the forward and backward covariant differences therefore gives

$$|(\mathcal{L}_{U_0} + \Delta_\eta)h| \le C\big(|A_0|\,|Dh| + (|DA_0| + |A_0|^2)|h|\big). \tag{BGH27}$$

One can verify this by first writing each covariant first difference as the ordinary difference plus $(T_b - I)/\eta$ times a neighbouring value. In the second difference the apparent η^{-1} terms combine into a first difference of h, a difference of the coefficient $(T_b - I)/\eta$, and a product of two such coefficients. These are exactly the three terms on the right of (BGH27); the factors there denote maxima on the fixed neighboring stencil. There is no estimate of an individual uncombined $\eta^{-1}A_0h$ term.

Assume on a fixed enlargement of the local-scale evaluation cube that

$$\begin{aligned} \xi|A_0| + \xi^2|DA_0| &\le a_0, \\ \xi|h| + \xi^2|\nabla_{U_0}h| + \xi^3|\mathcal{L}_{U_0}h| &\le B. \end{aligned} \tag{BGH28}$$

Formula (BGH26) first gives $\xi^2|Dh| \le C(1 + a_0)B$. Then (BGH27) implies

$$\xi^3|\Delta_\eta h| \le C(1 + a_0 + a_0^2)B. \tag{BGH29}$$

Applying (BGH20) at scale ξ yields $[Dh]_\alpha \le C_{\alpha,a_0}B\xi^{-2-\alpha}$. The difference between Dh and $\nabla_{U_0}h$ is a bounded coefficient times a neighbouring h. Its Hölder seminorm is controlled by the same local bounds, using the Lipschitz coefficient bound from DA_0 and the first-difference bound for h. Thus

$$[\nabla_{U_0}h]_\alpha \le C_{\alpha,a_0}B\xi^{-2-\alpha}. \tag{BGH30}$$

This initially compares gradient components in the chosen local gauge. If the assigned Hölder seminorm transports one component to the other along paths of length at most $C|x - x'|$, unitary product telescoping and (BGH28) give $\|\operatorname{Ad} U_0(\Gamma) - I\| \le Ca_0|x - x'|/\xi$. Multiplying by $|\nabla_{U_0}h| \le B/\xi^2$ adds at most $Ca_0B\xi^{-2-\alpha}|x - x'|^\alpha$ on the scale-ξ cube. Hence (BGH30) also holds in that transported seminorm. Its premise still includes the pointwise covariant-gradient bound in (BGH28); the argument removes an additional Hölder premise, rather than that gradient bound.

If the right-hand side of (BGH28) has the response factor $B = Ce^{-md(y,y_c)}|e|$, that factor persists on a fixed-size local-scale cube up to the fixed exponential comparison of distances within the cube. This gives the corresponding Hölder member of the response estimate. For the physical background response at $s \equiv 1$, we apply this implication after obtaining the local real gauge and the three readouts in (BGH28) from the two physical response systems above and their identification in mechanism (k). At a seam we must verify these readouts and compare the local frame and equation on their common two-layer determining neighborhood.

For arbitrary decoupled data s, a bound on the componentwise Laplacian is not among the declared operator norms. We do not use (BGH28) for that response. Its gradient Hölder estimate instead uses the Hölder member already retained in the norms of $\mathfrak{G}(s)$, $\widetilde{G}(s)$ and the datum lift: apply those operator bounds in the same contraction for the potential and its first differences. The force and reduced-current terms in that contraction involve only the declared first-order norms. This use of the physical Laplacian estimate therefore imposes no stronger second-order bound on the decoupled operator data.

23. The Hessian of a pointed member and its own action counterterm

The object in this section is the actual pointed total composed with the actual free response. It need not be a convolution. Its bulk sum will be a convolution, and the distinction is retained throughout. The result identifies its marginal quadratic term and its own action counterterm. The nonlinear remainder uses the localized response and field-domain hypotheses of Section 24.

The pointed object and the analytic hypotheses. Work in coarse lattice units on $\mathbb{Z}^4$. An independently oriented bond (x, μ) starts at x and ends at $x + e_\mu$. A real field $B_\mu(x)$ is a traceless Hermitian matrix. Choose a color basis t_a, $1 \le a \le 3$, orthonormal for $\langle A, C\rangle = \frac{1}{2}\operatorname{Tr}(AC)$. Thus $B = \sum_a B^a t_a$. Complexification is bilinear, not sesquilinear. The normalization gives the quadratic Wilson action

$$A(e^{itB}) = \frac{t^2}{2}\sum_p |dB(p)|^2 + O(t^3) \tag{PH1}$$

on every finite set of plaquettes. Here d is the oriented lattice curl and each coordinate-plane plaquette is counted once.

An individual member G has the pointed pair total, in its exponential chart,

$$\mathcal{G}_z(e^{i\eta A}, J) = \sum_{X \ni z} G(X; (e^{i\eta A}, J), z), \qquad z \in \mathbb{Z}^4. \tag{PH2}$$

The bare spacing η can differ from the coarse unit spacing. There are no stored spectators in a regular member. Assume the following analytic and geometric properties of this actual family.

The sum in (PH2) and its first two derivatives converge normally in the neighborhoods where they are used. Every summand is local in its pair variables and invariant under all local gauge transformations for which the arguments remain in its declared domain. In particular it is invariant under constant adjoint rotations. Let $\mathcal{P}(B)$ be the actual background pair formed from a sufficiently small finitely supported coarse field, with $\mathcal{P}(0) = (1, 0)$. It is twice differentiable, is gauge covariant, and has first response, expressed in this potential chart,

$$hB = D\mathcal{P}(0)B = (HB, D_1^* D_1 HB). \tag{PH3}$$

The precise linear map H is part of the background construction; it is not replaced by a convolution chosen for this proof. Put $F_z(B) = \mathcal{G}_z(\mathcal{P}(B))$. Assume the pointed totals are covariant under unit translations and the full hypercubic group, including reflections. These are properties of one individual member at all its base points. They do not require two different members chosen at different sites to be related.

Finally assume an absolute exponential Hessian moment bound. With $\delta^a_{x,\mu}$ the field of color a at one coarse bond, let

$$\mathcal{K}_z^{\mu\nu;ab}(x,y) = \sum_{X \ni z} D^2 G(X,(1,0),z)[h\delta^a_{x,\mu}, h\delta^b_{y,\nu}]. \tag{PH4}$$

All derivatives of a pair function are taken in the (A,J) chart of (PH2), including its factor η. For some $m > 0$ assume

$$\sum_{\mu,\nu,a,b} \sum_{x,y\in\mathbb{Z}^4} e^{m(|x-z|+|y-z|)} |\mathcal{K}_z^{\mu\nu;ab}(x,y)| \le C_m. \tag{PH5}$$

The corresponding derivative representation on finitely supported fields is assumed, as is usual when the normally convergent localized Hessians are composed with the free response. This bound and the actual background/core hypotheses are model inputs here. In particular, this section does not infer a uniformly bounded coefficient extractor from a potential norm on a shrinking field ball.

We first justify that (PH4) really is the Hessian of F_z. Constant adjoint invariance makes the first derivative of each summand of (PH2) zero at $(1,0)$: each coefficient of a potential or current coordinate would be an invariant linear functional on the adjoint color space, which has no fixed vector. This applies to its finitely many local coordinates and hence to every direction supplied by h. The normal convergence permits summation. In the second-order chain rule the term containing $D\mathcal{G}_z(1,0)D^2\mathcal{P}(0)$ is therefore zero. Consequently (PH4), with the displayed actual h, is exactly $D^2F_z(0)$.

The exponential moment hypothesis has a concrete sufficient localized form. Suppose the analytic core contains a usable pair norm ball of radius a_{core}, and on it $|G(X,\cdot,z)| \le ge^{-\kappa d(X)}$. Suppose also

$$\|h\delta^a_{x,\mu}\|_{1;X} \le Ce^{-\delta\,\mathrm{dist}(x,X)}, \qquad \#\{X \ni z : q \le d(X) < q+1\} \le Ce^{\kappa_D q}, \qquad \kappa > \kappa_D,$$

and $\mathrm{diam}(X) \le C_M(d(X)+1)$, with fixed localization cube size. These are the core, response and entropy conditions used in quadratic extraction. Two-variable Cauchy gives

$$|D^2G(X,0,z)[h\delta_x, h\delta_y]| \le Cga_{\mathrm{core}}^{-2} e^{-\kappa d(X)} e^{-\delta(\mathrm{dist}(x,X)+\mathrm{dist}(y,X))}.$$

Choose $m < \delta/2$ so small that $2mC_M < (\kappa-\kappa_D)/2$. Since $z \in X$, inserting the weight in (PH5) costs at most $e^{2mC_M(d(X)+1)}$ after retaining the two response distances. The remaining sums over x and y cost at most $C(d(X)+1)^8$: enclose X in a box of diameter $C_M(d(X)+1)$ and sum the exponentially decreasing distances outside it. The rooted domain count now gives

$$C_m \le Cga_{\mathrm{core}}^{-2} \sum_{q\ge 0} (q+1)^8 e^{-(\kappa-\kappa_D-2mC_M)q} \le C' ga_{\mathrm{core}}^{-2}. \tag{PH5a}$$

Thus (PH5) need not be introduced as a new physical estimate when these localized hypotheses have already been supplied. The factor a_{core}^{-2} is retained. The coordinate normalization and near-zero core inclusions needed to use this Cauchy estimate are treated in Section 14.

Color, symmetry, and the local gauge Ward identity. Constant adjoint invariance implies

$$\mathcal{K}_z^{\mu\nu;ab}(x,y) = \delta_{ab} K_z^{\mu\nu}(x,y), \qquad K_z^{\mu\nu}(x,y) = \frac{1}{3}\sum_{a=1}^{3} \mathcal{K}_z^{\mu\nu;aa}(x,y). \tag{PH6}$$

For a direct proof, fix the spatial indices and view the color coefficients as a 3×3 complex matrix. Adjoint rotations through π about the color axes force its off-diagonal entries to zero, and rotations interchanging the axes make its diagonal entries equal. The adjoint action of SU(2)

contains these rotations. This also proves the assertion for complex members. Equality of mixed derivatives gives

$$K_z^{\mu\nu}(x,y) = K_z^{\nu\mu}(y,x). \tag{PH7}$$

For a real member all these coefficients are real. Thus its bulk kernel is self-adjoint in addition to being bilinearly symmetric. No positivity of a Hessian is assumed.

For a finitely supported Lie-algebra-valued site function χ, write $(d\chi)_\mu(x) = \chi(x+e_\mu) - \chi(x)$. Gauge covariance of $\mathcal{P}$ and gauge invariance of the member give $F_z(B^g) = F_z(B)$, where B^g is the logarithmic coarse bond field after a small gauge transformation. Choose the gauge convention $U(x,\mu)^g = g(x)^{-1}U(x,\mu)g(x+e_\mu)$, so the derivative at the identity of $g = e^{it\chi}$ is $d\chi$. Differentiate this identity once in t and once in a finitely supported field direction C. The mixed derivative of the gauge coordinate itself is multiplied by $DF_z(0) = 0$. Therefore

$$D^2F_z(0)[d\chi, C] = 0. \tag{PH8}$$

The other sign convention for gauge transformations gives the same zero identity. The argument initially uses real directions and extends complex bilinearly. It uses neither a noncompact gauge transformation nor a polynomial field in a nonlinear field domain.

Polynomial extensions justified by absolute moments. For fields of polynomial growth define the extended Hessian by

$$\mathcal{H}_z(B,C) = \sum_{\mu,\nu,a} \sum_{x,y} K_z^{\mu\nu}(x,y) B_\mu^a(x) C_\nu^a(y). \tag{PH9}$$

Every such sum converges absolutely by (PH5). Compact cutoffs converge to it by dominated convergence, independently of the cutoffs. This defines an extension of the quadratic jet, not a value of the nonlinear member at an unbounded polynomial field.

The Ward identity extends to polynomial gauge parameters and polynomial second arguments. To prove it, let χ be a polynomial of degree q on sites and let $\chi_R(x) = \chi(x)\vartheta(x/R)$ with a fixed smooth compact cutoff equal to one near zero. Its forward differences converge pointwise to $d\chi$, and satisfy a bound

$$|d\chi_R(x)| \le C_\chi(1+|x|)^{\max(q-1,0)} \quad (R \ge 1).$$

Indeed away from the cutoff annulus the usual polynomial difference has this bound; on the annulus the extra difference of the cutoff costs R^{-1} and $|x|$ is comparable to R. Apply (PH8) with a single-bond second argument and pass to the limit using (PH5). Summing against any polynomial second argument is then legitimate by absolute convergence. The radius of the nonlinear gauge chart may depend on R: only its derivative at zero was used for each fixed R.

In particular the linear polynomial $\chi(x) = x_\mu t_a$ gives the separate row and column zeroth moments

$$\sum_x K_z^{\mu\nu}(x,y) = 0, \qquad \sum_y K_z^{\mu\nu}(x,y) = 0. \tag{PH10}$$

The second identity follows also from (PH7). Both hold for each fixed remaining edge and pair of directions, not only after a full trace.

For the moment at $z = 0$ set, initially in lower bond-start coordinates,

$$T_{\mu\nu,\kappa\lambda} = \sum_{x,y} K_0^{\mu\nu}(x,y) x_\kappa y_\lambda. \tag{PH11}$$

For a symmetric matrix S, the quadratic gauge polynomial $\chi(x) = \frac{1}{2}\sum_{\mu,\kappa} S_{\mu\kappa}x_\mu x_\kappa t_a$ has forward difference

$$(d\chi)_\mu(x) = \sum_\kappa S_{\mu\kappa}x_\kappa + \tfrac{1}{2}S_{\mu\mu}.$$

Use the polynomial Ward identity with second argument $C_\nu(y) = y_\lambda t_a$ at one direction ν. The constant term vanishes by (PH10). Since S is arbitrary symmetric, one obtains

$$T_{\mu\nu,\kappa\lambda} = -T_{\kappa\nu,\mu\lambda}, \qquad T_{\mu\nu,\kappa\lambda} = -T_{\mu\lambda,\kappa\nu}. \tag{PH12}$$

The second identity follows by interchanging the two Hessian arguments and repeating the first. These are exactly the two antisymmetries which remove symmetric gradients.

The bulk sum and its second moment. Unit translation covariance of this individual member gives $K_z^{\mu\nu}(x,y) = K_0^{\mu\nu}(x-z, y-z)$. The bulk quadratic kernel is

$$\Pi^{\mu\nu}(x,y) = \sum_z K_z^{\mu\nu}(x,y) = \pi_{\mu\nu}(y-x), \qquad \pi_{\mu\nu}(u) = \sum_x K_0^{\mu\nu}(x, x+u). \tag{PH13}$$

All sums converge absolutely. In fact, for $0 < m' \le m$,

$$\sum_u e^{m'|u|} |\pi_{\mu\nu}(u)| \le \sum_{x,y} e^{m'(|x|+|y|)} |K_0^{\mu\nu}(x,y)| < \infty. \tag{PH14}$$

For fixed x, y, translation of z in (PH13) is a sub-sum of the same absolute double sum. This proves both the bulk formula and every rearrangement below. Only this bulk object is a convolution. The original $K_0(x,y)$ continues to have two independent spatial arguments.

The zeroth and first bulk moments vanish by (PH10). Its second moment is

$$\begin{aligned} P_{\mu\nu,\kappa\lambda} &:= \sum_u \pi_{\mu\nu}(u) u_\kappa u_\lambda \\ &= \sum_{x,y} K_0^{\mu\nu}(x,y)(y_\kappa - x_\kappa)(y_\lambda - x_\lambda) \\ &= -T_{\mu\nu,\kappa\lambda} - T_{\mu\nu,\lambda\kappa}. \end{aligned} \tag{PH15}$$

The pure $y_\kappa y_\lambda$ term is zero after summing x by its row identity, and the pure $x_\kappa x_\lambda$ term is zero after summing y. This is the reason for both minus signs; no pointwise convolution identity for K_0 has been used.

The Fourier series $\widehat{\pi}_{\mu\nu}(k) = \sum_u \pi_{\mu\nu}(u) e^{ik\cdot u}$ is holomorphic for sufficiently small $|\operatorname{Im} k|$ by (PH14). In particular $\partial_\kappa \partial_\lambda \widehat{\pi}_{\mu\nu}(0) = -P_{\mu\nu,\kappa\lambda}$. This optional Fourier description concerns the bulk sum alone.

Bond midpoints and the hypercubic tensor. Under reflections, lower bond starts undergo a direction-dependent integer displacement. To apply tensor symmetry without dropping that displacement, move first to bond-midpoint coordinates $\widetilde{x} = x + e_\mu/2$ and $\widetilde{y} = y + e_\nu/2$. The value of T does not change: expanding $\widetilde{x}_\kappa \widetilde{y}_\lambda$ adds only terms of the forms $\sum_{x,y} K_0 y_\lambda$, $\sum_{x,y} K_0 x_\kappa$ and $\sum_{x,y} K_0$, all zero by (PH10). The displacement u changes by the constant $(e_\nu - e_\mu)/2$. Because the bulk zeroth and first moments are zero, its second moment P is also unchanged. Thus the lower-start extractor and the midpoint extractor agree exactly.

The midpoint bond sets are $\mathbb{Z}^4 + e_\mu/2$. A signed permutation w with $we_\mu = s_\mu e_{\sigma(\mu)}$ maps this set to the midpoint set of direction $\sigma(\mu)$; a negative orientation contributes the sign s_μ to a one-form. Pointed covariance at the fixed base point zero and the ordinary transformation of the midpoint coordinates therefore give

$$T_{\sigma(\mu)\sigma(\nu),\sigma(\kappa)\sigma(\lambda)} = s_\mu s_\nu s_\kappa s_\lambda T_{\mu\nu,\kappa\lambda}. \tag{PH16}$$

There is no affine correction left in this formula; the preceding moment calculation proved that it vanished.

We now classify this tensor directly. By (PH12) it is zero if $\mu = \kappa$ or $\nu = \lambda$. Reflecting one coordinate in (PH16) shows that each coordinate label must occur an even number of times among the four indices of a nonzero entry. The two antisymmetric pairs must consequently be the same

unordered pair. The only possibilities are $T_{\mu\mu,\kappa\kappa}$ and $T_{\mu\kappa,\kappa\mu}$ for $\mu \neq \kappa$. Permutations make all the first values equal to a number b; (PH12) makes the second values $-b$. Hence

$$T_{\mu\nu,\kappa\lambda} = b\big(\delta_{\mu\nu}\delta_{\kappa\lambda} - \delta_{\mu\lambda}\delta_{\kappa\nu}\big). \tag{PH17}$$

Full reflections are used here: four distinct indices, including a possible alternating tensor, are excluded by a single reflection. This is not merely an argument using proper rotations.

Combining (PH15) with (PH17) gives

$$P_{\mu\nu,\kappa\lambda} = b\big(\delta_{\mu\lambda}\delta_{\kappa\nu} + \delta_{\mu\kappa}\delta_{\lambda\nu} - 2\delta_{\mu\nu}\delta_{\kappa\lambda}\big), \qquad b = P_{\mu\nu,\mu\nu} \quad (\mu \neq \nu). \tag{PH18}$$

Thus the extractor is explicitly $b[G] = \sum_u \pi_{\mu\nu}(u) u_\mu u_\nu$ for any distinct μ, ν. It has no extra color factor, because (PH6) used the average of the three diagonal color Hessians. The quadratic term of the bulk symbol is $b(|k|^2\delta_{\mu\nu} - k_\mu k_\nu)$; higher lattice derivative terms have not been set to zero.

For fixed actual h, the extractor is complex linear in the member and ignores additive constants. It is continuous in the exponential Hessian moment norm: the absolute second moment is bounded by that norm times $\sup_{r\geq 0} r^2 e^{-mr}$, with a harmless choice of a smaller exponential weight. Normal convergence in that norm also permits termwise extraction and preserves holomorphy in a member parameter. These are Hessian-norm statements, and make no assertion that the Hessian norm is uniformly bounded by a potential norm on coupling-dependent shrinking cores.

The affine quadratic jet and the tent action. Let $B_\mu(x) = a_\mu + \sum_\kappa C_{\mu\kappa}(x - z)_\kappa$, using midpoint coordinates or lower starts. The difference is a constant one-form, which the Ward identity annihilates. Equation (PH10) removes every occurrence of a, and (PH17) proves

$$\frac{1}{2}\mathcal{H}_z(B, B) = \frac{b}{2}\sum_{\mu<\nu} |C_{\nu\mu} - C_{\mu\nu}|^2. \tag{PH19}$$

For complex fields the squared norm means the bilinear color pairing; for real fields it is the ordinary squared norm. The identity follows by expanding $\sum_{\mu,\kappa} |C_{\mu\kappa}|^2 - \sum_{\mu,\nu}\langle C_{\mu\nu}, C_{\nu\mu}\rangle$ into unordered pairs. In particular the axial affine choice $\ell_\mu(x) = \sum_{\kappa<\mu}(x - z)_\kappa F_{\kappa\mu}$ has curl F and quadratic jet $\frac{b}{2}\sum_{\mu<\nu} |F_{\mu\nu}|^2$.

To identify this with the member's own action counterterm, one more normalization of the actual free response is needed. Assume that on an affine coarse field ℓ of constant curl it reproduces the same affine bare potential modulo a lattice gradient, and its current component is zero. This is an affine-response statement about the actual averaging/background problem, not a consequence of the tensor classification. For this normalization argument assume also that, at each bare bond in a bounded coarse region, the matrix elements of H in the coarse input bond decay exponentially in its distance from that region. Their constants must be uniform in its finitely many bare bonds at fixed η. This is a precise sufficient free-response input. It makes $H(\ell\vartheta(\cdot/R))$ converge at every such bond to an absolutely convergent polynomially weighted matrix sum, by dominated convergence. Thus its affine reproduction statement is an equality of well-defined extended first responses, with no nonlinear affine field inserted outside its analytic domain.

This input also gives the moment bound needed to apply the tensor theorem to the pointed action itself. Its Hessian is a finite sum over the bare plaquettes in a unit tent of products of first differences of H. The exponential bounds at their finitely many bonds give a product bound $C_\eta e^{-m_H(|x-z|+|y-z|)}$. Summing in the two input bonds with any smaller exponential weight proves (PH5) for this action. This absolute convergence statement does not assert that C_η has a uniform cutoff bound; any such bound is a further quantitative input.

There is a useful stronger scope. Write $d_\eta H(p) = \eta^{-1}\sum_{b\in\partial p} H(b)$ with oriented signs. If the actual response satisfies the physical first-difference bound

$$|d_\eta H \delta^a_{x,\mu}(p)| \leq C e^{-m_H|x-z|} \qquad (h_z(p) > 0),$$

uniformly in η, then the moment constant for the pointed action is uniform in η as well. Indeed, its kernel contains

$$\sum_p \eta^4 h_z(p)(d_\eta H\delta_x)(p)\cdot(d_\eta H\delta_y)(p).$$

The exact quadrature proved below makes the weighted plaquette sum one in each orientation. Consequently this kernel is bounded by $Ce^{-m_H(|x-z|+|y-z|)}$, and any smaller exponential moment is summable uniformly. This explains precisely why physical first differences, rather than identity-only bounds on bare values, remove the possible C_η loss.

Let $\eta = 1/N_0$ for an integer $N_0 \geq 1$, and use the midpoint tents

$$h_z(p) = \prod_{r=1}^{4}(1-|p_r - z_r|)_+, \qquad A_z(U) = \sum_{\mu<\nu}\sum_{p_{\mu\nu}} h_z(p)A_p(U). \tag{PH20}$$

For each fixed plaquette orientation,

$$\eta^4 \sum_{p_{\mu\nu}} h_z(p) = 1. \tag{PH21}$$

Here its barycenters have half-mesh shifts in directions μ,ν and zero shifts in the other directions. The one-dimensional unshifted identity is $N_0^{-1}[1+2\sum_{r=1}^{N_0-1}(1-r/N_0)] = 1$. For the half-mesh shift it is $2N_0^{-1}\sum_{r=0}^{N_0-1}(1-(r+1/2)/N_0) = 1$. Multiplication of these four identities proves (PH21), including the coarse case $N_0 = 1$. Also $\sum_z h_z(p) = 1$, since the two neighboring one-dimensional tents sum to one in each coordinate.

For a bare affine potential of curl F, the first variation of a plaquette holonomy in the chart $U_b = e^{it\eta A_b}$ is $it\eta^2 F_{\mu\nu}$. Equation (PH1), gradient invariance and (PH21) therefore give

$$\frac{1}{2}D^2(A_z\circ\mathcal{P})(0)[\ell,\ell] = \frac{1}{2}\sum_{\mu<\nu}|F_{\mu\nu}|^2. \tag{PH22}$$

The affine-response hypothesis was used exactly in this line. Using the polynomial Ward identity to remove symmetric gradients and constants, polarization of (PH22) determines its affine tensor directly as (PH17) with $b = 1$. Writing K_z^A for its color-averaged composed Hessian, the same gauge Ward calculation gives its separate row and column zeroth moments. Applying the algebra of (PH15) to its absolutely convergent pointed contraction $\sum_x K_z^A(x, x+u)$ therefore gives $P^A_{\mu\nu,\mu\nu} = 1$ for $\mu \neq \nu$, the moment extractor $b[A_z] = 1$. Identifying this contraction with the full bulk sum over base points additionally requires translation covariance of $A_z\circ\mathcal{P}$; applying the full tensor theorem also requires its stated hypercubic covariance. Gauge covariance alone does not supply those spacetime properties. The direct identities (PH19) and (PH22) already give

$$D^2\big(F_z - b[G](A_z\circ\mathcal{P})\big)(0)[\ell,\ell] = 0 \tag{PH23}$$

for every affine constant-curvature field. Polarization gives the corresponding mixed affine identity. This proves the exact extractor-to-counterterm match with the normalization $A(e^{iB}) = \frac{1}{2}|dB|^2 + O(B^3)$.

Selecting different members at different sites. The proof can be applied separately to each member $G^{[z]}$ chosen at a site. Within that member the family at all hypothetical base points is translation covariant, so its own bulk extractor $b[G^{[z]}]$ is defined by (PH13) and (PH18). Its pointed total at the actual site z is canceled by that number times A_z, by (PH23). No translation covariance of the array $z \mapsto G^{[z]}$ has been used.

This distinction is necessary for an inhomogeneous scalar source: one may freeze it to a different constant-source member at each site, but must use the coefficient of that same member at that site. Equality of two unpointed totals alone does not identify their separate site weights. The

nonlinear third-order and Hölder remainder uses the actual localized response, uniform field-domain inclusion and analytic derivative bounds stated for the window.

24. The nonlinear remainder in a pointed window

The algebraic cancellation of a pointed Hessian concerns the second derivative at the identity. Its use at a nonlinear background requires a local Taylor estimate. We prove that estimate from the background hypotheses, retaining the domain assertions that make the compositions meaningful. In particular, the datum is large compared with its value close to the base point. A Taylor bound using only its global supremum would lose the required power of the scale ratio.

The background obligations and their coordinate radii. Work first in units in which the coarse spacing of a single-scale window is one. The background package A1–A3 is used in the following explicit form. On a box with collars there is a holomorphic potential/current representative $\Phi_\Lambda(B)$, with $\Phi_\Lambda(0) = 0$, for the actual constrained critical orbit. Its datum domain contains

$$\mathcal{B}_{r_B} = \{B : \text{determining bonds} \longrightarrow \mathfrak{su}(2)^{\mathbb{C}} : \sup_c |B(c)| < r_B\}. \tag{PW1}$$

Here $B(c)$ is a dimensionless logarithm of a group-valued coarse bond. This is a product of balls at bonds. It has no coupled derivative restriction. The orbit representative, its current, and the response estimates hold throughout this complex domain. Core and tower memberships under restriction and substitution are also part of A1; they are not inferred from the formal existence of a logarithm. All differentiations and coordinate sums below are first carried out on finite boxes. Their uniform bounds permit the stated coherent infinite-volume limits; no representation of a derivative on an infinite ℓ^∞ space by coordinate derivatives is assumed.

For clarity, the output norm over a top of local spacing ξ_y is

$$Z_y = (\xi_y H, \xi_y^2 \nabla H, \xi_y^3 J), \qquad |Z_y| = \max\{\xi_y|H|,\ \xi_y^2|\nabla H|,\ \xi_y^3|J|\}. \tag{PW2}$$

The first-response bound supplied by A3 is

$$|\partial_{B(c)} Z_y(B)[e]| \le C e^{-md(y,y_c)}|e|, \qquad B \in \mathcal{B}_{r_B}, \tag{PW3}$$

together with the stated spatial Hölder bound for ∇H. The distance is the local-scale path distance. For the window below all tops have the same scale; after rescaling, (PW3) uses ordinary lattice distance and all the output weights are one. Constants may depend on the fixed window size, the decay rate and the Hölder exponent, but not on the bare mesh, the torus or the number of preceding steps.

For the finite-box comparison we also use the background package's compatibility of the actual linearised equations and their boundary rows. In the natural force norms, the A3 inverse bound from one coarse pillar y' to y is $Ce^{-md(y,y')}$, including the output differences and current used here. A cutoff of an affine solution leaves a forcing of polynomial size supported near the boundary, or with an exponentially decaying boundary tail if a gauge row is exponentially local. This boundary-locality assertion must hold for the actual gauge and constraint operators. It follows by commuting the cutoff through their local enlarged saddle equations, or through their compatible exponentially local rows. It is not a consequence of (PW1)–(PW3) for unrelated maps. The single-pillar inverse estimate includes the integration or cell-volume weights in its natural norms; a bare entrywise bound without those weights would not control a fine-mesh sum.

Choose a fixed basis of the three-dimensional complex colour space. If $\sup|B| \le r_B/2$, varying one basis coordinate by $|s| < c_{\mathrm{col}} r_B/2$ stays in (PW1), where $c_{\mathrm{col}} > 0$ depends only on this choice of norm and basis. Thus there is an admissible coordinate Cauchy radius independent of the number of bonds. The factors ξ_y in (PW2) are output weights and commute with this Cauchy estimate. If instead a physical potential datum is written $a(c) = B(c)/\xi_c$, its coordinate radius is

$c_{\mathrm{col}} r_B/(2\xi_c)$, and $\partial_{a(c)} = \xi_c \partial_{B(c)}$. Contracting a second derivative against $a(c)a(c')$ cancels both input factors $\xi_c, \xi_{c'}$. No inverse local spacing is lost by using dimensionless bond coordinates.

Lemma 24.1 (Product decay for the second response)**.** *On $\mathcal{B}_{r_B/2}$, the first-response hypothesis implies*

$$|\partial_{B(c)}\partial_{B(c')} Z_y(B)[e, e']| \le \frac{C}{r_B} e^{-\frac{m}{2}(d(y,y_c)+d(y,y_{c'}))} |e|\,|e'|. \tag{PW4}$$

The same conclusion holds for the local first-order pair norm on a domain X, with $d(y, y_c)$ replaced by $\operatorname{dist}(c, X)$.

Proof. Apply the coordinate Cauchy estimate to $\partial_{B(c)} Z_y(B)[e]$, using a slightly smaller fixed fraction of the radius just specified. This gives $C r_B^{-1} e^{-m d(y,y_c)} |e|\,|e'|$. The finite-dimensional holomorphic map has symmetric second derivatives, so the same argument interchanging c, c' gives the corresponding bound with c'. The smaller of two positive bounds is at most their geometric mean, which proves (PW4). Taking the supremum over the output bonds, first differences and current components gives the stated local norm version. Colour indices cost only a fixed factor. □

This argument uses response decay at every complex datum in a ball. Decay of the linear response at $B = 0$ alone is insufficient.

Weighted polynomial data and the local expansion. Fix a base point z, a scale ratio $0 < \sigma \le 1$, and a window whose distance from z to its boundary is comparable with σ^{-1}. Write a for a radius of a regular member, a_{out} for the output field radius and

$$\rho = a_{\mathrm{out}}/a \ge 1, \qquad a \le C_0 r_B, \qquad a_{\mathrm{out}} \le c_0, \qquad 0 < \alpha \le 1. \tag{PW5}$$

The constant c_0 is chosen small enough for the matrix charts. A fixed constant C_{ax} in the axial datum estimate below is understood. For the local Taylor argument we require $C_{\mathrm{ax}} a_{\mathrm{out}} \sigma \le r_B/2$. This is a global input-domain condition, distinct from the local radius-ratio dichotomy used later. The ordinary age bound will deal with its failure at the finitely many young ages. A harmless enlargement of the radius ratio to at least one is allowed. For a domain X containing z, let $D = 2 + \operatorname{diam} X$, in the present units.

The local regularity conclusion of A1–A3 for the combined background and fluctuation gives, on the box before choosing its axial gauge, a logarithm B with

$$|B| \le C a_{\mathrm{out}} \sigma, \quad |\nabla B| \le C a_{\mathrm{out}} \sigma^2, \quad |\nabla B(x) - \nabla B(x')| \le C a_{\mathrm{out}} \sigma^{2+\alpha} |x - x'|^\alpha. \tag{PW6}$$

These are spatial regularity bounds, not consequences of holomorphy alone. All statements in this section require the actual pair and its restrictions to remain in the member's prescribed core. The small chart ball used for Taylor's formula must be contained in that core. The radius a is a fixed fraction of its curvature radius. Throughout the admitted core $|\partial \mathbf{U}(p) - 1| \le C a \eta_j^2$, so its own tent action obeys $|A(h_z, \mathbf{U})| \le C a^2$, by the determinant identity and exact tent plaquette count. This is a bound on the whole core, not just its contained chart ball.

In the axial gauge about z, the datum has a local form

$$\widehat{B}(c) = \ell_F(c) + r(c), \qquad \ell_{F,\mu}(x) = \sum_{\kappa<\mu} (x - z)_\kappa F_{\kappa\mu}, \tag{PW7}$$

where $|F| \le C a_{\mathrm{out}} \sigma^2$, and

$$|\widehat{B}(c)| \le C a_{\mathrm{out}} \sigma^2 (1 + |c - z|), \qquad |r(c)| \le C a_{\mathrm{out}} \sigma^{2+\alpha} (1 + |c - z|)^2. \tag{PW8}$$

The first bound also gives $\sup |\widehat{B}| \le C_{\mathrm{ax}} a_{\mathrm{out}} \sigma$, so the stated global input-domain condition places $\widehat{B}$ and its segment from zero in $\mathcal{B}_{r_B/2}$.

Here are details of (PW7)–(PW8). Fixing the transports along a coordinate comb to the identity expresses a remaining bond as a product of at most $C(1+|c-z|)$ transported plaquette holonomies. With constant plaquette strengths, the linear part of this product is exactly ℓ_F. Equation (PW6) shows that a plaquette logarithm differs from its value at z by at most

$$C\big[a_{\mathrm{out}}\sigma^{2+\alpha}(1+|c-z|)^{\alpha}+a_{\mathrm{out}}^2\sigma^3(1+|c-z|)\big].$$

The second term includes differentiation of the non-Abelian commutator. Conjugating a plaquette logarithm by a comb transport adds at most $Ca_{\mathrm{out}}^2\sigma^3(1+|c-z|)$. Since a_{out} is bounded and $\sigma^{1-\alpha}\le 1$, both errors are bounded by $Ca_{\mathrm{out}}\sigma^{2+\alpha}(1+|c-z|)$. Summing along the strip proves the second bound in (PW8).

For completeness, the finite-product estimate used here follows by differentiating $P(t)=\prod_i e^{tX_i}$ twice: if $\sum_i|X_i|\le c$, then $\sup_{0\le t\le 1}|P''(t)|\le C(\sum_i|X_i|)^2$. The analytic logarithm in this fixed matrix ball has bounded first two derivatives, so

$$\left|\log\prod_i e^{X_i}-\sum_i X_i\right|\le C\left(\sum_i|X_i|\right)^2.$$

Its error here is at most $Ca_{\mathrm{out}}^2\sigma^4(1+|c-z|)^2$, which is absorbed in (PW8). The unexpanded product bound gives the first bound of (PW8). All transports stay in a fixed matrix ball, because the original comb has length $O(\sigma^{-1})$ and its original bond logarithms are $O(a_{\mathrm{out}}\sigma)$.

We shall use the elementary weighted sum

$$\sum_{c\in\mathbb{Z}^4} e^{-\mu\,\mathrm{dist}(c,X)}(1+|c-z|)^k\le C_{\mu,k}D^{k+4},\qquad k\ge 0. \tag{PW9}$$

Indeed $|c-z|\le D+\mathrm{dist}(c,X)+C$. The shell at distance between q and $q+1$ is contained in a box with at most $C(D+q+1)^4$ bonds. Bounding each shell by this larger number and summing $e^{-\mu q}(D+q+1)^{k+4}$ proves (PW9). Only a polynomial in D is spent. Every power needed below is fixed.

Integrate the first response on the segment $t\widehat{B}$, and then use (PW3), (PW8) and (PW9). With a fixed integer s, increased below when necessary, this gives

$$\|\Phi_\Lambda(\widehat{B})\|_X\le CD^s a_{\mathrm{out}}\sigma^2,\qquad \|D\Phi_\Lambda(0)r\|_X\le CD^s a_{\mathrm{out}}\sigma^{2+\alpha}. \tag{PW10}$$

Taylor's formula with integral remainder for Φ_Λ, followed by (PW4) and the product of the two weighted sums, gives

$$\|\Phi_\Lambda(\widehat{B})-D\Phi_\Lambda(0)\widehat{B}\|_X\le\frac{CD^s}{r_B}a_{\mathrm{out}}^2\sigma^4. \tag{PW11}$$

The use of (PW9) in (PW11) is essential. Using only $\sup|\widehat{B}|=O(a_{\mathrm{out}}\sigma)$ would produce the wrong power σ^2 in its right-hand side.

The finite-box linear response. The free response reproduces an affine constant-curvature field, modulo a gauge direction, with zero linear current. This follows from the actual linearised equations: the affine field has constant curl, zero current and zero divergence; centred block averages reproduce it with the correct bond-length normalization. It therefore solves the stationarity, constraint and gauge equations. For the coherent response, the finite-box comparison below identifies its affine extension by exhaustion and coordinatewise dominated convergence; no uniqueness assertion in a class of unbounded fields is required. The response is applied only inside gauge-invariant Hessians, so its gauge direction has no effect.

We also explain the finite-box error instead of identifying a box response with an infinite response. The two responses solve the same local linearised equations in the interior. Multiply the affine solution by a cutoff equal to one on the interior and zero near the box boundary. Its first

two finite differences are bounded in the present scale units; it can be obtained by sampling a fixed smooth profile across a boundary layer of fixed coarse width. The cut-off affine pair satisfies the imposed homogeneous boundary conditions. Its discrepancy from the prescribed equations and constraints is supported in that boundary layer. The equations and their boundary conditions give a bound $C|F|(R+1)^s$ for that forcing, where R is the box radius. Here the constraint rows, possibly exponentially local gauge rows and their differences are measured in the local norms of A2–A3. Apply the inverse bound of A3 from this boundary layer to X. Decompose the forcing into coarse pillars, so the sum has unit-scale density rather than bare-mesh density. For an exponentially local boundary tail use $d(x,\partial\Lambda) \le d(x,y) + d(y,\partial\Lambda)$; splitting the two decay exponents in half and summing a polynomial unit-scale shell gives the same bound with a smaller fixed rate. When $\operatorname{diam} X \le \min\{R/4, \operatorname{dist}(z,\partial\Lambda)/2\}$, every point of X is at distance at least $\operatorname{dist}(z,\partial\Lambda)/2$ from the boundary. For $R \asymp \sigma^{-1}$, this distance is comparable with R, and the inverse estimate gives

$$\|D\Phi_\Lambda(0)\ell_F - L_F\|_X \le C|F|(R+1)^s e^{-cR}, \tag{PW12}$$

where L_F is the infinite affine response pair. Polynomial boundary counts, and the finitely many local differences used to form the current, are absorbed in s. This argument uses agreement of the actual local equations; bounds for unrelated kernels on two regions would not imply (PW12).

For $R \asymp \sigma^{-1}$, $(R+1)^s e^{-cR} \le C_\alpha \sigma^\alpha$. Combining (PW7), (PW10)–(PW12), we obtain

$$\begin{aligned} \Phi_\Lambda(\widehat{B}) &= L_F + E_X, \qquad \|L_F\|_X \le C D^s a_{\mathrm{out}} \sigma^2, \\ \|E_X\|_X &\le C D^s \left[a_{\mathrm{out}} \sigma^{2+\alpha} + r_B^{-1} a_{\mathrm{out}}^2 \sigma^4 \right]. \end{aligned} \tag{PW13}$$

Domains too large for the interior condition in (PW12) are put into the exponentially long tail. In applications their tree length is at least $c/\sigma - C$.

Taylor's formula for the actual member. Let G_X denote a regular pointed member on X, in its exponential potential/current chart. Assume its supremum on that core is $g_X = \mathtt{g} e^{-\kappa d_j(X)}$, and that its chart contains the ball of radius a. A fixed change of constants allows the use of a ball of radius $2a$ and estimates on its half ball; this is the specified room in the field domains. Constant gauge transformations act by the adjoint representation on both chart components. There is no invariant linear functional on $\mathfrak{su}(2)$, so $DG_X(0) = 0$. In particular the second derivative of the background does not enter the Hessian of this composition:

$$D^2(G_X \circ \Phi_\Lambda)(0)[B,B'] = D^2 G_X(0)[D\Phi_\Lambda(0)B, D\Phi_\Lambda(0)B']. \tag{PW14}$$

This assertion concerns regular members without stored colour spectators.

Theorem 24.2 (Pointed nonlinear window remainder)**.** *Under the background and domain hypotheses above, for the near domains there is a fixed integer s_1 such that*

$$\begin{aligned} G_X(\Phi_\Lambda(\widehat{B})) - G_X(0) &= \tfrac{1}{2} D^2 G_X(0)[L_F, L_F] + \mathcal{R}_X, \\ |\mathcal{R}_X| &\le C \mathtt{g} e^{-\kappa d_j(X)} D^{s_1} \left[\rho^2 \sigma^{4+\alpha} + \rho^3 \sigma^6 \right]. \end{aligned} \tag{PW15}$$

The same estimate holds after absorbing the member's own tent-action counterterm, with its own Hessian subtracted.

Proof. Put $t = D^s \rho \sigma^2$, increasing s to cover every polynomial factor in (PW10)–(PW13). First suppose t is less than a sufficiently small fixed constant. Equation (PW10) then places the composed pair in the half chart ball of G_X. Cauchy's formula and Taylor's theorem give

$$\left| G_X(P) - G_X(0) - \tfrac{1}{2} D^2 G_X(0)[P,P] \right| \le C g_X (\|P\|_X / a)^3, \qquad \|D^2 G_X(0)\| \le C g_X / a^2. \tag{PW16}$$

In (PW13), divide by a, use $a/r_B \le C_0$, and expand the difference of the two quadratic forms. The product of L_F/a and the first error term is bounded by $C D^{2s} \rho^2 \sigma^{4+\alpha}$. The product with the

second error term is bounded by $CD^{2s}\rho^3\sigma^6$. The square of the first error is smaller than the first of these bounds. The square of the second is bounded by Ct^4, which is at most a constant times t^3 in this case. The mixed product is bounded by the sum of the squares. Finally the Taylor remainder in (PW16) is at most Cg_Xt^3. Increasing s_1 proves (PW15).

If t exceeds that fixed small constant, use the actual core bound $|G_X(P) - G_X(0)| \le 2g_X$, rather than Taylor's formula at P. The Hessian estimate still bounds its polynomial value at L_F by Cg_Xt^2. For $t \ge c > 0$, $1 + t^2 \le C_ct^3$, and this proves (PW15) after increasing s_1 again. Thus large radius ratios do not impose an upper bound on the age.

The coefficient of a member obeys $|b(G)| \le C\mathsf{g}/a^2$ by its Hessian moments. The tent action on its fixed local support has norm at most Ca^2. Consequently its counterterm is another holomorphic family of size $C\mathsf{g}$, with bounded support. Apply the same proof to this family and subtract. The affine field and its curvature in the two quadratic terms are the same. □

The exact own-counterterm identity (PH23) of the pointed Hessian cancellation theorem now applies to this last, normalised family: its complete sum of pointed curvature tensors is zero. A sum cut to the window is the complete sum minus the omitted domains. The omitted domains have $d_j(X) \ge c/\sigma - C$. If the number through a cube of length q is at most $Ce^{\kappa_D q}$ with $\kappa > \kappa_D$, polynomial factors in $D \le CM(d_j(X) + 2)$ are summable. Thus (PW15), summed over near domains, gives

$$C\mathsf{g}\big[\rho^2\sigma^{4+\alpha} + \rho^3\sigma^6\big], \tag{PW17}$$

and completing the tensor sum costs at most $C\mathsf{g}\rho^2\sigma^4e^{-c/\sigma}$. If a far domain is not contained in any non-wrapping chart, its value has only the norm bound and contributes an additional $C\mathsf{g}e^{-c/\sigma}$. Such a term has no justified factor proportional to the actual curvature: a flat field around a hole or a torus cycle need not be a pure gauge. It must be retained when a field-sensitive tail estimate is claimed.

For the radius estimate used in a window, $\rho \ge 1$ permits both exponentially small tails to be absorbed in (PW17). Moreover

$$\rho^2\sigma^{4+\alpha} + \rho^3\sigma^6 \le 2\rho^3\sigma^{4+\alpha}, \qquad \rho \ge 1, \quad \sigma \le 1, \quad \alpha \le 1. \tag{PW18}$$

There are $O(\sigma^{-4})$ level-j base points in one window of the later scale. Its remaining factor is therefore $C\mathsf{g}\rho^3\sigma^\alpha$. With $\rho^2 \le C(1+n-j)$ and $\sigma = L^{-(n-j)}$, this is the summable ordinary weight $C\mathsf{g}(1+n-j)^{3/2}L^{-\alpha(n-j)}$. The global input-domain condition also holds at every sufficiently old age. Indeed

$$\frac{C_{\mathrm{ax}}a_{\mathrm{out}}\sigma}{r_B} \le C_{\mathrm{ax}}C_0\rho\sigma \le C(1+n-j)^{1/2}L^{-(n-j)}.$$

Choose a fixed age $a_\star$ beyond which this is at most $1/2$. At each of the finitely many younger ages, the given core membership bounds the member and its own counterterm by $C\mathsf{g}(1+\rho^2\sigma^4)$. Here ρ is uniformly bounded and $\sigma \ge L^{-a_\star}$, so this is bounded by a fixed constant times $\mathsf{g}\rho^3\sigma^{4+\alpha}$. This last argument uses the ordinary age bound; it is not valid for an arbitrary unrelated pair (ρ, σ). Each site may carry a different Euclidean member: cancellation and the estimate are performed at that site with that member's own coefficient.

Proposition 24.3 (Vacuum-only window bound)**.** *For an analytic gauge-invariant member without stored colour spectators, no lattice covariance is needed to obtain*

$$|G_X(\Phi_\Lambda(\widehat{B})) - G_X(0)| \le C\mathsf{g}e^{-\kappa d_j(X)}D^{s_2}\rho^2\sigma^4. \tag{PW19}$$

After the domains in a window have been summed, the bound is $C\mathsf{g}\rho^2\sigma^4$, with the same qualifications on remote, non-wrapping charts and young ages as above.

Proof. Gauge invariance under constant colour rotations again gives $DG_X(0) = 0$, independently of any spatial symmetry. By (PW10), with $t = D^s\rho\sigma^2$, the local pair lies in the half chart ball whenever $t < c$. Cauchy's estimate for the second-order Taylor remainder then gives $Cg_X t^2$. For $t \geq c$, use the core bound $2g_X \leq C_c g_X t^2$. This proves (PW19), and the domain entropy absorbs its fixed polynomial in D. A remote domain not admitting a chart contributes only $C\mathfrak{g}e^{-c/\sigma}$; it is bounded by $C\mathfrak{g}\rho^2\sigma^4$ for $\rho \geq 1$. For the finitely many young ages at which the global input-disc condition is unavailable, the core bound is absorbed since $\rho \geq 1$ and $\sigma \geq L^{-a_\star}$. Finally the $O(\sigma^{-4})$ sites in the window cancel σ^4, leaving the ordinary weight ρ^2 for a defect or a remainder. □

25. Comparison of the actual localized complex Gaussian functional

The finite regular expansion uses a normalized Gaussian functional on a local set of fluctuation coordinates. Its auxiliary Gaussian can have as many coordinates as the surrounding volume. We prove its comparison bound without paying for that auxiliary dimension. The argument permits sharp, measurable cutoffs; it never shifts their arguments into the complex domain.

The local matrices and the normalized functional. All matrix norms below are operator norms for the Euclidean complex space. Transposition is bilinear transposition, not conjugate transposition. Let $C = C^{\mathsf{T}}$ be an invertible complex m-by-m matrix and let Γ be a complex m-by-n matrix, where $m \geq 1$ and $n \geq 0$. Assume real reference matrices $C_0 = C_0^{\mathsf{T}} > 0$ and Γ_0, and one constant $K \geq 1$, such that

$$\|C_0\| + \|C_0^{-1}\| + \|\Gamma_0\| \leq K, \qquad \|C - C_0\| + \|\Gamma - \Gamma_0\| \leq \varepsilon. \tag{FG1}$$

The number ε will be smaller than a constant depending only on K. No dimension enters that choice. Define

$$D = C + C\Gamma\Gamma^{\mathsf{T}}C, \qquad D_0 = C_0 + C_0\Gamma_0\Gamma_0^{\mathsf{T}}C_0. \tag{FG2}$$

The real matrix D_0 is positive definite. In the small neighborhood in (FG1), normalized Gaussian densities are defined by

$$d\mu_D(b) = (2\pi)^{-m/2}(\det D)^{-1/2}e^{-b^{\mathsf{T}}D^{-1}b/2}\,db, \qquad b \in \mathbb{R}^m. \tag{FG3}$$

The square root is the holomorphic continuation of the positive root at the reference. We justify both convergence and normalization below.

The auxiliary representation of the localized functional is

$$\mathcal{J}_{C,\Gamma}[F] = \int d\mu_{I_n}(\xi)\, e^{-\xi^{\mathsf{T}}\Gamma^{\mathsf{T}}C\Gamma\xi/2} \int d\mu_C(b)\, e^{b^{\mathsf{T}}\Gamma\xi}F(b). \tag{FG4}$$

For $n = 0$ the outer integral is absent. An integrand F need only be Borel measurable on $\mathbb{R}^m$. Suppose

$$|F(b)| \leq F_0 e^{\rho|b|^2}, \qquad F_0 \geq 0, \quad \rho \geq 0. \tag{FG5}$$

Theorem 25.1 (Local Gaussian comparison)**.** *There are positive constants $\varepsilon_0(K), \rho_0(K)$ and A_K, independent of m, n, such that (FG1), $\varepsilon \leq \varepsilon_0$ and $\rho \leq \rho_0$ imply absolute convergence of (FG4),*

$$\mathcal{J}_{C,\Gamma}[F] = \int F\, d\mu_D, \qquad \mathcal{J}_{C,\Gamma}[1] = 1, \qquad |\mathcal{J}_{C,\Gamma}[F]| \leq F_0 e^{A_K(\varepsilon+\rho)m}. \tag{FG6}$$

The last estimate also bounds $\int |F|\,|d\mu_D|$. In particular its exponential cost uses the local dimension m, even when the auxiliary dimension n grows with the full volume.

Accretivity, integration and the determinant branch.

Proof. The joint precision in (FG4) is the symmetric block matrix

$$Q = \begin{pmatrix} C^{-1} & -\Gamma \\ -\Gamma^{\mathsf{T}} & I_n + \Gamma^{\mathsf{T}} C \Gamma \end{pmatrix}. \tag{FG7}$$

At the real reference it factors as

$$Q_0 = L_0^{\mathsf{T}} \begin{pmatrix} C_0^{-1} & 0 \\ 0 & I_n \end{pmatrix} L_0, \qquad L_0 = \begin{pmatrix} I_m & -C_0\Gamma_0 \\ 0 & I_n \end{pmatrix}. \tag{FG8}$$

Both $\|L_0\|$ and $\|L_0^{-1}\|$ are at most $1 + K^2$. Thus $Q_0 \geq c_K I$ for a positive c_K. The resolvent identity for $C^{-1} - C_0^{-1}$, followed by expanding the products in (FG7), gives $\|Q - Q_0\| \leq A_K \varepsilon$ for small ε; the constants depend only on K. For real (b, ξ), consequently,

$$\operatorname{Re}(b,\xi)^{\mathsf{T}} Q (b,\xi) \geq (c_K/2)(|b|^2 + |\xi|^2).$$

Choose $\rho_0 < c_K/8$. This proves absolute convergence of the joint integral with (FG5), hence Fubini's theorem applies. It does not yet give the desired dimension bound; taking its absolute value before integrating ξ would pay unnecessarily for n.

Set $A = I_n + \Gamma^{\mathsf{T}} C \Gamma$. It is a small perturbation of the real positive matrix $I_n + \Gamma_0^{\mathsf{T}} C_0 \Gamma_0$. Integrate ξ first. The normalized Gaussian exponential formula gives the factor

$$(\det A)^{-1/2} \exp\{b^{\mathsf{T}} \Gamma A^{-1} \Gamma^{\mathsf{T}} b/2\}.$$

This formula for a complex linear force does not require a contour change. One proof normalizes the complex Gaussian first and differentiates its exponential transform in the force. Integration by parts, justified by accretivity, gives $\nabla_\ell G = A^{-1}\ell G$, with $G(0) = 1$, and hence $G(\ell) = e^{\ell^{\mathsf{T}} A^{-1} \ell/2}$.

For completeness, the normalization used in that proof follows without a complex change of variables. On an accretive symmetric precision neighborhood the real-space integral is holomorphic by a fixed Gaussian majorant. It equals $(\det A)^{-1/2}$ at every real positive symmetric matrix there by real orthogonal diagonalization and the one-dimensional real Gaussian integral. Successive one-variable identity theorems in the independent symmetric matrix entries extend the equality to the complex neighborhood, with the root continued from its positive value. The real symmetric matrices have a full set of real coordinates for this argument; no identity is being extended from a lower-dimensional background graph.

The elementary inverse and determinant identities are

$$\begin{aligned} D^{-1} &= C^{-1} - \Gamma (I_n + \Gamma^{\mathsf{T}} C \Gamma)^{-1} \Gamma^{\mathsf{T}}, \\ \det D &= \det C \, \det(I_n + \Gamma^{\mathsf{T}} C \Gamma). \end{aligned} \tag{FG9}$$

The inverse identity follows by multiplication with (FG2). For the determinant identity, eliminate either off-diagonal block in $\left(\begin{smallmatrix} I_m & -C\Gamma \\ \Gamma^{\mathsf{T}} & I_n \end{smallmatrix}\right)$; this proves $\det(I_m + C\Gamma\Gamma^{\mathsf{T}}) = \det(I_n + \Gamma^{\mathsf{T}} C \Gamma)$. The roots in (FG9) agree at the real reference, and agree throughout the connected small perturbation neighborhood by continuity and nonvanishing. The remaining b integral is precisely (FG3). This proves the first identity in (FG6), including for Borel F, since no complex argument of F was introduced. The same normalization argument gives $\int d\mu_D = 1$.

The local-dimension estimate. The reference bounds imply $D_0 \geq C_0$, $\|D_0\| \leq A_K$ and $\|D_0^{-1}\| \leq K$. Expanding (FG2) yields

$$T = D_0^{-1/2}(D - D_0)D_0^{-1/2}, \qquad \|T\| \leq A_K \varepsilon =: \eta. \tag{FG10}$$

Shrink ε_0 so that $\eta \le 1/4$. Then $D^{-1} = D_0^{-1/2}(I+T)^{-1}D_0^{-1/2}$ and $\|(I+T)^{-1} - I\| \le \eta/(1-\eta)$. For the real symmetric matrix

$$R = D_0^{1/2}\big(\operatorname{Re} D^{-1} - 2\rho I_m\big)D_0^{1/2}$$

we obtain $\|R - I_m\| \le A_K(\varepsilon + \rho) =: u$. Shrink the constants further so that $u \le 1/2$. Taking the absolute value of the *marginal* density, and then integrating the real positive Gaussian, gives

$$\int e^{\rho|b|^2}|d\mu_D(b)| = |\det(I+T)|^{-1/2}(\det R)^{-1/2} \le (1-\eta)^{-m/2}(1-u)^{-m/2}. \tag{FG11}$$

Every singular value of $I+T$ is at least $1-\eta$, and every eigenvalue of R is at least $1-u$. Finally $-\log(1-v) \le 2v$ for $0 \le v \le 1/2$ proves (FG6). □

The matrices supplied by the regular step. We now identify the matrices to which the comparison applies. Let $P = (\mathbf{U}, \mathbf{J})$ be a regular complex pair. In the notation of Section 13, put $V = \mathcal{M}^k\mathbf{U}$ and

$$\Delta_k(P) = (Q\mathsf{G}_1Q^{\mathsf{T}})^{-1} - a - \mathsf{E}^{(2)}[\mathsf{H}_1^{\mathsf{T}}\mathbf{J}], \tag{25.1}$$

$$\mathsf{A}(P) = \mathsf{C}_V^{\mathsf{T}}\big(\mathrm{GF}_V^{(2)} + \Delta_k(P)\big)\mathsf{C}_V. \tag{25.2}$$

Here C_V eliminates the pivot bonds. The many-scale rows Q come from the straightened average. The quadratic form $\mathsf{E}^{(2)}$ comes from the plain one-step constraint on V. These two averages have different roles.

At a background pair $P_W = (U_+, J(U_+))$, the form in (25.2) is the Hessian on the fibre of $A \circ U_k + \mathrm{GF}_k$. Indeed, the multiplier in the background equation is the derivative of the critical action with respect to its datum. Differentiating that equation gives $(Q\mathsf{G}_1Q^{\mathsf{T}})^{-1} - a$. A curve in the fibre of the next average has a second derivative in its constraint; pairing it with the multiplier gives the last term in (25.1). Thus the Gaussian of the step is $C^{(k)} = \mathsf{A}(P_W)^{-1}$. Soft gauge fixing supplies the positive quadratic form on the orbit directions, on which the critical action has zero Hessian.

The comparison starts with the flat bound $\mathsf{A}(\mathbf{1}, 0) \ge \gamma_0 > 0$. Here and below the constants are independent of the cutoff and the volume. Fix ℓ_0, a sufficiently large fixed multiple of M_1. On each cube of side $O(\ell_0)$ the real factor has a real gauge in which its potential and first difference are bounded by a_{w}. The core supplies this gauge with $a_{\mathrm{w}} = CLMa_0$, where a_0, a_1 denote its two radii. Schwarz estimates first compare $\mathsf{A}(P)$ with $\mathsf{A}(U, 0)$ and then, in this gauge, with the flat operator. The walks that leave the cube cost $Ce^{-c\ell_0}$. For a partition $\sum_i h_i^2 = 1$ on these cubes, the localization formula is

$$\mathsf{A} = \sum_i h_i\mathsf{A}h_i + \frac{1}{2}\sum_i [h_i, [h_i, \mathsf{A}]], \qquad \Big\|\sum_i [h_i, [h_i, \mathsf{A}]]\Big\| \le C\ell_0^{-2}.$$

The double commutator estimate uses the exponential moment of the walk kernels and the slopes of the cutoffs. Consequently

$$\operatorname{Re}\mathsf{A}(P) \ge \gamma_0 - C\ell_0^{-2} - Ce^{-c\ell_0} - C(a_{\mathrm{w}} + a_0 + a_1) \ge \tfrac{1}{2}\gamma_0.$$

Choose ℓ_0, then a_{w}, then $M \ge C\ell_0$, then the radii with $CLMa_0 \le a_{\mathrm{w}}$. Every compression inherits this gap. The same exponential kernel bounds, by conjugation, give uniform decay of the compressed shifted resolvents. Integrating

$$\mathsf{A}^{-1/2} = \frac{1}{\pi}\int_0^\infty y^{-1/2}(y + \mathsf{A})^{-1}\,dy$$

produces the walk-chain expansion of the inverse root.

Let Z_0 be the fine support of an integrand and set $\chi = \chi_{Z_0}$. Only coarse cubes outside the collar $Z_0^\sharp$ carry parameters. Decouple the precision and its inverse root by their respective expansions, and write

$$C(Z_0; s) = (\chi \mathsf{A}(s) \chi)^{-1}, \qquad \Gamma(Z_0; s) = -\chi \mathsf{A}(s)(1 - \chi)\mathsf{r}(s). \tag{25.3}$$

The symbol $\mathsf{r}(s)$ denotes the decoupled inverse-root series; it does not denote $\mathsf{A}(s)^{-1/2}$. The minus sign in Γ matches (FG4). Equivalently one can use the opposite sign and replace the auxiliary white noise by its negative. This change leaves (FG2) and every comparison bound unchanged.

The reference matrices are obtained by setting $s = 0$ and replacing P by $(U, 0)$. They are real, and $C_0^{-1} \geq c\gamma_0$. For independent $|s_\Delta| \leq e^{\bar\kappa}$, a carrier that meets m parameter cubes contributes at most $e^{m\bar\kappa}$. The first 2^4 cubes may meet at a corner. Every further cube requires length proportional to M, whose exponential decay pays for its parameter once $cM \geq C\bar\kappa$. A carrier connecting a controlled row in Z_0 to a parameter must also cross the collar. Its bound contains $e^{C\bar\kappa - cM}$, which is $O(e^{-c'M})$ after M is chosen. The weighted Schur bounds and the resolvent identity therefore give

$$\|C - C_0\| + \|\Gamma - \Gamma_0\| \leq C(e^{-cM} + LMa_0 + a_1), \qquad \|C_0\| + \|C_0^{-1}\| + \|\Gamma_0\| \leq K.$$

The corresponding comparison also holds for C^{-1} and $\Gamma^{\mathsf{T}} C \Gamma$. Taking the radii small after M makes the perturbation as small as required in (FG1), uniformly on these independent discs. A complex coefficient ratio τ changes the structure to $(\tau^{-1}C, \tau^{1/2}\Gamma)$ and adds $C|\tau - 1|$ to the perturbation. The choice $|\tau - 1| \leq c(LM)^{-4}$ suffices for the Gaussian comparison; the free quadratic potential may impose a smaller radius.

Only rows near Z_0 have been compared. Far from Z_0 the mutilated form need not have a gap. The collar is essential even for the reference comparison: without it, the reference rows along a wall of Z_0 can vanish.

Finally set $s = 1$. The Schur complement gives

$$C + C\Gamma\Gamma^{\mathsf{T}} C = \chi \mathsf{A}(P)^{-1} \chi.$$

At $P = P_W$ this is the marginal covariance of the step Gaussian. The local comparison theorem consequently supplies its normalized functional, its holomorphy with fixed real cutoff sets, and the local-coordinate estimate used in (UM4)–(UM6). This construction uses the regular operator and core bounds; it does not assert a Gaussian law for the full nonlinear averaging disintegration.

Sharp failures, holomorphic parameters, and scope. Partition the real coordinates into disjoint bond blocks. If a set P_0 of distinct blocks fails the tests $|b_\beta| < \mathfrak{b}$, then $\sum_{\beta \in P_0} |b_\beta|^2 \geq |P_0|\mathfrak{b}^2$. Therefore, for $\gamma \geq 0$ with $\rho + \gamma/2 \leq \rho_0$, (FG11) gives

$$\int \prod_{\beta \in P_0} \mathbf{1}_{\{|b_\beta| \geq \mathfrak{b}\}} e^{\rho|b|^2} |d\mu_D| \leq \exp\{A_K(\varepsilon + \rho + \gamma/2)m - \gamma \mathfrak{b}^2 |P_0|/2\}. \tag{FG12}$$

This is a product-of-failures estimate, with one Gaussian tail per distinct tested bond. Multiplication by additional indicators only improves the bound.

If the matrices and the potential part of an integrand depend holomorphically on external parameters, and the sharp cutoff sets remain fixed, the same estimates give a parameter-independent Gaussian majorant on each smaller parameter neighborhood. Fubini and Morera then prove holomorphy of the integrals. No holomorphy of a characteristic function in its real field argument is asserted or required.

The rescaled structure $(\tau^{-1}C, \tau^{1/2}\Gamma)$ has effective covariance $\tau^{-1}D$. For real τ in a fixed compact subinterval of $(0, \infty)$, rescale the real reference in the same way; all constants remain uniform. A small complex displacement of τ is included in the matrix perturbation.

In the regular step, let $N_c(Z)$ be the number of coarse localization cubes. The local free-bond color-coordinate count obeys $m \le c(LM)^4 N_c(Z)$. Consequently (FG6) supplies the factor $\exp\{A_K(LM)^4(\varepsilon+\rho)N_c(Z)\}$, and (FG12) supplies the failed-cutoff tail. If a Gaussian comparison instead uses the symbol $N(Z)$ for the local coordinate count itself, its bound is $\exp\{A_K(\varepsilon+\rho)N(Z)\}$, with no second factor $(LM)^4$. These are two conventions for the same local cost. The matrix gap and kernel closeness are the actual A2–A3 obligations, which we must prove; this section proves the resulting finite-dimensional integration estimate.

Completeness is separate. Equality of this local functional with a marginal of one whole-torus Gaussian requires the actual completeness identity for its effective covariance. In the source construction that identity is supplied on complete physical pairs and their physical-data continuation. Arbitrary independent-current cores need only their own local holomorphic functionals and the bounds proved here. This theorem does not manufacture a global covariance on those cores, verify nonlinear background/core estimates, or establish a Gaussian law for actual averaging disintegrations.

26. The unmarked majorant for the actual regular polymer gas

We derive the activity and local logarithm bounds from the finite Mayer, cutoff and coarse-decoupling formulas. The Gaussian estimate is used at a potential budget larger than the physical one. No logarithm or nonvanishing assertion is needed at that larger budget. Only after the activities have been estimated do we apply the proved hard-core polymer theorem.

All bounds below are local on the specified complex potential/current cores. At complete physical pairs the completeness identity also identifies their gas with the regular fluctuation integral. No global Gaussian on an arbitrary independent-current pair is assumed.

The actual formulas and their analytic inputs. Fix a finite four-dimensional periodic cube lattice, with fine localization cubes of side M and coarse cubes of side LM. Both grids tile the torus. Lengths d_f and d_c are tree lengths in fine and coarse cube units. There are a fixed finite number of potential types: the free, intrinsic, already formed older, and optional inserted potential groups may each be combined within their type before the Mayer expansion. Thus this type count is independent of the number of older levels or source profiles. A tagged fine domain Y includes its type in its name; geometry ignores this tag. For the actual local potentials assume

$$|V(Y;P,B)| \le a_Y e^{-\lambda_V d_f(Y)} \left(1 + M^{-4}\sum_{b\subset Y}|B(b)|^2\right), \qquad a_Y \le a,$$
$$a = \sum_{\text{types}} \|\mathbf{V}^{\text{type}}\|_{\lambda_V}. \tag{UM1}$$

The potentials are holomorphic on their declared complex fluctuation boxes and external cores. If a potential is defined only on a box, the corresponding sharp cutoff is kept wherever that potential occurs. The free potential, when present, is defined on all real fluctuation coordinates.

Let $\mathbf{D}$ be a selected set of tagged domains. Write Y_0 for the union of the free-bond sets requiring a field box among those selections, and let P_0 be a set of distinct failed-cutoff bonds outside Y_0. The exact integrand is

$$F_{\mathbf{D},P_0} = (-1)^{|P_0|}\chi_{Y_0}\prod_{b\in P_0}(1-\chi_b)\prod_{Y\in\mathbf{D}}(e^{V(Y)}-1), \qquad \chi_b = \mathbf{1}_{\{|B(b)|<\mathfrak{b}\}}. \tag{UM2}$$

The cutoff product on Y_0 makes every selected factor well defined. Outside that product its integrand is set to zero. This convention introduces no complex argument into a characteristic function.

For any set S of fine cubes, let $\mathcal{N}_f(S)$ consist of S and its fixed first layer of neighbors in the given cubical adjacency. Define $S^\sharp$ to be the union of the coarse cubes containing at least one fine cube of $\mathcal{N}_f(S)$. This is a local cube union, not a rectangular bounding box. A bond uses its incident fine cubes in this definition. The collar preserves unions. A fixed number of additional neighboring cube layers is permitted only if it is part of the assigned collar in every structural identity; its fixed width is then included in the constants.

For the index $\iota = (\mathbf{D}, P_0)$, let $Z_0(\iota)$ be its actual fine support, including the globally defined selected potentials and the failed bonds. Its coarse collar is $Z_0^\sharp$. For $Z \supset Z_0^\sharp$, the actual terms and activities are

$$\begin{aligned} T(Z,\iota) &= \prod_{\Delta \in Z \setminus Z_0^\sharp} \int_0^1 ds_\Delta \, \partial_{s_\Delta} \, \mathcal{J}(Z_0; s_Z)[F_{\mathbf{D},P_0}], \\ H(Z) &= \sum_\iota T(Z,\iota). \end{aligned} \tag{UM3}$$

Here $s_Z = 1$ on $Z_0^\sharp$, has the displayed parameters on the remaining cubes of Z, and is zero elsewhere. Polymers Z are nonempty wall-connected coarse domains. The empty selection contributes only the gas's leading one. It is omitted from H: on a nonempty Z, its term has a derivative of the normalized constant functional $\mathcal{J}[1] = 1$, and is zero. Disconnected regions factor into their wall-components rather than being counted again as single polymers.

The local functional $\mathcal{J}$ has the actual matrix representation and reference bounds of the local Gaussian comparison theorem. Uniformly on independent coarse parameter discs

$$|s_\Delta| < R_s, \qquad R_s > 2, \tag{UM4}$$

its matrix perturbation is at most ε, its gap constant is K, and its local real scalar-coordinate dimension obeys

$$m(Z_0; s_Z) \le c_m (LM)^4 N, \qquad N = \#Z. \tag{UM5}$$

The auxiliary Gaussian dimension is unrestricted. For an integrand bounded by $F_* e^{\rho |B|^2}$, including distinct failed tests P_0, the proved comparison gives

$$|\mathcal{J}[F]| \le F_* \exp\bigl\{ A_K(\varepsilon + \rho + \gamma/2)m - \gamma \mathfrak{b}^2 |P_0|/2 \bigr\} \tag{UM6}$$

when $\gamma > 0$, $\varepsilon \le \varepsilon_0(K)$, and $\rho + \gamma/2 \le \rho_0(K)$. This is the marginal local-coordinate estimate; no factor using the exterior auxiliary dimension is allowed. The potentials, matrices and integrals are jointly holomorphic in the scalar potential couplings and s, with the fixed cutoff sets just described.

The remaining structural hypotheses are the actual heredity, normalized-functional and wall-factorization identities. They identify (UM3) with the finite expansion on complete physical pairs and define coherent local terms on every local complex core. These hypotheses concern the assigned operators and supports. They are not consequences of a numerical Gaussian bound.

The two geometric counts. Let $Y^\sharp$ be the locally defined coarse collar above. All neighborhood widths are fixed independently of L and of the diameter or shape of Y. There is a fixed geometric constant C_h such that

$$\#Y^\sharp \le C_h \left(1 + \frac{d_f(Y)}{L} \right). \tag{UM7}$$

To prove this, take a fine covering tree for Y, with length within one of its infimum, and traverse every edge twice. After adding a bounded initial segment, this gives a closed walk of length at most $C(d_f(Y) + 1)$ passing within a fixed fine-cube distance of every cube of Y. Split the walk into segments of length at most L. Each segment and its fixed fine collar meet a bounded number

of coarse cubes, independent of $L \geq 2$. The number of segments is at most $C(1 + d_f(Y)/L)$. This proves (UM7), also on the periodic graph. A failed bond has a coarse collar with at most a fixed number of cubes. Therefore, writing

$$r = \#\mathbf{D}, \qquad v = \#P_0, \qquad h = \#(Z \setminus Z_0^\sharp),$$

and increasing C_h once,

$$N \leq C_h \left[r + \frac{1}{L} \sum_{Y \in \mathbf{D}} d_f(Y) + v + h \right]. \tag{UM8}$$

There is no term proportional to the surrounding volume.

We also use the rooted fine-domain entropy bound. For every $\xi > \kappa_f$,

$$S_\xi := \sup_{\text{fine cube } q} \sum_{\text{tagged } Y \ni q} e^{-\xi d_f(Y)} < \infty. \tag{UM9}$$

One direct counting proof rounds a covering tree to the bounded-degree graph of fine cubes, traverses its edges twice starting at q, and records which neighboring cubes are selected in Y. The walk has length at most $C(d_f(Y) + 1)$. Its choices and the selection flags are bounded by a fixed constant to that power. This proves exponential rooted entropy, uniformly on the torus. The fixed number of type tags changes that constant only. In particular, if $\mathcal{Y}(Z) = \{Y : Y^\sharp \subset Z\}$,

$$\sum_{Y \in \mathcal{Y}(Z)} e^{-\xi d_f(Y)} \leq c_f L^4 S_\xi N. \tag{UM10}$$

This follows by charging a domain to any one of its fine cubes and summing over all fine cubes in Z; overcounting only enlarges the bound. The number of free bonds whose collars fit in Z is at most $n_b N$, where

$$n_b \leq c_b (LM)^4. \tag{UM11}$$

The bond count in (UM11) and the scalar-coordinate dimension in (UM5) are distinct counts. Neither is counted twice.

A Gaussian-only budget and repeated Schwarz. Choose a positive number W_0, independent of cutoff and volume, small enough for a Gaussian tilt, and choose $\lambda_0 > \kappa_f$. This is not yet the physical potential budget. For every selected tagged domain introduce its own coupling ζ_Y. For $a > 0$ use the radii

$$\mathcal{R}_Y = \frac{W_0}{a} e^{(\lambda_V - \lambda_0) d_f(Y)}. \tag{UM12}$$

We will require $a \leq W_0/2$, so every radius exceeds one. On the product of these discs, the local density bound (UM9) gives

$$\begin{aligned} \sum_{Y \in \mathbf{D}} |\zeta_Y V(Y; P, B)| &\leq W_0 \sum_{Y \in \mathbf{D}} e^{-\lambda_0 d_f(Y)} \left(1 + M^{-4} \sum_{b \subset Y} |B(b)|^2 \right) \\ &\leq C_p W_0 L^4 N + C_p W_0 M^{-4} |B|^2. \end{aligned} \tag{UM13}$$

The constant C_p includes the finite tags and their rooted sums, but no number of selected domains or global volume. The quadratic term uses the rooted sum at each bond. The constant term uses the fine-cube count (UM10).

Choose W_0, γ so that

$$C_p W_0 M^{-4} + \gamma/2 \leq \rho_0(K). \tag{UM14}$$

For complex ζ_Y the Mayer product is bounded by $2^r \exp(\sum_Y |\zeta_Y V(Y)|)$. Apply (UM6) with the tilt in (UM13). Uniformly on both parameter polydiscs its integral is at most

$$2^r e^{C_0 N}\, p^v, \qquad p = e^{-\gamma \mathfrak{b}^2/2}, \quad C_0 = C_p W_0 L^4 + A_K c_m (LM)^4 \big(\varepsilon + C_p W_0 M^{-4} + \gamma/2\big). \tag{UM15}$$

The failed tests are distinct bond blocks, so the tail is p^v. The Gaussian theorem was applied only to a fixed real integration space with measurable cutoffs. There is no zero-freeness assertion on this larger coupling polydisc.

The integral vanishes whenever any selected $\zeta_Y = 0$. Repeated Schwarz on the polydisc consequently divides the bound by $\prod_Y \mathcal{R}_Y$ at $\zeta_Y = 1$. For clarity, division by $\prod_Y \zeta_Y$ has removable singularities; apply the maximum principle on smaller polydiscs and let their radii increase to (UM12). This proves the product factor without a factorial. Mixed coarse Cauchy on circles centered at $s_\Delta \in [0,1]$, of radii tending to $R_s - 1$, then gives the literal term bound

$$|T(Z, \mathbf{D}, P_0)| \le e^{C_0 N} \left(\frac{2a}{W_0}\right)^r e^{-(\lambda_V - \lambda_0) \sum_{Y \in \mathbf{D}} d_f(Y)} p^v (R_s - 1)^{-h}. \tag{UM16}$$

The coupling derivatives and the coarse derivatives commute by joint holomorphy and the same Gaussian domination. If $a = 0$, all nonempty Mayer selections are zero. Apply the same estimate to the cutoff-only terms, with $r = 0$; this is also the limiting interpretation of (UM16).

The activity bound, including its linear amplitude. Fix a desired activity rate $\Lambda > 0$. For wall-connected coarse polymers there is a fixed c_ℓ such that

$$\ell(Z) := d_c(Z) + 5 \le c_\ell N. \tag{UM17}$$

A cube-adjacency spanning tree proves $d_c(Z) \le CN$; the added five is absorbed because $N \ge 1$. Choose

$$K_* = 2 + c_\ell \Lambda, \qquad \Theta = C_h (C_0 + K_*). \tag{UM18}$$

Choose one $\sigma_0 > \kappa_f$, and impose the explicit rate conditions

$$\lambda_V - \lambda_0 \ge \Theta/L + \sigma_0, \qquad \log(R_s - 1) \ge \Theta. \tag{UM19}$$

Using (UM8) to pay for $e^{(C_0 + K_*)N}$, (UM16) becomes

$$|T(Z, \mathbf{D}, P_0)| \le e^{-K_* N} \prod_{Y \in \mathbf{D}} \left(\frac{2a}{W_0} e^{\Theta} e^{-\sigma_0 d_f(Y)}\right) \prod_{b \in P_0} (p e^{\Theta}). \tag{UM20}$$

The unused factor $((R_s - 1)^{-1} e^{\Theta})^h$ is at most one. Every selection has thereby paid its own part of the hull cost. This is the place where the coarse hull estimate retains the favorable $d_f(Y)/L$.

At fixed Z, dropping the restriction $P_0 \cap Y_0 = \emptyset$ only enlarges the nonnegative sum. All selected domains and failed bonds still have collars inside Z, so (UM10)–(UM11) give

$$\begin{aligned} \sum_{\substack{\mathbf{D}, P_0 \\ (\mathbf{D}, P_0) \ne (\emptyset, \emptyset)}} \prod_{Y \in \mathbf{D}} u_Y \prod_{b \in P_0} q &\le \prod_{Y \in \mathcal{Y}(Z)} (1 + u_Y) \prod_{b : b^\sharp \subset Z} (1 + q) - 1 \\ &\le e^{\tau N} - 1, \end{aligned} \tag{UM21}$$

where

$$\begin{aligned} u_Y &= (2a/W_0) e^{\Theta} e^{-\sigma_0 d_f(Y)}, \qquad q = p e^{\Theta}, \\ p_\star &= n_b p, \qquad \tau = e^{\Theta} \left[\frac{2 c_f L^4 S_{\sigma_0}}{W_0} a + p_\star\right]. \end{aligned} \tag{UM22}$$

The products are finite in finite volume. In particular no factor from bonds outside Z has been inserted. The subtraction of one is essential: it retains a selected potential or failed cutoff, rather than creating a unit amplitude for every Z.

Take the physical smallness conditions

$$a \le W_0/2, \qquad \tau \le 1. \tag{UM23}$$

For $N \ge 1$, $e^{\tau N} - 1 \le \tau N e^{\tau N}$ and $N \le e^N$. Equations (UM20)–(UM23) prove

$$\begin{aligned} |H(Z)| &\le e^{-K_* N}(e^{\tau N} - 1) \le \tau e^{-(K_*-2)N} \\ &\le C_\tau (a + p_\star) e^{-\Lambda \ell(Z)}, \\ C_\tau &= e^{\Theta} \max\left\{ \frac{2 c_f L^4 S_{\sigma_0}}{W_0}, 1 \right\}. \end{aligned} \tag{UM24}$$

This is an estimate for the actual activities (UM3). Their smallness has not been assumed. The larger Gaussian budget W_0 was used only to bound the analytic integrals; the physical budget in (UM23) is smaller and is what makes the activity amplitude at most one.

The actual union filing and the local logarithm. Choose a desired logarithmic output rate $\mu > 0$, and set $g_* = \Lambda - \mu$. Let $N \le C_v(d_c(Z) + 1)$, and let the number of coarse polymers containing a fixed cube with $m \le d_c(Z) < m + 1$ be at most $C_D e^{\kappa_D m}$. These geometric bounds follow from bounded-degree cube covering walks as above. There is a fixed c_{adj} such that polymers incompatible with Z contain at least one cube in a set of at most $c_{\mathrm{adj}} N$ cubes: use Z and its wall neighbors.

For these counting estimates use the positive norm majorants

$$h_Z = \sup_{P_Z} |H(Z; P_Z)|, \qquad \widehat{w}_Z = h_Z e^{\mu \ell(Z)},$$

where the supremum is over the declared local core of Z, not over restrictions of one global field. At a fixed core pair P_X on X, actual activities are used only for polymers $Z \subset X$, with their values obtained by the assigned hereditary restrictions. Their weighted absolute values are bounded by $\widehat{w}_Z$. The bound with amplitude $\tau \le 1$ in (UM24) gives

$$\sup_q \sum_{Z \ni q} \widehat{w}_Z e^{\#Z} \le \tau\, C_D e^{2C_v} e^{-5g_*} \sum_{m \ge 0} e^{-(g_* - \kappa_D - C_v)m} =: \tau B_*. \tag{UM25}$$

Choose the rates so that

$$g_* > \kappa_D + C_v, \qquad c_{\mathrm{adj}} B_* \le 1. \tag{UM26}$$

The explicit factor e^{-5g_*} allows the last inequality by increasing the rate gap. Only the upper bound $\tau \le 1$ is needed here. It follows that

$$\sum_{Z' : Z' \not\sim Z} \widehat{w}_{Z'} e^{\#Z'} \le c_{\mathrm{adj}} N \tau B_* \le N. \tag{UM27}$$

This verifies, for the actual activities, the criterion of the already proved hard-core tree theorem, with the root weight $\#Z$. In particular it verifies that criterion on every hereditary subsystem $Z \subset X$, without assigning actual fields outside X.

If two polymers are incompatible, their covering trees can be joined at a common cube or wall with length at most five in four dimensions. For a connected incompatibility list this gives

$$\ell\left(\bigcup_{i=1}^{n} Z_i \right) \le \sum_{i=1}^{n} \ell(Z_i). \tag{UM28}$$

Indeed a spanning tree of the incompatibility graph has $n - 1$ joins, and $d_c(\cup_i Z_i) \le \sum_i d_c(Z_i) + 5(n-1)$. Thus the union filing has the same exponential weight.

Define the actual local logarithmic term by

$$E(X)=\sum_{n\ge1}\frac{1}{n!}\sum_{\cup_i Z_i=X}\phi(Z_1,\dots,Z_n)\prod_i H(Z_i), \tag{UM29}$$

where ϕ is the connected hard-core graph coefficient. For a fixed cube $q\in X$, mark a vertex whose polymer contains q. This only enlarges the absolute series. Symmetry of the ordered list changes its factor $1/n!$ into $1/(n-1)!$. When dropping the union restriction, retain $Z_i\subset X$ for the actual terms at P_X. Use (UM28), and then the rooted tree theorem for that subsystem, dominated by the numerical majorants $\widehat{w}$. Only the subsequent nonnegative majorant sum may be enlarged to all geometric polymers containing q. The result is

$$\begin{aligned} e^{\mu\ell(X)}|E(X)| &\le \sum_{\substack{Z\subset X\\ Z\ni q}}\widehat{w}_Z e^{\#Z}\le \tau B_* \\ &\le C(a+p_\star). \end{aligned} \tag{UM30}$$

There is no factor $\#X$ or ambient volume in this local estimate. It proves the unmarked potential-ball majorant used in the finite insertion theorem.

Uniform convergence on smaller external and coupling domains follows from these same positive majorants. The actual local functions are holomorphic, so their union-filed sum is holomorphic on each declared core. At complete physical pairs, the complete-Gaussian and wall-factorization identities identify the gas with the finite regular integral; the hard-core theorem then gives $\exp(\sum_X E(X))=\Xi\neq 0$. On general local cores the activity and logarithm bounds do not require that global Gaussian interpretation.

Parameter choices and the available insertion slot. All constants above concern fixed geometry and the actual matrix/core package. One must verify (UM4)–(UM6) and (UM14), (UM19), (UM23), (UM26) together. For example, it suffices that the matrix package control $A_K c_m(LM)^4\varepsilon$ by a fixed constant on the chosen decoupling discs. Choose W_0 so its two contributions to C_0 are bounded, and choose positive γ of order at most $(LM)^{-4}$ to bound its contribution. Then C_0 is bounded independently of exterior volume. One can choose the target rate gap, the coarse parameter radius and λ_V by (UM18)–(UM19), and finally make a small and $\mathfrak{b}$ large to obtain (UM23). This lists explicit sufficient inequalities; it does not assert them for a matrix family whose constants have not been controlled on the required enlarged discs.

For an explicit insertion budget, put

$$A_a=\frac{2c_f L^4 S_{\sigma_0}}{W_0},\qquad a_*=\min\left\{\frac{W_0}{2},\frac{1}{4e^{\Theta}A_a}\right\},\qquad p_\star\le\frac{1}{4e^{\Theta}}.$$

Then every total norm $a\le a_*$ has $\tau\le 1/2$ and satisfies (UM23), uniformly above this cutoff threshold. If the base total potential norm is at most $a_*/2$, reserve one additional potential slot of norm $w_\star\le a_*/2$. The entire slot stays within (UM23) and hence within (UM24)–(UM30). This supplies the local uninserted majorant uniformly on that slot, rather than presupposing a zero-free activity domain. The constants may depend on the fixed localization parameters and rate choices, but not on cutoff, torus size, old-level count or the auxiliary Gaussian dimension.

27. A source insertion in the finite regular fluctuation integral

We keep the calibration, the real action weight, the Gaussian structure, the cutoffs and the background maps fixed. The result below concerns the finite regular fluctuation integral. It does not enlarge the source domain of a large-field history operation. This distinction permits a small potential direction to be used at its actual norm. Localized terms and their insertion estimates are defined on the specified complex local cores, with potential and current independent there.

Their sum is identified with a complete Gaussian integral only at physical complete background pairs $P = p_W$. An identity on that graph is not extended to arbitrary independent currents.

The fixed-reference integral. Let $\theta \geq \theta_{\min}$, $g = \theta^{-1/2}$, and $\tau = x/\theta$, with $|\tau - 1| < \sigma_\star$. The fixed small number $\sigma_\star$, the allowed external pairs and their Gaussian gap are restricted by the ordinary Gaussian comparison and potential allowance below. In particular the decoupled precisions stay in their coercive regime and the free potential leaves the declared allowance; no assertion is made for arbitrary positive x/θ. The sourced run considered here keeps $x > 0$ fixed. At a physical complete background pair $P = p_W$, in the fluctuation coordinates B on the non-pivot bonds, the reference measure is the normalized Gaussian with covariance $\tau^{-1}C(P)$. The same identity holds on a holomorphic continuation in physical background data wherever that continuation and the ordinary Gaussian bounds have been supplied. It is not an identity on every complex core pair. Put

$$\Xi(P,z) = \int d\mu_{\tau^{-1}C(P)}(B) \prod_b \chi_b(B) \exp\left\{\sum_Y V(Y;P,B) + z\sum_Y v(Y;P,B)\right\}. \tag{SI1}$$

The parameter z enters only the potentials. In particular the covariance, the cutoff thresholds and the Gaussian determinant do not depend on z. Here the cutoffs are sharp indicators $\chi_b(B) = \mathbf{1}\{|B(b)| < \mathfrak{b}\}$ on free bonds. For a bond set A, write $\chi_A = \prod_{b\in A}\chi_b$ and $\chi_A^c = \prod_{b\in A}(1-\chi_b)$. Thus the latter denotes a product of complements. After the old action at the background has been carried separately, the new fluctuation increment is

$$\mathcal{E}(P,z) = \mathcal{N}[\theta;x](P) + \log\Xi(P,z). \tag{SI2}$$

The field-independent Gaussian number includes $-\frac{3}{2}|\mathfrak{X}|\log\tau$, as well as the Haar-coordinate, scaling and gauge-fixing counts. The localised normalizer $\mathbf{N}^\circ$ and the free-potential family $\mathbf{G}_\tau$ also use the fixed (θ, x). All these quantities have zero z-derivative. Moving z into x, and differentiating the resulting Gaussian determinant, would describe a different presentation.

For an inserted exponent H, the potential actually inserted is its fluctuation difference. Thus, when the fibre field is $U_P(B)$ and $U_P(0)$ is its background, at a physical complete pair,

$$\sum_Y v(Y;P,B) = H(U_P(B)) - H(U_P(0)). \tag{SI3}$$

Consequently the background subtraction in the step functional is already present in (SI1); it must not be subtracted a second time. Equivalently, the raw fibre integral with the additional exponent $zH(U_P(B))$ has the extra factor $e^{zH(U_P(0))}$ outside (SI1). That factor belongs to the separately carried old action, rather than to the newly generated cluster activities.

The ordinary Gaussian expansion used here. We state the analytic inputs at the level at which they are applied. On an arbitrary complex local core pair P, the actual localized Gaussian functionals $\mathcal{J}(Z_0;s;P)$, potentials and union-filed terms below are defined by their local rules. We do not assume that the complete values for different Z_0 all arise as marginals of one global Gaussian at that P. A potential has norm

$$|V(Y;P,B)| \leq \|\mathbf{V}\|_\lambda e^{-\lambda d(Y)}\left(1 + M^{-4}\sum_{b\subset Y}|B(b)|^2\right) \tag{SI4}$$

on its specified complex field box. The potential may be defined only there: the cutoff product on its domain is kept in every term in which it occurs. The ordinary Gaussian comparison estimate bounds a localized decoupled functional by $F_0 e^{CuN(Z)}$ when its integrand is bounded by $F_0 e^{\rho|B|^2}$, with $u = \widehat{\varepsilon} + \rho$. Here $N(Z)$ counts the affected real scalar Gaussian coordinates, of order $(LM)^4 N_c(Z)$, where $N_c(Z)$ is the number of coarse cubes. In the cube-count convention the same

bound is $F_0 e^{C(LM)^4 u N_c(Z)}$; the dimension factor is counted once. The Gaussian reference gap, radii and tilt are chosen so these factors are bounded per coarse cube. The decoupling parameters have a fixed enlarged complex disc. These are the ordinary background/Gaussian inputs; no source derivative is contained in them.

Choose a potential allowance $w_\star > 0$, independent of the cutoff and volume, small enough for the following ordinary expansion. The base potentials use at most a fixed number of slots, and leave one slot available for an additional potential of norm at most $w_\star$. The same Gaussian comparison, rooted domain count and decoupling majorant are valid throughout this allowance. The convergent hard-core expansion gives local logarithmic terms $E(X)$ and bounds

$$|E(X)| \le C_\star e^{-\lambda_{\mathrm{out}} d(X)}. \tag{SI5}$$

The constant is uniform in all allowed external pairs, decoupling parameters and potentials. This is the ordinary uninserted polymer majorant, not an assumed differentiability bound for a renormalization map.

Here is the explicit finite expansion to which that majorant applies. Let $\mathbf{D}$ be a set of selected interaction domains. Write Y_0 for the union of their free-bond sets on which a field box is required; a globally defined free potential does not contribute to this union. Let $P_0 \subset \mathfrak{X} \setminus Y_0$ be a set of failed fluctuation cutoffs. The indicator identities $\chi_b^2 = \chi_b$ and $\chi_b(1-\chi_b) = 0$ give exactly $\prod_b \chi_b = \chi_{Y_0} \sum_{P_0 \subset \mathfrak{X}\setminus Y_0} (-1)^{|P_0|} \chi^c_{P_0}$. Before decoupling, its integrand is

$$F_{\mathbf{D},P_0}(z) = (-1)^{|P_0|} \chi_{Y_0} \chi^c_{P_0} \prod_{Y \in \mathbf{D}} (e^{V(Y)+zv(Y)} - 1). \tag{SI6}$$

The empty choice $(\mathbf{D}, P_0) = (\emptyset, \emptyset)$ supplies the leading one in the gas and is omitted from its activities. All polymer domains Z are nonempty and wall-connected, so no $H(\emptyset)$ is defined. Disconnected regions are products of their wall-component activities, rather than additional single polymers. Equivalently, for a nonempty Z the empty choice has at least one decoupling derivative of the normalized constant functional $\mathcal{J}[1] = 1$ and contributes zero. When several potential origins have the same geometric support, their origin is included in the index Y; the product selects tagged local terms. The support and its tree length ignore this tag. For each support index $\iota = (\mathbf{D}, P_0)$ let $Z_0(\iota)$ be its support. The decoupled activities are formed by

$$\begin{aligned} T(Z,\iota;z) &= \prod_{\Delta \subset Z \setminus Z_0(\iota)^\sharp} \int_0^1 ds_\Delta \, \partial_{s_\Delta} \, \mathcal{J}(Z_0(\iota); s_Z)[F_{\mathbf{D},P_0}(z)], \\ H(Z,z) &= \sum_\iota T(Z,\iota;z). \end{aligned} \tag{SI7}$$

In this formula s_Z is one on $Z_0(\iota)^\sharp$, is the displayed free parameter on $Z \setminus Z_0(\iota)^\sharp$, and is zero outside Z. The normalized Gaussian functional $\mathcal{J}$ is independent of z. At a physical complete pair $P = p_W$, its complete-structure identity is

$$\mathcal{J}(Z_0; 1)[F] = \int d\mu_{\tau^{-1}C(P)}(B)\, F(B).$$

This physical identity, together with interpolation of every remaining coarse-cube parameter and the ordinary decoupling property, is what identifies the gas with (SI1). Two polymers are compatible when they have neither a cube nor a wall in common. Multiplicativity over their wall-components yields $\Xi = \sum_{\Gamma \text{ compatible}} \prod_{Z \in \Gamma} H(Z)$, with empty product one, at a physical complete pair. On general local cores, the same local formulas define activities and their union-filed terms

without asserting one global Gaussian representation. Those logarithmic terms are

$$E(X,z)=\sum_{n\geq 1}\frac{1}{n!}\sum_{\cup_{i=1}^{n}Z_i=X}\rho^T(Z_1,\dots,Z_n)\prod_{i=1}^{n}H(Z_i,z). \tag{SI8}$$

Here $\rho^T(Z_1,\dots,Z_n)$ is the sum over connected graphs on $\{1,\dots,n\}$ of the products of their edge weights; an edge has weight -1 for an incompatible pair and zero otherwise. For $n=1$, $\rho^T=1$. Every sum is first formed in finite volume; the logarithmic series and its differentiated series are justified by the same absolute majorants.

Proposition 27.1 (The inserted-potential estimate)**.** *Let $\varepsilon=\|\mathbf{v}\|_\lambda$. Under the ordinary local Gaussian/polymer hypotheses above, on every specified complex local core pair,*

$$|\partial_z E(X,z)|_{z=0}|\leq C\varepsilon e^{-\lambda_{\mathrm{out}}d(X)}. \tag{SI9}$$

At a complete physical pair $P=p_W$, its sum over X is the normalized expectation

$$\partial_z\log\Xi(P,z)|_{z=0}=\frac{\int d\mu\,\prod_b\chi_b\,e^{\sum_Y V(Y)}\sum_Y v(Y)}{\int d\mu\,\prod_b\chi_b\,e^{\sum_Y V(Y)}}. \tag{SI10}$$

There is no volume factor in the local estimate (SI9).

Proof. If $\varepsilon=0$, every inserted potential vanishes on its domain and the assertion is immediate. Otherwise take the actual auxiliary potential $z\mathbf{v}$ on $|z|<w_\star/\varepsilon$. This is a legitimate potential in the actual local Mayer and decoupling expressions, regardless of whether it would be an admissible physical source for a history construction. The ordinary Gaussian comparison and the polymer criterion hold on this disc. Every integrand and every activity is holomorphic there; Gaussian domination and the parameter-independent majorant give holomorphy of the union-filed series (SI8). This local argument does not require one global Gaussian at a general independent-current core pair. Apply Cauchy's estimate on $|z|=w_\star/(2\varepsilon)$ to (SI5). It gives (SI9) with constant $2C_\star/w_\star$, which depends only on fixed ordinary allowances.

The same argument gives a norm derivative, which is useful when the actual source family of potentials is nonlinear. On the potential space with its fixed declared domains, joint holomorphy on every two-dimensional affine plane makes the directional derivative complex linear: the degree-one term of $E_X(\mathbf{V}+z_1\mathbf{h}_1+z_2\mathbf{h}_2)$ is the sum of its two degree-one terms. Cauchy's estimate makes this linear map bounded. For $\|\mathbf{h}\|_\lambda<w_\star/2$, apply one-variable Cauchy to $t\mapsto E_X(\mathbf{V}+t\mathbf{h})$ on the disc $|t|<w_\star/\|\mathbf{h}\|_\lambda$. Summing its Taylor coefficients of degree at least two gives

$$|E_X(\mathbf{V}+\mathbf{h})-E_X(\mathbf{V})-DE_X(\mathbf{V})[\mathbf{h}]|\leq CC_\star\frac{\|\mathbf{h}\|_\lambda^2}{w_\star^2}e^{-\lambda_{\mathrm{out}}d(X)}.$$

All constants are uniform in the external core pair, so this is also the remainder bound in the unpointed supremum norm $\|F\|_{\mathrm{out}}=\sup_{X,P}e^{\lambda_{\mathrm{out}}d(X)}|F_X(P)|$. It does not assert a pointed or stronger rooted norm bound. Uniform local Lipschitz bounds on smaller potential balls give the unique continuous extension to the potential-norm completion. The displayed remainder passes to that extension and proves Fréchet differentiability there as well. If an actual ordinary source family satisfies $\mathbf{V}(w)=\mathbf{V}(0)+w\dot{\mathbf{V}}+o(|w|)$ in that norm, the remainder bound proves its chain rule $\partial_w E_X(\mathbf{V}(w))|_0=DE_X(\mathbf{V}(0))[\dot{\mathbf{V}}]$. Pointwise holomorphy of $\mathbf{V}(w)$ on its ordinary source disc, with a uniform potential-norm bound there, gives its norm Taylor expansion by the same uniform Cauchy estimates. Thus the auxiliary linear direction computes the actual first source jet; the nonlinear physical source family need not exist on the large auxiliary-potential disc.

This is also the marked-activity derivative of the explicit expansion. The derivative of (SI6) replaces precisely one selected factor by

$$v(Y)e^{V(Y)}\prod_{\substack{Y'\in\mathbf{D}\\ Y'\neq Y}}(e^{V(Y')}-1). \tag{SI11}$$

Domains with $V(Y)=0$ are included: their unmarked term is zero, but their derivative need not be zero. In a term with a marked potential the cutoff on its domain is retained. Derivatives commute with (SI7) by the same Gaussian comparison and with (SI8) by absolute convergence. Thus this proof differentiates the actual finite interaction formula, not a hypothetical normed map without a specified integral. The pinned cluster estimate is what keeps the local Cauchy bound (SI5) independent of volume.

At a complete physical pair, the sum rule for the ordinary cluster expansion gives $\exp(\sum_X E(X,z)) = \Xi(P,z) \neq 0$. Differentiating it and the finite integral proves (SI10). The denominator is this one regular Gaussian integral; no division by the full sourced partition function or by a history mass is used. □

The actual scalar action direction. Let $H_u(U)=A(u,U)$, where u is a bounded function on bare plaquettes. It is filed by assigning a plaquette to the union of the fixed cubes containing its corners. In the background field domain, $|\partial U(p)-1|\le C\alpha\eta^2$. The identity $a_p(U)=\frac12\det(1-\partial U(p))$ gives a raw local action member of norm at most $CM^4\alpha^2\sup|u|$. It has no coupling counterterm at this stage.

More precisely, at a complete physical pair the potential in (SI3) is a first difference of this action. Suppose the actual background response and its decoupled substitutions stay in their stated field domains and shift a plaquette by at most $C\delta\eta^2$, where $0<\delta\ll\alpha$. Along this shift, the plaquette remains at distance at most $C\alpha\eta^2$ from the identity. The derivative of the determinant of a 2×2 matrix is bounded by a constant times its size, so

$$|a_p(U_P(B))-a_p(U_P(0))|\le C\alpha\delta\eta^4. \tag{SI12}$$

This proof works at complex pairs. It uses the actual curvature and response bounds, not positivity. On a local core the same determinant bound applies to the two actual substituted plaquette matrices whenever the declared response bounds give these curvature and difference estimates; it does not require a global fibre map there.

We give the quantitative fine-response localization separately from the coarse Gaussian decoupling in (SI7). For a plaquette p, let $X_0(p)$ be the union of its fixed corner cubes. Its number of cubes is at most a dimensional constant C_{base}, and its tree length is zero. Write $\mathcal{U}(P')$ for the bare configuration used by the actual background evaluation of a pair P', and set $F_p(P')=u(p)a_p(\mathcal{U}(P'))$. Thus this local function is explicitly a function of a pair. The ordinary response-bracket hypotheses used here are the following. On each $Y\supset X_0(p)$ the same response rule gives

$$\Phi_{p,Y}(s,\nu)=F_p\big(P|_{X_0(p)}\dot{+}\,\nu\,\mathfrak{h}_Y(s;B)\big),\qquad |s_\Delta|<R=e^{\bar\kappa},\quad |\nu|<\rho,$$

holomorphic jointly on these independent discs, uniformly in the allowed complex external fields and fluctuation box, with

$$\rho\ge c_R\alpha/\delta>1,\qquad \sup|\Phi_{p,Y}|\le C\alpha^2\eta^4|u(p)|.$$

Here $c_R>0$ can contain the fixed loss $e^{-C\bar\kappa}$. In particular all substituted pairs must stay in their actual cores on the whole joint polydisc, not just for real interpolation parameters. The operation $\dot{+}$ shifts the potential and current by the actual response. These are quantitative background and membership hypotheses; the following Cauchy argument does not prove them.

For disjoint fine-cube sets K, N, define the mixed bracket by

$$\Psi[K|N] = \int_{[0,1]^K} \prod_{\Delta\in K} \partial_{s_\Delta} \Psi(s_K + 1_N)\, ds_K,$$

with all other parameters zero. Its scalar-action potential is the bracket of $\Psi(s) = \Phi_{p,Y}(s,1) - \Phi_{p,Y}(s,0)$ with $K = Y \setminus X_0(p)$, $N = X_0(p)$. At $\nu = 0$ the field is the unshifted restricted background. Schwarz's lemma in ν first gives $\sup_s |\Psi(s)| \le 2C\rho^{-1}\alpha^2\eta^4|u(p)|$. For every $s_\Delta \in [0,1]$, a coordinate Cauchy circle has radius arbitrarily close to $R-1$. Iterated Cauchy therefore bounds the bracket by

$$2C\rho^{-1}\alpha^2\eta^4|u(p)|(R-1)^{-|K|} \le C_R\alpha\delta\eta^4|u(p)|e^{-(\bar\kappa-1)|K|},$$

where $\bar\kappa \ge 1$. There is no factorial in this estimate, because each independent cube parameter is differentiated once. The ordinary rooting property says that a component containing a differenced cube and missing $X_0(p)$ makes the bracket zero. For the present choice $N = X_0(p)$, every component missing the base has such a differenced cube. No assertion about an arbitrary remote component consisting only of undifferenced parameters is being made. The local response rule agrees with the response built on the active wall-component. Between local ambient domains on which the substitutions are defined it has hereditary consistency, $\Phi_{p,Y}(s,\nu) = \Phi_{p,\Lambda}(s\,\mathbf{1}_Y,\nu)$, with the base parameters set to one. At a complete physical pair on the full lattice its endpoints are

$$\Phi_{p,\Lambda}(1,1) = u(p)a_p(U_P(B)), \qquad \Phi_{p,\Lambda}(1,0) = u(p)a_p(U_P(0)).$$

Apply the finite fundamental theorem of calculus in every fine-cube parameter outside $X_0(p)$, expanding evaluation at one as evaluation at zero plus the integral of the derivative. The resulting sum of brackets equals $\Phi_{p,\Lambda}(1,1) - \Phi_{p,\Lambda}(1,0)$. At a complete physical pair this is exactly $u(p)[a_p(U_P(B)) - a_p(U_P(0))]$. Wall-component consistency identifies each bracket with the corresponding $\Phi_{p,Y}$; rooting removes the disconnected terms. Thus the construction gives a uniformly bounded local extension on every declared core and localizes the actual difference (SI3) at complete physical pairs.

For a surviving bracket, the ordinary domain geometry gives $d(Y) \le C_{\mathrm{geo}}(|K|+1)$. At fixed Y there are at most $CM^4\eta^{-4}(|K| + C_{\mathrm{base}})$ possible plaquette anchors. Choose the fixed surplus $\bar\kappa$ so that $\bar\kappa - 1 > C_{\mathrm{geo}}\lambda + 2$, increasing it further if a subsequent rooted-domain entropy sum is included. Sum the displayed bracket bound over its plaquette anchors and use that a linear polynomial times $e^{-2|K|}$ is bounded. This proves

$$\|\mathbf{v}_u\|_\lambda \le CM^4\alpha\delta \sup|u| = CM^4\alpha^2 s \sup|u|, \qquad s = \delta/\alpha. \tag{SI13}$$

The factors η^4 and η^{-4} cancel exactly. The required membership and response estimates on all decoupling substitutions are background obligations that we must prove. The bracket argument neither creates those inclusions nor replaces them by bounds at real fields.

Combining (SI9) with (SI13) proves a one-step unpointed source response of size $C\alpha^2 s \sup|u|$ for the actual local regular family on every declared complex core. Its physical sum is the source response of the fixed-reference regular integral. The Gaussian normalizer contributes zero derivative. Its interaction vacuum value contributes $\partial_z E(X,(1,0),0)$, which is the source vacuum energy and obeys the same bound. Subtracting it leaves a factor at most two in the estimate.

Own counterterms and the remaining iteration. For a constant source, once the actual bond-density pointing is applied, the derivative member is Euclidean. Its own coefficient is $\partial_z b[E_z]|_0 = b[\partial_z E_z|_0]$, because the free response and the coefficient extractor are fixed and complex linear. The counterterm to remove from the derivative action at site y is therefore this number times $A(h_y, U)$. A coefficient belonging to another member, or a derivative of a Gaussian presentation with a different weight, would not supply that identity.

There are two further tasks before the one-step estimate is a complete multilevel source-response bound. First, renewing the actual pointing uses three bond-indexed interpolation legs. Differentiating a normalized leg expectation produces both the derivative of its leg insertion and a connected expectation of that insertion with $\sum_Y v(Y)$. A root-plus-insertion cluster bound and the path's admissible field room must control those terms before the same estimate is asserted for each pointed term.

Second, the old regular derivative members and the nonconstant source defects have to be inserted again with their own coefficients. A constant-source derivative of size $\alpha_j^2 s_j$ can be normalized by the pointed Hessian theorem. Write $\alpha_H > 0$ for the fixed spatial Hölder exponent, as distinct from an analytic field radius. On a non-wrapping local chart, retaining the *true* radius ratio $\rho_j = \delta_{\mathrm{out}}/\alpha_j$ in the near-domain Taylor proof, before any enlargement to $\max\{1, \rho_j\}$, would give after the fine-site count the contribution

$$C\alpha_j^2 s_j \left[\left(\frac{\delta_{\mathrm{out}}}{\alpha_j} \right)^2 L^{-\alpha_H (n-j)} + \left(\frac{\delta_{\mathrm{out}}}{\alpha_j} \right)^3 L^{-2(n-j)} \right]. \tag{SI14}$$

Its quadratic part sums to $C\delta_{\mathrm{out}}^2 s_{\mathrm{out}}$ when the ordinary flow and age weights apply. For $\alpha_{\mathrm{out}} \asymp x_{\mathrm{out}}^{-1/2} (\log x_{\mathrm{out}})^q$ and $\delta_{\mathrm{out}} \asymp x_{\mathrm{out}}^{-1/2} (\log x_{\mathrm{out}})^{p_0}$, this is

$$C x_{\mathrm{out}}^{-1} (\log x_{\mathrm{out}})^{3p_0 - q}. \tag{SI15}$$

Under this homogeneous local refinement, the arithmetic supplies the indicated scale bound. The derivative recurrence also uses the regular member and defect hypotheses of the linear bootstrap. In particular the defects must retain the factor $\alpha_j^2 s_j$ as well as the scaled Lipschitz gain. A fixed-disc source bound alone does not provide it. Marked and remote terms need their own uniform fixed-disc bounds. Field-independent tails around holes or torus cycles cannot be assigned an extra ρ_j^2 when $\rho_j < 1$; they require a non-wrapping chart or a separately small class estimate.

28. Renewing the actual bond pointing after one source insertion

An estimate for an unpointed localized logarithm is not by itself an estimate for a particular pointing of that logarithm. We retain the actual three interpolation legs and differentiate their bond densities. For a source filed as an older weight member, the Gaussian structure, free potential, cutoffs and numerical normalizers are fixed. Thus the first, free-potential leg and the second, cutoff leg are independent of this source. Only the interaction leg contributes below. This choice of presentation is essential to the statement.

The finite local logarithm and its actual locality. Fix a finite regular fluctuation integral, its Gaussian reference, its cutoffs and all geometric and external data. Fix nested full complex fluctuation boxes

$$\mathcal{B}_0 = \{B : \sup_b |B(b)| < \mathfrak{b}_0\}, \qquad \mathcal{B}_1 = \{B : \sup_b |B(b)| < (1+\nu)\mathfrak{b}_0\}, \qquad \mathfrak{b}_0 \geq 1, \quad \nu > 0.$$

The ordinary spare-potential estimate below accepts potentials holomorphic and bounded in its growth norm on $\mathcal{B}_0$; all its actual Gaussian cutoff supports lie there. The larger box $\mathcal{B}_1$ is reserved for the one coordinate Cauchy estimate on an inserted potential. Both boxes and the external-pair and decoupling domains are fixed once for this step. Ordinary inputs whose cutoffs fill half their analytic box leave this room by choosing an intermediate $\mathcal{B}_0$. This uses that existing room once, and is not a new shrinking at each source derivative. A local potential family has the Gaussian growth norm

$$|V(Y; P, B)| \leq \|\mathbf{V}\|_\lambda e^{-\lambda d(Y)} Q_Y(B), \qquad Q_Y(B) = 1 + M^{-4} \sum_{b \subset Y} |B(b)|^2. \tag{SP1}$$

There are finitely many bond variables but arbitrarily many of them as the volume grows. The norm and all constants used below are those of the ordinary regular expansion, uniform in that number. A fixed finite number of potential types can be combined by addition of their norms.

Let $E_X(\mathbf{V})$ denote the actual union-filed logarithmic term of the local Gaussian functionals, Mayer expansion and connected-cluster expansion. These local objects are defined on their independent-current pair cores. Away from complete physical data, their wall-component functionals need not be margins of one global Gaussian measure. Normalized global Gaussian expectations will be used only at a complete physical pair $P = p_W$, or by holomorphic continuation in the physical datum W where the same complete construction is defined. We use the following consequence of its uninserted estimate, as established for the fixed-reference integral in Section 27. At every base system used below there is an available potential allowance $w_\star > 0$ such that

$$|E_X(\mathbf{V}+\mathbf{U})| \le C_\star e^{-\lambda_{\mathrm{out}} d(X)}, \qquad \|\mathbf{U}\|_\lambda < w_\star. \tag{SP2}$$

The actual finite local Gaussian functionals and the absolutely convergent logarithmic series are holomorphic in every finite collection of potential parameters on this allowance. A single additional family is enough: several auxiliary directions are added together in $\mathbf{U}$, with the sum of their norm loads less than $w_\star$.

The locality needed here is part of this actual expansion, not a choice made after taking its logarithm. A contribution filed at X uses only selected interaction domains whose enlarged union is in X. In particular, if a family $\mathbf{a}_b$ vanishes except on domains containing $b^\sharp$, then

$$DE_X(\mathbf{V})[\mathbf{a}_b] = 0, \qquad D^2E_X(\mathbf{V})[\mathbf{a}_b, \mathbf{v}] = 0 \quad \text{if } b^\sharp \not\subset X. \tag{SP3}$$

The assertions include domains with a zero unmarked potential. In a Mayer derivative their zero factor can be replaced by the nonzero inserted factor, and the marked domain must then be included in the union. Omitting such terms would invalidate both locality and the insertion identity.

Lemma 28.1 (Two derivatives of the actual localized term)**.** *For potential families* $\mathbf{a}, \mathbf{v}, \mathbf{c}$*, the allowance* (SP2) *gives*

$$\begin{aligned} |D^2E_X(\mathbf{V})[\mathbf{a},\mathbf{v}]| &\le \frac{16C_\star}{w_\star^2}\|\mathbf{a}\|_\lambda\|\mathbf{v}\|_\lambda e^{-\lambda_{\mathrm{out}} d(X)}, \\ |DE_X(\mathbf{V})[\mathbf{c}]| &\le \frac{2C_\star}{w_\star}\|\mathbf{c}\|_\lambda e^{-\lambda_{\mathrm{out}} d(X)}. \end{aligned} \tag{SP4}$$

These derivatives preserve (SP3)*.*

Proof. If either direction in the first estimate has zero norm, the claim is immediate. Otherwise put the single auxiliary family $\zeta\mathbf{a} + z\mathbf{v}$ in the available slot. On the bidisc with radii $w_\star/(4\|\mathbf{a}\|_\lambda)$ and $w_\star/(4\|\mathbf{v}\|_\lambda)$ its total norm is at most $w_\star/2$. Apply the two-variable Cauchy formula to the actual holomorphic term in (SP2). For the second estimate use radius $w_\star/(2\|\mathbf{c}\|_\lambda)$. The support assertion follows from the union filing before differentiation. No radius for a full history construction has been enlarged in this argument. □

Thus a separate second-mark estimate is not needed if the actual ordinary bound is already uniform on the full potential allowance (SP2). If only a one-direction derivative bound were known, without this allowance and locality, the lemma would not follow from it.

The interpolation paths and the differentiated density. Let $\mathbf{V}_t$, $0 \le t \le 1$, be the actual interaction leg, with local bond directions $\mathbf{a}_{t,b}$ satisfying

$$\partial_t \mathbf{V}_t = \sum_b \mathbf{a}_{t,b}. \tag{SP5}$$

Every summand is supported on domains containing $b^\sharp$. At a finite volume the sum is finite. We must prove that the ordinary interpolation construction supplies the uniform bound

$$\sup_b \int_0^1 \|\mathbf{a}_{t,b}\|_\lambda \, dt \le A. \tag{SP6}$$

This is a property of the actual unmarked legs and their admissible field domains. It is not inferred merely from an endpoint norm. Uniform ordinary bounds and field room are required throughout the path, with the same allowance (SP2). The third-order vanishing of the intrinsic $V^{G,A}$ makes its t^{-2} radial path regular at zero. It does not by itself identify the actual bond split. We do not replace that split by an Euler decomposition or any other pointing to infer (SP6).

For the inserted source let

$$\mathbf{V}_t(s) = \mathbf{V}_t + s\mathbf{v}_t, \qquad \mathbf{a}_{t,b}(s) = \mathbf{a}_{t,b} + s\mathbf{c}_{t,b}, \qquad \partial_t \mathbf{v}_t = \sum_b \mathbf{c}_{t,b}, \tag{SP7}$$

with the same bond support and with $\mathbf{v}_0 = 0$. It is enough that these are the first source derivatives of holomorphic paths; the affine notation in s makes the formula easier to read. Suppose

$$\sup_{0\le t\le 1} \|\mathbf{v}_t\|_\lambda \le C_v\varepsilon, \qquad \sup_b \int_0^1 \|\mathbf{c}_{t,b}\|_\lambda \, dt \le C_c\varepsilon. \tag{SP8}$$

Choose $|s|C_v\varepsilon < w_\star/4$. The remaining allowance in (SP2) bounds the first derivative in direction $\mathbf{a}_{t,b} + s\mathbf{c}_{t,b}$ by a constant times $\|\mathbf{a}_{t,b}\|_\lambda + |s|\|\mathbf{c}_{t,b}\|_\lambda$. This is integrable by (SP6) and (SP8). On a smaller source disc the mixed Cauchy estimate similarly dominates its source derivative. Consequently the actual path integral is holomorphic there and may be differentiated under the integral.

The actual interaction-leg term carrying bond b is

$$E_s^\bullet(X,b) = \int_0^1 DE_X(\mathbf{V}_t(s))[\mathbf{a}_{t,b}(s)] \, dt. \tag{SP9}$$

Its source derivative contains two terms:

$$\dot{E}^\bullet(X,b) = \int_0^1 \left\{ D^2E_X(\mathbf{V}_t)[\mathbf{a}_{t,b}, \mathbf{v}_t] + DE_X(\mathbf{V}_t)[\mathbf{c}_{t,b}] \right\} dt. \tag{SP10}$$

The first is the connected root-plus-source term. The second is the direct change of the leg insertion. Neither can be dropped. Equations (SP4), (SP6) and (SP8) prove

$$|\dot{E}^\bullet(X,b)| \le C_\star \left(\frac{16AC_v}{w_\star^2} + \frac{2C_c}{w_\star} \right) \varepsilon e^{-\lambda_{\mathrm{out}} d(X)}. \tag{SP11}$$

It is zero unless $b^\sharp \subset X$.

Here is the exact normalized-expectation interpretation on a complete physical pair $P = p_W$. Write $\langle\cdot\rangle_t$ for the normalized, possibly complex, fixed-reference integral with interaction $\sum_Y V_t(Y)$. The denominator is nonzero by the same ordinary expansion. With $a_{t,b} = \sum_Y a_{t,b}(Y)$, $v_t = \sum_Y v_t(Y)$ and $c_{t,b} = \sum_Y c_{t,b}(Y)$,

$$\partial_s \langle a_{t,b}(s)\rangle_{t,s}\big|_{s=0} = \langle c_{t,b}\rangle_t + \langle a_{t,b}v_t\rangle_t - \langle a_{t,b}\rangle_t\langle v_t\rangle_t. \tag{SP12}$$

This follows by differentiating its numerator and denominator. It requires no positivity. On these complete data, summing (SP10) over X gives (SP12) by the physical sum rule for the actual connected expansion. The disconnected product cancels; an unnormalized insertion would give the wrong formula and could carry a volume factor. On an arbitrary independent-current core, (SP10) remains the local derivative identity and bound. It is not identified there with a covariance under an unavailable global Gaussian measure.

The older-weight path and its field room. An older weight enters through its fluctuation difference $v(Y;P,B)$, so $v(Y;P,0)=0$. Its interaction path is $v_t(Y;P,B)=v(Y;P,tB)$, and

$$c_{t,b}(Y;P,B)=\sum_{a=1}^{3} B_b^a \partial_{B_b^a} v(Y;P,tB). \tag{SP13}$$

The intrinsic path $t^{-2}V^{G,A}(tB)$ belongs to the terms with the action or gauge-fixing inverse-coupling coefficient. It must not be applied to this older weight. In particular, no artificial vanishing of its linear and quadratic fluctuation terms is assumed.

We spell out sufficient field room for (SP8). Suppose v is holomorphic and obeys (SP1) with norm ε on $\mathcal{B}_1$, for the actual external pairs and decoupling parameters. We prove the bound on the entire $\mathcal{B}_0$ required by (SP2), not merely on real integration points. For real $t\in[0,1]$, vary one coordinate at tB by a complex number of radius

$$r_b=c_\nu(1+|B_b^a|),$$

where $c_\nu>0$ is a sufficiently small fixed fraction of the available room and the fixed color norm. This disc lies in $\mathcal{B}_1$, uniformly for $B\in\mathcal{B}_0$ and in t,b and the number of variables. Moreover its growth factor is bounded by $C_\nu Q_Y(B)$: the one changed coordinate adds at most $C_\nu(1+|B_b^a|^2)$ to the sum defining Q_Y, with M fixed. Cauchy's formula therefore gives

$$|B_b^a \partial_{B_b^a} v(Y;P,tB)|\le C_\nu \varepsilon e^{-\lambda d(Y)} Q_Y(B). \tag{SP14}$$

Also $Q_Y(tB)\le Q_Y(B)$, so $\|\mathbf{v}_t\|_\lambda\le\varepsilon$ on the actual cutoff box. Summing the three color coordinates in (SP13) proves (SP8). The original B is used in this majorant, so no $1/t$ or $\mathfrak{b}_0^2$ loss occurs as t tends to zero. Thus the differentiated root is a potential on exactly $\mathcal{B}_0$, while the undifferentiated source used its one fixed larger box.

A complex disc for a common radial parameter alone would not justify (SP14). The individual bond legs require the independent-coordinate field room just used. Nor does an endpoint inserted-potential norm bound the variation of an arbitrary interpolation path. The prescribed path and its ordinary bounds (SP6) remain necessary.

Midpoint tents, entropy, and the exact sum rule. Use the actual geometric tents at bond midpoints. The collars in the union filing are assumed large enough that $h_z(b)\neq 0$ implies $z\in b^\sharp\subset X$ for a nonzero bond-root term. The fixed block ratio L gives

$$\sum_{z\in X} h_z(b)=1, \qquad \sum_b h_z(b)\le 4L^4. \tag{SP15}$$

These geometric identities are source independent. The renewed pointed source family is therefore

$$\dot F(X,z)=\sum_b h_z(b)\dot E^\bullet(X,b), \qquad \sup_{X,z} e^{\lambda_{\mathrm{out}} d(X)}|\dot F(X,z)|\le C\varepsilon. \tag{SP16}$$

The constant depends on L, the fixed ordinary allowances, A and the field room, and is independent of volume and field-variable count. No additional source-position factor is spent in this supremum norm.

For completeness, the geometric root cost also remains harmless when an application sums domains through z. If that count at length m is at most $C_D e^{\kappa_D m}$ and $\lambda_{\mathrm{out}}>\kappa_D$, then a smaller output rate costs only

$$4L^4 C\varepsilon\sum_{m\ge 0}(m+1)^q e^{-(\lambda_{\mathrm{out}}-\kappa_D)m}<\infty$$

for every fixed polynomial power q. The two-variable Cauchy estimate has already summed all cluster multiplicities and choices of source domains through the ordinary rooted majorant. No second unsummed count of them is justified or needed. Additional geometric position sums cost

a polynomial only when the actual domain geometry supplies that bound. This does not permit unrecorded extra domain types or an exponent loss larger than the available gap.

Finally sum (SP9) over b at a fixed localization domain X. The chain rule and (SP5) give $E_X(\mathbf{V}_1(s)) - E_X(\mathbf{V}_0(s))$. Because $\mathbf{v}_0 = 0$, its source derivative is $DE_X(\mathbf{V}_1)[\mathbf{v}_1]$. On every declared local core, (SP15) therefore proves the algebraic identity

$$\sum_{z\in X} \dot{F}(X,z) = DE_X(\mathbf{V}_1)[\mathbf{v}_1].$$

At a complete physical pair p_W, the physical Gaussian sum rule then gives

$$\sum_{X,z} \dot{F}(X,z) = \sum_{X} DE_X(\mathbf{V}_1)[\mathbf{v}_1] = \partial_s \log \Xi(\mathbf{V}_1 + s\mathbf{v}_1)\big|_0. \tag{SP17}$$

The same formula is valid on the permitted analytic continuation of the physical datum W. This is not continuation from the physical graph to an arbitrary independent current. All sums are finite-volume sums followed by absolutely convergent cluster sums. Coherent infinite-volume extensions of the local families use the same uniform local bounds and companion limits, when supplied.

If the actual reference, union filing and interpolation paths are covariant under lattice isometries and a constant source has that same covariance, these identities also hold after transporting X, b, P. The midpoint tent then gives $\dot{F}(\mathsf{r}X, \mathsf{r}z; \mathsf{r}P) = \dot{F}(X, z; P)$. The same reasoning preserves gauge invariance and reality. These properties are inherited from the actual construction, not inferred from a scalar norm estimate. A nonconstant source does not acquire Euclidean covariance merely from (SP16).

If the source entered a Gaussian structure or the cutoff leg, additional derivatives would be required. With a fixed reference, a cutoff root is in general a boundary functional $N_{t,b}$, not the bounded function $\partial_t\chi/\chi$. Its differentiated normalized density is $N_{t,b}(ve^V)/\Xi - (N_{t,b}(e^V)/\Xi)\langle v\rangle$, plus any direct variation of $N_{t,b}$. If the reference measure also varies, its score contributes to the denominator derivative as well. The retained Gaussian tail would have to be proved under insertion. The fixed-reference older-weight presentation avoids those additional terms here.

The conclusion is one finite-step pointed insertion estimate. It does not differentiate a full history class, prove closure of the response under iteration, supply the spatial defect gain of a nonconstant profile, or establish the clean-history Gaussian disintegration needed for an interacting two-point law.

29. From ordinary windows to local fluctuation potentials

The estimate in this section is an ordinary formation estimate. It concerns the actual functions obtained by rewriting old terms on box backgrounds, before their fine-cube bracket localization. The outer potential/current pair may have independent components. This causes no substitution of an arbitrary pair into a theorem about critical backgrounds: the inner window function first constructs its own constrained box background and its own current. The hypotheses below retain the domain and response assertions needed to perform that construction on the whole complex parameter set. We fix the window-first rule below as part of the localization: at one old base point, its near pointed sum is one window member before any fine bracket is applied.

The maps and geometric inputs. Fix a step from level k to level $k+1$. For an old level $j \le k$, put

$$a = k - j, \qquad \sigma_j = L^{-a}, \qquad \rho_j = \max\{1, \alpha_k/\alpha_j\}, \qquad s_k = \delta_k/\alpha_k. \tag{WF1}$$

The radii and the ordinary flow satisfy $\rho_j \le C(1+a)^{1/2}$. In particular

$$C_{\mathrm{age}} := \sup_{a\ge 0} C(1+a)^{1/2}L^{-a} < \infty, \qquad \rho_j\sigma_j \le C_{\mathrm{age}}. \tag{WF2}$$

Here δ_k is the tested fluctuation amplitude, whereas α_k is the analytic field radius. The ratio in (WF1) is an ordinary full-radius ratio. No small-field gain is being deduced by replacing a smaller ratio with one.

We spell out the ordinary background and geometry inputs. They must concern the actual assigned maps, not arbitrary functions with similar notation.

First, let P range over the local complex core used to form the new potentials. Let $\mathbf{H}_k(P;s,B')$ be the actual response to the tested free fluctuation coordinates, with the indicated fine-cube parameters. The plain nonlinear iterated block average is denoted $\mathcal{M}^j$. On a window $\square$, with a fixed partition function $\zeta_\square$, the datum used for an inner term is the restriction of

$$\mathcal{V}_{j,\square}(P;s,t,\nu) = \mathcal{M}^j\Big(e^{\,i\eta_k(t+\nu\zeta_\square)\mathbf{H}_k(P;s,B')}\,U_+(P)\Big), \qquad 0\le t\le 1. \tag{WF3}$$

The notation $U_+(P)$ denotes the configuration in the assigned background evaluation and η_k the conversion from its potential to a bare bond logarithm. Formula (WF3) uses the nonlinear average. Replacing it by its differential would not give the window identity. The box-background map applied to this datum returns a pair

$$P^{\mathrm{box}}_{j,\square}(P;s,t,\nu) = \big(U^{\mathrm{box}}_{j,\square}(\mathcal{V}_{j,\square}), J^{\mathrm{box}}_{j,\square}(\mathcal{V}_{j,\square})\big) \tag{WF4}$$

in which J^{box} is the current of that constrained critical configuration. The ordinary maps include their prescribed node and gauge representatives.

Second, all these maps are holomorphic on the full independent fine-parameter discs $|s_\Delta| < R = e^{\bar\kappa}$. The unamplified substitutions have a fixed positive amount of datum, chart and core room. If $r_{B,j}$ is the dimensionless datum radius, the background package supplies $\alpha_j/r_{B,j} \le C$. Its response and nonlinear-average estimates bound the change caused by the ν term, in every relevant normalized datum, chart and core norm, by

$$C_R|\nu|\,\delta_k\sigma_j/\alpha_j \le C_R|\nu| s_k\rho_j\sigma_j. \tag{WF5}$$

The constant C_R includes the fixed loss from the enlarged fine-parameter discs. These are estimates for the actual nonlinear average and response on their assigned analytic domains. For example, in a datum norm the estimate follows by the uniform first-response bound, composed with the derivative of the actual average, and integration along the perturbation. It must also hold for the other core inequalities and tower substitutions used in the inner, outer and removed terms. Existence and holomorphy on the corresponding datum balls are part of the background package.

It follows from (WF2)–(WF5) and the fixed room that there is a constant $c_\nu > 0$, independent of j,k, such that all substitutions are admissible on the common disc

$$|\nu| < \mathcal{R}_\nu = c_\nu/s_k. \tag{WF6}$$

Indeed choose c_ν so that $C_R c_\nu C_{\mathrm{age}}$ is smaller than half of every fixed room. Equation (WF5) then keeps each substitution inside its assigned domain. This argument verifies an inclusion using the actual input-domain estimates; it does not extend functions to unions of complex gauge orbits. We take $s_k < c_\nu/2$, so that $\mathcal{R}_\nu > 2$.

Third, the critical box-background maps in (WF4) obey the weighted first-response, Hölder and compatible boundary-comparison estimates used in the nonlinear pointed window theorem. The complete fields on (WF6), including the unperturbed field, have output size at most $C\alpha_k$. For old ages the axial datum is bounded by $C\alpha_k\sigma_j$. Consequently its global half-datum-ball condition holds when $C\rho_j\sigma_j$ is sufficiently small. By (WF2) this fails at most at a fixed finite set of young

ages. At those ages the actual core memberships and the whole-core bound for a member's own tent action remain available. Finite boxes are treated first; their linearized boundary comparison is uniform in the bare mesh. Periodic or distant tails are retained with their stated exponential lengths.

Finally, assume the actual window identities and elementary localization geometry. The partition $\sum_{\square} \zeta_{\square} = 1$ is inserted along the direction of differentiation. For complete physical pairs the old action difference is the sum over windows of $\int_0^1 \partial_\nu|_0$ of its window evaluations (WF3)–(WF4), plus the prescribed outer and removed terms. The same local formulas define their extensions at every local complex core. Equality with a global physical fibre integral is asserted only at complete physical pairs.

An inner window contains at most $C_M \sigma_j^{-4}$ old base points. At a fixed base point the number of old domains of tree length between m and $m+1$ is at most $C_D e^{\kappa_D m}$. Diameters are bounded by $C_M(1+m)$, and the input decay rate has a fixed surplus over κ_D. Thus every fixed polynomial diameter factor is summable. An outer field has the additional decay from its distance to the window, while a removed field has the factor e^{-cM/σ_j}. In the member's local pair norm in level-j physical units, whose analytic radius is α_j, their changes are bounded by

$$C_R|\nu|\delta_k\sigma_j e^{-cM/\sigma_j}. \tag{WF7}$$

The corresponding statement with the actual distance is used when summing outer terms. Both endpoints and the interpolation segment stay in the same member core, with a fixed fraction of its analytic radius available for Cauchy's inequality. Long input domains retain their original tree decay: one fraction supplies e^{-c_κ/σ_j}, and the remaining fraction supplies the declared output spatial rate. Here $c_\kappa > 0$ comes from tree length in M-cubes and the allocated input rate. When these long-domain tails are written together with removed-response tails as e^{-cM/σ_j}, the fixed c is allowed to depend on M and the input rate; it is not asserted independent of them. These are geometric and response hypotheses, not bounds on the value of an old member.

The room used by the substitutions. The displacement estimate must include the hereditary tower clauses. Their control uses the curvature of both factors, not a bound on the complex potential alone. Suppose that $\mathbf{U} = e^{i\eta A'}U$ and that $U, \mathbf{U}$ both have plaquettes bounded by $\alpha_0'\eta^2$, while $|A'|, |\nabla_U A'| \le \alpha_1'$. The local expansion is

$$\partial(e^{i\eta A'}U) = \big(1 + i\eta^2 D_U A' + O(\eta^2|A'|^2 + \eta|A'|\,|\partial U - 1|)\big)\partial U.$$

It follows that $|D_U A'| \le C(\alpha_0' + (\alpha_1')^2)$. Since U is real, its adjoint transports commute with the real conjugation. The real and imaginary parts of A' separately have this bound. Their real perturbations therefore have curvature of size $C(\alpha_0' + (\alpha_1')^2)$.

Let ϕ be a rescaled tower curvature or current. It is holomorphic in A' on a fixed ball of radius r. Its linear Taylor part is determined by the values at $0, A'_{\mathrm{R}}$ and A'_{I}. Consequently

$$|\phi(A')| \le C\big(|\phi(0)| + |\phi(A'_{\mathrm{R}})| + |\phi(A'_{\mathrm{I}})|\big) + C\sup|\phi|\,\|A'\|^2/r^2.$$

The real regularity estimate controls the three real values. Only the quadratic remainder uses the crude tower bound. At an old level, the complex potential gains one power of the scale ratio and its square gains the two powers required by the curvature normalization. Thus, uniformly in the depth, the tower bound is $K^\sharp(\alpha_0' + (\alpha_1')^2)$ in its own units. The relations $\alpha_1^2 \le C\alpha_0$ and $K^\sharp(\alpha_0' + (\alpha_1')^2) \le \alpha_0$ are the corresponding restrictions on the radii.

This also explains why a fixed core margin gives actual analytic room. Apply Schwarz's lemma to a tower quantity on a larger potential disc on which the preceding a priori bound holds. Its increment is bounded by a constant times the potential displacement measured in the allowed radius. The constant is uniform over the tower and its depth. A sufficiently small displacement therefore consumes at most a prescribed fraction of every tower margin. Clauses involving only the

real factor are unchanged by this shift. The independent current is shifted by its reduced-current response, whose bound requires only first differences of the potential.

The same argument is used on layered spaces with their geometric gaps. If ν layers separate a clause from the integrated field, the quotient of its response bound by its margin contains

$$2^{\nu}(1+\nu)^{1/2}e^{-c\delta_0 ML(\nu-1)}.$$

This is uniformly bounded after M is fixed. The factor 2^{ν} is the reciprocal gap, and $(1+\nu)^{1/2}$ is the largest running-radius loss. No sum over the old levels is introduced.

We apply this room estimate to the three substitutions as follows. The outer field is the response on the original domain. The inner field is the background of the enlarged window, evaluated at the datum obtained by the actual nonlinear average in (WF3). The two removed fields are the differences produced by replacing the original response and the original background by their box versions. Their bounds have the additional factor e^{-cM/σ_j}. Their analytic discs are chosen separately from the inner disc. All these fields use the assigned node factors. The inner pair in (WF4) carries the actual current of its generated box background, also for complex data on its analytic domain. Outer and removed substitutions use their prescribed reduced-current assignments away from the critical physical locus. At physical parameters these assignments agree with the currents of the corresponding critical configurations.

For the inner substitution, the datum has the form $\varsigma B + B'$ with $0 \le \varsigma \le 1$. Here B is the nonlinear average of the shifted box background and $|B'| \le c\min\{\alpha_{0,j-1}, \alpha_{0,j}\}$. The box background is taken in Landau form. Its lower-level restrictions are backgrounds over their own averages. The running-radius inequality and the two powers of scale in the curvature estimate put its triple in the core of every level at which it is used. The box gauge has bounded condition number; its distance from the identity is not required to be small when the real factor is an arbitrary pure gauge.

The window multiplier can therefore be amplified until its displacement uses a fixed fraction of this room. For a term of level j one obtains the radius

$$\varrho_F \asymp \sigma_j^{-1}\alpha_j\delta_k^{-1}e^{-C\bar{\kappa}} \ge c\sigma_j^{-1}(\rho_j s_k)^{-1}e^{-C\bar{\kappa}}.$$

Because $\rho_j \le C(1+k-j)^{1/2}$ and $\sigma_j = L^{-(k-j)}$, their product is uniformly bounded. It follows that $\varrho_F^{-1} \le Ce^{C\bar{\kappa}}s_k$, and that a common disc $|\nu| < c_\nu/s_k$ is available. This supplies the scale arithmetic in (WF5)–(WF6) and the corresponding normalized displacement estimate in (OA7). In applying it, the three substituted maps and their node and current conventions are kept fixed throughout the disc. The estimate is a statement about those maps, including all their tower clauses; it is not a consequence of a bound on an old activity's Hessian.

Renormalize the critical inner function first. Let an old regular member have norm g_j, be gauge invariant and Euclidean as a pointed total, and have its own coefficient b_j. At every base point remove its vacuum value and $b_j A(h_z, \cdot)$ before bracket interpolation. The pointed Hessian theorem fixes this coefficient by the unit normalization of the same tent action. Site-frozen members may differ at different base points: the argument uses each member's own hypothetical Euclidean family and its own coefficient. No translation invariance of that array is required.

For $j < k$, here is the quantitative window estimate on (WF4). The axial field near a base point has linear constant-curvature part of size $C\alpha_k\sigma_j^2$. The Hölder and nonlinear response errors have sizes $C\alpha_k\sigma_j^{2+\alpha_H}$ and $Cr_{B,j}^{-1}\alpha_k^2\sigma_j^4$, respectively. Cauchy's second and third derivatives of a member cost Cg_j/α_j^2 and Cg_j/α_j^3. The linear derivative vanishes by constant-color invariance. The quadratic value on the affine part is canceled on the whole pointed total at that base point by the member's own tent action. No coefficient is assigned to an individual domain X. The individual Taylor remainders have a fixed polynomial in $D = 2 + \operatorname{diam}_j X$ times their original

spatial decay; the domain count above sums these polynomials. Multiplying the remaining errors therefore gives, at one base point after this domain sum,

$$Cg_j\left[\rho_j^2\sigma_j^{4+\alpha_H}+\rho_j^3\sigma_j^6\right]. \tag{WF8}$$

This is the local Taylor argument on the constrained critical box pair, with its uniform weighted response sums. When its local Cauchy-size parameter is large, the whole-core bound and the own-action bound give the same estimate after increasing that summable polynomial, since $1+T^2\le CT^3$ for $T\ge 1$. The finite set of young ages failing the global half-datum-ball condition is handled by the same whole-core bounds; their scale ratios have a fixed positive lower bound. The resulting constant is independent of the number of old levels.

Now sum the $C_M\sigma_j^{-4}$ base points. Because $\rho_j\ge 1$, $0<\alpha_H\le 1$, and $\sigma_j\le 1$, the renormalized inner window function has bound

$$Cg_j\rho_j^3\sigma_j^{\alpha_H}. \tag{WF9}$$

Truncating the pointed total at the enlarged window contributes only the long-domain tail already set aside in the geometry. It does not replace the pointed total by an unnormalized individual term.

A defect or an aged remainder has no coupling counterterm. After its value at the identity is removed, constant-color invariance still eliminates the linear derivative, provided there are no fixed color spectators. The vacuum-only window argument gives at one base point, after the same polynomial domain sum, $Cd_j\rho_j^2\sigma_j^4$, where d_j is that origin's norm. The large local-size case uses $1\le CT^2$ for $T\ge 1$. After the base-point sum its inner bound is

$$Cd_j\rho_j^2. \tag{WF10}$$

Stored-field boundary origins are not included in this statement. Their spectators and their hull geometry need the separate boundary formation estimate.

At $j=k$ no age improvement is needed. The vacuum-normalized old member together with its own action has whole-core norm at most Cg_k, by the coefficient bound $|b_k|\le Cg_k/\alpha_k^2$ and the core bound on its tent action. The actual fluctuation change has size $C\delta_k$ within a core with fixed analytic room $c\alpha_k$. Cauchy's inequality therefore gives Cs_kg_k, with its original spatial decay. There is no fine-volume age factor at this level. This is the term with weight ω_1; no cancellation between different current-level domains is used for it.

The one small factor and the outer terms. Fix $t\in[0,1]$ and the fine parameters. Subtract the window function's value at $\nu=0$. It vanishes there and is holomorphic on (WF6). The bounds (WF9)–(WF10) hold on that whole disc. Schwarz's lemma, or Cauchy's formula for its derivative at zero, therefore supplies precisely one factor $1/\mathcal{R}_\nu\le Cs_k$. The actual window formula uses that derivative and its integral over t; the interval has length one. Thus an old regular inner term contributes $Cs_kg_j\rho_j^3\sigma_j^{\alpha_H}$, and a vacuum-only origin contributes $Cs_kd_j\rho_j^2$. No second amplification factor is extracted later.

Outer and removed terms need no marginal cancellation on a general independent-current pair. Use Cauchy's inequality in the actual member core and (WF7). Before its t-integration and the fine-site count, a difference is bounded by $Cg_j|\nu|(\delta_k/\alpha_j)\sigma_je^{-cM/\sigma_j}$. Equivalently, use the uniform disc (WF6) and then its one Schwarz or Cauchy factor. After the fine-site count the bound is

$$Cs_kg_j\rho_j\sigma_j^{-3}e^{-cM/\sigma_j}. \tag{WF11}$$

The same reasoning applies to a vacuum-only origin with g_j replaced by d_j. For $0<\sigma\le 1$ and $\rho\ge 1$,

$$\rho\sigma^{-3}e^{-cM/\sigma}\le C\rho^3\sigma^{\alpha_H},\qquad \rho\sigma^{-3}e^{-cM/\sigma}\le C\rho^2. \tag{WF12}$$

Indeed the suprema of $\sigma^{-3-\alpha_H}e^{-cM/\sigma}$ and $\sigma^{-3}e^{-cM/\sigma}$ are finite. Any fixed additional polynomial from spatial summation is absorbed in the same way, by decreasing the exponential rate once. Long-domain tails are treated with their retained tree decay and the same elementary inequality. This proves the ordinary outer and removed bounds without applying the pointed cancellation to an arbitrary pair (U, J).

Fine brackets and the resulting formation estimate. We now make the base of an inner bracket explicit. For $j < k$ and one old base point z, let $\square = \square_k(z)$ be its assigned current window and $R_z = X'_k(z)$ its fixed enlarged box. There are only a fixed number of direction-partition windows near this base window. Group their directions by linearity before applying the box-background replacement, and choose R_z to contain their fixed enlargements. In (WF3) their sum is the inner cutoff $\zeta_\square$; the remaining directions are outer terms. This is part of the fixed window-first rule. The derivative at zero is linear in these directions, so this grouping preserves the partition identity without changing any incoming pointed member. Let $\mathsf{A}_{j,z}(X,\cdot)$ be the fixed action-family filing with total $A(h_z^{(j)},\cdot)$, and put $\widetilde{G}_{j,z} = G_{j,z} - b_{j,z}\mathsf{A}_{j,z}$. The one inner window member is the function

$$\mathcal{W}_{j,z}(P') = \sum_{\substack{X\ni z\\ X\subset\square^{\sim 2}}} \big[\widetilde{G}_{j,z}\big(X, P^{\mathrm{box}}_{j,z}(P'), z\big) - \widetilde{G}_{j,z}\big(X,(1,0),z\big)\big]. \tag{WF13a}$$

Here P' is a local potential/current pair of the current resolved level k, on R_z. The map $P^{\mathrm{box}}_{j,z}$ first forms its actual nonlinear averaged datum and then takes the old-level constrained critical box pair of that datum. The same box-background map is used for every X in this sum. The own tent action is included in the near sum; its fixed support lies inside the assigned enlarged window. The declared domain of this entire member is R_z, and its fine-cube base is $B_z = \mathrm{Hull}_{\pi_k} R_z$. Its number of current fine cubes is bounded by a fixed constant C_{win}, independent of j, k, z. This is a presentation of an inner member before localization; we do not apply different old-X bracket operators to its summands.

Let $\mathcal{W}_{j,z,Y}(s,t,\nu)$ mean this one function evaluated by the assigned local response and the datum (WF3) on $Y \supset B_z$. The inner potential contributed by this member is

$$V^{\mathrm{in}}_{j,z}(Y) = \int_0^1 \big[\,\partial_\nu \mathcal{W}_{j,z,Y}(\,\cdot\,,t,\nu)|_{\nu=0}\big]\,[\,Y\setminus B_z \mid B_z\,]\,dt. \tag{WF13b}$$

The brackets in this formula are linear in the whole function (WF13a). Thus its cancellation and bound (WF8) are established before this single bracket is estimated. Different z's need not have a common base: there is no cancellation between them, and their bounds can be added. Hereditary restriction identifies the bracket on Y with the bracket of the same ambient interpolation function whose parameters outside Y are zero. Finite fundamental theorem of calculus therefore gives, at a complete physical pair,

$$\sum_{Y\supset B_z} V^{\mathrm{in}}_{j,z}(Y; p_W) = \int_0^1 \partial_\nu \mathcal{W}_{j,z,\Lambda}(1,t,\nu; p_W)|_{\nu=0}\,dt.$$

Every local term uses only the pair on Y and the input members on its declared domain $B_z \subset Y$. This proves chart-coherent locality and the physical sum identity for this fixed window-first rule. Summing the window identities and the retained outer and removed origins recovers the old action difference. No equality of individual output domains with a different filing is required. The current-level origins $j = k$, and the far and removed origins, retain their own declared bases and the bounds already proved. For vacuum-only near origins the same window-first grouping may be used without a marginal subtraction.

The assigned maps are hereditary on active wall-components, and complete at all parameters equal to one. For an interpolation function Ψ, its bracket is

$$\Psi[K|N] = \int_{[0,1]^K} \prod_{\Delta\in K} \partial_{s_\Delta} \Psi(s_K + 1_N)\, ds_K. \tag{WF13}$$

For (WF13b), $K = Y \setminus B_z$, $N = B_z$. For an origin retaining its own base X_0, use $K = Y \setminus X_0$, $N = X_0$. A component carrying a differenced cube and missing the base makes this bracket zero. Each coordinate has the disc $R = e^{\bar\kappa}$, so mixed Cauchy costs $(R-1)^{-|K|} \le e^{-(\bar\kappa-1)|K|}$, with no factorial. The common ν-disc and its one derivative commute with these brackets by joint holomorphy. The preceding amplitude bounds therefore retain this additional spatial factor.

Here is a quantitative rate allocation. For a retained old domain X and its current fine hull X_0, put $m = \#(Y \setminus X_0)$. The actual cube geometry and rooting give a fixed constant C_h such that

$$d_k(Y) \le C_h\big[d_j(X) + m + 1\big]. \tag{WF13c}$$

To see this, view a tree covering X on the old lattice in current units and cover it by current fine cubes. Its hull has a covering tree of length at most $C(d_j(X)+1)$. Each component of the additional cubes meets that hull, or its bracket vanishes. A rooted spanning forest adds at most a fixed number of edges per additional cube. This proves (WF13c). For an inner window member the same argument and $\#B_z \le C_{\mathrm{win}}$ give $d_k(Y) \le C_h(m+1)$, increasing C_h by a fixed factor if necessary.

Split the old input rate into positive parts $\kappa_{\mathrm{sum}} + \kappa_{\mathrm{far}} + \kappa_{\mathrm{ret}} \le \kappa$. Choose $\kappa_{\mathrm{sum}} > \kappa_D$ with enough strict gap to sum the fixed polynomial diameter factors, and use κ_{far} for the long-domain tail in (WF7)–(WF12). The retained old-domain factor is $e^{-\kappa_{\mathrm{ret}} d_j(X)}$. Put $\beta = \bar\kappa - 1$. Choose an output rate λ and a number $\epsilon > 0$ with

$$C_h(\lambda + \epsilon) < \min\{\kappa_{\mathrm{ret}}, \beta\}. \tag{WF13d}$$

All rates are chosen once, sufficiently large for the subsequent ordinary polymer criterion. Combining the retained old factor with $e^{-\beta m}$ and (WF13c) gives $Ce^{-(\lambda+\epsilon)d_k(Y)}$. For inner members the bounded base needs only the β part of this inequality. The number of possible current-window anchors at fixed Y is polynomial in $\#Y$; the old bases within one such window have already been counted by $C_M\sigma_j^{-4}$. The bound $\#Y \le C(d_k(Y)+1)$ and the remaining ϵ absorb this polynomial. This is a pointwise output-domain estimate; any later sum over output domains spends its separately declared entropy reserve.

The function $\rho_j^3\sigma_j^{\alpha_H}$ is bounded by

$$\omega_{k+1-j} := C(1+k-j)^{3/2} L^{-\alpha_H(k-j)}, \qquad \sum_{n\ge 1} \omega_n < \infty. \tag{WF14}$$

All age functions have the same enlarged fine-parameter domains and the same ν-disc. At fixed j, z, (WF13a)–(WF13b) provide the common-root function before any bracket. The resulting bounds for distinct base points and levels are then added absolutely. No bound for a sum of differently based bracket operators has been inferred from a sup norm of their unbracketed sum. The uniform bounds justify passage through the bracket integrals and to limits of the old-level sum.

We have proved the following formation estimate. Let g_j be regular-member norms, d_j defect norms and r_j aged-remainder norms. Under the explicit background, window and geometry inputs above, the actual old-origin potential satisfies

$$\|\mathbf{V}_k^{\mathrm{old}}\|_\lambda \le C s_k \left[\sum_{j\le k} \omega_{k+1-j} g_j + \sum_{j\le k} \rho_j^2 d_j + \sum_{j\in\mathcal{J}_k} \rho_j^2 r_j\right]. \tag{WF15}$$

These are local bounds on every specified complex core. At complete physical pairs, finite fundamental theorem of calculus, the window partition, and hereditary resummation of (WF13) identify their sum with the actual old action difference. There is no assertion of a single global off-graph Gaussian or a global torus background for an arbitrary local pair.

Adding a scalar Wilson-weight direction contributes at most $CM^4 s_k \alpha_k^2 \|u\|_\infty$, by the separate determinant/response-bracket argument. The formation rules are linear in their origins, so (WF15) applies to their first source derivatives. For a frozen regular member that direction is $f(p_z)\mathcal{L}_j[1]$ with its own $b_j'(0)$; the remaining direction is the source defect. The finite potential-ball Taylor remainder justifies the chain rule even when the actual ordinary source families are nonlinear. This gives the formation inequality used in the linear source bootstrap, with no additional unexplained source-response norm assumption.

Assume the joint complex window/core room, nonlinear averaged-datum estimates, Hölder response and compatible boundary comparison, and the outer/removed-field locality of the assigned maps. These are the analytic inputs used in the formation proof. The large-field history operation uses its separate history-induction hypotheses.

30. A first-source-response bootstrap on the ordinary domains

This section isolates a linear consequence of the ordinary local formation bounds. It uses the inserted-potential and renewed-pointing estimates at one step, rather than a larger domain of physical sources. In particular, it distinguishes the derivative of a newly generated activity from the old background action that is carried to the next scale. The formation and domain hypotheses below concern the actual local maps. Their statement is not an assertion that a nonlinear history construction satisfies them.

Meshes, norms, and the fixed ordinary run. Work first on a finite periodic lattice. Let $h_j = L^j h_0$ be the physical mesh of level j, with $L > 1$. A source profile is a bounded Lipschitz function on the physical torus; complex profiles are allowed by complexification. Put

$$\|f\|_{\sharp,j} = \max\{\|f\|_\infty, C_M h_j \operatorname{Lip} f\}. \tag{LB1}$$

At level j, a pointed localization domain X contains its base point z. Its enlarged source region $X^\sim$ and the fixed plaquette point p_z satisfy

$$\sup_{p \in X^\sim} |p - p_z|_{\mathrm{tor}} \le C_M h_j (1 + d_j(X)). \tag{LB2}$$

Distances and diameters in this section use the torus metric. The family norm with rate γ is the supremum of $e^{\gamma d_j(X)}|F(X,P,z)|$ over the actual specified complex field domain and all X, z. Companion families, when present, are included in the same norm. A bound with rate $\kappa + 1$ therefore leaves one unit of rate for a diameter.

The ordinary run fixes all Gaussian structures, background maps, thresholds, label spaces and reference operations. Sources change local exponent members only. Let

$$\alpha_j = \theta_j^{-1/2} (\log \theta_j)^q, \qquad \delta_j = \theta_j^{-1/2} (\log \theta_j)^{p_0}, \qquad s_j = \delta_j / \alpha_j, \qquad q > p_0 \ge 0. \tag{LB3}$$

Fixed multiplicative constants in these definitions can be included. Assume the ordinary flow bounds, with constants independent of the number of levels,

$$c(k-j) - C_0 \le \theta_j - \theta_k \le C(k-j) + C_0 \qquad (j \le k). \tag{LB4}$$

All θ_j are sufficiently large. These bounds imply, with a uniform constant C_F, that for $1 \le j \le k$,

$$\begin{aligned} \alpha_{j-1}^2 s_{j-1} &\le C_F \alpha_k^2 s_k, \\ \varrho_{k,j}^2 \alpha_{j-1}^2 s_{j-1} &\le C_F \alpha_k^2 s_k, \qquad \varrho_{k,j} = \alpha_k / \alpha_j, \\ \varrho_{k,j}^2 &\le C_F (1 + k - j). \end{aligned} \tag{LB5}$$

Indeed the functions $t^{-1}(\log t)^{2q}$ and $(\log t)^{p_0-q}$ are decreasing for sufficiently large t. The bounded displacement C_0 in (LB4) changes their ratios by at most a constant. The second line also uses the bounded consecutive ratio α_{j-1}/α_j. The upper flow bound and $\theta_k \geq 1$ prove the last line. These observations also give $s_j \leq C_F s_k$ when $j \leq k$.

Locality supplies the source-gradient gain. Let E_j^w be an ordinary sourced regular family, holomorphic in a neighborhood of $w = 0$. Define its first derivative by

$$\mathcal{L}_j[f] = \frac{d}{dt} E_j^{tf}\Big|_{t=0}, \qquad A_j = \sup_{f\neq 0} \frac{\|\mathcal{L}_j[f]\|_{\kappa+1}}{\|f\|_{\sharp,j}}. \tag{LB6}$$

The supremum is taken over the permitted profiles and the tori in the run. At a fixed finite lattice all differentiations are in finite-dimensional source coordinates. Locality means that the value indexed by X depends only on the values of f on $X^\sim$. To apply the following argument it suffices that the source class admit bounded Lipschitz truncations of a profile and constants, as does the class defined by (LB1). No density claim about an infinite-dimensional algebraic dual is used.

The derivative defect at a base point is exactly

$$\mathcal{D}_j[f](X,P,z) = \mathcal{L}_j[f](X,P,z) - f(p_z)\mathcal{L}_j[1](X,P,z). \tag{LB7}$$

We claim

$$\|\mathcal{D}_j[f]\|_\kappa \leq C_M A_j h_j \operatorname{Lip} f. \tag{LB8}$$

To prove it, fix X, z, and subtract $f(p_z)$ from f. On $X^\sim$ the resulting profile has modulus at most $C_M h_j(1 + d_j(X)) \operatorname{Lip} f$. For real f, clip its values outside an interval twice this large, using a Lipschitz scalar map that is the identity on that interval and has Lipschitz constant at most a fixed constant. For a complex profile clip the real and imaginary parts separately. The resulting global profile g agrees with $f - f(p_z)$ on $X^\sim$ and satisfies $\|g\|_{\sharp,j} \leq C_M h_j(1 + d_j(X)) \operatorname{Lip} f$. Locality and linearity give $\mathcal{D}_j[f](X,P,z) = \mathcal{L}_j[g](X,P,z)$. Apply (LB6) and $(1+d)e^{-d} \leq 1$ for $d \geq 0$ to obtain (LB8). Smooth profiles may instead use a smooth clipping map with the same fixed bounds. This proof applies uniformly at every external complex field point in the family norm. For a source measured at level $m \geq j$, it gives in particular

$$\|\mathcal{D}_j[f]\|_\kappa \leq C A_j L^{-(m-j)} \|f\|_{\sharp,m}. \tag{LB9}$$

Thus the derivative defect retains the response amplitude A_j and the scale-gradient gain together.

The precise one-step inputs. For clarity we list the ordinary inputs needed for the bootstrap. The first two are estimates for local formation and the finite regular gas, at fixed background maps; they are not estimates for complex history masses.

The formation of potentials is linear in each origin. When a source is differentiated at zero, the regular-origin direction at a site is $f(p_z)\mathcal{L}_j[1]$, accompanied by its own coefficient, and the remaining direction is (LB7). We require the following ordinary formation estimate on the actual localized complex pairs:

$$\begin{aligned}\|\dot{\mathbf{V}}_k[f]\|_\lambda \leq C s_k \Big[&\alpha_k^2 \|\dot{u}_k[f]\|_\infty + \sum_{j\leq k} \omega_{k+1-j} A_j \|f\|_\infty \\ &+ \sum_{j\leq k} \varrho_{k,j}^2 \|\mathcal{D}_j[f]\|_\kappa + \sum_{j\in\mathcal{J}_k} \varrho_{k,j}^2 \|\dot{R}_j[f]\|_\kappa \Big].\end{aligned} \tag{LB10}$$

Here $\omega_n \leq C(1+n)^C L^{-\alpha_H n}$ with $\alpha_H > 0$, and $\mathcal{J}_k$ contains the aged remainder origins that actually enter the regular part. Younger remainders remain in the remainder class. The factor s_k is the fluctuation-amplitude/field-radius ratio. In a formulation with additional positive powers of θ_k^{-1} in

this factor, they are bounded by a constant times s_k at weak coupling. The current scalar-origin term in (LB10) follows more sharply from its fluctuation-difference estimate $C\alpha_k\delta_k\|\dot{u}_k[f]\|_\infty$. All substitutions in (LB10) must lie in their actual specified cores; a norm inequality does not create those inclusions. In particular, a proof of the pointed Hessian cancellation and nonlinear window estimate on constrained critical backgrounds alone does not prove (LB10) for independent potential/current pairs used in localization. Passing from those background estimates to all the actual localized pairs remains part of this ordinary formation input.

The inserted-potential estimate for the actual finite Mayer/decoupling series, followed by its actual renewed bond pointing, gives

$$\|\mathcal{L}_{k+1}[f]\|_{\kappa+1} \le C_P\|\dot{\mathbf{V}}_k[f]\|_\lambda. \tag{LB11}$$

Its proof uses the ordinary spare-potential slot. The auxiliary potential parameter is not a physical source in a history box. The normalized Gaussian, all reference measures, and $\mathbf{N}^\circ, \mathbf{G}_\tau$ have zero source derivative. These facts are essential to (LB10)–(LB11).

The ordinary constant-source construction supplies numbers $b'_j(0)$ with $\sum_j |b'_j(0)| \le B_0$, uniformly in the cutoff. Its fixed tents and stage factors satisfy

$$\dot{u}_k[f](p) = f(p) + \sum_{j\le k} b'_j(0)\varphi_j(p)\sum_y h_y^{(j)}(p)f(p_y), \qquad \|\dot{u}_k[f]\|_\infty \le (1+B_0)\|f\|_\infty. \tag{LB12}$$

The last bound uses $0 \le \varphi_j \le 1$, $h_y^{(j)} \ge 0$, and $\sum_y h_y^{(j)} = 1$. The coefficient of $\mathcal{L}_j[1]$ is exactly $b'_j(0)$: the free response and coefficient extractor are fixed and complex linear. Thus no coefficient of another member enters (LB10), even when $f(p_z)$ varies with the site.

Finally suppose the ordinary remainder families are bounded by $C_R\theta_j^{-p}e^{-\kappa d_j(X)}$, with $p > 3$, on the fixed physical source disc $|t| < r/\|f\|_{\sharp,j}$ around zero. Cauchy's inequality at half that radius proves

$$\|\dot{R}_j[f]\|_\kappa \le (2C_R/r)\theta_j^{-p}\|f\|_{\sharp,j}. \tag{LB13}$$

This is the only source-disc estimate needed for the small remainder class. It is not used to produce an α_j^2 factor for regular activities.

The induction and its constants.

Proposition 30.1 (Linear regular-response bootstrap)**.** *Under (LB1)–(LB13), suppose the initial regular list is empty. There are $M_0 < \infty$ and $s_\star > 0$, independent of the number of levels, cutoff and volume, such that $s_j \le s_\star$ for all j implies*

$$A_j \le M_0\alpha_{j-1}^2 s_{j-1} \qquad (j \ge 1). \tag{LB14}$$

Proof. Divide (LB10) by $\|f\|_{\sharp,k+1}$. Use (LB9) with $m = k+1$, (LB11)–(LB13), and the monotonicity $\|f\|_{\sharp,j} \le \|f\|_{\sharp,k+1}$. This gives

$$A_{k+1} \le C_1 s_k\left[(1+B_0)\alpha_k^2 + \sum_{j\le k}\omega_{k+1-j}A_j + C\sum_{j\le k}\varrho_{k,j}^2 A_j L^{-(k+1-j)} + C_r\sum_{j\le k}\varrho_{k,j}^2\theta_j^{-p}\right]. \tag{LB15}$$

There is no sum over the number of source labels in this inequality. The norms of linear directions have already combined them into f. The flow bound and (LB5) imply

$$\sum_{j\le k}\varrho_{k,j}^2\theta_j^{-p} \le C\sum_{a\ge 0}(1+a)(\theta_k+ca)^{-p} \le C_p\theta_k^{2-p} \le C_p\alpha_k^2. \tag{LB16}$$

To check the middle bound, compare the sum with its first term and the integrals of $(\theta+ct)^{-p}$ and $t(\theta+ct)^{-p}$ on $[0,\infty)$. They are fixed constants times θ^{1-p} and θ^{2-p}, respectively; the latter integral is finite exactly when $p > 2$. The final bound uses $p > 3$, $q \ge 0$, and large θ_k.

Assume (LB14) through level k. Both sums involving A_j in (LB15) are at most

$$CM_0\alpha_k^2 s_k\left(\sum_{n\ge1}\omega_n+\sum_{n\ge1}L^{-n}\right)$$

by (LB5). Consequently, for constants C_2, C_3 independent of k,

$$A_{k+1}\le\alpha_k^2 s_k(C_2+C_3M_0s_k).$$

Choose $M_0\ge 2C_2$ and $s_\star\le(2C_3)^{-1}$. The right-hand side is at most $M_0\alpha_k^2 s_k$, as required. At the first step there are no old regular members, so the same argument starts the induction. The direct bottom-level scalar potential, if it occupies a separate initial slot, must satisfy the same fixed-reference fluctuation estimate used in (LB10). □

A consequence on the near, non-wrapping window class. Let the final fluctuation size be $\delta_n=\alpha_n s_n$, and consider only terms for which the actual local window chart, the pointed affine comparison and its complex domain inclusions hold without a field-independent wrapping or holonomy tail. In addition, the axial and Hölder bounds of the nonlinear window proof must hold for the *complete* near background-plus-fluctuation pair with output size $a_{\mathrm{out}}\le C\delta_n$. A bound on its fluctuation increment alone is insufficient: an unrestricted background of size α_n would require that larger size in the radius ratio. The intended near class is the one on which the actual small-field tests impose a fixed multiple of δ_n on the complete field. Keep the true ratio $\rho_j=\delta_n/\alpha_j$; it need not exceed one. The global half-datum-disc condition in the near-domain Taylor proof holds uniformly at all ages. The actual window-domain input includes $\alpha_j/r_{B,j}\le C$, where $r_{B,j}$ is the independent-coordinate datum radius. Hence, with $\sigma=L^{-(n-j)}$, the flow gives

$$\frac{C_{\mathrm{ax}}\delta_n\sigma}{r_{B,j}}\le C\rho_j\sigma\le Cs_n(1+n-j)^{1/2}L^{-(n-j)}.$$

Choose $s_\star$ smaller by a fixed factor so that the right-hand side stays below the required half-disc constant. This uses the near-domain proof for $\rho_j>0$ itself and avoids the young-age core-only fallback, which would lose a ρ_j^2 gain when $\rho_j<1$. After the four-dimensional fine-site count, the renormalized constant-member contribution at age $a=n-j$ is bounded by

$$CA_j\|f\|_\infty\left[\left(\frac{\delta_n}{\alpha_j}\right)^2L^{-\alpha_H a}+\left(\frac{\delta_n}{\alpha_j}\right)^3L^{-2a}\right]. \tag{LB17}$$

The value at the identity and this member's own $b_j'(0)A(h_z,\cdot)$ are removed here. By (LB14)–(LB5), $A_j/\alpha_j^2\le Cs_n$ and $\delta_n/\alpha_j\le Cs_n(1+a)^{1/2}$. The sum of (LB17) is therefore at most $C\delta_n^2 s_n\|f\|_\infty$.

For the defect, the vacuum-only window estimate instead gives

$$C\|\mathcal{D}_j[f]\|_\kappa\left(\frac{\delta_n}{\alpha_j}\right)^2\le C\delta_n^2 s_n h_j\,\mathrm{Lip}\,f. \tag{LB18}$$

Since $h_j=h_nL^{-(n-j)}$, its sum is at most $C\delta_n^2 s_n\|f\|_{\sharp,n}$. Thus these near regular and source-gradient terms together have size $C\delta_n^2 s_n\|f\|_{\sharp,n}$ per final window unit. With the radii (LB3), the prefactor is $\theta_n^{-1}(\log\theta_n)^{3p_0-q}$. The gain is obtained because the newly generated derivative member already has amplitude $\alpha_j^2 s_{j-1}$.

No field-independent far tail is included in (LB17)–(LB18). Such a tail cannot acquire a factor ρ_j^2 merely from a source derivative. Recent large-field terms, marked remainder windows and torus or hole windings still require their own ordinary source-disc bounds, domain comparisons or separately small-class estimates. Nor do the local implications in this section verify the nonlinear formation hypotheses for every history. They show exactly how the α^2 normalization and

the source-gradient gain propagate once those ordinary local hypotheses and the finite pointed estimate hold.

31. The Hessian of old activities in a run window

This section proves a conditional ordinary estimate. Its inputs are the complex domains and first-response estimates of the actual background maps, the pointed quadratic cancellation, and the window formation rule. The desired Hessian estimate is not one of the inputs. In particular, an estimate on the Gaussian cutoff box alone is not used as an analytic continuation theorem. We construct the larger fluctuation domain from the unscaled constraint chart and the datum domains of the background maps.

The conclusion concerns the regular old-origin part of a step exponent. The conditional exterior-force estimate, boundary origins with stored fields, and the identification of a clean-class conditional measure are separate issues.

Physical and Gaussian coordinates. Fix a step from level k to level $k+1$. Write

$$g_k = \theta_k^{-1/2}, \qquad \alpha_i \asymp \theta_i^{-1/2}(\log \theta_i)^q, \qquad \mathfrak{b}_k \asymp (\log \theta_k)^p, \qquad \delta_k = g_k \mathfrak{b}_k. \tag{OA1}$$

Here $q > p$, and α_i is the smaller of the usable physical radii in the datum, potential, curvature and current conditions. Their fixed ratios are included in the constants. A finite alteration of an index in α_i, as needed for adjacent input layers, has the same meaning. We must prove that the ordinary flow gives

$$\rho_i := \max\{1, C\alpha_k/\alpha_i\} \le C_\rho (1+k-i)^{1/2}, \qquad \sigma_i = L^{-(k-i)}, \qquad i \le k. \tag{OA2}$$

Constants depend on the fixed block and collar choices, but not on the number of old levels, the cutoff or the volume. In particular, $C_{\mathrm{a}} := \sup_{i \le k} \rho_i \sigma_i < \infty$.

The free Gaussian variable y_b has three real color coordinates at each nonpivot bond of the level-k lattice. The physical bond logarithm is of size $g_k y_b$. Thus a physical analytic radius α_k corresponds to a Gaussian radius of order α_k/g_k, rather than to α_k. For complex variables we use the Euclidean norm of the three complex components at a bond and its supremum over bonds.

Let I be the finite box of free bonds under consideration. Its outside free coordinates and other exterior data are fixed. Every real free coordinate appearing in the regular fields is bounded by $C_{\mathrm{w}} \mathfrak{b}_k$. The number of coordinates satisfies

$$m_I = 3\#I \le C(rL^{-a_*} + D)^4 \le C(r+D)^4. \tag{OA3}$$

This count will not enter the localized estimate below. It is relevant to the weaker estimate available without spatial locality.

The primitive analytic inputs. We specify the domain inputs used in the proof. These are statements about assigned maps and their first responses, not bounds on the Hessian of an old activity.

For the actual one-step average, at its actual background $V^{\mathrm{bg}}(W)$, put

$$q_W(a) = \frac{1}{i} \log\Big[\mathcal{M}(e^{ia} V^{\mathrm{bg}}(W)) W^{-1}\Big] = LQ_W a + \mathcal{R}_W(a). \tag{OA4}$$

The logarithm is taken on its assigned local branch. The fixed-pivot construction supplies linear maps C_W and h_W with

$$LQ_W \mathsf{C}_W = 0, \qquad LQ_W h_W = 1, \qquad \|\mathsf{C}_W\|_{\infty\to\infty} + \|h_W\|_{\infty\to\infty} \le C_L. \tag{OA5}$$

The map C_W is the pivot elimination map, not a Gaussian covariance. The maps in (OA4)–(OA5) are the ones used in the run. Uniformly on an unscaled physical ball $\|a\|_\infty < r_0$, the remainder is holomorphic and obeys

$$\|\mathcal{R}_W(a)\|_\infty \le C_0\|a\|_\infty^2, \qquad \|D\mathcal{R}_W(a)\|_{\infty\to\infty} \le C_0\|a\|_\infty. \tag{OA6}$$

Here r_0, C_0, C_L are independent of the box size. The same bounds, with fixed enlarged constants, apply to the assigned local charts on the independent fine-parameter discs $|s_\Delta| < e^{\bar\kappa}$. At physical parameters the local charts agree with the run chart by their restriction and completeness identities.

The ordinary background package supplies holomorphic maps on their declared datum balls and their declared core slices. The datum radii satisfy $r_{B,i} \ge c\alpha_i$. At zero fluctuation all substituted data and pairs have a fixed positive fraction ϵ_0 of room in these domains. This room includes the tower conditions, node representatives, current bounds, and the inner, outer and removed substitutions. These are local complex assertions; they do not assert the existence of a global Gaussian measure for independent currents.

We use the following precise first-response form of that package. Measure each defining datum or core displacement at old level i in units of its allowed radius. Along an admitted physical perturbation of supremum size u, its derivative in any of these normalized quantities is bounded by

$$C_R\sigma_i\alpha_i^{-1}u. \tag{OA7}$$

The bound holds throughout the declared complex domains, for all the fine parameters above. It is the integrated first-response and nonlinear-average estimate: the actual average, not its differential, is used to construct the datum. The constant C_R includes the fixed enlarged parameter discs and the chart derivative. For outer and removed derivatives there are additionally their assigned exponential distance factors. We must also prove a compatible boundary comparison and the weighted Hölder response estimates for the actual box backgrounds.

Finally, the field maps are hereditary on active wall components. After the parameters outside a domain Y are set to zero, a local potential uses only the free coordinates in Y. A fixed collar can instead be included in Y throughout; it changes the constants below. The number of domains of tree length in $[m, m+1)$ containing a fixed cube is at most $C_D e^{\kappa_D m}$, and

$$\operatorname{diam} Y \le C_M(1 + d_k(Y)). \tag{OA8}$$

The spatial rates have the strict reserves needed to sum these domains and the fixed polynomial diameter factors.

The actual chart on the larger ball. For clarity we give the contraction argument behind the radius claim. Put $z = g_k\mathsf{C}_W y$. The nonlinear correction is the solution of

$$d = \mathcal{R}_W(z - h_W d), \qquad a_W(y) = z - h_W d, \qquad V_W(y) = e^{ia_W(y)}V^{\mathrm{bg}}(W). \tag{OA9}$$

Choose $r_1 > 0$ so small that $2r_1 < r_0$ and $4C_0C_Lr_1 < 1/2$. For $\|z\|_\infty \le r_1$, the right side maps the ball $\|d\|_\infty \le 4C_0\|z\|_\infty^2$ to itself. Indeed $\|z - h_W d\|_\infty \le 2\|z\|_\infty$, and (OA6) then gives that bound. Its Lipschitz constant in d is at most $2C_0C_Lr_1 < 1/2$. Iteration from zero therefore converges uniformly to the unique solution, including locally uniformly in all complex parameters. Uniform limits of the holomorphic iterates are holomorphic. Differentiating the fixed-point equation, or its uniformly convergent derivative iteration on a smaller ball, gives

$$\|a_W(y)\|_\infty \le Cg_k\|y\|_\infty, \qquad \|D_y a_W(y)\|_{\infty\to\infty} \le Cg_k. \tag{OA10}$$

Moreover, $q_W(a_W(y)) = 0$, because $LQ_W z = 0$ and $LQ_W h_W d = d$. This is the actual constraint, with its nonlinear average. Uniqueness shows that this chart agrees on the operational cutoff box with the chart already used in the run.

There is consequently a fixed $c_* > 0$ for which all the old substitutions are defined and holomorphic on

$$\|y\|_\infty < R_k, \qquad R_k = c_* \alpha_k / g_k. \tag{OA11}$$

Here and below fixed exterior coordinates may be retained with norm at most $C_{\mathrm{w}} \mathfrak{b}_k$. To prove the assertion, first make α_k small enough that (OA10) stays in its physical chart. For any fixed complex y in (OA11), follow the segment ty, $0 \le t \le 1$. On the portion where a composition of assigned background maps is admitted, (OA7) and (OA10) bound each normalized displacement by

$$C_R g_k \|y\|_\infty \sigma_i / \alpha_i \le C_R c_* \rho_i \sigma_i \le C_R c_* C_{\mathrm{a}}. \tag{OA12}$$

Choose c_* so that the last expression is less than $\epsilon_0/4$. The segment cannot reach a domain boundary: every defining inequality still has at least half of its initial room. Atomic background maps are already given on those domains, so this is an inclusion argument for a composition of assigned holomorphic maps, not an assertion that a branch continues through an unknown singularity. The same argument gives the required joint holomorphy with the fine parameters. It also applies to local independent-current formulas without postulating an off-graph global Gaussian integral.

The window multiplier used to differentiate an inner direction can range on a fixed disc $|\nu| < c_1/c_*$, where c_1 is chosen from the same room estimates and $c_* < c_1/4$. In fact its normalized increment is bounded by $C_R |\nu| c_* \rho_i \sigma_i$. The unamplified part costs only $C_R c_* C_{\mathrm{a}}$; choose c_1 so that their sum still spends less than half the room. Thus this disc contains $\nu = 1$ and permits the window derivative at zero.

Since $q > p$, increasing the coupling floor gives $C_{\mathrm{w}} \mathfrak{b}_k < R_k/4$. Two coordinate Cauchy circles with radii cR_k, centered anywhere on the real product window, therefore fit in (OA11). There is no factor depending on m_I in this inclusion. No indicator function or cutoff is analytically continued. Only the activity functions are evaluated on the larger ball; the real measure, Gaussian scaling and all numerical choices of the run remain fixed.

An amplitude bound before the Gaussian-growth norm. Let the regular member at old level i have norm e_i, with $e_i \le E_0$. The norms include its infinite companion if one is required. Each pointed total has local gauge invariance, constant color invariance and the full site symmetry needed for the pointed Hessian theorem. At every old base point subtract its vacuum and its own extracted coefficient times its own tent action. The tent has unit quadrature; the action convention is $A(e^{iB}) = \frac{1}{2}|dB|^2 + O(B^3)$. This is the regular part of the actual ledger decomposition: its counterterms have been transferred to the running Wilson coefficient. A profile deficit is retained as a lag or boundary origin, rather than silently subtracted from the measure. The clean interior has no such deficit in the canonical redrawn-profile ledger. Thus the affine quadratic term cancels with coefficient one in that normalization. We do not replace the pointed object by a bulk convolution or assign an action coefficient to an individual domain.

We recall explicitly the value estimate supplied by this cancellation. Write $D_X = 2 + \operatorname{diam}_i X$. On an inner box background the affine part and its error have sizes bounded, with fixed powers of D_X, by

$$\alpha_k \sigma_i^2, \qquad \alpha_k \sigma_i^{2+\alpha_H} + r_{B,i}^{-1} \alpha_k^2 \sigma_i^4, \qquad 0 < \alpha_H \le 1. \tag{OA13}$$

These estimates concern the complete substituted field, not just its fluctuation increment. They hold on the whole disc just constructed. Second and third derivatives of a member cost $C e_i \alpha_i^{-2}$ and $C e_i \alpha_i^{-3}$, with its original tree decay. Its linear derivative at the trivial pair is zero by constant color invariance.

Set $T = CD_X^s \rho_i \sigma_i^2$. When T is small, Taylor's formula, the cancellation of the affine quadratic value on the pointed total, and $r_{B,i} \geq c\alpha_i$ bound the remaining terms by $Ce_i D_X^{s'}[\rho_i^2 \sigma_i^{4+\alpha_H} + \rho_i^3 \sigma_i^6]$. For example, the product of the affine part and its error, divided by α_i^2, gives exactly these two powers. Squares of the error and cubic terms are bounded by the same expression, since their normalized sizes are bounded by $C(T\sigma_i^{\alpha_H} + T^2)$ and CT^3. If T is not small, whole-core bounds for the member and its own action, and the comparison affine quadratic value, give $Ce_i(1 + T^2) \leq Ce_i C_T T^3$. The fixed powers of D_X are summable against the tree decay. The finite set of young ages for which the global half-datum-ball condition fails is also covered by the whole-core bound: their σ_i have a fixed positive lower bound. This is precisely the reason to use the ordinary full-radius ratio $\rho_i \geq 1$ here.

At one base point, after its domain sum, the bound is therefore

$$Ce_i \left[\rho_i^2 \sigma_i^{4+\alpha_H} + \rho_i^3 \sigma_i^6\right]. \tag{OA14}$$

A current window contains at most $C\sigma_i^{-4}$ old base points. Its value bound is at most $Ce_i \rho_i^3 \sigma_i^{\alpha_H}$. Terms extending beyond that window retain an exponentially small tree-length tail. Outer and removed terms retain their actual response factors. Even the cruder amplitude $Ce_i \sigma_i^{-4} e^{-c/\sigma_i}$, when only a long-domain value bound is used, is bounded by $Ce_i \rho_i^3 \sigma_i^{\alpha_H}$. This does not give such a tail a square of the actual small curvature; it is a full analytic-radius estimate.

It is important where the inner estimate is localized. At each old base point form one window function from the near pointed sum, its own action counterterm, and one common critical box map. This entire function has one fixed current-scale base, of uniformly bounded size. Only then apply the fine brackets of the window formation rule. Thus the cancellation in (OA14) occurs before a single common bracket. No sup norm of a sum is used to bound a sum of differently based bracket operators.

On the independent fine discs the bracket with m differenced cubes costs at most $(e^{\bar{\kappa}} - 1)^{-m}$, with no factorial. The derivative at $\nu = 0$ costs the reciprocal of its fixed disc, hence only a fixed constant. For outer origins retaining their original base, split the old input rate into the domain-sum, long-tail and retained parts. If Y is the resulting current domain, the hull and rooting geometry gives $d_k(Y) \leq C_h(d_i(X) + m + 1)$; for a bounded window base it gives $d_k(Y) \leq C_h(m + 1)$. Choose the retained rates and $\bar{\kappa} - 1$ larger than $C_h(\lambda + \varepsilon)$, after spending the old rooted entropy and diameter powers. The remaining ε absorbs the polynomial number of window anchors at a fixed Y. This proves the following pointwise analytic bound for the actual chosen window filing:

$$|V_{k,i}^{\mathrm{old}}(Y; y)| \leq Ce_i \omega_{k+1-i} e^{-\lambda d_k(Y)}, \qquad \|y\|_\infty < R_k, \quad \omega_{a+1} = C(1 + a)^{3/2} L^{-\alpha_H a}. \tag{OA15}$$

The current origin $i = k$ uses its direct whole-core Cauchy bound and $\omega_1 = O(1)$; it needs no age improvement. The proof was carried out on the larger ball using a physical fluctuation amplitude at most $c_* \alpha_k$, rather than the tested amplitude δ_k. It does not use the final potential bound with a Gaussian-growth factor $1 + M^{-4} \sum_{b \subset Y} |y_b|^2$. Using that weaker envelope on a radius R_k would introduce R_k^2 and erase the desired Cauchy gain.

Let

$$\mathcal{E}_k = \sum_{i \leq k} \omega_{k+1-i} e_i \leq CE_0. \tag{OA16}$$

Absolute convergence is uniform on every smaller ball in (OA11), by (OA15). It justifies summing old levels before taking two coordinate derivatives. All estimates can first be made with finitely many old levels and on a finite torus. Their constants do not change in either limit.

Cauchy, rooted counting and the weighted precision gap. Write $P^{\mathrm{old}}_{k,I}(y) = \sum_Y \sum_{i\le k} V^{\mathrm{old}}_{k,i}(Y;y)$, with exterior variables fixed. Additive terms depending only on the exterior can be omitted. By two coordinate Cauchy formulas, including the factor 2! for a repeated coordinate, (OA15) gives

$$\left|\partial_{y^a_b}\partial_{y^{a'}_{b'}} \sum_{i\le k} V^{\mathrm{old}}_{k,i}(Y;y)\right| \le C\mathcal{E}_k R_k^{-2} e^{-\lambda d_k(Y)} \tag{OA17}$$

throughout the real product window. The derivative is zero unless both bonds belong to the declared domain, including its fixed collar. For such a domain, $d(b,b') \le C_M(1+d_k(Y))$. Choose $\mu>0$ so small that $\lambda-\mu C_M > \kappa_D$, retaining a strict gap. Root the domains at a cube containing b; the fixed number of possible root cubes costs only a constant. Then

$$\sum_{Y\ni b,b'} e^{-\lambda d_k(Y)} \le Ce^{-\mu d(b,b')} \sum_{m\ge 0} e^{-(\lambda-\mu C_M-\kappa_D)m} \le C' e^{-\mu d(b,b')}. \tag{OA18}$$

Consequently the full old-activity Hessian, in the actual Gaussian coordinates, satisfies

$$\left\|D_b D_{b'} P^{\mathrm{old}}_{k,I}(y)\right\| \le C\mathcal{E}_k \left(\frac{g_k}{\alpha_k}\right)^2 e^{-\mu d(b,b')} \le CE_0(\log\theta_k)^{-2q} e^{-\mu d(b,b')}. \tag{OA19}$$

Color matrix norms cost a fixed factor only. The count of bonds at distance in $[r,r+1)$ is at most $C(1+r)^3$. Thus for every $0<\mu'<\mu$, both weighted row and column sums of the Hessian are at most $C\mathcal{E}_k(g_k/\alpha_k)^2$. In particular, Schur's inequality gives

$$\|T_\phi D^2 P^{\mathrm{old}}_{k,I}(y) T_\phi^{-1}\|_{2\to 2} \le C\mathcal{E}_k(g_k/\alpha_k)^2, \qquad |\phi(b)-\phi(b')| \le \mu' d(b,b'), \tag{OA20}$$

where T_ϕ multiplies the three coordinates at b by $e^{\phi(b)}$. This estimate is uniform in I.

For completeness, suppose the actual free precision A, after the prescribed constraint and gauge fixing, has a gap $A \ge a_-\mathbf{1}$ and uniform exponentially summable entries. This is a separate free-background input, concerning the constrained pullback Hessian in the coordinates C_W, not the ambient curl Hessian. Its Dirichlet compression A_{OO} has the same gap. For small Lipschitz weights its conjugation difference has norm at most $C\mu'$. Hence the conjugated inverse of A_{OO} has norm at most $2/a_-$ when $C\mu' < a_-/2$, by the Neumann series. Applying this to $\phi(b)=\mu' d(b,b')$ bounds each inverse entry by $(2/a_-)e^{-\mu' d(b,b')}$. This proof is uniform in O. It follows by two exponentially summable convolutions that

$$K_I = ((A^{-1})_{II})^{-1} = A_{II} - A_{IO}A_{OO}^{-1}A_{OI}$$

has exponentially summable entries at a possibly smaller rate. Furthermore, $\langle v, K_I v\rangle = \inf_w \langle (v,w), A(v,w)\rangle \ge a_-\|v\|^2$. Thus the inverse marginal covariance, with its Schur correction, retains the free gap. When exterior coordinates are fixed, the free conditional precision is instead A_{II}, which also retains that gap by compression. The Schur matrix K_I describes the free marginal obtained by integrating the exterior. Neither statement identifies an interacting marginal with the pointwise potential $P^{\mathrm{old}}_{k,I}$ at fixed exterior data. The conjugation difference of K_I also has norm at most $C\mu'$: use $|e^{\phi(b)-\phi(b')}-1| \le \mu' d(b,b') e^{\mu' d(b,b')}$ and the row and column sums. Choose μ' so that this norm is at most $a_-/4$. Then

$$\mathrm{Re}\langle v, T_\phi(K_I - D^2 P^{\mathrm{old}}_{k,I}) T_\phi^{-1} v\rangle \ge \left(\frac{3a_-}{4} - C\mathcal{E}_k(g_k/\alpha_k)^2\right)\|v\|^2. \tag{OA21}$$

For a sufficiently large coupling floor the old activities cost at most $a_-/4$, leaving a weighted gap of at least $a_-/2$. Other intrinsic or remainder Hessians must be charged separately. The reference precision is held fixed while differentiating in y; no derivative of a covariance with respect to y is introduced.

Remainders and the weaker finite-box alternative. A holomorphic aged remainder with no fixed color spectators has zero linear term at the trivial pair, but has no action counterterm. Its vacuum-only window bound has weight ρ_i^2 instead of ω_{k+1-i}. The same proof gives (OA19) with $\mathcal{E}_k$ augmented by $\sum_i \rho_i^2 r_i$, provided its actual maps have the same domains and locality. If $r_i \le C\theta_i^{-s}$, $s > 2$, and $\theta_i \ge \theta_k + c(k-i) - C_0$, this sum is at most

$$C\sum_{a\ge 0}(1+a)(\theta_k + ca)^{-s} \le C_s\theta_k^{2-s}. \tag{OA22}$$

To see the last inequality, compare the sum with its integral and put $a = \theta_k t$. The remaining integral of $(1+t)(1+ct)^{-s}$ is finite. This optional conclusion does not cover a boundary term retaining stored color variables or a term for which only an integrated norm is known.

If only a bound on the whole finite-box analytic function is available, a useful but weaker statement remains. Suppose $|P(y)| \le CE_0(r+D)^A$ on a Gaussian polydisc of radius $c\alpha_k/[g_k(r+D)^B]$. Then, on the concentric polydisc of half that radius, the same coordinate Cauchy estimate and $m_I \le C(r+D)^4$ give

$$\|D^2P\|_{2\to 2} \le CE_0(r+D)^{A+2B+4}(g_k/\alpha_k)^2. \tag{OA23}$$

Applying (OA23) throughout the operational real window requires that this window lie in that smaller polydisc; this inclusion is an additional hypothesis, especially when $B > 0$. Indeed each matrix entry has the bound without the last power $(r+D)^4$, and the row and column contain at most m_I entries. For a direct volume bound $A = 4, B = 0$, this is $CE_0(r+D)^8(\log\theta_k)^{-2q}$. When $r + D \le C(\log\theta_k)^{2p}$, it tends to zero if $q > 8p$. It preserves an unweighted gap. It does not imply (OA20) at a fixed positive spatial weight: conjugation could cost $e^{\mu' \operatorname{diam} I}$. The rooted localization in (OA15)–(OA18), rather than a count of window coordinates, is what proves the stronger weighted conclusion.

What the estimate does and does not settle. Under the stated ordinary analytic map hypotheses and the reviewed pointed and window inputs, the old regular-activity contribution to the localized Hessian is bounded by $CE_0(\log\theta_k)^{-2q}$, uniformly over old ages and box sizes. The analytic radius is in the physical-to-Gaussian coordinate conversion (OA11). A bare operational cutoff domain, or only a Gaussian-growth potential bound on that domain, would not prove it. The two required primitive verifications are the uniform unscaled pivot estimates (OA5)–(OA6) and the all-domain first-response and membership estimates (OA7), including towers and outer substitutions. They belong to the actual ordinary background analysis and have not been proved for the Yang–Mills maps merely by this deduction.

The statement has no automatic consequence for the change in force when exterior data are changed. For that one needs, separately, uniform estimates such as $\|D_b(P_e - P_{e_0})\| \le C\mathfrak{b}_k e^{-cd(b,O)}$ on the actual compatible exterior and stored data. A Hessian bound in the variables of I does not bound derivatives with respect to the exterior parameters. Nor does it identify a clean-class restriction with a Gaussian fibre kernel. Field-independent remote or holonomy bounds without the analytic domain assertion used above remain separate tails; they receive neither an invented curvature factor nor an unjustified Cauchy estimate.

32. Large-field histories and the induction

Proof outline. The supporting analytic results used in this section, including the obligations that we must still prove, are listed under C in the dependency catalogue. Their hypotheses and precise forms are retained at the points where the argument uses them.

2. Where the fields are wild: large-field regions, the factor paying for them, and the closed induction. Convergence of the expansion of Step 1 requires that the plaquette variables of the backgrounds lie within $\varepsilon_k\eta_k^2$ of $\mathbf{1}$, with $\varepsilon_k = g_k p_0(g_k)$, and that the fluctuation fields stay below $\delta_k = (A_1/A_0)\varepsilon_k$. The two thresholds carry the same power of the logarithm and differ by a small fixed ratio. Nothing guarantees these conditions. For this reason each step inserts decompositions of unity $1 = \sum \prod \chi \prod \chi^c$ built from local tests. At a place where a test fails, no expansion is made and the fields there are stored. The weights present at that place then supply a factor $e^{-c\,x_k\varepsilon_k^2} = e^{-c\,p_0(g_k)^2}$. For a failed plaquette test this weight is the Wilson weight. When an approximate fluctuation test fails, the weights present are the Wilson weight together with the kept gauge-fixing exponents. When a fluctuation test, a collar test or a rim test fails, the weight is only the Gaussian form of the stored fluctuation field. (The failure of one of the sharper tests of the large-field operation $\mathbf{R}$ of (2b) yields only $e^{-p_0(g_k)}$: 2a, 2c.) In this way the density becomes a sum over histories h, collected according to their labels $\mathcal{S} = (\Omega_j, \Lambda_j, S_j)_{j\le k}$, which consist of the regions of all levels. Besides the regions, a label carries the marks left by the operations $\mathbf{R}$, one for each island and for each drop-out that stands (2a); the labels are thus labels with marks. The exponential form given below holds for sums of histories and not for an individual history (2a, 2d).

Theorem 2.1 [inductive description; ultraviolet stability]. Suppose that the constants are chosen in the prescribed order, beginning with the exponents and a large L and ending with a small γ (2e). Suppose also that the torus is admissible up to level k, and up to K for the stability bounds; that is, $\ell L^{K-j} \ge 2L^{11}M\check{R}_j$ for $j \le k$, respectively for $j \le K$. This is a lower bound on ℓ, which is chosen after M and g. When $\ell \in \mathfrak{L}(g)$ and the datum is a first-passage datum, the condition follows from the upper flow bound up to K (3b). Assume further that $g_j \le \gamma$ and $\lambda_\sharp(j-i) - c_5 \le x_i - x_j \le B_\sharp(j-i)$ for $i \le j \le k$ (up to K for the stability bounds). When g is small, (3a) provides these conditions for any first-passage datum; they are conditions on how the couplings flow: β has no upper bound, and $K = K(\beta, g)$ is determined together with it. Then $\rho_k = (\mathbf{RT})^k\rho_0$ is defined, and

$$\rho_k(V_k) = \sum_{\mathcal{S}} \mathbf{T}'_k(\mathcal{S})\big[\chi_{k,\mathcal{S}} \exp A_{k,\mathcal{S}}(U_k(\mathbf{V}))\big],$$
$$A_{k,\mathcal{S}} = -A(x_k(\cdot), U) + \mathbf{E}_k + \mathbf{R}_k + \mathbf{B}_k - \mathcal{E}_{k,\mathcal{S}},$$

$$\|E^{(j)}\|_{j,\kappa} \le E_0,$$
$$\|R^{(j)}\|_{j,\kappa} \le g_j^{\kappa_0},$$
$$|B^{(j)}(X,\cdot)| \le \mathsf{B}\, e^{-\kappa d_j^\natural(\mathcal{S};X)},$$
$$\mathsf{M}(i,X) \le e^{-(2-c')p_0(g_i)},$$

where $-\mathcal{E}_{k,\mathcal{S}}$ is the constant term of the action of the label and B is a constant fixed before γ (2d, 2e); $d_j^\natural$ is the hull length of (2d), and $\mathsf{M}(i,X)$ is the mass of a component X that, at some level $i \le k$, has passed all those tests of the operation $\mathbf{R}$ that come before the last one. Such a component is either healed at that level or an island, which remains in the label. The mass is defined as follows. Let e^{Φ_ω} be the integrand of the operation $\mathbf{T}'_i(X)$ belonging to the component. This integrand contains the Wilson factor of the core (2b, 2c), and its exponent has the form $\Phi_\omega(\xi;\mathbf{p}) = \Phi^\circ_\omega(\xi) + \sum_f \Phi_{\omega,f}(\xi;\mathbf{p})$, which is Borel in ξ, analytic in $\mathbf{p}$ and real at real pairs; each factor $\Phi_{\omega,f}$ depends on ξ and on $\mathbf{p}$ only on a support $R_f \subset X^\sim$ (2d). Let ω run over the local histories of X, each with a frame in the hypercubic group W_4. Let ξ denote the stored variables

and μ_ω their measure. Let $\mathbf{p} = (\mathbf{U}, \mathbf{J})$ denote the complex pairs of the space $U_i^c(X^\sim, \alpha_{0,i}, \alpha_{1,i})$, which contains the pairs at which a term evaluates the factor. Let $\mathsf{v}(\omega)$ be the space-time volume of the patch, which counts $\log x_j$ for each stored site of level j (2c). Then

$$\mathsf{M}(i, X) := \frac{1}{|W_4|} \sum_{\omega} e^{c\,\mathsf{v}(\omega)} \int \sup_{\mathbf{p}} \left| e^{\Phi_\omega(\xi;\mathbf{p})} \right| \mu_\omega(d\xi).$$

The operation of a drop-out contains no Wilson factor of its own. For a drop-out the same bound holds in a modified form, with two changes. First, the Wilson weight over X of the carrying term is included with the operation; the carrying term is a term of some level $k \geq i$ in which X stands. Second, the supremum is taken over the evaluations at ξ. These are the real configurations that are compatible with ξ, regular layer by layer, and at which every test listed by ω, with the exception of the test whose failure made X a drop-out, takes its recorded value (the full list of these conditions is given in (2c)). This form follows from the induction and is not one of its clauses. The mass bound holds for every $c \geq 0$ in the weight $e^{c\,\mathsf{v}}$ and for every small $c' > 0$ in its exponent, provided the ceiling on the couplings is chosen after both c and c'; the reserve constant c_{r} of (2e) is one such c. Apart from the core factors of the islands in Z (2b), the operation $\mathbf{T}_k(Z)$ contains no Wilson weight. The factor coming from a failed plaquette test sits in $e^{-A(x_k(\cdot), U_k)}$, which is the function the operation acts on. In the sum over the labels, every component C of Z_k keeps a factor $e^{-c\,p_0(g_k) - \kappa_1 |C|_k}$. The constant $\kappa_1 > 0$ outweighs the number C^n of connected sets of n cells that contain a given cell (2c). Each term is a non-negative measure. Neither $E^{(j)}$ nor $R^{(j)}$ carries a label index. On non-wrapping domains they are the same functions on every torus and at every cutoff: they are the values, at the lifts of the domains, of a single family on $\mathbb{Z}^4$. (The constant $\mathcal{E}_{k,\mathcal{S}}$, by contrast, depends on the label and on the torus.) Finally, let $\chi_k = \chi_k(T_\ell)$ denote the product, taken over every cube of the torus, of the tests $\chi_k(\square)$. Then, uniformly, there is a constant E_- such that

$$\chi_k \, e^{-x_k A(U_k(V_k)) - E_- |\mathbb{T}^{(k)}|} \leq \bar{\rho}_k,$$

$$\int \bar{\rho}_k \, dV_k \leq e^{C(1 + \log x_k) |\mathbb{T}^{(k)}|},$$

$$\int \rho_k \, dV_k = Z_{K,\ell,\beta}.$$

The theorem is the case $\theta = x(\beta)$ of an induction carried out along an arbitrary frame θ (the opening of Step 3). That induction runs on all admissible tori simultaneously. The tube is a clause of the class $\mathfrak{C}_k(\theta)$ of (2e), and the step by itself does not yield this clause again (Theorem 2.11). For complex field arguments, each term depends on a triple $(U, A', \mathbf{J})$, in which U is G-valued and $A', \mathbf{J}$ are complex. For a term of level j, the pair of a triple is $\mathbf{p} = (\mathbf{U}, \mathbf{J})$ with $\mathbf{U} = e^{i\eta_j A'} U$, and the triple is called a splitting of its pair. The domains of analyticity are defined by conditions on triples. One of the statements propagated by the induction is that a term takes the same value at any two splittings of one pair. The splitting is therefore a coordinate and not an argument, and a term is a function of the pair. (In Step 1 the values coincide on a complex gauge orbit, with a margin left in the radii. Here what is asserted concerns two splittings of a single pair, and it holds on the domain itself.)

The core of the theorem is due to Bałaban: Comm. Math. Phys. 119 (1988) 243, Theorem 1 and Corollary 3, supplemented by 122 (1989) 175 and 122 (1989) 355, Theorem 1. For the idea of $\mathbf{R}$, he cites Gallavotti and co-workers as the closest earlier work. He states, for one torus, that neither the regular terms nor the part of the remainders produced by the fluctuation integral depend on the label (119, pp. 278–279). He also names a second choice of the thresholds and

sets it aside; in that choice only the logarithm is evaluated on a bound of the flow known a priori (p. 246). The upper bound he gives is pointwise, $\varrho_k(V_k) \le e^{E_+ |T_\eta|}$, stated for his density and his volume ((2.50)), and Theorem 2.1 gives the integrated form in its place. The extension adds the following, which belong to the Package C statements that we must prove:

- that independence with a range covering the remainders in their entirety, simultaneously on all admissible tori and cutoffs, by means of one family on $\mathbb{Z}^4$ (for the counterterm, the paper 119, p. 281, uses "$\mathbf{E}^{(j)}(U_j, z)$ defined on the whole lattice $L^{-j}Z^d$");
- the positivity of each term, as a clause of the statement (in 122 (1989) 175, p. 176, Bałaban announces that the densities $\rho(Z'', V)$ are positive);
- measure and formula as two separate layers;
- the class taken as a set of presentations along a frame;
- the flow inequalities, hypotheses of Theorem 2.1, obtained as conclusions of the induction, carried out jointly with the flow (3a);
- covariance of the R-terms under translations only, term by term, and covariance of their totals by way of measures;
- the mass, with the complex supremum inside the integral and with a reserve weight;
- the hull length;
- the three possible fates of a component, two of which are recorded as marks of the label (in 122 (1989) 175, pp. 181–183, Bałaban removes from the region on which $\mathbf{R}$ acts any component in which a test of some stage fails, and in 122 (1989) 355, (1.102), he retains the operation $\mathbf{T}_k(Y)$ where the last test fails), and the two sequences of regions attached to a label;
- an order of the constants that contains no circle;
- exactness for each partition of the histories.

The route also makes use of the following:

- the collar and the rim of (2a), which ensure that in the interior the mean of the conditional Gaussian measure stays under the thresholds set by the tests;
- the factor depending on two levels in the sum over the scales of the R-terms (2e);
- the terms $V(Y)$, which are not of small amplitude, with the consequent removal of the empty family before the exponentiation (2d);
- in (2c), a single count over the genealogy of a component, which replaces the double induction of 122 (1989) 355, pp. 384–387.

In that induction, the case in which two regions join rests on a sentence on p. 387. That sentence states that after n_1 further hull operations, where n_1 is a fairly small number, for instance $n_1 < 10$, the domain obtained from the region whose wait is the longer of the two contains the domain obtained from the other region. The count never compares two such regions. A joining of regions costs it no separate payment, because the count charges the joining to a created cell (2c). The later steps rely on the theorem and on its sourced form (4a), which contains the insertion formula.

(2a) The tests, the nested regions, and the content of a history. **The tests of step** $t = k + 1$ are applied to a term of ρ_k whose live region is $Y = \mathcal{Z}_k$. First, outside a margin of Y, the new field V_t undergoes the plaquette test $\chi_t(\square)$, and the cubes where this test fails make up P_t. The factor $\mathfrak{G}_k$ is inserted away from $P_t \cup Y$. By the Faddeev–Popov argument, its integral over the gauge group of the blocks equals 1. Next comes the approximate fluctuation test $\chi'_k(\square) = \mathbf{1}\{|V_k(\mathcal{M}^k U_{t,\square}(V_t))^{-1} - \mathbf{1}| < 2\delta_k\}$. This test is evaluated at the bonds of $\square^{\sim 2}$ other than the pivot bonds (1d), and the third test below is evaluated at the same bonds. Here one takes from the orbit $U_{t,\square}(V_t)$ an element whose k-th average lies in the block axial gauge, and every such element gives the same tested number. The cubes where this test fails make up Q_t. These sets determine Ω_t, the region on which $U_t(\mathbf{V})$ and $\mathcal{A}_k$ exist. The backgrounds and the propagators on sequences

of regions are taken from (1b), (1c), together with their regularity and their decay in the local scale. A label's sequence starts at $\Omega_0 = \Omega_1^{\sim}$, the first region enlarged by one layer of level-1 cells, and not at T_ℓ; Theorem 4.13 and Lemma 4.15 require this. After that the test $\chi^{(k)}(\square)$ is made, and its failures make up R_t. Hulls $(', \widehat{\cdot})$ and layers are taken in level-t cells. The boxing map bx replaces each component contained in a cube of $O(1)$ cells by its bounding box, and repeats this until nothing changes. With these conventions,

$$\Omega_t^c = Q_t'^{\sim 2} \cup P_t'^{\sim 4} \cup \widehat{Y}^{\sim 8},$$

$$Z^{[0]} = \mathsf{bx}\big((\Omega_t^c)^{\sim 2} \cup R_t'^{\sim}\big),$$

$$Z^{[n+1]} = \mathsf{bx}\big(Z^{[n]} \cup (\text{cells of the failures of round } n)^{\sim}\big),$$

where the rounds are those of the recursion described below. The set $Z_t = \Lambda_t^c$ is the final $Z^{[n]}$, and $S_t = (\Omega_t \setminus \Omega_t^{\sim -1}) \cup R_t$. The numbers of layers are those of Bałaban. If the boxing map is omitted, $Z^{[0]}$ coincides with his Λ_t^c (Comm. Math. Phys. 119 (1988) 243, (3.16), (3.20)). The order of these numbers matters: an older set receives more layers, and each move is made one layer inside the region of the move before it. The creation set $\mathcal{W}_t^{\wedge}$ is the union of the hulls of every failure of the step: P_t', Q_t', R_t', the cells of the collar bonds that fail, and the rim cells that fail. If nothing is created, the step reduces to the free evolution

$$\Phi_t(Y) = \mathsf{bx}\big(\widehat{Y}^{\sim 10}\big);$$

in general

$$Z_t \subset \widehat{Y}^{\sim 10} \cup (\mathcal{W}_t^{\wedge})^{\sim O(1)}$$

up to boxing

That is, one coarsens, adds a margin and boxes; this is the $S(Z)$ of (2c). The size $\check{R}_k$ is the smallest power of L that is not less than $(\log \min_{i \le k} x_i)^r$, with $r \ge 2$. Because it is a power of L, all partitions are nested. Because it is formed from the running minimum, it is non-increasing (Bałaban forms it with g_k itself), and therefore $t \mapsto t - \mathsf{N}_t$ is strictly increasing. Moreover $\check{R}_k \le L\check{R}_{k+1}$, and two consecutive drops never occur, because x changes by $O(1)$ per step and γ is small. A test cube $\square$ has side of order $M_2\check{R}_t$. Here M_2 is a power of L with $L^2 M_2 \le M$ and $c\,\delta_0 M_2 \ge 1$. The last condition is a floor on M_2, and δ_0 is fixed together with L, which comes earlier. Hence a test cube lies well inside a cell of its own level.

What factorises, and why. The tests χ_t and χ_k' are built from localised backgrounds. They depend on V_t, V_k only within $O(1)$ test cubes. The third test, and also the collar and rim tests, depend on $\mathcal{A}_k = \frac{1}{i}\log(V_k(V_k^{\mathrm{bg}})^{-1})$, which is formed with the true averaged background on Ω_t. These tests are therefore local in the new variable $\mathcal{A}_k$, but not in (V_k, V_t). Once $\mathcal{A}_k$ has become the variable, the surviving χ_k' (on S_t) compare $e^{i\mathcal{A}_k}V_k^{\mathrm{bg}}$ with $\mathcal{M}^k U_{t,\square}$. They therefore depend on V_t over the whole of Λ_t. Four facts together ensure that everything attached to Z_t factorises over its components:

- The backgrounds are localised.
- The substitution on the strip (2b) is made. After it, every indicator kept in $\Omega_t \cap \Lambda_t^c$ depends on V_t in Λ_t^c alone, apart from the factors that equal 1 on the supports.
- The conditional quadratic form is cut (2b).
- The pivot values are dealt with as follows. An old test $\chi_k(\square)$ of a cube $\square \subset \Omega_t^c$ adjacent to Ω_t also depends on V_k at pivot bonds (1d) inside Ω_t. There V_k is not a stored variable, and through $e^{i\mathcal{A}_k}V_k^{\mathrm{bg}}$ a test of one component depends on the fields stored in another. At

such a bond, V_k is given the value that solves, for the pivot bond, the averaging constraint $\mathcal{M}(V_k)(c) = V_t(c)$ of the single coarse bond c. Bałaban (122 (1989) 175, (1.3)) retains only those old tests whose cubes lie outside $\Omega_t^{\sim}$; the tests that he leaves out equal 1 on the supports by Lemma 2.2 (a), whose margin is this layer.

Inside $\mathbf{R}$, the tests are built from backgrounds localised in the component, never from the background of the torus. The induction carries the following property: each indicator stored in $\mathbf{T}_k(Z)$ depends only on those field variables that are localised in Z.

Totality. By definition, a test is the indicator function of the set where all objects occurring in it are defined and where its inequalities hold. These objects are the localised minimal orbit, which must exist and be a single orbit, the contour variables, and the logarithms. Consequently $\chi^c = 1$ wherever some object is undefined. A factor of $\mathfrak{G}_k$ equals 1 where $V(y,x)$ is undefined, by a separate convention for that factor. Hence $\int \mathfrak{G}_k(V^u)\,du = 1$ for every V, and each insertion $1 = \sum \prod \chi \prod \chi^c$ is an identity that needs no assumption. The tests are not empty conditions. The averaging constraint transports smallness: $|\partial V_k - \mathbf{1}| < 2\varepsilon_k$ implies $|\partial V_t - \mathbf{1}| < CL^2\varepsilon_k$. Therefore $U_{t,\square}(V_t)$ exists on the carried supports, and the new test further restricts a field that is regular already (Comm. Math. Phys. 119 (1988) 243, p. 246). The cost is that χ^c contains configurations that are not large in any quantitative sense. These configurations must also produce the small factor (2c).

Collar and rim: this device is an addition to the step of Comm. Math. Phys. 119 (1988) 243, Sect. 3. The stored part of the fluctuation field moves the mean of the conditional Gaussian measure of (2b). The thresholds δ_k are stated in the unscaled normalisation, and in that normalisation $\mathcal{A}_k|_{\Lambda_t}$ has the mean

$$\mathbf{m} = -\,C^{(k)}(\Lambda_t)\,\Delta^{(k)}\,\mathcal{A}_k\big|_{\Omega_t\setminus\Lambda_t}$$

This map is linear, so in terms of the scaled fields A, A' of (2b) it is the same map, with each threshold divided by g_k; only $|\mathbf{m}|$ enters. This shift must remain well below the least threshold among the inner tests; that threshold is $\theta_c\delta_k$ on bonds that cross $\partial\Lambda_t$, and δ_k further in. A recursion therefore keeps enlarging the region until, along its boundary, $\mathcal{A}_k$ satisfies two sharper tests:

- $|\mathcal{A}_k| < \delta_k/(4\Theta_1)$ on the collar, the set of stored bonds whose distance from Λ_t, measured in level-k units, is at most ϖ_T. With a qualifying word added, the same name serves for three further objects. In (2b), the collar of a stage means the shell over which that stage integrates. In (2c), the waiting collar is the set that keeps $X \cap \Omega_{m_\flat}$ apart from the floor. In (2d), (2e), the collar of a component X means the enlarged component $X^{\sim}$, that is, X together with one layer of M-cubes of its level.
- $|\mathcal{A}_k| < \theta_c\delta_k$ on the rim, the set of cells of level t of Z_t that touch Λ_t.

At greater distance, all that is known is $|\mathcal{A}_k| < C\delta_k$. Every round examines bonds and cubes that no earlier round examined. Hence the recursion terminates, and the admissible outcomes it produces constitute a decomposition of unity. The kernel of $\mathbf{m}$ is bounded, with exponential decay. The contributions of the collar, the rim and the remaining part give

$$|\mathbf{m}| \le C\big(\Theta_1^{-1} + \theta_c e^{-c\varpi_T} + e^{-cM\check{R}_t}\big)\delta_k < \tfrac{1}{2}\theta_c\delta_k$$
$$\text{if } \theta_c\Theta_1 \ge C,\ \varpi_T \ge C,\ \gamma \text{ small.}$$

The rim fraction $\theta_c \in (0,1)$, the collar width ϖ_T and the collar factor Θ_1 serve this purpose.

Lemma 2.2 [the new tests imply the old; the form reproduces itself]. (a) On $\Omega_t^{\sim}$, that is, on Ω_t enlarged by a margin of one layer of cells, the indicators in $\mathfrak{G}_k$ and the tests $\chi_k(\square)$ that the term carries all equal 1. This holds on the support of the new tests together with the tests

and averaging constraints that the term still carries alongside them. (b) The test $\chi^{(k)}$ is made only on cubes $\square \subset \Omega_t^{\sim -1}$. For such a cube, if $\square \not\subset R_t$, then $\chi^{(k)}(\square)$ implies $\chi'_k(\square)$. Consequently, within Ω_t, the test χ'_k remains only on S_t. (c) After the step, the tests that stand outside the integral operations are exactly $\chi_{t,\mathcal{S}} := \chi_t(\Omega_t) = \prod_{\square \subset \Omega_t} \chi_t(\square)$. They are formed with the action sequence defined below, and with χ_t^c on an island less a margin.

For (a): up to a gauge transformation,

$$U_{k,\square} = \exp\big[i\eta_k \mathbf{H}_{k,\square}\big(\tfrac{1}{i}\log[V_k(\mathcal{M}^k U_{t,\square'})^{-1}]\big)\big] U_{t,\square'}$$

. Here $\mathbf{H}_{k,\square}$ is the localised response $\mathbf{H}_k$ of (1b), so the argument has the form displayed only off a boundary layer of the cube; inside that layer the argument is built from the lift of V_k and is bounded by $C\varepsilon_k$. It follows that

$$|\partial U_{k,\square} - \mathbf{1}| \le (1+c)\varepsilon_t \eta_t^2 + C\big(\delta_k + e^{-c\,\delta_0 M_2 \check{R}_k}\varepsilon_k\big)\eta_k^2 \le \big(CL^{-2} + CA_1/A_0 + Ce^{-\check{R}_k}\big)\varepsilon_k \eta_k^2 < \varepsilon_k \eta_k^2.$$

Three mechanisms appear repeatedly:

- Curvature scales by the factor $\eta_t^2/\eta_k^2 = L^{-2}$.
- The ratio $\delta_k/\varepsilon_k = A_1/A_0$ is small.
- The artificial boundary conditions of localised minimisers act only through $e^{-c\delta_0\,\mathrm{dist}}$ (1b, 1c). On cubes whose side is proportional to $\check{R}_k \approx (\log x_k)^r$, this quantity is smaller than every power of g_k, because $r > 1$. In the present estimate it needs to beat only constants; on the strip (2b) it must beat a negative power of g_k (Bałaban, Comm. Math. Phys. 119 (1988) 243, (3.2)–(3.8)).

For (b):

$$|V_k^{\mathrm{bg}}(\mathcal{M}^k U_{t,\square})^{-1} - \mathbf{1}| \le C\tfrac{A_0}{A_1} e^{-c\check{R}_k}\delta_k < \delta_k$$

((3.17)–(3.19) of the same paper). The factor 2 between the two thresholds gives exactly the room needed for this. For (c): a test cube of step t serves also as a test cube of the density of level t, carrying the same background and the same threshold. The old tests inside Ω_t^c pass into the integral operation. The same ladder of implications holds inside $\mathbf{R}$ (2b). On the support of the tests of a stage, the following tests equal 1: those of the preceding stage, and those of the single stored step whose fields the stage integrates (the plaquette tests of its two levels, the gauge indicators, the approximate fluctuation tests). Further, consider any point at which a term is evaluated: the triple $(U, 0, J(U))$, where U is the background of the label, at fields where no characteristic function of the term vanishes. At such a point, $\chi_q(\square)(V_q) = 1$ holds on every cube $\square$ of level q that lies in the layer $\Omega_{q+1}^{\sim} \setminus \Omega_{q+1}$ and not in a drop-out. Also at such a point, the current of the bare field, formed from the plaquettes inside Ω_0 only, stays within the bound that the domains of analyticity impose on their current (the size $|\mathbf{J}|$ is smaller than the first radius of the domain, measured in the units of the local scale). This bound holds also at the bonds of Ω_0 that have a plaquette leaving Ω_0.

Proposition 2.3 [geometry]. The regions are nested, $T_\ell \supset \Omega_1 \supset \Lambda_1 \supset \cdots \supset \Lambda_k$, with gaps of at least $cM\check{R}_j$ in level-j units. Every large-field region and every creation set is a union of cells of its own level. Both statements hold for the regions that the steps produce (119, p. 256). Each operation $\mathbf{R}$ has stage sequences and prepared sequences (namely the regions that the label has after m of its stages, and after all N_k of its stages; see (2b)), on which the backgrounds of (2b), (2d) are formed, and drop-outs keep nests. In these sequences and nests the boxes are only unions of M-cubes, and the only lower bound on the gaps is one M-cube of the later level:

$$\operatorname{dist}_\infty(Z_j, \Omega_{j+1}) \geq LM,$$
$$\Omega_{j+1} \subset \Lambda_j^{\sim -1},$$
$$Z_j^\sim \cap \Omega_{j+2} = \emptyset$$
(level-j units; layers of M-cubes of level j),

the first of these inequalities being an equality in the upper part of a nest, where $\Lambda_j = \Omega_j$; the reason is that a step places eight layers around the old region, and the preparation removes most of them again. Consequently the estimates of (1b), (1c) must hold with a separation of only one M-cube. A smallness $e^{-c\check{R}_j}$ from a separation is available only at regions produced by steps. In each decomposition exactly one outcome equals 1, and hence $\rho_k\, dV_k = \sum_h \tau_h$ holds exactly. A history records every outcome: P_t, Q_t, R_t, the collars and rims, and the outcomes inside $\mathbf{R}$. It determines the label, the creations and the ages. There are two kinds of creation:

- Creations by step t are the cells $\mathfrak{c}$ of $\mathcal{W}_t^\wedge$, which are $M\check{R}_t$-cubes of level t. They are counted one cell at a time, and each carries one factor $e^{-c\, p_0(g_t)^2}$.
- Creations by an operation are its marked components (next paragraph); the tests of an operation are decompositions of unity component by component. Each marked component counts as one creation and carries one factor $e^{-p_0(g_t)}$, which must pay for $\mathsf{N}_t + 1$ levels of one small box.

All later statements of locality rest on this.

Three fates, marked components, two sequences of regions (Bałaban declines "a systematic description of all possible cases": 122 (1989) 175, p. 183). We set $t_\flat := t - \mathsf{N}_t$. In each component X of its class (2b), the operation $\mathbf{R}$ of level t prepares a nest $Z_l''(X)$, $t_\flat < l \leq t$. The nest consists of increasing boxes in $\Omega_l^c \cap X$. Each box is what remains of the large-field region of scale l after the stage that integrates the stored fields of level $l-1$. A component has one of three fates:

- *Healed* (every test passed): $\Omega_l, \Lambda_l, S_l \mapsto \Omega_l \cup X, \Lambda_l \cup X, S_l \setminus X$ for $l \leq t$, and X leaves the label.
- *Island*, also called second class (only the last test fails, namely χ_t of the new field on X less a margin): the same replacement is made for $l < t$ and for Ω_t, S_t, but Λ_t is left unchanged. Thus $X \subset Z_t$ remains, as a component without inner regions. On it, $\mathbf{T}_t'(X)$ is kept together with its variables, its measure and its factors (2d).
- *Drop-out* (some test fails after $n \leq \mathsf{N}_t$ stages): X remains in Z_t and retains, as its integral operation (2b), the operations carried out up to that point. Moreover

$$\Omega_l \cap X = X \setminus Z_l''(X)$$
$$(t_\flat < l \leq t_\flat + n),$$
$$\Lambda_l \cap X = X \setminus Z_l''(X),$$
$$S_l \cap X = \emptyset$$
(the same l, $l < t$),

all other entries unchanged, so $\Lambda_t \cap X = \emptyset$.

(For $n = \mathsf{N}_t$, then, the entry $S_t \cap X$ keeps its previous value; on the support of the tests of the last stage, the tests retained on it equal 1, by the ladder of implications inside $\mathbf{R}$ that is stated in the argument for Lemma 2.2.)

Islands and drop-outs are the *marked components*. A mark persists until a later healed component or island swallows it. The word also denotes two other objects: the mark $\Theta^{(k)}$ below, which is a weight on histories, and a marked term of a cluster expansion (2e). A drop-out's mark keeps the regions $\Omega_l^\circ = \Omega_l \cap X$, and likewise $\Lambda_l^\circ, S_l^\circ$, which X possessed before the operation. For $0 \le \boldsymbol{\nu}(X) \le n(X)$ we set

$$\Omega_l^{\boldsymbol{\nu}} \cap X = X \setminus Z_l''(X) \ (t_\flat < l \le t_\flat + \boldsymbol{\nu}(X)),$$
$$\Omega_l^{\boldsymbol{\nu}} \cap X = \Omega_l^\circ \ (t_\flat + \boldsymbol{\nu}(X) < l \le t_\flat + n(X)).$$

The choice $\boldsymbol{\nu} = n$ gives the *determining sequence*, which describes the field in its present state. It governs the determining sets, $U_k(\mathbf{V})$ and the propagators; the analyticity spaces involve all the $\Omega^{\boldsymbol{\nu}}$. The choice $\boldsymbol{\nu} = 0$ gives the *action sequence*, the regions for which the terms were constructed. It governs the index sets of $\mathbf{E}_k, \mathbf{R}_k, \mathbf{B}_k$, the profiles φ_j and hence $x_k(\cdot)$, the carried tests, and the constants $\mathcal{E}_{k,\mathcal{S}}$. No single sequence can play both roles. A stored region in a determining position requires variables that have already been integrated. A stage region in the sums re-sorts terms that were made for other regions. Three facts make the device workable. *Marks wait*: a mark of level s lying inside a class component of level t satisfies $s < t_\flat$. Hence standing marks that share a part are nested, their windows $(s_\flat, s]$ are disjoint, and at any one place and level at most one drop-out mark is active. Next, each $\Omega^{\boldsymbol{\nu}}$ is the label of an actual history, the one in which the operations stop after $\boldsymbol{\nu}$ stages. Its steps, classes and nests coincide with those of the history at hand, and so Proposition 2.3 applies to it: *the later geometry does not depend on how far an interrupted operation had progressed*. Finally, a term created at a moment μ (following the step, or the operation, of a level $j \le t$) retains permanently the space in which it was made:

$$\Omega_l^{\boldsymbol{\nu}_\mu}(\mathcal{S}_t) = \Omega_l(\mathcal{S}^\mu) \cup Z_{[j,t]}$$
$$(l \le j),$$

where $\mathcal{S}_t$ denotes the level-t label as it stands once the operation $\mathbf{R}$ has been performed, $\boldsymbol{\nu}_\mu = n$ on each mark older than μ and is 0 on each later mark, $\mathcal{S}^\mu$ is the label at that moment, and $Z_{[j,t]}$ is the union of all components that were healed or became islands at levels $j, \dots, t$.

Locality in the history. If the large-field regions of two histories never touch, the two histories combine into a single history $h^1 \cup h^2$. There are two reasons for this. (i) *A margin beats a reach*: two regions made of level-t cells that do not touch are separated by at least one cell. One cell exceeds the largest range of a bond test, of a pivot bond and of a collar (in the case of the collar, this is the floor $M \ge \varpi_T$ of (2e)). Taking hulls and adding layers commute with forming unions, and boxing depends on a single component only. A step therefore acts on each component separately. (ii) The class of $\mathbf{R}$ depends on the past: it depends on every object whose creation falls in the window $(t_\flat, t]$, or which carries a mark at a level in $[t_\flat, t)$. Moreover, the recursion is not monotone: a later region may cover a healed place once more. *Window lemma*: no creation within the window of t has been healed. Indeed, healing at a level $i < t$ something created at step s requires $s \le i - \mathsf{N}_i$, and for something marked at level s it requires $s < i - \mathsf{N}_i$. On the other hand $i - \mathsf{N}_i < t - \mathsf{N}_t \le s$, since $\mathsf{N}_t = \check{R}_t$ is non-increasing. Within the window, therefore, the creations and the marks of each history still lie in that history's own region. It follows that each history is the superposition of its clusters, and that clusters may be interchanged between histories. Each cluster has a root. The root is of one of two kinds: a component that remains live at the last level, or a healed component, considered at the level where its healing took place. The cluster is made up of the components of earlier levels that flow into the root, together with everything the history records in them. Each component, at whatever level, lies in the cluster

of precisely one root, namely the last of its successors; thus nothing recorded about the healed components is lost.

A hard-core gas; the standing part. Erasing clusters of healed components from a history yields a history with the same label. The *standing part* $h^\flat$ of h is what remains after the clusters of all its healed components are removed. For a string s of outcomes of the next step one has $(h,s)^\flat = (h^\flat, s)^\flat$. Let $\bar h$ be a history of level k in which no component is healed at any level. In the rest of this paragraph each history is taken without the clusters healed below level k. It stands for the class of histories that yield it once these clusters are erased. Consider the histories that reduce to $\bar h$ once one erases the clusters belonging to those components that were healed at level k. They are in bijection with the pairs $(\mathfrak{X}, (\varsigma_X)_{X\in\mathfrak{X}})$. Here $\mathfrak{X}$ is a finite collection of possible healed components which touch neither one another nor $Z_k(\bar h)$. Further, $\varsigma_X \in \mathcal{H}^1(X)$ ranges over the histories of X alone in which X is healed and nothing is healed before it, and the choices for different components are independent. All these histories share one label. An individual τ_h is a measure produced by the moves, and it has no exponent of the form in Theorem 2.1. The part of $R^{(k)}$ produced by $\mathbf{R}$ arises only through the sum over $(\mathfrak{X},\varsigma)$, a sum which $\mathbf{R}$ itself performs (Proposition 2.10 [outlined below; Section 32]). This sum is a polymer sum. The reasons are the gas property and two facts of locality in the history: $\mathbf{T}'_k(X)$ is the same in all the summands containing X, and a healed patch leaves no trace outside itself (2d).

Traces and charts. Let W be a region, and let W_i^+, the *window* of level i, denote the enlargement of this region by a single layer of level-i cells. The trace of a label is

$$\operatorname{tr}_W \mathcal{S} := \big(\Omega_i^c \cap W_i^+,\, Z_i \cap W_i^+,\, S_i \cap W_i^+\big)_{i\le k},$$

with two additions. The first addition is the marks of the label. They enter collectively rather than individually. For every level, every kind of mark and, for drop-outs, every number of stages carried out, the trace records the union of all components carrying such a mark, together with the unions of their stored regions and of their nests. Hence the trace does not show which pieces in W_i^+ belong to the same component. The second addition consists of three separate sets of stored bonds of level $i-1$ contained in W_i^+: those in the collar, those at a rim cube, and those in the cut-off zone of the strip. The cut-off zone consists of those bonds of the strip that lie near the set where the fluctuation field is large and carry no kept small-field condition of that field; there the substitution of (2b) uses the cut-off ϑ_{co}. Where a bond carries a kept condition, the variables are left unchanged, while on the remaining bonds of the strip the substitution proceeds with no cut-off. The terms depend on these three sets, and the regions in W_i^+ do not fix them: to decide if a cube is a rim cube one has to look at its neighbours, and Ω_i^c enters up to two layers away. The trace $\operatorname{tr}_W h$ of a history is formed in the same way from everything that h records, including the labels of its earlier moments. It is finer: knowing the trace of a label does not fix the traces of the histories that carry it. A *chart* $\iota : W \to W'$ is a translation whose vector belongs to the lattice of level-t cells. It acts either within one torus or, through a common lift of a non-wrapping region, from one torus and cutoff to another. It leaves invariant the lattices $\mathbb{T}^{(i)}$, $i \le t$, and also the partitions of all levels up to t into cubes and cells. Neither a shift by a single site nor a mere isometry of the windows does so. Suppose that $\iota \operatorname{tr}_W h = \operatorname{tr}_{W'} h'$, that the increments and the earlier terms in W correspond, and that the bare weight corresponds on the window W_t^+ (this window contains $W^{\sim O(1)}$, since the cells of level t are $M\check R_t$-cubes). Then the two histories produce corresponding terms with domains in W, including their natural domains of definition. Here a term is regarded as a partial map on the fields on W, and one of two corresponding terms is defined exactly where the other is. The integration of V_k over the box is the move (3) of (2b) and forms part of the main step of $\mathbf{R}$. Consider the part of the regular terms on a domain of level m that this integration takes. It depends only on the trace through the

levels $i \geq m$, whatever the values of k, of $i-m$, of the size of the domain and of the volume. The standing operations, that is the $\mathbf{T}'_i(X)$ of the islands carried by a term, are transported likewise, measures included and one factor at a time (2e). For the analyticity spaces more is needed. The spaces of two corresponding terms are mapped onto one another if two further conditions hold. First, the bonds of the bare lattice carrying their fields must form corresponding sets. This is needed because whether a bond leaving the domain ends in $\Omega_1^\sim$ is decided outside the window; for a domain in Ω_1 there is no condition. Second, at each level, two unit cubes of the bare lattice lying in the domain must belong to a common drop-out mark of that level if and only if their images do. This is needed because a domain that is not convex may cut a mark into two pieces; where no mark meets the domain there is no condition. On the infinite lattice no sum over labels is formed. The only terms formed there are those of the never-large label, the label at which no region stands; domain by domain, these terms contain the sums over healed components that $\mathbf{R}$ performs (Proposition 2.10 [outlined below; Section 32]). A term whose window W_t^+ lies in Λ_t coincides with the corresponding term of that label (the terms belong to the standing part $h^\flat$, and everything that this part records lies in Z_t: 2e). If the domain of such a term does not wrap, it is one and the same function on every torus and at every cutoff, and in this way $E^{(j)}$ and b_j are computed (3b). It is this that links the geometry to (3b) and (4b).

The net form of the exponent. Each function $\mathfrak{t}$ of the exponent has a fixed place, given by its level, its domain and the way it is made. The label enters only through the sorting into $\mathbf{E}_k, \mathbf{R}_k, \mathbf{B}_k$, the profile and one number:

$$A_{k,\mathcal{S}}(U) = -\beta A(U) + \sum_{\mathfrak{t}} \mathfrak{t}(U) + \mathrm{const}(\mathcal{S}),$$

because

$$x_k(\cdot) + \sum_{j \leq k} b_j \varphi_j(\cdot) = \beta$$

and $c \mapsto A(c, U)$ is linear. Consequently a change of label changes no local term.

The law of the histories. The set $\mathfrak{H}_k$ of histories of level k is finite. A run has the datum $\mathfrak{d} = (\beta, m, g)$ and takes place on T_{ℓ_m}, $\ell_m = 2L^m$. The mass of a history $h \in \mathfrak{H}_k$ is $\mathfrak{t}(h) := \tau_h(\mathcal{V}_k) \geq 0$. Since the steps are exact (2e), $\sum_{h \in \mathfrak{H}_k} \mathfrak{t}(h) = Z_{K,\ell,\beta}$ at each level. The law of the histories of the last level, and in the same way the law of the histories of any level, is

$$\tilde{\nu}_{K,\ell,\beta}(\mathsf{E}) := Z_{K,\ell,\beta}^{-1} \sum_{h \in \mathsf{E}} \mathfrak{t}(h),$$

where events of histories are written in sans-serif letters, $\mathsf{E}_k \subset \mathfrak{H}_k$. The creation set of step j is $\mathcal{W}_j^\wedge(h)$, as before. Without the superscript, $\mathcal{W}_K(g)$ denotes the first-passage window. The live region $\mathcal{Z}_k(h)$ is $Z_k(h)$ with the components healed at step k removed; it is the input region of step $k+1$. The event that the cell $\mathfrak{c}$ is live at level k, that is $\mathfrak{c} \subset \mathcal{Z}_k$, is written $\mathfrak{w}_k(\mathfrak{c})$. For a cell z of level n, the survival event $\mathcal{E}_z^{(n)}(k)$ is the event that $z \subset \mathcal{Z}_i$ for $n \leq i \leq k$. The statements in which a run is continued past K are (3d), (4d) and (7e). For these statements we fix an anchor coupling $\gamma_\star \in (0, \gamma_J]$, which is determined by the constants up to c_a and by γ_J, and put $X_\star = \gamma_\star^{-2}$. The anchor level is $K_\star = K(\beta, \gamma_\star)$; it goes to infinity as β does. In these statements the reference coupling obeys the ceiling $g \leq \gamma_\star$, and the bounds on the flow hold up to $K_\star$. Also, at a given β, the run passes the window of g no later than the window of $\gamma_\star$, $K(\beta, g) \leq K_\star$; this follows from $g \leq \gamma_\star$ and the monotonicity of the first passage.

Lineages and marks. Let X be a component of $Z_k(h)$, at any level, that is either live at level k or healed at step k; a component live at $k = K$ is a terminal component. Its lineage regions

are $\mathsf{Z}_k(X) := X$ and, going down the levels, $\mathsf{Z}_j(X) :=$ the union of all components that are live at level j and contained in $\mathsf{Z}_{j+1}(X)$. Its lineage consists of these components (j, C) together with the creations (j, W) inside the $\mathsf{Z}_j(X)$. The weight of a level-j cell is $\mathsf{r}_j := M^4 \check{R}_j^5$. We write $|C|_j$ for the number of level-j cells in C. For comparison, $\#_j(Y)$ counts the sites of $\mathbb{T}^{(j)}$ in Y, and a cell contains $(M\check{R}_j)^4$ of them. Finally, for a component W of the set that step j creates in the lineage, $d'_j(W)$ is its tree length measured in these cells; the marks set by the operations **R** do not enter Dep. Then

$$\mathrm{Vol}_k(X; h) := \sum_{(j,C) \text{ in the lineage}} 2\,\mathsf{r}_j\,|C|_j,$$

$$\mathrm{Dep}_k(X; h) := \sum_{(j,W) \text{ created in the lineage}} c\,p_0(g_j)^2\big(d'_j(W) + 1\big).$$

The mark at level k has two dials, $\mathsf{w} \in [0, 1)$ and $c_a \geq 0$, the second also called the age constant. The mark and the end mark $\Theta := \Theta^{(K)}$ are

$$\Theta^{(k)}(h) := \sum_{X \text{ live at level } k} \big[\mathsf{w}\,\mathrm{Dep}_k(X; h) + c_a\,\mathrm{Vol}_k(X; h)\big] \geq 0.$$

The mark keeps back the fraction w of the small factor of each creation in such a lineage. It also charges $2c_a$ for every cell, multiplied by the weight of the cell, and for every level. The constant c in Dep is a single number. It is chosen after the fractions of the fluctuation threshold, namely those at which a step evaluates its tests: $1/(4\Theta_1)$ on the collar and θ_c on the rim. It is subject to $c \leq c' \min\{\Theta_1^{-2}, \theta_c^2\}$, where c' is chosen after (A_1, A_0) and contains the ratio $(A_1/A_0)^2$ of δ_k^2/g_k^2 to $p_0(g_k)^2$. No ceiling on g can take the place of this choice, since such a test has a deposit and a gain that are both of degree $2\mathfrak{p}_0$ in $\log x_j$.

Three ages are attached to a component C that is live at level k. (1) The age in the label is determined by the oldest region of the label within C: $\mathrm{age}^{\mathrm{lab}}(C) = k - \min\{t : C \cap Z_t \neq \emptyset\}$, where Z_t denotes the large-field region of the label $\mathcal{S}(h)$ at scale t. (2) The age of the lineage is $\mathrm{age}_k(C; h) = k - j_0$, with j_0 the earliest level that has a non-empty lineage region $\mathsf{Z}_j(C)$. The events $\mathfrak{lv}_k^{=a}(\mathfrak{c})$ and $\mathfrak{lv}_k^{\geq a}(\mathfrak{c})$ say that at level k the cell lies in a live component whose lineage has age a, respectively at least a. (3) The seed age is k minus the seed level. The seed level is the least t such that C contains, as a point set, some region created at step t. The corresponding events are written $\mathfrak{sd}_k^{=a}(\mathfrak{c})$. The first age is at most the second, and the second is at most the third. The second and the third can differ. Suppose that a region was healed at some earlier step and that a later live region again covers its place. Such a region counts for the seed level but is not part of the lineage. If the seed level is defined using the regions that no step before k has healed, the two ages coincide. Payment is made for the age of the lineage alone (2e). Age has to be paid for with creations. In a lineage with n creations, those made by operations included, a creation made at step t satisfies $k - t \leq Cn(\mathsf{N}_t + \log n)$. Here a creation is a created cell or a mark. By the waiting bound and the count of (2c), this bound holds for every lineage; hence an old lineage contains many creations. These objects enter (2e) and (4d). Through the age series of (4d) they underlie both the two-point law (7a)–(7c) and the four-point law (7d), namely clauses (c)(3), (c)(4), (c)(5).

(2b) The large-field operation: which variables are frozen, which are integrated out, and what a patch retains. **Measure and formula: two layers.** At the level of measures, a step acts as a finite sequence of moves on states $\sigma = (h, \xi)$, where h records the outcomes obtained up to that point and ξ the variables stored up to that point. Each move is given by positive kernels $\Pi^{(o)}$,

finitely many and one for each outcome o, such that $\sum_o \Pi^{(o)}(\sigma,\cdot)$ has total mass one. A move is one of four kinds. It may be a test, that is, an indicator whose outcome is added to h. It may be a map. It may be a draw, that is, the image of a probability law, as for gauge fixings, hypercubic frames $r \in W_4$ and reference laws. Or it may delete a stored variable, which amounts to an integration. Everything expressed by a formula, such as Gaussian weights, Jacobians, expansions, resummations or the replacement of a test by 1, only recasts an integrand in another form. A gauge fixing is a draw as a move. For it to be the factor that the formula writes, the measure must be gauge invariant. Let ν_0 be any complex bare measure, let $\mu_0 \geq 0$, and let $\mathfrak{w}$ be any bounded Borel function of the bare field. Denote by K_h the kernel of the history h, obtained by composing the kernels $\Pi^{(o)}$ of its moves along its outcomes. Then

$$\tau_h[\nu_0] = \int \nu_0(dU)\, \mathsf{K}_h(U,\cdot),$$

$$\sum_h \mathsf{K}_h(U,\mathcal{V}_k) = 1 \;\forall U,$$

$$\sum_h |\tau_h|(\mathcal{V}_k) \leq |\nu_0|(\mathcal{V}_0),$$

$$\big|\tau_h[\mathfrak{w}\mu_0]\big| \leq \sup|\mathfrak{w}|\, \tau_h[\mu_0].$$

By totality (2a; a reference law whose normalisation does not lie in $(0,\infty)$ is not drawn, and in that case the component drops out; this case does not arise on the supports, see (2e), "Reality and positivity"), every kernel is defined at every field, and no smallness condition is needed. They depend on the lattices and the constants. Of the frame θ (2e: the real sequence from which the choices are formed; in the present case $\theta = x$, and it is only through this frame that the kernels then vary with β; it is unrelated to the hypercubic frames r), they use only the gauge-fixing law, the coupling of the reference law, the cube sizes and the thresholds. Apart from their dependence through the frame, the kernels do not depend on any measure, on β, on a source, on any term of an effective action, or on whether some series converges. Exactness and positivity therefore come at no cost, because a move does nothing but split mass. The sourced densities come from $\nu_0 = \rho_0^w\, dU$ with the same kernels (4a), and runs whose choices are frozen are runs of that same procedure (3c). The estimates of (2c)–(2e) add only one further statement: the formulas are formed on the supports and there coincide with the measures. *Passengers*: divide the stored variables into those, $\mathfrak{v}$, that a move uses and the remaining ones, $\mathfrak{q}$. In step $t = k+1$ the operation $\mathbf{T}$ uses the fields of level k and, from the lower levels, only the data of its backgrounds; the operation $\mathbf{R}$ uses variables only in the components of its class and next to them. Any variable not used by a move of the step is carried along unchanged,

$$\Pi^{(o)}((h,(\mathfrak{v},\mathfrak{q})),\cdot) = \Pi^{\flat(o)}((h,\mathfrak{v}),\cdot) \otimes \delta_{\mathfrak{q}}$$

, in which $\Pi^{\flat(o)}((h,\mathfrak{v}),\cdot)$ is a measure on the states $((h,o),\mathfrak{v}')$. This is the locality of a step, in measure-theoretic form.

T on a label (Bałaban, Comm. Math. Phys. 119 (1988) 243, Sect. 3) comprises the following, carried out on the regions fixed by the tests of (2a). First the multi-scale background $U_{j+1}(\mathbf{V})$ is formed, on the determining set $\mathbf{B}[\Omega_\bullet]$ (1b); the third test of (2a) is formed with this background. Next, a change of variables is made on the strip. Then a conditional Gaussian integral is taken over Λ_{j+1}. Finally, the logarithm of this integral is split into E, R and B. The operation sits between (1d), in which no regions occur, and the operation $\mathbf{R}$.

The strip. The tests kept on $S_{j+1} \subset \Omega_{j+1} \setminus \Lambda_{j+1}$ depend on V_{j+1} non-locally, by way of V_j^{bg}. Take a background localised to $\Lambda_{j+1}^c \cap \Lambda_j$ and denote its average by V_j^{loc}. Then

$$V_j^{\mathrm{bg}} (V_j^{\mathrm{loc}})^{-1} = e^{i\mathbf{H}^{(j)}},$$
$$|\mathbf{H}^{(j)}| \le C\varepsilon_j e^{-\check{R}_j};$$
$$V_j(b) = W' e^{-i\vartheta_{\mathrm{co}}(W')\mathbf{H}^{(j)}(b)} e^{i\mathbf{H}^{(j)}(b)}\, V_j^{\mathrm{loc}}(b),$$

where the second relation is a substitution made bond by bond, with new variable $W' = e^{i\mathcal{A}'(b)}$. In it, ϑ_{co} is a smooth cut-off in $|\log W'|$. It equals 0 up to slightly above δ_j and 1 from slightly below $2\delta_j$ onwards. This cut-off is applied on the cut-off zone of the strip ((2a)); at the remaining bonds where the substitution is made, $\vartheta_{\mathrm{co}} \equiv 1$. Consequently, on the cut-off zone, a test whose threshold does not exceed δ_j is the same function of the new variable as of the old one, while a test at $2\delta_j$ is now made against the localised background. Because $|\mathbf{H}^{(j)}| \ll \delta_j$, a contraction argument shows that the map on a bond is a diffeomorphism of G. Its Jacobian equals 1 wherever ϑ_{co} is locally constant, and it contributes to the action as a boundary term. Once the substitution has been made, the factors of the kept tests at those bonds next to Λ_{j+1} where no substitution takes place are omitted. On the supports these factors equal 1, and where $\vartheta_{\mathrm{co}} < 1$, a test at $2\delta_j$ has a factor equal to 1 whichever background is used. Moreover, at a pivot bond of Ω_{j+1} the value of V_j is taken from the constraint $\mathcal{M}(V_j) = V_{j+1}$. With this, the tests depend on V_{j+1} only in Λ_{j+1}^c, and the integral operations therefore factorise ((1.21)–(1.25), (3.22) there).

What is frozen and what is integrated. On Ω_{j+1}^c, step $j+1$ stores V_j together with its constraint. On Ω_{j+1} the action is expanded around $U_{j+1}(\mathbf{V})$. The only variable integrated there is $A = \mathcal{A}_j^{\mathrm{s}}|_{\Lambda_{j+1}}$, conditionally on the stored field $A' = \mathcal{A}_j^{\mathrm{s}}|_{\Omega_{j+1}\setminus\Lambda_{j+1}}$. The measure is Gaussian with covariance $C^{(j)}(\Lambda_{j+1})$, which is the inverse of the restriction of the form $\Delta^{(j)}$ to the bonds of Λ_{j+1}, and its linear term is $-\langle A, \Delta^{(j)} A'\rangle$. The coupling flows only where this integration is carried out, $\varphi_{j+1} \in C_0^\infty(\Lambda_{j+1})$, and it stops on the whole of Z_{j+1}, including $\Omega_{j+1}\setminus\Lambda_{j+1}$. Once the square is completed, the conditional weight appears as a factor inside $\mathbf{T}_k(Z) = \prod_{j<k} \mathbf{T}^{(j)}$, with the regions cut by the component Z ((2.21), (3.23) there; this operation is the product shown for a component on which $\mathbf{R}$ has never acted, and in general the operation is defined by a recursion over the marks):

$$\mathbf{T}^{(j)} = \int dV_j\big|_{\Omega_{j+1}^c} \delta\big(\mathcal{M}(V_j) V_{j+1}^{-1}\big)\, \mathfrak{G}_j\, \chi\chi^c\, z^{(j)} \int dA'\, \chi\chi^c\, e^{-\frac{1}{2}\langle A', [\Delta^{(j)} - \Delta^{(j)} C^{(j)}(\Lambda_{j+1}) \Delta^{(j)}] A'\rangle}.$$

This operation contains no Wilson weight. That weight belongs to $\exp A_{k,\mathcal{S}}$, the function on which the operation acts; $z^{(j)}$ is explained below. The old R-terms, $\mathbf{R}'_j = \sum_{i\le j_1} \mathbf{R}^{(i)}$, appear alongside $\mathbf{E}_j$ in the regular action $\widetilde{\mathbf{E}}_j$ of the step (2e). Here $\mathbf{R}^{(i)} = \sum_X [R^{(i)}(X,\cdot) - R^{(i)}(X,\mathbf{1})]$ is the summand of level i of $\mathbf{R}_j$, and j_1 is the index k_1 of (2d) at level j. The young ones, $\mathbf{R}''_j = \sum_{j_1 < i \le j} \mathbf{R}^{(i)}$, are handled separately. Put $\delta F := F(U_j) - F(U_{j+1})$. Attach a parameter t to the terms present only next to Z_j, and a parameter s to $\mathbf{R}''_j$ ((3.23)–(3.28) and pp. 270–271 there). Let $z^{(j)} = \prod_c z^{(j)}(c)$ be the factors that come from eliminating the delta functions. In $\mathbf{T}^{(j)}$ their product runs over those coarse bonds c of Ω_{j+1} that lie outside Λ_{j+1}; under the logarithm it runs over those of Λ_{j+1}. Then

$$\log\Big[z^{(j)} \int dA\, \chi^{(j)} e^{-\frac{1}{2}\langle A, \Delta^{(j)} A\rangle - \langle A, \Delta^{(j)} A'\rangle + \cdots}\Big] - \tfrac{1}{2}\big\langle A', \Delta^{(j)} C^{(j)}(\Lambda_{j+1}) \Delta^{(j)} A'\big\rangle$$

$$= \int_0^1 dt\, \langle \delta \mathbf{B}_j - \delta A(x_j(\cdot) - x_j, \cdot) + \cdots \rangle_{s=1,t} + \int_0^1 ds\, \langle \delta \mathbf{R}_j'' \rangle_{s,t=0} + \Big(\log[\cdots] - \tfrac{1}{2} \big\langle A', \Delta^{(j)} C^{(j)}(\Lambda_{j+1}) \Delta^{(j)} A' \big\rangle \Big)_{s=t=0}.$$

In this identity $\langle \cdot \rangle_{s,t}$ denotes the normalised mean with respect to the measure that the parameters s, t interpolate. The left-hand side is the unnormalised integral. Its second term is present since the square, once completed, was moved into $\mathbf{T}^{(j)}$. Inside the integral over t, the dots stand for two kinds of terms. Those of the first kind are the terms of the strip, namely the Jacobians together with the differences of all quantities that depend on a shifted variable. Those of the second kind are the differences, taken between the two backgrounds, of the exponents of the integral operations carried by the term. The coupling difference carries the sign it has in $A_{j,\mathcal{S}} = -x_j A - A(x_j(\cdot) - x_j, \cdot) + \dots$. A cluster expansion as in (1d) is applied to each part. The Gaussian normaliser is contained in the last part; it is that part of $E^{(j+1)}$ which carries b^0_{j+1}. On a label it is a determinant: that of a multi-scale operator, compressed to Λ_{j+1} and taken in the background of that label; so by definition it is non-local. It is converted into local terms,

$$\sum_c \log z^{(j)}(c) - \tfrac{1}{2} \log \det \big(\mathsf{C}^{\mathsf{T}} \Delta^{(j)} \mathsf{C} \big|_{\Lambda_{j+1}} \big) = \text{number} + \sum_X \mathbf{N}(X, U_{j+1}),$$

$$|\mathbf{N}(X)| \le C e^{-cM d_{j+1}(X)},$$

which Bałaban does by means of a resolvent integral and walks (1d), whereas here it is done by sums with alternating signs, taken over the sub-regions $Y \subset X$, whose terms are the diagonal entries of the logarithm of the operator built on Y. Möbius inversion yields the sum rule, and the decay is due to a sign-reversing involution, which cancels everything except the chains of walks that visit all of X. Only the last two parts give terms with domain $X \subset \Lambda_{j+1}$. These terms are built from walks and fields inside X, so the regions do not affect them. They are of two kinds. The first is $R^{(j+1)}$; it arises from the part with s, to which what $\mathbf{R}$ leaves is added (2d). The second is $E^{(j+1)}$, which comes from the last part, where $\mathbf{E}_j$ stands with the old R-terms. These terms are the same for every label. For the regular terms this holds for every such X. For the remainders it holds for X one cell inside Λ_{j+1}, the only place where a label shows them, and this is the margin of (2e). Thus the young R-terms go into $R^{(j+1)}$, and the old ones go into $E^{(j+1)}$ and b_{j+1} (2d, 2e). Everything else makes up $B^{(j+1)}$; among it is the part of the quadratic form that is rejected (next paragraph). It amounts to $O(1)$ for each M-cube of level $j+1$ next to Z_{j+1}; this cost is covered by (2c), and for that reason no renormalisation of it is required.

What a one-step operation keeps. The conditional form is a Schur complement. It is a sum over chains of operators with connected carriers, where a chain is either a walk term of $\Delta^{(j)}$ or a sequence of a walk, a resolvent walk and a walk. A chain is kept if the cells of level $j+1$ that it meets intersect $\Omega_{j+1} \cap Z_{j+1}$ in a non-empty connected set and remain within one layer of that set; a kept chain is then counted with the component $\mathfrak{k}$ of $\Omega_{j+1} \cap Z_{j+1}$ that contains this set:

$$\big\langle A', [\Delta^{(j)} - \Delta^{(j)} C^{(j)}(\Lambda_{j+1}) \Delta^{(j)}] A' \big\rangle = \sum_{\mathfrak{k}} \langle A', \Delta^{\mathrm{loc}}_{\mathfrak{k}} A' \rangle - 2 \sum_X \mathsf{B}^{\times}(X),$$

$$|\mathsf{B}^{\times}(X)| \le C p_0(g_j)^2 e^{-cM\check{R}_{j+1}} e^{-\kappa d_{j+1}(X)},$$

$$\langle A', \Delta^{\mathrm{loc}}_{\mathfrak{k}} A' \rangle \ge c \sum_{b \text{ in } \mathfrak{k}} |A'(b)|^2 :$$

(the bound on $\mathsf{B}^{\times}$ is for stored fields in range, $|A'(b)| \le C p_0(g_j)$, which they are on the support of the kept tests) a rejected chain has length greater than the size of a cell, and the size $p_0(g_j)^2$ of the stored field squared is outweighed by $e^{-cM\check{R}_{j+1}}$. The rule of Bałaban, by which the form

"couples only the fields A_j in the same component" (119, p. 258), fails to be local: whether two bonds belong to a common component is settled at a distance from them, and two histories having the same trace can differ on it. The kept form is not a Schur complement but a truncated series. Its lower bound is therefore not inherited and has to be proved. The proof localises to small cubes (Ismagilov–Morgan–Simon) and compares with a genuine Schur complement. *Carriage*: in an integral operation that a term carries, the tests, the gauge factors and the delta functions depend on no background, whereas the forms $\Delta_{\mathfrak{k}}^{\mathrm{loc}}$ and the factors $z^{(j)}$ do. These forms and factors are given as functions of a pair $(\mathbf{U}, \mathbf{J})$ near $\mathfrak{k}$, and they are constructed once and for all. They are evaluated at the current background, which in a term of level k is $U_k(\mathbf{V})$. The background changes with every later step and with every stage of $\mathbf{R}$. Each change produces differences

$$-\tfrac{1}{2}\langle A', [\Delta_{\mathfrak{k}}^{\mathrm{loc}}(U^{\sharp}) - \Delta_{\mathfrak{k}}^{\mathrm{loc}}(U^{\flat})]A'\rangle$$

, where $U^{\flat}$ is the background after the move and $U^{\sharp}$ the one before it, both written in the new variables. Localisation turns these differences into boundary terms. For a piece of a kept form whose carrier is R, the oscillation with respect to the pair is bounded by $Cp_0(g_j)^2 e^{-\kappa d^{\natural}_{j+1}(R)}$, measured in the hull length of the carrier; this bound is large, being the square of a field at its threshold. The difference of such a piece at the two backgrounds of a stage of level i is localised as well, and each of its localised pieces, on a domain Y, has modulus at most $C(\log x_i)^{-(q-3\mathfrak{p}_0-Cr)} e^{-c\, d^{\natural}_k(Y)}$. This is a negative power, because q is chosen after $\mathfrak{p}_0$ and r, with $q > 3\mathfrak{p}_0 + Cr$.

The idea of R (Comm. Math. Phys. 122 (1989) 175, (0.3)–(0.4)). We split $Z = Z' \cup Z''$. Following that paper (p. 176), Z'' is the union of the components "for which the small factors connected with large field control some number of next steps", and Z' is the union of all the other components (the rule by which the present construction sorts the components into these two sets is given under "Where it acts", below). The density is a sum $\rho(V) = \sum_Z \rho(Z, V)$ over the large-field regions Z ((0.2) of that paper), and $\rho(Z'', V)$ denotes the density belonging to the region Z''. Then

$$(\mathbf{R}\rho)(V) = \sum_Z \rho(Z'', V)\, \frac{\int dV|_{Z'}\, \rho(Z, V)}{\int dV|_{Z'}\, \rho(Z'', V)},$$

$$\int \mathbf{R}\rho = \int \rho.$$

In Z' the law of the field is replaced by the small-field law, and the mass is preserved. This idealised form is not linear in ρ. The form actually used is linear, because its reference law contains the Wilson weight and nothing else.

Where it acts. Let $k_\flat = k - \mathsf{N}_k \geq 1$ (*depth*); this restriction excludes nothing at $k = \mathsf{N}_k$, since by the second condition below a member of the class is then a component of Z_k in which no step has created anything, and no such component exists. A component X of Z_k belongs to the class when three conditions hold. First, for $k_\flat \leq i \leq k$ the set $X \cap Z_i$ is a single component of Z_i and is small, that is, contained in a cube of $O(1)$ level-i cells (*single boxes*, and at $i = k$ *size*). Second, for $k_\flat < i \leq k$, step i creates nothing in it, and it does not absorb a second component, $X \cap Z_i = \Phi_i(X \cap Z_{i-1})$ (*quiet*). Third, it has not been acted on by any operation whose level lies in $[k_\flat, k)$, the lower end point included (*quarantine*). Bałaban imposes two conditions, that the component be small and quiet (ibid., p. 177); to these two, the exclusion of a joining is added here as a separate condition, named explicitly. Two small quiet boxes that join within the window form a small component without creation, yet its trace one scale lower has two components: a joining does not count as a creation. Let H_i denote the hull, in level-i cells, of $X \cap Z_{i-1}$, and let $B^{[r]}$ be the box B with each face pushed outward by r such cells. Then

$$\Omega_i^c \cap X = H_i^{[8]},$$

$$Z_i \cap X = H_i^{[10]}$$

$$(k_\flat < i \le k) :$$

the boxes are nested, the two of each level being concentric, and the number of cells along a side of X is comparable to the fixed point of the free evolution (2c). As long as $k \le \mathsf{N}_k$ the class is empty, so $\mathbf{R}$ does nothing at the first levels. For $k > \mathsf{N}_k$ the class is non-empty: a single cell created at step $k - \mathsf{N}_k$, with nothing else, belongs to it. Up to level k the torus is admissible, $\frac{1}{2}\ell L^{K-i} \ge CM\check{R}_i$ for $i \le k$, as the opening of Theorem 2.1 requires. For a first-passage datum, $x_K \le g^{-2} + B_\sharp$, this condition follows from $\ell \in \mathfrak{L}(g)$ by way of $\check{R}_i \le L^{K-i}\check{R}_K$, a consequence of the upper flow bound; it is a lower bound on ℓ. Hence no cell, no box and no neighbourhood of a box wraps around the torus.

The operation. (0) *The frame sum*, described first although it belongs inside (3). In each component the identity $1 = |W_4|^{-1}\sum_{r\in W_4} 1$ is inserted after the integration over the box and before the tree gauge of (3). Consequently no frame sum is attached to a component that leaves at (2) or during the stages. In the summand of r, every object of (3), (4) and (2d) that belongs to X is formed in the frame r: the reference law with its denominator, the two trees, the test on B, and the last-shell test. If a component leaves after the tree gauge, its frame sum stays with it, and an island keeps, in each summand, the cut-off and the delta function belonging to the frame of that summand. Formed instead in a single frame r_0, these make the bracket of X integrate to $|W_4|^{-1}\sum_r \mathfrak{a}_r\, \mathsf{D}_X^{(r_0)}/\mathsf{D}_X^{(r)}$ instead of $|W_4|^{-1}\sum_r \mathfrak{a}_r$. Here $\mathfrak{a}_r$ denotes that part of the summand of r which is independent of the box, and $\mathsf{D}_X^{(r)}$ denotes the denominator of $\mathfrak{s}_X$ formed in the frame r. The reason for keeping the frame sum is that neither the test on B nor the last-shell test is gauge invariant. Each of them enters only by way of decompositions of unity performed within a single gauge-fixed measure. Let m be the measure before the tree gauge, and let $\mathsf{a}_r, \mathsf{b}_r$ be its parts that pass, respectively fail, these tests in the frame r. Then $\mathsf{a}_r + \mathsf{b}_r = \mathsf{m}$ for every r, and hence $|W_4|^{-1}\sum_r(\mathsf{a}_r + \mathsf{b}_r) = \mathsf{m}$. With the drop-outs kept instead in a single fixed frame r_0, the total is $\mathsf{m} + (|W_4|^{-1}\sum_r \mathsf{a}_r - \mathsf{a}_{r_0})$. The average is taken so as to obtain covariance under the lattice isometries; trees break this covariance. (1) *Stages*: for $j = k_\flat, \dots, k-1$, one value after another, the stored variables of the outer shell of $Z_{j+1} \cap X$ are integrated out as in a small-field step. Before this, tests at the sharper threshold $\delta_j' := g_j A_1 p_1(g_j)$ are made, formed with backgrounds localised in X. What remains is a box Λ^{pr} of side at most CM: it is the set $Z_k''(X)$ of the nest of (2a) enlarged by one more layer of M-cubes, and this layer is the shell of level k. In it the levels $k_\flat, \dots, k$ are stacked in thin shells, the shell of level j carrying the field V_j. Together these fields form the datum V'' attached to the prepared determining set $\mathbf{B}''$; its background is U_k''. At the innermost position lies the core $C_{k_\flat} = X \cap Z_{k_\flat}$, whose side, measured in the unit of level k, is at most $CM\check{R}_{k_\flat}L^{-\mathsf{N}_k}$. (2) The test asks whether the exterior field V_k admits an extension into Λ^{pr} whose curvature is less than $2\varepsilon_k$. If an exterior field satisfies $|\partial V_k - \mathbf{1}| < \varepsilon$, then it does admit an extension, one with $|\partial V_k - \mathbf{1}| < CM^2\varepsilon$. To see this, set the field equal to $\mathbf{1}$ inside after choosing an axial gauge on the surface $\partial^+\Lambda^{\mathrm{pr}}$. Hence "no regular extension" means that some exterior plaquette has $|\partial V_k - \mathbf{1}| \ge cM^{-2}\varepsilon_k$, and this pays as in (2c) (p. 193 there). (3) V_k is integrated over Λ^{pr}. Taken together, the parts from the test (2) up to the division into the two classes (2d) form the *main step* of the operation; the stages that precede it and the exponentiation that follows it are counted separately. This integration is the first move to alter the density. Integrals against any bounded F that is independent of V_k on the box are left unchanged

by it. Level by level, from the top down, it resolves the bare chain $\prod_j \int dV_j\, \delta(\mathcal{M}(V_j)V_{j+1}^{-1})$, $\int dV_i(c)\, \delta(\mathcal{M}(V_{i-1})(c)V_i(c)^{-1}) = 1$. What remains is a free Haar integral,

$$\chi_k\, \chi_{k,\Lambda} \int dV_{k_\flat}\big|_{C_{k_\flat}} \chi\, \mathbf{T}_{k_\flat}(C_{k_\flat}) \int dV''\big|_{\mathbf{B}''\cap(\Lambda^{\mathrm{pr}}\setminus C_{k_\flat})}\, \chi_k''\, e^{A_k''(U_k'')}$$

Here $\chi_{k,\Lambda}$ is the test of (2); χ stands for the plaquette tests of level $k_\flat$ carried by the term on $\Omega_{k_\flat} \cap C_{k_\flat}$; and χ_k'', A_k'' are the tests and the action that (1) leaves. The integral is regarded as a density that is constant in V_k on the box. The field that is expanded is V''. It is expanded around the averages of the minimiser U_0 below, each average taken at its own level:

$$V'' = e^{ig_kB}V_0 \ \text{ on the fluctuation bonds } \mathbb{B}_0 = \mathbf{B}'' \cap (\text{shells}),$$

$$V_0 = \big(\mathcal{M}^j U_0 \text{ on the shell of level } j\big)_{k_\flat \le j \le k}.$$

Thus B is a multi-scale field on bonds of every level from $k_\flat$ to k, with a single scaling g_k and a single test $|B| < \delta_k'/g_k$. This is the field B that occurs in Proposition 2.8. The gauge of this field is fixed along a tree T_0 in $\mathbb{B}_0$, whose bonds come from all these lattices, one lattice for each shell. For two nested boxes $P_1 \supset P_2$, the tree is made up of axial contours which leave from a single initial site and go around P_2. The family of trees has two properties (pp. 195–196 there). First, the family is stable. Suppose one enlarges P_2 to a box $P_2' \subset P_1$ and deletes the contour of each site of $P_2' \setminus P_2$; then what is left is the tree of $P_1 \setminus P_2'$. Bałaban describes this property as important, and boxes are needed because of it. Second, the tree of a shell uses up the gauge freedom at every site of the shell except one. For this reason, for each inner box on which a gauge is fixed, one external bond crossing its boundary is added, and one more is added for the outermost box. Through these extra bonds the trees of the shells become a single tree, and it fixes the gauge entirely, with Jacobian one. The core operation $\mathbf{T}_{k_\flat}(C_{k_\flat})$, which contains the failed tests, is at no point unfolded. One further test, the *last-shell test*, compares with the vacuum and constrains $V_{k_\flat}$ between the core and $\mathbb{B}_0$. It requires

$$|\partial U_{k_\flat,\square}((\mathbf{1}, V_{k_\flat})) - \mathbf{1}| < \tfrac{1}{2}\varepsilon_{k_\flat}\eta_{k_\flat}^2$$

on the test cubes of that region, where $(\mathbf{1}, V_{k_\flat})$ equals $\mathbf{1}$ on the side of $\mathbb{B}_0$. (4) For each term τ, its mass $m_\tau = \int \tau\, dV_k|_{\Lambda^{\mathrm{pr}}}$ multiplies the reference law. The level-k variable $V_k' := V_k V_\Lambda^{-1}$, where V_Λ is the element of the critical orbit of Proposition 2.4 that equals $\mathbf{1}$ on the tree of δ_{ax}, occurs only in this law and in its denominator. Let $U_{k,X}$ denote the background localised in X, and let $\chi(\Lambda^{\mathrm{pr}})$ be the cut-off $\mathbf{1}\{|\frac{1}{i}\log V'(b)| < M_0\varepsilon_k$ on the bonds of the box$\}$, in which M_0 is a constant chosen large after M. Let δ_{ax} be the tree gauge of $\mathbb{T}^{(k)} \cap \Lambda^{\mathrm{pr}}$, taken on a tree that contains all the sites of this set together with the external bond of T_0 into the box; that bond is part of the tree because, without it, a constant gauge transformation of the box is left unfixed. Let ϕ_X be a cut-off function that equals 1 well inside X. Then

$$\mathfrak{s}_X(V_k) = \frac{\delta_{\mathrm{ax}}(V_k')\, \chi(\Lambda^{\mathrm{pr}})(V_k')\, e^{-x_k A(\phi_X, U_{k,X}(V_k'V_\Lambda))}}{\int dV'|_{\Lambda^{\mathrm{pr}}}\, \delta_{\mathrm{ax}}(V')\, \chi(\Lambda^{\mathrm{pr}})(V')\, e^{-x_k A(\phi_X, U_{k,X}(V'V_\Lambda))}}.$$

If one of the tests of (1)–(3) fails in a component, that component drops out (2a): it remains live, and the failure counts as a creation. This can happen to the same component repeatedly. Each occurrence counts as a creation and is followed by a wait of at most $\mathsf{N}_k + 1$ steps.

The purpose of the stages, and the reason for a sharper threshold. Continuing the field by least action costs a constant factor: for exterior data with $|\partial V_k - \mathbf{1}| < \varepsilon$ one has

$|\partial V_\Lambda - \mathbf{1}| \le C(M)\varepsilon$, and $|\partial U_0 - \mathbf{1}| \le C(M)\varepsilon_k \eta_k^2$ near the box. For every value of the coupling, the stages reduce the component from side $M\check{R}_k$ to a box of side $O(M)$. Hence $C(M)$ is a constant fixed before γ. On the shell of level j it is overcome by the factor $L^{-(k-j)}$ by which a level-k field scales when viewed from level j. On the outermost shells it is overcome by $\check{R}_k^{-c}$, because $1+\log_L \check{R}_i$ levels are compressed into the box, namely the levels from the last level i with $L^{k-i} = \check{R}_i$ up to level k. Consequently U_0 belongs to the domain of the carried tests. It also belongs to the analyticity domains of the terms, insofar as the domain of a term lies inside the component or at a distance from the box; the remaining terms are treated in (2d). In the words of 122 (1989) 175, p. 195, "it is the reason why we have done the preliminary integrations". As in Lemma 2.2, the tests made at a stage have to imply the earlier ones. Here this must happen $\mathsf{N}_k = \check{R}_k$ times in succession, and the losses together amount to a fixed fraction. If two backgrounds satisfy the same constraints on a region, their difference is the response to the difference of their data, and this response decays. Let $U_{k,X}^{(n)}$ denote the background after n stages, localised in X. Put $j = k_\flat + n$, and let β_{s} be a fixed small fraction, written β by Bałaban. Then, on the support of the tests of stage $n+1$ and on the shell of level $i \le j$,

$$|\partial U_{k,X}^{(n)} - \mathbf{1}| \le |\partial U_{k,X}^{(n+1)} - \mathbf{1}|\big(1 + c\beta_{\mathrm{s}} 2^{-(j-i)}\big) + c\beta_{\mathrm{s}} 2^{-(j-i)} \varepsilon_i \eta_i^2,$$

where the error term is

$$C\delta'_j e^{-\delta_0 \operatorname{dist}} \eta_i^2 \le C\, \tfrac{\delta'_j}{\varepsilon_j} (1 + (j-i)^{1/2}) e^{-c(M/M_1)(j-i)} \varepsilon_i \eta_i^2$$

. For M large, the decay $e^{-c(M/M_1)(j-i)}$ across the shells dominates $1 + (j-i)^{1/2}$, and $2^{-(j-i)}$ remains. The ratio $\delta'_j/\varepsilon_j = (A_1/A_0)(\log x_j)^{-(\mathfrak{p}_0 - \mathfrak{p}_1)}$ dominates the fixed constants, and at this point smallness of A_1/A_0 suffices as well. After n stages the test requires $|\partial U_{k,X}^{(n)} - \mathbf{1}| < \big(1 - \beta_{\mathrm{s}}(1 - 2^{-(j-i+1)})\big)\varepsilon_i \eta_i^2$ on that shell. On the outermost shells, where the collars of the stages overlap, this requirement is relaxed by a factor $L_\flat^2$ for each overlap, where $L_\flat$ is an integer with $2 \le L_\flat < \frac{1}{2}L$ that is chosen together with L; such a choice requires $L \ge 5$. Let $k_\sharp + 1$ be the last of the levels at which an $M\check{R}$-cube, measured in units of level k, has side precisely M; in formulas, $M\check{R}_{k_\sharp+1} L^{k_\sharp+1-k} = M$. The collars of the stages with $j \le k_\sharp$ are disjoint, and the collars of the later stages overlap. On the shell i the factor is $L_\flat^{2(i-k_\sharp-1)_+}$. It depends on the shell only, not on the stage, and it becomes larger with each successive shell until it reaches $\check{R}_k^c$. In the region not covered by the shells, the bound is expressed in the scale j of the stage and equals $(1 - \frac{1}{2}\beta_{\mathrm{s}})\, c_l\, L_\flat^{2e} \varepsilon_j \eta_j^2$, where $c_l = 3$ adjacent to Ω_k at the last stage and $c_l = 1$ elsewhere. Far from the box, the curvature of the least-action continuation U_0 can be as large as $(2+c)\varepsilon_k \eta_k^2$, and the test of the last stage must be implied there as well. Here $e = 0$ for $j \le k_\sharp$. For $j > k_\sharp$ one takes $e = (j - l - 1)_+$ on $\Omega_l \setminus \Omega_{l+1}$, $k_\sharp < l \le j$, and $e = j - k_\sharp - 1$ between the shells and $\Omega_{k_\sharp+1}$. This is the form of 122 (1989) 175, (1.24), where the level is denoted k_0 and the base L_0. On a layer $l < j$ this bound remains below $\varepsilon_l \eta_l^2$, the scale of that layer. In passing from the scale $j+1$ to the scale j one gains L^{-2} and pays $L_\flat^2$. Hence the bound of each stage implies the bound of the preceding stage. The margin between consecutive stages pays for the error, which decays geometrically. The sharper threshold enters in two places. In the first, the new fluctuation test has to imply the old fluctuation tests of step $j+1$, at δ_j and $2\delta_j$. This holds because $CL\delta'_j = CL(\log x_j)^{-(\mathfrak{p}_0 - \mathfrak{p}_1)}\delta_j \le \frac{1}{2}\delta_j$ for γ small, and it puts no new condition on the constants chosen earlier. In the second, at the main step, δ'_k has to remain a small fraction of $\varepsilon_{k_\flat}$, which is the curvature threshold of the core level. In this estimate the thresholds of two levels may differ by $\varepsilon_j/\varepsilon_i \le C(1 + (j-i)^{1/2})$, the bound valid for every pair of levels (2e). Restrict

to the common support of the carried tests χ_k, the tests of (2) and (3), and the tests and delta functions that stand under the core operation; the latter make the data stored in the core regular. On this set, and provided $C(M)\check{R}_k^{-c} \le c$ on the outermost shells, every test that the stages leave behind equals 1 when read on X minus one layer of level-k cells. The equality does not reach ∂X: adjacent to X^c, the background localised in X is constrained bond by bond on the bare lattice. There the test therefore demands $|\partial V_k - \mathbf{1}| < C\varepsilon_k \eta_k^2$ of the block field itself, while the carried tests provide only $2\varepsilon_k$. All of this holds under the condition

$$C\,\mathsf{N}_k^{1/2}\,\frac{p_1(g_{k_\flat})}{p_0(g_{k_\flat})} + C(M)\,L^{-\mathsf{N}_k}\mathsf{N}_k^{1/2} \le c\,:$$

the first term is δ_k' measured relative to $\varepsilon_{k_\flat}$, including the allowance $\mathsf{N}_k^{1/2}$. Since $\mathsf{N}_k = \check{R}_k \approx (\log x_k)^r$, it is a condition on the exponents, $\mathfrak{p}_1 < \mathfrak{p}_0 - Cr$, together with a ceiling on g_k. The second term is a level-k field viewed from level $k_\flat$, so N_k is taken large.

How a patch recovers its coupling. Each stage gives back to the collar it integrates the increments that this collar did not receive. After m stages the Wilson term is formed with

$$x^{(m)}(\cdot) := \beta - \sum_{i\le k} b_i\,\varphi\big[\Lambda_i^{(m)}\big](\cdot),$$

$$\Lambda_i^{(m)} = \Lambda_i \cup (X \setminus Z_i'') \ (k_\flat < i \le k_\flat + m),$$

$$\Lambda_i^{(m)} = \Lambda_i \text{ otherwise}$$

where $\varphi[\Lambda]$ is the profile of the region Λ. The difference $A(x^{(m)} - x^{(0)}, \cdot)$ is carried separately, as a Wilson piece (2d). In this way, at each bare plaquette the weights of all the Wilson terms in the exponent still sum to $-\beta$ (2a). Hence $x_k'' = x^{(\mathsf{N}_k)}$ coincides with x_k on the part of X outside the box, and equals $x_k + O(1 + k - j)$ on the shell of level j. This is what makes the three Wilson terms of Proposition 2.8 [outlined below; Section 32] cancel; two of them carry the constant x_k.

What a patch retains. A component that has passed every test carries the operation below. It is (1.71) of 122 (1989) 355, in which the summation extends over r and over all sequences of regions contained in X, with the prefactor $1/(2^d d!)$ in front. The notation is that of (2d), and χ' denotes the test $|B| < \delta_k'/g_k$.

$$\mathbf{T}_k'(X)F = \frac{1}{|W_4|}\sum_{\text{histories in } X,\,r}\int dB\,\delta_{rT_0}(B)\,\chi'\int dV_{k_\flat}\,\chi\,\mathbf{T}_{k_\flat}(C_{k_\flat})\,e^{-\frac12\|\phi^{1/2}D\mathsf{H}B\|^2 - A(x_k''\phi_{\mathrm{c}},U''(e^{ig_kB},V''))+V(X^\sim,U_k)+\text{numbers}}\,F.$$

A component is healed when the new V_k is regular in X. Then nothing is retained once the sum over the families has been exponentiated (2d), and X is removed from the label. For an island, X retains

$$\chi_k^c\,\chi_{k,\Lambda}\,\chi(\Lambda^{\mathrm{pr}})\,\delta_{\mathrm{ax}}(V_kV_\Lambda^{-1})\,\mathbf{T}_k'(X)$$

as a single block under its frame sum, for as long as it stands ((1.72), (1.104)). The cut-off $\chi(\Lambda^{\mathrm{pr}})$ is absent from (1.102). A drop-out after n stages retains the following: the one-step operations of the levels that were not reached; the bare integrals $\int dV_l\,\delta(\mathcal{M}(V_l)V_{l+1}^{-1})$ of the levels already treated; the core; and the functions present together with the failing factor $1-\chi$. The terms produced up to that point are added to the boundary terms.

Proposition 2.4 [minimiser in a box; Comm. Math. Phys. 122 (1989) 175, Proposition 1]. There exist a radius $a_\Lambda > 0$ and a constant $C(M)$, both chosen after M and before γ, with the following property. Suppose the exterior data satisfy $|\partial V_k - \mathbf{1}| < \varepsilon$ with $C(M)\varepsilon \le a_\Lambda$.

In the application $\varepsilon \le C\varepsilon_k$, so this is a ceiling on γ. Take the class of continuations with $|\partial V_k - \mathbf{1}| < a_\Lambda$ at those level-k plaquettes whose corners all lie within distance 1 of $\Lambda^{\rm pr}$. On this class the background $U_{k,X}(V_k)$ exists. Within the class, the map $V_k|_{\Lambda^{\rm pr}} \mapsto A(U_{k,X}(V_k))$, which is a function of V_k on the level-k bonds having an end point in $\Lambda^{\rm pr}$, has exactly one critical gauge orbit V_Λ. This orbit is the set of its minima in the class, it is analytic in the data, and it satisfies $|\partial V_\Lambda - \mathbf{1}| \le C(M)\varepsilon$. Bałaban states it for the function on fields of mild regularity and for ε sufficiently small, with the bound $CM^5\varepsilon$. The orbit of this class is the one used in (3) and (4). Moreover, $U_0 := U_{k,X}(V_\Lambda)$ satisfies $J(U_0) = 0$ in $\Lambda^{\rm pr}$. The proof is a contraction argument in a ball whose radius depends on M, which is why a class of continuations is fixed. In axial gauge the Hessian is bounded below by $c(M) > 0$; Bałaban obtains the lower bound cM^{-5}. Indeed, the Hessian is at least $c\|\partial B\|^2$ less a small multiple of $\|B\|^2$. In a tree gauge one has the Poincaré inequality

$$\sum_b |B(b)|^2 \le C(M) \sum_p |\partial B(p)|^2$$

(122 (1989) 355, (1.7)–(1.9), with CM^5). The corresponding inequality for a single bond of the nest, $|B(b)|^2 \le C(M\check{R}_k)^C \sum_p |\partial B(p)|^2$, yields the factor of the test in (3). It is (1.78) there, with every bond and every plaquette taken in its own lattice. The power of $\check{R}_k$ has two origins: the side of a shell in its own lattice, which is at most $CML\check{R}_k$ sites, and the number $\mathsf{N}_k + 1$ of shells, where $\mathsf{N}_k = \check{R}_k$. The first form serves the level-k variable V_k' and the uniqueness of V_Λ. The form for one bond serves the multi-scale field B.

A gauge for the configuration generated in a healing. The configuration generated in a healing must belong to the analyticity spaces. To this end it is put in the form $e^{i\eta_k A^{\rm g}} U$, where $A^{\rm g}$ and its covariant differences $\nabla_U A^{\rm g}$ (∇_μ^U as in the Notation) are smaller than $(1 + c\beta_{\rm s})\alpha_{1,k}$ (2d). On boxes of side $O(1)$ around the sites y there are local gauges $\hat{u}_y$, with potentials $A^{(y)}$. They are glued together by a partition of unity (χ_y) taken in the quaternions,

$$u^{\rm g}(x) = \mathcal{N}\Big(\sum_y \chi_y(x)\hat{u}_y(x)\Big),$$

$$\mathcal{N}(m) = (\det m)^{-1/2} m,$$

and then $A^{\rm g} = \sum_y \chi_y A^{(y)}$ is a mean whose weights sum to 1. This holds apart from quadratic terms and from terms comparable to the differences $\hat{u}_y \hat{u}_{y'}^{-1} - \mathbf{1}$ between local gauges at neighbouring sites. In $\nabla_U A^{\rm g}$, the term $\sum_y (\partial\chi_y) A^{(y)}$ has weights summing to 0. It is therefore comparable in size to the differences between the potentials of neighbouring sites, which have a leading part in common. The square root is the branch on $|\det m - 1| < 1$ that takes the value 1 at $\det m = 1$. The mean in the display lies in this domain. It is the product of one local gauge $\hat{u}_y$ with $\mathbf{1}$ plus an average of the small differences $\hat{u}_{y'}\hat{u}_y^{-1} - \mathbf{1}$ between local gauges at neighbouring sites. The real linear span of SU(2) is $\mathbb{H}$. There $\det m = |m|^2$, and $\mathcal{N}(m) = m/|m| \in G$ for $0 < |m| < \sqrt{2}$. Wherever the root is defined, $\mathcal{N}$ is holomorphic and bi-covariant: $\mathcal{N}(u_1 m u_2) = u_1 \mathcal{N}(m) u_2$ whenever $u_1, u_2 \in G^c$. Hence the same construction applies to complex fields.

(2c) The rarity of wild patches: a small factor for each creation set against a cost for each site.
Lemma 2.5 [one factor per creation]. The following holds on the support of a term, at step $t = k + 1$. (i) When the plaquette test of step t fails at $\square$, each real U that has the stored averages satisfies $\sum_{p \text{ near } \square} a_p(U) \ge c\,\varepsilon_t^2$. (ii) When the approximate fluctuation test fails, this sum, increased by the part of GF_k located at the blocks near $\square$, is bounded below by $c\,\delta_k^2$. (In

what follows, $\mathrm{GF}_j(X;V_j)$ denotes the part of $\mathrm{GF}_j(V_j)$ located at the blocks in X.) (iii) One has $\langle B,\Delta^{(k)}B\rangle \geq c\|B\|^2$, and the bound extends to the kept conditional forms (2b). Consequently a failed fluctuation test yields

$$\tfrac{1}{2}\langle \mathcal{A}_k^{\mathrm{s}},\Delta^{(k)}\mathcal{A}_k^{\mathrm{s}}\rangle \geq c\,\delta_k^2/g_k^2$$

. The same is true of a failed collar test or rim test, because the threshold of either differs from δ_k by a constant factor, namely $1/(4\Theta_1)$ or θ_c (2a). Each of the three is proved in contrapositive form, and this form also covers the configurations at which some object entering a test is not defined. Suppose that the quantity which pays lies below the bound and below a constant. Then the data are regular, so (1b) provides the localised orbit and the contour variables, and the inequalities of the test are satisfied. In (iii), if the logarithm of the tested field is undefined, the variable of integration must satisfy $|\mathcal{A}'| \geq \delta_k$ (on the cut-off zone of the strip, the shift vanishes when $\mathcal{A}'$ is small). Since the coupling at that place is x_{t-1}, every created cell produces the factor $e^{-c\,p_0(g_t)^2}$. Inside $\mathbf{R}$ the tests work with the threshold δ_j', and the gauge trees they use have length at most $C\check{R}_j^C$. When a bond variable is large in a tree gauge, all that follows is that some plaquette exceeds $c\check{R}_j^{-C}\delta_j'$. For this reason these tests give no more than $e^{-c\check{R}_j^{-C}p_1(g_j)^2} \leq e^{-p_0(g_j)}$, and this inequality requires $2\mathfrak{p}_1 - Cr > \mathfrak{p}_0$.

The energy of the averages. To prove (i), apply (1b) in the reverse direction: it gives $|\partial V_t - \mathbf{1}| \geq c\,\varepsilon_t$ somewhere nearby. Moreover, a configuration of small action has flat averages, $\sum|\partial\mathcal{M}^tU-\mathbf{1}|^2 \leq C\sum a_p(U)$, and this inequality is scale invariant in dimension four. Stated separately, the bound reads as follows. Let U be any real configuration, with no regularity required, so that the whole constraint set is covered. Let S be any set of plaquettes of $\mathbb{T}^{(t)}$, and let $\mathcal{N}_0(S)$ denote the bare plaquettes lying beneath it, namely those all of whose corners lie in the union of the blocks $B^t(y)$, taken over the corners y of a single $p' \in S$. Then

$$\sum_{p\in\mathcal{N}_0(S)} a_p(U) \;\geq\; \min\Big\{\tfrac{1}{4}\sum_{p'\in S}\big|\partial\mathcal{M}^tU(p')-\mathbf{1}\big|^2,\ \tfrac{1}{2}e_*\Big\},$$

$$e_* = e_*(L) > 0.$$

Put $f_l = |\partial\mathcal{M}^lU-\mathbf{1}|$. A single averaging step yields $f_{l+1} \leq \mathsf{W}f_l + C\sum f_l^2$, in which W is an averaging matrix whose row sums are L^2 and whose column sums are L^{-2}. By Schur's test, $\|\mathsf{W}\|_{2\to2} \leq L^{(4-d)/2} = 1$ in dimension four. The bound $\|\mathsf{W}\|_{1\to1} \leq L^{-2}$ makes the quadratic errors sum geometrically, so no constant builds up as t grows. (Bałaban gives (i) at the background and under regularity: in 122 (1989) 355, p. 380, the factor of a failed plaquette test is derived from Theorem 1 of 102 (1985) 277 and from (71) of 102 (1985) 255. Here, by contrast, the bound is valid for all real U, with constants independent of how many steps have been taken.)

No integration is undone. For the superrenormalisable model (Comm. Math. Phys. 86 (1982) 555, Sect. 3.A and Proposition 3.1), the conditional integrations are reversed at the end. This recovers a single positive quadratic form over many scales, and the factors of all the characteristic functions are then extracted from it simultaneously. Here the operations $\mathbf{T}_k(Z)$ keep the integrals, and every factor is obtained at the place where it sits. It comes from the Wilson weight, through the display above; or from the Wilson weight together with the kept gauge-fixing exponents (Lemma 2.5 (ii)); or from the conditional form, whose lower bound is c. (Taken as a whole, the conditional form inherits the bound because it is a Schur complement; the kept part of this form is treated in (2b).) With this, the cost of the stored Gaussian integrals is $e^{C\#_j(Z_j\cap\Omega_j)}$ (122 (1989) 355, p. 383).

One action and many claimants. Gains are added up in the ledger that follows, and this addition is legitimate: nothing is spent twice and the integrals that remain converge. A single Wilson term, a single set of gauge-fixing factors and a single set of stored Gaussians pay for every failed test at every level. A *piece* is a pair (t, D) in which D is a component of the region produced by step t on which no $\mathbf{R}$ has acted up to level k. We write $\mathsf{w}_{t,D}$ for the weight of its one-step operation. This weight consists of the gauge-fixing factors with their normalisers, the Jacobians that arise when the averaging constraint is solved for the pivot bonds, and the conditional Gaussian. We write $\xi_{t,D}$ for the variables that belong to the piece. Let the shares of the action be as defined in (1), and extend U_k beyond Ω_0 by means of the stored bare field. Then at each point of the support of the term of a history one has $A(x_k(\cdot), U_k) \geq A_{\mathrm{cr}} := \sum_{(t,D)} x_{t-1} \sum_{p \in \text{share of } (t,D)} a_p(U_k) \geq 0$, so that the only part of the action spent is the Wilson sums over the shares. Moreover

$$e^{-A_{\mathrm{cr}}} \prod_{(t,D)} \mathsf{w}_{t,D} \leq \exp\Big[- c \sum_{\text{created cells } (t,\mathfrak{c}) \text{ in the pieces}} p_0(g_t)^2 \Big] \prod_{(t,D)} \bar{\mathsf{w}}_{t,D}(\xi_{t,D}),$$

$$\int \bar{\mathsf{w}}_{t,D} \leq e^{C \#_t(D)},$$

where each $\bar{\mathsf{w}}_{t,D}$ is a Borel function of the variables of its own piece only (the background no longer enters it), and $C = C(L)$ carries no $\log x$ per site. (1) *The action is divided among the pieces, not held in common.* The share of a piece consists of the bare plaquettes that lie within one cell of those of its cells at which a plaquette test or an approximate fluctuation test has failed. Distinct pieces have disjoint shares, because the tests of a step are carried out several cells away from the hull of the older region. On a share one has $x_k(\cdot) = x_{t-1}$, since the older profiles have completed their transition and the newer ones have not yet begun it. Every other plaquette contributes ≥ 0. (2) *The tests concern averages of the configuration that appears in the action.* The tests were built from localised backgrounds of step t, whereas the Wilson term contains U_k. The two are connected as follows. The identity $\mathcal{M}^{t-1} U_k = V_{t-1}$ on Γ_{t-1} is the constraint that defines the background, and on the same set $V_t = \mathcal{M}(V_{t-1})$ is imposed by the corresponding delta function. Lemma 2.5 holds for every U with these averages, so it applies to U_k. (3) *Each exponent is split into two halves*: one half is used for payment and the other is kept for the integration. The only normaliser that depends on the coupling is integrated against half of its own exponent,

$$\int du\, e^{-\frac{1}{2} x (1 - \frac{1}{2} \operatorname{Re} \operatorname{tr} u)} \leq C \int du\, \mathbf{1}\{|u - \mathbf{1}| < \epsilon\}\, e^{-x(1 - \frac{1}{2} \operatorname{Re} \operatorname{tr} u)}$$

$$(\epsilon \leq 1,\ x\epsilon^2 \geq 1),$$

and both sides are of order $x^{-3/2}$. On the cut-off zone of the strip the shift (2b) vanishes for small $\mathcal{A}'$; consequently a large tested field is a large variable of integration. (The approximate test requires GF for the following reason. Take a gauge transformation at a site of $\mathbb{T}^{(t-1)} \setminus \mathbb{T}^{(t)}$. It moves V_{t-1} away from $\mathcal{M}^{t-1} U_{t,\square}(V_t)$, the average of the localised background (2a), which is in the block axial gauge (1a). It does so at no cost in action, since it alters neither the action nor V_t.) (4) *Within a patch on which the* $\mathbf{R}$ *of level* m *acts*, a test that fails may not be paid for by the large field from which the patch arose. Whatever is older than the wait was paid for by plaquettes that lie in the *floor*. The floor is made up of those cells of level $m_\flat$ of the old core, $\Omega^c_{m_\flat} \cap X$, which do not touch $\Omega_{m_\flat}$; this old core is a part of the core $C_{m_\flat} = X \cap Z_{m_\flat}$ of (2b), because $\Lambda_i \subset \Omega_i$. The gauge-fixing factors and stored Gaussians that paid for it are all of levels below $m_\flat$. Let X be a drop-out, and consider stored variables with two properties: each test carried by X takes its recorded value, and the plaquette variables of the datum in the old core, taken layer by layer (V_l on Γ_l), are within $C \check{R}_l^2 \varepsilon_l$ of $\mathbf{1}$. (These two properties hold on the supports. They also hold at the

points of evaluation that occur below in the mass of a later drop-out that contains X. Within the mass of X itself, no use is made of the test whose failure turned it into a drop-out.) For every real U compatible with stored variables of this kind, one has

$$c\,x_m \sum_{p\subset X^{+},\ \text{no corner in the floor}} a_p(U) + \sum_{j=m_\flat}^{m-1} x_j \mathrm{GF}_j(X;V_j) + c\sum_{j=m_\flat}^{m-1} \|\mathcal{A}^{\mathrm{s}}_j\|^2_X \;\geq\; (\log x_m)^{2\mathfrak{p}_1 - Cr} \;\geq\; p_0(g_m),$$

where X^{+} denotes X enlarged by two layers of unit cubes of level m and U is compatible on X^{+}. This follows from a single dichotomy: if the left-hand side is the smaller, then each test of the operation takes the value 1. In every case the bridge runs along the same chain. It starts from the action of a compatible configuration and passes to the plaquettes of the stored fields (by the energy of the averages). From there it passes to the regularity of the background used by the test, and then to the bond variables. That last link comes either from the rigidity of a single averaging step or from loops in the shells. In a shell of side $s \leq CM\check{R}_m$, a loop composed of n straight pieces has a deviation from $\mathbf{1}$ that is bounded by $C\mathsf{n}s^2$ times the plaquette bound, and this is where the loss $\check{R}_m^{-C}$ comes from. The waiting collar screens off the unknown large field in the floor; this collar is what separates the floor from the region $X \cap \Omega_{m_\flat}$ in which the operation acts, and its width is at least one cell in the local units. Where the operation acts, the background of any stage has curvature at most $C(\varsigma + \check{R}_m^2 \varepsilon_m e^{-\delta_0 M\check{R}_m})\eta_j^2$ on the layer j, which is the local scale. Here ς is the size of the plaquettes of the stored fields that follows from the smallness of the left-hand side. A drop-out uses no plaquette that has a corner in its floor. By contrast, each plaquette used beneath it lies inside the floor, and each gauge-fixing factor and stored Gaussian used beneath it is of a level below $m_\flat$. The windows of nested operations are disjoint.

The cost. Carrying Z_n through step n incurs three costs. The first is a factor $e^{C\log x_{n-1}}$ for each site of $\mathbb{T}^{(n)}$ lying in $T_\ell \setminus \Omega_n$. The logarithm in question is $-\frac{3}{2}\log x_{n-1}$ times the number of free bonds less the number of gauge-fixed sites. It is produced where the fluctuation field is introduced and rescaled, $\mathcal{A}_{n-1} = g_{n-1}\mathcal{A}^{\mathrm{s}}_{n-1}$, and where the gauge is fixed, that is on Ω_n, and not where the field is integrated; the normalisation removes it everywhere. The second is a factor e^{C} for each site of $\mathbb{T}^{(n)}$ lying in the rest of Z_n enlarged by a layer. It comes from $\mathbf{E}_n, \mathbf{R}_n, \mathbf{B}_n$ and from the stored Gaussians, which are written in the scaled variable. The third is the entropy of the subregions. Altogether, for each component Z of Z_n, the cost is $\exp[CM^4\check{R}_n^5(d'_n(Z)+1)]$, where d'_n is the tree length measured in cells. Of the five powers of $\check{R}_n$, four account for volume and one for $\log x_{n-1} \leq C\check{R}_n$. In the count below, the number $|Z_n|_n$ of level-n cells of the region takes the place of Bałaban's linear size d'_n, because a region that consists of a single cell has $d'_n = 0$ and still incurs a cost.

Proposition 2.6 [the count, the ledger, the credit]. Let a history and one of its components be given. Then the count and the ledger displayed below are valid. If moreover the component is neither healed nor marked, the credit inequality displayed below is valid for it as well. In all of these the constants depend neither on the history nor on the number of steps. The ledger and the credit are statements of Bałaban (Comm. Math. Phys. 122 (1989) 355, (1.79)–(1.81), (1.89)), here with volumes counted in cells. The proof does not follow his induction. It rests instead on counting once, globally, along the whole genealogy of a component.

Shrinking and waiting. We denote by $S(Z)$ the free evolution of (2a), that is, the hull of Z in the next partition together with $O(1)$ further layers, boxed. Every step that is not a drop of the cube size $\check{R}$ divides linear sizes, counted in cells, by the factor L. When a step is a drop, the partitions before and after it coincide, so that nothing shrinks; and of two consecutive steps, at most one is a drop. Consequently

$$d'_n(S^{n-j}Z) \le CL^{-\nu} d'_j(Z) + C,$$

$$\nu \ge \lfloor (n-j)/2 \rfloor \text{ the number of the steps } j+1,\dots,n \text{ that are not drops}$$

(this display replaces the scaling of the tree length in Bałaban, Comm. Math. Phys. 122 (1989) 355, p. 384, which in the present notation, with $Z^{(n-j)}$ denoting the hull of Z in level-n cells, reads: $d'_n(Z^{(n-j)}) \le L^{-\frac{1}{2}(n-j)} d'_j(Z) \le 2^{-(n-j)} d'_j(Z)$ over $n-j>1$ steps, and $d'_n(S^{n-j}Z) \le C\, d'_n(Z^{(n-j)})$, or $\le C$ when the right-hand side vanishes). The added layers do not pile up, because every change of scale contracts what the earlier levels added. It follows that within $\mathsf{k} \le \mathsf{N}_j + C\log(d'_j(Z)+2)$ steps a component Z of Z_j does one of three things, provided the history continues up to level $j+\mathsf{k}$: it meets a creation, it joins another component, or it enters the class of $\mathbf{R}$.

Why a quiet component is bound to enter the class. Shrinking makes the component small, but smallness alone does not place it in the class. Consider a quiet line, that is, the sequence of components v_i, of the levels i, into which a single component is carried as long as nothing is created in it and nothing joins it. Such a line develops deterministically, $v_i = \Phi_i(v_{i-1})$, where Φ_i is the rule of the step for a lone component, and its side in cells obeys $\alpha_{i+1} \le \alpha_i + C$, $\alpha_{i+2} \le \alpha_i/L + 2C$. This is an affine contraction over two levels; its fixed point is less than half of the threshold of smallness, so that room is left for a drop. Apart from the depth $k_\flat \ge 1$, which is a matter of the arithmetic of the cube sizes, a line of this kind acquires on its own precisely the conditions of the class (2b). Moreover, a condition that fails on account of an event in the window also fails at every intermediate level, because $n \mapsto n - \mathsf{N}_n$ is strictly increasing. Hence the carrier of the event cannot have been healed in between. The waiting bound is founded on this property of the class (in Bałaban, a single sentence, p. 385).

The count. An unhealed component of level $n-1$ is a predecessor of the component (n, C) of Z_n that contains its hull. For a component u_*, the genealogy $\mathrm{Gen}(u_*)$ is the tree formed by all $u = (n(u), C(u))$ that equal u_* or are iterated predecessors of it. It is the same object as the two of (2a). The cluster of a root is its genealogy together with everything that the history records within it. The lineage of a live component is made up of the members of its genealogy, regarded as regions Z_j, together with the creations in them. None of these three contains a component that was healed earlier and whose place a later region covers again; such a component is the root of a cluster of its own. Write $\mathsf{n}_t^{\mathrm{cr}}(u_*)$ for the number of created cells of level t in $\mathrm{Gen}(u_*)$. Write $\mathsf{n}_t^{\mathrm{mk}}(u_*)$ for the number of its marked components of level t other than u_* itself. These are the drop-outs and islands, that is, the creations by an operation of (2a), and they are counted separately here because, for each of them, the count uses only the factor e^{-p_0}, which is all that a drop-out contributes. Set $\mathsf{T}_t := C\check{R}_t$; this number bounds the length of a wait and does not increase with t. Then for non-increasing costs $\hat{\mathsf{r}}_1 \ge \hat{\mathsf{r}}_2 \ge \cdots \ge 0$,

$$\sum_{u \in \mathrm{Gen}(u_*)} \hat{\mathsf{r}}_{n(u)}\, |C(u)|_{n(u)} \;\le\; C \sum_t \mathsf{n}_t^{\mathrm{cr}}(u_*)\, \mathsf{T}_t \hat{\mathsf{r}}_t + C \sum_t \mathsf{n}_t^{\mathrm{mk}}(u_*)\, \mathsf{T}_t \hat{\mathsf{r}}_t.$$

In particular, for each history and without any bound on the number of steps,

$$\sum_{n \le k} \#\{\text{cells of } Z_n\} \le C \sum_{\text{creations of both kinds, at steps } t} (\mathsf{N}_t + 1).$$

The proof consists of three moves, and nothing is carried through an induction over the levels. (1) Contraction. The cells are counted by way of the occupied cells of a component, which are the hulls of the created cells beneath it. These form a skeleton that commutes with taking hulls, and the component never lies more than $O(1)$ cells away from it. Over a fixed number s_0 of levels, at

least half of the occupied cells of a component that has many of them disappear, whereas a light component (one with at most C of them) retains at most C. Let o_n be the number of occupied cells of level n in $\mathrm{Gen}(u_*)$, and let l_n be the number of its light components of level n. Then

$$\mathsf{o}_n \le \tfrac{1}{2}\mathsf{o}_{n-s_0} + C\, l_{n-s_0} + \sum_{n-s_0<t\le n} \mathsf{n}_t^{\mathrm{cr}}$$

(quantities of levels ≤ 0 are 0),

whence

$$\sum_n \hat{\mathsf{r}}_n \mathsf{o}_n \le C \sum_n \hat{\mathsf{r}}_n l_n + C \sum_t \hat{\mathsf{r}}_t \mathsf{n}_t^{\mathrm{cr}},$$

where the second inequality is obtained from the first by multiplying by $\hat{\mathsf{r}}_n$ and summing, since the costs are non-increasing. (2) Stretches. A member of $\mathrm{Gen}(u_*)$ is called an event if it contains a created cell of its own level, or comes after a marked component, or has two or more predecessors (in which case it is a joining). The genealogy is a disjoint union of stretches. Each stretch starts at an event and continues through components that have a single predecessor and contain nothing new. A stretch that starts at level t has at most T_t light members. Indeed, after $O(1)$ levels a light component is contained in a cube of $O(1)$ cells, and once a further N_t quiet levels have passed it has entered the class, in which it is healed or else marked. (3) For each of the three kinds, the events are paid for through an injective assignment, and no created cell is charged more than twice. An event that contains a created cell is paid by that cell, the successor of a mark is paid by that mark, and a joining is paid by a created cell at a leaf of the tree. For joinings, select at each member that has predecessors one of them as its first predecessor. To a joining v, assign the leaf reached by starting from a predecessor of v other than the first and always moving to the first predecessor. No two joinings receive the same leaf, since on the way up from that leaf the first member entered otherwise than from its first predecessor is v (the leaves of a tree outnumber its branchings). Moreover, the leaf lies at a lower level, at which $\mathsf{T}_t\hat{\mathsf{r}}_t$ is no smaller (because $\mathsf{T}_t = C\check{R}_t$ is non-increasing in t), and each leaf contains a created cell. Thus a joining requires no payment of its own.

The ledger. Each created cell brings the gain $\mathsf{g}_t = c\,p_0(g_t)^2$ and each mark the gain $\mathsf{p}_t = p_0(g_t)$ (Lemma 2.5). Against these stands, for every cell and every level, the cost $\mathsf{r}_n^{\mathrm{c}} = C\log x_n\,(M\check{R}_n)^4$ found above, which is bounded by the non-increasing sequence $\hat{\mathsf{r}}_n = C\mathsf{r}_n = CM^4\check{R}_n^5$; here r_n is the weight of a cell, (2a). With these choices,

$$\begin{aligned}\mathsf{F}(u_*) &:= \sum_t \mathsf{g}_t \mathsf{n}_t^{\mathrm{cr}}(u_*) + \sum_t \mathsf{p}_t \mathsf{n}_t^{\mathrm{mk}}(u_*) - \sum_{u\in\mathrm{Gen}(u_*)} \mathsf{r}_{n(u)}^{\mathrm{c}} |C(u)|_{n(u)} \\ &\ge\ 2\sum_t \mathsf{p}_t \mathsf{n}_t^{\mathrm{cr}}(u_*)\ \ge\ (2-c)\,p_0(g_{n(u_*)}),\end{aligned}$$

provided that $\mathsf{g}_n \ge 2\mathsf{p}_n + C\mathsf{T}_n\hat{\mathsf{r}}_n$ and $\mathsf{p}_n \ge C\mathsf{T}_n\hat{\mathsf{r}}_n$. An event costs $\mathsf{T}_n\hat{\mathsf{r}}_n \le CM^4\check{R}_n^6 \approx (\log x_n)^{6r}$. This cost is made up of $\log x_n \le C\check{R}_n$ for each site, $(M\check{R}_n)^4$ sites in a cell and a wait of length $\check{R}_n$. Since $\mathfrak{p}_0 \ge 6r+1$, the quantity $p_0(g_n) = A_0(\log x_n)^{\mathfrak{p}_0}$ exceeds this cost, and the extra power of the logarithm absorbs the constants. The final inequality holds for the following reason. The genealogy has a leaf, and so it has a created cell at some level $t \le n(u_*)$. Moreover $p_0(g_t) \ge (1-c)p_0(g_{n(u_*)})$, because, up to a bounded error, the flow is monotone: $x_t \ge x_{n(u_*)} - C$. When volumes are counted in cells, $e^{-\mathsf{F}(u_*)}$ coincides, up to two modifications, with the quantity whose supremum is taken on the right-hand side of (1.79). First, the $2p_0(g_j)$ that Bałaban reserves for

each created region is taken from $c\,p_0(g_j)^2$ (the first of the two conditions). Second, Bałaban's cost of a cell, which in the notation of this text reads $CM^4\check{R}_j^5$, is replaced by the finer form $C\log x_j\,(M\check{R}_j)^4$. Components that consist of a single cell are included: such a component costs $\mathsf{r}_n^{\mathrm{c}}$ at every level, and this cost is paid through the light members.

The credit, as a corollary. Let $z=(j,Z)$ be a component that is neither healed nor marked. We write its factor as $e^{-\mathsf{F}(z)}=e^{-\kappa_j(Z)-2p_0(g_{j(Z)})}$, where $j(Z)$ is the level of the first creation occurring in its genealogy. We extend the history of z by the fictitious continuation $S(Z),\dots,S^{\mathsf{k}}(Z)$. Nothing is created in this continuation, and it runs at the levels $n\le K$, where the flow bounds are available; beyond K it is cut off. We mark the last member of this continuation and apply the ledger to the extended history. The extended history has the same created cells as z, and apart from that last member it has the same marked components. The ledger therefore gives

$$\mathsf{F}(z)-\sum_n \mathsf{r}_n^{\mathrm{c}}|S^{n-j}Z|_n \ge 2\sum_t \mathsf{p}_t\mathsf{n}_t^{\mathrm{cr}}(z)\ge 2\mathsf{p}_{j(Z)},$$

which is to say

$$\kappa_j(Z)\ \ge\ \sum_{n=j+1}^{\min\{j+\mathsf{k},K\}} \mathsf{r}_n^{\mathrm{c}}\,\big|S^{n-j}Z\big|_n,$$

that is, (1.80), in which Bałaban has K in place of k and the summand

$$O(1)M^dR_n^{d+1}d'_n(S^{n-j}(Z)),$$

where R_n is his size, built from g_n itself. Thus the credit $\kappa_j(Z)$ pays for the entire wait. By the shrinking, the sum is bounded by $CM^4L^5\check{R}_j^6(d'_j(Z)+1)$ ((1.81)).

Who pays. At a creation, the deposit $c\,p_0(g_j)^2(d'_j+1)$ has degree $2\mathfrak{p}_0$ in $\log x_j$. It exceeds the total cost $CM^4L^5\check{R}_j^6(d'_j+1)$, whose degree is $6r$, so that per cell p_0^2 covers $\check{R}_j^6$. Now suppose a test fails inside $\mathbf{R}$. All that is known of the old credit is $\kappa\ge 0$, so the renewed wait of a small region, $CM^4\check{R}_j^6$, must be paid by the new factor $e^{-p_0(g_j)}$ on its own. For this reason that factor has to be at least as good as e^{-p_0}. A joining neither pays anything nor releases anything.

Joining: Bałaban's induction and the count. Bałaban proves (1.80) by induction on j and, within a step, by induction on the number of regions that join (pp. 384–387). When X and Y join to form Z, he sets free the reserve $2p_0(g_{j(X)})+2p_0(g_{j(Y)})-2p_0(g_{j(Z)})\ge c\,p_0(g_j)$ to pay for the renewed wait. This step relies on the claim of p. 387 that, once S has been applied n_1 further times, the domain built from Y includes the domain built from X; there n_1 is called a rather small number, with $n_1<10$ given as an instance. The count uses no containment of this kind. For the hull operation of this part, whether such a containment holds depends on where the two regions lie relative to the grids. In a single coordinate the hull map is $\mathsf{h}(y)=\lfloor(y+\frac12(L-1))/L\rfloor$, and S, taken with ten layers, maps an upper end y to $\mathsf{h}(y)+10$. Set $y_{m_*}=10$ and $y_m=L(y_{m+1}-10)+\frac12(L-1)$. Then

$$\mathsf{h}(y_m)=y_{m+1}-10,$$
$$\mathsf{h}(y_m+l)=y_{m+1}-9$$
$$(0\le m<m_*,\ 1\le l\le L).$$

Suppose that Y ends at y_0 and that X, a block of side 5, begins at y_0+1. The two regions touch. The upper end of $S^m(Y)$ is y_m and the upper end of $S^m(X)$ is y_m+1 (for $m=0$ it is

$y_0 + 5$), so $S^m(X) \not\subset S^m(Y)$ for every $m \le m_*$, where m_* is arbitrary. At each of these steps the two partitions have ratio L, so that none of the steps is a drop. The count never compares $S^m(X \cup Y)$ with $S^m(Y)$.

Proposition 2.7 [the mass bound; where the small factor stands]. Consider a component X that has passed every test of the operation $\mathbf{R}$ of some level $i \le k$ before the last decomposition $1 = \chi_i + \chi_i^c$ (a healed component or an island). The operation of (2b) on it is written

$$\mathbf{T}_i'(X)F\,(\mathbf{p}) = \frac{1}{|W_4|}\sum_{\omega}\int e^{\Phi_\omega(\xi;\mathbf{p})}F\,\mu_\omega(d\xi),$$

$$\mathbf{p} = (\mathbf{U},\mathbf{J})\,\text{a pair on } X^{\sim},$$

where ω ranges over the local histories of X, that is, the results of every decomposition of unity inside X together with a frame in W_4 of the gauge tree, ξ ranges over the stored variables, and e^{Φ_ω} contains the core Wilson factor $\exp[-A(x_i''\phi_{\mathrm{c}}, U''(\xi))]$. (a) For a healed component or an island, in which case $\mathsf{k} = 0$,

$$\mathsf{M}(i,X) = \frac{1}{|W_4|}\sum_{\omega} e^{c\,\mathsf{v}(\omega)}\int \sup_{\mathbf{p}\in U_i^c(X^{\sim},\alpha_{0,i},\alpha_{1,i})}\left|e^{\Phi_\omega(\xi;\mathbf{p})}\right|\mu_\omega(d\xi) \le e^{-(2-c')p_0(g_i)},$$

$$\mathsf{v}(\omega) = \sum_{u\ \text{listed by}\ \omega}\log x_{n(u)}\,\#_{n(u)}(C(u))\,:$$

here the coupling is taken at the level of the action, and the current coupling then follows from $p_0(g_i) \ge (1-c')p_0(g_k)$; the supremum stands inside the integral; and a weight growing exponentially with the space-time volume of the patch is paid for in advance. It is this bound that the induction carries forward. Bałaban has (1.89) instead, at each fixed pair ((1.73)). (b) Take a single history and a single component Z of Z_k on which no $\mathbf{R}$ has acted. For every value of the exterior variables (in total variation),

$$\left\|\mathbf{T}_k(Z)\big[\chi_{k,\mathcal{S}}\,e^{-A(x_k(\cdot),U_k)}\big]\right\| \le \prod_{\text{created cells}\ (t,\mathfrak{c})\subset Z} e^{-c\,p_0(g_t)^2}\cdot\exp\Big[C\sum_{t\le k}\#_t(Z_t\cap Z)\Big],$$

with each factor taken at the coupling of its own step. Bounds of this kind do not multiply over the components, since the background couples the components. The pointwise majorant given above does multiply (Tonelli). Consequently, after integration and for an entire label, there exists a history h carrying the label $\mathcal{S}$ for which the bound below holds, where $(t,\mathfrak{c})$ ranges over the created cells of h lying outside the marked X. (The created cells are not determined by the label: a row of three failing cells and that row with its middle cell removed yield the same regions.)

$$\int \mathbf{T}_k'(\mathcal{S})\big[\chi_{k,\mathcal{S}}\,e^{-A(x_k(\cdot),U_k)}\big]dV_k \le e^{C\sum_j \#_j(Z_j)}\prod_{(t,\mathfrak{c})\ \text{outside the marked}\ X} e^{-c\,p_0(g_t)^2}\prod_{(i,X)\ \text{marked, outermost}}\mathsf{M}(i,X).$$

(The bound holds a fortiori when each site carries $C(1+\log x_j)$ and each created cell carries $e^{-c\,p_0(g_t)}$. This is the form used in Proposition 2.13 [outlined below; Section 32], the statement that carries the logarithm, and in (4b).) (c) Consider the sum over the labels, with the costs paid by Proposition 2.6 and the nested sequences inside each component summed. In this sum every component C of Z_k keeps the factor $\exp[-c\,p_0(g_k) - \kappa_1\#_k(C)]$, in which the volume is counted in sites (and so, a fortiori, the factor with $\kappa_1|C|_k$, in which it is counted in cells; this is the form in the introduction to this step). Since $d_k(C) \le \#_k(C)$, this per-site factor includes the term $-\kappa_1 d_k(C)$,

which is what p. 387 asserts for an arbitrary region. The decay is one further cost $O(1)$ per site; like the other costs, it is paid for by the events (Proposition 2.6). If failed plaquette tests have produced a component, then $\mathbf{T}_k(Z)1 \le \exp[C\sum_{t\le k}\#_t(Z_t\cap Z)]$ (119, (2.21), and p. 264, where it is said of the Wilson term that it "yields the small factors for large field characteristic functions").

The bound has to reproduce itself. (i) *Supremum under the integral.* In a term of level $k \ge i$, the factor remaining on X is evaluated at $(U_k, J(U_k))|_{X^\sim}$, and U_k depends on the integration variables of every component. Such an integral is not controlled by a bound at each fixed pair, and bounds that are merely uniform in the other variables do not multiply. For example, take $\mathbf{1}\{|\xi_1-\xi_2|<\epsilon\}$ on $[0,T]^2$: each integral in one variable is $\le 2\epsilon$, whereas the double integral is about $2\epsilon T$. What does multiply is a majorant given by a Borel function that depends only on the variables of X. The integral of the supremum itself is understood as an upper integral. Such a majorant can be obtained because only two factors depend on the pair: $\Phi_\omega = \Phi^\circ_\omega(\xi)+[-\frac12 q_X+V(X^\sim)](\xi;\mathbf{p})$, where $q_X = \|\phi^{1/2}D\mathsf{H}B\|^2 \ge 0$ is formed at a real background, and $|V(X^\sim)| \le C\,\#\{\text{stored sites}\}$, with every site counted at its own level. The real background here is the critical configuration of the prepared sequence, taken with $B=0$ and at the datum of the main step, and it is built with the G-valued factor of the triple. Through that factor, q_X depends on the splitting, whereas the exponent Φ_ω, taken in its entirety, is determined by the pair (see clause (vi) of the class of (2e)). (ii) *A fixed set.* If the supremum is taken over the arguments that occur at level k, the bound is not stable, since the next step replaces U_k by U_{k+1}. The supremum is therefore taken over a larger fixed set, built only from the data of the levels $\le i$. This set contains every later argument, since on the supports the following bounds hold, with nothing lost in the age $k-j$ of the layer $\Omega_j\setminus\Omega_{j+1}$,

$$|\partial U_k - \mathbf{1}| < C\check R_j^2\varepsilon_j\eta_j^2,$$

$$|J(U_k)| < C\check R_j^2\varepsilon_j$$

$$\text{on } \Omega_j\setminus\Omega_{j+1} \text{ (the current in level-}j\text{ units)},$$

$$C\check R_j^2\varepsilon_j < \alpha_{0,j} \text{ for all } j \ (q \text{ is chosen after } \mathfrak{p}_0 \text{ and } r, \text{ the ceiling on the couplings after } L)$$

(the last inequality holds since the quotient is a negative power of $\log x_j$, since $q > \mathfrak{p}_0+2r$, and since the couplings stay below a ceiling; moreover, a layer at index distance a has an influence that decays like $e^{-(a-1)_+}$, while its threshold grows like $(1+a)^{1/2}$). Hence the mass seen at level k does not exceed the mass of level i, and a single step **RT** gives back the bound with unchanged constants. (iii) *The weight* $e^{c\mathsf{v}(\omega)}$ has as its only cost a larger constant in the ledger. Since it carries $\log x_j$ per site, it absorbs every perturbation of size $O(1)$ per site of the patch, such as the factor e^{Ψ} of the sources, (4b). In Proposition 2.13 [outlined below; Section 32] the weight covers whatever the label still exhibits inside a drop-out, since the small factors produced there have already been consumed by the mass; this is why the masses are required with the weight raised. The argument requires a volume that counts each pair (level, cell) only a bounded number of times. A volume that counts the past again at every drop-out may exceed the total of all the gains; an example is a row of cells all created at the first step, with a drop-out at each completed wait. (iv) *Drop-outs.* The factors kept by a drop-out contain no Wilson action, and no inner background has been formed; the kept factors are evaluated at the background of whichever later term contains X. On the other hand, the configuration enters the Wilson term of the carrying term, and the failed plaquette tests listed by ω, which are the tests of the creations in X, bound plaquettes formed from averages of this configuration. That weight, over the bare plaquettes all of whose corners lie in X, is therefore counted with the operation. The pair is replaced by an *evaluation* at ξ. By this we mean a real U on X with four properties. First, it is compatible with

ξ, that is, $\mathcal{M}^j U = V_j(\xi)$ on the bonds of Γ_j in X. Second, it is regular layer by layer, in the same sense as on the supports: on the layer j one has $|\partial U - \mathbf{1}| < C\check{R}_j^2 \varepsilon_j \eta_j^2$, while the plaquette variables of the stored fields differ from $\mathbf{1}$ by less than $C\check{R}_j^2 \varepsilon_j$. Third, at this configuration each test listed by ω takes its recorded value, with the exception of the test whose failure made X a drop-out; here the fields not formed by ξ, including V_i on X, are read as averages of U. Fourth, $(U, J(U))|_{Y^\sim} \in U_m^c(Y^\sim, \alpha_{0,m}, \alpha_{1,m})$ for each island (m, Y) in X. Together with this configuration, the supremum also runs over further free parameters subject to one-sided bounds. These are the weight υ of the Wilson term, with $\upsilon \geq c\, x_t$ on Z_t, and the kept forms and Jacobians, which are evaluated at a background (2b), with form $\geq c$ and Jacobian $\leq C$:

$$\frac{1}{|W_4|} \sum_{\omega} e^{c\, \mathsf{v}(\omega)} \int \sup_{\text{evaluations at } \xi} e^{-A(\upsilon \mathbf{1}_X, U)} \,(\text{kept factors})\, \mu_\omega(d\xi) \leq e^{-(2-c')p_0(g_i)}$$
$$(\upsilon \geq c\, x_t \text{ on } Z_t).$$

(Over the empty set the supremum is 0.) Take a term of some level $k \geq i$ in which X occurs, and a point of its support. Put $\upsilon = x_k(\cdot)$ and take the forms and Jacobians at the background of that term. Then the configuration that the Wilson term of that term carries on X is an evaluation at those coordinates of the point that belong to X. Hence, as in (ii), the set contains all later arguments. For a drop-out, the left-hand side is its mass $\mathsf{M}(i, X)$. Where X has no sum over frames, nothing depends on the frame listed by ω, by the identity $1 = |W_4|^{-1} \sum_r 1$. This bound does not use the test whose failure made X a drop-out. The gain $p_0(g_i)$ of that test is reserved for the renewed wait, and it is drawn on once X occurs as a mark within a subsequent component (Proposition 2.6). The supremum is taken over a set fibred over ξ. Over a product set it selects a flat U, at which $e^{-A} = 1$, and the factor is lost. In a toy case, take $\Phi = -T(1 - \cos p)$ together with the test $|\xi| \geq \epsilon$: its mass is $e^{-cT\epsilon^2}$ on the graph $p = \xi$, but approximately 1 over the product set. For this reason each inequality of Lemma 2.5 is required to hold at every configuration of this set, and not merely at backgrounds. These inequalities do hold for all real configurations having the prescribed averages. Nothing is assumed inductively at a drop-out; the inductive bound is the one for the islands. (v) *Islands inside X.* The count and the ledger are carried out on the outer genealogy of X, that is, on the genealogy cut off at the islands (m, Y) in X. Each such island is a minimal node, and all of its cells, of which there are $O(1)$, are counted as created cells of level m. These cells are paid for in the same way as created cells. The factor χ_m^c of the new field on Y less a margin is present in the term. Consequently there is a test cube of level m in which the test fails at $V_m(\xi)$; this is the test χ_m of a step, and not one of the sharper tests of the stages. Lemma 2.5 (i) then extracts the gain $c\, p_0(g_m)^2$ from the core Wilson factor of X, on plaquettes that are used by no other gain; when X is a drop-out, the gain is extracted instead from the Wilson weight that (iv) counts together with its operation, and this holds at every evaluation. This test does not enter the mass of Y, since that mass is formed before the last decomposition. One evaluates the factor of Y at a real pair of $U_m^c(Y^\sim, \alpha_{0,m}, \alpha_{1,m})$, where it is bounded by its majorant. Since v is additive, the sum over the local histories of Y separates and is at most $\mathsf{M}(m, Y) \leq 1$. Every leaf of the cut genealogy contains a created cell, and the ledger leaves $2 \sum_t \mathsf{p}_t \mathsf{n}_t^{\mathrm{cr}}$, just as it does on the full genealogy.

(2d) Cancellation, localization and the sum over healed components. The main operation has two tasks. It must replace the prepared background by the healed background without leaving a large Wilson action on the shells. It must then sum the histories inside the healed components while keeping the exterior data fixed. The local terms produced by these two tasks have different bounds. A term confined to the collar of a component has a bound by its stored volume. A term

extending beyond that collar decays in the boundary hull length. The volume term stays inside the component integral, where the large-field mass pays for it.

Write U'' for the prepared background, $\bar{U}$ for the healed background, and U_0 for the real critical configuration at the centre of the reference law. Let ϕ_X cut off inside the treated component X, let ϕ_{c} cut off its old core, and put $\phi = \phi_X(1-\phi_{\mathrm{c}})$. At a real point of the tests the Wilson part satisfies

$$\begin{aligned} &- x_k A(\phi_X, U_{k,X}(e^{ig_k B}V_\Lambda)) - A(x_k''(\cdot), U'') + x_k A(\phi_X, U_0) \\ &\qquad = -A(x_k(\cdot), \bar{U}) - A(x_k''\phi_{\mathrm{c}}, U'') - \tfrac{1}{2}\|\phi^{1/2} D\mathsf{H}B\|^2 + \mathsf{R}_X. \end{aligned}$$

The positive term in the first line comes from the denominator of the reference law. The linear term in B vanishes because U_0 is critical on the fibre. The second term on the right retains the Wilson weight which pays for old large plaquettes. The quadratic form is nonnegative.

To estimate R_X, insert the localized backgrounds of the successive prepared sequences. The ten resulting terms are differences of neighboring Wilson actions, Wilson actions with small weights, and the remainder beyond the quadratic Taylor term. For each comparison the two determining sets and their data agree in a collared region. The plaquettes being compared lie at weighted distance at least $cM\check{R}_k$ from its boundary. Exponential locality therefore gives an error bounded by

$$C(1+\bar{c})(\log\theta_k)^{2q+Cr} e^{-c\delta_0 M(\log\theta_k)^r}.$$

This tends to zero uniformly in the level and in the torus. The cubic Taylor terms contain a positive power of g_k; the terms from the shrinking core contain $L^{-c\mathsf{N}_k}$. The weight difference on a shell is $O(k-j)$ rather than $O(x_k(k-j))$. Its growth is absorbed by the weighted-distance decay across the layers.

The inner-core term has a useful complex continuation. Move the complex part into the response from data supported outside the core. At the stored real variables ξ one obtains

$$A(x_k''\phi_{\mathrm{c}}, U''(\mathbf{U})) = A(x_k''\phi_{\mathrm{c}}, U''(\xi)) + O(1).$$

The first term on the right is the action of a G-valued configuration. It is retained in the component measure. The response error has the factor $L^{-\mathsf{N}_k}$ and is bounded even though the old shell weights increase with the depth.

The remainders and the changes in earlier activities are localized by one interpolation. Each walk in the response carries a parameter $s(\Delta)$, $|s(\Delta)| < e^{\bar{\kappa}}$, for every cube it visits. Setting a parameter to zero prevents a walk from crossing that cube. For a term f of bound b_f, interpolate its argument by $\lambda(\mathbf{H}, \mathsf{j})(s)$. The response and hereditary-domain estimates give an analytic interpolation on $|\lambda| < \mathsf{a}^{-1}$, where a is the size of the response divided by the room in the domain of f. Its values at $(s,\lambda) = (\mathbf{1}, 1)$ and at $s = 0$ are the old and new evaluations. Brackets in the s variables and Schwarz's lemma in λ then give

$$f(U^{\mathrm{old}}) - f(U^{\mathrm{new}}) = \sum_{Y \supset X} f_Y, \qquad |f_Y| \le 2\mathsf{a} b_f e^{-(\bar{\kappa}-1)\mathfrak{m}(X,Y)}.$$

Here $\mathfrak{m}$ counts the new cubes and the cubes needed to join them to both the original term and the integrated field. The new argument, its gauge factor and its stored data are formed together. In particular, a stored field is not transported to the top of a tower whose depth increases at every later step.

At a stage the amplitude is at most $C\mathsf{N}_k\delta_j/\min_\iota \alpha_{\iota,j}$; at the main step away from the box it is at most $Cg_k^{-1}e^{-cM\check{R}_k}$. Inside a healed box the new field is evaluated through its own generated background at its averages. The critical-configuration identity identifies the arrival with the old background. The corresponding gauge has leading potential given by an average of the argument's

potential. Thus it obeys the potential bound required by the old term; an axial representative without this control would not suffice.

Near a sharp test no analyticity in the stored field is claimed. Instead the term is the value at $\mu_b = 1$ of a function analytic for $|\mu_b| < \lambda^{\mathrm{st}}$, with the switch properties that it does not read the field when $\mu_b = 0$ and does not read μ_b when the field is zero. Block differences therefore gain $(\lambda^{\mathrm{st}} - 1)^{-1}$ per group of variables. Choosing $\lambda^{\mathrm{st}} = (\log \theta_j)^{6r+1}$ pays for the $O(\check{R}_k^4)$ cubes in a component and the N_k stages. Once a stage integrates a stored field, its carried terms are evaluated at field zero and its switch is removed. This preserves their bounds.

The sum of the stage logarithms is consequently bounded by

$$\sum_{\nu<\mathsf{N}_k} \sup |\mathbb{C}_k^{(\nu+1)}(Y)| \le C_{\mathbf{S}} e^{-(1+3\beta_{\mathrm{s}})\kappa d_k^\sharp(\mathcal{S};Y)}.$$

The constant does not grow with the number of stages. Every older term in a stage occurs through a difference, whose amplitude beats N_k; the stage normalizers have summable amplitudes.

Before summing healed components, complete the action at the output label. If an old slot disappears, use the term formed from the history with the whole treated cluster erased. Superposition makes this a history of the construction. The erased history agrees with the original one wherever the latter is carried unchanged. In this way the completed action depends on the exterior data alone. The correction $V_{\mathfrak{X}}(Y)$ depends on the healed family $\mathfrak{X}$ only through its components inside Y, their histories and the label near Y.

Fix these exterior data. The remaining expression is

$$\sum_{\mathfrak{X}\in\mathfrak{F}} \prod_{C\in\mathfrak{X}} \mathbf{T}'_k(C) \exp \sum_Y V_{\mathfrak{X}}(Y).$$

The operations act also on the exponential. Each integrates the stored variables of its component, and $V(Y)$ reads those variables whenever the component lies in Y. The component measure is the same in every summand containing that component. If $\mathfrak{X} \subset \mathfrak{X}'$ and $\mathfrak{X}'(Y) = \mathfrak{X}(Y)$, then $V_{\mathfrak{X}'}(Y) = V_{\mathfrak{X}}(Y)$.

Factor out the empty family. The Mayer expansion then has a healed component in every nonempty polymer. Its mass supplies $e^{-(2-c)p_0(g_k)}$, while the remaining local factors cost at most exponentials of the component volume. The mass estimate below has already prepaid those costs. The connected expansion converges and gives one family, consistently on all restricted domains,

$$\log \mathsf{K} = \sum_Y \big(V_\emptyset(Y) + \mathbf{R}'_\sharp(Y)\big), \qquad |\mathbf{R}'_\sharp(Y)| \le C e^{-(1+c)p_0(g_k)} e^{-(1+\beta_{\mathrm{s}}/2)\kappa d_k^\sharp(\mathcal{S}';Y)}.$$

The same bound holds near standing islands. The part with no standing object in its domain agrees with the term formed at the never-large label and enters $R^{(k)}$; the rest is a boundary term.

The successive rates are

$$(1 + 4\beta_{\mathrm{s}})\kappa > (1 + 3\beta_{\mathrm{s}})\kappa > (1 + 2\beta_{\mathrm{s}})\kappa > (1 + \tfrac{1}{2}\beta_{\mathrm{s}})\kappa > \kappa.$$

Each resummation spends only its indicated gap. An older boundary constant enters a difference, a long domain, or the collar-volume estimate under the component integral. It therefore changes the final coupling ceiling, rather than the returned boundary constant.

Stored operations and the norm propagated by the induction. We give the mass argument with its integration variables visible. This also specifies the bound used in the preceding sum over healed components. Fix a frame (θ_j) and put

$$\mathsf{l}_j = \log \theta_j, \qquad g_j = \theta_j^{-1/2}, \qquad R_j = \check{R}_j, \qquad P_j = p_0(g_j) = A_0 \mathsf{l}_j^{\mathfrak{p}_0}. \tag{32.1}$$

The constants below are independent of the torus, cutoff and height of the history. They can depend on the geometric constants chosen earlier. Whenever a prescribed extra volume weight is introduced, its coefficient is fixed before the coupling ceiling is lowered.

For a component X treated at level i, write the operation after using its surviving averaging delta functions as

$$\mathbf{T}_i'(X)F(\mathbf{p}) = \frac{1}{|W_4|}\sum_{\omega}\int_{S_\omega} e^{\Phi_\omega(\xi;\mathbf{p})}F\,d\mu_\omega(\xi). \tag{32.2}$$

Here ω records the gauge-tree frame, all outcomes in the outer genealogy, and recursively a local history for each inner island. The variables ξ are the fields still free after a surviving delta function has been read as the definition $V_{j+1}(c) = \mathcal{M}(V_j)(c)$. Their measure is a product of Haar and Lebesgue measures. A Gaussian normalizer, gauge normalizer or elimination determinant occurring in the operation remains a factor of its integrand. It is not absorbed into a new probability measure.

The pair $\mathbf{p} = (\mathbf{U}, \mathbf{J})$ belongs to the complex domain on $X^{\sim}$ formed at level i. To read its exponent one may use an admitted splitting $\mathbf{U} = e^{i\eta_j A'}U$, with U real on the layer j. The action calculation above gives

$$\Phi_\omega(\xi;\mathbf{p}) = \Phi_\omega^\circ(\xi) - \tfrac{1}{2}q_X(\xi;U,A',\mathbf{J}) + V_X(\xi;U,A',\mathbf{J}), \qquad q_X \geq 0, \tag{32.3}$$

where $q_X = \|\phi^{1/2}D\mathsf{H}B\|^2$ is formed at the real glued background. The two last summands can depend separately on the splitting. Their sum, and hence the full exponent, depends only on the pair. The defining formula of the operation makes that exponent holomorphic on the pair domain. We apply Cauchy estimates to the complete integrand; the splitting-dependent terms are used for its modulus bound.

The part Φ_ω° contains the unexpanded inner Wilson factor

$$\exp\{-A(x_i''\phi_{\mathrm{c}}, U''(\xi))\}, \tag{32.4}$$

the stored Gaussian and gauge weights, their normalizers, chart densities, elimination factors, constants and the inner operations at their real reads. The background $U''(\xi)$ in this formula is the real inner background, obtained from the truncated prepared sequence and its stored innermost datum. It is distinct from the glued background used in q_X. Only the part outside the old core was expanded in obtaining (32.3). Thus the old large plaquettes remain in (32.4) with their full positive action.

The localization estimates for the main step and the earlier activities give, on every admitted splitting,

$$|V_X| \leq CN_0(\omega), \qquad \sup_{\mathbf{p}}|e^{\Phi_\omega(\xi;\mathbf{p})}| \leq e^{\Phi_\omega^\circ(\xi)+CN_0(\omega)}. \tag{32.5}$$

Here $N_0(\omega) = \sum_j \#_j(\omega)$ counts the stored sites at their own levels, including the assigned collars and the operation's fixed coordinate multiplicities. Write $N_0^\circ(\omega)$ for the same count in the outer record, before inserting its inner histories; thus $N_0(\omega) = N_0^\circ(\omega) + \sum_y N_0(\omega_y')$. The bound in (32.5) follows by sorting the localized terms into those confined to $X^{\sim}$, for which the rooted sum is a constant per stored site, and those that leave that collar, for which the boundary hull decay is summable. The response differences in the inner operations are included in V_X before this estimate is taken. In particular an inner operation is read at a real admissible pair in Φ_ω°.

The right side of (32.5) depends only on the component's own stored variables. That fact permits composition at much later backgrounds. Taking an integral at a fixed pair first would lose it: a pair at a later stage depends on the stored fields of other components. For an ordinary

component put

$$m_\omega(\xi) = 1_{S_\omega}(\xi) \sup_{\mathbf{p}} |e^{\Phi_\omega(\xi;\mathbf{p})}|,$$
$$m_\omega^{\mathbb{R}}(\xi) = 1_{S_\omega}(\xi) \sup_{\mathbf{p}\ \text{real}} e^{\Phi_\omega(\xi;\mathbf{p})},$$

where both suprema use the admitted pair domain. The mass is

$$\mathsf{M}_c(i,X) = \frac{1}{|W_4|} \sum_\omega e^{c\mathrm{v}^\natural(\omega)} \int^* m_\omega(\xi)\, d\mu_\omega(\xi). \tag{32.6}$$

The star denotes the infimum of the integrals of nonnegative Borel majorants. For an ordinary fixed finite-dimensional core the supremum is itself Borel: use continuity in the pair and a countable dense set, and then take a countable supremum of Borel functions of ξ. The upper integral also covers the varying evaluation fibres below. All products are estimated by explicitly constructed Borel majorants before Tonelli's theorem is applied.

The evaluations used by a drop-out. A drop-out has no new inner background and no new core Wilson factor. Its retained operation contains the tests, integrations, gauge factors, complete kept forms and elimination factors left when it stopped. The required action is supplied by the term carrying it. Transfer to its factor the Wilson weight on the bare plaquettes all of whose corners lie in X. This is an exact redistribution of factors in the same integrand.

For fixed ω, ξ the admissible evaluations form a set $\mathfrak{K}^\sharp(\omega,\xi)$. An evaluation includes a real field U on X satisfying

$$\mathcal{M}^j U(b) = V_j(\xi)(b) \quad \text{on every determining bond whose blocks lie in } X. \tag{32.7}$$

At the bare level these equations use the stored bare field. The field is regular layer by layer, as on the support of a history; each recorded test has its recorded value, except for the last test whose failure stopped this particular operation. Fields not supplied by ξ are read as the specified averages of U. On each inner island (m,Y) the pair $(U, J(U))|_{Y^\sim}$ belongs to the real domain formed at its own level. The evaluation also includes the kept complete forms and Jacobian factors, with their proved one-sided bounds, and a real Wilson weight υ with $\upsilon \geq c\theta_t$ on the action layer owning a plaquette of level t.

The drop-out version of (32.6) uses

$$\mathsf{E}_\omega(\xi;\mathfrak{e}) = e^{-A(\upsilon 1_X, U)} \mathsf{K}_\omega(\xi;\mathfrak{e}), \qquad \sup_{\mathfrak{e}\in\mathfrak{K}^\sharp(\omega,\xi)} \mathsf{E}_\omega(\xi;\mathfrak{e}) \tag{32.8}$$

in place of the pair supremum: for a drop-out both m_ω and $m_\omega^{\mathbb{R}}$ denote this fibre supremum multiplied by 1_{S_ω}. Its value is zero when the fibre is empty. The fibre depends on ξ through (32.7) and its tests. In particular a flat field cannot be chosen independently of stored nontrivial averages. The Wilson field in (32.8) stays real. Complexifying it would destroy the positive energy comparison used here.

There are two uses of the last failed test. It is excluded from the mass of the operation that it stops. When this drop-out occurs inside a later component, its record includes that test, and its failure supplies the gain for the renewed wait. The latter evaluation fibre is a restriction of the former one. No root gain is charged in the earlier mass and then charged again in the later one. If a particular operation carries no gauge-tree frame, insert $1 = |W_4|^{-1} \sum_{r\in W_4} 1$; this keeps the same normalization in (32.6) for both kinds of mark.

These domains are fixed at the birth level. To see that every later read lies in them, use the support estimates, in local level-j units,

$$|\partial U_k - 1| \leq C R_j^2 \varepsilon_j \eta_j^2, \qquad |J(U_k)| \leq C R_j^2 \varepsilon_j, \qquad C R_j^2 \varepsilon_j < \alpha_{0,j}. \tag{32.9}$$

The last inequality follows from $q > \mathfrak{p}_0 + 2r$ after the coupling ceiling is lowered. It has no loss in $k - j$. Influences across layers at index distance a carry $e^{-(a-1)_+}$, whereas the threshold comparison costs at most $(1 + a)^{1/2}$. Thus their sum is bounded. The action sequence gives the retained Wilson weight, and the determining sequence gives its compatible field. Using either sequence in both places would not prove (32.9) for the actual operation.

The action seen by the tests. We recall why failed tests of averages can be paid for by the action of any compatible real field. This is needed for the whole drop-out fibre, not only for one minimizing background. For a real field W put $f_W(p) = |\partial W(p) - 1|$, using the matrix operator norm. In SU(2),

$$f_W(p)^2 = 2a_p(W). \tag{32.10}$$

On the four blocks at the corners of a coarse plaquette, a comb gauge and the discrete Stokes formula give

$$f_{\mathcal{M}W} \le \mathsf{P} f_W + \mathsf{r}(W), \qquad \|\mathsf{r}(W)\|_1 \le C\|f_W\|_2^2. \tag{32.11}$$

This formula is used while the plaquettes in those blocks lie in the fixed averaging chart. The matrix P is nonnegative. Its row sums are L^2 and its column sums are L^{-2} in four dimensions. The row count is the area of a fine spanning surface; the column count includes the L^{-4} average over block origins. For j successive levels write $\mathsf{P}^{[j]} = \mathsf{P}_{j-1} \cdots \mathsf{P}_0$, with each factor acting from its fine plaquette grid to the next coarse grid. Each factor has these same row and column bounds. Thus Schur's inequality gives

$$\|\mathsf{P}\|_{2\to 2} \le 1, \qquad \|\mathsf{P}\|_{1\to 1} \le L^{-2}, \qquad \|\mathsf{P}^{[j]} h\|_2 \le L^{-2j}\|h\|_1. \tag{32.12}$$

In the last inequality use $\|v\|_2 \le \|v\|_1$ after applying the column bound j times. Restrictions to the relevant ancestor blocks can only decrease these positive row and column sums.

Here is the uniform bootstrap. Let E_0 be the sum of f_U^2 over those ancestor blocks. Iterating (32.11) along them and using (32.12) gives

$$\|f_{\mathcal{M}^t U}\|_2 \le E_0^{1/2} + C \sum_{l<t} L^{-2(t-1-l)} E_l, \qquad E_l = \|f_{\mathcal{M}^l U}\|_2^2. \tag{32.13}$$

Suppose successively that $E_l \le 4E_0$ for $l < t$. The sum in (32.13) is then at most $4CE_0/(1 - L^{-2})$. For E_0 below a fixed threshold it is at most $E_0^{1/2}/2$. The bound $E_t \le 4E_0$ follows with room to spare. The same threshold keeps every intervening plaquette inside its averaging chart, since its size is at most $E_l^{1/2}$. This starts at level zero and proves the assertion at every finite depth. Retaining the sharper right side also gives $E_t^{1/2} \le E_0^{1/2} + C'E_0$.

Consequently, for a set S of tested plaquettes of level t,

$$\sum_{p \in \mathcal{N}_0(S)} a_p(U) \ge \min\left\{ c \sum_{p' \in S} |\partial \mathcal{M}^t U(p') - 1|^2, e_* \right\}. \tag{32.14}$$

Here $\mathcal{N}_0(S)$ is the union of the ancestor corner blocks and $c, e_* > 0$ depend on L, not on t. If the left side is below e_*, the bootstrap proves the first bound; otherwise the second alternative applies. No regularity of U was assumed. The small-energy alternative proves the regularity required to apply the averaging charts. This is what makes the bound usable at every compatible evaluation in (32.8).

For a failed plaquette test at step t, linear regularity first shows that some nearby plaquette of V_t has size $c\varepsilon_t$. Otherwise the localized background exists and passes the test. Equation (32.14) therefore supplies $c\varepsilon_t^2$ from the compatible field's action. If the approximate fluctuation test fails,

the corresponding estimate is

$$A_{\mathfrak{c}^\sim}(U) + \mathrm{GF}_{t-1}(\mathfrak{c}^\sim; V_{t-1}) \geq c\delta_{t-1}^2. \tag{32.15}$$

Indeed, smallness of its left side makes the coarse plaquettes and the contour gauge variables $O(\sqrt{\sigma})$. One-step rigidity, using the average and that gauge, then bounds the difference from the localized background by $C\sqrt{\sigma}$. This contradicts failure when $\sigma < c\delta_{t-1}^2$. The gauge exponent is essential here: a gauge transformation inside a block can move its fine field while preserving both its Wilson action and its block average.

For a failed fluctuation, collar or rim test, the stored Gaussian supplies the energy. Choose $\kappa_{\mathrm{pos}} > 0$ below both the uniform real lower bound for the complete precision (25.2) and the lower bound for its kept conditional form in (2b), in the stored scaled coordinates. If a is the stored part and B the part already integrated, the complete conditional form satisfies

$$\langle a, \Delta_{\Upsilon\prime} a\rangle = \min_B \langle a + B, \mathsf{A}(a + B)\rangle \geq \kappa_{\mathrm{pos}} \|a\|^2. \tag{32.16}$$

The free coordinate groups are orthogonal, so the last inequality follows from $\mathsf{A} \geq \kappa_{\mathrm{pos}}$. The actual kept form has the stated lower bound by the separate cut-form comparison in (2b). A failed test forces a stored scaled coordinate of size at least $c\delta_{t-1}/g_{t-1}$, with the stated fixed collar or rim fraction included in c. On the cut-off strip the shift vanishes when the integration variable is small. Thus the same implication holds where the tested logarithm is not defined: a small integration variable would put that logarithm in its chart and make the test pass. Multiplication by θ_{t-1} in (32.15), or insertion in (32.16), gives a gain cP_t^2 for a created cell. The energy pieces used by distinct certificates are assigned on the cell neighborhoods fixed in (2a). Their bounded overlaps are included in c before the gains are summed.

A new mark pays outside the old floor. Let a component be treated at level m with preparation bottom $m_\flat$. Its old floor consists of the level-$m_\flat$ cells in $\Omega_{m_\flat}^c \cap X$ that do not touch $\Omega_{m_\flat}$. The plaquettes spent before the wait lie in this floor. The gauge and stored Gaussian factors spent then have levels below $m_\flat$. The waiting collar separates the floor from the part on which the new operation acts.

We check a renewed test by contraposition. Suppose the sum of the unspent Wilson, gauge and stored Gaussian energies in the new window is below $c\mathsf{l}_m^{2\mathfrak{p}_1 - 8r}$. The energy of averages and the gauge bounds make all corresponding stored plaquettes and contour variables at most

$$\varsigma = C g_m \mathsf{l}_m^{\mathfrak{p}_1 - 4r}. \tag{32.17}$$

The unknown field in the floor does not enter this estimate. For a stage background $\widehat{U}$, the background locality estimate through the waiting collar gives, outside the floor,

$$|\partial \widehat{U}(p) - 1| \leq C\big(\varsigma + R_m^2 \varepsilon_m e^{-\delta_0 M R_m}\big) L^{-2j(p)}. \tag{32.18}$$

The index $j(p)$ is the local layer of p. The second term is smaller than the first after the coupling ceiling is lowered. This uses a collar of fixed positive width in its own local units and the weighted-distance background estimate. It does not require a small field inside the floor.

The most restrictive bond test is the main-step test on V'. On a shell between two boxes, with side at most CMR_m, a loop of n straight pieces has holonomy within $Cns^2\varpi$ of the identity if the shell plaquettes are within ϖ. To obtain this estimate choose the shell reference paths, move the loop across plaquettes in the shell, and use telescoping products of unitary matrices. The required spanning surfaces have at most Cns^2 plaquettes. The inner box sits strictly inside the outer one in every direction, so the paths and surfaces remain in the shell. The same reference gauge extends a one-site-thick shell into the inner box by setting its new bonds to the identity and transforming back; the resulting plaquettes have size at most $Cs^2\varpi$.

Apply this to the averages of the compatible field and the basic background. They agree on the gauge tree and outside the central box. The tested $V'(b)$ is conjugate to the quotient of their

two loop holonomies, and hence

$$|\tfrac{1}{i}\log V'(b)| \le CM^2R_m^2\varsigma < \delta'_m. \tag{32.19}$$

Two powers of R_m pay for the shell area, and the other two in (32.17) absorb its fixed constant. The other stage tests use one-step rigidity or the same shell comparison with a smaller loss. Existence and extension tests also pass: small exterior plaquettes give the extension just described, and the localized background theorem then supplies their background and logarithm charts. Thus failure of any test provides, outside the floor, an energy witness of size at least

$$c\mathsf{l}_m^{2\mathfrak{p}_1-8r} \ge P_m. \tag{32.20}$$

The exponent hierarchy $2\mathfrak{p}_1 - 8r > \mathfrak{p}_0$ gives the last inequality at sufficiently small coupling.

For this root witness the compatible field may be needed on X^+, two unit layers beyond X. We do not claim the witness in the mass whose evaluation only supplies a field on X. It is used in the later containing component, where that field and the last failed test are present. Nested operation windows are disjoint. Their gauge and Gaussian witnesses have different levels, and their Wilson witnesses avoid the earlier floors. This verifies the allocation needed to add the gains without spending the same action twice.

A volume with bounded multiplicity. A node $u = (n, C)$ is a component of the region made at level n. An unhealed node has one successor, and the predecessors of a node are disjoint. Let $\mathcal{G}(x)$ be the genealogy below $x = (i, X)$, and cut it at the inner islands to form $\mathcal{G}^\circ(x)$. The cut nodes are leaves of the outer tree and roots of their own inner trees. Define

$$\begin{aligned} \mathrm{vol}(u) &= \mathsf{l}_{n(u)}(MR_{n(u)})^4|C(u)|_{n(u)}, \\ \mathsf{V}(x) &= \sum_{u\in\mathcal{G}(x)} \mathrm{vol}(u), \qquad \mathsf{V}^\circ(x) = \sum_{u\in\mathcal{G}^\circ(x)} \mathrm{vol}(u), \\ \mathsf{v}^\natural(x) &= \mathsf{V}^\circ(x) + \sum_{y \text{ inner}} \mathsf{v}^\natural(y). \end{aligned} \tag{32.21}$$

Expanding this finite recursion counts every uncut node once. A non-root clearing node occurs twice, as the leaf of the outer tree and as the root of its inner tree. No third outer tree meets it. Consequently

$$\mathsf{V}(x) \le \mathsf{v}^\natural(x) \le 2\mathsf{V}(x). \tag{32.22}$$

Keeping a drop-out does not make another copy of all its older nodes. A component healed earlier and covered again later has its own root; its old genealogy is not inserted in the new one merely because their physical locations overlap.

The site count without logarithms, with these same multiplicities, is at most $\mathsf{v}^\natural/\mathsf{l}_{\min}$. A fixed coarse-$M$ read collar multiplies the site count of an owning MR_j-cell by at most $(1 + C/R_j)^4$. At large R_j this is at most two. Overlaps only decrease the union count. Thus the number $n^\sharp$ of stored and owned read sites satisfies

$$n^\sharp(\omega) \le \frac{2}{\mathsf{l}_{\min}}\mathsf{v}^\natural(\omega). \tag{32.23}$$

Bond directions and color multiplicities are included in the constant of a per-site energy estimate. The same ownership shows that the part of a label still visible below a drop-out obeys

$$\mathsf{V}_{\mathrm{in}}(i, X; \mathcal{S}) := \sum_{j<i} \mathsf{l}_j \#_j(Z_j^{\mathrm{a}} \cap X) \le \mathsf{v}^\natural(\omega_X). \tag{32.24}$$

The regions on the left are action regions. The determining regions may have redone some of these levels and cannot replace them in this count.

Contraction and the count over generations. The geometric count is the step that converts the individual event gains into a uniform mass. Let $O(u)$ consist of the images, in the cell grid of u, of its earlier creation cells. The actual hull, enlargement and bounded-size boxing rules give

$$O(u) \subset C(u) \subset O(u)^{[e]}, \qquad O(u) \text{ is } h_0\text{-connected}, \qquad |C(u)|_{n(u)} \le C_e |O(u)|_{n(u)}. \tag{32.25}$$

All constants are fixed. The layers added at earlier levels have a bounded total reach in current cell units: every two levels contract the earlier reach by L^{-1}, since size drops are not consecutive. The sum is a geometric series. Boxing affects only components of bounded side, and adds a further fixed reach. Mergers take unions of these sets with touching collars. These facts prove (32.25); a large rectangular hull of an arbitrary thin component is not taken.

For an h_0-connected set A with N cells, its image under a coarsening of ratio q has at most

$$C_0(1 + h_0 N/q) \tag{32.26}$$

cells, including a fixed coarse collar. Traverse a spanning tree of A in both directions, joining neighboring cell centers. The walk has length at most $2h_0(N-1)$. Divide it into pieces of length at most q. Each piece meets a fixed number of coarse cells in four dimensions. Their number is at most $1 + 2h_0 N/q$, proving (32.26) with its collar.

Choose a fixed number s_0 of levels so that their net coarsening makes the coefficient of N at most $1/4$, and choose $\nu_* \ge 4C_0$. A node with at most ν_* occupied cells is called light. A heavy component propagated for s_0 levels without new material loses at least half its occupied cells. A light component contributes only a fixed number. If o_n is the total occupied count of a genealogy at level n, l_n its number of light nodes, and n_t^{cr} its number of created cells, we obtain

$$o_n \le \tfrac{1}{2} o_{n-s_0} + C_1 l_{n-s_0} + C_1 \sum_{n-s_0 < t \le n} n_t^{\mathrm{cr}}. \tag{32.27}$$

Unions at a merger only decrease this count. At a cut through an inner island (m, Y), the last ordinary plaquette test χ_m fails at threshold ε_m. The energy-of-averages estimate supplies cP_m^2 from the outer component's retained core Wilson action or, for a containing drop-out, from its carrying-term Wilson weight at every compatible evaluation. The plaquettes have not been spent on another witness. This test is excluded from the mass of Y. The uniformly bounded number $|Y|_m \le N_*$ of cut cells is counted as creations. Choose c_G small and the floor large so that $N_* G_m + P_m \le c_{\mathrm{w}} P_m^2$, where c_{w} is the reserved witness constant. This pays the cut cells and any mark credit from this single witness. Keeping drop-outs are not cut: their renewed marks retain the credit P_m already assigned above.

For any nonnegative non-increasing sequence r_n, multiply (32.27) by r_n and sum over the finite levels. The shifted first sum is at most half of $\sum_n r_n o_n$. A creation occurs in at most s_0 of the last summands, with weights at most its birth weight. Thus

$$\sum_n r_n o_n \le 2C_1 \sum_n r_n l_n + 2C_1 s_0 \sum_t r_t n_t^{\mathrm{cr}}. \tag{32.28}$$

No term proportional to the starting height occurs.

It remains to count light nodes. A quiet chain has one predecessor at each step, no new creation, and no preceding mark. Its side lengths satisfy

$$\alpha_{n+1} \le \alpha_n + A, \qquad \alpha_{n+2} \le L^{-1}\alpha_n + 2A. \tag{32.29}$$

The fixed point $2A/(1 - L^{-1})$ lies strictly below the small-component threshold. A light node has bounded diameter by (32.25); after a fixed number b_0 of quiet levels it is therefore a single small box. After another preparation depth, its window lies wholly after the event that began the chain. The depth, quietness, quarantine and single-box trace conditions now hold, and the

dynamic class includes the component. It cannot be withheld. The number of light nodes on a chain starting at level t is consequently at most

$$T_t = b_0 + \overline{N}_t + 3 \le C_T R_t, \tag{32.30}$$

where T_t has a non-increasing choice. Heavy nodes before this light segment have already been paid for in (32.28).

A quiet chain starts at a creation, after a mark, or at a merger. The first two are charged to their creation or mark. For a merger choose a non-first predecessor and descend through first predecessors until reaching a leaf. Different mergers receive different leaves. For incomparable mergers this follows from disjointness of their predecessor trees. For comparable mergers, a first-predecessor descent from the higher merger cannot take the non-first branch selected at the lower merger. Each leaf contains a created cell. A creation pays once for its own chain and at most once for an assigned merger.

Since $T_t r_t$ is non-increasing, payment at the earlier assigned event only increases the available weight. Combining this assignment with (32.28) gives

$$\sum_{u \in \mathcal{G}(x)} r_{n(u)} |C(u)|_{n(u)} \le C_2 \sum_t T_t r_t \big(n_t^{\mathrm{cr}} + n_t^{\mathrm{mk}}\big). \tag{32.31}$$

The root's own mark is excluded from n_t^{mk}, since it has not yet begun a renewed wait. On a torus too small for a required small-box collar to embed, a bounded periodic cell side leaves only a bounded number of admissible further coarsenings. Increasing the fixed part of T_t includes these terminal levels. This alternative uses no wrapped-box background.

The scalar reserve. Let c_t be the complete cost per level-t cell, including the normalizer, local energy, counting and prescribed volume weight prices. A non-increasing majorant is

$$c_t \le r_t = C_3 c_1 M^4 R_t^5. \tag{32.32}$$

Indeed the cost before this enlargement is $c_1 \mathsf{l}_t (M R_t)^4$, and $\mathsf{l}_t \le C R_t$. Use a creation credit $G_t = c_G P_t^2$ and a conservative mark credit P_t. Equation (32.31) shows that the balance

$$\mathfrak{F}(x) = \sum_t G_t n_t^{\mathrm{cr}} + \sum_t P_t n_t^{\mathrm{mk}} - \sum_{u \in \mathcal{G}(x)} c_{n(u)} |C(u)|_{n(u)} \tag{32.33}$$

satisfies

$$\mathfrak{F}(x) \ge 2 \sum_t P_t n_t^{\mathrm{cr}} \quad \text{if} \quad G_t \ge 2P_t + C_2 T_t r_t, \qquad P_t \ge C_2 T_t r_t. \tag{32.34}$$

The mark can spend its conservative credit on the new wait; the remaining creation credit supplies the small mass.

Here the hierarchy is effective. Since $T_t \le C R_t$ and $R_t \asymp \mathsf{l}_t^r$, the cost on the right is at most $C c_1 M^4 \mathsf{l}_t^{6r}$. Take $\mathfrak{p}_0 \ge 6r + 1$, as permitted by the exponent order. Dividing by P_t gives a bound $C' c_1 \mathsf{l}_t^{-(\mathfrak{p}_0 - 6r)}$. Therefore a floor

$$\mathsf{l}_t \ge K(1 + c_1), \qquad c_G A_0 \mathsf{l}_t \ge 4 \tag{32.35}$$

with sufficiently large fixed K implies both inequalities in (32.34). These conditions are one ceiling on the coupling, imposed after c_1. They are uniform over all levels. In particular the coefficient of an additional fixed volume weight only changes the floor by an affine bound; no height-dependent ceiling is needed.

Every nonempty genealogy has a creation leaf. The bounded-error monotonicity of the frame gives, for $t \le i$, $P_t \ge (1 + \beta_0)^{-1} P_i$ at a sufficiently high common floor. Thus (32.34) leaves at least $2(1 + \beta_0)^{-1} P_i$. For an unfinished unmarked root $z = (j, Z)$ let S denote one step of deterministic quiet propagation of its cell set. Continue this geometry for q levels, stopping strictly before a

new creation, merger or completed wait, and at the terminal level if necessary. Only the cell costs change. Applying (32.34) to the extended genealogy, whose first creation level is t_0, gives

$$\mathfrak{F}(z) - 2P_{t_0} \geq \sum_{n=j+1}^{j+q} c_n |S^{n-j} Z|_n \geq 0.$$

This is the credit for the future quiet evolution of a live component. It uses neither a new test nor the admission of a new event.

The entropy of the literal histories. The count must include the level and position of an earlier event. Make a graph of cell addresses together with their levels. Join neighboring cells at the same level and each cell to its parent at the next. Its degree is at most

$$D = L^4 + 3^4 + 2. \tag{32.36}$$

At a size drop a parent has one child; otherwise it has at most L^4. The active addresses of an outer component form a connected set rooted at a fixed cell of X. A connected set of N vertices through that root has at most $D^{2(N-1)}$ choices: order the neighbors, choose its canonical spanning tree and follow the depth-first walk that traverses every tree edge twice. The visited set recovers the original set. Downward paths record earlier birth levels and positions, so neither the height nor the torus volume supplies another factor.

For a fixed active set the remaining code has at most $e^{b\mathsf{V}^\circ}$ choices. At level j there are $O((MR_j)^4)$ local sites and bonds per cell, with a fixed finite list of test outcomes at each. The stage position and failure type have at most a fixed power of $R_j + 1$ choices; their logarithm is $O(\mathsf{l}_j)$. Stored variable types, seam factors and class entrances use the same local addresses. The normalized frame sums contribute one, including the root frame and the frames inside the outer record. Since $\mathsf{V}^\circ \geq N$, choosing

$$c_{\mathrm{e}} \geq b + 2\log D + \log 2 \tag{32.37}$$

gives

$$\sum_{\omega^\circ} |W_4|^{-1-f(\omega^\circ)} e^{-c_{\mathrm{e}} \mathsf{V}^\circ(\omega^\circ)} \leq \sum_{N \geq 1} D^{2(N-1)} e^{-(c_{\mathrm{e}} - b)N} \leq 1. \tag{32.38}$$

The integer f counts the additional outer frames. The component is nonempty, which explains the starting value $N = 1$. The identity operation for an empty component has no rare-mass assertion. The bound by one, rather than by an unspecified constant, is useful in the induction.

The common envelope and the mass induction. We now combine the field estimates with this count. First bound every nonquadratic term of the complete integrand on the support by its proved cost per owned site. This includes the boundary terms involving stored fluctuation fields. Divide the positive action, gauge and Gaussian exponents in fixed proportions. After decreasing the fixed c_G and raising the coupling floor, the witness share supplies a credit $\Gamma_e = G_t = c_G P_t^2$ for each outer creation at level t, and $\Gamma_e = P_t$ for each outer renewed mark. The root's own mark is excluded, as in (32.33). The creation and mark certificates above prove these bounds on their owned pieces. A separate fixed share supplies the integrable reference envelope; it is not spent again on the event gains.

The scalar normalizers have the appropriate local prices. For the truncated gauge factor,

$$z(\theta, \epsilon) = \int_G 1_{\{|u-1|<\epsilon\}} e^{-\theta(1 - \frac{1}{2}\operatorname{tr} u)}\, du, \quad c\theta^{-3/2} \leq z(\theta, \epsilon) \leq C\theta^{-3/2} \tag{32.39}$$

when $\epsilon \leq 1$ and $\theta\epsilon^2 \geq 1$. Integrating a Lie-algebra ball of radius $c\theta^{-1/2}$ proves the lower bound; the radial Haar density and the quadratic lower bound on $1 - \frac{1}{2}\operatorname{tr} u$ prove the upper bound. The same upper bound with $\theta/2$ shows that half the exponent integrates against the original normalizer at a bounded cost. An elimination determinant and its chart Jacobian cost a fixed factor per owning

coordinate; their inverse determinants are those of the actual pivot charts. Rescaling g_j and the remaining normalizers cost at most $C\mathsf{l}_j$ per level-j site.

For stored Gaussian blocks, use the complete kept forms and the actual free coordinates of the constraint chart. Their coercivity and the local reference-coordinate comparison in (2b) leave an envelope g_{ω° whose conditional integral satisfies

$$\sup_{\xi_{\mathrm{in}}} \int g_{\omega^\circ}(\xi^\circ;\xi_{\mathrm{in}})\, d\mu^\circ(\xi^\circ) \le e^{C_5 \mathsf{V}^\circ}. \tag{32.40}$$

One can see the local cost directly in the triangular stored-coordinate representation. At fixed later variables replace each earlier free block b_j by its actual shifted coordinate A_j. The inverse Jacobian is bounded by $J_0^{d_j}$ on that chart. A translation has $J_0 = 1$. Dropping restrictions then gives the full Euclidean Gaussian integral,

$$\int e^{-\beta \sum_j |A_j|^2}\, d\mu^\circ \le J_0^{\sum d_j} (\pi/\beta)^{\frac{1}{2}\sum d_j} \le e^{C_5 \mathsf{V}^\circ}. \tag{32.41}$$

The dimension count is the count of stored scalar coordinates, not the number of all exterior coordinates used to construct their covariance. This retains uniformity in the exterior volume. For a localized marginal the same estimate follows from its local Gaussian comparison. In both forms it is the actual complete conditional form, including its compensating gauge potential, that is bounded. A flat covariance with the same formal name would not establish (32.40).

At every inner island replace the real admissible read by its strong real majorant. With the scalar and local energy prices just listed, the complete outer factor is bounded by

$$\begin{aligned} m_\omega(\xi) &\le |W_4|^{-f(\omega^\circ)} e^{C_7 \mathsf{V}^\circ - \sum_e \Gamma_e} g_{\omega^\circ}(\xi^\circ;\xi_{\mathrm{in}}) \prod_y \frac{m^{\mathbb{R}}_{\omega'_y}(\xi'_y)}{|W_4|}, \\ \sup_{\xi_{\mathrm{in}}} \int g_{\omega^\circ}\, d\mu^\circ &\le e^{C_5 \mathsf{V}^\circ}. \end{aligned} \tag{32.42}$$

For a drop-out the identical envelope holds at every evaluation in its fibre. Its compatible fields obey the same certificates (32.14)– (32.20); its complete kept forms and Jacobians obey the same one-sided bounds. The envelope is therefore formed before the fibre supremum. Replacing the inner suprema by nonnegative Borel majorants makes its right side Borel on the full product of stored coordinates. The upper integrals permit these majorants to be chosen with arbitrarily small integral excess. Tonelli is applied to those majorants, without asserting that a varying-fibre supremum is Borel. The integral excesses are sent to zero after the outer integration. This is stronger than separate integral bounds at each exterior argument.

Fix ω°. Its inner islands, domains and levels are now fixed, and their local histories can be summed independently. Multiply (32.42) by $e^{c\mathsf{v}^\natural}$. Additivity in (32.21) and Tonelli allow the outer variables to be integrated first. The remaining integrals and history sums are the earlier strong real masses, including their own frame normalizations. If these masses are at most one, the result is

$$\frac{1}{|W_4|} \sum_{\omega \text{ over } \omega^\circ} e^{c\mathsf{v}^\natural(\omega)} \int^* m_\omega\, d\mu_\omega \le |W_4|^{-1-f(\omega^\circ)} e^{(C_5+C_7+c)\mathsf{V}^\circ - \sum_e \Gamma_e}. \tag{32.43}$$

No measurable maximizing pair or maximizing evaluation has been selected. The supremum has not been moved through an integral.

Apply the scalar reserve with $c_1 = C_5 + C_7 + c + c_{\mathrm{e}}$, enlarged for its fixed coordinate multiplicities. The cut leaves' new tests supply their outer creation witnesses, while their old histories have just

been integrated in their own masses. The reserve therefore gives

$$\sum_{e} \Gamma_e - (C_5 + C_7 + c + c_{\mathrm{e}})\mathsf{V}^{\circ} \geq 2 \sum_{t} P_t n_t^{\mathrm{cr}}. \tag{32.44}$$

There is a creation leaf. Comparing its level with i leaves $\mathsf{a}_* P_i$, where $\mathsf{a}_* = 2(1+\beta_0)^{-1} = 2 - c'$. Sum (32.43) over its outer records and use (32.38). We obtain

$$\mathsf{M}_c(i, X) \leq e^{-\mathsf{a}_* P_i}. \tag{32.45}$$

This proves the complex-pair bound from the earlier strong real bounds. Restricting the supremum to real pairs returns the induction hypothesis at level i. At the first level the product of inner masses is empty. The drop-out estimate uses the same proof and needs no additional inductive drop-out hypothesis. Constants are returned unchanged at every level.

Restriction and the healed histories. We also justify the history product used before the mass estimate. Include in the read region of a move every stored variable, determining datum, action plaquette, gauge factor and reference normalizer that it uses. A localized background formed on a box reads that box; the short path of one of its kernel terms does not reduce this read region. Class membership reads the entire preparation window and its quarantine marks.

If two clusters never touch, their cell margin exceeds the fixed reach of every test, pivot and collar. Hulls and enlargement commute with their union, and boxing acts on its separate components. For the preparation window there is one further point. If an event created at level s lies in the window of a later level t, it has not healed at an intermediate $i < t$. Such healing would require $s \leq i - \mathsf{N}_i$. Since $\mathsf{N}_t = \check{R}_t$ is non-increasing, however,

$$i - \mathsf{N}_i < t - \mathsf{N}_t \leq s. \tag{32.46}$$

For a mark the first inequality is strict at the lower endpoint as well. Thus the reads that decide the class are still present in their own cluster. The actual class contains all eligible components; there is no global choice withholding one of two disjoint eligible components.

Order the moves by their production, reading and deletion of stored slots, including the just described class reads. Independent moves have disjoint read regions, product reference measures and disjoint output substitutions. Positive kernels commute by Tonelli; their admitted complex continuations commute by Fubini using (32.42). A test is unchanged by a move that changes none of its inputs. Swapping adjacent independent moves therefore preserves the whole operation. Induction along this order, using the hereditary background identities, shows that restriction to a union of full clusters produces those same local moves and their same normalizers.

Erasing healed clusters is consequently an actual history, and inserting finitely many compatible clusters recovers an actual history with the same standing part. Connected components of the erased record give the inverse correspondence. At fixed exterior data its local operations are exactly (32.2). This proves the product and sum used above to form $\mathbf{R}'_{\sharp}$, including their measures. It supplies no global pointwise gas over arbitrary final fields; the remaining standing operations must still be integrated and counted.

Action shares and the loaded label sum. For the latter count first cut a history at its outermost marks. Above the cuts, a piece (t, D) is a component made at step t on which no $\mathbf{R}$ has acted. It retains its one-step variables $\xi_{t,D}$ and their gauge, Gaussian and elimination factors $\mathsf{w}_{t,D}$. Assign to it the bare plaquettes within one cell of its failed plaquette and approximate fluctuation tests. Write this set as $\mathcal{P}(N_{t,D})$.

These sets are disjoint. A piece untouched by $\mathbf{R}$ stays in the live region at every later step, and new tests are made only beyond the prescribed margin of that older region. No outermost mark meets these action shares. On a share the older profiles have completed their transition and the newer ones have not begun theirs, so its action weight is exactly x_{t-1}. Extending U_k by the

stored bare field gives

$$A(x_k(\cdot),U_k)\ge\sum_{(t,D)}x_{t-1}\sum_{p\in\mathcal{P}(N_{t,D})}a_p(U_k)\ge 0. \tag{32.47}$$

At these plaquettes the background constraints and the surviving averaging delta functions give the same averages as the tested fields. Hence the all-compatible field estimates apply to this U_k as well. After reserving the event payments and integrating the remaining half-exponents, one obtains Borel functions of each piece's own variables with

$$\begin{aligned} e^{-A_{\mathrm{cr}}}\prod_{(t,D)}\mathsf{w}_{t,D}&\le e^{-c\sum_{(t,\mathfrak{c})\text{ created}}P_t^2}\prod_{(t,D)}\overline{\mathsf{w}}_{t,D}(\xi_{t,D}),\\ \int\overline{\mathsf{w}}_{t,D}\,dm_{t,D}&\le e^{C\#_t(D)}. \end{aligned} \tag{32.48}$$

The Gaussian integration uses the Schur minimum in (32.16). The gauge normalizer uses half its own exponent as in (32.39). The resulting C has no $\log\theta_t$ factor. All logarithmic prices will enter with the vacuum constants.

For a marked component replace its actual later read by the majorant in (32.6). The groups $\xi_{t,D}$ and ξ_X are disjoint. Their majorants therefore multiply by Tonelli, even though U_k depends on all of them. Summing the local histories at the marks produces their masses. The restrictions of histories that give the same label have at most a fixed exponential number of choices per stored site; that is the local code already counted in (32.38). We do not identify a label with one prescribed set of failed cells.

Put $N_k=|\mathbb{T}^{(k)}|$ and define the required load

$$\mathscr{L}_c(\mathcal{S})=c\sum_{n\le k}|\Gamma_n|+\sum_{j\le k}(C_{\log}\log\theta_{j-1}+C_{\mathrm{loc}}+c)\#_j(Z_j^+). \tag{32.49}$$

The first term counts determining bonds; Z_j^+ is the collared action defect. Assign each bond to an endpoint in its determining layer, and then to that site's owned cell. There are only finitely many incident bond directions. For an unchanged layer, charge the resulting site address to its action cell; for a redone layer, charge it to the local history of the mark that owns the redone slot, at its original level. A one-step conversion costs at most L^4, and collars have bounded multiplicity. For nested marks the inner record already accounts for the earlier history. Thus each owned level-site address receives a bounded number of charges, independently of the number of subsequent reads. The fixed incidence and collar multiplicities are included in the cell price and in $c^\sharp(c)$ below. Equations (32.22)– (32.24) show that the load inside a mark is paid by $e^{(c^\sharp-c)\mathsf{v}^\natural}$ for a fixed affine $c^\sharp(c)>c$.

The vacuum constants have the same assignment. The regular one-step number on the full torus is a sum of one common local number over its sites. Subtracting the label's number cancels the sites where that label still uses it. Only the missing or redone sites remain. The scalar gauge and Gaussian normalizers bound each by $C_{\log}\log\theta_{j-1}+C_{\mathrm{loc}}$. Stage numbers are assigned to their own integration sites. The cancellation uses the actual same local number, with its actual sign, on both lists; bounds for unrelated volume constants would not suffice.

After this normalization the total unmarked local energy is at most $C_{\mathrm{sf}}N_k$ plus the assigned defect prices. The summation over old levels uses the regular member's own vacuum and quadratic subtraction, as in the earlier pointed cancellation; it is not a sum of a constant times every fine-lattice volume. The unspent real coarse Wilson factor is at most one and final Haar measure is normalized. Inside outermost marks use (32.45) at weight $c^\sharp$. Above them use (32.48) and the scalar reserve. For unfinished live trees use their future-wait credit after (32.35).

It remains to sum the positive records. Root every carried component at a cell of the fixed M-partition of the last lattice. There are at most N_k/M^4 such addresses. Earlier healed components are already in the local energies and supply no additional final roots. After adding the load price to c_1, the connected-set code and (32.38) give integrated weights $a_\gamma \geq 0$ with

$$\sum_{\gamma \text{ rooted at } z} a_\gamma \leq 1. \tag{32.50}$$

Dropping compatibility from a positive sum then gives

$$\sum_{\mathfrak{F} \text{ allowed}} \prod_{\gamma \in \mathfrak{F}} a_\gamma \leq \prod_{\gamma} (1 + a_\gamma) \leq \exp\left(\sum_{\gamma} a_\gamma \right) \leq e^{N_k/M^4}. \tag{32.51}$$

This also includes all local outcome codes leading to a given label. For any finite cutoff the histories are finite; increasing finite partial sums give the same positive bound if a countable local representation is used.

Combining the estimates gives

$$\sum_{\mathcal{S}} |e^{\mathcal{E}_k(\mathcal{S})}| \tau^{\mathrm{abs}}_{\mathcal{S},c}(\mathcal{V}_k) e^{\mathscr{L}_c(\mathcal{S})} \leq e^{C(c)N_k}. \tag{32.52}$$

Here each outer retained operation in τ^{abs} has had its integrand replaced by its modulus times $e^{c\mathsf{v}^\natural}$. The constant depends on c and the earlier construction parameters, and not on the cutoff, torus or last coupling. The surplus mass weight pays the lower levels still visible inside a drop-out after their creation gains have already been spent in its mass.

The source excess uses the same reserve. The ordinary complex source keeps all sharp tests, reference measures and background choices at their unsourced values. It changes the complete exponent by an excess Ψ_ω. At a main operation its new part is

$$\Psi^{\mathrm{new}}_\omega = A(\zeta'' \phi_{\mathrm{c}}, U''(\xi)) + V_X^{;w} - V_X, \qquad \Psi_\omega = \Psi^{\mathrm{new}}_\omega + \sum_y \Psi_{\omega'_y}(\xi'_y; \mathbf{q}_y(\xi)). \tag{32.53}$$

The running source profile ζ'' is distinct from the core cutoff ϕ_{c}. The first action is at the real inner background, and the second difference is formed by the same localized rule as the unsourced potential. If an interpolation cuts an inner island, that rule forms its excess recursively on the admitted data box. It does not keep an unrelated partial sum of its carrier factors. For a drop-out there is no new inner background or new potential from a completed main step. Its retained excess is the sum of its inner excesses. The source in the Wilson factor supplied by its carrier stays in that carrier's action comparison. This keeps the two types of operation in the same fixed unsourced representation.

For completeness, the per-site estimate is obtained before summing the nesting. Assign a new local member ψ_f to its owned carrier R_f, and bound both its value at a fixed reference pair and its oscillation on the admitted carrier core. Rooted decay with rate strictly above the incidence entropy gives

$$\sum_{f \colon x \in R_f} \frac{\sup |\psi_f|}{|R_f|} \leq C(\varrho). \tag{32.54}$$

Indeed, sum by tree length m; the count is $C_0 C_1^m$, and the bound is $Ce^{-\kappa m}$ with $\kappa > \log C_1$. Any polynomial root multiplicity is absorbed by a smaller retained decay rate. The reference value is controlled by the same vacuum and scalar-action readouts; an oscillation bound alone would miss an arbitrarily large constant.

For the Wilson member in (32.53) use $0 \leq a_p(U'') \leq 2$ where no small-field information is present, and the curvature radius of its own layer where it is present. Each bare plaquette is

charged to its actual owned level; it is not counted as just one last-level plaquette. More explicitly, on the layer n, $|\partial U'' - 1| \le C R_n^2 \varepsilon_n \eta_n^2$. The factor $R_n^2 \varepsilon_n$ is bounded at small coupling. Equation (32.10) then bounds the bare action density by $C\eta_n^4$. There are at most $C\eta_n^{-4}$ such bare plaquettes per site of that layer. The two powers cancel, leaving $O(\varrho)$ per owned level-n site, uniformly in its depth. On the stored bare layer the compact-group bound supplies the same conclusion without a curvature estimate. Together with (32.54), the formed estimates for $V_X^{\prime w} - V_X$ give

$$|\Psi_\omega^{\mathrm{new}}| \le f(\varrho) N_0^\circ(\omega), \qquad |\Psi_\omega| \le f(\varrho) n^\sharp(\omega). \tag{32.55}$$

For the second inequality expand the recursion in (32.53). Each new part uses its own sites and each inner history occurs once. The same constant therefore returns at every depth; it is not added once per ancestor.

Choose the mass weight so that $c\mathsf{l}_j \ge 2f(\varrho)$ on every relevant level. Then (32.23) gives

$$\sup |e^{\Phi_\omega + \Psi_\omega}| \le e^{c\nu^\natural(\omega)} \sup |e^{\Phi_\omega}|. \tag{32.56}$$

This bound is taken inside every integral. The same fixed Borel envelope works uniformly on a compact source polydisc. Holomorphy then follows from Fubini and Morera, with no parameter dependence of the test support or reference measure. The source contributes an ordinary mass price; no comparison of derivatives of two different histories is involved.

Finally remove from each sourced term its own vacuum constant. The change of the remaining small-field exponent is bounded by $C(1+\varrho)$ per determining-layer site, and the differences between its vacuum number and the common one have the defect price in (32.49). Apply (32.52) at the weight enlarged for (32.56). It supplies the actual total-variation sum needed in (OH2)–(OH3), at the volume N_k of the last lattice. The partition floor remains a separate positive-label argument. No location of the complex zeros of the full partition function is used in this mass or label calculation.

(2e) The induction closed: the order of the constants, the sum over histories, exactness. **The class.** Let a frame $\theta_0, \dots, \theta_k$ be fixed (see the opening of Step 3). Each choice of level j that depends on the coupling is made with θ_j. These choices are the thresholds, the radii, the scaling $\mathcal{A}_j = \theta_j^{-1/2} \mathcal{A}_j^{\mathrm{s}}$, the gauge-fixing weight, the Gaussian forms of the stages of $\mathbf{R}$ and the reference laws. The cube size and the waiting depth $\check{R}_j = \mathsf{N}_j$ are made instead with the running minimum $\min_{i \le j} \theta_i$ (a frame can go up by as much as c_5, and taking the minimum ensures that $\check{R}_j$ does not increase). Consequently the tests, the regions, the histories, the domains of analyticity and the measures of the integral operations depend on θ only, and on no entry of the density. In the bounds of this part we put $g_j := \theta_j^{-1/2}$, so that $g_j \le \gamma$ is the floor $\theta_j \ge \gamma^{-2}$ of a frame (at $\theta = x$ this is the $g_j = x_j^{-1/2}$ of Theorem 2.1). In this way the circle "regions depend on couplings, couplings on activities, activities on regions" is broken. The elements of $\mathfrak{C}_k(\theta)$ are presentations rather than densities: the families are not determined by the density, and locality in the history is a property of how the terms are placed. Each element is an array, specified simultaneously on every admissible torus and cutoff,

$$\mathfrak{P} = \Big((x_j);\, (E^{(j)}, \widehat{E}^{(j)}, b_j);\, (R^{(j)}, \widehat{R}^{(j)});\, \big(B_{\mathcal{S}}^{(j)}(X)\big);\, (\Phi_\omega) \Big)_{j \le k},$$

$$b_j = x_{j-1} - x_j = b[\widehat{E}^{(j)}]$$

$$(1 \le j \le k;\ \text{at } j = 0 \text{ the one entry } x_0),$$

whose density $\rho[\mathfrak{P}]$ has the form given in Theorem 2.1. In it, $\widehat{E}^{(j)}, \widehat{R}^{(j)}$ denote single families on $\mathbb{Z}^4$ that coincide with the families on the torus on lifts of non-wrapping domains; $b[\cdot]$ denotes

the coefficient of (1e); and $\Phi_\omega = \Phi^\circ_\omega(\xi) + \sum_f \Phi_{\omega,f}(\xi; \mathbf{p}|_{R_f})$ are the exponents of the integral operations attached to the islands. Membership in the class requires the following. (i) *the tube*: $|\mathsf{z}_j| \le \mathsf{t}_j \approx c\,\theta_j(\log\theta_j)^{-C}$, where $\mathsf{z}_j = \theta_j - x_j$. This relates the frame, which is given, to the weights of the array, and the choice of the frame ensures it. The tube is wide in absolute terms, but narrow enough that the discrepancy between the weight and the choices, regarded as a potential, is small: $|\mathsf{z}_j|\alpha_{0,j}\delta_j \le c(\log\theta_j)^{-c}$. (ii) The bounds of Theorem 2.1 on $E^{(j)}, R^{(j)}, B^{(j)}$ hold, together with analyticity, reality, and invariance under a single fixed gauge action on the stored data. A boundary term is analytic as well in the switches μ_b that belong to those of its stored bonds lying close to a test, and in those of its stored fields that lie far from every test. Any two splittings of one pair give each term the same value. If the tests of a term take the value 1, then each boundary term and each exponent has in its domain the triple $(U, 0, J(U))$ formed from the background U. (iii) *identification*: $E^{(j)}$ coincides with the family produced from the earlier entries, in each of its two chains (see below), by the small-field step at the pair (θ_{j-1}, x_{j-1}), carried out on the whole lattice. (iv) *lattice covariance* holds in the forms stated below. Above all, the pointed totals are covariant under every isometry of $\mathbb{T}^{(j)}$, unit translations included (119, (2.29)). (v) *coherence*: whenever a chart (2a) identifies, scale by scale, what two histories record within one cell of X, it identifies the terms at X. This single clause contains locality in the history, covariance under translations and independence of the torus. For a standing operation, the chart identifies μ_ω together with the Borel set carrying it, as well as its set of factors and Φ°_ω. It also identifies the factors one by one: $\Phi_{\iota\omega,\iota f}(\iota\xi; \iota\mathbf{p}) = \Phi_{\omega,f}(\xi; \mathbf{p})$, for arbitrary components and histories, provided that the chart ι transports both the record of the history at R_f and ξ on R_f. (vi) For an island, μ_ω is a Borel measure, positive and defined on a Borel set; Φ_ω depends in a Borel way on ξ, is analytic in $\mathbf{p}$, and takes real values whenever the pair is real. Moreover, gauge invariance holds pointwise in ξ and for each factor separately. The bounds of *mass* (Proposition 2.7, with a reserve constant $c_{\mathrm{r}} \ge 0$ in the weight) and of *oscillation* (2d) hold, the coupling in them being taken at the level at which the island was made. The class imposes no bound on the exponent itself, which is large and negative, and no clause concerns a drop-out. (vii) Each term, taken together with its stored variables, is a non-negative measure $\hat\tau_h$. The form and the bounds alone do not give a class that a step preserves. The gain $L^{-c(n-j)}$ of the regular terms does not follow from $\|E^{(j)}\|_{j,\kappa} \le E_0$, and neither do $|\mathbf{E}_k| \le C\sum_n |\Gamma_n| + C\sum_j \#_j(Z_j^+)$ (below) and the input of the next step, which depend on that gain. Norm bounds alone do not imply age decay, because they allow marginal action contributions; the example below exhibits one before its own correct subtraction. Take the tents $h_z^{(j)}$ of (1d); then $\mathsf{Q}_z := c\,\alpha_{0,j}^{-2} A(h_z^{(j)}, \cdot)$ defines an activity whose norm does not exceed 1, and

$$\sum_z \mathsf{Q}_z = c\,\alpha_{0,j}^{-2} A(U),$$

$$\alpha_{0,j}^{-2} = c\,\theta_j(\log\theta_j)^{-2q}\,:$$

this is the inverse coupling θ_j itself, up to the factor $c(\log\theta_j)^{-2q}$. For a regular family satisfying the full hypotheses of Section 23, its own extracted coefficient b_j and its own tent action remove this marginal part. The Ward identities, full hypercubic tensor symmetry and response hypotheses identify its affine quadratic jet at each base point with that multiple of the tent action. Take the format of the complex class below, in which each site has its own member. In that format one can find arrays which obey all the bounds of clause (ii) and which, at each level, add no less than the same multiple of the action, with no part of it decaying as the age grows. Uncancelled marginal contributions of this kind are excluded by those structural hypotheses together with each member's own correct subtraction. Unit-translation covariance alone neither forces isotropy nor removes a marginal action term.

Theorem 2.11 [the step preserves the class]. Suppose the following:

- $\mathfrak{P} \in \mathfrak{C}_k(\theta)$;
- θ_{k+1} prolongs the frame;
- all $g_j \le \gamma$, and they lie below a ceiling that depends on the reserve constant c_{r} and does not increase with it;
- the next weight $x_{k+1} := x_k - b[\widehat{E}^{(k+1)}]$ lies in the tube (here $\widehat{E}^{(k+1)}$ is formed first, by clause (iii), at the pair (θ_k, x_k), and its formation involves no choice of level $k+1$).

Then each move of step $k+1$ is defined. The identity $\mathbf{RT}\rho[\mathfrak{P}] = \rho[\mathfrak{P}']$ holds separately for each standing part (2a), every part keeping its stored variables, and $\mathfrak{P}' \in \mathfrak{C}_{k+1}(\theta)$, the constants being unchanged. Moreover, $\mathfrak{P}'$ prolongs $\mathfrak{P}$. An array belongs to the class of level $k+1$ exactly when its cut at k belongs to the class of level k and, in addition, it satisfies the conditions of level $k+1$. Finally, the array whose single entry is $x_0 = \beta$, with $|\beta - \theta_0| \le \mathtt{t}_0 \le c\,\theta_0(\log\theta_0)^{-C}$, belongs to $\mathfrak{C}_0$. *Not returned*: the hypothesis that the next weight belongs to the tube. At every step this hypothesis is provided by how the frame is chosen. For $\theta = x$ the deviation is 0, and the requirement is that x_{k+1} prolong the frame, which is the content of the flow inequalities (3a) and of the nesting of the cube sizes, a consequence of the upper flow inequality (3b). For the reference line it is supplied by the drift bound of (3c), below. (In Bałaban, Comm. Math. Phys. 119 (1988) 243, the second remark after Theorem 1 concerns the space of "all densities satisfying the conditions of the inductive assumption", and these conditions include (2.29), (2.32), (2.33).)

Why the induction closes. (a) Room: fix a fraction β_{s}. The terms produced by $\mathbf{T}$ (2b) are defined on the radii $(1+\beta_{\mathrm{s}})\alpha_{i,k}$ and decay at the rate $(1+4\beta_{\mathrm{s}})\kappa$. Moreover, $e^{-\kappa d_j(X)} \le e^{-c\kappa d_j(X)} e^{-(1+3\beta_{\mathrm{s}})\kappa d_k(\widehat{X})}$ for $j < k$, where $\widehat{X}$ denotes the union of all level-k cubes meeting X, taken without any added layer. The operation $\mathbf{R}$ uses up this room and nothing beyond it. (b) Drift: for $m < n$ one has $g_m \le (1+c)g_n \le C(n-m)^{1/2} g_m$. This does no harm over the N_k levels that $\mathbf{R}$ spans, because every threshold changes only by powers of $\log x_k$. (c) Order. Every constant is of one of four kinds:

- free upward: its range has no upper bound (the odd integers for L, the powers of L for M);
- free downward: its range is some $(0, c]$;
- confined to a window;
- determined by earlier constants: a function of those alone.

The text states finitely many conditions on the constants. Each condition is assigned to the last constant on which it depends, after every determined constant in it has been expressed through the earlier ones. For every admissible choice of the earlier constants, the condition is then, at that constant, of one of three kinds:

- a floor on a constant free upward, which holds from some value onward;
- a ceiling on a constant free downward, which holds on some $(0, c]$;
- an endpoint of a non-empty window.

A condition that follows from such conditions is not counted separately, with one exception: the requirement that a window be non-empty. That requirement is a floor or a ceiling on some earlier constant, and it counts at that earlier constant. Two further stipulations apply:

- Every required comparison of thresholds must retain its stated margin. Lowering a threshold changes both its passing set and the gain available from its failure, so it cannot be treated as preserving all the estimates automatically.
- Some families of conditions are indexed by a quantified number, for instance the floor $c_a \ge C(b)\log L$, $C(b) \to \infty$, with an integer $b \ge 1$. No single c_a satisfies this for all b. Such a family belongs to the statement in which the number is quantified, and it is imposed only at finitely many values of that number, which are fixed before the constant.

Thus, apart from the constant it bounds, each of these conditions depends only on earlier constants, in the order

$$(r, \mathfrak{p}_0, \mathfrak{p}_1, q, \kappa_0, \beta_{\mathrm{s}}, \theta_{\mathrm{s}}, \theta_c) \prec L \prec (M_1, R_1) \prec \kappa \prec (\bar\kappa, \kappa_1) \prec M_1' \prec M_2 \prec (\Theta_1, \varpi_T) \prec M$$
$$\prec \sigma_\star \prec a_\Lambda \prec (A_1, A_0, C_0, C_1, \alpha_1, \alpha_0, E_0, M_0, \alpha', \mathsf{B}) \prec (\underline\lambda, c_5, \bar B) \prec \gamma \prec (\varrho, c_a) \prec (g, \ell, \beta, K)$$

Some of these constants require comment:

- $\underline\lambda = \frac14 c_0$ and $c_5 = 2c_2$, where c_2 is the constant of Proposition 3.3, are fixed together with L; in the display they appear as members of the triple that defines the frames.
- $\bar\kappa$ is the exponent of the decoupling parameters of (1d). The factor $e^{-\bar\kappa}$ of a link of the cluster expansion comes from the decay of the covariance across an M-cube.
- $\delta_0 M \gg \bar\kappa \gg L\kappa$: rates are chosen before the cube sizes M_1', M_2, M. These are floors on $\bar\kappa$ and then on M, both chosen after L and κ.
- M_1 depends on L alone, and R_1 depends on L and M_1.
- $\Theta_1 \geq C/\theta_c$ and $\varpi_T \geq C$, and both are free upward.
- $M \geq \max\{\varpi_T, L^2 M_2\}$ and $A_0 \gg A_1$.
- The later ceilings on g may depend on c_a and on the dial $\mathsf{w} \in [0,1)$. The dial is a fraction that depends on no constant and is fixed together with the exponents.
- The reserve constant $c_{\mathrm{r}} \geq 0$ of (vi) is free upward and comes before the ceiling that depends on it.

Induction along this order now gives two facts, because finitely many floors hold together from some value onward, finitely many ceilings hold together on some $(0, c]$, and the finitely many endpoints of a window leave it non-empty owing to the conditions that count at earlier constants. Every admissible choice made up to a given position can be extended, and no later condition places a restriction on a constant chosen earlier. Adding one more floor or ceiling of the same kind leaves both facts true. The choice is therefore not circular, and a condition arising in the construction fits the order if, at the last of the constants on which it depends, it is of the kind named. A bound may contain a number that a statement returns, or a number given as a function of a whole group of constants. Any such number must be fixed before the constant that it bounds; the cycle cut below is a case of this. A representative condition on M or on γ asserts that e^{-cM}, $e^{-c\check R_k}$ or a power of $\log x_k$ dominates constants fixed earlier. *The supremum form*: a condition imposed on all couplings of all runs reduces to a single ceiling on γ, namely $\sup_{x \geq \gamma^{-2}} h(x) \leq c$ with $h \to 0$, because the scale appears in it solely by way of x_k (122 (1989) 175, p. 186: the conditions "do not depend on any scale"). *The cycle that is cut*: $\lambda_\sharp$ and $B_\sharp$ depend on γ through $\sup|\mathsf{q}|$ (here q_i is the part of the increment b_i contributed by the remainders; see below), whereas the ceiling γ depends on the flow constants. Frames are therefore defined with constants $(\underline\lambda, c_5, \bar B)$ fixed before γ: $\underline\lambda(j-i) - c_5 \leq \theta_i - \theta_j \leq \bar B(j-i)$ for $i \leq j$. Hence a frame with first member θ_0 has length at most $(\theta_0 - \gamma^{-2} + c_5)/\underline\lambda$, in the same way that K is bounded in terms of β in (3a). In addition, $\underline\lambda \leq \lambda_\sharp$ and $B_\sharp \leq \bar B$. *No caps*: there is no upper bound on $L, M, \kappa, q, \mathfrak{p}_0, c_a$, on the cutoff, the volume, an age or a number of steps, and no lower bound on a coupling. *The role of each exponent*:

- $\mathfrak{p}_0 - \mathfrak{p}_1 > Cr$, for comparing the ratio δ_j'/δ_j with the square root of the waiting depth, and for the accumulation of the stages ((2b), (2d));
- $2\mathfrak{p}_1 - Cr > \mathfrak{p}_0$, so that a failed stage test still gives e^{-p_0};
- $\mathfrak{p}_0 > 3Cr$, so that p_0 dominates $\kappa\check R^4$ and the cost $\check R^6$ of a wait, and so that the window left for $\mathfrak{p}_1$ by the two preceding conditions is non-empty;
- $q - \mathfrak{p}_0 > Cr$, so that $\mathsf{N}_k \mathsf{s}_k \to 0$;

- $q > C\mathfrak{p}_0$, so that the product of an amplitude and an oscillation of degree $2\mathfrak{p}_0$ remains bounded;
- $\kappa_0 > C$, for the sum over the scales.

(d) *No circle.* New terms are built from old ones, and the old ones are bounded with the same constants that the new ones must satisfy. The circle is avoided because an old term enters again in one of three ways:

- unchanged;
- through a difference, which carries a small factor. In **T** this factor is s_k, the factor that the Cauchy estimate gives for a first difference in the fluctuation field (2d).
- as a whole, in the main step of **R**. In that case there are two possibilities. Either the domain of the term must stretch across $c\check{R}_k L^{k-j}$ cubes of the term's own level $j < k$, in which case its decay makes the term small. Alternatively, the domain lies inside an enlarged component (a collar $X^\sim$ of (2d)). There only a volume bound $C\sum_{j\le k} \#_j(Z_j \cap X)$ is available, each scale counted in its own unit, and its constant is chosen later. The list is given below under *Boundary terms*.

Uniformity of the input. A term of level j is defined on the radii at which it was created, and step $n+1$ evaluates it up to the output radius of that step. Let $w_{n,j}$ be the square of the ratio of these two radii. Let $\mathsf{f}_\nu \approx \nu^C L^{-c\nu}$ be forgetting weights; in (1d) they are written ω_ν. Then

$$w_{n,j} \approx \frac{\theta_j}{\theta_n}\Big(\frac{\log\theta_n}{\log\theta_j}\Big)^{2q} \le C(n+2-j),$$

$$\sum_{j\le n} \mathsf{f}_{n+1-j}\|E^{(j)}\|_{j,\kappa} + \sum_{j\le n} w_{n,j}\|R^{(j)}\|_{j,\kappa} \le N_\star$$

uniformly in n and in the frame. This is everything that the small-field step requires of its input. The two factors in the approximation above are never unfavourable at the same time: the upper flow bound gives $\theta_j/\theta_n \le 1+(n-j)$, and the lower one gives $\theta_n \le \theta_j + c_5$. The regular terms are of size $O(1)$ at each level and gain $L^{-c(n-j)}$ through their two counterterms. The remainders gain nothing; they are summed by means of the coupling. *Regular terms*: let $\tau_k = x_k/\theta_k$, and let E_{N} be the bound on the localised Gaussian normaliser. Then

$$\|E^{(k+1)}\|_{k+1,\kappa} \le E_{\mathsf{N}} + C\big(|1-\tau_k| + \delta_k + \mathsf{s}_k N_\star + e^{-c\,p_0(g_k)^2}\big) \le E_0,$$

so E_0 is determined by the one-loop normaliser alone, and C depends neither on E_0 nor on $N_\star$. *Boundary terms*: B is the maximum of the bounds coming from their origins (2d), and none of these bounds depends on B. This holds because data bounded by B enter the step in four ways:

- In **T**, they enter as differences in the fluctuation field, with $C\mathsf{B}\mathsf{s}_k \le 1$.
- In the stages, they enter as differences in the collar field, taken one block of bonds at a time. Each block that is switched off contributes a negative power of $\log\theta_k$, with an exponent larger than the one belonging to (number of stages)$\times$(M-cubes of level k of a component).
- In the main step, outside the enlarged components, they enter either through the difference under the substitution, or as undifferenced functions of a level $j < k$. In the second case the domain must extend over $c\check{R}_k L^{k-j}$ cubes, and $\mathsf{B}e^{-c\check{R}_k} \le 1$.
- Inside an enlarged component there is no smallness, and they enter only through the volume bound on $V(X^\sim)$. The constant of this bound is chosen after B, and the bound is paid by the ledger of (2c).

(e) *No growth with the level.* A carried term is literally unchanged: it has the same function, the same domain of analyticity and the same tree length. The bounds of a standing island refer to the level at which it was made and do not involve k. Indeed, the supremum in the mass bound is taken over a fixed set, built from the data of the levels up to that level, and this set contains the arguments of all later levels. Hence the mass at level k does not exceed the mass at the level at which the island was made ((2c), "The bound has to reproduce itself", (ii)). The sizes that are returned are maxima rather than sums.

33. Ordinary source formation and volume bounds

This section separates three estimates that have different mathematical content: forming local interactions, exponentiating a polymer gas, and integrating the resulting expression. A convergent polymer expansion supplies the second estimate. To apply it to a renormalisation construction one must first verify its activity bounds for the interactions that the construction actually forms. The elementary implications below specify the ordinary source estimates needed for that verification.

Site-dependent members and the summation over old levels. Fix an odd integer $L \geq 3$. Let $(\theta_j)_{0\leq j\leq n}$ satisfy

$$\theta_j \geq Y, \qquad \lambda(n-j) - c_5 \leq \theta_j - \theta_n \leq B(n-j), \qquad Y \geq 2(c_5+1), \tag{OSF1}$$

where $\lambda > 0$ and $c_5 \geq 0$. A pointed family at level j is a collection $G(X,P,z)$, where X is a localisation domain, $z \in X$ is a base point and P belongs to a specified complex field domain. Its norm is

$$\|G\|_{j,\kappa} = \sup_{X,z,P} e^{\kappa d_j(X)} |G(X,P,z)|. \tag{OSF2}$$

Any companion families used to define its quadratic coefficient are included in this supremum. The field domains and restrictions are part of the data; holomorphy does not assert that a function extends beyond these domains.

At every site z, permit a different family $G_{j,z}$. Each such family comes with its own quadratic coefficient $b(G_{j,z})$. The cancellation invoked here uses the full hypotheses of Section 23, including gauge invariance, the actual normalized response, absolute Hessian moments and covariance of its *pointed* total under unit translations and the full hypercubic group. Under those hypotheses, subtracting its own coefficient times its own tent action cancels its marginal affine quadratic term at that base point. No covariance relating the chosen members at different sites is required. This distinction permits a scalar source to be frozen at the value belonging to each site.

Here is a precise quantitative formulation of the input used in this argument. Put $\epsilon_{jn} = L^{-(n-j)}$, and let ρ_{jn} be the ratio between the field radius required at level n and the radius on which the member of level j is given, enlarged to be at least one. Suppose $\rho_{jn}^2 \leq C_\rho(1+n-j)$. Assume a linear localisation rule with the following estimates, including summability over its rooted domains. For one base point of level j, a normalised regular member contributes at most

$$C\,\|G_{j,z}\|_{j,\kappa}\,\rho_{jn}^3 \epsilon_{jn}^{4+\alpha} \qquad (0<\alpha<1) \tag{OSF3}$$

to a window of level n, after its vacuum and quadratic subtractions. A member with its vacuum value alone subtracted contributes at most

$$C\,\|F_{j,z}\|_{j,\kappa}\,\rho_{jn}^2 \epsilon_{jn}^4. \tag{OSF4}$$

The constants include the permitted loss of spatial decay and the count of domains through the base point. The rule must evaluate every member within its stated field domain for every allowed decoupling and window parameter. This domain inclusion is a separate hypothesis of (OSF3)–(OSF4).

The factor ϵ_{jn}^4 has a direct dimensional meaning. A window of level n contains at most $C\epsilon_{jn}^{-4}$ base points of level j. Thus vacuum subtraction balances the number of base points in four dimensions, whereas the quadratic subtraction must supply the additional ϵ_{jn}^{α} to make old regular members summable.

Proposition 33.1 (Uniform ordinary input)**.** *Suppose the regular members obey*

$$\sup_z \|G_{j,z}\|_{j,\kappa} \le G_\star.$$

Suppose defects and remainders obey, respectively,

$$\|D_j\|_{j,\kappa} \le L^{-(n+1-j)}, \qquad \|H_j\|_{j,\kappa} \le C_H \theta_j^{-p}, \qquad p > 2. \tag{OSF5}$$

Under (OSF3)–(OSF4), their combined ordinary input at level n is bounded by

$$CG_\star + C\sum_{a=0}^{n}(1+a)L^{-(a+1)} + CC_H\sum_{a=0}^{n}(1+a)(\theta_n + \lambda a - c_5)^{-p} \le CG_\star + C + CC_H\theta_n^{2-p}. \tag{OSF6}$$

The constant is independent of n, the number of sites and the choice of the member at each site.

Proof. Sum (OSF3) over the base points in a window. The inverse volume factor cancels ϵ_{jn}^4, leaving

$$CG_\star \sum_{a\ge 0}(1+a)^{3/2}L^{-\alpha a} < \infty.$$

This uses only the supremum of the member norms, so choosing a different covariant member at each site has no effect on the bound. The same cancellation in (OSF4) gives the second sum in (OSF6) for defects. That sum is bounded by a convergent geometric series and its first derivative.

For remainders use (OSF1) and put $y = \theta_n - c_5 \ge \theta_n/2$. For $a \ge 0$, $1 + a \le C(y + \lambda a)$, since $y \ge 1$. Comparison with the integral of the decreasing function $(y + \lambda t)^{1-p}$ gives

$$\sum_{a\ge 0}(1+a)(y+\lambda a)^{-p} \le C\left(y^{1-p} + \int_0^\infty (y+\lambda t)^{1-p}\,dt\right) \le Cy^{2-p}. \tag{OSF7}$$

This proves the stated, deliberately nonoptimal, bound. □

For $SU(2)$, a bounded scalar action weight supplies another admissible ordinary input. If $W \in SL(2,\mathbb{C})$, then

$$1 - \tfrac{1}{2}\operatorname{tr} W = \tfrac{1}{2}\det(1 - W). \tag{OSF8}$$

On a field domain where each plaquette satisfies $|W_p - 1| \le \alpha_n\eta_n^2$, its action density is bounded by $C\alpha_n^2\eta_n^4$. A local cube contains at most $CM^4\eta_n^{-4}$ plaquettes. Filing each plaquette in a domain containing its corners therefore gives a weight member of norm at most $CM^4\alpha_n^2 \sup|u|$. This proof uses the determinant identity and the count of plaquettes; it does not use covariance of u. With $\alpha_n = O(\theta_n^{-1/2}(\log\theta_n)^q)$, any fixed bound on $\sup|u|$ is eventually admissible.

These summations do not prove (OSF3). That estimate contains the renormalisation assertion: covariance of the pointed Hessian, the correct counterterm, regularity of the background within the window and uniform membership after substitution must be proved for the actual formation rule. They cannot be replaced by a bound on the full total or by holomorphy in the source alone.

Exponential reserves and generated local perturbations. Let a local history ω have a positive measure μ_ω on S_ω, a reference exponent $\Phi_\omega(\xi;P)$, and a site count N_ω. Suppose its reserve volume V_ω is additive over the older histories swallowed by this history and satisfies

$$V_\omega \ge AN_\omega, \qquad \sum_\omega \int_{S_\omega} e^{c_\star V_\omega} \sup_P |e^{\Phi_\omega(\xi;P)}|\, d\mu_\omega(\xi) \le \varepsilon, \qquad c_\star \ge 1. \tag{OSF9}$$

The supremum in (OSF9) is assumed measurable, or is replaced by a measurable majorant throughout. If a holomorphic perturbation satisfies $|\Psi_\omega| \le fN_\omega$ and $A \ge f$, then for bounded F

$$\left| \sum_\omega \int F e^{\Phi_\omega + \Psi_\omega}\, d\mu_\omega \right| \le \varepsilon \|F\|_\infty. \tag{OSF10}$$

Indeed, $|e^{\Psi_\omega}| \le e^{V_\omega}$, so take absolute values inside every integral and apply (OSF9). If each integrand is holomorphic in the parameter and all reference measures and integration sets are parameter independent, a common integrable majorant on compact parameter sets also proves holomorphy by Fubini and Morera. This is an ordinary complex mass estimate. It requires no derivative with respect to the number of source zones.

Suppose a formation rule produces a new excess exponent in the explicit form

$$\Psi_\omega^{\mathrm{new}} = A(\widehat{c}\, \upsilon 1_X, U''(\xi)) + V(X^\sim; \mathfrak{L}) - V(X^\sim; \mathfrak{L}^0), \tag{OSF11}$$

where $U''(\xi)$ is real, $0 \le \upsilon \le 1$ and $\sup|\widehat{c}| \le \bar{c}$. To apply (OSF10) it is sufficient to prove the actual formation bounds

$$|A(\widehat{c}\, \upsilon 1_X, U''(\xi))| \le C_W \bar{c}\, N_\omega^0, \qquad |V(X^\sim; \mathfrak{L})| + |V(X^\sim; \mathfrak{L}^0)| \le 2c_{\mathrm{int}}(1+\bar{c})N_\omega^0. \tag{OSF12}$$

If $\Psi_\omega = \Psi_\omega^{\mathrm{new}} + \sum_{\omega'} \Psi_{\omega'}$ and $N_\omega = N_\omega^0 + \sum_{\omega'} N_{\omega'}$, these inequalities give

$$|\Psi_\omega| \le \max\{C_W \bar{c} + 2c_{\mathrm{int}}(1+\bar{c}), f\}\, N_\omega. \tag{OSF13}$$

The maximum, rather than a sum of the old and new constants, is what permits unbounded nesting. Preserving the carrier oscillation norm needed in the next formation step is an additional estimate; it does not follow from (OSF13).

There is a genuine reason to generate local perturbations through this formation rule. A bound on an exponent and on the oscillation of each carrier does not bound arbitrary partial sums of carrier factors. On a one-point integration space, write $\Psi^\circ(\lambda) = -N\lambda$ and $\psi_1(\lambda; P) = N\lambda$, with a second, unused carrier if desired. Then $\Psi^\circ + \psi_1 = 0$, both carrier oscillations in P are zero, and the full mass is unchanged for every λ. Interpolating only ψ_1, while leaving Ψ° at its value at zero, instead gives the mass $e^{N\lambda}$. No bound uniform in N survives on a fixed complex disc. This example refutes that partial-interpolation argument, not a construction which defines its localized exponents recursively by (OSF11) and proves (OSF12) on the whole complex data box.

What a pointwise polymer logarithm proves. Consider a finite hard-core polymer gas Ξ_y, depending on an exterior field y. Suppose its activities $h_y(C)$ satisfy a uniform Kotecký–Preiss criterion, including absolute convergence of the logarithmic series and the bound

$$|\log \Xi_y| \le \sum_C |h_y(C)| e^{a(C)} \le c\, N. \tag{OSF14}$$

If the activities are real for real y, then $\Xi_y > 0$. One proof is that the absolutely convergent logarithmic series is real. Equivalently, replace every activity by $t h_y(C)$, $0 \le t \le 1$. The same criterion applies, so $\Xi_y(t)$ is real, continuous and nonzero. Its value at zero is one, proving positivity at one. Individual activities need not be positive. Moreover $e^{-cN} \le \Xi_y \le e^{cN}$.

If a final representation really has the form $Z = \int q(y) \Xi_y\, d\nu(y)$, with $q \ge 0$, then

$$e^{-cN} \int q\, d\nu \le Z \le e^{cN} \int q\, d\nu. \tag{OSF15}$$

Thus even a pointwise polymer logarithm requires control of the remaining integral. It does not turn a restricted Haar integral into a probability integral. For example, take the empty gas and let q be the indicator of a Haar set of measure e^{-N^2}. The polymer criterion is automatic, but a lower bound e^{-CN} for Z is false.

In a representation that sums over standing labels, the useful upper estimate instead has the form

$$\sum_{\mathcal{S}} \left|e^{\mathcal{E}_{\mathcal{S}}(w)}\tau_{\mathcal{S}}^{w}\right|(\mathcal{V}_n)e^{W(\mathcal{S})} \le e^{cN_n}. \tag{OSF16}$$

Here the absolute value is total variation, W pays for missing vacuum contributions and the measures inside the retained integral operations are also majorised before they are integrated. A pointwise bound for one gas does not supply this sum: arbitrarily many positive labels can have the same individually bounded gas. The counting of labels and the exponential reserve in their operations are essential.

For the lower bound at zero source, positivity permits retaining the all-small-field label. Suppose its density, after vacuum normalisation, is at least $\chi(y)\exp[-xA(U(y)) - CN_n]$. Assume the ball in which every one of the $4N_n$ link variables has distance less than s from the identity is contained in $\{\chi = 1\}$, and that $A(U(y)) \le Cs^2N_n$ there. Normalized Haar measure on $SU(2)$ assigns this link ball mass at least $(cs^3)^{4N_n}$. It follows directly that

$$e^{-E}Z(0) \ge \exp[-(C + 4\log(1/(cs^3)) + Cxs^2)N_n].$$

Taking $s = c_0x^{-1/2}$ gives $e^{-E}Z(0) \ge \exp[-(C + 6\log x)N_n]$. The nontrivial model input here is that the link ball passes every background test and that the action of its background has the asserted bound. No zero-free statement for the sourced full partition function is needed.

34. First passage and the arithmetic of a marginal flow

Assume the coupling recursion and uniform increment bounds below. We derive their first-passage and scale-summation consequences, allowing individual increments to be negative. The RG realization, torus and cutoff compatibility, and nonconstant-source estimates are included in the separate flow and source hypotheses.

34.1. **Hypotheses and first passage.** Fix numbers $c_0 > 0$, $C_0 \ge 0$, $h > 0$, and $0 \le \epsilon < c_0/4$. For each initial value $\beta \ge h + B$ consider a real run with $x_0 = \beta$. Assume that a next step exists whenever $x_j > h$, and that every such step has the form

$$x_j = x_{j-1} - b_j,$$
$$b_j = b_j^0 + r_j.$$

For every interval of defined steps assume

$$\left|\sum_{l=i+1}^{j}(b_l^0 - c_0)\right| \le C_0,$$
$$|r_j| \le \epsilon$$
$$\text{if } x_{j-1} \ge h.$$

All constants are independent of β and of the length of the run. The existence of the next step is an assumption: an inequality about hypothetical increments cannot establish that an underlying analytic construction is defined. Set

$$\lambda = c_0 - \epsilon > 0,$$
$$\Lambda = c_0 + \epsilon,$$
$$B \geq c_0 + C_0 + \epsilon.$$

The one-step case of the interval estimate gives $b_j \leq B$. This is an upper bound, not an assertion $b_j \geq 0$.

Define

$$K(\beta) = \min\{k \geq 0 : x_k \leq h + B\},$$
$$\mathcal{W}_k = \{\beta \geq h + B : K(\beta) = k\}.$$

The minimum is initially a possible stopping index; its finiteness will be proved. At the endpoint $\beta = h + B$ it equals zero. When $\beta > h + B$ it is positive, so the last increment is an actual step of the run.

Proposition 34.1 (Block drift and a protected terminal window)**.** *The first-passage time is finite. For defined segments up to that time,*

$$\lambda(j - i) - C_0 \leq x_i - x_j \leq \Lambda(j - i) + C_0$$
$$(0 \leq i \leq j \leq K).$$

If $K > 0$, then $h < x_K \leq h + B$. Moreover

$$\max\left\{0, \frac{\beta - h - B - C_0}{\Lambda}\right\} \leq K \leq \frac{\beta - h + C_0}{\lambda}.$$

Proof. Summing the recursion over a segment gives

$$x_i - x_j = c_0(j - i) + \sum_{l=i+1}^{j} (b_l^0 - c_0) + \sum_{l=i+1}^{j} r_l.$$

The two assumed bounds give the block drift estimate. If the run never reached $h + B$, all its values would remain above h, so it could be extended for every finite number of steps. The lower drift bound would then imply $x_j \leq \beta - \lambda j + C_0$, contradicting $x_j > h + B$ for sufficiently large j. Thus first passage occurs.

For $K > 0$, minimality gives $x_{K-1} > h + B$. Since $b_K \leq B$, $x_K = x_{K-1} - b_K > h$. The upper endpoint follows from the definition of K. The extra width B is exactly what prevents the final step from overshooting the lower reference value h.

Finally $\beta - x_K \leq \Lambda K + C_0$ and $x_K \leq h + B$ give the lower bound on K. Conversely $\lambda K - C_0 \leq \beta - x_K < \beta - h$ gives the stated upper bound for $K > 0$. The endpoint case $K = 0$ also satisfies both inequalities. □

In particular $\beta \to \infty$ forces $K(\beta) \to \infty$. Conversely a sequence with $K \to \infty$ has $\beta \to \infty$, by the lower drift estimate and the terminal lower bound. Since these stopping times are integers, the sets $\mathcal{W}_k$ are nonempty for infinitely many different k. Nothing here says that every one is nonempty, that one is an interval, or that the terminal values in it coincide. Such assertions would require additional information about the dependence of the analytic recursion on β.

For example, $b_j^0 = c_0 + (-1)^j q$, with $q > c_0$ and $r_j = 0$, satisfies the interval hypothesis with $C_0 = q$. Every other increment is negative. Nevertheless the block drift is positive and the preceding proof applies. Replacing block drift by a silently assumed monotonicity would discard this perfectly admissible numerical example.

34.2. **Vanishing mean error and the leading logarithm.** Assume in addition that

$$|r_j| \le \omega(x_{j-1}),$$
$$\omega_*(M) := \sup_{x \ge M} \omega(x) \longrightarrow 0$$
$$(M \to \infty),$$

where ω is nonnegative on $[h, \infty)$. The earlier global bound by ϵ remains in force. No monotonicity of ω is needed.

Lemma 34.2 (Time spent below a fixed inverse coupling)**.** *For $M \ge h$, the number N_M of steps $1 \le j \le K$ with $x_{j-1} \le M$ is bounded by*

$$N_M \le 1 + \frac{M - h + C_0}{\lambda}.$$

This bound does not depend on the initial value.

Proof. There is nothing to prove if no such step occurs. Otherwise let i be the smallest index in $\{0, \dots, K-1\}$ with $x_i \le M$. Every counted step has its preceding index between i and $K-1$, so $N_M \le K - i$. The block drift from i to K gives $\lambda(K-i) - C_0 \le x_i - x_K < M - h$. The displayed, slightly weaker bound therefore follows. Excursions back above M cause no problem: all subsequent indices are included in this bound on the entire remaining time. □

Theorem 34.3 (Leading first-passage asymptotics)**.** *With h and the other constants fixed,*

$$\frac{1}{K(\beta)} \sum_{j=1}^{K(\beta)} |r_j| \longrightarrow 0,$$
$$\frac{\beta}{K(\beta)} \longrightarrow c_0$$
$$(\beta \to \infty).$$

If $a_K = L^{-K}$ for a fixed $L > 1$, then $\beta / \log(a_K^{-1}) \to c_0 / \log L$.

Proof. Split the steps according to whether $x_{j-1} \le M$. The lemma gives

$$\frac{1}{K} \sum_{j=1}^{K} |r_j| \le \frac{\epsilon}{K} \left(1 + \frac{M - h + C_0}{\lambda} \right) + \omega_*(M).$$

First let $\beta \to \infty$ at fixed M, using $K \to \infty$, and then let $M \to \infty$. This proves the first limit. Summing the whole recursion gives

$$\beta = x_K + c_0 K + e_K + \sum_{j=1}^{K} r_j,$$

$$|e_K| \le C_0.$$

The terminal value belongs to a fixed bounded interval, so division by K proves the second limit. Finally $\log(a_K^{-1}) = K \log L$ exactly. □

If the analytic one-loop calculation establishes $c_0 = 11N^2 \log L/(12\pi^2)$ in the convention $\beta = g_0^{-2}$, this arithmetic yields $\beta/\log(a_K^{-1}) \to 11N^2/(12\pi^2)$. The one-loop coefficient is an input to this implication, not a consequence of the stopping argument. The result is uniform over any family of runs satisfying the same constants and error envelope.

34.3. **A quantitative optional error bound.** Suppose $h \ge e$, $p \ge 0$, and $\omega(x) \le Cx^{-1/2}(\log x)^p$. Then

$$|\beta - c_0 K| \le h + B + C_0 + C'\sqrt{\beta + C_0 + 2}\,[\log(\beta + C_0 + 2)]^p.$$

To prove it, partition $[h, \infty)$ into unit intervals. If two preceding indices $i < j < K$ have values in the same interval, then $x_i - x_j < 1$ and the lower drift gives $j - i < (1 + C_0)/\lambda$. A unit interval is therefore visited at most $J = 2 + \lceil (1 + C_0)/\lambda \rceil$ times. Also $x_i \le \beta + C_0$ by the lower drift from zero. Summing the supremum of the error envelope in each visited interval consequently gives

$$\sum_{j=1}^{K} |r_j| \le CJ[\log(\beta + C_0 + 2)]^p \sum_{n=0}^{\lceil \beta + C_0 - h \rceil} (h + n)^{-1/2}.$$

The last sum is at most a constant times $\sqrt{\beta + C_0 + 2}$, by comparison with the integral of $x^{-1/2}$. Substitute this bound in the summed recursion. For fixed h this is an $O(\sqrt{\beta}(\log \beta)^p)$ error. It supplies no second logarithmic coefficient, convergence rate for observables, or summable comparison between neighboring cutoffs.

34.4. **Depth envelopes and sums over scales.** The same block estimate gives useful bounds measured from the terminal scale. For $0 \le s \le K$, apply it between $K - s$ and K to obtain

$$h + \lambda s - C_0 < x_{K-s} \le h + B + \Lambda s + C_0.$$

This estimate records the possible bounded backward excursions without requiring the inverse couplings to be monotone. It also distinguishes the fixed reference value h from the terminal value x_K, which is allowed to range over its window.

For an explicit scale-summation corollary suppose $h \ge \max\{1, 2C_0\}$. Then the lower bound becomes $x_{K-s} \ge h/2 + \lambda s$. Write $g_j = x_j^{-1/2}$. For every $q > 2$, comparison with a decreasing integral gives

$$\sum_{j=0}^{K} g_j^q \le \sum_{s=0}^{\infty} (h/2 + \lambda s)^{-q/2} \le (h/2)^{-q/2} + \frac{(h/2)^{1-q/2}}{\lambda(q/2 - 1)} \le C(q, \lambda) h^{1-q/2}.$$

The bound is uniform in the number of steps. The first term of the sum is bounded separately, and the integral accounts for all later terms; this explains both the strict condition $q > 2$ and the displayed dependence on the reference inverse coupling.

At the marginal exponent $q = 2$ the integral instead grows logarithmically. Using the two depth envelopes gives, for example,

$$\frac{1}{\Lambda}\log\left(1+\frac{\Lambda(K+1)}{h+B+C_0}\right)\le\sum_{j=0}^{K}g_j^2\le\frac{2}{h}+\frac{1}{\lambda}\log\left(1+\frac{2\lambda K}{h}\right).$$

For the lower bound integrate the decreasing function $(h+B+C_0+\Lambda s)^{-1}$ from zero to $K+1$. For the upper bound separate its counterpart's first term and integrate from zero to K. These elementary estimates are often the numerical part of a renormalization argument: sufficiently high powers of the running coupling are summable, whereas the marginal power is not. They cannot replace the analytic cancellation or renormalization that produces those powers in a particular model.

The extra lower bound on h belongs only to this corollary. For a smaller fixed h, separate the finitely many terminal depths $s<2C_0/\lambda$, bound them using $x_j>h$, and apply the same integral comparison to the remaining depths. The resulting constants may depend on h and C_0.

34.5. **An exact change of lattice units.** There is also an identity that requires no coupling-flow assumption. Let $\Lambda_{K,\ell}=(L^{-K}\mathbb{Z}/\ell\mathbb{Z})^4$ be a compatible finite lattice and let b be an integer with $K-b\ge 0$. Dilation $D_bx=L^bx$ maps it bijectively to $\Lambda_{K-b,L^b\ell}$. Transfer a bond variable by $U'_{D_be}=U_e$. The map bijects positively oriented bonds and plaquettes, preserves their incidence, and satisfies $U'_{D_bp}=U_p$. Product normalized Haar measure is therefore unchanged, as is the Wilson action $A(U)=\sum_p[1-\frac{1}{2}\operatorname{Re}\operatorname{tr}U_p]$.

For a test function on T_ℓ, put $f^{[b]}(x')=f(L^{-b}x')$ on $T_{L^b\ell}$. Plaquette barycenters obey $x(D_bp)=D_bx(p)$, so

$$\mathcal{O}_{K-b,L^b\ell}(f^{[b]})(U')=\sum_p f(L^{-b}D_bx(p))a_p(U)=\mathcal{O}_{K,\ell}(f)(U).$$

No spacing prefactor appears in this observable. Consequently the two Wilson partition functions at the same β are equal, and all joint moments and cumulants satisfy the exact identity

$$S^T_{n;K,\ell,\beta}(f_1,\dots,f_n)=S^T_{n;K-b,L^b\ell,\beta}(f_1^{[b]},\dots,f_n^{[b]}).$$

This is one finite-dimensional integral written in two units, including the same number of sites in each direction. It is not a comparison between different bare measures.

For $\|f\|_\sharp=\max(\|f\|_\infty,c_M\operatorname{Lip}f)$ the exact norm transformations are

$$\|f^{[b]}\|_\infty=\|f\|_\infty,$$
$$\operatorname{Lip}f^{[b]}=L^{-b}\operatorname{Lip}f.$$

Thus $\|f^{[b]}\|_\sharp\le\|f\|_\sharp$ for $b\ge 0$, and it is at most $L^{|b|}\|f\|_\sharp$ for $b<0$. All support distances and support radii are multiplied by L^b. The same bond bijection carries a fixed-spacing DLR specification to its dilated specification, since it carries each finite integration set and all plaquettes meeting that set to their counterparts. It therefore also transports infinite-lattice Gibbs states and their local moment identities.

To identify this unit change with two initial portions of a renormalization run requires more: the blocking rules, gauges, auxiliary choices and extraction of increments must commute with the bond bijection. Likewise, torus-independent increments require a separately defined common local construction. Neither property follows from the scalar recursion or the exact dilation identity proved here.

35. The running coupling and first-passage arithmetic

Proof outline. The supporting analytic results used in this section, including the obligations that we must still prove, are listed under B, C, and E in the dependency catalogue. Their hypotheses and precise forms are retained at the points where the argument uses them.

(3a) The run, its coefficient, and the analytic requirements. The numerical running coupling and the auxiliary frame have different roles. This distinction is needed even for a real bare coupling. A frame fixes the tests, cubes, Gaussian references, coordinate choices and large-field operations. The weight of the Wilson action is another entry of the construction. A derivative with the frame fixed differentiates that weight; it does not differentiate a characteristic function whose threshold is moving. We first state the actual analytic requirements on these objects. The arithmetic consequences below are conditional theorems under those requirements.

Fix odd $L > 1$, the block parameter M, and the remaining geometric constants. In the ordinary-source application use its actual $M = L^{m_0}$, fixed collar layers and compatible cofinal tori; admissibility below refers to those assigned lattices. The group is SU(2); in the coefficient formulas we retain $N = 2$. The pointed families, hereditary complex potential/current cores, midpoint tents, companions and Euclidean list norms are those of Section 13. In particular a pointed family of level j has terms $E(X, P, z)$, where X is a connected localization domain and z is a unit lattice site in X. Its norm is a supremum in z, not a sum in z. The inequality

$$|E(X, P, z)| \le \mathcal{A} e^{-\kappa d_j(X)} \tag{RF1}$$

is required on its declared core. Summing domains requires the separate rooted-domain entropy estimate. A companion agrees on every common nonwrapping lift, with the required collar, for all admissible tori with the same frame prefix and initial weight. Local analyticity does not establish that compatibility.

Write $P_\circ = (\mathbf{1}, 0)$, and let $\mathfrak{h}_\infty$ be the infinite-volume first response of the background potential/current pair at $P_\circ$. Its existence, gauge covariance, affine reproduction modulo gauge and exponential decay belong to the background hypotheses. For the composed physical Hessian, average over an orthonormal color basis t_a, $1 \le a \le 3$, and set

$$\begin{aligned} \Pi(x, y) &= \frac{1}{3} \sum_{a=1}^{3} \sum_{z} \sum_{X \ni z} D^2 E(X, P_\circ, z)[\mathfrak{h}_\infty \delta_x^a, \mathfrak{h}_\infty \delta_y^a], \\ b(E) &= \sum_{u \in \mathbb{Z}^4} \Pi_{\mu\nu}(u) u_\mu u_\nu, \qquad \mu \ne \nu. \end{aligned} \tag{RF2}$$

The second line is used only after translation covariance and independence of the choice $\mu \ne \nu$ have been justified. Hessians are in the pair exponential chart, not in an unspecified coordinate system. Their directions have norm $\max\{\|A\|_\infty, \|\nabla A\|_\infty, \|J\|_\infty\}$. For $a' = \min\{a_1, c, a_0/C\}$, a ball of radius ca' in this norm must lie in the core. A ball controlled by $\|A\|_\infty$ alone does not supply the uniform mesh normalization.

The coefficient and pointed-cancellation theorem in Section 23 applies under its full response, summability and symmetry hypotheses. Its consequences used here are absolute convergence, vanishing zeroth and first moments, and

$$\begin{aligned} &\sum_{u} \Pi_{\mu\nu}(u) u_\rho u_\sigma = b(E)(\delta_{\mu\rho}\delta_{\nu\sigma} + \delta_{\mu\sigma}\delta_{\nu\rho} - 2\delta_{\mu\nu}\delta_{\rho\sigma}), \\ &|\Pi(x, y)| \le C(a')^{-2} \mathcal{A} e^{-c|y_- - x_-|}, \qquad |b(E)| \le C(a')^{-2} \mathcal{A}. \end{aligned} \tag{RF3}$$

The extraction is linear, holomorphic in admitted parameters and real on real families. It depends on the total, so two coherent localizations of the same physical functional have the same coefficient. A constant total has coefficient zero. At absolute radii the factor $(a')^{-2}$ is a constant depending

on M; at running radii it can cost $Cx_j(\log x_j)^{-2q}$. Neither the kernel nor its coefficient depends on which admitted radii were used to estimate it.

The pointed Wilson family assigns the action to the same tents as the regular member. Its norm is Ca_0^2, so subtracting its own extracted coefficient preserves a constant multiple of the family norm. This is the subtraction used below. The stronger statement that the remaining translation-invariant quadratic kernel is divisible by three lattice derivatives, with exponentially decaying quotient kernels on a fixed polyannulus, is a separate analytic target when that representation is used. The moment identity alone is not presented as a proof of such division.

Frame, weight, tube and box. The functions used for the auxiliary choices are

$$\begin{aligned} \varepsilon(y) &= A_0 y^{-1/2}(\log y)^{\mathfrak{p}_0}, & \delta(y) &= A_1 y^{-1/2}(\log y)^{\mathfrak{p}_0}, \\ \delta'(y) &= A_1 y^{-1/2}(\log y)^{\mathfrak{p}_1}, & \alpha_\iota(y) &= C_\iota y^{-1/2}(\log y)^{q}, \\ R(y) &= \min\{L^b : L^b \ge (\log y)^r\}, & q &> \mathfrak{p}_0 + 2r, \quad \mathfrak{p}_1 < \mathfrak{p}_0. \end{aligned} \tag{RF4}$$

The fluctuation scaling is $y^{-1/2}$. Gauge-fixing weights, large-field stage scalings, stage Wilson constants and reference laws also use the frame. Every occurrence of a coupling is assigned either to such a choice or to the action weight. No occurrence is assigned to both roles. Cube sizes and waiting depths use the running minimum of the frame.

A finite reference sequence $\theta_0, \dots, \theta_n$ is real, above its floor, and obeys

$$\lambda_\flat(j-i) - c_5 \le \theta_i - \theta_j \le \bar{B}(j-i), \qquad \check{R}_j = R(\min_{i\le j}\theta_i), \quad \check{R}_j \le L\check{R}_{j+1}. \tag{RF5}$$

In the run along this sequence,

$$x_0^\theta = \beta, \qquad x_{j+1}^\theta = x_j^\theta - b_{j+1}^\theta, \qquad z_j = \theta_j - x_j^\theta, \qquad \tau_j = x_j^\theta/\theta_j. \tag{RF6}$$

The frame and weights are separate lists. Put $\mathsf{t}^\sharp(y) = \varpi y(\log y)^{-2q_{\mathrm{t}}}$. A tube has positive, nonincreasing radii t_j, with $\mathsf{t}_j \le \mathsf{t}^\sharp(\theta_{j+1})$ for $j < n$. For a one-member sequence require $\mathsf{t}_0 \le 2\mathsf{t}^\sharp(\theta_0)$. The constants ϖ, q_{t} are fixed before the coupling ceiling, with $2\varpi \le \min\{\varpi_0, \sigma_\star\}$ and $q_{\mathrm{t}} \ge q_0$. The pair is alive through k if its coefficients through k are defined and $|z_j| < \mathsf{t}_j$. It has room r' when $|z_j| + r' \le \mathsf{t}_j$ at every required level.

The box of level k is the full inductive array: weights, regular families with companions and coefficients, remainders, boundary terms and standing large-field operations. It includes the bounds $|E^{(j)}| \le E_0 e^{-\kappa d_j}$, $|R^{(j)}| \le \theta_j^{-\kappa_0/2} e^{-\kappa d_j}$, the covariance and positivity clauses, identification with the two chains below, and the drift bounds. Its reserve constant c_{m}, also denoted c_{r}, enters the large-field mass bound; increasing it shrinks the box and does not change the step. An admissible torus must satisfy

$$\tfrac{1}{2}\ell L^{K-i} \ge C(L) M \check{R}_i \qquad (i \le k). \tag{RF7}$$

All box statements are simultaneous statements on these tori and on their common companions. The positivity of a label measure $\tau_{\mathcal{S}} = \sum_{\mathcal{S}(h)=\mathcal{S}} \tau_h$ must not be confused with the numerical ratio τ_j.

The Gaussian and one-loop requirements; statement 3.3. Let $\Phi_l = \log Z^{(l-1)} - \log Z^{(l-1)}(\mathbf{1})$. The localized Gaussian family is required to satisfy

$$\Phi_l(V) = \sum_X \sum_{z\in X} N^{(l)}(X, P_V, z), \qquad |N^{(l)}| \le E_{\mathsf{N}} e^{-\delta_{\mathsf{N}} M d_l(X)}, \qquad b(N^{(l)}) = b_l^0. \tag{RF8}$$

The constants depend only on L. No coupling occurs in these terms. Nonwrapping terms agree on all common lifts, and their pointed totals are covariant under every unit lattice isometry.

For precision, the localization meant here is the following actual one. Set $k = l - 1$, let $A_Y = C^{(k)}(Y)^{-1}$ be the supplied fluctuation form constructed inside Y and compressed to free

bonds at depth cM, and define

$$f_b(Y) = \operatorname{tr}(\log A_Y)(b,b),$$
$$E_{\mathsf{N}}(X;b) = \sum_{\substack{Y\subset X\\ b \text{ at depth } cM \text{ in } Y}} (-1)^{|X|-|Y|} f_b(Y). \tag{RF9}$$

The free-bond contribution to $N^{(l)}$ is $-[E_{\mathsf{N}}(X;P,b) - E_{\mathsf{N}}(X;P_\circ,b)]/2$, smeared by the midpoint tents. The pivot contribution is the logarithm of its actual elimination factor minus its value at $P_\circ$, smeared by the same tents. The required spectral gap and complex continuation of A_Y, the locality of its walk expansion and the c/M radius ceiling are part of this Gaussian-localization input. Finite Gaussian integration estimates alone do not prove them.

The distinction between an on-shell pair and a critical pair matters. On shell, the explicit current form expresses gauge covariance. Identifying it with the Hessian on a constrained fibre also requires the multiplier equation and actual criticality. The four tempered localization properties used in the one-loop analysis are its sum formula, common lifts, exponential domain decay on the shrinking supremum polydisc of radius $c\eta_l^2$, and invariance under constant gauge transformations. Uniform Hessian Cauchy estimates instead use the pair-direction norm preceding (RF3).

For obligation B3 we must prove, with the fixed momentum convention $(2\pi)^{-4}dq$,

$$c_0 = \frac{11N^2}{12\pi^2}\log L, \qquad \left|\sum_{l=1}^{k} b_l^0 - kc_0\right| \le c_2, \tag{RF10}$$
$$|b_{l+1}^0 - b_l^0| \le C_\alpha L^{-\alpha l} \quad (0<\alpha<1).$$

Thus $b_l^0 \to c_0$. The standard convention is $\beta = 2N/g_{\mathrm{std}}^2$, giving the increment $(11N/(24\pi^2))\log L$ of g_{std}^{-2} per factor L. The numerical value in (RF10) requires the actual lattice trace calculation. It is not deduced by quoting a continuum perturbative coefficient.

We retain the precise auxiliary targets of that calculation. A tempered functional is real analytic near the trivial field on arbitrarily large tori, gauge and lattice-isometry invariant, normalized at that field, and has a plane-wave Hessian per unit volume converging uniformly in momentum to a bounded holomorphic symbol on a fixed strip. For such functionals the background substitution is required to preserve the coefficient. If $\widehat{\pi}^F$ is the limiting bond Hessian symbol with lower-endpoint Fourier convention, its midpoint form obeys

$$e^{i(p_\mu - p_\nu)/2}\widehat{\pi}^F_{\mu\nu}(p) = b[F](|p|^2\delta_{\mu\nu} - p_\mu p_\nu) + O(|p|^4),$$
$$\widehat{\pi}^{F\circ p_k}(p) = L^2\widehat{\pi}^F(p/L) + O(|p|^3). \tag{RF11a}$$

The second formula is an actual background/average requirement; its coefficient $L^2 = L^{-1}L^{-1}L^4$ records two form scalings and the volume ratio. Uniform analytic control of the aliased modes is part of that requirement. The first formula follows from the moment identity and midpoint lattice symmetries. The constrained Gaussian logarithms Ψ_k then obey $\Psi_k = \Phi_k + \Psi_{k-1}\circ p_k$, where p_k is the preceding background map, and $\sum_{l\le k} b_l^0 = b[\Psi_k]$. The finite Schur representation uses

$$\mathbb{S}(X,Q) = \begin{pmatrix} X & Q^{\mathsf{T}} \\ Q & 0 \end{pmatrix}. \tag{RF11}$$

Here Q is the covariant form average, R is the covariant site average, D is the covariant gradient, $\Delta = D^{\mathsf{T}}D$ is the site Laplacian, Δ^{W} is the Wilson-action Hessian on forms, and $\mathcal{K} = \Delta^{\mathrm{W}} + DD^{\mathsf{T}}$, $\Xi = \mathcal{K} - D(1-P)D^{\mathsf{T}} + Q^{\mathsf{T}}Q$, where P is the orthogonal projection onto $\ker R$. On real backgrounds the representation to be justified is

$$\Psi_k = -\tfrac{1}{2}\log|\det\mathbb{S}(\Xi - Q^{\mathsf{T}}Q + \mathrm{j}[J,\mu], Q)| + \log|\det\mathbb{S}(\Delta, R)| - (\text{the same expression at } \mathbf{1}). \tag{RF12}$$

Here $U = U_k(V)$ is the actual critical background, $J = \nabla_{\mathrm{fine}} A(U) = J_{\eta_k}(U)$, and $\mu = \nabla_{\mathrm{block}}(A \circ U_k)(V)$. The convention is the bilinear color pairing $\langle B, C\rangle = \frac{1}{2}\operatorname{tr}(BC)$; fine-bond inner products have weight η_k^4, and block-bond inner products have unit weight. Explicitly,

$$\begin{aligned} \frac{d}{dt} A(e^{i\eta_k t B} U)\Big|_0 &= \eta_k^4 \sum_b \langle J(b), B(b)\rangle, \\ \frac{d}{dt} (A \circ U_k)(e^{itC} V)\Big|_0 &= \sum_c \langle \mu(c), C(c)\rangle. \end{aligned} \tag{RF12a}$$

All gradients and adjoints in (RF12) use those fixed conventions. The form $\mathrm{j}[J, \mu]$ is the current/multiplier correction obtained by second variation of the constrained background problem. Its explicit form and the cancellation of its quadratic momentum coefficient are established in (RF12b)–(RF12c). For complex continuation the absolute determinants in this real identity must be replaced by coherent logarithmic branches; an absolute value is not holomorphic.

The mass calculation uses finite determinants before any thermodynamic limit. Work here on the unit bare lattice. For an expanding box $\Lambda_R = [-R, R]^4 \cap \mathbb{Z}^4$, let ι_R^0 and ι_R^1 be zero extension from its sites and its positive bonds with both endpoints in the box. Put

$$\begin{aligned} U_f(x,1) = \exp(-ifx_2 t_3), \quad U_f(x,\nu) = I \quad (\nu \neq 1), \quad \tfrac{1}{2}\operatorname{tr}(t_3^2) = 1, \\ \Delta_{f,R} = \iota_R^{0*} \Delta_{U_f} \iota_R^0, \qquad \mathcal{K}_{f,R} = \iota_R^{1*} \mathcal{K}_{U_f} \iota_R^1. \end{aligned} \tag{RF13a}$$

These are Dirichlet compressions of the scalar and vector operators appearing in (RF12), before its averaged-constraint corrections. The vector Wilson Hessian is defined by second variation against finitely supported bond fields, so no infinite total action is differentiated. The oriented 12-plaquette is $\exp(ift_3)$. Adjoint charged colors therefore carry fluxes $2f$ and $-2f$, while the third color is neutral.

For each fixed $z > 0$ and finite box, take the real logarithms near $f = 0$, where the massive matrices are positive, and set

$$\begin{aligned} \mathcal{D}_R(f, z) &= \log\det(\Delta_{f,R} + zI) - \tfrac{1}{2}\log\det(\mathcal{K}_{f,R} + zI), \\ \widetilde{\beta}(z) &= \lim_{R\to\infty} \frac{1}{|\Lambda_R|} \frac{\partial^2}{\partial f^2} \mathcal{D}_R(f, z)\Big|_{f=0}. \end{aligned} \tag{RF13b}$$

These identifications can be obtained directly from the fixed-mass decay used in the calculation below. Centre a gauge at the point where the differentiated trace density is read. The two field derivatives then cost polynomial factors in relative distances, while the massive kernels decay exponentially. Terms affected by the boundary are therefore summable and their contribution per unit volume tends to zero along the stated box exhaustions. This gives the bulk coefficient and its independence of the exhaustion. The order is the derivative at fixed positive mass, then volume, then $z \downarrow 0$.

For the finite-volume normalized Hessian kernels the Bloch-symbol calculation below gives the translation-invariant form $\mathcal{L}(z)$. Decay makes the Bloch integrands holomorphic on a common momentum strip; their torus Riemann sums converge to the bulk integrals. The comparison concerns $\mathcal{L}^\sharp = \mathcal{L}(1) - \mathcal{L}'(1) + \mathcal{L}''(1)/2$, not $\mathcal{L}(1)$ alone, uniformly on the prescribed external-momentum strip. The required ultraviolet formulas for $N = 2$ are

$$\begin{aligned} \widetilde{\beta}(z) &= \frac{11}{6\pi^2}\log(1/z) + O(1), \\ |z\widetilde{\beta}'(z)| + |z^2\widetilde{\beta}''(z)| &\le C, \\ \sum_{l=1}^{k} b_l^0 &= \widetilde{\beta}(\eta_k^2) - \eta_k^2\widetilde{\beta}'(\eta_k^2) + \tfrac{1}{2}\eta_k^4\widetilde{\beta}''(\eta_k^2) + O(1). \end{aligned} \tag{RF13}$$

All field expansions are at fixed positive mass before taking that mass to zero. The vector form need not be nonnegative without its mass; the small-field hypothesis must make the massive form coercive. The comparison with the averaged constraints and gauge determinants can be carried out by the stopped resolvent expansion below. Its two local checks are the mixed energy bounds for the differentiated factors and the grouping of the finite magnetic remainder before mass differentiation. We give the estimates that make both checks uniform.

The geometric-convergence target can equivalently be organized through the Taylor triple $\mathfrak{J}_k = (\Delta_k, S_3^{(k)}, S_4^{(k)})$ of $\mathcal{S}_k = A \circ U_k$. With $\Lambda(B,\upsilon) = S(B) + \langle \upsilon, m(B)\rangle$, $\mathbb{G} = (D^2\Lambda(0))^{-1}$, $V_i = D^i\Lambda(0)$ and $\dot z = \mathbb{G}(0,C) = (HC, -\Delta_{k+1}C)$, the identity is

$$\begin{aligned}\Phi''_{k+1}(C,C') = {}& \tfrac{2}{3}\langle C, C'\rangle - \tfrac{2}{3}\langle HC, HC'\rangle \\ & - \tfrac{1}{2}\operatorname{Tr}(\mathbb{G}V_4[\dot z, \dot z']) + \tfrac{1}{2}\operatorname{Tr}(\mathbb{G}V_3[\dot z]\mathbb{G}V_3[\dot z']).\end{aligned} \tag{RF14}$$

Here $\eta_k = L^{-k}$ in the mass comparison, $S(B) = \mathcal{S}_k(e^{iB}) + \mathrm{GF}_k(e^{iB})$, $m(B) = i^{-1}\log\mathcal{M}(e^{iB})$, $\Delta_k = \mathcal{S}_k''(\mathbf{1})$, and $S_j^{(k)}$ is its degree-j Taylor form in the same chart. The vector HC is the linear constrained background response. Its uniform weighted-norm continuity and $\|\mathfrak{J}_{k+1} - \mathfrak{J}_k\| \le C_\alpha L^{-\alpha k}$ are the additional actual-background comparison hypotheses used here. These statements preserve the geometric comparison used at equal depth in the stronger matched-cutoff route.

The two-coupling Gaussian factor. At level k, write the fibre form as $\Delta = \Delta_A + \Delta_{\mathrm{GF}}$. The frame fixes the field scaling and gauge-fixing weight; the action weight supplies $\tau\Delta_A + \Delta_{\mathrm{GF}}$. The actual finite-dimensional hypothesis is that Δ_A vanishes on the block gauge directions and Δ_{GF} acts only on those directions, with a positive nondegenerate form on each complementary integration space. Under this splitting the two ordinary Gaussian integrals factor, giving

$$\begin{aligned}&\int dB\,\delta(QB)e^{-\langle B,(\tau\Delta_A+\Delta_{\mathrm{GF}})B\rangle/2} = \tau^{-d_l/2}Z^{(k)}(V),\\ &\int d\mu_{\tau^{-1}C^{(k)}}(B)e^{-(1-\tau)\langle B,\widehat G B\rangle/2} = \tau^{n_{\mathcal O}/2},\\ &\qquad n_{\mathcal O} = 3\#(\mathbb{T}^{(k)}\setminus\mathbb{T}^{(k+1)}),\quad d_l = 3|\mathfrak{X}_k| - n_{\mathcal O},\quad l = k+1.\end{aligned} \tag{RF15}$$

Here Q is the one-step average on the fibre, not the k-fold average, $\mathfrak{X}_k$ is the free-bond set, and $\widehat G$ is the gauge-fixing form in those coordinates. For positive real τ, the formula follows by rescaling only the non-gauge coordinates. Analytic continuation to the right half-plane uses the branch of $\log\tau$ real on the positive axis. Both dimensions are independent of the background.

Consequently the total of the extra Gaussian family G_τ is $n_{\mathcal O}\log\tau/2$. Its physical second derivatives vanish and $b(G_\tau) = 0$. Its norm may nevertheless be $K_{\mathbf M}|1-\tau|e^{-(1+c)\kappa d}$, with no small-coupling gain. This cancellation is needed to avoid a spurious accumulation of slopes of order θ_k^{-1}. Formula (RF15) presupposes the actual gauge splitting; it is not a consequence of a bound for a general complex Gaussian.

The complete regular family has the form $E^{(k+1)} = N^{(k+1)} + G_{\tau_k} + F$. Its physical step functional, up to field-independent normalizations, is

$$\begin{aligned}\mathcal{E}^{(k+1)}[\theta_k; x; S](V) = {}& \log\Big(\mathbf{T}_k^{\mathrm{sf}}[\theta_k]\, e^{-xA(U_k(\cdot))+S(U_k(\cdot))}\Big)(V)\\ & + xA(U_{k+1}(V)) - S(U_{k+1}(V)).\end{aligned} \tag{RF16}$$

The operator uses the whole torus as its small-field region, and S is the action of the incoming list. Equality between this physical functional and localized totals is asserted on their admitted backgrounds. It is not an identity of a global Gaussian at arbitrary independent-current pairs. The local formulas on such pairs have their own complex cores.

Lists, the two chains, and the increment as a function. A list contains regular families $G^{(j)}$ with their own action counterterms and unpointed remainders $H^{(j)}$. At the output radii $\bar{a}_\iota$ its required load is

$$\begin{aligned} \mathcal{N}(\mathfrak{f}) &= \sum_{j=1}^{k} \omega_{k+1-j} \|G^{(j)}\| + \sum_{j \text{ aged}} [\bar{a}/a_j]^2 \|H^{(j)}\|, \\ \omega_n &= (n+1)^{3/2} L^{-\alpha_{\mathrm{H}}(n-1)}, \\ |F(X)| &\le K_{\mathbf{M}} \{\delta(\theta_k) + \mathsf{s}\mathcal{N}(\mathfrak{f}) + e^{-c\theta_k \delta(\theta_k)^2}\} e^{-(1+c)\kappa d(X)}. \end{aligned} \tag{RF17}$$

The radius ratio is the maximum of the two coordinate-radius ratios; $\mathsf{s} = \delta(\theta_k)/\bar{a}_0$. The first, second and third terms pay respectively for the interaction beyond quadratic order, old-family differences, and failed Gaussian tests. The parameter choices used to infer a power bound for the last term must actually satisfy that power bound.

The regular part has a summable age weight because its own affine quadratic action is removed before localization. A remainder has no such counterterm, so its squared-radius load is not given an age decay. Sections 24, 27, 28, 29 and 30 prove the corresponding finite implications under their actual background, bond-split, potential, core and boundary hypotheses. Their general polynomial age factors do not automatically give the exact exponent 3/2 in (RF17). One must either verify that sharper estimate or allocate a smaller positive exponential rate and prove the resulting comparison.

A Euclidean list has one companion, the correct coefficients of its regular members, covariant pointed totals, and invariant remainder totals on the declared covariance class. Its radii of covariance obey

$$e_{j'} \le L^{2(j'-j)} e_j \qquad (j \le j'). \tag{RF18}$$

Closure means that the actual output again has these properties. Restriction to smaller cores does not prove covariance. The backgrounds of fluctuating fields must first lie in a common covariance class on an open real fluctuation ball; continuation then requires a connected analytic domain on which both expressions exist. The actual domain and naturality hypotheses remain part of closure.

The unmarked chain uses absolute radii and starts with the empty list. The marked chain uses running radii $\check{\alpha}_{\iota,j} = \min\{\alpha_\iota(\theta_{j-1}), \alpha_\iota(\theta_j)\}$ and inserts a remainder only when $M\check{R}_j \le L^{m+1-j}$ at step $m+1$. This age is distinct from the waiting depth of a large-field operation. Before that age the remainder lies outside the regular exponent and contributes to the remainder and boundary outputs.

Restriction from absolute to running radii must restrict the same actual functions. Under that compatibility the marked difference cancels N and G_τ. The required recurrence is

$$\begin{aligned} u_{m+1} &\le K_{\mathbf{M}\mathsf{s}}(\theta_m) \left(\sum_{i=1}^{m} \omega_{m+1-i} u_i + C_{\mathfrak{h}} \theta_m^{-p} \right), \\ p &= (\kappa_0 - 2)/2 > 1, \\ \sum_{j \le m} \frac{\alpha_0(\theta_m)^2}{\check{\alpha}_{0,j}^2} \theta_j^{-\kappa_0/2} &\le C_{\mathfrak{h}} \theta_m^{-p}, \\ C_{\mathfrak{h}} &\le C \left(1 + \frac{1}{\lambda_{\mathfrak{b}}(p-1)} \right). \end{aligned} \tag{RF19}$$

The lower flow bound pays one inverse-coupling power in this sum. With a constant frame its load can grow with the number of levels. Smallness of $K_{\mathbf{M}\mathsf{s}}$, relative to the summable kernel and

$C_{\mathfrak{h}}$, closes a weighted supremum induction. This is an actual marked-chain requirement, not a conclusion of an unmarked finite source derivative.

The resulting increment requirement, with independent deviations before imposing their recursion, is

$$\begin{aligned} b_{k+1} &= b^0_{k+1} + f_{k+1}(z_0,\dots,z_k) + q_{k+1},\\ \sup|f_{k+1}| &\le C\theta_k^{-1/2}(\log\theta_k)^{\mathfrak{p}_0} =: \epsilon_k,\\ \sup|q_{k+1}| &\le C_v\theta_k^{-p_R} =: \epsilon'_k, \qquad\qquad p_R = (\kappa_0-4)/2. \end{aligned} \tag{RF20}$$

For f, the supremum is on the whole product $|z_i| < \sigma_\star\theta_i$, with $0 < \sigma_\star \le 1/2$, and requires neither a lower flow bound nor remainders. For q, the frame obeys the weak flow bounds, $|z_i| < \varpi_0\theta_i(\log\theta_i)^{-2q_0}$ with $\varpi_0 \le \sigma_\star/2$, and the admitted aged remainder data range over their stated holomorphic parameter domain. Their bounds are $4\theta_j^{-\kappa_0/2}e^{-\kappa d_j}$, with (RF18). The functions are real on real data. Constants are independent of the frame length, torus, cutoff and admitted parameter; the weak triple affects their threshold.

There is also a weaker sufficient marked-multiplier estimate. If the marked exponent has norm $\mathfrak{n}_k \le Cg_k^{\kappa_0-5}$ and the actual step is holomorphic in its multiplier on a disc of radius comparable to $1/\mathfrak{n}_k$, Schwarz yields

$$|E^{(k+1),\mathfrak{R}}| \le C\mathsf{s}(x_k)\mathfrak{n}_k e^{-(1+c)\kappa d}, \qquad |b^{\mathfrak{R}}_{k+1}| \le Cg_k^{\kappa_0-7}. \tag{RF21}$$

This preserves an alternative sufficient error for the real flow; it does not prove the stronger exponent in (RF20).

Well-founded induction and statement 3.1. The substantive construction input says that, for an admitted box and frame prefix through k, $E^{(k+1)}$ and its coefficient are formed before any choices using θ_{k+1}. They use the creation radii $(1+\beta_{\mathrm{s}})\alpha_\iota(\theta_k)$. The terminal covariance radius is $e_{k+1} = c\varepsilon(\theta_k)$, so it also needs no next frame entry. Once the computed next weight is admitted, the actual **RT** operation closes the next box, consistently on every admissible torus and its companion. The standing-history kernel identity is $(m(\kappa\otimes 1))|_{\mathsf{C}} = (m|_{\mathsf{A}})\kappa$ for a measure on $\mathsf{A}\times\mathsf{B}$ and a kernel from A to C. Here κ denotes a kernel, not a decay rate. This identity preserves the history sum; it does not supply its quantitative mass or locality bounds.

Choose $c_5 = 2c_2$, $C_\beta \ge c_0 + 2c_2 + 1$, and first fix

$$\lambda_\flat = c_0/4, \quad \bar B = C_\beta + 1, \qquad \bar\varsigma = C'\gamma^{\kappa_0-8}, \quad \lambda_\sharp = c_0/2 - \bar\varsigma, \quad B_\sharp = C_\beta + \bar\varsigma. \tag{RF22}$$

The coupling ceiling is chosen so $\bar\varsigma \le \min\{c_0/4, 1\}$. The error exponent and other parameters must permit this ceiling; $\kappa_0 > 8$ is a sufficient choice for the displayed power of γ. An error estimate making that ceiling impossible cannot be used in this parameter selection. The marked-chain constants are computed at the weak triple before this choice. Then choose $Y_H = \gamma_H^{-2}$ above every required floor, with $\epsilon(y) + \epsilon'(y) \le \min\{c_0/4, 1\}$ and $\epsilon'(y) \le \bar\varsigma$ for $y \ge Y_H$. In particular $\gamma_H \le (\gamma^{-2} + c_5 + 1)^{-1/2}$. Increasing a reserve constant beyond $[0,1]$ is allowed only after its stated floor is increased.

Put $s_k = kc_0 - \sum_{l\le k} b^0_l$; then $|s_k| \le c_2$. For any formed prefix in the weak class,

$$\begin{aligned} x_i - x_j - (j-i)c_0 &= s_i - s_j + \sum_{i<l\le j}(b_l - b^0_l),\\ \tfrac{3}{4}c_0(j-i) - c_5 \le x_i - x_j &\le B_\sharp(j-i). \end{aligned} \tag{RF23}$$

The last upper bound uses the individual coefficient bound $|b^{\mathfrak{A}}| \le C_\beta$, $|b^{\mathfrak{R}}| \le \bar\varsigma$. The lower bound is stronger than the strong triple, which in turn implies the weak triple. Above a further floor, a bounded one-step drop changes the smallest power $R(y)$ by at most one factor L. Thus the computed next weight extends the frame while it remains above the floor. The telescoping

estimate supplies the numerical part of the induction. Each next step is formed under the separate analytic formation and closure hypotheses.

For $h = g^{-2} \geq Y_H$, define

$$K(\beta, g) = \min\{k : x_k(\beta) \leq h + B_\sharp\}, \qquad \mathcal{W}_k(g) = \{\beta \geq h + B_\sharp : K(\beta, g) = k\}. \tag{RF24}$$

The conditional flow statement 3.1 is that the run exists through this index, has the same coefficients on all admissible tori, and

$$\begin{aligned} \lambda_\sharp(j-i) - c_5 \leq x_i - x_j \leq B_\sharp(j-i) \qquad & (i \leq j \leq K), \\ h < x_K \leq h + B_\sharp, & \\ (\beta - h)/B_\sharp - 1 \leq K \leq 1 + (\beta - h - B_\sharp + c_5)/\lambda_\sharp. & \end{aligned} \tag{RF25}$$

At $\beta = h + B_\sharp$, $K = 0$; the terminal inequality still holds. For $K > 0$, the previous value exceeds $h + B_\sharp$ and the last decrement is at most $B_\sharp$, proving its strict lower endpoint. The lower drift forces finite exit. The two bounds for K follow by applying the upper drift to K and the lower drift to $K - 1$. Arbitrarily large β give arbitrarily large K, hence infinitely many nonempty $\mathcal{W}_k$. No stepwise positivity of b_k was used.

At depth s, write $x_{[s]} = x_{K-s}$. The exact envelopes are

$$h + \max\{0, \lambda_\sharp s - c_5\} < x_{[s]} \leq h + B_\sharp(s+1). \tag{RF26}$$

Put $\underline{x}_s = h + \lambda_\sharp s - c_5$, $\bar{x}_s = h + B_\sharp(s+1)$, and $g_-(g) = (g^{-2} + B_\sharp)^{-1/2}$. A diagonal subsequence of any first-passage sequence with $K \to \infty$ therefore converges at every fixed depth, to values in the closed versions of these intervals. Its couplings converge as well, with terminal value in $[(h+B_\sharp)^{-1/2}, g]$. A whole-sequence limit along tuned runs requires the later matched-cutoff input; compactness alone gives no such conclusion.

The arithmetic section, Section 34, proves the leading logarithm from vanishing mean errors, including bounded backward excursions. In the present notation its uniform form is

$$\begin{aligned} |x_{[s]} - x_{[0]} - s c_0| \leq \Delta_g(s), \qquad & \Delta_g(s)/s \longrightarrow 0, \\ \left| \frac{\beta}{K \log L} - \frac{11N^2}{12\pi^2} \right| \leq \frac{\Delta_g(K) + h + B_\sharp}{K \log L} & \longrightarrow 0. \end{aligned} \tag{RF27}$$

Here $K \geq 1$, $a_K = L^{-K}$. Denote the displayed relative error bound by δ_K, distinct from the fluctuation threshold δ_j. The function Δ_g can be chosen nondecreasing. Under the stated power/log error bound, with the marked exponent sufficient for the same estimate, one may take $\Delta_g(s) \leq C + C(h + c_0 s/2)^{1/2}(\log(h + c_0 s/2))^{\mathfrak{p}_0}$. The smaller ceiling $\gamma_{\mathrm{d}} \leq \gamma_H$ imposes the monotonicity and envelope floors for this quantitative bound; its constant is independent of g and s. For fixed g the relative error is $O_g(K^{-1/2}(\log K)^{\mathfrak{p}_0})$. For example $p_R > 1$, used below, suffices for the marked sum. The qualitative limit needs only an error tending to zero.

For the uniform depth statement, apply the arithmetic argument to the segment from $K - s$ to K. Its shifted one-loop partial sums have the same bound $2c_2$. If $\omega_*(y)$ is a nonincreasing envelope for the error bound on $x \geq y$, one may explicitly choose

$$\Delta_g(s) = 2c_2 + \sum_{u=0}^{s-1} \omega_*(h + \max\{0, \lambda_\sharp u - c_5\})\,. \tag{RF27a}$$

This is independent of the initial coupling and cutoff, nondecreasing, and sublinear by Cesaro averaging. Integral comparison for the square-root/log envelope gives the quantitative bound above.

The sharper two-sided logarithmic comparison retained in statement 3.1 is

$$\begin{aligned} &h - 2c_2 + \tfrac{1}{2}c_0(K-k) < x_k \le h + B_\sharp + 2c_2 + \tfrac{3}{2}c_0(K-k),\\ &|x_k - x_K - c_0(K-k)| \le \Delta_g(K-k). \end{aligned} \tag{RF28}$$

These are the logarithmic inequalities since $\log(L^k a_K)^{-1} = (K-k)\log L$. The upper slope in the first line uses the uniform error bound $|b_l - b_l^0| \le c_0/2$ and the partial-sum bound, not an assumption on the sign of each one-loop increment.

Scale comparisons and layered evaluation. These estimates retain the distinction between a numerical flow sum and an estimate on an actual field. Write $t_p(x) = x^{-1/2}(\log x)^p$, $p \ge 0$. If $-c \le x_m - x_n \le B(n-m)$ for $m < n$, then, above a floor depending on a freely chosen $\beta_0 > 0$,

$$\begin{aligned} t_p(x_n) &\le (1+\beta_0)(n-m)^{1/2} t_p(x_m),\\ t_p(x_m) &\le (1+\beta_0)^2 (n-m)^{\beta_0} t_p(x_n),\\ \check R_j/\check R_{j+1} &\in \{1, L\},\\ \sum_{j=0}^{k} g_j^p &\le C_p \left(1 + \frac{1}{\lambda_\sharp(p-2)}\right) g_k^{p-2} \qquad (p > 2). \end{aligned} \tag{RF29}$$

The last line also assumes the lower drift bound. The first two follow by comparing the square roots and logarithms above their monotonicity floor; a bounded backward excursion is absorbed by the chosen $1+\beta_0$. The third uses the running minimum and the bounded one-step drop. For the last, use $x_j \ge x_k - c_5 + \lambda_\sharp(k-j)$, absorb c_5 by the floor, and compare the decreasing power with its integral. The exponent $p = 2$ is excluded: that integral is logarithmic.

For the field estimate fix an admitted label through k, with layers $\Gamma_n = \Omega_n \setminus \Omega_{n+1}$, $\Omega_0 = T_\ell$ and $\Omega_{k+1} = \emptyset$. The determining sequence starts at $\Omega_1^\sim$. Let Λ_j^+ be Λ_j together with its level-j islands. Base points are the unit sites lying one layer of $M\check R_j$-cells inside this set. Let $\mathsf Y$ be any set of these points and $\mathsf Y_n = \mathsf Y \cap \Gamma_n$. The notation $|D|_n$ counts sites of $\mathbb T^{(n)}$ in D; it never counts cells.

Assume the actual window/core and boundary hypotheses for these backgrounds, Euclidean regular members of norm at most E_0, remainders of norm at most $g_j^{\kappa_0}$, and the stated boundary-domain decay. In particular the actual boundary terms, indexed by their recorded origins o, must satisfy

$$|B_{\mathcal S}(j, X, o)| \le C_B e^{-\kappa d_j^\natural(\mathcal S; X)}. \tag{RF30a}$$

The metric is the one in (R23) of Section 13: the minimizing tree represents each chosen standing-island hull by one member, rather than filling it with all of its cubes. It is not ordinary d_j-decay

across an island. The retained layered requirements are

$$\begin{aligned}
&\left|\sum_{z\in\mathsf{Y}}\left\{\sum_{\substack{X\ni z\\ X\subset\Lambda_j^+}}[E^{(j)}(X,U_k,z)-E^{(j)}(X,\mathbf{1},z)]-b_jA(h_z,U_k)\right\}\right|\\
&\qquad\le E_1\sum_{n=j}^{k}(1+n-j)L^{-\beta_T(n-j)}L^{-4(n-j)}\#\mathsf{Y}_n,\\
&\sum_{\substack{X\in\mathcal{R}_j\\ X\cap\Omega\neq\emptyset}}|R^{(j)}(X,U_k)-R^{(j)}(X,\mathbf{1})|\\
&\qquad\le R_1g_j^{\kappa_0}\sum_{n=j}^{k}(1+n-j)|\Gamma_n\cap\Omega^{\sim}|_n.
\end{aligned}\tag{RF30}$$

Here $0<\beta_T\le\alpha_{\mathrm{H}}<1$ is fixed, $\mathcal{R}_j$ consists of domains outside Z_j^+, and Z_j^+ is the stored large-field region enlarged by one cell layer. The set Ω is a union of level-k localization cubes and $\Omega^{\sim}$ has one level-j localization layer added. The constant R_1 in this display is not the geometrical gap parameter. Constants may depend on β_T, L, N and the construction parameters, but not on j,k,K,ℓ, the label or the positions.

For clarity, the local estimate behind the first line, when its actual background hypotheses have been verified, is

$$\begin{aligned}
&\left|\sum_{X\in\mathcal{X}}[G(X,U,z)-G(X,\mathbf{1},z)]-b(G)A(h_z,U)\right|\\
&\qquad\le C\mathsf{g}\left\{(\varepsilon_z/a_0)^2\sigma^{4+\beta_T}+e^{-\kappa\min\{\mathsf{D},c/\sigma\}/2}\right\},\qquad \sigma=L^{-(n-j)}.
\end{aligned}\tag{RF31}$$

The member has norm g on radii $a_0\le a_1$, with $a_0\le 1$; the datum is ε_z-regular near z, $\varepsilon_z\le c$, and $\varepsilon_z\sigma\le ca_0$. Every term is evaluated in its actual core. The collection $\mathcal{X}$ contains all domains within D layers of z. With complete near coverage to distance c/σ, the right side is bounded by $C\mathsf{g}(1+(\varepsilon_z/a_0)^2)\sigma^{4+\beta_T}$. The local cancellation and nonlinear remainder sections supply the analytic implication under their full hypotheses, including the boundary-response compatibility and the own tent counterterm. Checking those hypotheses on this layered background remains part of the actual construction.

The tail in (RF31) has no factor $(\varepsilon_z/a_0)^2$. For example a wrapping holonomy term, multiplied by its allowed exponentially small domain weight, can have zero nonwrapping companion and coefficient zero while its normalized value is nonzero on a flat torus field. A curvature amplitude cannot control that topological contribution.

To pass to (RF30) one also needs the actual geometry: $\varepsilon_z\le C\check{R}_n^2\varepsilon_n\le c\alpha_{0,n}$, the comparison $\alpha_{0,n}\le C(2+n-j)^{1/2}\check{\alpha}_{0,j}$, and the available collar depth. The condition $q>\mathfrak{p}_0+2r$ pays the first comparison. The factor $L^{-4(n-j)}$ is used entirely by converting level-j sites to level-n sites. For remainders the corresponding near-domain Cauchy bound is

$$|R^{(j)}(X,U_k)-R^{(j)}(X,\mathbf{1})|\le C(d_j(X)+3)^2L^{-4(n-j)}\left(\frac{\check{\alpha}_{0,n-1}}{\alpha_{0,j}}\right)^2g_j^{\kappa_0}e^{-\kappa d_j(X)}.\tag{RF32}$$

Far domains use their norm, not this amplitude estimate. The second line of (RF30) also holds with $1+n-j$ replaced by $\varpi_{jn}=\max\{1,(x_j/x_n)^2\}$. Both dominate the required squared-radius ratio up to fixed constants. The two resulting summations give $|\mathbf{R}_k(U_k)|\le C\sum_n g_n^{\kappa_0-6}|\Gamma_n|_n$.

The regular counterterm in (RF30) is the sum of its own tents. An arbitrary smooth weight equal to one on Ω differs from that sum on its collar; it produces a new action term without an age gain. For the actual profile φ_j, $\varphi_j h_z = h_z$ at every base point. The remaining rim terms obey $0 \le A(\varphi_j h_y, U_k) \le C\alpha_{0,j}^2$. The needed geometrical count is $\sum_j \#_j(Z_j^+) \le C\sum_n |\Gamma_n|_n$. It includes the stored collars of healed components, not just the re-done small-field regions. This window and history count is a separate geometric hypothesis.

Here $\mathbf{R}_k$ denotes the sum of the normalized remainder differences on the label. The net regular action $\mathbf{E}_k$ includes its profile counterterms and rim terms. With those counts and boundary sums, $|\mathbf{E}_k(U_k)| \le C\sum_n |\Gamma_n|_n$. For admitted complex perturbations of total coefficient size $\bar{c}$, regular norms at most $E_0 + 1$, remainder norms at most $4g_j^{\kappa_0}$, and the same boundary hull bounds, the sum of the elementary moduli is at most $C(1+\bar{c})\sum_n |\Gamma_n|_n$. This is a pointwise estimate on the admitted arrays; it is not an assertion that arbitrary such arrays are produced by the RG map. Its constants and evaluation floor need no new dependence on $\bar{c}$. The variable action coefficient is

$$x_k(y) = \beta - \sum_{j\le k} b_j \varphi_j(y), \qquad |x_k(y) - x_k| \le B_\sharp \#\{j \le k : \varphi_j(y) < 1\}. \tag{RF33}$$

It equals x_k one coarse cell inside all the small-field regions. Let $A_{k,\mathcal{S}}$ be the ACTUAL label exponent and $\mathcal{E}_k$ its once-counted field-independent number. The sign convention is $\mathcal{E}_k = -E_{(k)}$, where $E_{(k)}$ is the accumulated log vacuum normalizer of (CP1). Thus the constant part of the exponent is $E_{(k)}$. The exact density format, which is a construction input, identifies the sum of the elementary regular, remainder, rim and boundary terms with the expression on the left of

$$|A_{k,\mathcal{S}}(U_k) + \mathcal{E}_k + A(x_k(\cdot), U_k)| \le C(1+\bar{c})\sum_n |\Gamma_n|_n. \tag{RF33a}$$

Their modulus sum proves this consequence on the admitted backgrounds. Defining an exponent by a bound would not prove the required exact density format.

35.1. **The constrained determinant and the logarithmic coefficient.** The one-loop sum is computed before choosing a localisation of its logarithm. This separates the coefficient from the individual polymer terms and keeps the averaging and gauge determinants in the calculation.

Let Ψ_k be the normalized logarithm of the Gaussian integral over the linearized fibre of the k-fold average at the actual critical background. With every delta-function Jacobian retained, Gaussian integration in stages is the Schur-complement identity

$$\Psi_k = \Phi_k + \Psi_{k-1} \circ p_k.$$

Here p_k is the preceding critical block field. The linearized average annihilates the aliased modes at zero momentum, while its unaliased branch reproduces a constant form. Its two responses and the volume ratio give

$$\widehat{\pi}^{F\circ p_k}(p) = L^2 \widehat{\pi}^F(p/L) + O(|p|^3), \qquad L^2 = L^{-1}L^{-1}L^4.$$

Taking the quadratic coefficient therefore gives $b[F\circ p_k] = b[F]$ and $\sum_{l\le k} b_l^0 = b[\Psi_k]$. The uniform strip estimates below justify the Hessians, the infinite-volume limit and the alias expansion in this calculation.

The constrained Gaussian integral of a quadratic form T with linear constraint M is, up to a numerical power of 2π,

$$\left|\det \begin{pmatrix} T & M^{\mathsf{T}} \\ M & 0 \end{pmatrix}\right|^{-1/2}.$$

The formula is valid when M is onto and T is positive on its kernel. It follows by splitting off $\ker M$ and changing variables on its orthogonal complement; the determinant of that change is exactly the delta-function Jacobian. The same splitting, with the residual gauge directions separated

first, proves the linear Faddeev–Popov identity. In this identity the constrained integral, the determinant along residual gauge directions and the determinant compensating a coarse gauge displacement of the constraint must be changed together. The axial gauge is triangular along the levels and its gauge determinant is one. Changing to the covariant background gauge and straightening the constraint consequently gives the saddle determinant in (RF12).

There are two current terms to retain. Gauge invariance at a field with nonzero current relates the Wilson Hessian on a gauge direction to a commutator paired with that current. Curvature of the averaging fibre contributes the second differential of the average paired with its multiplier. Equivalently, one first takes the Hessian of the actual constrained Lagrangian, rather than restricting the Wilson Hessian to a linear space, and then changes gauge. This is the origin of $\mathfrak{j}[J,\mu]$ in (RF12). Its value must be the one produced by these operations.

At the trivial field the residual gauge space is $\ker R$. The projection in its gauge term is the orthogonal projection P onto this space; it is not the Landau projection onto $\Delta\ker R$ used in the propagator construction. The flat operator

$$\Xi_0 = \Delta^{\mathrm{W}} + DPD^{\mathsf{T}} + Q^{\mathsf{T}}Q$$

has a uniform comparison with $\Delta^{\mathrm{F}}+1$, where $\Delta^{\mathrm{F}} = \Delta^{\mathrm{W}} + DD^{\mathsf{T}}$:

$$c(\Delta^{\mathrm{F}}+1) \le \Xi_0 \le C(\Delta^{\mathrm{F}}+1).$$

The curl controls the transverse modes. On gradients the identity $QD = D_{\mathrm{block}}R$ and the projected gauge term control, respectively, the block mean and the zero-mean part. The only remaining constant mode is detected by Q. This is also the comparison needed for the site-field saddle determinant.

In level-k units take Bloch symbols, whose fine-cell dimension is $O(\eta_k^{-4})$. A complex momentum shift is conjugation by an exponential weight. The averages are bounded under this conjugation, and the sandwiched operator $(\Delta^{\mathrm{F}}+1)^{-1/2}\Xi_0(\Delta^{\mathrm{F}}+1)^{-1/2}$ is bounded on a fixed strip. Its variation under an imaginary shift is $O(|\mathrm{Im}\,p|)$. The real coercivity therefore gives invertibility on a smaller strip of width independent of k. The symbols of the linearized background have summable Fourier modes. Covariant averages at a small complex potential obey

$$\|Q_k(U)\| \le \prod_{i<k}(1 + CL^{i-k}\|A\|_\infty) \le e^C.$$

Thus differentiating their finite jet does not introduce a factor growing with the number of levels. Fourier summation on a torus then differs from the Bloch integral by $O(e^{-c\mathfrak{s}})$, where $\mathfrak{s}$ is its side in these units.

The current and multiplier correction. We compute its coefficient at each fixed depth k. Write $(\cdot,\cdot)_{\mathrm{f}}$ and $(\cdot,\cdot)_{\mathrm{c}}$ for the fine and coarse bond pairings in (RF12a), and retain R_U for the site average. The Landau projection and the projection of forms to its slice are

$$\begin{aligned} G'_U &= (\Delta_U + R_U^{\mathsf{T}}R_U)^{-1}, \\ \mathcal{R}_U &= 1 - G'_U R_U^{\mathsf{T}}(R_U G'^2_U R_U^{\mathsf{T}})^{-1} R_U G'_U, \\ \lambda_B &= G'_U \mathcal{R}_U D_U^{\mathsf{T}} B, \qquad\qquad \Pi_U B = B - D_U\lambda_B. \end{aligned}$$

Here $\mathcal{R}_U$ projects onto $\Delta_U\ker R_U$; it is different from P_U, which projects onto $\ker R_U$ in (RF12). In particular, $R_U\lambda_B = 0$ and $Q_U\Pi_U = Q_U$. Let $C_U^{(2)}$ be the quadratic homogeneous part of the actual k-fold straightened average, normalized so that its linear part is Q_U. All its accumulated node factors are removed before taking this Taylor expansion.

For a site field λ, let $\lambda_+(b)$ denote its value at the positive endpoint transported to the negative endpoint by $U(b)$. With $\lambda = \lambda_B$, put

$$E_U(B) = i[\lambda, B] - i[B, \lambda_+] - i[\lambda, D_U\lambda],$$

where every bracket is taken at the negative endpoint. The second variation of gauge invariance gives the exact identity

$$(\Pi_U B, \Delta_U^{\mathrm{W}} \Pi_U B)_{\mathrm{f}} - (B, \Delta_U^{\mathrm{W}} B)_{\mathrm{f}} = -(E_U(B), J(U))_{\mathrm{f}}.$$

The covariant difference contains the fine spacing: the transported endpoint difference is $\lambda_+ - \lambda = \eta_k D_U \lambda$. Thus the same identity follows from the mixed gauge variation

$$(B, \Delta_U^{\mathrm{W}} D_U \lambda)_{\mathrm{f}} = \tfrac{i}{2}(J(U), [\lambda + \lambda_+, B])_{\mathrm{f}};$$

applying it to B and $D_U\lambda$ gives the displayed difference of Hessians because $[\lambda_+ - \lambda, D_U\lambda] = 0$.

The curved constraint contributes $-2(\mathsf{H}_U C_U^{(2)}(\Pi_U B), J(U))_{\mathrm{f}}$. On the critical branch, $J(U) = Q_U^{\mathsf{T}}\mu$ and $Q_U \mathsf{H}_U = 1$, so this term is $-2(C_U^{(2)}(\Pi_U B), \mu)_{\mathrm{c}}$. The first identity also holds for the straightened average: its linear part differs from the plain one by a coarse gradient, which pairs to zero with μ since $D^{\mathsf{T}}_{\mathcal{M}^k U}\mu = 0$. Consequently the actual correction, extended to independent J and μ, is the symmetric form defined by polarization of

$$(B, \mathfrak{j}_U[J,\mu]B)_{\mathrm{f}} = -(E_U(B), J)_{\mathrm{f}} - 2(C_U^{(2)}(\Pi_U B), \mu)_{\mathrm{c}}. \tag{RF12b}$$

This is the invariant extension of the Wilson Hessian followed by the second variation of the averaging constraint. The exact change of gauge then allows the soft gauge term $D_U P_U D_U^{\mathsf{T}}$ in (RF12); it does not replace $\mathcal{R}_U$ by P_U in Π_U.

For the coefficient, it is essential to keep the two currents independent until after differentiating. Set $K_U = \Delta_U^{\mathrm{W}} + D_U P_U D_U^{\mathsf{T}}$ and $z = (J, \mu)$. On each finite torus define, near $(\mathbf{1}, 0)$,

$$F_k(U, z) = -\tfrac{1}{2} \log \frac{\det \mathbb{S}(K_U + \mathfrak{j}_U[z], Q_U)}{\det \mathbb{S}(K_U, Q_U)}.$$

The logarithm is its analytic branch with $F_k(U, 0) = 0$. The site determinant cancels in this difference. At fixed k, invertibility at the trivial field gives analyticity on each finite torus. The kernels of local variations have thermodynamic limits and decay exponentially. These kernels define the normalized plane-wave jets; no total infinite-volume logarithmic determinant is needed.

Every term in (RF12b) is covariant under a background gauge transformation, also for independent J, μ. Indeed, the parallel transports, the removed node factors, G'_U, $\mathcal{R}_U$ and Π_U conjugate under that transformation; taking the quadratic part preserves the same rule for $C_U^{(2)}$. Both saddle matrices therefore conjugate on their fine and coarse spaces. Their determinants, and hence F_k, are invariant. This uses background gauge covariance, which also holds for K_U without the current correction.

Define $w_J(U)$ and $w_\mu(U)$ by

$$D_z F_k(U, 0)[j, \nu] = (w_J(U), j)_{\mathrm{f}} + (w_\mu(U), \nu)_{\mathrm{c}}.$$

These derivatives are the actual determinant traces:

$$D_z F_k(U, 0)[\zeta] = -\tfrac{1}{2} \operatorname{Tr}\left[\mathbb{S}(K_U, Q_U)^{-1} \begin{pmatrix} \mathfrak{j}_U[\zeta] & 0 \\ 0 & 0 \end{pmatrix} \right].$$

They transform as adjoint fields at the corresponding endpoints. At $U = \mathbf{1}$, constant gauge transformations fix the background. There is no nonzero adjoint-invariant vector in $\mathfrak{su}(2)$, so $w_J(\mathbf{1}) = w_\mu(\mathbf{1}) = 0$. Differentiating their covariance under a local gauge transformation now gives, for either field w,

$$D_U w(\mathbf{1})[d\lambda] = 0.$$

This is the gauge identity on the other leg of the mixed current–background derivative. The gauge variation of the current variable contributes only a commutator with $w(\mathbf{1})$, which is zero.

At fixed k, exponential decay allows a linear site function λ in this identity by cutoff approximation. Its gradient is an arbitrary constant fine form. The mixed kernel thus annihilates constant

forms as well as gradients. The zero-momentum linear background of a constant coarse form is constant modulo a gradient. Its mixed symbol consequently vanishes at $p = 0$; analyticity makes the composed symbol $O_k(|p|)$.

The first variations of the physical current and multiplier are $O_k(|p|^2)$. This holds in every fine cell, including the aliased modes. In the linearized problem at $\mathbf{1}$, the symmetric averages reproduce affine potentials, whose minimizers have zero current and multiplier. Their linearized current kernels therefore annihilate affine coarse potentials, so both their constant and linear moments vanish. A change of the fine gauge does not alter this conclusion, since $J(\mathbf{1}) = 0$ and $DJ(\mathbf{1})d\lambda = 0$.

Put Ψ_k^0 equal to (RF12) with $\mathfrak{j}$ omitted and substitute the physical currents only now. In the second-variation chain rule for $\Psi_k - \Psi_k^0$, every second variation of a current is paired with $w(\mathbf{1}) = 0$. Terms with two first current variations are $O_k(|p|^4)$; mixed terms have orders $O_k(|p|)O_k(|p|^2) = O_k(|p|^3)$. There is no term with two background variations because $F_k(U, 0) = 0$ for every U. Hence

$$\widehat{\pi}^{\Psi_k - \Psi_k^0}(p) = O_k(|p|^3), \qquad b[\Psi_k] = b[\Psi_k^0]. \tag{RF12c}$$

The thermodynamic limit and the coefficient are taken at fixed k. The equality of coefficients is exact for every k; it requires no bound uniform in k on the discarded cubic remainder. The uniform massive comparison is applied to Ψ_k^0 after this cancellation.

To compare the current-free saddle determinant with the massive determinant pair, interpolate in the averaging perturbation and in the mass. The perturbation is

$$Q^{\mathsf{T}}Q - D(1 - P)D^{\mathsf{T}},$$

with the corresponding site-field perturbation. Put $\mathcal{W}_{\mathrm{F}} = (\Delta^{\mathrm{F}} + 1)^{1/2}$. The flat comparison above controls the interpolated inverse between its two energy weights. Its continuation to the common smaller momentum strip follows from the same bounded-conjugation argument.

After shrinking the common strip, the linearized background has uniformly summable Fourier modes there. Its plane-wave variations therefore enter the fixed potential ball with a bound independent of the bare spacing. The average factors must be differentiated before estimating the trace. Their bound on the fixed complex potential ball above gives, by Cauchy's formula on a smaller ball of radius $h > 0$,

$$\|D_A^j Q_k\| \le C_j h^{-j}, \qquad j = 0, 1, 2,$$

uniformly in k; the site average is treated in the same way. This differentiates one bounded holomorphic map, so it avoids a loss for each of its k factors. The factors from the nonlinear averages carry L^{i-k}, whose sum is bounded. For the inverse use $D(A^{-1}) = -A^{-1}(DA)A^{-1}$ and its second derivative. In these products every differentiated precision is paired with the adjacent $\mathcal{W}_{\mathrm{F}}^{-1}$ factors. We must check the resulting finite list of mixed energy bounds on this same strip, with the actual average and constraint jets. This is the norm check needed in addition to the coarse-rank count.

A trace containing an average can then be taken over the coarse cell, whose dimension is bounded independently of k. The remaining resolvent expansion is stopped when it has three free propagators. Its four-dimensional Schatten majorant is

$$\int \|\widehat{G}(q)\|_{\mathfrak{S}^3}^3 dq \le C \int_{\mathbb{R}^4} (1 + c|q|^2)^{-3} dq < \infty, \qquad G = (\Delta^{\mathrm{F}} + 1)^{-1}.$$

The lower powers are not uniformly trace class as the fine spacing tends to zero. Retain them exactly as the first two mass derivatives of the determinant pair. Thus the explicit part is

$$\mathcal{L}^\sharp = \mathcal{L}(1) - \mathcal{L}'(1) + \tfrac{1}{2}\mathcal{L}''(1).$$

The coarse traces and the stopped remainder give a bounded holomorphic difference on the common external-momentum strip, once the indicated mixed bounds are inserted. Cauchy's estimate then bounds its quadratic coefficient uniformly in k.

The divergent coefficient is computed at fixed positive mass on a constant Abelian field. Write $\psi(q) = \sum_\mu (2 - 2\cos q_\mu)$ and $\Upsilon = (\psi + z)^{-1}$. For sufficiently small plaquette flux relative to z the vector operator plus z is positive: its possibly negative part is bounded by $C|\sin f|$. Differentiation at $f = 0$ is therefore legitimate at each finite volume. Massive decay controls the linearly growing vector potential and removes boundary contributions after division by the volume.

The neutral color contributes no flux dependence. The charged colors carry fluxes $\pm\varphi$, with $\varphi = 2f$, and their covariant shifts satisfy $u_1 u_2 = e^{i\varphi} u_2 u_1$. The second-order magnetic product is

$$a \star b = ab - \tfrac{i}{2}\varphi\{a, b\} - \tfrac{1}{8}\varphi^2\{\{a, b\}\} + O(\varphi^3).$$

At zero flux the vector symbol is ψI. Solving for its resolvent through flux degree two uses only Υ, momentum derivatives of ψ, and the first two flux coefficients of the finite-range Wilson symbol. These coefficients are smooth trigonometric matrices independent of z. The cross terms from the first magnetic bracket cancel because Υ is a scalar function of ψ. Integrating the mass derivative of the logarithm and then integrating by parts in the periodic momentum groups the second bracket into terms with a smooth numerator times Υ^2; the other terms have a bounded numerator times Υ, or are the vector spin term with Υ^2.

Isolate the numerator at $q = 0$ in every Υ^2 term. The remaining numerator is $O(|q|)$, and hence its absolute integral is bounded uniformly for $0 < z \le 1$, since

$$\sup_{0<z\le 1} \int_{[-\pi,\pi]^4} \left\{ |q|(\psi + z)^{-2} + (\psi + z)^{-1} \right\} dq < \infty.$$

Here $\psi(q)$ is comparable to $|q|^2$ near its only zero. The constant numerators give

$$\widetilde{\beta}(z) = 8(4 - \tfrac{1}{3}) \int_{[-\pi,\pi]^4} \frac{dq}{(2\pi)^4} (\psi(q) + z)^{-2} + r(z) = \frac{11}{6\pi^2} \log(1/z) + O(1).$$

The 4 is the spin contribution from the vector Hessian, read from its intertwining with the covariant gradient on the critical constant field. The 1/3 is the orbital contribution of the second magnetic bracket at $q = 0$. To finish this finite calculation, we must match the actual vector numerators to the grouping just described, retaining its absolute integral bounds.

The same integral bounds control the mass derivatives of r. Indeed, for a mass-independent numerator $F(q)$,

$$z\partial_z\{F(\psi + z)^{-m}\} = -m \frac{z}{\psi + z} F(\psi + z)^{-m},$$
$$z^2\partial_z^2\{F(\psi + z)^{-m}\} = m(m + 1) \left(\frac{z}{\psi + z}\right)^2 F(\psi + z)^{-m}.$$

Both extra ratios are bounded. If a rearrangement introduces explicit polynomial powers of z, the finite Leibniz rule gives the same absolute integral bounds. Differentiating the grouped integrals therefore gives $|zr'(z)| + |z^2 r''(z)| \le C$ as well as $|r(z)| \le C$. For the leading integral each mass derivative adds a resolvent, and $\int \Upsilon^j dq \le C z^{2-j}$ for $j = 3, 4$. Consequently $|z\widetilde{\beta}'(z)| + |z^2\widetilde{\beta}''(z)| \le C$. For $z = \eta_k^2$, these bounds, the exact cancellation (RF12c) and the uniform current-free remainder give (RF13), and hence the bounded telescoping error in (RF10).

(3b) A common companion and the torus floor.

Statement 3.4 and its compatibility hypotheses. For each fixed frame and weight prefix there must be a single family $\widehat{E}^{(j)}(\widehat{X}, P, z)$ on the infinite level-j lattice, and likewise a companion for $R^{(j)}$. For every admissible (ℓ, K'), $K' \geq j$, and every label, it agrees with the torus term when the projection of $\widehat{X}$ is injective with its margin and lies one full level-j cell layer inside Λ_j. A sufficient margin places $\widehat{X}$ in a rectangle with sides less than half the corresponding torus side. The admissibility used here is

$$\ell L^{K'-i} \geq 2L^{11} M \check{R}_i \qquad (i \leq j). \tag{RF34}$$

The family has the same analytic bounds and cube-translation covariance on every such lift. Its coefficient b_j is a function only of the frame and weights through level $j-1$.

Uniqueness follows because every bounded domain, collar included, embeds in arbitrarily large admissible tori. Existence requires more. First, the walk, propagator, localized background and gauge choices must be made once by a rule natural under covering maps, cube translations and the later dilations. Independently chosen objects on separate tori cannot be compared. Second, each local formation formula must use only its supplied neighborhood, earlier terms restricted there and the common numbers x_i, b_i. The induction is simultaneous over tori of unbounded side; it does not define b_j using only domains that happen to fit a particular torus.

Third, formation must be local in the history trace on the prescribed one-layer windows X_i^+. A chart identifying these traces, the bare weight, prefix coefficients and incoming local terms identifies the newly formed term on the common local core. In the one-layer interior the standing history has empty trace: healed components have already been absorbed into local terms. This actual standing-history property is what makes that interior term independent of the label. It is not implied by an inequality for the modulus of an activity.

The order $E^{(j)}, b_j, x_j, R^{(j)}$ avoids a circular definition. In particular the new large-field weight x_j does not enter the preceding extraction of b_j. We must prove this compatibility for the full interacting construction. The finite formulas proved elsewhere can verify pieces of it; their holomorphy alone does not prove it.

Under these conditions, the coefficient of the physical totals, taken in the large-torus limit, equals the companion coefficient in (RF2). Use the all-small-field totals for their unit-lattice symmetries; a truncated label total has no such invariance in general. On nonwrapping domains the Hessians agree by compatibility and the torus response converges to $\mathfrak{h}_\infty$. Domains escaping a large fixed box carry an exponential tree-length tail. The rooted entropy is absorbed by the chosen κ; response decay controls remote differentiations. Dominated convergence therefore passes the differentiated invariant totals to the companion kernel. This proves the sum-rule identification, including that a constant total has coefficient zero. It requires the uniform response convergence and domain bounds just stated.

Wrapping domains have $d_j(X) \geq c\ell L^{K-j}/M$. Their exponential decay is used directly and does not define an additional renormalized coefficient. For a self-calibrated run, the cube ratio shows that (RF34) at the terminal level implies it at all preceding levels. Since $x_K \leq h + B_\sharp$, it suffices that

$$\ell \geq \ell_\star(g) := 2L^{11} M R(g^{-2} + B_\sharp). \tag{RF35}$$

Write $\mathfrak{L}(g)$ for the compatible torus sides above this floor. The same floor works on the reference line, whose terminal value is h. The side is selected after g; the cutoff K remains unrestricted. This statement is a uniform construction requirement, not an assertion of infinite-volume uniqueness.

(3c) Frozen choices, slopes, and tuned bare couplings. The reference line for a run of length K is

$$\theta_k = h + c_0(K-k), \qquad h = g^{-2}. \tag{RF36}$$

Equal depth in runs of lengths K and $K+1$ gives equal frame numbers, lattices, partitions, radii and cube sizes. It does not give equal tests, since their backgrounds live over different bare lattices, and it identifies no labels or histories between those runs. The line beginning at depth s is the reference line of $g_{[s]} = (h + c_0 s)^{-1/2}$.

For fixed choices, every inserted characteristic function is the same real Borel function of its integration fields. The actual operations act linearly on the initial density. If their required dominated analytic estimates hold, applying them to $e^{-(\beta-\zeta)A}$ gives holomorphic measures and local families in ζ. No derivative of a cutoff is taken. Continuity of the real run follows from the actual fixed-frame input-list Lipschitz estimate while the arrays retain their tube room. It is continuity, not a contraction. When each bare coupling chooses its own frame, the tests and domains can change discontinuously and this argument does not apply.

The exact decompositions of unity and normalized gauge factors imply that the bare measure, partition function and observable integrals are independent of admitted auxiliary choices. Intermediate densities after a large-field operation need not coincide. Comparing their extracted couplings requires E4. The retained self-calibrated versus reference-run target is

$$\max_{k\le K} |x_k(\beta) - x_k^{\natural}(\beta)| \le C g (\log g^{-2})^{3\mathfrak{p}_0+1} \le 1. \tag{RF37}$$

It is required for first-passage data with $\varrho \ge B_\sharp + 1$ and $0 < g \le \gamma_{U'} \le \gamma_U$. Threshold and large-field changes contribute at most $C\theta_k^{-(\kappa_0-4)/2}$ per step, and gauge-fixing changes at most $C\theta_k^{-3/2}(\log\theta_k)^{3\mathfrak{p}_0+1}$. Proving those actual comparisons remains necessary. Equality of bare integrals does not prove (RF37).

The small-field coupling polydisc has relative radius $\sigma_\star\theta_j$. The large-field tube is narrower. On a level-j unit cube, an action-weight change has size

$$|x - \theta_j|\alpha_{0,j}^2 \le C\mathtt{t}_j\theta_j^{-1}(\log\theta_j)^{2q}. \tag{RF38}$$

A logarithmic tube factor at least $(\log\theta_j)^{-2q}$ keeps this bounded. A fluctuation difference gains a further $\delta_j/\alpha_{0,j}$. This explains the different roles of the full small-field polydisc and the large-field tube. It does not justify extending the entire interacting history construction to the larger polydisc. Complex bare couplings in the latter construction are obtained near admitted real runs through the fixed-source analytic domain.

Statement 3.5: the weaker marked-disc slope route. For this paragraph require $p_R > 1$. The unmarked term has the independent-variable bound in (RF20). Along the actual run the marked term is allowed a different analytic input: on a disc in the bare variable about each admitted real centre, of radius at least $c_\beta\varrho$, it is holomorphic and bounded by ϵ_k'. Bounds are uniform on centres whose required prefixes retain the prescribed tube room. On a smaller concentric disc, Cauchy therefore gives $C\epsilon_k'/\varrho$. This hypothesis includes the actual remainder dependence on that bare variable. It does not assert that this dependence is a function of independent deviations alone.

The dimension-free polydisc lemma, Lemma A.1, gives on the half polydisc

$$\sum_i r_i|\partial_i F| \le \tfrac{4}{3}\sup|F|, \qquad |F(z) - F(z')| \le \tfrac{4}{3}\sup|F| \max_i |z_i - z_i'|/r_i. \tag{RF39}$$

One proves it by sending a single unit disc through the point with coordinate disc automorphisms, choosing their phases to align the derivative terms, and applying the one-variable Schwarz inequality. Integrating the derivative bound on the segment gives the second formula. Neither bound contains the number of earlier levels. Neither implies age forgetting: the bounded function $F(z) = Bz_0/(\sigma_\star\theta_0)$ already has a nondecaying weighted derivative in its first coordinate.

Here is the needed forced derivative estimate, including the marked term. Let $b_i' = db_i/d\beta$, $M_i = \max_{0\le j\le i}|x_j'|$, with $x_0' = 1$. The frame is fixed, so $z_j' = -x_j'$. The tube constants and coupling

floor put these evaluation points in the half polydisc required in (RF39). Since $\min_{j<i}\theta_j \ge \theta_{i-1} - c_5 \ge \theta_{i-1}/2$ above the floor, (RF39) and the separate bare-disc Cauchy estimate imply

$$\begin{aligned} &|b_i'| \le a_i M_{i-1} + d_i, \\ &a_i \le C\theta_{i-1}^{-3/2}(\log\theta_{i-1})^{\mathfrak{p}_0}, \quad d_i \le C\varrho^{-1}\theta_{i-1}^{-p_R}. \end{aligned} \tag{RF40}$$

The θ^{-1} factor in a_i is the unmarked independent-polydisc gain. The factor $1/\varrho$ in d_i comes from the actual bare-variable disc and must not be replaced by $1/\theta_i$.

Put $V_i = 1 + \sum_{l\le i}|b_l'|$. The differentiated recursion gives $M_i \le V_i$, and hence

$$V_i \le (1+a_i)V_{i-1} + d_i, \qquad \sum_{i=1}^{K}|b_i'| \le e^{A_K}(1+D_K) - 1, \quad A_K = \sum_{i=1}^{K} a_i, \quad D_K = \sum_{i=1}^{K} d_i. \tag{RF41}$$

Indeed iterating the scalar inequality gives $V_i \le \prod_{l\le i}(1+a_l) + \sum_{l\le i} d_l \prod_{m=l+1}^{i}(1+a_m)$; each product is at most e^{A_i}. No triangular estimate for the physical field response is hidden in this one-dimensional calculation.

On the reference line, integral comparison yields, uniformly in K,

$$A_K \le Ch^{-1/2}(\log h)^{\mathfrak{p}_0}, \qquad D_K \le C\varrho^{-1}h^{1-p_R}. \tag{RF42}$$

The condition $p_R > 1$ is exactly what makes the second bound uniform over all depths for a fixed bare-disc radius. For the original full Lipschitz numbers one may now take $\mathsf{a}_{[K-i]} = (4/3)a_i + d_i$, once the uniform slope bound below is imposed. Thus

$$\sum_{s<K}\mathsf{a}_{[s]} \le C\{h^{-1/2}(\log h)^{\mathfrak{p}_0} + \varrho^{-1}h^{1-p_R}\}. \tag{RF42a}$$

The sharper original $O(g(\log g^{-2})^{\mathfrak{p}_0})$ rate follows when $p_R \ge 3/2$, in particular in the intended large-κ_0 regime. For merely $p_R > 1$, retain the general forced rate displayed here. Choose h large enough that $e^{A_K}(1+D_K) - 1 \le 1/3$. Then $|x_j' - 1| \le 1/3$ at every admitted prefix. On real intervals, $2/3 \le x_j' \le 4/3$. The statement is uniform only where the analytic inputs and room conditions hold. It does not identify different frames.

The stronger independent marked-coupling polydisc route in the comparison appendix remains available separately. Under its actual hypotheses the marked row has order $\theta^{-p_R-1}\log^C\theta$; it is not the premise used in (RF40)–(RF42). Both routes require their actual domains. In particular a fixed physical-source disc alone does not supply the unmarked independent-variable gain.

Canonical tube and tuning theorem 3.2. For a general finite reference sequence of length $n \ge 1$, the largest permitted tube is

$$\mathsf{t}_j^* = \min_{1\le t\le \min\{j+1,n\}} \mathsf{t}^\sharp(\theta_t). \tag{RF43}$$

This is just the combination of all preceding upper bounds and the requirement that tube radii do not increase with the level. The one-member case retains its separate convention. A constant-tube criterion follows by setting $\mathsf{K}(a) = \sup_{y\ge e} y^{-1/4}(\log y)^a < \infty$. If the frame floor is $x_\star \ge e$ and $\mathsf{K}(2q_{\mathrm{t}})t \le \varpi x_\star^{3/4}$, then

$$t \le \frac{\varpi y^{3/4}}{\mathsf{K}(2q_{\mathrm{t}})} \le \varpi y(\log y)^{-2q_{\mathrm{t}}} \qquad (y \ge x_\star). \tag{RF43a}$$

Thus the constant sequence of radius t is a tube; its criterion uses the floor rather than the frame length. On the line (RF36), set

$$\iota_k = \epsilon_k + \epsilon'_k, \qquad T_K = \varrho + c_2, \qquad T_k = T_K + \sum_{j=k}^{K-1} \iota_j, \tag{RF44}$$
$$F_k(\beta) = \theta_k - x_k^\natural(\beta) + s_k, \qquad \widehat{\mathfrak{t}}_k = T_k + c_2 + 2R', \qquad R' = 4\varrho.$$

This is the canonical tube: in depth notation $T_{[s]} = T_{[0]} + \sum_{u<s} \iota_{[u]}$. The correction by s_k removes the bounded one-loop oscillation, giving $|F_{k+1} - F_k| \le \iota_k$. The accumulated drift satisfies $\sum_{j=k}^{K-1} \iota_j \le C\theta_k^{1/2} (\log \theta_k)^{\mathfrak{p}_0}$ in the stated power regime. For sufficiently large h, uniformly over K and k, this and the fixed additive terms are smaller than $\varpi\theta_{k+1}(\log \theta_{k+1})^{-2q_{\mathfrak{t}}}$. The same holds at the terminal convention. Thus $\widehat{\mathfrak{t}}$ is an admitted tube. The choice is made after ϱ, not by shrinking the prescribed offset range as K grows.

We spell out the domain premise needed by the interval argument. At a prefix satisfying $|F_j| \le T_j$, the actual array is defined with room $2R' = 8\varrho$. The next coefficient is formed using that prefix before the new large-field operation. The uniform bounds and the marked bare-disc hypothesis above apply on a neighborhood of every such prefix. Whenever the computed next value satisfies $|F_{k+1}| \le T_{k+1}$, the next array closes with the same room. These are the actual prefix formation/closure hypotheses, not consequences of the scalar intermediate value theorem.

Define $I_0 = \{\beta : |F_0(\beta)| \le T_0\}$. Since $F_0 = \theta_0 - \beta$, this is a compact interval mapped decreasingly onto $[-T_0, T_0]$. Choose the floor so all its initial weights are admitted. Inductively suppose I_{k-1} has this property for F_{k-1}, and every earlier prefix lies in its required room. The preceding domain premise defines b_k, hence F_k, on a neighborhood of I_{k-1}, even before restricting its new output to the target window. Equations (RF40)–(RF42) make $F'_k = -x'_k$ strictly negative there.

At the left endpoint, $F_{k-1} = T_{k-1} = T_k + \iota_{k-1}$, so $F_k \ge T_k$. At the right endpoint the corresponding inequality is $F_k \le -T_k$. Continuity supplies unique points with values T_k and $-T_k$, and strict monotonicity orders them. Their intervening compact interval

$$I_k = \{\beta \in I_{k-1} : |F_k(\beta)| \le T_k\} \tag{RF45}$$

is mapped decreasingly onto the whole target interval. On it $|\theta_j - x_j| \le T_j + c_2$ for $j \le k$, leaving room $2R'$ in (RF44). The actual closure premise therefore permits the induction to continue. This proves existence and interval connectedness rather than presuming them from an implicit tuning equation.

At K, $x_K = h + s_K - F_K$. Since $|s_K| \le c_2$, its image contains every x with $|x - h| \le \varrho$. Each such value has exactly one preimage in I_K. Let $\gamma_U \le \gamma_J \le \gamma_H$ impose the preceding conditions for the fixed ϱ, with γ_J also imposing the admitted source analytic domain. For every K and $0 < g \le \gamma_U$, the tuning theorem under these hypotheses is

$$x_K^\natural(\beta_K^\natural) = g^{-2}, \qquad \tfrac{1}{2}c_0 K - c_5 \le \beta_K^\natural - g^{-2} \le \tfrac{3}{2}c_0 K + c_5. \tag{RF46}$$

The common companion makes this the same number on every admissible torus. Uniqueness has been proved in the admitted canonical interval, not among arbitrary disconnected presentations at all bare couplings. Also $|\beta_K^\natural - h - c_0 K| \le 2c_2 + \sum_k \iota_k$. The single-run spacing estimate is

$$\tfrac{1}{2}c_0(j-i) - c_5 \le x_i^\natural - x_j^\natural \le \tfrac{3}{2}c_0(j-i) + c_5 \qquad (0 \le i \le j \le K). \tag{RF46a}$$

It follows by summing $|b_l - b_l^0| \le c_0/2$. It gives (RF46), but gives no bound between the tuned numbers for two different cutoffs.

The bare offset t and terminal coordinate x are distinct: $x_K^\natural(\beta_K^\natural - t) = h - t + O(c|t|)$ when the slope error is at most c. The reference frame is independent of β; the self-calibrated frame

$\theta_j = x_j(\beta)$ and its frozen shifted run are other applications of the uniform prefix induction. For a constant source ζ, the frozen shifted weights are $x_j(\beta)-\zeta_j$, and their slope normalization is $x_j'(\beta)$. Identifying this derivative with a particular F3 witness still requires that witness's admitted run and comparison hypotheses.

A finite nontriangular comparison lemma. The stronger two-run route uses an estimate different from (RF41). Fix integers $n \ge 1$, $0 \le r_0 < n$, numbers $0 < \rho < 1$, $\Pi, \mathsf{M} \ge 0$, nonnegative y_w for $r_0 < w \le n$, and nonnegative D_u, $k_{u,w}$, h_u, h' for the finite indicated ranges. Put

$$S_w = \frac{\rho^{n-w+1}}{1-\rho} + (w - r_0)\Pi. \tag{RF47}$$

Suppose, for $r_0 \le s < n$ and $r_0 \le u < n$,

$$\begin{aligned} y_{s+1} &\le \sum_{u=r_0}^{s} D_u, \\ D_u &\le \mathsf{M}(\rho^{n-u} + \Pi) + \sum_{w=u+1}^{n} k_{u,w} y_w, \\ \sum_{w=u+1}^{n} k_{u,w} S_w &\le h_u S_{u+1} + h'\Pi, \qquad \sum_{u=r_0}^{n-1} h_u + h' \le \tfrac{1}{2}. \end{aligned} \tag{RF48}$$

Then $y_{s+1} \le 2\mathsf{M} S_{s+1}$. Indeed $S_w > 0$ and increases with w. Let $Z = \max_{r_0 < w \le n} y_w / S_w$, a finite number. The forcing sum is at most $\mathsf{M} S_{s+1}$. The kernel sum is at most

$$Z\left(\sum_{u=r_0}^{s} h_u S_{u+1} + (s - r_0 + 1)h'\Pi\right) \le \tfrac{1}{2} Z S_{s+1}. \tag{RF49}$$

Taking the maximum gives $Z \le \mathsf{M} + Z/2$, as claimed. The sums couple a difference to later depths, so this is weighted-max absorption, not a triangular recursion. Its actual profile-kernel bound in (RF48) remains a substantive two-run input. Bounds on increments alone do not establish that input.

Statement 3.6: real presentations of the same measure. A real presentation over a frame is an actual inductive array whose weights remain in the tube. At each level it carries one Euclidean member, with one scalar coefficient common to all sites and no defect, and one shift c_j. These shifts obey the admitted $\bar{c}$ bound, with the frame floor chosen in terms of $\bar{c}$. Members, companions and coefficients are independent of the admissible torus and real at real fields. The all-small-field label is the label whose regions fill the torus. In its exponent the action coefficient is $x_j - \mathsf{c}_j$. The weight actually used in the Gaussian step is x_j, and $\mathsf{c}_j A$ is carried in the local action. A spatially varying source array with defects is not a presentation of this special kind.

Write $b_j = x_{j-1} - x_j$, let $\widehat{b}_j$ be the coefficient actually extracted at step j, and set $u = \mathsf{c}_0$. Coefficient bookkeeping gives

$$\mathsf{c}_j = u + \sum_{l=1}^{j} (\widehat{b}_l - b_l), \qquad x_j - \mathsf{c}_j = x_{j-1} - \mathsf{c}_{j-1} - \widehat{b}_j. \tag{RF50}$$

In particular $\widehat{b}_j$ need not equal the planned decrement b_j when the shifts vary with the level.

The comparison hypothesis is equality of the all-small-field measures at every level, on every admissible torus, for two real presentations over the same frame. Require hereditary open windows $\mathfrak{O}_m \ni \mathbf{1}$ of block fields: their backgrounds lie in every needed core, all the relevant tests equal one, and $\mathcal{M}^i U_m(V) \in \mathfrak{O}_i$ when $V \in \mathfrak{O}_m$ and $i < m$. Also require the common connected analytic

fluctuation domain on which continuation from such a window to the next actual integral is used. These are properties of the actual maps and tests.

Under these hypotheses the conclusions are

$$\begin{aligned} x_j - \mathsf{c}_j &= x'_j - \mathsf{c}'_j, \\ \widehat{b}_j &= \widehat{b}'_j, \\ \sum_X (E^{(j)} - E'^{(j)})(X,U) &= \tfrac{3}{2}|\mathfrak{X}_{j-1}| \log(x_{j-1}/x'_{j-1}). \end{aligned} \tag{RF51}$$

The first line holds through level n, and the coefficient equality through $n+1$ only when both presentations admit that next step. The total identity is evaluated at backgrounds $U = U_m(V)$, $j \le m \le n$, of the common windows. On those backgrounds the remainder totals agree, and the vacuum constants agree. When the total suppresses z, $E^{(j)}(X,U) = \sum_{z\in X} E^{(j)}(X, P_U, z)$. No assertion identifies individual localized terms of the two presentations.

Here is the induction. At the bare level equality identifies the effective weight $\beta - u$, since the action is not a constant function. On an open small-field window the positive continuous densities agree everywhere if they agree almost everywhere, because product Haar measure has full support on the open window. Their logarithmic exponents are real, so their equality involves no integer ambiguity. Analytic continuation on the admitted fluctuation domain therefore identifies the next unnormalized integral. The normalized Gaussian in each presentation uses that presentation's own weight. Its normalization has the term $-3|\mathfrak{X}_{j-1}| \log(x_{j-1}/\theta_{j-1})/2$. It follows that the next regular totals differ by the field-independent constant displayed in (RF51). The common-companion sum rule makes their extracted coefficients equal. Equation (RF50) then identifies the next effective weights. The same physical formation identities on the hereditary windows identify remainder totals and the once-counted vacuum constants. This completes the induction. It uses all-torus compatibility to extract a coefficient from a constant total; one finite torus alone is insufficient.

Apply the result to a real β whose prefix along θ has room 2ϱ, and a real $|\zeta| < \varrho$ for which both constructed presentations are admitted. One presentation starts at $\beta - \zeta$ with no shift; the other starts at β with constant source ζ and running shift ζ_k. The exact identity is

$$x_k^{\theta}(\beta - \zeta) = x_k^{\theta}(\beta) - \zeta_k. \tag{RF52}$$

The slope bound keeps the running shift at most $4|\zeta|/3$ in its stated range. For $\theta = x(\beta)$, this describes the frozen shifted run, not the self-calibrated run of $\beta - \zeta$. For the reference line it identifies the two presentations of $x_k^{\natural}(\beta - \zeta)$.

Complex gluing also uses one frame. First restrict the real centres to the canonical interval I_K constructed above, or to one of its admitted prefix intervals. Each source disc is supplied by the actual analytic source construction. Where a real point in that disc also belongs to the canonical interval, (RF52) identifies its continuation with the real run. If two discs about such centres overlap, their real overlap contains an open interval in I_K: the segment between the centres lies in I_K, and the open real disc intervals overlap along that segment. For coincident centres at an endpoint, a one-sided open interval inside the nondegenerate I_K suffices. The identity theorem therefore identifies the coefficient functions on the full complex overlap.

The union carries a single holomorphic bare-coupling function. Remainder totals on common windows glue likewise; regular totals glue after accounting for their explicit Gaussian constant. This proves uniqueness of the supplied continuations, not their original bounds or radii.

The broader target for arbitrary admitted real centres is retained conditionally: every overlap used for gluing must contain a common open interval of ADMITTED real runs, or the actual prefix/source-disc construction must prove real admission there. That extra obligation is substantive if the real admitted set has disconnected components. Geometric intersection of two discs alone does not settle it. With the stated admission property the same identity theorem

proof applies. Nothing here identifies runs made with different frames, individual localized terms about different centres, or functions on independent-current cores from equality only on a physical graph.

(3d) Units, common prefixes and change of reference.
Statement 3.7: the finite dilation identity. For an integer b with $0 \le b \le K$, the map $D_b x = L^b x$ identifies the bare lattices at (K, ℓ) and $(K-b, L^b\ell)$. The exact result of Section 34 is

$$\begin{aligned} Z_{K,\ell,\beta}(w) &= Z_{K-b,L^b\ell,\beta}(w^{[b]}), \\ S^T_{n;K,\ell,\beta}(\underline{f}) &= S^T_{n;K-b,L^b\ell,\beta}(\underline{f}^{[b]}), \quad f^{[b]}(x) = f(L^{-b}x). \end{aligned} \tag{RF53}$$

There is no spacing prefactor in $\mathcal{O}(f) = \sum_p f(x(p))a_p$. The map transports bonds, plaquettes and Haar variables bijectively; both sides are the same finite integral. The partition identity is entire in its finite source variables and bare coupling. Logarithmic derivatives use its nonzero domain. The same transport identifies periodic Gibbs states on the two infinite lattices. Radii and distances multiply by L^b, $\operatorname{Lip} f^{[b]} = L^{-b} \operatorname{Lip} f$, and $\|f^{[b]}\|_\sharp \le \|f\|_\sharp$. The free bilinear form is invariant. This proves neither convergence nor agreement of two RG constructions.
Statement 3.8: what the step reads. Construction naturality is the additional assertion that step k reads only its actual tower through k, partitions, frame and weight prefixes, earlier outputs and fixed supply rules. Those rules are made for the lattice, not for its unit of length. For two presentations of the same bare lattice, their images under D_b therefore identify the histories and their measures, regions, healing data and local nonwrapping terms through level $K-b$, provided the prefixes agree. This is required for both fixed-frame and self-calibrated runs. Separate arbitrary choices on the two lattices would not establish it.

For a self-calibrated run this permits continuation from first passage at g to first passage at g', provided the prolonged run remains below the anchor ceiling and the rescaled torus is admissible. Its history law through the earlier level is unchanged. For reference runs, $\theta_{[r+s]}(g) = \theta_{[r]}(g_{[s]})$; the prefix at g, down to depth s, is the reference run at $g_{[s]}$ of length $K-s$, on $T_{L^s\ell}$. Its deviations, increments and tube are the same. The implication $\ell \in \mathfrak{L}(g) \Rightarrow L^s\ell \in \mathfrak{L}(g_{[s]})$ uses the stated torus floor and parameter choices.

Keep the beyond-first-passage scope separate. If $\beta \in \mathcal{W}_{K'}(g)$ and $K = K' + q$, $q \ge 0$, then (RF53) identifies the cutoff-K measure with the first-passage measure on $T_{L^q\ell}$. Estimates for it require $L^q\ell \in \mathfrak{L}(g)$. A torus volume factor becomes $\exp[nc_V(L^q\ell)^4]$, whereas a per-ball estimate has its radius and separation changed to L^qR, L^qd and no side dependence in its constant:

$$|S^T_{n;K,\ell,\beta}(\underline{f})| \le A(L^qd; L^qR)B(L^qR)^n(n!)^{\varkappa_\star} \prod_i \|f_i\|_\sharp. \tag{RF54}$$

The stated lower torus floor for that bound also applies. The factorial finite-torus estimate (RF54) is an additional conditional target: its use requires that estimate for the actual finite-torus source construction. Theorem 39.1 proves its stated graded bound for periodic Wilson DLR states; it does not by itself supply (RF54), nor a polynomial separation prefactor of degree independent of the number of tests. The finite dilation identity transports a supplied bound and does not prove this additional target. For $q < 0$, use the identity in reverse, require $L^q\ell \in \mathfrak{L}(g)$, and allow a norm cost at most $L^{|q|}$ per test. Applying this to tuned offsets requires the separate comparison $|K(\beta_K^\natural - t, g) - K| \le C(\varrho)$. It does not follow from tuning at a frozen frame.
Statement 3.9: all four reference-change targets. Let

$$S^{[t]}_\ell[g] = \lim_K S^T_{\cdot;K,\ell,\beta_K^\natural(g)-t}, \qquad \mathsf{J} = [-\varrho/4, \varrho/4]. \tag{RF55}$$

Existence along the whole sequence, its common complex offset strip and moving-offset continuity require E3 and the terminal-coordinate comparison in Appendix A. The strip width and the cutoff

floor may depend on the fixed torus. The coordinates are

$$\Xi_K(t) = x_K^\natural(\beta_K^\natural - t), \qquad \Xi_{\infty,s}(t) = \lim_K x_{K-s}^\natural(\beta_K^\natural - t). \tag{RF55a}$$

The depth limit requires the matched comparison. Terminal matching uses $n_K = \Xi_{K+1}^{-1} \circ \Xi_K$ on its admitted common strip. Its near-identity and summable-error properties are part of obligation E3, which we must prove; they are not consequences of the existence of individual tuning inverses. First choose ϱ large compared with $c_2, B_\sharp + 1$ and a bound for the tuned increments; then choose g small. The bound for those increments is a two-run comparison requirement fixed before ϱ, not a consequence of one-run spacing.

(i) At fixed depth s, the same bare coupling has two offset labels:

$$\begin{aligned} \widehat{t}_{K,s} &= \beta_{K-s}^\natural(g_{[s]}) - \beta_K^\natural(g), \\ \beta_K^\natural(g) - t &= \beta_{K-s}^\natural(g_{[s]}) - (t + \widehat{t}_{K,s}). \end{aligned} \tag{RF56}$$

Require the actual two-run estimate $|\widehat{t}_{K,s}| \le C(c_2 + \sum_{K-s\le j<K} \iota_j)$, its summable cutoff increments, and the comparison domain on which both presentations exist. The initial comparison requires $2c_2 + \sum_{K-s\le j<K} \iota_j \le \varrho$; a smaller fixed fraction of that bound keeps the offsets in the inner window. For g small in terms of the fixed s, these offsets lie in $\frac{1}{2}\mathsf{J}$ and converge to $\widehat{t}_s$. The whole-sequence moving-offset limit then gives

$$\begin{aligned} S_\ell^{[t]}[g](\underline{f}) &= S_{L^s\ell}^{[t+\widehat{t}_s]}[g_{[s]}](\underline{f}^{[s]}), \\ \widehat{t}_{s+s'}(g) &= \widehat{t}_s(g) + \widehat{t}_{s'}(g_{[s]}). \end{aligned} \tag{RF57}$$

Here $t \in \frac{1}{2}\mathsf{J}$; each depth condition must hold at its displayed reference coupling. The condition on g is an upper bound, never a lower bound. The terminal coordinate on the right is the limiting depth coordinate $\Xi_{\infty,s}(t)$. Its existence belongs to the matched comparison.

(ii) For arbitrary sufficiently small g, g' and $t \in \mathsf{J}$, retain the stronger target that some $t' \in \mathsf{J}$ and integer b satisfy

$$S_\ell^{[t]}[g](\underline{f}) = S_{L^b\ell}^{[t']}[g'](\underline{f}^{[b]}). \tag{RF58}$$

Its torus conditions are both $\ell \ge \ell_\star(g)$ and $\ell \ge L^{(|g'^{-2}-g^{-2}|+C)/\lambda_\sharp}\ell_\star(g')$. The sign of b is that of $g'^{-2} - g^{-2}$, up to a bounded ambiguity. The number of intervening passage levels lies between $|g'^{-2} - g^{-2}|/B_\sharp - 1$ and $(|g'^{-2} - g^{-2}| + C)/\lambda_\sharp$; $|b|$ differs from it by $O(1)$. This assertion needs recognition of the same bare run in both calibrated families, not merely (RF53).

(iii) Choose once an anchor $0 < \gamma_\star \le \gamma_J$, sufficiently small in terms of the fixed large-field window dials w, c_a. For any proposed $\mathbb{R}^4$ limit at $g \le \gamma_\star$, retain the prolonged-run target: after a subsequence, $K(\beta_j, \gamma_\star) - K_j = s_\star$ is constant, of order $g^{-2} - \gamma_\star^{-2}$, and

$$S_n(\underline{f}) = S'_n(\underline{f}^{[-s_\star]}). \tag{RF59}$$

The anchor limit uses the same bare-coupling sequence. This preserves the target for arbitrary defining sequences carrying the required construction and recognition hypotheses. It does not give an automatic mirror/cone theorem for unrestricted growing-torus limits. Dilation-invariant properties need be checked at only one reference coupling once this target is established.

(iv) The bare tuned increments $\alpha_K = \beta_{K+1}^\natural - \beta_K^\natural$ are a new notation, unrelated to the field radii. The retained conclusion is

$$\alpha_K \longrightarrow c_0, \qquad S_\ell^{[t]}(\underline{f}) = S_{L^b\ell}^{[t-bc_0]}(\underline{f}^{[b]}) \quad (b \ge 1,\ t, t - bc_0 \in \mathsf{J}). \tag{RF60}$$

For completeness the shape convention is $\mathfrak{T}_{\mathbb{R}^4}(d;\vartheta)$: both profiles are nonnegative nonzero Lipschitz functions supported in radius-d balls, with support distance in $[d, 6d]$, and each obeys $\int f \ge \vartheta d^4 \|f\|_\infty$ and $\operatorname{Lip} f \le \|f\|_\infty/(\vartheta d)$. Clause (iv) retains the condition $\mathfrak{T}_{\mathbb{R}^4}(1;\vartheta) \ne \emptyset$ and

$g \le \gamma_W(\vartheta)$. The proof also needs the actual uniform smooth-profile envelopes on the pairs used in its iteration; class nonemptiness by itself is not an envelope estimate. The comparison appendix proves this implication with the actual two-run depth/profile estimates, summable comparison errors and uniform two-point envelopes. Those envelopes are uniform in the offsets and tori needed for repeated blocking; the fixed shape class must be nonempty. Two distinct cluster increments would give an offset period by analyticity, and repeated blocking would contradict the uniform short-distance envelopes. The unique increment limit is then c_0 by the averaged one-loop law. The same argument yields $\beta^{\natural}_{K+b} - \beta^{\natural}_{K} \to bc_0$ for every fixed integer b. The exact input hypotheses remain in force; finite dilation and weak numerical inequalities do not establish them.

Vacuum constants and the interfaces used later. The vacuum recursion is (CP1) of Appendix A. Each assigned step number, Gaussian local number and regular/remainder value at $P_\circ$ is counted once. Using the normalized N requires adding its removed local numbers; using N° does not. At fixed calibration these numbers have zero source derivative. They are not additional freely chosen normalizations.

The further vacuum targets retained here are

$$\begin{aligned} &\log(\mathsf{c}_k(\theta_k) Z^{(k)}(\mathbf{1})) = \mathsf{N}^{(k)} + \sum_X \sum_{z \in X} \mathsf{n}^{(k+1)}(X, z), \\ &\mathsf{n}_k^{\mathrm{tot}} = 36 N_{k+1} \log L - \tfrac{1}{2} \log\det(\mathsf{C}^{\mathsf{T}} \Delta^{(k)} \mathsf{C})|_{\mathbf{1}}, \quad |\mathsf{n}_k^{\mathrm{tot}}| \le C N_{k+1}, \\ &\mathsf{n}_0^{\mathrm{tot}} \le -3(L-1)^4 N_1 \log 2 < 0. \end{aligned} \tag{RF61}$$

Here N_j counts sites, c_k is the actual change-of-variable normalization and $\mathsf{N}^{(k)}$ is the free-bond/coarse-site count, with $-3 \log \theta_k/2$ per free bond. The map C is the actual linear embedding of nonpivot coordinates into full constrained fluctuations; this determinant is on the free color coordinates. It is not the covariance $C^{(k)}$. The local numbers are Gaussian values before normalization, real and coupling independent, with common nonwrapping lifts and the Gaussian domain bound. Their coefficients vanish. The count alone is not the entire step number. The determinant estimates and first-step sign in (RF61) remain actual Gaussian targets, beyond the once-only accounting identity.

The explicit leading vacuum requirement is

$$\left| E_{K,\ell,\beta} + \tfrac{3}{2}(N^2-1)(L^4-1) \sum_{j=0}^{K-1} (\log x_j) N_{j+1} \right| \le C N_0, \tag{RF62}$$

uniform in the admitted runs, tori and cutoffs. The estimate (RF62) is the quantitative hypothesis on the vacuum term used here.

The order of choices is retained. Fix the geometric exponents and L, with the Gaussian constants and one-loop data; then choose κ above the domain entropy, the required lower block scales and M. Choose $\sigma_\star$, absolute radii below c/M, $K_{\mathbf{M}}$, and $E_0 \ge \max\{2E_{\mathsf{N}}, 4K_{\mathbf{M}}\sigma_\star\}$. Next choose $N_\star$, the ceiling on s, the unmarked constants, C_β and the weak triple. Compute the marked-load and tube constants at that triple before choosing γ, the strong triple, γ_H and the smaller drift ceiling. The source radius and comparison windows have their stated place before the final smallness of g. Only then are ℓ, β, K varied. Each claimed uniformity depends on respecting this order.

The later uses retain their distinct scopes. Beyond-first-passage data use (RF54); prolonged runs use statement 3.8; fixed-depth reference prefixes and terminal coordinates use (RF57). The germ/rigidity route in the comparison appendix also carries its own shape and second-difference hypotheses. In the anchor unit its correlation constants acquire $L^{-(4-n)s_\star}$. Blocking and distinct-member comparisons use (RF60), not just a relabeling of K. The sub-unit cells of the angular route

keep their separate moment, mirror and ultraviolet interfaces. None of these preserved stronger targets changes the conditional status of the ordinary interacting source construction, the actual clean-class two-point extraction, the external reconstruction theorem or the identification with the full gauge theory.

36. Sources, local energies, and separated correlation bounds

Proof outline. The supporting analytic results used in this section, including the obligations that we must still prove, are listed under C and D in the dependency catalogue. Their hypotheses and precise forms are retained at the points where the argument uses them.

Source domains and the exact running-shift identity. We define the source variables and prove their finite-dimensional identities. Their use for effective energies takes the multiscale estimates from the source propositions stated separately.

Fix K, a torus T_ℓ, an odd $L > 1$, and positive constants M, c_M with $c_M \geq M$. The depth s corresponds to physical spacing L^{-s}; level j of a run of length K corresponds to depth $K - j$. A source triple is

$$w = (c, (f_1, \dots, f_n), (z_1, \dots, z_n)),$$
$$w(x) = c + \sum_i z_i f_i(x),$$

where f_i are real Lipschitz functions, c, z_i are complex, and the family may be empty. Put

$$\|f\|_{\sharp,s} = \max\{\|f\|_\infty, c_M L^{-s} \operatorname{Lip} f\},$$
$$\|w\|_{\flat,s} = |c| + \sum_i |z_i| \|f_i\|_{\sharp,s}.$$

The class $\mathcal{W}_r^{(s)}$ means the triples with $\|w\|_{\flat,s} < r$. It is a union of finite-dimensional source balls, one for each fixed real family. Holomorphy on this class means holomorphy in (c, z) on each such ball. The number $\|w\|_{\flat,s}$ belongs to the chosen triple and can change when its function is represented differently. The physical partition function and every source-local energy are required to agree for two triples that have the same values on all bare plaquettes on which that quantity depends. This representation-independence is a separate part of the source format.

Always $\|w\|_\infty \leq \|w\|_{\flat,s}$ and $c_M L^{-s} \operatorname{Lip} w \leq \|w\|_{\flat,s}$. Thus, if $m = K - s$ and $j \leq m$,

$$c_M a_{K-j} \operatorname{Lip} w = L^{-(m-j)} c_M a_{K-m} \operatorname{Lip} w \leq L^{-(m-j)} \|w\|_{\flat,s},$$
$$a_t = L^{-t}.$$

This is the exact scale gain used when a source is frozen on an earlier block. It follows from the norm definition, not from an analyticity theorem.

Here is an explicit algebraic version of the running profiles. Let P_K be the finite set of bare plaquettes, with barycentres $x(p)$. For each level j choose a finite set Y_j of base points, anchors $p_y \in P_K$, and nonnegative functions $h_y^{(j)}$ on P_K such that

$$\sum_{y \in Y_j} h_y^{(j)}(p) = 1,$$
$$h_y^{(j)}(p) \neq 0 \implies |x(p) - x(p_y)| \leq C_h M a_{K-j}.$$

Choose also profiles $0 \le \varphi_j(p) \le 1$. The grid, anchors, partition and profiles are fixed independently of the source throughout the analytic family. They are supplied by the geometric construction; the identities below hold for every finite choice satisfying these conditions.

Let b_j be holomorphic on $D(0,\rho)$ and define

$$d_j = \sup_{0<|z|<\rho} \frac{|b_j(z) - b_j(0)|}{|z|},$$

$$\sum_{j\le k} d_j \le \frac{1}{3}.$$

For triples whose function has $\|w\|_\infty < \rho$, set

$$x_k(p) = \beta - \sum_{j\le k} b_j(0)\varphi_j(p),$$

$$c_j(p;w) = \varphi_j(p) \sum_{y\in Y_j} h_y^{(j)}(p) b_j(w(x(p_y))),$$

$$\zeta_k(p;w) = w(x(p)) + \sum_{j\le k} \varphi_j(p) \sum_{y\in Y_j} h_y^{(j)}(p)[b_j(w(x(p_y))) - b_j(0)].$$

Then the exact bookkeeping identity is

$$x_k(p) - \zeta_k(p;w) = \beta - w(x(p)) - \sum_{j\le k} c_j(p;w).$$

To prove it, expand both sides and use $\sum_y h_y^{(j)} = 1$ in each constant term $b_j(0)\varphi_j$. Consequently the Wilson weights in a sourced exponent can be added or subtracted without changing their total bare coefficient. This assertion concerns that total coefficient; it does not assert equality of two effective measures unless the exact integral transport has also been proved.

The same definitions give

$$|\zeta_k(p;w) - w(x(p))| \le \tfrac{1}{3}\|w\|_\infty,$$

$$|\zeta_k(p;w)| \le \tfrac{4}{3}\|w\|_\infty.$$

Each value is holomorphic on every source ball with supremum norm below ρ, because only finitely many evaluations and compositions occur. It depends on the source only inside distance $C_h M a_{K-k}$ from p, including the value at p itself; here $a_{K-j} \le a_{K-k}$ for $j \le k$.

For a constant source $w \equiv z$, in a region where every $\varphi_j(p) = 1$, this reduces to

$$\zeta_k(z) = z + \sum_{j\le k}[b_j(z) - b_j(0)],$$

$$\zeta_0(z) = z,$$

$$\zeta_{k+1}(z) = \zeta_k(z) + b_{k+1}(z) - b_{k+1}(0).$$

Differentiating at zero gives $\zeta_k'(0) = 1 + \sum_{j\le k} b_j'(0)$ and $|\zeta_k'(0) - 1| \le 1/3$. The derivative bound follows from the definition of d_j by taking the limit at zero. For spatially varying sources, freezing the function w itself costs at most $\operatorname{Lip} w$ times the anchor distance. To freeze $b_j \circ w$ one must

work on a smaller source disc. Indeed $h_j(z) = [b_j(z) - b_j(0)]/z$ extends holomorphically to zero and is bounded by d_j. For $|z| \le \rho' < \rho$, Cauchy's estimate gives

$$|b_j'(z)| = |h_j(z) + zh_j'(z)| \le d_j \frac{\rho}{\rho - \rho'}.$$

Thus on the half-disc the additional factor is at most $2d_j$. Apply the mean value integral on the segment between the two source values to bound their composed difference. The displayed depth identity then supplies the factor $L^{-(m-j)}$, with the fixed anchor constant $C_h M/c_M$. The radial bound d_j alone is not a Lipschitz bound on the whole open disc.

All smallness, radius and representation-independence conditions used here must be retained when invoking the sourced RG format. They do not follow from the entire dependence of the bare lattice partition function alone.

4. Transporting the observables: sources, analyticity, and bounds independent of the cutoff and of the volume. The measure $\mu_{K,\ell,\beta}$ is described at the unit scale in Steps 1–3, and the present step gives the corresponding description of its correlations. Because $\mathcal{O}(f) = A(f, U)$, a source $w = c + \sum_i z_i f_i$ acts as a complex shift of the bare inverse coupling that depends on position:

$$\rho_0^w = e^{-A(\beta - w, U)},$$

$$Z(w) = \int \rho_0^w \, dU,$$

$$S^T_{n;K,\ell,\beta}(f_1, \dots, f_n) = \partial_{z_1} \cdots \partial_{z_n} \log Z(w)\big|_{w=0}.$$

The renormalisation steps are applied to ρ_0^w. Whatever the source leaves behind is gathered into local numbers, the source energies $e_j(X; w)$ of (4c). We set $\mathcal{E}(w) := \sum_{1 \le j \le K} \sum_{X \in \mathcal{D}_j} e_j(X; w)$, $\widetilde{Z}(w) := e^{-E - \mathcal{E}(w)} Z(w)$ and $C_g := C(1 + \log g^{-2})$.

Theorem 4.1 (The construction with sources). Let $0 < g \le \gamma_J$. For each first-passage datum (K, β), each $\ell \in \mathfrak{L}(g)$ and each source satisfying $\|w\|_\flat < 2\varrho$, the pair $(E, (e_j(X; \cdot)))$ forms a system of localised source energies. That is, each $e_j(X; w)$ is holomorphic in (c, z) (analyticity). If the f_i are real, it takes real values whenever (c, z) is real. It depends on w only through its values on $X^{\sim 1}$, and it is zero at $w = 0$ (locality). Moreover, uniformly,

$$|e_j(X; w)| \le C e^{-\kappa_e d_j(X)},$$

$$|\widetilde{Z}(w)| \le e^{C_g \ell^4},$$

$$\widetilde{Z}(0) \ge e^{-C_g \ell^4}$$

(these are, in order: the exponential bound, whose rate $\kappa_e := c\kappa$, $c < 1$, exceeds the entropy rate of the localisation domains through a cube; upper stability; the floor). The same conclusions hold along a reference run, with every choice frozen at θ_k, as long as $x_k^\natural$ remains in a tube around θ_k with a margin of $C\varrho$. For each K this requirement is a condition on β only. When $g \le \gamma_U$, it is met on the interval I_K of (3c), an interval that contains the tuned $\beta_K^\natural$. The conclusions also hold for the run stopped at any level k, where the sums are stopped at $j \le k$, the vacuum constant $E_{(k)}$ of the first k steps replaces E, and $C(1 + \log x_k)$ per site of $\mathbb{T}^{(k)}$ replaces $C_g \ell^4$ (4b).

Theorem 4.2 (Correlation bounds: Theorem 0 (c)(1)–(2)). Take g, K, β, ℓ as above and $n \ge 2$. Let the f_i be real Lipschitz functions whose supports lie at mutual distances $\ge \mathsf{d} > 0$. For the second and the third bound, let each support also be contained in a ball of radius R.

The third bound below concerns periodic Wilson DLR states. It assumes real C^1 tests and the ordinary source hypotheses on the compatible cofinal torus family with the actual rooted geometry specified by D1. These are hypotheses of the graded deduction used below. Then the following estimates hold in their stated scopes:

$$|S_n^T| \le C_n(\mathsf{d}) e^{n c_V \ell^4} \prod_i \|f_i\|_\sharp;$$

$$|S_n^T| \le C_n^\infty(\mathsf{d}; R) \prod_i \|f_i\|_\sharp \ (\ell \ge Cn(R+1));$$

$$|S_n^T[\nu]| \le [2B(R)]^n (n!)^{\varkappa_\star} \max\{1, \mathsf{d}^{-5n}\} \prod_i \|f_i\|_\sharp,$$

uniformly, where every constant may also depend on g and $C_n^\infty(\mathsf{d}; R)$ does not increase with d. In the third bound, $\varkappa_\star$ is one finite integer and $B(R) = C(R+1)^4$, as in Theorem 39.1 under the actual rooted geometry and analytic hypotheses of D1 and G1. The second bound also holds for $S_n^T[\nu]$ in each periodic Gibbs state ν at β on $\mathbb{L}_K$, with no condition on the volume. The third bound is the stated graded theorem for such periodic Wilson DLR states; it is not a finite-torus factorial conclusion. A uniform finite-torus factorial estimate remains an additional target. A stronger bound with a single polynomial separation prefactor $A(\mathsf{d}; R)$ of degree independent of n remains an additional target. It does not follow from the graded estimate displayed here and is not used as a proved consequence of D2.

† Theorem 4.13 is the third main result. Under the law of the histories, the probability that a cell is live, or that it is created, is $\le Ce^{-cp_0(g_k)}$, uniformly in the volume. The short-distance laws rest on this result. (7b) treats the two-point law, clauses (c)(3), (c)(4). It uses these bounds on a single cell and, from their age-resolved form (Proposition 4.14), the age series at a rate $b \ge 8$. (7d) treats the four-point law, clause (c)(5), and uses the series at a rate $b > 16$.

The same tools also give two further results that are used later. The first is the one-point transfer (Proposition 4.12 [proved here; Section 38]), $|\nu(\mathcal{O}(f)) - \nu'(\mathcal{O}(f))| \le C_1(R)\|f\|_\sharp$, which compares periodic Gibbs states taken at the same first-passage datum (K, β). With it, one can centre a pure phase at the means of that periodic Gibbs state whose decomposition contains the phase, the coordinates of (10c) become compact, and the one-point numbers $\mathfrak{c}_\xi$ of Theorem 0'(3)(ii) are bounded. It also enters the growth constants of (b)(v), so that a single set of constants covers these components as well. The second consists of the insertion bound and the plate bound (Propositions 4.17 [proved here; Section 46], 4.18), which are the a priori estimates on which (10b) rests.

(5b) uses only the bounds of Theorem 4.2. (5c) (for the re-centring), (6a), (6c) and (10c)–(10e) also use the one-point transfer of Proposition 4.12. The tuned couplings need not be first-passage data. (5a) and Step 8 apply Theorem 4.1 along reference runs, and this gives the tuned limits the torus bound only. The format underlying Theorem 4.1 is the starting point of (7a)–(7b). The difficulty lies in two places. The first is Theorem 4.1: the source term is marginal, since a step leaves it no smaller, just as it leaves the action no smaller; it takes complex values, so positivity is no longer available; and it must be carried through the large-field operations. The second is Theorem 4.13, in which the ratio of the stability bounds loses $e^{C_g \ell^4}$. The finite Cauchy and Gibbs-state transfer implications are proved under their explicit analytic inputs in the corresponding sections. The history witnesses, their locality and replication, and their age-resolved weights in Lemma 4.15 remain construction requirements; reflection positivity by itself does not verify them.

(4a) The curvature observable as a source, and the reason the choices of the construction stay fixed. **Objects.** The norm $\|f\|_\sharp = \max\{\|f\|_\infty, CM \operatorname{Lip} f\}$ controls how much f can vary within

any M-cube of the unit lattice. The sourced densities ρ_k^w arise from ρ_0^w through the operations of Steps 1–2, and every choice in them is the one made in the run at $w = 0$. These choices are the tests χ, the thresholds ε_k, δ_k, the radii, the sizes $\check{R}_k$, the backgrounds $U_k(\mathbf{V})$ (which minimise the action with no source), the scaling $\mathcal{A}_k = g_k \mathcal{A}_k^{\mathrm{s}}$, the covariances $C^{(k)}$, the gauge-fixing weight and the reference laws of $\mathbf{R}$.

Proposition 4.3 (Kernels; same histories; exactness). To each history h of level k there corresponds a positive kernel $\mathsf{K}_h(U, dV_k)$, built from the choices of the run and from nothing else. For Φ bounded and gauge invariant, the operations carry $\Phi\rho_0$ to

$$\sum_h \tau_h[\Phi],$$

$$\tau_h[\Phi] = \int \Phi(U)\, \rho_0(U)\, \mathsf{K}_h(U, \cdot)\, dU,$$

$$\sum_h \tau_h[\Phi](\mathcal{V}_k) = \int \Phi\rho_0\, dU.$$

For the choice $\Phi = e^{A(w,U)}$ one obtains $\rho_k^w\, dV_k = \sum_h \tau_h^w$, where the sum runs over the histories of ρ_k. Each τ_h^w is a complex measure which, as a function of (c, z), is entire, and $\tau_h^0 = \tau_h \geq 0$. Moreover $\int \rho_k^w\, dV_k = Z(w)$ for $k \leq K$.

Each step consists of finitely many moves. The first is the insertion of $1 = \int dV\, \delta(\mathcal{M}(U)V^{-1})$. The second is a family of decompositions of unity $1 = \sum \prod \chi \prod \chi^c$. The third is a gauge-fixing factor, of total integral 1 over the gauge group; its insertion is harmless since $\Phi\rho_0$, like each a_p, is gauge invariant. The fourth is a set of changes of variables. The last occurs inside $\mathbf{R}$: a term is replaced by the product of a fixed probability law and its integral over a component. Every move is linear in the density and leaves the total integral unchanged, and no move requires positivity. The tests depend on the fields only, so the regions, labels and histories are the same as at $w = 0$.

Proposition 4.4 (Exact insertion of block observables). Given a history h of level k, denote by $Y(h)$ the union of all components that $\mathbf{R}$ has acted upon at levels not exceeding k. A bond c of $\mathbb{T}^{(k)}$ is said to be far from $Y(h)$ when neither the two blocks $B^k(c_\pm)$ nor the bare bonds that join them meet $Y(h)^\sim$. There exists a probability measure P_k on triples (U, h, V), each made of a bare field, a history and a block field, with four properties. First, its marginal on U is $\mu_{K,\ell,\beta}$. Second, under its marginal on (h, V) each bounded gauge-invariant $\Phi(h, V)$ has integral $Z(0)^{-1} \sum_h \int \Phi(h, V)\, \tau_h(dV)$. Third, it is invariant under $(U, h, V) \mapsto (U^u, h, V^{u|})$, where u is a gauge transformation of the bare lattice and $u|$ is its restriction to the sites of $\mathbb{T}^{(k)}$. Fourth, under it $V = \mathcal{M}^k U$ holds on every bond far from $Y(h)$. Consequently, setting $F^{(h)}(V) := \mathbb{E}_{\mathsf{P}_k}[F(\mathcal{M}^k U) \mid h, V]$, which is the conditional expectation with respect to P_k, taken in a version that is invariant in V, one has for each bounded gauge-invariant F on $\mathcal{V}_k$

$$\big\langle F(\mathcal{M}^k U) \big\rangle_{K,\ell,\beta} = \frac{\sum_h \int F^{(h)}\, d\tau_h}{\sum_h \int d\tau_h},$$

$$|F^{(h)}| \leq \sup |F|,$$

$$F^{(h)} = F \ \text{ if } F \text{ reads only bonds far from } Y(h);$$

in particular, $F^{(h)} = F$ for any history in which $\mathbf{R}$ has acted on none of the components. In precisely this sense ρ_k represents the block averages of the bare field. The mechanism has three parts. First, $\mathbf{T}$ integrates $F \circ \mathcal{M}$ exactly against the density one level lower. Second, on each component on which it acts, $\mathbf{R}$ puts a reference law in place of the law of the block field; this

alters the joint distribution of $\mathcal{M}^k U$ and V_k only in a neighbourhood of that component, and it leaves every mass as it was. Third, the steps are glued in succession, each being regarded as a transport of the gauge-averaged terms. All the assertions are identities between measures and involve neither a constant nor a smallness condition. The role of the proposition is to fix what ρ_k means; nothing is derived from it. In the two-point law (7b), a single scale is not isolated by this coupling, because, inside the region it heals, the law of the block field is replaced by $\mathbf{R}$. It is isolated by the exactness with the sources read at depth s, that is, by $\sum_h \tau_h^w(\mathcal{V}_{K-s}) = Z(w)$ (Proposition 4.3, together with (4b)).

Why the frozen choices still suffice. Bałaban's theorems concern a single real coupling at each step (from these numbers the profiles build the function $g_k^2(x)$ of 119 (1988), (2.24), which coincides with g_k^2 on the last domain), a background that minimises the action appearing in the exponent, and a single coupling-constant counterterm in each step. For this reason the source is not built into that framework but is carried alongside it. In terms of the running shift $\zeta_k(\cdot; w)$ of (4b),

$$-A\big(x_k(\cdot) - \zeta_k(\cdot; w), U\big) = -A\big(x_k(\cdot), U\big) + \sum_{y \in \mathbb{T}^{(k)}} A\big(h_y^{(k)} \zeta_k(\cdot; w), U\big),$$

where $h_y^{(k)}$ denote Bałaban's tent functions (Comm. Math. Phys. 119 (1988), (3.40), (3.65)). On the set where $|\partial \mathbf{U} - \mathbf{1}| < \alpha_{0,k} \eta_k^2$ one has

$$\big|A(h_y^{(k)} \zeta_k(\cdot; w), \mathbf{U})\big| \le C \alpha_{0,k}^2 \sup |\zeta_k| \le C \varrho \, g_k^2 (\log x_k)^{2q} \ll 1 :$$

it is merely an additional local term, small and analytic, once g is chosen small after ϱ. When the fields are real, $|e^{-A(x_k(\cdot) - \zeta_k, U)}| \le e^{-(1 - C\varrho g^2) A(x_k(\cdot), U)}$; hence a failed test still costs $e^{-c p_0 (g_k)^2}$.

Proposition 4.5 (Reality and positivity at real sources). All the choices are real, and so are the f_i; consequently each move takes real data to real data. Hence, at real (c, z), each of the following quantities is real: the boundary terms, the increments, the activities $E^{(j),w}, R^{(j),w}$ (at real fields), the vacuum constants, the running shifts, and also the $e_j(X; w)$. The exponents are therefore real at real block fields. Suppose moreover that the source is a real constant with $|\zeta| < 2\varrho$. Then, in the operations $\mathbf{T}_k(Z)$ and $\mathbf{T}_i'(X)$ of (4b), every exponential factor is strictly positive at real pairs, and

$$x_k(\cdot) - \zeta_k(\cdot; \zeta) > 0,$$
$$\tau_h^{\zeta} \ge 0 :$$

thus the stability bounds of the run rely on a positivity that is lost at complex w and is still available at real shifts. Furthermore, ρ_0^{ζ} coincides with the bare density at $\beta - \zeta$, written with the choices belonging to β: these are two presentations of a single measure (3c). The choices of a reference run do not depend on β. Take a real β whose run has room 2ϱ in the tube, and a real ζ with $|\zeta| < \varrho$; along such a run one then obtains $x_j^{\natural}(\beta - \zeta) = x_j^{\natural}(\beta) - \zeta_j(\zeta)$ and $b_j^{\natural}(\beta - \zeta) = b_j(\zeta)$. These facts enter the runs with real shifts in (3c) and in Step 8, as well as the monotone map that sends the bare coupling to the renormalised coupling (8b). The conjugated copies of (4d) and (4e) make no use of these facts; holomorphy is all they require (Lemma 4.11 [to be proved; Package D]).

The scope of the mechanism. Because $\mathcal{O}(f) = A(f, U)$ is a weighted action, its source is a coupling shift. A different bounded gauge-invariant local observable Φ need not be a weighted plaquette action. The exact insertion identity can transport $e^{z\Phi}$, but does not put it in the

variable-Wilson format or supply its counterterms, source stability or uniform correlation estimates. Those require a separately closed source construction.

Exact insertion of block observables: a coupling of bare field, history and block field. **(i) The setting and the objects.**

Purpose of the block. The effective density of level k decomposes into a finite sum of positive measures indexed by the histories h of level k: $\rho_k\,dV_k = \sum_h \tau_h$. A single application of $\mathbf{T}$ integrates $F \circ \mathcal{M}$ exactly against the full density one level lower, so after k applications one has $\int F\,d(\mathbf{T}^k\rho_0) = \int F(\mathcal{M}^k U)\,\rho_0(U)\,dU$. In the case where $\mathbf{R}$ acts nowhere, $\rho_k\,dV_k$ is the image of $\rho_0\,dU$ under $\mathcal{M}^k$, and the assertion of this block then amounts to the definition of an image measure. Where $\mathbf{R}$ acts on a component, a reference law takes the place of the law of the block field inside that component. This modifies how $\mathcal{M}^k U$ and V_k are jointly distributed on bonds close to that component, and no mass changes. What this block asserts is that the replacements fit together into a single probability measure, defined on triples (bare field, history, block field), with respect to which, away from the components acted upon, the block field coincides with the block average of the bare field. No small factor and no constant enters Statements 1 to 6.

Spaces. Each $\mathcal{V}_j$ is a finite product of copies of the compact group G, and is therefore a compact metrisable space. Finite sets are given the discrete topology, and all measures are Borel measures.

Terms as the measures produced by moves. In parts (4a) and (4c) the construction is regarded as a chain of moves, a move being a finite family of positive kernels with one kernel for each outcome. The outcomes of the moves of the steps $1, \dots, j$, listed in order, form a string, and such a string is called a history of level j. We write $\mathsf{K}_h(U, \cdot)$ for the kernel obtained by composing along the string h; it runs from $\mathcal{V}_0$ to the states reached after step j. The admissible histories of level j, namely the strings for which $\mathsf{K}_h \neq 0$, form the finite set $\mathfrak{H}_j$, and $\mathfrak{H}_0 = \{\varnothing\}$. The term of a history $h \in \mathfrak{H}_j$ is defined as the measure into which its string transforms the bare measure, taken as a measure in the block field. Its mass is $\mathsf{t}(h)$:

$$\tau_h(\mathcal{B}) := \int \rho_0(U)\,\mathsf{K}_h\big(U, \{V_j \in \mathcal{B}\}\big)\,dU,$$

$$\tau_\varnothing = \rho_0\,dU,$$

$$\mathsf{t}(h) := \tau_h(\mathcal{V}_j).$$

The statements and the proofs below use the terms only through this definition.

The two stages of a history. Given $h' \in \mathfrak{H}_{j+1}$, we write $h'|_j \in \mathfrak{H}_j$ for the sub-list of those entries of h' that come from the steps and from the large-field operations of the levels $\leq j$. The strings at the stage reached once the moves of $\mathbf{T}$ of step $j+1$ have been made form a finite set, $\mathfrak{H}^{\mathbf{T}}_{j+1}$. Its elements are the lists $a = (h; P, Q, \Xi)$. Each is formed by some $h \in \mathfrak{H}_j$ together with the outcomes of those three moves of $\mathbf{T}$ which are built from tests:

- $P = P_{j+1}$, the outcome of the plaquette tests (at step 1, the pair (P_0, P_1));
- $Q = Q_{j+1}$, the outcome of the approximate fluctuation tests;
- $\Xi = \Xi_{j+1}$, the outcome of the fluctuation tests together with the recursion along the boundary, which determines $\Omega_{j+1}, \Lambda_{j+1}, S_{j+1}$.

For such an a we set $a|_j := h$. We let $\tau^{\mathbf{T}}_a$ be the measure produced by the string a, taken in the variable V_{j+1}, so that it is a measure on $\mathcal{V}_{j+1}$. The class of the large-field operation of level $j+1$ is denoted $\mathcal{Q}_{j+1}(a)$. It is determined by the entries of a and consists of every component of Z_{j+1} that satisfies the class conditions (part (2b)). The union of its members is

$$\mathsf{Z}(a) := \bigcup_{X \in \mathcal{Q}_{j+1}(a)} X \subset T_\ell.$$

The components that leave the class in the course of the operation (the drop-outs) are contained in $\mathsf{Z}(a)$, and so are the second-class components, although both kinds remain live. For $h' \in \mathfrak{H}_{j+1}$, the sub-list $h'^{\mathbf{T}} \in \mathfrak{H}_{j+1}^{\mathbf{T}}$ consists of $h'|_j$ together with the outcomes P, Q, Ξ of step $j+1$. Every other entry of h' is an outcome or a class produced within the large-field operation of level $j+1$.

The region of action. Let $h \in \mathfrak{H}_j$. We denote by $Y(h) \subset T_\ell$ the union of those components of the large-field regions of h which have been acted on by the large-field operation at one of the levels $\le j$; equivalently, it is the union of the class components of these levels. In particular $Y(\varnothing) = \emptyset$, and

$$Y(h') = Y(h'|_j) \cup \mathsf{Z}\big(h'^{\mathbf{T}}\big) \qquad (h' \in \mathfrak{H}_{j+1}).$$

Clear and far bonds. To a bond c of $\mathbb{T}^{(j)}$ we attach the set of bare sites $\mathrm{Bl}_j(c) := B^j(c_-) \cup B^j(c_+)$, where $B^0(y) = \{y\}$. We say that the bond c is clear of a set Y when $\mathrm{Bl}_j(c) \cap Y = \emptyset$. A bond c of $\mathbb{T}^{(k)}$ is called far from $Y(h)$ when neither $\mathrm{Bl}_k(c)$ nor any bare bond joining its two blocks meets $Y(h)^\sim$. This set is the union of the enlarged sets $X^\sim$, one for each of the class components X whose union is $Y(h)$. The enlargement adds to such a component one layer of cubes of $\pi_{j(X)}$, and $j(X)$ is the level at which the operation acted on X.

Gauge averages. The G-valued gauge transformations of $\mathbb{T}^{(j)}$ form the group $\mathcal{G}_j$. It is a finite product of copies of G and carries the Haar probability measure dv. It acts through $V \mapsto V^v$, with $(V^v)^{v_0} = V^{v_0 v}$. Given $u \in \mathcal{G}_i$ with $i \le j$, we write $u|_j \in \mathcal{G}_j$ for the gauge transformation obtained from it by restriction to the sites of $\mathbb{T}^{(j)}$. A function on $\mathcal{V}_j$, or a measure on it, is called invariant if this action leaves it invariant. The same convention applies on the configurations over a set of bonds closed under reversal, and on $\mathfrak{H} \times \mathcal{V}_j$, where the group acts trivially on the finite set $\mathfrak{H}$. Given a bounded Borel function F and a finite positive measure τ on $\mathcal{V}_j$, we define

$$\bar F(V) := \int F(V^v)\, dv,$$

$$\int F\, d\bar\tau := \int \bar F\, d\tau.$$

(ii) The statements. Throughout, β is real, $0 \le s \le K$ and $k = K - s$. The run is assumed to satisfy, through level k, the inductive description of the effective densities (part (2e)), with $x_0, \dots, x_k \ge \gamma^{-2}$, together with the two-sided flow bound (part (3a)). Of all these data, the proofs use only the fact that the moves of the steps $1, \dots, k$ are formed, that is, they use nothing beyond a reference sequence and a torus that is large compared with the cubes of the levels $\le k$.

1. The step, taken term by term. Let $0 \le j < k$ and $h \in \mathfrak{H}_j$. Then each $\tau_a^{\mathbf{T}}$ with $a|_j = h$ is a finite non-negative measure on $\mathcal{V}_{j+1}$. Moreover, every bounded Borel invariant F on $\mathcal{V}_{j+1}$ satisfies the following identity (the transport identity of the stage $\mathbf{T}$):

$$\sum_{a \in \mathfrak{H}_{j+1}^{\mathbf{T}} :\, a|_j = h} \int F\, d\tau_a^{\mathbf{T}} = \int (F \circ \mathcal{M})\, d\tau_h.$$

Now let a be such that $a|_j = h$, and let F be a bounded Borel invariant function on $\mathcal{V}_{j+1}$ which does not depend on the variables $V(b)$ for which b is a bond of $\mathbb{T}^{(j+1)}$ having at least one end point in $\mathsf{Z}(a)$. Then (the transport identity of the stage $\mathbf{R}$)

$$\sum_{h' \in \mathfrak{H}_{j+1} \colon h'^{\mathbf{T}} = a} \int F \, d\tau_{h'} = \int F \, d\tau_a^{\mathbf{T}}.$$

After summation over $h \in \mathfrak{H}_j$, the first identity expresses the exactness of $\mathbf{T}$ on the whole density. Taking $F \equiv 1$ in both identities gives

$$\sum_{h' \in \mathfrak{H}_{j+1} \colon h'|_j = h} \mathsf{t}(h') = \mathsf{t}(h) :$$

thus the step divides the mass of a history among its continuations. The law of the histories (part (4d)) is based on this identity.

2. Gauge averages. Suppose a compact group $\mathcal{G}$ with Haar probability dv acts continuously on a compact metrisable space $\mathcal{X}$. Define $\bar{F}(x) = \int F(v \cdot x) \, dv$ and $\int F \, d\bar{\tau} = \int \bar{F} \, d\tau$. (a) The function $\bar{F}$ is bounded, Borel and invariant, with $\sup |\bar{F}| \le \sup |F|$, and $\bar{F} = F$ when F is invariant. The measure $\bar{\tau}$ is finite, positive and invariant. It satisfies $\int F \, d\bar{\tau} = \int F \, d\tau$ for invariant F, and $\bar{\tau}(N) = \tau(N)$ for invariant Borel sets N. (b) Every invariant finite positive measure ν satisfies $\int \bar{F} \, d\nu = \int F \, d\nu$. Hence, if two invariant finite positive measures give the same integral to every bounded Borel invariant function, they are equal. (c) Let $r : \mathcal{G} \to \mathcal{G}'$ be surjective, and let $q : \mathcal{X} \to \mathcal{X}'$ be a Borel map with $q(v \cdot x) = r(v) \cdot q(x)$. Then the image under q of any invariant measure is invariant, and $f \circ q$ is invariant whenever f is invariant.

3. Gluing. Let $\mathcal{X}_1, \mathcal{X}_2, \mathcal{Y}$ be compact metrisable spaces and let $p_i : \mathcal{X}_i \to \mathcal{Y}$ be Borel maps. Let λ_i be finite positive measures on $\mathcal{X}_i$ such that $(p_1)_* \lambda_1 = (p_2)_* \lambda_2$. Then $\mathcal{X}_1 \times \mathcal{X}_2$ carries a finite positive measure λ with the following two properties: its marginals are λ_1 and λ_2, and $p_1(x_1) = p_2(x_2)$ for λ-almost every (x_1, x_2).

4. Clear bonds. (a) One has $c_-, c_+ \in \mathrm{Bl}_j(c)$. (b) Let c be a bond of $\mathbb{T}^{(j+1)}$, and let b be a bond of $\mathbb{T}^{(j)}$ whose two end points lie in the double block $B(c_-) \cup B(c_+)$. Then $\mathrm{Bl}_j(b) \subset \mathrm{Bl}_{j+1}(c)$. (c) Every bond of $\mathbb{T}^{(k)}$ that is far from $Y(h)$ is clear of $Y(h)$.

5. A single step as a transport of the averaged terms. Let $0 \le j < k$. Then $(\mathfrak{H}_j \times \mathcal{V}_j) \times (\mathfrak{H}_{j+1} \times \mathcal{V}_{j+1})$ carries a finite positive measure m_{j+1} with the following properties: (a) the marginal on $\mathfrak{H}_j \times \mathcal{V}_j$ equals $\sum_{h \in \mathfrak{H}_j} \delta_h \otimes \bar{\tau}_h$; (b) the marginal on $\mathfrak{H}_{j+1} \times \mathcal{V}_{j+1}$ equals $\sum_{h' \in \mathfrak{H}_{j+1}} \delta_{h'} \otimes \bar{\tau}_{h'}$; (c) for m_{j+1}-almost every $((h, V), (h', V'))$ both of the following hold: $h'|_j = h$, and $V'(c) = \mathcal{M}(V)(c)$ for every bond c of $\mathbb{T}^{(j+1)}$ that has no end point in $\mathsf{Z}(h'^{\mathbf{T}})$.

6. Exact insertion of block observables. We write $Z = Z_{K,\ell,\beta} = Z(0)$ and $\mu = \mu_{K,\ell,\beta}$, and $\mathbb{E}_{\mathsf{P}}$ denotes the expectation with respect to a probability measure P. (a) On $\mathcal{V}_0 \times \mathfrak{H}_k \times \mathcal{V}_k$ there exists a probability measure P with the following four properties. Its marginal on $\mathcal{V}_0$ equals μ. Its marginal on $\mathfrak{H}_k \times \mathcal{V}_k$ is built from the averaged terms:

$$\bar{\nu}_k := Z^{-1} \sum_{h \in \mathfrak{H}_k} \delta_h \otimes \bar{\tau}_h.$$

Hence this marginal assigns to every bounded Borel function $\Phi(h, V)$ that is invariant in V the integral $Z^{-1} \sum_h \int \Phi(h, V) \, \tau_h(dV)$, and the law of the histories of level k is the marginal of P on $\mathfrak{H}_k$: for $\mathsf{E}_k \subset \mathfrak{H}_k$, $\mathsf{P}(h \in \mathsf{E}_k) = Z^{-1} \sum_{h \in \mathsf{E}_k} \mathsf{t}(h) = \tilde{\nu}(\mathsf{E}_k)$. It is invariant under the maps $(U, h, V) \mapsto (U^u, h, V^{u|_k})$, $u \in \mathcal{G}_0$. Conditionally on (U, h), the third coordinate agrees with $\mathcal{M}^k U$ on every bond that is clear of $Y(h)$, and in particular on every bond far from $Y(h)$. (b) For each bounded Borel invariant function F on $\mathcal{V}_k$, let

$$F^{(h)}(V) := \mathbb{E}_{\mathsf{P}}\big[F(\mathcal{M}^k U) \,\big|\, h, V \big]$$

be chosen among the versions of the conditional expectation as one that is invariant in V. Versions with this invariance exist, and for each h any two of them coincide τ_h-almost everywhere. Each of them satisfies

$$\big\langle F(\mathcal{M}^k U)\big\rangle_{K,\ell,\beta} = \frac{\sum_h \int F^{(h)}\, d\tau_h}{\sum_h \int d\tau_h},$$

$$|F^{(h)}| \le \sup|F| \ \tau_h\text{-almost everywhere},$$

$$\sum_h \int d\tau_h = Z.$$

(c) Suppose F depends only on the variables of bonds that are clear of $Y(h)$; this is the case in particular when its dependence is confined to bonds far from $Y(h)$. Then one has $F^{(h)} = F$ almost everywhere with respect to τ_h. In particular, for every history in which the large-field operation has acted on none of the components, $F^{(h)} = F$. Without this restriction one has, in general, $F^{(h)} \neq F$. Here the depth s is arbitrary, $0 \le s \le K$, and $k = K - s$.

7. Bounded block observables. Take F as in Statement 6 (b) and level-k cells $\mathfrak{c}_1, \dots, \mathfrak{c}_n$, and denote by $\mathsf{E}_k := \bigcup_{i \le n} \mathfrak{lv}_k(\mathfrak{c}_i) \subset \mathfrak{H}_k$ the event that at least one of these cells is contained in the live region of level k. Then

$$\Big|\big\langle F(\mathcal{M}^k U)\big\rangle_{K,\ell,\beta} - \frac{\sum_{h \notin \mathsf{E}_k} \int F^{(h)}\, d\tau_h}{\sum_h \int d\tau_h}\Big| = \Big|\frac{\sum_{h \in \mathsf{E}_k} \int F^{(h)}\, d\tau_h}{\sum_h \int d\tau_h}\Big|$$

$$\le \sup|F|\ \tilde{\nu}(\mathsf{E}_k) \le n\, C\, \sup|F|\, e^{-c_{\mathrm{lv}} p_0(g_k)}.$$

The final inequality is valid whenever the data satisfy the hypotheses of the one-cell bound of part (4d), with C and $c_{\mathrm{lv}} > 0$ the constants of that bound (written C and c in Theorem 4.13). It holds uniformly in K, β, ℓ and the positions of the cells. In this form the bound applies to an observable of bounded modulus that depends on the block variables of a bounded number of cells at its own scale. In the expectation of such an observable, the part contributed by the histories of E_k is smaller than any power of g_k.

(iii) Proofs.

On 1. The two transport identities and the specified locality of each move are properties of the actual history operations that we must prove: obligation C1 of Section 9. The following argument explains their consequences for specified kernels and observables. It does not replace their verification for the Yang–Mills construction by a reference to (4c). Positivity. All kernels of all moves are non-negative. Hence the measure into which a string turns $\rho_0\, dU \ge 0$ is finite and non-negative. The stage **T**. In its first move, **T** sets $V_{j+1} := \mathcal{M}(V_j)$, and no later move of **T** alters V_{j+1}. The reason is that the Faddeev–Popov draw uses gauge transformations supported on the sites $B(y) \setminus \{y\}$, and these act on V_j without touching V_{j+1}. Hence at this stage the identity is valid for every bounded Borel F. As an equality of measures, $\sum_{a|_j = h} \tau_a^{\mathbf{T}} = \mathcal{M}_* \tau_h$. The stage **R**. Given a class component X, denote by $\Lambda^{\mathrm{pr}} \cap X$ its box. This box lies well inside X, and **R** integrates the block field over it. Denote by $\mathfrak{b}_X$ the set of bonds of $\mathbb{T}^{(j+1)}$ that have an end point in that box. The moves of **R** alter V_{j+1} on the bonds of the sets $\mathfrak{b}_X$ and nowhere else. Hence at this stage the identity is valid for every bounded Borel F that does not depend on these bonds. Each bond of $\mathfrak{b}_X$ has both end points in X. Therefore every function that depends on no bond with an end point in $\mathsf{Z}(a)$ belongs to this class. The property that matters is that F does not depend on $\mathfrak{b}_X$. Gauge invariance by itself is not enough: the plaquette function of (i) is a counterexample. Why history by history. Each move, once summed over its outcomes,

is a probability kernel at each state. For a move consisting of tests, exactly one outcome takes the value 1. For the gauge fixing the property holds because the gauge fixing is carried out as a procedure, namely as a draw. Let $\mathfrak{G} \geq 0$ be normalised by $\int \mathfrak{G}(V^u)\, du = 1$. Then the measure

$$(\tau\kappa_{\mathfrak{G}})(\mathcal{B}) = \int \tau(dV) \int du\, \mathfrak{G}(V^u)\, \mathbf{1}_{\mathcal{B}}(V^u)$$

has the same mass as τ, for every measure τ. The product $\mathfrak{G}\,\tau$, by contrast, has mass $\int \mathfrak{G}\, d\tau$. If τ is invariant under the group with respect to which $\mathfrak{G}$ is normalised, this coincides with the mass of τ, and then $\tau\kappa_{\mathfrak{G}} = \mathfrak{G}\,\tau$. Suppose that the bare weight is invariant under gauge transformations and that the representatives transform covariantly. Then, history by history, the term on which a gauge-fixing move acts has this invariance (see the treatment of the steps as stochastic kernels at the beginning), and for such a term the draw coincides with the product. The invariance meant here is invariance under the group of the move. A term need not be invariant under the full gauge group of the level. For a measure without this invariance, replacing the draw by the product does not conserve the mass in general; this already happens in exactly computable models over finite groups. The masses considered from here on are the ones produced by the draw, which preserves them for any bounded Borel weight. Likewise, the axial frames are drawn before the tests that use them. The test that follows the gauge fixing along the tree lacks gauge invariance. Therefore the mass a_r that passes it depends on the frame r. Suppose the components that leave kept a single frame e while the survivors were averaged over the frames. Then the mass after the reference laws would be off by $\bar{a} - a_e$, with $\bar{a}$ the average of the a_r, and already for $F \equiv 1$ the identity of the stage $\mathbf{R}$ would not hold. The actual history construction must verify both points: the frame draw precedes its tests, and the product formula represents that draw on the required invariant terms. Why it does not suffice to have the identities in their form summed over the histories. Let $\mathcal{V}_0 = \mathcal{V}_1 = \mathcal{V}_2 = \{0,1\}^2$, where bond 1 is exterior and bond 2 is the component. Let $\mathcal{M} = \mathrm{id}$, let the gauge group be trivial, and let μ be the uniform law on the diagonal. Level 1 has two outcomes a_1, a_2, with $\tau^{\mathbf{T}}_{a_1} = \tau^{\mathbf{T}}_{a_2} = \mu/2$, and $\mathbf{R}$ acts on bond 2 in a_2 alone. Thus $\tau_{a_1} = \mu/2$, while τ_{a_2} is the product of the marginal of $\mu/2$ on bond 1 with the uniform law on bond 2. At level 2 no $\mathbf{R}$ acts, and every history has a single continuation, a_i' for a_i. We set $\tau_{a_1'} := \tau_{a_2}$, $\tau_{a_2'} := \tau_{a_1}$. Every term is non-negative. The totals are transported exactly, and the masses are conserved history by history, $\mathsf{t}(a_i') = \mathsf{t}(a_i)$. Moreover, $\mathbf{R}$ replaces a term by the product of its exterior marginal with a probability law. On $\{h = a_1'\}$, however, $Y(a_1') = \emptyset$ forces $V = U$, which is a point of the diagonal. Yet $\tau_{a_1'}$ assigns mass $\frac{1}{4}$ to the complement of the diagonal. Hence no P exists. The failure lies in the identity of the stage $\mathbf{T}$ applied to the one history a_1 by itself. Below, the two identities are applied to invariant F alone.

Proof of 2. (a) The map $(v, x) \mapsto F(v \cdot x)$ is bounded and Borel, so Fubini's theorem shows that $\bar{F}$ is Borel. Right invariance of Haar measure gives $\bar{F}(v_0 \cdot x) = \int F(vv_0 \cdot x)\, dv = \bar{F}(x)$. The functional $F \mapsto \int \bar{F}\, d\tau$ is positive and continuous under monotone limits. It therefore defines a measure $\bar{\tau}$. With $F_{v_0}(x) = F(v_0 \cdot x)$, left invariance yields $\overline{F_{v_0}} = \bar{F}$, and this is the invariance of $\bar{\tau}$. The remaining assertions follow from $\bar{F} = F$ for invariant F, applied to indicators as well. (b) If ν is invariant, Fubini's theorem gives $\int \bar{F}\, d\nu = \int dv \int F(v \cdot x)\, \nu(dx) = \int F\, d\nu$. Hence ν is determined by the values it assigns to the integrals of the invariant functions $\bar{F}$. (c) This follows by a change of variables. By this device, the proof converts the two identities of Statement 1 into equalities of measures, namely equalities between the gauge averages of the terms. The proof applies these identities to invariant functions only.

Proof of 3. We apply the disintegration theorem in the following form. Let $\mathcal{X}$, $\mathcal{Y}$ be compact metrisable, let $p : \mathcal{X} \to \mathcal{Y}$ be Borel, let λ be a finite positive measure on $\mathcal{X}$, and let $\nu = p_*\lambda$. Then there exist probability measures $\kappa(y; \cdot)$ on $\mathcal{X}$, Borel in y, such that $\lambda(B) = \int \nu(dy)\, \kappa(y; B)$

and $\kappa(y; p^{-1}\{y\}) = 1$ for ν-almost every y. Set $\nu = (p_i)_*\lambda_i$, and let $\kappa_i(y;\cdot)$ be a disintegration of this kind of λ_i along p_i. Define

$$\lambda(B) := \int \nu(dy)\,\big(\kappa_1(y;\cdot)\otimes\kappa_2(y;\cdot)\big)(B).$$

The integrand is Borel in y, because the sets B with this property form a Dynkin class that contains the products of Borel sets. We have $\lambda(B_1\times\mathcal{X}_2) = \int\nu(dy)\kappa_1(y;B_1) = \lambda_1(B_1)$, and the second marginal is treated in the same way. The set $\{p_1(x_1) = p_2(x_2)\}$ is Borel and contains $p_1^{-1}\{y\}\times p_2^{-1}\{y\}$. For ν-almost every y, the latter set has full measure under $\kappa_1(y;\cdot)\otimes\kappa_2(y;\cdot)$.

Proof of 4. (a) Every site lies in its own block. (b) For a site y of $\mathbb{T}^{(j+1)}$, its cube is the union, over the sites $x\in B(y)$ of $\mathbb{T}^{(j)}$, of their cubes. After intersection with the bare lattice this becomes $B^{j+1}(y) = \bigcup_{x\in B(y)} B^j(x)$. Consequently $B^j(b_\pm)\subset B^{j+1}(c_-)\cup B^{j+1}(c_+)$. (c) This follows from $Y(h)\subset Y(h)^\sim$.

Proof of 5. The measures τ_h, $\tau_{h'}$, $\tau_a^{\mathbf{T}}$ are all finite and non-negative. Consequently the gauge average of each of them is a finite, positive and invariant measure. Step 1 (the stage **T**: Radon–Nikodym). Let $h\in\mathfrak{H}_j$ be fixed. The map $\mathcal{M}$ is covariant, $\mathcal{M}(V^v) = \mathcal{M}(V)^{v|_{j+1}}$, as well as Borel measurable, and $v\mapsto v|_{j+1}$ is a surjection of $\mathcal{G}_j$ onto $\mathcal{G}_{j+1}$. Therefore $\mathcal{M}_*\bar\tau_h$ is invariant, and $F\circ\mathcal{M}$ is invariant whenever F is. Let F be any bounded Borel function on $\mathcal{V}_{j+1}$. Applying the transport identity of the stage **T** to the invariant function $\bar F$, we obtain

$$\sum_{a|_j=h}\int F\,d\bar\tau_a^{\mathbf{T}} = \sum_{a|_j=h}\int \bar F\,d\tau_a^{\mathbf{T}} = \int(\bar F\circ\mathcal{M})\,d\tau_h = \int(\bar F\circ\mathcal{M})\,d\bar\tau_h = \int \bar F\,d(\mathcal{M}_*\bar\tau_h) = \int F\,d(\mathcal{M}_*\bar\tau_h).$$

Hence $\sum_{a|_j=h}\bar\tau_a^{\mathbf{T}} = \mathcal{M}_*\bar\tau_h$. In particular $0\le\bar\tau_a^{\mathbf{T}}\le\mathcal{M}_*\bar\tau_h$. Therefore $\bar\tau_a^{\mathbf{T}} = f_a\cdot\mathcal{M}_*\bar\tau_h$ for Borel functions $0\le f_a\le 1$ with $\sum_{a|_j=h} f_a = 1$ almost everywhere. Define $\mathsf{m}_a^{\mathbf{T}}$ as the image of $(f_a\circ\mathcal{M})\,\bar\tau_h$ under $V\mapsto(V,\mathcal{M}(V))$. It is a measure on pairs $(V,W)\in\mathcal{V}_j\times\mathcal{V}_{j+1}$. The marginals of this measure are $(f_a\circ\mathcal{M})\bar\tau_h$ and $\bar\tau_a^{\mathbf{T}}$, and almost surely $W = \mathcal{M}(V)$. What the identity transports is the sum over the outcomes and nothing more. It does not fix which portion of τ_h goes to which a, and the only splitting it supplies is the one by $f_a\circ\mathcal{M}$. Accordingly, under the coupling just constructed, the outcome a is drawn with probability $f_a(\mathcal{M}(V))$, and it is not determined by the tests. Step 2 (the stage **R**: restriction off the class). Fix a. Let $\mathrm{Bd}(a)$ denote the set of bonds of $\mathbb{T}^{(j+1)}$ having no end point in $\mathsf{Z}(a)$. Let $\mathcal{V}(a)$ be the space of configurations on $\mathrm{Bd}(a)$, and let $\mathrm{res}_a:\mathcal{V}_{j+1}\to\mathcal{V}(a)$ be the restriction map, which commutes with the gauge action. Set $\lambda_a := \sum_{h'^{\mathbf{T}}=a}\delta_{h'}\otimes\bar\tau_{h'}$. Let f be a bounded Borel invariant function on $\mathcal{V}(a)$. Then $f\circ\mathrm{res}_a$ is invariant and depends on no bond with an end point in $\mathsf{Z}(a)$. The transport identity of the stage **R** therefore gives

$$\sum_{h'^{\mathbf{T}}=a}\int(f\circ\mathrm{res}_a)\,d\bar\tau_{h'} = \sum_{h'^{\mathbf{T}}=a}\int(f\circ\mathrm{res}_a)\,d\tau_{h'} = \int(f\circ\mathrm{res}_a)\,d\tau_a^{\mathbf{T}} = \int(f\circ\mathrm{res}_a)\,d\bar\tau_a^{\mathbf{T}}.$$

The measure $(\mathrm{res}_a)_*\bar\tau_a^{\mathbf{T}}$ and the image of λ_a under $(h',V')\mapsto\mathrm{res}_a(V')$ are both invariant, so Statement 2 (b) shows that they are equal. The identity says nothing about the bonds with an end point in $\mathsf{Z}(a)$. For this reason the two measures are glued over $\mathcal{V}(a)$. The result is a measure $\mathsf{m}_a^{\mathbf{R}}$ on $\mathcal{V}_{j+1}\times(\mathfrak{H}_{j+1}\times\mathcal{V}_{j+1})$ with marginals $\bar\tau_a^{\mathbf{T}}$ and λ_a. Almost surely under it, $\mathrm{res}_a(V') = \mathrm{res}_a(W)$ and $h'^{\mathbf{T}} = a$. Step 3 (composition). On the middle field W, the marginal of $\mathsf{m}_a^{\mathbf{T}}$ and the marginal of $\mathsf{m}_a^{\mathbf{R}}$ are both equal to $\bar\tau_a^{\mathbf{T}}$. We glue over $\mathcal{V}_{j+1}$ and discard the middle field. We then attach to V the label $h = a|_j$ and sum over a. The resulting measure is m_{j+1}. (a): $\sum_{a|_j=h}(f_a\circ\mathcal{M})\bar\tau_h = \bar\tau_h$. (b):

$\sum_a \lambda_a = \sum_{h'} \delta_{h'} \otimes \bar{\tau}_{h'}$, because each h' has exactly one first stage. (c): almost surely $W = \mathcal{M}(V)$, $\mathrm{res}_a(V') = \mathrm{res}_a(W)$ and $h'^{\mathbf{T}} = a$, so that $h'|_j = a|_j = h$.

Proof of 6. For $0 \le j \le k$ we define

$$\mathsf{A}_j := \big\{(U,h,V) \in \mathcal{V}_0 \times \mathfrak{H}_j \times \mathcal{V}_j : V(b) = (\mathcal{M}^j U)(b) \text{ for every bond } b \text{ of } \mathbb{T}^{(j)} \text{ clear of } Y(h)\big\},$$

which is a Borel set. Step 1 (induction on the level). We show that, for each $0 \le j \le k$, the space $\mathcal{V}_0 \times \mathfrak{H}_j \times \mathcal{V}_j$ carries a probability measure P_j with the following three properties. Its projection to $\mathcal{V}_0$ is μ. Its marginal on $\mathfrak{H}_j \times \mathcal{V}_j$ equals $Z^{-1} \sum_{h \in \mathfrak{H}_j} \delta_h \otimes \bar{\tau}_h$, which is the family of averaged terms. Finally, $\mathsf{P}_j(\mathsf{A}_j) = 1$. To begin the induction: the measure $\rho_0\, dU$ is invariant, so $\bar{\tau}_\varnothing = \tau_\varnothing$, and we take for P_0 the push-forward of μ under $U \mapsto (U, \varnothing, U)$. Passage from j to $j+1$: by the second property together with Statement 5 (a), the measures P_j and $Z^{-1}\mathsf{m}_{j+1}$ induce the same marginal on $\mathfrak{H}_j \times \mathcal{V}_j$. We glue the two over this space and project the result onto (U, h', V'), which gives P_{j+1}. The first two properties carry over, and Statement 5 (b) is what carries over the second. As for the third, with probability one we have $(U,h,V) \in \mathsf{A}_j$, $h'|_j = h$, and $V'(c) = \mathcal{M}(V)(c)$ outside $\mathsf{Z}(h'^{\mathbf{T}})$. Take a bond c clear of $Y(h') = Y(h) \cup \mathsf{Z}(h'^{\mathbf{T}})$. Both of its end points belong to $\mathrm{Bl}_{j+1}(c)$, so neither lies in $\mathsf{Z}(h'^{\mathbf{T}})$, and therefore $V'(c) = \mathcal{M}(V)(c)$. For each bond b of the double block of c we have $\mathrm{Bl}_j(b) \subset \mathrm{Bl}_{j+1}(c)$. Hence such a bond is clear of $Y(h)$, and $V(b) = (\mathcal{M}^j U)(b)$. Because $\mathcal{M}(V)(c)$ depends on V only through these bonds, we obtain

$$V'(c) = \mathcal{M}(V)(c) = \mathcal{M}(\mathcal{M}^j U)(c) = (\mathcal{M}^{j+1} U)(c).$$

Step 2 (symmetrisation). The group $\mathcal{G}_0$ acts on triples by $(U,h,V) \mapsto (U^u, h, V^{u|_k})$. Averaging P_k over this action gives a measure P, and it is invariant under the action. Since μ and $\bar{\nu}_k = Z^{-1}\sum_h \delta_h \otimes \bar{\tau}_h$ are both invariant, P still has the first two properties. Covariance gives $\mathcal{M}^k(U^u) = (\mathcal{M}^k U)^{u|_k}$, so the condition $V(b) = (\mathcal{M}^k U)(b)$ holds exactly when $V^{u|_k}(b) = (\mathcal{M}^k(U^u))(b)$ does. Thus the set A_k is invariant, and $\mathsf{P}(\mathsf{A}_k) = \mathsf{P}_k(\mathsf{A}_k) = 1$. Let Φ be a function and N a Borel set, both invariant in V, with sections N_h. Then Statement 2 (a) gives

$$\int \Phi\, d\bar{\nu}_k = Z^{-1} \sum_h \int \Phi(h,V)\, \tau_h(dV),$$

$$\bar{\nu}_k(N) = Z^{-1} \sum_h \tau_h(N_h).$$

This is why the statement is restricted to invariant Φ and F. The marginal of P consists of the averaged terms, and the averaged terms coincide with the terms only on invariant functions and invariant sets. Step 3 (assertion (a)). The marginals and the invariance were obtained in Step 2. A function of h alone is invariant in V, and this yields the law of the histories. Because $\mathsf{P}(\mathsf{A}_k) = 1$, for almost every (U,h) a regular conditional distribution of V given (U,h) is concentrated on the configurations that agree with $\mathcal{M}^k U$ on every bond clear of $Y(h)$, and so in particular on every far bond. Step 4 (invariant versions). Suppose that F is invariant, and set $\mathsf{f}(U) := F(\mathcal{M}^k U)$. This function is invariant under $\mathcal{G}_0$. Take an arbitrary version Ψ of $\mathbb{E}_{\mathsf{P}}[\mathsf{f}(U)|h,V]$ with $|\Psi| \le \sup|F|$, so that $\mathbb{E}_{\mathsf{P}}[\mathsf{f}(U)\phi(h,V)] = \int \Psi\phi\, d\bar{\nu}_k$ for every bounded Borel ϕ. Each $v \in \mathcal{G}_k$ is of the form $u|_k$, where $u \in \mathcal{G}_0$ takes the value $v(y)$ on the block $B^k(y)$. Using the invariance of P, of f and of $\bar{\nu}_k$, we get

$$\mathbb{E}_{\mathsf{P}}[\mathsf{f}(U)\phi(h,V)] = \mathbb{E}_{\mathsf{P}}[\mathsf{f}(U^u)\phi(h,V^v)] = \mathbb{E}_{\mathsf{P}}[\mathsf{f}(U)\phi(h,V^v)] = \int \Psi(h,V)\phi(h,V^v)\, d\bar{\nu}_k = \int \Psi(h,V^{v^{-1}})\phi(h,V)\, d\bar{\nu}_k.$$

The first equality holds because P is jointly invariant. We integrate over v, apply Fubini's theorem and use the fact that Haar measure is invariant under inversion. It follows that $F^{(h)}(V) := \int \Psi(h, V^v)\, dv$ is also a version, and that it is Borel, invariant in V and bounded by $\sup|F|$. Any two invariant versions differ only on an invariant Borel set N with $\bar\nu_k(N) = 0$. Since the terms are non-negative, this gives $\tau_h(N_h) = 0$ for every h. Step 5 (assertion (b)). Take an invariant version. Then

$$\big\langle F(\mathcal{M}^k U)\big\rangle = \int \mathsf{f}\, d\mu = \mathbb{E}_{\mathsf{P}}[\mathsf{f}(U)] = \int F^{(h)}(V)\, \bar\nu_k(dh, dV) = Z^{-1} \sum_h \int F^{(h)}\, d\tau_h.$$

The last equality uses the invariance of $F^{(h)}$ in V. Taken by itself, the defining property of a conditional expectation yields the integral against $\bar\nu_k$ and nothing more. No use can be made at this point of a non-invariant version. Consider a term in which some component has been put into the second class by the large-field operation at level k. By part (4c), such a term carries, in the level-k block field, a Dirac distribution on the gauge tree of the box. Consequently τ_h and $\bar\tau_h$ are mutually singular for such a term. A version is determined $\bar\tau_h$-almost everywhere, and in this case the integral against τ_h of a version that is not invariant is then arbitrary. Taking $\Phi \equiv 1$ gives $\sum_h \int d\tau_h = Z$. For every h, any invariant version coincides τ_h-almost everywhere with the version constructed in Step 4, and that version is bounded by $\sup|F|$. Step 6 (assertion (c)). Fix h_0, and suppose that F depends only on bonds clear of $Y(h_0)$. On $\mathsf{A}_k \cap \{h = h_0\}$ we then have $F(\mathcal{M}^k U) = F(V)$. It follows that

$$\mathbb{E}_{\mathsf{P}}[\mathsf{f}(U)\mathbf{1}_{\{h=h_0\}}\phi(V)] = \mathbb{E}_{\mathsf{P}}[F(V)\mathbf{1}_{\{h=h_0\}}\phi(V)]$$

for every bounded Borel ϕ. In other words, $F^{(h_0)} = F$ $\bar\nu_k$-almost everywhere on $\{h_0\} \times \mathcal{V}_k$. The set on which this equality fails is invariant in V and is therefore a τ_{h_0}-null set. When the operation has acted on no component of h_0, we have $Y(h_0) = \emptyset$, and then every bond is clear.

Proof of 7. The equality comes from splitting the numerator of Statement 6 (b) into the part over E_k and the part over its complement. We have $\tau_h \geq 0$, and $|F^{(h)}| \leq \sup|F|$ holds τ_h-almost everywhere. Hence $|\int F^{(h)}\, d\tau_h| \leq \sup|F|\, \mathsf{t}(h)$. Summing this over $h \in \mathsf{E}_k$ and dividing by $Z = \sum_h \mathsf{t}(h)$ gives $\sup|F|\, \tilde\nu(\mathsf{E}_k)$. Subadditivity together with the one-cell bound $\tilde\nu(\mathfrak{lv}_k(\mathfrak{c})) \leq C\, e^{-c_{\mathrm{lv}} p_0(g_k)}$ of part (4d) yields $\tilde\nu(\mathsf{E}_k) \leq n\, C\, e^{-c_{\mathrm{lv}} p_0(g_k)}$. For a history outside E_k, no one of the cells lies in a large-field region that is live at level k.

The proofs above use gauge averaging, the disintegration of measures and their gluing, the Radon–Nikodym theorem, an induction on the level, and the symmetrisation of a coupling.

(iv) What is used.

From other parts. The following results of part (1a) are used. The map $\mathcal{M}$ is Borel. It is gauge covariant, $\mathcal{M}(V^v) = \mathcal{M}(V)^{v|}$. It is local: $\mathcal{M}(V)(c)$ depends on V only through the bonds of the double block of c. From part (4c) we use three things. The first is the description of the construction as a chain of moves. The second is that every kernel is non-negative, so that every τ_h and every $\tau_a^{\mathbf{T}}$ is a finite non-negative measure. The third is the pair of transport identities of Statement 1. From part (2b) we use the class of the large-field operation, which enters through the recursion $Y(h') = Y(h'|_j) \cup \mathsf{Z}(h'^{\mathbf{T}})$. Statements 1 to 6 use nothing further from the construction: no bound, no analyticity, and no form of a term. Only Statement 7 draws in addition on part (4d), through the one-cell bound there on the law of the histories.

What reads it. No argument in another part draws its conclusion from Statement 6. Within this part Statement 6 has a single consequence, Statement 7, and the proof of Statement 7 combines it with the one-cell bound of part (4d). Of the results of this block, the other parts rely only on the last line of Statement 1, $\sum_{h'|_j = h} \mathsf{t}(h') = \mathsf{t}(h)$. This identity makes the laws of the histories of the

levels $j \le K$ a consistent family of probability measures. That family is the law $\tilde{\nu}$ of part (4d). The bounds on large fields rest on it, and through them so do the laws at one scale. For $k \ge 1$ the observables of the theorem do not have the form $F(\mathcal{M}^k U)$. Indeed, $\mathcal{O}(f) = \sum_p f(x(p))\, a_p(U)$ is a function of the bare plaquettes, and Statement 6 cannot be applied to it. These observables are carried as sources (first block of this part). A single scale is isolated by exactness with sources at depth s:

$$\sum_{h \in \mathfrak{H}_{K-s}} \tau_h^w(\mathcal{V}_{K-s}) = Z(w),$$

whence, at $c = 0$,

$$\Big\langle \exp \sum_i z_i\, \mathcal{O}(f_i) \Big\rangle_{K,\ell,\beta} = \frac{Z(w)}{Z(0)}$$

and this quantity is an expectation, taken over the histories, the block fields and the stored variables at level $K - s$, of the exponential of the sourced exponent minus the unsourced one.

Against Bałaban's papers. Bałaban presents the operation $\mathbf{R}$ in a simplified form in the Introduction of Comm. Math. Phys. 122 (1989) 175, formula (0.3) on p. 176:

$$(\mathbf{R}\rho)(V) = \sum_Z \rho(Z'', V) \int dV|_{Z'}\, \rho(Z, V) \Big/ \int dV|_{Z'}\, \rho(Z'', V).$$

This form has the normalisation property $\int dV\, (\mathbf{R}\rho)(V) = \int dV\, \rho(V)$ ((0.4)). In the procedure as actually carried out, the denominators integrate parts of the densities that are localised near Z' (p. 177), and the normalisation property takes the form (1.102), p. 201. The objects treated in Bałaban's papers are densities. On observables, he states that his Theorem 1 "is the basis for many other applications, for example, for an analysis of expectation values of physical observables, like loop variables, averaged loop variables", and calls these problems that "deserve detailed analysis and further publication" (Comm. Math. Phys. 122 (1989) 355, p. 356).

(4b) Analytic dependence on the source; locality and decay in tame and in wild regions. **Objects.** We write $w(p) := w(x(p))$. When the source is a constant shift $w \equiv \zeta$, the increments $b_j(\zeta)$ are computed on the infinite lattice in the same way as b_j. They therefore do not depend on the torus, and they are real for real ζ. The running shift is $\zeta_k = \zeta + \sum_{j \le k} [b_j(\zeta) - b_j(0)]$, and the inverse couplings are $x_k - \zeta_k$. For a general source, let p_y denote a bare plaquette at y. Then

$$\zeta_k(p; w) = w(p) + \sum_{j \le k} \varphi_j(p) \sum_{y \in \mathbb{T}^{(j)}} h_y^{(j)}(p)\, \big[b_j(w(p_y)) - b_j(0) \big] :$$

in other words, the coupling counterterm of level j is applied locally, and its coefficient is taken at the local value of the source.

Proposition 4.6 (The format with sources). The density ρ_k^w has the form of (2e), with $x_k(\cdot)$ replaced by $x_k(\cdot) - \zeta_k(\cdot; w)$. Its activities $E^{(j),w}, R^{(j),w}, B^{(j),w}$ are holomorphic in (c, z), depend on w only through its values on $X^{\sim 1}$, and obey the bounds valid at $w = 0$ up to constants. Moreover $|\zeta_k(p; w)| \le \frac{4}{3} \sup |w|$, and for every constant $|\zeta_*| < 2\varrho$

$$\big| E^{(j),w}(X, \cdot) - E^{(j),\zeta_*}(X, \cdot) \big| \le E_0\, \varrho^{-1} \mathsf{m}_j\, e^{-\kappa d_j(X)},$$

$$\mathsf{m}_j := \max \Big\{ \sup_{X^{\sim 1}} |w - \zeta_*|,\, CML^{j-K} \operatorname{Lip} w \Big\}.$$

Strategy: induction on the levels, on a complex polydisc of data. On a complex polydisc of data (activities, boundary terms, shifts) centred at the real run, a single step (the cluster expansion of (1d) together with **R** of (2b)–(2d)) is local, analytic and uniformly bounded. This is the step on the complex class of (2e). Its small-field core is Proposition 1.6 [to be proved; Package A–H] of (1d), which is the one-step theorem stated as a holomorphic map on a class of families of activities. The sourced data remain inside the polydisc, so holomorphy follows by composition. Locality follows because an activity at X depends on the data, and on the outcomes of the history, only near X (locality in the history, (2a)).

The marginal term. When the source is constant, the step preserves the symmetries of (1e), so once $b_{k+1}(\zeta)A$ has been extracted only irrelevant terms remain. The estimate is the one of (3c). There is a remainder $\le C g_k^{\kappa_0-4}$, coming from the $R^{(j)}$, which is holomorphic in ζ on a disc of radius $C\varrho$. Apart from this remainder, the increment b_{k+1} is holomorphic in the shifts ζ_j of the earlier levels on discs of radius cx_j, where $c \le \sigma_\star$. These discs make up the polydisc of (1d), taken along the run's own couplings. On these discs its non-constant part, less the remainder, is bounded by $C g_k (\log x_k)^{\mathfrak{p}_0}$. Cauchy's estimate together with $x_k \ge g^{-2} + c(K-k)$ yields

$$|b_{k+1}(\zeta) - b_{k+1}(0)| \le \mathsf{a}_{[K-k-1]}|\zeta|,$$
$$\sum_{s<K} \mathsf{a}_{[s]} \le C \sum_k \left[g_k^3 (\log x_k)^{\mathfrak{p}_0} + \varrho^{-1} g_k^{\kappa_0-4} \right] \le \tfrac{1}{3},$$

where the quantity $\mathsf{a}_{[s]}$ carries a depth index and denotes, for the step that arrives at depth s, the Lipschitz number of its increment (the same letter is used in (3c) and (8b)). At zero source, reality and the displayed radial bound give $\zeta_k'(0) \in [\frac{2}{3}, \frac{4}{3}]$. Ordered derivative bounds on other admitted real constant-source intervals use the full slope estimate of (3c). On X a general source is compared with $\zeta_* = w(p_*)$, where p_* is a bare plaquette of $X^{\sim 1}$. When $\mathsf{m}_j < \varrho$, the slope displayed above comes from the Schwarz lemma, applied along a line of data and not along a line of sources. This line is a family, holomorphic in t, of data of the level below; at $t = 0$ it coincides with the data of ζ_*, and at $t = 1$ it coincides on $X^{\sim 1}$ with those of w. For $|t| < \varrho/\mathsf{m}_j$ the family remains in the convex polydisc. When $\mathsf{m}_j \ge \varrho$, the slope is simply the sup bound. Subtract from each of the two activities the normaliser part $\mathbf{N}^{(j)}$ of (1d), which is common to both; what is left of each is at most $\frac{1}{2} E_0 e^{-\kappa d_j(X)}$. The slope is thus a clause that the induction carries along. The induction carries similar slopes for $R^{(j)}$ and $B_S^{(j)}$, and also, among the operations treated below, for the excess exponents Ψ_ω. The slope does not follow, up to the boundary of the class of sources $\{\|w\|_\flat < 2\varrho\}$, from holomorphy in w, from the bound, from locality and from the agreement at constant sources. Moreover, $\mathsf{m}_j \le e^{d_j(X)} L^{-(K-j)} \|w\|_\flat$. The part that no counterterm renormalises therefore carries a factor $L^{-(K-j)}$, which is summable over the scales. In short, a gradient of the source is irrelevant.

Wild regions. Two kinds of operations act on the sourced exponent. The first kind consists of the operations $\mathbf{T}_k(Z)$ on the components on which **R** has not acted. These are built from the choices of the run only, namely delta functions, tests, gauge-fixing factors and conditional Gaussian weights, and they contain no Wilson weight (2b). On such components the source appears in the function on which these operations act, that is, in the Wilson weight and in the boundary terms. Taken on its own, $\mathbf{T}_k(Z)1$ carries no small factor: for a component produced by failed plaquette tests, this quantity is $\le \exp[C \sum_{t \le k} \#_t(Z_t \cap Z)]$. Three places carry the small factor (2c): the Gaussian forms of the fluctuation fields, the kept gauge-fixing exponents and the Wilson weight. Only the last of these depends on the source; the first two are built from the choices of the run. The pointwise majorant of (2c) gives one factor $e^{-c p_0(g_t)^2}$ for each creation, at the coupling of that creation's own step, and one constant for each site. It is used in the form it

has at $w = 0$; the sourced exponent and the unsourced one are compared further below, on the supports of the tests.

The second kind concerns a component X on which $\mathbf{R}$ has acted at some level i (whether it was healed, remained as an island, or dropped out because a test of a stage or of the main step failed). For such a component the source enters inside its operation $\mathbf{T}'_i(X)$. Except for a drop-out, this operation contains the Wilson factor of the core. When the component is healed or is an island, the operation coincides with that of Bałaban, Comm. Math. Phys. 122 (1989) 355, (1.71). Let ω run over the local histories of X, let ξ be the stored variables and $\mu_\omega \geq 0$ their measures, and let $\mathbf{p} = (\mathbf{U}, \mathbf{J})$ be the pairs on $X^\sim$ at which a term evaluates the factor of X (2c). For each ω, e^{Φ_ω} is replaced by $e^{\Phi_\omega + \Psi_\omega}$, with an *excess exponent*

$$\Psi_\omega = A\big(\hat{\zeta}\,\phi_{\mathrm{c}} \mathbf{1}_X, U''(\xi)\big) + \big[V^w - V^0\big](X^\sim) + \sum_{\omega'} \Psi_{\omega'} :$$

its summands are the Wilson term, weighted by the shift $\hat{\zeta}$ of the prepared regions ($|\hat{\zeta}| \leq C\varrho$, and $0 \leq \phi_{\mathrm{c}} \leq 1$ is the cut-off on the old core, (2d)) and evaluated at the background $U''(\xi)$ of the inner region, which is constructed from the stored fields; the difference between the interaction terms V of $\mathbf{R}$ with and without the source; and the exponents of the inner histories ω', namely those of the swallowed islands and drop-outs. The operation modified in this way is written $\mathbf{T}'_i(X; w)$. (In the case of a drop-out, the operation (2c) is taken to include the Wilson weight that the carrying term has over X. The operations that are kept are built from the choices of the run only, and none of their factors depends on the source, which appears in that Wilson weight and in the boundary terms. The exponent is then $\sum_{\omega'} \Psi_{\omega'}$ alone, the contribution of the swallowed islands, and it obeys the bound given below.) The exponent Ψ_ω is holomorphic in (c, z), is real for real pairs and real sources, vanishes at $w = 0$, and depends on w only on $X^{\sim 2}$. The background U'' is real, and its regularity holds only at the scale on which the respective layer was stored. Where $\phi_{\mathrm{c}} \neq 0$, one has $|\partial U'' - \mathbf{1}| \leq C\check{R}_n^2 \varepsilon_n \eta_n^2 \leq C\eta_n^2$ on $\Omega_n \setminus \Omega_{n+1}$, so the bound $a_p \leq C\eta_n^4$ balances the $C\eta_n^{-4}$ plaquettes per site of level n (on the bare layer $a_p \leq 2$, since G is compact). Each layer therefore contributes $O(\varrho)$ for each site of its own scale:

$$|\Psi_\omega| \leq \bar{f}\,\#(\omega),$$
$$\bar{f} = C(1 + \varrho),$$

where, as in (2e), $\#(\omega)$ is the number of stored sites, each being counted at its own level. Since $\#(\omega)$ is additive over the inner histories, the constant after a step is the maximum of $\bar{f}$ and the constant of the new term, not their sum. Hence the constant does not increase with the level. The perturbation is extensive and must therefore be paid in advance. This is done by the mass bound of (2c), (2e), which the induction maintains with a reserve weight $e^{c\mathsf{v}(\omega)}$, where $c = c_{\mathrm{r}}$ is the reserve constant of (2e), one fixed number (the same constant stands in every weight $e^{c\mathsf{v}}$ of this part except the raised one in the bound of the whole label, whereas in the exponents that contain p_0 the letter c denotes a generic constant). Here the space-time volume $\mathsf{v}(\omega)$ counts $\log x_j$ for each stored site of level j:

$$\mathsf{M}(i, X) := \frac{1}{|W_4|} \sum_{\omega} e^{c\mathsf{v}(\omega)} \int \sup_{\mathbf{p}} \big|e^{\Phi_\omega(\xi; \mathbf{p})}\big|\, \mu_\omega(d\xi) \leq e^{-(2-c')p_0(g_i)},$$

with the supremum over the complex pairs taken inside the integral and the sum. This form applies to a healed component and to an island. In the case of a drop-out, the integral contains the Wilson weight of the carrying term over X. The supremum is then taken over the evaluations

at ξ, that is, over the real configurations which are compatible with the stored variables, are regular layer by layer, and give the listed tests their recorded values, with the exception of the test whose failure made X a drop-out; these evaluations form a set fibred over ξ (2c). A product set contains a flat $\mathbf{U}$, at which $e^{-A} = 1$, and the supremum over such a set removes the factor. Bałaban's bound is $\mathbf{T}'_k(X)1 \le e^{-(2-c')p_0(g_k)}$ (ibid., (1.89), where Bałaban has $2(1+\beta_0)^{-1}$ in place of $2 - c'$), for each fixed pair and without the weight. If g is small enough once ϱ has been fixed, then $c \log x_j \ge 2\bar{f}$. Consequently $|e^{\Psi_\omega}| \le e^{c\mathsf{v}(\omega)}$, and for every bounded function F of the stored variables and every evaluation of this kind

$$\big|\mathbf{T}'_i(X;w)F\big| = \Big|\frac{1}{|W_4|}\sum_{\omega}\int F\, e^{\Phi_\omega+\Psi_\omega}\, d\mu_\omega\Big| \le \sup|F|\,\mathsf{M}(i,X) \le \sup|F|\, e^{-(2-c')p_0(g_i)}.$$

Under a positive measure, the modulus of a complex factor serves as a bound for it; in the case of the small complex term produced by a complex pair, this is the device of Bałaban, Comm. Math. Phys. 122 (1989) 355, (1.73). The estimate just displayed is made on the moduli, and a sourced component retains the small factor of the unsourced one. That $\mathbf{T}'_i(X;w)F$ is holomorphic follows from dominated convergence together with the theorems of Morera and Fubini. For a healed component, the exponentiation of (2d) produces its local terms from this bound, each at most $e^{-p_0(g_i)}e^{-\kappa d_i(Y)}$ on a domain Y, just as at $w = 0$.

The whole label. Bounds on the individual components cannot be multiplied together, even though each is uniform in the variables outside its own component, because the background couples the components. (For $\mathbf{1}\{|\xi_1 - \xi_2| < \epsilon\}$ on $[0,T]^2$, every integral in one variable is $\le 2\epsilon$, while the double integral equals $2\epsilon T - \epsilon^2$.) By Tonelli, what does multiply is a product of functions of disjoint groups of variables; here the factors of this one product are the pieces of the pointwise majorant together with the integrands of the masses. For this reason the supremum is placed inside the integral. The result is the integrated bound of (2c), with one factor for each creation and one mass for each outermost component on which $\mathbf{R}$ has acted. If every e^{Φ_ω} in $\mathbf{T}'_k(\mathcal{S})$ is replaced by $e^{c\mathsf{v}(\omega)}|e^{\Phi_\omega}|$, then

$$\int \mathbf{T}'_k(\mathcal{S})\Big[\chi_{k,\mathcal{S}}\, e^{-A(x_k(\cdot),U_k(\mathbf{V}))}\Big]dV_k \le \exp\Big[C\sum_{j\le k}(1+\log x_j)\,\#_j(Z_j)\Big]\prod_{(t,\mathfrak{c})} e^{-cp_0(g_t)}\prod_{(i,X)}\mathsf{M}(i,X),$$

where the first product runs over those creation cells, outside these components, that belong to a single history with the label $\mathcal{S}$, namely the history for which this product is largest (the creation cells are not determined by the label, (2c)). The second runs only over the outermost ones among the components on which $\mathbf{R}$ has acted and which remain in the label. (In (2c) the bound carries $p_0(g_t)^2$ and has no logarithm. The sum over the labels needs only the present weaker form, because $CM^4\check{R}_t^5 \log x_t \le \frac{1}{2}c\,p_0(g_t)$ (Proposition 2.13 [outlined below; Section 32]): half of the exponent pays the cost and the other half is kept. Inside a drop-out the mass has consumed the factors of the creations, and whatever the label still shows there is paid for by the weight. The masses are required at a weight that exceeds the $e^{c\mathsf{v}(\omega)}$ spent by the sources by a constant; this constant is determined by the per-site cost of the part of the label visible inside the drop-out, and its choice comes after that of ϱ and before the ceiling on g is set (the mass bound is valid at every weight $c \ge 0$, (2c), and the masses $\mathsf{M}(i,X)$ in the display are understood at the raised weight). The surplus comes from that part of $\mathsf{M}(i,X)$ which is made up of the local histories admitted by the label; on these, $\mathsf{v}(\omega)$ is bounded below by the volume shown (2c), (2e).) Once its vacuum constant is set aside, the sourced term of the label is dominated by the left-hand side multiplied by $e^{C(1+\varrho)}$ for each bond of the determining set, for three reasons. The moduli are taken inside the integrals. In the operations, $|e^{\Psi_\omega}| \le e^{c\mathsf{v}(\omega)}$. On the supports of the tests, what

remains of the sourced exponent once the Wilson term $-A(x_k(\cdot),U_k)$ and the vacuum constant are set aside is at most $C(1+\varrho)$ for each bond of the determining set (4c). Consider the weighted sum over the labels of (2e), that is, the total masses summed over $\mathcal{S}$, where the credit pays the costs. In it every component C of Z_k keeps a factor

$$\exp\big[-cp_0(g_k)-\kappa_1|C|_k\big],$$

where $|C|_k$ is measured in cells. In (2c) the factor is taken per site, with $\#_k(C)\ge d_k(C)$ in place of $|C|_k$. This contains the form "with the additional term $-\kappa_1 d_k(X)$", which Bałaban gives for a general region (ibid., p. 387).

The cost of the reserve. A factor $e^{c\log x_j}$ for each site amounts to at most $e^{cM^4\check R_j^5}$ for each cell. Its form is that of the volume costs $CM^4\check R_n^5$ per cell and per step, which, until $\mathbf{R}$ acts on a component, are covered by the credit $\kappa_j(Z)$ of that component: an amount $p_0(g_j)^2$ for each cube, deposited at its creation, pays for $\check R_j^6$ for each cube (ibid., (1.79)–(1.81) and (1.89); these bound $\mathbf{T}'_k(X)1$ for the operation of (1.71), which contains the Wilson factor). Bałaban already pays $O(1)\log g_j^{-2}$ for each site of Z_j, on account of the normalisation constants (ibid., p. 380). The spare weight, whose constant c is free, holds once g is small enough, the constant c being chosen first (2c).

Proposition 4.7 (Sources along a frame). Let $(\theta_j)_{j\le n}$ be a frame such that $\theta_j\ge\gamma_J^{-2}$ throughout, and let (x_j) be the run along it starting from a real β, as in the opening of Step 3. The cube sizes are $\check R_j=R(\min_{i\le j}\theta_i)$. All the other auxiliary choices of level j (the reference laws, the thresholds, the gauge-fixing weight and the scaling of the fluctuation field) are made with θ_j, whereas the weight of the action obeys its own recursion. The reference line is a frame. A first-passage run, that is, the run starting from a first-passage datum (K,β), is the run taken along its own couplings. The run is said to stay in the tube if $|\theta_j-x_j|\le\mathsf{t}_j$ for all j, and to have room r if $|\theta_j-x_j|+r\le\mathsf{t}_j$. Suppose that the run has room $2r$ for $r=\varrho$ or for $r=2\varrho$. Then, for all sources with $\|w\|_\flat<r$ and for the steps of the frame, Propositions 4.3, 4.5, 4.6 and Theorem 4.1 are valid, with upper stability and the floor holding with $C(1+\log\theta_n)$ for each site of $\mathbb{T}^{(n)}$. They hold on every admissible pair (ℓ,K) of a torus and a cutoff, admissibility meaning $K\ge n$ and $\ell L^{K-i}\ge CM\check R_i$ for $i\le n$, as in (3b); all of this is uniform in ℓ, in K, in n, in β and in the frame. Moreover

$$\mathsf{z}_j(\zeta):=(\theta_j-x_j)+\zeta_j(\zeta)\in D(0,\mathsf{t}_j),$$
$$\sum_{j\le n}\sup_{0<|\zeta|<r}\frac{|b_j(\zeta)-b_j(0)|}{|\zeta|}\le\tfrac13.$$

The mechanism is as follows. A step depends separately on the choice θ_j and on the weight $x_j-\zeta_j$. For given remainders, its small-field part is holomorphic in their difference $\mathsf{z}_j(\zeta)$ on $D(0,\sigma_\star\theta_j)\supset D(0,\mathsf{t}_j)$ (Proposition 1.6 [to be proved; Package A–H]); this difference is the deviation $\mathsf{z}_j=\theta_j-x_j$ displaced by the running shift. The full step is formed at a real deviation in the tube, together with a complex shift about it of modulus at most $\bar c$ (the complex class of (2e)). Consequently, as far as the increments are concerned and with the remainders $R^{(i)}$ held fixed, a deviation from the frame and a constant source are the same variable. The remainders depend on the source as well, and the activities in the two presentations of Proposition 4.5 need not coincide term by term. The room is what keeps the complex excursion $|\zeta_j|\le\frac43|\zeta|$ inside the disc. The objects constructed do not depend on the tube. A single statement therefore serves the first-passage runs, the comparison of two cutoffs in (8a), and the runs with frozen choices that

occur in (3c) and (5a); it is used with $r = \varrho$, where the room is 2ϱ, and with $r = 2\varrho$, where the room is 4ϱ (which is the room $C\varrho$ of Theorem 4.1).

Reading at depth s. For the run stopped at level $K - s$, everything above holds with the following changes. The norm $\|f\|_{\sharp,s} = \max\{\|f\|_\infty, CML^{-s} \operatorname{Lip} f\}$ replaces $\|f\|_\sharp$, the Lipschitz constant being measured at scale L^{-s}. The sums are restricted to $j \le K - s$. Either normaliser may be used, each with exponents of its own: E of the whole run, or the vacuum constant $E_{(K-s)}$ of the first $K - s$ steps. The number of sites of $\mathbb{T}^{(K-s)}$ replaces ℓ^4, and $L^{-(K-s-j)}$ replaces $L^{-(K-j)}$. The exponents of the two global bounds also depend on s (7b).

(4c) Finite ordinary sources and the positive-label floor.
A finite source neighbourhood without loss of radius. Fix the choices of the real run and put $r = 2\varrho$. At depth s use

$$\|f\|_{\sharp,s} = \max\{\|f\|_\infty, CML^{-s} \operatorname{Lip} f\}, \qquad \|w\|_{\flat,s} = |c| + \sum_i |z_i| \|f_i\|_{\sharp,s}.$$

The constant-source disc is doubled. Constant sources are constructed first. The increment is holomorphic on the coupling polydisc of radii proportional to θ_j. A single Schwarz estimate moving every coordinate simultaneously bounds its slope by the maximum of the coordinate displacements divided by these radii. Thus the slope gains θ_j^{-1}; it does not contain a sum of reciprocals over all earlier levels. The lower flow bound makes the resulting slopes summable. For a holomorphic increment put

$$\mathsf{d}_\rho[b] = \sup_{0<|\zeta|<\rho} \frac{|b(\zeta) - b(0)|}{|\zeta|}.$$

For a sufficiently small coupling,

$$\sum_j \mathsf{d}_{2r}[b_j] \le \tfrac{1}{3}, \qquad |\zeta_j(\zeta)| \le \tfrac{4}{3}|\zeta|.$$

This leaves the same fixed margin at every level.

For a general source, freeze its value near a base point. The difference between the actual regular activity and the one at the frozen constant is a defect with no coupling counterterm. At an old level $j \le m$ it has size $L^{-(m-j)}$. This gain must hold on the whole source class. Holomorphy on that class alone would give a derivative estimate only after shrinking the class, which cannot be repeated indefinitely.

Instead interpolate the data of a step. We specify the distance used for this interpolation. Identify a region Y with the bare plaquettes having a corner in it, and use the torus distance between their barycentres to set

$$\operatorname{Lip}(Y; w) = \sup_{\substack{p,p' \in Y \\ x(p) \ne x(p')}} \frac{|w(p) - w(p')|}{d_{T_\ell}(x(p), x(p'))}, \qquad \mathsf{l}_j(Y; w) = CML^{-(K-j)} \operatorname{Lip}(Y; w),$$

$$\mathsf{m}_j(Y; w, \zeta_0) = \max\left\{ \sup_{p \in Y} |w(p) - \zeta_0|, \mathsf{l}_j(Y; w) \right\}.$$

An empty supremum is zero. These quantities are monotone in Y, and $\mathsf{m}_j \le \mathsf{m}_{j+1}$ on a fixed region.

For a union W of cubes of the output partition of step $k + 1$, take $\|w\|_{\flat,K-k-1} < r$, fix one constant $\zeta_0 \in D(0, r)$, and put $\mathsf{m} = \mathsf{m}_{k+1}(W; w, \zeta_0)$. Keep the unsourced tests, backgrounds, coordinate maps and reference measures fixed. At sites in W, interpolate the local weight and each Euclidean member, together with its own coefficient, from the data of ζ_0 to those of w. Interpolate defects, remainders and boundary terms only when their complete read regions lie in W; treat a boundary sum site by site. Outside W use the constant-source data. Write $\mathfrak{L}_t^W$ for this

line of data. Its complex parameter satisfies $|t|\mathsf{m} < r$. It need not be the system of one physical source function. If $\mathsf{m} = 0$, locality gives the desired zero differences directly; the disc below is used for $\mathsf{m} > 0$.

The input slopes admit the entire line to the larger data box. Its weight size is at most $3r$ and its regular norm at most $E_0 + 1$. For every older read $Y \subset W$ of level $j \le k+1$,

$$\mathsf{l}_j(Y; w) \le L^{-(k+1-j)} \mathsf{l}_{k+1}(W; w) \le L^{-(k+1-j)} \mathsf{m}.$$

Together with vanishing at the constant source, this admits the defects with their required older-level factors. The other members retain their own majorants. The one-step formation theorem is used on this whole box and on the original common field cores.

For the regular output indexed by X, take $W = X^{\sim}$. This enlargement contains its complete input read region, including the finer read enlargements, so locality identifies the line's endpoints with the two source outputs below when $t = 1$ is admitted. Subtract the common Gaussian normalizer from the regular output. The cluster contribution and the thin-tube weight discrepancy bound what remains by $\frac{1}{2} e^{-\kappa' d(X)}$, with $\kappa' \ge \kappa + 1$, throughout $|t| < r/\mathsf{m}$. Schwarz's lemma therefore gives

$$|E^{(k+1),w}(X, \cdot, z) - E^{(k+1),\zeta_0}(X, \cdot, z)| \le \frac{\mathsf{m}}{r} e^{-(\kappa+1) d_{k+1}(X)}.$$

For $r/\mathsf{m} \le 1$, use the two ordinary endpoint bounds instead of evaluating the data line at $t = 1$. The factor $1/2$ pays for the factor two in Schwarz's estimate and returns the defect constant 1. Taking $\zeta_0 = w(p_z)$ and using the spare decay to sum the diameter of X gives

$$|D^{(j)}(X, \cdot, z)| \le L^{-(m-j)} e^{-\kappa d_j(X)}$$

at the terminal level m. The slopes of the other regular activities and boundary terms follow with their own majorants. The same ζ_0 is used for every object in the enclosing region.

For an operation cut by W, its exponent on $\mathfrak{L}_t^W$ is formed by the earlier steps on that line. Affine interpolation of a partial sum of its factors is not used. This recursion is well-defined: an output partition region is also a region for the preceding partition, the next source class is contained in the preceding class, and $\mathsf{m}_k \le \mathsf{m}_{k+1}$. A line needed at level $k+1$ has therefore already been admitted at level k. At $t = 0$ the formed exponents are those of the constant source. At $t = 1$, when this point is admitted, a factor satisfies $\psi_f(1) = \psi_f^w$ if $R_f^{\sim} \subset W$; agreement of the scalar part $\Psi^{\circ}(1) = \Psi^{\circ,w}$ requires $X^{\sim 2} \subset W$. Agreement on one carrier does not identify the whole scalar exponent.

Let $N_0^{\circ}(\omega)$ count the operation's own sites, at their respective levels, and define its total source size recursively by

$$N_0^{\circ}(\omega) = \sum_j \#_j(Z_j(\omega) \cap X), \qquad \mathsf{n}(\omega) = N_0^{\circ}(\omega) + \sum_{\omega' \in \mathrm{In}(\omega)} \mathsf{n}(\omega').$$

Each inner history enters once. At a new completed operation,

$$\Psi_\omega(t) = \Psi_\omega^{\mathrm{new}}(t) + \sum_{\omega'} \Psi_{\omega'}(t),$$

$$\Psi_\omega^{\mathrm{new}}(t) = A(\hat{c}(t) \upsilon \mathbf{1}_X, U''(\xi)) + V(X^{\sim}; \mathfrak{L}_t^W) - V(X^{\sim}; \mathfrak{L}^0).$$

Here $\mathfrak{L}^0$ is the fixed unsourced system, $|\hat{c}(t)| \le 3r$, $0 \le \upsilon \le 1$, and $U''(\xi)$ is real. Its layer bounds cost $3C_W r$ per newly owned site for the Wilson term; the bare layer is counted in bare sites. The two collar potentials cost $2c_{\mathrm{int}}(1 + 3r)$ per such site. Put

$$c_{\mathrm{new}} = 3C_W r + 2c_{\mathrm{int}}(1 + 3r), \qquad c_\Phi = 2c_{\mathrm{new}}.$$

Additivity of n returns the ordinary exponent constant as $\max(c_{\mathrm{new}}, c_\Phi) = c_\Phi$ and its formed-line bound as $\max(c_{\mathrm{new}}, 2c_\Phi) = 2c_\Phi$. On $W = X^{\sim 2}$, Schwarz gives slope $2c_{\mathrm{new}}\mathsf{m}/r = c_\Phi \mathsf{m}/r$ per newly

owned site. Each inner exponent retains its previous slope toward this same ζ_0, and its local distance is at most the enclosing distance. The size recursion thus returns c_Φ, with no new price per ancestor. A drop-out has only the sum of its inner excesses.

For the carrier oscillations this argument must be made in the sum norm. For a carrier R born at level i, put

$$M_R = (\log \theta_i)^{2\mathfrak{p}_0} e^{-\kappa d_i^\flat(\mathcal{S};R)}, \qquad \operatorname{osc}_{\mathbf{p}} h = \sup_{\mathbf{p},\mathbf{p}'} |h(\mathbf{p}) - h(\mathbf{p}')|.$$

At fixed stored variables the local formation bound is

$$\sum_{f:R_f=R} \operatorname{osc}_{\mathbf{p}} \psi_f(t) \le c_\times(3r) M_R, \qquad |t|\mathsf{m} < r.$$

The pair cores in this bound are fixed common admitted cores for the entire line. A carried mark retains the bound; at a new mark the same one-step estimate returns it.

Apply the data-line construction at the carrier's formation level i, with $W = R^\sim$ and now $\mathsf{m} = \mathsf{m}_i(R^\sim; w, \zeta_0)$. The case $\mathsf{m} = 0$ again follows by locality. First suppose $T = r/\mathsf{m} > 1$. For finitely many factors choose fixed admitted pairs $\mathbf{p}_f, \mathbf{p}'_f$ and fixed unimodular numbers σ_f so that the increments between zero and one of $\sigma_f[\psi_f(t;\mathbf{p}_f) - \psi_f(t;\mathbf{p}'_f)]$ are nonnegative real numbers. Their sum $H(t)$ is holomorphic and bounded by $c_\times(3r)M_R$ on $D(0,T)$. Schwarz gives $|H(1) - H(0)| \le 2c_\times(3r)M_R/T$. Take the pair suprema after this estimate, then increase the finite factor sets. It follows that

$$\sum_{f:R_f=R} \operatorname{osc}_{\mathbf{p}}[\psi_f^w - \psi_f^{\zeta_0}] \le 2c_\times(3r)\frac{\mathsf{m}_i(R^\sim; w, \zeta_0)}{r} M_R.$$

For $T \le 1$ the two ordinary oscillation bounds give the same result. The phases test already formed functions of one parameter; they do not introduce independent carrier coefficients or local sources. This proves the slope bound for the sum of suprema without treating that sum itself as holomorphic.

The formed-line bounds propagate with the same ordinary radius r. Their inputs are the actual one-step formation estimate, hereditary core inclusions and collar-potential bound used above. The mass reserve absorbs the resulting exponent; it does not prove those formation estimates. This argument does not assert the stronger weighted discs for independently marked zone uses.

The coupling ceiling is chosen after r and these exponent constants. On each component, $|e^{\Psi_\omega}| \le e^{C(r)\mathsf{n}(\omega)}$ is absorbed by its weighted unsourced mass. All estimates are made in total variation before the operations are composed. With $c_{\mathrm{L}} = C(r)$ this gives the actual term comparison

$$|e^{\mathcal{E}_k(\mathcal{S};w)}\tau_{\mathcal{S}}^w|(\mathcal{V}_k) \le e^{\mathcal{E}_k(\mathcal{S})}\tau_{\mathcal{S},c_{\mathrm{L}}}^{\mathrm{abs}}(\mathcal{V}_k)e^{c_{\mathrm{L}}\sum_l |\Gamma_l|}.$$

The weighted label sum applies at the raised mass weight required by c_{L}. The source-energy share missing from a label is bounded by $\sum_j (C\log\theta_{j-1} + C)\#_j Z_j^+$. The same weight absorbs it. Thus

$$\left| e^{-E_{(k)} - \sum_{j\le k,X} e_j(X;w)} Z(w) \right| \le e^{[C\log\theta_k + C(r)]N_k}.$$

No source-dependent denominator has entered this argument.

At zero source all terms are positive measures. Keep the never-large label and restrict every last-level bond to a ball of radius $s = c\theta_k^{-1/2}$ about $\mathbf{1}$. To check the tests, lift the last block field to each earlier lattice by taking the field equal to $\mathbf{1}$ inside its blocks and putting the block bonds on the crossings. A lifted plaquette is $\mathbf{1}$, a cancelling product VV^{-1}, or a plaquette of V; its curvature is at most $4s$. The local variational background estimate on each test cube then

gives the required small-field tests and $A(U_k) \le Cs^2 N_k$. Thus $x_k A(U_k) \le C N_k$ on this ball. If $D = \dim \mathrm{SU}(2) = 3$, its product Haar measure costs at most $\frac{1}{2} dD \log \theta_k + C$ per site. Hence

$$e^{-E_{(k)}} Z(0) \ge \exp[-(C + \tfrac{1}{2} dD \log \theta_k) N_k].$$

At $k = K$, $N_K = \ell^4$ and first passage confines θ_K to a fixed window above g^{-2}. The upper bound and the floor therefore have constants depending on g alone. They verify the two history criteria and the positive-label criterion in Section 37 for this construction.

All formations read a source only in their enlarged domain and begin with the function $w(p)$ at bare plaquettes. The resulting energies therefore depend on that function, independently of its presentation as a finite linear combination. Their exponential rate exceeds the domain entropy. At a chessboard continuation the copies vanish near cell faces; joining them increases neither their sup norm nor their Lipschitz constant. Splitting even and odd copies costs only the fixed factor two already allowed by the doubled source radius.

The energies just formed also have the rooted geometry used in Section 39. The sum over a regular point $z \in X$ and the remainder at X use the same domains of π_j; summing histories introduces no new independent domain index. At depth $s = K - j$ these are connected unions of cubes of side $M a_s$. With $M = L^{m_0}$ the number of such cubes along a torus side is $2L^{m - m_0 + s}$, hence even on the admitted tori. Their complete source-dependence region is a fixed-layer enlargement. For a domain of cubical edge-tree length $m(X)$, the sum over z costs at most $C_M(m(X) + 1)$. Combining the fixed origin groups and using the strict entropy-rate surplus absorbs this polynomial into the exponential bound. The rooted energy majorant therefore has precisely the geometry used in (GS1)–(GS3), with fixed constants independent of the cutoff.

The common vacuum density. The vacuum constants are sums of elementary Gaussian count constants and the values of the regular and remainder terms at $\mathbf{1}$. The Gaussian count constants per physical volume agree on two tori: their couplings and choices agree, and each count is proportional to that volume. Distribute $R^{(j)}(X, \mathbf{1})$ equally among its sites. The remaining density at a site is a sum over domains through it. Domains contained in a box of side less than one quarter of the smaller torus have a unique common lift and give identical densities.

At depth $u = K - j$, every remaining domain has $d_j(X) \ge c\ell_\wedge L^u / M - C$. Its exponential decay, summed against the domain entropy, costs $Ce^{-c\ell_\wedge L^u / M}$ per site. A unit physical volume contains L^{4u} sites. Consequently

$$\left| \frac{E_{K,\ell,\beta}}{\ell^4} - \frac{E_{K,\ell',\beta}}{\ell'^4} \right| \le C \sum_{u \ge 0} L^{4u} e^{-c\ell_\wedge L^u / M} \le C.$$

The bound is uniform in the cutoff. It uses the exact companion formation on nonwrapping domains, not a comparison of unspecified families at fixed norm. Reflection-positive subadditivity then moves a mirror-periodized source to the fixed reference torus. Upper stability is used there, while the larger torus contributes its unsourced floor and the displayed vacuum-density comparison. Thus the common normalizer can be formed on the reference torus itself.

(4d) Rarity, witnesses and ages. The history propositions below rest on obligation D3 of Section 9. In particular, we must prove witness locality, narrow-window estimates and replication with all required weights for the actual history operations. The argument below explains how these estimates enter the rarity and age bounds.

Theorem 4.13 (Large fields are rare uniformly in the volume). (Live regions, creation sets, lineages, marks and ages, as well as the law $\tilde{\nu}$ of the histories, are defined in (2a), and the prepaid bounds are stated in (2e). They have here the meaning given there.) For g small, for every level $0 \le k \le K$ (with $\mathcal{Z}_0 = \emptyset$) and every level-$k$ cell $\mathfrak{c}$, one has uniformly

$$\tilde{\nu}\big(\mathfrak{lv}_k(\mathfrak{c})\big) = \tilde{\nu}\big(\mathfrak{c} \subset \mathcal{Z}_k\big) \le Ce^{-cp_0(g_k)},$$
$$\tilde{\nu}\big(\mathfrak{c} \text{ is created at step } k\big) = \tilde{\nu}\big(\mathfrak{c} \subset \mathcal{W}_k^{\wedge}\big) \le Ce^{-cp_0(g_k)}.$$

The second bound ($k \ge 1$) is a consequence of the first, because creation cells are live. Indeed, $\mathbf{R}$ touches a component only if no creation has occurred in it during the last N_k steps. Hence a component of Z_k containing a creation cell of step k is not healed at step k, which gives $\mathcal{W}_k^{\wedge} \subset \mathcal{Z}_k$. The same inclusion also bounds the event that $\mathfrak{c}$ lies in a live component in which a creation occurs at step k. The theorem relies on the following point. A label's determining set starts from a neighbourhood of its first region, $\Omega_0 = \Omega_1^{\sim}$, and not from the whole torus. Its background therefore requires the bare field to be regular only on that neighbourhood (see under Lemma 4.15). If regularity is required at every bare plaquette of T_ℓ, then a single rough plaquette, wherever it lies, leaves the step-1 background undefined. By totality (2a), every test formed with that background then fails at every place, so that Z_1 equals the whole torus and is never healed. Moreover, at fixed (β, g), the probability that such a plaquette is present tends to 1 as $\ell \to \infty$. The theorem concerns only the law of the histories of the run; it gives no bound on the law that $\mu_{K,\ell,\beta}$ induces on the bare plaquettes inside a box of fixed physical size.

The proof is a first-moment argument. For a level-n cell $\mathfrak{z}$, let $\mathcal{E}_{\mathfrak{z}}^{(n)}(k)$ be the survival event of (2a), namely $\mathfrak{z} \subset \mathcal{Z}_i$ for $n \le i \le k$. The event $\mathcal{E}_{\mathfrak{c}}^{(k)}(K) \subset \mathfrak{lv}_k(\mathfrak{c})$ is the never-healed event. (i) If healing takes place at a level $j > k$, the factor of $\mathbf{R}$, $Ce^{-cp_0(g_k)-cp_0(g_j)}$, appears, and it is summable in j. (ii) We use Vol_i, Dep_i, the mark $\Theta^{(i)}$ with its dials w, c_a, and the end mark $\Theta = \Theta^{(K)}$, all as in (2a). For a component live at level i, along its lineage Vol_i charges every live cell, at every level $j \le i$, with twice the weight $\mathsf{r}_j = M^4\check{R}_j^5$, while Dep_i contributes $cp_0(g_j)^2(d'_j(W)+1)$ for every component W that is created. By (2e), the upper bound of (4c) stays valid once the mark is prepaid: for the run stopped at an arbitrary level when the dial is $\mathsf{w} = 0$, and at the last level when the dial is $\mathsf{w} > 0$. In the form needed here, the prepaid bound takes up the large-field bound one lineage at a time and consists of two statements, independent of each other. *Volumes are paid by events.* Let $\mathsf{r}_j = M^4\check{R}_j^5$. For every history, and for every component X of Z_k, whether live or healed at step k, one has

$$\sum_{(j,C)} \mathsf{r}_j\,|C|_j \le c_G \sum_{(j,C)} \mathsf{r}_j \check{R}_j\, e(j,C),$$
$$e(j,C) := \big(|\operatorname{pred}| - 1\big)^+ + \big|\operatorname{pred} \cap \mathrm{Act}_{j-1}\big| + \sum_W \big(d'_j(W)+1\big),$$

where both sums run over the components of the lineage of X. Here pred denotes the set of components live at level $j-1$ that are contained in C; Act_{j-1} is the set of those components that are live even though $\mathbf{R}$ of level $j-1$ has acted on them; W ranges over those components of the step-j creation set that lie in C; and the constant c_G is determined by L, M and the remaining constants fixed ahead of the ceiling γ imposed on the couplings, and by nothing else. The coupling $\gamma_\star$ is taken small. The inequality is a statement on the geometry of the regions alone; neither a field nor a measure enters. At levels where nothing happens, a component, measured in cells of its own level, shrinks geometrically until it becomes small; and once a component has stayed small over $\check{R}_j + 1$ levels without any event, it belongs to the class of $\mathbf{R}$. *Masses by lineage traces.* The lineage trace $\mathfrak{l}$ of X is the list of the regions $\mathsf{Z}_j(X)$, of the sets created in them, and of the sets Act_j. We set

$$\mathrm{wt}(\mathfrak{l}) := \exp\big[C_{\mathrm{chg}}\,\mathrm{Vol}_k - \mathrm{Dep}_k - \sum_{(j,W)} 2p_0(g_j) - \sum_j |\,\mathrm{Act}_j\,|\,p_0(g_j)\big]$$

, where C_{chg} is the charge constant; it is defined by the requirement that, for each cell of level j, the sums over the admissible regions, the Gaussian integrals, the normalisation constants and the vacuum-energy counterterms contribute a factor no larger than $e^{2C_{\mathrm{chg}}\mathsf{r}_j}$. This constant is determined by the parameters of the construction alone, and it is fixed before $\gamma_\star$. Let $\mathscr{L}$ be any set of lineage traces of level k, and sum over those histories of level k whose live components carry the lineage traces $\mathscr{L}$. Then

$$e^{-E_{(k)}} \sum_h \mathsf{t}(h) \le e^{C|\mathbb{T}^{(k)}|} \prod_{\mathfrak{l}\in\mathscr{L}} \mathrm{wt}(\mathfrak{l}) :$$

indeed, one unfolds the majorants of (2c), passing through the components that $\mathbf{R}$ has acted on, as far as the creations, and one sums the entropy with the lineage trace held fixed. No factor arises from a joining, since a creation at a leaf of the lineage pays for it (Proposition 2.6, move (3)). The relation $\mathsf{r}_j\check{R}_j = M^4\check{R}_j^6 \ll p_0(g_j)$ is valid for all couplings not exceeding the anchor coupling $\gamma_\star$. This anchor coupling is fixed small only once L, M, the dials w and c_a, and the rate c' introduced below (a constant of this proof only, different from the small c' of the mass bound) have been fixed, and it does not exceed γ_J; g is subsequently taken small depending on $\gamma_\star$. Consequently the volumes, the entropy of the lineage traces and the mark are all paid for by the events. Precisely, take the rate $c' \ge 0$, which is fixed before $\gamma_\star$ and may be given any value; the value $c' = 1$ already outweighs the number $C^{|X|_k}$ of connected sets X of cells containing a given cell, because $\mathsf{r}_k \ge M^4$. Then for every set of lineage traces ending in X,

$$\sum_{\mathfrak{l}} \mathrm{wt}(\mathfrak{l})\, e^{\mathsf{w}\,\mathrm{Dep}_k + c_a\,\mathrm{Vol}_k} \le e^{-c'\mathsf{r}_k|X|_k}\, e^{-2\min p_0(g_{j_0})}$$

, where j_0 denotes the first level of a lineage. It is this bound that prepays the mark. There are two reasons for carrying out the argument at the last level, with the end mark. The first reason applies already at $\mathsf{w} = 0$. The gain obtained from the first moment in (iii) is the quotient of two quantities: the exponent in the moment bound, equal to $\mathsf{B}(y) = C(1 + \log y)$ for every site of the lattice on which the run stops, and the product of $c_a\mathsf{r}_k$ with the count of level-k cells. When the run is stopped at level k, the quotient equals $\mathsf{B}(x_k)(M\check{R}_k)^4/(c_a\mathsf{r}_k) \approx \log x_k/(c_a\check{R}_k)$, and this carries no smallness of order $e^{-cp_0(g_k)}$. At the last level there are fewer sites, by the factor $L^{-4(K-k)}$. The second reason concerns the alternative of (iv) in which the torus shrinks. There a single deposit is weighed against the volume, and this requires $\mathsf{w} > 0$. Whatever value the dial takes, the end mark registers only lineages of components still live at the last level, that is, only the never-healed lineages. This is why the healed part is split off in (i). Together with the floor one obtains $\mathbb{E}_{\tilde\nu} e^{\Theta} \le e^{C_g\ell^4}$, hence $\mathbb{E}_{\tilde\nu}\Theta \le C_g\ell^4$ by Jensen's inequality. On the other hand Vol_K charges every never-healed cell, so $\sum_{\mathfrak{c}} \mathbf{1}_{\mathcal{E}_{\mathfrak{c}}^{(k)}(K)} \le \Theta/(c_a\mathsf{r}_k)$, where $c_a > 0$. (iii) Translations by level-k cells commute with the steps up to level k; therefore $\tilde\nu(\mathfrak{lv}_k(\mathfrak{c}))$ is the same for every $\mathfrak{c}$. The event $\mathcal{E}_{\mathfrak{c}}^{(k)}(K)$, by contrast, depends on coarser partitions and is not covariant. Averaging gives

$$\tilde\nu\big(\mathfrak{lv}_k(\mathfrak{c})\big) \le \max_{\mathfrak{c}'} \tilde\nu\big(\mathfrak{lv}_k(\mathfrak{c}') \setminus \mathcal{E}_{\mathfrak{c}'}^{(k)}(K)\big) + \frac{C_g\ell^4}{c_a\mathsf{r}_k\,\#\{\mathfrak{c}'\}} \le Ce^{-cp_0(g_k)} + C_gL^{-4(K-k)}.$$

(iv) To make $K - k$ large, the run is extended on the same bare lattice up to a larger reference coupling, not exceeding the anchor coupling $\gamma_\star$. This is a fixed upper limit for the reference

couplings. It is chosen small once the dials w and c_a are fixed, in such a way that (i), (ii) and the comparison made in the first alternative below are valid at each reference coupling up to $\gamma_\star$; only afterwards is g chosen small, depending on $\gamma_\star$. This continuation leaves the law up to level k unchanged (3d). Steps (i)–(iii) are applied to the continued run, with Θ the mark at its last level. There are two alternatives. In the first, the torus shrinks to polylogarithmic size. Every never-healed lineage then contains a creation, so $\Theta \geq \mathsf{w}\, cp_0(g_k)^2$ on $\mathcal{E}_{\mathfrak{c}}^{(k)}(K)$, and by Markov's inequality this single deposit outweighs $C_g \ell^4$. In the second, at least $c\, g^{-2}$ steps are gained, and $L^{-4(K-k)}$ outweighs $e^{cp_0(g_k)}$. Between them, these two alternatives cover all data.

Proposition 4.14 (Ages). Denote by C the component live at level k that contains $\mathfrak{c}$, by $\mathsf{Z}_j(C)$ its lineage regions, and by j_0 the least level at which these regions are non-empty. The three ages of C are the age read in the label, the age of the lineage, $\mathrm{age}_k(C;h) = k - j_0$, and the seed age. These ages and the events $\mathfrak{lv}_k^{=a}(\mathfrak{c})$, $\mathfrak{lv}_k^{\geq a}(\mathfrak{c})$, $\mathfrak{sd}_k^{=a}(\mathfrak{c})$ are defined as in (2a). The age read in the label refers to the oldest region of the label that appears in C: $\mathrm{age}^{\mathrm{lab}}(C) = k - \min\{t : C \cap Z_t \neq \emptyset\}$, where $Z_t = \Lambda_t^c$ is the large-field region of scale t in the label $\mathcal{S}(h)$ of the history. The proposition states that the three ages are ordered,

$$\mathrm{age}^{\mathrm{lab}}(C) \;\leq\; \mathrm{age}_k(C;h) \;\leq\; \text{seed age},$$

and it gives a decay in the second of them. For the first inequality, note that every point of $C \cap Z_t$ belongs to the live region at each level between t and k. It therefore belongs to $\mathsf{Z}_t(C)$, and so $t \geq j_0$. For the second inequality, no component of $\mathsf{Z}_{j_0}(C)$ contains a component that is live at the preceding level. Each such component is therefore created at step j_0, and it is contained in C. The second inequality may be strict. Suppose a region created at a step $t < j_0$ is contained in C. Then it was healed at some step before k, and the live region of some later level covers its place anew. As a point set it counts for the seed level. It contributes no factor, however, since no cell of it stays live through all the levels from t to k. In terms of events, $\mathfrak{sd}_k^{=a}(\mathfrak{c}) \subset \bigcup_{a' \leq a} \mathfrak{lv}_k^{=a'}(\mathfrak{c})$. Let $b \geq 1$, let $c_a \geq C(b) \log L$ (a floor on c_a, which is chosen after L), and let g be small depending on b and c_a. Then, uniformly, with constants C on the right-hand sides that may depend on b and c_a,

$$\tilde{\nu}\big(\mathfrak{lv}_k^{=a}(\mathfrak{c})\big) \leq C e^{-cp_0(g_k)} (2L^b)^{-a},$$

$$\tilde{\nu}\big(\mathfrak{lv}_k^{=a}(\mathfrak{c}) \cap \mathcal{E}_{\mathfrak{c}}^{(k)}(K)\big) \leq C (2L^b)^{-a-(K-k)}.$$

Both bounds decay in the age of the lineage, and so does the age series given further below. Suppose the seed level is formed only from regions that are healed at no step before k. Then the two bounds and the age series hold in the seed age as well, that is, for $\mathfrak{sd}_k^{=a}(\mathfrak{c})$: the seed level equals j_0, the two ages coincide, and $\mathfrak{sd}_k^{=a}(\mathfrak{c}) = \mathfrak{lv}_k^{=a}(\mathfrak{c})$. When the seed level is formed from all regions, the price of survival yields no decay. Two facts show this. First, a level-k cell contains of the order of $L^{4(k-t)}$ cells of level t, whereas the bound per creation has an exponent that is merely polylogarithmic in $k - t$. Second, take $k = K$ and $a = K - 1$. Applied there, the bound gives at most $C(2L^b)^{-(K-1)}$, at fixed g and for every K, for the probability that a fixed cell at the final level is live and occupies the place of a region created at step 1. Yet the tests of the last step depend only on $x_{K-1}, x_K \approx g^{-2}$. A first-moment argument yields only a density, and never a factor per level for a single cell. For this reason the chessboard estimate is now applied to the large fields, which is a second use of reflection positivity.

Lemma 4.15 (Chessboard for the law of the histories). Put $q_a = e^{-c_a}$, where $c_a \geq c^{-1} \log(2L^4)$, equivalently $L^4 q_a^c \leq \frac{1}{2}$. In this condition, in the two bounds below and wherever these bounds are quoted, the constant is $c = \frac{1}{16}$; it is fixed before c_a, and it arises as the root,

taken over the tiles, of the bound for a pattern that is repeated in one tile out of 2^4. Fix $\epsilon > 0$ and take g small depending on c_a, ϵ. A *tiling* is a translate, by a multiple of the side length of a level-k cell, of the partition of the torus into cubes of side $\mathsf{l}_\square$, called *tiles*, whose faces lie in mirror planes that pass through midpoints of bonds. In each tile, the cube with the same centre and with side $\frac{3}{4}\mathsf{l}_\square$ is called its *core*. A cell, of whatever level, is called *deep* in a tile when the level-k cell containing it lies in the core of that tile. Then for each level $k \le K$ there is a tile side $\mathsf{l}_\square \approx M\check{R}_k$, measured in level-$k$ units, which is a power of L and spans at least sixteen level-k cells, with the following property. Let $n \le k$, let $\mathfrak{z}$ be a level-n cell situated deep inside a tile $\square_0$ of a tiling, and let $\square_1, \dots, \square_r$ be further tiles, all distinct. Then

$$\tilde{\nu}\big(\mathcal{E}^{(n)}_{\mathfrak{z}}(k)\big) \le \mathsf{l}_\square^{-4}\, q_a^{c(k-n+1)},$$

$$\sum_{\mathfrak{c}_1,\dots,\mathfrak{c}_r} \tilde{\nu}\Big(\mathcal{E}^{(n)}_{\mathfrak{z}}(k) \cap \bigcap_{i\le r} \mathfrak{fv}_k(\mathfrak{c}_i)\Big) \le \mathsf{l}_\square^{-4}\, q_a^{c(k-n+1)}\, \epsilon^r,$$

where the sum runs over the level-k cells $\mathfrak{c}_i$ lying deep inside $\square_i$. The first bound is valid for every level-n cell on every torus, including tori too small to carry a tiling with tiles of this side. The bounds are independent of how many tiles the torus contains. Survival costs a factor q_a^c per level, and every further tile that contains a deep live cell costs a factor ϵ. The proof has three steps. *Price of survival:* take the dial $\mathsf{w} = 0$, so that only the volumes are prepaid. For the run stopped at level k, the moment bound of (2e) then applies. Its exponent per site is $\mathsf{B}(x_k) = C(1 + \log x_k)$, which has degree one in the logarithm. Markov's inequality gives, for each event $\mathsf{E}_k \subset \mathfrak{H}_k$,

$$\tilde{\nu}(\mathsf{E}_k) \le \exp\Big[\mathsf{B}(x_k)|\mathbb{T}^{(k)}| - \inf_{\mathsf{E}_k} \Theta^{(k)}\Big].$$

Consider a cell that is live at each of the levels $n, \dots, k$. At every one of these levels it belongs to the lineage of the component that is live at level k and contains it. Thus Vol_k charges it once for each level. Consequently the joint survival of several cells is bounded by $e^{\mathsf{B}(x_k)|\mathbb{T}^{(k)}|}$, multiplied by a factor $q_a^{\mathsf{r}_j}$ for each level j and each distinct cell encountered at that level. The other cells of a pattern, that is, of a tuple of cells of one level in a tile as defined below, are paid for by other payers. Take a cell that has no large-field cell within two cells of it and at one of whose bonds a lower threshold is passed by the scaled fluctuation field of the step. The price of such a cell is borne by the conditional law of the fluctuation field in its box. If the component of a cell is absorbed into a component healed at a step $\le k$, the lineage of that component pays for the cell. *Witnesses:* the chessboard estimate concerns functions of the bare field, whereas $\tilde{\nu}$ is a law on histories. A history is determined by the bare field together with independent auxiliary variables, namely one uniform variable for every possible application of a probability kernel in the steps up to k, each of them attached to a location. By the tile data of a run we mean the restriction of the bare field to the tile, together with the auxiliary variables attached to locations in it. The estimate rests on a locality property, the *agreement of two runs inside a tile*. Consider two runs with identical tile data. Assume that in the first run, up to some level $J' \le k$, two things hold in the tile. First, no tested scaled fluctuation value lies in a narrow window around its threshold (at a test of step j of a value scaled by g_i, $i \le j$, the window has width $2\varepsilon_j^{\mathrm{dec}}/g_i$, with $\varepsilon_j^{\mathrm{dec}} = Ce^{-\delta D_j}\varepsilon_j$; here δ is the rate at which two backgrounds that have the same data in a region approach each other within that region, and $D_j = \delta^{-1}[C + 4\log \mathsf{l}_\square + (4\log L + \log 2)(k-j) + \dots]$, where the dots denote terms of order $\mathsf{N}_j \log L + \log p_0(g_j)$. Thus the width decreases like $\mathsf{l}_\square^{-4}(2L^4)^{-(k-j)}$, which is set against the $\mathsf{l}_\square^4 L^{4(k-j)}$ sites of level j in a tile. For the decoupling, a separation of CD_j lattice units of level j is needed, which amounts to $\lceil CD_j/(M\check{R}_j)\rceil$ layers of level-j cells. Measured in units of level k, these cells shrink like $L^{-(k-j)}$, so the layers consumed over all the levels together

amount to a fraction of the tile side). Second, no long cluster of cells of a level $j \le J'$ that lie in Ω_j^c or carry a tested bond at which a lower threshold is passed by the scaled fluctuation field of the step (a trigger) appears. Then, at the end of step J', the two runs coincide in the inner layers of the tile: their block fields, both current and stored, agree there, and so do the regions of the label. Moreover, every live component that meets these layers is contained in the tile and is common to both runs, and the same is true of its label, of its inner history and of its stored block fields. The recorded collar test is centred at the background localized in the component, as in the stages above. Its comparison with the true averaged background supplies the collar bound, rather than an equality of the two sharp indicators. The common data do not include the stored fluctuation fields. On the bonds examined by a passed fluctuation test (2a) they are built from the averaged background, which is not local. Among the later moves, a single test of $\mathbf{R}$ depends on them, and on those bonds that test is passed in every run. The hypotheses concern only one of the two runs; no condition is placed on the other run outside the tile. On the runs that meet the two hypotheses, this property makes every survival event read deep inside a tile a function of the tile data only. As a result, products over distinct tiles can be integrated against the Wilson measure. Two cases occur for the survival of a cell deep in a tile. Either the tile data force it, or, at the first level with a trigger, it is accompanied by the event of a pattern, an event that is likewise forced by the tile data. By a pattern we mean a tuple of $m = 1 + 2(k - J)$ cells of that level J in the tile, any two of which are at least $2c_{\mathrm{adj}}(J) + 3$ cells apart and each of which lies within $m_0(J)/2$ cells of the first cell of the tuple. Every cell of the tuple either lies within two cells of some cell of the large-field region of that level, or carries a bond at which a lower threshold is passed by the scaled fluctuation field of the step. Here $c_{\mathrm{adj}}(J) := \lceil CD_J/(M\check{R}_J)\rceil + C$, the number of layers that the decoupling takes at level J plus a constant, is the largest step, measured in cells, by which a cluster advances; and $m_0(J) := 6(c_{\mathrm{adj}}(J) + 1)m$ is the diameter, measured in cells, beyond which a cluster counts as long, the factor 6 leaving room for the m cells. The widths $m_0(J)$ belong to the layers consumed at the levels $J \le k$, and these layers together amount to at most $\mathsf{l}_\square/32$. The proof is an induction on the level and, within one level, over the stages of $\mathbf{R}$. Locality requires that the determining set of a label start from a neighbourhood of the first region ($\Omega_0 = \Omega_1^{\sim}$), not from the whole torus. In Bałaban, Comm. Math. Phys. 119 (1988), the determining set has as its support a small neighbourhood of Ω_1 (p. 255), and the region $\Omega_0 = \Omega_1^{\sim}$ appears in a localisation property of the boundary terms (pp. 261–262). If regularity is required of all the bare plaquettes of the torus, then a single rough plaquette anywhere turns off the averaged background on the whole torus, and nothing is local. Hence every survival event read deep inside a tile is bounded above by a finite family of witness functions $f_\iota \in [0,1]$, each depending only on the bare field in that tile. The cost is C per tile, which pays for the windows: under the conditional law of a tested fluctuation value, a narrow window has small probability. Thus

$$\tilde{\nu}\Big(\bigcap_{i\le r} \mathcal{E}_{\mathfrak{d}_i}^{(n_i)}(k)\Big) \le C^{r+1} \sum_{\iota_0,\dots,\iota_r} \Big\langle \prod_i f_{\iota_i} \Big\rangle_{K,\ell,\beta} \le \prod_{i\le r} C \sum_{\iota_i} \|f_{\iota_i}\|_{\mathrm{ch}},$$

where the second inequality follows from reflection positivity of the Wilson measure in the faces of the tiles. *Root and surplus:* a witness has norm bounded by a root of the probability that its pattern occurs, copied across the whole torus by reflections; among these copies only the translates are retained, and they lie in one tile out of 2^4, which gives $c = \frac{1}{16}$. There are three payers for a replicated pattern. Consider first a cell such that no large-field cell lies within two cells of it and that has a bond at which a lower threshold is passed by the scaled fluctuation field of the step; it is paid for by the conditional law of the fluctuation field in its box. A cell whose component remains live up to level k is paid for by the mark $\Theta^{(k)}$ of the run stopped at level k. If the component of a cell is absorbed into a component healed at a step $\le k$, the cell is paid for by the lineage of that

component. Taking the root turns the volume factor into $e^{\mathsf{B}(x_k)\mathsf{l}_\square^4}$ per tile and turns q_a into q_a^c. Since $\mathsf{l}_\square^4 \log x_k \ll \mathsf{r}_k = M^4 \check{R}_k^5$, the surplus $q_a^{c\mathsf{r}_k}$ of one level absorbs this factor, together with the number of level-k cells in a tile and ϵ^{-1}. The patterns are summed one level at a time. A pattern of level J pays $q_a^{\mathsf{r}_J}$ for each of its $m = 1 + 2(k-J)$ cells, and survival from level n up to level $J-1$ pays q_a for each of those levels. A tile contains at most $\mathsf{l}_\square^4 L^{4(k-J)} (m_0(J)+1)^{4(m-1)}$ patterns of level J, where $m_0(J) \le C(x_k + k - J)^3$ is the length from which onward a cluster counts as long. Half the weight of each cell beats this number, except for the positions of the first cell: $Cm \log(x_k + k - J)$ is set against $c\, c_a \mathsf{r}_J m$, where $\mathsf{r}_J \ge M^4 (\log(x_k + c(k-J) - C))^{5r}$ and $r \ge 2$ is the exponent of $\check{R}_k$. This is a logarithm against a power of a logarithm, uniformly in $k-J$, because x_J grows linearly in $k-J$. Finally, the inequality $(J-n)^+ + 1 + 2(k-J) \ge (k-n+1) + (k-J)$ retains $q_a^{c(k-n+1)}$ and leaves a further factor $q_a^{c(k-J)}$. This factor is set against the $L^{4(k-J)}$ positions, and together they give $(L^4 q_a^c)^{k-J} \le 2^{-(k-J)}$. At this point ℓ disappears.

Proof of Proposition 4.14. Anchor: choose any cell $\mathfrak{z}$ of the lineage region $\mathsf{Z}_n(C)$ at the first level $n = j_0 = k - a$ of the lineage. The lineage regions are nested and contained in the live regions. Hence $\mathfrak{z}$ stays live throughout the levels $n, \dots, k$, so that the event $\mathcal{E}_{\mathfrak{z}}^{(n)}(k)$ occurs, and the cell belongs to the component of $\mathfrak{c}$. This is the point at which the age of the lineage enters: a region healed in the meantime provides no such cell. *Sum over anchors:* a box of side $\mathsf{l}_\square$ contains at most $\mathsf{l}_\square^4 L^{4(k-n)}$ anchors, and each costs $\mathsf{l}_\square^{-4} q_a^{c(k-n+1)}$. For an anchor at a distance of $i \ge i_0$ boxes, the live component is connected and therefore passes through a chain of at least i distinct adjacent boxes, each of which contains a live cell. These cells need not be deep. Consider the 2^4 tilings whose face planes are displaced by $\pm\lfloor n_\square/4 \rfloor$ cells from a corner of the level-k cell containing $\mathfrak{z}$, where $n_\square \ge 16$ is the number of level-k cells along a side of a tile. Every level-k cell is deep in a tile of one of these tilings, and $\mathfrak{z}$ is deep in all of them. By the pigeonhole principle a single tiling serves at least $i/16$ of the cells. These cells lie in at least $i/16^2$ distinct tiles, since a tile meets at most 2^4 boxes, and at least i/i_0 of these tiles, where $i_0 = 2 \cdot 16^2$, are different from the tile of $\mathfrak{z}$. By the second bound of the lemma and the choice $\epsilon = (2C_{\mathrm{ent}})^{-i_0}$, where C_{ent} is the entropy per link of the chain, such an anchor costs an additional factor $16\, C_{\mathrm{ent}}^i \epsilon^{i/i_0} = 16 \cdot 2^{-i}$, which is summable against the $(2i+3)^4$ boxes of a shell. Finally $L^4 q_a^c \le (2L^b)^{-2}$, and this is exactly the floor $c_a \ge C(b) \log L$ imposed in the statement of the proposition. Therefore $\tilde{\nu}(\mathfrak{lw}_k^{=a}(\mathfrak{c})) \le C(2L^b)^{-2a}$. The geometric mean of this bound and that of Theorem 4.13 is the first bound. Summing it gives the age series

$$\sum_{a \ge 0} L^{ba}\, \tilde{\nu}\big(\mathfrak{lw}_k^{\ge a}(\mathfrak{c})\big) \le C e^{-c p_0(g_k)},$$

which two later parts use. In (7b) the series is needed with rate $b \ge 8$, because for two points a live component of age a carries a weight of up to L^{8a}; there the series underlies the two-point law, Theorem 0 (c)(3), (c)(4). In (7d) the series enters with rate $b > 16$, on each torus, since a weight L^{16a} arises for four points; this concerns clause (c)(5). The expansions of (7b) and (7d) require the age up to which a live component stores fields. This is the age read in the label. For a component holding an island, the block of which retains the fields of its core (2b), it is at most the age of the lineage. Here, as in (7d), the series is formulated with the age of the lineage, and this age is never smaller. The series therefore covers what these expansions require:

$$\{\mathfrak{c} \subset C,\ \mathrm{age}^{\mathrm{lab}}(C) \ge a\} \subset \mathfrak{lw}_k^{\ge a}(\mathfrak{c})$$

. Applying the lemma at level K in place of k yields the second bound.

A second route to decay in the age: the two-term age bound. This bound complements Lemma 4.15 and does not replace it. The prepaid bounds of (2e) alone give decay in the age of the

lineage. They need no tiles, no auxiliary variables and no second run, and therefore no windows and no triggers. As in (i) and (ii) of the proof of Theorem 4.13, the lineage of the live component containing $\mathfrak{c}$ is classified by its fate. Suppose it is healed at a level $j > k$. Then it contributes the factor of $\mathbf{R}$, and the sum over lineage traces is restricted to traces of age at least a. Suppose instead it is never healed. Then the moment bound is applied at the last level, at the dial $\mathsf{w} = 0$, to the histories that contain such a component. Throughout this event the mark collects at least $\bar{c}_a := 2c_aM^4$ for each level at which the lineage has a member, and these contributions add up over the levels. Consequently no sum over positions of an anchor arises, and the linear gain of the oldest creation remains. Let $c_a > 0$ and take $\gamma_\star$ small depending on c_a. Then on each torus of the family, for each level k, each level-k cell $\mathfrak{c}$ and each $a \geq 0$,

$$\tilde{\nu}\big(\mathfrak{w}_k^{\geq a}(\mathfrak{c})\big) \leq Ce^{-\bar{c}_a a}\, e^{-cp_0(g_k)} + C\exp\Big[\mathsf{B}(x_{K'})\,|\mathbb{T}^{(K')}| - 2p_0(g_k) - \bar{c}_a\,(a + K' - k + 1)\Big],$$

where $K' \geq K$ is the last level of an arbitrary prolongation of the run (3d), the run itself counting as one of its prolongations. The first term contains no volume, and no condition ties k, a, K and ℓ together. The consequences of the bound follow from its second term, case by case; none of the cases is a hypothesis of any statement. (1) There is a side $\ell_V(g)$, tending to infinity as $g \to 0$, with the following property. For every torus with $\ell \leq \ell_V(g)$, the prolongation of (iv) terminates in the alternative where the torus is reduced to polylogarithmic size. In that case the gain $2p_0(g_k)$, of degree $\mathfrak{p}_0$ in the logarithm, dominates the volume term, of degree $4r + 1$. The bound is then $Ce^{-\bar{c}_a a}e^{-cp_0(g_k)}$ at every level. (2) The last lattice of the reference torus $T_{\ell_\star(g)}$ of Proposition 7.3 has polylogarithmic size even without prolongation. Hence that torus satisfies the same bound at all levels. (3) If $\ell > \ell_V(g)$, the torus satisfies the same bound at those levels at which $\bar{c}_a(K'-k+1) \geq \mathsf{B}(x_{K'})|\mathbb{T}^{(K')}|$. This is a floor on the depth that depends on the torus; once it is met at one level, it is met at every lower level. At all of these levels, if $\bar{c}_a \geq b\log L + \log 2$, summing the bound gives the age series at the rate b. Thus this road alone supplies what (7b) uses, namely the age series at the rate $b \geq 8$, at all levels of the reference torus, and hence Theorem 0 (c)(3), (c)(4). For (7d) it supplies the series at $b > 16$ on each torus. On the tori of (1) it holds at all levels; on the tori of (3) it holds at the levels below one that depends on ℓ. Theorem 7.5 [to be proved; Package F] states clause (c)(5) in exactly this form: $\mathsf{d} \leq \mathsf{d}_4$ if $\ell \leq \ell_4(g)$, and $\mathsf{d} \leq \mathsf{d}_4(\ell)$ otherwise, where $\ell_4(g) := \max\{\ell_V(g), \ell_\star(g)\}$. The contribution of Lemma 4.15, through its tiles, windows and triggers, is decay at all levels and on all tori, with constants independent of ℓ. This is uniformity as $\ell \to \infty$. Because of it, one threshold suffices for every torus, and the four-point law carries over to the limits on $\mathbb{R}^4$ (7d).

The age constant. In (2e) two roles of c_a are described; the one at work here is role (A), the dial of the mark. Its effect on the law of the histories is $C(2L^b)^{-a-(K-k)}$. Role (B), the age rate q_a of the aged load of Proposition 2.12 [to be proved; Package C], concerns the pairs of lists belonging to a class of the kind given there (2e), and it is used in (8a). The floors on c_a listed in (2e) concern histories, rotations and age-resolved rarity. Proposition 4.14 imposes the third of these.

(4e) The available source-transfer consequences. Section 38 proves the transfer of the ordinary analytic source input to periodic Gibbs states, including the one-point difference bound. Theorem 39.1 gives the graded bound used in Theorem 4.2, with its d^{-5n} separation loss. Section 46 proves the insertion implication under its stated source and mirror hypotheses. These results do not verify the ordinary interacting source input or the history-persistence requirements of (4d).

† **Cutoffs beyond first passage.** The dilation identity of (3d) gives

$$S^T_{n;K,\ell,\beta}(f_1,\dots,f_n) = S^T_{n;K-q,\,L^q\ell,\,\beta}\big(f_1^{[q]},\dots,f_n^{[q]}\big),$$
$$\|f^{[q]}\|_\sharp \le \|f\|_\sharp,$$

since $\mathcal{O}$ carries no power of the spacing. The same relation holds between $\mathbb{L}_K$ and $\mathbb{L}_{K-q}$. Let $\beta \ge g^{-2} + B_\sharp$ and $q \ge 0$, and suppose that $K - q = K(\beta, g)$. The bounds of Theorem 4.2 are then valid at the cutoff K, with $(L^q\mathsf{d}; L^qR; L^q\ell)$ in place of $(\mathsf{d}; R; \ell)$.

37. Ordinary stability from a normalized history representation

Assume the normalized positive history representation and weighted label bounds stated below. We derive upper stability and the unsourced partition floor, retaining each label-dependent integral operation. The proof uses the total-variation label sum and the regular-label lower bound as separate estimates.

Let the final four-dimensional periodic lattice have N sites and $4N$ independently oriented bonds. Haar measure on each SU(2) bond is normalized. Let w range over a specified ordinary complex source neighborhood containing zero. Assume there are complex measures $\tau_S(w)$, indexed by a finite set of labels, such that

$$Z(w) = \sum_S \tau_S(w)(\mathcal{V}), \qquad \tau_S(0) \ge 0. \tag{OH1}$$

Assume all normalizers and scalar quantities below are well defined; E is the real vacuum constant of the unsourced all-small-field label. Put

$$\mathcal{E}(w) = \sum_{j,X} e_j(X; w), \qquad \mathcal{E}(0) = 0, \qquad \widetilde{Z}(w) = e^{-E-\mathcal{E}(w)}Z(w).$$

At each finite cutoff the sums are finite. Their locality and uniform estimates, rather than convergence of a sum over cutoffs, are what matter here.

A total-variation criterion for upper stability. Let $\mathcal{E}_S(w)$ denote the vacuum constant in the exponent of the term carrying label S. Suppose there are nonnegative weights W_S and real label costs $q(S)$ satisfying

$$\|e^{\mathcal{E}_S(w)}\tau_S(w)\|_{\mathrm{TV}} \le W_S, \qquad \mathrm{Re}[-E - \mathcal{E}(w) - \mathcal{E}_S(w)] \le C_0N + q(S), \tag{OH2}$$

and the weighted label bound

$$\sum_S e^{q(S)}W_S \le e^{C_1N}. \tag{OH3}$$

The constants are independent of cutoff, volume and w in the specified source neighborhood. Then

$$|\widetilde{Z}(w)| \le e^{(C_0+C_1)N}. \tag{OH4}$$

Indeed multiply each measure in (OH1) by the common normalization, insert and remove its own $e^{\mathcal{E}_S(w)}$, take total variations, apply (OH2), and sum by (OH3). This proof takes no logarithm of $Z(w)$ and asserts no absence of its complex zeros.

In a history construction, W_S is obtained by bounding the unsourced integrations after inserting a reserve factor such as $e^{cv(\omega)}$ on every local history. The ordinary source changes an island exponent by an extensive amount, $|\Psi_\omega(w)| \le C_r\, n(\omega)$, where sites are counted in their own scales. A sufficient reserve is $cv(\omega) \ge C_r n(\omega)$, uniformly in the history and the exterior arguments. This is an ordinary mass reserve for one presentation, and is distinct from estimates comparing marked or weighted derivatives of two different presentations.

It is essential that the majorant used for each component be a function of its own integration variables. A bound on its integral at each fixed exterior field is insufficient in a coupled product. For example, let both component factors be $f_1(x,y) = f_2(x,y) = \mathbf{1}_{|x-y|<\epsilon}$ on $[0,T]^2$. Each integral of a factor in its own variable is at most 2ϵ, whereas the double integral of their product has order ϵT, not ϵ^2. A bound with the supremum over the admissible exterior pair *inside* the local integral supplies a component majorant; Tonelli then permits multiplication when the integration-variable groups are disjoint. The construction must also specify which Wilson factors are allotted to those majorants. An operation with no Wilson factor of its own has no independent plaquette-tail gain.

Neither (OH2) nor (OH3) is a consequence of the last triangle inequality. They encode ordinary renormalization estimates: suppression of creation events, control of their genealogies, the allocation of Wilson and Gaussian factors without reuse, and payment of the missing vacuum normalizers of every large-field label. In particular, estimates with the volume of each fine lattice cannot simply be summed to obtain a bound with the volume N of the final lattice.

The floor from one positive label. Let $x \geq 1$ be the last inverse coupling and set $r = cx^{-1/2}$ with a fixed $0 < c \leq 1$. Write

$$\mathcal{B}_r = \{V : \|V_b - 1\|_{\mathrm{op}} < r \text{ for every bond } b\}.$$

Suppose the all-small-field label S_0 has a density with respect to final product Haar measure satisfying

$$\text{(OH5)}\quad d\tau_{S_0}(0)(V) \geq e^E \chi(V) \exp[-x\mathcal{A}(V) - C_I N]\, dV, \quad \chi = 1 \text{ on } \mathcal{B}_r, \quad \mathcal{A}(V) \leq C_A r^2 N \text{ on } \mathcal{B}_r.$$

Here $\mathcal{A}$ may be the action of a minimizing background. If so, inclusion in the characteristic set and the displayed action bound are background estimates which must be verified; they do not follow merely by renaming $\mathcal{A}$ as a coarse action.

For normalized Haar measure there is an absolute $c_H > 0$ such that the operator-norm ball of radius $r \leq 1$ about the identity has measure at least $c_H r^3$. To prove this write $U = \cos\theta\, 1 + i \sin\theta\, n\cdot\sigma$, so $\|U-1\|_{\mathrm{op}} = 2\sin(\theta/2)$ and normalized Haar measure is $(2\pi^2)^{-1}\sin^2\theta\, d\theta\, d\Omega(n)$. The interval $0 \leq \theta \leq r$ lies in this ball. Since $\sin\theta \geq 2\theta/\pi$ on $[0,1]$, its measure is at least $8r^3/(3\pi^3)$. Decreasing the constant handles the irrelevant choice of an open instead of a closed ball.

Positivity in (OH1) permits restriction to S_0, then to $\mathcal{B}_r$. Independence of the $4N$ Haar bonds gives

$$\text{(OH6)}\qquad \begin{aligned} e^{-E} Z(0) &\geq e^{-(C_I + C_A c^2)N} (c_H c^3 x^{-3/2})^{4N} \\ &\geq \exp[-(C + 6\log x)N]. \end{aligned}$$

Since $\mathcal{E}(0) = 0$, this is the required floor for $\widetilde{Z}(0)$. The constant depends on the fixed choices and on C_I, C_A, but not on cutoff or volume.

For the Wilson action of the coarse bond field itself, the action bound in (OH5) is elementary. Telescoping the product of four unitary matrices gives $\|V_p - 1\|_{\mathrm{op}} \leq 4r$. For SU(2), $1 - \frac{1}{2}\operatorname{Re}\operatorname{Tr} V_p = \frac{1}{2}\|V_p - 1\|_{\mathrm{op}}^2 \leq 8r^2$. There are $6N$ plaquettes, so one may take $C_A = 48$. This calculation does not prove the corresponding assertion for a non-Abelian minimizing background or for the tests at all its earlier levels; those are the additional parts of (OH5) in a multiscale construction.

Local energy bounds and the first-passage window. Suppose the source families are actual functions of the source values in a fixed-layer enlargement X^+, holomorphic in its finite-dimensional source parameters and real at real arguments. Suppose their differences at the trivial field obey

$$|\Delta E_j(X, z; w)| \leq C_E e^{-\kappa d_j(X)}, \qquad |\Delta R_j(X; w)| \leq C_R e^{-\kappa d_j(X)}.$$

These are source-minus-unsourced differences: assume explicitly $\Delta E_j(X, z; 0) = \Delta R_j(X; 0) = 0$. Let the number of possible base points $z \in X$ be at most $C_M(d_j(X) + 1)$, as follows for

domains covered by the vertices of a cubical spanning tree. For $e_j(X;w) = \sum_{z\in X} \Delta E_j(X,z;w) + \Delta R_j(X;w)$,

$$|e_j(X;w)| \le C_\delta e^{-(\kappa-\delta)d_j(X)} \qquad (\delta > 0). \tag{OH7}$$

This uses only $(m+1)e^{-\delta m} \le C_\delta$. If the rooted entropy rate is $\kappa_D < \kappa$, choose $0 < \delta < \kappa - \kappa_D$. The resulting energy decay still exceeds entropy. Locality, holomorphy, realness and vanishing at zero pass through the finite base-point sum. Dependence on the source function itself is a structural hypothesis here; it is not implied by a bound on a chosen list of source coefficients.

If a first-passage argument supplies $g^{-2} < x \le g^{-2} + B_\sharp$, then the constants in (OH4) and (OH6), when bounded by $C(1+\log x)$, are uniform over the cutoff data at this fixed g. This is the final elementary step producing constants of the form C_g. It supplies no proof that an admissible run reaches that window.

What a pointwise polymer criterion would supply. There is a different sufficient mechanism when a construction actually has a pointwise hard-core gas. Let its polymers be nonempty subsets of a set of N sites. Suppose, uniformly in a real coarse field V, its real activities obey

$$\sum_{\gamma' \not\sim \gamma} |z_{\gamma'}(V)| e^{a(\gamma')} \le a(\gamma), \qquad \sup_x \sum_{\gamma \ni x} |z_\gamma(V)| e^{a(\gamma)} \le b. \tag{OH8}$$

These inequalities remain valid after multiplication of all activities by any $t \in [0,1]$. The convergent cluster expansion makes its partition function $\Xi_V(t)$ nonzero and continuous on this interval. It is real there and $\Xi_V(0) = 1$; hence $\Xi_V(1) > 0$. The absolute cluster theorem bounds its logarithm by $\sum_\gamma |z_\gamma| e^{a(\gamma)} \le bN$: sum the rooted cluster bound and then use the nonempty site supports in (OH8). The branch is the one continued from $t = 0$. It is real on the interpolation, and therefore gives

$$e^{-bN} \le \Xi_V(1) \le e^{bN}.$$

For complex source activities, the corresponding uniform criterion gives $|\Xi_V(w)| \le e^{bN}$. With a verified common Haar/Wilson integral representation and its action estimates, these facts could be used to prove stability and a floor.

This lemma is not a license to replace a label/history representation by that integral. Local polymer exponentiations on small-field regions, or exponentiation of the sum over healed components, do not by themselves provide a pointwise gas over all final coarse fields. For the history representation the concrete sufficient conditions are (OH2), (OH3), and (OH5); verification of those ordinary conditions remains the upstream work.

38. Transferring ordinary source bounds to all periodic states

The transfer in this section uses reflection positivity and ordinary complex source bounds on finite tori. It does not use geometric forgetting estimates for derivatives of inherited activities. Nor does it assume moment bounds for Gibbs states of a perturbed interaction. The original arbitrary periodic state enters through its DLR equation; the only measures to which a source estimate is applied are finite Wilson torus measures.

Fix an odd integer $L \ge 3$, $a = L^{-K} \le 1$, a coupling $\beta > 0$, and $\mathcal{O}_a(f) = \sum_p f(x(p))a_p$. All sums defining this observable are finite when f has compact support. Although its supremum norm may grow as a^{-4}, it is bounded at each fixed spacing. Every periodic Wilson DLR state is invariant under all lattice isometries and positive at all lattice mirrors by Theorem 7.1.

An exact domination lemma, before source estimates. Use a cubic board of cells of side $l = L^j \geq a$, whose face planes cut links in half. Take cubic tori of side $\ell = 2L^m$ with $m \geq \max(0, j)$, having an even number of cells in every direction. Every sequence of sides tending to infinity in this family eventually accommodates any fixed coarser board of allowed side. All sequences in this section are taken in this family. When localization cubes have side Ma_s with the construction convention $M = L^{m_0}$, restrict to the cofinal subfamily $m \geq m_0$. Then $\ell/(Ma_s) = 2L^{m-m_0+s}$ is an even integer at every scale, and all localization partitions are retained. The cell algebra consists of bounded functions of bonds lying wholly inside the open cell; crossing bonds belong to no cell algebra. Reflected copies are conjugated when the reflecting isometry has negative determinant. For a cell function F define

$$p_{l,\ell}(F) = \left\langle \prod_C F^{[C]} \right\rangle_{a,\ell,\beta}^{1/N},$$

$$N = (\ell/l)^4. \tag{ST1}$$

The number inside the root is nonnegative. To see this, split the board at one face and the opposite face. The disseminated product is $\Theta\Phi\,\Phi$ for the product of its copies in one half. Finite-torus reflection positivity applies.

We will use, including for complex functions,

$$\left|\left\langle \prod_C F_C \right\rangle\right| \leq \prod_C p_{l,\ell}(F_C),$$

$$p(F+G) \leq p(F) + p(G),$$

$$p(\bar{F}) = p(F),$$

$$p(1) = 1. \tag{ST2}$$

Here are details extending the finite chessboard argument to this form. In one direction transport all slab profiles to one reference slab, using conjugated reflection for odd copies. For an arbitrary filling of the n slabs, n even, reflection Cauchy–Schwarz bounds the squared absolute partition integral by the two integrals obtained by retaining either half and reflecting it. Normalize label j by $(Z_j^{\mathrm{dis}} + \epsilon)^{1/n}$, where $Z_j^{\mathrm{dis}} \geq 0$ is its fully disseminated integral and $\epsilon > 0$. The normalized filling of maximum absolute value exists. Both reflected fillings also maximize whenever that maximum is positive: each is nonnegative, bounded by the same maximum, and their product bounds its square. A constant run in a maximizing filling can be doubled by retaining the half starting at that run. Once its length reaches $n/2$, a constant filling results. Every constant normalized filling is at most one. Thus the maximum is at most one; the zero maximum is trivial. Letting $\epsilon \downarrow 0$ proves the one-direction bound even when some disseminated integral is zero. Applying this argument successively to the four coordinate directions proves the first inequality in (ST2): coordinate reflections commute, and each successive dissemination copies the preceding pattern literally in the other coordinates. The resulting exponent for each original cell is $1/N$.

Expanding the disseminated product for $F + G$ and applying the first inequality term by term bounds it by $\sum_{k=0}^{N} \binom{N}{k} p(F)^k p(G)^{N-k}$. Its nonnegative Nth root proves the triangle inequality. Exactly half the copies are conjugated, giving $p(zF) = |z|p(F)$; conjugating the entire nonnegative disseminated integral proves $p(\bar{F}) = p(F)$. In particular $p(\mathrm{Re}(e^{-i\phi}F)) \leq p(F)$.

Domination. For any periodic Wilson DLR state ρ, any finite collection of occupied cells, and any sequence of compatible tori $\ell_k \to \infty$,

$$\left|\rho\left(\prod_C F_C\right)\right| \le \liminf_{k\to\infty} \prod_C p_{l,\ell_k}(F_C). \tag{ST3}$$

Only finitely many factors differ from 1.

First consider a bounded real function H of one cell. Sum its mirror copies over a cube of side ℓ. Reflection invariance of ρ makes the expectation of the sum equal to $N\rho(H)$. Conditional Jensen bounds that expectation by the supremum, over exterior configurations, of the logarithm of its conditional exponential moment. Replace the exterior condition by periodic bonds. After adjoining dummy independent Haar variables if necessary, the two integrals use the same variables. Only boundary plaquettes change; each changes the Hamiltonian by at most 2β, and there are $O((\ell/a)^3)$ such plaquettes. Source factors do not cross the outer cell faces and therefore agree in the two integrals. It follows that

$$\rho(H) \le \log p_{l,\ell}(e^H) + C\beta l^4 a^{-3}\ell^{-1}. \tag{ST4}$$

This estimate is uniform in the exterior configuration, not in a; the volume limit is taken with a fixed.

For bounded real G, put $H = \log(1+tG)$ with $0 < t < (2\|G\|_\infty)^{-1}$. The elementary inequalities $\log(1+tG) \ge tG - Ct^2\|G\|_\infty^2$ and $\log p(1+tG) \le tp(G)$ turn (ST4), first as $\ell_k \to \infty$ and then as $t \downarrow 0$, into $\rho(G) \le \liminf_k p_{l,\ell_k}(G)$. For complex F choose ϕ so that $\rho(\mathrm{Re}(e^{-i\phi}F)) = |\rho(F)|$ and use the last consequence of (ST2).

Finally enclose the finitely many occupied fine cells strictly inside one larger coarse board cell. The one-cell inequality applies to $G = \prod_C F_C$. On every compatible torus the fine-board chessboard inequality gives $p_{\mathrm{coarse},\ell}(G) \le \prod_C p_{l,\ell}(F_C)$: disseminate the coarse cell, apply (ST2) on the fine board, and take the coarse root. Choose the larger cell as a sufficiently large allowed integral power of the fine side, so that both boards are compatible along the chosen torus sequence. This proves (ST3). The lower limit along *any* sequence is essential below.

The perturbation $\log(1+tG)$ in this proof is arbitrary and need not be an admissible linear source. No renormalization-group bound was applied to it or to its Gibbs states. It enters only an exact Jensen and boundary-comparison argument and the elementary seminorm inequality.

The precise ordinary torus hypothesis. Here is one sufficient source hypothesis, stated to expose the actual input to the transfer. Fix constants $r, A, C_\pm > 0$, an entropy rate κ_0, and $\kappa > \kappa_0$, uniformly over the selected cutoff–coupling data (a,β) and allowed tori. Let $\|f\|_\sharp = \max(\|f\|_\infty, c_M\|\nabla f\|_\infty)$. For every finite family of real Lipschitz torus functions require the ordinary complex source bound, on $\sum_i |z_i|\|f_i\|_\sharp < 2r$,

$$|Z_{a,\ell}(W)| \le \exp\left[V_{a,\ell} + \mathrm{Re}\, U_{a,\ell}(W) + \mathrm{Re}\sum_{s=0}^{K}\sum_D e_{a,\ell,s,D}(W) + C_+\ell^4\right],$$

$$Z_{a,\ell}(0) \ge e^{V_{a,\ell} - C_-\ell^4}. \tag{ST5}$$

The terms in this expression have the following meaning and bounds. The source-independent vacuum number $V_{a,\ell}$ is real.

The point counterterm is $U_{a,\ell}(W) = \sum_p u_{a,\ell,p}(W(x(p)))$, where each u is holomorphic and zero at zero. For each fixed cutoff its norm on smaller discs is bounded uniformly in p and ℓ; this bound may depend on the cutoff. The domain term $e_{a,\ell,s,D}$ is holomorphic in the ordinary source variables, is zero at zero, depends only on $W|_{D^+}$, and satisfies $|e_{a,\ell,s,D}| \le Ae^{-\kappa m(D)}$. Put

$a_s = L^{-s}$ and require $\operatorname{diam} D^+ \le C_0 a_s (m(D)+1)$. The domain entropy bound, in a convenient localized form, is

$$\sum_{D:\, D^+\cap B\neq\varnothing,\, m(D)\ge t} e^{-\kappa m(D)} \le C(R+1)^4 a_s^{-4} e^{-ct} \tag{ST6}$$

for every radius-R ball B and $t \ge 0$, uniformly in its position. A count per unit cube bounded by $e^{\kappa_0 m}$ implies this form after absorbing a fixed polynomial in m into a smaller entropy gap. All domain sums used are normally convergent on smaller source balls. Source representation independence is required: local terms depend on the source function on their indicated region, not on the particular list representing that function.

Condition (ST5) is an upper bound and a floor, not a claim that $Z(W)$ is zero-free on the whole source ball. Its ordinary source class includes the global real Lipschitz profiles constructed by mirror periodization. This last inclusion matters. For $w = \sum_i z_i f_i$ in one cell, the periodized source is

$$W^{\mathrm{ch}} = \sum_i z_i H_i^+ + \sum_i \bar z_i H_i^-. \tag{ST7}$$

The even and odd copy profiles $H_i^\pm$ are real. If each support is at distance $b > a$ from the faces, the profiles vanish near the faces and their Lipschitz norms are at most $\|f_i\|_\sharp$. Hence $\|W^{\mathrm{ch}}\|_\flat \le 2\sum_i |z_i| \|f_i\|_\sharp$, independently of the number of cells. The ordinary radius $2r$ in (ST5) therefore supplies the radius r used below. One first regards the two lists of coefficients as independent complex variables; evaluating one at the conjugate of the other does not require a new analytic extension.

Holomorphic cell normalizers from the ordinary hypothesis. Suppose m radius-R support balls lie at distance at least a fixed $b > 1$ from the faces of a cell C of side l. For each torus form a normalizer $E_{C,\ell}(z)$ as follows. Retain in $U + \sum e$ only the point terms and domains whose complete dependence region lies in one open board cell. Evaluate each retained term at the source copy in that cell, conjugate its value if that cell has odd parity, sum over cells, and divide by N. The result is holomorphic in z, even though (ST7) itself involves $\bar z$: on an odd cell the retained term has the form $g(\bar z)$, and $\overline{g(\bar z)}$ is holomorphic. Its real part is unchanged by this outer conjugation. Also $E_{C,\ell}(0) = 0$.

Any discarded source-dependent domain meets a support ball and reaches a cell face. Its diameter is at least b, so $m(D) \ge b/(C_0 a_s) - 1$. Counting once per ball, (ST6) shows that the discarded contribution divided by N is at most

$$B_{\mathrm{face}}(m,R,b) \le CAm(R+1)^4 \sum_{s\ge 0} a_s^{-4} e^{-c' b/a_s} < \infty. \tag{ST8}$$

The sum is independent of the cutoff. Overcounting domains meeting several balls only enlarges this upper bound. Point counterterms whose source is nonzero lie strictly inside cells, so none are discarded. Combining (ST1), (ST5) and (ST8) proves

$$p_{l,\ell}(e^{\mathcal{O}_a(w)}) \le \exp[\operatorname{Re} E_{C,\ell}(z) + B_C],$$
$$B_C = (C_+ + C_-) l^4 + B_{\mathrm{face}}(m,R,b). \tag{ST9}$$

At a fixed cutoff the normalizers are locally bounded uniformly in ℓ. For the point terms this follows from their fixed-cutoff bound and $O(l^4 a^{-4})$ plaquettes per cell. For the domain terms

use (ST6) with $t = 0$ and the finite sum of a_s^{-4} over $s \le K$. These bounds may grow with K, which is harmless for the next step. Montel's theorem therefore gives a subsequence on which $E_{C,\ell}$ converges locally uniformly to a holomorphic E_C. Explicitly, Cauchy estimates on smaller source balls give equicontinuity there; Arzela–Ascoli and a diagonal selection over a countable compact exhaustion give the subsequence, and Cauchy's formula proves holomorphy of its limit. For finitely many occupied cells take one common subsequence. It is chosen from the torus normalizers alone, independently of every infinite-volume state.

There is a second uniform bound that does not grow with K. If $n \ge 2$ sources in the same cell have pairwise support distance at least d, a point counterterm has zero mixed derivative, because a point cannot belong to two of the supports. A contributing domain must meet all supports; in particular its diameter is at least d. Cauchy's formula and (ST6), with the first support ball as anchor, give

$$|\partial_1 \cdots \partial_n E_C(0)| \le D_n(d;R) \prod_i \|f_i\|_\sharp,$$

$$D_n(d;R) = CA(R+1)^4 \max(1, d^{-4})(n/r)^n. \tag{ST10}$$

Indeed the remaining scale sum is $\sum_s a_s^{-4} e^{-c'd/a_s} \le C \max(1, d^{-4})$. This follows by splitting at $a_s \approx d$; the tail becomes a convergent sum $Cd^{-4} \sum_{j\ge 0} L^{4j} e^{-c''L^j}$. The same estimate holds before the Montel limit, and hence after it.

All-state generating functions, cumulants and mean differences.

Theorem 38.1 (Ordinary torus source estimates imply all-state bounds)**.** *Assume the ordinary torus source hypothesis (ST5)–(ST6), including the stated locality, point counterterms and uniformity, and the proved lattice symmetries. Then (ST13), (ST14) and (ST15) hold for every periodic Wilson DLR state, uniformly in the cutoff, the states and the support positions. Their constants depend only on the displayed source constants, the number of entries, the support radius and, for higher cumulants, the support separation.*

Let $f_1, \dots, f_n$ have radius-R supports at arbitrary positions. A zero test makes every corresponding multilinear assertion trivial, so such tests may be omitted. Choose an allowed side $l \le C_L n(R + b + 1)$ and a board offset that puts each support ball at distance at least b from every face. For each coordinate, the forbidden offsets occupy at most $2n(R + b + a)$ of a period; choosing l larger than a fixed multiple of this number leaves an allowed link-plane offset. Thus at most n cells are occupied. Put $w_C = \sum_{i:\,\mathrm{supp}\, f_i \subset C} z_i f_i$. Apply (ST3) to $F_C = e^{\mathcal{O}_a(w_C)}$ along the common Montel subsequence in (ST9). For every periodic DLR state ρ this gives

$$\left| e^{-E(z)} \rho\left(e^{\sum_i z_i \mathcal{O}_a(f_i)} \right) \right| \le Q,$$

$$E = \sum_C E_C,$$

$$Q = e^{\sum_C B_C},$$

$$\sum_i |z_i| \|f_i\|_\sharp < r. \tag{ST11}$$

The same E and Q serve all states at the datum. The number Q depends on n, R and the fixed source constants, but not on the cutoff, state, or support positions. This conclusion uses the lower limit along the selected sequence, rather than an upper limit over all tori.

Write the normalized holomorphic function in (ST11) as h_ρ. It has $h_\rho(0) = 1$ and $|h_\rho| \le Q$. A logarithm is needed only near zero, where it exists by continuity. For any nonempty subset $B \subset \{1, \dots, n\}$, Cauchy's formula, with all other variables zero, gives

$$|\partial_B h_\rho(0)| \le Q(|B|/r)^{|B|} \prod_{i \in B} \|f_i\|_\sharp.$$

The finite derivative identity for a logarithm now implies

$$|\partial_1 \cdots \partial_n \log h_\rho(0)| \le r^{-n} \sum_\pi (|\pi| - 1)! Q^{|\pi|} \prod_{B \in \pi} |B|^{|B|} \prod_i \|f_i\|_\sharp. \tag{ST12}$$

The sum is over set partitions. No uniform zero-free polydisc is needed for this identity at zero.

For $n \ge 2$, the cumulant is the sum of the mixed derivatives of E and $\log h_\rho$. If more than one cell is occupied, the mixed derivative of $E = \sum_C E_C$ is zero because every summand lacks a variable. If one cell is occupied, use (ST10). Equations (ST10)–(ST12) prove the all-state separated cumulant bound

$$|\kappa_n^\rho(\mathcal{O}_a(f_1), \dots, \mathcal{O}_a(f_n))| \le C_n(d; R) \prod_i \|f_i\|_\sharp. \tag{ST13}$$

Here C_n is explicit as (ST12), with the uniform choice of Q above, plus D_n. This proves a finite constant for each n; no factorial growth assertion beyond these formulas is being made.

For one radius-R test choose one cell with $l \le C_L(R + b + 1)$ and the corresponding common normalizer E_f. Cauchy directly gives

$$|\rho(\mathcal{O}_a(f)) - E_f'(0)| = |h_\rho'(0)| \le (Q_R/r)\|f\|_\sharp.$$

Consequently, for every pair of periodic states at the same datum,

$$|\rho(\mathcal{O}_a(f)) - \nu(\mathcal{O}_a(f))| \le (2Q_R/r)\|f\|_\sharp. \tag{ST14}$$

The common derivative $E_f'(0)$ may diverge with the cutoff. The estimate controls its differences from the means and hence differences between the means. The argument also covers separated reflected copies: their support radii, distances and test norms are unchanged.

If desired the one-point constant can be chosen to grow only as $(R+1)^4$. Choose a fixed smooth partition of unity by translates of compactly supported functions of uniformly bounded derivative. Its members have supports in balls of a fixed radius R_0, only $C(R+1)^4$ meet a radius-R support, and $\|f\psi_j\|_\sharp \le C\|f\|_\sharp$. Apply (ST14) to these pieces with the fixed constant Q_{R_0} and add. Such a partition is obtained by dividing a nonnegative compact bump, positive on a unit cube, by the positive periodic sum of its integer translates.

Finally put $b_1 = C_1(R)$ and $b_m = C_m(d; R)$ for $m \ge 2$. The moment–cumulant formula and invariance of higher cumulants under constant shifts give

$$\left| \rho \prod_{i=1}^k [\mathcal{O}_a(f_i) - \nu(\mathcal{O}_a(f_i))] \right| \le \left(\sum_{\pi \in \mathrm{Part}\{1,\dots,k\}} \prod_{B \in \pi} b_{|B|} \right) \prod_i \|f_i\|_\sharp. \tag{ST15}$$

Singleton blocks use (ST14); the other blocks use (ST13). This is the centered moment estimate required when a component is centered at a different state or mixture, with the same uniformity in both states.

Thus (ST5)–(ST6), source locality, the stated ordinary holomorphy, and the proved lattice symmetries imply the bounds used under G4. The transfer itself introduces no weighted analytic-extension obligation. The analytic construction must still prove the actual ordinary source bound, entropy gap, upper stability and floor. A torus statement controlling only separated mixed derivatives of a logarithm does not state the cell estimate (ST9) or a common one-point normalizer. Those concrete conditions must be retained when replacing an all-state input by the transfer theorem.

39. Factorial order bounds from rooted domain counting

We strengthen the spatial counting used in the ordinary-source transfer. The resulting estimates have factorial growth in the number of entries and a separation singularity of order proportional to that number. Only the ordinary finite-torus source estimates are used. In particular, neither weighted forgetting nor source bounds for perturbed Gibbs states are added.

Retain the ordinary source upper bound and floor (ST5), the point-local counterterms, and the proved chessboard domination (ST3). Let $L \geq 3$ be odd, $a = L^{-K}$, and $a_s = L^{-s}$, $0 \leq s \leq K$. All constants below are uniform over the chosen cutoff–coupling data and all large allowed tori. The physical source norm is $\|f\|_\sharp = \max(\|f\|_\infty, c_M \|\nabla f\|_\infty)$.

The stronger count and its geometric origin. The localization parameter obeys $M = L^{m_0}$ for a fixed integer $m_0 \geq 0$. Use the cofinal family of original tori $\ell = 2L^m$, $m \geq m_0$. At every actual scale, $\ell/(Ma_s) = 2L^{m-m_0+s}$ is an even integer, so all localization partitions tile the torus; no scale terms are omitted. Every fixed power-of-L board also tiles all sufficiently large tori in this family, with even cell counts. At scale s use a cubical partition of side $\rho_s = Ma_s$. A localization domain X is a union of these cubes. It admits a finite connected cubical edge tree meeting every cube of X, and $m(X)$ is the minimum number of edges of such a tree. The source dependence region is contained in a fixed number of neighbouring cube layers around X. This includes source values at plaquette barycentres whose plaquette has a corner in a one-layer enlargement of X, because a bare plaquette has diameter $O(a) \leq O(a_s)$.

For a fixed cube Q the elementary rooted entropy estimate is

$$\#\{X : Q \subset X,\, m(X) = m\} \leq C e^{\kappa_D m}. \tag{GS1}$$

Here is a proof for the stated domain indexing. A minimizing tree with m edges has at most $m+1$ vertices. Every cube meeting an edge of the cubical grid contains an endpoint of that edge; hence every cube of X contains a tree vertex. Choose a tree vertex at a corner of Q in at most 16 ways. Traversing each tree edge twice gives a closed walk of length $2m$ from that corner. There are at most 8^{2m} such walks. At most $16(m+1)$ cubes contain a vertex of the walk, and a subset of these specifies X. Thus

$$\# \leq 16\, 8^{2m} 2^{16(m+1)}.$$

This covers all admissible domains; restricting the domain class can only reduce the count. The same upper bound applies on the finite cubical quotient.

Suppose $|e_{s,X}(w)| \leq A e^{-\kappa m(X)}$, where $\kappa > \kappa_D$. Summing (GS1) gives a positive entropy gap $c > 0$. If the dependence region sees a radius-u ball, some cube of X lies within $C\rho_s$ of that ball. There are at most $C(1 + u/a_s)^4$ choices of this root cube. Consequently

$$\sum_{X:\, X^+ \text{ sees } B,\, m(X) \geq t} e^{-\kappa m(X)} \leq C(1 + u/a_s)^4 e^{-ct}. \tag{GS2}$$

There is no factor a_s^{-4} when $u \ll a_s$. This improvement uses the fixed-layer dependence geometry and rooted indexing; a bound only by $C(u+1)^4 a_s^{-4}$ would not suffice for the argument below. Also, summing over every possible root cube, with harmless overcounting, gives the global estimate

$$\sum_X e^{-\kappa m(X)} \le C\ell^4 a_s^{-4}. \tag{GS3}$$

The tree representation implies $\operatorname{diam} X^+ \le C a_s (m(X)+1)$, as needed in these estimates.

A polynomial bound for one finely supported cell source. Let a board have side $\eta \le 1$ and one real test v supported in a ball of radius $\delta \le \eta$, lying at distance at least δ from the faces. Assume $\delta \ge 16a$. Periodize zv by reflected copies, conjugating odd copies. Its ordinary source norm is at most $2|z|\|v\|_\sharp$. Thus ordinary source radius $2r$ allows $|z| < r/\|v\|_\sharp$, independently of the number of cells.

Construct the interior-cell holomorphic normalizer $E_{v,\ell}(z)$ as in (ST9). The discarded terms are the source-dependent domains whose full dependence region is not contained in one cell. Such a domain sees a copy of the support ball and reaches a face, so $m(X) \ge c_0\delta/a_s - C_0$. Let $N = (\ell/\eta)^4$. The discarded sum divided by N has three scale ranges.

For $a_s \ge \eta$, discard even the source restriction and use (GS3). The cost is at most $CA(\eta/a_s)^4$ per scale and per cell; its sum is bounded by a geometric series. For $\delta \le a_s < \eta$, (GS2) gives a bounded cost per scale, and there are at most $C(1+\log(\eta/\delta))$ such scales. For $a_s < \delta$, (GS2) and the forced domain size give a cost at most

$$CA(1+\delta/a_s)^4 e^{-c'\delta/a_s}.$$

The sum is bounded uniformly in δ by comparison with $\sum_{j\ge 0}(1+L^j)^4 e^{-c''L^j}$. All ranges may be truncated at the actual cutoff, which only decreases the bounds. The point-local counterterms are retained: their nonzero sources are strictly inside cells. Their possibly large first derivatives cause no face error.

The stability and floor cost per cell is $(C_+ + C_-)\eta^4 \le C_+ + C_-$. We obtain

$$p_{\eta,\ell}(e^{z\mathcal{O}_a(v)}) \le \exp[\operatorname{Re} E_{v,\ell}(z)]\, Q_0(\eta/\delta)^p \tag{GS4}$$

for constants $Q_0 \ge 1$, $p \ge 0$ independent of a, η, δ and the support position. Fixed-cutoff local boundedness and Montel selection supply a limiting holomorphic univariate normalizer E_v. For any finite list of pieces use one common subsequence. It is chosen from the torus data and is independent of the infinite-volume states.

Board offsets and a genuine product source polydisc. Consider n tests supported in radius-R balls, with distinct supports at distance at least $d > 0$. Put $\bar d = \min(d, 1)$, and choose an allowed board side, a power of L, such that

$$\frac{\bar d}{16L} < \eta \le \frac{\bar d}{16},$$
$$\delta = \frac{\eta}{64n}. \tag{GS5}$$

First suppose $a \le \delta/16$; thus this board side is a multiple of the mesh. A fixed smooth partition of unity at mesh proportional to δ splits each f_i into pieces $v_{i,k}$ with

$$\#\{k : v_{i,k} \neq 0\} \le C(1 + R/\delta)^4,$$
$$\operatorname{supp} v_{i,k} \subset B_{i,k}(\delta),$$
$$\|v_{i,k}\|_\sharp \le C\delta^{-1}\|f_i\|_\sharp. \tag{GS6}$$

For example, scale a smooth compact partition obtained from translates of a bump divided by their positive periodic sum. Scaling its bounded first derivatives gives the last inequality. A zero piece contributes nothing and is omitted.

Choose one piece from each original test. In any coordinate the offsets for which one enclosing radius-δ ball comes within δ of a face occupy an interval of length at most 4δ modulo η. On the discrete link-plane offsets, one such interval contains at most $4\delta/a+2$ candidates. The n intervals exclude at most $n(4\delta/a + 2) < \eta/a$ candidates by (GS5) and the fine-mesh assumption. There is an allowed offset in each coordinate. Hence all chosen balls have the required margin. Different original tests occupy different cells: a four-dimensional side-η cube has Euclidean diameter $2\eta < d$, whereas their nonempty supports are d apart.

Write these chosen pieces as $v_1, \dots, v_n$ and their different cells as $C_1, \dots, C_n$. By (GS4) each has a univariate normalizer E_i and a bound

$$Q_i \le Cn^p,$$
$$r_i = r/\|v_i\|_\sharp. \tag{GS7}$$

Product domination (ST3) is applied to the separate cell factors. It therefore gives, for every subset S and every periodic state ρ,

$$\left|\exp\left[-\sum_{i\in S} E_i(z_i)\right]\rho\left(e^{\sum_{i\in S} z_i\mathcal{O}_a(v_i)}\right)\right| \le \prod_{i\in S} Q_i$$
$$(|z_i| < r_i, i \in S). \tag{GS8}$$

This is a product polydisc; no bound on its combined source norm is needed. Each chessboard factor in (ST3) uses only one periodized piece and its own ordinary source disc. Inactive factors are exactly 1, with $p(1) = 1$, so (GS8) contains the product over the active subset, not the product over all n indices. The same normalizers work for every state, including a second centering state ν.

Common centering and factorial order. Define $X_i = \mathcal{O}_a(v_i) - E_i'(0)$. Differentiating the normalized function in (GS8) once in each active variable, and using product Cauchy, gives

$$|\rho\prod_{i\in S} X_i| \le \prod_{i\in S}(Q_i/r_i). \tag{GS9}$$

Only the first derivative of each univariate E_i occurs; no bound on its higher derivatives is required. Applying the one-variable case to ν gives $|\nu(\mathcal{O}_a(v_i)) - E_i'(0)| \le Q_i/r_i$. Expanding the product centered at ν over its active subsets and using (GS9) therefore proves

$$\left|\rho\prod_{i=1}^{n}[\mathcal{O}_a(v_i) - \nu(\mathcal{O}_a(v_i))]\right| \le 2^n \prod_i (Q_i/r_i). \tag{GS10}$$

This explains both the necessity of subset bounds in (GS8) and the role of a common normalizer. Uniform moments for auxiliary perturbed Gibbs states are not being presumed.

Sum (GS10) over the piece choices from (GS6), using linearity of centering. The board may depend on that finite piece choice; every constant is independent of that choice. Since $\delta \le 1$,

$$\begin{aligned}\left|\rho \prod_{i=1}^{n} [\mathcal{O}_a(f_i) - \nu(\mathcal{O}_a(f_i))]\right| &\le \left[C(R+1)^4 n^p \delta^{-5}\right]^n \prod_i \|f_i\|_\sharp \\ &\le B_0(R)^n n^{(p+5)n} \bar{d}^{-5n} \prod_i \|f_i\|_\sharp.\end{aligned} \tag{GS11}$$

If $a > \delta/16$, dispense with the partition and the board. A radius-R ball contains at most $C(1 + R/a)^4$ plaquette barycentres. Since $0 \le a_p \le 2$,

$$\|\mathcal{O}_a(f) - \nu(\mathcal{O}_a(f))\|_\infty \le C(R+1)^4 a^{-4} \|f\|_\sharp \le C(R+1)^4 \delta^{-4} \|f\|_\sharp.$$

Its product bound is absorbed by (GS11), because δ is comparable to $\bar{d}/n$, $p \ge 0$ and $\bar{d} \le 1$. This handles every cutoff, including the case when the prospective side η is smaller than the mesh.

Choose the fixed integer $k = \lceil p \rceil + 5$. The elementary inequality $n! \ge (n/e)^n$, obtained by integrating $\log x$, converts (GS11) to

$$\left|\rho \prod_{i=1}^{n} [\mathcal{O}_a(f_i) - \nu(\mathcal{O}_a(f_i))]\right| \le B(R)^n (n!)^k \max(1, d^{-5n}) \prod_i \|f_i\|_\sharp. \tag{GS12}$$

Theorem 39.1 (Graded moments and cumulants)**.** *Under the ordinary torus source hypotheses and the rooted domain geometry stated above, (GS12) holds for every pair of periodic Wilson DLR states at the same datum. For $n \ge 2$ their cumulants satisfy*

$$|\kappa_n^\rho(\mathcal{O}_a(f_1), \dots, \mathcal{O}_a(f_n))| \le [2B(R)]^n (n!)^k \max(1, d^{-5n}) \prod_i \|f_i\|_\sharp. \tag{GS13}$$

The constants are independent of the cutoff, the states and support positions. The exponent k is one fixed finite integer. One may take $B(R) = C(R+1)^4$ with a constant fixed by the source and domain parameters.

To verify the cumulant assertion without an unnecessary extra factorial, center at ρ and apply the moment–cumulant formula. For a partition with block sizes $n_1, \dots, n_m$, use (GS12) on each block and $\prod_j n_j! \le n!$. The remaining combinatorial bound is

$$\sum_\pi (|\pi| - 1)! \prod_{D \in \pi} (|D|!)^k \le (n!)^{k-1} \sum_\pi (|\pi| - 1)! \prod_{D \in \pi} |D|! \le 2^{n-1} (n!)^k.$$

For exactly m blocks, the middle sum equals $\frac{n!}{m}\binom{n-1}{m-1}$: order the blocks, sum over positive compositions of n, and cancel their factorial denominators. Summing over m proves the inequality and (GS13).

The separation singularity in this theorem is d^{-5n}, not a single d^{-5} prefactor independent of n. This is a factorial-growth estimate at each fixed separation, with diagonal order proportional to n. It is sufficient for extensions whose derivative order may grow linearly with the number of entries. A stronger separation law requires an additional argument. The ordinary analytic source and stability hypotheses are those stated in D1.

One may also recover a presentation with a separation prefactor independent of n, at the cost of one factorial power. Put $x = \max(1, d^{-5})$. The exponential series gives $x^n \le n!e^x$, so (GS12) implies

$$\left|\rho \prod_{i=1}^{n} [\mathcal{O}_a(f_i) - \nu(\mathcal{O}_a(f_i))]\right| \le e^{\max(1,d^{-5})} B(R)^n (n!)^{k+1} \prod_i \|f_i\|_\sharp. \tag{GS14}$$

Thus $A(d; R) = e^{\max(1,d^{-5})}$ is one possible prefactor in that notation. Its separation behavior is much worse than a polynomial. The quantitative extension uses the sharper graded form (GS12), rather than attempting to infer diagonal order from this rewritten bound.

40. Compactness on separated tests and the lattice Gibbs limit

The conclusion of this section is subsequential compactness. A sequence of cutoff data with uniform source estimates is an input. Existence of Wilson Gibbs states at each fixed datum is proved below, but does not produce the cutoff data or prove those uniform estimates. No matching of different cutoffs, tuning, or uniqueness of their continuum limits is asserted.

The fixed-cutoff Gibbs measure argument. Fix a spacing $a > 0$, a finite inverse coupling $\beta \ge 0$, and the compact group $G = \mathrm{SU}(2)$. Choose one orientation of each nearest neighbor bond of $a\mathbb{Z}^4$ and denote this countable set by E. A configuration is $U \in \mathcal{X} = G^E$; the opposite bond has matrix U_e^{-1}. The product topology is compact and metrizable. For example, enumerating the bonds gives a metric $\sum_{r\ge1} 2^{-r} \min(1, \|U_{e_r} - V_{e_r}\|)$. Continuous functions depending on finitely many bonds are uniformly dense in $C(\mathcal{X})$. Indeed, replacing all coordinates after the first N by a fixed configuration changes the metric by at most 2^{-N}, and uniform continuity proves this approximation.

Let $A_p(U) = 1 - \frac{1}{2}\operatorname{Re}\operatorname{Tr} U_p \in [0, 2]$ be the Wilson plaquette action. For a finite bond window W, let $\mathcal{P}(W)$ be the plaquettes meeting W and let ∂W be their bonds outside W. Define

$$\gamma_W F(U) = \frac{\int_{G^W} F(VU) \exp[-\beta \sum_{p\in\mathcal{P}(W)} A_p(VU)]\, dV}{\int_{G^W} \exp[-\beta \sum_{p\in\mathcal{P}(W)} A_p(VU)]\, dV}. \tag{CC1}$$

Here VU replaces the variables in W, and Haar measure is normalized. The denominator lies between $e^{-2\beta|\mathcal{P}(W)|}$ and one. The normalized density is therefore a continuous function of the finite variables $(V, U_{\partial W})$. If F is continuous local, $\gamma_W F$ is continuous local, depending only on the remaining variables of F and on ∂W. This proves the Feller property directly; integration of a uniformly continuous function over the compact finite product preserves continuity. By uniform approximation it also holds for $F \in C(\mathcal{X})$.

Take Wilson probability measures on periodic tori with integer periods $N \to \infty$, and lift them periodically to $\mathcal{X}$. The finite partition function is finite and strictly positive because the integrand is continuous and positive on a compact space. The lifted measures have a weakly convergent subsequence. One direct compactness argument is as follows. The space $C(\mathcal{X})$ is separable: on a compact metric space finite approximations by distance functions with rational coefficients form a countable uniformly dense family. Successively extract convergence of the integrals on this family. Uniform approximation then defines a bounded positive functional on all of $C(\mathcal{X})$, with value one

at the constant one. Its Riesz probability measure is the weak limit. This argument simultaneously gives convergence against every continuous function, not merely the chosen countable family.

The lifted torus measure is not itself an infinite-volume DLR state: remote periodic copies can determine an inside variable. The identity needed for passage to the limit is a local one. Fix W and a continuous local F. For all sufficiently large periods, the finite bonds involved in W, ∂W and F embed injectively into the torus, and the plaquettes meeting W are precisely their usual torus counterparts. Integrating the variables in that torus window first gives exactly

$$\widetilde{\mu}_N(\gamma_W F) = \widetilde{\mu}_N(F). \tag{CC2}$$

Both integrands are continuous local. Their weak limits satisfy $\mu(\gamma_W F) = \mu(F)$. Cylinder approximation and $\|\gamma_W F\|_\infty \le \|F\|_\infty$ extend this identity to all continuous F. Equivalently the Borel probability measures $\mu\gamma_W$ and μ agree. The identity consequently holds for every bounded Borel F.

It is also the conditional-law formulation of DLR. For bounded outside-measurable g, the kernel is proper and satisfies $\gamma_W(gF) = g\gamma_W F$. Hence $\mu(gF) = \mu(g\gamma_W F)$, and $\gamma_W F$ is outside measurable. Thus (CC1) is the conditional expectation given the bonds outside W. This holds for every finite window.

The torus lifts are invariant under lattice translations and the hypercubic group. These transformations preserve Haar measure and the Wilson action, including when a chosen bond orientation is reversed. Weak convergence preserves their invariance, so μ is an invariant Wilson DLR state. In particular it is periodic, with the whole lattice as a period group. The same Feller argument makes the class of translation-invariant DLR states weakly closed, hence compact, and its defining identities make it convex. No compactness of the union over arbitrary period groups is being asserted.

If a source argument uses a prescribed compatible cofinal family of tori, take that family in this proof. No measure estimate beyond compactness is needed. Reflection positivity and the stronger symmetry theorems for every periodic DLR state are separate results; this measure argument proves neither their all-state estimates nor any uniformity as $a \downarrow 0$.

The continuum hypothesis. Fix an odd $L \ge 3$ and a sequence $a_j = L^{-K_j} \downarrow 0$. At the same datum (a_j, β_j) let ρ_j, ν_j be periodic Wilson DLR states. The all-mirror lattice theorem supplies the following properties used here: both states are invariant under unit lattice translations and the hypercubic group, and ρ_j is reflection positive at coordinate and diagonal lattice mirrors. The centering state ν_j is invariant at those mirrors as well. These properties are lattice inputs to the compactness argument.

These are limits through states on an infinite lattice. They include volume-first limits when the states are torus limit points. A direct limit on growing tori, or a spacing-first limit on fixed tori followed by removal of volume, is a different construction until a separate recognition theorem identifies it with this class. In particular this section does not assign diagonal reflection positivity to an arbitrary growing cubic-torus limit. Tuned fixed-torus limits, when obtained by matched-cutoff estimates, remain separate from the subsequence theorem.

For $f \in C_c^1(\mathbb{R}^4)$ write

$$\mathcal{O}_j(f) = \sum_p f(x_p) A_p, \qquad X_j(f) = \mathcal{O}_j(f) - \nu_j(\mathcal{O}_j(f)), \qquad m_{j,n}(\mathbf{f}) = \rho_j \prod_{i=1}^{n} X_j(f_i). \tag{CC3}$$

There is one plaquette for each coordinate plane and lattice site, so six orientations are counted once each; x_p is its barycenter. There is no power of a_j in the observable. At a fixed cutoff these are bounded continuous random variables with finite bond dependence. Set $m_{j,0} = 1$ and complexify multilinearly.

Let $\mathfrak{S}_n$ be the tuples of compact C^1 tests with pairwise disjoint supports. A zero entry gives value zero and can be set aside. The separation is the minimum distance of the supports, not that of balls containing them. Every support may lie in its own radius-R ball; these balls may overlap. Use the fixed physical norm

$$\|f\|_\sharp = \max(\|f\|_\infty, c_M\|\nabla f\|_\infty), \qquad c_M > 0 \text{ fixed.}$$

The ordinary source hypothesis is used only through the following proved implications of it. At each fixed order there is a bound

$$|m_{j,n}(\mathbf{f})| \le C_n(d;R)\prod_i \|f_i\|_\sharp \tag{CC4}$$

when the supports have distance at least $d > 0$ and the stated radius bound. Bounds holding eventually on each fixed support neighborhood suffice for extraction. The one-point version is included: for different centering and integrating states it is a uniform bound on their mean difference, not on either unsubtracted mean.

For the quantitative conclusion assume the graded bound, with one finite integer $k \ge 1$ and $B(R) < \infty$ independent of j and the two states,

$$|m_{j,n}(\mathbf{f})| \le B(R)^n (n!)^k \max(1, d^{-5n}) \prod_i \|f_i\|_\sharp. \tag{CC5}$$

For $n = 1$ take $d = 1$. A harmless change of B includes complex entries. These are all-state bounds at the indicated data and also apply to the pair (ρ_j, ρ_j) used for canonical recentering. The ordinary-source theorem obtains this bound from D1 and its actual rooted domain geometry, including the fixed-layer dependence and strict entropy gap. A weaker counting assertion alone does not supply this hypothesis. Those source hypotheses, and existence of the given sequence of cutoff data with common constants, are not proved by this section.

A countable support-adapted family. Let $\mathcal{U}$ be the bounded finite unions of open boxes with rational endpoints. This is countable. Every finite collection of separated compact sets K_i has neighborhoods $U_i \in \mathcal{U}$ such that $K_i \subset U_i$ and the closures $\overline{U}_i$ are pairwise separated. Cover each compact set by sufficiently small rational boxes inside a neighborhood of radius less than one fourth of its distance from the other sets, then take a finite subcover. The neighborhoods can also stay in any prescribed open support regions.

For each $U \in \mathcal{U}$ we now exhibit a countable family dense in $C^1_c(U)$ for the C^1 norm, with every approximant supported in U. Fix a nonnegative smooth unit-mass bump ϱ in the unit ball. For integers $m \ge 1$ let

$$\chi_{U,m} = \mathbf{1}_{\{x:\operatorname{dist}(x,U^c)>1/(2m)\}} * \varrho_{1/(8m)}.$$

It is smooth, compactly supported in U, takes values in $[0,1]$, and equals one where $\operatorname{dist}(x, U^c) \ge 1/m$. Use finite rational linear combinations of

$$\chi_{U,m}(x)\cos(2\pi q\cdot x/N), \qquad \chi_{U,m}(x)\sin(2\pi q\cdot x/N), \quad q \in \mathbb{Z}^4,\ N \in \mathbb{N}, \tag{CC6}$$

and take their rational span over all indices. Call this real countable family D_U.

To verify density, fix $f \in C^1_c(U)$ and choose m such that $\chi_{U,m} = 1$ on a neighborhood of its support. Extend f by zero, smooth it on a sufficiently small scale, and view the resulting smooth compact function as periodic on a large integer-side box. Smoothing converges to f in C^1 because f and its gradient are uniformly continuous. The Fourier partial sums of this smooth periodic function converge with their first derivatives: repeated integration by parts makes the differentiated Fourier series absolutely summable. Approximate their finitely many real sine and cosine coefficients by rationals and multiply by $\chi_{U,m}$. The multiplication is bounded in C^1, and $\chi_{U,m} f = f$. This produces the required approximants. Complexification gives the countable dense family $D_U + iD_U$ for complex tests.

List all tuples $(g_1, \dots, g_n)$, for every $n \geq 1$ and every ordered list of pairwise separated $\overline{U}_i$, with $g_i \in D_{U_i} + iD_{U_i}$. This is one countable list. By (CC4) the values of each coordinate form a bounded sequence; finite initial values do not matter if a bound starts eventually. Extract an infinite subsequence for convergence of the first coordinate, then a subsequence of it for the second, and so on. If I_r is the rth nested infinite index set, choose increasing $j_r \in I_r$. For each fixed coordinate its values along (j_r) eventually lie in the convergent subsequence for that coordinate. All listed values therefore converge along this single subsequence.

Extension to every separated tuple. The following telescoping estimate explains why that one extraction is enough. If $\operatorname{supp} f_i \cup \operatorname{supp} g_i$ lies in $\overline{U}_i$, let $d_U > 0$ be a lower bound for their pairwise distances, taking $d_U = 1$ when $n = 1$, and choose a common finite radius bound R_U. Multilinearity and (CC4) give, uniformly in the relevant tail,

$$|m_{j,n}(\mathbf{f}) - m_{j,n}(\mathbf{g})| \leq C_n(d_U; R_U) \sum_{i=1}^{n} \|f_i - g_i\|_\sharp \prod_{h \neq i} \max(\|f_h\|_\sharp, \|g_h\|_\sharp). \tag{CC7}$$

Every mixed tuple in this telescoping sum is separated because the neighborhoods, rather than just the two endpoint tuples, carry the separation.

Given any $\mathbf{f} \in \mathfrak{S}_n$, choose such rational neighborhoods and approximate each entry in C^1 by the corresponding countable family. The right side of (CC7) tends to zero uniformly in j. For a fixed approximating tuple its values converge along (j_r). Comparing two indices through this tuple shows that $m_{j_r,n}(\mathbf{f})$ is Cauchy. Define its limit by $\sigma_n(\mathbf{f})$. This definition is independent of all neighborhoods and approximants because it is the limit of the original sequence at that tuple. The subsequence has not been chosen again for $\mathbf{f}$; convergence now holds on every separated compact C^1 tuple simultaneously. Set $\sigma_0 = 1$.

Passing in (CC7) proves the same local continuity estimate for σ_n. On compact subsets of each product of the fixed support spaces, convergence is uniform: choose a finite C^1 net in that compact set, apply pointwise convergence at its centers, and use (CC7) at both ends. Its constants are uniform on a norm-bounded set.

Multilinearity passes whenever all tuples in the proposed identity are admissible. In particular the union of the alternative supports in a single entry must stay separated from the other entries. Permutation symmetry, conjugation and reality pass directly from their exact lattice identities. The bounds (CC4) and (CC5) pass with their original constants at every fixed tuple, since absolute-value inequalities are closed under convergence. Their radius and separation parameters concern the original supports, not the larger neighborhoods used to construct the limit.

Moving arguments, real symmetries, and rounded mirrors. Suppose $f_{i,j} \to f_i$ in C^1 and all their supports lie in fixed pairwise separated compact neighborhoods. Equation (CC7) and fixed-tuple convergence give

$$m_{j_r,n}(f_{1,j_r}, \dots, f_{n,j_r}) \longrightarrow \sigma_n(f_1, \dots, f_n). \tag{CC8}$$

This is also valid for every finite combination of tuples. For translations the needed C^1 convergence has the explicit modulus

$$\|\tau_v f - \tau_w f\|_\sharp \leq \max\{\|\nabla f\|_\infty |v - w|, c_M \omega_{\nabla f}(|v - w|)\} \longrightarrow 0, \tag{CC9}$$

where $\omega_{\nabla f}$ is the uniform modulus of continuity of the compactly supported continuous gradient. No second derivative is required. The chain rule gives the corresponding convergence for a convergent sequence of Euclidean isometries.

For any fixed real translation v choose lattice vectors $v_j \in a_j\mathbb{Z}^4$ with $v_j \to v$. Both states are invariant under v_j, so $m_{j,n}(\tau_{v_j}\mathbf{f}) = m_{j,n}(\mathbf{f})$. The moving-test statement proves real translation invariance of σ_n. Hypercubic invariance is exact at every spacing and passes without rounding.

For clarity the reflection limit also works at arbitrary affine positions. A coordinate mirror $x_\mu = c$ can be approximated by link mirrors with $c_j \in a_j(\mathbb{Z}+1/2)$ and $c_j \to c$. A diagonal mirror $x_\mu \pm x_\lambda = c$ is approximated by $c_j \in a_j\mathbb{Z}$. In unit-normal coordinates its offset grid is $a_j\mathbb{Z}/\sqrt{2}$. These are different conventions for the same diagonal mirror; no factor $\sqrt{2}$ is discarded. The reflected tests converge in C^1: the linear reflection part is fixed and the translation parts differ by $2(c_j - c)e_\mu$ in the coordinate case, and by $(c_j - c)(e_\mu \pm e_\lambda)$ in the diagonal case.

Let F be a finite complex combination of separated tuples whose entries lie strictly on one open side of the limiting mirror. Its compact support has positive distance from the mirror. For all fine cutoffs the barycenters sampled by those entries and all the bonds of their plaquettes lie on the same side of the rounded mirror: a plaquette has diameter $O(a_j)$, while that distance is fixed. Form its centered lattice polynomial F_j. Reflection invariance of ν_j makes centering commute with the lattice reflection. Lattice reflection positivity gives

$$0 \le \rho_j(\overline{\theta_j F_j} F_j) = m_j(\theta_j \overline{F} \cdot F).$$

Every tuple on the right has a common positive support separation for large j: internal separations are fixed and the two sides are separated by the mirror gap. Apply (CC8) to get

$$\sigma(\theta \overline{F} \cdot F) \ge 0. \tag{CC10}$$

Reflection invariance itself passes in the same way. This proves coordinate and diagonal reflection positivity, including their affine translates. It proves no new reflection direction.

The one-point limit and the connected functions. The first moment is determined more precisely by the plaquette means. Unit translations and hypercubic symmetry make

$$\delta_j = \rho_j(A_p) - \nu_j(A_p)$$

independent of the plaquette position and its coordinate plane. Define

$$R_j(f) = \frac{a_j^4}{6} \sum_p f(x_p), \qquad c_j = 6a_j^{-4}\delta_j. \tag{CC11}$$

Then $m_{j,1}(f) = c_j R_j(f)$. Each orientation samples a translate of $a_j\mathbb{Z}^4$. Comparing its sample value to the integral on the corresponding side-a_j cube gives

$$R_j(f) \longrightarrow \int f, \qquad |R_j(f) - \textstyle\int f| \le C_R a_j \|\nabla f\|_\infty \tag{CC12}$$

for a test supported in a fixed radius-R ball. Indeed the sample and any point of its cube are at distance at most $2a_j$; only $C(R+1)^4 a_j^{-4}$ cubes can contribute. Average the six such estimates. The same bound is independent of the center of that ball.

Choose one real smooth compact bump f_* with $\int f_* = 1$. Its limiting moment exists by the preceding extraction, and $R_j(f_*) \to 1$. Consequently along the extracted subsequence

$$c_j = \frac{m_{j,1}(f_*)}{R_j(f_*)} \longrightarrow c \in \mathbb{R}, \qquad \sigma_1(f) = c \int f. \tag{CC13}$$

In particular the coefficients are bounded on that tail. This bound is uniform over both states: $|c_j| \le 2B(R_*)\|f_*\|_\sharp$ once $R_j(f_*) \ge 1/2$, and $|c| \le B(R_*)\|f_*\|_\sharp$. This argument uses the uniform one-point comparison and the common plaquette means; it is not a deduction from formal continuum translation invariance alone. If $\rho_j = \nu_j$, then $c = 0$.

All connected limits are finite algebraic consequences, requiring no common disc of analyticity for an exponential. For a fixed n work in the algebra with $z_i^2 = 0$ and let

$$M_j(z) = 1 + \sum_{\varnothing \neq A \subset [n]} m_{j,|A|}(\mathbf{f}_A) \prod_{i \in A} z_i.$$

The logarithm is the finite series for $\log(1 + (M_j - 1))$. Its coefficient of $\prod_{i=1}^n z_i$ is

$$K_{j,n}(\mathbf{f}) = \sum_{\pi \in \mathrm{Part}[n]} (-1)^{|\pi|-1}(|\pi| - 1)! \prod_{A \in \pi} m_{j,|A|}(\mathbf{f}_A). \tag{CC14}$$

In fact the term of degree r in the logarithm counts the $r!$ orders of each partition with r blocks and divides by r, giving $(r-1)!$. Every subtuple is separated. Hence these finite polynomials converge to the same formula with σ in place of m_j; denote it by S_n. Conversely the finite exponential series gives $\sigma_n = \sum_\pi \prod_{A \in \pi} S_{|A|}$, since its factor $1/r!$ cancels the orders of the blocks.

Adding constants to the $X_j(f_i)$ multiplies M_j by $\exp(\sum_i b_i z_i)$, so changes its logarithm only in degree one. Thus for $n \geq 2$, $K_{j,n}$ is the usual cumulant of the raw variables $\mathcal{O}_j(f_i)$ under ρ_j, independent of the centering state. The graded cumulant bound passes as well:

$$|S_n(\mathbf{f})| \leq [2B(R)]^n (n!)^k \max(1, d^{-5n}) \prod_i \|f_i\|_\sharp, \qquad n \geq 2. \tag{CC15}$$

One may obtain it either from the uniform lattice cumulant bound, or directly from (CC14): if block sizes are n_h, $\prod_h n_h! \leq n!$, and

$$\sum_\pi (|\pi| - 1)! \prod_{A \in \pi} (|A|!)^k \leq (n!)^{k-1} \sum_\pi (|\pi| - 1)! \prod_{A \in \pi} |A|! \leq 2^{n-1} (n!)^k.$$

For exactly r blocks the last unweighted sum is $\frac{n!}{r}\binom{n-1}{r-1}$, by ordered compositions of n.

There is a canonical centered functional distinct from the original two-state moments. Center at ρ_j instead of ν_j; expanding each product and using convergence of $m_{j,1}$ gives along the same subsequence

$$\sigma_n^0(\mathbf{f}) = \sum_{A \subset [n]} \sigma_{|A|}(\mathbf{f}_A) \prod_{i \notin A} \big(-c \textstyle\int f_i\big) = \sum_{\pi \in \mathrm{Part}_{\geq 2}[n]} \prod_{A \in \pi} S_{|A|}. \tag{CC16}$$

It has empty value one, singleton zero, the same higher connected functions, and all the bounds and positivity properties above. For the bounds use the lattice pair (ρ_j, ρ_j) before passing to the limit, so no uncontrolled constant from recentering is introduced.

The quantitative flat extension and closed inequalities. For completeness the topology retained by this compactness statement is explicit. Let $\Delta_n = \bigcup_{i<h} \{x_i = x_h\}$ and let $\mathcal{S}_{\Delta,n}$ be the Schwartz functions whose every jet vanishes on Δ_n. Use

$$p_s(\Phi) = \max_{|\alpha| \leq s} \sup_x (1 + |x|^2)^{s/2} |\partial^\alpha \Phi(x)|.$$

On unit-radius supports, (CC5) and $\|f\|_\sharp \leq \max(1, c_M) \max(\|f\|_\infty, \|\nabla f\|_\infty)$ give the hypothesis of the proved quantitative flat-extension theorem, with separation exponent five. That theorem supplies unique extensions $\Sigma_{j,n}$ and Σ_n of the lattice and limiting compact forms and a common bound

$$|\Sigma_{j,n}(\Phi)| + |\Sigma_n(\Phi)| \leq 2C^n (n!)^{k+32} p_{20n}(\Phi). \tag{CC17}$$

Its separated smooth tensors are dense in this Schwartz subspace. The same conclusions apply to σ^0 and, after changing C, to the connected forms. The theorem is an extension on this specified flat space, not a claim of a canonical extension across the diagonals on all Schwartz functions.

There is convergence on the whole flat space along the already chosen subsequence. Given Φ, approximate it by a finite sum T of smooth separated tensors in the seminorm p_{20n}. Equation (CC17) controls both $\Sigma_{j,n}(\Phi - T)$ and $\Sigma_n(\Phi - T)$ uniformly in j. On T convergence has already been proved. Taking first the cutoff limit and then the approximation error to zero proves

$$\Sigma_{j_r,n}(\Phi) \longrightarrow \Sigma_n(\Phi). \tag{CC18}$$

The positive-time version of the density theorem uses tensors strictly above the mirror. Their reflected products are flat on all cross diagonals, including the time boundary. Thus the limit's symmetry, reality, real translations and reflection positivity extend by this density and the continuity of tensor products in Schwartz seminorms. For a coordinate or diagonal mirror the corresponding positive half-space is used. It follows by an orthogonal change of variables in the flat-space density construction; it does not assume invariance of the functional under that change of variables.

More generally any closed inequality involving finitely many fixed tuple evaluations and continuous functions of them passes to the limit. The statement allows moving tests as in (CC8), convergent scalar coefficients and eventual uniform bounds. Its proof is simply convergence of those finitely many numbers followed by continuity and closedness of the inequality. It does not turn strict inequalities into uniform strict bounds or provide estimates whose constants diverge with the cutoff.

Theorem 40.1 (Subsequential continuum compactness)**.** *For every input sequence satisfying the lattice properties and two-state uniform bounds above, there is one subsequence whose moments converge on every separated compact C^1 tuple. The limiting functional is admissibly multilinear, symmetric and real, invariant under real translations and the hypercubic group, and reflection positive at every coordinate and diagonal affine mirror. It retains the graded bound and its quantitative diagonal-flat extension. Its first moment is $c\int f$, its higher connected functions are the limits of the integrating states' cumulants, and* (CC16) *is their canonical centered functional. Moving arguments and fixed-tuple closed inequalities pass as stated above.*

The preceding sections prove every assertion. The fixed-cutoff measure argument ensures that invariant Wilson DLR states are available at each given datum. It supplies neither a sequence of admissible renormalization data with $K_j \to \infty$ nor D1's uniform source bounds. It also supplies no identification of two different extracted limits, no matched-cutoff convergence rate, and no claim of a full Yang–Mills observable theory.

41. A quantitative extension to diagonal-flat Schwartz tests

Assume the graded compact separated-moment bound below. We construct the diagonal-flat extension and control its constants and derivative orders in the number of arguments. In the gauge-theory application the graded-source theorem supplies this bound under D1 on its actual GS geometry and G1. The proof uses explicit Fourier series on product boxes to retain that quantitative control.

41.1. **The hypothesis and the quantitative conclusion.** Use $\|f\|_1 = \max(\|f\|_\infty, \|\nabla f\|_\infty)$ on $C^1_c(\mathbb{R}^4;\mathbb{C})$. Suppose S_n, $n \geq 1$, are complex multilinear forms on tuples with pairwise separated compact supports. Multilinearity is required whenever the supports in the identity remain separated. Fix numbers $1 \leq b < \infty$, $\kappa \geq 0$ and $\alpha \geq 0$, independent of n, such that

$$|S_n(f_1,\dots,f_n)| \leq b^n (n!)^\kappa \max(1, d^{-\alpha n}) \prod_i \|f_i\|_1 \tag{QF1}$$

if every support lies in a ball of radius one, placed anywhere, and different supports have Euclidean distance at least $d > 0$. For $n = 1$ only the radius and norm conditions are imposed. A

centered first moment equal to zero is included. Real forms may first be complexified, changing b by an absolute factor. Neither invariance nor positivity is needed for the extension.

Put $A = \lceil \alpha \rceil$, $k_* = A + 15$, and $\Delta_n = \bigcup_{i<j} \{x_i = x_j\}$. Let $\mathcal{S}_{\Delta,n}$ be the Schwartz functions on $\mathbb{R}^{4n}$ whose derivatives of every order vanish on Δ_n. For $n = 1$ this is the ordinary Schwartz space. Write

$$p_k(\varphi) = \max_{|\beta| \le k} \sup_x (1 + |x|^2)^{k/2} |\partial^\beta \varphi(x)|. \tag{QF2}$$

Theorem 41.1 (Quantitative diagonal-flat extension)**.** *There is $C < \infty$, depending on b, α and fixed numerical cutoff choices but not on n, and a unique continuous linear functional Σ_n on $\mathcal{S}_{\Delta,n}$ agreeing with S_n on smooth separated tensors. It satisfies*

$$|\Sigma_n(\varphi)| \le C^n (n!)^{\kappa+32} p_{k_* n}(\varphi), \qquad n \ge 1. \tag{QF3}$$

Finite sums of smooth compact separated tensors are dense in this diagonal-flat space in its Schwartz topology. In its positive-time subspace, the tensors may be chosen with supports compactly contained in $\{x_4 > 0\}$. These are constrained density assertions, not density in the unrestricted Schwartz space across the diagonals.

The conclusion is an H1 growth estimate with linear derivative order and a fixed factorial exponent; its constants are intentionally not optimized. For example, $\alpha = 5$ gives the sufficient order $20n$. The probability-theory normalization $\Sigma_0(1) = 1$ may be added separately.

The radius-one formulation loses no information about a hypothesis stated as

$$B(R)^n (n!)^\kappa \max(1, d^{-\alpha n})$$

with constants independent of support positions. Choose a fixed smooth unit-scale partition in each slot. A function supported in a radius-R ball is a sum of at most $C(1+R)^4$ supported pieces of radius at most one, each with C^1 norm at most $C\|f\|_1$. Supports of pieces from different slots retain the original separation. Multilinearity and (QF1) therefore imply the same bound with

$$B_{\mathrm{eff}}(R) \le Cb(1+R)^4. \tag{QF4}$$

Thus only $B(1)$ is used, even if the supplied large-radius upper bound was exponential. This assertion concerns a separate ball for each entry, not one ball enclosing the entire configuration.

41.2. **Cutoffs with controlled derivatives.** We fix all cutoffs once. They can be chosen in Gevrey class two, with

$$0 \le \eta \le 1, \qquad |\partial^\beta \eta| \le C^{|\beta|+1} (|\beta|!)^2. \tag{QF5}$$

Here is an explicit construction and a justification of this bound. Set $a(t) = e^{-1/t}$ for $t > 0$ and $a(t) = 0$ for $t \le 0$. On the complex circle $|z - t| = t/2$ one has $\mathrm{Re}(1/z) \ge 2/(9t)$. Cauchy's formula bounds $|a^{(m)}(t)|$ by $m!(2/t)^m e^{-2/(9t)}$. Maximizing $t^{-m} e^{-c/t}$ and using $(m/e)^m \le m!$ gives $C^{m+1}(m!)^2$. The step

$$s(t) = \frac{a(t)}{a(t) + a(1-t)}$$

is set equal to zero for $t \le 0$ and one for $t \ge 1$. Its denominator on $[0,1]$ is at least e^{-2}. The same derivative bound holds for its reciprocal and hence for s. For precision, differentiate $D(1/D) = 1$ inductively; after division by $(m!)^2$, the convolution coefficients are $1/\binom{m}{r} \le 1$. If $|D^{(r)}| \le C_D^{r+1}(r!)^2$, a derivative constant $C_D(1 + 2e^2 C_D)$ for the reciprocal closes the resulting geometric sum. Products obey the same bound by Leibniz's formula. This proves (QF5), including flatness at the join points.

Choose an even η_0 equal to one on $[-1,1]$, zero outside $[-2,2]$, and nonincreasing on $[0,\infty)$, using $s(2-|t|)$ in the transition. Put $\psi(t) = \eta_0(2t)$ and $D(t) = \sum_{z\in\mathbb{Z}} \psi(t-z)$. At most three summands occur locally and $D \ge 1$. Then $\chi_1 = \psi/D$ is supported in $[-1,1]$, has (QF5), and $\sum_z \chi_1(t-z) = 1$. Define

$$\chi(u) = \prod_{a=1}^{4} \chi_1(u_a),$$

$$\chi^+(u) = \prod_{a=1}^{4} \eta_0(u_a),$$

$$\zeta(u) = 1 - \chi^+(u). \tag{QF6}$$

Thus $\sum_{z\in\mathbb{Z}^4} \chi(u-z) = 1$, $\chi^+ = 1$ on the support of χ, and ζ is zero for $|u|_\infty \le 1$ and one for $|u|_\infty \ge 2$. All these functions lie between zero and one and have (QF5), with an absolute constant. In particular, undifferentiated factors cost one.

We will repeatedly use the following elementary product estimate. For N factors of this kind at scale h, every derivative of order q is bounded by

$$(CN)^q (q!)^2 h^{-q}. \tag{QF7}$$

An absolute prefactor is allowed at $q = 0$. Indeed, distribute the multiindex among the factors by Leibniz. The sum of the multinomial coefficients is N^q, and the product of the factorials of the assigned derivative orders is at most $q!$. Derivatives of pair differences $x_i - x_j$ introduce only signs. This argument produces no factor C^N when most factors are undifferentiated.

41.3. **An explicit local functional on a product box.** First, tensor relations cause no ambiguity. In a finite relation between separated tensors, let d be smaller than every separation. Localize all entries by one finite smooth partition on sets of diameter less than $d/8$. A product of slot sets with two sets closer than $d/2$ has a zero factor in every original tensor: otherwise two original supports would be closer than d. On every remaining product the slot sets are separated, and the form is an ordinary multilinear map on their supported function spaces. It factors through their algebraic tensor product. The latter embeds injectively in functions on the Cartesian product: for the finite-dimensional function spans in any relation, finitely many point evaluations give invertible matrices in each slot. A zero tensor therefore has zero value. Summing the localization proves independence of the tensor presentation.

Here is a quantitative local extension. Let $0 < h \le 1/16$, $\ell = h/64$, and let slot centers be $c_i \in \mathbb{R}^4$. Suppose the cubes $Q_i^+ = c_i + [-2\ell, 2\ell]^4$ have pairwise distance at least $h/2$. They lie in balls of radius one. For a smooth function u supported in $\prod_i (c_i + [-\ell,\ell]^4)$, take its Fourier series on the product of period-8ℓ cubes centered at these points. Write its coefficients as $c_{k_1,\dots,k_n}$, $k_i \in \mathbb{Z}^4$. Multiplying the series by $\prod_i \chi^+((x_i - c_i)/\ell)$ gives a series of separated tensors equal to u everywhere. There is no boundary ambiguity: u vanishes near the boundaries of the period cubes. Explicitly, with $Q = \prod_i (c_i + [-4\ell, 4\ell]^4)$, the coefficients are

$$c_{k_1,\dots,k_n} = (8\ell)^{-4n} \int_Q u(x) \exp\left[-\frac{i\pi}{4\ell} \sum_i k_i \cdot (x_i - c_i)\right] dx.$$

In the scaled variables $y_i = (x_i - c_i)/(4\ell)$, apply $(1 - \pi^{-2}\Delta_i)^4$ in every four-dimensional slot. Integration by parts gives

$$|c_{k_1,\dots,k_n}| \le C^n J_{8n,\ell}(u) \prod_i (1 + |k_i|^2)^{-4},$$

$$J_{q,\ell}(u) = \max_{|\beta| \le q} \ell^{|\beta|} \|\partial^\beta u\|_\infty. \tag{QF8}$$

The sum of absolute coefficients in the product differential operator is at most C^n; no factor depending unspecifiedly on dimension is hidden here. The localized exponential in slot i has C^1 norm at most $C\ell^{-1}(1 + |k_i|)$. Since

$$\sum_{k \in \mathbb{Z}^4} (1 + |k|)(1 + |k|^2)^{-4} < \infty,$$

(QF1) makes the series of its moment values absolutely convergent and gives a local functional $T(u)$ with

$$|T(u)| \le (Cb)^n (n!)^\kappa h^{-(\alpha+1)n} J_{8n,\ell}(u). \tag{QF9}$$

For a tensor supported in the inner cubes it agrees with S_n: its Fourier coefficients factor slot by slot, the partial sums converge in C^1, and (QF1) permits passage through the multilinear form.

These local extensions glue to a unique functional T on $C_c^\infty(\mathbb{R}^{4n} \setminus \Delta_n)$. To check this explicitly, in an overlap choose smaller product boxes enclosing the compact support under consideration. Fourier expansions with cutoffs inside the overlap converge with every fixed number of derivatives and consist of separated tensors. Two local functionals agreeing on those tensors therefore agree there, by their finite-order bounds (QF9). A finite partition of unity on a compact off-diagonal set now defines T independently of the cover. For original separated tensors of larger radius, first partition each entry into sufficiently small pieces and use multilinearity. This also shows that the glued T agrees with the given form on all smooth compact separated tensors, not just on the boxes used in its construction. This gluing argument is used for consistency, not for the numerical bound: the latter uses only the fixed explicit grid and (QF9).

41.4. **Dyadic configuration layers and the factorial accounting.** Set $h_j = 2^{-j}/16$ and define

$$P_h(x) = \prod_{i<l} \zeta((x_i - x_l)/h),$$

$$D_0 = P_{h_0},$$

$$D_j = P_{h_j} - P_{h_{j-1}}$$

$$(j \ge 1). \tag{QF10}$$

For $n = 1$ the empty product is one and $D_j = 0$ for $j \ge 1$. Off the diagonals $\sum_{j \ge 0} D_j = 1$. A fine layer has support

$$h_j \le \min_{i<l} |x_i - x_l|_\infty \le 4h_j. \tag{QF11}$$

The statements about support also hold for its derivatives. Monotonicity of the one-variable cutoffs makes $0 \le D_j \le 1$, although positivity of these differences is not needed for the estimates.

At level j set $\ell_j = h_j/64$ and use the product grid partition

$$\chi_{j,z}(x) = \prod_{i=1}^{n} \chi(x_i/\ell_j - z_i),$$
$$z \in (\mathbb{Z}^4)^n. \tag{QF12}$$

Retain a box if $\chi_{j,z} D_j$ is not identically zero. At one point of that box every pair has distance greater than h_j; its centers therefore have pair distance at least $h_j - 2\ell_j$. The expanded cubes $\ell_j z_i + [-2\ell_j, 2\ell_j]^4$ have pair distance at least $h_j - 6\ell_j > h_j/2$. Thus (QF9) applies to $u_{j,z} = \chi_{j,z} D_j \varphi$. A box with only boundary contact can be ignored if its smooth localized function is zero.

There are at most Cn^2 pair and grid factors in such a cutoff. Using (QF7), Leibniz's formula and $q = 8n$, we obtain the uniform scaled-derivative cost

$$(Cn^2)^{8n}((8n)!)^2 \le C^n (n!)^{32}. \tag{QF13}$$

Here $n^{16n} \le e^{16n}(n!)^{16}$, while $(8n)! \le 8^{8n}(n!)^8$. This accounts for every dependence on the number of pair cutoffs; the $\binom{n}{2}$ undifferentiated factors do not create C^{n^2}.

On a fine layer choose a pair with $|x_i - x_l|_\infty \le 4h_j$ and replace x_i by x_l. The resulting point y lies on a diagonal and $|x - y| \le 8h_j$. Every derivative of φ vanishes there. Taylor's integral remainder along this segment gives, for $|\beta| \le 8n$ and any positive integer m,

$$|\partial^\beta \varphi(x)| \le \frac{(Ch_j)^m}{m!} C^{k_* n} (1 + |x|^2)^{-k_* n/2} p_{k_* n}(\varphi) \tag{QF14}$$

when $8n + m \le k_* n$. The derivative of order m along a vector uses at most four coordinate directions; its multinomial sum is bounded by the m-th power of that vector's ℓ^1 norm. Also $|x - y| \le 1/2$, so the weight at every point of the segment is comparable with that at x, at cost $C^{k_* n}$. These are exponential, not factorial, costs in n.

Take

$$m = (A + 5)n + 2,$$
$$8n + m = (A + 13)n + 2 \le (A + 15)n. \tag{QF15}$$

In differentiating $u_{j,z}$, a derivative on a cutoff costs h_j^{-1}, which its scaled-derivative factor in J_{8n,ℓ_j} cancels. Derivatives remaining on φ are bounded by (QF14). Since $h_j \le 1$, unused positive powers of h_j can be discarded. Equations (QF7), (QF13)–(QF15) consequently imply

$$J_{8n,\ell_j}(u_{j,z}) \le C_A^n (n!)^{32} h_j^m p_{k_* n}(\varphi) \sup_{x \in \operatorname{supp} \chi_{j,z}} (1 + |x|^2)^{-k_* n/2}. \tag{QF16}$$

For $j = 0$ use the same estimate without h_j^m, by the ordinary Schwartz derivative bound; all powers of the fixed h_0 are absorbed in C_A^n.

The grid sum has exactly the needed size. Since $1 + \sum_i |x_i|^2 \ge \prod_i (1 + |x_i|^2)^{1/n}$, and grid points in one slot differ from its box by a fixed distance less than one,

$$\sum_z \sup_{x\in\operatorname{supp}\chi_{j,z}} (1+|x|^2)^{-k_*n/2} \le C_A^n \prod_{i=1}^{n} \sum_{z_i\in\mathbb{Z}^4} (1+|\ell_j z_i|^2)^{-k_*/2} \le C_A^n h_j^{-4n}. \tag{QF17}$$

The one-slot sum is bounded by $C_A\ell_j^{-4}$ by comparison with the integral of $(1+|u|^2)^{-k_*/2}$ on $\mathbb{R}^4$; it is finite because $k_*>4$. No number of configurations depending exponentially on n^2 is being counted.

Combine (QF9), (QF16) and (QF17). Since $\alpha\le A$, the sum of absolute values at every fine level is at most

$$C^n(n!)^{\kappa+32}p_{k_*n}(\varphi)h_j^{m-(\alpha+5)n} \le C^n(n!)^{\kappa+32}p_{k_*n}(\varphi)h_j^2. \tag{QF18}$$

The coarse level has the same bound without h_j^2. The dyadic sum $\sum_j h_j^2$ is finite. We may therefore define

$$\Sigma_n(\varphi) = \sum_{j\ge0}\sum_z T(\chi_{j,z}D_j\varphi). \tag{QF19}$$

This absolutely convergent sum proves (QF3) and continuity in its stated seminorm. It agrees with T on compact off-diagonal tests, because there only finitely many layers and boxes occur and their partition sums to one. In particular it has the required tensor values.

41.5. **Density, the time boundary, and independence of choices.** We give the density details also needed for uniqueness. First cut off infinity by $Q_R(x)=\prod_i\chi^+(x_i/R)$. For a fixed Schwartz seminorm p_k, an extra L powers of decay give

$$p_k((1-Q_R)\varphi) \le C_{n,k,L}R^{-L}p_{k+L}(\varphi), \qquad R\ge1. \tag{QF20}$$

This follows by Leibniz, since the nonzero terms have at least one $|x_i|\ge R$ and cutoff derivatives cost R^{-1}. Next cut out all pairwise diagonals with P_h. Where a derivative of $1-P_h$ can contribute, some pair is within $2h$. Applying the same Taylor argument with arbitrarily many extra derivatives gives

$$p_k((1-P_h)\varphi) \le C_{n,k,M}h^{M-k}p_{k+M}(\varphi). \tag{QF21}$$

For each fixed n,k choose $M>k$. Thus compact tests supported away from every diagonal are dense in $\mathcal{S}_{\Delta,n}$. For example both displayed density constants may be bounded by $C^{k+M+L+1}(n^2+1)^k(k!)^2$, with unused parameters omitted. This follows from (QF7) and the same directional Taylor remainder; these constants are not used in place of the uniform estimate (QF18). On each such compact set, use a finite product-box partition and the Fourier expansions of (QF8). Their partial sums converge with every derivative and have a common compact support. They consequently converge in the Schwartz topology and are finite sums of separated smooth tensors. This proves the asserted constrained tensor density. Choosing the truncations and Fourier partial sums successively for the first $1,2,\dots$ seminorms gives a single convergent sequence.

For positive time define $\mathcal{S}_{\Delta,+,n}$ by requiring all derivatives to vanish outside $\{x_{1,4} > 0, \dots, x_{n,4} > 0\} \setminus \Delta_n$. In addition to the preceding cutoffs, set

$$B_h(x) = \prod_{i=1}^{n} s(x_{i,4}/h - 1). \tag{QF22}$$

It is zero if some time is at most h and one if all are at least $2h$. Where $1 - B_h$ contributes to a positive-time test, a time is within $2h$ of zero. Taylor expansion to that forbidden hyperplane, where every jet vanishes, proves the analogue of (QF21). Hence $Q_R P_h B_h \varphi$ approximates every positive-time test in the full Schwartz topology. Product boxes for this approximation can be expanded while staying strictly above time zero. Their tensor approximants are therefore strictly positive-time separated tensors.

There is also a uniform quantitative version: replace P_h throughout (QF10) by $P_h B_h$. A fine layer then has all pair distances and all times at least h, while one pair distance or one time is at most $4h$. The expanded grid boxes retain separation and positive time. There are still at most Cn^2 cutoff factors. Taylor to the nearest forbidden pair diagonal or time hyperplane supplies (QF14), so exactly the same k_* and factorial exponent apply. This verifies explicitly that the time-boundary constraint introduces no hidden growth in n.

Uniqueness now follows without a choice of a particular partition. Any two continuous extensions agreeing on separated tensors agree on a dense subspace and hence on all of $\mathcal{S}_{\Delta,n}$. Thus the Fourier boxes, cutoffs, dyadic grids and orders of their absolutely convergent sums do not change the resulting functional. The same statement holds on the positive-time subspace.

If the compact moments are symmetric, real or invariant under a specified Euclidean motion, their extensions inherit that identity by density and continuity of the test-function transformations. If the compact moments are reflection positive, take finite positive-time Schwartz sequences and approximate their components by the tensors just constructed. For a negative-time and positive-time factor, their tensor product is flat on all cross diagonals: away from the time boundary one factor already vanishes, and at time zero its jets vanish. Internal diagonals are handled by their own factors. The tensor product is continuous in Schwartz seminorms. Thus the reflected quadratic forms converge, and their nonnegativity passes to these full sequence domains. This proves the distribution-space implication under the graded moment hypothesis (QF1).

The graded-source limit and phase consequence. Under Theorem 39.1, every pointwise limit on separated tuples of its two-state centered moments retains (GS12): for each fixed tuple the inequality is closed and its constants are uniform in cutoff and in both states. The phase-interface theorem realizes each of its component forms by such sequences of pure lattice states, centered at the specified reference states. Therefore the same bound holds for those components, including their possibly nonzero first moments; their one-point shifts are covered by the all-state comparison bound. The fixed equivalence between $\|\cdot\|_\sharp$ and $\|\cdot\|_1$ is absorbed in the radius-one constant b. Thus (QF1) applies with $\alpha = 5$ and the graded factorial exponent k. These limits and components consequently have the unique diagonal-flat extensions with order $20n$ and factorial exponent $k + 32$, uniformly over the stated phase realization. This uses that realization rather than assuming an unrelated component decomposition has the same bounds. It supplies neither contact terms across the diagonals nor full rotational covariance.

41.6. **Why the precise hypotheses matter.** Position uniformity cannot be replaced by unrestricted exponential growth in an enclosing radius. For example, $S_1(f) = \int e^{|x|^2} f(x)\, dx$ has finite bounds for compact tests in every fixed ball about the origin, with an exponential radius cost, but has no continuous extension to Schwartz space agreeing with those values. Translated nonnegative compact bumps, damped by $e^{-|x|^2/2}$, converge to zero in every Schwartz seminorm while their values diverge. Their supports can be chosen in successively more distant unit balls.

This example violates the position-uniform radius-one hypothesis. Under that actual hypothesis (QF4) rules out this obstruction.

Nor may a prefactor such as e^{cn^2} be silently called factorial growth of a fixed order. A lognormal random constant field has moments $S_n(f_1,\dots,f_n) = e^{n^2/2}\prod_i \int f_i$. It is translation invariant and reflection positive, but violates every estimate (QF3) with fixed linear seminorm order and fixed factorial exponent. To see this, take unit-integral Gevrey-two bumps in n separated unit balls with centers at distance $O(n)$ from the origin. Their tensor is diagonal-flat. For every fixed k, its seminorm p_{kn} grows at most $C^n(n!)^{C_k}$, by (QF5) and the polynomial location weights. Its moment grows like $e^{n^2/2}$, which is larger than $C^n(n!)^q$ for every fixed q. Thus the graded, fixed-exponent assumption in (QF1) has mathematical content; reflection positivity alone does not supply it.

42. From separated moments to Euclidean evolution

This section proves a conditional reconstruction on compactly supported separated test tuples. The time contraction follows from a bound which is uniform in the positions of finitely many supports. No estimate on arbitrary Schwartz sequences, and no bound uniform in the number of fields, enters that argument. A further lattice approximation hypothesis gives a spectral cone with a cubical spatial section. The round cone and continuous rotations require additional input.

42.1. **The precise compact test domain.** Put $\mathcal{D} = C_c^1(\mathbb{R}^4;\mathbb{C})$ and

$$\|f\|_1 = \max\{\|f\|_\infty, \|\nabla f\|_\infty\}.$$

A separated tuple $X = (f_1,\dots,f_n)$ has compact supports at pairwise positive distance. The empty tuple is denoted by 1. Write $\mathscr{V}$ for the complex vector space of finite formal linear combinations of these tuples. Multilinear and permutation relations may be imposed at the outset; equivalently they will be null relations in the forms below. Concatenation $X \cdot Y$ is used only if the supports of all its entries remain separated. There is no claim that $\mathscr{V}$ is an algebra under unrestricted concatenation.

Our translation convention is $\tau_v f(x) = f(x-v)$, so its support is translated by v. For the time reflection $\theta(x_1,x_2,x_3,x_4) = (x_1,x_2,x_3,-x_4)$ put $\theta f = f\circ\theta$. These maps act entrywise on tuples and linearly on formal combinations. The bar conjugates both the coefficients and the test functions. Let $\mathscr{V}_+$ consist of combinations whose entries have supports compactly contained in $\mathbb{E}_+ = \{x_4 > 0\}$.

Assume normalized symmetric multilinear moments S_n on this separated domain. Write $\sigma(X) = S_n(f_1,\dots,f_n)$, $\sigma(1) = 1$, and extend linearly. They obey the following hypotheses. They preserve conjugation, are translation invariant, and are invariant under the coordinate reflections. Their reflection-positive form satisfies

$$q(F,F) := \sigma(\overline{\theta F}\cdot F) \ge 0 \qquad (F \in \mathscr{V}_+). \tag{HB1}$$

This is positivity for every finite linear combination, including the empty tuple, rather than just for individual monomials. The concatenations in (HB1) and its polarized form are separated: reflected supports lie strictly below time zero, and the others strictly above it.

For every integer $n \ge 1$, $R < \infty$ and $d > 0$, assume a finite number $M_n(d,R)$ such that

$$|S_n(f_1,\dots,f_n)| \le M_n(d,R)\prod_{i=1}^n \|f_i\|_1, \tag{HB2}$$

whenever each support is in a ball of radius R, centered anywhere, and their pairwise distances are at least d. The centers need not lie in a common bounded region. For one entry the separation condition is empty. These are moment bounds, not merely cumulant bounds. Bounds on cumulants give (HB2) after the finite moment–cumulant sum, if the needed one-point bound is included. There is no restriction here on the growth of M_n as n increases or d decreases.

Theorem 42.1 (Compact-tuple Euclidean reconstruction)**.** *Under (HB1)–(HB2) and the stated invariances, the completion of $\mathscr{V}_+$ by its reflection form is a separable Hilbert space $\mathcal{H}$. The empty tuple has class Ω of norm one. The formulas*

$$T_t[F] = [\tau_{te_4} F] \quad (t \geq 0),$$
$$U_x[F] = [\tau_{(x,0)} F] \quad (x \in \mathbb{R}^3) \tag{HB3}$$

define, respectively, a strongly continuous semigroup of positive self-adjoint contractions and a strongly continuous unitary representation. They commute and fix Ω. There is a unique nonnegative self-adjoint operator H with $T_t = e^{-tH}$. Each T_t, $t > 0$, is injective. The spatial generators P_i and H commute strongly.

42.2. **The quotient and the contraction proof.** Positivity of the quadratic form gives Hermiticity by polarization and the Cauchy–Schwarz inequality $|q(F,G)|^2 \leq q(F,F)q(G,G)$. Thus $\mathscr{N} = \{F : q(F,F) = 0\}$ is a linear subspace orthogonal to everything. The quotient $\mathscr{V}_+/\mathscr{N}$, with inner product q, is pre-Hilbert; its completion defines $\mathcal{H}$. In particular $\|\Omega\|^2 = \sigma(1) = 1$ and $\langle \Omega, [F] \rangle = \sigma(F)$.

Time and spatial translations in (HB3) preserve their positive-time domain. Translation invariance and $\theta\tau_{te_4} = \tau_{-te_4}\theta$ give

$$q(F, \tau_{te_4} G) = q(\tau_{te_4} F, G),$$
$$q(\tau_{(x,0)} F, \tau_{(x,0)} G) = q(F,G). \tag{HB4}$$

The second identity proves that spatial translations preserve null vectors and induce isometries with inverse U_{-x}. For the first identity, boundedness of the induced time maps remains to be proved; symmetry alone would not suffice.

Fix $F = \sum_\alpha c_\alpha X_\alpha \in \mathscr{V}_+$. There are $\delta > 0$, $d > 0$ and $R < \infty$ such that every entry has time coordinate at least δ, every individual tuple has internal separation at least d, and each entry has support in a ball of radius R. For every $s \geq 0$, every concatenation $\overline{\theta\tau_{se_4} X_\alpha} \cdot \tau_{se_4} X_\beta$ has internal separation at least $d_* = \min\{d, 2\delta\} > 0$. Indeed its two groups lie below $-\delta - s$ and above $\delta + s$. Moving the groups apart changes their centers but not their radii, norms or internal separations. Applying (HB2) to the finitely many pairs gives

$$\sup_{s \geq 0} \|\tau_{se_4} F\|_q \leq C_F < \infty. \tag{HB5}$$

The same assertion covers empty entries and constants by normalization. Its essential hypothesis is that the bound is independent of support positions. A bound with a growing enclosing-volume factor would not prove (HB5).

Write $a(s) = \|\tau_{se_4} F\|_q$. Symmetry and Cauchy–Schwarz give

$$a(t)^2 = q(F, \tau_{2te_4}F) \le a(0)a(2t).$$

Iteration yields

$$a(t) \le a(0)^{1-2^{-k}} a(2^k t)^{2^{-k}} \le a(0)^{1-2^{-k}} C_F^{2^{-k}}. \tag{HB6}$$

If $a(0) = 0$, the first inequality already proves $a(t) = 0$. Otherwise let $k \to \infty$ to obtain $a(t) \le a(0)$. Consequently time translations preserve $\mathscr{N}$ and extend to contractions on $\mathcal{H}$. Their symmetry in (HB4) makes each bounded extension self-adjoint. Moreover

$$q(F, \tau_{te_4}F) = q(\tau_{(t/2)e_4}F, \tau_{(t/2)e_4}F) \ge 0,$$

so every T_t is positive. The semigroup law is inherited from the exact translation law on the dense domain.

42.3. **Continuity, generators and elementary consequences.** For $f \in C_c^1$, uniform continuity of ∇f gives

$$\|\tau_h f - f\|_1 \le \max\left\{ |h| \|\nabla f\|_\infty, \sup_{|x-y|\le|h|} |\nabla f(x) - \nabla f(y)| \right\} \longrightarrow 0. \tag{HB7}$$

This statement need not hold in the Lipschitz norm for a merely Lipschitz function. To apply it to a moment, enclose the old and slightly moved supports of each entry in one common neighborhood. For sufficiently small shifts these neighborhoods retain half their original positive separation. The multilinear telescoping identity and (HB2) then bound the change of a moment by a constant times the sum of the C^1 changes of its entries. Differences of functions are evaluated only on those fixed neighborhoods.

Expanding $\|[\tau_h F] - [F]\|^2$ into its finitely many moment terms and using this observation proves convergence to zero for $h = te_4$, $t \downarrow 0$, and for spatial $h \to 0$. The contraction and unitary bounds extend this continuity from the dense quotient domain to all of $\mathcal{H}$. The semigroup and group laws then give strong continuity at every parameter.

Here is a construction of the generator which also rules out a hidden zero-transfer subspace. If $T_1 v = 0$, positivity and $T_1 = T_{1/2}^2$ give $T_{1/2} v = 0$. Repeated halving and strong continuity at zero imply $v = 0$. Thus the spectral measure of T_1 has no atom at zero. Define the densely defined self-adjoint operator $H = -\log T_1$ by functional calculus; its spectrum is nonnegative. Uniqueness of positive roots gives $T_{m/n} = T_1^{m/n}$ for nonnegative rationals, and strong continuity gives $T_t = e^{-tH}$ for all $t \ge 0$. Functional calculus also proves uniqueness of H and injectivity of every T_t, $t > 0$. Since translations fix the empty tuple, $H\Omega = 0$ and $U_x\Omega = \Omega$.

The commuting spatial unitaries have self-adjoint generators $U_x = e^{i\sum_i x_i P_i}$. They commute with T_1 and hence with its spectral projections and with H. The joint spectral theorem, equivalently the spectral theorem for the strongly continuous abelian unitary group $(s, x) \mapsto e^{isH}U_x$, supplies one projection-valued measure E on $[0, \infty) \times \mathbb{R}^3$. Here and below strong commutation means commutation of these spectral projections. For $F, G \in \mathscr{V}_+$, $t \ge 0$ and $x \in \mathbb{R}^3$ the dictionary is

$$\begin{aligned} \sigma(\overline{\theta F}\, \tau_{(x,t)} G) &= \langle [F], e^{-tH} U_x [G] \rangle \\ &= \int e^{-tE + ix\cdot p} \langle [F], \mathsf{E}(dE\, dp)[G] \rangle. \end{aligned} \tag{HB7a}$$

For completeness, the quotient is separable. Cover compact positive-time supports by finite unions of rational boxes with closures still in $\mathbb{E}_+$; choose these unions separately for each entry so that they remain separated. There are countably many such finite systems. On each fixed bounded union, the C^1 space is separable: restriction of a function and its four derivatives embeds it isometrically in a finite product of separable continuous-function spaces. Choose countable dense subsets inside the supported C^1 spaces, and rational complex coefficients for formal combinations. Telescoping with (HB2), on the fixed separated neighborhoods, shows that the resulting countable set is total in $\mathcal{H}$.

The formal description does not create extra vectors from zero tensors. For a finite homogeneous relation $\sum_\alpha c_\alpha f_{\alpha,1} \otimes \cdots \otimes f_{\alpha,n} = 0$ whose tuples are separated, choose $d > 0$ below all their separations. Localize every entry with one finite smooth partition covering their supports, whose sets have diameter less than $d/8$. In a product of partition sets, either two sets have distance less than $d/2$, in which case every original tensor has a zero factor there, or all the slot sets remain separated. In the latter case the moment is multilinear on the product of their supported function spaces and hence is a linear functional on their algebraic tensor product. The map from this tensor product into functions on the Cartesian product is injective: on the finitely many function spans involved, independent functions can be distinguished by finitely many point evaluations, and the corresponding evaluation matrices invert each slot. Thus the localized zero tensor has zero moment. Summing proves the assertion. Applying it also after concatenation with a reflected test shows that a zero tensor is a reflection-null vector. This is an algebraic compact-support fact, distinct from the Schwartz density input below.

Any realization with a dense image of the same test domain and the same scalar products is therefore canonically unitarily equivalent, by sending each test class to its other realization; the generators intertwine by density. The construction can use any coordinate as time if its RP hypothesis is supplied. For $c \geq 0$ and F supported strictly above time c, the exact identity is $[F] = T_c[\tau_{-ce_4}F]$.

Any further linear symmetry of the test functions, acting entrywise on tuples, which preserves the admissible positive-time domain and the moments and commutes with θ and conjugation acts isometrically on this quotient. In particular, if the moments also have the spatial signed-permutation symmetry W_3, then $V_w[F] = [wF]$ is unitary, $V_w T_t = T_t V_w$ and $V_w U_x V_w^{-1} = U_{wx}$. Uniqueness of the joint spectral measure gives the corresponding covariance under $(E, p) \mapsto (E, wp)$ and hence W_3 symmetry of the joint spectrum. This optional finite symmetry is not continuous rotation invariance.

Spectral dominated convergence gives $T_t \to P_0 = 1_{\{0\}}(H)$ strongly as $t \to \infty$. Hence the large-time reflected correlations equal the matrix elements of P_0. In particular, $\ker H = \mathbb{C}\Omega$ exactly when their limits, or their Cesàro limits, equal $\overline{\sigma(F)}\sigma(G)$ for every positive-time F, G. If $\sigma(f) = 0$ and $S_2(\theta\bar{f}, f) > 0$ for one positive-time test, then $[(f)]$ is a nonzero vector orthogonal to Ω, so $\dim \mathcal{H} \geq 2$; this requires an actual positive two-point input.

A shift of the fields does not change this representation. More precisely, let m be a complex linear, translation- and reflection-invariant functional on $\mathcal{D}$, satisfying $m(\bar{f}) = \overline{m(f)}$ and bounded on each fixed compact support. Define

$$C_m(f_1, \ldots, f_n) = \sum_{A \subset \{1,\ldots,n\}} \prod_{i \notin A} [-m(f_i)]\, (f_i)_{i \in A},$$

$$\sigma^{\mathrm{c}}(F) = \sigma(C_m F).$$

Then C_{-m} is the inverse of C_m, and on every admissible concatenation $C_m(FG) = C_m(F)C_m(G)$. It commutes with the bar, reflections and translations. Therefore $[F]_{\sigma^{\mathrm{c}}} \mapsto [C_m F]_\sigma$ is a unitary intertwining the two evolutions. Its finite expansion also transfers (HB2), since local

boundedness and translation invariance bound m uniformly on balls of each fixed radius. This algebraic statement does not estimate the size of the one-point difference between two cutoff states.

In the particular unweighted Wilson-density normalization, the one-point calibration can be written explicitly. Let two states ρ_j, ν_j on spacing a_j be translation and hypercubic invariant, and put $\mathcal{O}_j(f) = \sum_p f(x(p))a_p$. Their plaquette mean difference $d_j = \rho_j(a_p) - \nu_j(a_p)$ is independent of p. Define

$$R_j(f) = \frac{a_j^4}{6} \sum_p f(x(p)),$$

$$c_j = 6a_j^{-4} d_j.$$

Then $\rho_j(\mathcal{O}_j(f)) - \nu_j(\mathcal{O}_j(f)) = c_j R_j(f)$ exactly, and the six shifted Riemann sums give $R_j(f) \to \int f$. If these one-point differences converge to $m(f)$ on all compact C^1 tests, choose one f_0 with $\int f_0 = 1$. The ratio at f_0 shows $c_j \to m(f_0) = c$, and consequently $m(f) = c \int f$ for every test. A uniform one-point bound at f_0 bounds $|c|$; this bound is an input, not a consequence of translation invariance alone. When the two states coincide, $c_j = c = 0$.

42.4. **Coarse lattice inequalities and the continuum cube cone.** The spectral cone needs additional information. Assume infinite-lattice probability measures ν_j at spacings $a_j \downarrow 0$, invariant under lattice translations and reflection positive on the closed side of all coordinate and diagonal lattice mirrors. Their moment forms will be denoted by $S_n^{(j)}$. For periodic Wilson DLR states these hypotheses are proved in Theorem 7.1. For other lattice measures they remain explicit assumptions. The normal transfer T_j and tangential translation $V_{j,i}$ in their reflection Hilbert spaces then have the carrier established in the plateau argument:

$$\sinh(E/2) \geq |\sin(p/2)|,$$

$$T_j = e^{-E},$$

$$V_{j,i} = e^{ip}. \tag{HB8}$$

Here E is dimensionless and may be infinity on $\ker T_j$, while p is a circle variable. Only this lattice carrier, rather than a round relativistic cone, is taken from that argument.

One lattice distinction is useful. For spacing a, the site mirror at zero and the link mirror at $-a/2$ have the same positive-side lattice algebra. Their forms obey $q_{\mathrm{link}}(A, B) = q_{\mathrm{site}}(A, TB)$, by translation invariance. Consequently $[A]_{\mathrm{link}} \mapsto T^{1/2}[A]_{\mathrm{site}}$ is an isometry onto the closure of $\operatorname{Ran} T^{1/2}$, namely $(\ker T)^\perp$. It intertwines the two normal transfers, since this holds on the common local domain. Thus the link transfer is injective, even if the site transfer has a kernel. This proves the relation between the two pictures without discarding site-kernel contributions in (HB10).

For every integer $k \geq 1$, convexity of sinh and the elementary inequality $|\sin(kv)| \leq k|\sin v|$ give

$$\sinh(kE/2) \geq k \sinh(E/2) \geq k|\sin(p/2)| \geq |\sin(kp/2)|. \tag{HB9}$$

The sine inequality follows by induction from the addition formula and $|\cos v| \leq 1$. Thus functional calculus gives the bounded operator inequality

(HB10) $$(I - T_j^k)^2 \geq T_j^k(2I - V_{j,i}^k - V_{j,i}^{-k}).$$

At $T_j = 0$ this inequality holds directly, so no division by a transfer eigenvalue is required. Its quadratic form is a five-point inequality. If $s = ka_j$ and $\Phi_{j,F}(v) = \nu_j(\overline{\theta F_j}\,\tau_v F_j)$, it reads

(HB11) $$\Phi_{j,F}(0) - 4\Phi_{j,F}(se_4) + \Phi_{j,F}(2se_4) + \Phi_{j,F}(se_4 + se_i) + \Phi_{j,F}(se_4 - se_i) \geq 0.$$

To pass this statement to the limit, specify the approximation rather than identify Hilbert spaces at different cutoffs. Assume linear local lattice bounded Borel representatives $\iota_j(f)$ for compact C^1 tests, preserving conjugation: $\iota_j(\bar{f}) = \overline{\iota_j(f)}$. Their supports lie within distance $\epsilon_j \to 0$ of $\operatorname{supp} f$. They commute with coordinate reflection and with translations by $a_j\mathbb{Z}^4$. Let F_j be the product representative of each tuple, extended linearly, and let $S_n^{(j)}$ be its moments. Assume convergence on every fixed separated compact tuple to S_n, and assume (HB2) for these lattice moments with constants independent of j once j is sufficiently large for the fixed n, d, R. For action-density representatives their supremum norms may depend on j; the asserted uniform control is on these moments.

If $v_j \to v$, the supports of $\tau_{v_j} f$ and $\tau_v f$ lie in fixed slightly enlarged sets. Whenever the limiting tuple is separated, these sets retain a positive separation. Telescoping and the uniform moment bound give

(HB12) $$S_n^{(j)}(f_{1,j}, \ldots, f_{n,j}) - S_n^{(j)}(f_1, \ldots, f_n) \longrightarrow 0$$

for such translated tests, by (HB7). Together with fixed-tuple convergence, this proves convergence at the moving arguments.

The same argument also explains inheritance of the Hilbert hypotheses from these approximants, when it is needed. A continuum translation is approached by lattice vectors v_j; exact lattice covariance and (HB12) pass its moment identity to the limit. Coordinate reflections through the origin need no rounding. For any fixed $F \in \mathscr{V}_+$ the representatives F_j lie on the positive side for large j, so their finite quadratic RP inequalities pass directly to (HB1). Normalization, conjugation, multilinearity and the uniform moment bounds pass by their finite defining relations. The lattice mirror hypotheses themselves are supplied by Theorem 7.1 in the periodic Wilson application.

There is also a finite-torus scalar-product statement, with a narrower conclusion. If fixed separated moments converge from growing periodic tori and satisfy uniform C^1 bounds on the fixed neighborhoods under consideration, RP at the link mirror $x_4 = a_j/2$ gives

$$q(F, G) = \lim_j \nu_j(\overline{\theta_{a_j/2} F_j}\, G_j).$$

Indeed $\theta_{a_j/2} f = \tau_{a_j e_4} \theta f$, whose difference from θf tends to zero in C^1, and the same telescoping estimate applies. This compares scalar products as numbers; it asserts no maps between the cutoff Hilbert spaces. Finite-torus positivity of this kind does not by itself supply the global lattice semigroup or (HB8).

Fix $t > 0$ and choose integers $k_j \geq 1$ with $k_j a_j \to t$. For positive-time F, its lattice representative is supported on the positive side of the fixed site mirror for large j. All five moment expressions in (HB11) are separated uniformly: the negative reflected group and every translated positive group retain their original positive-time margin. Internal separations are unchanged. Therefore (HB12) passes (HB11) to its continuum version. In $\mathcal{H}$ this is

$$B_{t,i} := (I - T_t)^2 - T_t(2I - U_{te_i} - U_{-te_i}) \geq 0. \tag{HB13}$$

Initially its form is nonnegative on the dense compact-tuple domain; since $B_{t,i}$ is bounded and self-adjoint, the inequality holds on all of $\mathcal{H}$. No differentiation of a cutoff correlation at $t = 0$ has been used. The fixed-time coarse inequality is passed first.

In the joint functional calculus its multiplier is

$$b_{t,i}(E,p) = 1 - 4e^{-tE} + e^{-2tE} + 2e^{-tE}\cos(tp_i) = 2e^{-tE}[\cosh(tE) + \cos(tp_i) - 2]. \tag{HB14}$$

Take a countable sequence $t_m \downarrow 0$, for example $t_m = 1/m$. Positivity of (HB13) means that the joint projection-valued measure gives zero mass to $\{b_{t_m,i} < 0\}$. Remove the countable union of those sets for the three spatial indices. At every remaining spectral point,

$$0 \leq \lim_{m\to\infty} \frac{\cosh(t_m E) + \cos(t_m p_i) - 2}{t_m^2} = \tfrac{1}{2}(E^2 - p_i^2).$$

Since $E \geq 0$, this proves

$$\mathsf{E}\{(E,p) : E < \max_i |p_i|\} = 0. \tag{HB15}$$

In particular $\operatorname{Dom} H \subset \operatorname{Dom} P_i$ and $\|P_i\psi\| \leq \|H\psi\|$ on that domain. The corresponding form-domain statement follows by integrating $|p_i| \leq E$ rather than their squares. Zero-energy vectors are spatially invariant. Consequently vectors fixed by all Euclidean time and spatial translations are exactly $\ker H$. This statement concerns invariant Hilbert vectors; identifying them with an ergodic decomposition of correlation states needs the separate phase argument. The cone (HB15) has a cubical spatial section. Nothing here proves $E \geq |p|_2$ or supplies an O(4) action. Finite periodic torus laws cannot simply be substituted for the infinite lattice hypotheses at arbitrarily large separations; any required infinite-volume reflection-positive state must first be obtained.

42.5. **What the full Schwartz sequence domain additionally needs.** The compact-tuple construction above does not establish the claimed Schwartz-domain growth property. Here is a precise sufficient additional input for that extension. For $n \geq 2$ put $\Delta_n = \bigcup_{i<j}\{x_i = x_j\}$ and let $\mathcal{S}_{\Delta,n}$ be the Schwartz functions on $\mathbb{R}^{4n}$ whose derivatives of every order vanish on Δ_n. Use $\mathcal{S}_{\Delta,0} = \mathbb{C}$ and the usual Schwartz space for $n = 1$. Define the seminorms

$$p_k(\varphi) = \max_{|\alpha|\leq k} \sup_x (1 + |x|^2)^{k/2} |\partial^\alpha \varphi(x)|.$$

The positive-time subspace $\mathcal{S}_{\Delta,+,n}$ consists of functions all of whose derivatives vanish outside $\{x_{1,4} > 0, \ldots, x_{n,4} > 0\} \setminus \Delta_n$. This includes flatness at the time boundary, as well as at the diagonals.

The separate continuity/growth input, called H1 here, supplies continuous linear moment functionals Σ_n on these full diagonal-flat spaces, agreeing with the compact smooth separated tensors, and constants $c, C < \infty$, $k \in \mathbb{N}$, $\varkappa < \infty$ such that

$$|\Sigma_n(\varphi)| \leq cC^n (n!)^{\varkappa} p_{kn}(\varphi). \tag{HB16}$$

This is a condition on moments, not just connected functions. It is not a consequence of having a finite $M_n(d,R)$ for every fixed triple of parameters. The Density lemma of Section 54 proves that finite linear combinations of smooth compact separated tensors are dense in $\mathcal{S}_{\Delta,n}$, and those with strictly positive-time supports are dense in $\mathcal{S}_{\Delta,+,n}$, in their Schwartz subspace topologies.

These assertions include the necessary cutoffs at infinity, at the time boundary, and near the diagonals. They cannot be replaced by ordinary tensor density without those constraints.

Using H1 and that density lemma, the finite Schwartz sequence space $\bigoplus_{n\ge 0}\mathcal{S}_{\Delta,+,n}$ has form $\sum_{m,n}\Sigma_{m+n}(\overline{\theta F_m}\otimes G_n)$. The tensor here lies in the diagonal-flat space: an internal coincidence is killed by the corresponding factor, and a coincidence between a negative and a positive time argument forces the common time to zero, where the factors are flat. Tensor multiplication is continuous in the Schwartz topologies. Thus H1 and tensor density transfer the compact reflection positivity and show that its quotient completion is canonically the same $\mathcal{H}$. Compact C^1 tuples themselves are approximated by compact smooth ones inside the same separated positive-time neighborhoods, using (HB2). The already proved contraction and unitary groups then extend by this identification. We use H1 for continuity on the enlarged domain. For reconstruction in a quantitative growth class, we also use the growth bound (HB16). The constrained tensor-density argument and the quantitative extension estimates are given in the cited extension section.

43. The short-distance coefficient and nontriviality

Proof outline. The supporting analytic results used in this section, including the obligations that we must still prove, are listed under F in the dependency catalogue. Their hypotheses and precise forms are retained at the points where the argument uses them.

7. Non-triviality of the limit: the two-point law of asymptotic freedom, a positive four-point function on the tori, and short-distance universality. At short distances, the connected two-point function behaves to leading order like that of N^2-1 free fields. On the tori the same holds for the connected four-point function, restricted to nearly collinear quadruples, which form the class $\mathfrak{T}^{(4)}$ introduced in (7d). The strength of these free fields equals the running coupling evaluated at the observation scale; it is non-zero and tends to zero only logarithmically.

Objects. The test pairs are $(f,h)\in\mathfrak{T}_{\mathbb{E}}(\mathsf{d};\vartheta)$, that is, pairs of non-negative Lipschitz bumps, each supported in a ball of radius d, placed at a distance from one another that lies between d and $6\mathsf{d}$ ((7a)). The free kernel is $\mathcal{K}_{\mathrm{free}}(f,h)=\iint f(x)h(y)|x-y|^{-8}dx\,dy$. We take the coupling $g(\mathsf{d})=g_{K-s(\mathsf{d})}(\beta)$, as well as the normalisation $\zeta_K(\mathsf{d})=\zeta'_{K-s(\mathsf{d})}(\beta)$, at the level whose spacing is smaller than d by a power of $\log g(\mathsf{d})^{-2}$.

Theorem 7.1 [non-triviality; Theorem 0(c)(3)-(5)]. Assume $g\le\gamma_W(\vartheta)$. Then there exist an integer $n_0\ge 1$, a threshold $\mathsf{d}_0=\mathsf{d}_0(g,\vartheta)\in(0,1)$, a function $\bar\eta:(0,\mathsf{d}_0]\to[0,\frac13]$ that is non-decreasing with $\bar\eta(\mathsf{d})\to 0$ as $\mathsf{d}\to 0$, and, for each $\ell\in\mathfrak{L}(g)$, a function $\bar\varepsilon_\ell:(0,\mathsf{d}_0]\to[0,\frac12]$ that is non-decreasing with $\bar\varepsilon_\ell(\mathsf{d})\to 0$ as $\mathsf{d}\to 0$, with the following properties whenever $\mathsf{d}\le\mathsf{d}_0$. (i) Take any first-passage datum with $K-s(\mathsf{d})\ge n_0$, any $\ell\in\mathfrak{L}(g)$ and any pair in $\mathfrak{T}_{T_\ell}(\mathsf{d};\vartheta)$. Then

$$\langle\mathcal{O}(f);\mathcal{O}(h)\rangle_{K,\ell,\beta}=\zeta_K(\mathsf{d})^2\,\tfrac{3(N^2-1)}{\pi^4}\,g(\mathsf{d})^4\,\mathcal{K}_{\mathrm{free}}(f,h)\,(1+\varepsilon),$$
$$|\varepsilon|\le\tfrac12,$$

and moreover $|\varepsilon|\le\bar\varepsilon_\ell(\mathsf{d})$, the inequality $|\frac{11N^2}{12\pi^2}g(\mathsf{d})^2\ln(1/\mathsf{d})-1|\le\bar\eta(\mathsf{d})$ holds, and $|\zeta_K(\mathsf{d})-1|$ is bounded by the same right-hand side; all of these bounds are uniform in (K,β). The same identity is also valid in every periodic Gibbs state at β on $\mathbb{L}_K$, with the same $\zeta_K(\mathsf{d})$ and $g(\mathsf{d})$ and with an error satisfying $|\varepsilon|\le\frac12$ ((7a)). (ii) On pairs of class C^1, every continuum limit obtained from (5b) on the torus T_ℓ, as well as every one obtained from (5c) on $\mathbb{R}^4$, obeys

$$S_2(f,h)=\tfrac{3(N^2-1)}{\pi^4}\Big(\tfrac{12\pi^2}{11N^2}\Big)^2\frac{\mathcal{K}_{\mathrm{free}}(f,h)}{(\ln(1/\mathsf{d}))^2}\,(1+\eta),$$
$$c\le 1+\eta\le C,$$

and on each T_ℓ one has $\eta \to 0$ as $\mathsf{d} \to 0$. (iii) On the tori, let $\vartheta \le \vartheta_\square$, the number from Theorem 0(c)(5), which depends only on the fixed data. Then, by Theorem 7.5 [to be proved; Package F], there exist an integer $n_4 \ge n_0$, a number $\mathsf{d}_4 \in (0, \mathsf{d}_0]$, a side $\ell_4(g) \in \mathfrak{L}(g)$, together with, for each $\ell \in \mathfrak{L}(g)$ with $\ell > \ell_4(g)$, a threshold $\mathsf{d}_4(\ell) \in (0, \mathsf{d}_4]$, with the following property. Suppose $K - s(\mathsf{d}) \ge n_4$. Suppose $\mathsf{d} \le \mathsf{d}_4$ when $\ell \le \ell_4(g)$, and $\mathsf{d} \le \mathsf{d}_4(\ell)$ otherwise. Then for every quadruple in $\mathfrak{T}^{(4)}_{T_\ell}(\mathsf{d}; \vartheta)$ ((7d)), the following bounds hold uniformly in (K, β). Here c, C are constants that are allowed to depend on g as well as on the shape ϑ.

$$c\,(N^2-1)\,g(\mathsf{d})^8 \prod_i \|f_i\|_\infty \le S^T_{4;K,\ell,\beta}(f_1, \dots, f_4) \le C\,(N^2-1)\,g(\mathsf{d})^8 \prod_i \|f_i\|_\infty.$$

Clause (iii) is a statement about the lattice functions on the tori. For four points, Theorem 0 asserts nothing about the limits, and nothing about $\mathbb{R}^4$.

† The sign of S^T_4 for quadruples of arbitrary shape at scale d, at every place where the box kernel lies above a floor, and, for nearly collinear quadruples and for g below a further threshold, the positivity of this function both in the limits on $\mathbb{R}^4$ and on a set of ergodic components having positive measure (7d); the rigidity of the short-distance germ and its relative form (7e).

(7a) The short-distance two-point law and its asymptotic-freedom normalisation. **Objects.** The class $\mathfrak{T}_{\mathbb{E}}(\mathsf{d}; \vartheta)$ consists of the pairs of non-zero Lipschitz functions $f, h \ge 0$ satisfying

$$\begin{gathered}
\operatorname{supp} f \subset B(x, \mathsf{d}), \\
\operatorname{supp} h \subset B(y, \mathsf{d}), \\
\mathsf{d} \le \operatorname{dist}(\operatorname{supp} f, \operatorname{supp} h) \le 6\mathsf{d}, \\
\textstyle\int f \ge \vartheta \|f\|_\infty \mathsf{d}^4, \\
\operatorname{Lip} f \le \tfrac{\|f\|_\infty}{\vartheta \mathsf{d}},
\end{gathered}$$

where the final two conditions are required of h as well. It follows that $c\vartheta^2 \|f\|_\infty \|h\|_\infty \le \mathcal{K}_{\text{free}}(f, h) \le C \|f\|_\infty \|h\|_\infty$. On $\Lambda^2 \mathbb{R}^4$ the free curvature propagator is given by

$$\begin{gathered}
\mathbb{K}(u) = \pi^{-2} |u|^{-4}\, \Lambda^2 \mathsf{I}(\hat{u}), \\
\mathsf{I}(\hat{u}) = 1 - 2\hat{u}\hat{u}^{\mathsf{T}}, \\
\hat{u} = u/|u|,
\end{gathered}$$

which is the two-point tensor of dA when A is a one-form with covariance $\delta_{\mu\nu}(4\pi^2|x|^2)^{-1}$. Since $\Lambda^2 \mathsf{I}$ is an involution, $\operatorname{tr} \mathbb{K}(u)^2 = 6\pi^{-4}|u|^{-8}$. On $\Lambda_{K,\ell}$, let $F = dA$ be the free lattice curvature, where d denotes the lattice coboundary; this curvature is the standard Gaussian on exact 2-cochains with a single colour and unit coupling, and $\mathbb{E}_{\text{free}}$ denotes the expectation with respect to it. Its background part at depth s is $F_B := \mathbb{E}_{\text{free}}[F \mid Q_{K-s}(\mathbf{1})A$ modulo closed forms], and we set

$$\begin{gathered}
C^{\text{free}}_K = C^{\text{bg}}_{K,s} + C^{\text{fl}}_{K,s}, \\
C^{\text{bg}}_{K,s}(p, q) = \mathbb{E}_{\text{free}}\, F_B(p) F_B(q), \\
\mathcal{K}_{K,s}(f, h) = \sum_{p,q} f(x(p)) h(x(q))\, C^{\text{bg}}_{K,s}(p, q)^2.
\end{gathered}$$

Let Q_j^{pl} be the average of plaquettes, that is, of 2-cochains, over blocks of order j; it is the counterpart of the straight average of 1-forms. Because $d\,Q_j(\mathbf{1}) = d\,Q_j^{\mathrm{st}}(\mathbf{1}) = Q_j^{\mathrm{pl}} d$, the conditioning above is conditioning on $Q_{K-s}^{\mathrm{pl}} F$. The $C_{K,K-j}^{\mathrm{bg}}$ are therefore kernels of orthogonal projections whose ranges are nested, and the slices

$$\mathcal{C}_{(j)} := C_{K,K-j+1}^{\mathrm{bg}} - C_{K,K-j}^{\mathrm{bg}}$$
$$(C_{K,K}^{\mathrm{bg}} = C_K^{\mathrm{free}}),$$
$$C_{K,s}^{\mathrm{fl}} = \sum_{1\le j\le K-s} \mathcal{C}_{(j)},$$

are the covariances of mutually independent pieces of F. For the free field, the renormalisation group thus reduces to an orthogonal decomposition.

Statement. The error in Theorem 7.1(i) satisfies $|\varepsilon| \le \frac{1}{2}$ and, in addition, $|\varepsilon| \le \bar{\varepsilon}_\ell(\mathsf{d}) \to 0$ on each torus of the family, uniformly in the cutoff, the bare coupling, the pair and its position. The same identity, with the same $\zeta_K(\mathsf{d})$ and $g(\mathsf{d})$ and with $|\varepsilon| \le \frac{1}{2}$, holds for $\langle \mathcal{O}(f); \mathcal{O}(h)\rangle_\nu$ in every periodic Gibbs state ν at β on $\mathbb{L}_K$. The constant arises as follows: $a_p \approx \frac{1}{2}|F(p)|^2$; for each colour, Gaussian half-squares have covariance $\frac{1}{2}(\text{covariance})^2$; and $\frac{N^2-1}{2}\cdot\frac{6}{\pi^4} = \frac{3(N^2-1)}{\pi^4}$.

† **Two variants.** (i) *The one-loop coupling.* Define $\mathsf{g}(\mathsf{d}) := \big(g^{-2} + \frac{11N^2}{12\pi^2}\ln\frac{1}{\mathsf{d}}\big)^{-1/2}$, which depends on d and g only. Lemma 7.2, together with $0 \le s\log L - \ln\frac{1}{\mathsf{d}} - \ln C_W(s) < \log L$, gives

$$\big|g(\mathsf{d})^{-2} - \mathsf{g}(\mathsf{d})^{-2}\big| \le o(s) + C + \tfrac{c_0}{\log L}\ln C_W(s) = o(s(\mathsf{d})),$$

uniformly in (K,β). Hence $g(\mathsf{d})/\mathsf{g}(\mathsf{d}) \to 1$, and if $\mathsf{g}(\mathsf{d})$ is put in place of the running coupling $g(\mathsf{d})$ of the construction, the law still holds below a different threshold, with error functions that differ but are of the same type. (ii) *Cutoffs off first passage.* Let $K' = K(\beta, g)$, and let $b = K - K'$, which may be positive, zero or negative. The dilation identity of (3d) then takes the form

$$\langle \mathcal{O}(f); \mathcal{O}(h)\rangle_{K,\ell,\beta} = \langle \mathcal{O}(f^{[b]}); \mathcal{O}(h^{[b]})\rangle_{K',L^b\ell,\beta},$$

and the map $f \mapsto f^{[b]}$ sends $\mathfrak{T}_{T_\ell}(\mathsf{d};\vartheta)$ into $\mathfrak{T}_{T_{L^b\ell}}(L^b\mathsf{d};\vartheta)$ and leaves $\mathcal{K}_{\mathrm{free}}$ invariant. Consequently, whenever $L^b\ell \in \mathfrak{L}(g)$, $L^b\mathsf{d} \le \mathsf{d}_0$ and $K' - s(L^b\mathsf{d}) \ge n_0$, the law at $(K', L^b\ell, \beta)$ with separation $L^b\mathsf{d}$ determines the behaviour of $\mu_{K,\ell,\beta}$ at separation d. For K large and under the conditions of clause (d)(1), (3d) and (8b) give $|b| \le C(\varrho)$ for the tuned data and for each of their offsets $t \in \mathsf{J}$. When $b < 0$, the identity is to be read in the direction from $(K', L^b\ell)$ to (K, ℓ).

Argument. (1) The local analysis is carried out on a single torus, the smallest one, $T_{\ell_\star(g)}$. On that torus the law holds by Proposition 7.3 of (7b), and its kernel there is $\frac{N^2-1}{2}\mathcal{K}_{K,s}$. Reflection positivity then carries it from this torus to the remaining volumes and to the Gibbs states. For the Gibbs states this is the only route available, because the expansion deals with torus measures. The local analysis can also be carried out on the other tori. (7d) does this, with four sources, on every torus of the family, but only below a threshold on d which may decrease with ℓ. The transfer, by contrast, yields a single threshold d_0 common to all of them. (2) Comparison of lattice and continuum. Let u be the shortest vector representing $x(p) - x(q)$ on T_ℓ, with $|u| \approx \mathsf{d}$, let $\sigma_{pq} := a_K^4 \pi^{-2}|u|^{-4}$, and let $\mathbb{K}(\cdot)_{pq}$ be the entry belonging to the planes of p, q. Then

$$\big|C_K^{\mathrm{free}}(p,q) - a_K^4\,\mathbb{K}(u)_{pq}\big| \le C\Big[\Big(\frac{a_K}{|u|}\Big)^{1/2} + \Big(\frac{|u|}{\ell}\Big)^4\Big]\sigma_{pq}$$

$$(10a_K \le |u| \le \ell/10),$$
$$\big|C^{\mathrm{fl}}_{K,s}(p,q)\big| \le o(1)\,\sigma_{pq}.$$

The first inequality is the proved free comparison of Section 5; it compares second differences of the Green's function on the lattice torus with those of the continuum Green's function. Because $a_K/\mathsf{d} \le L^{-n_0}/C_W(s)$, the first term on its right is small by the choice of n_0, and it tends to 0 with d uniformly in K, since the room increases. On the reference torus the second term is at most $C(\mathsf{d}/\ell_\star)^4$, which is small with d_0 and tends to 0 with d. The second inequality is (7b)(c). Approximating by a Riemann sum, at a relative cost of $Ca_K/(\vartheta^3\mathsf{d})$, yields $\mathcal{K}_{K,s} = 6\pi^{-4}\mathcal{K}_{\mathrm{free}}(1+o(1))$. (3) Lemma 7.6 [outlined here; Section 44.3] of (7e) transfers the law to every T_ℓ and every ν at a cost of $C_2\mathsf{d}^2\|f\|_\infty\|h\|_\infty$, in which $C_2 = C_2(g,\vartheta)$ denotes the constant of that lemma at $n=2$. Take $\mathsf{d} \le \mathsf{d}_0(g,\vartheta)$; there Lemma 7.2 gives $\frac{11N^2}{12\pi^2}g(\mathsf{d})^2\ln(1/\mathsf{d}) \ge \frac{2}{3}$. In this range the cost is a fraction of the main term. That fraction goes to 0 as d does, and so do the $o(1)$ of (2) and the R of Proposition 7.3. The threshold d_0 is chosen at the point where these three errors combined give $|\varepsilon| \le \frac{1}{2}$. The main term is $\ge c\vartheta^2(\ln(1/\mathsf{d}))^{-2}\|f\|_\infty\|h\|_\infty$. Consequently, on every torus, the error goes to 0 as d does.

(† The same convergence holds in every periodic Gibbs state and on $\mathbb{R}^4$.)

(4) The quantities $\zeta_K(\mathsf{d})$ and $g(\mathsf{d})$ are controlled by Lemma 7.2.

(7b) Isolating a single scale: coupling and normalisation at the scale d. **Objects.** We set $s = s(\mathsf{d})$ and $k_0 := K - s$, the witness level. We also write $x := x_{k_0}(\beta) = g(\mathsf{d})^{-2}$. The separation measured in level-k_0 units is $\hat{\mathsf{d}} := L^s\mathsf{d} \in [C_W(s), LC_W(s))$. This inclusion holds for $\mathsf{d} < 1$, in which case $s(\mathsf{d}) \ge 1$, and it follows from the minimality of $s(\mathsf{d})$ and the monotonicity of C_W. We write $\underline{x}_s := g^{-2} + \lambda_\sharp s - c_5$ and $\bar{x}_s := g^{-2} + B_\sharp(s+1)$ for the ends of the envelope of (3a); they do not depend on (K,β), and they are non-decreasing in s. The room $C_W(s) = C_{W,0}\,p_0(\bar{x}_s^{-1/2})^2$ contains a constant $C_{W,0}$, chosen large after L, M and ϑ and before g. The lower bounds placed on this constant are $\vartheta\,C_W(s) \ge CM$, which is used in the Strategy, and those required in (b) and (c) below. The room tends to infinity, but $\log C_W(s) = o(s)$.

Lemma 7.2 [coupling and normalisation]. Uniformly in (K,β), and with no torus entering (3b), one has

$$|x_{K-s} - x_K - c_0 s| \le o(s),$$
$$\zeta'_k = 1 + \sum_{j\le k} b'_j(0) \in [\tfrac{2}{3}, \tfrac{4}{3}],$$
$$|\zeta'_{K-s} - 1| \le C g_{K-s}\,p_0(g_{K-s}).$$

The first bound is the flow statement of (3a). Because $s(\mathsf{d})\log L/\ln(1/\mathsf{d}) \to 1$, it yields $g(\mathsf{d})^{-2}/\ln(1/\mathsf{d}) \to c_0/\log L = \frac{11N^2}{12\pi^2}$. For the other two bounds, note that a constant source $w \equiv \zeta$ acts as a shift of β in which every auxiliary choice is held frozen. The increment $b_{j+1}(\zeta)$ is holomorphic for $|\zeta| < 2\varrho$, and its slope is $\le C\big[g_j^3(\log x_j)^{\mathfrak{p}_0} + \varrho^{-1}g_j^{\kappa_0-4}\big]$ (4b). Since $x_j \ge x_k + \lambda_\sharp(k-j) - C$, the slopes summed over the levels $j<k$ are at most $C\big[g_k(\log x_k)^{\mathfrak{p}_0} + \varrho^{-1}g_k^{\kappa_0-6}\big]$. This bound is a function of x_k alone, it tends to 0, and for $\kappa_0 \ge 7$ it is $\le C g_k\,p_0(g_k)$.

Proposition 7.3 [the law on the reference torus]. Let $g \le \gamma_W(\vartheta)$, let (K,β) be a first-passage datum, and suppose that $\mathsf{d} \le \mathsf{d}_0$, $K - s(\mathsf{d}) \ge n_0$ and $(f,h) \in \mathfrak{T}_{T_{\ell_\star(g)}}(\mathsf{d};\vartheta)$. Then on the torus $T_{\ell_\star(g)}$, uniformly in (K,β) and in the pair, one has

$$\langle \mathcal{O}(f); \mathcal{O}(h)\rangle = \zeta'^2_{k_0}\,\tfrac{N^2-1}{2}\,x^{-2}\,\mathcal{K}_{K,s}(f,h)\,(1+R),$$

$$|R| \le \tfrac{1}{4},$$
$$R \to 0\,(s \to \infty).$$

Strategy. In this proof the pair (f,h) is also written (f_1,f_2). We set $k_1 := k_0 + \mathsf{n}$ with $\mathsf{n} := \lceil 2\log_L \bar{x}_s \rceil$. Here g is fixed first. Then d_0 is chosen, small enough that $\mathsf{n} \le s(\mathsf{d})$ whenever $\mathsf{d} \le \mathsf{d}_0$. This gives $k_1 \le K$. Moreover $x_j \approx x$ for $k_0 \le j \le k_1$, because $B_\sharp \mathsf{n} \ll x$. The sourced construction (4a) is carried out up to a level $k \le k_1$, with the sources measured at depth $K-k$. At depth s one has $\|f\|_{\sharp,s} = \|f\|_\infty$, because $\vartheta\, C_W(s) \ge CM$ for $\mathsf{d} \le \mathsf{d}_0(g,\vartheta)$. Above the witness level, $\|f\|_{\sharp,K-k} \le L^{k-k_0}\|f\|_\infty$, so the disc in z becomes smaller. The identities for the derivatives at $z=0$ remain valid, and in the bounds each derivative in a source costs a factor L^{k-k_0}. One then has $Z(z)/Z(0) = \mathbb{E}_{\mathsf{P}^{(k)}}\, e^{\Phi^{(k)}(z)}$. In what follows we write $\mathsf{P} = \mathsf{P}^{(k_0)}$, which is a law distinct from the law P_{k_0} of Proposition 4.4 that appears further on, and $\Phi = \Phi^{(k_0)}$. Everything stated about these two objects in this paragraph holds at each level $k \le k_1$, with k in place of k_0 and $K-k$ in place of s. P is a probability law on triples. Each triple consists of a class $[h]$ of histories of level k_0 sharing one standing part $h^\flat$ (2a), a block field $V \in \mathcal{V}_{k_0}$, and the stored fields. The law is determined by the unsourced sums $\tau_{[h]} := \sum_{h' \in [h]} \tau_{h'}$. Here $\tau_{[h]}(dV)$ is an integral over the stored fields, which are the fields left unintegrated at level k_0. Its integrand is the exponential of the exponent of the sum, integrated against a positive measure composed of the gauge-fixing factors, the characteristic functions, the delta functions of the averaging constraints and Haar measures. Dividing this integrand by $Z(0)$ yields the law on triples. Φ is the difference between the sourced and the unsourced exponent of $\tau_{[h]}$. An exponent is attached to such a sum and not to an individual τ_h (2a), (2e). The marginal of P in $([h],V)$ equals $Z(0)^{-1}\tau_{[h]}(dV)$. Tested on gauge-invariant functions, this marginal agrees with the push-forward under $h \mapsto [h]$ of the (h,V)-marginal of P_{k_0}, the law on triples (bare field, history, block field) of Proposition 4.4 (4a). Projected further onto $[h]$, the marginal is the image of $\tilde{\nu}$. Both laws therefore extend one and the same law of $[h]$ and of the gauge orbit of V, but they add different variables. P_{k_0} adds the bare field and h, the particular member of the class. P adds the stored fields, on which the part of Φ attached to the live components depends. With the notation $\Phi_B^{(k)} := \prod_{i\in B} \partial_{z_i} \Phi^{(k)}|_{z=0}$ we have

$$\langle \mathcal{O}(f_1); \mathcal{O}(f_2) \rangle = \langle \Phi_1^{(k)}; \Phi_2^{(k)} \rangle_{\mathsf{P}^{(k)}} + \mathbb{E}_{\mathsf{P}^{(k)}} \Phi_{12}^{(k)}$$
$$(1 \le k \le k_1).$$

All of these are expressions for one and the same quantity. At $k = k_0$ the first term on the right is the main term and the second term is an error ((a), (b)). At $k = k_1$ the roles are exchanged, and the law is obtained at that level ((c)). Both $\Phi_i^{(k)}$ and $\Phi_{12}^{(k)}$ are derivatives with respect to the sources. The argument uses conservation of mass alone, $\sum_h \tau_h^w(\mathcal{V}_k) = Z(w)$. In other words, it uses exactness with the sources measured at depth $K-k$, which is the depth s when $k = k_0$ (Proposition 4.3 of (4a), together with (4b)). This is what isolates the scale; no conclusion is drawn from P_{k_0}, the coupling with the bare field. The law of $\mathcal{M}^{k_0}U$ is not identified with the marginal of P in V, because inside what $\mathbf{R}$ heals it replaces the law of the block field. Under P_{k_0}, the equality $V = \mathcal{M}^{k_0}U$ is valid only on bonds at a large distance from the components that $\mathbf{R}$ has acted on. For this reason the sources are carried through the steps, and no observable is conditioned.

(a) *Locality.* The source enters the exponent in three ways. *Pointed pieces:* away from the live components, the coupling term is a sum of functions, each of which depends on a single value of w; this holds because the running shift at a plaquette is a finite sum of functions of this kind. As $f_1 f_2 = 0$, these pieces give no mixed derivative. *Regular local terms:* these are indexed by

$j \le k_0$ and $X \in \mathcal{D}_j$, and they depend on w only on $X^{\sim}$ (4b). Every domain that meets both supports satisfies $d_j(X) \ge c\,\hat{\mathsf{d}}/M$. By Cauchy bounds on the source disc, the contribution of these terms to Φ_{12} is bounded pointwise by $C\hat{\mathsf{d}}^4 e^{-c\kappa\hat{\mathsf{d}}/M}\prod\|f_i\|_\infty$. This is of order $e^{-cp_0^2}$, where, in this step, p_0 without an argument stands for $p_0(g_{k_0})$. It is therefore smaller than any power of $g(\mathsf{d})$. *The part attached to the live components:* only a cruder bound is used for this part. Consider a class $[h]$ in which a live component is in contact with a support, that is, lies within distance $c\,\mathsf{d}$ of it. The label is common to the class, and hence each live component, its lineage and the age of that lineage are common to the class as well; the members of the class differ only in healed components. Let $a(h)$ denote the largest lineage age among such components (Proposition 4.14 of (4d)). On such a class the mixed derivative of this part is bounded by

$$(\log x)^C\Big[e^{-c\,\hat{\mathsf{d}}} + L^{C\check{R}_{k_0}}L^{8a(h)}\Big]\prod\|f_i\|_\infty.$$

It is the mean that is small: $|\mathbb{E}_{\mathsf{P}}\Phi_{12}| \le o(1)\,x^{-2}\prod\|f_i\|_\infty$. This follows from the age series of (4d) (Proposition 4.14 and its proof), taken with rate $b \ge 8$. The growth of the amplitude and the decay of the series are governed by one and the same age, namely the age of the lineage. The age recorded in the label restarts wherever $\mathbf{R}$ leaves a component standing as an island, and it is not larger. The number of cells of level k_0 in contact with the supports is $\le C\hat{\mathsf{d}}^4$, so

$$\mathbb{E}_{\mathsf{P}}\Big[\mathbf{1}\{\text{contact}\}\,L^{8a(h)}\Big] \le C\hat{\mathsf{d}}^4\sum_{a\ge 0}L^{8a}\max_{\mathfrak{c}}\tilde{\nu}\big(\mathfrak{lw}_{k_0}^{\ge a}(\mathfrak{c})\big) \le C\hat{\mathsf{d}}^4 e^{-c\,p_0(g(\mathsf{d}))},$$

and this bound overcomes $(\log x)^C L^{C\check{R}_{k_0}}x^2$ because $\mathfrak{p}_0 \gg r$. *At level k_0 the two-point function draws on this series twice*, both times at a rate $b \ge 8$. At level k_1, in (c), it draws on the series once, at the same rate. The first use is the present one, the bound on the mean of the live part's mixed derivative. The second occurs in (b), in the covariances of the live first derivatives; there the product of two such derivatives carries $L^{4a}\cdot L^{4a}$. At level k_0 these two uses are the source of the weight L^{8a} for two points. At level k_1 its source is (c1). Of the growth, only one property enters: as a function of the age, it is no faster than geometric. A different exponent, or an additional factor polynomial in a, is handled by a larger rate b, that is, by a larger c_a, a constant that may be increased freely. Hence the theorem on the ages (Proposition 4.14) underlies Theorem 0 (c)(3) and (c)(4). It also underlies (c)(5), for which (7d) employs this series with a larger rate.

(b) *The marginal part.* Here the coupling carries the source. Let $h_y^{(j)} \ge 0$, with $\sum_y h_y^{(j)} = 1$, be the tent functions of (4a), (4b). These weights have range CMa_{K-j} and serve to localise the counterterm of step j. They are written h_z in (0b) and in Step 1, in which the level is held fixed. Let p_y be a bare plaquette at y. At points where every profile φ_j, $j \le k_0$, takes the value 1, that is, off the live components, the running shift satisfies, by its formula in (4b),

$$\partial_{z_i}\zeta_{k_0}(p;w)\big|_0 = f_i(x(p)) + \sum_{j\le k_0}b_j'(0)\sum_{y\in\mathbb{T}^{(j)}}h_y^{(j)}(p)\,f_i(x(p_y)) =: \zeta_{k_0}'\tilde{f}_i(p),$$

$$|\tilde{f}_i - f_i| \le \tfrac{CM}{\vartheta\hat{\mathsf{d}}}\sum_j|b_j'(0)|\,\|f_i\|_\infty;$$

the factor $\zeta_K(\mathsf{d})^2$ arises here. Differentiating $-A(x_{k_0}(\cdot) - \zeta_{k_0}(\cdot\,;w), U_{k_0})$ gives

$$\Phi_i = \text{const} + \zeta_{k_0}'X_i + (\text{irrelevant}) + (\text{live}) + (\text{far}),$$

$$X_i(V) := \sum_p \tilde{f}_i(p)\, \chi_{k_0}(\square(p))\, a_p\big(U_{k_0,\square(p)}(V)\big).$$

In this formula $\square(p)$ is the test cube containing $x(p)$, and $U_{k_0,\square(p)}$ is the background localised about that cube, hence near $\operatorname{supp} f_i$. The test $\chi_{k_0}(\square(p))$ acts as a window: a plaquette contributes only if its localised background passes the plaquette test. Consequently X_i depends on V only near the support and is the same for every history. It also satisfies $|X_i| \le C x^{-1} p_0^2\, \hat{\mathsf{d}}^4 \|f_i\|_\infty$. Away from the large-field regions, the global background differs from the localised one by an amount exponentially small in the radius of localisation; this difference is the term (far). The size of the irrelevant terms is $\le C(\log x)^{3\mathfrak{p}_0-q} x^{-1} \hat{\mathsf{d}}^4 \|f_i\|_\infty$. Per cube they are of size $\alpha^2(\varepsilon_{k_0}/\alpha)^3$, where α is of the order of the analyticity radii $\alpha_{0,k_0}, \alpha_{1,k_0}$. This is a third-order Taylor remainder multiplied by the magnitude that the source term (w times a_p) attains when the fields range over their analyticity domain. The factor x^{-1} cannot be dispensed with. The law uses $p_0^2\, \hat{\mathsf{d}}^8 (\log x)^{3\mathfrak{p}_0-q} \to 0$ as the coefficient of x^{-2}, and Cauchy bounds over the source radius ϱ yield no power of x. The factor comes from the bound on the regular family produced by a step ((2e), "Dependence of the regular family on the array"). That bound also holds for shifts of the coupling that vary with position, and it weights a difference of shifts by $CM^4\alpha_{0,k}^2$. We apply it to the sourced and the unsourced arrays of (4b). The remainders enter through the slope towards the constants carried by the induction of (4b), and they are small because of their size. The result is a linear recursion for the slopes in the source, and the small factor in front of the bound closes this recursion over the levels. Its solution is $\approx x_j^{-1} (\log x_j)^{\mathfrak{p}_0+q}$, and this takes the place of ϱ^{-1}. The oscillating part of the source, namely what remains after subtracting from the source its value at a single point of the domain, has a slope bounded in the same way, with an extra factor $L^{-(k_0-j)}$. The live terms become larger as the age increases. If a live component has $\mathrm{age}^{\mathrm{lab}} = a$, the fields stored in it are those of every level from $k_0 - a$ onwards, together with those of the next finer level. For a component holding an island, the block of which retains the fields of its core (2b), the stored fields begin at the first level of its lineage. Over one cube of level k_0, the plaquette sums of these fields contain at most $L^{4(a+1)}$ terms. For the live terms, only their first two moments on the events of contact enter, so that, as in (a), it suffices that their growth be geometric in the age. Let h be a history in which a live component is in contact with $\operatorname{supp} f_i$. On such a history the live contribution to Φ_i is bounded by $(\log x)^C \big[e^{-c\,\hat{\mathsf{d}}} + L^{C\check{R}_{k_0}(a(h)+2)^C L^{4a(h)}}\big] \|f_i\|_\infty$. The power of the age in this bound is the one of (7d). In the bound $Ce^{-cp_0(g_j)}(2L^b)^{-a}$ of Proposition 4.14, the factor 2^{-a} absorbs this power, so the age series is used at the same rate. This live contribution has two covariances. One is with the live contribution of the other factor, with weight $L^{8a(h)}$. The other is with those functions of V that occur in the other factor, with weight $L^{4a(h)}$, taken against their supremum bound. By the age series of (4d), as in (a), both of these covariances are $o(1)\, x^{-2} \prod \|f_i\|_\infty$.

(c) *Gaussian extraction: the free template.* Denote by $\mathcal{H}_j B$ the exact bare 2-cochain of smallest norm that satisfies $Q_j^{\mathrm{pl}} \mathcal{H}_j B = dB$. It is the curvature of the linear minimiser, namely $d\,\mathsf{H}_j$ at $U = \mathbf{1}$, a point at which $J = 0$ and the three Green's functions of (1c) agree. We then have

$$\mathcal{C}_{(j)} = \mathcal{H}_{j-1} C_j^{\mathrm{tr}} \mathcal{H}_{j-1}^*,$$

$$|\mathcal{H}_j(p,b)| \le C\eta_j^2 e^{-\delta|x(p)-x_b|/a_{K-j}},$$

$$|\mathcal{C}_{(j)}(p,q)| \le C\eta_{j-1}^4 e^{-\delta|x(p)-x(q)|/a_{K-j+1}}.$$

The decay of $\mathcal{H}_j$ comes from Bałaban's work, apart from the passage to centred blocks and to level-j units described below. In that work the linear minimiser is given by $H_k = GQ^*(QGQ^*)^{-1}$ (Comm. Math. Phys. 95 (1984) 17, (1.103); at $U = \mathbf{1}$ the straight average coincides with the

Q_k of that paper and of 96 (1984) 223), and both G and ∇G decay exponentially, at a rate that depends on the dimension alone (95, Proposition 1.2 [cited from Bałaban; Package A–H], (1.110)). In 95 each block has y as one of its corners ((1.6) there); from 109 onwards Bałaban's blocks are centred at y ((0.3) there), like the blocks of the present construction. For centred blocks the same bounds follow from three devices, once one passes to level-j units, and this passage is the origin of the factor η_j^2. (i) A spectral gap that is uniform in the fine spacing η_j, equivalently in the number L^{4j} of fine sites in a block: the operator built from the mass of the averages (which depends on the block means alone), the Landau gauge term (a non-local projection) and the curl form is $\geq c(-\Delta+1)$. (ii) A Combes-Thomas conjugation with the exponential weights $e^{\delta|\cdot - x_b|}$, which yields the decay of the minimiser only in weighted L^2. (iii) A bootstrap that upgrades control in weighted L^2 to control of the first differences in weighted sup norms, as the pointwise bound on $\mathcal{H}_j$ requires. Uniformly in η_j, the kernel of $(-\Delta+1)^{-1}$ and its first differences belong to weighted L^p, the former for $p<2$ and the latter for $p<\frac{4}{3}$ (in four dimensions the singularities are $|z|^{-2}$ and $|z|^{-3}$). Young's inequality is then applied twice, $L^2 \to L^8 \to L^\infty$. What results is elliptic regularity on the lattice with bounds independent of the fine spacing. The symbol C_j^{tr} denotes the single-step covariance in a tree gauge,

$$C_j^{\mathrm{tr}} := \iota \left(\iota^* \mathcal{H}_{j-1}^* \mathcal{H}_{j-1}\, \iota\right)^{-1} \iota^*,$$
$$d'\, C_j^{\mathrm{tr}}\, d'^* = C_j^{\mathrm{curv}},$$

where ι embeds into the 1-forms B of level $j-1$ the subspace of those that satisfy $B=0$ on one contour for each site (a tree in each block, imposed sharply) together with $Q^{\mathrm{st}}B=0$. In this formula Q^{st} denotes the straight average over the blocks of the single step from level $j-1$ to level j, at $U=\mathbf{1}$. It is thus the covariance, on that subspace, of the quadratic form $\langle B, \mathcal{H}_{j-1}^*\mathcal{H}_{j-1}B\rangle = \|\mathcal{H}_{j-1}B\|^2$, which equals twice the free action of the minimiser with datum B. On the subspace one has $\|\mathcal{H}_{j-1}B\|^2 \geq c\|d'B\|^2 \geq c_L\|B\|^2$. It differs from the covariance $C^{(j-1)}$ of (1c)-(1d) at the trivial background in two ways. The gauge differs: there it is imposed softly, on the means over all contours. The constraint also differs: the straight average is not the linearisation of $\mathcal{M}$. The two agree in the curvature covariance $d'\cdot d'^*$, where d' is the coboundary of level $j-1$. Indeed, given its block averages, the level-$(j-1)$ curvature $Q_{j-1}^{\mathrm{pl}}F$ has conditional covariance C_j^{curv}. Since $\mathcal{H}_{j-1}B$ depends on B only through $d'B$, it determines $\mathcal{C}_{(j)}$, and a_p depends on nothing else. Summing over $j \leq k_0$, where the coarsest slice gives the dominant contribution, one obtains the following bound. Here $t = |x(p)-x(q)|/a_s \geq \frac{1}{2}C_W(s)$, and $\delta_{\mathrm{fl}}>0$ is a rate formed from the rates δ of the slices, depending only on L and the dimension.

$$\left|C_{K,s}^{\mathrm{fl}}(p,q)\right| \leq C\left(\tfrac{a_K}{a_s}\right)^4 e^{-\delta_{\mathrm{fl}}t} = C\pi^2 t^4 e^{-\delta_{\mathrm{fl}}t}\,\sigma_{pq}:$$

the fluctuation part is small because the curvature of the linear minimiser decays. Because $t^4e^{-\delta_{\mathrm{fl}}t}$ is decreasing only for $t \geq 4/\delta_{\mathrm{fl}}$, the room is given the floor $C_{W,0}A_0^2 \geq 8/\delta_{\mathrm{fl}}$. Over $j>k_0$ the sum of the exponentials reconstitutes the power law: $\sum_i L^{-4i}e^{-tL^{-i}} \leq Ct^{-4}$.

The interacting measure. We apply the formula of the Strategy at level k_1, whose backgrounds are flat on the scale of the pair. Only the classes $[h]$ occur, and no finer class of histories: the sourced exponent of level k_1 is the one that (4b) provides. (c1) *The first member is an error term.* Consider the terms of the levels $k_0<j\leq k_1$ and a source that is rough on their scale. Their slopes in such a source remain $\leq Cx^{-1}(\log x)^{\mathfrak{p}_0+q}$, measured against $\|f_i\|_\infty$ (the norm $\|f_i\|_{\sharp,K-j}$ of the level of the term can exceed it by a factor of up to L^{j-k_0}). The recursion for these slopes closes since $\mathsf{n} \leq C\log x$ (its size is set through $\bar{x}_s \leq Cx$). Hence $\mathsf{n}^2(\log x)^{\mathfrak{p}_0-q} \leq C(\log x)^{2+\mathfrak{p}_0-q}$ is small ($q-\mathfrak{p}_0>2$, g small). Suppose first that no large-field region touches the supports at level k_1. Then

$|\Phi_i^{(k_1)} - c_i| \le o(1)\, x^{-1} \|f_i\|_\infty$ for a constant c_i, where $o(1)$ refers to $\mathsf{d} \to 0$, uniformly in the datum. In the other case $|\Phi_i^{(k_1)} - c_i|$ grows no faster than $L^{\mathsf{n}} L^{4a} \|f_i\|_\infty$ in the age a of the touching region. The age series of (4d), taken at the single level k_1 with $b \ge 8$, beats this growth: $L^{\mathsf{n}} \le L \bar{x}_s^2$ is a power of x, while the series lies below every power. Consequently $|\langle \Phi_1^{(k_1)}; \Phi_2^{(k_1)} \rangle_{\mathsf{P}^{(k_1)}}| \le o(1)\, x^{-2} \prod \|f_i\|_\infty$. (c2) *The second member becomes one number.* The quantity $\Phi_{12}^{(k_1)}$ is the sum of two parts. The first is the mixed derivative of the sum $\sum_{j \le k_1} \sum_X e_j(X; w)$ of localised source energies from (4c). The second is a part that depends on the fields and is bounded only crudely, by a quantity of order $L^{2\mathsf{n}}$ for each bond of the determining set; by (4c), the size of that set is counted by the unit volumes of level k_1 together with the sites $\sum_{j \le k_1} \#_j(Z_j)$ of the large-field regions (every source derivative costs a factor L^{n}). We translate the pair by the points v of a grid $\mathfrak{P} \subset \Lambda_{s,\ell}$ whose spacing is comparable to d, and we write $f_i^v := f_i(\cdot - v)$ for the translated sources. The left-hand side of the formula is unchanged by this. The sum over v of the mixed derivatives equals the mixed derivative evaluated at the two array sources $\sum_v \epsilon_v f_i^v$, whose norms are those of f_i, averaged over the signs $\epsilon \in \{\pm 1\}^{\mathfrak{P}}$ (this is the device of Lemma 7.6 [outlined here; Section 44.3] of (7e)). The crude bound is therefore divided by the number $\approx L^{4\mathsf{n}} \hat{\mathsf{d}}^{-4}$ of pairs contained in a unit volume of level k_1. What survives is $L^{-2\mathsf{n}} \hat{\mathsf{d}}^4 \le \bar{x}_s^{-4} \hat{\mathsf{d}}^4$, and this requirement fixes n. There remains the number

$$c_{12} := |\mathfrak{P}|^{-1} \sum_{v \in \mathfrak{P}} \partial_{z_1} \partial_{z_2} \sum_{j \le k_1} \sum_X e_j(X; z_1 f_1^v + z_2 f_2^v)|_0$$

. (c3) *The number is evaluated.* The number c_{12} is the value at a single point of the mean of $\Phi_{12}^{(k_1)}$ over $\mathfrak{P}$, namely of the term without large-field regions, taken at the trivial block field. It satisfies an exact single-step recursion at the flat background, leading from level k to level $k+1$ for $k_0 \le k < k_1$. The measures there are positive, so a remainder is carried through a step by its supremum alone. For each step, two inequalities on a single fluctuation integral (1d) are then required. That integral belongs to the small-field step taken over the whole lattice, with a background that is real and regular and with coupling $x_j \approx x$. The first inequality concerns the covariance of two sums of background plaquette energies. The second concerns the mean value of a bilinear form whose arguments are the background plaquette angles. The expansion of (1d), together with Cauchy bounds on a large polydisc, removes the interaction. A Gaussian computation remains, and Taylor's formula compares it with the flat background. At the flat background, the object that agrees with the free template is the curvature covariance C_j^{curv} introduced above. It determines $\mathcal{C}_{(j)}$, and a_p depends on nothing else. Summed over $k_0 < j \le k_1$, the $\mathcal{C}_{(j)}$ give $C_{K,s}^{\mathrm{bg}}$ minus $C_{K,s-\mathsf{n}}^{\mathrm{bg}}$, and the subtracted term is $\le C L^{-4k_1}$. Three facts are used here. The source arrives at level k_0 in the form $\zeta'_{k_0} \tilde{f}_i$ ((b)). Next, $a_p \approx \frac{1}{2}|F(p)|^2$, where $F(p)$, an element of the Lie algebra, is the plaquette angle at p of $U_{k_0,\square}(V)$ (the same angle is written Y_p in (7d)). Finally, half-squares of Gaussian variables have covariance $\frac{1}{2}(\text{covariance})^2$ for each colour. We obtain

$$\Big| \langle \mathcal{O}(f_1); \mathcal{O}(f_2) \rangle - \zeta'^2_{k_0} \tfrac{N^2-1}{2}\, x^{-2}\, \mathcal{K}_{K,s}(\tilde{f}_1, \tilde{f}_2) \Big| \le o(1)\, x^{-2} \prod \|f_i\|_\infty .$$

The formula at level k_0, combined with (a) and (b), gives the same estimate for $\zeta'^2_{k_0} \langle X_1; X_2 \rangle_{\mathsf{P}}$ in the average over $\mathfrak{P}$. *Positions.* The cumulants are invariant under the translations of $\Lambda_{K,\ell}$. A translation $v \in \Lambda_{s,\ell}$ carries blocks onto blocks, which finer translations do not do. Hence

$$C_{K,s}^{\mathrm{bg}}(p + v, q + v) = C_{K,s}^{\mathrm{bg}}(p, q),$$

$$\mathcal{K}_{K,s}(f,h)=\frac{1}{|\Lambda_{s,\ell}|}\sum_{v\in\Lambda_{s,\ell}}\mathcal{K}_{K,s}\big(f(\cdot-v),h(\cdot-v)\big)\,:$$

since both sides of the law are invariant, it is enough to make the comparison in the average over such positions.

(d) *Wild histories.* For two points, the relevant event is that a large-field region touches the supports at the single level k_1 ((c1)). The weight of these histories is $\le C(M\check R_{k_1})^4 e^{-c\,p_0(g_{k_1})}$, and no union over the levels is taken. In this bound the prefactor counts the sites of level k_1 in one cell of that level, and $O(1)$ cells touch the supports. For four points (7d), the unclean histories are the ones that matter. At each level j of the segment of (7d), $k_0-O(L\check R_{k_0})\le j\le k_0+r_4$, the number of cells of level j near the supports, that is, within $C\hat{\mathsf d}$ cells of level k_0, is $\le C\hat{\mathsf d}^4L^{4\max\{k_0-j,0\}}$; altogether there are $\le e^{CL\check R_{k_0}}$ such cells. A component on which $\mathbf{R}$ acts at a level j of the segment contains a component that is live at level $j-1$. The one-cell bounds of (4d) give $\tilde\nu(\mathfrak{c}\subset\mathcal{Z}_j)\le Ce^{-c\,p_0(g_j)}$, and the same bound for creation at step j. Moreover $p_0(g_j)\ge c\,p_0(g(\mathsf d))$, because $x_j\ge x^{1/2}$. Hence $\tilde\nu(\text{unclean})\le e^{CL\check R_{k_0}-c\,p_0(g(\mathsf d))}$, and this is small because $\mathfrak{p}_0\gg r$. Even here these one-cell bounds must hold uniformly in the volume (prolonged run, translation covariance up to the level, first moment of the prepaid end mark: (4d)). The reason is that $T_{\ell_\star}$ contains $(\ell_\star/Ma_s)^4$ cubes of level k_0, and this number is unbounded.

What must be overcome. The natural size of a factor is $x^{-1}\|f_i\|_\infty$. The bound on X_i is larger by $p_0^2\,\hat{\mathsf d}^4=(\log x)^{C\mathfrak{p}_0}$, and the irrelevant terms are controlled by sup bounds only. This is the reason for $q\gg\mathfrak{p}_0$ and for the super-polynomial smallness in (a), (d). The effective action is that of Bałaban (Comm. Math. Phys. 109 (1987) 249; 122 (1989) 175, 355); the sources and the extraction are added to it.

The setting and the objects. In (7a) the two-point law is stated for two bumps f,h whose size and mutual separation are both $\approx\mathsf d$. They are placed on a lattice of spacing $a_K\ll\mathsf d$ inside a volume of side $\ell\gg\mathsf d$. Isolating the scale $\mathsf d$ has two aspects. *From below*: $K-s(\mathsf d)$ renormalisation steps integrate out the scales between a_K and $\mathsf d$. In the leading term these scales survive only through two numbers of the run: $g(\mathsf d)$, the coupling, and $\zeta_K(\mathsf d)$, the normalisation. *From above*: comparing T_ℓ with a single fixed torus eliminates the scales from $\mathsf d$ up to ℓ, together with the volume itself. This comparison costs $O(\mathsf d^2)$, which is of lower order than the main term $\approx(\ln\frac{1}{\mathsf d})^{-2}$. This part defines the two numbers, bounds them uniformly in the cutoff and in the bare coupling, and carries out the comparison of volumes. The Gaussian main term is extracted in (7a), at the level $K-s(\mathsf d)$ on the fixed torus. There the cumulant is written in terms of a law on resummed histories and block fields (Lemmas 7a.1.8 [to be proved; Package F], 7a.1.9). Resummed histories are classes of histories that become equal once their entries inside healed components are deleted. The large-field operation $\mathbf{R}$ sums over the outcomes inside the components it heals, so an exponent is attached to such a class rather than to an individual history; the class is the index $\mathsf h$ of Remark 7b.11′ (ii).

(7c) The law passed to the limits. **Corollary 7.4 [Theorem 0(c)(4)].** Let S be a continuum limit, either on T_ℓ of (5b) or on $\mathbb{R}^4$ of (5c). Let $\mathsf d\le\mathsf d_0$, and let $(f,h)\in\mathfrak{T}_{\mathbb{E}}(\mathsf d;\vartheta)$ be a pair of class C^1. Then the conclusion of Theorem 7.1(ii) is valid with

$$(1-\bar\varepsilon)\big(1-\bar\eta(\mathsf d)\big)^4\le 1+\eta\le(1+\bar\varepsilon)\big(1+\bar\eta(\mathsf d)\big)^4,$$

in which $\bar\eta$ and $\bar\varepsilon_\ell$ denote the functions of Theorem 7.1. Their values lie in $[0,\frac13]$ and $[0,\frac12]$ respectively, and both tend to 0 with $\mathsf d$. The quantity $\bar\eta(\mathsf d)$ is a common bound for $|\zeta_K(\mathsf d)-1|$ and for $|\frac{11N^2}{12\pi^2}g(\mathsf d)^2\ln\frac{1}{\mathsf d}-1|$. On T_ℓ one has $\bar\varepsilon=\bar\varepsilon_\ell(\mathsf d)$, and on $\mathbb{R}^4$ one has $\bar\varepsilon=\frac12$. Consequently $S_2(f,h)>0$, and on $\mathbb{R}^4$ we have $\dim\mathcal{H}(S)\ge2$.

Argument. The argument follows from the uniformity in Theorem 7.1(i). In the lattice law, the only quantities that depend on (K,ℓ,β) are $\zeta_K(\mathsf{d})=1+\eta''$, $g(\mathsf{d})^2\propto(1+\eta')/\ln(1/\mathsf{d})$ and ε. Therefore

$$\frac{S^T_{2;K,\ell,\beta}(f,h)\,(\ln(1/\mathsf{d}))^2}{\frac{3(N^2-1)}{\pi^4}(\frac{12\pi^2}{11N^2})^2\mathcal{K}_{\mathrm{free}}(f,h)}=(1+\eta'')^2(1+\eta')^2(1+\varepsilon)$$

takes its value in the closed interval of the display above, for every datum. For fixed d, we have $K_j-s(\mathsf{d})\ge n_0$ once j is large. Every pair on $\mathbb{R}^4$ is contained in some ball of radius $5\mathsf{d}$. Hence, once transported to T_{ℓ_j}, it belongs to the class and keeps the same free kernel. A closed condition is preserved in the limit. The bound on ε that clause (c)(4) records on $\mathbb{R}^4$ is the only one in Theorem 7.1(i) that is free of any side, namely $|\varepsilon|\le\frac12$.

The Hilbert space. Let f_{d} be a C^1 bump whose support is the ball $B(2\mathsf{d}e_4,\mathsf{d})$. Then $(\theta f_{\mathsf{d}},f_{\mathsf{d}})$ is a test pair. Its shape ϑ_1 is fixed and does not depend on d, because the bumps are dilates of a single profile. When $\vartheta>\vartheta_1$, the pair may fail to belong to $\mathfrak{T}_{\mathbb{R}^4}(\mathsf{d};\vartheta)$. We take the ceiling $\gamma_W(\vartheta)$ smaller than the ceiling under which Theorem 7.1(i) is valid at the shape ϑ_1. That theorem is applied there with the witness depth $s_1(\cdot)$, which is built from the room constant $C_{W,0}$ of that shape, and with the thresholds of the same shape. Once d is below its threshold, the three inequalities $\zeta'_{K-s_1(\mathsf{d})}\ge\frac23$, $1+\varepsilon\ge\frac12$ and $g^{-2}_{K-s_1(\mathsf{d})}\le\bar{x}_{s_1(\mathsf{d})}$ bound $S^T_{2;K_j,\ell_j,\beta_j}(\theta f_{\mathsf{d}},f_{\mathsf{d}})$ from below by a positive number that does not depend on j. Consequently (6d) gives

$$\big\|[(f_{\mathsf{d}})]\big\|^2=\sigma_S(\theta f_{\mathsf{d}},f_{\mathsf{d}})=S_2(\theta f_{\mathsf{d}},f_{\mathsf{d}})>0,$$
$$\langle\Omega,[(f_{\mathsf{d}})]\rangle=\sigma_S((f_{\mathsf{d}}))=0.$$

Periodic Gibbs states. The expansion concerns torus measures. The law is carried over to these states by Lemma 7.6. That lemma relies on Lemma 4.10 [proved here; Section 38] (4d), according to which the chessboard dominates the periodic Gibbs states. Precisely: let F be a function of a single board cell of side $\mathsf{l}=O(1)$, and on T_ℓ let the chessboard seminorm be denoted $\|\cdot\|_{\mathrm{ch},\ell}$. Then, along any compatible sequence of tori (ST3),

$$|\nu(F)|\le\liminf_{\ell\to\infty}\|F\|_{\mathrm{ch},\ell}.$$

This is a separate lemma. It is not the chessboard estimate, whose subject is torus measures. Its proof, taken in the form that involves all mirror copies, combines four ingredients: the reflections of the cells leave ν invariant (10a); Jensen's inequality is used together with the Gibbs property; the boundary condition is changed at the cost of a surface term, which is weighed against a volume; and finally the seminorm enters.

Ergodic components. At (K_j,β_j), each periodic Gibbs state obeys the law. A closed condition that all of these states satisfy is inherited by the components (10c). This yields Theorem 0'(3)(ii): for a volume-first limit, each of its ergodic components satisfies the law in the form it takes on $\mathbb{R}^4$, with d_0 and $\bar\eta$ unchanged.

43.1. **The conditional measure in a clean fluctuation box.** We explain the conditional measure used in the Gaussian comparison. Put $k_0=K-s$, $x=x_{k_0}$ and $k_1=k_0+n$, where $n\asymp\log x$. On this segment $x_j\asymp x$. The supports are separated by r spacings of level k_0, with $\mathsf{r}\asymp(\log x)^{2\mathfrak{p}_0}$. At step j choose a read set S_j containing every free bond read by the two insertions, and a box I_j with $d(S_j,I_j^c)\ge D$. Both insertions lie in the same box, and

$$N_j:=\#I_j\le C(\mathsf{r}L^{-(j-1-k_0)}+D)^4\le C(\mathsf{r}+D)^4.$$

The box is contained in the tame region throughout the segment. Let $\mathcal{N}$ be the neighbourhood of the two supports of radius $C(\mathsf{r}+\check{R}_{k_0})a_{K-k_1}$, enlarged to contain the boxes and all their determining bonds. It is chosen not to wrap. Cleanliness means that the live region misses $\mathcal{N}$ at every level in the finite window

$$\mathcal{I}_{\mathrm{cl}} = \{t : \max(0, k_0 - C\check{R}_{k_0}) \le t \le k_1\}.$$

A new creation is live at its creation level. A component acted on after its wait contains a component live at the preceding level. Thus this condition excludes creations and large-field operations near the boxes on the segment, including operations hidden by a merge. The waiting factor is $\Upsilon = L^{C\check{R}_{k_0}} = \exp[O((\log x)^r)]$.

Here we use the convention in which healing redraws every lower small-field profile and compensates its change separately at each plaquette. Thus the sum of the Wilson weights is still $-\beta$ at each bare plaquette. A healed component leaves its deterministic activity $R^{(i)}$; the discrepancy $x_j(\cdot) - x_j$ is supported on the live region. It consequently vanishes in a clean box. This convention is needed below: a bound on the average lag of individual histories would not be a property of a resummed clean class.

The histories of a class are summed before taking a logarithm. The following comparison is made with the same standing part, stored variables and exterior fields on both sides. Mark the cubes near the clean neighbourhood with a factor e^u. For a fixed cutoff the identity of measures for one step is a polynomial identity in e^u. Its coefficient of degree zero deletes precisely the inner histories forbidden by cleanliness. The pointwise bracket estimate gives

$$d\tau_{\mathrm{cl}} = e^{-\Delta_j} d\tau_{\mathrm{res}}, \qquad 0 \le \Delta_j \le \epsilon_j, \qquad \epsilon_j \le C \sum_{t \le j} c_t e^{-(1+c)p_0(g_t)}.$$

Here the equality is on the common space of reference and stored variables, before projection. Set $c_t = 0$ for $t \le \min \mathcal{I}_{\mathrm{cl}}$. For $\min \mathcal{I}_{\mathrm{cl}} < t \le k_1$, c_t counts the cubes of level t within $(200\check{R}_t + 1)M$ level-t units of $\mathcal{N}$. It is bounded by $C(\mathsf{r} + \check{R}_{k_0})^4 L^{4(k_1 - t)}$. Consequently the accumulated flag error is bounded by $CL^{4n}\Upsilon(1+n)(\mathsf{r} + \check{R}_{k_0})^4 e^{-cp_0(g)}$. No sum over all earlier ultraviolet levels is taken. It is the pointwise bracket estimate from the large-field construction that justifies this display. A bound on integrated masses alone would not justify it. Positivity preserves its two-sided comparison under all subsequent operations. The conditional expectation at each clean step is therefore a reference conditional expectation with a normalized weight in $[e^{-\epsilon_{j-1}}, e^{\epsilon_{j-1}}]$.

Their product dominates the reference segment kernel with weight

$$P_0 = \exp\Big(- \sum_{k_0 < j \le k_1} \epsilon_{j-1} \Big).$$

Splitting off this common subkernel refines the resolved coupling by one Bernoulli flag. On its reference branch no comparison weight remains. The other branch has mass at most $n\epsilon_{k_1}$, in addition to the mass of histories failing cleanliness. Both masses are included in the unclean error. This construction uses the actual resolved coupling of the run; an arbitrary disintegration having the right first marginal would give no such conclusion.

The fluctuation law with an insertion. At one tame step the shift to the critical background, the linearization of the constraint, pivot elimination and rescaling give, with a common prefactor $\mathfrak{N}(W)$,

$$\begin{aligned} \int_{\mathcal{M}(V)=W} & F(V)\chi(V) e^{-x_{j-1}[A(U_{j-1}(V)) + \mathrm{GF}_{j-1}(V)]} \\ & \times e^{\mathbf{E}_{j-1}(U_{j-1}(V))}\, dV \\ & = \mathfrak{N}(W) \int_{\mathcal{W}_j} F(\mathfrak{v}(y;W)) e^{P_j(y;W)} d\gamma_j(y), \end{aligned}$$

where

$$\mathfrak{v}(y;W)=e^{i[g_{j-1}\mathsf{C}y-\widetilde{\mathsf{D}}(g_{j-1}\mathsf{C}y)]}V_{j-1}^{\mathrm{bg}}(W),\qquad \mathcal{W}_j=\prod_{b\ \text{free}}\{|y_b|<\mathsf{b}_j\},\qquad \mathsf{b}_j\asymp p_0(g_{j-1}).$$

The map $\widetilde{\mathsf{D}}$ starts at degree two and changes only pivot variables. Hence the window remains a product on the free variables. The Gaussian γ_j has covariance $C^{(j-1)}(W)$. Its exponent P_j contains the cubic action and gauge terms, the Jacobian and Haar density, and the increment of the old activities. All transformations retain the factor F unchanged. For gauge-invariant F, inserting or removing the gauge fixing also retains its integral. Dividing by the formula for $F=1$ identifies the normalized conditional expectation. Applied to a clean class, this identity uses the common reference branch just constructed. Operations outside the box and merges replace the exterior datum by a mixture; their weights remain nonnegative. □

For completeness we give the estimate that removes that exterior. It concerns the density on the read set, so it also applies to bounded insertions without estimating their derivatives.

Lemma 43.1. *Let I be finite, $\mathcal{W}=\prod_{b\in I}K_b$, with K_b convex bounded neighbourhoods of zero, and put*

$$H_a(y)=\tfrac12\langle y,A_ay\rangle+\langle l_a,y\rangle-P_a(y),\qquad \nu_a=Z_a^{-1}\mathbf{1}_{\mathcal{W}}e^{-H_a}dy.$$

For multiplication by $e^{\varphi(b)}$ write T_φ. Suppose

$$\langle v,T_\varphi A_0T_\varphi^{-1}v\rangle\ge\lambda_A|v|^2,\qquad \sup_{\mathcal{W}}\|T_\varphi P_0''T_\varphi^{-1}\|\le\varepsilon_0<\lambda_A,\qquad \lambda=\lambda_A-\varepsilon_0.$$

If $|\partial_b(H_1-H_0)|\le\delta_b$ and $|G(y)-G(y')|\le\ell_G|T_\varphi(y-y')|$, then

$$|\nu_1G-\nu_0G|\le\frac{\ell_G}{\lambda}\Big(\sum_b e^{2\varphi(b)}\delta_b^2\Big)^{1/2}.$$

No convexity assumption on H_1 is needed. Suppose the same coercivity bounds hold for every φ with $\operatorname{Lip}\varphi\le\mu$. For $S\subset I$, $R=I\setminus S$, let m_a be the marginal density on $\prod_{s\in S}K_s$. Then

$$e^{-\Xi}\le\frac{m_1}{m_0}\le e^{\Xi},\qquad \Xi=\sum_{s\in S}\operatorname{diam}(K_s)\left(\delta_s+\frac{L_sD_s}{\lambda}\right),$$

where

$$L_s^2=\sup_{\mathcal{W}}\sum_{b\in R}e^{2\mu d(b,s)}\|(A_0-P_0'')(b,s)\|^2,\qquad D_s^2=\sum_{b\in R}e^{-2\mu d(b,s)}\delta_b^2.$$

Proof. Shrink each K_b slightly, extend P_a smoothly by a retraction that is the identity on the smaller product, and approximate the walls by smooth convex one-bond potentials. Their Hessians commute with T_φ and are nonnegative. Choosing the walls sufficiently steep preserves the displayed coercivity for the extended H_0. The wall terms cancel in $\nabla(H_1-H_0)$. These approximations have integrable Gaussian majorants, so the walls and then the shrinking can be removed by dominated convergence.

For the smooth problem couple the Hamiltonian flows of H_0,H_1 using the same initial momentum, refreshing that momentum after each interval of length t. Each marginal Gibbs law is preserved. For $v=T_\varphi(q_1-q_0)$ the difference equation is

$$\ddot v=-M(s)v-T_\varphi\nabla(H_1-H_0)(q_1),\qquad \dot v(0)=0,\qquad \langle v,M(s)v\rangle\ge\lambda|v|^2.$$

If $\|M\|\le\Lambda$, Taylor's formula bounds its homogeneous contraction by $1-\lambda t^2/2+\Lambda^2t^4$. Iterating and then letting $t\downarrow0$ gives $\mathbb{E}|v|\le\nu_1(|T_\varphi\nabla(H_1-H_0)|)/\lambda$. The Lipschitz bound proves the first assertion. The order of these limits removes Λ before the walls are removed.

For the second assertion fix the variables on S. The remaining integration domain is independent of them because the window is a product. Differentiate its marginal density:

$$\partial_s \log(m_1/m_0) = -\nu_1^u(\partial_s(H_1 - H_0)) - [\nu_1^u(\partial_s H_0) - \nu_0^u(\partial_s H_0)].$$

Use the first assertion on R with $\varphi(b) = -\mu d(b,s)$ to bound the bracket by $L_s D_s/\lambda$. Integration along segments bounds the oscillation by Ξ. Both densities integrate to one, which fixes their ratio between $e^{-\Xi}$ and e^{Ξ}. □

43.2. **The four exterior forces.** Freeze an exterior datum e for step j, including its stored fields, and write $O = I_j^c$ for the remaining free bonds. On the reference branch the conditional density on $I = I_j$ is proportional to $\mathbf{1}_{\mathcal{W}_I} e^{-H_e}$, where

$$H_e(y) = \tfrac{1}{2}\langle y, Q_{II}^e y\rangle + \langle Q_{IO}^e y_O(e), y\rangle - P_e(y).$$

Choose e_0 with the same new field near the box, set $\Gamma = (Q^{e_0})^{-1}$, and let P_0 be the exponent at e_0, normalized at zero. The reference law on the box is the full Gaussian marginal tilted by e^{P_0}, not the Gaussian obtained by simply discarding its exterior. Its precision is

$$K = (\Gamma_{II})^{-1} = Q_{II}^{e_0} - Q^{\mathrm{sc}}, \qquad Q^{\mathrm{sc}} = Q_{IO}^{e_0}(Q_{OO}^{e_0})^{-1} Q_{OI}^{e_0}.$$

Consequently its discrepancy from H_e has exactly four forces:

$$\nabla(H_e - H_0) = Q^{\mathrm{sc}} y + Q_{IO}^e y_O + (Q_{II}^e - Q_{II}^{e_0}) y - \nabla(P_e - P_0).$$

Uniform positivity and decay of the actual quadratic forms and their inverses, the bound $|y_O| \le C\mathsf{b}_j$, and locality of the background bound each force at b by $C\mathsf{b}_j e^{-cd(b,O)}$. For the last force one uses the localized form of the old activities and boundary terms. A term whose domain reaches O has size at least $c\, d(b,O)/M$; half its decay pays for that distance and the other half sums the domains. For a domain contained in I, only the background changes, and its difference is exponentially small in the distance to O. The same argument applies to the actual localized quadratic form. All these statements require the common cores for the entire interpolation and uniform bounds on its exterior data.

The old-activity subtraction is essential for the small Hessian of P_0. Before summation the second-order part is marginal. After the coefficient under each tent is subtracted, it has the factor $L^{-(4+\alpha)(j-1-i)}$ at level i, against $L^{4(j-1-i)}$ sites. The remaining powers of the ratio of radii grow only polynomially in $j - i$. The sum is therefore uniform. For $R^{(i)}$ there is no coefficient subtraction; instead use $\sum_{i<j} x_i^{1-\kappa_0/2} \le C x^{2-\kappa_0/2}$. On a box this yields

$$\sup|P_0| \le C N_j \left[g\mathsf{b}_j^3 + (\log x)^{2(\mathfrak{p}_0 - q)} + g^{\kappa_0 - O(1)} \right], \qquad \|\partial_b \partial_{b'} P_0\| \le \varepsilon_P e^{-cd(b,b')},$$

where $g = x^{-1/2}$ and $\varepsilon_P \to 0$. There is no lag term under the redrawn-profile convention above. The analyticity radii in y exceed b_j by a fixed margin, so Cauchy's estimate preserves decay in this Hessian bound. Choosing q sufficiently large compared with $\mathfrak{p}_0$ makes the exponent and its Hessian small despite $N_j = O((\log x)^{8\mathfrak{p}_0})$.

The preceding lemma now gives, on S_j, a density ratio in $[e^{-\Xi_j}, e^{\Xi_j}]$, with

$$\Xi_j \le C N_j \mathsf{b}_j^2 e^{-cD}.$$

It holds uniformly for the exterior data, hence also for their positive mixtures. This proves the needed exterior removal for bounded insertions. It uses the actual localized force; an internal Hessian bound alone would not imply it.

43.3. **Completing the Gaussian segment comparison.** The remaining comparison can be made on a generating function. For each step, first remove the exterior by the density estimate, then the small tilt, then the Gaussian tails outside the product window, and finally compare its covariance on the read set with the flat covariance. Relative covariance comparison costs $CN_j\varpi_j$, where $\varpi_j \le C(\mathsf{r}gp_0 + e^{-cD})$; the actual gap and the analytic comparison of the quadratic form give this estimate. The total one-step variation is bounded by

$$T_j = (e^{\Xi_j} - 1) + C\theta_j + Ct_j + CN_j\varpi_j, \qquad \theta_j = \sup|P_0|,$$

with t_j its Gaussian-tail error. The flat covariances are isotropic in color. Deterministic nesting and localization of the minimizers, in one common box gauge, give the successive curvature decomposition, with quadratic remainder $d_{\mathrm{st}} \le C(\mathsf{r}gp_0)^2$.

We give the source normalization and the carried functions explicitly. There are two insertions, so put $m = 2$, keeping $n = k_1 - k_0$ for the length of the segment. Write ϕ_a, $a = 1, 2$, for the level-k_0 profiles $\widetilde{f}_a$ in (7b)(b), and O_a for the corresponding windowed insertions X_a. Set

$$|\lambda_a| \le \Lambda_a := \frac{cx}{m\mathsf{r}^4 p_0^2 \|\phi_a\|_\infty}, \qquad p_0 = p_0(g), \qquad g = x^{-1/2}.$$

The deterministic window bound gives $\Lambda_a \le c'/(m \sup|\mathsf{O}_a|)$ after reducing c. Thus the exponential of the two normalized insertions is uniformly bounded, including on the unclean part of the measure.

For $k_0 \le j \le k_1$, let $\mathsf{X}^{(j)}[A]$ be the Landau representative of the localized minimizer with datum e^{iA} in the box around the pair, and put

$$\mathsf{P}_p^{(j)}(A) = \mathsf{X}^{(j)}[A](\partial p) - \mathbf{1}.$$

The read sets contain these boxes and their determining bonds. The actual linear curvature response $\mathcal{H}_j$ and the localized minimizer estimates give

$$D\mathsf{P}_p^{(j)}(0)[A] = i(\mathcal{H}_j A)(p), \qquad |(\mathcal{H}_j A)(p)| \le C\eta_j^2 \|A\|,$$

$$|\mathsf{P}_p^{(j)}(A) - i(\mathcal{H}_j A)(p)| \le C\eta_j^2 \|A\|^2.$$

On the clean window, $\mathsf{O}_a = \frac{1}{2}\sum_p \phi_a(x(p))|\mathsf{P}_p^{(k_0)}(A)|^2$. Here η_j is the fine spacing in level-j units, as in the notation appendix. In particular, $\eta_{j-1}^2/\eta_{k_0}^2 = L^{-2(j-1-k_0)}$.

Let γ_j^0 be the flat Gaussian law on the read set, with covariance $C^{(j-1)}(\mathbf{1})$, and let C^0 eliminate the flat pivot bonds. For independent flat copies $\xi_{j'}$ define

$$\mathsf{m}_{\le j}(p;\xi) = i \sum_{k_0 < j' \le j} g_{j'-1}(\mathcal{H}_{j'-1}\mathsf{C}^0\xi_{j'})(p),$$

$$\mathsf{Z}_j(\mathsf{p}) = \int \prod_{k_0 < j' \le j} d\gamma_{j'}^0(\xi_{j'} \mid \mathcal{W}) \exp\left\{\frac{1}{2}\sum_{a=1}^m \lambda_a \sum_p \phi_a(x(p))|\mathsf{m}_{\le j}(p;\xi) + \mathsf{p}(p)|^2\right\}.$$

If $A^{\mathrm{ax}}(V)$ is a box axial potential for V, the carried function is

$$Y_j(V) = \int_{\mathrm{SU}(2)} d\mathsf{g}\, \mathsf{Z}_j\big((\mathsf{P}_p^{(j)}(\mathrm{Ad}_{\mathsf{g}}\, A^{\mathrm{ax}}(V)))_p\big).$$

The flat laws are isotropic in color and $\mathcal{H}_j$ acts on each color separately. The constant gauge average therefore removes the residual box frame. In particular Y_j is a local Borel gauge-invariant function, so the insertion identities apply to it. At $j = k_0$ it is the original exponential. Integrating the next flat copy transforms Z_{j-1} into Z_j.

The polydisc just chosen bounds Z_j uniformly and gives

$$|\mathsf{Z}_j(\mathsf{p}') - \mathsf{Z}_j(\mathsf{p})| \le \frac{C}{\eta_{k_0}^2 g p_0} \max_p |\mathsf{p}'(p) - \mathsf{p}(p)|.$$

Apply this to the deterministic one-level curvature identity, whose remainder is at most $\eta_{j-1}^2 d_{\mathrm{st}}$, and then apply the read-set variation bound. This yields

$$|\mathbb{E}_j Y_{j-1} - Y_j| \le C\left[L^{-2(j-1-k_0)} d_{\mathrm{st}}/(g p_0) + T_j\right].$$

The outer conditional expectations are positive and normalized. Consequently they telescope this estimate by summing its suprema; no derivative of an accumulated remainder is taken. At the top, reflection of all the flat copies $\xi \mapsto -\xi$ removes the linear term in the remaining background. Its contribution is $O(L^{-4n})$.

Write t for the Gaussian probability that at least one of the flat copies leaves its window. The union bound gives $t \le C\sum_j N_j e^{-c p_0^2}$. Removing those windows by Cauchy–Schwarz costs $Ct^{1/2}$. Thus on the source polydisc the conditional generating function differs from its flat Gaussian value by at most

$$\delta_{\mathrm{cl}} \le C\left[\sum_{j=k_0+1}^{k_1} \left(L^{-2(j-1-k_0)} d_{\mathrm{st}}/(g p_0) + T_j\right) + L^{-4n} + t^{1/2}\right].$$

In each color the resulting Gaussian covariance is

$$\mathcal{C}^x = \sum_{k_0 < j \le k_1} x_{j-1}^{-1} \mathcal{C}_{(j)},$$

and its generating function is

$$\mathsf{Z}^0(\lambda) = \det\left(1 - \sum_{a=1}^{m} \lambda_a \phi_a \mathcal{C}^x\right)^{-(N^2-1)/2}.$$

Its trace-log uses the branch equal to zero at $\lambda = 0$. The diagonal bound on $\mathcal{C}^x$ and the definition of Λ_a give

$$\left\| (\mathcal{C}^x)^{1/2} \left(\sum_a \lambda_a \phi_a\right) (\mathcal{C}^x)^{1/2} \right\|_1 \le C c p_0^{-2}.$$

Hence Z^0 has no zeros and $c \le |\mathsf{Z}^0| \le C$ uniformly on a fixed smaller polydisc. It is independent of the clean class and of the top field.

Let ϵ_{uc} include both the unclean histories and the nonreference flag. Their masses are $e^{-c p_0}$ times powers of L^n, the waiting factor $\Upsilon = L^{C\check{R}_{k_0}}$, and polylogarithms of x. The same deterministic insertion bound controls their generating functions. After averaging over the classes and the top field, the true generating function Z therefore satisfies

$$|\mathsf{Z}/\mathsf{Z}^0 - 1| \le C(\delta_{\mathrm{cl}} + \epsilon_{\mathrm{uc}}).$$

When this bound is small, the logarithm of this ratio is bounded by $C(\delta_{\mathrm{cl}} + \epsilon_{\mathrm{uc}})$. Cauchy's estimate on the fixed smaller polydisc then gives

$$|\partial_{\lambda_1}\partial_{\lambda_2}(\log \mathsf{Z} - \log \mathsf{Z}^0)(0)| \le C(\delta_{\mathrm{cl}} + \epsilon_{\mathrm{uc}})\Lambda_1^{-1}\Lambda_2^{-1}.$$

The relative loss in units of $x^{-2}\|\phi_1\|_\infty \|\phi_2\|_\infty$ is

$$\mathsf{L}_2 := x^2 \prod_{a=1}^{2} (\Lambda_a \|\phi_a\|_\infty)^{-1} \le (C\mathsf{r}^4 p_0^2)^2.$$

It remains to pay this loss, rather than merely to show that δ_{cl} tends to zero. The only error with exclusively logarithmic decay is the contribution of the old regular activities. After multiplication by L_2 it is at most

$$Cn\mathsf{r}^{12}p_0^4(\log x)^{2(\mathfrak{p}_0-q)},$$

which tends to zero when q is chosen sufficiently large in the stated order of constants. Terms carrying a positive power of g beat every polylogarithmic loss. Choose the box margin D and $n \asymp \log x$ so that the exponentially small exterior and top errors also survive L_2. The window and exceptional masses are negligible because $\mathfrak{p}_0 > \max\{1, r\}$. Consequently

$$\mathsf{L}_2(\delta_{\mathrm{cl}} + \epsilon_{\mathrm{uc}}) \longrightarrow 0,$$

which gives the mixed logarithmic derivative with error $o(x^{-2}\|\phi_1\|_\infty\|\phi_2\|_\infty)$.

The Gaussian covariance in each color is $\sum_{k_0<j\le k_1} x_{j-1}^{-1}\mathcal{C}_{(j)}$. Since $|x_{j-1} - x| \le Cn$, replacing its coefficients by x^{-1} has relative error $O(n/x)$. Wick's formula then gives $\frac{1}{2}(N^2-1)x^{-2}\mathcal{K}_{K,s}$ up to the segment's top tail. Transporting the source to level k_0 multiplies this by ${\zeta'_{k_0}}^2$. Its profile error is $CM\mathfrak{b}_{k_0}/(\vartheta\mathsf{r})$, and the summable slope bound makes $\zeta'_{k_0} = 1 + o(1)$. These are the interacting comparisons needed before the free kernel can be substituted.

44. Covariance and transfer estimates for interacting extraction

The finite insertion, renewed pointing, window formation and linear bootstrap sections derive the ordinary regular first response under their explicit unmarked, background, domain and slope inputs. They retain the potential normalization through local integration and give the near bound $C\delta^2 s$ on complete nonwrapping fields of size δ. The remainder below must also include the separately bounded marked, remote and boundary contributions. None of these estimates proves an interacting Gaussian law.

44.1. **Covariance bookkeeping with the retained source normalization.** A source-disc bound alone cannot replace the potential estimate. Indeed $H(z, B) = (z/\rho)(B/\alpha)^4$ is bounded by one on its product unit balls, but its source derivative at $B = \epsilon$ is $\rho^{-1}(\epsilon/\alpha)^4$. Relative to $\alpha^2(\epsilon/\alpha)^3$, this is $x(\log x)^{p-3q}/\rho$ when $\alpha = x^{-1/2}(\log x)^q$ and $\epsilon = x^{-1/2}(\log x)^p$; it diverges at fixed ρ. The finite construction supplies its response by differentiating the inserted potential, with the stated unmarked hypotheses.

Here is the corresponding covariance bookkeeping. Let λ be any probability measure, put $n_i = \|f_i\|_\infty$, fix $p > 0$, and suppose, uniformly in $x \to \infty$,

$$\begin{aligned} \|\mathsf{O}_i\|_\infty &\le Cx^{-1}r^4(\log x)^{2p}n_i, \\ \|\mathsf{I}_i\|_\infty &\le Cx^{-1}r^4[(\log x)^{3p-q} + x^{-a}]n_i, \qquad a > 0, \\ r &\le C_r(\log x)^{2p}, \qquad q > 21p. \end{aligned} \tag{ET3}$$

Then

$$\begin{aligned} &|\operatorname{Cov}_\lambda(\mathsf{O}_1, \mathsf{I}_2)| + |\operatorname{Cov}_\lambda(\mathsf{I}_1, \mathsf{O}_2)| + |\operatorname{Cov}_\lambda(\mathsf{I}_1, \mathsf{I}_2)| \\ &\qquad \le C'x^{-2}\big[(\log x)^{21p-q} + (\log x)^{18p}x^{-a}\big]n_1n_2 = o(x^{-2}n_1n_2). \end{aligned} \tag{ET4}$$

To prove it use $|\operatorname{Cov}(A, B)| \le 2\|A\|_\infty\|B\|_\infty$. The product of the two support counts costs $r^8 \le C(\log x)^{16p}$, and the a priori insertion costs another $(\log x)^{2p}$. For large x the bracket defining I_i is at most one, so its square is absorbed by itself. This gives (ET4). The numerical inequality $q > 21p$ is a sufficient bound for this particular sup-norm step; it does not claim to be the optimal requirement for the full extraction.

44.2. **A scalar transfer estimate with a genuinely common normalizer.** Let ρ, τ be two probability measures with real bounded observables $\mathcal{O}(f)$ linear in a real compactly supported Lipschitz function f. They may be measures on different configuration spaces. Write

$$D(f,h) = \mathrm{Cov}_\rho(\mathcal{O}(f),\mathcal{O}(h)) - \mathrm{Cov}_\tau(\mathcal{O}(f),\mathcal{O}(h)).$$

Assume that D is invariant under joint translations by $a\mathbb{Z}^4$. Set $\|f\|_\sharp = \max(\|f\|_\infty, c_M \operatorname{Lip} f)$ with c_M fixed.

Suppose there are $r_0 > 0$ and $Q \geq 1$ such that, for every pair F_1, F_2 supported in one common unit ball, there is one holomorphic function $E(z_1, z_2)$, $E(0) = 0$, serving both measures, for which

$$\left| e^{-E(z)} \omega\Big(e^{z_1 \mathcal{O}(F_1) + z_2 \mathcal{O}(F_2)} \Big) \right| \leq Q, \qquad \omega \in \{\rho, \tau\}, \tag{ET5}$$

on $|z_1|\|F_1\|_\sharp + |z_2|\|F_2\|_\sharp < r_0$. The constants are uniform in the ball position and in the measures. The tests in this hypothesis may overlap and have either sign.

Put $c_i = \|F_i\|_\sharp / r_0$ and denote the normalized holomorphic function in (ET5) by h_ω. It has $h_\omega(0) = 1$. Schwarz's lemma along rays gives $|h_\omega(z) - 1| \leq (Q+1)\sum_i c_i |z_i|$. On the bidisc with radii $[4(Q+1)c_i]^{-1}$ its principal logarithm has modulus at most $\log 2$. Thus Cauchy's formula gives

$$|D(F_1, F_2)| \leq 32 \log 2\, (Q+1)^2 r_0^{-2} \|F_1\|_\sharp \|F_2\|_\sharp. \tag{ET6}$$

The derivatives of E cancel because it is the same function for both measures. If an entry is zero the conclusion is immediate. No global zero-free statement for the original generating functions is needed.

Suppose now $a \leq d$ is sufficiently small, and f, h have supports in one common ball of radius $5d$, with $\operatorname{Lip} f \leq \|f\|_\infty / (\vartheta d)$ and the analogous condition for h. Choose a lattice multiple $r' \in [5d, 10d)$ and $N \geq c d^{-4}$ translates of that ball, with centers in $4r'\mathbb{Z}^4$, all lying inside a common unit ball. For each choice of signs $\varepsilon_v = \pm 1$ let $F^\varepsilon = \sum_v \varepsilon_v \tau_v f$ and $H^\varepsilon = \sum_v \varepsilon_v \tau_v h$. The translated balls are disjoint. Since the functions vanish on their ball boundaries, these signed sums do not increase either supremum or Lipschitz norm. Independent signs with the same sign used in the two arguments give

$$\mathbb{E}_\varepsilon D(F^\varepsilon, H^\varepsilon) = N D(f,h).$$

Apply (ET6) and divide by N. The result is

$$|D(f,h)| \leq C(Q, r_0, c_M) \vartheta^{-2} d^2 \|f\|_\infty \|h\|_\infty. \tag{ET7}$$

The exponent is $2 = 4 - 2$: the number of copies gains four powers and the two Lipschitz norms lose one power each.

If a reference law has a main term bounded below by $c_\vartheta x^{-2} \|f\|_\infty \|h\|_\infty$, its relative transfer error is at most $C_\vartheta d^2 x^2$. In the actual short-distance shape class, each nonnegative entry also has the explicit mass lower bound $\int f \geq \vartheta d^4 \|f\|_\infty$, and the two supports are at distance between d and $6d$. These requirements give $\iint f(x) h(y) |x-y|^{-8}\, dx\, dy \geq 10^{-8} \vartheta^2 \|f\|_\infty \|h\|_\infty$. The mass requirement is additional to the Lipschitz bound in (ET7). At a witness depth with $x \leq \bar{x}_s = O(s)$ and $d < L C_W(s) L^{-s}$, where C_W is polylogarithmic in s, this tends to zero uniformly. For a fixed $\vartheta = 1/12000$ all constants remain finite; their dependence on that fixed shape may affect the permitted coupling and distance thresholds.

The common-normalizer hypothesis (ET5) is not a consequence of having unrelated bounded normalizers for the two measures. There are two different ways of supplying it. If the scalar stability construction already provides one common cell normalizer for every periodic infinite-volume Gibbs state at the same datum, then (ET5) holds for any two such states. This includes a reference state obtained by taking finite-torus limits, provided its Gaussian estimate was uniform along arbitrarily large tori for each fixed cutoff and pair. The source $z_1 F_1 + z_2 F_2$ is a single cell source, even when the supports overlap. This route needs no comparison with a fixed finite torus.

If, instead, the reference Gaussian estimate has only been proved on one prescribed finite torus, a comparison of that torus with the infinite-volume states remains necessary. Compatible local energies and vacuum-density/floor bounds give one scalar chessboard route to (ET5). Bounds valid on each torus with a distance threshold tending to zero as its side grows do not by themselves furnish an infinite reference with the needed fixed-pair estimate. In either route, a typed Ward insertion estimate is not needed for (ET6)–(ET7).

A local companion criterion for vacuum-density compatibility. For the finite-reference route just described, the following criterion can remove a separate vacuum-pressure comparison assumption. It is not needed for comparison of infinite-volume states when their common cell normalizer is already available. It does not follow from upper stability and a partition floor alone. Fix $L > 1$ and a cutoff K. In the depth convention $a_s = L^{-s}$, level j has spacing $a_{K-j} = L^{j-K}$, $0 \le j \le K$. Take compatible side lengths ℓ for which ℓ/a_{K-j} is integral. Suppose the vacuum constant has the form

$$E_{\ell,K} = \sum_{j=0}^{K} \left[(\ell/a_{K-j})^4 c_{j,K} + \sum_{z \in (a_{K-j}\mathbb{Z}/\ell\mathbb{Z})^4} \sum_{D \in \mathcal{D}_{\ell,j}(z)} v_{\ell,j}(D,z) \right]. \tag{ET8}$$

The numbers $c_{j,K}$, the coupling sequence and every auxiliary choice are the same for all ℓ. There is a translation-covariant infinite companion $v_{\infty,j}$ and an integer length $m(D) \ge 0$. For both families, assume that the number of domains of length m at a root is at most $C_D e^{\kappa_D m}$ and $|v_{\ell,j}(D,z)|, |v_{\infty,j}(D,z)| \le C_E e^{-\kappa_E m}$ with $\kappa_E > \kappa_D$. The constants are uniform in j, K, ℓ. Assume that rooted domains with $m < \ell/(C_0 a_{K-j})$ correspond bijectively to companion domains, with equal energy values. This is an explicit locality and nonwrapping assumption, with C_0 fixed. It is stronger than simply asserting that the two models use the same formula for their energies.

Absolute convergence defines $\bar{v}_j = \sum_{D \in \mathcal{D}_{\infty,j}(0)} v_{\infty,j}(D,0)$ and $A_K = \sum_{j=0}^{K} a_{K-j}^{-4}(c_{j,K} + \bar{v}_j)$. At each root all terms below the stated threshold cancel in the difference from the companion. Sum the two tails of the geometric series with rate $\delta = \kappa_E - \kappa_D > 0$. Division by the physical volume therefore gives

$$\left| \ell^{-4} E_{\ell,K} - A_K \right| \le C \sum_{j=0}^{K} a_{K-j}^{-4} e^{-c\ell/a_{K-j}}. \tag{ET9}$$

For every fixed $\ell_0 > 0$ and $\ell \ge \ell_0$, this is bounded by $C_{\ell_0} e^{-c\ell/2}$, independently of K. Indeed, with $b = K - j$,

$$\sum_{b \ge 0} L^{4b} e^{-c\ell L^b} \le e^{-c\ell/2} \sum_{b \ge 0} L^{4b} e^{-(c\ell_0/2)L^b} \le C_{\ell_0} e^{-c\ell/2}. \tag{ET10}$$

Thus two compatible tori obey the same bound with $2C_{\ell_0} e^{-c \min(\ell,\ell')/2}$. The divergent common number A_K need not converge as the cutoff is removed; it cancels exactly in the comparison.

In a localized source argument, equality of the corresponding nonwrapping source energies also supplies the same interior-cell counterterm for all tori. Bounds on the energies crossing the cell faces and the appropriate partition floor are still needed to obtain (ET5). This lemma addresses its vacuum-density datum only. If the coupling sequence or local normalizer numbers differ between tori, or if nonwrapping compatibility has not been established for the actual formation rule, (ET9) has not been proved for that construction.

44.3. **A common normalizer for the finite reference torus and all states.** We now obtain (ET5) with the prescribed finite reference torus as one of the two measures. Fix one first-passage datum and write $T_* = T_{\ell_*}$ for the smallest admissible torus. Choose a fixed cell side l_c such that $n_* = \ell_*/l_c$ is even, and place the support ball a fixed distance from the cell faces. All larger

admissible sides are integer multiples of ℓ_*. The constants below may depend on the reference coupling, on l_c and on this fixed distance. They are independent of the cutoff, the larger side and the periodic Gibbs state.

The ordinary source construction gives the upper bound and the floor in (ST5). Write V_ℓ for its source-independent vacuum number and $\mathcal{E}_*(W)$ for the complete source energy on T_*, including the point terms U and the local domain terms. Thus

$$|Z_*(W)| \le \exp\{V_* + \operatorname{Re}\mathcal{E}_*(W) + C_+\ell_*^4\}, \qquad Z_\ell(0) \ge \exp\{V_\ell - C_-\ell^4\}.$$

The common companions of the regular and remainder families identify the vacuum terms on every nonwrapping lifted domain. The explicit Gaussian constants per site depend only on the common coupling run. Consequently (ET8)–(ET10), applied to those families, give

$$\left|\ell^{-4}V_\ell - \ell_*^{-4}V_*\right| \le \delta_{\mathrm{vac}}, \qquad \ell \ge \ell_*. \tag{44.1}$$

Only a uniform bound is needed here. The common divergent vacuum density cancels before the cutoff is removed.

Let $w_z = z_1F_1 + z_2F_2$, where the real Lipschitz functions F_i are supported in the chosen ball. Reflect this source into all the cells, conjugating its coefficients in every odd cell, and denote the result by W_z. The two real profiles of even and odd copies of F_i have the same supremum and Lipschitz bounds as F_i. They vanish near the faces. Hence

$$\|W_z\|_\flat \le 2\sum_i |z_i|\|F_i\|_\sharp.$$

We use half the ordinary source radius to pay for this factor two. No differentiability across a cell face is required; the joined profiles are globally Lipschitz.

Moving the repeated source to the reference torus. Put $M_\ell(z) = \langle e^{\mathcal{O}(w_z)}\rangle_\ell$ and $n_\ell = \ell/l_c$. The finite chessboard inequality gives

$$|M_\ell(z)| \le Z_\ell(W_z)^{1/n_\ell^4} Z_\ell(0)^{-1/n_\ell^4}.$$

The numerator is nonnegative. To compare it between volumes, cut in one coordinate direction. The source is zero near the cutting faces, so the crossing operator is the positive Wilson operator. The operators through two reflected adjacent slabs are adjoints, including the conjugation of the source. Their product is a positive trace-class operator Y. Repeating the pair q times replaces $\operatorname{Tr} Y$ by $\operatorname{Tr} Y^q$. Its eigenvalues are nonnegative, and $\operatorname{Tr} Y^q \le (\operatorname{Tr} Y)^q$ for integer $q \ge 1$. Apply this successively in the four directions, allowing rectangular tori at the intermediate stages. It follows that

$$0 \le Z_\ell(W_z) \le Z_*(W_z)^{(\ell/\ell_*)^4}. \tag{44.2}$$

This argument uses the exact finite lattice measure; the local source-energy representation is required only on T_*.

Combine this inequality with stability on T_*, the floor on T_ℓ, and (44.1). Since $\ell^4/n_\ell^4 = l_c^4$, both ℓ and ℓ_* satisfy

$$|M_\ell(z)| \le \exp\{n_*^{-4}\operatorname{Re}\mathcal{E}_*(W_z) + (C_+ + C_- + \delta_{\mathrm{vac}})l_c^4\}. \tag{44.3}$$

In particular the large-volume factor has become a fixed cell factor.

The holomorphic part common to the two volumes. For each cell rC, retain the source-energy terms on T_* whose complete dependence region lies in its interior. Evaluate a retained term at the single reflected copy w_z^r in that cell. If the cell is odd, conjugate this value once more. Define $E^\circ(z)$ to be the sum of these values, divided by n_*^4. Include the point terms in the same rule. In an

even cell a summand is holomorphic in z; in an odd cell it has the form $\overline{\phi(\bar{z})}$, with ϕ holomorphic, and is holomorphic as well. We have $E^\circ(0) = 0$. Locality gives

$$\operatorname{Re} E^\circ(z) = n_*^{-4} \sum_{\text{interior terms}} \operatorname{Re} e_*(W_z).$$

This definition averages over all cells. It requires no covariance of the individual localized terms under the cell group.

The omitted domain terms meet a cell face and also reach the support of a source copy. At depth $i = K - j$, such a domain has length at least $[cL^i/M - C]_+$. It contains one of at most $(CL^i/M + C)^4$ anchor cubes near the support ball. If $\kappa_e > \kappa_{\mathcal{D}}$ are the energy and entropy rates, the sum per cell is bounded by

$$C_{\text{face}} = C \sum_{i \geq 0} (CL^i/M + C)^4 e^{-(\kappa_e - \kappa_{\mathcal{D}})[cL^i/M - C]_+} < \infty. \tag{44.4}$$

There are only finitely many coarsest levels without exponential suppression. At finer levels the exponential in L^i pays for the anchor count. Thus this bound is uniform in the cutoff. Point terms on the faces vanish, since the source is zero there. We obtain

$$\left| n_*^{-4} \operatorname{Re} \mathcal{E}_*(W_z) - \operatorname{Re} E^\circ(z) \right| \leq C_{\text{face}}.$$

Together with (44.3), this proves

$$|e^{-E^\circ(z)} M_\ell(z)| \leq Q, \qquad Q = \exp\{C_{\text{face}} + (C_+ + C_- + \delta_{\text{vac}}) l_c^4\}, \quad \ell \geq \ell_*. \tag{44.5}$$

The very same E° serves the finite reference torus and all larger tori. Interior source energies on the larger tori have not been compared term by term with those on T_*.

Periodic states and the short-distance gain. For an arbitrary periodic DLR state, apply the exact domination lemma of Section 38 to the cell observable $e^{\mathcal{O}(w_z)}$. That lemma bounds its expectation by the limsup of the finite chessboard seminorms. The preceding calculation bounds each such seminorm by $|e^{E^\circ(z)}|Q$. Hence (44.5) also holds for every periodic state, with the same normalizer and the same Q. At fixed cutoff the observable is bounded and its generating function is entire, so there is no additional interchange of an unbounded source with an infinite-volume limit.

This supplies (ET5) for the finite reference and every periodic state. The elementary logarithm estimate (ET6) then cancels E°, and the average over signed translates in (ET7) gives

$$|S_2^T[\nu](f,h) - S_2^T[T_*](f,h)| \leq C_\vartheta d^2 \|f\|_\infty \|h\|_\infty.$$

The covariance difference is translation invariant under $a\mathbb{Z}^4$, as required for that averaging. A reference main term of size $x^{-2}\|f\|_\infty\|h\|_\infty$ therefore has relative transfer error at most $C_\vartheta d^2 x^2$, which tends to zero at the witness depths in (ET7). The order is: finite reference extraction, this uniform transfer at each cutoff, and then the continuum and component limits. No assertion that every pure Gibbs state is a limit of unperturbed torus measures is used.

45. Sharp cutoff boundaries in the classical Gaussian law

The boundary term in the conditional integration-by-parts identity need not be small merely because a Gaussian precision has a gap. The event that every bond is small can have a probability tending to zero with the volume. Dividing a whole-space tail estimate by that probability would therefore give no uniform conditional estimate. In the centered Gaussian reference law the symmetry and convexity of all the other cutoffs give a different argument: their conditional probability decreases on every radial ray. A radial boundary density can then be compared with an inner radial integral in the same direction. The normalization cancels before directions or bonds are summed.

This section proves that comparison for possibly singular physical bond covariances. It derives the resulting second through eighth moment errors, including the covariance and third and fourth connected errors of localized quadratic curvature observables. The actual flat pivot chart and contour gauge supply a concrete fixed-step application. The measure here is classical Gaussian. Haar and coarea factors, old activities, nonlinear action weights and conditional exterior forces are additional terms in the complete conditional law of Section 44.

Physical blocks, actual covariance entries and the cutoff law. Let Y be a centered Gaussian vector in a finite-dimensional real space, with covariance $G \geq 0$, and let $Z = CY$. The blocks Z_b belong to $\mathbb{R}^3$; repeated or linearly dependent blocks are allowed. All covariances in this section are the actual matrix entries

$$\Gamma_{bc} = \mathbb{E}[Z_b Z_c^{\mathsf{T}}] = C_b G C_c^{\mathsf{T}}, \qquad \Gamma = CGC^{\mathsf{T}}. \tag{GCB1}$$

For every tested block choose $b_b > 0$, put

$$\mathcal{A} = \bigcap_b \{|Z_b| < b_b\}, \qquad \nu(P) = \frac{\mathbb{E}[P(Z)1_{\mathcal{A}}]}{\mathbb{P}(\mathcal{A})}, \qquad d_b = \operatorname{rank}\Gamma_{bb} \leq 3, \tag{GCB2}$$

and choose an upper marginal scale $\sigma_b > 0$ with $\Gamma_{bb} \leq \sigma_b^2 I$. There is no lower eigenvalue assumption. A sufficiently small neighborhood of zero in the latent Gaussian support is contained in $\mathcal{A}$, so its probability is positive. If $d_b = 0$, then $Z_b = 0$ almost surely and that constraint has no boundary contribution. All blocks occurring in a polynomial to be bounded below are included among the tested blocks.

For a finite volume no lower bound on $\mathbb{P}(\mathcal{A})$ uniform in the volume is assumed or needed. The estimates concern these normalized finite-volume laws. They do not define conditioning on a simultaneous infinite collection of cutoff events, whose intersection could have probability zero.

Marginal log-concavity from one-dimensional transport. We first prove the integration fact needed for the conditional cutoff probability. Suppose nonnegative integrable functions f, g, h on $\mathbb{R}$, positive in the interiors of their interval supports, satisfy, for $0 < t < 1$,

$$h((1-t)x + ty) \geq f(x)^{1-t} g(y)^t. \tag{GCB3}$$

It is enough initially that the functions be continuous and the positive densities be smooth on their supports, as will hold in the application. Write $F = \int f$, $G_1 = \int g$, and assume both are positive. The increasing quantile map T from density f/F to g/G_1 is continuously differentiable in the interior and satisfies $T'(x) = f(x)G_1/[Fg(T(x))]$. The increasing map $T_t(x) = (1-t)x + tT(x)$ has derivative at least $T'(x)^t$, by the weighted arithmetic-geometric mean inequality. Changing variables on its image and using (GCB3) gives

$$\begin{aligned} \int h &\geq \int h(T_t(x)) T_t'(x)\, dx \\ &\geq \int f(x)^{1-t} g(T(x))^t T'(x)^t\, dx = F^{1-t} G_1^t. \end{aligned} \tag{GCB4}$$

One first integrates over compact subintervals of the positive support and then lets them exhaust it. All terms are nonnegative. This also handles unbounded interval endpoints and densities vanishing at the boundary. If either integral is zero the conclusion is immediate.

The same assertion holds in $\mathbb{R}^q$. Here is the induction needed to avoid importing a stronger integration theorem. Split $x = (x', u)$, $y = (y', v)$. For fixed x', y', apply the one-dimensional assertion in u, v, with the slice of h at $(1-t)x' + ty'$. The resulting marginals obey the same inequality in $q - 1$ dimensions. Apply the induction hypothesis to those marginals and use Tonelli. Slices of zero integral cause no problem. In the application all positive supports are open convex sets and all slices have these interval supports. Products of smooth compact cutoffs

with a Gaussian have integrable dominating functions for their derivatives on every compact set of external parameters. Their marginals are therefore continuous, smooth where required, and positive on the interiors of their convex supports. These facts justify each induction step directly in this class of functions.

In particular, if $k(z,w)$ is log-concave jointly in z,w, and its w slices are of the preceding integrable class, then

$$w_0(z)=\int k(z,w)\,dw \quad \text{is log-concave in } z. \tag{GCB5}$$

Indeed use the joint log-concavity inequality for the slices at z, z' and $(1-t)z+tz'$, then integrate by (GCB4) and its induction. The function k need not be integrable over the external variable z. If $k(-z,-w)=k(z,w)$, its marginal is even. An even log-concave function satisfies $w_0(rn)\le w_0(sn)$ whenever $0\le s\le r$: write sn as a convex combination of rn and $-rn$. This ray monotonicity, rather than a volume estimate for the complete cutoff probability, is the property used below.

Smooth compact approximants with an inner plateau. For $0<\delta\le 1/2$, construct a radial cutoff as follows. Put $\phi(t)=0$ for $t\le 0$, and on $0<t<1$ set

$$\begin{aligned}\phi''(t)&=\frac{e^{-1/t^2}}{(1-t)^4}, \qquad \phi(t)=\int_0^t (t-s)\frac{e^{-1/s^2}}{(1-s)^4}\,ds,\\ \psi_{b,\delta}(z)&=\begin{cases} e^{-\phi((|z|/b-1+\delta)/\delta)}, & |z|<b,\\ 0, & |z|\ge b.\end{cases}\end{aligned} \tag{GCB6}$$

The function ϕ is convex, nondecreasing, flat at zero and diverges at least as a positive multiple of $(1-t)^{-2}$ near one. Consequently $\psi_{b,\delta}$ is smooth on all of $\mathbb{R}^3$, including the sphere $|z|=b$, is even and log-concave, and equals one on $|z|\le(1-\delta)b$. The smoothness at zero follows because it is constant near zero; the smoothness at the outer sphere follows from the exponential decay of $e^{-\phi}$ faster than every inverse power of $1-t$. Convexity of $z\mapsto\phi((|z|/b-1+\delta)/\delta)$ follows from convexity of the norm and monotonic convexity of ϕ, extended by $+\infty$ at and beyond one.

As $\delta\downarrow 0$, these cutoffs increase pointwise to $1_{\{|z|<b\}}$. To check monotonicity inside the ball, the argument of ϕ is $1-(1-|z|/b)/\delta$, which decreases as δ decreases. Define

$$\chi_\delta(Z)=\prod_b \psi_{b_b,\delta}(Z_b), \qquad \nu_\delta(P)=\frac{\mathbb{E}[P(Z)\chi_\delta(Z)]}{\mathbb{E}\chi_\delta(Z)}. \tag{GCB7}$$

Every denominator is positive, and Gaussian dominated convergence gives $\nu_\delta(P)\to\nu(P)$ for every polynomial P. No independence of the tested coordinates enters this statement.

The plateau in (GCB6) is essential to the estimate: the derivative is supported in the outer annulus $[(1-\delta)b,b]$. A smooth cutoff whose derivative remained nonzero close to zero would not by itself yield the exponential boundary tail established next.

Conditional Gaussian support and monotonicity on rays. Fix a block b of positive rank and let $V_b=\operatorname{ran}\Gamma_{bb}$, of dimension $d=d_b$. Orthogonal decomposition of a whitened latent Gaussian gives, when conditioning on $Z_b=z\in V_b$,

$$Z_a=\Gamma_{ab}\Gamma_{bb}^{+}z+D_aW \quad (a\ne b), \qquad w_{b,\delta}(z)=\mathbb{E}_W\prod_{a\ne b}\psi_{b_a,\delta}(\Gamma_{ab}\Gamma_{bb}^{+}z+D_aW). \tag{GCB8}$$

Here W is a standard Gaussian independent of Z_b, possibly in dimension zero, and Γ_{bb}^{+} is the inverse on V_b, extended by zero. For completeness, write the latent standard Gaussian as its orthogonal projection onto the row space of the map defining Z_b and the perpendicular projection. The first projection is determined by z, and the second is an independent standard Gaussian on its

subspace. This proves (GCB8), including singular full covariances. There is no bound asserted on $\Gamma_{ab}\Gamma_{bb}^{+}$.

The integrand defining $w_{b,\delta}$, multiplied by the Gaussian density of W, is jointly log-concave in (z,W) and unchanged under $(z,W)\mapsto(-z,-W)$. Formula (GCB5) therefore proves that $w_{b,\delta}$ is even and log-concave on V_b. Its value decreases along every ray. It is positive near zero, although it may vanish elsewhere because of exact dependencies between blocks. Such zeros strengthen the inequalities below and require no separate boundary trace convention.

A normalized boundary flux with no lower marginal variance. Define the normalized smooth flux

$$\beta_{b,\delta}=\frac{\mathbb{E}[|\nabla\psi_{b_b,\delta}(Z_b)|\prod_{a\neq b}\psi_{b_a,\delta}(Z_a)]}{\mathbb{E}\chi_\delta(Z)}. \tag{GCB9}$$

For $d_b=0$, set $\beta_b^*=0$; the flux is zero because the cutoff is constant near zero. In positive rank the radial gradient lies in V_b, so the ambient Euclidean norm in (GCB9) is also its intrinsic gradient norm. Set $B=b_b$, $a=(1-\delta)B$, and use polar coordinates $z=rn$ in V_b. Up to a constant independent of r,n, the intrinsic Gaussian density is $\exp[-q(n)r^2/2]$, with $q(n)=n^{\mathsf{T}}(\Gamma_{bb}|_{V_b})^{-1}n\geq\sigma_b^{-2}$.

For a fixed direction let

$$\begin{aligned}D_n&=\int_0^B\psi_{B,\delta}(rn)w_{b,\delta}(rn)r^{d-1}e^{-q(n)r^2/2}\,dr,\\ I(q,a)&=\int_0^a s^{d-1}e^{-qs^2/2}\,ds.\end{aligned} \tag{GCB10}$$

For every $r\in[a,B]$, ray monotonicity and the inner plateau give $D_n\geq w_{b,\delta}(rn)I(q(n),a)$. The integral of the nonnegative radial derivative $-\partial_r\psi_{B,\delta}$ on $[a,B]$ is one. Thus the corresponding numerator in direction n is at most

$$D_n\,\frac{B^{d-1}e^{-q(n)a^2/2}}{I(q(n),a)}. \tag{GCB11}$$

This inequality remains true if $D_n=0$, because the preceding pointwise inequality then forces the numerator to vanish. In particular it does not divide by a possibly zero directional weight.

There are constants depending only on $d\leq 3$ such that $I(q,a)\geq c_d\min(a^d,q^{-d/2})$. If $a\leq q^{-1/2}$, integrate over $[0,a]$, where the exponential is at least $e^{-1/2}$. Otherwise integrate over $[0,q^{-1/2}]$. Since $B/2\leq a\leq B$, the ratio in (GCB11) is bounded by $C_dB^{-1}[1+(qB^2)^{d/2}]e^{-qB^2/8}$. Taking the supremum over all $q\geq\sigma_b^{-2}$ gives

$$\boxed{\beta_{b,\delta}\leq\beta_b^*:=\frac{C_d}{b_b}\left[1+\left(\frac{b_b}{\sigma_b}\right)^d\right]e^{-c_db_b^2/\sigma_b^2}} \qquad (1\leq d=d_b\leq 3). \tag{GCB12}$$

To justify the last maximization explicitly, put $x=qB^2$, $x_0=B^2/\sigma_b^2$. The product $(1+x^{d/2})e^{-x/16}$ is bounded on $[0,\infty)$. For $x\geq x_0$, the remaining factor is at most $e^{-x_0/16}$; increasing the constant gives the displayed bound, with for example a smaller fixed c_d. Now integrate (GCB11) over directions. The determinant normalization of the intrinsic Gaussian and the total integral of D_n cancel. This proves (GCB12) uniformly in δ, the volume, the number of other constraints, their dependencies and every lower marginal eigenvalue.

For $b_b\geq\sigma_b$ the useful simpler consequence is

$$\beta_b^*\leq C_db_b^{d-1}\sigma_b^{-d}e^{-c_db_b^2/\sigma_b^2}. \tag{GCB13}$$

The qualification is necessary. At small radius the normalized radial flux can be of order d/b_b; an estimate proportional to $b_b^{d-1}\sigma_b^{-d}$ alone would be false there. The proof has retained that regime in (GCB12).

Gaussian integration by parts before the sharp limit. For a polynomial P in the ambient physical blocks, ordinary Gaussian integration by parts, applied to a whitened latent vector, gives

$$\begin{aligned}\nu_\delta[Z_i P] &= \sum_j \Gamma_{ij}\nu_\delta[\nabla_j P] + \sum_b \Gamma_{ib}\mathcal{L}_{b,\delta}(P),\\ \mathcal{L}_{b,\delta}(P) &= \frac{\mathbb{E}[P(Z)\nabla\psi_{b_b,\delta}(Z_b)\prod_{a\neq b}\psi_{b_a,\delta}(Z_a)]}{\mathbb{E}\chi_\delta(Z)}.\end{aligned} \tag{GCB14}$$

The gradients are column vectors and Γ_{ib} are the corresponding 3 by 3 blocks. Degeneracy of Γ does not alter the formula: in the latent variables its two chain-rule factors are exactly the factors producing CGC^{T}. Gaussian domination justifies integration by parts even when the cutoffs have no compact support in null directions of C.

Let $\|P\|_{\mathcal{B}}$ be the supremum on the product of closed balls $\mathcal{B} = \prod_b\{|z_b| \le b_b\}$. Then

$$|\mathcal{L}_{b,\delta}(P)| \le \beta_b^*\|P\|_{\mathcal{B}}. \tag{GCB15}$$

For $L = \sum_i \ell_i^{\mathsf{T}} Z_i$, define its canonical sharp defect by

$$\mathcal{D}_L(P) = \nu[LP] - \sum_{i,j}\ell_i^{\mathsf{T}}\Gamma_{ij}\nu[\nabla_j P]. \quad |\mathcal{D}_L(P)| \le \left(\sum_{i,b}|\ell_i|\,\|\Gamma_{ib}\|\beta_b^*\right)\|P\|_{\mathcal{B}}. \tag{GCB16}$$

Indeed the left side is the limit of the left side of (GCB14) contracted with ℓ, while (GCB15) bounds the right side uniformly. This proves the sharp identity and its estimate without assigning a meaning to a product of a boundary delta function with another sharp indicator on the same surface.

For repeated or dependent constraints the separate sharp limits of $\mathcal{L}_{b,\delta}$ need not be intrinsic: the allocation between coincident constraint surfaces can depend on the smoothing schedule. The total defect in (GCB16) is intrinsic, because its definition uses only sharp conditional polynomial expectations. This is the only boundary object needed for the moment estimates. When a classical smooth-boundary free-coordinate trace exists, $\nabla 1_D = -n\,dS$, so this defect is the negative of the covariance contracted with the outward boundary functional in the conditional-law identity. This also fixes the sign convention.

The actual weighted Schur accounting. Choose positive amplitude scales a_b and positive cell weights μ_b. Assume the following explicit two-sided bound, using the covariance entries of (GCB1):

$$S = \max\left\{\sup_i\sum_j \mu_j\frac{\|\Gamma_{ij}\|}{a_i a_j}, \sup_j\sum_i \mu_i\frac{\|\Gamma_{ij}\|}{a_i a_j}\right\} < \infty, \qquad \varepsilon_\partial = \sup_b\frac{a_b\beta_b^*}{\mu_b}. \tag{GCB17}$$

The two sums give the usual bounded operator on weighted counting ℓ^2 after normalizing by a_b: weighted Cauchy in each row and then the column bound give norm at most S. For the boundary estimate the more specific computation is

$$\frac{1}{a_i}\sum_b\|\Gamma_{ib}\|\beta_b^* = \sum_b \mu_b\frac{\|\Gamma_{ib}\|}{a_i a_b}\frac{a_b\beta_b^*}{\mu_b} \le S\varepsilon_\partial. \tag{GCB18}$$

Thus, with $U(L) = \sum_i a_i|\ell_i|$,

$$|\mathcal{D}_L(P)| \le \delta(L)\|P\|_{\mathcal{B}}, \qquad \delta(L) = S\varepsilon_\partial U(L). \tag{GCB19}$$

There is no number of ambient coordinates in this estimate. The factor μ_b^{-1} in ε_∂ cannot be dropped: coordinate Gaussian integration by parts uses the ordinary partial derivative with respect to that coordinate. A weighted kernel convention does not change that derivative.

For example, if a precision is written as a self-adjoint operator K in pairing $\sum_b \mu_b y_b \cdot z_b$, its coordinate density has Euclidean precision MK, where $M = \operatorname{diag}(\mu_b I_3)$. Its covariance matrix is $(MK)^{-1}$. If $(K^{-1}f)_i = \sum_j \mu_j \mathsf{G}(i,j) f_j$, then $K^{-1} = \mathsf{G}M$, and $(MK)^{-1} = \mathsf{G}$. Accordingly $\mathsf{G}(i,j)$, rather than $\mu_j \mathsf{G}(i,j)$, is the covariance entry used here. This calculation prevents either suppressing or inserting a second cell-volume factor during the physical coordinate conversion.

All linear marginals remain Gaussian dominated. There is a useful moment bound independent of the cutoff radii. For a real linear statistic L, put $v(L)^2 = \mathbb{E}L^2 = \ell^{\mathsf{T}}\Gamma\ell$. If this is zero then $L = 0$ almost surely. Otherwise condition the latent Gaussian on $L = x$. The argument of (GCB8), now including every cutoff, gives an even log-concave weight $w(x)$ for its one-dimensional Gaussian density, first for ν_δ and then by the sharp limit. Therefore w decreases as $|x|$ increases. For any nonnegative increasing function f of $|x|$,

$$\nu[f(|L|)] \le \mathbb{E}f(|v(L)N|), \qquad N \sim N(0,1). \tag{GCB20}$$

To see the normalization explicitly, let R, R' be independent absolute Gaussian variables of this variance. The product $(f(R) - f(R'))(w(R) - w(R'))$ is nonpositive. Its expectation is twice $\mathbb{E}[fw] - \mathbb{E}[f]\mathbb{E}[w]$. Divide by $\mathbb{E}[w] > 0$, and truncate f followed by monotone convergence if necessary. This proves the assertion without a lower bound on the cutoff probability.

In particular the law is centered and, for $m \ge 1$,

$$\nu[L^{2m}] \le (2m-1)!!\, v(L)^{2m}, \qquad \nu[e^{tL}] \le e^{t^2 v(L)^2/2}, \qquad \operatorname{Cov}_\nu(Z) \le \Gamma. \tag{GCB21}$$

The exponential assertion follows by symmetry and the increasing function $\cosh(t|L|)$. The covariance ordering follows by testing every linear functional. Every joint moment of an odd number of linear statistics is zero, because the entire law is invariant under $Z \mapsto -Z$.

Covariance and fourth-order defects of linear statistics. For linear statistics L_r, write

$$\begin{aligned} &\Gamma^L_{rs} = \mathbb{E}[L_r L_s], \quad \Sigma_{rs} = \nu[L_r L_s], \quad E_{rs} = \Sigma_{rs} - \Gamma^L_{rs}, \\ &B_r = \sum_i b_i |\ell_{r,i}|, \quad \delta_r = S\varepsilon_\partial U(L_r), \quad v_r = v(L_r). \end{aligned} \tag{GCB22}$$

Here B_r bounds $|L_r|$ on the entire cutoff box, whereas v_r is its untruncated Gaussian standard deviation. These two quantities have different uses. Taking $P = L_s$ in (GCB16) gives

$$|E_{rs}| \le \min(\delta_r B_s, \delta_s B_r), \qquad |\Sigma_{rs}| + |\Gamma^L_{rs}| \le 2 v_r v_s. \tag{GCB23}$$

The second estimate follows separately for its two terms from (GCB21) and Gaussian Cauchy–Schwarz. Every third joint moment of linear statistics is exactly zero.

Apply the same identity to $P = L_2 L_3 L_4$. The derivative of this product produces exactly three pairings, and hence

$$\nu[L_1 L_2 L_3 L_4] = \sum_{j=2}^{4} \Gamma^L_{1j} \Sigma_{\{2,3,4\}\setminus\{j\}} + \mathcal{D}_{L_1}(L_2 L_3 L_4). \tag{GCB24}$$

In this formula $\Sigma_{\{r,s\}} = \Sigma_{rs}$. Subtracting the three Gaussian pairings gives an error at most $\delta_1 B_2 B_3 B_4 + \sum_{j=2}^4 |\Gamma^L_{1j}||E_{\{2,3,4\}\setminus\{j\}}|$. Subtracting instead the three pairings with the actual covariance Σ gives the fourth connected moment and the explicit bound

$$\begin{aligned} |\kappa_\nu(L_1, L_2, L_3, L_4)| \le \delta_1 \big(& B_2 B_3 B_4 + B_2 v_3 v_4 \\ & + B_3 v_2 v_4 + B_4 v_2 v_3 \big). \end{aligned} \tag{GCB25}$$

Indeed replace $\Gamma^L_{1j} - \Sigma_{1j}$ by $-E_{1j}$, use $|E_{1j}| \le \delta_1 B_j$, and bound each remaining Σ by its two Gaussian standard deviations. Any of the four indices can be chosen as the first one. The identity, rather than an assumed small fourth cumulant, proves the connected estimate.

For example, $\mathrm{Cov}_\nu(L_1L_2, L_3L_4) = \kappa_\nu(L_1, L_2, L_3, L_4) + \Sigma_{13}\Sigma_{24} + \Sigma_{14}\Sigma_{23}$. Together with (GCB23) this gives the covariance error of two factored quadratic observables relative to their untruncated Gaussian Wick contractions. The higher degree estimates below also control three and four such observables.

The moment recursion through degree eight. Let a collection of localized linear statistics obey

$$B_r \le R\,U_r, \qquad v_r \le v\,U_r, \qquad U_r = U(L_r), \qquad D = S\varepsilon_\partial. \tag{GCB26}$$

One may take $R = \max_{i\in\mathcal{S}} b_i/a_i$ and $v = \max_{i\in\mathcal{S}} \sigma_i/a_i$, where $\mathcal{S}$ is the union of the supports of these statistics. The variance bound follows from $\|\Gamma_{ij}\| \le \sigma_i\sigma_j$, itself a consequence of Gaussian Cauchy–Schwarz. Thus neither R nor v requires a maximum over coordinates outside their supports.

Let $M_\nu(1,\dots,n) = \nu[\prod_{r=1}^n L_r]$ and let M_0 denote the corresponding untruncated Gaussian moment. For even n, integration by parts gives the exact recurrence

$$M_\nu(1,\dots,n) - M_0(1,\dots,n) = \sum_{j=2}^{n} \Gamma^L_{1j}(M_\nu - M_0)(\{2,\dots,n\}\setminus\{j\}) + \mathcal{D}_{L_1}\Big(\prod_{r=2}^{n} L_r\Big). \tag{GCB27}$$

For $n = 2$ the empty moment difference is zero. Odd moments vanish for both laws. Consequently

$$|M_\nu(1,\dots,2m) - M_0(1,\dots,2m)| \le d_{2m}\prod_{r=1}^{2m} U_r, \tag{GCB28}$$
$$d_2 = DR, \qquad d_n = DR^{n-1} + (n-1)v^2 d_{n-2}.$$

This is a direct induction: each derivative pairing contributes at most $v^2U_1U_j$, and the boundary defect contributes at most $DU_1\prod_{r=2}^n B_r$. In particular

$$\begin{aligned} d_2 &= DR,\\ d_4 &= D(R^3 + 3v^2R),\\ d_6 &= D(R^5 + 5v^2R^3 + 15v^4R),\\ d_8 &= D(R^7 + 7v^2R^5 + 35v^4R^3 + 105v^6R). \end{aligned} \tag{GCB29}$$

All constants are independent of the total number of Gaussian coordinates. The factors U_r retain the actual localization and scale of each inserted statistic.

Absolute mixed moments have a second bound useful when subtracting products of moments. Hölder with $2m$ equal exponents and (GCB20) gives, for either law,

$$\Big|\mathbb{E}_*\prod_{r=1}^{2m} L_r\Big| \le (2m-1)!!\prod_{r=1}^{2m} v_r \le (2m-1)!!\,v^{2m}\prod_{r=1}^{2m} U_r, \qquad \mathbb{E}_* \in \{\nu, \mathbb{E}\}. \tag{GCB30}$$

No independence of the L_r is used. The simultaneous use of the support bound in the defect and the variance bound in unaffected moment factors avoids an unnecessary cutoff radius in every term.

Localized quadratic observables and their connected errors. Write each real quadratic observable in a finite factored form

$$Q_j = \sum_\alpha c_{j,\alpha} L_{j,\alpha,1} L_{j,\alpha,2}, \qquad \mathcal{A}(Q_j) = \sum_\alpha |c_{j,\alpha}| U(L_{j,\alpha,1}) U(L_{j,\alpha,2}). \tag{GCB31}$$

Choose common R, v for this finite list as in (GCB26). Every product of $p \le 4$ quadratic observables is a sum of words of degree $2p$. Equations (GCB29) and (GCB30) can therefore be summed with absolute coefficients, with factor $\prod_j \mathcal{A}(Q_j)$. This step counts the inserted coefficients, not all coordinates of the integration space.

For clarity define the joint cumulant by the finite partition formula

$$\kappa_*(Q_1, \dots, Q_p) = \sum_{\pi \in \mathfrak{P}_p} (-1)^{|\pi|-1} (|\pi|-1)! \prod_{A \in \pi} \mathbb{E}_* \prod_{j \in A} Q_j. \tag{GCB32}$$

Write $\Delta_p = \kappa_\nu(Q_1, \dots, Q_p) - \kappa_0(Q_1, \dots, Q_p)$. The following explicit estimates result:

$$\begin{aligned} |\nu Q_1 - \mathbb{E} Q_1| &\le d_2 \mathcal{A}(Q_1), \\ |\Delta_2| &\le (d_4 + 2v^2 d_2) \mathcal{A}(Q_1) \mathcal{A}(Q_2), \\ |\Delta_3| &\le (d_6 + 3v^2 d_4 + 15 v^4 d_2) \prod_{j=1}^{3} \mathcal{A}(Q_j), \\ |\Delta_4| &\le (d_8 + 4v^2 d_6 + 30 v^4 d_4 + 156 v^6 d_2) \prod_{j=1}^{4} \mathcal{A}(Q_j). \end{aligned} \tag{GCB33}$$

Here the subscript zero always refers to the original untruncated Gaussian with covariance Γ. In particular its quadratic third and fourth cumulants are generally nonzero. They are the Gaussian Wick values, not quantities being set equal to zero.

We give the product bookkeeping in (GCB33). By multilinearity it is enough to use one factored monomial for each Q_j. For covariance the full moment difference is bounded by d_4; replacing the product of its two means costs $2v^2 d_2$. At order three the three terms of partition type $(2,1)$ cost at most $3(v^2 d_4 + 3v^4 d_2)$. The product of three means, with coefficient two, costs at most $6v^4 d_2$. Together with d_6, these give the third line. At order four, the four partitions of type $(3,1)$ cost $4(v^2 d_6 + 15 v^6 d_2)$; the three of type $(2,2)$ cost $18 v^4 d_4$. The six of type $(2,1,1)$, with coefficient two, cost $12 v^4 d_4 + 72 v^6 d_2$. The final product of four means, with coefficient six, costs $24 v^6 d_2$. Their sum gives exactly the fourth line. Each estimate uses the telescoping difference of products, bounding unchanged factors by (GCB30). This establishes the constants without an assumed connected-moment estimate.

Substituting (GCB29), the three connected error polynomials multiplying D are respectively

$$\begin{aligned} \mathcal{P}_2(R, v) &= R^3 + 5v^2 R, \\ \mathcal{P}_3(R, v) &= R^5 + 8v^2 R^3 + 39 v^4 R, \\ \mathcal{P}_4(R, v) &= R^7 + 11 v^2 R^5 + 85 v^4 R^3 + 411 v^6 R. \end{aligned} \tag{GCB34}$$

The same method covers mixed linear and quadratic observables: expand each quadratic into its two factors, use (GCB27) for each resulting word, and then the finite partition formula. An odd total linear degree vanishes before taking absolute values. No additional probabilistic assumption is required for this mixed case.

The reference Gaussian Wick values themselves can be calculated without a nonsingular physical covariance. If $Q_j = Z^{\mathsf{T}} A_j Z$ with real symmetric A_j, whitening the latent Gaussian gives, for

$p \geq 1$,

$$\kappa_0(Q_1, \dots, Q_p) = \frac{2^{p-1}}{p} \sum_{\pi \in S_p} \operatorname{tr}(\Gamma A_{\pi(1)} \cdots \Gamma A_{\pi(p)}). \tag{GCB35}$$

Indeed, near zero the logarithm of the Gaussian exponential generating function is

$$\frac{1}{2} \sum_{k \geq 1} \frac{2^k}{k} \operatorname{tr} \left[\left(\Gamma \sum_j t_j A_j \right)^k \right].$$

Taking the coefficient of $t_1 \cdots t_p$ gives (GCB35). This also identifies precisely the contractions to which the sharp-cutoff cumulants are compared.

The fixed-step flat chart realizes the covariance hypothesis. We now specialize to the actual first plain-average fibre at the identity, with fixed odd block size L, in current unit-spacing coordinates. Let P_L be the plain derivative of the nonlinear average, C_0 its pivot inclusion with identity free restriction, and $\mathcal{T}$ the internal contour matrix. The definitions and the actual first-step precision are

$$\begin{aligned} &\operatorname{ran} \mathsf{C}_0 = \ker P_L, \qquad (\mathsf{C}_0 u)|_{\text{free}} = u, \\ &\mathsf{K}_L = \mathsf{C}_0^{\mathsf{T}} (d_1^{\mathsf{T}} d_1 + \mathcal{T}^{\mathsf{T}} \mathcal{T}) \mathsf{C}_0, \quad Y \sim N(0, \mathsf{K}_L^{-1}), \quad Z = \mathsf{C}_0 Y. \end{aligned} \tag{GCB36}$$

The quadratic form here is exactly the first-step form, with no previous finer-level minimization. This is the classical reference with action weight $x = \theta$, so $\tau = x/\theta = 1$. The canonical normalization $\theta g^2 = 1$ then gives coefficient one both to the Wilson curl form and to the prescribed actual soft gauge. In particular $\mathcal{T}$ is not a divergence gauge or a freely chosen projector.

The pivot matrix has finite range with at most C_L nonzero blocks in each row and column and entries bounded by C_L. The column assertion follows by the same double-block support as the row assertion: a free bond belongs to only finitely many double blocks when L is fixed. The flat gap and the exponential inverse estimate of Section 20 give $\|\mathsf{K}_L^{-1}(b,c)\| \leq C_L e^{-\mu_L d(b,c)}$, uniformly on the admitted tori. Consequently

$$\Gamma = \mathsf{C}_0 \mathsf{K}_L^{-1} \mathsf{C}_0^{\mathsf{T}}, \quad \max \left(\sup_b \sum_c \|\Gamma_{bc}\|, \sup_c \sum_b \|\Gamma_{bc}\| \right) \leq S_L, \quad \Gamma_{bb} \leq \sigma_L^2 I. \tag{GCB37}$$

To check volume uniformity, a four-dimensional shell of distance m has at most $C_L(m+1)^3$ bond indices in block coordinates. Summing $e^{-\mu_L m}$ over these shells is finite; periodic image sums have the same upper bound. Multiplying the finite row and column sums of C_0, its transpose and the inverse proves the two sums in (GCB37). The diagonal bound follows, for example, from either sum. Increase σ_L to at least one if convenient.

Thus (GCB17) is realized with $a_b = \mu_b = 1$, $S = S_L$, independent of the number of blocks. This is a statement at a fixed current spacing and fixed L. Pointwise decay in a multilevel physical metric would not alone give its unweighted analogue: the number and normalization of coordinates at different scales must still be charged in the actual weighted norm (GCB17).

Why the actual contour gauge leaves no physical quadratic residue. We prove the needed range statement rather than assuming that an arbitrary soft gauge can be integrated away. Let $\mathcal{S}_0$ be the real site fields vanishing at every block center. For any target internal contour data $s = (s_{y,x})$, $x \in B(y) \setminus \{y\}$, define the site field Rs by $(Rs)(x) = s_{y,x}$ and $(Rs)(y) = 0$. Blocks partition the sites, so this is well defined. The plain-gradient identity and the internal path

telescoping identity give exactly

$$\begin{aligned} &P_L d_0 Rs = 0, \qquad d_1 d_0 Rs = 0, \qquad (\mathcal{T} d_0 Rs)_{y,x} = s_{y,x}, \\ &\|Rs\|_2 = \|s\|_2, \qquad \|d_0 Rs\|_2 \le 4\|s\|_2. \end{aligned} \tag{GCB38}$$

Here d_0 is the forward site gradient and d_1 the plaquette circulation. The last norm estimate follows by summing $|f(x+e_\mu)-f(x)|^2 \le 2|f(x+e_\mu)|^2 + 2|f(x)|^2$ over four directions. Internal contour means average paths, but on a gradient every one of those paths gives the same difference $f(x)-f(y)$; hence no path count or L^{-4} factor occurs in the third identity. The constraint identity is $P_L d_0 f = d_L(f|_{\text{centers}})$, not a site-average gradient identity.

In particular the image of $\mathcal{T}$ on $\ker d_1 \cap \ker P_L$ is the entire internal target space. It equals the image of $\mathcal{T}$ on $\ker P_L$. This is the precise range condition that prevents a leftover physical gauge penalty. If an arbitrary alternative gauge failed this condition, integrating its gauge directions could instead leave a positive quadratic form in curvature.

Put $\mathcal{E}_C = d_1 \ker P_L$. The linear map

$$\ker P_L \longrightarrow \mathcal{E}_C \oplus \mathcal{S}_0, \qquad a \longmapsto (d_1 a, \mathcal{T} a) \quad \text{is a bijection.} \tag{GCB39}$$

It is onto because any initial lift of a prescribed curvature in $\mathcal{E}_C$ can be adjusted to any prescribed contour data by adding $d_0 Rs$ from (GCB38). It is one-to-one because the flat gap on $\ker P_L$ says that both $d_1 a = 0$ and $\mathcal{T} a = 0$ force $a = 0$. Equivalently use the actual internal-bond and crossing-face zero-mode argument of Section 18.

Apply the constant linear change of variables $Y \mapsto (F, s) = (d_1 \mathsf{C}_0 Y, \mathcal{T} \mathsf{C}_0 Y)$. Its Jacobian is a nonzero constant between the indicated Euclidean spaces. The quadratic exponent in (GCB36) becomes exactly $\frac{1}{2}\|F\|^2 + \frac{1}{2}\|s\|^2$. The constant Jacobian and the integral in s cancel from normalized curvature expectations. Therefore

$$F = d_1 Z \text{ is a standard Gaussian on } \mathcal{E}_C, \qquad d_1 \Gamma d_1^{\mathsf{T}} = P_{\mathcal{E}_C}. \tag{GCB40}$$

The same calculation with any prescribed positive quadratic form in s changes only the gauge normalization.

If the action weight is instead $x = \tau\theta$, $\tau > 0$, multiply the curl term in (GCB36) by τ, leaving the reference soft gauge coefficient one. The same bijection gives exponent $\frac{\tau}{2}\|F\|^2 + \frac{1}{2}\|s\|^2$, so the corresponding covariance satisfies $d_1 \Gamma_\tau d_1^{\mathsf{T}} = \tau^{-1} P_{\mathcal{E}_C}$. All numerical reference applications below use $\tau = 1$. No uniform covariance bound as $\tau \downarrow 0$ is inferred.

The precise relation with the full free-curvature comparison. Let $\mathcal{E} = \operatorname{ran} d_1$ on the same finite periodic lattice, with each positive plaquette counted once. Generally $\mathcal{E}_C$ is a proper subspace of $\mathcal{E}$. With $\mathcal{E}_\perp = \mathcal{E} \cap \mathcal{E}_C^\perp$, ordinary orthogonal decomposition gives

$$P_{\text{ex}} = P_{\mathcal{E}} = P_{\mathcal{E}_C} + P_{\mathcal{E}_\perp}, \qquad F_{\text{full}} = F_C + F_\perp, \tag{GCB41}$$

where the two summands on the right can be taken independent standard Gaussians on their respective subspaces. This proves that the law of F_{full} is the full exact-cochain Gaussian used in the free-kernel comparison. Conditional on $F_\perp$, its centered fluctuation has exactly the curvature law (GCB40). Conversely, integrating this complementary Gaussian supplies the missing curvature covariance.

The decomposition is an exact classical Gaussian identity. It does not identify the law of a coarse mean in an interacting run. In particular it does not permit replacing $P_{\mathcal{E}_C}$ by P_{ex} in a single fixed-fibre Gaussian estimate. Fine fluctuations alone omit the second term of (GCB41); exterior or coarse averaging is a separate covariance contribution in the complete conditional-law identity.

There is a finite-dimensional expression for this distinction which needs no chosen coarse-coordinate parametrization. Let Q_C be any matrix whose kernel inside $\mathcal{E}$ is $\mathcal{E}_C$. Then

$$P_{\mathcal{E}_C} = P_{\mathrm{ex}} - P_{\mathrm{ex}} Q_C^{\mathsf{T}} (Q_C P_{\mathrm{ex}} Q_C^{\mathsf{T}})^{+} Q_C P_{\mathrm{ex}}. \tag{GCB42}$$

To prove the formula, set $B = Q_C P_{\mathrm{ex}}$ restricted to $\mathcal{E}$. Singular-value decomposition shows that $B^{\mathsf{T}}(BB^{\mathsf{T}})^{+}B$ is the orthogonal projection onto $(\ker B)^{\perp}$ within $\mathcal{E}$. Subtract it from the identity on $\mathcal{E}$, then extend by zero on $\mathcal{E}^{\perp}$. This yields exactly (GCB42). The identity is not an estimate that the correction is small. Such an estimate on a particular physical window would require its own quantitative proof.

At fine physical spacing η, write the canonical raw curvature as F. The physical curvature of this canonically normalized field is $\eta^{-2}F$. The full exact-cochain covariance is therefore $\eta^{-4}P_{\mathrm{ex}}$, precisely the normalization of the free-curvature comparison in Section 5. Its plaquette components are sampled at their own barycentres. For three independent color copies and $Q_f = \frac{1}{2}\sum_p f(x_p)|F_p|^2$, Gaussian Wick contraction gives

$$\mathrm{Cov}_0(Q_f, Q_h) = \frac{1}{2}\sum_{p,q} f(x_p)h(x_q)\|\mathbb{E}[F_p F_q^{\mathsf{T}}]\|_{\mathrm{HS}}^2. \tag{GCB43}$$

The physical Riemann weight η^4 multiplying the squared physical curvature cancels its η^{-4} scaling in Q_f. Thus the $\mathrm{tr}(F^2)$ readout is the unweighted plaquette sum displayed here; inserting an additional η^4 into that sum would change the observable.

For the full law in (GCB41), (GCB43) has the full free-curvature kernel and its proved separated-smearing comparison. For the first-step fluctuation it has $P_{\mathcal{E}_C}$ instead. The cutoff estimates compare each specified Gaussian reference with its own Wick kernel. They transfer that comparison only after the reference used in the application has actually been identified with, or compared to, the full exact-cochain reference. The decomposition records how this works in the free Gaussian theory without asserting it for a conditional interacting measure.

Uniform cutoff errors for the curvature polynomial. Take all physical tangent bond radii equal to $B \geq \sigma_L$ in (GCB36), and retain counting weights and unit amplitudes. From (GCB13), including all ranks from one through three and enlarging the fixed constants,

$$\sup_b \beta_b^* \leq \beta_L(B) := C_L B^2 e^{-c_L B^2}, \qquad D \leq S_L \beta_L(B), \qquad R = B, \quad v \leq \sigma_L. \tag{GCB44}$$

The bounds remain valid if some physical blocks are deterministic. For a sampled test function let $W_f = \sum_p |f(x_p)|$, and define the canonical quadratic curvature polynomial

$$Q_f(Z) = \frac{1}{2}\sum_p f(x_p)|d_1 Z(p)|^2, \qquad \mathcal{A}(Q_f) \leq 24 W_f. \tag{GCB45}$$

Indeed a fixed color component of a plaquette curl has four coefficients of absolute value one, hence $U \leq 4$. There are three colors and the coefficient is $f/2$, so $(1/2)\cdot 3 \cdot 4^2 = 24$. Each positive direction pair is already counted once in the sum over p. This calculation requires no estimate on the number of unobserved bonds.

For $p = 2, 3, 4$, equations (GCB33)– (GCB44) now imply, with $\Delta_p = \kappa_\nu(Q_{f_1}, \ldots, Q_{f_p}) - \kappa_0(Q_{f_1}, \ldots, Q_{f_p})$,

$$\begin{aligned} |\nu Q_f - \mathbb{E}Q_f| &\leq C_L \beta_L(B) B W_f, \\ |\Delta_p| &\leq C_{L,p}\beta_L(B)(B^{2p-1} + B^{2p-3} + \cdots + B)\prod_{j=1}^{p} W_{f_j}. \end{aligned} \tag{GCB46}$$

Here all expectations are normalized, including the sharp product cutoff law. The third and fourth lines implicit in the second formula use sixth and eighth underlying linear moments. The

vanishing of odd linear moments alone would not establish them. The Gaussian values on the right are the actual cycle contractions (GCB35), with the constrained curvature projection of (GCB40) in this particular application.

If each test is supported in a ball of current radius $r \geq 1$, then $W_f \leq C(r+1)^4 \|f\|_\infty$. Thus the covariance error has the expected two localized volume factors $C(r+1)^8 \|f\|_\infty \|h\|_\infty$, rather than a factor proportional to the volume of the torus. At orders three and four the factors are $(r+1)^{12}$ and $(r+1)^{16}$.

The nonlinear Wilson readout on the actual joint chart. At the flat background let $a = \Phi_0(u)$ be the actual nonlinear fibre chart of Section 17, so u is the free raw logarithm and $D\Phi_0(0) = \mathsf{C}_0$. Its domain contains $\|u\|_\infty < r_L$, with $r_L = cL^{-9}$ after reducing the fixed constant. The finite local support and the uniform holomorphic bounds on that joint domain give, on a smaller real ball,

$$\begin{aligned} \|\Phi_0(u) - \mathsf{C}_0 u\|_\infty &\leq C_L \|u\|_\infty^2, \\ A_p(e^{i\Phi_0(u)}) &= \frac{1}{2} |d_1 \mathsf{C}_0 u(p)|^2 + R_p(u), \\ |R_p(u)| &\leq C_L \|u\|_\infty^3. \end{aligned} \tag{GCB47}$$

For the second assertion use the holomorphic extension $1 - \frac{1}{2} \operatorname{tr} U_p$, which equals the real Wilson readout on SU(2). Its constant and linear terms at the identity are zero. Differentiating the four bond products shows its quadratic term is $\frac{1}{2}|d_1 \mathsf{C}_0 u(p)|^2$, with Pauli pairing $|X|^2 = \frac{1}{2} \operatorname{tr} X^2$. To bound the remainder, apply one-variable Cauchy to the actual local function of tu on a disc of radius proportional to $r_L / \|u\|_\infty$. The joint chart bounds this function uniformly; summing the Taylor terms of order at least three on a smaller disc gives the displayed cubic estimate. The first assertion follows by the same argument at order two. A plaquette and its pivot dependencies occupy a fixed number of double blocks, so neither remainder constant contains the lattice volume.

Set $u = gY$. Under the full physical tangent cutoff $|\mathsf{C}_0 Y(b)| < B$, the identity free restriction implies $\|Y\|_\infty \leq B$. Thus $gB < r_L/2$ places every field of this cutoff event in the actual chart, and for $O_f = \sum_p f(x_p) A_p$,

$$O_f(e^{i\Phi_0(gY)}) = g^2 Q_f(Z) + \mathcal{R}_f(Y), \quad |\mathcal{R}_f| \leq C_L g^3 B^3 W_f, \quad |Q_f| \leq C_L B^2 W_f. \tag{GCB48}$$

This uses the literal sampled $\operatorname{tr}(F^2)$ sum, with no extra fine-cell volume factor. The probability measure in this identity is still the Gaussian cutoff measure, rather than the nonlinear action measure.

The finite partition definition of a cumulant gives $|\kappa(X_1, \ldots, X_p)| \leq C_p \prod_j \|X_j\|_\infty$. Expand each slot of (GCB48). If $gB \leq 1$, the terms with at least one remainder have total absolute value at most $C_{L,p} g^{2p+1} B^{2p+1} \prod_j W_{f_j}$. Indeed a remainder replaces one factor $g^2 B^2$ by $g^3 B^3$; the finite sum over subsets costs at most a constant times $gB(1+gB)^{p-1}$. Together with (GCB46), this proves

$$\begin{aligned} &|\kappa_\nu(O_{f_1}, \ldots, O_{f_p}) - g^{2p} \kappa_0(Q_{f_1}, \ldots, Q_{f_p})| \\ &\quad \leq C_{L,p} g^{2p} \left[\beta_L(B)(B^{2p-1} + B^{2p-3} + \cdots + B) + gB^{2p+1} \right] \prod_{j=1}^{p} W_{f_j}, \qquad p = 2, 3, 4. \end{aligned} \tag{GCB49}$$

The same reasoning for one insertion gives mean error $C_L g^2 [\beta_L(B) B + gB^3] W_f$.

Literal radius conversions and comparison of cutoff sets. At physical fine spacing η, raw bond logarithm and physical potential obey $a_b = \eta A_b$. In canonical tangent variables $a_b = g Z_b$.

Consequently the conversions are

$$|a_b| < \alpha_b \iff |Z_b| < \alpha_b/g, \qquad |\ell_b A_b| < r_b \iff |Z_b| < \frac{\eta r_b}{g\ell_b}. \tag{GCB50}$$

The second relation refers to the tangent physical potential; ℓ_b is the actual scale multiplying it in that field bound. For the nonlinear chart use (GCB47): its raw field differs from gZ by at most $C_L g^2 B^2$ on the event in question. Any comparison with a cutoff on the nonlinear field must charge this error and preserve a stated radius margin.

A free-coordinate box and the complete physical tangent product box are not equal events. With $c_L = \|\mathsf{C}_0\|_{\infty\to\infty} \geq 1$, one has only

$$\{\|Y\|_\infty < B/c_L\} \subset \{\|\mathsf{C}_0 Y\|_\infty < B\} \subset \{\|Y\|_\infty < B\}. \tag{GCB51}$$

The first inclusion uses the actual row norm of the pivot map; the second uses its identity free restriction. Inclusion alone does not compare the corresponding normalized conditional expectations. The proof in this section works directly with the middle, complete physical event, so requires no false identification of these boxes. If an actual integration rule instead cuts only free coordinates, apply the abstract theorem to those actual independent-coordinate blocks and carry the observable coefficient map separately; it is a different cutoff law with its own explicitly specified covariance and radii.

For growing canonical radius $B = c_*(\log\theta)^s$, $s > 0$, and $g = \theta^{-1/2}$, every fixed power of $\log\theta$ times $\beta_L(B)$ tends to zero, as does every fixed such power times g. The chart condition $gB < r_L/2$ holds eventually. Thus for support radii bounded by any fixed power of $\log\theta$, the product of the bracket and localized volume factors in (GCB49) tends to zero for test functions with bounded sup norms, uniformly in torus volume. This is an absolute error relative to the displayed insertion norms. Converting it into a relative error for a specified two-point coefficient also requires a proved lower bound for that coefficient in the same test class.

Local limits and the conditional integration-by-parts identity. The finite-volume cutoff laws have Gaussian-dominated moments for every fixed local linear statistic by (GCB20). In the fixed-step setting their marginal variances are bounded uniformly by (GCB37). These bounds give tightness on each finite collection of bond coordinates. A diagonal subsequence on the countable lattice therefore has compatible finite-dimensional limits. Uniform higher local moments give uniform integrability, so the local polynomial estimates above pass to every such limit. This conclusion does not assert uniqueness of the limit, and does not identify it with conditioning an infinite-volume Gaussian on a global event of positive probability. It only uses uniform finite-volume bounds.

The boundary proof also identifies its precise obstruction in a general conditional law. If in actual free coordinates, with $G > 0$, its complete density is $Z_e^{-1}\chi(CY)\exp[-\frac{1}{2}Y^{\mathsf{T}}G^{-1}Y + R_e(Y)]$, integration by parts has, in addition to the boundary defect, the force term

$$\mathbb{E}_{\nu_e}[P(CY)\,(CG\nabla R_e(Y))_i]\,. \tag{GCB52}$$

This follows by differentiating the entire density, with its actual normalizer; all integrability and chart-domain conditions must hold. An arbitrary exterior linear force can translate the law toward the boundary, and an arbitrary old activity can destroy the even log-concavity used in (GCB8). Neither its small Hessian alone nor the Gaussian boundary theorem bounds these two effects.

The same radial argument would remain valid for a further even log-concave factor in the physical variables, because that factor can be included in the joint log-concave integrand defining $w_{b,\delta}$. Its new force term still has to be bounded in the complete integration-by-parts identity. No such convexity, symmetry or force bound is asserted for the resummed interacting conditional density.

The interacting application additionally requires the complete density, exterior averaging and old-activity force estimates of Section 44, and comparison with the full free-curvature reference.

46. A one-field insertion bound uniform in the outside fields

An estimate for each correlation function does not automatically give an estimate in the Hilbert norm of an arbitrary polynomial of fields. The argument below obtains that stronger norm by repeated reflections at a fixed lattice spacing, before taking a continuum limit. Its analytic input is the ordinary scalar source upper bound and floor, not a typed insertion estimate.

Assume the all-mirror invariance and reflection positivity of every periodic Wilson DLR state, the ordinary torus source hypothesis (ST5), and the fixed-layer rooted domain geometry (GS1)–(GS3). The localization size is $M = L^{m_0}$, with L odd. Torus limits use the cofinal compatible family $\ell = 2L^m$, $m \geq m_0$. Every scale cube Ma_s then tiles the torus, since $\ell/(Ma_s) = 2L^{m-m_0+s}$. Every fixed board of power-of-L side, and every fixed coarser such board, tiles all sufficiently large tori in this family. No scale terms are omitted.

A raw-algebra reflection argument. Fix the spacing $a = L^{-K}$ and a periodic state ρ. Take a board cell C of side $l = L^j \geq a$, centered at a lattice site, with lower time face t. Its face planes are link mirrors: l/a is odd. Its middle time plane is a site mirror. Both are supplied by the all-mirror theorem. For bounded finite-support functions above t put

$$\langle B, A\rangle_t = \rho(\overline{\theta_t B} A), \qquad \|A\|_t^2 = \langle A, A\rangle_t. \tag{UI1}$$

For a bounded complex cell function D, let $q_D = \limsup_k p_{l,\ell_k}(D)$ along a compatible cofinal sequence. Define, on the raw half-space algebra,

$$T_D A = D\, \tau_{l e_4} A, \qquad D^* = \overline{\theta_{t+l/2} D}.$$

Reflection and translation invariance give the formal adjoint identity $\langle B, T_D A\rangle_t = \langle T_{D^*} B, A\rangle_t$. No bounded operator on a quotient has been assumed.

Put $K_D = T_{D^*} T_D$ and $m_j = \langle A, K_D^j A\rangle_t$. Formal adjoints and reflection Cauchy–Schwarz imply

$$m_{2r} = \|K_D^r A\|_t^2, \qquad m_{2r+1} = \|T_D K_D^r A\|_t^2, \qquad m_j^2 \leq m_{j-1} m_{j+1}. \tag{UI2}$$

For odd $j = 2r+1$ the last inequality uses the two vectors $K_D^r A, K_D^{r+1} A$; for even $j = 2r \geq 2$ it uses $T_D K_D^{r-1} A, T_D K_D^r A$. Thus it is a raw semidefinite-form calculation, not an application of a spectral theorem to an operator whose boundedness is still to be proved. If $m_0 = 0$, Cauchy–Schwarz already gives $m_1 = 0$. Otherwise log-convexity gives

$$m_1^N \leq m_0^{N-1} m_N. \tag{UI3}$$

The expression for m_N consists of the left endpoint $\overline{\theta_t A}$, a column of $2N$ alternating translated copies of D^* and D, and the right endpoint $\tau_{2Nl e_4} A$. Choose a coarse side $l' = L^r l$ large enough to contain A in a coarse half-space cell above t, spatially aligned with C. The oddness of L^r makes all coarse faces fine-board faces. Take N through sufficiently large multiples of L^r.

On this coarse grid retain fixed endpoint zones containing both endpoint supports and, if convenient, a fixed number c_A of the adjacent column factors. Bound the complete endpoint functions by their fixed-cutoff supremum norms. This yields a finite constant $C(a, A, D)$ independent of N. In the middle, every occupied coarse cell is a product of fine cell functions only. Fine-to-coarse chessboard refinement and product domination (ST3) apply there. Grid-preserving translates and reflected conjugates of D have the same fine seminorm. Therefore, for every $\epsilon > 0$,

$$m_N \leq C(a, A, D)(q_D + \epsilon)^{2N - c_A}. \tag{UI4}$$

The volume limit is taken first for each fixed N. The estimate uses no assertion $p(AD) \leq \|A\|_\infty p(D)$: a mixed endpoint block, if present, is bounded in its entirety by its supremum

norm. The endpoint zones contain only a bounded number of coarse cells; their factor count and supremum bound stay fixed as the column grows.

Taking the Nth root in (UI3)–(UI4), then sending $N \to \infty$ and finally $\epsilon \downarrow 0$, gives

$$\|T_D A\|_t \le q_D \|A\|_t, \qquad |\rho(\overline{\theta_t B} D \tau_{le_4} A)| \le q_D \|B\|_t \|A\|_t. \tag{UI5}$$

The same order of limits proves this when $q_D = 0$; no negative power of q_D is introduced. It also proves that null vectors are mapped to null vectors. Only now may the map be regarded as a bounded operator on the Hilbert quotient. Endpoint constants may depend on the number of fields inside A, but they disappear before (UI5). The final constant does not depend on that number, on B, or on their polynomial coefficients.

One ordinary source and a common centering. Let h be real and supported in the ball of radius $l/4$ around the cell center. For $a \le l/64$, apply the three-range count in (GS4) with board side $\eta = l$, support radius $\delta = l/4$ and the same face margin. Since $\eta/\delta = 4$, its face cost is bounded uniformly as $l \downarrow 0$. Coarse scales use the global count divided by the number of cells, not a separate root count at every source copy. Upper stability and the floor cost only Cl^4. Thus, for $l \le 1$, there are holomorphic univariate normalizers with

$$E_{h,\ell}(0) = 0, \qquad p_{l,\ell}\big(e^{z\mathcal{O}_a(h) - E_{h,\ell}(z)}\big) \le Q_0, \qquad |z| \|h\|_\sharp < r_0, \tag{UI6}$$

where $Q_0, r_0 > 0$ are independent of the spacing and cell side. They are also independent of the surrounding fields. At a fixed cutoff these normalizers are locally bounded uniformly in the torus, so a compatible Montel subsequence gives one E_h, independently of every infinite-volume state.

The remaining coarse mesh ratios can also be included. If $3 \le l/a < 64$, the ball and its contributing plaquettes lie wholly inside the cell, and $\|\mathcal{O}_a(h)\|_\infty \le C 64^4 \|h\|_\infty$. Set $E_{h,\ell} = 0$ and enlarge Q_0 in (UI6). The only smaller allowed ratio is $l/a = 1$; then a radius-$a/4$ ball about a lattice site contains no plaquette barycentre, so $\mathcal{O}_a(h) = 0$. Consequently (UI6) covers every allowed ratio, with the same constants.

For fixed z put $D_z = e^{z\mathcal{O}_a(h) - E_h(z)}$. Homogeneity of the seminorm and the locally uniform convergence of the normalizers give $q_{D_z} \le Q_0$. Apply (UI5) and then Cauchy's formula on the one source disc. The expectation is holomorphic, because at fixed cutoff all observables are bounded. We obtain

$$\big|\rho\big(\overline{\theta_t B}[\mathcal{O}_a(h) - E_h'(0)]\tau_{le_4} A\big)\big| \le (Q_0/r_0)\|h\|_\sharp \|B\|_t \|A\|_t.$$

Taking $A = B = 1$ in a second arbitrary periodic state ν bounds $|\nu(\mathcal{O}_a(h)) - E_h'(0)|$ by the same expression without the outside norms. The case $D = 1$ of (UI5) handles the scalar error. Hence

$$\big|\rho\big(\overline{\theta_t B}[\mathcal{O}_a(h) - \nu(\mathcal{O}_a(h))]\tau_{le_4} A\big)\big| \le C\|h\|_\sharp \|B\|_t \|A\|_t. \tag{UI7}$$

Both states are arbitrary. Large cutoff-dependent normalizer derivatives cancel; uniform moments of perturbed Gibbs states have not been assumed. In using (UI5) for D_z, endpoint supremum bounds may depend on z and the cutoff. They disappear pointwise before the uniform bound $q_{D_z} \le Q_0$ is used in Cauchy's formula.

Continuum passage with fixed support gaps. Suppose $a_j \downarrow 0$ and two periodic states ρ_j, ν_j at each datum give convergent centered separated moments

$$\sigma_j(f_1, \dots, f_n) = \rho_j \prod_i [\mathcal{O}_{a_j}(f_i) - \nu_j(\mathcal{O}_{a_j}(f_i))] \longrightarrow \sigma(f_1, \dots, f_n).$$

Use the uniform all-state local moment bounds already proved to obtain equicontinuity under small C^1 test changes. In particular, on each fixed separated tuple, $\sigma_j(\tau_{v_j} X) \to \sigma(X)$ whenever $v_j \to 0$. This follows directly by multilinear telescoping and $\|\tau_v f - f\|_\sharp \to 0$ for compact C^1 functions. The constants in this passage may depend on the fixed tuple and its length; they do not enter the constant in (UI7). Approximating a fixed real translation by lattice translations

also proves translation invariance of σ, since both states in each datum are lattice-translation invariant.

Let F, G be arbitrary finite complex linear combinations of field products whose entries have compact supports strictly in positive time and are separated within each product. Define $\|[F]\|_\sigma^2 = \sigma(\theta\overline{F} \cdot F)$. This is nonnegative as a limit of the lattice reflection forms. For a fixed power $l \le 1$ and a real h supported in $\bar{B}((\vec{y}, 0), l/4)$, choose the nearest spatial lattice point $\vec{y}_j$ and put

$$v_j = (\vec{y}_j - \vec{y}, a_j/2), \quad t_j = a_j/2, \quad x_{*,j} = (\vec{y}_j, l/2 + a_j/2).$$

For all sufficiently fine cutoffs l/a_j is odd. Therefore $x_{*,j}$ is a lattice site and the cell faces are link mirrors. The shifted inserted test is explicitly $h_j = \tau_{v_j + (l/2)e_4} h$, supported around that cell center. The identity $\theta_{t_j} \tau_{v_j} = \tau_{v_j} \theta_0$ identifies the lattice reflected norm tuples with the corresponding small translations of the continuum norm tuples. No half-mesh translation invariance of ν_j is being assumed.

Apply (UI7) to the centered lattice polynomials associated to $\tau_{v_j} F$ and $\tau_{v_j} G$. All insertion and norm tuples have fixed positive support gaps for large j. Their values converge by the preceding equicontinuity, including when a limiting Hilbert norm is zero. Thus

$$|\sigma(\theta\overline{G} \cdot (\tau_{(l/2)e_4} h) \cdot \tau_{le_4} F)| \le C \|h\|_\sharp \|[G]\|_\sigma \|[F]\|_\sigma. \tag{UI8}$$

The constant is independent of the numbers of entries and the finite polynomial coefficients of F, G. Arbitrary centering states ν_j are retained throughout.

For completeness time translations are contractions in these norms. At fixed spacing, reflection Cauchy–Schwarz gives $\|\tau_u A\|_t^2 \le \|A\|_t \|\tau_{2u} A\|_t$ for every positive lattice time u. Iteration against $\|\tau_{2^N u} A\|_t \le \|A\|_\infty$ proves contraction. The same fixed-gap limit passage proves it for every positive real time in the continuum. It uses no bounded insertion operator in advance.

A C^1 insertion estimate near the time mirror. We record the form needed independently of any plate or angular argument. Let $0 < \eta \le 1$, and let $f \in C_c^1$ be supported in $\{\eta \le x_4 \le 3\eta\}$, with support diameter at most R. Take the largest power l of L below η and partition f at mesh cl into pieces in radius-$l/4$ balls. There are at most $C_L(1 + R)^3 \eta^{-3}$ pieces: only the three spatial directions have unbounded length in units of η. Since f vanishes on both time faces, $\|f\|_\infty \le \eta \|\partial_4 f\|_\infty$. This cancels the $1/l$ from the partition derivative, because $\eta/l \le L$. Each piece has sharp norm at most $C\|f\|_{C^1}$.

Translate a piece to the middle of its side-l time slab and use (UI8). Its center time lies between $\eta - l/4$ and $3\eta + l/4$; both outside factors therefore retain a positive margin of at least $\eta - 3l/4$. Time contraction absorbs those margins. Summing gives

$$|\sigma(\theta\overline{G} \cdot (f) \cdot \tau_{4\eta e_4} F)| \le C(1 + R)^3 \eta^{-3} \|f\|_{C^1} \|[G]\|_\sigma \|[F]\|_\sigma.$$

Taking $G = (f) \cdot \tau_{4\eta e_4} F$ and dividing its norm proves the field-operator estimate; splitting real and imaginary parts handles complex f. We may weaken η^{-3} to η^{-4} and write

$$\|[(f) \cdot \tau_{4\eta e_4} F]\|_\sigma \le C(R) \eta^{-4} \|f\|_{C^1} \|[F]\|_\sigma. \tag{UI9}$$

Theorem 46.1 (Uniform one-field insertion)**.** *Under the stated ordinary source, rooted-domain and lattice-symmetry hypotheses, every limit just described satisfies (UI8)–(UI9). Choose a fixed $0 < c < 1/64$ and let $0 < \eta \le 1$. If a real $k \in C_c^1$ is supported in a ball of radius $c\eta$ centered at y, $|y_4| \le \eta/8$, and all entries of F, G lie strictly above $\eta/4$, then*

$$|\sigma(\theta\overline{F} \cdot (k) \cdot G)| \le C\eta^{-4} \|k\|_\sharp \|F\|_{(\eta/4)} \|G\|_{(\eta/4)}, \quad \|F\|_{(\eta/4)} = \|[\tau_{-(\eta/4)e_4} F]\|_\sigma. \tag{UI10}$$

The constant depends only on the ordinary source and localization constants and the fixed geometric choices. It is independent of the defining sequence, the states, support positions, and the numbers of fields in the surrounding finite complex polynomials.

Indeed put $\xi = c\eta$ and translate the insertion in time by $2\xi - y_4$, placing it in $[\xi, 3\xi]$. After this translation the left and right polynomials are, respectively, $\tau_{y_4-2\xi}F$ and $\tau_{-y_4-2\xi}G$, the latter followed by $\tau_{4\xi e_4}$. Each is a positive time translate of its $\eta/4$-downshift by at least $\eta/8 - 2\xi > 0$. Apply (UI9), reflection Cauchy–Schwarz and contraction, absorbing the fixed factor c^{-4}. This proves (UI10). The theorem asserts the C^1 insertion bound only. It does not assert a plate density theorem, an L^1 insertion bound, or full rotational covariance.

47. Analytic insertion densities and the plate bound

This section derives an L^1 insertion bound from the proved C^1 Hilbert insertion bound. All operators are regularized by positive Euclidean time. No unsandwiched point field is presumed to exist. The exact spectral cube cone is used, but rotational covariance is not.

Let σ be a symmetric, real, time-reflection-invariant, translation-invariant functional on admissible separated tests, with its reflection-positive Hilbert reconstruction. Write $\mathcal{D}$ for the dense classes of finite complex positive-time polynomials and $\Omega = [1]$. Use the probability normalization $\sigma(1) = 1$, so Ω is a unit vector fixed by all translations. Positive time translation is e^{-tH}, spatial translation is $U(\mathbf{x}) = e^{i\mathbf{x}\cdot\mathbf{P}}$, and these operators strongly commute. Assume their joint spectral measure is carried by

$$E \geq 0, \qquad |q_j| \leq E \quad (j = 1, 2, 3). \tag{PD1}$$

Also assume local C^1 continuity of the separated moments, as proved from the ordinary source bounds. This continuity is used for fixed finite polynomials only; none of its field-number constants enter the uniform bounds below.

The insertion input has fixed constants $C_I, c_I > 0$, $p \geq 0$, with $c_I < 1/64$. For $0 < \eta \leq 1$, let

$$I(F; h; G) = \sigma(\theta\overline{F} \cdot (h) \cdot G), \qquad \|F\|_{(\eta/4)} = \|[\tau_{-(\eta/4)e_4}F]\|.$$

Whenever the entries of F, G lie strictly above $\eta/4$ and h is real with support in a radius-$c_I\eta$ ball centered at y, $|y_4| \leq \eta/8$, the input is

$$|I(F; h; G)| \leq C_I\eta^{-p}\|h\|_\sharp \|F\|_{(\eta/4)}\|G\|_{(\eta/4)}. \tag{PD2}$$

The norm is $\|h\|_\sharp = \max(\|h\|_\infty, c_M\|\nabla h\|_\infty)$. Splitting real and imaginary parts permits complex h, at a fixed factor two. The preceding insertion theorem supplies $p = 4$; keeping p general makes the additional five powers transparent.

Bounded forms, real translations and holomorphy. For h supported in $\bar{B}(0, c_I\eta)$, define the form on raw positive-time polynomials by

$$b_{h,\eta}(A, B) = I(\tau_{(\eta/4)e_4}A; h; \tau_{(\eta/4)e_4}B).$$

By (PD2) it descends through null vectors and extends to a bounded sesquilinear form on the Hilbert space. Riesz representation gives a unique bounded operator $B_{h,\eta}$, with

$$\|B_{h,\eta}\| \leq C\eta^{-p}\|h\|_\sharp, \quad B_{h,\eta}[A] = [(\tau_{(\eta/4)e_4}h) \cdot \tau_{(\eta/2)e_4}A]. \tag{PD3}$$

The vector identity follows by testing against the dense polynomial classes and translating the insertion form. Its displayed product is admissible: the insertion is strictly positive in time and is separated from the shifted entries of A. Thus this identity proves the action of the bounded operator; it does not postulate a point field on the quotient.

For complex $z = (z_4, \mathbf{z})$ put

$$T(z) = \int e^{-z_4E + i\mathbf{z}\cdot\mathbf{q}}\, d\Pi(E, \mathbf{q}).$$

The spectral cone gives $\|T(z)\| \leq 1$ if $\operatorname{Re} z_4 \geq \sum_j |\operatorname{Im} z_j|$. It is operator-norm holomorphic in the strict interior. Indeed, on a neighbourhood with margin $\epsilon > 0$, derivatives of total order m of

the integrand are bounded by $E^m e^{-\epsilon E}$. Its supremum is at most $m!\epsilon^{-m}$. The exponential Taylor series consequently converges uniformly in spectral variables on smaller complex neighbourhoods, so its spectral integrals converge in operator norm and may be differentiated there.

Now take h supported in $\bar{B}(0, c_I\eta/2)$, and set $\Phi = [\tau_{-(\eta/4)e_4}F]$, $\Psi = [\tau_{-(\eta/4)e_4}G]$. For real x, $|x_4| < \eta/8$, direct use of (PD3) gives

$$I(F;\tau_x h;G) = \langle \Phi, M_{h,\eta}(x)\Psi\rangle, \quad M_{h,\eta}(z) = T(\tfrac{\eta}{8}e_4 + z)B_{h,\eta/2}T(\tfrac{\eta}{8}e_4 - z). \tag{PD4}$$

For clarity, the real time factors in this formula are $e^{-(\eta/8+x_4)H}$ and $e^{-(\eta/8-x_4)H}$; the spatial factors are $U(\mathbf{x})$ and $U(-\mathbf{x})$. Substitution in (PD3), followed by one common time translation, recovers exactly $\theta\overline{F}\cdot(\tau_x h)\cdot G$. Both factors in (PD4) are contractions on

$$\mathcal{O}_\eta = \{z : |\operatorname{Re} z_4| + \textstyle\sum_{j=1}^3 |\operatorname{Im} z_j| < \eta/8\}. \tag{PD5}$$

Hence $M_{h,\eta}$ is operator-norm holomorphic there and $\|M_{h,\eta}(z)\| \le C\eta^{-p}\|h\|_\sharp$. There is no restriction on real spatial translations.

A scaled compact parametrix, with its distribution proof. In four dimensions let $G(x) = -(64\pi^2)^{-1}|x|^2\log|x|$, with value zero at the origin. It is C^1. For $x \neq 0$,

$$\Delta(|x|^2\log|x|) = 8\log|x| + 6, \quad \Delta\log|x| = 2|x|^{-2}, \quad \Delta|x|^{-2} = 0.$$

The first two identities also hold as distributions: Green's formula on $\{|x| > \epsilon\}$ has boundary errors tending to zero, respectively of orders $O(\epsilon^4|\log\epsilon|)$ and $O(\epsilon^2)$, up to smaller terms. In the third identity the boundary term is nonzero. Against a compact smooth test f, the inner normal is $-\partial_r$, so the term containing $f\partial_n|x|^{-2}$ tends to $-2|S^3|f(0) = -4\pi^2 f(0)$; the other boundary term is $O(\epsilon)$. Thus

$$\Delta|x|^{-2} = -4\pi^2\delta_0, \qquad \Delta^3 G = \delta_0. \tag{PD6}$$

Choose real $\chi \in C_c^\infty(B(0,1))$ equal to one near zero. Put $n = \chi G$ and $s = \delta_0 - \Delta^3 n$. Since $\chi = 1$ near the singularity, (PD6) cancels its delta term. Every remaining Leibniz term contains a derivative of χ supported away from zero, where G is smooth. Therefore $s \in C_c^\infty(B(0,1))$. For $0 < r \le 1$ define

$$n_r(x) = r^2 n(x/r) = -\frac{\chi(x/r)|x|^2\log(|x|/r)}{64\pi^2}, \qquad s_r(x) = r^{-4}s(x/r). \tag{PD7}$$

Distributional scaling in four dimensions gives $\Delta^3 n_r = \delta_0 - s_r$: six derivatives contribute r^{-6}, the prefactor contributes r^2, and the scaled delta contributes r^4. Ordinary differentiation gives

$$\|n_r\|_\sharp \le Cr, \qquad \|s_r\|_\sharp \le Cr^{-5}. \tag{PD8}$$

Both supports lie in $\bar{B}(0,r)$. The logarithm is $\log(|x|/r)$, not $\log|x|$ after simply cutting off G. The subtracted quadratic polynomial removes a spurious $|\log r|$ in (PD8).

For smooth compact k, convolution with (PD6)–(PD7) gives

$$k = (\Delta^3 k) * n_r + k * s_r. \tag{PD9}$$

This can be verified directly against compact smooth tests by Fubini and distributional integration by parts. Alternatively, the compact C^1 kernel n_r is integrable, so all derivatives of its convolution with k may be placed on k under the integral; the distribution identity then yields (PD9). Both sides are C^1, so their distributional equality is also pointwise equality.

A bounded analytic density and the L^1 bound. Fix $\epsilon = \min(c_I/8, 1/1024)$ and put $r = \epsilon\eta$. Both n_r and s_r fit the source ball used in (PD4). Define the operator-valued holomorphic function

$$V_\eta(z) = \Delta_z^3 M_{n_r,\eta}(z) + M_{s_r,\eta}(z). \tag{PD10}$$

On the smaller domain $|\operatorname{Re} z_4| + \sum_j |\operatorname{Im} z_j| < \eta/32$, a polydisc of radius $\eta/128$ lies in $\mathcal{O}_\eta$. Cauchy's inequalities bound the six derivatives by $6!(128/\eta)^6$ times the bound in (PD4). The sum of coefficients of Δ^3 is 4^3. Equations (PD8) therefore give

$$\|V_\eta(z)\| \le C\eta^{-p}(\eta^{-6}r + r^{-5}) \le C\eta^{-(p+5)}. \tag{PD11}$$

Every constant is independent of the surrounding polynomials and their numbers of entries.

Let $k \in C_c^\infty$ have support in $|x_4| < \eta/64$. For a continuous compact ϕ and a compact C^1 kernel q, Riemann sums for $\phi * q = \int \phi(x)\tau_x q\, dx$ converge in C^1, with common compact support. Local C^1 continuity of the insertion form consequently implies $I(F; \phi * q; G) = \int \phi(x) I(F; \tau_x q; G)\, dx$. All convolution supports here stay strictly between the two outside time supports. Applying this observation to (PD9), then integrating six derivatives off k onto the analytic matrix element in (PD4), proves

$$I(F; k; G) = \int k(x)\langle \Phi, V_\eta(x)\Psi\rangle\, dx. \tag{PD12}$$

The integrations by parts have no boundary terms, since k is compactly supported in the open strip. This proves the density identity, not just an inequality.

Mollification carries (PD12) to compact C^1 tests in the strip: the left side converges by local C^1 continuity and the right side by L^1 convergence and (PD11). Thus it has the unique continuous extension to every L^1 function on that strip, with

$$|I(F; k; G)| \le C\eta^{-(p+5)}\|k\|_{L^1}\|F\|_{(\eta/4)}\|G\|_{(\eta/4)}. \tag{PD13}$$

For an L^1 test this formula defines the extended insertion; no pointwise field interpretation is required. The operator weak integral $\int k(x)V_\eta(x)\, dx$ is well-defined by its bounded sesquilinear forms and has the same norm bound. For compact C^1 tests it agrees with the regularized field product in (PD3). If two choices of η admit the same outside polynomials and test strip, their forms agree first on compact smooth tests and then on the common L^1 domain by density. Thus the extension is compatible on intersecting pulled-down domains.

Different compact parametrices give the same density on the real strip: (PD12) makes their continuous matrix-element differences zero against every compact smooth test. They therefore agree pointwise. Their holomorphic continuations agree on connected overlapping domains, by uniqueness from an open set of real points, first in scalar matrix elements and then as bounded operators.

Plates and compatible point-plate operators. Split $x = (t, u, x_\perp)$, with $x_\perp \in \mathbb{R}^2$. For $\varphi \in L^1(\mathbb{R}^2)$ set

$$V_{\varphi,\eta}(t, u) = \int \varphi(x_\perp)V_\eta(t, u, x_\perp)\, dx_\perp. \tag{PD14}$$

The weak integral is a bounded operator, of norm at most $C\eta^{-(p+5)}\|\varphi\|_1$. It is operator-norm holomorphic for $|\operatorname{Re} t| + |\operatorname{Im} u| < \eta/32$. Indeed the integrands are uniformly bounded and holomorphic on smaller neighbourhoods, so their locally uniformly convergent Taylor series may be integrated. For an L^1 plane profile α supported in $|t| < \eta/64$, Tonelli and (PD12)–(PD13) give

$$|I(F; \alpha \otimes \varphi; G)| \le C\|\varphi\|_1\eta^{-\wp}\|\alpha\|_1\|F\|_{(\eta/4)}\|G\|_{(\eta/4)}, \qquad \wp = p + 5. \tag{PD15}$$

Profiles may extend arbitrarily far in the spatial coordinate u. For smooth compact profiles and labels this is the original insertion; all other stated L^1 cases are its continuous extension.

For $0 < a \le 1/4$ define the regularized point-plate operator

$$B_{\varphi;a} = V_{\varphi,4a}(0,0), \qquad \|B_{\varphi;a}\| \le Ca^{-\wp}\|\varphi\|_1. \tag{PD16}$$

It is the operator-norm limit of the regularized insertions of $\alpha_\epsilon \otimes \varphi$, where α_ϵ is a unit-integral smooth plane approximate identity at zero. This follows from the operator-norm continuity in (PD14). For $0 < a < b \le 1/4$,

$$B_{\varphi;b} = e^{-(b-a)H} B_{\varphi;a} e^{-(b-a)H}. \tag{PD17}$$

To prove it, first use a compact smooth plate supported in the central strips of both scales. The vector formula (PD3), or its dense matrix-element identity, proves the relation for its two regularized operators. Approximate the plane point by α_ϵ and pass in operator norm. Finally approximate an L^1 label by smooth compact labels using (PD15). This proves compatibility on all the stated domains rather than assuming an unbounded field common to the scales.

The same calculation gives real translation compatibility. For $0 < a' < a \le 1/4$ and $|t| < a/16$,

$$V_{\varphi,4a}(t,u) = e^{-(a-a'+t)H} U(ue_1) B_{\varphi;a'} U(-ue_1) e^{-(a-a'-t)H} \tag{PD18}$$

whenever $a - a' > |t|$. One proves this first for a sufficiently narrow smooth approximate identity, using the real translation calculation of (PD4), and then passes in operator norm. The right side also supplies the corresponding holomorphic continuation. These identities are suitable for composing independently regularized plate insertions.

Theorem 47.1 (Analytic density and plate bound)**.** *Under (PD1)–(PD2) and the stated local continuity, the insertion has the analytic operator density (PD10)–(PD12), the uniform L^1 bound (PD13), and the plate bound (PD15) with exponent $\wp = p + 5$. The point-plate operators (PD16) obey the compatibility and translation identities (PD17)–(PD18). All constants are independent of the numbers of outside fields. No additional typed source estimate is used.*

As a useful spectral consequence, suppose φ is real. Reflection of real, time-even approximate identities proves $B^*_{\varphi;a} = B_{\varphi;a}$. Let $v_{\varphi,a} = B_{\varphi;a}\Omega$ and let $\mu_{\varphi,a}$ be its finite positive joint spectral measure for (H, P_1). Since $H\Omega = 0$, (PD17) gives $\mu_{\varphi,b} = e^{-2(b-a)E}\mu_{\varphi,a}$. The positive measure $\mu_\varphi = e^{2aE}\mu_{\varphi,a}$ is thus independent of a, is finite on bounded-energy sets, and may have infinite total mass. For $t > 0$ choose $0 < a < \min(t/2, 1/4)$ and put

$$K_\varphi(t,u) = \langle v_{\varphi,a}, e^{-(t-2a)H + iuP_1} v_{\varphi,a}\rangle = \int e^{-tE+iuq}\, d\mu_\varphi(E,q). \tag{PD19}$$

It is independent of a, finite, and its spectral measure is supported on $E \ge |q|$. The integral is holomorphic on $\operatorname{Re} t > |\operatorname{Im} u|$: on smaller neighbourhoods all differentiated integrands are bounded by a polynomial in E times a strictly decaying exponential, integrable by (PD19) at a smaller positive real time.

To identify this as the original two-plate kernel, take real plane approximate identities even in time, first with a smooth compact real label φ. The vectors $[(\tau_{ae_4}\alpha_\epsilon) \otimes \varphi]$ converge to $v_{\varphi,a}$ by (PD16). Their matrix elements in (PD19) are precisely the two-plate moments at time separation t, with the narrow profiles, by reflection and common translation. For compact profiles with a fixed positive time gap, choose a smaller than half that gap. Convolution approximates those profiles in C^1, while the approximating matrix elements are uniformly bounded by the two vector norm bounds. Fubini and dominated convergence therefore recover the integral of K_φ against the original profiles. Smooth compact approximation of an L^1 label uses the uniform bounds above. The kernel here is the full two-point moment kernel; for connected moments one applies the construction to the centered functional. This is a statement in the chosen coordinate frame. No diagonal-frame cone, contact value at $t = 0$, or rotational covariance is asserted here.

48. Analytic planar kernels and the finite angular band

This section constructs the analytic objects used by the angular argument. Its quantitative input is a plate estimate uniform in the numbers of outside fields. It proves neither that estimate nor the short-distance two-point estimate. In particular, the conclusions below do not by themselves establish rotation invariance of the Euclidean functional.

The assumptions and the meaning of a product. Write a point of Euclidean space as (y, z), where $y = (y^0, y^1) = (x_4, x_1)$ and $z = (x_2, x_3)$. Put $\theta(y^0, y^1) = (-y^0, y^1)$. A tuple is a finite ordered list of compactly supported C^1 test functions with pairwise disjoint supports. Products of tuples mean concatenation, whenever the concatenation still has disjoint supports. We use finite complex linear combinations of such tuples, including the empty tuple 1. The notation (h) denotes the tuple with its single entry h.

Assume the following properties of a centered functional σ. It is multilinear in the entries, symmetric, real on real entries, and invariant under simultaneous real translations and the hypercubic group. Its values satisfy $\sigma(1) = 1$ and $\sigma((h)) = 0$. On every fixed collection of separated compact support sets it is continuous for the C^1 norms, with a bound independent of the positions of those sets when their diameters and mutual distances are fixed. No bound on the growth of these constants with the number of entries is assumed in this section.

Let $\mathscr{A}_+$ denote the combinations of tuples supported strictly in $\{x_4 > 0\}$. Reflection positivity and the reconstruction give a Hilbert space $\mathcal{H}$, a dense class map $F \mapsto [F]$, and $\Omega = [1]$, such that

$$\langle [F], [G] \rangle = \sigma(\theta\overline{F} \cdot G), \qquad \|\Omega\| = 1. \tag{AK1}$$

The inner product is antilinear in the first entry. The reconstructed time and planar spatial translations have the actions

$$e^{-tH}[F] = [\tau_{te_4} F], \qquad e^{ixP}[F] = [\tau_{xe_1} F], \qquad t \geq 0, \tag{AK2}$$

where H, P are strongly commuting self-adjoint operators, $H \geq 0$, and

$$H \geq |P|, \qquad H\Omega = P\Omega = 0. \tag{AK3}$$

Only this one planar consequence of the exact cube cone is used.

For F supported above a height a define the pulled-down norm by

$$\|F\|_{(a)} = \|[\tau_{-ae_4} F]\|.$$

A label is a real $\varphi \in C_c^1(\mathbb{R}^2)$ in the transverse variables. A profile is a complex $\alpha \in C_c^1(\mathbb{R}^2)$. The corresponding plate is $\alpha \otimes \varphi$. Fix constants $0 < c < 1/4$ and $\wp > 0$. The plate input is that, for each label, a constant $C_\varphi < \infty$ satisfies

$$\begin{aligned} &|\sigma(\theta\overline{F} \cdot (\alpha \otimes \varphi) \cdot G)| \\ &\qquad \leq C_\varphi \eta^{-\wp} \|\alpha\|_1 \|F\|_{(\eta/4)} \|G\|_{(\eta/4)} \end{aligned} \tag{AK4}$$

for $0 < \eta \leq 1$, profiles supported in $|y^0| < c\eta$, and F, G supported strictly above $\eta/4$. This includes arbitrary finite combinations and arbitrary numbers of entries in F, G. The constant C_φ is the same for all these choices and all η. Every concatenation in this inequality has separated supports. The plate theorem obtained from the ordinary-source insertion estimate supplies precisely this input, with a fixed small c and $\wp = p + 5$.

Bounded plates and their padding. For a profile in the strip of (AK4) put $u = \eta/4$. On the dense classes define

$$B_\varphi[\alpha; \eta][G] = \big[(\tau_{ue_4}(\alpha \otimes \varphi)) \cdot \tau_{2ue_4} G\big]. \tag{AK5}$$

The plate on the right lies at times strictly between 0 and $2u$; the other entries lie strictly above $2u$. Thus the right side is an actual class in $\mathcal{H}$.

To prove that it depends only on $[G]$, test it against $[F]$ and translate the full tuple down by u. The resulting expression is

$$\langle [F], B_\varphi[\alpha;\eta][G]\rangle = \sigma\big(\theta\overline{\tau_{ue_4}F}\cdot(\alpha\otimes\varphi)\cdot\tau_{ue_4}G\big).$$

The two pulled-down norms on the right are $\|[F]\|$ and $\|[G]\|$. By (AK4), density and Hilbert duality,

$$\|B_\varphi[\alpha;\eta][G]\| \le C_\varphi \eta^{-\wp}\|\alpha\|_1\|[G]\|. \tag{AK6}$$

In particular null classes map to null classes. There is a unique bounded extension to the whole Hilbert space, with the same norm bound. No domain containing an unspecified number of fields remains in this statement.

The definitions also fix all padding conventions. If $\eta' \le \eta$ and the profile is supported in $|y^0| < c\eta'$, then, with $d = (\eta-\eta')/4$,

$$B_\varphi[\alpha;\eta] = e^{-dH}B_\varphi[\alpha;\eta']e^{-dH}. \tag{AK7}$$

Indeed both sides applied to $[G]$ give the right side of (AK5); density proves equality of the bounded operators. Realness, symmetry, reflection invariance and a translation by $2u$ similarly give

$$B_\varphi[\alpha;\eta]^* = B_\varphi[\overline{\alpha}\circ\theta;\eta]. \tag{AK8}$$

For example, conjugate the expression defining $\langle [G], B_\varphi[\alpha;\eta][F]\rangle$, reflect all its entries, translate by $2u$, and permute the entries. The resulting expression is $\langle [F], B_\varphi[\overline{\alpha}\circ\theta;\eta][G]\rangle$. The label remains unchanged because it is real and the reflection acts only on the plane.

The tube and its bounded translations. For $v = (v^0, v^1) \in \mathbb{C}^2$ use the linear coordinates $v_\pm = v^0 \pm iv^1$, and set

$$\mathbb{V} = \{\operatorname{Re} v_+ > 0, \operatorname{Re} v_- > 0\} = \{\operatorname{Re} v^0 > |\operatorname{Im} v^1|\}, \qquad v^* = (\overline{v^0}, -\overline{v^1}). \tag{AK9}$$

This is an open convex cone. Its real points are the half-plane $\mathbb{H} = \{v^0 > 0\}$. Both entrywise conjugation and the star preserve it, and $\operatorname{Re} v \in \mathbb{V}$ whenever $v \in \mathbb{V}$. These assertions follow directly from the two strict real linear inequalities in (AK9). Notice the distinct identities

$$(v^*)_\pm = \overline{v_\pm}, \qquad v^* = -\theta v \quad \text{for real } v.$$

The joint spectral measure of (H, P) is carried by $\{(E,q) : E \ge |q|\}$. Define on the closed tube

$$T(v) = \int e^{-v^0E + iv^1q}\, d\mathsf{P}(E,q). \tag{AK10}$$

Writing $E_\pm = (E \pm q)/2 \ge 0$ gives $-v^0E + iv^1q = -v_+E_- - v_-E_+$. Consequently

$$\|T(v)\| \le 1, \quad T(v)T(w) = T(v+w), \quad T(v)^* = T(v^*), \quad T(v)\Omega = \Omega. \tag{AK11}$$

The last assertion uses $H\Omega = P\Omega = 0$. Dominated convergence in the spectral measure of each vector proves strong continuity on the closed tube.

The map T is holomorphic in operator norm on $\mathbb{V}$. Here is a direct proof, which also records why the exact cone is appropriate. At a point with $\operatorname{Re} v_\pm \ge r > 0$, expand in increments $\Delta_\pm$. The coefficient of $\Delta_+^j\Delta_-^k$ is the bounded functional-calculus operator with multiplier

$$\frac{(-E_-)^j(-E_+)^k}{j!k!}e^{-v_+E_- - v_-E_+}.$$

Its norm is at most r^{-j-k}, since $x^je^{-x}/j! \le 1$ for $x \ge 0$. The resulting two power series converge in operator norm when $|\Delta_\pm| < r$. For real v with $v^0 \ge 0$, the reconstruction gives

$$T(v)[F] = [\tau_{v^0e_4 + v^1e_1}F]. \tag{AK12}$$

Let R_z be the complex rotation with the usual matrix

$$R_z = \begin{pmatrix} \cos z & -\sin z \\ \sin z & \cos z \end{pmatrix}.$$

In light coordinates $(R_z v)_\pm = e^{\pm iz} v_\pm$. In particular

$$R_{is}\mathbb{V} = \mathbb{V}, \qquad (R_{is}v)^* = R_{is}(v^*) \quad (s \in \mathbb{R}). \tag{AK13}$$

These are algebraic identities: the multipliers $e^{\mp s}$ are positive real numbers. For a tuple $v = (v_i)$ in the tube define

$$d_{\mathbb{V}}(v) = \min\{1, \min_{i,\pm} \operatorname{Re}(v_i)_\pm\}.$$

Then its exact transformation and a useful lower bound are

$$d_{\mathbb{V}}(R_{is}v) = \min\{1, \min_i e^{-s}\operatorname{Re}(v_i)_+, \min_i e^{s}\operatorname{Re}(v_i)_-\} \geq e^{-|s|} d_{\mathbb{V}}(v). \tag{AK14}$$

The same assertions hold in every quarter-turn frame $\mathbb{V}_\nu = R_\nu \mathbb{V}$, $\nu \in (\pi/2)\mathbb{Z}$. The rotations commute; R_ν is real; and the real points of $\mathbb{V}_\nu$ are $\{e_\nu \cdot v > 0\}$, where $e_\nu = R_\nu(1,0)$.

We will use the following elementary uniqueness principle. A holomorphic function on a connected open subset of $\mathbb{C}^N$ which vanishes on a nonempty relatively open subset of $\mathbb{R}^N$ vanishes identically. At an interior real point all real derivatives are zero, so its complex Taylor series is zero; overlapping balls then propagate the equality along a path. The same proof applies to Hilbert-valued holomorphic functions after testing against each vector. An antiholomorphic variable is treated by conjugating that variable first. This last use will always take place on domains stable under conjugation.

Products with smooth plane profiles. For $n \geq 1$ define the open convex domain

$$\mathbb{D}^+_{n,\eta} = \{\mathbf{x} : x_1 - \eta e_0/4 \in \mathbb{V},\ x_{i+1} - x_i - \eta e_0/2 \in \mathbb{V}\ (i < n)\}. \tag{AK15}$$

For profiles α_i in the strip $|y^0| < c\eta$ and labels φ_i, abbreviate $B_i = B_{\varphi_i}[\alpha_i; \eta]$ and put

$$\begin{aligned} \Psi_\eta[\vec{\alpha}](\mathbf{x}) = T(x_1 - \eta e_0/4)B_1 T(x_2 - x_1 - \eta e_0/2)B_2 \cdots \\ T(x_n - x_{n-1} - \eta e_0/2)B_n \Omega. \end{aligned} \tag{AK16}$$

It is Hilbert-valued holomorphic, and

$$\|\Psi_\eta[\vec{\alpha}](\mathbf{x})\| \leq \prod_{i=1}^n C_{\varphi_i} \eta^{-\wp} \|\alpha_i\|_1. \tag{AK17}$$

The definitions remain valid on the closed domain using strong continuity of T. When all x_i are real and the displayed non-strict time inequalities hold, direct substitution gives

$$\Psi_\eta[\vec{\alpha}](\mathbf{x}) = \Big[\prod_{i=1}^n \tau_{x_i}(\alpha_i \otimes \varphi_i)\Big]. \tag{AK18}$$

For clarity this is an induction from the right. The rightmost operator first puts its plate at time $\eta/4$. The preceding translation moves it by $x_n - x_{n-1} - \eta e_0/2$; the next plate operator translates that entry forward by $\eta e_0/2$ and inserts its own plate at $\eta e_0/4$. The two centers are therefore $\eta e_0/4$ and $\eta e_0/4 + x_n - x_{n-1}$. Iteration and the leftmost translation put all centers at the x_i. Spatial shifts follow the same addition rule. Time gaps and $c < 1/4$ ensure that every tuple in this induction is separated.

Define the smoothed scalar kernel on $U_\eta = (\eta e_0/2 + \mathbb{V})^{n-1}$ by

$$K_{n,\eta}[\vec{\alpha}](v) = \langle \Omega, B_1 T(v_1 - \eta e_0/2)B_2 \cdots T(v_{n-1} - \eta e_0/2)B_n \Omega\rangle. \tag{AK19}$$

It is holomorphic and bounded by the right side of (AK17). If the y_i are real and $y^0_{i+1} - y^0_i \geq \eta/2$, then

$$\sigma\Big(\big(\tau_{y_i}(\alpha_i \otimes \varphi_i)\big)_{i=1}^n\Big) = K_{n,\eta}[\vec{\alpha}](v(\mathbf{y})), \qquad v(\mathbf{y})_i = y_{i+1} - y_i. \tag{AK20}$$

Translate the first center to $(\eta/4, 0)$ and use (AK18); taking its scalar product with Ω proves the formula. Translation invariance then restores the original centers. Set $K_0 = 1$ and $K_1 = 0$, consistently with the empty and centered one-point moments.

Removing the plane smoothing. Fix η and choose real radial nonnegative smooth bumps $\beta_{i,j}$, each of mass one, whose support radii tend to zero. For all large j the bumps fit in $D(0, c\eta)$. The kernels $K_{n,\eta}[\vec{\beta}_j]$ have a common bound

$$C_\eta = \prod_i C_{\varphi_i} \eta^{-\wp n} \quad \text{on } U_\eta. \tag{AK21}$$

The bumps need not have the same radii or shapes in different entries. This freedom is needed later for mixed inner products.

Here are the details identifying every limit. Let b_i be C^1 profiles whose successive support time gaps exceed η. For all large j, their convolutions with $\beta_{i,j}$ have separated supports in fixed compact sets. The identity

$$b_i * \beta_{i,j} = \int b_i(y) \tau_y \beta_{i,j} \, dy$$

holds also as a C^1 Riemann integral. Indeed, for fixed j, the integrand and its first derivatives in the test variable are uniformly continuous on a common compact set. Riemann sums converge there in C^1, with their supports in the same compact neighborhood. Multilinearity, the assumed separated C^1 continuity, and successive Riemann summation in all entries give

$$\sigma\big(((b_i * \beta_{i,j}) \otimes \varphi_i)_i\big) = \int \prod_i b_i(y_i) \, K_{n,\eta}[\vec{\beta}_j](v(\mathbf{y})) \, d\mathbf{y}. \tag{AK22}$$

The integrands are continuous and compactly supported; thus Fubini and the successive integrals are legitimate.

A locally bounded family of holomorphic functions has locally uniformly convergent subsequences here. One can see this directly: Cauchy estimates on smaller polydiscs bound first derivatives, Arzelà–Ascoli gives convergence there, and a diagonal choice on a countable exhaustion gives convergence on all compact subsets. The Cauchy integral formula shows that the limit is holomorphic. Apply this observation to (AK21). Along a subsequence let the limit be K'. Since $b_i * \beta_{i,j} \to b_i$ in C^1, passage in (AK22) gives

$$\sigma\big((b_i \otimes \varphi_i)_i\big) = \int \prod_i b_i(y_i) K'(v(\mathbf{y})) \, d\mathbf{y}. \tag{AK23}$$

All differences in this integral lie in a compact subset of U_η, so locally uniform convergence suffices.

There is at most one holomorphic K' with this property. If two continuous densities give the displayed integrals, choose the b_i to be unit-mass bumps in successively smaller discs around any fixed real configuration with gaps $> \eta$. Their integrals converge to the values of the two densities there. These values coincide. Such differences contain the nonempty relatively open set $\{v \text{ real} : v_i^0 > \eta\}$ in the connected set U_η; the uniqueness principle proves equality on U_η. Consequently the entire smoothing family converges, not merely a subsequence. Otherwise a sequence staying away on a compact set would have a further subsequence with the same uniquely determined limit. Denote it by $K_{n,\eta}$.

The argument proves a useful strengthened conclusion: for any choice of unit-mass bumps in the different entries with maximal support radius tending to zero, the corresponding kernels

converge to the same $K_{n,\eta}$, locally uniformly on U_η. No relation between their rates of shrinkage is required.

If $\eta' < \eta$, the two limiting kernels agree on U_η. Both satisfy (AK23) for gaps $> \eta$, and the same uniqueness proof applies. Since $\bigcup_{\eta>0} U_\eta$ equals $\mathbb{V}^{n-1}$, this constructs a unique holomorphic function

$$K_n(v;\vec{\varphi}) \quad \text{on } \mathbb{V}^{n-1}.$$

It represents σ on every time-ordered family of profiles:

$$\sigma\big((\alpha_i \otimes \varphi_i)_i\big) = \int \prod_i \alpha_i(y_i) K_n(v(\mathbf{y});\vec{\varphi})\, d\mathbf{y}. \tag{AK24}$$

Choose η below all the support gaps to obtain this formula. It also proves reality at every real point of the tube, by testing against real profiles concentrated near the chosen points.

The bound retains exactly one factor per field:

$$|K_n(v;\vec{\varphi})| \le \Big(\prod_i C_{\varphi_i}\Big) d_{\mathbb{V}}(v)^{-\wp n}. \tag{AK25}$$

Indeed $v \in U_\eta$ whenever $\eta < \min\{1, 2\min_{i,\pm} \mathrm{Re}(v_i)_\pm\}$. Take the limit up to this upper endpoint in (AK21); that endpoint is at least $d_{\mathbb{V}}(v)$. Neither a derivative of a shrinking bump nor an estimate whose constant depends on outside field counts has entered this step.

Real overlap identities and the four frames. For a quarter turn ν define $K_n^{(\nu)}(v;\vec{\varphi}) = K_n(R_{-\nu}v;\vec{\varphi})$ on $\mathbb{V}_\nu^{n-1}$. Hypercubic invariance and the change of variables $y_i = R_\nu y_i'$ show that (AK24) holds with $K_n^{(\nu)}$ whenever the profiles are ordered in the direction e_ν. The rotations act only on profiles and leave the labels unchanged.

It is useful to account for permutations as well. For a permutation π of $\{1,\dots,n\}$ let

$$\mathcal{U}_{\nu,\pi} = \{\mathbf{z} \in (\mathbb{C}^2)^n : R_{-\nu}(z_{\pi(i+1)} - z_{\pi(i)}) \in \mathbb{V}\ (i < n)\}. \tag{AK26}$$

On it use K_n with these ordered differences and the correspondingly permuted labels. Each domain is open, convex, nonempty, invariant under simultaneous complex translations, and satisfies $\mathrm{Re}\,\mathbf{z} \in \mathcal{U}_{\nu,\pi}$ whenever $\mathbf{z} \in \mathcal{U}_{\nu,\pi}$.

Two of these kernels agree on every nonempty intersection. The intersection is convex and hence connected. It contains the real part of each of its points, so contains a relatively open real ball. At any real configuration in the intersection take small discs about its points which preserve both strict orders. The moment of profiles supported in these discs has both continuous kernel densities, by (AK24), the frame identity and symmetry of σ. Testing with unit-mass profiles shrinking to the points shows equality of the two densities at that configuration. The holomorphic uniqueness principle now gives equality on the whole complex intersection. This proves the overlap identity, rather than presupposing an arbitrary-angle rotation symmetry.

Thus the ordered kernels glue to a single holomorphic function $\mathbf{K}_n(\mathbf{z};\vec{\varphi})$ on

$$\mathcal{U}_n^{\mathrm{ord}} = \bigcup_{\nu,\pi} \mathcal{U}_{\nu,\pi}. \tag{AK27}$$

This union is connected. To check this explicitly, a real configuration can be assigned any prescribed strict time ordering and, independently, any prescribed strict abscissa ordering. Hence each 0-frame permutation domain intersects every $\pi/2$-frame permutation domain. These convex domains form a connected union. The other two frames give the same domains with their orderings reversed, so add no disconnected components. Its real points form

$$\mathrm{Ord}_n = \{\mathbf{y} : \text{all times are distinct or all abscissae are distinct}\}.$$

The function is translation invariant, symmetric in the labelled points, real on real configurations, and invariant under the quarter turns. These properties follow first from the corresponding moment densities on real open sets and then from uniqueness. Imaginary rotations preserve each domain $\mathcal{U}_{\nu,\pi}$ and hence preserve their union.

Point product vectors and their Gram kernel. Put

$$\mathbb{D}_n^+ = \{\mathbf{x} : x_1 \in \mathbb{V},\ x_{i+1} - x_i \in \mathbb{V}\ (i < n)\}. \tag{AK28}$$

This is the union of the nested $\mathbb{D}_{n,\eta}^+$ as η decreases to zero. For $n = 0$ there is one empty configuration and its vector will be Ω.

We prove strong convergence of the smooth product vectors, rather than analytically continuing an inequality. Fix η, choose a real radial unit-mass bump β_ϵ supported in $D(0,\epsilon)$, and take $\epsilon < c\eta$. Use it in every entry of (AK16), writing the result as $\Psi_{\eta,\epsilon}(\mathbf{x};\vec{\varphi})$.

If $\mathbf{x} \in \mathbb{D}_{n,\eta}^+$ and $\mathbf{y} \in \mathbb{D}_{m,\eta}^+$, define the list of differences

$$J(\mathbf{x},\mathbf{y}) = \big((x_n - x_{n-1})^*, \dots, (x_2 - x_1)^*, x_1^* + y_1, y_2 - y_1, \dots, y_m - y_{m-1}\big). \tag{AK29}$$

If $n = 0$, omit the left list and the central entry. If $m = 0$, omit the central entry and the right list. The labels of this list are $(\varphi_n, \dots, \varphi_1, \psi_1, \dots, \psi_m)$. Every entry belongs to $\eta e_0/2 + \mathbb{V}$: for the central one subtracting $\eta e_0/2$ gives the sum $(x_1 - \eta e_0/4)^* + (y_1 - \eta e_0/4)$ in the open cone. The other entries follow from star invariance.

For different radii ϵ, ϵ' the exact identity is

$$\begin{aligned} &\langle \Psi_{\eta,\epsilon}(\mathbf{x};\vec{\varphi}), \Psi_{\eta,\epsilon'}(\mathbf{y};\vec{\psi})\rangle \\ &\quad = K_{n+m,\eta}[\underbrace{\beta_\epsilon, \dots, \beta_\epsilon}_{n}, \underbrace{\beta_{\epsilon'}, \dots, \beta_{\epsilon'}}_{m}](J(\mathbf{x},\mathbf{y})), \end{aligned} \tag{AK30}$$

with the reversed left labels specified above. For real positions this is just (AK18), the Hilbert inner product, symmetry, and (AK20). Radiality and realness are important: $\theta\overline{\tau_x\beta_\epsilon} = \tau_{\theta x}\beta_\epsilon$ for real x. The left side is antiholomorphic in $\mathbf{x}$ and holomorphic in $\mathbf{y}$; the right side has exactly the same properties because the star is conjugate linear. The product domains are convex, conjugation stable and have real interior points. Mixed holomorphic uniqueness therefore proves (AK30) for complex positions too. For total length one both sides are zero; for length zero both equal one.

The strengthened smoothing conclusion after (AK23) shows that the right side converges to $K_{n+m}(J(\mathbf{x},\mathbf{y}))$ when both radii tend to zero, with no restriction on their relative rate. Convergence is uniform on compact sets of the position domains. In particular, for a compact subset of $\mathbb{D}_{n,\eta}^+$, expand

$$\|\Psi_{\eta,\epsilon}(\mathbf{x}) - \Psi_{\eta,\epsilon'}(\mathbf{x})\|^2$$

as its four mixed inner products. All four tend uniformly to the same diagonal kernel value; their signed sum tends to zero. This proves a locally uniform strong Cauchy property. The limit $\Psi_\eta(\mathbf{x};\vec{\varphi})$ is Hilbert-valued holomorphic: uniform convergence on smaller polydiscs passes the vector Cauchy formula and its power series to the limit.

These vectors are independent of η on overlaps. For any fixed sufficiently small physical bump and real positions in an overlap, both definitions give the same class in (AK18). Vector holomorphic uniqueness on the connected overlap proves equality for complex positions, and then the strong limits agree. We have therefore constructed

$$\Psi(\mathbf{x};\vec{\varphi}) \in \mathcal{H}, \qquad \mathbf{x} \in \mathbb{D}_n^+, \tag{AK31}$$

holomorphically, with the bound $\|\Psi(\mathbf{x};\vec{\varphi})\| \le \prod_i C_{\varphi_i}\eta^{-\wp}$ for every admissible η. On the domain where $v \in \overline{\mathbb{V}}$, the semigroup rule also gives

$$\Psi(\mathbf{x} + v;\vec{\varphi}) = T(v)\Psi(\mathbf{x};\vec{\varphi}). \tag{AK32}$$

It holds before removing the bumps and hence afterward.

For labelled positive configurations define

$$\mathrm{Gr}(\mathsf{x},\mathsf{y}) = K_{n+m}(J(\mathbf{x},\mathbf{y});\varphi_n,\dots,\varphi_1,\vec{\psi}).$$

Use the empty and one-point conventions already specified. The construction proves the exact equality

$$\mathrm{Gr}(\mathsf{x},\mathsf{y}) = \langle \Psi(\mathsf{x}), \Psi(\mathsf{y})\rangle. \tag{AK33}$$

Thus this is a positive semidefinite kernel at complex positions, for arbitrary finite families and different lengths, including the empty configuration.

Finally $R_{is}\mathbb{D}_n^+ = \mathbb{D}_n^+$, and (AK13) gives $J(R_{is}\mathbf{x}, R_{is}\mathbf{y}) = R_{is}J(\mathbf{x},\mathbf{y})$. Consequently

$$\begin{aligned}\mathrm{Gr}_s(\mathsf{x},\mathsf{y}) &:= K_{n+m}(R_{is}J(\mathbf{x},\mathbf{y}); \text{the same labels})\\ &= \langle \Psi(R_{is}\mathsf{x}), \Psi(R_{is}\mathsf{y})\rangle\end{aligned} \tag{AK34}$$

is positive semidefinite for every real s. This is an actual Gram representation; positivity has not been inferred from analytic continuation of a positive scalar function.

Narrow pencils and the entire angle function. Fix $0 < \phi_0 < \pi/4$ and let

$$\mathrm{Sec} = \{v \in \mathbb{R}^2 \setminus \{0\} : |\arg(v^0 + iv^1)| < \phi_0\}.$$

For $n \geq 2$, a pencil is a configuration with every consecutive difference in this sector. The cases $n = 0, 1$ have their separate constant kernels fixed above. Write $r_{\mathbf{y}} = \min_i |y_{i+1} - y_i| > 0$. For a pencil with differences v_i and labels $\vec{\varphi}$ put, on the vertical strip

$$S_\nu = \{z = t + is : |t - \nu| < \pi/2 - \phi_0\}, \qquad \nu \in (\pi/2)\mathbb{Z},$$

the function

$$a_{\mathbf{y}}^{(\nu)}(z) = K_n(R_{z-\nu}v;\vec{\varphi}). \tag{AK35}$$

It is well defined and holomorphic. In fact, writing $(v_i)_\pm = |v_i|e^{\pm i\phi_i}$ with $|\phi_i| < \phi_0$ gives

$$\mathrm{Re}(R_{t+is-\nu}v_i)_\pm = |v_i|e^{\mp s}\cos(\phi_i + t - \nu) > 0. \tag{AK36}$$

Only neighboring strips intersect. Their intersection is exactly

$$S_\nu \cap S_{\nu+\pi/2} = \{t + is : \nu + \phi_0 < t < \nu + \pi/2 - \phi_0\},$$

which is nonempty and connected. At a real angle there, all the rotated differences lie strictly in the quadrant between the two frame normals. Small discs about the rotated points are ordered in both frames. The proved real overlap identity therefore identifies the two values in (AK35) on this real interval. The one-variable identity theorem identifies the holomorphic functions throughout the intersection.

The strip width $\pi - 2\phi_0$ exceeds their spacing $\pi/2$. They cover the complex plane and glue to a unique entire function $a_{\mathbf{y}}(z;\vec{\varphi})$. The construction immediately gives

$$a_{\mathbf{y}}(z + \pi/2) = a_{\mathbf{y}}(z). \tag{AK37}$$

Indeed use the patch with center $\nu + \pi/2$ for the shifted argument and the patch with center ν for the original argument; the arguments of K_n are identical. For real t it equals $\mathbf{K}_n(R_t\mathbf{y};\vec{\varphi})$, since a suitable frame orders this rotated pencil. In particular it is real on the real axis, and $a_{\mathbf{y}}(is) = K_n(R_{is}v;\vec{\varphi})$.

For every real t choose a nearest quarter turn ν, so $|t - \nu| \leq \pi/4$. Set $c_0 = \cos(\phi_0 + \pi/4) > 0$. By (AK36), every light-coordinate margin is at least $c_0 r_{\mathbf{y}} e^{-|s|}$. Hence

$$|a_{\mathbf{y}}(t + is)| \leq \Big(\prod_i C_{\varphi_i}\Big) \min\{1, c_0 r_{\mathbf{y}} e^{-|s|}\}^{-\wp n} \leq A_{\mathbf{y},\vec{\varphi}} e^{\wp n|s|}, \tag{AK38}$$

where $A_{\mathbf{y},\vec{\varphi}} = \prod_i C_{\varphi_i} \max\{1, (c_0 r_{\mathbf{y}})^{-\wp n}\}$. This proves the claimed exponential type with its linear dependence on n; an uncontrolled prefactor depending on n would not change that conclusion.

The finite band and analytic harmonic coefficients. Define

$$c_m(\mathbf{y};\vec{\varphi}) = \frac{2}{\pi}\int_0^{\pi/2} a_{\mathbf{y}}(t;\vec{\varphi}) e^{-4imt}\,dt. \tag{AK39}$$

Cauchy's theorem on a rectangle of height s and periodicity give

$$c_m = \frac{2}{\pi}\int_0^{\pi/2} a_{\mathbf{y}}(t+is) e^{-4im(t+is)}\,dt, \qquad |c_m| \le A_{\mathbf{y},\vec{\varphi}} e^{4ms+\wp n|s|}.$$

The vertical sides cancel exactly. Sending s to $-\infty$ for $4m > \wp n$, and to $+\infty$ for $4m < -\wp n$, proves

$$c_m = 0 \quad \text{if } 4|m| > \wp n, \qquad a_{\mathbf{y}}(z) = \sum_{4|m|\le \wp n} c_m e^{4imz}. \tag{AK40}$$

For the equality, periodicity makes $a(z)$ a single-valued holomorphic function of $w = e^{4iz}$ on $\mathbb{C}\setminus\{0\}$. Its Laurent coefficients are the displayed c_m, so its Laurent series is the finite sum after the vanishing just proved. Equivalently one may use the uniformly convergent Fourier series of the smooth periodic function on the real axis and then the identity theorem. Reality implies $c_{-m} = \overline{c_m}$.

We next construct coefficients beyond pencils, without assuming a continuity assumption across different ordering charts. Put $N = \lfloor \wp n/4 \rfloor$ and on planar configurations introduce the angular vector field

$$\mathcal{L} = \sum_{i=1}^{n} (-y_i^1 \partial_{y_i^0} + y_i^0 \partial_{y_i^1}).$$

On translation-invariant functions it is the same angular vector field in the consecutive-difference coordinates. Let

$$P(X) = \prod_{m=-N}^{N} (X - 4im), \qquad p_m(X) = \prod_{\substack{-N\le j\le N \\ j\ne m}} \frac{X - 4ij}{4im - 4ij}.$$

The finite-band identity gives $P(\mathcal{L})\mathbf{K}_n = 0$ on the relatively open real set of pencils. The left side is holomorphic on the connected domain $\mathcal{U}_n^{\mathrm{ord}}$. Uniqueness therefore proves this equation everywhere on that domain. Define there

$$c_m = p_m(\mathcal{L})\mathbf{K}_n \quad (|m| \le N). \tag{AK41}$$

On pencils these are the Fourier coefficients in (AK39): each polynomial p_m selects its corresponding distinct exponential. The polynomial identities $\sum_m p_m = 1$ and $(X - 4im)p_m(X) = \text{constant}\, P(X)$ give

$$\sum_m c_m = \mathbf{K}_n, \qquad \mathcal{L}c_m = 4imc_m. \tag{AK42}$$

The coefficients are holomorphic on the ordered domain, symmetric in labelled points and invariant under simultaneous translations. On its real points $c_{-m} = \overline{c_m}$: the angular vector field has real coefficients, the density is real there, and $\overline{p_m(X)} = p_{-m}(X)$ for real X. For translation invariance, note that applying $\mathcal{L}$ to a translation-invariant function again gives one: the extra term under a common shift is a linear combination of the total derivatives $\sum_i \partial_{y_i^\mu}$, which vanish.

Since imaginary rotations preserve the domain, the last equation in (AK42) can be integrated along them:

$$c_m(R_{is}\mathbf{z}) = e^{-4ms}c_m(\mathbf{z}), \qquad \mathbf{K}_n(R_{is}\mathbf{z}) = \sum_m e^{-4ms}c_m(\mathbf{z}). \tag{AK43}$$

The same equation gives $c_m(R_t\mathbf{y}) = e^{4imt}c_m(\mathbf{y})$ on any real interval whose rotated configurations remain in Ord_n.

These coefficients extend uniquely and real-analytically to every configuration of distinct real planar points. Here is the full gluing argument. Let $\mathcal{G}_n$ consist of configurations with distinct points and with no two nonzero difference vectors perpendicular. This is open and dense: its complement, in addition to the collisions, is a finite union of zero sets of nonzero real polynomials. Each polynomial is nonzero, as can also be checked by taking distinct points on one line. Every real rotation of a configuration in $\mathcal{G}_n$ is ordinary. Failure of ordinariness would produce a time tie and an abscissa tie, whose nonzero difference vectors are perpendicular. Thus the preceding angular ordinary differential equation proves covariance for every real angle on $\mathcal{G}_n$.

For an arbitrary distinct configuration $\mathbf{y}$, choose an angle t at which all its rotated times are distinct. Only finitely many angles are forbidden. In a neighborhood of $\mathbf{y}$ define the extension by

$$\widetilde{c}_m(\mathbf{u}) = e^{-4imt}c_m(R_t\mathbf{u}). \tag{AK44}$$

This is real-analytic. Two choices agree on the dense set $\mathcal{G}_n$ in every overlap, by the covariance already proved there; continuity makes them agree everywhere in the overlap. They consequently glue, agree with the original coefficients on ordinary configurations, and obey exact covariance

$$\widetilde{c}_m(R_t\mathbf{y}) = e^{4imt}\widetilde{c}_m(\mathbf{y}) \quad (t \in \mathbb{R}). \tag{AK45}$$

This last equality, permutation symmetry, translation invariance and $\widetilde{c}_{-m} = \overline{\widetilde{c}_m}$ follow by continuity from $\mathcal{G}_n$. Uniqueness follows because Ord_n is dense in the collision-free set. The collision-free set itself is connected: one may move its points successively along polygonal paths avoiding the finitely many positions occupied by the other points, with small preliminary moves if necessary. Removing finitely many points does not disconnect the plane.

Set $\overline{m}(0) = \overline{m}(1) = 0$; for $n \geq 2$ let $\overline{m}(n)$ be the largest $|m|$ occurring for any pencil and any real labels, or zero if none occurs. It is a finite integer and

$$\overline{m}(n) \leq \wp n/4. \tag{AK46}$$

Coefficients with $|m| > \overline{m}(n)$ vanish identically on the ordered domain by uniqueness, and then on every collision-free configuration by (AK44). If $\overline{m}(n) > 0$, some pencil and labels have $c_{-\overline{m}(n)} \neq 0$, by reality.

Boost positivity without time ordering and the top kernel. For arbitrary labelled configurations of k and l distinct points in the real open half-plane, define

$$\widetilde{\mathrm{Gr}}_s(\mathsf{x},\mathsf{y}) = \sum_m e^{-4ms}\widetilde{c}_m^{(k+l)}(\theta\mathbf{x},\mathbf{y};\vec{\varphi},\vec{\psi}). \tag{AK47}$$

All points in this combined configuration are distinct: the two groups lie on opposite sides of the mirror. Empty tuples have the conventions $\widetilde{c}_m^{(0)} = \delta_{m0}$ and $\widetilde{c}_m^{(1)} = 0$.

This kernel is positive semidefinite for every real s. First suppose each individual configuration has distinct times. Sort those times. The combined reflected configuration is then ordinary, and (AK43), symmetry and (AK34) identify (AK47) with the proved Gram kernel. For a finite family with time ties, perturb the positions slightly inside the positive half-plane to remove all such ties. The reflected combined configurations remain collision free. Every coefficient is continuous there, so their finite matrices converge entrywise to the desired matrix. Positive semidefiniteness

is closed under this limit. This proves the assertion with time ties, arbitrary lengths and the empty configuration included.

In particular, fix $k \geq 1$, put $m_* = \overline{m}(2k)$, and set $c_* = \widetilde{c}^{(2k)}_{-m_*}$. On k-point configurations the kernel

$$(\mathsf{x}, \mathsf{y}) \longmapsto c_*(\theta\mathbf{x}, \mathbf{y}; \vec{\varphi}, \vec{\psi}) \tag{AK48}$$

is positive semidefinite: multiply the matrix in (AK47) by $e^{-4m_* s}$ and send $s \to +\infty$. All the other finitely many terms tend to zero.

There is also an exact formulation across an arbitrary line $\mathfrak{l}$. Let $b \in \mathfrak{l}$ and let $R_\nu e_0$ be its normal pointing into a chosen side $\mathfrak{l}_+$. The map $r(y) = R_{-\nu}(y - b)$ sends that side to $\mathbb{H}$ and intertwines its reflection $\theta_\mathfrak{l}$ with θ. Using translation invariance and (AK45), (AK48) becomes positivity of

$$(\mathsf{x}, \mathsf{y}) \longmapsto e^{4im_*\nu} c_*(\theta_\mathfrak{l}\mathbf{x}, \mathbf{y}; \vec{\varphi}, \vec{\psi}) \tag{AK49}$$

on k-point configurations in $\mathfrak{l}_+$. The compensating unit phase is essential. This is a statement about the top coefficient, and does not assume reflection positivity of the original functional at every line.

The cut inequality, including the empty tuple. The preceding constructions imply

$$2\overline{m}(k + l) \leq \overline{m}(2k) + \overline{m}(2l) \qquad (k, l \geq 0). \tag{AK50}$$

If the left top frequency is zero there is nothing to prove. Otherwise choose a pencil and labels whose negative top coefficient is nonzero. When $k, l \geq 1$, translate the midpoint between y_k, y_{k+1} to the origin, and put

$$x_i = \theta y_{k+1-i} \ (1 \leq i \leq k), \qquad z_j = y_{k+j} \ (1 \leq j \leq l),$$

with the correspondingly reversed left labels. Both are positive configurations. Their first points and all consecutive differences are in Sec: reflection followed by a minus sign changes the sign of a real spatial coordinate and preserves this sector. The first points are the two reflected halves of the cut difference.

The combined reflected configuration for x, z is exactly the original pencil. The doubled configurations for x, x and z, z are also pencils. Their central differences are respectively $2x_1^0 e_0$ and $2z_1^0 e_0$, and all other differences are sector vectors or their stars. The Gram Cauchy–Schwarz inequality gives

$$|a_\mathbf{y}(is)|^2 \leq a_{(\theta x_k, \ldots, \theta x_1, x_1, \ldots, x_k)}(is) a_{(\theta z_l, \ldots, \theta z_1, z_1, \ldots, z_l)}(is).$$

The left side has leading rate $e^{8\overline{m}(k+l)s}$ as $s \to +\infty$, with nonzero coefficient. Each right factor has rate at most $e^{4\overline{m}(2k)s}$ or $e^{4\overline{m}(2l)s}$ by its finite band. Comparison proves (AK50).

If $k = 0$, translate a nonzero pencil so that its first point is in Sec. Apply Gram Cauchy–Schwarz to the empty vector Ω and its product vector. The empty diagonal entry is one, so the same comparison gives $2\overline{m}(l) \leq \overline{m}(2l)$. This is exactly the required inequality with $\overline{m}(0) = 0$. The case $l = 0$ follows by symmetry, and the centered one-point and fully empty cases are immediate. Thus the initial convexity information has not been lost by omitting the vacuum vector.

The two-point interface and its positive plane measure. For two points, Ord_2 is the entire real collision-free set: two distinct points cannot have both coordinates equal. Translation invariance therefore gives a real-analytic function $G_{\varphi,\psi}(r)$ for every real $r \neq 0$, equal to $K_2(r; \varphi, \psi)$

when $r^0 > 0$. Its global coefficients obey

$$\begin{aligned} G_{\varphi,\psi}(r) &= \sum_m \widetilde{c}_m(r;\varphi,\psi), \\ \widetilde{c}_m(R_t r) &= e^{4imt}\widetilde{c}_m(r), \\ K_2(R_{is}r) &= \sum_m e^{-4ms}\widetilde{c}_m(r) \qquad (r^0>0). \end{aligned} \tag{AK51}$$

The last identity holds for every real s, not merely near zero; it is (AK43). Every sum is finite, with $|m| \le \overline{m}(2) \le \wp/2$. The kernel in (AK47) for one-point configurations is positive semidefinite for every s. A quarter-turn frame with projection at least $|r|/\sqrt{2}$ also gives the useful bound

$$|G_{\varphi,\psi}(r)| \le C_\varphi C_\psi \min\{1, |r|/\sqrt{2}\}^{-2\wp}. \tag{AK52}$$

The density formula for two points also holds for arbitrary disjoint compact plane-profile supports, even if those whole supports have no strict coordinate ordering. Let their distance be $d > 0$ and partition each profile into finitely many pieces supported in sets of diameter less than $d/(10\sqrt{2})$. For each pair of pieces, two chosen points in their supports differ in at least one coordinate by $d/\sqrt{2}$ or more. The small diameters then make the two entire piece supports strictly ordered in that coordinate, in one of the four frames. Apply (AK24) and the frame identity to this pair. Summing the finitely many identities by bilinearity proves

$$\sigma(\alpha\otimes\varphi, \beta\otimes\psi) = \iint \alpha(y)\beta(y')G_{\varphi,\psi}(y'-y)\,dy\,dy'$$

for every such pair of profiles. All partitions can be smooth; their products with the original C^1 profiles remain C^1 and have supports in the original separated sets.

There is a positive spectral plane measure for equal real labels. Let $\zeta_s = \Psi((s,0);\varphi)$, $s > 0$. By (AK32), $\zeta_{s'} = e^{-(s'-s)H}\zeta_s$ if $s' > s$. Let $\mu_s(B) = \langle \zeta_s, \mathsf{P}(B)\zeta_s\rangle$, using the joint spectral measure of (H, P). The locally finite positive measure

$$d\mu_\varphi(E,q) = e^{2sE}\,d\mu_s(E,q) \tag{AK53}$$

is independent of s, because the preceding vector identity multiplies μ_s by $e^{-2(s'-s)E}$. It is carried by $E \ge |q|$ and satisfies

$$K_2(v;\varphi,\varphi) = \int e^{-v^0E + iv^1q}\,d\mu_\varphi(E,q) \qquad (v\in\mathbb{V}). \tag{AK54}$$

For real $v = (t,x)$ choose $0 < s < t/2$ and use (AK33) and (AK32): the left side is $\langle \zeta_s, T(t-2s,x)\zeta_s\rangle$, which is the displayed integral. In particular $\int e^{-tE}\,d\mu_\varphi < \infty$ for every $t > 0$. On a compact sub-tube the multiplier is bounded by e^{-dE} for some $d > 0$. Its derivatives are controlled by a smaller exponential weight because $|q| \le E$. The integral is therefore holomorphic there, and uniqueness extends its equality from real points to all of $\mathbb{V}$.

No uniqueness of the vacuum or absence of a zero-energy mass is asserted here. A positive plane measure in a diagonal-reflection picture requires the reflection-positivity input for that picture; it is not inferred from the coordinate cone. Likewise, eliminating the nonzero two-point harmonics requires a separate ultraviolet estimate and the diagonal-frame argument. The polygon and integer arguments can use (AK49) and (AK50), but are not contained in this section. Finally, the analytic extension of coefficients to distinct planar points does not identify the original moment functional on every exceptional support arrangement. That distributional passage, in particular arrangements with coincident planar projections and separated transverse supports, is a separate obligation.

49. Ultraviolet elimination of the two-point angular harmonics

This section proves an implication for the two-point function. Its inputs are a continuous planar kernel with its axis-frame spectral realization, the exact cone in that frame, one diagonal reflection, and a uniform short-distance bound on a specified class of positive shapes. It constructs the diagonal spectral measure and proves its cone support; a diagonal cone and continuous rotation covariance are not assumed. The angular continuation needed at order two is proved directly. In particular this argument does not require an all-orders highest-harmonic theorem.

The kernel and Hilbert realization are supplied by the planar-kernel construction under its insertion, continuity and reconstruction hypotheses. We use these together with the interacting-source and short-distance hypotheses stated below.

49.1. **The precise two-point interface.** Write a point of $\mathbb{R}^4$ as (y,z), with $y=(y^0,y^1)=(x_4,x_1)$ and $z=(x_2,x_3)$. On the plane let $e_0=(1,0)$, let R_t be the rotation

$$R_t(u,v)=(u\cos t-v\sin t,\,u\sin t+v\cos t),\qquad \theta(u,v)=(-u,v).$$

The same notation denotes these maps on $\mathbb{R}^4$, leaving z fixed. Put $n=R_{\pi/4}e_0$. The reflection at $n^\perp$ is $\theta_n=R_{\pi/4}\theta R_{-\pi/4}$.

Let S_2 be a real symmetric bilinear form on pairs of real $C^1_c(\mathbb{R}^4)$ tests with disjoint supports, complexified bilinearly. It is translation- and W_4-invariant and satisfies coordinate reflection positivity. It also satisfies

$$S_2(\theta_n\overline{f},f)\ge 0\quad \big(f\in C^1_c(\{n\cdot y>0\};\mathbb{C})\big). \tag{TU1}$$

Positivity at any one diagonal normal in four dimensions implies this choice by hypercubic conjugation. All scalar products below are conjugate linear in the first variable.

The kernel interface is the following explicit assumption. For real $\varphi,\psi\in C^1_c(\mathbb{R}^2)$ there are continuous functions $G_{\varphi,\psi}$ on $\mathbb{R}^2\setminus\{0\}$ such that

$$S_2(\alpha\otimes\varphi,\beta\otimes\psi)=\iint \alpha(y)\beta(y')G_{\varphi,\psi}(y'-y)\,dy\,dy' \tag{TU2}$$

whenever the compact supports of $\alpha,\beta\in C^1_c(\mathbb{R}^2;\mathbb{C})$ are disjoint. There is a Hilbert space with strongly commuting self-adjoint operators H,P, with $H\ge|P|$, and vectors $\zeta_{\varphi,a}$, $a>0$, such that

$$\begin{aligned}\zeta_{\varphi,b}&=e^{-(b-a)H}\zeta_{\varphi,a}\quad (b>a),\\ G_{\varphi,\psi}(t,x)&=\langle\zeta_{\varphi,a},e^{-(t-2a)H+ixP}\zeta_{\psi,a}\rangle\quad (t>2a).\end{aligned} \tag{TU3}$$

This is a point-kernel interface, not a bound on unsmeared field operators. At order two every distinct planar pair is in an ordinary coordinate frame. Consequently the two-point specialization of the planar kernel theorem supplies (TU2)–(TU3) without an exceptional planar set.

When S_2 is the connected function of a functional with unit vacuum vector Ω, apply the interface to $\zeta_{\varphi,a}-\langle\Omega,\zeta_{\varphi,a}\rangle\Omega$. The vacuum is translation invariant, so its orthogonal projection commutes with H,P; this subtraction preserves (TU3), positivity and the cone. Thus the interface here concerns the connected kernel when the ultraviolet hypothesis concerns the connected function. No uniqueness of the vacuum is required.

Kernel uniqueness follows by testing (TU2) with small nonnegative approximate identities at two distinct points. The same test shows that the kernels inherit reality, bilinearity in the labels, and all symmetries of S_2 that preserve the plane. In particular, for $G_\varphi=G_{\varphi,\varphi}$,

$$G_\varphi(R_{\pi/2}y)=G_\varphi(y),\qquad G_\varphi(y^0,-y^1)=G_\varphi(y),\qquad G_\varphi(y)\in\mathbb{R}. \tag{TU4}$$

49.2. **The axis measure and the plate norm.** Fix a real label φ. Let P be the joint spectral measure of (H, P) and set

$$\mu_{\varphi,a}(B) = \langle \zeta_{\varphi,a}, \mathsf{P}(B)\zeta_{\varphi,a}\rangle, \qquad d\mu_\varphi(E,q) = e^{2aE}\, d\mu_{\varphi,a}(E,q). \tag{TU5}$$

The first measure is finite and positive. Equation (TU3) gives $d\mu_{\varphi,b} = e^{-2(b-a)E} d\mu_{\varphi,a}$, so the second measure is independent of a. It is positive and carried by

$$\mathcal{C} = \{(E,q) \in \mathbb{R}^2 : E \geq |q|\}.$$

For every $t > 0$ choose $a < t/2$ in (TU3). Then

$$G_\varphi(t,x) = \int e^{-tE+ixq}\, d\mu_\varphi(E,q), \qquad \Gamma_\varphi(t) = \int e^{-tE}\, d\mu_\varphi = G_\varphi(t,0) < \infty. \tag{TU6}$$

The measure is locally finite and σ-finite: on $\{E \leq N\}$ its mass is at most $e^{tN}\Gamma_\varphi(t)$. The function Γ_φ is nonnegative and nonincreasing. By a quarter turn, any $y \neq 0$ has a positive first coordinate at least $|y|/\sqrt{2}$. Therefore

$$|G_\varphi(y)| \leq \Gamma_\varphi(|y|/\sqrt{2}). \tag{TU7}$$

This bound will give the diagonal reconstruction its bounded time orbits. It uses no decay at large distance. Define also

$$\mathbb{V} = \{v \in \mathbb{C}^2 : \operatorname{Re} v^0 > |\operatorname{Im} v^1|\}, \qquad K_\varphi(v) = \int e^{-v^0 E + iv^1 q}\, d\mu_\varphi. \tag{TU7a}$$

On a compact subset of this tube the absolute multiplier is at most e^{-cE} for some $c > 0$. Its derivatives are dominated by a smaller exponential because $|q| \leq E$. Thus K_φ is holomorphic and agrees with G_φ on the real positive-time half-plane.

On positive-time tests put $\|[f]\|^2 = S_2(\theta\overline{f}, f)$; reflection positivity and polarization give the associated Hilbert seminorm and its triangle inequality. For α supported in $\{y^0 > 0\}$, (TU2) and (TU6) give

$$\|[\alpha \otimes \varphi]\|^2 = \int \left| \iint \alpha(t,x) e^{-tE+ixq}\, dt\, dx \right|^2 d\mu_\varphi(E,q). \tag{TU8}$$

Indeed, if $t \geq \delta > 0$ on the support, the absolute integral before interchanging the variables is bounded by $\|\alpha\|_{L^1}^2 \Gamma_\varphi(2\delta)$. This proves Fubini's applicability, including when the unweighted measure has infinite mass.

49.3. **The ultraviolet class and fixed shapes.** For $d > 0$ and $\vartheta > 0$, let $\mathfrak{T}(d;\vartheta)$ consist of pairs of nonzero, nonnegative $C_c^1(\mathbb{R}^4)$ tests f, h, each supported in a ball of radius d, satisfying

$$\begin{gathered} d \leq \operatorname{dist}(\operatorname{supp} f, \operatorname{supp} h) \leq 6d, \\ \int f \geq \vartheta \|f\|_\infty d^4, \qquad \operatorname{Lip} f \leq \|f\|_\infty/(\vartheta d), \end{gathered} \tag{TU9}$$

and the analogous conditions for h. The balls may be placed anywhere. The ultraviolet input is a finite nonnegative function u, with $u(d) \to 0$ as $d \downarrow 0$, such that for $0 < d \leq d_0$,

$$0 \leq S_2(f,h) \leq u(d)\|f\|_\infty \|h\|_\infty \quad \big((f,h) \in \mathfrak{T}(d;\vartheta_*)\big). \tag{TU10}$$

Here the universal $\vartheta_* > 0$ is chosen below, before S_2. A hypothesis at a smaller shape parameter also suffices. A hypothesis only on a narrower positive-shape class must be checked against this choice; that inclusion is part of the ultraviolet premise.

The following explicit shapes fix all regularity constants. Choose $a \in C_c^1((2,3))$, between zero and one, equal to one on $[9/4, 11/4]$, zero outside $[17/8, 23/8]$, with cubic smoothstep ramps on

the two intervening intervals. Then $\int a = 5/8$ and $\|a'\|_\infty \le 12$. Let $I = [-1/4, 1/4]$, $\tau = \mathbf{1}_I * \mathbf{1}_I$, and

$$b = 12(\tau * \tau), \qquad c_b = \int b = 3/4. \tag{TU11}$$

This is a nonnegative C^2_c spline, supported in $[-1, 1]$, with maximum one and $\|b'\|_\infty = 2$. Explicitly it is $1 - 6s^2 + 6|s|^3$ for $|s| \le 1/2$, and $2(1 - |s|)^3$ for $1/2 \le |s| \le 1$. With the Fourier convention $\widehat{b}(q) = \int b(s)e^{isq}\,ds$, convolution gives

$$\widehat{b}(q) = c_b \left(\frac{\sin(q/4)}{q/4}\right)^4, \tag{TU12}$$

where the quotient has value one at zero.

For completeness a transverse partition can use $q(s) = b(4s/3)$, $D(s) = \sum_{j\in\mathbb{Z}} q(s - j)$ and $\chi(s) = q(s)/D(s)$. There are at most two nonzero summands, $D \ge 2/27$ by choosing a nearest integer, and $\sum_j \chi(s - j) = 1$. The bounds $\|q'\|_\infty \le 8/3$ and $\|D'\|_\infty \le 16/3$ give $\|\chi'\|_\infty \le 1008$. Put $\psi(z) = \chi(z_1)\chi(z_2)$. It takes values in $[0, 1]$, has support in $[-3/4, 3/4]^2$, and its integer translates sum to one. Let $\widetilde{\psi}(z) = \rho(z_1)\rho(z_2)$, where $\rho = 1$ on $[-1, 1]$, $\rho = 0$ outside $[-3/2, 3/2]$, and on each intervening interval use the cubic smoothstep. Thus

$$\int \widetilde{\psi} = 25/4, \quad \|\nabla\widetilde{\psi}\|_\infty \le 6, \quad \|\nabla\psi\|_\infty \le 2016, \quad \psi \le \widetilde{\psi}.$$

All these functions are globally C^1. The numerical choice

$$\vartheta_* = 1/12000 \tag{TU13}$$

will suffice for all tests constructed next.

49.4. **Dilation removes the spline zeros.** For $h > 0$ and $1 \le \lambda \le 2$ put

$$\alpha^\lambda_h(t, x) = a(t/h)b(4x/(\lambda h)), \qquad f^\lambda_h = \alpha^\lambda_h \otimes \varphi.$$

The time transform and spatial transform are

$$A_h(E) = \int a(t/h)e^{-tE}\,dt, \qquad B_{h,\lambda}(q) = \frac{c_b\lambda h}{4}\left(\frac{\sin(\lambda hq/16)}{\lambda hq/16}\right)^4.$$

For $E \ge 0$, $A_h(E) \ge (h/2)e^{-3hE}$. There is an absolute $c > 0$ such that

$$\int_1^2 |B_{h,\lambda}(q)|^2\,d\lambda \ge ch^2(1 + h^2q^2)^{-4}. \tag{TU14}$$

Here is a proof that includes every zero of the transform. Set $x = h|q|/16$. For $0 \le x \le 1$, the sinc in the display is bounded below on $0 \le \lambda x \le 2$, giving a fixed positive lower bound after division by h^2. For $x \ge 1$, that divided integral is at least

$$cx^{-8}\int_1^2 \sin^8(\lambda x)\,d\lambda = cx^{-9}\int_x^{2x} \sin^8 v\,dv.$$

The number $c_1 = \min_{r\in[0,\pi]} \int_r^{r+1} \sin^8 v\,dv$ is positive: the integral is continuous, periodic and never zero. An interval of length $x \ge 1$ contains $\lfloor x\rfloor \ge x/2$ successive intervals of length one. The last display is therefore at least cx^{-8}. Both regimes imply (TU14), with constants independent of h, q.

The finite number $C_* = \sup_{r\ge0} e^{-r}(1 + r^2)^4$ and $|q| \le E$ imply

$$e^{-6hE}(1 + h^2q^2)^{-4} \ge C_*^{-1}e^{-7hE}.$$

Using (TU8), (TU14) and nonnegative Fubini yields

$$\int_1^2 \|[f_h^\lambda]\|^2\, d\lambda \ge c h^4 \Gamma_\varphi(7h). \tag{TU15}$$

The cone is essential here. Without it, arbitrarily large $|q|$ at small E could make the spline factor too small.

49.5. **Positive pedestals put the plate in the ultraviolet class.** Assume $\varphi \neq 0$, choose $r_1 > 0$ with $\operatorname{supp}\varphi \subset B(0, r_1)$, and take $0 < h \le \min(1, r_1, 2d_0/5)$. Put $A = \|\varphi\|_\infty + h\|\nabla\varphi\|_\infty > 0$ and

$$\psi_i(z) = \psi(z/h - i), \quad \widetilde{\psi}_i(z) = \widetilde{\psi}(z/h - i), \quad \varphi_i^- = 2A\widetilde{\psi}_i, \quad \varphi_i^+ = 2A\widetilde{\psi}_i + \psi_i\varphi.$$

Only indices in $\mathcal{I} = \{i : \psi_i\varphi \not\equiv 0\}$ are needed. Its size is at most $(2r_1/h + 3)^2 \le 25 r_1^2 h^{-2}$. For a fixed λ let $h_i^\pm = \alpha_h^\lambda \otimes \varphi_i^\pm$. Then

$$f_h^\lambda = \sum_{i\in\mathcal{I}} (h_i^+ - h_i^-), \qquad A\widetilde{\psi}_i \le \varphi_i^\pm \le 3A\widetilde{\psi}_i. \tag{TU16}$$

In particular these tests are nonnegative, nonzero, and have sup norm between A and $3A$.

We check (TU9), rather than invoking a positive decomposition without shape control. Each support lies in the ball of radius $d = 5h/2$ about $(5h/2, 0, hi)$: its four coordinate half-widths are at most $h/2, h/2, 3h/2, 3h/2$, whose squared sum is $5h^2 < d^2$. The reflected ball also has radius d. The distance between the support and its time reflection is at least $4h$ and at most $6h$, hence between d and $6d$.

Moreover, direct product integration gives

$$\int h_i^\pm \ge A h^4 \frac{\int a \int b \int \widetilde{\psi}}{4} = \frac{375}{512} A h^4.$$

Since $\|h_i^\pm\|_\infty \le 3A$, the integral condition holds for every $\vartheta \le 1/160$. The product derivative rule gives

$$\operatorname{Lip} h_i^\pm \le \frac{A}{h}\,[3(12 + 4\cdot 2) + 2\cdot 6 + 2016 + 1] = 2089 A/h.$$

As $\|h_i^\pm\|_\infty \ge A$, (TU13) gives the Lipschitz condition: $2089A/h \le \|h_i^\pm\|_\infty/(\vartheta_* d)$. The reflected tests have the same integrals and norms. Thus $(\theta h_i^\pm, h_i^\pm) \in \mathfrak{T}(5h/2; \vartheta_*)$.

By (TU10), reflection positivity and the Hilbert triangle inequality,

$$\|[f_h^\lambda]\| \le \sum_i \big(\|[h_i^+]\| + \|[h_i^-]\|\big) \le 6\,\#\mathcal{I}\, A\sqrt{u(5h/2)}.$$

Combining its square with (TU15) proves

$$\Gamma_\varphi(7h) \le C r_1^4 (\|\varphi\|_\infty + \|\nabla\varphi\|_\infty)^2 h^{-8} u(5h/2). \tag{TU17}$$

Consequently

$$t^8 \Gamma_\varphi(t) \longrightarrow 0 \quad (t \downarrow 0). \tag{TU18}$$

The zero label is immediate. The two factors of h^{-4} in (TU17) have different origins: normalization in (TU15), and the square of the number of transverse cells. No uniform bound on a compact profile's Fourier transform from below was used.

49.6. **Entire angular continuation and the critical Fourier modes.** For $r > 0$, define on $|\operatorname{Re} z| < \pi/2$

$$g_r^{[0]}(z) = \int \exp[-rE\cos z + irq\sin z]\, d\mu_\varphi(E,q). \tag{TU19}$$

Writing $z = u + is$, the absolute integrand is $\exp[-r\cos u(E\cosh s + q\sinh s)]$. On the cone this is at most $\exp[-r\cos u\, e^{-|s|}E]$. On a compact substrip its derivatives are dominated by a polynomial in E times a smaller exponential. Thus the integral is holomorphic. Its real restriction is $G_\varphi(r\cos u, r\sin u)$.

On $|\operatorname{Re} z - k\pi/2| < \pi/2$ define $g_r^{[k]}(z) = g_r^{[0]}(z - k\pi/2)$. On real overlaps these functions agree by (TU4). The complex overlaps are connected vertical strips with real intervals, so the identity theorem glues them. Their union is the complex plane. The resulting g_r is entire and $\pi/2$-periodic. For every real u, s choose k with $|u - k\pi/2| \le \pi/4$ to obtain

$$|g_r(u+is)| \le \Gamma_\varphi(re^{-|s|}/\sqrt{2}) = o(e^{8|s|}). \tag{TU20}$$

The last little-o, uniform in u, is exactly (TU18).

Define $a_m(r) = \frac{2}{\pi}\int_0^{\pi/2} g_r(u)e^{-4imu}\,du$. Cauchy's theorem on a rectangle of height s, with cancellation of its vertical sides by periodicity, gives

$$|a_m(r)| \le e^{4ms} \sup_{u\in[0,\pi/2]} |g_r(u+is)|.$$

For $m \ge 2$ take $s \to -\infty$; for $m \le -2$ take $s \to +\infty$. Equation (TU20) gives $a_m(r) = 0$ in both cases, including $|m| = 2$. A mere $O(t^{-8})$ bound on Γ would not eliminate that boundary mode. Fourier uniqueness for a continuous periodic function now gives, first on the real line and then everywhere by the identity theorem,

$$g_r(z) = a_0(r) + a_1(r)e^{4iz} + a_{-1}(r)e^{-4iz}. \tag{TU21}$$

One elementary uniqueness proof uses the positive Fejer kernels: their integrals are one, their mass outside every fixed neighbourhood of zero tends to zero, and their convolution converges uniformly on continuous periodic functions. A function with all Fourier coefficients zero has all these convolutions zero. Thus no convergence assumption on a formal Fourier series is implicit in (TU21).

49.7. **The positive measure in the diagonal frame.** Put $D(t,x) = G_\varphi(R_{\pi/4}(t,x))$ for $t > 0$. We construct its positive measure without assuming a diagonal cone. For $y = (t,x)$, $t > 0$, introduce a formal vector e_y with

$$\langle e_y, e_{y'}\rangle_D = D(t+t', x'-x). \tag{TU22}$$

This kernel is positive semidefinite. To see it, approximate each planar point $R_{\pi/4}y$ by a nonnegative radial smooth mollifier supported strictly on the positive side of the diagonal mirror, tensor it with φ, and apply (TU1) to an arbitrary complex linear combination. Formula (TU2) and continuity away from zero give (TU22) in the limit. Thus its quotient and completion define a Hilbert space.

On finite sums define $T_s e_{(t,x)} = e_{(t+s,x)}$, $s \ge 0$, and $U_b e_{(t,x)} = e_{(t,x+b)}$, $b \in \mathbb{R}$. The U_b preserve the form and have inverse U_{-b}. The T_s are symmetric and commute with the U_b. If $v = \sum c_i e_{(t_i,x_i)}$ and $\delta = 2\min_i t_i > 0$, (TU7) gives a finite bound on $\|T_s v\|$ for every $s \ge 0$, namely its square is at most $\Gamma_\varphi(\delta/\sqrt{2})(\sum|c_i|)^2$. On the unquotiented vector space Cauchy–Schwarz gives

$$\|T_s v\|^2 = \langle v, T_{2s}v\rangle_D \le \|v\|\,\|T_{2s}v\|.$$

If $\|v\| = 0$ this preserves the null space. Otherwise iterate and use the bounded orbit to get $\|T_s v\| \le \|v\|$. Therefore T_s descend to self-adjoint contractions. They are positive because

$T_s = T_{s/2}^2$. Continuity of D implies strong continuity of both families on point vectors, hence on their completion.

There are strongly commuting generators $H_D \geq 0$ and P_D with $T_s = e^{-sH_D}$ and $U_b = e^{ibP_D}$. For specificity, T_1 is injective: $T_1 v = 0$ implies $T_{1/2} v = 0$, then all its dyadic roots annihilate v, and strong continuity gives $v = 0$. Define $H_D = -\log T_1$ by spectral calculus. Uniqueness of positive roots proves the semigroup identity first at rational times and then at all times. The spectral theorem for the commuting unitary group supplies P_D and the joint spectral measure. This is the elementary Hilbert reconstruction for (TU22), not an OS analytic-continuation theorem.

Let $v_a = e_{(a,0)}$. If P_D is that joint spectral measure, define

$$\nu_a(B) = \langle v_a, \mathsf{P}_D(B) v_a \rangle_D, \qquad d\nu(E,q) = e^{2aE}\, d\nu_a(E,q). \tag{TU23}$$

As $v_b = T_{b-a} v_a$ for $b > a$, ν is independent of a. It is a positive, locally finite, σ-finite measure on $[0,\infty) \times \mathbb{R}$, and for $t > 0$

$$D(t,x) = \int e^{-tE+ixq}\, d\nu, \qquad \int e^{-tE}\, d\nu = D(t,0) < \infty. \tag{TU24}$$

Indeed choose $a < t/2$ and evaluate $\langle v_a, T_{t-2a} U_x v_a \rangle_D$. Nothing so far asserts $|q| \leq E$ for ν.

49.8. **Fourier–Laplace uniqueness and the boost identity.** We shall use the following uniqueness fact. Positive Borel measures ρ_1, ρ_2 on $[0,\infty) \times \mathbb{R}$ with finite $\int e^{-tE} d\rho_i$ for every $t > 0$ are equal if their transforms $\int e^{-tE+ixq} d\rho_i$ agree for all $t > 0, x \in \mathbb{R}$. For each real x these transforms are holomorphic in t on $\operatorname{Re} t > 0$, by exponential domination, so the identity theorem gives equality at $t = a + i\tau$. These are precisely the joint Fourier transforms of the finite measures $e^{-aE}\rho_i$. Finite Fourier transforms determine their measures: convolve their difference with a Gaussian, apply the Gaussian Fourier integral and Fubini to get zero, and let the Gaussian width tend to zero against compactly supported continuous functions. Hence the weighted measures, and then the original measures, agree. The same proof applies to finite positive linear combinations on either side of an identity.

For $s \in \mathbb{R}$ define the Lorentz map of spectral variables

$$L_s(E,q) = (E\cosh s + q \sinh s, E \sinh s + q \cosh s), \qquad \mu_s = (L_s)_* \mu_\varphi. \tag{TU25}$$

This preserves $\mathcal{C}$ and $L_s L_t = L_{s+t}$. Its energy satisfies $E_s \geq e^{-|s|} E$, so $\int e^{-tE} d\mu_s \leq \Gamma_\varphi(t e^{-|s|})$. Define $F_s(t,x) = \int e^{-tE+ixq} d\mu_s$ for $t > 0$. Direct multiplication of the rotation matrix by is gives

$$F_s(v) = K_\varphi(R_{is} v) = g_r(\theta + is) \quad \big(v = R_\theta(re_0),\ v^0 > 0\big), \tag{TU26}$$

where the middle expression uses (TU7a). In fact $R_{is}(t,x) = (t\cosh s - ix \sinh s, x \cosh s + it \sinh s)$, and its exponential multiplier is exactly $e^{-tE_s + ixq_s}$. Thus (TU26) holds on the whole real half-plane.

Write $C = \cosh 4 > 1$. The three terms in (TU21) obey the algebraic identity

$$(C-1) g_r(\theta + \pi/4) + g_r(\theta + i) + g_r(\theta - i) = (C+1) g_r(\theta).$$

Using (TU6), (TU24) and (TU26) and then transform uniqueness gives the identity of positive measures

$$(C-1)\nu + \mu_1 + \mu_{-1} = (C+1)\mu_\varphi. \tag{TU27}$$

Every measure has the required finite Laplace weights. In particular

$$0 \leq \nu \leq \frac{C+1}{C-1} \mu_\varphi. \tag{TU28}$$

The diagonal measure is therefore carried by the exact cone. This conclusion was deduced from (TU27), not inserted as an additional reflection-positivity consequence.

For $r > 0$ and real s the function $w_s(E,q) = \exp[-r(E\cosh s + q\sinh s)]$ is positive on the cone and at most $e^{-re^{-|s|}E}$. It is integrable against all four measures in (TU27); for $\mu_{\pm1}$ use $L_sL_{\pm1} = L_{s\pm1}$. Integrating (TU27) against w_s yields

$$(C-1)\int w_s\,d\nu + F_{s+1}(r,0) + F_{s-1}(r,0) = (C+1)F_s(r,0).$$

Substitute (TU21) into this identity to obtain

$$\begin{aligned} F_s(r,0) &= a_0(r) + a_1(r)e^{-4s} + a_{-1}(r)e^{4s} \ge 0,\\ \int w_s\,d\nu &= a_0(r) - a_1(r)e^{-4s} - a_{-1}(r)e^{4s} \ge 0. \end{aligned} \tag{TU29}$$

The first positivity comes from the axis measure, the second from the diagonal measure. Divide by e^{4s} and let $s \to +\infty$: the two limits give $a_{-1}(r) \ge 0$ and $a_{-1}(r) \le 0$. Then let $s \to -\infty$ to get both signs for $a_1(r)$. The limits also show these coefficients are real, if that was not already deduced from (TU4). Thus $a_1(r) = a_{-1}(r) = 0$ for every $r > 0$.

Theorem 49.1 (Two-point angular elimination)**.** *Under (TU1)–(TU3) and the short-distance hypothesis (TU9)–(TU13), every equal-real-label plane kernel is radial:*

$$G_\varphi(R_t y) = G_\varphi(y) \quad (y \neq 0,\ t \in \mathbb{R}).$$

Every nonzero angular coefficient of the two-point kernel vanishes, including coefficients with mixed transverse labels.

Proof. The preceding argument proves the first assertion. Symmetry of S_2 gives $G_{\varphi,\psi}(y) = G_{\psi,\varphi}(-y)$; invariance under the planar half turn gives $G_{\psi,\varphi}(-y) = G_{\psi,\varphi}(y)$. Therefore the mixed kernel is symmetric in its real labels and

$$G_{\varphi,\psi} = \tfrac{1}{4}(G_{\varphi+\psi} - G_{\varphi-\psi}).$$

Both terms on the right are radial. Complex labels follow by bilinear extension. Uniqueness of Fourier coefficients identifies this conclusion with vanishing of every $m \neq 0$ coefficient in any finite-angular-band construction of the same two-point kernel. In its notation, $\overline{m}(2) = 0$. □

This order-two conclusion enters the all-orders harmonic-closure argument. The passage to arbitrary separated four-dimensional tuples uses its separately stated extension hypotheses. The premises here are the stated ultraviolet law and kernel/spectral interface. The Ward and infrared routes retain their separate hypotheses.

50. From the two-point angular law to full Euclidean covariance

This section closes the all-orders and distributional parts of the angular argument. Its inputs are the analytic kernels and boost Gram identities of Section 48, and the two-point elimination of Section 49. In particular, the two-point ultraviolet estimate remains part of the hypotheses of the combined implication. No rotational Ward identity is used here.

Work first with the centered scalar hierarchy. Compact separated moments are symmetric, translation invariant and hypercubic invariant, have position-uniform local C^1 bounds, and are reflection positive in every coordinate picture. In each such picture the Hilbert generators have the exact cube cone $H_j \geq |P_{j,r}|$ for every tangential coordinate $r \neq j$. Their off-diagonal distributions are those constructed in Section 41.

50.1. **The harmonic data used below.** The chosen plane has coordinates (x_4, x_1), and R_u denotes its rotation through u. A plate label is a real smooth compact function of the two transverse coordinates. The analytic kernel construction supplies, for each n, real-analytic functions

$$c_m(y_1, \dots, y_n; \varphi_1, \dots, \varphi_n), \qquad y_i \in \mathbb{R}^2, \quad y_i \neq y_j, \tag{HC1}$$

multilinear in the labels, with only $|m| \leq Cn$ possibly nonzero. Their sum is the moment density on the ordinary ordering charts. They are symmetric under simultaneous permutations of positions and labels, translation invariant in the plane, and satisfy

$$c_m(R_u \mathbf{y}; \vec{\varphi}) = e^{4imu} c_m(\mathbf{y}; \vec{\varphi}), \qquad c_{-m} = \overline{c_m} \quad \text{for real labels.} \tag{HC2}$$

An ordinary chart means that all time coordinates are distinct, or that all abscissae are distinct; a fixed ordering is imposed on its connected component. The functions (HC1) already extend over the nongeneric planar configurations with distinct points. This is the analytic-projector conclusion of the kernel construction, not a density assertion about distributions on the exceptional set.

Let b_n be the greatest absolute index of a harmonic that is not identically zero for some choice of real labels. Set $b_n = 0$ if there is no nonzero nonconstant harmonic, and put $b_0 = 0$ for the vacuum kernel. Thus b_n are nonnegative integers and $b_n \leq Cn$. The boost Gram identity gives

$$2b_{k+l} \leq b_{2k} + b_{2l}, \qquad k, l \geq 0. \tag{HC3}$$

For clarity, this inequality compares global degrees. A nonzero coefficient can be tested on an open configuration set split into k negative and l positive times: real analyticity and connectedness prevent a nonzero coefficient from vanishing on that whole open set. Cauchy–Schwarz for the boosted product vectors bounds its largest exponential by the two diagonal ones. The empty product is the vacuum vector, so the cases $k = 0$ or $l = 0$ are included.

For $m_* = b_{2k}$ the coefficient c_{-m_*} is a positive kernel on reflected pairs of k-point configurations in a half-plane. This follows by multiplying the positive boost Gram matrix by $e^{-4m_* s}$ and taking $s \to +\infty$. Covariance (HC2) transports it to every oriented line: after multiplication by the fixed unit phase for that line it is again positive. We use the zero-row consequence of this positivity, which is unchanged by that phase. All these statements have explicit product-vector proofs in the preceding kernel section; analytic continuation of a nonnegative real-valued function alone would not prove them.

50.2. **The polygon divisibility argument, with label release.** Fix $k \geq 1$, put $m_* = b_{2k}$, and abbreviate $H = c_{-m_*}$. We prove

$$k \mid 2m_*. \tag{HC4}$$

Suppose otherwise. For any real plate label φ, place that same label at every vertex of a regular $2k$-gon. Rotation through π/k permutes the labelled vertices. Symmetry and (HC2) imply

$$H(\mathbf{y}; \varphi, \dots, \varphi) = e^{-4\pi i m_*/k} H(\mathbf{y}; \varphi, \dots, \varphi) = 0. \tag{HC5}$$

The polygon may be translated; rotation about its center has the same effect by translation invariance. The assumption that (HC4) fails is exactly what makes the phase in (HC5) different from one.

Call a labelled set S a zero set if H vanishes on every $2k$-point extension of it, with all remaining real labels arbitrary. A full polygon with the labels above is a zero set, since no completion is then needed. Here is the reduction of such a set. Suppose it is contained, with its labels, in $\theta Q \cup Q$, where Q has $j \leq k$ points strictly on one side of a line, and the reflected labels agree. Fill Q by any $k - j$ additional points A on that side, with arbitrary real labels. The positive kernel has zero diagonal at $Q \cup A$, because its diagonal configuration contains S. Its entire row is therefore zero. Indeed, positivity of the 2×2 Gram matrix gives $|K(X, Y)|^2 \leq K(X, X)K(Y, Y)$; a zero diagonal makes the right side zero. Consequently

$$H\big(\theta(Q \cup A) \cup B\big) = 0 \tag{HC6}$$

for every k-point B on the other side of the reflected set, with arbitrary real labels.

For each fixed choice of these free labels, (HC6) holds on a nonempty open set of all the free positions: the reflected points of A and the points of B can be varied independently in their respective half-planes. The configuration space of these $2k - j$ free points, avoiding each other and the fixed reflected Q, is connected. It is the complement of a finite union of affine sets of codimension two. One elementary proof is to join two configurations through a generic intermediate point; the intermediate points whose segment meets one of these affine sets form a finite union of measure-zero cones. Real analyticity and the identity principle extend (HC6) to this whole configuration space. It follows that θQ itself is a zero set, with no restriction on the remaining labels or half-planes. This is the label-release step; it is stronger than just showing that a special diagonal configuration has zero value.

Apply this reduction to the polygon. Reflection in the perpendicular bisector of an edge pairs its vertices and has no vertex on its line. It first reduces the full polygon to a consecutive arc of k vertices. A consecutive arc of $j \leq k$ vertices fits into the reflected union of an arc of $\lceil j/2 \rceil$ vertices. To check the odd case, index the vertices by integers modulo $2k$. For the arc $0, \dots, j - 1$, reflect indices by $i \mapsto j - i$ when j is odd; the union has $j + 1$ vertices, paired without a fixed vertex. For even j, use $i \mapsto j - 1 - i$ and the arc itself is paired. These are edge-bisector reflections, and

the selected half has the stated size. All its labels are still φ. Repeated reduction reaches a single labelled point.

Translation covariance makes every point with label φ a zero set. Since φ was arbitrary, every labelled $2k$-point configuration has zero value: choose any one of its labels as that single fixed label and use the unrestricted completion. Thus H is identically zero, contradicting the definition of m_*. This proves (HC4). When $m_* = 0$ the divisibility is automatic; no contradiction argument is needed.

50.3. **Discrete convexity reduces every order to two points.** Set $a_k = b_{2k}$, $a_0 = 0$. Equation (HC3) with $k-1, k+1$ gives

$$2a_k \le a_{k-1} + a_{k+1}, \qquad 0 \le a_k \le C'k. \tag{HC7}$$

The integer increments $d_k = a_k - a_{k-1}$ are nondecreasing, initially nonnegative, and bounded. If one increment exceeded the linear-growth constant, all later increments would do so, contradicting (HC7). Hence they are eventually a fixed integer d. For large k write

$$a_k = dk - D, \qquad D = \sum_{j \ge 1} (d - d_j) \ge 0, \tag{HC8}$$

where only finitely many summands are nonzero. By (HC4), $k \mid 2a_k$ for every k, so every sufficiently large k divides $2D$. Thus $D = 0$. Each summand in (HC8) is nonnegative, and every increment equals $d = a_1$. Therefore

$$b_{2k} = kb_2, \qquad 2b_n \le b_{2n} = nb_2. \tag{HC9}$$

The second inequality is the empty-tuple case of (HC3). Convexity and linear growth without (HC4) would not suffice: $a_k = \max(0, k-1)$ is a counterexample. The polygon argument is what removes that defect.

The two-point elimination theorem gives $b_2 = 0$ under its ultraviolet shape estimate. Equations (HC9) then give $b_n = 0$ for every n. Consequently the ordinary-chart kernel is invariant under rotations in the chosen plane. The following argument upgrades this fact to the actual off-diagonal distributions, including their exceptional supports.

50.4. **A local analytic translation lemma.** Fix a coordinate j and divide the entries of a compact separated tuple into two groups whose supports lie on opposite sides of a coordinate hyperplane, with a positive gap. Their scalar moment is a matrix element in the coordinate-j Hilbert picture. After removing some unused gap, it has the form

$$\sigma(F_- \cdot \tau_z F_+) = \langle \Phi, e^{-(d+z_j)H_j} U_{z_\parallel} \Psi \rangle, \qquad d > 0, \tag{HC10}$$

for real z sufficiently small. Here the left group has been reflected to define Φ, both groups have been shifted away from their mirror, and d is the remaining normal separation. This is exactly the compact-tuple translation dictionary, applied to products in each group. Its vectors need not be point fields, and no coincident product is used.

Let v be any real four-vector. The right side with $z = sv$ is holomorphic for small complex s. The precise sufficient domain is

$$d + v_j \operatorname{Re} s > \sum_{r \ne j} |v_r| \, |\operatorname{Im} s|. \tag{HC11}$$

Indeed the joint spectral cone $|P_{j,r}| \le H_j$ bounds the spectral exponential by one there. On every smaller domain a positive damping margin controls all powers of H_j and $P_{j,r}$, so difference quotients converge in operator norm. This proves holomorphy and local uniform bounds, rather than formally translating test functions to complex arguments. Translating the negative group instead gives the same conclusion with the corresponding signs. In particular, any specified half

of a strict coordinate cut may be moved in any real direction, with a holomorphic parameter near zero.

50.5. **Removing the exceptional distributional support.** For $n \geq 2$ put

$$\mathcal{C}_n = \{(x_1, \dots, x_n) \in (\mathbb{R}^4)^n : x_i \neq x_l \ (i \neq l)\}.$$

Let T_n be the off-diagonal moment distribution. For a fixed rotation R_u define its difference by

$$D_u(f_1 \otimes \cdots \otimes f_n) = \sigma(f_1, \dots, f_n) - \sigma(R_u f_1, \dots, R_u f_n), \qquad (R_u f)(x) = f(R_u^{-1} x). \tag{HC12}$$

Both terms are well-defined distributions on $\mathcal{C}_n$.

First suppose all original and all rotated time coordinates at a configuration are distinct. Choose small product support boxes keeping both time orderings strict. In those boxes approximate each smooth test in C^1 by finite sums $\alpha(x_4, x_1)\varphi(x_2, x_3)$, with supports staying inside the boxes. For example, a product Fourier expansion with fixed compact cutoffs gives these approximants with all first derivatives. The rotation-invariant ordinary-chart kernel gives equality for the plate approximants. The local moment bounds pass it to the original tests. Local product-tensor density then says that D_u vanishes on this ordinary open set. This step alone would not rule out a distribution concentrated at the remaining time ties.

Fix any $y \in \mathcal{C}_n$. For each coordinate take all threshold hyperplanes halfway between successive distinct coordinate levels of the finite set $\{y_1, \dots, y_n\}$. Each threshold divides the indices into two nonempty sets, and these finitely many cuts together separate every pair of indices. No point lies on one of their hyperplanes. There exist $u_0 > 0$ and a product neighborhood W of y such that, for every $|u| < u_0$, each cut has the same index partition for x and for $R_u x$, for all $x \in W$, with a uniform positive gap in both configurations. To see this, take the minimum distance of the finitely many original points from these finitely many hyperplanes. Rotation is continuous and the points are bounded, so their displacement and then the size of W can be made smaller than a quarter of that minimum. Also take $u_0 < \pi/3$. The cuts and their partitions are kept fixed throughout the following local argument.

Fix $|u| < u_0$. At $x \in W$ let

$$q(x) = \#\{i < l : x_{i,4} = x_{l,4}\} + \#\{i < l : (R_u x_i)_4 = (R_u x_l)_4\}. \tag{HC13}$$

The set $\{q \leq q_0\}$ is open: every inequality between distinct time coordinates remains an inequality in a sufficiently small neighborhood. We prove by induction that D_u vanishes on this open set for each integer q_0. The case $q_0 = 0$ is the ordinary-chart argument above.

Take a point with $q = q_0 > 0$. Choose a tied pair in either frame and one of the fixed common cuts that separates it. Let A be the positive side of that cut. Translate every entry in A by se_4. Original cross-group time differences have slope 1 or -1 in s; rotated cross-group differences have slope $\cos u$ or $-\cos u$. All slopes are nonzero. Internal differences in either group are unchanged. For all sufficiently small nonzero real s, no previously nonzero difference becomes zero, while every previously zero cross difference becomes nonzero. At least the chosen tied pair is of this kind, so the number in (HC13) decreases.

Choose one such s_* and a small real interval I about it. Shrink the product neighborhood V of the original point so that its translates stay in W, all previously nonzero differences stay nonzero, and all cross-group differences in both frames are nonzero for every $s \in I$. Every configuration in the translated product boxes then has $q \leq q_0 - 1$. By induction, D_u vanishes there as a distribution.

Now take arbitrary smooth slot tests supported in V. Their difference (HC12), after the group translation, is a holomorphic function of s near the real segment joining zero to I. For the first moment term this is (HC10)–(HC11) with $v = e_4$. For the rotated term the same common cut separates the same two groups, and the direction is $v = R_u e_4$. Strict gaps persist on the chosen segment; shrinking the segment and its complex neighborhood supplies (HC11) in both cases.

This is why both configurations were required to share the cut. The holomorphic function is zero on I, hence is zero at zero. Thus D_u vanishes on every such product tensor in V. The explicit local Fourier tensor density for distributions makes it vanish on all tests in V.

This proves the induction step. Since $q \le n(n-1)$, the induction terminates and D_u vanishes on W. It excludes a singular part on any of the exceptional time-tie sets, including configurations with coincident plane projections but separated transverse positions. At no point was the kernel integral assumed analytic across those sets.

50.6. **All angles, all planes, and the Schwartz domain.** We have proved local invariance for every sufficiently small u on a neighborhood of each point of $\mathcal{C}_n$. Let a compactly supported off-diagonal test be given, and fix a finite angle v. The union of the supports of its rotations between zero and v is a compact subset of $\mathcal{C}_n$. Cover it by finitely many neighborhoods just constructed, and take the minimum of their positive angle bounds. Partition v into finitely many smaller steps. On each step the difference distribution is zero on the current support, so composition gives invariance through v.

Thus all separated smooth moments are invariant in the chosen plane. Compact C^1 tests follow by smooth approximation on fixed separated neighborhoods. The hypercubic group conjugates this circle of rotations to rotations in every coordinate plane. Givens elimination of the columns of an orthogonal matrix expresses every member of $\mathrm{SO}(4)$ as a finite product of such plane rotations: successively zero the entries below each diagonal position, using a rotation in the corresponding two-coordinate plane, and finish with diagonal signs. A coordinate reflection supplies the other component of $\mathrm{O}(4)$.

For $n = 1$, the translation-invariant one-point functional is $c \int f$ and is already invariant. Restoring such a mean to a centered field uses a finite subset expansion on each compact tuple; this commutes with every rotation because $\int Rf = \int f$. Finally rotations act continuously on Schwartz space, preserve every pairwise coincidence diagonal, and preserve flatness there. The diagonal-flat extension and its constrained tensor density therefore transfer the same invariance to that full test domain.

Theorem 50.1 (All-orders angular closure)**.** *Under the moment, cone, analytic-kernel and two-point hypotheses stated above, the full scalar hierarchy is* $\mathrm{O}(4)$ *invariant, both on separated compact* C^1 *tuples and on its unique diagonal-flat Schwartz extension. The polygon, discrete-convexity and exceptional-support steps require no further analytic rotation assumption.*

This closes a conditional Euclidean-covariance implication. It neither proves the ordinary interacting source estimates nor the ultraviolet two-point law, and it does not identify the scalar cyclic theory with a complete family of Yang–Mills observables.

51. Conditional measures on a compact metric configuration space

The conditional laws used in the phase construction take values in the Borel probability measures on a compact metric configuration space. The conditioning field may be completed and need not be countably generated as a literal collection of sets. It is the Borel field of the *target* space that is countably generated.

Theorem 51.1 (Compact-metric conditional law)**.** *Let X be a nonempty compact metric space, let μ be a Borel probability measure on X, and let $\overline{\mathcal{B}}_\mu$ be the μ-completion of its Borel field $\mathcal{B}(X)$. Suppose*

$$\mathcal{N}_\mu \subset \mathcal{H} \subset \overline{\mathcal{B}}_\mu, \tag{CD1}$$

where $\mathcal{H}$ is a σ-field and $\mathcal{N}_\mu$ denotes all subsets of Borel μ-null sets. There exists a map $K : X \to \mathcal{P}(X)$, where $\mathcal{P}(X)$ carries its weak topology, with the following properties.

The map K is both Borel measurable and $\mathcal{H}$-measurable. For every bounded Borel $f : X \to \mathbb{C}$,

$$Kf(x) := \int_X f(y)\, K(x, dy)$$

(CD2) *is a version of $E_\mu[f \mid \mathcal{H}]$.*

In particular $x \mapsto K(x, B)$ is $\mathcal{H}$-measurable for every $B \in \mathcal{B}(X)$, and

$$\int_H K(x, B)\, \mu(dx) = \mu(H \cap B)$$

(CD3) $(H \in \mathcal{H}, B \in \mathcal{B}(X))$.

The measure on the right is understood on the completion. If K' is another probability-valued kernel satisfying (CD2), then there is a single μ-null set N such that

$$K(x, \cdot) = K'(x, \cdot)$$

(CD4) $(x \notin N)$

as Borel probability measures. One may take N to be a Borel null set. Consequently the law $K_*\mu$ on $\mathcal{P}(X)$ is independent of the choice of conditional kernel.

Borel representatives and countably many tests. We first record why completion causes no obstacle to the Borel version in this statement. Every bounded $\overline{\mathcal{B}}_\mu$-measurable real function u has a Borel representative. Indeed, for each rational r replace $\{u > r\}$ by a Borel set differing from it inside a Borel null set. Outside the union of those countably many null sets, the rational level cuts reconstruct u. Their supremum, clipped to a bound for u, is a Borel function equal to u there. If u was $\mathcal{H}$-measurable, this representative remains $\mathcal{H}$-measurable: the change is confined to a Borel null set, all of whose subsets belong to $\mathcal{H}$ by (CD1). This observation will be applied only countably many times.

Choose a countable dense set $\{x_j\}$ in X. Let $D \subset C(X; \mathbb{R})$ be the collection generated by the constant 1 and the functions $y \mapsto d(y, x_j)$ under rational linear combinations, multiplication, pointwise minimum and pointwise maximum. It is countable, is a rational vector lattice and algebra, and separates points: if $x \neq y$, a dense point sufficiently near x gives different distances from x and y. Its uniform closure is a real algebra containing the constants and separating points, so the Stone–Weierstrass theorem makes D uniformly dense in $C(X; \mathbb{R})$.

For each $f \in D$ take a conditional expectation u_f of f given $\mathcal{H}$. These conditional expectations can be obtained as the orthogonal projections of f onto the closed subspace $L^2(\mathcal{H}, \mu)$ of $L^2(\mu)$. The projection identity against 1_H, $H \in \mathcal{H}$, gives

$$\int_H u_f \, d\mu = \int_H f \, d\mu. \tag{CD5}$$

Testing the sets on which $u_f < 0$, or u_f exceeds an upper bound for f, proves positivity and the usual L^∞ bound. Linearity and uniqueness follow from the same identity. Choose for each u_f a bounded representative that is both Borel and $\mathcal{H}$-measurable, using the preceding observation.

There are only countably many assertions

$$u_1 = 1,$$

$$u_{rf+sg} = r u_f + s u_g$$

$$(f, g \in D, r, s \in \mathbb{Q}),$$

$$u_f \geq 0$$

$$(f \in D, f \geq 0 \text{ on } X). \tag{CD6}$$

Each holds almost surely. Their exceptional sets therefore have a single Borel, $\mathcal{H}$-measurable null union N_0. This is the simultaneous exceptional set needed for the construction. No intersection over all continuous or all Borel functions is taken.

Pointwise Riesz representation. For $x \notin N_0$ define $L_x(f) = u_f(x)$ on D. It is rational linear, positive on nonnegative elements of D, and $L_x(1) = 1$. If rational $r > \|f\|_\infty$, then $r \pm f \in D$ are nonnegative, so

$$|L_x(f)| \leq r.$$

Letting rational r decrease to $\|f\|_\infty$ proves $|L_x(f)| \leq \|f\|_\infty$. Uniform density consequently gives a unique continuous extension to $C(X; \mathbb{R})$. The extension is real linear: approximate a real scalar by rationals and use the norm bound. It is positive: if $f \geq 0$ is continuous, choose $f_j \in D$ with $\|f_j - f\|_\infty < 1/j$; then $f_j + 1/j \geq 0$ and $L_x(f_j) \geq -1/j$. It still takes the value one at 1.

The Riesz representation theorem supplies a unique regular Borel probability measure $K(x, \cdot)$ representing L_x. For $x \in N_0$ set $K(x, \cdot) = \delta_{x_*}$ at one fixed $x_* \in X$. Thus $K(x, \cdot)$ is a probability measure at *every* point. Its values on the null set are a choice, with no conditional interpretation required.

We check the probability-valued measurability rather than inferring it from uncountably many pointwise representations. Enumerate D as $\{d_j\}$ and define

$$I : \mathcal{P}(X) \longrightarrow \mathbb{R}^{\mathbb{N}},$$

$$I(\rho) = (\rho(d_j))_j. \tag{CD7}$$

The map is continuous and injective, by uniform density and uniqueness in Riesz representation. The weak probability space $\mathcal{P}(X)$ is metrized by $\sum_j 2^{-j} |\rho(d_j) - \rho'(d_j)|/(1 + 2\|d_j\|_\infty)$: uniform density identifies convergence in this metric with weak convergence. It is compact. Indeed, compactness follows by taking a diagonal subsequence of integrals on D, extending the resulting positive normalized functional as above, and applying Riesz again. Hence I is a homeomorphism onto its compact image in the Hausdorff space $\mathbb{R}^{\mathbb{N}}$.

Every coordinate of $I \circ K$ equals u_{d_j} outside N_0 and the constant $d_j(x_*)$ on N_0. It is both Borel and $\mathcal{H}$-measurable. The countable product map is therefore measurable for both fields; composing with the inverse of I proves the asserted measurability of K. This also supplies an explicit countable determining family for uniqueness.

All bounded Borel tests and uniqueness. For $f \in D$, the equality $Kf = u_f$ holds outside N_0; thus (CD5) holds with Kf in place of u_f. Uniform approximation extends it to every real continuous f, and Kf is measurable as a uniform limit of the corresponding measurable integrals.

Here are the remaining monotone-class details. The collection of bounded real Borel functions f for which Kf is measurable for both fields and

$$\int_H Kf \, d\mu = \int_H f \, d\mu$$

$$\text{for every } H \in \mathcal{H} \tag{CD8}$$

is a vector space containing $C(X;\mathbb{R})$. It is closed under uniformly bounded increasing limits, by monotone convergence for each probability $K(x,\cdot)$ and for μ; subtracting a constant reduces to nonnegative functions. The bounded functional monotone-class theorem therefore makes it all bounded Borel functions, since the continuous algebra generates $\mathcal{B}(X)$. One can see the latter directly from a countable dense family of distance functions and the countable base of metric balls. Complex functions follow by their real and imaginary parts. This proves (CD2), and indicators give (CD3). In particular it also verifies directly that K is a probability kernel, including measurability of every Borel-set evaluation.

For uniqueness, apply uniqueness of conditional expectation to Kd_j and $K'd_j$ for the countably many $d_j \in D$. Outside a single null set all these integrals agree. Uniform density then gives equality on every continuous function, and Riesz uniqueness gives (CD4). The null set belongs to the completed Borel field and hence is contained in a Borel null set, as asserted. Changing a map on a null set does not change its probability-valued law. The law has barycenter μ, because (CD8) with $H = X$ gives

$$\mu(f) = \int_{\mathcal{P}(X)} \rho(f)\,(K_*\mu)(d\rho) \qquad \text{(CD9)}$$

(f bounded Borel).

The integrand is Borel on $\mathcal{P}(X)$: this is immediate for continuous f, and the same bounded monotone-class argument proves it for every bounded Borel f. □

The completed invariant field and the exact null-set scope. For a countable group of Borel, μ-preserving transformations, the field of events in $\overline{\mathcal{B}}_\mu$ invariant modulo μ under every group element is a σ-field containing $\mathcal{N}_\mu$. Thus it satisfies (CD1). In particular the theorem applies to the completed translation-invariant field of a compact metric lattice configuration space. The conditional measures are Borel probabilities on that configuration space; no countable generation of the completed invariant field is needed.

The uniqueness assertion is simultaneous for the resulting probability measures. The assertion that Kf is a conditional-expectation version holds separately for each bounded Borel f and, of course, for any preselected countable collection on one common conull set. It does not assert equality outside one universal exceptional set to an arbitrary previously selected version for *every* Borel function.

Nor does the theorem assert pointwise properness $K(x,H) = 1_H(x)$ simultaneously for all conditioning events. Even when those events are Borel, such a demand is false for completed fields. For example let μ be nonatomic and let $\mathcal{H} = \{H \in \overline{\mathcal{B}}_\mu : \mu(H) \in \{0,1\}\}$ be the null/co-null field. The conditional kernel is $K(x,\cdot) = \mu$ almost surely. If simultaneous properness held outside one null set, then at any point x of that conull set the conditioning event $\{x\} \in \mathcal{H}$ would require $K(x,\{x\}) = 1$, whereas $\mu(\{x\}) = 0$. This does not contradict (CD2): for a fixed Borel $H \in \mathcal{H}$, it gives $K(x,H) = 1_H(x)$ almost surely, with an exceptional set that may depend on H. Equation (CD3) is the corresponding integral identity. The theorem concerns the countably generated Borel target $\mathcal{B}(X)$; no regular-conditional-law assertion for an arbitrary non-countably-generated target field is being made.

52. A phase construction from upstream lattice estimates

Under the lattice estimates specified below, we construct the canonical phase measure from explicit probabilistic objects. The finite-spacing law G3 is proved in Section 8; it is used before any continuum component is constructed. The all-state moment and one-point comparison estimate G4 is derived in Section 38 from the ordinary torus source obligation D1. Conditional probability

measures are constructed in Section 51. The argument here deduces the canonical phase measure from those precise ingredients.

In this section a pure state means an extreme point of the translation-invariant DLR class $\mathcal{G}_j$, equivalently an ergodic state for lattice translations. Extremality in the larger class of all, possibly noninvariant, DLR states is not used and is not identified with this property without an additional argument.

Put $a_j = L^{-K_j} \downarrow 0$, with L an integer, and let $\Omega_j = \mathrm{SU}(2)^{\mathcal{B}_j}$ be the space of bond configurations on $a_j\mathbb{Z}^4$. The set $\mathcal{B}_j$ contains one orientation of each bond. Its countability makes Ω_j compact and metrizable: enumerate the bonds and use the sum of 2^{-m} times bounded metrics on the group factors. The Wilson specification γ_W integrates finitely many bonds W with density proportional to $\exp[-\beta_j \sum_{p:p\cap W\neq\varnothing} a_p]$. The denominator is positive, and compactness of the integration space shows that $\gamma_W F$ is continuous and local whenever F is. Define $\mathcal{G}_j$ to consist of the translation-invariant probability measures with $\rho(\gamma_W F) = \rho(F)$ for every finite W and continuous local F. The equations are weakly closed, so $\mathcal{G}_j$ is compact and convex. Its nonemptiness follows from the fixed-spacing torus-limit construction, and a centering state $\nu_j \in \mathcal{G}_j$ is fixed. G1 supplies the lattice symmetries and mirror positivity used in G3.

Let $\mathfrak{T}$ be the set of finite tuples of real C^1 compactly supported functions with pairwise disjoint supports, including the empty tuple. Concatenation is used only when its resulting supports are disjoint. For $X = (f_1, \dots, f_k)$ define the bounded continuous local function

$$X_j = \prod_{i=1}^{k} [\mathcal{O}_j(f_i) - \nu_j(\mathcal{O}_j(f_i))],$$

$$\varnothing_j = 1.$$

Assume $\nu_j(X_j) \to \sigma(X)$ on $\mathfrak{T}$. For support radii R and separation d, put $b_1 = C_1(R)$ and $b_n = C_n(d; R)$ for $n \geq 2$, and define

$$M_k(d; R) = \sum_{\pi \in \mathrm{Part}(\{1,\dots,k\})} \prod_{J\in\pi} b_{|J|},$$

$$M_0 = 1.$$

The ordinary moment–cumulant identity, applied under any $\rho \in \mathcal{G}_j$, gives

$$|\rho(X_j)| \leq M_k(d; R) \prod_i \|f_i\|_\sharp.$$

Indeed, a singleton block contributes the difference between the means in ρ and ν_j, and a larger block contributes the corresponding cumulant under ρ. This is why a bound on one-point differences is needed. Telescoping a difference of two products gives equicontinuity with the same constants whenever all supports remain in fixed separated containers.

52.1. **Decomposition at one spacing.** Let $\mathcal{I}$ denote the completed invariant σ-field for a fixed $\nu \in \mathcal{G}_j$. Theorem 51.1 gives a probability kernel $\omega \mapsto \nu^\omega$ satisfying $\nu^\omega(F) = E_\nu[F \mid \mathcal{I}](\omega)$ for bounded Borel F. The following details ensure that its law is a decomposition into the required states, rather than into arbitrary probability measures.

First, a bounded invariant density preserves the DLR equations. To prove it, approximate such a density h in $L^1(\nu)$ by bounded continuous local functions h_n. This approximation is available because cylinder functions are uniformly dense in continuous functions on the compact product space, and continuous functions are dense in L^1 of a Borel probability measure. Translate h_n so

its support misses W. Invariance of h and ν leaves the approximation error unchanged, while $\gamma_W(h_nF) = h_n\gamma_W F$ for the translated function. Passing to the limit proves $\nu(h\gamma_W F) = \nu(hF)$.

Consequently $\nu^\omega(\gamma_W F) = \nu^\omega(F)$ almost surely. Likewise $\nu^\omega(\tau_v F) = \nu^\omega(F)$ almost surely for every lattice translation v. These statements can be imposed on one common full-measure set: finite bond sets and translations form countable families, and continuous functions on the compact configuration space have a countable dense family of continuous cylinder functions. Supremum-norm continuity then extends the equations to all continuous local functions. A monotone-class argument extends the integral identities to bounded Borel functions. Thus $\nu^\omega \in \mathcal{G}_j$ almost surely.

The law G3 proves that almost every such state is ergodic. Fix a continuous local A. Applied to ν^ω and then integrated, its far-translation limit gives

$$\int \mathrm{Pl}_{\nu^\omega}(A, A)\, d\nu(\omega) = \mathrm{Pl}_\nu(A, A) = \int |\nu^\omega(A)|^2\, d\nu(\omega).$$

The first passage to the limit uses the bound $\|A\|_\infty^2$ and dominated convergence. The difference of the integrands on the two sides is the nonnegative variance of the invariant conditional expectation of A in ν^ω. It is therefore zero almost surely. Impose this on a countable dense family of continuous local A. Density in $L^2(\nu^\omega)$ then says that every invariant L^2 function is constant, which is ergodicity.

Write $\mathcal{E}_j$ for these ergodic states. They are exactly the extreme points of $\mathcal{G}_j$. A nontrivial invariant event decomposes a state into two DLR states by the invariant-density argument. Conversely, the Radon–Nikodym density in a convex decomposition of an ergodic invariant state is invariant, and hence constant. Moreover $\mathcal{E}_j$ is closed. If $\rho_n \in \mathcal{E}_j$ tends weakly to ρ, G3 gives a decay estimate uniform in n. Passing first to this weak limit and then to far translations identifies $\mathrm{Pl}_\rho(A, A)$ with $|\rho(A)|^2$. The same dense-set argument proves that ρ is ergodic. Thus $\mathcal{E}_j$ is compact.

Let D_j be the law of ν_j^ω under ν_j. It is supported on $\mathcal{E}_j$, has barycenter ν_j, and does not depend on versions of the kernel. For this last assertion choose one countable determining family of continuous cylinder functions. Conditional expectations are unique almost surely for that entire family simultaneously; equality on the family implies equality of the probability measures. This also establishes measurability of the probability-valued map and of every evaluation used below.

Uniqueness among all decompositions supported on $\mathcal{E}_j$ requires one more observation. For any such law with barycenter ν_j, the uniform law G3 identifies the integral of a product of finitely many state evaluations with successive far-translation limits of ν_j. These moments are therefore the same as those of D_j. Evaluations of continuous local functions are continuous and separate measures in the compact space $\mathcal{E}_j$. Their polynomial algebra is dense by Stone–Weierstrass, so equal mixed moments imply equal laws.

52.2. **Continuum evaluation coordinates.** Choose countable dense test sets as follows. Let $\mathcal{Q}$ contain all finite unions of closed boxes with rational corners. The space of C^1 functions vanishing off $Q \in \mathcal{Q}$, with its C^1 norm, is separable: embed it by its value and four first derivatives in a finite product of separable continuous-function spaces. Choose a countable dense set in each space and enlarge their union by the countable group of L-adic translations and by W_4. Let $\mathfrak{T}^\circ$ consist of the separated tuples from this union. It is countable. Every separated tuple can be approximated by such tuples with supports in fixed separated containers: cover each original support by a finite union of sufficiently small rational boxes inside its chosen open container and use the dense set for that union.

For each $X \in \mathfrak{T}^\circ$ choose once and for all a moment bound M_X from the preceding estimate. Define the compact metrizable product

$$\mathsf{K} = \prod_{X \in \mathfrak{T}^\circ} [-M_X, M_X],$$
$$e_j(\rho) = (\rho(X_j))_{X \in \mathfrak{T}^\circ},$$
$$\lambda_j = (e_j)_* D_j.$$

The evaluation map is continuous for each fixed j. In particular λ_j is a well-defined probability measure on K.

Theorem 52.1 (Conditional canonical decomposition)**.** *Under G1, the finite-spacing law G3, the estimates G4, and the stated convergence of $\nu_j(X_j)$, the measures λ_j converge weakly to one probability measure λ. Put $\mathsf{X} = \operatorname{supp}\lambda$. Every $\xi \in \mathsf{X}$ defines a unique functional σ_ξ on $\mathfrak{T}$, obtained as a limit along pure states $\rho_j \in \mathcal{E}_j$ centered at ν_j. The maps $\xi \mapsto \sigma_\xi(X)$ are continuous and separate points. Furthermore*

$$\sigma(X) = \int_{\mathsf{X}} \sigma_\xi(X)\, d\lambda(\xi).$$

Their joint moments are the successive far-translation limits of σ. More precisely, for $q \geq 2$,

$$\int_{\mathsf{X}} \prod_{i=1}^{q} \sigma_\xi(X^i)\, d\lambda(\xi) = \lim_{|v_q|\to\infty} \cdots \lim_{|v_2|\to\infty} \sigma(X^1 \cdot \tau_{v_2} X^2 \cdots \tau_{v_q} X^q).$$

The innermost limit is taken first. At each stage the remaining groups must have disjoint supports; every sufficiently distant next translation is then admitted. Each of these intermediate limits exists. These properties determine the compact space, measure and functionals uniquely among compact metrizable full-support representations with continuous point-separating evaluations and the displayed mixed-moment property, up to the homeomorphism preserving all evaluations. Each component clusters with a bound $C(X,Y)|v|^{-2}$ for sufficiently distant separated products, where $C(X,Y)$ and the distance threshold are uniform in ξ.

Proof. The crucial uniformity comes from reflection sizes rather than supremum norms of X_j. Choose an integer physical half-side ϖ so that all supports under consideration lie in $(-\varpi + 1, \varpi - 1)^4$. This also leaves room for the corners of plaquettes whose barycenters are in those supports. Then $r_j = \varpi/a_j$ is an integer. Each expression defining $\mathcal{N}_{\rho,r_j}(X_j)$ is the moment of two separated reflected copies of X. G4 bounds it uniformly in j and ρ by a constant determined by $M_{2|X|}$ and the test norms. For a displacement v common to the lattices,

$$\frac{r_j}{|v|_\infty/a_j - 2r_j} = \frac{\varpi}{|v|_\infty - 2\varpi}.$$

Thus G3 gives, in every pure state, factorization of the two separated products with error at most $C(X,Y)|v|^{-2}$, uniformly in the cutoff. The norms of the individual representatives, which can diverge, do not enter this bound.

For finitely many coordinates $X^1, \ldots, X^q$, integrate this factorization over D_j. Choose v_q first, then v_{q-1}, down to v_2, with each new vector sufficiently far from the already fixed union. In the factorization at that stage the enclosing support box and its constant may depend on the fixed outer translations, but not on j. Repeating the estimate and multiplying errors only by the uniform individual moment bounds makes the total error smaller than any prescribed ϵ, uniformly in j. At that one fixed finite configuration the remaining single moment converges by the hypothesis on ν_j. The mixed coordinate moments of λ_j are therefore Cauchy. Their precise

iterated-limit representation will follow after the components and their uniform clustering have been constructed. Polynomials in finitely many coordinates separate points of K and contain constants. By Stone–Weierstrass they are uniformly dense in its continuous functions. Compactness therefore supplies a weak limit, and the convergent mixed moments make that limit unique along the full sequence.

If ξ belongs to its support, every neighborhood of ξ has positive λ measure. Portmanteau implies that its preimage has positive D_j measure for all sufficiently large j. Use a nested countable neighborhood base and a diagonal choice to obtain $\rho_j \in \mathcal{E}_j$ with $e_j(\rho_j) \to \xi$. The uniform telescoping estimate extends convergence from $\mathfrak{T}^\circ$ to every tuple, giving σ_ξ. It also proves independence from the choice of ρ_j and continuity in ξ, by uniformly approximating a given evaluation with dense-set evaluations. Multilinearity, symmetry and the moment bounds pass to this limit.

Barycenters and higher mixed moments pass to the limit first on the countable set and then on all tuples by the same uniform approximation. The evaluations separate points because their countable subfamily consists of the coordinates of K. If another compact full-support representation has the same mixed moments, embed both spaces by *all* their tuple evaluations into $\mathbb{R}^{\mathfrak{T}}$. Their images have compact union. Polynomials in finitely many of these coordinates are dense in the continuous functions on that union, again by Stone–Weierstrass. Equality of mixed moments therefore gives equality of the two image measures. Each compact image equals the support of its image measure. The images consequently coincide, and injectivity and compactness identify the two spaces by a homeomorphism preserving the measure and every evaluation. This argument does not presume that a countable subfamily separates an arbitrary competing representation.

Finally pass the pure-state factorization estimate to the limit for L-adic translations. Continuity of C^1 tests in the norm $\|\cdot\|_\sharp$ extends it to real translations, preserving the uniform constants after a fixed enlargement of the support margin. This proves the asserted clustering. Finally hold $v_3, \dots, v_q$ fixed and let $|v_2|$ tend to infinity in the barycenter identity. Uniform component clustering separates X^2 from the fixed union of all the other groups. The uniform moment bounds justify passing this limit through the integral. Then let $|v_3|$ tend to infinity, and continue in that order. This proves existence of every intermediate limit and the displayed mixed-moment formula, with the innermost limit taken first. □

53. The invariant Hilbert space of a canonical phase measure

The canonical decomposition is a statement about correlation forms. Its relation to the zero-energy Hilbert space is a further implication, proved here with the required density argument. No assertion that a linear span of phase evaluations is automatically an algebra is needed.

Let X be a compact metrizable phase space, let λ be a Borel probability measure of full support on X, and let σ_ξ be its component correlation forms. They act on finite separated tuples of compactly supported test functions and their complex linear span, including the empty tuple 1. Assume that

$$f_F(\xi) := \sigma_\xi(F) \in C(X),$$

$$\sigma(F) = \int_X f_F \, d\lambda,$$

$$f_1 = 1. \tag{PV1}$$

Assume that the functions f_F separate points of X, and that the components preserve conjugation, time reflection, and translations. Concatenation is used only when all its supports are separated.

The quantitative phase construction supplies uniform clustering: for any fixed separated tuples F, G, once their translated supports are disjoint,

$$\sup_{\xi\in X} |\sigma_\xi(F\,\tau_{te_4}G) - \sigma_\xi(F)\sigma_\xi(G)| \longrightarrow 0 \qquad (t \longrightarrow \infty). \tag{PV2}$$

Finite linear combinations inherit this property by summation. Uniform separated moment estimates ensure that all functions displayed in (PV1)–(PV2) are bounded, for each fixed tuple. The phases need not have the same one-point mean.

Assume that the reflection-positive quotient has been constructed and completed to a Hilbert space $\mathcal{H}$, with dense classes $[F]$ of separated tuples whose compact supports lie strictly in $x_4 > 0$, and their linear combinations. Suppose positive time translations induce a strongly continuous contraction semigroup e^{-tH} with a nonnegative self-adjoint generator H and

$$\langle [F], e^{-tH}[G]\rangle = \sigma(\overline{\theta F}\,\tau_{te_4}G), \qquad t > 0. \tag{PV3}$$

These are the explicit Hilbert-space hypotheses; this section does not derive them from qualitative convergence alone. Write $P_0 = 1_{\{0\}}(H)$ for the zero-energy projection.

Theorem 53.1 (The zero-energy representation)**.** *Under (PV1)–(PV3), there is a unique unitary*

$$U : \ker H \longrightarrow L^2(X, \lambda)$$

such that

$$UP_0[F] = f_F. \tag{PV4}$$

The vacuum vector [1] *maps to the constant function* 1*. If X has exactly $m < \infty$ points, then* $\dim\ker H = m$*. If X is infinite, then* $\ker H$ *has countably infinite Hilbert dimension.*

Proof. The spectral theorem and dominated convergence give $e^{-tH} \to P_0$ strongly as $t \to \infty$. For fixed F, G, use (PV1), reflection invariance, conjugation, and (PV2) in (PV3) to obtain

$$\begin{aligned}\langle P_0[F], P_0[G]\rangle &= \lim_{t\to\infty} \int_X \sigma_\xi(\overline{\theta F}\,\tau_{te_4}G)\,\lambda(d\xi) \\ &= \int_X \overline{f_F(\xi)} f_G(\xi)\,\lambda(d\xi).\end{aligned} \tag{PV5}$$

The limit is justified by uniform clustering, applied to the finite linear combinations as well as to individual tuples. In particular, if $P_0[F] = 0$, then $f_F = 0$ in L^2. Thus the rule in (PV4) is well defined and isometric on the span of the displayed vectors. That span is dense in $\ker H$, since the classes $[F]$ are dense in $\mathcal{H}$ and P_0 is a bounded projection. Its extension is an isometry with closed range. It remains to prove surjectivity.

Let $\mathcal{V}$ be the complex linear span in $C(X)$ of all the phase evaluations f_F with F supported at positive times. Any separated tuple can be moved into this half-space by a positive time translation, which leaves its phase evaluations unchanged. Consequently $\mathcal{V}$ still separates points of X. It contains 1 and is closed under complex conjugation.

For positive-time tuples F, G, the concatenation $F\,\tau_{te_4}G$ is a positive-time separated tuple for all sufficiently large t. By (PV2),

$$f_F \tau_{te_4} G \longrightarrow f_F f_G$$

uniformly on X.

It follows first for generators, and then for their finite linear combinations, that products of elements of $\mathcal{V}$ belong to its uniform closure. The uniform closure is itself closed under products: approximate each of two factors uniformly by elements of $\mathcal{V}$ and use boundedness of the approximants. It is therefore a unital conjugation-closed subalgebra separating points. The complex Stone–Weierstrass theorem makes that closure equal to $C(X)$. Since continuous functions are dense in $L^2(X, \lambda)$, the image in (PV4) is dense and closed, hence is the whole space.

The stated vector correspondence determines the unitary on a dense set, proving uniqueness. It sends $[1] = P_0[1]$ to 1, since translation fixes the empty tuple.

For a finite phase space, full support gives positive mass to each of its m points, so L^2 has dimension m. For an infinite compact metric space, L^2 is separable. One way to see this is to take a countable uniformly dense subset of $C(X)$ and then its rational complex linear span, which is dense in L^2. Its dimension cannot be finite: for every integer m choose m distinct points and disjoint open neighbourhoods of them. Full support gives each neighbourhood positive measure, so their indicator functions are m nonzero mutually orthogonal vectors. The dimension is therefore countably infinite. This proves all assertions. □

In particular, a single phase gives a one-dimensional zero-energy space under these hypotheses. Several phases can produce several vacua, even when each component has a unique vacuum. An infinite phase set is recorded by the extended natural count ∞ in dimension statements; its set-theoretic cardinality is not equated with Hilbert dimension. Nothing in this identification gives a strictly positive lower spectral bound on $(\ker H)^{\perp}$.

54. Reconstruction of the action-density sector

Reconstruction: hypotheses, consequences, and scope. All growth constants below are positive, the factorial exponents are nonnegative, and derivative-order coefficients are positive integers. The D2 and H1 identifiers are retained for the conditional graded-moment and diagonal-flat growth conclusions. G2 now denotes the derived scalar rotation requirement under the explicit ordinary-source, mirror, cone and ultraviolet hypotheses described below.

The passage from Euclidean correlation functions to a quantum field requires several logically separate inputs. Compactness of separated correlations gives an off-diagonal distribution. Reflection positivity gives a Hilbert space. Uniform growth in the number of arguments is needed for the corrected reconstruction theorem. Full Euclidean covariance and clustering have to be established in their own right. Keeping these steps separate prevents a qualitative convergence statement from carrying quantitative information it does not contain.

The Euclidean test domain. Write

$$D_n = \{(x_1, \dots, x_n) \in (\mathbb{R}^4)^n : x_i \neq x_j \text{ whenever } i \neq j\}.$$

Let $\mathcal{S}_{\neq,n}$ be the closed subspace of the Schwartz space $\mathcal{S}(\mathbb{R}^{4n})$ consisting of functions whose every derivative vanishes on the complement of D_n. These functions are flat on all coincidence diagonals. Let $\mathcal{S}_{+,n}$ consist of those elements that are also flat outside the region in which all time coordinates are positive. Set $\mathcal{S}_{\neq,0} = \mathcal{S}_{+,0} = \mathbb{C}$. The time reflection is $\theta(x_1, x_2, x_3, x_4) = (x_1, x_2, x_3, -x_4)$, and

$$(\Theta\overline{\varphi})(x_1, \dots, x_n) = \overline{\varphi(\theta x_1, \dots, \theta x_n)}.$$

Symmetry of the functions makes reversal of the order of the arguments immaterial in this scalar setting.

These are valid test domains for the historical Osterwalder–Schrader reconstruction problem. They do not assert that the Euclidean functions have a preferred extension to every Schwartz test. A distribution supported on a coincidence diagonal vanishes on $\mathcal{S}_{\neq,n}$. Such a distribution therefore cannot be recovered from separated correlations. Any later use of contact terms, composite operator identities at coincident points, or local Ward identities has to specify the additional extension that it uses. This observation does not obstruct reconstruction on the stated domain.
Density lemma. Finite sums of tensors $f_1 \otimes \cdots \otimes f_n$, with smooth compactly supported f_i having pairwise disjoint supports, are dense in $\mathcal{S}_{\neq,n}$. The corresponding assertion holds in $\mathcal{S}_{+,n}$ when all supports lie at positive times.

The proof, including the simultaneous infinity, diagonal and positive-time cutoffs and the product Fourier approximation, is given in Section 41.

The lemma is a statement about the flat subspace. It gives no density claim in the full Schwartz space, where evaluation on the diagonal is a continuous functional that vanishes on every such tensor.

What convergence preserves. Let σ_j be centred Euclidean moment functionals at finite cutoffs. Suppose that for every separated smooth tuple X the numbers $\sigma_j(X)$ converge to $\sigma(X)$. Suppose also that each σ_j is reflection positive for finite families of internally separated positive-time tuples. Different tuples in a family may overlap. This last qualification is essential: reflection positivity is a matrix condition, not a condition on one tuple at a time.
Closed positivity lemma. For positive-time tuples $X_1, \dots, X_r$ and complex coefficients $c_1, \dots, c_r$,

$$\sum_{i,k=1}^{r} \overline{c}_i c_k \, \sigma(\theta\overline{X_i} \cdot X_k) \geq 0.$$

Proof. Each X_i is internally separated. Its conjugated reflected entries have negative times, so $\theta\overline{X_i} \cdot X_k$ is a separated tuple for every pair (i,k). The displayed expression is the limit of the corresponding finite sum for σ_j, all of which are nonnegative. □

Suppose, in addition, that the limiting functionals extend continuously to $\mathcal{S}_{\neq,n}$. Reflection positivity then holds on finite Schwartz sequences $\Phi = (\varphi_0, \dots, \varphi_N)$ with $\varphi_n \in \mathcal{S}_{+,n}$:

$$\sum_{n,m=0}^{N} S_{n+m}(\Theta\overline{\varphi}_n \otimes \varphi_m) \geq 0.$$

Indeed, the mixed tensor belongs to $\mathcal{S}_{\neq,n+m}$. Coincidences within one factor are killed by its flatness; a coincidence across the two factors has time coordinate zero or lies outside the allowed half-space of one factor, so is killed there. Approximate each φ_n by the tensors in the density lemma, using a seminorm strong enough for all the finitely many mixed terms, and pass to the limit. Permutation symmetry and invariance under any fixed Euclidean isometry pass in exactly the same way once they hold on separated tensors.

There is one further step when translations are initially known only for vectors in a dense subgroup, for example the union of nested lattice translation groups. Given an arbitrary vector v, choose subgroup vectors $v_k \to v$. Translation is continuous on the smooth tests, and the local bounds permit passage from $S_n(\tau_{v_k}\varphi) = S_n(\varphi)$ to $S_n(\tau_v\varphi) = S_n(\varphi)$. This argument does not generate rotations from the finite hypercubic group.

The quantitative reconstruction hypothesis. Fix the seminorms

$$|\varphi|_k = \max_{|\alpha|\le k} \sup_x (1+|x|^2)^{k/2} |\partial^\alpha \varphi(x)|.$$

For reconstruction, assume a sequence of symmetric functionals $S_n \in \mathcal{S}'_{\neq,n}$, with $S_0 = 1$, and constants $A, B, k_0, p < \infty$, independent of n, such that

$$|S_n(\varphi)| \le A B^n (n!)^p |\varphi|_{k_0 n}. \tag{RC1}$$

Assume full Euclidean invariance and reflection positivity on the domains just described. Assume also spatial clustering on compact positive-time sequences: for every such finite pair F, G and every nonzero spatial vector v,

$$\sigma(\theta\overline{F} \cdot \tau_{\lambda(v,0)} G) \longrightarrow \overline{\sigma(F)}\sigma(G) \qquad (\lambda \longrightarrow \infty).$$

Here each homogeneous term of a sequence is a finite sum of internally separated smooth compact tuples. The Hilbert-space approximation below extends this cluster identity to the full positive-time test domain.

The imported result is Theorem $E' \to R'$ of [OS75, §IV.1, p. 287]. Its condition $E0'$ is the bound by a single Schwartz seminorm of order sn and constants of factorial growth, on the diagonal-flat space. This is exactly the form of (RC1): take an integer $s \ge \max(1, k_0)$ and $\sigma_n = AB^n(n!)^p \le Ae^B (n!)^{p+1}$. Euclidean invariance, the full-sequence reflection-positive form, permutation symmetry and the cluster identity supply its remaining conditions $E1$–$E4$, respectively. Thus the theorem gives a Hermitian scalar Wightman field with positive energy, locality, Poincaré covariance and a unique vacuum in its cyclic representation. Its analytic-continuation proof is imported; the interacting estimates needed to obtain a hierarchy with these properties remain separate premises.

Condition (RC1) must not be replaced by the weaker assertion that every S_n is separately continuous with unspecified constants. The growth of the distribution order and the constants in n matters to the existence of tempered boundary values in the analytic continuation. Likewise, a local off-diagonal kernel theorem alone does not establish (RC1).

The explicit rooted-source and dyadic extension route. There is now a quantitative route to (RC1), rather than an unspecified fixed-n Whitney estimate. Theorem 39.1 derives moments directly from the ordinary source obligation D1 and the proved G1 chessboard domination, using the actual domain geometry in (GS1)–(GS3). At scale $a_s = L^{-s}$, domains are unions of Ma_s cubes, indexed by those unions, and their source dependence occupies only a fixed number of neighboring cube layers. Their size is the minimum cubical edge-tree length meeting all cubes, with source-energy decay above its rooted entropy rate. Here $M = L^{m_0}$, and the cofinal tori $\ell = 2L^m$, $m \ge m_0$, tile every actual scale with even cube counts. No scale is discarded for a failure to tile. The weak abstract (ST6) count alone does not furnish the sharper rooted and global counts needed for this proof.

For real compact C^1 tests, the theorem gives one fixed finite integer k, $B(R) = C(R+1)^4$, and the two-state centered bound

$$\left| \rho \prod_{i=1}^{n} [\mathcal{O}_a(f_i) - \nu(\mathcal{O}_a(f_i))] \right| \le B(R)^n (n!)^k \max\{1, d^{-5n}\} \prod_i \|f_i\|_\sharp. \tag{RC2}$$

Each support lies in its own radius-R ball, placed anywhere; the separation is d. The same factorial exponent bounds higher cumulants. Uniformity lets (RC2) pass to the separated continuum limits and their component limits. This is requirement D2 conditional on the actual D1 source representation and G1, not a new independent analytic estimate.

Theorem 41.1 uses only the unit-radius version, with the fixed equivalence of the sharp and C^1 norms absorbed in $B(1)$. Its product Fourier construction uses $8n$ derivatives; Gevrey-two pair/grid cutoffs cost at most $C^n(n!)^{32}$. Dyadic Taylor flatness to order $10n + 2$ pays the local h^{-6n} cost and the h^{-4n} configuration grid count, leaving a summable dyadic square. It yields (RC1) with

$$k_0 = 20,$$
$$p = k + 32.$$

The proof includes independence of tensor presentations, overlap gluing, cutoffs at infinity and every pairwise diagonal, and the positive-time boundary. Its constrained tensor density supplies the full-sequence reflection-positive form. Thus H1 is a proved implication of D2 here; the ordinary analytic RG source estimates in D1 are still assumptions. Neither separate continuity with arbitrary order constants nor weak (ST6) counting alone may be substituted for this GS/QF route.

Hilbert space and clustering. Section 42 proves the quotient construction on compact positive-time tuples from their invariant reflection-positive moments and position-uniform per-ball bounds. Fixed separation of the reflected supports bounds the translated orbits before iterated Cauchy–Schwarz proves contraction. The resulting time semigroup is strongly continuous, positive and self-adjoint, and spatial translations are strongly continuous unitaries. The stated lattice approximation also gives the cube spectral cone. The GS/QF growth and density argument then identifies its completion with the full positive-time Schwartz sequence quotient. These deductions retain their explicit hypotheses; positivity alone is not being substituted for the continuity and growth checks.

For completeness, clustering on a dense class of vectors extends to the whole Hilbert space. Let $U(v)$ denote the spatial unitary translations, and suppose that for dense vectors ξ, η,

$$\langle \xi, U(v)\eta \rangle \longrightarrow \langle \xi, \Omega \rangle \langle \Omega, \eta \rangle$$
$$(|v| \to \infty).$$

Choose dense approximants $\xi_k \to \xi$ and $\eta_k \to \eta$. Unitarity bounds the difference of the matrix elements, uniformly in v, by

$$\|\xi - \xi_k\| \, \|\eta\| + \|\xi_k\| \, \|\eta - \eta_k\|.$$

The vacuum matrix elements obey the same estimate. First let $|v| \to \infty$ at fixed k, then let $k \to \infty$. The argument proves the qualitative cluster limit. It does not preserve a numerical decay rate on arbitrary vectors without further quantitative control of the approximations.

Phases and full rotational covariance. A symmetry of a mixture need not be a symmetry of each component. For example, the uniform probability measure on a sphere is rotation invariant, while its decomposition into point masses is permuted by rotations. Therefore an invariant Euclidean limit and an ergodic decomposition do not, by themselves, supply an invariant component. A construction that reconstructs every phase must prove covariance and positivity for those phases, or realize each phase as a limit of cutoff states for which the properties and uniform estimates

have already been established. The phase-interface construction provides this realization for the G1 mirror properties and the D1/GS/QF moment bounds. The state-uniform version of the two-point extraction must also hold for every approximating state. Under that ultraviolet input, the angular chain proves covariance for each realized component. One first subtracts its own mean $c \int f$; the finite subset expansion and convergent cutoff one-point differences identify the self-centered limit, and the same expansion restores the mean afterwards. It is not inferred merely from a symmetric mixture.

Similarly, invariance under the hypercubic group, coordinate reflection positivity, and translation invariance do not by themselves imply invariance under O(4). The finite group is not dense in the rotation group. Any route through diagonal mirrors, a sharp energy–momentum cone, bounded insertions, or the exclusion of nonconstant angular harmonics needs an additional theorem. The following chain now supplies that theorem conditionally, before the external reconstruction theorem is invoked.

Theorem 50.1 completes the scalar covariance implication. Its preceding sections prove, in order, the uniform Hilbert insertion bound, the analytic L^1 density, the product kernels and continued Gram identities, and the ultraviolet elimination of the two-point harmonics. Polygon divisibility and common coordinate cuts then remove all higher harmonics and exceptional distributional support. The proof works with finite support boxes; density of generic point configurations alone would not exclude a singular part.

The ultraviolet premise is exactly (TU10) on the full positive-shape class with $\vartheta_* = 1/12000$, or a broader class, uniformly in the approximating states and positions. The dependency catalogue proves its implication from F2–F3 by the separated-kernel bound and logarithmic coupling decay. A narrower shape class or a torus-only law cannot be substituted. The resulting G2 deduction applies to periodic-state limits and their realized components carrying the mirror and exact cone interface; arbitrary growing cubic-torus limits require that interface to be justified separately. Typed Ward, distant-shell flux and matched-run estimates remain scoped to their separate stronger routes.

The actual Gaussian, history and finite-source estimates and the interacting extraction are now developed as proof outlines in Sections 32, 36, 43.1–43.3 and 44.3, with the Gaussian and substitution estimates cited there. The OS analytic-continuation theorem is still imported, and the identification with full Yang–Mills observables remains separate.

What the reconstructed field establishes. When the limiting observable is the centred scalar action density, the theorem above constructs its scalar observable sector. A nonzero two-point function on separated tests excludes a field that is a multiple of the identity. A nonzero higher cumulant excludes Gaussianity of that scalar field. Neither conclusion alone identifies a full Yang–Mills observable algebra. In particular, a Wick square of a free Gaussian field already has nonzero higher cumulants. Establishing a Yang–Mills theory in a stronger sense requires the stated additional observables, identities, and compatibility relations that define that sense.

Nor does polynomial clustering establish a mass gap. The positive measure $Ee^{-E}\,dE$ on $[0,\infty)$ has Laplace transform

$$\int_0^\infty Ee^{-E}e^{-tE}\,dE = (1+t)^{-2}.$$

Its support reaches zero, although its transform decays quadratically. A positive spectral gap requires an estimate that excludes spectral support in an entire interval above the vacuum. That question remains separate from the existence and reconstruction of the scalar sector described here.

Appendix A. Coupling coordinates and optional comparison implications

This section preserves the conditional targets of the coupling, matched-run and Ward arguments. It supplies their elementary analytic implications without treating a proposed interacting estimate as a proof. The pointed families, companions, physical Hessian extractor, midpoint tents, list action and radius weights are defined in Section 13. The pointed cancellation, nonlinear window estimate and finite formation sections have their stated background and domain hypotheses. None computes the one-loop trace or proves a matched-cutoff or typed Ward estimate.

Vacuum accounting and the stronger coupling inputs. Write $P_\circ = (\mathbf{1}, 0)$. With $E_{(0)} = 0$, the accumulated vacuum normalization is defined by

$$\begin{aligned} E_{(k+1)} &= E_{(k)} + \mathsf{e}_k, \qquad E = E_{(K)}, \\ \mathsf{e}_k &= \mathfrak{n}_k + \sum_X \left[\sum_{z \in X} E^{(k+1)}(X, P_\circ, z) + R^{(k+1)}(X, P_\circ) \right]. \end{aligned} \tag{CP1}$$

Here $\mathfrak{n}_k$ is the assigned field-independent step number. For $\tau = x/\theta$, its Gaussian scaling part is $-3|\mathfrak{X}_k| \log \tau/2$; $\mathfrak{X}_k$ is the set of non-pivot bonds, with three real color coordinates per bond. Haar and gauge-fixing counts are included in that assigned number. The family in (CP1) contains the unnormalized local Gaussian numbers: $\mathbf{N}^\circ = \mathbf{N} + \mathbf{N}^\circ(P_\circ)$. Those numbers must not also be added to $\mathfrak{n}_k$. Equivalently, using $\mathbf{N}$ requires adding the removed local numbers once separately. This convention counts every constant once. At fixed calibration, $\mathfrak{n}_k, \mathbf{N}^\circ, \mathbf{G}_\tau$ have zero source derivative. Source energies are the differences of the displayed E- and R-values between sourced and unsourced runs. More explicitly, the localized source energy of level j is

$$e_j(X; w) = \sum_{z \in X} \{E^{(j),w}(X, P_\circ, z) - E^{(j),0}(X, P_\circ, z)\} + R^{(j),w}(X, P_\circ) - R^{(j),0}(X, P_\circ).$$

The source vacuum exponent is the sum of these values over levels and local domains. This definition introduces no additional arbitrary normalizer. The zero source derivative of the Gaussian pieces refers to the frozen potential presentation. Moving the source into the Gaussian action weight changes the determinant and is another presentation, whose numbers must be compared by its own sum rule.

Requirement B3 retains an actual one-loop input:

$$b_i = b_i^0 + f_i + q_i, \qquad c_0 = \frac{11N^2}{12\pi^2} \log L, \qquad \left| \sum_{i=1}^{r} (b_i^0 - c_0) \right| \le C_0. \tag{CP2}$$

Here $N = 2$, and b_i^0 is independent of the bare coupling and admissible torus. A summable estimate $|b_i^0 - c_0| \le CL^{-\sigma i}$, $\sigma > 0$, suffices. The asserted normalization requires the actual lattice determinant/trace calculation and its zero-mode convention. Gaussian integration alone does not establish it.

Some later slope conclusions require more than (CP2). Fix a calibration $\theta_0, \dots, \theta_K$, independent of the varied bare coupling. Put $\xi_j = \theta_j - x_j$. Require the actual unmarked f_i and marked q_i, for $1 \le i \le K$, to be jointly holomorphic on the independent earlier-variable polydiscs with radii

$$\begin{aligned} r_j^{\mathrm{u}} &= c\theta_j, & \sup |f_i| &\le C\theta_i^{-1/2} (\log \theta_i)^p, \\ r_j^{\mathrm{m}} &= c\theta_j (\log \theta_j)^{-C_*}, & \sup |q_i| &\le C\theta_i^{-p_R}, \qquad 0 \le j < i. \end{aligned} \tag{CP3}$$

The constants $p, C_* \ge 0$, $p_R > 0$, and $c > 0$ are fixed. The actual runs must remain in the half polydiscs, with common localization, counterterms and admissible domains. These are stronger actual coupling inputs, including the marked tube. They are not inferred from holomorphy of the bare partition function, nor from a running-radius activity bound followed by two field derivatives.

The marked exponent is often $p_R = (\kappa_0 - 4)/2$, but any later use must retain the actual proved loss and tube radius. All norms and source Taylor expansions here use fixed declared domains; shrinking the field domains while differentiating would be a different claim. Compatibility of the companion and earlier cutoff prefix also remains part of the construction.

A dimension-free polydisc lemma and its precise consequence.

Lemma A.1 (The bounded-polydisc estimate, code 1*e*.3.7)**.** *If F is holomorphic on $\prod_{j=1}^{d} D(0, r_j)$ and $|F| \le B$, then*

$$\sum_j r_j \left(1 - \frac{|\xi_j|^2}{r_j^2}\right) |\partial_j F(\xi)| \le B. \tag{CP4}$$

On the half polydisc this implies

$$\begin{aligned} \left|\sum_j v_j \partial_j F(\xi)\right| &\le \tfrac{4}{3} B \max_j |v_j|/r_j, \\ |F(\xi) - F(\xi')| &\le \tfrac{4}{3} B \max_j |\xi_j - \xi'_j|/r_j. \end{aligned} \tag{CP5}$$

No constant depends on d.

Proof. The case $B = 0$ is immediate. Set $a_j = \xi_j/r_j$ and $\phi_a(z) = (a + z)/(1 + \bar{a}z)$. Choose unit complex numbers ϵ_j so the terms in the derivative at zero of

$$H(z) = F\big(r_j \phi_{a_j}(\epsilon_j z)\big)_{j=1}^{d}$$

have the same argument. Since $\phi_a'(0) = 1 - |a|^2$, the modulus of that derivative is the left side of (CP4). The one-variable Schwarz lemma, applied after a disc automorphism sending $H(0)/B$ to zero, gives $|H'(0)| \le B(1 - |H(0)/B|^2) \le B$. The Schwarz lemma itself follows by applying the maximum principle to a holomorphic function vanishing at zero divided by z, first on smaller circles and then letting their radius increase to one. An attained boundary value makes H constant. On the half polydisc, (CP4) gives $\sum_j r_j |\partial_j F| \le 4B/3$. This proves the first estimate of (CP5); integration along the segment from ξ to ξ', which stays in that half polydisc, proves the second. □

Assume $\theta_j \ge \theta_i$ for $j < i$, with all centers sufficiently large that $t/(\log t)^{C_*}$ is increasing. Applying the lemma separately in (CP3) bounds the absolute row sum by

$$a_i := \sum_{j<i} |\partial_{\xi_j}(f_i + q_i)| \le C \left[\theta_i^{-3/2} (\log \theta_i)^p + \theta_i^{-p_R - 1} (\log \theta_i)^{C_*}\right]. \tag{CP6}$$

Suppose also that the calibration has backward linear drift $\theta_i \ge h + c_1(K - i)$, $c_1 > 0$. Comparison with integrals gives

$$A := \sum_{i=1}^{K} a_i \le C \left[h^{-1/2} (\log h)^p + h^{-p_R} (\log h)^{C_*}\right] \longrightarrow 0.$$

Indeed substitute $t = hu$ in each integral; the integrals of $u^{-1-\delta}(1 + |\log u|)^r$ are finite for $\delta > 0$. A drift with a fixed additive error is equivalent after increasing h.

For the actual differentiable recursion $x_0(\beta) = \beta$, $x_i(\beta) = \beta - \sum_{r \le i} b_r(\beta)$, put $M_i = \max_{j \le i} |x_j'|$. Calibration is held fixed, so $|b_i'| \le a_i M_{i-1}$ and $M_i \le (1 + a_i) M_{i-1}$. Consequently

$$M_K \le \prod_i (1 + a_i) \le e^A, \qquad \sum_i |b_i'| \le \prod_i (1 + a_i) - 1 \le e^A - 1. \tag{CP7}$$

The middle estimate telescopes the successive products. In particular

$$|x'_K(\beta) - 1| \le \sum_i |b'_i| \le e^A - 1 = O\Big(h^{-1/2}(\log h)^p + h^{-p_R}(\log h)^{C_*}\Big).$$

On witness prefixes with terminal $h \to \infty$, under the same uniformly admitted hypotheses, this makes the constant-source normalization tend to one with the displayed error. Identifying that prefix and derivative with the actual F3 witness remains an input; the estimate does not establish that identification. At weak coupling $A \le \log(4/3)$, hence the last sum is at most $1/3$. Constant sources varying β to $\beta - \zeta$ have the same absolute slopes. This supplies the slope premise in (LB12) under (CP3) and the admitted drift/domain conditions. It does not prove those conditions. Reality gives $2/3 \le x'_K(\beta) \le 4/3$ on an admitted real interval. Integrating the derivative proves the uniform terminal bi-Lipschitz bound. If the same row estimates hold on a whole complex offset disc, write $\Xi_K(t) = h - t + R_K(t)$; then $R_K(0) = 0$ and $\sup |R'_K| \le 1/3$. One can construct the matching inverse without any global uniqueness assertion. For $|t| < \varrho$, solve

$$s = h - \Xi_K(t) + R_{K+1}(s)$$

on the closed disc $|s| \le 2\varrho$, assuming the functions are defined on a neighborhood of that disc. Its right side maps the disc strictly inside itself and has Lipschitz constant at most $1/3$. Iteration converges geometrically to its unique solution, uniformly on smaller t-discs. The iterates are holomorphic, so the solution is holomorphic; reality gives a real solution at real t. The bounds $|\Xi_K(t) - h| \le 4|t|/3$ and $|\Xi_{K+1}(s) - h| \ge 2|s|/3$ give $|\mathfrak{n}_K(t)| \le 2|t|$. This supplies the coordinate part of E2 when its actual full-disc premises are available, rather than proving the existence of those premises. Nor does a polydisc supremum give forgetting in $i-j$: $F_i(\xi) = B\xi_0/r_0$ is bounded by B for every i, with unchanged first derivative.

The fixed-torus coordinate and the matched limit. Fix a torus T_ℓ, $h = g^{-2}$, and a separated tuple of compact Lipschitz tests, with the fixed norm $\|\cdot\|_\sharp$ of the source convention. Every bound below includes their product norm; its constant may depend on order and positive separation. Let $\beta_K^\natural$ be its torus-independent tuned bare center and write depth as $[s] = K - k$. Define

$$\begin{aligned} \mathsf{J} &= [-\varrho/4, \varrho/4], \qquad \Xi_K(t) = x_K^\natural(\beta_K^\natural - t), \\ F_{K,n}(t;\mathbf{f}) &= S^T_{n;K,\ell,\beta_K^\natural - t}(\mathbf{f}), \qquad \mathfrak{n}_K = \Xi_{K+1}^{-1} \circ \Xi_K. \end{aligned} \tag{CP8}$$

The inverse is the specified holomorphic matching branch. For obligations E2–E3 we must prove its existence on a common neighborhood, reality on the real axis, uniform upper and lower real Lipschitz bounds for Ξ_K, $\Xi_K(0) = h$, and $|\Xi_K(t) - h + t| \le |t|/3$. On a fixed smaller complex neighborhood of J, we must prove

$$\begin{aligned} &|\mathfrak{n}_K - t| + |\mathfrak{n}'_K - 1| + |\Xi_{K+1} - \Xi_K| \le C\eta_K, \\ &|F_{K,n}(t) - \mathfrak{n}'_K(t)^n F_{K+1,n}(\mathfrak{n}_K(t))| \le C_{n,\ell,\mathbf{f}}\epsilon_K, \qquad \sum_K (\eta_K + \epsilon_K) < \infty. \end{aligned} \tag{CP9}$$

The $F_{K,n}$ must be uniformly bounded and holomorphic on a slightly wider strip, for example $\mathcal{N}_\ell = \{|\operatorname{Re} t| < \varrho/2,\ |\operatorname{Im} t| < \upsilon_\ell\}$, where $\upsilon_\ell = c\varrho e^{-C(g)\ell^4} > 0$. Matching paths stay in that strip after shrinking it. Constants and cutoff floors may depend on this fixed ℓ. Often $\eta_K = \Upsilon_K \le CK^{-3/2}(\log(2+K))^{p+1}$; a separate remainder error Π_K is part of ϵ_K, not a polarization tensor.

Proposition A.2 (The conditional terminal-coupling limit)**.** *These inputs imply uniform convergence $F_{K,n} \to S^{[t]}_{\ell,n}$ on the smaller strip and $\Xi_K \to \Xi_\infty$. The latter is a real bi-Lipschitz homeomorphism from J onto $I \supset [h - \varrho/6, h + \varrho/6]$. Set*

$$S^{(\ell,x)} = S_\ell^{[\Xi_\infty^{-1}(x)]}, \qquad x \in I.$$

If $t_K \in \mathsf{J}$ and $\Xi_K(t_K) \to x$, then $F_{K,n}(t_K) \to S_n^{(\ell,x)}$, without a subsequence.

Proof. Cauchy's estimate on the wider strip bounds $F'_{K,n}$ uniformly. Thus replacing $\mathfrak{n}_K(t)$ by t in (CP9), and its derivative factor by one, costs $C\eta_K$. The resulting increments are summable, uniformly on the smaller strip. Their limit is holomorphic and continuous on J. The coordinate increments are summable too. Its real Lipschitz bounds pass to the limit; the endpoint bounds give the asserted interval. Uniform coordinate convergence gives $\Xi_\infty(t_K) \to x$; the lower Lipschitz bound forces $t_K \to \Xi_\infty^{-1}(x)$. Uniform convergence of F_K and continuity of its limit finish the proof. □

This preserves target $0(d)$ under its comparison premises. It proves no infinite-volume uniqueness. The depth coordinate $\Xi_{\infty,s}(t) = \lim_K x^\natural_{K-s}(\beta^\natural_K - t)$ requires the corresponding fixed-depth comparison. If its envelope is

$$\lambda s - C \le \Xi_{\infty,s}(t) - \Xi_{\infty,0}(t) \le \Lambda s + C,$$

then $\Xi_{\infty,s}(t)^{-1/2}$ decreases in order as $s^{-1/2}$; at length L^{-s} this is logarithmic running. The reference value $g_{[s]} = \theta_{[s]}^{-1/2}$ is a calibration, not this arbitrary member's running coupling. Common-prefix matching must also control $\hat{t}_{K,s} = \beta^\natural_{K-s}(g_{[s]}) - \beta^\natural_K(g)$. If it converges to $\hat{t}_s$, exact finite dilation passes to $S_\ell^{[t]}[g] = S_{L^s\ell}^{[t+\hat{t}_s]}[g_{[s]}](\mathbf{f}^{[s]})$, wherever both offsets are in their windows. Obligation E4 separately compares self-calibrated and frozen runs, places their first-passage offsets in J, and supplies $|K(\beta^\natural_K - t, g) - K| \le C(\varrho)$. Neither assertion follows from the new finite source derivative. Exact lattice dilation uses $f^{[b]}(y) = f(L^{-b}y)$, without an observable prefactor, and $T_\ell \mapsto T_{L^b\ell}$. Conditional offset limits and depth shifts pass through that identity only when the offsets converge and both torus floors hold. The change-of-reference claims retain these matching premises. The bare tuned-increment target must also be retained explicitly:

(CP9a) $$\beta^\natural_{K+1}(g) - \beta^\natural_K(g) \longrightarrow c_0, \qquad \beta^\natural_{K+b}(g) - \beta^\natural_K(g) \longrightarrow bc_0 \quad (b \in \mathbb{Z} \text{ fixed}).$$

The second assertion follows by finite telescoping from the first. Here is a conditional proof preserving the stronger comparison route. Write $x_{[s]}^{[i]}$ for the tuned depth-s coupling in the run of length i, with $x_{[0]}^{[i]} = h$ and $x_{[i]}^{[i]} = \beta_i^\natural$. Require $x_{[v]}^{[i]} - x_{[u]}^{[i]}$, for $u < v$, to lie in $[\lambda(v-u)-C, \Lambda(v-u)+C]$, and

$$|x_{[s]}^{[i+1]} - x_{[s]}^{[i]}| \le C\big(L^{-c(i-s)} + s\Pi_i\big), \qquad \sup_{s\ge1} s \sum_{i\ge s} \Pi_i < \infty.$$

These constants and the last bound are fixed before ϱ. Comparing two bare centers at the same depth K', then telescoping in the run length, bounds their extra discrepancy by $C\sum_{i\ge K'} L^{-c(i-K')} + CK'\sum_{i\ge K'}\Pi_i \le C'$. The spacing bounds therefore give a uniform bound A on $\alpha_K = \beta^\natural_{K+1} - \beta^\natural_K$. Choose $\varrho > 20A$.

Each cluster value α of α_K gives $S_\ell^{[t]} = S_{L\ell}^{[t-\alpha]}(\mathbf{f}^{[1]})$ by exact finite dilation and uniform moving-offset convergence. Two cluster values $\alpha^- < \alpha^+$ give an offset period $\omega = \alpha^+ - \alpha^-$ on an open interval. Holomorphy in the offset extends that identity throughout the overlapping strips. Repeated blocking can then restore every offset modulo ω into $[0, \omega)$; all intermediate offsets stay in J by the choice of ϱ. Require, additionally, one fixed positive shape Ψ, a floor ℓ_0, and $r_0 > 0$, such that

$$0 < \mathfrak{m}(r) \le S_{\ell,2}^{[t]}(\Psi_r) \le \mathfrak{M}(r), \quad \mathfrak{M}(r) \longrightarrow 0, \qquad t \in \mathsf{J},\ \ell \ge \ell_0,\ 0 < r \le r_0.$$

Normalization by the two fixed supremum norms is understood. The required uniformity in offset, shape placement and arbitrarily large tori is part of the unverified tuned two-point comparison. For $r_m = L^{-m}r_0$, blocking gives $\mathfrak{m}(r_0) \le \mathfrak{M}(r_m) \to 0$, a contradiction. Thus α_K has one

cluster value and converges. Finally (CP2)–(CP3), backward drift and $x_K^\natural = h$ make the average nonlinear increment error tend to zero, so $\beta_K^\natural/K \to c_0$. Cesaro averaging identifies the unique cluster value with c_0. Without these two-run and uniform-envelope premises, (CP9a) remains an additional comparison obligation. With (CP9a), exact finite dilation and the moving-offset limit give

$$S_\ell^{[t]}(\mathbf{f}) = S_{L^b\ell}^{[t-bc_0]}(\mathbf{f}^{[b]}) \qquad (b \geq 1),$$

when both offsets lie in their windows and the torus floors hold. Indeed the same bare coupling has the second offset $t-[\beta_{K+b}^\natural - \beta_K^\natural]$, which converges to $t-bc_0$. Thus the original blocking target is preserved without claiming it follows from one-step potential estimates.

Distinct offsets need the additional 7*e* germ-dilation premise. Fix a nonempty class of nonnegative C^2 pairs: each test is supported in a radius-r ball, the two supports have distance in $[r, 6r]$, and each test satisfies $\int f \geq \vartheta r^4 \|f\|_\infty$ and $\operatorname{Lip} f \leq \|f\|_\infty/(\vartheta r)$. Define $\sigma \approx_2 \sigma'$ by an error $Cr^2\|f\|_\infty\|h\|_\infty$, uniformly on this class at all sufficiently small r. Set $(D_\mu\sigma)_2(f,h) = \sigma_2(f(\cdot/\mu), h(\cdot/\mu))$. Require $\sigma_t \approx_2 D_\mu\sigma_{t'}$, with $\mu = \exp((t-t')\log L/c_0)$, and uniform two-point upper and lower envelopes of order $(\log(1/r))^{-2}$. A germ cannot be fixed by D_μ, $\mu \neq 1$. For $\mu > 1$, apply its error bound to one fixed rescaled shape, with normalized value $Q(r)$, at radii $r\mu^{-a}$. The upper envelope gives $Q(r\mu^{-a}) \to 0$, so

$$Q(r) \leq C\sum_{a\geq 1}(r\mu^{-a})^2 = \frac{Cr^2}{\mu^2-1}.$$

This contradicts the logarithmic lower envelope for arbitrarily small r. For $\mu < 1$, invert the dilation. Thus equality of two different-offset limits would contradict rigidity. The comparison retains its shape, placement, offset and torus-floor quantifiers; it is an additional premise, not a consequence of matching alone.

Why a geometric relabelling needs another construction. Take three test tuples of the same fixed correlation order, for example three pairs, with observable values

$$F_K(t) = (1,\, 1+t,\, 1+\Pi_K) = \alpha + \beta t + \gamma\Pi_K, \qquad \Pi_K = K^{-q}, \quad q > 1.$$

Here $\alpha = (1,1,1)$, $\beta = (0,1,0)$, $\gamma = (0,0,1)$, with determinant one. On $|t| \leq 1/4$ the values are positive and uniformly bounded. With $\Xi_K(t) = h - t$ and $\mathfrak{n}_K = \mathrm{id}$, all summable matching inequalities hold; their increments are $O(K^{-q-1})$. Suppose a reparametrization $t \mapsto \tau_K(t)$ and one common normalization $a_K(t)$, bounded above and away from zero, made their increments geometrically small. At a fixed t, geometric summability forces a_K, the first component, to converge geometrically to a nonzero limit. The third component minus the first is $a_K\Pi_K$, tends to zero, and would also be geometrically small. This contradicts its polynomial lower bound. The same argument allows arbitrary reparametrizations: their value does not enter either the first component or the third-minus-first. Only a common, nonvanishing field normalization is required. This is a counterexample to the proposed implication from numerical inequalities, not an assertion that these three numbers already form a physical Yang–Mills state. Thus those inequalities alone do not supply the stronger second-coordinate claim. A proposed coordinate $\hat{x}_K = x_K^\natural - \sum_{i:\, i+\log_L(M\check{R}_i)\leq K} \operatorname{coeff}_A(\operatorname{tot} R^{(i)})$ needs an actual compatible remainder-transfer and pointing theorem. The finite scalar insertion results do not establish that theorem.

The optional Ward target on arbitrary defining sequences. In this section $S_{j,n}$ and S_n, $n \geq 2$, are *connected* forms. Require convergence on every smooth compact separated tuple on $\mathbb{R}^4$, along the specified defining sequence. On every fixed collection of separated compact support neighborhoods require uniform local multilinear C^1 bounds. Mollification inside slightly enlarged separated neighborhoods then extends the forms and their convergence to compact C^1 tuples: first

approximate each entry, use the uniform telescoping bound, and then pass to the limit for the smooth approximants. Assume full signed-permutation symmetry W_4 of the limit. The specified lattice probability laws realize these connected forms; the mixed cumulants below use those laws. For torus approximants a fixed R must fit in one common chart for all sufficiently large j. For $\omega \in \mathfrak{so}(4)$, set $\delta_\omega f = -(\omega x)\cdot \nabla f$ and

$$D^s_{j,\omega} f(x) = \sum_\rho (\omega x)_\rho \frac{f(x + a_j e_\rho) - f(x - a_j e_\rho)}{2a_j}.$$

Here are actual additional Ward inputs, independent of the angular route. Choose smooth $\chi_R = 1$ on B_R, zero outside B_{2R}, with derivatives bounded by $C_r R^{-r}$. For every j, ω, R, supply fixed annular local functions

$$\mathfrak{B}_{j,\omega,R} = \sum_{\Delta \in \mathcal{A}_{j,R}} \mathfrak{B}^\Delta_{j,\omega,R}.$$

There are at most CR^4 physical unit boxes; each summand is a specified gauge-invariant Borel function of links in a fixed enlargement of its box, linear in ω, supported near $R - Ca_j \le |x| \le 2R + Ca_j$. The functions may contain the divergent Wilson weight. They are supplied by the actual change of variables, independently of the tests, not defined as their desired rotation defect. Require the exact finite identity

$$\sum_i S_{j,n}(f_1, \dots, D^s_{j,\omega} f_i, \dots, f_n) = \mathfrak{U}_{j,\omega,R}(\mathbf{f}) + \langle \mathfrak{B}_{j,\omega,R}; \mathcal{O}_j(f_1); \dots; \mathcal{O}_j(f_n)\rangle^T. \tag{CP10}$$

In (CP10), $\mathcal{O}_j(\mathbf{f})$ denotes the separate list of cumulant arguments, not their coincident product. The bulk functional $\mathfrak{U}$ must also be the specified correlation from that same Ward construction; it is not an arbitrary error chosen after the boundary functions. For every fixed sufficiently large R, require $\mathfrak{U}_{j,\omega,R} \to 0$ and the analogous sum with $D^s_{j,\omega} f_i + \delta_\omega f_i$ to tend to zero. These include the actual typed anisotropy and volume-transfer estimates where needed, and are not proved by scalar source bounds.

Multilinear telescoping and local C^1 continuity differentiate the limit's rotation orbit on smooth tuples. Applying the assumed convergence at the fixed derivative tests in (CP10), without differentiating a cutoff limit, gives

$$\lim_j \langle \mathfrak{B}_{j,\omega,R}; \mathcal{O}_j(\mathbf{f})\rangle^T = -\delta_\omega S_n(\mathbf{f}). \tag{CP11}$$

It is independent of R. Consequently target 0(e)'s condition

$$\liminf_{R\to\infty} \limsup_{j\to\infty} \left|\langle \mathfrak{B}_{j,\omega,R}; \mathcal{O}_j(\mathbf{f})\rangle^T\right| = 0 \tag{CP12}$$

is equivalent to vanishing of this derivative. The inner limit always has R fixed.

For completeness, one nonzero generator suffices. Assume also the invariant one-point convention $c\int f$ and zeroth moment one, or use the centered functional. Let G be the closed subgroup of orthogonal transformations preserving the connected forms. Closedness follows from C^1 continuity on separated tuples. Equivalently, define G from the centered moment forms obtained by the finite sum over partitions of these cumulants; the inverse partition formula gives the same group. The invariant one-point function $c\int f$ makes the uncentered moment group the same. Let $\mathcal{A}$ consist of skew matrices whose entire one-parameter group lies in G. It is closed under real scalar multiplication and addition: the finite-matrix product formula

$$e^{t(A+B)} = \lim_{r\to\infty} (e^{tA/r} e^{tB/r})^r$$

follows by expanding each factor to second order and telescoping the product error, which is $O(r^{-1})$. Closedness of G then applies. It is also stable under conjugation by W_4. For $a < b$, sign

matrices $D_\epsilon = \operatorname{diag}(\epsilon)$ give the projection

$$2^{-4} \sum_{\epsilon \in \{\pm 1\}^4} \epsilon_a \epsilon_b D_\epsilon A D_\epsilon = A_{ab} E_{ab}.$$

Here $E_{ab} = e_a e_b^{\mathsf{T}} - e_b e_a^{\mathsf{T}}$ is the rotation generator of the (a, b) coordinate plane; the identity uses that A is skew. Thus any nonzero element of $\mathcal{A}$ supplies a coordinate-plane generator; permutations supply all of them. Successive plane rotations reduce any determinant-one orthogonal matrix to the identity column by column. Hence G contains $\mathrm{SO}(4)$, and its coordinate reflections give all $\mathrm{O}(4)$. Condition (CP12) for a nonzero ω, on every smooth tuple, gives its whole one-parameter group by applying the zero derivative also at each rotated tuple. Conversely full invariance makes (CP11) zero for every ω. No Lie-subgroup theorem is needed. Indeed an infinite closed G has nonidentity $g_r \to I$; the normalized skew logarithms have a nonzero subsequential limit A. Integer powers $g_r^{\lfloor t/\|\log g_r\| \rfloor} \to e^{tA}$ supply the required one-parameter group. Therefore such a W_4-invariant group is finite or all of $\mathrm{O}(4)$.

Infrared meaning and the Hilbert-space consequence. A uniform mixed-insertion estimate $|\langle \mathfrak{B}^{\Delta}; \mathcal{O}_j(\mathbf{f})\rangle^T| \le CD^{-4-\delta}$, $\delta > 0$, at distance D from the fixed supports, implies (CP12): its shell sum is at most $CR^4R^{-4-\delta}$. Decay D^{-4} alone yields only a bounded sum. It also suffices that the sum of absolute mixed insertions tend to zero along one sequence $R_l \to \infty$, after $\limsup_j$. Ordinary per-ball density bounds do not give either estimate for these different, weight-bearing insertions. If an older survival argument is used, its event $\mathcal{L}_{\mathfrak{c},k,a,K}$ means that cell $\mathfrak{c}$ is live at level k, its component lineage starts at $k-a$, and it remains live at every level $k,\dots,K$. Its weighted law is an additional history input.

Assume the relevant centered or recentered moment functional also satisfies the HB quotient hypotheses. Once full scalar rotation invariance is supplied, spatial $R \in \mathrm{O}(3)$ acts on the HB quotient by $V_R[F] = [(R \oplus 1)F]$. It preserves the positive form and null space, has inverse $V_{R^{-1}}$, fixes Ω, commutes with time evolution and conjugates a spatial translation by x to one by Rx. Local C^1 continuity proves strong continuity on dense test vectors, and unitarity extends it to the completion. The joint spectral measure is consequently covariant under $(E,p) \mapsto (E,Rp)$. For limits for which HB's cube inequalities have been established, they give $E \ge |p|$: use a countable dense set of rotations, remove their common null set, and approximate a rotation carrying any given p to $|p|e_1$. Centering is the map C_m proved in Section 42 and changes none of these conclusions. In the alternative notation, $\mathsf{sh}_\xi = C_{-m_\xi}$, $m_\xi(f) = c_\xi \int f$, so its inverse is C_{m_ξ}. No operator for a general time-mixing Euclidean rotation is asserted. External OS reconstruction, F4 identification of the full theory and any mass-gap estimate remain separate. The periodic-state angular proof retains its own mirror, cone and ultraviolet interface; the optional Ward premises above do not silently enlarge that state class.

Appendix B. Notation and units

This index records the meanings and units of the principal symbols. Definitions of analytic domains and history data are given in the technical sections. When a construction requires an auxiliary map not yet instantiated, its type and required estimates are inputs; a symbol in this index does not establish their existence. Hand-numbered references to the notation in candidate arguments refer to this index and the corresponding detailed definition.

Lattices, fields, and action.

$G, \mathfrak{g}, N, d_G$**:** The gauge group is $G = \mathrm{SU}(2)$; $\mathfrak{g}$ is represented by Hermitian traceless matrices with $\langle X, Y\rangle = \frac{1}{2}\operatorname{tr}(XY)$. Here $N = 2$ and $d_G = N^2 - 1 = 3$. In statements retaining N, the corresponding matrix group and trace convention must be specified.

L, M**:** $L > 1$ is the fixed odd blocking integer. M is the fixed side of a localization cube in the units of its own lattice, with $M = L^{m_0}$ for a fixed integer $m_0 \ge 0$. The graded ordinary-source argument uses the cofinal compatible tori $\ell = 2L^m$, $m \ge m_0$, so $\ell/(Ma_s)$ is an even integer at every actual scale. The other geometric parameters M_1, M_2 and the buffer constants are fixed before the cutoff, fields, and sources are chosen.

K, j, s, a_s, η_j**:** K is the number of ultraviolet blocking steps. Level j has depth $s = K - j$. Physical spacing at depth s is $a_s = L^{-s}$. The fine spacing measured in level-j units is $\eta_j = L^{-j}$; it is not the physical spacing a_j.

$\Lambda_{K,\ell}, \mathbb{T}^{(j)}$**:** $\Lambda_{K,\ell} = (L^{-K}\mathbb{Z}/\ell\mathbb{Z})^4$ is a compatible finite torus lattice. $\mathbb{T}^{(j)} = \Lambda_{K-j,\ell}$ is the level-j lattice in physical units. A fixed-spacing infinite lattice is $a\mathbb{Z}^4$. The chosen torus family may restrict ℓ further.

$U_b, U_p, \partial U$**:** $U_b \in G$ is a positively oriented bond variable; $U_{\bar{b}} = U_b^{-1}$. U_p or $\partial U(p)$ is the ordered transport around a plaquette. $x(p)$ is its barycentre. Gauge transformations act by $U_b \mapsto h_{b_-} U_b h_{b_+}^{-1}$.

$a_p, A(w,U), \beta$**:** $a_p = 1 - \frac{1}{2}\operatorname{Re}\operatorname{tr} U_p$, $A(w,U) = \sum_p w(x(p))a_p$, and $A(U) = A(1,U)$. Wilson's weight is $e^{-\beta A(U)}$. The inverse coupling convention is $\beta = g_0^{-2}$; with the usual Wilson convention it equals $2N/g_{\mathrm{std}}^2$.

$\mathcal{O}(f), S_n^T$**:** $\mathcal{O}(f) = \sum_p f(x(p))a_p$ has no additional spacing prefactor. S_n^T is its joint connected correlation, defined by differentiating the logarithm of the sourced partition function. Centering changes only the first cumulant.

Averages, Gaussian coordinates, and propagators.

$\mathcal{M}, Q_k, Q_k^\sharp$**:** $\mathcal{M}$ is the chosen Borel group-valued block average, analytic near regular real configurations. Q_k is the linearization of its plain iterate. $Q_k^\sharp$ is the straightened linearization after the coarse gauge factor is removed. They are different operators at a nontrivial background.

Q_k^{st}, Q_k'**:** Q_k^{st} is a straight covariant bond average. Q_k' averages site functions. An adjoint always uses the inner products of both source and target lattices.

d_0, d_1, d_η**:** d_0 and d_1 denote the lattice gradient and circulation in the cochain convention of the relevant theorem. $d_\eta = \eta^{-1}(\mathrm{shift} - 1)$ is the physical scalar difference in unit blocks. The unscaled combinatorial difference is ηd_η. The free curvature comparison is unchanged by a common scaling of d_1, because it uses the projection onto its range.

Q, P, A_η, G'**:** In the scalar block calculation Q is the arithmetic mean, the fine inner product is $\eta^4 \sum_x$, and the coarse inner product is the unweighted sum. Then Q^* lifts constants, $P = Q^*Q$, $A_\eta = d_\eta^* d_\eta + P$, and $G' = A_\eta^{-1}$. With plain inner products on both lattices the arithmetic mean's adjoint is different. The unitary conversion is proved in the block-normalization section.

$R, \mathbf{R}, R^{(j)}$**:** In scalar Landau-gauge formulae R is an orthogonal projection, for example onto $\ker(QG')$. Bold $\mathbf{R}$ is the large-field operation. $R^{(j)}$ is a local remainder activity. These three uses are distinguished by their types, not interchangeable.

$P_{\mathrm{ex}}, K, \mathsf{K}$**:** P_{ex} is the orthogonal projection onto exact lattice two-forms. Its physical kernel is $a^{-4}P_{\mathrm{ex}}(p,q)$. K or K denotes the six-by-six continuum curvature covariance at unit coupling. Away from zero it equals $(\pi^2|x|^4)^{-1}\Lambda^2(I - 2\widehat{x}\otimes\widehat{x})$. The Fourier distributions also contain contact terms.

$U_j(V), \mathsf{G}, \mathsf{H}$**:** $U_j(V)$ is a constrained regular background with specified coarse averages. The symbols G and H denote a constrained inverse and a response operator in that background. Existence, analytic domains, gauges and uniform bounds are the background obligations A1–A4, which we must prove; the proved scalar model is only part of them.

$\Phi_\Lambda(B), r_B, Z_y$**:** In Section 24, B is a dimensionless coarse bond logarithm and $\Phi_\Lambda(B)$ the actual potential/current background pair on a collared box. The response bound holds throughout the product datum polydisc $\sup_c |B(c)| < r_B$, not merely at zero. The local output norm uses $Z_y = (\xi_y H, \xi_y^2 \nabla H, \xi_y^3 J)$. Finite-box comparison also uses the actual compatible equations and boundary rows in natural force norms, including their cell weights.

Running quantities and source domains.

x_j, g_j, b_j**:** $x_0 = \beta$, $x_j = x_{j-1} - b_j$, and $g_j = x_j^{-1/2}$. The actual increment includes all contributions assigned to the marginal action coefficient. Positivity of every individual b_j is not assumed in the arithmetic theorem.

c_0, b_j^0**:** b_j^0 is a designated one-loop increment. The proposed leading slope is $c_0 = 11N^2 \log L/(12\pi^2)$ in this normalization. The numerical first-passage theorem assumes its interval sum law and an error envelope; it does not calculate this coefficient.

$\theta_j, x_j^\natural$**:** $\theta_j = g^{-2} + c_0(K-j)$ is a prescribed frozen reference sequence. In the technical frame estimates $\bar{g}_j = \theta_j^{-1/2}$ denotes its auxiliary coupling, distinct from the running coupling $g_j = x_j^{-1/2}$. The superscript $\natural$ denotes quantities computed with the auxiliary choices frozen on that sequence. Holomorphy of this calibrated coordinate and its comparison across cutoffs are additional analytic assertions.

$K(\beta, g), \mathcal{W}_K(g)$**:** $K(\beta, g)$ is the first scale at which x_j enters the reference window of width $B_\sharp$ above g^{-2}. $\mathcal{W}_K(g)$ is the set of bare couplings with that first passage. The terminal value can range over the window.

w, c, z, f_i**:** A source is a triple consisting of a constant c, a finite real test family (f_i), and complex coefficients (z_i), with function $w = c + \sum_i z_i f_i$. Analytic bookkeeping uses the triple; physical quantities must depend only on its relevant plaquette values.

$\|\cdot\|_{\sharp,s}, \|\cdot\|_{\flat,s}$**:** $\|f\|_{\sharp,s} = \max(\|f\|_\infty, c_M L^{-s} \operatorname{Lip} f)$, where c_M absorbs the fixed block constant. The source-triple norm is $|c| + \sum_i |z_i| \|f_i\|_{\sharp,s}$. The index zero is usually omitted. $\mathcal{W}_r^{(s)}$ is the corresponding open source ball on each slice.

$h_y^{(j)}, \varphi_j, \zeta_j$**:** $h_y^{(j)}$ is a partition over base points, with its anchor support and normalization stated in the source domain section. φ_j is a fixed coupling profile. The running shift $\zeta_j(p; w)$ is defined there by the exact counterterm identity; it is not an additional freely chosen bare coupling.

$\varrho, \rho, \alpha_{0,j}, \alpha_{1,j}$**:** ϱ and ρ are specified source or coupling-disc radii in their respective statements. $\alpha_{0,j}, \alpha_{1,j}$ are radii of complex field domains. A radius used for coefficient extraction cannot silently be replaced by a smaller running radius.

$p_0, \check{R}_j, \varepsilon_j, \delta_j$**:** $p_0(g) = A_0(\log g^{-2})^{\mathfrak{p}_0}$ is a deposit scale. $\check{R}_j$ is the specified localization and waiting scale, rounded to a power of L. The curvature and fluctuation tests use the frame thresholds ε_j, δ_j defined in the presentation section; they are fixed throughout a source family.

Local families, histories, and analytic requirements.

$E^{(j)}, R^{(j)}, B^{(j)}$**:** These are, respectively, regular local activities, small remainder activities, and boundary activities. A pointed family also has a base-point index. Its companion is a specified family on the infinite lattice; a finite-torus total alone does not define it.

$\mathsf{b}(E), e_j, \mathcal{E}$**:** $\mathsf{b}(E)$ is the scalar marginal-coefficient extraction defined for an actual regular pointed member by (PH18), under Requirement B2's conditional hypotheses. Lowercase $e_j(X; w)$ denotes a localized source energy, and $\mathcal{E}(w)$ their sum. The vacuum constant E without a domain index is a different scalar normalization.

$d_j(X), \kappa, \kappa_{\mathcal{D}}$**:** $d_j(X)$ measures the size or spanning length of the localization domain in its level's cubes. κ is a decay exponent; $\kappa_{\mathcal{D}}$ is a domain-counting exponent. Mixed derivatives spanning separated source supports can be summed only when the declared decay beats that entropy.

Ω_j, Λ_j, Z_j**:** These are the nested regular region, its buffered inner part, and its large-field complement in a history. Λ_j^+ may also include islands left by the operation. Exact inclusions, buffers, tower maps and their domains belong to the declared geometric data; the names alone impose no such conditions.

$h, \underline{h}, \omega$**:** h records all outcomes of a history. $\underline{h}$ records its standing data after absorbed inner outcomes are summed out. ω often records a local history inside one component. A measure associated with a standing history need not be the law of a predetermined block variable after a healing operation.

$\Xi_\omega, \mu_\omega, \Phi_\omega, \Psi_\omega$**:** Ξ_ω is a stored integration space and μ_ω its positive reference measure. Φ_ω is the exponent depending on stored and external variables; Ψ_ω is an additional source

exponent. Their suprema and oscillations are taken on the stated domains under the mass integral when that is what the induction requires.

$\Theta(h), q_a, \Sigma_k$**:** $\Theta(h)$ is a prepaid history weight for deposits and space-time volume. $q_a = e^{-c_a}$ is an age-survival rate; here the subscript is not a lattice spacing. Σ_k is the aged load defined in the weighted-domain section. A survival probability belongs in an output bound and does not by itself enlarge a complex input radius.

$\mathfrak{C}_k, \mathcal{U}_j^c$**:** $\mathfrak{C}_k$ denotes a class of complete presentations and $\mathcal{U}_j^c$ a complex pair domain. Membership requires all declared structural clauses, maps and estimates. An independent current is a coordinate in the pair, not automatically the current evaluated on its field coordinate. No closure theorem follows from writing these symbols without the formation data.

$\mathcal{G}_z, \mathcal{K}_z, A_z$**:** $\mathcal{G}_z$ is one member's complete pointed pair total at base point z, and $\mathcal{K}_z$ its Hessian after composition with the actual free response. It need not be a convolution. A_z is that base point's midpoint tent action in (PH20); its extractor is one. The identity (PH23) uses the coefficient of that same member. The full spatial array may choose different members at different sites.

$\rho, \sigma, a, a_{\mathrm{out}}$ **in the window proof:** Here $\rho = \max(1, a_{\mathrm{out}}/a)$ is a ratio of field radii and $\sigma = L^{-(n-j)}$ a scale ratio; this ρ is not a Gibbs state. The local Hölder exponent called α in that proof is distinct from an analyticity radius $\alpha_{i,j}$. The ordinary age bound $\rho^2 \le C(1+n-j)$ gives the finite young-age fallback. Actual whole-core membership is still required.

OSF and OH: Section 33 treats the given regular members, defects, remainders and reserved histories. Section 37 treats a sum of complex label measures $\tau_S(w)$, positive at zero. Its site count N is unrelated to the gauge-group parameter $N = 2$. Its weighted label sum and all-small-field action bound are inputs to its stability/floor implication, not supplied by the final sum.

$\mathscr{T}, W_f, \alpha^2$ **in the response lemma:** $\mathscr{T}$ is a holomorphic formation map on a potential-norm ball; W_f is the inserted source potential, of norm $O(\alpha^2\|f\|_\infty)$ on the complex field domain. This potential norm is used in the finite insertion and the response bootstrap of Sections 27 and 30, with the actual analytic room described in Section 29. The ordinary coefficient bound on a bare weight u is a different norm and does not imply this potential bound.

$E(z), E_{\ell,K}, A_K$ **in the transfer lemma:** $E(z)$ is a single cell normalizer common to the states being compared. $E_{\ell,K}$ is a finite-torus vacuum constant; A_K is its hypothetical common companion density in (ET8)–(ET10), which may diverge with K. The companion comparison is one finite-reference route, unnecessary when an infinite reference already shares D1's all-state normalizer.

Limits, phases, and reconstruction.

$\mathfrak{S}, \sigma$**:** $\mathfrak{S}$ is the set of separated test tuples, including the empty tuple. Its span is a vector space; concatenation is used only when the resulting supports are separated. σ is a moment functional, not the connected family itself.

$\mathfrak{V}^1(A), \mathsf{d}(X), \mathfrak{D}_L$**:** $\mathfrak{V}^1(A)$ consists of real compactly supported C^1 functions whose supports lie in A. A separated-tuple chart is a product of these spaces on sets at positive mutual distance. $\mathsf{d}(X)$ is the minimum support distance; an empty support contributes infinite distance and a zero entry makes a multilinear value zero. The dense translation subgroup is $\mathfrak{D}_L^4$, with $\mathfrak{D}_L = \bigcup_K a_K\mathbb{Z}$.

σ_S**, two-state centering:** If $S = (S_n)_{n\ge 2}$ is the connected family, σ_S is the partition sum with all blocks of size at least two, with empty value one and singleton value zero. The

different expression $\rho\prod_i[\mathcal{O}_K(f_i)-\nu(\mathcal{O}_K(f_i))]$ is centered in ν, not necessarily in its integrating state ρ. Its limiting first moment can be $c\int f$; its cumulants of order at least two are those of the integrating state.

$\mathcal{S}_{\neq}, |\varphi|_k$**:** $\mathcal{S}_{\neq}(\mathbb{R}^{4n})$ consists of Schwartz functions flat with all derivatives on every coincidence diagonal. One seminorm is the supremum of $(1+|x|^2)^{k/2}|\partial^\alpha\varphi(x)|$ for $|\alpha|\le k$. This space is different from all of $\mathcal{S}(\mathbb{R}^{4n})$.

GS geometry, $m(X), X^+$**:** At scale a_s, localization domains are indexed by unions of Ma_s cubes. The size $m(X)$ is the least number of cubical tree edges needed to meet all their cubes. The source dependence X^+ lies in a fixed number of neighboring cube layers. The rooted, local ball-root and global weighted counts are (GS1)–(GS3). Their sharper geometry is not implied by the weak abstract (ST6) count.

D2, $k, B(R)$**:** D2 is the graded all-state moment and cumulant requirement derived in Theorem 39.1 under D1 in that actual geometry and G1. One may take $B(R)=C(R+1)^4$ and one fixed integer factorial exponent k. The separation factor is $\max(1, d^{-5n})$, explicitly depending on the number of fields.

H1, $k_{\mathrm{E}}, \kappa_{\mathrm{E}}$**:** H1 is the diagonal-flat growth requirement derived from D2 by Theorem 41.1. The sufficient constants are $k_{\mathrm{E}}=20$ and $\kappa_{\mathrm{E}}=k+32$, with seminorm $|\varphi|_{20n}$. The QF proof includes constrained tensor density at every diagonal, infinity and the positive-time boundary. It does not prove the ordinary analytic source representation D1 or extend tests across diagonals.

UI, PD, $\mathsf{p}, \wp$**:** UI is Theorem 46.1: an insertion into a positive-time gap has a Hilbert operator bound with exponent p independent of the number of outside fields. PD is Theorem 47.1; its analytic L^1 density has exponent $\wp=\mathsf{p}+5$ and compatible padded point operators. These follow from the ordinary repeated-source input, not from graded moment bounds alone.

$G_{\varphi,\psi}, \mu_\varphi, \Gamma_\varphi, L_s$**:** $G_{\varphi,\psi}$ is the two-point kernel after smearing the two transverse variables with real compact labels. For an equal label its coordinate spectral measure is the positive measure μ_φ on $\{E\ge|q|\}$, with $G_\varphi(t,x)=\int e^{-tE+ixq}\,d\mu_\varphi$ and $\Gamma_\varphi(t)=\int e^{-tE}\,d\mu_\varphi<\infty$ for $t>0$. $L_s(E,q)=(E\cosh s+q\sinh s, E\sinh s+q\cosh s)$ is the spectral boost used in the proof. The local letter ν in that proof denotes a diagonal-frame spectral measure, rather than a Gibbs state; its cone support is proved from the identity of measures (TU27).

$\vartheta_*, u(d), c_m, b_n$**:** The ultraviolet theorem fixes $\vartheta_*=1/12000$ and the full positive-shape class (TU9). Its input is (TU10), with $u(d)\to 0$, uniform over that class, positions and approximating states. The class includes the explicit mass condition $\int f\ge\vartheta_* d^4\|f\|_\infty$ for each nonnegative radius-d test, in addition to the Lipschitz and separation bounds. It is not inferred from the Lipschitz condition alone. F2–F3 imply it only with those same quantifiers. The angular coefficients transform as $c_m(R_t\mathbf{y})=e^{4imt}c_m(\mathbf{y})$. b_n is the largest absolute nonzero nonconstant frequency at order n, with value zero when there is none. Polygon divisibility and convexity give $b_{2k}=kb_2$; the ultraviolet argument gives $b_2=0$.

G2 and its scope: G2 is now the derived scalar rotation requirement supplied by UI–PD–AK–UV–HC under D1 in its actual rooted geometry, G1, the exact cube cone and the stated ultraviolet law. It applies to the periodic-state limits and canonical components carrying that interface. Arbitrary growing cubic-torus limits require their own mirror/cone justification. Typed Ward and matched-cutoff estimates still belong to the stronger retained routes that use them.

$\mathcal{G}^{\mathrm{per}}, \mathcal{G}^{\mathrm{tor}}$**:** At fixed spacing and coupling the first is the union of Wilson DLR classes with finite-index translation period groups; the second consists of weak limits of periodically lifted torus measures as volume tends to infinity. $\mathcal{G}^{\mathrm{ex}}$ denotes the translation-ergodic members of the periodic class, not unrestricted DLR extremality. A periodic lift before

this limit is not asserted to be an infinite-volume DLR measure. These classes, growing-torus limits and conditionally tuned full sequences have distinct definitions in Section 40.

$f^{(\ell)}$**, the common periodization chart:** If the union of all supports under consideration lies in one Euclidean ball of radius $R < \ell/4$, periodization is defined by $f^{(\ell)} \circ \pi_\ell = \sum_{k\in\mathbb{Z}^4} f(\cdot+\ell k)$. On this common chart it preserves the relevant pairwise support distances and norms. Independent small balls placed arbitrarily in $\mathbb{R}^4$ do not give this conclusion modulo ℓ. Their separate radius bound is the different, position-uniform convention in the moment estimates.

$\mathcal{G}_j, \mathcal{E}_j, D_j$**:** $\mathcal{G}_j$ is the compact translation-invariant DLR class at fixed spacing and coupling. $\mathcal{E}_j$ is its ergodic, equivalently class-extreme, subset. D_j is the decomposition law obtained by conditioning the centering state on the invariant sigma-field. Full DLR extremality without invariance is not identified with this notion here.

$\mathcal{I}, \mathrm{Pl}_\nu, \mathcal{N}_{\nu,r}$**:** $\mathcal{I}$ is the full translation-invariant sigma-field. $\mathrm{Pl}_\nu(A, B)$ is the inner product of their conditional expectations on that field. $\mathcal{N}_{\nu,r}$ is the reflected size in the upstream lattice law G3; it replaces divergent supremum norms of cutoff field products in the phase limit.

$\mathsf{X}, \lambda, \sigma_\xi$**:** X is the compact full-support phase-coordinate space constructed from the weak limits of lattice evaluation laws, λ its probability measure, and σ_ξ its continuous component evaluations. These are conclusions of the conditional phase theorem, not assumed continuum data.

$\theta, \mathcal{H}, H, P, \Omega$**:** θ is the specified Euclidean reflection. $\mathcal{H}$ is the completed quotient of its positive form. Ω is the class of the empty tuple, H the nonnegative time generator, and P the spatial momentum tuple when the corresponding continuous groups have been constructed. P_0 projects onto $\ker H$.

$\mathfrak{U}, \mathfrak{B}, W_4$**:** $\mathfrak{U}$ denotes a rotation defect or limiting flux in the Ward argument; $\mathfrak{B}$ is its insertion. W_4 is the group of signed coordinate permutations. Its invariance alone does not imply invariance under O(4).

Appendix C. Index of arguments and proof obligations

These code families organize the proof across the book. They locate retained statements, detailed proofs of referenced results, and the statements that we must still prove. Their exact forms are given at the cited statement and in the dependency catalogue. Outlined supporting arguments remain distinct from those obligations.

Named statements in the proof outline. 0, 1, 1a.1.1, 1a.1.10, 1a.1.11, 1a.1.12, 1a.1.13, 1a.1.14, 1a.1.2, 1a.1.3, 1a.1.4, 1a.1.5, 1a.1.6, 1a.1.7, 1a.1.9, 1a.2.1, 1a.2.10, 1a.2.2, 1a.2.3, 1a.2.4, 1a.2.7, 1a.2.8, 1a.2.9, 1a.3.1, 1a.3.11, 1a.3.12, 1a.3.13, 1a.3.3, 1a.3.4, 1a.3.6, 1a.3.7, 1a.3.8, 1a.3.9, 2, 2.1, 2.11, 2.2, 2.3, 2.4, 2.5, 2.6, 2.7, 3, 3.1, 3.2, 3.3, 3.4, 3.5, 3.6, 3.7, 3.8, 3.9, 4, 4.1, 4.13, 4.14, 4.15, 4.2, 4.3, 4.4, 4.5, 4.6, 4.7, 5, 6, 7.1, 7.2, 7.3, 7.4.

Detailed proofs of referenced statements. 4.10: chessboard domination for arbitrary periodic states, Section 38.
4.12: uniform one-point differences, Section 38.
4.17: uniform one-field Hilbert insertion, Section 46.
4.18: analytic insertion density and the plate-bound implication under its stated hypotheses, Section 47.

Statements still to be proved, by package. The ordinary history and source tasks 2.8, 2.10, 2.13 and 4.11 have proof outlines in Sections 32 and 36. The extraction and transfer tasks 7.5,

7.6 and 7a.1.8 are developed in Sections 43.1–43.3 and 44.3. The stronger aged comparison 2.12 remains separate. The one-loop calculation is in Section 35.1.

In Package E, the code 8 refers both to Bałaban's Theorem 8, which he states without proof and which we must prove, and to Proposition 8 of Commun. Math. Phys. 102 (1985) 277–309, which is cited. In Package A–H, the code 1.2 is Proposition 1.2 of Commun. Math. Phys. 95 (1984) 17–40, which is cited. The following list retains the task identifiers; the preceding paragraph gives the locations of the supplied outlines.
Package A.. 1a.3.2, 1a.3.5.
Package C.. 2.10, 2.12, 2.13, 2.8.
Package D.. 4.11.
Package E.. 8.
Package F.. 7.5, 7.6, 7a.1.8.
Package G.. 9.1, 9.10, 9.11, 9.3, 9.7.
Package A–H.. 1.2, 1.6.

Formula and statement navigation. The first component of a hand-numbered code records its mathematical theme: 1a–1c backgrounds and propagators; 1d–1e local interactions; 2 histories; 3 flow; 4 sources; 5 limits; 6 Euclidean structure; 7 short distance; 8 comparison; 9 rotations; 10 phases and reconstruction. These codes retain their meaning when a theme occupies separate book sections. A grouped reference is read with the hypotheses and scope of its stated result.

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